Theory and Applications of Transport in Porous Media

Alexander H.-D. Cheng

Poroelasticity

INTERPORE

Springer

Theory and Applications of Transport in Porous Media

Volume 27

Series editor:
S. Majid Hassanizadeh, Department of Earth Sciences, Utrecht University, The Netherlands

Founding series editor:
Jacob Bear

More information about this series at http://www.springer.com/series/6612

Alexander H.-D. Cheng

Poroelasticity

Springer

Alexander H.-D. Cheng
University of Mississippi
Oxford, Mississippi, USA

ISSN 0924-6118 ISSN 2213-6940 (electronic)
Theory and Applications of Transport in Porous Media
ISBN 978-3-319-25200-1 ISBN 978-3-319-25202-5 (eBook)
DOI 10.1007/978-3-319-25202-5

Library of Congress Control Number: 2016937748

© Springer International Publishing Switzerland 2016
This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, express or implied, with respect to the material contained herein or for any errors or omissions that may have been made.

Printed on acid-free paper

This Springer imprint is published by Springer Nature
The registered company is Springer International Publishing AG Switzerland

To Nady
(Mme M. A. Biot)
for her devotion to
Dr. M. A. Biot



With Mme Biot in her Bruxelles apartment (2008)



During a visit to Leuven in 2008, I took the photo of the street scene across from the famous Library of Katholieke Universiteit Leuven without knowing that I had accidentally captured the Biot family house from 1924 to 1940 (see arrow). (Compare with Fig. F.1(b))

Preface

Porous materials are found in nature as inanimate objects, such as soils and rocks, in living bodies, such as plants, human flesh, and bones, and as man-made materials, such as polyurethane foams and nanofiltration filters. The pores, or voids, are distributed all over the space the porous body occupies, as cavities, fractures, fissures, vugs, capillary channels, and other interstitial and intergranular spaces. The pores are many, in various shapes and sizes, and are randomly oriented and distributed. For modeling purposes, it is either undesirable, or impossible, to describe their exact geometry and location. As a result, only the averaged, or "homogenized," material properties and mechanical responses are considered.

When the interstitial space is impregnated with one or more fluids, the mechanical interaction between the solid and the fluid can be interesting and complex, particularly in time-dependent behaviors, both in the quasi-static and the dynamic range. Time-dependent behaviors can manifest in many ways, as the macroscopic porous medium flow in interconnected channels, as grain scale squeeze film lubrication flow, as intergranular frictional sliding of particles, or as solid intrinsic viscoelasticity. In addition to the mechanical forces, a number of other physico-chemical factors can come into play, such as thermal, electrical, electromagnetic, and chemical (osmotic) forces.

The modeling of a full range of porous material responses, from quasi-static to dynamic, from linear to nonlinear, and from partial uncoupling to fully coupled, can be complex and unwieldy. The goal of this book is limited. It focuses largely on the linear theories, as in the classical linear elasticity and porous medium flow (Darcy flow); hence the book is titled *poroelasticity*, rather than a more general term of *poromechanics*. Although the word "Introduction" does not appear in the book title, by virtue of the above-stated scope, the book is essentially an *introduction to linear poroelasticity*.

The book contains 12 chapters. In the "Introduction," the reader is presented with a set of basic poroelastic mechanisms, such as drained and undrained responses, effective stress, time scale, etc., explained as intuitive physical-mechanical concepts, without introducing the governing equations and mathematical rigor. It is then

demonstrated how these poroelastic mechanisms manifest in a wide range of physical phenomena in geophysics, geomechanics, acoustics, and biomechanics, such as soil consolidation and land subsidence, slope stability, fault slippage, pumping- and injection-induced seismicity, outburst of coal formation, hydraulic fracturing, water wave and sediment interaction, aquifer response to tidal and barometric loading, viscous damping and squeeze film lubrication of articular cartilage, thermal fracturing and burst of cylinder, swelling of clay and shale, and seismoelectric and seismoelectromagnetic phenomena.

For a quick construction of the theory for application, Chap. 2 lays out a continuum theory of poroelasticity, in terms of bulk constitutive equations, using a phenomenological approach, as an ad hoc extension of elasticity. Hooke's law-type linear constitutive equations are semiempirical and are presented through simple mathematical logic, without a deep examination of the underlying physics. The constitutive constants are then correlated to the macroscopically observable phenomena and measured responses.

Chapter 3 examines the micromechanics of porous materials, which isolates the mechanical properties of the solid and the fluid phases, as well as the pore space. Association is made with the phenomenological theory to rationalize its approach and to bring physical insights into Hooke's law coefficients as composite material constants incorporating multiple mechanisms. Special and idealized models are introduced, which provide additional insight into these mechanisms, and lead to bounds for material constants. Laboratory setups that can measure not only the bulk material but also the micromechanical constants are demonstrated.

Chapter 4 gives an even more rigorous examination of the poroelasticity theory, by applying the variational energy minimization principle to the composite material averaging theory. A product of this analysis is a set of intrinsic material constants directly associated with the three fundamental deformation modes of a porous solid: a geometrically similar and porosity-preserving volumetric deformation, a solid shape- and volume-preserving particle rearrangement induced pore space change, and a solid volume-preserving shape change induced pore change. This model gives a clear interpretation of the often used microhomogeneity and microisotropy (ideal porous medium) assumptions. Links between the intrinsic and the micromechanical constants are established. Limiting material behaviors are examined using the asymptotic values of these constants. Laboratory technique that can measure these intrinsic material constants is discussed.

Chapter 5 gives an examination of the material anisotropy, at both the bulk continuum and the micromechanical level. The generalized models are reduced to various models with special symmetry, such as orthotropy, and transverse isotropy. Under the ideal porous medium assumption, the micromechanical analysis shows that not all the bulk continuum general anisotropic material constants are independent to each other. The introduction of one micromechanical constant of solid bulk modulus causes the reduction of six bulk continuum constants, bringing a constraint to the continuum model that cannot be obtained at the bulk continuum level.

All the theories presented so far are constitutive laws that relate the externally applied forces to the deformation of a body. Chapter 6 lays out other physical laws that are required for the modeling of poroelasticity, including the force equilibrium, the fluid mass conservation, and Darcy's law for fluid flow. These laws are assembled into a set of governing equations for mathematical solution. Various physical contact conditions leading to the mathematical boundary conditions are discussed. Provided with a proper set of initial and boundary conditions, a well-posed initial and boundary value problem can be formed, which guarantees the existence of a unique mathematical solution. For the convenience of mathematical solution, several methodologies that reduce the physical variables, such as displacements, stresses, and pore pressure, to a set of "potentials," known as stress and displacement functions, are presented.

Chapter 7 derives a set of analytical solutions, ranging from the one-dimensional geometry to axial and spherical symmetry and from plane strain to generalized plane strain conditions. These simplified problem geometries allow the analytical solutions to be found. A range of problems, many with important practical applications, including soil consolidation, seabed deformation and mass exchange in response to water wave, stability of excavated borehole, in situ stress determination, drill core retrieval, burst of hollow cylinder, and other problems, are investigated.

Chapter 8 presents the singular solutions due to concentrated loads, such as point force, dislocation, fluid source and dipole, etc., in infinite space, also known as fundamental solutions or free space Green's functions. A closely related subject, the singular integral equations of various forms, particularly those that represent the solution of the governing equations, such as the Somigliana, the stress discontinuity, and the displacement discontinuity integral equations, is also presented.

The dynamic problems are addressed in Chap. 9 as poroelastodynamics. The variational Lagrangian formulation is utilized to interpret the inertia and added mass effects. The frequency-dependent dynamic permeability is examined based on conceptual models. Governing equations are laid out in the time and the frequency domain. The wave propagation phenomenon and the phase velocities and attenuation are analyzed, based on the Biot and the extended models, such as the squirt flow model. One-dimensional analytical solutions and solutions of plane wave reflection and refraction on free surface and water and porous medium interface are examined. Fundamental solutions and singular integral equation representation are also derived.

Chapter 10 presents the viscoelastic effects. A brief review of viscoelasticity and the correspondence principle set the stage for the generalization to poroviscoelasticity. In a physically consistent manner, the spring-dashpot models are applied to the micromechanical and intrinsic material constants, which lead to the non-monotonic time evolution behavior of some bulk continuum material constants. Borehole and cylinder problems are examined based on the poroviscoelastic constitutive behaviors. The theory is also extended to poroviscoelastodynamics.

In Chap. 11 the thermal effect is taken into consideration and the theory is extended to porothermoelasticity. The constitutive and governing equations are

derived based on the rigor of the first and second laws of thermodynamics. The irreversible processes of nonequilibrium thermodynamics and transport laws are constructed based on Onsager's reciprocal principle. The fully coupled and partially coupled porothermoelasticity theories are presented. Analytical solutions of one-dimensional geometry, and with axial and spherical symmetry, as well as the fundamental solutions, are derived.

Finally, in Chap. 12 the chemical effect is incorporated to present the theory of porochemoelasticity. The electrochemical effect is introduced as Gibbs chemical potential and free energy. The porous material is considered in three phases, the solid, the fluid, and the bound fluid absorbed and adsorbed to the solid. The chemical effects of the two fluid phases are separately modeled; mechanically, however, the bound fluid is considered as a part of the solid phase. Onsager's reciprocal principle is again used to construct the transport laws. Analytical solutions are provided.

The book also contains extensive appendices, which include a biography section giving biographical sketches of pioneers on whose shoulders the theory of poroelasticity is built.

Oxford, Mississippi, USA Alexander H.-D. Cheng
2015

Acknowledgments

This acknowledgment is a personal story of how I stumbled into the subject of poroelasticity, and the journey of learning, dragged along all the way by friends and collaborators.

The year was 1980, at the end of my second year as a doctoral student at Cornell University. I was supported as a research assistant by a project on boundary integral equation method (or more commonly known as the boundary element method), so I was obliged to do my dissertation on that subject. I completed my assigned task of solving two problems, one on locating the unknown free surface of a sluice gate and a spillway flow and the other on the moving interface between a freshwater and a saltwater in porous medium, both of which were later published [15, 28]. I was pondering what would be the next step.

I went to Professor James Liggett, my advisor, and told him that I could extend the work to solve some problems in three-dimensional geometry, but I hardly thought it was fundamental enough to be worthy of a doctoral dissertation. Prof. Liggett said that he had just submitted a proposal with Prof. Anthony Ingraffea on hydraulic fracturing and asked whether I would like to have a look. I had never heard of hydraulic fracturing, but was willing to give it a try. He handed me the proposal.

Being a hydraulics and hydrology major in civil engineering, I had few graduate courses in solid mechanics. To understand what hydraulic fracturing was about, I began auditing Ingraffea's fracture mechanics course. But largely I started reading from the reference list of the proposal. I was led to Cleary's 1978 paper [16] on applying moving singularity to solve propagating hydraulic fracture. It brought me to Rice and Cleary's 1976 work [34] on reformulating poroelasticity, which in turn led me to the "source," M. A. Biot's pioneering work. I was busy in the library photocopying a bunch of Biot papers [9–12] and gulped as much as I could.

I also had never heard of poroelasticity. But I was deeply intrigued by the papers. I went to Prof. Liggett said that I would like to do my dissertation on the formulation and numerical implementation of the boundary integral equation method for poroelasticity. He said that it was fine, but he still needed me to do a hydraulic fracturing problem. A compromise came at a later time. After I completed

the formulation and solved a soil consolidation problem [13], I also solved a moving hydraulic fracture problem [14] using the same methodology, for my dissertation.

Prof. Liggett pretty much bid me good luck, and I was on my own. I was probably more foolish than courageous, for I lacked the training and skill. First I needed to derive the integral equations for poroelasticity as an extension to elasticity. Next I needed to derive the fundamental solutions. Only then I could apply my original skill of numerical modeling to write a computer program to solve it. I ran into one stumbling block after another, but somehow I always overcame. I was lucky to be able to complete the dissertation within 1 year and graduate on time, before Prof. Liggett had to leave for Australia for his sabbatical. I stayed one extra year at Cornell as an acting assistant professor from 1981 to 1982.

The next turning point of my journey came in 1985, when I was moving from Columbia University as an assistant professor to the University of Delaware, accepting an associate professor appointment. Upon arriving at my new office, the first thing I saw was a note on my desk. Someone named Emmanuel Detournay tried to contact me at Columbia as I was leaving, and the message was forwarded. I had no idea who he was. I called him back at Dowell-Schlumberger at Tulsa. That started a fruitful collaboration that lasted over a decade.

Emmanuel at that time was asked by his supervisor Jean-Claude Roegiers to look into the application of poroelasticity to borehole and hydraulic fracturing problems. Of so many people who worked on these problems, I didn't know why he chose me. During a more intensive period, between 1987 and 1991, we published seven journal papers, including a highly cited on the analytical solution of poroelastic borehole problems [22]. Other papers were on poroelastic fracture [23], hydraulic fracturing [25, 35], displacement discontinuity method [21], and in situ stress determination [26]. We also wrote a chapter on "Fundamentals of Poroelasticity" for a *Comprehensive Rock Engineering* handbook [24]. It was supposed to pave the way for a coauthored book. My apology here that more than 20 years later, I single authored this book.

Emmanuel pushed me to learn poroelasticity from a physical point of view. Coming from my numerical method background, poroelasticity was a set of elegant mathematical equations. I derived integral equations, found fundamental solutions, and solved boundary value problems with great satisfaction. Emmanuel always engaged me for a debate at the physics level. I had to force myself to learn and to think in order to catch up with him. I was indebted to him for bringing me up to speed with the physics.

My next stage of collaboration was with Younane Abousleiman, my former Ph.D. student, working at the University of Oklahoma. He was not intimidated by my initial reluctance, and we got close and started an intensive period of collaboration on research and application of poroelasticity. We extended the borehole solution to allow its inclination with respect to the in situ stress field [18] and applied it to mud weight design for wellbore stability [17], which eventually became an industry software [5], used the micro and mini hydraulic fracturing pressure transient to determine formation permeability [2], revisited and extended the classical Mandel

problem [1], presented the micromechanically consistent poroviscoelasticity model [3, 4], and applied the finite element method [19, 20].

While at the University of Delaware, I was acquainted with Mohsen Badiey, a colleague at the College of Marine Studies. He was well connected with the ocean acoustics program at the Office of Naval Research, which provided the needed funding. After the Cold War, the focus of ocean acoustics shifted from deep water to shallow water, and the interaction with the porous sea bottom became important. Through several projects, we worked on ocean and seabed acoustics. In a few years we published a sequence of papers in JASA [6–8, 33].

Sergei Lopatnikov and I started collaborating around 1997 on magnetoporoelasticity [29]. At my invitation for a visit in the USA, he boarded an airplane in Moscow. The plane sat on the runway for hours for unknown reason, and then all passengers were discharged and sent home. By the time Sergei was able to rebook a flight, he was barred from entering the USA due to his pending immigration application. Two years later, I got a call from Sergei. He was in a Greyhound station in Miami. Sergei had successfully immigrated and was boarding a bus to visit me in Delaware. We started a fruitful period of collaboration. Sergei was intent on educating me on the "correct" formulation of poroelasticity. We debated a great deal. Colleagues passing by often heard us shouting at each other. He finally educated me and we developed the theory [30, 31] that formed the basis of Chap. 4. We also wrote a short biography honoring Sergei's hero, Yacov Il'ich Frenkel [32], who was the first to formulate the dynamic poroelasticity theory and predicted the second compressional wave and explained the seismoelectromagnetic phenomenon [27].

I was lucky to have the opportunity to honor my own hero, Maurice A. Biot. In 1991 I took a sabbatical leave at Schlumberger Cambridge Research in Cambridge, UK, hosted by Emmanuel. One day over lunch, we were talking about organizing a conference on poroelasticity. I said that I greatly admired the work of Biot, and we should organize a conference to honor him. However, I had no information on Biot. It had been 50 years since the 1941 paper on poroelasticity [9] was authored. Was he still alive? It so happened that Emmanuel, of the same country origin (Belgium) as Biot, did know something—Biot had passed away a few years ago, in 1985.

In 1996, as a precursor of the dedicated conference, a Poroelasticity Symposium was organized under the auspices of the ASCE Engineering Mechanics Conference at Fort Lauderdale, Florida. The idea of contacting Biot's widow to seek authorization of using Biot's name arose among Emmanuel, Younane, and I. Younane was the first to seek out Mme Biot and visited her in Bruxelles. With Emmanuel's introduction, the group expanded to include Oliver Coussy at Laboratoire Central des Ponts et Chaussées, France, and Jean-François Thimus at Université catholique de Louvain, Belgium. The five of us had organizational meetings in Delaware and in Louvain-la-Neuve, Belgium, with a visit to Mme Biot. It was during these times the word *poromechanics* was coined.

With Mme Biot's blessing, the Biot Conference on Poromechanics was convened at Université catholique de Louvain, in 1998. Subsequent conferences were held every 4 years in Université Joseph Fourier, France (2002), University of Oklahoma

(2005, Biot Centennial), Columbia University (2009), and Technische Universität Wien, Austria (2013). A future one is scheduled at Ecole des Ponts ParisTech, France (2017).

Since 1995, I joined as a member of the Engineering Mechanics Division of American Society of Civil Engineers (ASCE), which later became Engineering Mechanics Institute of ASCE. In 2002, I formed a Poromechanics Committee within the organization and served as its first chair. In 2002, a donation was made by Mme Biot to set up an endowment for a Maurice A. Biot Medal in ASCE to award "an individual who has made outstanding research contributions to the mechanics of porous materials." Oliver Coussy became the first recipient of the medal in 2003.

It was tragic that Olivier passed away in 2010 at the height of his career, at age 56. It was at the memorial symposium honoring Coussy in 2011 at École des Ponts ParisTech, France, that I reflected upon myself: time and tide wait for no man; it is high time to start this long-planned book. So I embarked on this endeavor that summer.

It was not easy to write a book that kept on growing in size while carrying on full academic and administrative duties. I had to abandon weekend yard work and occasional housework for a few years, relying on my dear wife Daisy for the toil. I thank her for her understanding, as I had to admit that I enjoyed the toil of writing this book more than yard work.

So I had stumbled into the field by accident and was dragged through by friends and collaborators. They motivated, pushed, and educated me. I would like to use this opportunity to thank them for their interest and patience in working with me. Without them, this book would not exist.

Alexander H.-D. Cheng

References

1. Abousleiman Y, Cheng AHD, Cui L, Detournay E, Roegiers JC (1996) Mandel's problem revisited. Géotechnique 46(2):187–195
2. Abousleiman Y, Cheng AHD, Gu H (1994) Formation permeability determination by micro or mini hydraulic fracturing. J Energy Resour Technol ASME 116(2):104–114
3. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1993) A micromechanically consistent poroviscoelasticity theory for rock mechanics applications. Int J Rock Mech Mining Sci 30(7):1177–1180
4. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1996) Poroviscoelastic analysis of borehole and cylinder problems. Acta Mech 119(1–4):199–219
5. Abousleiman Y, Cui L, Ekbote S, Zaman M, Roegiers JC, Cheng AHD (1997) Applications of time-dependent pseudo-3D stress analysis in evaluating wellbore stability. Int J Rock Mech Mining Sci 34(3–4):paper no. 1
6. Badiey M, Cheng AHD, Jaya I (1996) Deterministic and stochastic analyses of acoustic plane-wave reflection from inhomogeneous porous seafloor. J Acoust Soc Am 99(2):903–913
7. Badiey M, Cheng AHD, Mu YK (1998) From geology to geoacoustics—evaluation of biot-stoll sound speed and attenuation for shallow water acoustics. J Acoust Soc Am 103(1):309–320

8. Badiey M, Jaya I, Cheng AHD (1994) A shallow water acoustic/geoacoustic experiment near the New Jersey Atlantic Generating Station site. J Acoust Soc Am 96(6):3593–3604
9. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2):155–164
10. Biot MA (1955) Theory of elasticity and consolidation for a porous anisotropic solid. J Appl Phys 26(2):182–185
11. Biot MA (1956) General solutions of the equations of elasticity and consolidation for a porous material. J Appl Mech ASME 78:91–96
12. Biot MA, Willis DG (1957) The elastic coefficients of the theory of consolidation. J Appl Mech ASME 24:594–601
13. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to soil consolidation. Int J Numer Methods Eng 20(2):255–278
14. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to fracture propagation. Int J Numer Methods Eng 20(2):279–296
15. Cheng AHD, Liggett JA, Liu PLF (1981) Boundary calculations of sluice and spillway flows. J Hydraul Div ASCE 107(10):1163–1178
16. Cleary MP (1978) Moving singularities in elasto-diffusive solids with applications to fracture propagation. Int J Solids Struct 14(2):81–97
17. Cui L, Abousleiman Y, Cheng AHD, Roegiers JC (1999) Time-dependent failure analysis of inclined boreholes in fluid-saturated formations. J Energy Resour Technol ASME 121(1):31–39
18. Cui L, Cheng AHD, Abousleiman Y (1997) Poroelastic solution for an inclined borehole. J Appl Mech ASME 64(1):32–38
19. Cui L, Cheng AHD, Kaliakin VN, Abousleiman Y, Roegiers JC (1996) Finite element analyses of anisotropic poroelasticity: a generalized Mandel's problem and an inclined borehole problem. Int J Numer Anal Methods Geomech 20(6):381–401
20. Cui L, Kaliakin VN, Abousleiman Y, Cheng AHD (1997) Finite element formulation and application of poroelastic generalized plane strain problems. Int J Rock Mech Mining Sci 34(6):953–962
21. Detournay E, Cheng AHD (1987) Poroelastic solution of a plane-strain point displacement discontinuity. J Appl Mech ASME 54(4):783–787
22. Detournay E, Cheng AHD (1988) Poroelastic response of a borehole in a non-hydrostatic stress field. Int J Rock Mech Mining Sci 25(3):171–182
23. Detournay E, Cheng AHD (1991) Plane strain analysis of a stationary hydraulic fracture in a poroelastic medium. Int J Solids Struct 27(13):1645–1662
24. Detournay E, Cheng AHD (1993) Fundamentals of poroelasticity. In: Fairhurst C (ed) Comprehensive rock engineering: principles, practice and projects, vol II, analysis and design method. Pergamon, Oxford/New York, pp 113–171
25. Detournay E, Cheng AHD, McLennan JD (1990) A poroelastic PKN hydraulic fracture model based on an explicit moving mesh algorithm. J Energy Resour Technol ASME 112(4):224–230
26. Detournay E, Cheng AHD, Roegiers JC, McLennan JD (1989) Poroelasticity considerations in *in-situ* stress determination by hydraulic fracturing. Int J Rock Mech Mining Sci 26(6):507–513
27. Frenkel J (1944) On the theory of seismic and seismoelectric phenomena in moist soil. J Phys USSR 13(4):230–241
28. Liu PLF, Cheng AHD, Liggett JA, Lee JH (1981) Boundary integral equation solutions to moving interface between two fluids in porous media. Water Resour Res 17(5):1445–1452
29. Lopatnikov SL, Cheng AHD (1998) A thermodynamically consistent formulation of magneto-poroelasticity. Int J Solids Struct 35(34–35):4637–4657
30. Lopatnikov SL, Cheng AHD (2002) Variational formulation of fluid infiltrated porous material in thermal and mechanical equilibrium. Mech Mater 34(11):685–704
31. Lopatnikov SL, Cheng AHD (2004) Macroscopic Lagrangian formulation of poroelasticity with porosity dynamics. J Mech Phys Solids 52(12):2801–2839

32. Lopatnikov SL, Cheng AHD (2005) If you ask a physicist from any country: a tribute to Yacov Il'ich Frenkel. J Eng Mech ASCE 131(9):875–878
33. Mu YK, Badiey M, Cheng AHD (1999) Parameter uncertainty analysis on acoustic response in fluid filled poroelastic media. J Acoust Soc Am 106(1):151–163
34. Rice JR, Cleary MP (1976) Some basic stress diffusion solutions for fluid-saturated elastic porous media with compressible constituents. Rev Geophys 14(2):227–241
35. Vandamme L, Detournay E, Cheng AHD (1989) A two-dimensional poroelastic displacement discontinuity method for hydraulic fracture simulation. Int J Numer Anal Methods Geomech 13(2):215–224

Contents

1 Introduction 1
- 1.1 Porous Material 2
- 1.2 Physical Mechanism 4
  - 1.2.1 Drained and Undrained Responses 5
  - 1.2.2 Time and Length Scale 9
  - 1.2.3 Skempton Pore Pressure Effect 10
  - 1.2.4 Effective Stress for Volumetric Deformation 12
  - 1.2.5 Effective Stress for Pore Collapse 16
  - 1.2.6 Fluid Storage 17
  - 1.2.7 Thermoelasticity Analogy 18
  - 1.2.8 Coupled Versus Uncoupled Diffusion 21
- 1.3 Poroelastic Phenomena 22
  - 1.3.1 Borehole Failure 22
  - 1.3.2 Mandel-Cryer Effect 26
  - 1.3.3 Noordbergum Effect 27
  - 1.3.4 Land Subsidence 29
  - 1.3.5 Slope Stability and Fault Slippage 30
  - 1.3.6 Fluid Induced Seismicity 32
  - 1.3.7 Outburst of Coal 33
  - 1.3.8 Hydraulic Fracturing 35
  - 1.3.9 Water Wave and Seabed Interaction 38
  - 1.3.10 Tidal and Barometric Efficiency 40
  - 1.3.11 Biomechanics 42
  - 1.3.12 Poroviscoelasticity and Anelastic Strain Recovery 43
  - 1.3.13 Porothermoelasticity and Thermal Fracturing 44
  - 1.3.14 Poroelastodynamics and Seismoelectric Effect 45
  - 1.3.15 Swelling of Clay and Shale 47
  - 1.3.16 Nanoporous Material 47
- References 48

2 Constitutive Equation 61
2.1 Physical Versus Phenomenological Approach 62
2.2 Stress and Strain of Porous Medium 64
2.2.1 Stress 64
2.2.2 Strain 66
2.3 Poroelastic Constitutive Equation 67
2.3.1 Isotropic Elastic Material 67
2.3.2 Isotropic Poroelastic Material 69
2.3.3 Reciprocal Work Theorem 70
2.3.4 Stress-Strain Relation 73
2.3.5 Strain-Stress Relation 73
2.4 Bulk Material Constant 75
2.4.1 Drained and Undrained Constant 75
2.4.2 Effective Stress Coefficient 77
2.4.3 Pore Pressure Coefficient 78
2.4.4 Storage Coefficient 79
References 81

3 Micromechanics 83
3.1 Micromechanical Analysis 84
3.1.1 Solid and Pore Volumetric Strain 84
3.1.2 Fluid Volumetric Strain 87
3.1.3 Link Among Material Constants 88
3.2 Ideal Porous Medium 89
3.3 Effective Modulus 92
3.3.1 Mackenzie Model 92
3.3.2 Walsh Model 93
3.3.3 Budiansky and O'Connell Model 94
3.3.4 Bounds on Material Constants 95
3.4 Nonlinear Model 97
3.4.1 Effective Stress Dependent Pore Compressibility 98
3.4.2 Compaction Induced Permeability Change 100
3.5 Laboratory Test 102
3.5.1 Drained Test 103
3.5.2 Undrained Test 104
3.5.3 Unjacketed Test 104
3.6 Table of Poroelastic Constants 105
References 109

4 Variational Energy Formulation 113
4.1 Internal and External Stress and Strain 114
4.1.1 Porosity 115
4.1.2 Volume and Surface Averaging of Elastic Material 117
4.1.3 Volume and Surface Averaging of Porous Material 120
4.1.4 Linkage Between Internal and External Strains 123

4.2 Thermodynamic Principles 125
4.3 Variational Formulation 129
4.3.1 Virtual Work 130
4.3.2 Internal Energy 130
4.3.3 Porosity Equilibrium 132
4.4 Constitutive Equation 133
4.4.1 Linear Material Model 133
4.4.2 Linear Model 135
4.5 Intrinsic Material Constant 137
4.5.1 Effective Solid Bulk Modulus 137
4.5.2 Fundamental Deformation Mode 138
4.5.3 Microisotropy and Microhomogeneity: Ideal Porous Medium 139
4.6 Link with Phenomenological Model 141
4.6.1 Link with Bulk Continuum Model 142
4.6.2 Link with Micromechanics Model 146
4.7 Deviation from Ideal Porous Medium 147
4.8 Limiting Material Properties 151
4.8.1 Ideal Porous Medium 151
4.8.2 Granular Material 152
4.8.3 Soil Mechanics Model: Saturated 154
4.8.4 Soil Mechanics Model: Nearly Saturated 155
4.8.5 Highly Compressible Solid 156
4.8.6 Highly Compressible Fluid 156
4.9 Material Stability and Energy Diagram 157
4.10 Semilinear Model 160
4.10.1 Geometric Nonlinearity 161
4.10.2 Structural Nonlinearity 161
4.11 Laboratory Measurement of Intrinsic Constant 163
References 166

5 Anisotropy 171
5.1 Anisotropic Constitutive Equation 171
5.1.1 Elasticity 172
5.1.2 Poroelastic Stress-Strain Relation 173
5.1.3 Poroelastic Strain-Stress Relation 175
5.2 Material Symmetry 176
5.2.1 Orthotropy 177
5.2.2 Transverse Isotropy 179
5.2.3 Isotropy 180
5.3 Micromechanics 181
5.4 Ideal Porous Medium 183
5.5 Example 185
References 186

6 Governing Equation 189
6.1 Darcy's Law 190
6.1.1 Darcy's Empirical Law 190
6.1.2 Homogenization Theory 192
6.1.3 Intrinsic Permeability and Mobility Coefficient 195
6.1.4 Irreversible Thermodynamics Process 196
6.2 Other Physical Laws 199
6.2.1 Mass Conservation 199
6.2.2 Force Equilibrium 201
6.3 Governing Equation 203
6.3.1 Navier-Cauchy Equation 204
6.3.2 Diffusion Equation 205
6.3.3 Compatibility Equation 206
6.3.4 Harmonic Relation 207
6.3.5 Orthotropy 208
6.3.6 Transverse Isotropy 209
6.4 Degenerated Governing Equation 209
6.4.1 Drained and Undrained State 209
6.4.2 Soil Mechanics Model 210
6.4.3 Irrotational Displacement Field 210
6.4.4 Uncoupling of Diffusion Equation 212
6.5 Boundary Value Problem 213
6.5.1 Existence and Uniqueness 214
6.5.2 Boundary Condition 216
6.6 Field Equation 218
6.6.1 Biot Function 218
6.6.2 Biot Decomposition 220
6.6.3 McNamee-Gibson Displacement Function 222
References 226

7 Analytical Solution 229
7.1 Review of Early Work 229
7.2 Uniaxial Strain 233
7.2.1 Isotropy 233
7.2.2 Transverse Isotropy 235
7.3 One-Dimensional Consolidation Problem 236
7.3.1 Terzaghi's Consolidation Problem 236
7.3.2 Loading by Fluid Pressure 242
7.3.3 Variable Rete Loading 244
7.3.4 Harmonic Excitation 246
7.4 Plane Strain 250
7.4.1 Orthotropy 251
7.4.2 Isotropy 252
7.4.3 Volumetric Strain and Rotation Formulation 253

7.5 Generalized Plane Strain 254
7.5.1 Definition of Generalized Plane Strain 254
7.5.2 Pure Shear 257
7.5.3 Warping 257
7.5.4 Torsion 260
7.5.5 Plane Strain 261
7.5.6 Axial Strain 261
7.5.7 Pure Bending 262
7.6 Pure Bending of Plate 264
7.6.1 Bending of Cantilever Plate 265
7.6.2 Buckling of Axially Loaded Plate 272
7.7 Mandel Problem 274
7.8 Water Wave Over Seabed 284
7.9 Spherical Symmetry 295
7.10 Cryer Problem 297
7.11 Spherical Cavity 306
7.11.1 Pressurized Cavity 306
7.11.2 Excavated Cavity 309
7.11.3 Pore Pressure Meter Problem 310
7.12 Axial Symmetry 316
7.13 Cylinder Problem 319
7.13.1 Solid Cylinder 319
7.13.2 Hollow Cylinder 331
7.14 Borehole Problem 336
7.14.1 Plane Strain Borehole Problem 337
7.14.2 Inclined Borehole Problem 350
7.15 Borehole and Cylinder Application Problems 356
7.15.1 Retrieval of Cylindrical Core 356
7.15.2 Excavated Borehole 358
7.15.3 Fluid Extraction and Injection 363
7.15.4 Borehole Breakdown Pressure 365
7.15.5 Borehole Stability Analysis 367
7.16 Moving Load on Half Plane 371
7.17 Plane Strain Half Space and Layered Problem 379
7.17.1 General Solution for Layered Problem 380
7.17.2 Plane Strain Half Space Problem 381
7.18 Axial Symmetry Half Space Problem 385
References 389

8 Fundamental Solution and Integral Equation 397
8.1 Reciprocal Theorem 398
8.1.1 Green's Second Identity 398
8.1.2 Betti-Maxwell Reciprocal Theorem 399
8.1.3 Reciprocal Theorem of Poroelasticity 400

8.2 Somigliana Integral Equation 404
8.2.1 Green's Third Identity 404
8.2.2 Elasticity 405
8.2.3 Poroelasticity 405
8.3 Fredholm Integral Equation 411
8.3.1 Potential Problem 411
8.3.2 Elasticity 413
8.3.3 Poroelasticity 413
8.4 Stress Discontinuity Method 415
8.5 Displacement Discontinuity Method 418
8.6 Dislocation Method 420
8.7 Galerkin Integral Equation 422
8.8 Fundamental Solution 424
8.8.1 Elementary Fundamental Solution 424
8.8.2 Elasticity Fundamental Solution 426
8.9 Poroelasticity Fundamental Solution 431
8.10 Fluid Source 435
8.10.1 Continuous Source 436
8.10.2 Instantaneous Source 441
8.11 Fluid Dipole 444
8.11.1 Continuous Dipole 445
8.11.2 Instantaneous Dipole 448
8.12 Fluid Dilatation 450
8.12.1 Continuous Fluid Dilatation 450
8.12.2 Instantaneous Fluid Dilatation 451
8.13 Fluid Force 453
8.13.1 Continuous Fluid Force 453
8.13.2 Instantaneous Fluid Force 454
8.14 Fluid Dodecapole 454
8.15 Total Force 456
8.15.1 Continuous Total Force 457
8.15.2 Instantaneous Total Force 460
8.16 Solid Quadrupole and Hexapole 461
8.17 Solid Center of Dilatation 463
8.18 Displacement Discontinuity 465
8.19 Edge Dislocation 467
8.20 Fundamental Solution Relation Based on Reciprocity 469
References 471

9 Poroelastodynamics 475
9.1 Dynamic Equilibrium Equation 476
9.2 Dynamic Permeability 486
9.3 Governing Equation 493
9.4 Wave Propagation 497
9.4.1 Elastic Wave 497
9.4.2 Poroelastic Wave 499

9.5 Phase Velocity and Attenuation 503
9.5.1 Phase Velocity 504
9.5.2 Attenuation 509
9.5.3 Extended Biot Models 512
9.6 One-Dimensional Wave Problem 517
9.6.1 Half Space 520
9.6.2 Finite Thickness Layer 521
9.7 Thermoelasticity Analogy 527
9.8 Poroelastodynamics Fundamental Solution 529
9.8.1 Elastodynamics Fundamental Solution 531
9.8.2 Helmholtz Decomposition 531
9.8.3 Three-Dimensional Point Force Solution 536
9.8.4 Three-Dimensional Fluid Source Solution 542
9.8.5 Two-Dimensional Fundamental Solution 544
9.9 Integral Equation Representation 545
9.10 Plane Wave Reflection and Refraction 547
9.10.1 Plane Strain Wave Solution 547
9.10.2 Reflection on Free Surface—Non-Dissipative Medium 549
9.10.3 Reflection on Free Surface—Dissipative Medium 556
9.10.4 Impermeable Surface 560
9.10.5 Fluid and Porous Medium Interface 562
References 567

10 Poroviscoelasticity 573
10.1 Viscoelasticity 575
10.1.1 Spring and Dashpot Model 575
10.1.2 Correspondence Principle 577
10.2 Poroviscoelasticity 580
10.2.1 Correspondence Principle 580
10.2.2 Intrinsic Models 581
10.3 Borehole Problem 585
10.4 Cylinder Problem 588
10.5 Poroviscoelastodynamics 594
References 596

11 Porothermoelasticity 599
11.1 Constitutive Equation 600
11.1.1 Thermodynamic Formulation 600
11.1.2 Linear Model and Intrinsic Material Constant 602
11.1.3 Ideal Porous Medium 605
11.1.4 Link with Bulk Continuum Model 606
11.1.5 Bulk Continuum Constitutive Equation 611
11.1.6 Porothermoelastic Physical Mechanism 614

11.2 Balance Law 616
11.3 Nonequilibrium Thermodynamics and Transport Law 618
11.3.1 Onsager's Reciprocal Relation 618
11.3.2 Thermal-Hydraulic Transport Phenomena 620
11.3.3 Irreversible Thermodynamics Process 621
11.3.4 Effective Thermal Conductivity 624
11.4 Governing Equation 627
11.4.1 Thermally Coupled Model 627
11.4.2 Porothermoelastic Constant 632
11.4.3 Irrotational Displacement Field 634
11.4.4 Thermally Uncoupled Model 635
11.4.5 Complete Model 637
11.5 Uniaxial Strain 638
11.6 Heating of a Half Space 639
11.6.1 Constant Temperature, Permeable Boundary 640
11.6.2 Constant Temperature, Impermeable Boundary 646
11.6.3 Constant Heat Flux, Permeable Boundary 647
11.6.4 Constant Heat Flux, Impermeable Boundary 649
11.7 Axial Symmetry 650
11.8 Borehole Problem 653
11.8.1 Constant Temperature and Pore Pressure Problem 654
11.8.2 Constant Heat and Fluid Flux Problem 659
11.9 Cylinder Problem 661
11.10 Spherical Symmetry 668
11.11 Cavity Problem 670
11.11.1 Constant Temperature Problem 671
11.11.2 Constant Heat Flux Problem 674
11.12 Sphere Problem 680
11.13 Porothermoelasticity Fundamental Solution 684
11.13.1 Biot Decomposition 686
11.13.2 Continuous Heat Source 688
11.13.3 Continuous Fluid Source 694
11.13.4 Continuous Fluid and Heat Dipole 696
11.13.5 Continuous Force 696
References 698

12 Porochemoelasticity 703
12.1 Electrochemical Effect 704
12.1.1 Thermodynamic Potential 704
12.1.2 Clay as a Chemically Active Material 710
12.1.3 Chemical Potential and Osmotic Pressure for Dilute Solution 714
12.2 Micromechanical Analysis 718

12.3 Constitutive Equation 727
12.4 Balance and Transport Law 734
12.4.1 Balance Law 734
12.4.2 Transport Law 736
12.5 Governing Equation 740
12.6 Uniaxial Strain Problem 745
12.6.1 One-Dimensional Governing Equation 745
12.6.2 Half Space Problem 746
12.7 Axial Symmetry Problem 754
12.7.1 General Solution 754
12.7.2 Borehole Problem 756
12.7.3 Cylinder Problem 761
12.8 Spherical Symmetry Problem 766
12.8.1 General Solution 766
12.8.2 Cavity Problem 767
12.8.3 Sphere Problem 768
References 770

Appendix A List of Symbols 775

Appendix B Poroelastic Constants 787
B.1 Relations Among Bulk Poroelastic Constants 787
B.2 Relations Among Bulk and Micromechanical Constants 789
B.3 Relations Among Bulk and Intrinsic Constants 792
B.4 Relations Among Micromechanical and Intrinsic Constants 796
B.5 Relations Among Biot-Willis and Micromechanical Constants 797
B.6 Biot Notations 797
B.7 Porothermoelasticity Constants 798
B.8 Porochemoelasticity Constants 801
References 804

Appendix C Cartesian and Curvilinear Coordinate System 807
C.1 Cartesian Coordinate System 807
C.2 Cylindrical Coordinate System 809
C.3 Spherical Coordinate System 811
Reference 812

Appendix D Laplace Transform and Inversion 813
D.1 Laplace Transform 813
D.2 Inverse Laplace Transform 816
D.3 Approximate Inverse Laplace Transform 817
References 818

Appendix E Poroelasticity Fundamental Solution 821
E.1 Continuous Fluid Source 821
E.2 Instantaneous Fluid Source 822
E.3 Continuous Fluid Dipole 823
E.4 Instantaneous Fluid Dipole 825
E.5 Continuous Fluid Dilatation 826
E.6 Instantaneous Fluid Dilatation 827
E.7 Continuous Fluid Force 828
E.8 Instantaneous Fluid Force 828
E.9 Continuous Fluid Dodecapole 829
E.10 Continuous Total Force 830
E.11 Instantaneous Total Force 832
E.12 Continuous Solid Quadrupole and Hexapole 833
E.13 Continuous Solid Dilatation 834
E.14 Continuous Displacement Discontinuity 835
E.15 Instantaneous Displacement Discontinuity 836

Appendix F Biography 839
F.1 Leonardo da Vinci (1452–1519) 839
F.2 Robert Hooke (1635–1703) 840
F.3 Jean Baptiste Joseph Fourier (1768–1830) 842
F.4 Thomas Young (1773–1829) 843
F.5 Claude Louis Marie Henri Navier (1785–1836) 844
F.6 Augustin-Louis Cauchy (1789–1857) 845
F.7 George Green (1793–1841) 846
F.8 Gabriel Lamé (1795–1870) 848
F.9 Henry-Philter Gaspard Darcy (1803–1858) 849
F.10 Achille Ernest Oscar Joseph Delesse (1817–1881) 850
F.11 Josiah Willard Gibbs (1839–1903) 851
F.12 Karl von Terzaghi (1883–1963) 853
F.13 Yacov Il'ich Frenkel (1894–1952) 855
F.14 Maurice Anthony Biot (1905–1985) 857
References 861

Index 867

Chapter 1
Introduction

The Earth is a living body. Its soul is its ability to grow. This soul, which also provides the Earth with its bodily warmth, is located in the inner fires of the Earth, which emerge at several places as baths, sulfur mines or volcanoes. Its flesh is the soil, its bones are the strata of rock, its cartilage is the tufa, its blood is the underground streams, the reservoir of blood around its heart is the ocean, the systole and diastole of the blood in the arteries and veins appear on the Earth as the rising and sinking of the oceans.

—Leonardo da Vinci (*Codex Leicester*, 1506–1510)

In *Codex Leicester* [61], da Vinci (see Sect. F.1 for a biography) drew the analogy between the Earth and the human body, to argue that Earth too, is a living body. He compared soils, rocks, and tufa (volcanic rock composed of fused detritus) to flesh, bones, cartilages, heart, arteries, and veins. We may ask, what do these materials have in common? They are all porous materials, that is, solid materials containing void space (pores) in them. Particularly, the pores can be occupied by a fluid, such be water, air, oil, methane gas, blood, body fluid, or a mixture of these.

This book is dedicated to the study of mechanics of porous materials, especially those infiltrated by a fluid. Porous materials can be found in nature as inanimate objects such as sand, soil, and rock, as living bodies such as plant tissue and animal and human flesh and bones, or as man-made materials for various industry or biomedical applications. These materials can look much different in their appearances due to their origin, but the underlying physical principles governing their mechanical behaviors can be the same. We are interested in the static and dynamic responses of these materials subject to mechanical as well as other type of forces, such as those of thermal and chemical origin. These studies are generally known as *poromechanics*.[1]

The modeling of a full range of porous material responses, ranging from quasi-static to dynamic, from linear to nonlinear, and from partial uncoupling to full coupling, can be complex and unwieldy. The goal of this book is limited. It focuses

[1]The term "poromechanics" was first created for the Biot Conference on Poromechanics [242].

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_1

largely on the linear theories, known as *poroelasticity*,[2] as in the linear theory of elasticity and Darcy flow; hence the book is of an introductory nature.

1.1 Porous Material

Porous material takes many shapes and forms. It can be a granular material, or a porous solid. Figure 1.1a shows a *granular material* as a pile of sand, which is a collection of particles of quartz, feldspar, mica, and other minerals of various sizes, ranging from 0.1 to 1 mm. Due to the irregular shape and random packing, a large percentage of the space is void, into which a fluid can penetrate and flow freely.

Figure 1.1b displays a section of sandstone under microscope. Sandstones are sand particles cemented together by calcite, clay and silica during the geological processes of sedimentation and compaction by overburden pressure. Unlike loose sand, sandstone is a *porous solid*; that is, a connected solid material containing pore space in it, shown in the figure as the areas without feature. The size of the pores for this sandstone is about 0.1 mm, and fluid can reside in it. In fact, rock can contain so much fluid such that it can be a water bearing and transmitting formation, known as *aquifer* in groundwater hydrology, or as an oil or gas bearing formation, called reservoir rocks in petroleum engineering.

Figure 1.1c shows a pumice, a volcanic rock. It contains large pores of millimeter or even centimeter sizes. The pores are formed by gas bubbles from exsolving volatiles during the cooling of lava. Figure 1.1d shows an outcrop of a fractured rock formation. Rock fractures can be caused by weathering or the change of underground stress environment. Fractured rock formations can be highly permeable. Even though the rock itself may not conduct water, a few large connected fractures can transmit a large quantity of water, and sometimes become the most productive aquifer.

The above are geomaterials occurring in nature. Porous materials can be manufactured. Concrete is a material used for the construction of buildings, roadways and ports by Romans since 2000 years ago. Concrete has relatively low porosity. Figure 1.1e however demonstrates a *pervious concrete* [266] with large porosity. The cement paste of this concrete is much reduced in order to create a large amount of interconnected pore space, yet it needs to be in sufficient amount to retain a large portion of its strength. Pavement made from this type of concrete can permeate and store water to reduce the surface runoff in the event of a storm.

Another artificial material is the ubiquitous polyurethane foam, seen under a microscope in Fig. 1.1f. The foam is lightweight and possesses good sound proofing, thermal insulating, and shock absorbing properties. Its structure can be open-cell or closed-cell. Figure 1.1f shows a cross section of a closed-cell foam. Figure 1.1g shows a *foam metal*, also known as structural foam. It can be created from various

[2]The term "poroelasticity" was first used by Deresiewicz and Skalak [69].

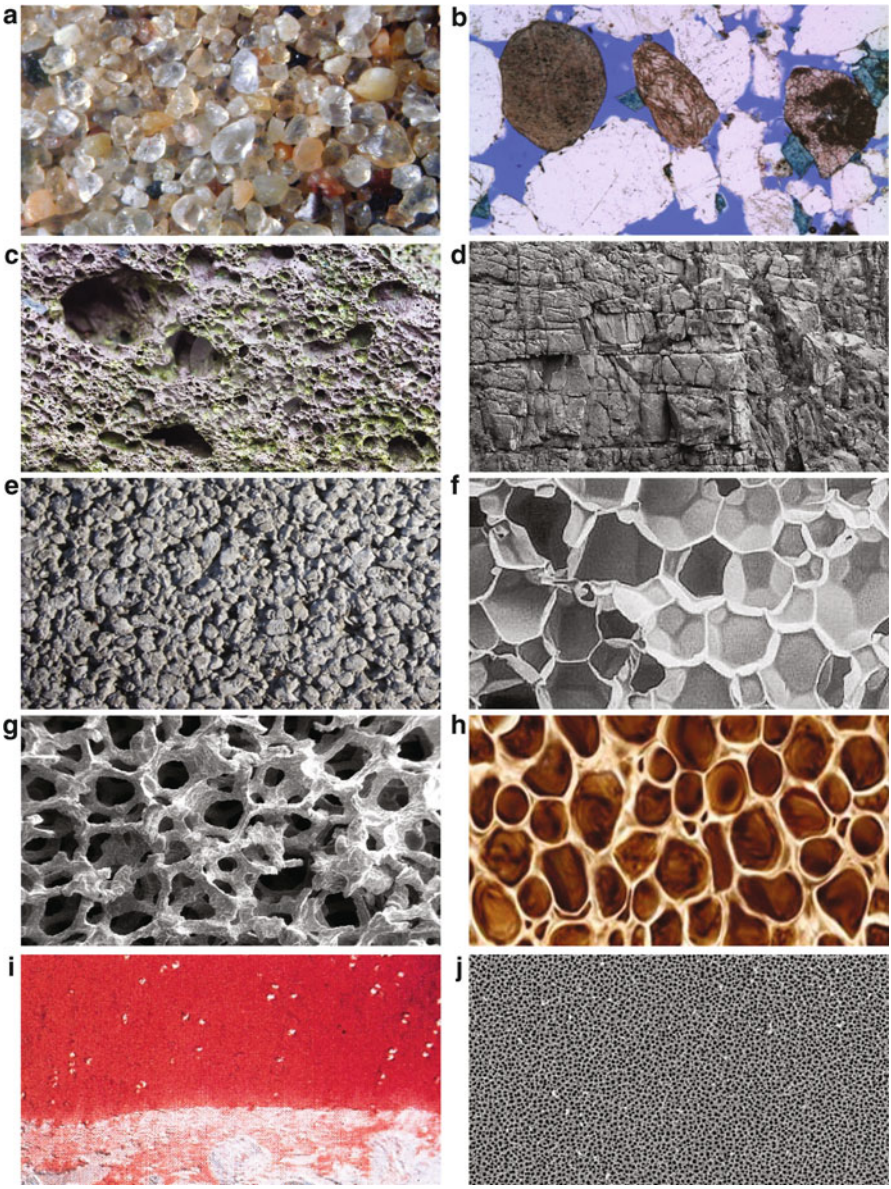


Fig. 1.1 Various porous materials. (**a**) Sand. (**b**) Sandstone. (**c**) Volcanic rock. (**d**) Fractured rock. (**e**) Pervious concrete. (**f**) Polyurethane foam. (**g**) Metal foam. (**h**) Bone with osteoporosis. (**i**) Articular cartilage. (**j**) Nanoporous alumina

metallic, glass, and carbon materials. It also can have closed or open cell structure. The one shown in the figure is open-cell. Such material is light weight and can be used for various purposes such as impact energy absorption, vibration damping, thermal and sound insulation, catalyst support, and metal and gas filtration [8].

Biological materials are typically porous. Figure 1.1h gives the cross section of a human bone. The pores of a living bone are filled with marrow. When the porosity of the bone increases for reasons such as aging or calcium deficiency, it leads to the disease called osteoporosis (means porous bone) [222]. The bone cross section in Fig. 1.1h is in fact a diseased bone with loss of bone mass. Figure 1.1i shows a healthy articular cartilage that lines the joint between two bones. The cartilage is a soft and porous material imbibed with fluid [170]. The viscous motion of fluid in the pores acts as a damper for shock loading. When the cartilage is compressed, fluid oozes out to provide a film of lubrication to reduce friction. Also, since the cartilage does not contain blood vessel, it relies on the squeezing motion of the fluid to bring nutrient to the cells. Hence the design and manufacturing of an artificial articular cartilage needs to take some of the above mechanisms into consideration [179, 180].

As observed above, pore space comes in various sizes, and in fact, pores can be as small as nano size. Nanoporous materials abound in nature, both in biological systems and in natural minerals. By definition, nanopores are of the size between 1 to 100 nm. Nanoporous solids have been made out of a wide variety of substances, including carbon, silicon, silicates, various polymers, ceramics, metals, metallic minerals, and organic materials [154]. Figure 1.1j shows the scanning electron micrograph of a nanoporous alumina membrane. Like the foam materials discussed above, the structure of nanoporous material can be open-cell, or closed-cell. In functional applications, open-cell porous materials are needed for adsorption, catalysis, sensing, separation, and filtration; and closed-cell materials are useful in acoustic, thermal insulation, and lightweight structural applications.

1.2 Physical Mechanism

Before a formal construction of the mathematical theory of poroelasticity, it is important to understand the basic mechanics of a fluid infiltrated porous material through *physical intuition*. To reflect on this view, we shall quote Maurice A. Biot (see Sect. F.14 for a biography), generally recognized as the "father of poroelasticity", from his acceptance speech of the Timoshenko Medal [22].

First, Biot expressed the honor of being associated with the name of Timoshenko, the great *Engineer and Scientist*. He praised Timoshenko's "*tradition of clarity, simplicity, intuitive understanding, unpretentious depth, and a shunning of the irrelevant*". Biot continued to say,

> *There is, of course, no merit in sophistication for its own sake. . . . We should not overlook the importance of simplicity combined with depth of understanding, not only for its cultural value, but as a technological tool. It leads to quantitative predictions without laborious and costly calculations; it suggests new inventions and simple solutions of engineering problems. . . . Deeper physical insight combined with theoretical simplicity provides the short-*

cuts leading immediately to the core of extremely complex problems and to straightforward solutions. This cannot be achieved by methods which are sophisticated and ponderous even in simple cases. The process of thought which is involved here may be described as cutting through the scientific red tape and bypassing the slow grinding mills of formal scientific knowledge. Of course, formal knowledge is essential but, as for everything in life, the truth involves a matter of balance. . . . it is essential to the make-up of a competent engineer.

It is in this spirit of "cut the red tape" of mathematical sophistication that in the sections to follow we shall discuss the mechanical responses of fluid infiltrated porous material by directly appealing to physical intuition.

1.2.1 Drained and Undrained Responses

For a fluid infiltrated porous material, its mechanical strength resisting compression is derived from three different sources. It is somewhat obvious that the rigidity of the solid and the fluid both contribute to the strength of material. It may be less obvious that the pore space, either carved in a piece of solid (porous solid), or in the intergranular space of stacked particles (granular material), has a strength that is dependent on the size, shape, orientation, and distribution of the pores.

By assigning an (apparent) strength to the pore space, which is a void, we are having in mind a composite material that various components contribute to the overall strength according to their volume fraction. The strength of the pore space is obviously smaller than the bounding solid that creates it, such that it weakens the material comparing to one that is totally solid without void. The strongest pore is that of spherical shape, with its strength characterized by the shear modulus of the solid and the porosity, and the weakest is a slit [59]. In fact if the slit has its orientation perpendicular to the applied stress direction, it readily closes offering no strength at all.

For granular materials, the major contribution of compressibility (inverse of rigidity) comes from the fluid and the pore structure, and not from the solid. For example, consider a pile of loose sand without fluid in it. We can make the sand particles (largely quartz) as rigid as they can be, yet they have little effect on the strength of the sand pile, which is rather weak subjected to compression. The strength of a granular material is largely determined by the denseness of the packing of the particles, not by the solid made of it. Fluid in liquid phase, on the other hand, can play an important part, particularly when it is trapped in the pores. In this subsection, we shall discuss the contribution of fluid in the rigidity of porous material under drained and undrained conditions.

Drained and undrained conditions are well-known concept in soil mechanics [146, 239]. Consider a porous solid with connected interstitial space that is saturated with a liquid. We can wrap this specimen in an impermeable but flexible membrane, and subject it to an all-around incremental compressive stress of magnitude ΔP. If we insert a tube through the membrane to reach the inside of the specimen, we can measure a pore pressure rise Δp. This experimental setup is called *undrained test* because the fluid is prevented from leaving the porous frame by the sealing membrane.

In a different experimental setup, we can leave the surface of the specimen exposed to atmosphere; or, we can utilize the tube inserted into the wrapped specimen and open its end to the atmosphere. In either of the above setup, fluid can flow out of the specimen. The flow process will continue, until everywhere in the specimen the pressure becomes equal to the atmospheric pressure, and $\Delta p = 0$. When a measurement is conducted in this state, it is called a *drained test*.

In both tests, we can measure the volumetric deformation of the specimen ΔV and compare it to the original volume V. The stiffness of the porous material can be characterized by the ratio of the applied load ΔP to the percentage change of volume $\Delta V/V$, known as volumetric strain. Following the concept of solid mechanics [93, 246], we can define a *material constant* characterizing the stiffness against volumetric deformation. For the drained test, we define

$$K = -\left.\frac{\Delta P}{\Delta V/V}\right|_{\text{drained}} \tag{1.1}$$

where K is called *drained bulk modulus*. (In the above, a negative sign is present because a positive compressive stress causes a decrease in volume.) Similarly, we can define an *undrained bulk modulus* K_u as

$$K_u = -\left.\frac{\Delta P}{\Delta V/V}\right|_{\text{undrained}} \tag{1.2}$$

In the following, we shall discuss the physical significance of these material constants.

For the drained test, the fluid is equilibrated with the atmospheric pressure, so it does not participate in the load bearing process. The drained bulk modulus K is then a measure of the stiffness of the *porous solid frame* without fluid. By porous solid frame we refer to the spatial structure consisting of the solid, as well as the enclosed voids (pores). When a frame is deformed, the reduction in volume is contributed both by the compression of the solid phase and by the compaction of empty space due to the structural rearrangement of the frame.

We may partition the total volume of the specimen V into two parts, a solid volume V_s, and a pore volume V_p, such that $V = V_s + V_p$. Then the change of total volume ΔV can also be partitioned into a solid and a pore part, such that $\Delta V = \Delta V_s + \Delta V_p$. If we consider that each part has its own stiffness against deformation, we then define these two apparent bulk moduli, respectively for the solid and the pore space, as

$$K_1 = -\frac{\Delta P}{\Delta V_s/V} \tag{1.3}$$

$$K_2 = -\frac{\Delta P}{\Delta V_p/V} \tag{1.4}$$

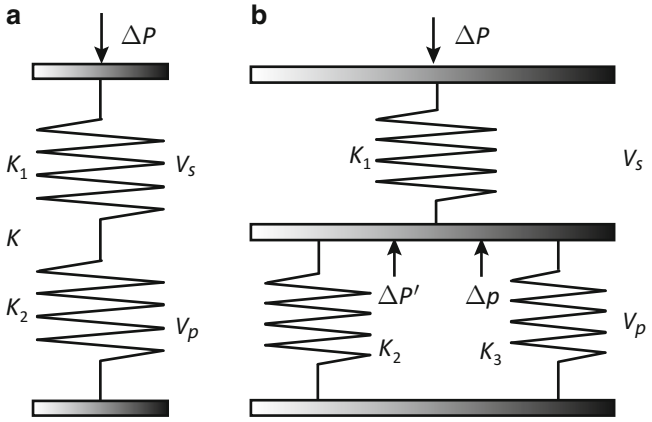


Fig. 1.2 Spring analogy: (**a**) Series springs for drained bulk modulus, and (**b**) Parallel springs for undrained bulk modulus

Utilizing the above, we can then express

$$\frac{\Delta V}{V} = \frac{\Delta V_s}{V} + \frac{\Delta V_p}{V} = -\Delta P \left(\frac{1}{K_1} + \frac{1}{K_2} \right) \tag{1.5}$$

Comparing with (1.1), it is apparent that

$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} \tag{1.6}$$

The above relation is similar to finding the equivalent spring constant of two coil springs arranged in a series. We hence can draw an analogy and present the relation among the rigidity of solid grain K_1, and of porous structure K_2, and the apparent rigidity of the frame K, as the series springs shown in Fig. 1.2a. If we consider a granular material, such as sand, with highly rigid solid grain (quartz), $K_1 \to \infty$, but weak porous structure, then the frame rigidity is dominated by the weak part, the pore structure, and the rigid solid has no effect on the value of K.

In Sect. 4.8.1, under an ideal porous medium condition (a porous medium made of homogeneous and isotropic solid material), we derive that

$$\frac{1}{K} = \frac{1}{K_s} + \frac{\phi}{K_p} \tag{1.7}$$

where K_s is the bulk modulus of the solid, K_p is the pore volume bulk modulus, and ϕ is porosity. Hence we confirm the series spring analogy.

Next, we examine the undrained condition in which the fluid is entrapped in the pores. When a load ΔP is applied and the pore volume is compressed, not only the porous structure, but also the entrapped fluid, will resist. If the former gives

a reaction stress $\Delta P'$ and the latter a pore pressure Δp, then $\Delta P = \Delta P' + \Delta p$. Particularly, the reaction stress provided by the pore structure, $\Delta P'$, is called an *effective stress*.

We may consider the above situation analogous to the parallel springs as shown in Fig. 1.2b. If we consider that $\Delta P'$ causes the pore volume to deform in the amount ΔV_p, with a bulk modulus K_2, and Δp causes the fluid volume to deform ΔV_f, with a bulk modulus K_3, then

$$K_2 = -\frac{\Delta P'}{\Delta V_p / V} \tag{1.8}$$

$$K_3 = -\frac{\Delta p}{\Delta V_f / V} \tag{1.9}$$

Summing the partial stresses $\Delta P'$ and Δp, and realizing that $\Delta V_f = \Delta V_p$ as the fluid occupies the pore space, we obtain

$$\Delta P = \Delta P' + \Delta p = -(K_2 + K_3)\frac{\Delta V_p}{V} \tag{1.10}$$

The above shows that we can define an equivalent bulk modulus

$$K_4 = K_2 + K_3 \tag{1.11}$$

just as the equivalent spring constant for parallel springs. However, we need to exercise some caution in the above argument. In order to arrive at a simple model, we have neglected an important effect that the pore pressure Δp can cause the solid volume V_s to deform too; so the actual model is more complex than the combined series-parallel spring system depicted in Fig. 1.2b.

Nevertheless, the above simplified model allows us to gain some basic physical understanding, and we may consider some limiting cases. For example, if the solid grains are relatively rigid, $K_1 \to \infty$, then the compressibility of the undrained material is dominated by the parallel springs system with bulk modulus K_4. In fact, in Sect. 4.8.2, it is demonstrated that for the special case that $K_s \to \infty$, the undrained bulk modulus is given by

$$K_u = K + \frac{K_f}{\phi} \tag{1.12}$$

The above equation confirms our intuitive model based on parallel spring analogy. Also, as the undrained bulk modulus is the sum of the drained bulk modulus and fluid bulk modulus, it is obvious that $K_u > K$. The above result shows that a porous material is stiffer under undrained condition than drained, as both the solid frame and the fluid are reacting to the load. Furthermore, we can consider the case that the fluid is much less compressible than the porous medium frame, and in fact incompressible, $K_f \to \infty$. In this case, the entire system is incompressible under undrained condition, which is the model for clay consolidation considered by Terzaghi [235, 237].

Tables 3.1 and 3.2 give a list of typical values of K, K_u, K_p, K_s, and K_f for a range of natural and man-made materials.

1.2.2 Time and Length Scale

In the above section we have described the undrained condition as one that is created by wrapping the sample in an impermeable membrane. This is in fact not a necessary condition for the undrained behavior. Fluid is a *viscous* material such that it takes a finite amount of time for fluid to move a finite distance. If a loading is instantly applied, fluid is trapped in the pore space even if the sample surface is open for drainage. Hence a porous material is in an undrained state responding to an instantly, or rapidly, applied load, whether its surface is sealed or exposed to atmosphere.

Given time, however, a small amount of the fluid will be expelled from the solid frame if the specimen is not completely sealed. This allows the fluid to expand, thus giving relief to the pressure. Eventually the pressure will equilibrate with the ambient or atmospheric pressure, and the state is referred to as drained. The stiffness of a porous material therefore gives the appearance of time dependency—it is stiffer (undrained) at small times, and continues to soften (drained) at large times.

In the above, when we refer to small or large time, we do not mean absolute time. Unlike some solid materials that exhibit time dependent behavior (viscoelasticity), which is a material property with absolute time, this time is relative to the *time scale* for the completion of the physical process of a given problem. This time scale is dependent not only on the solid and fluid material properties, but also on the size and geometry of a particular problem.

Karl Terzaghi (see Sect. F.12 for a biography) was the first to model the creep phenomenon of saturated clay as a *soil consolidation* process [234, 237]. He recognized that when a uniform surcharge is suddenly applied to the top of a clay layer, the soil is initially undrained and an initial excess pore water pressure p_o is generated. The excess pore water pressure will dissipate through the top surface following a one-dimensional diffusion equation

$$\frac{\partial p}{\partial t} - c\,\frac{\partial^2 p}{\partial x^2} = 0 \tag{1.13}$$

where c is a *consolidation coefficient* with dimension of [L^2/T], which is a combined property of the permeability of the clay and the elasticity of the solid skeleton.

To obtain a solution that is independent of the physical dimensions of the problem, and of the magnitude of the loading, we can perform nondimensionalization to (1.13) by introducing a *characteristic length* L_c, and the dimensionless quantities

$$p^* = \frac{p}{p_o}, \quad x^* = \frac{x}{L_c}, \quad t^* = \frac{c\,t}{L_c^2} \tag{1.14}$$

to obtain

$$\frac{\partial p^*}{\partial t^*} - \frac{\partial^2 p^*}{\partial x^{*2}} = 0 \tag{1.15}$$

We observe that (1.15) is free from any material coefficient.

The definition of the dimensionless time in (1.14) suggests a *characteristic time* $t_c = L_c^2/c$, and $t^* = t/t_c$. For the consolidation problem, the characteristic length is the thickness of the layer, that is, the distance that the pressure relief wave front needs to travel to reach the farthest region of pore pressure buildup. The characteristic time t_c is then the time it takes for the layer to be significantly drained. With this definition, a *small time* is referred to as $t \ll t_c$, and a *large time* is $t \gg t_c$. It is of interest to note that t_c is dependent on both the material properties, and the physical size of the problem. Particularly, it is proportional to the square of the characteristic length of the problem.

The time scale and length scale can manifest themselves in many different ways. For example, consider a case of cyclic loading created by a vibrating machinery on top of a soil layer, with a stress that is sinusoidal in time $P = P_o \sin \omega t$. In this case, there exists a characteristic frequency $\omega_c = c/L_c^2$ for diffusion, such that if the vibration is at high frequency, $\omega \gg \omega_c$, the layer behaves as an undrained material, and at low frequency, $\omega \ll \omega_c$, the layer is characterized by the drained property.

Now consider the case of high frequency. Although the layer is generally undrained, there exists a region near the top where the soil is drained, enforced by the boundary condition. In this case, we find a characteristic length $\delta_\ell = \sqrt{c/\omega}$ such that within this distance, the soil is largely drained. This thickness δ_ℓ is called a *boundary layer thickness* [165], within which the diffusion process is important, and outside which the undrained elastic behavior dominates. (It should be noted that the discussion in this section is limited to the quasi-static poroelasticity, with diffusion effect only, and without the dynamic wave propagation effects.)

Or, we can consider a moving load. The moving load can be a wheel traveling at a speed v on the surface of a soil layer [35], a drilling bit advancing and excavating the material ahead of it [6], or a propagating fracture [49]. For the moving wheel, if the surface of the soil is drained, then there is a region directly underneath the wheel where the soil is largely drained. Farther away, however, the speed of the wheel can be too fast to allow time for the induced pore pressure to dissipate. A length scale near the wheel within which the soil is drained is characterized by $L_c = c/v$. When the velocity is faster, the region is smaller, and the overall material response is close to undrained. When the velocity is slower, the material approaches the drained behavior.

1.2.3 Skempton Pore Pressure Effect

Let us consider the undrained condition. As the fluid is trapped, it will resist a compressive confining stress of magnitude ΔP and give rise to a pore pressure

increment Δp. Generally speaking, we anticipate that Δp is smaller than the applied load, $0 \leq \Delta p \leq \Delta P$. Skempton [226] measured this pore pressure rise and defined the relation

$$\Delta p = B\Delta P \tag{1.16}$$

in which B is the *Skempton pore pressure coefficient B*. It is anticipated that $0 \leq B \leq 1$, though a value $B > 1$ is possible.

Equation 1.16 is defined in three dimensions, and

$$\Delta P = -\frac{\sigma_{xx} + \sigma_{yy} + \sigma_{zz}}{3} \tag{1.17}$$

is the average compressive stress, where σ_{xx}, σ_{yy}, and σ_{zz} are stress tensor components, defined *positive for tension*. Under *plane strain* condition, that is, the body is confined from movement in one of the spatial dimensions, say z, then we define on the x-y plane

$$\Delta P = -\frac{\sigma_{xx} + \sigma_{yy}}{2} \tag{1.18}$$

and find the following relation for the Skempton effect [42]

$$\Delta p = \frac{2B(1 + \nu_u)}{3}\Delta P \tag{1.19}$$

where ν_u is the *undrained Poisson ratio*. Furthermore, if the deformation is allowed in one dimension only, say x, and a load $\sigma_{xx} = -\Delta P$ is applied, then the uniaxial Skempton effect is characterized by this pressure rise

$$\Delta p = \frac{B(1 + \nu_u)}{3(1 - \nu_u)}\Delta P \tag{1.20}$$

The above proportionality constant is also called the *tidal efficiency* [126]

$$\gamma_{te} = \frac{\Delta p}{\Delta P} = \frac{B(1 + \nu_u)}{3(1 - \nu_u)} = \frac{3BK_u}{4G + 3K_u} \tag{1.21}$$

where G is shear modulus. It is a constant associated with the ratio of water level rise in water wells resulting from the rise of ocean tide, a phenomenon that will be discussed in Sect. 1.3.10.

Referring to the parallel spring analogy in Fig. 1.2b, it is easy to realize that the share of the reaction force between $\Delta P'$ and Δp is proportional to their respective spring constants K_2 and K_3; that is, the stiffer the spring is, the more load that spring

takes. Using (1.8) and (1.9) to eliminate the volume terms, and substituting in (1.10) for the total load ΔP, we find that

$$\frac{\Delta p}{\Delta P} = \frac{K_3}{K_2 + K_3} = \frac{K_3}{K_4} \tag{1.22}$$

Assuming that the solid constituent is incompressible, we can utilize the relation (1.12) and (1.16) to obtain

$$B = \frac{K_f}{\phi K + K_f} \tag{1.23}$$

which is presented as (4.166).

Based on the above relation, we can examine some limiting cases. If the stiffness of the skeleton is much smaller than that of fluid, $K \ll K_f$, which is the case for soil saturated with water, then the fluid is bearing most of the load. In the limit, we expect $\Delta p \to \Delta P$ and $B \to 1$. This is the case that Terzaghi [234] considered in the theory of soil consolidation. Indeed, the Terzaghi theory has the inherent assumption that both the fluid and the solid are incompressible as compared to the soil skeleton, such that $K_f/K \to \infty$ and $K_s/K \to \infty$, which in turn leads to the condition $K_u/K \to \infty$. Hence saturated clay is considered incompressible at the instant of loading. Another special case to consider is that the fluid is not a liquid, but a gas. As gas is highly compressible, $K \gg K_f$, and there is little or no pressure rise; hence $\Delta p \to 0$ and $B \to 0$.

A more curious case is that, if the solid is soft, and softer than the liquid occupying the pore space, then the pore pressure rise can be greater than the load applied; that is, $\Delta p > \Delta P$, and $B > 1$. Such situation can exist for polyurethane foam saturated with a liquid. In that case, the solid part is in effective tension. This effect is more difficult to visualize, but can indeed be predicted by the micromechanical analysis (see Sect. 4.8.5), and observed in laboratory. Typical values of B are given in Table 3.2, and we do see that B can be greater than 1.

1.2.4 Effective Stress for Volumetric Deformation

Consider a porous medium under drained condition. When there exists a confining compressive stress ΔP, it is obvious that the load is entirely supported by the solid, and a volumetric deformation ΔV will result. On the other hand, if the fluid is not drained and there is fluid pressure Δp above the ambient pressure, then the pore pressure will counteract the applied load ΔP, such that the effective load applied to the solid frame is reduced. In Sect. 1.2.1 we argued that such effective load is simply $\Delta P' = \Delta P - \Delta p$. This is however not true. As we have commented, in order to reach such relation, we neglected the solid compressibility by assuming $K_s \to \infty$. The question is, as the solid is compressible, and the pressure has an

effect of compressing the solid, what is the effective reduction of the load applied to the solid? In other words, what is the *effective stress* applied to solid frame that can correctly predict the deformation of the frame? We shall answer this question in this section.

Putting the above question into an equation, it becomes

$$\frac{\Delta V}{V} = -\frac{1}{K}(\Delta P - \alpha \Delta p) \tag{1.24}$$

in which the left hand side is the volumetric strain of the frame, K is the drained bulk modulus of the frame, and α is the coefficient in question. The coefficient α defines the weighted contribution of pore pressure to the load reduction, and is called the effective stress coefficient. It has the value between 0 and 1, where 0 means that the pore pressure has no effect on the load reduction, and 1 means it has the full effect.

Based on an extensive historical study, de Boer [63–65] called the attention to a controversy between Karl von Terzaghi, *father of soil mechanics*, and Paul Fillunger, *father of mixture theory*, on the correct form of the effective stress. According to de Boer, the controversy is still not settled [26, 66, 145].

Terzaghi was more an engineer than a theoretician. Based on his physical intuition and laboratory observation, Terzaghi [234, 236] proposed the following effective stress law (see a historical review by Skempton [227]),

$$\frac{\Delta V}{V} = -\frac{1}{K}(\Delta P - \Delta p) = -\frac{\Delta P'}{K} \tag{1.25}$$

where

$$\Delta P' = \Delta P - \Delta p \tag{1.26}$$

is the *Terzaghi effective stress*. The above effective stress law has been widely adopted in soil mechanics, as well as man-made materials such as concrete. Fillunger [86], on the other hand, was more theoretically oriented. He derived the following effective stress law based on the forerunner of modern day mixture theory [66],

$$\frac{\Delta V}{V} = -\frac{1}{K}(\Delta P - \phi \Delta p) \tag{1.27}$$

in which ϕ is the porosity. It should be mentioned that in arriving at the above results, both Terzaghi and Fillunger assumed that the solid constituent is incompressible [66], while the solid frame is compressible.

The two effective stress laws, (1.25) and (1.27), clearly do not agree. To find out which of the above is correct, we can subject these relations to a *thought experiment*. Let us consider an *ideal porous medium*; that is, a porous medium made of a homogeneous and isotropic solid [97]. We can submerse this porous specimen in a chamber filled with a fluid. If we raise the fluid pressure in the chamber by Δp,

it is clear that there is an incremental compressive stress applied to the specimen as $\Delta P = \Delta p$. If we wait long enough for the fluid pressure inside the porous medium to equilibrate with the chamber pressure, then the pore pressure rise is also Δp. This loading condition implies that the Terzaghi effective stress $\Delta P' = 0$.

Under this test condition, the entire solid surface that is in contact with the fluid, internal and external, is subjected to the same normal stress Δp. Due to the material homogeneity and isotropy, a simple elasticity solution shows that every point inside the solid is under uniform compression, and there is no shear stress. Since both Terzaghi and Fillunger consider the solid constituent as incompressible, the solid simply does not deform! This easily leads to the conclusion that $\Delta V = 0$, and that the Terzaghi effective stress law (1.25) is correct. Fillunger's equation (1.27), on the other hand, indicates that the frame will deform by the amount $\Delta V/V = -(1/K)(1-\phi)\Delta p$. This is certainly erroneous, and the error is the largest for small porosity materials.

In Sect. 1.1 we discussed two types of porous material, a granular material and a porous solid. For a granular material such as sand, shown in Fig. 1.1a, it is clear that the pore space is much more compressible than the individual sand particles; hence the *incompressible solid constituent* assumption used above is a reasonable one. For a porous solid with low porosity and spherical pores, however, the pore space is well protected, and its compressibility is comparable to that of the solid. In that case, the solid compressibility must be considered together with the pore compressibility; hence the Terzaghi's effective stress law needs to be modified.

Now let us consider the solid constituent as compressible and reexamine the above thought experiment. First, it is convenient to rewrite (1.24) into the following form:

$$\frac{\Delta V}{V} = -\frac{1}{K}\left[\Delta P' + (1-\alpha)\Delta p\right] \tag{1.28}$$

where P' is defined in (1.26). Under the present loading condition, $\Delta P' = 0$ and the above equation becomes

$$\frac{\Delta V}{V} = -\frac{1}{K}(1-\alpha)\Delta p \tag{1.29}$$

Hence we can directly relate the volume change of the specimen to the pressure change in the chamber.

Following the same argument as above, when the fluid pressure in the chamber is raised by Δp and the system reaches an equilibrium, the homogeneous and isotropic specimen is under uniform compression everywhere. As a consequence, the entire solid deforms in a geometrically similar fashion. This deformation will preserve not only the shapes of the solid, but also that of the pore space. This condition then leads to the following proportions

$$\frac{\Delta V_s}{V_s} = \frac{\Delta V_p}{V_p} = \frac{\Delta V}{V} \tag{1.30}$$

where V_s, V_p and V, as defined in Sect. 1.2.1, are the solid, the pore, and the total volume, respectively. The solid deformation is given by

$$\frac{\Delta V_s}{V_s} = -\frac{\Delta p}{K_s} \tag{1.31}$$

where K_s is the bulk modulus of the solid. Substituting (1.30) and (1.31) into (1.29), we clearly see that in order for the effective stress law to pass this thought experiment, we must have

$$\alpha = 1 - \frac{K}{K_s} \tag{1.32}$$

This is indeed what proposed by Biot [16, 23]. We hence define the *Biot effective stress* $\Delta P''$ as

$$\Delta P'' = \Delta P - \alpha \Delta p \tag{1.33}$$

such that

$$\frac{\Delta V}{V} = -\frac{\Delta P''}{K} \tag{1.34}$$

which is the correct effective stress law. The coefficient α is referred to as the *Biot effective stress coefficient*. We also notice that the *incompressible constituent model* is given by $K_s \to \infty$, which means that $\alpha = 1$, and (1.24) reduces to (1.25). We then conclude that the Terzaghi effective stress law is a special case of the Biot effective stress law. In the other limit, it can be proven that the composite bulk modulus of the porous frame (solid and pores) is always smaller than that of the solid, $K \leq K_s$, known as the Hashin-Shtrikman bound [111], and $K \to K_s$ only if $\phi \to 0$ [155]. Hence $\alpha = 0$ for a pure solid, which is an obvious conclusion. Typical values of α for a range of different materials are given in Table 3.2.

It may be of interest to also examine the result derived from the modern mixture theory [26, 145, 231]:

$$\begin{aligned} \frac{\Delta V}{V} &= -\frac{1}{K}\left\{\Delta P - \left[1 - (1-\phi)\frac{K}{K_s}\right]\Delta p\right\} \\ &= -\frac{1}{K}\left[\Delta P - (\alpha + \phi - \alpha\phi)\Delta p\right] \end{aligned} \tag{1.35}$$

The above effective stress law approaches that of Biot's if $\phi \to 0$. But this law obviously does not pass the above thought experiment, so one needs to be very careful about its use.

We should note that the above presentation is based on the ideal porous medium assumption. However, one only needs to take a look at the material of sand, such as

the one shown in Fig. 1.1a, to realize that porous materials are hardly homogeneous at the grain size scale. For a real porous medium, it can be shown that the Biot effective stress law (1.24) is still valid; but the solid bulk modulus K_s appearing in (1.32) needs to be interpreted differently. This more general porous medium theory is presented in Chaps. 3 and 4.

1.2.5 Effective Stress for Pore Collapse

When a porous material is subjected to a large deformation, leading to the failure of the material, it is observed that the threshold for failure is predicted not directly by the total stress applied, but by a certain difference between the total compressive stress and the pore pressure—in other words, an effective stress. In the above section we have demonstrated that the volumetric deformation of poroelastic material is related to the Biot effective stress. For failure, however, experimental evidences have shown that the threshold is largely associated with the Terzaghi effective stress, and not the Biot effective stress [78, 96, 109]. This phenomenon can be explained as follows.

For a porous material, the strength of the pore structure is always weaker than the strength of the solid that surrounds it. A theoretical proof of this is called the Hashin-Shtrikman bound [111], as mentioned in the above section. When a porous material fails, either by compression or shear, it is likely that the pore structure is destroyed first, known as *pore collapse* [263], before the solid constituent fails. Furthermore, we can argue that it is not the change of pore volume alone, but the *disproportional* changes of pore volume versus solid volume that will cause the pore structure to fail. To visualize this, let us again construct a thought experiment using the example of an ideal porous medium submersed in a fluid chamber. As discussed in the preceding section, when the fluid pressure in the chamber is raised, the porous medium deforms in a geometric similar fashion. In other words, the pore space is reduced in the same proportion as the solid constituent, and its shape is preserved. When the chamber pressure is continuously increased, the porous material will not fail by pore collapse, until it reaches a point that the solid constituent fails. From the above description, it is clear that the failure is more strongly related to the relative change between the pore volume and the total volume, that is, the change in porosity, than the change in pore volume or total volume alone.

From this reasoning, we need to construct the constitutive law between the change of porosity, $\Delta\phi$, and the applied stresses, ΔP and Δp. We may assume that there exists a linear relation,

$$\Delta\phi = a\,\Delta P - b\,\Delta p \tag{1.36}$$

Or, we can put that relation in a different form

$$\Delta\phi = a\,\Delta P' - b'\,\Delta p \tag{1.37}$$

where $\Delta P'$ is the Terzaghi effective stress. If we subject an ideal porous medium to the above-stated experiment, then the loading conditions gives $\Delta P' = 0$. Porosity is defined as

$$\phi = \frac{V_p}{V} \tag{1.38}$$

and its variation is

$$\Delta\phi = \Delta\left(\frac{V_p}{V}\right) = \frac{V\Delta V_p - V_p\Delta V}{V^2} \tag{1.39}$$

Utilizing (1.30), it is easy to show that under this test condition, we must have $\Delta\phi = 0$. We hence conclude that $b' = 0$, and (1.37) becomes

$$\Delta\phi = a\,\Delta P' \tag{1.40}$$

Hence the porosity change is dependent on the Terzaghi effective stress only. In Sect. 3.2, we shall demonstrate such constitutive law

$$\frac{\Delta\phi}{1-\phi} = -\frac{\Delta P'}{K_\varphi} \tag{1.41}$$

where K_φ is the *bulk modulus of porosity*. We therefore confirm the laboratory observation that the failure of porous material is largely associated with the Terzaghi effective stress.

1.2.6 Fluid Storage

In groundwater and petroleum engineering applications, fluid can be extracted from water and oil bearing formations by lowering the fluid pressure in a cavity to induce a porous medium flow into it for collection (production). The effect of lowering the fluid pressure in the porous medium causes the fluid to expand and the formation to compact, thus releasing the fluid.

In a reverse application, fluid can be injected into the formation. For groundwater applications, it can be the recharging of treated waste water for later reclamation and reuse; or, for petroleum engineering, for the disposal of oil field brine. The re-injection of fluid into the formation can reverse the trend of land subsidence caused by the past excessive extraction. In present day, to combat the carbon dioxide greenhouse effect, the capturing of CO_2 emission and the *geoengineering* (also called *climate engineering*) of storing it in deep underground formation has been used as a climate remediation technique [85]. With these applications, it is of interest to learn about the fluid storage capacity of a porous medium.

The storage capacity of a porous medium can be quantified by the definition of a *storage coefficient*, given as the volume of fluid stored into or extracted from a unit volume of porous medium per unit increase or decrease of fluid pressure. Depending on the constraint imposed on the porous medium during measurement, two storage coefficients can be defined: S_e is a storage coefficient measured under the condition of constant sample volume (constant strain), and S_σ is measured under constant confining stress.

There are several ways that fluid can be accommodated into or expelled from the porous frame. One way is for the solid frame to expand or to compact; the second is for the solid constituent to dilate or to contract; and the third way is for fluid itself to be compressed or to expand. Hence we expect the storage coefficients to be dependent on the three bulk moduli, K, K_s, and K_f, as well as the porosity. Indeed, it is demonstrated in Sects. 2.4.4 and 3.2 that

$$S_e = \frac{1}{M} = \frac{\phi}{K_f} + \frac{1-\phi}{K_s} - \frac{K}{K_s^2} \tag{1.42}$$

$$S_\sigma = C = \frac{\phi}{K_f} + \frac{1}{K} - \frac{1+\phi}{K_s} \tag{1.43}$$

where M and C are poroelastic constitutive constants.

Typically, the solid constituent is less compressible than both the fluid or the frame, $K_s \gg (K, K_f)$; hence (1.42) and (1.43) can be approximated as

$$S_e = \frac{\phi}{K_f} \tag{1.44}$$

$$S_\sigma = \frac{\phi}{K_f} + \frac{1}{K} \tag{1.45}$$

Another storage coefficient of interest is that used in groundwater aquifer theory [13, 41]. The specific storativity S_s is defined as the volume of water released from storage per unit volume of porous medium per unit decline of piezometric head under uniaxial strain and constant overburden condition [13]. Or, adjusting the above definition from per unit decline of head to per unit decline of pressure, we obtain the constant stress uniaxial storage coefficient S as

$$S = \frac{S_s}{\gamma_f} = \frac{\phi}{K_f} + \frac{3}{3K + 2G} \tag{1.46}$$

where γ_f is the specific weight of water.

1.2.7 Thermoelasticity Analogy

The *coupled theory of thermoelasticity* was established by Biot [17] on the basis of irreversible thermodynamic processes [177]. In the same paper, Biot also

demonstrated the mathematical equivalence between the governing equations of poroelasticity and thermoelasticity.

For poroelasticity, the following governing equations are presented in Sect. 6.3:

$$\mu\nabla^2\vec{u} + (\lambda + \mu)\,\nabla e - \alpha\nabla p = 0 \tag{1.47}$$

$$\frac{\partial p}{\partial t} - \kappa M\nabla^2 p = -\alpha M\frac{\partial e}{\partial t} \tag{1.48}$$

where $\vec{u}$ is the displacement vector, $e = \nabla\cdot\vec{u}$ is the volumetric strain (dilatation), λ and μ are Lamé constants, κ is the permeability coefficient, and $1/M = S_e$ as defined in the above section. For thermoelasticity, the governing equations are [17, 20, 177]

$$\mu\nabla^2\vec{u} + (\lambda + \mu)\,\nabla e - \alpha_T\nabla T = 0 \tag{1.49}$$

$$\frac{\partial T}{\partial t} - \frac{k_T}{c_v}\nabla^2 T = -\frac{\alpha_T\mathcal{T}_o}{c_v}\frac{\partial e}{\partial t} \tag{1.50}$$

where T is the temperature increment from a reference temperature $\mathcal{T}_o$, k_T is the *thermal conductivity*, c_v is the *specific heat at constant strain*, and

$$\alpha_T = (3\lambda + 2\mu)\,\beta_l \tag{1.51}$$

is a thermoelastic constitutive constant, with β_l the *coefficient of linear thermal expansion*. Comparing the two sets of equations, (1.47) and (1.48) with (1.49) and (1.50), we observe the mathematical equivalence between pore pressure and temperature $p \leftrightarrow T$, and the coefficients $\alpha \leftrightarrow \alpha_T$, $\kappa \leftrightarrow k_T/\mathcal{T}_o$, and $M \leftrightarrow \mathcal{T}_o/c_v$ [43, 98, 176].

For thermoelasticity, it can be shown that the *entropy density* s satisfies the diffusion equation [17]

$$\frac{\partial s}{\partial t} - \frac{k_T(\lambda + 2\mu)}{c_v(\lambda + 2\mu + \alpha_T^2\mathcal{T}_o/c_v)}\nabla^2 s = 0 \tag{1.52}$$

which can be compared with the diffusion equation for poroelasticity,

$$\frac{\partial\zeta}{\partial t} - c\nabla^2\zeta = 0 \tag{1.53}$$

where ζ is the *variation in fluid content*, defined as the amount of fluid volume entering the solid frame per unit volume of solid frame, and c is the consolidation coefficient, given by

$$c = \frac{\kappa M(\lambda + 2\mu)}{\lambda + 2\mu + \alpha^2 M} \tag{1.54}$$

Here we see the analogy $\zeta \leftrightarrow s$. In the following, we shall use the above established analogies to visualize a few poroelastic effects.

Envision a solid body whose sides are not constrained by forces and are free to move. If the temperature of the solid is raised, it will expand, a phenomenon known as *thermal expansion*. Similarly, given a poroelastic body that is unconstrained, if the pore pressure inside the body is raised by injecting fluid into it, the frame will likewise expand. This can be viewed as a mathematical analogy with thermoelasticity; or, we may consider this as a result of effective stress. As the total stress is zero, a positive pore pressure gives rise to a negative effective compressive stress; that is, the body is under effective tension, hence it expands.

Consider a different situation. The solid body is constrained from movement on all sides and the temperature is raised. A *thermal stress* develops, sets the body into compression. Similarly, for poroelasticity, when fluid is injected into a constrained body to raise its pressure, a compressive stress is observed.

In Sect. 1.2.3 we discussed that when an poroelastic body is suddenly subjected to a compressive confining stress, a pore pressure rise is observed, known as the Skempton pore pressure effect. By analogy, when an elastic solid is suddenly compressed, we anticipate to observe a temperature rise. While this is indeed the case, the magnitude of the rise, however, is typically small, thus negligible. For this reason, the coupled thermoelasticity is often simplified to the *theory of thermal stress* [27], by ignoring the last term in (1.50), which is a heat generation effect related to the rate of solid deformation. The dropping of the heat generation term makes the theory of thermal stress *uncoupled*; that is, (1.50) can be solved independent of (1.49). This is certainly not the case for poroelasticity.

The above analogy can be further compared through the following experimental setup. Suppose that we test a thermoelastic and a poroelastic body for their bulk moduli based on formula provided in Sect. 1.2.1. For the thermoelastic body, we can conduct the test under the *adiabatic* condition, that is, by insulating the sample from heat flow, or under the *isothermal* condition, that is, by maintaining constant temperature throughout the test. Experience tells us that the two material constants measured, an *adiabatic bulk modulus* and an *isothermal bulk modulus*, are basically indistinguishable. For poroelasticity, we can identify the adiabatic condition with the undrained condition, and the isothermal with the drained, based on the above-mentioned analogy. Following the discussion in Sect. 1.2.1, we know that the drained and undrained bulk moduli are drastically different; hence the uncoupling of the thermoelasticity theory may be proper for most applications, the uncoupling of poroelasticity, on the other hand, should not be attempted.

As discussed above, the heating of an elastic body generates a thermal stress, and equivalently, the flooding (injection of fluid mass) of a poroelastic body generates a *poroelastic stress*. Similar to the concept of a material constant known as thermoelastic stress coefficient, we can define its counterpart as a *poroelastic stress coefficient*. Under certain conditions, there exists a simple relation between the pressure increment and the generated poroelastic stress. This condition exists in an unbounded (large) domain that is initially under a constant confining stress and pore pressure. When a portion of the domain is flooded by fluid injection to

create a distribution of pore pressure increment p, the corresponding generation of poroelastic confining stress is given by [42]

$$-\frac{\sigma_{xx}+\sigma_{yy}+\sigma_{zz}}{3}=\frac{4}{3}\eta p \tag{1.55}$$

in which η is a poroelastic stress coefficient, given by

$$\eta=\frac{\alpha(1-2\nu)}{2(1-\nu)} \tag{1.56}$$

where ν is the *Poisson ratio*. The coefficient has the value $0 \leq \eta \leq 0.5\alpha \leq 0.5$.

Or, in the case of a *plane strain* condition, that is, the body is confined from movement in one of the directions, $u_z = 0$, the poroelastic stress generated becomes

$$-\frac{\sigma_{xx}+\sigma_{yy}}{2}=\eta p \tag{1.57}$$

And if the displacement is allowed only in one direction, elected as x, we observe

$$-\sigma_{xx}=2\eta p \tag{1.58}$$

Typical values of η are given in Table 3.2.

1.2.8 Coupled Versus Uncoupled Diffusion

As mentioned in Sect. 1.2.2, Terzaghi's original consolidation theory [234] was characterized by a one-dimensional diffusion equation given as (1.13). To deal with multi-dimensional consolidation problems, the theory has been generalized in an ad hoc fashion to a three-dimensional form by Rendulic [200] and Terzaghi [237], as the following

$$\frac{\partial p}{\partial t}-c\nabla^2 p=0 \tag{1.59}$$

The Biot theory [16], however, shows that the simple diffusion equation is satisfied not by p, but by ζ, the relative volumetric strain between the porous medium frame and the fluid, as given by (1.53). Based on its definition, ζ is dependent not only on the pore pressure, p, but also on the volumetric strain, e. Converting (1.53) to these variables, we obtain (1.48), which shows that the pressure diffusion is coupled with the solid deformation, and cannot be independently solved. The right hand side of (1.48) shows that the rate of change of volumetric strain serves as a source term for pore pressure; that is, as the pore pressure is being dissipated, the solid continuously deforms (due to the change in effective stress), and new pore pressure

is continuously generated. Or, another way to view the pore pressure diffusion equation is to express it in the following form

$$\frac{\partial p}{\partial t} - \frac{\kappa KM}{K_u}\nabla^2 p = -\frac{\alpha M}{3K_u}\frac{\partial \sigma_{kk}}{\partial t} \tag{1.60}$$

where $\sigma_{kk} = \sigma_{xx} + \sigma_{yy} + \sigma_{zz}$. The above equation shows that the rate of change of compressive stress also serves as a pore pressure production term.

In conclusion, the poroelasticity theory shows that pore pressure diffusion is a process coupled with the solid deformation, and the Terzaghi-Rendulic three-dimensional uncoupled diffusion equation (1.59) is missing an important physical mechanism, hence is incorrect. We should however mention that Terzaghi's one-dimensional consolidation equation (1.13), as a special case, is in fact consistent with (1.48) and (1.60), if the soil stratum is restricted to deform only in the vertical direction, and the surcharge on top of soil remains constant with time.

1.3 Poroelastic Phenomena

With the basic understanding of the physical mechanisms underlying the poroelastic effects, we are ready to examine a few observed poroelastic phenomena and to explore their physical interpretations.

1.3.1 Borehole Failure

In deep geological formations, rock is subjected to a large compressive stress, which is either caused by the weight of rock mass above it, or by tectonic movement. If a circular hole is drilled through the formation, and left empty, elasticity solution predicts a maximum stress located on the borehole wall in the circumferential direction, which is twice that of the in situ stress. If this maximum compressive stress exceeds the ultimate compressive strength of the rock, it will fail instantly. The collapse of a borehole, however, is sometimes found to be time-delayed; that is, not happening at the time of the drilling, but at some later time [83, 183, 196]. Elasticity theory cannot explain such phenomenon. For shale, which is characterized by extremely low permeability and high chemical activity, this delay can be days to weeks. While this delay can be attributed to a number of factors, such viscoelastic creep and chemical reaction of shale, poroelastic effects certainly play an important role. We shall examine one such scenario of delayed borehole failure below.

Assume that a circular borehole is instantly drilled in an infinite domain initially under a uniform stress, $\sigma_{xx} = \sigma_{yy} = -P_o$, and constant pore pressure $p = p_o$ (Fig. 1.3). At the borehole wall, $r = a$, such conditions need to be satisfied: $\sigma_{rr} = 0$ and $p = 0$. For the elastic response, the described problem is just the classical Lamé problem [147], whose solution in terms of stresses is given by [129, 246]:

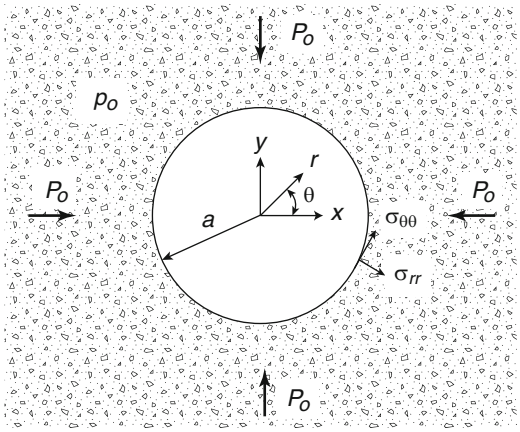


Fig. 1.3 Borehole in an uniform stress field

$$\sigma_{rr} = -P_o\left(1 - \frac{a^2}{r^2}\right) \tag{1.61}$$

$$\sigma_{\theta\theta} = -P_o\left(1 + \frac{a^2}{r^2}\right) \tag{1.62}$$

where σ_{rr} and $\sigma_{\theta\theta}$ are the normal stresses in the radial (r) and circumferential (θ) directions, respectively. We can easily observe from the above solution that as $r \to \infty$, $\sigma_{rr} = \sigma_{\theta\theta} = -P_o$, and as $r \to a$, $\sigma_{rr} = 0$, fulfilling the stated boundary conditions. Of interest to observe is that at the borehole wall a stress concentration is developed in the circumferential direction,

$$-\sigma_{\theta\theta}|_{r=a} = 2P_o \tag{1.63}$$

which is twice the magnitude of the original compressive stress.

As discussed in Sect. 1.2.1, the poroelastic body at the instant of loading is undrained, and behaves as an elastic body. Hence the above solution, (1.61) and (1.62), is indeed the stress state for a poroelastic body at $t = 0^+$. In Sect. 1.2.3, we also showed that at that instant, there is a pore pressure rise proportional to the increment of average compressive stress, given by (1.19) for plane strain condition. There is, however, a curiosity about the present solution, (1.61) and (1.62). It is easily shown that

$$-\frac{\sigma_{rr} + \sigma_{\theta\theta}}{2} = -\frac{\sigma_{xx} + \sigma_{yy}}{2} = P_o \tag{1.64}$$

So there is no change in the *average* compressive stress everywhere in the formation, before and after the excavation of the borehole. This means that there is no new pore pressure generated, and it stays everywhere as p_o. With these results, we are ready

to conclude that at the instant of the excavation, there is a Terzaghi effective hoop stress of the magnitude

$$-\sigma'_{\theta\theta}\big|_{r=a,t=0+} = 2P_o - p_o \tag{1.65}$$

developed around the borehole wall, which is increased from the original formation effective stress of $P_o - p_o$.

Next, we consider the stress condition around the borehole at large time. As time progresses, two mechanisms are taking place. One is more obvious: the pore pressure at and near the wall is dissipated, $p \to 0$; hence the protection provided by pore pressure against effective compression no longer exists. A less obvious mechanism can be explained using the thermoelastic analogy discussed in Sect. 1.2.7. The dissipation of pore pressure is equivalent to the cooling of solid around a hole. As the temperature at larger distance remains high, a tensile stress develops around the hole. In the poroelastic case, a relief of the compressive stress in the amount ηp_o results (see (1.57)). Hence the final effective hoop stress at the borehole is

$$-\sigma'_{\theta\theta}\big|_{r=a,t\to\infty} = 2P_o - \eta p_o \tag{1.66}$$

Comparing (1.65) with (1.66), we observe an increase of effective compressive stress near borehole wall of the magnitude $(1 - \eta)p_o$, which can be a cause for the delayed failure [42, 70].

In a drilling operation, to prevent the borehole from collapsing, the hole is filled with a drilling mud. The hydrostatic pressure of the mud pushes back the formation to provide support on the wall. However, when the mud weight becomes too large, it can more than compensate the inward compression, and push the borehole wall into tension. If the tension exceeds the tensile strength of the rock, fracture can initiate at the wall, and propagate into the formation. This is generally an unwanted result.

However, there are occasions that fractures are intentionally created in an operation called *hydraulic fracturing* [116]. In such application, an uncased section of the borehole is sealed off at top and bottom using packers. Fluid is injected into this interval to raise the pressure to initiate a fracture, and then continuously to drive it into the formation. The artificially created large fracture can be used to increase the hydraulic conductivity of reservoir to stimulate the hydrocarbon production. In such operations, it is of interest to predict the *borehole breakdown pressure*, p_b, that is, the minimum borehole pressure that can initiates a fracture.

As early as 1953, Scott et al. [214] observed in a laboratory setting simulating hydraulic fracturing that thick-walled hollow cylindrical cores could be ruptured using penetrating fluids at less than half the pressures required as compared to using non-penetrating fluids. As the difference between these two cases is fluid penetration into the rock specimen, it is apparent that poroelastic effects are at play.

Based on only elasticity consideration, the breakdown pressure for a borehole in a non-hydrostatic (anisotropic) stress field can be predicted by

$$p_b = 2P_o - 4S_o + T \tag{1.67}$$

where P_o and S_o are respectively the far-field mean stress and stress deviator, and T is the tensile strength of the rock. To take into consideration the effective stress in the presence of pore pressure, Hubbert in 1957 [122] modified the above formula, assuming non-penetration fluid and that the pressure at the borehole wall is that of the formation pore pressure p_o, to the following

$$p_b = 2P_o - 4S_o + T - p_o \tag{1.68}$$

As the pore pressure at the wall is generally not that of the virgin pore pressure, Detournay and Cheng [70] presented the upper bound of breakdown pressure as

$$p_{bu} = 2P_o - 4S_o + T - 2\eta p_o - (1 - 2\eta)p_i \tag{1.69}$$

where p_i is the pore pressure at the borehole wall, $p_o > p_i > 0$, and the coefficient η arises from the poroelastic stress effect, as explained in Sect. 1.2.7. Haimson and Fairhurst [106, 107] considered the penetrating fluid case to give

$$p_{bl} = \frac{2P_o - 4S_o + T - 2\eta p_o}{2(1 - \eta)} \tag{1.70}$$

where p_{bl} stands for the lower bound of breakdown pressure. Depending on the mud used for fracturing, the true breakdown pressure should be between these two limits, p_{bl} and p_{bu}.

Small scale hydraulic fracturing can be used as a way to measure the in situ stress of a formation [82, 139]. In such practices, pressure is continuously raised in a section of the borehole, until a sudden drop-off of pressure is observed, reflecting a situation where the rate of fluid loss, due to fracture extension and fluid leakoff, becomes greater than the pumping rate. Further injection of fluid causes the fracture to propagate, and reorient in the direction perpendicular to the minimum in situ stress. After pumping is stopped and the well is *shut in*, the pressure in the sealed-off interval decays because of further fracture propagation and fluid leakoff into the formation. At some stage, the newly-created fracture will close. Interpretation of in situ stress from hydraulic fracturing tests relies on the *fracture closure pressure* measured during shut-in, and also on the *breakdown pressure*, measured during the first pressurization cycle, or the *reopening pressure*, measured during subsequent pressurization cycles [105, 108, 112]. Poroelastic effects in the determination of in situ stress using the above information have been investigated [72, 73].

In the above we discussed borehole failure when the stress level exceeds its compressive or tensile strength. Borehole can also fail by shear, when the ratio of shear to normal stress exceeds the shear strength of the material, based on the *Mohr-Coulomb* failure theory. This is often observed as borehole *breakout* in deep wells; that is, the spalling of rock on two symmetric sides of the wall, making the hole into an elongated shape. Elasticity solution indicates that in the presence of an anisotropic stress field, breakouts are aligned in the minimum principal in situ stress direction; hence breakout observation has been used as a way to determine the in situ

stress directions [100, 189, 276, 278]. Elasticity solution also predicts an initiation of shear failure right at the borehole wall. Poroelasticity solution, however, shows that the shear failure initiation can take place at a small distance, about 5–10 % the borehole radius, inside the borehole wall [42, 70], which may explain the finite size rock debris, and the progressive failure in time, found during breakout events.

1.3.2 Mandel-Cryer Effect

To demonstrate the coupled effect of pore pressure diffusion and solid deformation in the poroelasticity theory, in contrast to the uncoupled diffusion in the Terzaghi-Rendulic three-dimensional consolidation theory (see Sect. 1.2.8), Mandel in 1953 [160] and Cryer in 1963 [55] each presented analytical solution that exhibits the curious response of non-monotonic pore pressure diffusion, characterizing the coupled theory. Such phenomenon of pore pressure continues to rise after its initial generation, before it declines to zero due to diffusion, has been called the Mandel-Cryer effect [211].

The Mandel problem is illustrated in Fig. 1.4a, which shows a poroelastic specimen sandwiched between two rigid, impermeable platens with frictionless surfaces. A force F is suddenly applied to these rigid platens, resulting in a uniform pore pressure rise p_o throughout the specimen, based on the Skempton effect. Because the sides of the specimen are exposed to atmosphere, this pressure will drive a horizontal seepage flow toward the drainage surfaces, and dissipates to zero with time. If the pore pressure obeys the uncoupled diffusion equation (1.59), the decline will be monotonic, from p_o to zero. The Mandel solution, however, shows that the pore pressure will first rise above the initially generated value, before it falls. This phenomenon can be explained as follows.

As the porous specimen is homogeneous, at the instant of the applied force, the load is uniformly distributed, giving a constant vertical compressive stress $-\sigma_{zz}$

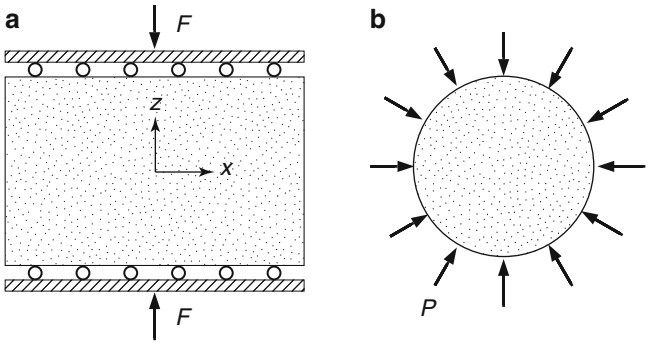


Fig. 1.4 (**a**) Mandel problem, and (**b**) Cryer problem

throughout the specimen. As the pore pressure starts to dissipate near the left and right edges, the material effectively softens as the fluid is taking less load. Effectively, the material becomes inhomogeneous with a harder core and softer edges. This will cause the load to redistribute, resulting in a higher $-\sigma_{zz}$ near the center. Eventually, the entire pore pressure will dissipate, and σ_{zz} will return to uniform again.

As demonstrated in (1.60), the rate of increase of compressive stress is a pore pressure generation mechanism. Hence at any point inside the specimen, the pore pressure is experiencing two opposing trends, a diffusion process making it to monotonically decline, and a pore pressure generation making it to rise. As the diffusion process takes time, at any point inside the specimen, it will first experience an elastic load transfer due to the diffusion taking place nearer to the edges, before the arrival of the diffusion front to dissipate the pressure. Hence we observe a non-monotonic behavior.

The Cryer problem is similar to the Mandel problem, but with a different geometry. Cryer solved the problem of a saturated spherical soil specimen suddenly compressed by an inwardly pointing uniform radial stress on its surface (see Fig. 1.4b). At $t = 0^+$, a uniform compressive stress and pore pressure develop everywhere inside the specimen. At the surface of the sphere, water can escape freely, so the specimen will consolidate in time. Cryer's analysis showed that during the early stages of consolidation the pore water pressure at the center of the sphere could rise almost 60 % above the initial pressure, before it starts to decline.

The physical explanation for the non-monotonic response is similar to that of the Mandel problem. As the pore pressure near the surface starts to dissipate, effectively the sphere becomes an inhomogeneous material with a softer outer shell and a stiffer inner core. This will result in an increase of total compressive stress in the inner part of the sphere. (Or, consider the opposite: a harder outer shell can protect an inner softer core, subjecting it to a lower stress.) This increase of compressive stress again serves as a pore pressure generation mechanism, giving the non-monotonic pressure behavior. This phenomenon predicted by the Cryer solution [55] was soon demonstrated in the laboratory [99, 255].

1.3.3 Noordbergum Effect

Verruijt in 1969 [256] reported a phenomenon in the well field of the small village of Noordbergum, in the Netherlands. When pumps were turned on to extract water from the upper, confined aquifer of a two-layer leaky aquifer system, it was anticipated that the water level (piezometric head) will drop in both the pumped upper aquifer and the unpumped lower aquifer, based on the leaky aquifer theory [110]. It was however curiously observed that at a distance from the pumping wells, water level in monitoring wells located in the lower aquifer actually rose for a period of time, before dropping. Verruijt [256] provided an approximate analytical solution based on poroelasticity theory, and demonstrated such reversal effect, which has

been called the *Noordbergum effect* [205]. Verruijt explained such phenomenon by referring to poroelastic effects as follows: "*Due to the decreasing pore water pressures in the upper layer; this layer will attract water from the upper part of the lower layer. The loss of water involves a decrease in volume of the upper part of the lower layer. All elements of a circular ring around the origin will decrease in volume, thus producing a radial displacement towards the center to keep the ring closed. For the lower parts of the layer, this radial displacement presents a loading due to which a tendency for compression will be present. Since the pore water opposes volume compression of the soil, this results in an increase of the pore water pressure*." Verruijt compared the solution with pumping tests conducted at the polder Vierbannen, on the island of Schouwen-Duiveland, the Netherlands [253], and obtained good agreement.

Wolff [262] pointed out that the "abnormal" water level responses after pumping activities had been observed in many occasions since 1936 [9]. These phenomena were generally referred to as *reverse groundwater level fluctuations*. Langguth and Treskatis [148] reported a similar phenomenon observed in the well field of Rhade Lowland near Dorsten, in southern Münster basin, Germany. In a well installed with a multilevel piezometer to monitor piezometric head in the aquifer as well as the overlain aquitard, when pumping started in the aquifer, the Noordbergum effect was observed in the aquitard. When the pump is shut off, the piezometric head in the aquitard dropped, rather rapidly, for a short while, before it recovered to the original level before pumping. Hence we may refer the decline in head in the adjacent aquifer or aquitard at the beginning of groundwater pumping the *Noordbergum effect*, and the drop in head after the end of pumping the *Rhade effect*. Based on numerical solutions, Kim and Parizek [141, 142] were able to simulated both effects using poroelasticity theory.

The above discussion involves the observation of reverse water level in aquifers and aquitards adjacent to the pumped aquifer. In fact, this type of behavior can exist in the pumped aquifer as well. The Theis solution of groundwater flow [13, 41] predicts that as pumping starts in a confined aquifer, the head in the aquifer declines everywhere. Numerical simulation based on poroelasticity, however, shows that at a distance from the well, the head rises first, before it falls [104, 272].

In Fig. 1.5 we provide a simple illustration of such a physical process, but in reversal—that is, rather than extracting from the aquifer, water is injected into it. The aquifer is assumed to be confined from above and below by impermeable layers. The overburden layer on top is assumed to have a certain degree of flexural rigidity, that is, a non-zero shear strength. As the formation is being flooded by the injected water, there is a cylindrical region surrounding the well with significant head rise such that the overburden formation is pushed up, like a subsidence in reverse. With a finite flexural rigidity, the formation arches up like a dome. In the region ahead of the significant pressure rise, the pressure first drops due to the uplift of the overburden layer, until the pressure front eventually reaches the region.

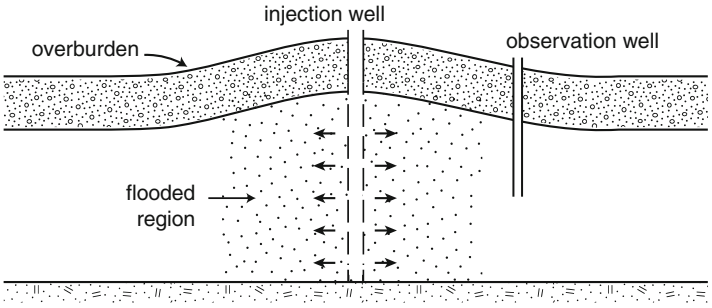


Fig. 1.5 Noordbergum-like effect due to uplift of overlain formation

1.3.4 Land Subsidence

Land subsidence is the settling of ground surface over large areas, due to the compaction of subsurface materials, primarily clay layers or lenses, usually as a consequence of pumping from underlying aquifers. It can also be the result of producing oil and gas from underground hydrocarbon reservoirs. The term *consolidation* is used for the similar land settlement phenomenon, when it occurs over relatively small areas, especially as a result of loading ground surface, e.g., by new structures.

The turn of the twentieth century witnessed an explosion in oil production and large scale irrigated agriculture. These activities very soon led to noticeable environmental effects including land subsidence. The first observed land subsidence due to subsurface fluid withdrawal was reported by Pratt and Johnson in 1926 [192] from the Goose Creek oil field, near Galveston, Texas. The first recorded land subsidence due to groundwater withdrawal was fortuitously discovered around San Jose, California in 1932, by the U.S. Coast and Geodetic Survey, during repeat precision-leveling [174, 198].

There exist many examples of subsidence over large areas as a result of pumping groundwater [190]. Perhaps, the most spectacular one in an urban area is in Mexico City, where almost the entire metropolitan area has subsided more than 3 m (with up to 8 m at some locations). As a result of excessive withdrawal of groundwater for irrigation, the San Joaquin Valley in California has experienced subsidence at a rapid rate of 30–40 cm per year, reaching a total subsidence of 9 m at some locations. Another well-known case is Venice, Italy, where all pumping has ceased in an effort to stop further land subsidence. Many measures have been suggested to prevent Venice from "sinking". One method suggests pumping seawater into deep aquifers to uplift the land surface [95, 232].

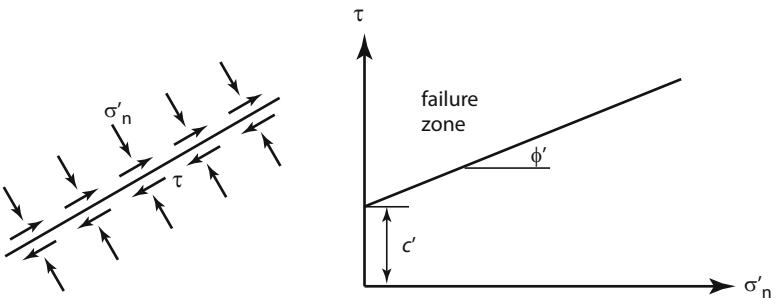


Fig. 1.6 Mohr-Coulomb shear failure criterion

1.3.5 Slope Stability and Fault Slippage

Mohr-Coulomb theory is a widely accepted criterion of shear failure for brittle and granular materials, or materials for which the compressive strength far exceeds the tensile strength. Most porous materials fall into this category. The criterion is a linear relation between the shear stress τ and compressive normal stress σ_n (for convenience, compression is considered positive in this section only), given as

$$\tau = c' + \sigma_n \tan \phi' \tag{1.71}$$

where c' is an internal strength of the material, known as *cohesion*, and ϕ' is the *angle of internal friction*. The above equation defines a *failure envelope* on the σ_n-τ plane (see Fig. 1.6). When a pair of normal and shear stress falls above the envelope, the shear stress is then too large for the strength of the material to sustain, and the material will fail.

Terzaghi [238] has modified the above equation by replacing the normal stress with the effective normal stress, to become the following

$$\tau = c' + (\sigma_n - p) \tan \phi' \tag{1.72}$$

which has been successfully used to predict the slope stability of soil embankment. Although there exists the more sophisticated *critical state* soil failure theory [212, 213], for the present purpose, we shall use the Mohr-Coulomb theory to explain the role of pore pressure in the failure of porous materials.

It is well known in soil mechanics that soil saturation and the rise of water table tend to destabilize embankment, increasing the risk of landslide. When water table rises, it has two effects: first, it adds to the overburden weight on the underlying soil, increasing its normal stress; and second, it develops a hydrostatic pore water pressure, which reduces the effective normal stress according to (1.72). The normal stress added by the weight of water is proportional to the water density, the elevation of the water table, and the porosity. The reduction of normal stress, however, is

dependent only on the water density and elevation of water table, and independent of porosity. Effectively, there is a reduction of normal stress that is equivalent to subjecting the solid to a buoyancy force by water. As the shear stress is unaffected by the water saturation, the reduction in normal stress tends to shift the stress state on the σ_n-τ plane to the left, such that it may cross the failure envelope, leading to sliding failure.

In a partially saturated (capillary) zone, the pore pressure is negative, due to the surface tension of water meniscus. Based on the effective stress concept, there is an increase in the effective normal stress, which tends to stabilize the soil. This is how one can build sand castle using wet (partially saturated) sand, and not dry sand. On the other hand, if the castle is submerged in water and become saturated, it collapses.

Similar stability issue exists for fractures or faults found in geological formations. Pre-existing fractures and faults are discontinuity surfaces that have little or no tensile strength perpendicular to the surface. Its strength against sliding is dependent on the material strength as well as the roughness of the discontinuity surface. A normal force can reduce the potential of sliding, with a stability criterion similar to that of Mohr-Coulomb law. Increasing pore pressure around a fault causes reduction in effective normal stress and can induce fault movement.

In 1959 Hubbert and Rubey [121, 209] introduced such concept to explain the puzzling paradox of the *overthrust faulting* found in Earth's crust. In a thrust fault, a large block of rock sheet is thrust to slide against another rock sheet, and is moved over a great distance. In these provocative papers written at that time, Hubbert and Rubey calculated the force needed to thrust one rock sheet to overcome the friction between the two sheets and to slide, and concluded that the force would far exceed the crushing strength of the rock in the typical overthrust faults found in Canadian Rockies and Appalachians. They then hypothesized the role of fluid pressure in the effective stress concept. They argued that if fluid pressures within the faulting plane became sufficiently high, the effective friction strength could dramatically decrease, enabling the transport of enormous rock blocks over long distances with smaller thrusting force.

To allow people to visualize the effect, Hubbert and Rubey [121] experimented on the "beer can experiment", originally suggested by M. A. Biot. In such experiment, an emptied beer can is placed upside down over a clean glass plate wetted with water. The plate is raised to a certain angle to allow the can to slide down the plane due to gravity pull. Hubbert and Rubey reported that the inclination angle needed was about 17° (for the heavier beer can in those days). The beer can is then chilled in a freezer, taken out, and the experiment is repeated. A small amount of time is allowed for the air entrapped in the can to warm and for the air pressure to increase. The beer can is then able to slide down the plane at less than 1° inclination. The reason is that the air pressure produces an uplift force that offsets the normal stress exerted on the glass plate. Nowadays the concept of pore pressure and effective stress is well received in structural geology [60].

1.3.6 Fluid Induced Seismicity

An geological event that is of man-made origin is the fluid induced seismicity [220]. It has been reported that during the injection of liquid nuclear waste into deep formation at a site near Denver, Colorado, an M5.5 earthquake was generated [113, 118]. Similarly, water injection at Cogdell Canyon Reef, Texas, for the purpose of secondary oil recovery, induced a M4.6 earthquake [62]. In the hot dry rock geothermal project at Fenton Hill, New Mexico, hydraulic fracturing is used to create flow path to enhance water circulation. During the hydraulic fracturing process, hundreds of micro earthquakes were detected [184, 187].

In yet another man-made situation, the impoundment of reservoirs can induce seismic events, which was reported as early as 1945 by Carder [36], and many more evidences were collected by Simpson [225]. The underground stress environment is changed not only by the weight of impounded water, but also by the invading pore pressure induced by seepage, which can alter the effective normal stress of pre-existing fracture. A number of reservoir impoundment induced earthquakes exceeding M6 have been recorded all around the world [102]. The 1967 Koyna, India earthquake of M6.3 is so far the largest and most damaging reservoir triggered earthquake [103]. A poroelastic modeling of the fault stability due to cyclic water level fluctuation has been conducted [206].

The pore pressure effect can manifest itself in many different ways. Not only the injection of fluid into formation, but also the extraction of fluid from it, can induce seismic events. As discussed in Sect. 1.2.7, the flooding of a smaller region contained within a larger (infinite) domain with a uniform pressure p induces an apparent increase of far-field confining stress of the magnitude $4\eta p/3$. At the same time, within the flooded region, the pore pressure also causes a reduction in compressive effective stress of the magnitude p. Hence the net effect is to destabilize a fracture within the flooded region. Extracting fluid has the opposite effect. Within the region of pressure decline, it experiences a reduction of far-field confining stress. This reduction, however, can extend a few distances beyond the size of the fluid decline region, with diminishing magnitude, as controlled by the elastic behavior. Outside the pressure decline region, there is no pore pressure to change the effective stress, hence the reduction in normal stress can destabilize a fracture outside the extraction region and cause it to slip. Fluid extraction induced earthquakes in fact have been observed [185, 216, 268], and modeled as a poroelastic effect [217, 229].

In the above we have discussed events triggered by man-made actions perturbing the underground environment. Pore pressure can also play an important role in many naturally occurring geophysical events, such as earthquakes [52, 91]. Nur and Booker [28, 178] were among the first to suggest that the time delayed *aftershock* events following a major earthquake can be explained by the poroelastic effects. Large shallow earthquakes alter the stress in the surrounding crust. While some parts of the earth may be relaxed, other parts are compressed. This stress redistribution can induce changes in the fluid pore pressure with magnitude that are comparable to stress drops on faults. The subsequent fluid flow and pore pressure redistribution can

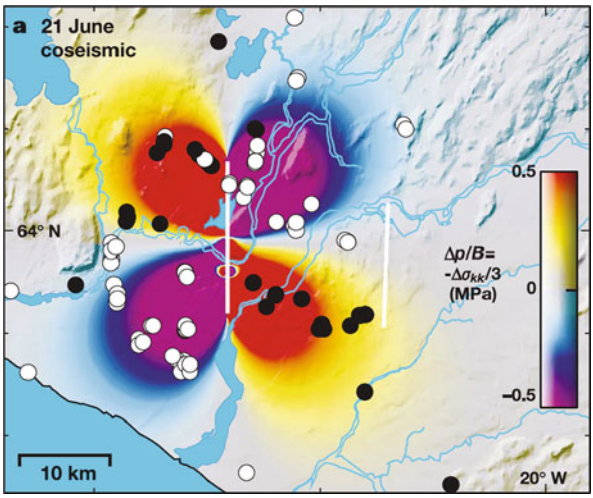


Fig. 1.7 Coseismic water level changes in geothermal wells in South Iceland. Water level increase is shown in black dots and decrease in white dots (From Jónsson et al. [136], with permission)

cause the pressure to decrease in the compressed region, and increase in the dilated region. Many smaller faults and fractures that are momentarily at rest after the major event can be at the critical state against sliding. The invasion of pore pressure can reduce the effective normal stress locally and trigger the aftershock events [218].

Other poroelasticity effects include the earthquake precursory effects due to pore fluid stabilization of a weakening fault zone [202], and the pore water diffusion to the dilating tip of the shear rupture zone (*shear band*) to stabilize it against rapid propagation in over-consolidated clay [181].

In Fig. 1.7 we present an interesting seismic observation. The South Iceland seismic zone is a left-lateral transform zone, where east-west transform motion is accommodated by north-south-oriented right-lateral faults. The earthquakes on June 17 and 21, 2000, ruptured two of these faults. During the earthquake period, water level changes were observed in numerous geothermal wells. The sign of the water level changes exhibits a quadrant pattern, as shown in Fig. 1.7 [136]. In Chap. 8 we shall derive the poroelastic solution of a *slip point dislocation*. A point dislocation is a *displacement discontinuity* at a point where the formation breaks apart and moves in opposite directions. The motion can be in an opening mode or a slipping mode. In Fig. 1.8 we plot the contour lines of pore pressure response created by the slipping motion. We can see the resemblance of the pattern of pore pressure field as compared to the field observed water level after the slippage of fault, shown in Fig. 1.7.

1.3.7 Outburst of Coal

Underground coal gasification is an in situ gasification process carried out in low grade coal seams that are not economical to mine, and the coal is burned underground without sufficient oxygen to produce methane gas. Injection wells are

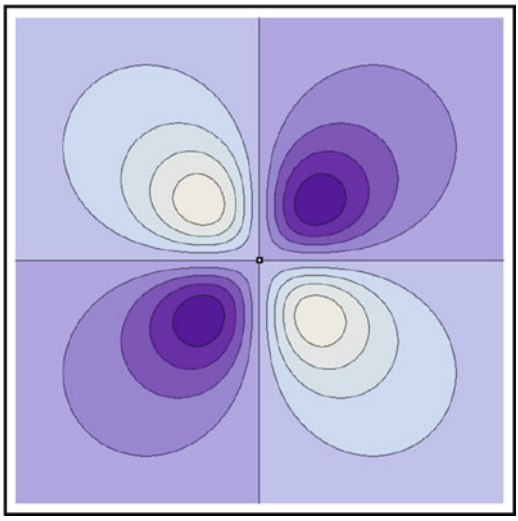


Fig. 1.8 Contour plot of pore pressure generated by a slipping displacement discontinuity

used to supply the oxidants (air, oxygen, or steam) to ignite and fuel the underground combustion process. Separate production wells are used to extract the product gas to surface.

In order to create an initial surface large enough to start the burning process, either hydraulic fracturing or the *cavitation* technique has been be employed [153, 163]. For the cavitation technique, in a process called *open-hole completion*, air is injected into coal seams through an uncased wellbore interval with high pressure. The pressure is sustained for a period of time to allow it to permeate into the formation, and then suddenly released. The sudden drop in pressure induces tensile failure of the coal in various orientations around the wellbore. As the weak, friable coal sloughs in, the wellbore is enlarged. (See Sect. 7.15.2 for a theoretical exposition.) This process can be repeated several times. In fact, this process can be a byproduct during production in a *dynamic open-hole completion* technique, in which air or an air-water mixture is repeatedly injected into an open-hole interval, followed by immediate release of wellbore pressure, and the cavity is continuously being enlarged during production.

This cavitation phenomenon can be attributed to the poroelastic effects. When the wellbore is pressurized for a period of time, a high gas pressure region is built up surrounding the borehole. Just before venting, the hole is stable as there is a radial stress pushing against the wall. However, when the air in the hole is instantly vented, the radial compressive stress on the wall is reduced to zero. Inside the formation, due to the presence of an overpressure, the effective radial stress becomes in tensile range. Since the tensile strength of coal is very small, the wall collapses by tension. This phenomenon has been simulated as a consequence of the poroelastic mechanism [37, 38, 42].

In fact, in coal mine operation, it is well known that when excavation reaches regions over-pressurized by methane gas, and the surface is exposed and

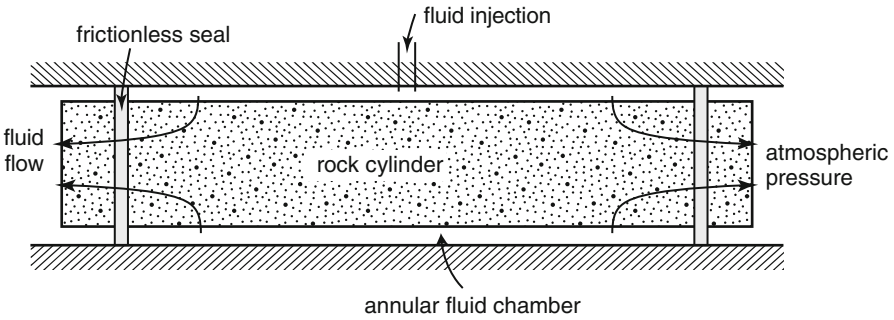


Fig. 1.9 Pinch-off test to break a rock cylinder by tension

unsupported, an explosive type of failure can take place, known as *outburst* [12]. Hence it is important to drill pilot holes ahead of mining excavation to vent methane gas, for the purpose of preventing not only the ignition of the combustible gas, but also outburst and cave-in. The poroelastic interpretation of this phenomenon has been investigated [182, 275].

The above described phenomenon may sound curious—there is no external physical force pulling the coal, yet it breaks off by tension! Is it possible? In fact, this effect can be demonstrated in laboratories using the so called "pinch-off" test of Bridgman [32], and modified by Jaeger and Cook [128]. Cylindrical rock core can be placed in a fluid chamber as illustrated in Fig. 1.9. Radial seals are positioned near the ends to confine the fluid in the chamber. Seal lubricant is applied to minimize the friction resistance in the cylinder axial direction [128]. Water is injected into the annular chamber between the core and vessel. The ends of the core are open to atmosphere so that there is no axial stress applied to the core, and fluid is allowed to flow out at the ends by permeation [34]. Fluid pressure in the chamber is slowly increased until the core ruptures by tension. The breakage takes place when the fluid pressure is of the order of, and slightly larger than, the tensile strength of the rock core [128]. A simple explanation is that although the axial stress σ_{zz} is zero, the Terzaghi effective axial stress σ'_{zz} is in tension with the magnitude of the pore pressure [127].

1.3.8 Hydraulic Fracturing

Hydraulic fracturing, a well stimulation technique first suggested by Clark in 1949 [47], has been widely used in oil and gas industry for enhancing underground hydrocarbon recovery. The mechanics of hydraulic fracturing was first rigorously examined by Hubbert in 1957 in his classic paper [122]. Later development in the theory and technology is well covered in a few comprehensive books [79, 116, 250, 269].

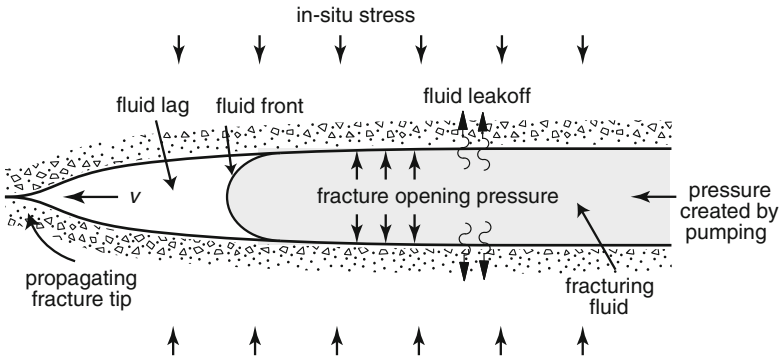


Fig. 1.10 A horizontal cross-section of a vertical hydraulic fracture

The concept of hydraulic fracturing is to greatly increase the production surface area of porous medium, where oil or gas can be collected, from that of a borehole wall to a massive fracture driven into the formation. The fracture is created by pumping at high pressure a fracturing fluid into a section of the wellbore to first break the formation by tension, and then continuously driving a contained fracture, typically vertical in orientation, into the *pay zone* (hydrocarbon containing formation). Containment of the fracture is important because penetration into the adjacent formations may mean producing a mixture of water and oil that is difficult to separate, or contaminating the water bearing aquifers. With careful design and field control, fractures can reach several hundred feet in distance.

To drive a fracture, a significant amount of pumping pressure is needed to overcome not only the formation in situ stress pushing to close the fracture, but also the large flow resistance caused by a number of factors, including the narrowness of the aperture, the roughness of the fracture surface, and the high viscosity of the fracturing fluid (see Fig. 1.10). Using a model combining solid fracture mechanic and non-Newtonian fluid flow in the aperture, it is possible to predict the pumping pressure required to create a designed fracture. It was found that these models generally underestimate the pumping pressure needed to create a given fracture, even if the fluid loss into the formation was taken into consideration [175, 228].

One mechanism proposed to account for the extra pumping pressure needed is the poroelastic stress effect, otherwise known as the *backstress* in hydraulic fracturing terminology [50, 51, 73]. As the fluid pressure in the fracture is higher than the formation pressure, a loss of fracturing fluid through infiltration into the formation will take place, known as *leakoff*. With a rise of pore pressure, the flooded region will expand, as explained by the thermoelasticity analogy in Sect. 1.2.7. This expansion tends to close the fracture, as if there exists an extra confining stress at the far-field; hence it is called a “backstress”.

Other poroelastic effects considered include the gradual opening of a stationary hydraulic fracture due to the large time, drained response of the formation in the far-field, making the material effectively more compliant. This is then offset by

the backstress effect due to the fluid penetration in the near-field, resulting in a combined large time fracture response of gradual decrease in fracture width [71].

The small and large time (undrained and drained) response of poroelastic medium also causes a time dependent stress concentration on the fracture tip [71]. Based on the same concept, the speed of a propagating fracture tends to have a stabilization effect, as the higher the speed, the more undrained (stiffer) the material behaves, hence resisting the propagation at an even higher speed [44, 203, 224]. If the stress intensity factor falls between the drained and undrained limits, a stable, constant speed propagating fracture can be maintained.

In another utilization of poroelastic effects, it is possible to attract a propagating fracture toward a target well by creating a favorable propagation path. In many applications, it is highly desirable to connect two wells with a fracture. Under normal condition, it is nearly impossible to control the fracture path other than utilizing the in situ minimum principal stress direction. With poroelastic effect, it is possible to pressurize a target well ahead of time to flood the region surrounding it, in order to reduce the effective compressive stress, and to attract the fracture initiated at another well toward the target well [15].

In present day, hydraulic fracture is used not only in oil and gas industry, but also in a number of other applications, such as injecting liquid nuclear waste into geological formations for isolation and disposal [67, 68], using water [77, 90, 149] or CO_2 [33] as injection fluid to create fracture to circulate water in enhanced geothermal systems (hot dry rock), creating horizontal fracture as a means of enhancing pump-and-treat, soil vapor extraction, and in situ environmental remediation in shallow soil, for pollutants such as heavy metals and hydrocarbon waste or spills [89, 173, 221], and stimulating groundwater production by connecting naturally occurring fractures in rock formations [247].

The hydraulic fracturing discussed above refers to the technology that creates a single, or a few, large fractures initiated mostly from a vertical well, and occasional from a horizontal well. With the advancement of the technology of directional drilling, and the need for new energy resources, a new technique, popularly known as *fracking*, has become wide spread in the oil and gas industry in the last decade. A vertically drilled well can only intersect a small length of a horizontal pay formation. Directional drilling allows a well to trace the formation in the horizontal direction to create a large length of wellbore section for hydrocarbon collection. The hydraulic fracturing technique now takes a different aim—instead of creating a single, large fracture from a vertical well, it aims at the creation of a large number of shorter, closely spaced, and interconnecting fractures from a horizontal well. The technique allows the production from the *unconventional reservoir* such as shale gas formation, also known as *tight gas reservoir*. "Tight" means very low permeability such that production using conventional technique is not economically feasible. This fracking technology opens the door for production of the vast reserve of shale gas and shale oil, thus extending the world's energy prospect. The practice of fracking, however, is controversial as it creates many environmental concerns [117].

1.3.9 Water Wave and Seabed Interaction

Sea bottom sediment is a porous medium. The rising and falling of water waves exerts a pressure on the bottom that drives a seepage flow into and out of the seabed, as well as deforms it. This seepage is a viscous flow; hence it dissipates the energy of the propagating water wave, causing the wave to dampen. This problem has been investigated by assuming that the flow in the seabed satisfies Darcy's law [152, 195, 199].

Based on the simplest model [195], the dynamic pressure at the sea bottom is given by the linear water wave theory of a harmonic, small amplitude wave propagating over constant water depth, as

$$p = P_o \cos(\ell x - \omega t) \tag{1.73}$$

where P_o is the pressure magnitude, ℓ is the wave number, and ω is the wave frequency. This cycling pressure, changing from positive to negative, is added to the hydrostatic pressure proportional to the water depth. The flow in the seabed is modeled as a Darcy flow of incompressible fluid, governed by the Laplace equation

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial z^2} = 0 \tag{1.74}$$

where z is pointing downward in the depth direction. Solution of (1.74), subject to the boundary condition (1.73) prescribed at $z = 0$, is

$$p(x, z, t) = P_o \, e^{-\ell z} \cos(\ell x - \omega t) \tag{1.75}$$

The above equation shows that the pore pressure in the seabed decays exponentially with depth, with a decay length characterized by the wavelength of the water wave; that is, the longer the water wave, the deeper the pore pressure penetration.

The curiosity of the above solution is that the pore pressure response is not dependent on any physical parameters of the seabed, not even the permeability. Also, the periodic response is completely in phase with the water wave at any depth. These are not realistic conditions. In other attempts, the seabed was modeled either as a fluid (mud) [58, 94], or as an elastic material without fluid [159]. These models are also physically deficient.

In a better approach, the seabed has been modeled as a poroelastic material. The resultant pore pressure is found to be dependent on a number of seabed physical parameters, such as permeability, compressibility of pore water, as well as that of the porous skeleton [156, 265]. Interestingly, it is demonstrated that the solution (1.75) is the limiting case of the poroelastic solution, if the pore water is considered to be incompressible ($C_f = 0$) [265], or if the permeability coefficient is taken to be infinite ($\kappa \to \infty$) [45]. In fact, (1.75) is the upper bound of pore pressure responses. That is, with a finite permeability or fluid compressibility, the pore pressure response is always smaller than that predicted by (1.75). Also, the cyclic response is increasingly off-phase going into depth.

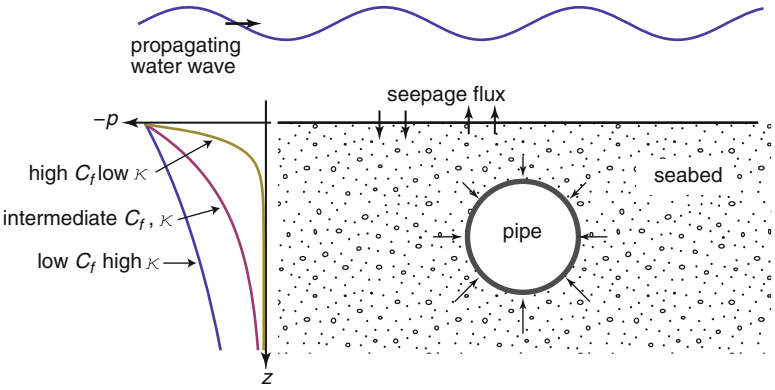


Fig. 1.11 Uplift force on a pipeline buried in sea bed

The poroelastic solution also shows that for small permeability or large pore water compressibility, the pressure penetration is limited to a surface layer; that is, there is a sharp drop-off within a small distance. It is of interest to comment that although seawater is considered nearly incompressible as compared to the sediment frame, a very small amount of entrapped gas can make it highly compressible. It is observed that recently deposited sediment often contains organic contents that can generate undissolved gas. For example, as much as 10 % by volume of undissolved gas was found in the submarine sediment of Mississippi Delta [14]; so the apparent pore water compressibility can be very large.

Next, consider a pipeline buried in the seabed (Fig. 1.11). The wave induced pore pressure has a gradient; hence creates a pressure differential on the upper and lower part of the pipe, resulting a net force. This dynamic buoyancy force pushes the pipe up and down, as well as left and right. Particularly, at the passage of wave trough, there is a uplift force. During a storm event, the upward dynamic buoyancy force can be large enough to overcome the overburden weight of the backfill to cause flotation of the pipeline [45, 131, 157]. Such force needs to be taken into design considerations.

As mentioned above, the penetration of pore pressure is strongly affected by the permeability of sediment, κ, and the (apparent) compressibility of water, C_f. In Fig. 1.11 we illustrate three pore pressure profiles corresponding to the large κ or small C_f, small κ or large C_f, and the intermediate κ and C_f case. If permeability is small or compressibility of water is large (right profile), the dynamic pore pressure drops off rapidly such that it does not penetrate to the depth of the pipeline; hence the pipe experiences very little uplift force. For large permeability and small pore water compressibility (left profile), the pore pressure response is the largest. However, buoyancy is generated not by the pressure itself, but by its differential at the top and bottom of the object. If the pressure profile is flat around where the pipeline is,

the buoyancy is also small. These cases lead to the interesting conclusion that the uplift force is largest at certain intermediate permeability and water compressibility values [45].

Other seabed related poroelastic phenomena include submarine slope instability and landslide [11, 114, 197], liquefaction [123, 210, 215], stability of breakwater [120, 130, 132], lifting of objects resting on seabed [87], and flux exchange between ocean and sediment as a mechanism of pollutant transport [133, 172, 193]. Another sediment related phenomenon is the damping of earthquake excitation in a dam-reservoir system by bottom sediment [40, 76].

1.3.10 Tidal and Barometric Efficiency

Gaius Plinius Secundus (AD 23–79), better known as Pliny the Elder, seems to be the first to observe that the water level in a coastal well responds to the rise and fall of ocean tide. In his book *Historia Naturalis* (Natural History), he stated that near the temple of Hercules in Cadiz "*there is a closed source similar to a well which occasionally rises and falls with the ocean*" [167].

In 1902, the United States Geological Survey reported that water level oscillations in wells in Atlantic City, New Jersey, were synchronous with the tidal cycle. The sea bottom pressure driven porous medium flow was ruled out as the cause, because the transmission of pore pressure by such flow would have a large time lag. Hence this effect was attributed to the weight of seawater compressing on the aquifer below, resulting an instant pore pressure rise, forcing the water level in the wells to go up [166].

Earth tide is the motion of the Earth's surface at periods of about 12 h and longer, due to the pull by gravity of the Moon and the Sun. Klonne in 1880 [144] reported on water level fluctuations in a flooded coal mine near Duchov, Czechoslovakia, and theorized that it was caused by the dilation produced by the Earth tide. According to Grablovitz [101] (translated):

> *It is known that the terrible catastrophe occurred February 10, 1879 at Dux, under which five large coal mines were flooded almost completely.*
>
> *He was reflecting on the means best adapted to repair that damage, when the director Mr. F. W. Klonne, mountain engineer, was made aware that the water level was not increasing continuously, as he had first thought, but decreased, and sometimes arrested for a few hours, and then increased with greater vigor.*
>
> *Being warned of this, the intelligent director did not take long to recognize that the fluctuations were due no less than the movement of ebb and flow, just as in the seas, of the lunar-solar attraction. He was so impressed that it did not take long for him to undertake, with a fervor worthy of anyone's praise, regular hourly observations, which began at 6 am of April 8 following the disaster, announced at the same time to the Academy of Sciences in Vienna, . . . and observations continued until 6 am of September 11, the day when the work on the extraction of water from flooded mines began . . .*

Other early observations of Earth tide induced water level fluctuation in wells include Young [273], Robinson [204], and Theis [241].

King in 1892 [143] reported that a heavily loaded moving train had the power of disturbing the water level in a well 140 ft away from the train station at Whitewater, Wisconsin. It was observed that with the passage of a train, there was "*a rapid but gradual rise of the water, which is followed by only a slightly less rapid fall again to the normal level, there being nothing oscillatory in character*." He reasoned that this phenomenon could not be explained by an elastic shock wave, because "*there is one single rise and fall, with no trace whatever in the curve of a repetition, as a true wave implies*."

In the same report, King also pored over the phenomenon of groundwater level fluctuation coinciding with the barometric pressure changes. He stated: "*a rise in the barometer is very nearly coincident in time with a fall of the water in wells and with a diminished rate of discharge of water from the ground, and vice versa*."

The ocean tide can also exert a periodic loading on the seafloor. Jupp and Schultz [137] presented a theory that the magnitude and phase of the temperature and flow rate of the seafloor *hydrothermal vent* are modulated by the changing weight of water column loaded on the seafloor.

From the poroelasticity theory, we realize that the ratio of the water surface rise in the well to the tidal level rise is given by the uniaxial Skempton pore pressure coefficient defined in (1.21), and is called the tidal efficiency [75, 126], denoted as γ_{te}. Utilizing the relations (1.12), (1.21), (1.23), and (1.46), and eliminating a few parameters, we obtain

$$\gamma_{te} = 1 - \frac{\phi}{SK_f} \tag{1.76}$$

The above equation suggests that if we obtain tidal efficiency from observation, and know the values of K_f and ϕ, we can estimate the storage coefficient S, which in turn is tied to the specific storativity of the aquifer as $S_s = \gamma_f S$ (see Sect. 1.2.6).

Similarly, we can define a *barometric efficiency* as the ratio of the pressure rise in the water well, calculated as $\Delta p = \gamma_f \Delta h$, to the change in atmospheric pressure, as

$$\gamma_{be} = \frac{\phi}{SK_f} \tag{1.77}$$

The reason that the above expression is different from (1.76) is because the atmospheric load exerts a pressure not only on the aquifer, but also on the water column in the well. So based on the observation of water level fluctuation in water well penetrating into an artisan aquifer, and the barometer record, the specific storativity of the aquifer can be estimated.

There are many applications that utilize the ocean, atmospheric, and Earth tide for the estimation of formation properties. Making the assumption that stresses associated with Earth tides act horizontally, and that the Earth is free to deform in the vertical direction, Bredehoeft [29] provided a method for estimating the storage coefficient of the aquifer from the magnitude of the water level change in a well, if the Poisson ratio of the formation is known. Further utilizing the phase shift of the water level response, the aquifer transmissivity can be determined [119].

Analyzing both the Earth and atmospheric tide effects, van der Kamp and Gale [252] presented formulas for the determination of the shear and bulk modulus, as well as porosity, of the formation. By filtering and separating the tidal data into high, intermediate, and low frequency responses, various aquifer parameters, such as well radius, lateral and vertical diffusivity, storativity, thickness of the unsaturated zone, vertical pneumatic diffusivity, etc., can be estimated [207].

Using the simple aquifer theory, Ferris in 1952 [84] suggested that the combined aquifer parameter T/S, where T is transmissivity and S storativity, can be determined from the data of water level fluctuation in wells, caused by the cyclic ocean tidal or stream stage loading. The theory was tested on stream stage fluctuation. The utilization of the ocean tidal effect for the determination of aquifer parameters was attempted by Erskine [81], as well as others [135, 169]. Similar technique was developed by Pinder et al. [188] using the recession limb of river flood stage.

1.3.11 Biomechanics

As cited in the prologue of this chapter, Leonardo da Vinci drew an analogy between the Earth and the human body, both as porous materials: "*Its flesh is the soil, its bones are the strata of rock, its cartilage is the tufa, its blood is the underground streams, the reservoir of blood around its heart is the ocean, the systole and diastole of the blood in the arteries and veins appear on the Earth as the rising and sinking of the oceans*". Biological materials, such as human and animal flesh, bones, cartilages, blood vessels, and plant tissues, are porous solid impregnated with fluid; hence should be studies as poroelastic materials, rather than solid or viscoelastic solid.

Compared with the lifeless rock, live cortical (compact) bone is a much more dynamic structure since it is under constant remodeling [274], meaning that bone tissue is removed from the skeleton and new bone tissue is formed, a process that can be stimulated by the stress loading. This seemingly dense matrix actually has several levels of porosity, which are filled with fluid. The fluid in the bone serves many functions. It transports nutrients to, and carries waste from, the bone cells (osteocytes) buried in the bony matrix. It is involved in the transport of mineral ions to the bone tissue for storage, and then for retrieval when they are needed by the body [54]. Bone deformation induced fluid flow causes the bone fluid to flow over the bone cell membrane. The shear stress of the flowing fluid is sensed by the cell, which has been suggested to have a role in bone's mechanosensory system [261].

Cartilage is a flexible connective tissue found in many parts of human body, such as in the joints between bones. It is a porous soft tissue impregnated with fluid. The viscous fluid flow in the pores gives the articular cartilage a viscoelastic-like behavior that serves as a damper to impact. It also provides a squeeze film lubrication with synovial fluid to prevent the solid surface from touching. The fluid exchange is a mechanism for nutrient transport [170, 171, 219].

Artery wall is the fibrous and muscular wall of vessels that carry oxygenated blood from the heart to structures throughout the body. A diseased vessel can get clogged by the deposit of low density lipoprotein (LDL), forming plaques,

and eventually becoming hardened, constricting the blood flow, and leading to the disease known as atherosclerosis. The inside of the artery wall is lined with a single layer of cells, known as endothelium. The onset of LDL deposit on the wall is often found at lesion site where the endothelial cells are torn off.

The artery wall is highly elastic [257], and it can inflate or contract with the high or low blood pressure inside the vessel. The higher intraluminal pressure drives a fluid flow through the wall, bringing nutrient, oxygen and macromolecules to the tissue [151, 230]. As the wall itself is infiltrated with fluid, the stretching and collapsing of the wall can respectively cause pressure drop and rise inside the wall. Particularly, with a sudden drop of intraluminal pressure, the contraction of the wall causes a sudden fluid pressure rise inside the artery wall, which in turn drives a flux both ways, toward the tissue, and toward the lumen. This reversal of flux toward lumen is suspected to be large enough to dislodge the endothelial cell, to be torn away by the shear force of the blood flow [46, 140]. Other scenarios of poroelastic effect on blood vessels include the rupture of abdominal aortic aneurysm when the artery wall stress exceeds that of the local wall strength [191].

Other biological tissues modeled as poroelastic material include ligament [7, 270], intervertebral disk [5, 223], cardiac tissue [39, 267], subcutaneous tissue [158, 264], and human brain [164].

1.3.12 Poroviscoelasticity and Anelastic Strain Recovery

The hydraulic fracturing technique described in the preceding section is the most reliable method for the determination of the magnitude of in situ stress. It is, however, costly to perform. Also, as the breakdown of borehole is initiated when the minimum in situ stress is reached, this technique can only measure the minimum, and not the maximum, in situ stress.

An alternative technique can be devised based on the strain relaxation phenomenon of cylindrical cores retrieved at great depths. Once the confining stress is removed from a retrieved core, it starts to slowly expand, typical of a *viscoelastic* material. This creep behavior can be attributed to several physical mechanisms, such as the intergranular frictional sliding, the intrinsic solid grain creep deformation, and the solid-fluid interactions in fissures at the sub-granular scale [21, 48]. However, sometimes the core is observed to contract, before it expands. This non-monotonic behavior can only be explained by the interaction between the poroelastic and viscoelastic effects.

This *anelastic strain recovery* (ASR) method [56, 258] has been used as one of the lower cost techniques for determining the in situ stresses. In the practice, a drill core is quickly retrieved to the ground (within several hours), immediately instrumented with displacement gages, and then monitored for strain relaxation for 1–3 days. With a valid viscoelastic constitutive model, this technique can be used to determine not only the directions, but also the magnitude of in situ stresses [24, 240, 259].

When we construct a constitutive model for the retrieved cores, we need to be aware that rocks are not only viscoelastic, but also poroelastic. As both of these mechanisms are time dependent, when we rely on the transient data to determine the in situ stress, a model cannot ignore one or the other. These two mechanisms each have their own response time. The characteristic time of viscoelastic behavior is a material property. In fact, as described above, there exist several transient physical mechanisms, such that several characteristic times may be needed to even model the viscoelastic responses. However, for a particular application, such as ASR, only characteristic times falling within a range of time of interest, such as from a few hours to a few days, need to be considered.

The poroelastic response, although transient, should not be modeled as a viscoelastic one [30], as the poroelastic characteristic time is dependent not only on the material itself, but also on the geometry of the specimen. As discussed in Sect. 1.2.2, the characteristic time of a poroelastic specimen can be estimated as $t_c = L_c^2/c$, where c is the consolidation coefficient, and L_c is largest distance that the pore pressure diffusion front needs to travel, originating from a drainage surface to reach to the innermost part of the specimen. So the characteristic time is sensitive to the square of the size of the specimen. The consolidation coefficient is a material property; and particularly, it is directly proportional to the *intrinsic permeability* of the rock, k, which can range from 10^{-1} darcy for sandstone to 10^{-7} darcy for shale.

While the solid viscoelastic behavior predicts that material will expand after the removal of the confining stress, the actual core behavior is more complex. Some cores were observed to contract, and some contract and then expand, or the other way around [31, 74, 80, 260]. The contraction of the core can be easily explained by the poroelastic effect. Once the core is retrieved, if there is still a residual pore pressure within the core, it will have to dissipate with time. Based on the thermoelasticity analogy discussed in Sect. 1.2.7, pore pressure is equivalent to temperature. As a cylinder cools down, it contracts; the same effect is found in pore pressure dissipation. Another way to view this phenomenon is by the effective stress. As the cylinder is free from an external compressive stress, the existence of pore pressure effectively makes the core in tension. Once that tension disappears, the core contracts. The coexistence of the viscoelastic and poroelastic effects can create the non-monotonic behaviors observed in the drill cores. Such phenomena have been modeled using the poroviscoelasticity theory [1, 2].

1.3.13 Porothermoelasticity and Thermal Fracturing

In Sect. 1.2.7 we discussed that heat on an elastic solid is analogous to pore pressure on a poroelastic material assuming it is under isothermal condition, in terms of their deformation and stress responses. When heat is applied to a saturated poroelastic material, we have more complicated physical mechanisms to consider, as there exists the interaction of heat effect on two constituents, solid and fluid, which have different thermal expansion coefficients. The relative magnitude of these coefficients has an important consequence on material failure by effective stress.

Consider a porous specimen free from external stresses and subject to a uniform temperature rise. The sample will expand just as an elastic solid. For geomaterials saturated with water, the fluid thermal expansion coefficient is typically larger than that of the solid. If the fluid is not allowed or does not have time enough to escape, a pressure will develop inside the specimen due to the smaller expansion of the pore space. As there is no externally applied confining stress, this internal pressure pushes on the porous structure making it effectively in tension, as predicted by Terzaghi effective stress. Hence heating a saturated concrete to high temperature can make it to explode, known as *pore pressure spalling* [10, 57, 138, 186].

Now consider a different scenario—the porous specimen is restrained from deformation and is cooled. Both solid and fluid will contract. The contraction of solid makes it in tension. The fluid however contracts more, and a negative pressure will develop, which compensates the effective stress, making the solid less in tension. At large time, the pore pressure is dissipated, and the solid is in maximum tension. If the material is cooled enough, the solid can crack. This effect has been utilized as a *thermal fracturing* technique by the industry. In a geological formation cold fluid can be injected into a borehole or a fracture to lower the temperature on the wall to induce cracks. These thermally induced fracture zone can enhance the permeability of the formation for enhanced oil recovery, unconventional gas extraction, nuclear waste disposal, geothermal energy extraction, and CO_2 geosequestration [271].

1.3.14 Poroelastodynamics and Seismoelectric Effect

As early as in 1930s, geophysicists in the US have observed that the passage of a seismic wave, such as one that is generated by an underground explosion, creates a disturbance in an electric field [25]; that is, the current strength in the soil, in the presence of a constant potential difference between two electrodes, fluctuates under the influence of elastic vibrations. Thompson in 1936 [245] provided an explanation to the phenomenon as a change in earth resistivity due to the elastic deformation. This has been called as the *seismoelectric effect of the first kind*, or the *J-effect*.

In 1939, Ivanov [124, 125] in Russia reported that in moist soil, there is a *seismoelectric effect of the second kind*, or the *E-effect*. The second kind effect is characterized by an electrical potential generated within the earth without an externally applied electric field, by the passage of seismic waves, Also, this electric potential travels ahead of the seismic waves, in contrast with the effect of the first kind. In 1944, the famous Russian physicist Frenkel (see Sect. F.13 for a biography) proposed a mechanism based on the relative movement of solid and fluid containing electrolytes during the passage of a seismic wave. In that effort, he formulated the first theory of poroelastodynamics [92]. The theory was later refined by Biot [18, 19] (see Chap. 9).

Martner and Sparks in 1959 [161] suggested that the arrival of the seismoelectric pulse at ground level far away from the source of explosion is instant; hence

it must be due to the propagation of an electromagnetic wave. The detection of such electromagnetic wave in the field was reported by Thompson and Gist [243]. The governing equations for the seismoelectric-electromagnetic phenomenon were presented by Pride [194], which include the Maxwell equations. In fact, not only a seismic event can generate electric and electromagnetic fields, but also electric and electromagnetic pulses can generate seismic waves [244], which is called the *electroseismic effect*.

It is well known that in an elastic medium there exist two propagating waves, a compressional wave, also known as the P-wave, and a shear wave, known as the S-wave. In porous medium, it has been demonstrated by Frenkel [92] and Biot [18, 19] that due to the presence of a fluid phase, two compressional waves exist—a *compressional wave of the first kind* characterized by the in-phase motion of solid and fluid, and a *compressional wave of the second kind* due to the relative motion between the solid and fluid. The first wave is similar to an elastic wave, and called a *fast wave*, with a small amount of attenuation. The second wave is with a much slower propagation speed and highly dissipative, and is called a *slow wave*. The phase velocities of both waves are frequency dependent; hence these waves are *dispersive waves*.

Figure 1.12 illustrates the propagation of a second wave in which the solid and fluid move in opposite directions. In soils and geological formations the fluid often contains dissolved electrolytes. The positive charge on the solid surface and the negative charge in the fluid form electric double layers. Their relative motion generates local electric currents, which is the E-effect. The fluctuation of the electric current further generates electromagnetic waves. The seismoelectric and the electroseismic effects can be used in many applications, such as the detection of hydrocarbon in formation [244], exploration of fractures [277] and permeable zones [168] in boreholes, detection of horizontal bedding plane fractures in groundwater exploration [88], and tracking the underground NAPL contaminant front [201].

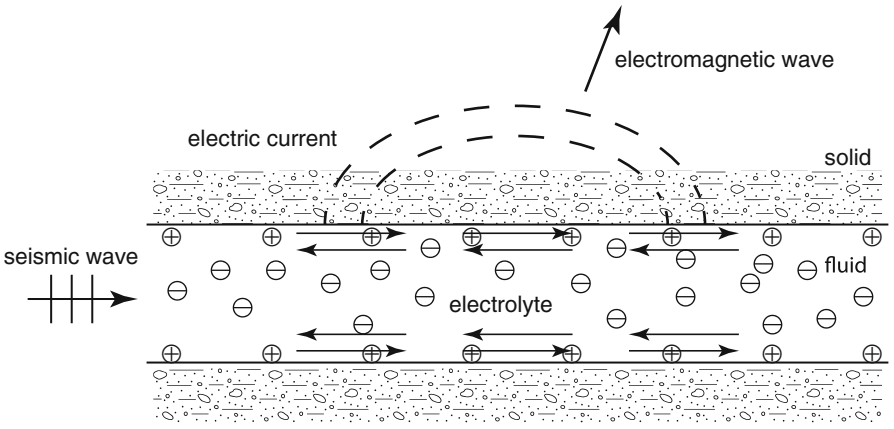


Fig. 1.12 Seismoelectric and seismoelectromagnetic effect

1.3.15 Swelling of Clay and Shale

It is well known that clay can swell by a large amount when it becomes wet, and contract and crack when it is dried. This drastic expansion and shrinkage behavior causes many construction problems, such as cracking of foundations and walls, and heaving of roadways. On the other hand, clay properties can also be used for beneficially purposes such as liner materials in the containment of waste [208] and catalytic materials [249].

The swelling of clay consists of two phases. In the *innercrystalline swelling*, water is taken up in a process called *hydration* for dry or partially saturated clay. Clay particles are in the form of platelets that stack together in multiple layers. In the fully dry montmorillonite, a dominant type of clay, the layers lie so close together that they are almost in contact. When in contact with water, the water molecules enter the interlayer space and bond with the cations and order themselves into single or multiple layers that repel the clay platelet, causing the swelling [115].

Once the clay becomes saturated, a second type of swelling can result from the difference in the ion concentration, and principally in the cation concentration, at the surface of the clay layers and in the pore water, known as *osmotic swelling*. The clay surface is negatively charged, which attracts positive ions surrounding it to form a diffuse electric double layer. This electric cloud behaves as a semi-permeable membrane is an osmosis process that allows the passage of water molecules yet blocks the charged ions. With the adsorbed water, the clay particles repel each other making it to swell, but to a much less extent than that of the hydration process.

Swelling of shale is of major problem for petroleum engineering. Shale makes up more than 75 % of drilled formations and causes at least 90 % of wellbore-stability problems [50]. Drilling in shale can result in a variety of problems, ranging from bit balling, sloughing, or creep, to washout and complete hole collapse. Although the use of oil based drilling mud can reduce the chemical effect, its use is much restricted due to the environmental concern of its disposal [254]. For the water based drilling mud, its chemistry can be controlled to minimize the osmotic effect, or even to utilize it for the stability of the shale. The difference in solute concentration between the drilling mud and the formation can create an osmotic pressure in the shale that plays a role in the extended effective stress concept, thus affecting the stability of the borehole wall. Some of these effects can be models by the *porochemoelasticity* theory as presented in Chap. 12.

1.3.16 Nanoporous Material

Nanoporous material is defined as materials with pore size of the order 100 nm or smaller [154]. Such materials are abundant in nature both in biological systems, such as cell membranes, and in natural minerals, such as activated carbon and zeolites. Nanoporous materials have been used in industry for various purposes for a long time. For example, zeolites are widely used for water purification and as catalysts.

However, only the recent emergence of nanotechnology allows us to realize the nanoscale nature of these materials, and the physico-chemical mechanisms that make them efficient. The invention of scanning tunneling microscope and atomic force microscope also enables us to see and manipulate materials at the nanoscale, opening the opportunity for designing and manufacturing new nanoporous materials for industrial, electronic, and biomedical applications.

Nanoporous materials can be classified as membrane and bulk materials. Membrane material works as a molecular sieve. Its nano sized pores allow the passage of only smaller size molecules, and blocks the larger ones. Its biological and medical applications involve sorting, sensing, isolating and releasing biological molecules [3]. The bulk materials are utilized for their large surface to volume ratios. For the same porosity, if we make the pores smaller and smaller, the pore surface area increases linearly as the pore size decreases. For example, one gram of activated carbon with pore size of a few nanometers can have as much as several hundred to one thousand square meters of surface area. Hence nanoporous materials can be highly effective for use as adsorbents and ion-exchangers in chemical industry, and for drug delivery in medical applications. The agricultural product biochar (a charcoal used for soil amendment) has been considered to have the potential as a carbon sequestration tool for reversing climate change, by adsorbing CO_2 onto its pore surface [150].

Many macroporous structural materials contain porosity at multiple scales. For example, cement is found to contain a hierarchy of pores, from mortar scale (10^{-3} m) to cement paste (10^{-4} m), to C-S-H (Calcium-Silicate-Hydrate) matrix (10^{-6} m), to C-S-H solid (17 nm), and to globules (2.5 nm or less) [53, 134]. Particularly, at the globules scale, the nanoporosity is estimated to be about 18 % [134, 233]. The multiscale concrete has been modeled as a microporomechanical material [248]. Through nanoindentation, it has been found that the packing density of the C-S-H is largely responsible for the strength and the large time creep of the concrete [162, 251]. Nano particles, such as graphene, have been added to modify cement's mechanical, thermal, and electrical properties [4]. This is indeed a developing area of poromechanics.

References

1. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1993) A micromechanically consistent poroviscoelasticity theory for rock mechanics applications. Int J Rock Mech Mining Sci 30(7):1177–1180
2. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1996) Poroviscoelastic analysis of borehole and cylinder problems. Acta Mech 119(1–4):199–219
3. Adiga SP, Jin C, Curtiss LA, Monteiro-Riviere NA, Narayan RJ (2009) Nanoporous membranes for medical and biological applications. Wiley Interdiscip Rev Nanomed Nanobiotechnol 1(5):568–581
4. Alkhateb H, Al-Ostaz A, Cheng AHD, Li XB (2013) Materials genome for graphene-cement nanocomposites. J Nanomech Micromech ASCE 3(3):67–77

5. Argoubi M, Shirazi-Adl A (1996) Poroelastic creep response analysis of a lumbar motion segment in compression. J Biomech 29(10):1331–1339
6. Atkinson C (1986) Steady pore pressure profiles associated with a moving trench. Int J Eng Sci 24(10):1681–1695
7. Atkinson TS, Haut RC, Altiero NJ (1997) A poroelastic model that predicts some phenomenological responses of ligaments and tendons. J Biomech Eng ASME 119(4):400–405
8. Banhart J (2001) Manufacture, characterisation and application of cellular metals and metal foams. Prog Mater Sci 46(6):559–632
9. Barksdale HC, Sundstrom RW, Brunstein MS (1936) Supplementary report on the groundwater supplies of the Atlantic City region. New Jersey State Water Policy Commission Special Report 6, 139pp
10. Bažant ZP, Thonguthai W (1979) Pore pressure in heated concrete walls–theoretical prediction. Mag Concr Res 31(107):67–76
11. Bea RG, Wright SG, Sircar P, Niedoroda AW (1983) Wave-induced slides in south pass block 70, Mississippi Delta. J Geotech Eng ASCE 109(4):619–644
12. Beamish BB, Crosdale PJ (1998) Instantaneous outbursts in underground coal mines: an overview and association with coal type. Int J Coal Geol 35(1–4):27–55
13. Bear J, Cheng AHD (2010) Modeling groundwater flow and contaminant transport. Springer, Dordrecht/London, 834pp
14. Bennett RH, Faris JR (1979) Ambient and dynamic pore pressures in fine-grained submarine sediments: Mississippi Delta. Appl Ocean Res 1(3):115–123
15. Berchenko I, Detournay E (1997) Deviation of hydraulic fractures through poroelastic stress changes induced by fluid injection and pumping. Int J Rock Mech Mining Sci 34(6):1009–1019
16. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2):155–164
17. Biot MA (1956) Thermoelasticity and irreversible thermodynamics. J Appl Phys 27(3):240–253
18. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid. 1. Low-frequency range. J Acoust Soc Am 28(2):168–178
19. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid. 2. Higher frequency range. J Acoust Soc Am 28(2):179–191
20. Biot MA (1959) New thermomechanical reciprocity relations with application to thermal stress analysis. J Aerosp Sci 26(7):401–408
21. Biot MA (1962) Mechanics of deformation and acoustic propagation in porous media. J Appl Phys 33(4):1482–1498
22. Biot MA (1963) Science and the engineer. Appl Mech Rev 16(2):89–90
23. Biot MA, Willis DG (1957) The elastic coefficients of the theory of consolidation. J Appl Mech ASME 24:594–601
24. Blanton TL (1983) The relation between recovery deformation and *in-situ* stress magnitudes. In: SPE/DOE low permeability gas reservoirs symposium, Denver
25. Blau LW, Statham L (1936) Method and apparatus for seismicelectric prospecting. U.S. Patent 2054067
26. Bluhm J, de Boer R (1996) Effective stresses—a clarification. Arch Appl Mech 66(7):479–492
27. Boley BA, Weiner JH (1960) Theory of thermal stresses. Wiley, New York, 586pp
28. Booker JR (1974) Time dependent strain following faulting of a porous medium. J Geophys Res 79(14):2037–2044
29. Bredehoeft JD (1967) Response of well-aquifer systems to Earth tides. J Geophys Res 72(12):3075–3087
30. Brereton NR, Chroston PN, Evans CJ (1995) Pore pressure as an explanation of complex anelastic strain recovery results. Rock Mech Rock Eng 28(1):59–66

31. Brereton NR, Chroston PN, Evans CJ, Hudson JA, Whitmarsh RB (1992) Anelastic strain recovery and elastic properties of oceanic basaltic rocks. In: Gradstein FM, Ludden JN et al (eds) Proceedings of the ocean drilling program. Scientific results, vol 123. Ocean Drilling Program, College Station, pp 469–491
32. Bridgman PW (1912) Breaking tests under hydrostatic pressure and conditions of rupture. Philos Mag 24(139):63–80
33. Brown DW (2000) A hot dry rock geothermal energy concept utilizing supercritical CO_2 instead of water. In: Proceedings of the 25th workshop on geothermal reservoir engineering, SGP-TR-165, Stanford University, Stanford
34. Bruno MS, Nakagawa FM (1991) Pore pressure influence on tensile fracture propagation in sedimentary rock. Int J Rock Mech Mining Sci 28(4):261–273
35. Burke M, Kingsbury HB (1984) Response of poroelastic layers to moving loads. Int J Solids Struct 20(5):499–511
36. Carder DS (1945) Seismic investigations in the Boulder Dam area, 1940–1944, and the influence of reservoir loading on earthquake activity. Bull Seismol Soc Am 35:175–192
37. Chan DYC, Hughes BD, Paterson L (1993) Transient gas flow around boreholes. Trans Porous Media 10(2):137–152
38. Chan DYC, Hughes BD, Paterson L (1993) Tensile stresses around boreholes due to transient fluid flow. Int J Numer Anal Methods Geomech 17(9):659–667
39. Chapelle D, Gerbeau JF, Sainte-Marie J, Vignon-Clementel IE (2010) A poroelastic model valid in large strains with applications to perfusion in cardiac modeling. Comput Mech 46(1):91–101
40. Cheng AHD (1986) Effect of sediment on earthquake-induced reservoir hydrodynamic response. J Eng Mech ASCE 112(7):654–665
41. Cheng AHD (2000) Multilayered aquifer systems—fundamentals and applications. Marcel Dekker, New York/Basel, 384pp
42. Cheng AHD, Abousleiman Y, Roegiers JC (1993) Review of some poroelastic effects in rock mechanics. Int J Rock Mech Mining Sci 30(7):1119–1126
43. Cheng AHD, Badmus T, Beskos DE (1991) Integral equation for dynamic poroelasticity in frequency domain with BEM solution. J Eng Mech ASCE 117(5):1136–1157
44. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to fracture propagation. Int J Numer Methods Eng 20(2):279–296
45. Cheng AHD, Liu PLF (1986) Seepage force on a pipeline buried in a poroelastic seabed under wave loadings. Appl Ocean Res 8(1):22–32
46. Cheng AHD, Skalak R, Weinbaum S, Ni JCY (1984) Consolidation of poroelastic arterial tissue. In: Boresi AP, Chong KP (eds) Engineering mechanics in civil engineering, proceedings of the 5th ASCE/EMD specialty conference. University of Wyoming, Laramie, pp 891–894
47. Clark JB (1949) A hydraulic process for increasing the productivity of wells. Trans Am Inst Mining Metall Eng 186(1):1–8
48. Cleary MP (1978) Elastic and dynamic response regimes of fluid-impregnated solids with diverse microstructures. Int J Solids Struct 14(10):795–819
49. Cleary MP (1978) Moving singularities in elasto-diffusive solids with applications to fracture propagation. Int J Solids Struct 14(2):81–97
50. Cleary MP (1980) Analysis of mechanisms and procedures for producing favourable shapes of hydraulic fractures. In: SPE annual technical conference and exhibition, Dallas
51. Cleary MP (1980) Comprehensive design formulae for hydraulic fracturing. In: SPE annual technical conference and exhibition. Dallas
52. Cocco M, Rice JR (2002) Pore pressure and poroelasticity effects in Coulomb stress analysis of earthquake interactions. J Geophys Res Solid Earth 107(B2):2030
53. Constantinides G, Ulm FJ (2004) The effect of two types of C-S-H on the elasticity of cement-based materials: results from nanoindentation and micromechanical modeling. Cem Concr Res 34(1):67–80
54. Cowin SC (1999) Bone poroelasticity. J Biomech 32(3):217–238
55. Cryer CW (1963) A comparison of the three-dimensional consolidation theories of Biot and Terzaghi. Q J Mech Appl Math 16(4):401–412

56. Dally JW, Riley WF (1978) Experimental stress analysis. McGraw-Hill, New York, 571pp
57. Dal Pont S, Colina H, Dupas A, Ehrlacher A (2005) An experimental relationship between complete liquid saturation and violent damage in concrete submitted to high temperature. Mag Concr Res 57(8):455–461
58. Dalrymple RA, Liu PLF (1978) Waves over soft muds: a two-layer fluid model. J Phys Oceanogr 8(6):1121–1131
59. David EC, Zimmerman RW (2011) Compressibility and shear compliance of spheroidal pores: exact derivation via the Eshelby tensor, and asymptotic expressions in limiting cases. Int J Solids Struct 48(5):680–686
60. Davies GH, Reynolds SJ (1996) Structural geology of rocks and regions, 2nd edn. Wiley, New York, 776pp
61. da Vinci L (1506–1510) Codex leicester
62. Davis SD, Pennington WD (1989) Induced seismic deformation in the Cogdell oil-field of West Texas. Bull Seismol Soc Am 79(5):1477–1494
63. de Boer R (2000) Theory of porous media, highlights in the historical development and current state. Springer, Berlin/New York, 618pp
64. de Boer R (2005) The engineer and the scandal: a piece of science history. Springer, Berlin/Heidelberg, 293pp
65. de Boer R, Ehlers W (1988) A historical review of the formulation of porous media theories. Acta Mech 74(1–4):1–8
66. de Boer R, Ehlers W (1990) The development of the concept of effective stresses. Acta Mech 83(1–2):77–92
67. de Laguna W (1966) Disposal of radioactive wastes by hydraulic fracturing: part I. General concept and first field experiments. Nucl Eng Design 3(2):338–352
68. de Laguna W (1966) Disposal of radioactive wastes by hydraulic fracturing: part II. Mechanics of fracture formation and design of observation and monitoring wells. Nucl Eng Design 3(3):432–438
69. Deresiewicz H, Skalak R (1963) On the uniqueness in dynamic poroelasticity. Bull Seismol Soc Am 53(4):783–788
70. Detournay E, Cheng AHD (1988) Poroelastic response of a borehole in a non-hydrostatic stress field. Int J Rock Mech Mining Sci 25(3):171–182
71. Detournay E, Cheng AHD (1991) Plane strain analysis of a stationary hydraulic fracture in a poroelastic medium. Int J Solids Struct 27(13):1645–1662
72. Detournay E, Cheng AHD (1992) Influence of pressurization rate on the magnitude of the breakdown pressure. In: Tillerson JR, Wawersik WR (eds) Rock mechanics, proceedings of the 33rd U.S. symposium, Balkema, pp 325–333
73. Detournay E, Cheng AHD, Roegiers JC, McLennan JD (1989) Poroelasticity considerations in *in-situ* stress determination by hydraulic fracturing. Int J Rock Mech Mining Sci 26(6):507–513
74. Dey TN, Kranz RL (1988) State of stress and relationship of mechanical properties to hydrothermal alteration at valles caldera core hole 1, New Mexico. J Geophys Res Solid Earth Planets 93(B6):6108–6112
75. Domenico PA (1983) Determination of bulk rock properties from ground-water level fluctuations. Bull Assoc Eng Geol 83(3):283–287
76. Domínguez J, Gallego R, Japón BR (1997) Effects of porous sediments on seismic response of concrete gravity dams. J Eng Mech ASCE 123(4):302–311
77. Dreesen DS, Nicholson RW (1985) Well completion and operations for MHF of Fenton Hill well EE-2. Geotherm Resour Counc Trans Part 2 9:105–110
78. Dropek RK, Johnson JN, Walsh JB (1978) Influence of pore pressure on mechanical properties of Kayenta sandstone. J Geophys Res 83(NB6):2817–2824
79. Economides MJ, Nolte KG (eds)(2000) Reservoir stimulation, 3rd edn. Wiley, Chichester/New York, 856pp
80. El Rabaa AW, Meadows DL (1986) Laboratory and field application of the strain relaxation method, SPE 15072, SPE 56th California regional meeting, Oakland, pp 259–272

81. Erskine AD (1991) The effect of tidal fluctuation on a coastal aquifer in the UK. Ground Water 29(4):556–562
82. Fairhurst C (1964) Measurement of *in situ* rock stresses, with particular reference to hydraulic fracturing. Rock Mech Eng Geol 2:129–147
83. Fam MA, Dusseault MB, Fooks JC (2003) Drilling in mudrocks: rock behavior issues. J Pet Sci Eng 38(3–4):155–166
84. Ferris JG (1952) Cycle fluctuation of water levels as a basis for determining aquifer transmissibility, Groundwater Note 1, U.S. Geological Survey, Washington, DC
85. Figueroa JD, Fout T, Plasynski S, McIlvried H, Srivastava RD (2008) Advances in CO2 capture technology—the U.S. Department of Energy's carbon sequestration program. Int J Greenh Gas Control 2(1):9–20
86. Fillunger P (1936) Erdbaumechanik? (Earthwork Mechanics?). Wien: Selbstverlag des Verfassers (Vienna: self published by author)
87. Foda MA (1982) On the extrication of large objects from the ocean bottom (the breakout phenomenon). J Fluid Mech 117:211–231
88. Fourie F (2006) Aspects of the lateral and vertical resolution of surface electroseismic data with implications for groundwater exploration in fractured Karoo rocks. S Afr J Geol 109(4):571–584
89. Frank U, Barkley N (1995) Remediation of low permeability subsurface formations by fracturing enhancement of soil vapor extraction. J Hazard Mater 40(2):191–201
90. Franke PR, Nunz GJ (1985) Recent developments in the hot dry rock geothermal energy program. Geotherm Resour Counc Trans Part 2 9:95–98
91. Freed AM (2005) Earthquake triggering by static, dynamic, and postseismic stress transfer. Ann Rev Earth Planet Sci 33:335–367
92. Frenkel J (1944) On the theory of seismic and seismoelectric phenomena in moist soil. J Phys USSR 13(4):230–241
93. Fung YC (1977) A first course in continuum mechanics, 2nd edn. Prentice-Hall, Englewood Cliffs, 351pp
94. Gade HS (1957) Effects of a non-rigid, impermeable bottom on plane surface waves in shallow water. J Marine Res 16:61–82
95. Gambolati G, Teatini P (2014) Venice shall rise again: engineered uplift of Venice through saltwater injection. Elsevier, London/Waltham, 83pp
96. Garg SK, Nur A (1973) Effective stress laws for fluid-saturated porous rocks. J Geophys Res 78(26):5911–5921
97. Gassmann F (1951) Über die elastizität poröser medien (On elasticity of porous media). Veirteljahrsschrift der Naturforschenden Gesellschaft in Zürich 96:1–23
98. Geertsma J (1957) A remark on the analogy between thermoelasticity and the elasticity of saturated porous media. J Mech Phys Solids 6(1):13–16
99. Gibson RE, Knight K, Taylor PW (1963) A critical experiment to examine theories of three dimensional consolidation. In: Proceedings, European conference on soil mechanics and foundation engineering, vol 1, Weisbaden, pp 69–76
100. Gough DI, Bell JS (1982) Stress orientations from borehole wall fractures with examples from Colorado, east Texas, and northern Canada. Canad J Earth Sci 19(7):1358–1370
101. Grablovitz G (1880) Sul fenomeno di marea osservato nelle miniere di Dun in Bohemia (On the phenomenon of tide observed in the mines of Dun in Bohemia). Bull della Soc Adriat di Sci Nat Trieste 6:24–50
102. Gupta HK (1992) Reservoir-induced earthquakes. Elsevier, Amsterdam/New York, 364pp
103. Gupta HK (2002) A review of recent studies of triggered earthquakes by artificial water reservoirs with special emphasis on earthquakes in Koyna, India. Earth Sci Rev 58(3–4):279–310
104. Gutierrez MS, Lewis RW (2002) Coupling of fluid flow and deformation in underground formations. J Eng Mech ASCE 128(7):779–787
105. Haimson BC (1978) The hydrofracturing stress measuring method and recent field results. Int J Rock Mech Mining Sci 15(4):167–178

106. Haimson B, Fairhurst C (1967) Initiation and extension of hydraulic fractures in rocks. Soc Pet Eng J 7:310–318
107. Haimson B, Fairhurst C (1969) Hydraulic fracturing in porous-permeable materials. J Pet Technol 21:811–817
108. Haimson BC, Rummel F (1982) Hydrofracturing stress measurements in the Iceland research drilling project drill hole at Reydarfjordur, Iceland. J Geophys Res 87(NB8):6631–6649
109. Handin J, Hager RV, Friedman M, Feather JN (1963) Experimental deformation of sedimentary rock under confining pressure: pore pressure effects. AAPG Bull 47:717–755
110. Hantush MS, Jacob CE (1955) Non-steady radial flow in an infinite leaky aquifer. Trans Am Geophys Union 36(1):95–100
111. Hashin Z, Shtrikman S (1961) Note on a variational approach to the theory of composite elastic materials. J Frankl Inst Eng Appl Math 271(4):336–341
112. Hayashi K, Haimson BC (1991) Characteristics of shut-in curves in hydraulic fracturing stress measurements and determination of *in situ* minimum compressive stress. J Geophys Res 96(B11):18311–18321
113. Healy JH, Rubey WW, Griggs DT, Raleigh CB (1968) Denver earthquakes. Science 161(3848):1301–1310
114. Henkel DJ (1970) Role of waves in causing submarine landslides. Géotechnique 20(1):75–80
115. Hensen EJM, Smit B (2002) Why clays swell. J Phys Chem B 106(49):12664–12667
116. Howard GC, Fast CR (1970) Hydraulic fracturing. Society of Petroleum Engineers, New York, 203pp
117. Howarth RW, Ingraffea A, Engelder T (2011) Natural gas: should fracking stop? Nature 477(7364):271–275
118. Hsieh PA, Bredehoeft JD (1981) A reservoir analysis of the Denver earthquakes—a case of induced seismicity. J Geophys Res 86(NB2):903–920
119. Hsieh PA, Bredehoeft JD, Farr JM (1987) Determination of aquifer transmissivity from Earth tide analysis. Water Resour Res 23(10):1824–1832
120. Hsu JRC, Jeng DS, Tsai CP (1993) Short-crested wave-induced soil response in a porous seabed of infinite thickness. Int J Numer Anal Methods Geomech 17(8):553–576
121. Hubbert MK, Rubey WW (1959) Role of fluid pressure in mechanics of overthrust faulting. 1. Mechanics of fluid-filled porous solids and its application to overthrust faulting. Geol Soc Am Bull 70(2):115–166
122. Hubbert MK, Willis DG (1957) Mechanics of hydraulic fracturing. Trans Am Inst Mining Eng 210:153–166
123. Ishihara K, Yamazaki A (1984) Analysis of wave-induced liquefaction in seabed deposit of sand. Soil Found 24(3):85–100
124. Ivanov AG (1939) Effect of electrization of earth layers by elastic waves passing through them. Dokl Akad Nauk SSSR 24(1):42–45
125. Ivanov AG (1940) The electroseismic effect of the second kind. Izv Akad Nauk SSSR Ser Geogr Geofiz 5:699–727
126. Jacob CE (1940) On the flow of water in an elastic artesian unconfined aquifer. Trans Am Geophys Union 21:574–586
127. Jaeger JC (1963) Extension failures in rocks subject to fluid pressure. J Geophys Res 68(21):6066–6067
128. Jaeger JC, Cook NGW (1963) Pinching-off and disking of rocks. J Geophys Res 68(6):1759–1765
129. Jaeger JC, Cook NGW, Zimmerman C (2007) Fundamentals of rock mechanics, 4th edn. Wiley-Blackwell, Malden, 488pp
130. Jeng DS (1997) Wave-induced seabed instability in front of a breakwater. Ocean Eng 24(10):887–917
131. Jeng DS (2001) Numerical modeling for wave-seabed-pipe interaction in a non-homogeneous porous seabed. Soil Dyn Earthq Eng 21(8):699–712
132. Jeng DS (2001) Mechanism of the wave-induced seabed instability in the vicinity of a breakwater: a review. Ocean Eng 28(5):537–570

133. Jeng DS, Barry DA, Li L (2001) Water wave-driven seepage in marine sediments. Adv Water Resour 24(1):1–10
134. Jennings HM (2000) A model for the microstructure of calcium silicate hydrate in cement paste. Cem Concr Res 30(1):101–116
135. Jiao JJ, Tang Z (1999) An analytical solution of groundwater response to tidal fluctuation in a leaky confined aquifer. Water Resour Res 35(3):747–751
136. Jónsson S, Segall P, Pedersen R, Björnsson G (2003) Post-earthquake ground movements correlated to pore-pressure transients. Nature 424(6945):179–183
137. Jupp TE, Schultz A (2004) A poroelastic model for the tidal modulation of seafloor hydrothermal systems. J Geophys Res Solid Earth 109(B3):B03105
138. Kalifa P, Menneteau F-D, Quenard D (2000) Spalling and pore pressure in HPC at high temperatures. Cem Concr Res 30(12):1915–1927
139. Kehle RO (1964) The determination of tectonic stresses through analysis of hydraulic well fracturing. J Geophys Res 69(2):259–273
140. Kenyon DE (1979) Mathematical model of water flux through aortic tissue. Bull Math Biol 41(1):79–90
141. Kim JM, Parizek RR (1997) Numerical simulation of the Noordbergum effect resulting from groundwater pumping in a layered aquifer system. J Hydrol 202(1–4):231–243
142. Kim JM, Parizek RR (2005) Numerical simulation of the Rhade effect in layered aquifer systems due to groundwater pumping shutoff. Adv Water Resour 28(6):627–642
143. King FH (1892) Fluctuations in the level and rate of movement of ground-water on the Wisconsin Agricultural Experiment Station Farm and at Whitewater, Wisconsin. Bulletin No. 5. U.S. Department of Agriculture, Washington, DC, 75pp
144. Klonne FW (1880) Die periodischen schwankungen des wasserspiegels in den inundieten kohlenschachten von Dux in der period von 8 April bis 15 September 1879 (The periodic fluctuations of water levels in the flooded coal mine at Dux in the period 8 April to 15 September 1879). Sitzungsberichte Kaiserliche Akademie der Wissenschaften in Wien
145. Lade PV, de Boer R (1997) The concept of effective stress for soil, concrete and rock. Géotechnique 47(1):61–78
146. Lambe TW, Whitman RV (1969) Soil mechanics. Wiley, New York, 553pp
147. Lamé G (1852) Leçons sur la théorie mathématique de l'élasticité des corps solides (Lectures on the mathematical theory of elasticity of solid bodies). Bachelier, Paris, 335pp
148. Langguth HR, Treskatis C (1989) Reverse water level fluctuations in semiconfined aquifer systems – "Rhade effect". J Hydrol 109(1–2):79–93
149. Legarth B, Huenges E, Zimmermann G (2005) Hydraulic fracturing in a sedimentary geothermal reservoir: results and implications. Int J Rock Mech Mining Sci 42(7–8):1028–1041
150. Lehmann J (2007) Bio-energy in the black. Front Ecol Environ 5(7):381–387
151. Lin SJ, Jan KM, Weinbaum S, Chien S (1989) Transendothelial transport of low-density lipoprotein in association with cell mitosis in rat aorta. Arterioscler Thromb Vasc Biol 9(2):230–236
152. Liu PLF (1973) Damping of water waves over porous bed. J Hydraul Div ASCE 99(HY12):2263–2271
153. Logan TL, Mavor MJ, Khodaverdian M (1993) Western cretaceous coal seam project. Q Rev Methane Coal Seam Technol 10(April):6–12
154. Lu GQ, Zhao XS (eds)(2004) Nanoporous materials: science and engineering. Imperial College Press, London, 897pp
155. Mackenzie JK (1950) The elastic constants of a solid containing spherical holes. Proc Phys Soc Lond Sect B 63(361):2–11
156. Madsen OS (1978) Wave-induced pore pressures and effective stresses in a porous bed. Géotechnique 28(4):377–393
157. Magda W (1997) Wave-induced uplift force on a submarine pipeline buried in a compressible seabed. Ocean Eng 24(6):551–576

158. Mak AFT, Huang LD, Wang QQ (1994) A biphasic poroelastic analysis of the plow dependent subcutaneous tissue pressure and compaction due to epidermal loadings—issues in pressure sore. J Biomech Eng ASME 116(4):421–429
159. Mallard HH, Dalrymple RA (1977) Water waves propagating over a deformable bottom. In: 9th offshore technology conference, Houston, pp 141–146
160. Mandel J (1953) Consolidation des sols (etude mathematique) (Consolidation of soils (mathematical study)). Géotechnique 3:287–299
161. Martner S, Sparks N (1959) The electroseismic effect. Geophysics 24(2):297–308
162. Masoero E, Del Gado E, Pellenq RJM, Yip S, Ulm F-J (2014) Nano-scale mechanics of colloidal C-S-H gels. Soft Matter 10(3):491–499
163. Mavor MJ, Logan TL (1994) Recent advances in coal gas-well openhole well completion technology. J Pet Technol 46(7):587–593
164. Mehrabian A, Abousleiman Y (2011) General solutions to poroviscoelastic model of hydrocephalic human brain tissue. J Theor Biol 291:105–118
165. Mei CC, Foda MA (1981) Wave-induced responses in a fluid-filled poro-elastic solid with a free surface—a boundary layer theory. Geophys J R Astron Soc 66(3):597–631
166. Meinzer OE (1928) Compressibility and elasticity of artisan aquifers. Econ Geol 23:263–291
167. Melchior PJ (1983) The tides of the planet Earth, 2nd edn. Pergamon, Oxford/New York, 653pp
168. Mikhailov OV, Queen J, Toksöz MN (2000) Using borehole electroseismic measurements to detect and characterize fractured (permeable) zones. Geophysics 65(4):1098–1112
169. Millham NP, Howes BL (1995) A comparison of methods to determine *K* in a shallow coastal aquifer. Ground Water 33:49–57
170. Mow VC, Holmes MH, Lai WM (1984) Fluid transport and mechanical properties of articular cartilage—a review. J Biomech 17(5):377–394
171. Mow VC, Kuei SC, Lai WM, Armstrong CG (1980) Biphasic creep and stress relaxation of articular cartilage in compression—theory and experiments. J Biomech Eng ASME 102(1):73–84
172. Mu YK, Cheng AHD, Badiey M, Bennett R (1999) Water wave driven seepage in sediment and parameter inversion based on pore pressure data. Int J Numer Anal Methods Geomech 23(13):1655–1674
173. Murdoch LC, Chen J-L (1997) Effects of conductive fractures during *in-situ* electroosmosis. J Hazard Mater 55(1–3):239–262
174. Narasimhan TN (2005) Hydrogeology in North America: past and future. Hydrogeol J 13(1):7–24
175. Nierode DE (1985) Comparison of hydraulic fracture design methods to observed field results. J Pet Technol 37(11): 1831–1839
176. Norris A (1992) On the correspondence between poroelasticity and thermoelasticity. J Appl Phys 71(3):1138–1141
177. Nowacki W (1986) Thermoelasticity, 2nd edn. Pergamon, Oxford/New York, 566pp
178. Nur A, Booker JR (1972) Aftershocks caused by pore fluid flow? Science 175(4024):885–887
179. Oka M, Noguchi T, Kumar P, Ikeuchi K, Yamamuro T, Hyon SH, Ikada Y (1990) Development of an artificial articular cartilage. Clin Mater 6(4):361–381
180. Oka M, Ushio K, Kumar P, Ikeuchi K, Hyon SH, Nakamura T, Fujita H (2000) Development of artificial articular cartilage. Proc Inst Mech Eng Part H J Eng Med 214(1):59–68
181. Palmer AC, Rice JR (1973) Growth of slip surfaces in progressive failure of over-consolidated clay. Proc R Soc Lond Ser A Math Phys Eng Sci 332(1591):527–548
182. Paterson L (1986) A model for outbursts in coal. Int J Rock Mech Mining Sci 23(4):327–332
183. Paul P, Zoback M (2008) Wellbore stability study for the SAFOD borehole through the San Andreas Fault. SPE Drill Complet 23(4):394–408
184. Pearson C (1981) The relationship between microseismicity and high pore pressures during hydraulic stimulation experiments in low permeability granitic rocks. J Geophys Res 86(B9):7855–7864

185. Pennington WD, Davis SD, Carlson SM, Dupree J, Ewing TE (1986) The evolution of seismic barriers and asperities caused by the depressuring of fault planes in oil and gas fields of South Texas. Bull Seismol Soc Am 76(4):939–948
186. Phan LT (2008) Pore pressure and explosive spalling in concrete. Mater Struct 41(10):1623–1632
187. Phillips WS, House LS, Fehler MC (1997) Detailed joint structure in a geothermal reservoir from studies of induced microearthquake clusters. J Geophys Res Solid Earth 102(B6):11745–11763
188. Pinder, G. F., J. D. Bredehoeft, and H. H. Cooper (1969), Determination of aquifer diffusivity from aquifer response to fluctuations in river stage. Water Resour Res 5(4):850–855
189. Plumb RA, Cox JW (1987) Stress directions in eastern North America determined to 4.5 km from borehole elongation measurements. J Geophys Res Solid Earth Planets 92(B6):4805–4816
190. Poland JF (ed)(1984) Guidebook to studies of land subsidence due to ground-water withdrawal. UNESCO, Paris, 305pp
191. Polzer S, Gasser T, Markert B, Bursa J, Skacel P (2012) Impact of poroelasticity of intraluminal thrombus on wall stress of abdominal aortic aneurysms. BioMed Eng OnLine 11(1):1–13
192. Pratt WE, Johnson DW (1926) Local subsidence of the Goose Creek oil field. J Geol 34(7):577–590
193. Precht E, Huettel M (2004) Rapid wave-driven advective pore water exchange in a permeable coastal sediment. J Sea Res 51(2):93–107
194. Pride S (1994) Governing equations for the coupled electromagnetics and acoustics of porous media. Phys Rev B 50(21):15678–15696
195. Putnam JA (1949) Loss of wave energy due to percolation in a permeable sea bottom. Trans Am Geophys Union 30:349–356
196. Rahman MK, Naseby D, Rahman SS (2000) Borehole collapse analysis incorporating time-dependent pore pressure due to mud penetration in shales. J Pet Sci Eng 28(1–2):13–31
197. Raman-Nair W, Sabin GCW (1991) Wave-induced failure of poroelastic seabed slopes—a boundary element study. Proc Inst Civil Eng Part 2 Res Theory 91:771–794
198. Rappleye HS (1933) Recent areal subsidence found in releveling. Eng News Rec 110:845
199. Reid RO, Kajuira K (1957) On the damping of gravity waves over a permeable seabed. Trans Am Geophys Union 30:662–666
200. Rendulic L (1936) Porenziffer und porenwasserdruck in tonen (Void ratio and pore pressure in clays). Der Bauing 17:559–564
201. Revil A, Mahardika H (2013) Coupled hydromechanical and electromagnetic disturbances in unsaturated porous materials. Water Resour Res 49(2):744–766
202. Rice JR, Rudnicki JW (1979) Earthquake precursory effects due to pore fluid stabilization of a weakening fault zone. J Geophys Res 84(NB5):2177–2193
203. Rice JR, Simons DA (1976) Stabilization of spreading shear faults by coupled deformation-diffusion effects in fluid-infiltrated porous materials. J Geophys Res 81(29):5322–5334
204. Robinson TW (1939) Earth-tides shown by fluctuations of water-levels in wells in New Mexico and Iowa. Trans Am Geophys Union 20:656–666
205. Rodrigues JD (1983) The Noordbergum effect and characterization of aquitards at the Rio-Maior mining project. Ground Water 21(2):200–207
206. Roeloffs EA (1988) Fault stability changes induced beneath a reservoir with cyclic variations in water level. J Geophys Res Solid Earth Planets 93(B3):2107–2124
207. Rojstaczer S (1988) Determination of fluid flow properties from the response of water levels in wells to atmospheric loading. Water Resour Res 24(11):1927–1938
208. Rowe RK (2005) Long-term performance of contaminant barrier systems. Géotechnique 55(9):631–677
209. Rubey WW, Hubbert MK (1959) Role of fluid pressure in mechanics of overthrust faulting. 2. Overthrust belt in geosynclinal area of Western Wyoming in light of fluid-pressure hypothesis. Geol Soc Am Bull 70(2):167–205

210. Sakai T, Hatanaka K, Mase H (1992) Wave-induced effective stress in seabed and its momentary liquefaction. J Waterw Port Coast Ocean Eng ASCE 118(2):202–206
211. Schiffman RL, Chen AT-F, Jordan JC (1969) An analysis of consolidation theories. J Soil Mech Found Div ASCE 95(SM1):285–312
212. Schofield AN (2005) Disturbed soil properties and geotechnical design. Telford, London, 216pp
213. Schofield AN, Wroth CP (1968) Critical state soil mechanics. McGraw-Hill, New York, 310pp
214. Scott PP, Bearden WG, Howard GC (1953) Rock rupture as affected by fluid properties. Trans Am Inst Mining Metall Eng 198:111–124
215. Seed HB, Rahman MS (1978) Wave-induced pore pressure in relation to ocean-floor stability of cohesionless soils. Marine Geotechnol 3(2):123–150
216. Segall P (1989) Earthquakes triggered by fluid extraction. Geology 17(10):942–946
217. Segall P, Fitzgerald SD (1998) A note on induced stress changes in hydrocarbon and geothermal reservoirs. Tectonophysics 289(1–3):117–128
218. Segall P, Rice JR (1995) Dilatancy, compaction, and slip instability of a fluid-infiltrated fault. J Geophys Res Solid Earth 100(B11):22155–22171
219. Setton LA, Zhu WB, Mow VC (1993) The biphasic poroviscoelastic behavior of articular cartilage—role of the surface zone in governing the compressive behavior. J Biomech 26(4–5):581–592
220. Shapiro SA (2015) Fluid-induced seismicity. Cambridge University Press, Cambridge, 289pp
221. Siegrist RL, Lowe KS, Murdoch LC, Case TL, Pickering DA (1999) In situ oxidation by fracture emplaced reactive solids. J Environ Eng ASCE 125(5):429–440
222. Siffert RS, Luo GM, Cowin SC, Kaufman JJ (1996) Dynamic relationships of trabecular bone density, architecture, and strength in a computational model of osteopenia. Bone 18(2):197–206
223. Simon BR, Wu JSS, Carlton MW, Evans JH, Kazarian LE (1985) Structural models for human spinal motion segments based on a poroelastic view of the intervertebral disk. J Biomech Eng ASME 107(4):327–335
224. Simons DA (1977) Boundary-layer analysis of propagating mode II cracks in porous elastic media. J Mech Phys Solids 25(2):99–115
225. Simpson DW (1976) Seismicity changes associated with reservoir loading. Eng Geol 10(2–4):123–150
226. Skempton AW (1954) The pore pressure coefficients A and B. Géotechnique 4(4):143–147
227. Skempton AW (1960) Terzaghi's discovery of effective stress. In: Bjerrum L, Casagrande A, Peck RB, Skempton AW (eds) From theory to practice in soil mechanics. Wiley, New York, pp 42–53
228. Smith MB (1985) Stimulation design for short, precise hydraulic fractures. Soc Pet Eng J 25(3):371–379
229. Soltanzadeh H, Hawkes CD (2009) Assessing fault reactivation tendency within and surrounding porous reservoirs during fluid production or injection. Int J Rock Mech Mining Sci 46(1):1–7
230. Speziale S, Tenti G, Sivaloganathan S (2008) A poroelastic model of transcapillary flow in normal tissue. Microvasc Res 75(2):285–295
231. Šuklje L (1969) Rheological aspects of soil mechanics. Wiley Interscience, London/New York, 571pp
232. Teatini P, Castelletto N, Ferronato M, Gambolati G, Tosi L (2011) A new hydrogeologic model to predict anthropogenic uplift of Venice. Water Resour Res 47:W12507
233. Tennis PD, Jennings HM (2000) A model for two types of calcium silicate hydrate in the microstructure of Portland cement pastes. Cem Concr Res 30(6):855–863
234. Terzaghi K (1923) Die Berechnung der Durchlässigkeitsziffer des Tones aus dem Verlauf der hydrodynamischen Spannungserscheinungen (A method of calculating the permeability of clay from the history of hydrodynamic stress variation). Sitzungsber d Akad d Wiss Wien Math-Naturwiss Kl Abt IIa 132(3/4):125–138

235. Terzaghi K (1925) Erdbaumechanik auf Bodenphysikalischer Grundlage (Soil Mechanics on Soil Physical Basis). F. Deuticke, Leipzig u. Wien, 399pp
236. Terzaghi K (1936) The shearing resistance of saturated soils and the angle between the planes of shear. In: First international conference on soil mechanics, vol 1. Harvard University, Cambridge, pp 54–56
237. Terzaghi K (1943) Theoretical soil mechanics. Wiley, New York, 528pp
238. Terzaghi K, Peck RB (1948) Soil mechanics in engineering practice. Wiley, New York, 566pp
239. Terzaghi K, Peck RB, Mesri G (1996) Soil mechanics in engineering practice, 3rd edn. Wiley, New York, 592pp
240. Teufel LW (1983) Determination of *in-situ* stress from anelastic strain recovery measurements of oriented core, SPE 11649, SPE/DOE low permeability gas reservoirs symposium, Denver
241. Theis CV (1939) Earth tides as shown by fluctuations of water level in artesian wells in New Mexico. International Union of Geodesy and Geophysics, Washington Assembly, 11pp
242. Thimus JF, Abousleiman Y, Cheng AHD, Coussy O, Detournay E (eds)(1998) Poromechanics–a tribute to Maurice A. Biot. Balkema, Rotterdam/Brookfield, 648pp
243. Thompson A, Gist G (1993) Geophysical applications of electrokinetic conversion. Leading Edge 12(12):1169–1173
244. Thompson AH, Hornbostel S, Burns J, Murray T, Raschke R, Wride J, McCammon P, Sumner J, Haake G, Bixby M, Ross W, White BS, Zhou M, Peczak P (2007) Field tests of electroseismic hydrocarbon detection. Geophysics 72(1):N1–N9
245. Thompson R (1936) The seismic electric effect. Geophysics 1(3):327–335
246. Timoshenko SP, Goodier JN (1970) Theory of elasticity, 3rd edn. McGraw-Hill, New York, 567pp
247. Todd DK (1980) Groundwater hydrology, 2nd edn. Wiley, New York, 535pp
248. Ulm FJ, Constantinides G, Heukamp FH (2004) Is concrete a poromechanics material?—a multiscale investigation of poroelastic properties. Mater Struct 37(265):43–58
249. Vaccari A (1999) Clays and catalysis: a promising future. Appl Clay Sci 14(4):161–198
250. Valkó P, Economides MJ (1995) Hydraulic fracture mechanics. Wiley, Chichester, 318pp
251. Vandamme M, Ulm FJ (2013) Nanoindentation investigation of creep properties of calcium silicate hydrates. Cem Concr Res 52:38–52
252. van der Kamp G, Gale JE (1983) Theory of Earth tide and barometric effects in porous formations with compressible grains. Water Resour Res 19(2):538–544
253. van Eyden WAA, Kuper HW, Santema PW (1963) Some methods used in the geohydrological survey of the south-western deltaic area in the Netherlands. Proc Gen Assem Berkeley IUGG Int Assoc Sci Hydro Publ No 63:528–557
254. van Oort E, Hale AH, Mody FK, Roy S (1996) Transport in shales and the design of improved water-based shale drilling fluids. SPE Drill Complet 11(3):137–146
255. Verruijt A (1965) Discussion. In: Proceedings, sixth international conference on soil mechanics and foundation engineering, vol 3, Montreal, pp 401–402
256. Verruijt A (1969) Elastic storage of aquifers. In: DeWiest RJM (ed) Flow through porous media. Academic, New York, pp 331–376
257. Vito RP, Dixon SA (2003) Blood vessel constitutive models—1995–2002. Ann Rev Biomed Eng 5:413–439
258. Voight B (1968) Determination of the virgin state of stress in the vicinity of a borehole from measurements of a partial anelastic strain tensor in drill cores. Felsmech und Ing 6:201–215
259. Warpinski NR, Teufel LW (1989) A viscoelastic constitutive model for determining *in-situ* stress magnitudes from anelastic strain recovery of core. SPE Prod Eng 4(3):274–279
260. Warpinski NR, Teufel LW (1989) Author's reply to Discussion of "A viscoelastic constitutive model for determining *in-situ* stress magnitudes from anelastic strain recovery of core". SPE Prod Eng 4(3):287–289
261. Weinbaum S, Cowin SC, Zeng Y (1994) A model for the excitation of osteocytes by mechanical loading-induced bone fluid shear stresses. J Biomech 27(3):339–360
262. Wolff RG (1970) Relationship between horizontal strain near a well and reverse water level fluctuation. Water Resour Res 6(6):1721–1728

263. Wong TF, David C, Zhu WL (1997) The transition from brittle faulting to cataclastic flow in porous sandstones: mechanical deformation. J Geophys Res Solid Earth 102(B2):3009–3025
264. Wu JZ, Dong RG, Rakheja S, Schopper AW, Smutz WP (2004) A structural fingertip model for simulating of the biomechanics of tactile sensation. Med Eng Phys 26(2):165–175
265. Yamamoto T, Koning HL, Sellmeijer H, Van Hijum E (1978) On the response of a poro-elastic bed to water waves. J Fluid Mech 87:193–206
266. Yang J, Jiang GL (2003) Experimental study on properties of pervious concrete pavement materials. Cem Concr Res 33(3):381–386
267. Yang M, Taber LA, Clark EB (1994) A nonlinear poroelastic model for the trabecular embryonic heart. J Biomech Eng ASME 116(2):213–223
268. Yerkes RF, Castle RO (1976) Seismicity and faulting attributable to fluid extraction. Eng Geol 10(2–4):151–167
269. Yew CH (1997) Mechanics of hydraulic fracturing. Gulf Professional Publishing, Houston, 182pp
270. Yin L, Elliott DM (2004) A biphasic and transversely isotropic mechanical model for tendon: application to mouse tail fascicles in uniaxial tension. J Biomech 37(6):907–916
271. Yin S (2013) Numerical analysis of thermal fracturing in subsurface cold water injection by finite element methods. Int J Numer Anal Methods Geomech 37(15):2523–2538
272. Yin S, Dusseault MB, Rothenburg L (2007) Analytical and numerical analysis of pressure drawdown in a poroelastic reservoir with complete overburden effect considered. Adv Water Resour 30(5):1160–1167
273. Young A (1913) Tidal phenomena at inland boreholes near Craddock. Trans R Soc South Afr 31:61–106
274. Zhang DJ, Weinbaum S, Cowin SC (1998) On the calculation of bone pore water pressure due to mechanical loading. Int J Solids Struct 35(34–35):4981–4997
275. Zhao YS, Hu YQ, Wei JP, Yang D (2003) The experimental approach to effective stress law of coal mass by effect of methane. Transp Porous Media 53(3):235–244
276. Zheng ZQ, Kemeny J, Cook NGW (1989) Analysis of borehole breakouts. J Geophys Res Solid Earth Planets 94(B6):7171–7182
277. Zhu Z, Toksöz M (2005) Seismoelectric and seismomagnetic measurements in fractured borehole models. Geophysics 70(4):F45–F51
278. Zoback MD, Moos D, Mastin L, Anderson RN (1985) Well bore breakouts and *in situ* stress. J Geophys Res 90(B7):5523–5530

Chapter 2
Constitutive Equation

> *The Power of any Spring is in the same proportion with the Tension thereof: That is, if one power stretch or bend it one space, two will bend it two, and three will bend it three, and so forward. Now as the Theory is very short, so the way of trying it is very easie.*
>
> —Robert Hooke (1678)

A *constitutive equation* is a mathematical relation between two or more physical quantities. To define such mathematical relations, coefficients that are specific to a material, or to a composite material, known as *material constants*, are needed.

As quoted above, Robert Hooke (see Sect. F.2 for a biography) was the first to establish a constitutive relation for elastic bodies by observing that the elongation of a coil spring, a spiral spring, a wire string, and also the bending of a straight piece of wood, are directly proportional to the weight attached to them [13]. When this observation is used to establish linear relations between the *stress* and *strain* of an elastic body, it is called *Hooke's law*, to honor the original contribution by Hooke.

Since Hooke's observation, it took almost 130 years, until 1807, that a material constant was explicitly defined by Thomas Young [29] (see Sect. F.4 for a biography), and he stated:

> The modulus of the elasticity of any substance is a column of the same substance, capable of producing a pressure on its base which is to the weight causing a certain degree of compression, as the length of the substance is to the diminution of its length.

In the above, Young first defined a *weight of modulus of elasticity*, which was dependent on the cross-sectional area of the column. When the weight of modulus is divided by the cross-sectional area to yield a *height of modulus*, then it becomes a material property only, which in present-day terminology is called the *Young's modulus*. In fact, it was quite amazing that Young was able to determine the Young's modulus of a steel to be 2.9×10^7 psi, which is the same as the present-day determined value, by finding the frequency of vibration of a tuning fork [26, 29]!

Other examples of constitutive equations include Newton's law of viscosity, relating the rate of fluid shear strain to applied shear stress, through a material constant called *viscosity*, and Darcy's law of porous medium, relating the fluid

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_2

specific flux to the applied head gradient, through a constant called *hydraulic conductivity*, which is a combined property of the porous medium void space geometry and the fluid residing in it.

These constitutive equations (laws) are often combined with other physical laws of more general nature, such as mass, momentum, and energy conservation, which are not specific to a material, to form a set of *governing equations* to predict the physical response of matters subject to disturbances (see Chap. 6).

In this chapter, we shall construct the constitutive equations for poroelasticity. Particularly, we are interested in the deformation of the porous solid and the fluid, subjected to applied forces and pressure.

2.1 Physical Versus Phenomenological Approach

In general, there are two ways to construct constitutive laws—one is based on the *first principles*, known as the *theoretical approach*, and the other based on *phenomenology*, known as the *phenomenological approach*. First principle means established fundamental laws of physics, such as Newton's law of motion, Maxwell's equations of electromagnetic waves, etc. Phenomenology, on the other hand, implies an empirically constructed theory that relates an observed phenomenon to its cause (causality), using certain simple mathematical expressions (usually linear or polynomial), based on the regression of observed data.[1] The resultant equations are consistent with the fundamental laws, but only in the qualitative sense.

In principle, the theoretical approach is preferred as it is based on the fundamental laws of nature. In practice, however, the phenomenological approach is the likely choice, because it may not be practical, or even desirable, to derive theories based on the first principles. The reasons could be that the underlying mechanisms of a phenomenon are not yet fully understood, or the physical interactions are too complex to construct, even if they are known. Another reason could be that the mathematical system to describe the observations is too difficult to solve. On these occasions, empirical relations analogous to some known physical laws are constructed to approximately fit the observed data. The complicated or unknown physical mechanisms are lumped into certain empirical coefficients that are dependent on a specific material, known as the *material constants*.

As an example, we may look at Darcy's law for flow through porous medium, which is a process of a viscous fluid flow through a complex channel geometry driven by pressure differential. In principle, we can solve this problem by solving

[1]The *Concise Dictionary of Physics* [25] defines phenomenological theory as: A theory that expresses mathematically the results of observed phenomena without paying detailed attention to their fundamental significance.

Navier-Stokes equation in the exact geometry of the flow channels with proper boundary conditions. In reality, however, the channel geometry of a porous medium, given by cracks, fissures, and the interstitial space between solid grains, is either unknown, or too complex to be of interest to find out. Hence, rather than attempting to solve the exact problem, an empirical law is applied, which relates the fluid specific flux to the gradient of hydraulic head, in a fashion similar to gradient laws of heat conduction or mass diffusion. The complex physics behind it, which includes the effect of fluid viscosity and density, the channel geometry, and gravity, is lumped into a coefficient called *hydraulic conductivity*.

We may argue from a philosophical viewpoint that the boundaries between the fundamental and the phenomenological laws are fuzzy and relative, as Fourier law of heat conduction itself is a law of empirical nature [6]. Without engaging in a philosophical discussion, we shall assume for purposes of the present book that there exist two levels of laws. On one level, we may take the empirical view that a fluid saturated porous medium is a single material whose deformation is related to not only the stresses, but also the pore pressure, as indicated by observations. Hooke's law like constitutive equations can be constructed based on the assumption. We shall call this approach of construction a poroelasticity theory a *phenomenological theory*. In the second approach, we shall take a closer view of the underlying physics of the two interacting material phases, based on the laws of thermodynamics and the equations of state (constitutive equations) of the phase materials. We shall call this more fundamental approach the *physically-based theory*, or simply the *physical theory*.

In this chapter, we shall take the phenomenological approach to quickly construct a set of linear constitutive equations for poroelastic materials. Without looking into the individual material components, the physical quantities modeled are the total stress, pore pressure, the deformation of the porous frame, and the fluid volume expelled from the frame. The material constants that relate these *bulk* quantities are called *bulk material constants*. This was the approach adopted by Terzaghi in 1923 [22], as well as Biot in 1941 [2], when they first presented the phenomenological theories of soil consolidation and poroelasticity, respectively. This approach presents a practical theory for engineering and other applications, and will be referred to as the *bulk continuum model*.

To gain a deeper understanding of the bulk continuum model, we shall look into the mechanical responses of the individual components, including the solid and the fluid phases, and the pore space, in Chap. 3, as the *micromechanics model*. These micromechanical constitutive laws are then assembled to relate to and to justify the bulk continuum model.

The micromechanics model is in fact still a phenomenological model. To set these models on a firmer theoretical foundation, and to allow a rational extension of the ad hoc, linear theories into the nonlinear range, a more rigorous approach based on the thermodynamics laws and the variational energy principles, together with the averaging theory of heterogeneous (composite) materials, is introduced in Chap. 4, which will be referred to as the physical model.

2.2 Stress and Strain of Porous Medium

2.2.1 Stress

As a first step to construct the mathematical equations that describe the mechanical behavior of a porous body infiltrated with a fluid, we examine how the continuum mechanics concept of *stress* and *strain* can be applied to porous materials. In solid mechanics, stress on a surface is defined as force per unit area. Following the concept of a *continuum* [9], the definition of stress can be extended to be associated with a *point* in the interior of a body as a second rank symmetric tensor σ_{ij}, known as the *Cauchy stress tensor*, a concept first introduced by Augustin Cauchy (see Sect. F.6 for a biography). In such case, the stress is a continuous function of space (and time) within a solid body, $\sigma_{ij}^s = \sigma_{ij}^s(x, y, z)$, where the superscript ^{s} denotes the solid.

For porous materials, the concept of stress at a point is more complicated. For an illustration, we show in Fig. 2.1 a cross section of a porous material. We observe that porous materials are highly heterogeneous even if it is made of the same homogeneous and isotropic material, when examined up close at the grain scale. That is, when we traverse across the space of a porous material, we will alternately encounter solid and void within a short distance. Consequently, stresses can vary rapidly within such distance. To demonstrate, we draw in Fig. 2.1 a path marked by four points A–D that spans the distance of about a grain size. In a granular material, forces are transmitted through the medium by points of contact among the grains. At a contact point (actually a small contact surface), such as point A, high stress concentration is observed. At point B, inside a solid grain, a much smaller stress than that at Point A is found. Point C is a surface point, which is free from surface tractions if the void is empty (without fluid). Point D is in the void, and there is no stress.

When we study the mechanics of porous materials, we are in fact not interested in such rapid stress variation, and the localized deformation at the grain level. Modeling porous materials in such detail requires the knowledge of the geometry of the pore structure, as well as the inhomogeneous material properties of the individual grains, which is undesirable and impractical. To have a practical model, we typically interpret the stresses and deformation as quantities averaged over a volume, known as the *Representative Elementary Volume* (REV) [1], surrounding

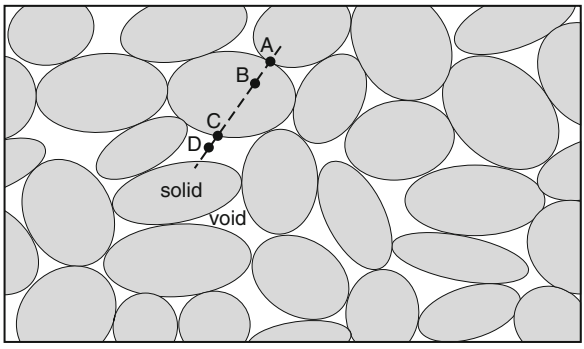


Fig. 2.1 A cross section of porous material. The four points *A* through *D* mark the path of rapid stress change from a high stress concentration at a solid-solid contact point to zero stress inside a void, within the distance of a grain size

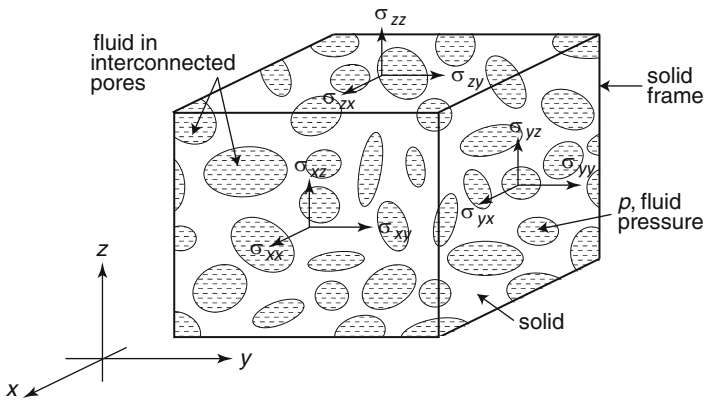


Fig. 2.2 Total stress tensor on an REV

the point of interest; see Fig. 2.2. The REV is typically taken to be one to two orders of magnitude greater than the grain size, in order to obtain a smooth and stable limit of the averaged quantities.

We shall examine such concept in further detail in Chap. 4. In the present chapter we shall simply accept the existence of a Cauchy stress tensor, in the form of a *total stress tensor*,[2] at every point of the porous material, as a continuous function, $\sigma_{ij} = \sigma_{ij}(x, y, z)$. Here we used the word "total" to denote the sum of the solid and fluid phases; that is, the total stress on a surface is defined as the sum of forces applied to the solid and the fluid phases, divided by the total surface area of the cross section, without differentiating the contribution by the individual phases.

To account for the presence of a fluid phase in a fluid infiltrated porous medium, we also introduce a fluid stress. Unlike a solid, fluid does not have the capability to resist shear stress under static equilibrium. In other words, when a shear stress is applied to a fluid element, it will continue to deform without rest. (According to Newton's law of viscosity, the *time rate* of fluid shear strain is proportional to the shear stress applied.) Also, based on Pascal's law, for a fluid in static equilibrium, its normal stress is the same in all directions. Hence, for the fluid phase, the stress tensor reduces to a single component, that is, a scalar, known as *pore pressure*, p (positive for compression).

By examining the cross section of a porous medium, such as that shown in Fig. 2.1, we observe that part of it is occupied by solid, and part by fluid. Hence we may decompose the total stress into two partial components, associated with the material phases; that is, a *solid partial stress* σ^s_{ij} and a fluid pressure p, given by the following relation

$$\sigma_{ij} = (1 - \phi)\, \sigma^s_{ij} - \phi\, \delta_{ij}\, p \tag{2.1}$$

[2]Following the convention of elasticity, stress is defined as a positive quantity for tension, and negative for compression.

where δ_{ij} is the Kronecker delta, and ϕ is the *areal porosity*. (We shall formally define porosity in Sect. 4.1.1.) In the above we note that the term containing fluid pressure is subtracted, rather than added, due to the different sign convention for stress and pressure.

We observe from (2.1) that the solid partial stress σ_{ij}^s is an interpreted, rather than a measured, quantity. Unlike pore pressure, which is constant under equilibrated state within a small neighborhood of interconnected pores, and can be measured, stress in solid varies rapidly at the grain/pore scale, as illustrated in Fig. 2.1. Hence the solid partial stress as defined in (2.1) cannot be directly measured. (We shall define in Chap. 4 the solid partial stress as a solid volume averaged stress.) Also, its interpretation based on (2.1) requires the measurement of porosity; so it is undesirable to explicitly model the solid partial stress in the constitutive equations that will be constructed.

2.2.2 Strain

We next turn our attention to deformation of porous medium, expressed in terms of *displacement*, or *strain*. Similar to the stress discussed in the preceding section, we shall only look at the bulk quantities, and avoid the partial components that are not directly observable, such as the strain of the solid phase.

Figure 2.3 shows a reference frame of a porous medium, which, for simplicity, shall be referred to as the *solid frame* henceforth. The frame is a volume element that incorporates not only the solid phase, but also the void space contained within the structure formed by the solid. When a porous medium is deformed, the deformation

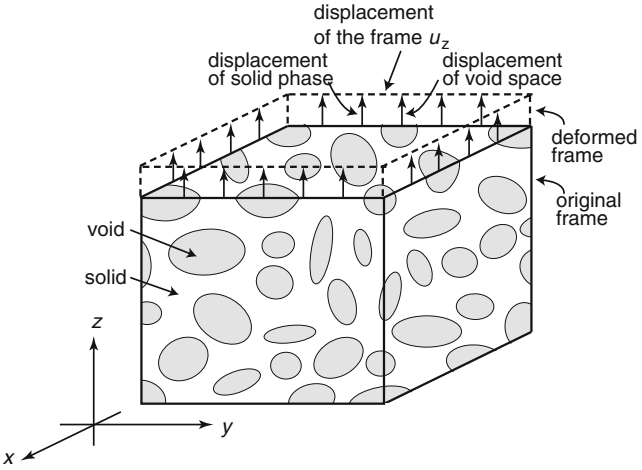


Fig. 2.3 Displacement of the porous medium frame

of the frame is defined by the displacement of the outer surface of the frame, including the part occupied by the solid, and that by the void. Figure 2.3 gives an illustration of the frame deformation in the z-direction.

Similar to solid mechanics, we shall define a solid displacement vector $\vec{u} = (u_x, u_y, u_z)$ with reference to the solid frame (Fig. 2.3). Based on this displacement, we can define the *total strain tensor* e_{ij} as

$$e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right) \tag{2.2}$$

In the above we have adopted the tensorial notation that a comma in the subscript followed by an index indicates partial differentiation with respect to spatial coordinate in that direction; that is, $u_{i,j} = \partial u_i/\partial x_j$, etc. The total volumetric strain (dilatation) is given by

$$e = e_{ii} = e_{xx} + e_{yy} + e_{zz} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = \nabla \cdot \vec{u} \tag{2.3}$$

where the repeated index means Einstein summation, and $\nabla\cdot$ is the divergence operator. In the above, we have used the word "total" to emphasize the fact that this is the apparent strain of the frame, which include both the solid and the void, and is not the strain of the solid phase.

As there exists a fluid phase, it is necessary to introduce a fluid strain as well. Similar to the situation with the fluid stress, in which only a scalar, fluid pressure p, is introduced, here only a scalar, fluid volumetric strain, will be needed. Fluid volumetric strain, however, is not an easily observable quantity, though it can be interpreted from the change in fluid pressure and the fluid compressibility. To define a quantity that is easy to observe and measure, Biot [2] introduced a fluid related strain as the *variation in fluid content* ζ, defined as the amount of fluid volume *entering* the solid frame per unit volume of solid frame. Figure 2.4 gives an illustration of such concept. The figure shows that both the solid frame and the fluid are deformed inward. However, as the solid frame is deformed more, the fluid actually escapes from the frame, giving a negative value of ζ.

2.3 Poroelastic Constitutive Equation

2.3.1 Isotropic Elastic Material

Before we construct the constitutive equations for poroelastic materials, we shall first examine the constitutive equations for linear, isotropic, elastic materials.

For elasticity, we seek to relate the stresses σ_{ij} to the strains e_{ij} through some linear relations with certain proportionality constants (material constants). We can construct a few simple relations, based on some easy to build laboratory

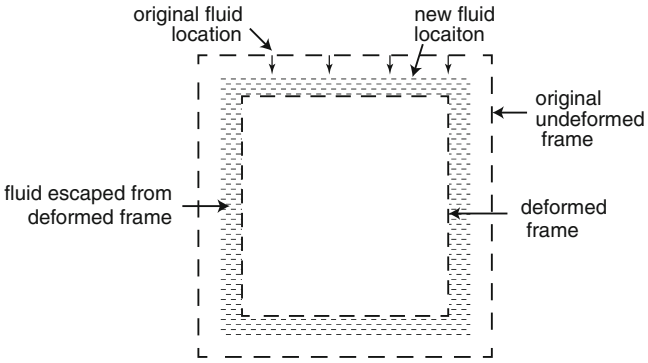


Fig. 2.4 Variation in fluid content: the shaded region indicates the volume of fluid escaped from the deformed frame as the negative variation in fluid content

experiments. For example, if we apply a uniform tension to a prismatic bar in its axial (z) direction, and observe its elongation, we find that its axial stress σ_{zz} is proportional to the axial strain e_{zz}, with a proportionality constant E, known as *Young's modulus*. This relation is also known as Hooke's law. Or, if we apply torsion to a prismatic bar, we will find that the torque is proportional to the angle of twist, with a proportionality constant G, known as the *shear modulus*. Or, if we apply normal stress in all three directions, then the average normal stress $(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})/3$ is proportional to the volumetric strain e, with a constant K, known as the *bulk modulus*. Other material constants can be defined through various testing configurations. Then the question arises: are these material constants independent, or are they related to each other? If these constants are related, then how many of them can be selected as independent, and the rest can be derived from the minimum set?

The above question in fact was not an easy one to answer in the development of theory of strength of materials [15]. The early efforts of finding the fundamental laws for elastic materials were largely based on the "molecule theory" of Newton, which assumed that all the interactions between parts of solid bodies could be reduced to attracting and repelling forces between the ultimate molecules. Navier (see Sect. F.5 for a biography) was the first to investigate the general equations of equilibrium and vibrations of elastic solids. He set out from the hypothesis that elastic reactions arise from variations in the intermolecular forces, consequent upon changes in the positions of molecules.[3] Based on his mathematical derivation, Navier deduced that there could be only one material constant for isotropic material [16]. Cauchy, who was the first to introduce the concept of stress and strain tensors at a point, and also to demonstrate the existence of principal directions,

[3]Modern day molecular mechanics theory shows that there exist multiple atomic bond potentials, including stretching, bending, torsion, etc. [17].

correctly showed that there should be two independent material constants for isotropic materials [7]. This result was further confirmed by Lamé (see Sect. F.8 for a biography) [14], among others.

Without going into the detail, which can be found in many elasticity textbooks [5, 19, 21, 27], the stress-strain relation for linear, isotropic elasticity can be shown in modern day notation as:

$$\sigma_{ij} = \lambda\, \delta_{ij} e + 2\mu\, e_{ij} \tag{2.4}$$

where λ and μ are two material constants known as *Lamé constants*. By setting up the various experimental configurations, we can demonstrate the following relations between the Lamé constants and the various "engineering constants" as

$$\lambda = \frac{2G\nu}{1-2\nu} = K - \frac{2G}{3} = \frac{E\nu}{(1+\nu)(1-2\nu)} \tag{2.5}$$

$$\mu = G = \frac{E}{2(1+\nu)} = \frac{3K(1-2\nu)}{2(1+\nu)} \tag{2.6}$$

In the above, ν is the *Poisson ratio*. With these relations, we can choose any two of the engineering constants as independent constants and rewrite (2.4). For example, if G and K are chosen, we can express the constitutive equation as

$$\sigma_{ij} = \left(K - \frac{2G}{3}\right) \delta_{ij} e + 2G e_{ij} \tag{2.7}$$

Now we have established elastic constitutive relation, we are ready to construct a similar relation for poroelasticity based on the phenomenological approach.

2.3.2 Isotropic Poroelastic Material

For poroelasticity, our goal is to relate the dynamic quantities (stresses), σ_{ij} and p, to the kinematic quantities (strains), e_{ij} and ζ. We can raise the following question: what is the most general relation that can be constructed in the spirit of Hooke's law (linear relation between stresses and strains)? We shall answer this question in full in Chap. 5 for a *general anisotropic* material. In the present chapter, we only consider an *isotropic* material.

For isotropic poroelastic material, our starting point is the elastic constitutive equation given by (2.7). With an extra kinematic quantity, the poroelastic (total) stress tensor σ_{ij} is dependent not only on the total strain tensor e_{ij}, but also on the variation in fluid content ζ. Hence the only possible linear relation is

$$\sigma_{ij} = \left(K_u - \frac{2G}{3}\right) \delta_{ij} e + 2G e_{ij} - A\, \delta_{ij} \zeta \tag{2.8}$$

where A is a new constitutive constant. In the above, the negative sign in front of the constant A is to ensure a positive material constant, and we have attached a subscript u to bulk modulus K, to designate an *undrained bulk modulus*. The reason for such designation will become clear in the next section, when the physical meaning of the material constants is explored.

For the dynamic quantities, in addition to the stress tensor, there also exists a pore pressure; hence a constitutive relation for pore pressure is also needed, which can be constructed as

$$p = -A'e + M\zeta \tag{2.9}$$

where A' and M are two additional material constants. We notice in (2.9) that as p is a scalar, only the volumetric strain e (a scalar), and not the full strain tensor e_{ij}, can be admitted to the right hand side.

The material constants appeared in (2.8) and (2.9) are not necessarily all independent. In fact, similar to the derivation of elasticity constitutive equation [5, 19], the constitutive relation must also satisfy certain energy principle, which imposes constraint to these constants. This is explored in the following section.

2.3.3 Reciprocal Work Theorem

In the absence of kinetic energy, that is, for a body in static equilibrium, and ignoring the thermal effects, the work done to an elastic body is stored as an internal energy, known as *strain energy*. Expressed as energy per unit volume, and in incremental form, this energy principle can written as

$$d\mathcal{W} = d\mathcal{U} \tag{2.10}$$

where $\mathcal{U} = \mathcal{U}(e_{ij})$ is the strain energy per unit volume, or the *strain energy density function*, and $\mathcal{W}$ is the work done to per unit volume of the body. (With the inclusion of kinetic energy, *Poroelastodynamics* will be introduced in Chap. 9; and with the consideration of thermal energy, *porothermoelasticity* will be presented in Chap. 11.)

Elasticity Consider only external mechanical forces, the work done per unit volume of body can be expressed as [21]

$$d\mathcal{W} = \sigma_{ij}\, de_{ij} \tag{2.11}$$

The fact that (2.10) and (2.11) are exact differentials suggests that

$$\sigma_{ij} = \frac{\partial \mathcal{U}}{\partial e_{ij}} \tag{2.12}$$

The above relation ensures that the deformation process is *reversible*, thus *elastic*.

If we aim at constructing a set of *linear* constitutive relations, that is, σ_{ij} is a linear function of e_{ij}, then $\mathcal{U}$ must be a *quadratic form* of e_{ij}. For isotropic elasticity, it is given as [21, 27]

$$\mathcal{U} = \frac{1}{2}\lambda\, e^2 + \mu\, e_{ij}e_{ij} \tag{2.13}$$

We also conclude that

$$\mathcal{U} = \frac{1}{2}\sigma_{ij}e_{ij} \tag{2.14}$$

The above equation is known as the Clapeyron formula [21]. From the above, we can further deduce that

$$\sigma_{ij}\, de_{ij} = e_{ij}\, d\sigma_{ij} \tag{2.15}$$

Or, put it in a different form,

$$\sigma_{ij}^{(1)} e_{ij}^{(2)} = \sigma_{ij}^{(2)} e_{ij}^{(1)} \tag{2.16}$$

where the superscripts $^{(1)}$ and $^{(2)}$ denote two independent system of stress and strain states. Equation (2.16) is known as the *Betti-Maxwell reciprocal relation*.

Poroelasticity The purpose of the above presentation is to show that for the constructed constitutive model to represent a reversible process, it must conform to the reciprocal energy principle (2.16). The elasticity constitutive equation (2.4) indeed satisfies the above principle. The question is: whether the newly constructed poroelasticity constitutive equations, (2.8) and (2.9), also satisfy such principle.

To answer the above question, we first need to define the external work applied to a poroelastic body infiltrated with a fluid. When an incremental external stress and pore pressure is applied, the total work performed to the body is the sum of the work done to the solid phase and the fluid phase. In the present phenomenological model, however, we do not explicitly model the solid and fluid phases and their associated partial stresses and strains; hence it is difficult to explicitly construct that expression. Using the available variables in the current phenomenological model, Biot [2] presented the following expression for work for poroelasticity, as an extension to (2.11),

$$d\mathcal{W} = \sigma_{ij}\, de_{ij} + p\, d\zeta \tag{2.17}$$

At a first glance, it is not clear why the above equation is the correct expression for work, as σ_{ij} is the total stress (combination of solid and fluid stresses), and e_{ij} is the strain of the solid frame (combination of solid phase and void space). For the fluid portion, p is the stress applied to the fluid, and ζ is the relative strain between the fluid and the solid frame. We shall demonstrate, however, using the physical

approach in Chap. 4, that this ad hoc constructed expression is indeed the correct work for a poroelastic body, as shown in (4.121). Hence, for the moment, we shall accept (2.17) as the correct statement for work on an poroelastic body, without a derivation.

Similar to elasticity, we can define a strain energy density function $\mathcal{U} = \mathcal{U}(e_{ij}, \zeta)$, such that

$$\sigma_{ij} = \frac{\partial \mathcal{U}}{\partial e_{ij}} \tag{2.18}$$

$$p = \frac{\partial \mathcal{U}}{\partial \zeta} \tag{2.19}$$

Within the framework of a linear constitutive model, we can further deduce the following results: the strain energy density function is given by

$$\mathcal{U} = \frac{1}{2}\left(\sigma_{ij} e_{ij} + p\,\zeta\right) \tag{2.20}$$

Its quadratic form is

$$\mathcal{U} = \frac{1}{2}\lambda_u\, e^2 + \mu\, e_{ij} e_{ij} - \alpha M e\,\zeta + \frac{1}{2} M \zeta^2 \tag{2.21}$$

in which the material constants will be defined in the sections below. Its differential form is

$$d\mathcal{U} = \sigma_{ij}\, de_{ij} + p\, d\zeta = e_{ij}\, d\sigma_{ij} + \zeta\, dp \tag{2.22}$$

leading to

$$\sigma_{ij}^{(1)} e_{ij}^{(2)} + p^{(1)} \zeta^{(2)} = \sigma_{ij}^{(2)} e_{ij}^{(1)} + p^{(2)} \zeta^{(1)} \tag{2.23}$$

as the extended Betti-Maxwell reciprocal relation.

To check the consistency of the poroelastic constitutive model with the energy principle, we can substitute (2.8) and (2.9) into (2.23). After eliminating terms, we find

$$A\,\zeta^{(1)} e^{(2)} + A' e^{(1)} \zeta^{(2)} = A\,\zeta^{(2)} e^{(1)} + A' e^{(2)} \zeta^{(1)} \tag{2.24}$$

It is clear that for the above to be an identity, we must have

$$A = A' \tag{2.25}$$

Hence, rather than the five material constants as shown in (2.8) and (2.9), for linear poroelasticity, we can admit only 4 independent material constants.

2.3.4 Stress-Strain Relation

With the above insight, and identify the undrained Lamé constant λ_u in (2.21) as

$$\lambda_u = K_u - \frac{2G}{3} \tag{2.26}$$

we can rewrite (2.8) and (2.9) as

$$\sigma_{ij} = \left(K_u - \frac{2G}{3}\right)\delta_{ij}e + 2Ge_{ij} - \alpha M \delta_{ij}\zeta \tag{2.27}$$

$$p = M\left(-\alpha e + \zeta\right) \tag{2.28}$$

We note that in the above we have written A in a special way, as the product of α and M. This change of notation only alters the definitions of the constitutive constants, and does not change the fundamental fact that there exist four independent constitutive constants, defined here as G, the *shear modulus*, K_u, the *undrained bulk modulus*, α, the *Biot effective stress coefficient*, and M, the *Biot modulus*. Equations (2.27) and (2.28) are then the *stress-strain relations* of linear poroelasticity.

An alternative form of stress-strain relation can be obtained by substituting (2.28) into (2.27) to eliminate ζ. We then find the following expression:

$$\sigma_{ij} = \left(K - \frac{2G}{3}\right)\delta_{ij}e + 2Ge_{ij} - \alpha\delta_{ij}p \tag{2.29}$$

in which we have define a new material constant K, the *drained bulk modulus*, with the following relation:

$$K = K_u - \alpha^2 M \tag{2.30}$$

In Sect. 2.4 we shall explore the physical significance of these material constants.

2.3.5 Strain-Stress Relation

The stress-strain relations, (2.27) and (2.28), can be inverted to give the *strain-stress relations*. Rather than using the previous set of material constants, we choose to write them in the following form

$$e_{ij} = \left(\frac{1}{9K} - \frac{1}{6G}\right)\delta_{ij}\,\sigma_{kk} + \frac{1}{2G}\,\sigma_{ij} + \frac{1}{3}CB\,\delta_{ij}\,p \tag{2.31}$$

$$\zeta = C\left(\frac{B}{3}\sigma_{kk} + p\right) \tag{2.32}$$

where $\sigma_{kk} = \sigma_{xx} + \sigma_{yy} + \sigma_{zz}$. In the above, we have defined four material constants, $\{G, K, B, C\}$, in which B is the *Skempton pore pressure coefficient B*, and $C = S_\sigma$ is a *constant stress storage coefficient* (Sect. 1.2.6). The physical interpretations of these bulk material constants are discussed in the next section.

Similar to the alternative expressions between (2.27) and (2.29), we can use (2.32) to eliminate p in (2.31), to obtain

$$e_{ij} = \left(\frac{1}{9K_u} - \frac{1}{6G}\right)\delta_{ij}\,\sigma_{kk} + \frac{1}{2G}\,\sigma_{ij} + \frac{1}{3}B\,\delta_{ij}\,\zeta \tag{2.33}$$

in which we have expressed

$$\frac{1}{K} = \frac{1}{K_u} + CB^2 \tag{2.34}$$

as the counterpart of (2.30).

As the strain-stress relations, (2.31) and (2.32), are simply the inverse of (2.27) and (2.28), it is necessary that the material constants are related. To find relations among the constitutive constants $\{\alpha, M\}$ and $\{B, C\}$, we can express (2.27), (2.29), (2.31), and (2.33) in terms of their deviatoric and volumetric components. First, defining the *deviatoric stress* and *deviatoric strain* as

$$\langle\sigma_{ij}\rangle = \sigma_{ij} - \frac{1}{3}\,\delta_{ij}\,\sigma_{kk} \tag{2.35}$$

$$\langle e_{ij}\rangle = e_{ij} - \frac{1}{3}\,\delta_{ij}\,e \tag{2.36}$$

we simply observe from the stress strain relations that

$$\langle\sigma_{ij}\rangle = 2G\,\langle e_{ij}\rangle \tag{2.37}$$

which is the same relation as elasticity.

For the volumetric responses, we can express the constitutive equations in terms of the average normal stress and volumetric strain, as

$$\sigma_{kk} = 3K_u\,e - 3\,\alpha M\zeta \tag{2.38}$$

$$\sigma_{kk} = 3Ke - 3\,\alpha\,p \tag{2.39}$$

$$e = \frac{1}{3K_u}\,\sigma_{kk} + B\,\zeta \tag{2.40}$$

$$e = \frac{1}{3K}\,\sigma_{kk} + CB\,p \tag{2.41}$$

By comparing (2.38) with (2.40), and (2.39) with (2.41), we observe the following relations:

$$\alpha M = K_u B \tag{2.42}$$

$$\frac{\alpha}{C} = KB \tag{2.43}$$

Utilizing (2.30) or (2.34) to eliminate either K or K_u in the above, we can express one set of material constants, $\{B, C\}$, into another set, $\{\alpha, M\}$, as

$$B = \frac{\alpha M}{K_u} = \frac{\alpha M}{K + \alpha^2 M} \tag{2.44}$$

$$C = \frac{1}{M} + \frac{\alpha^2}{K} = \frac{K_u}{M(K_u - \alpha^2 M)} \tag{2.45}$$

or vice versa

$$\alpha = KCB = \frac{K_u CB}{1 + K_u CB^2} \tag{2.46}$$

$$M = \frac{1}{C} + K_u B^2 = \frac{1}{C(1 - KCB^2)} \tag{2.47}$$

A comprehensive listing of the conversion formulae among these and other bulk material constants are summarized in Sect. B.1.

2.4 Bulk Material Constant

2.4.1 Drained and Undrained Constant

A key to the understanding of the basic response of fluid-infiltrated porous material is to recognize the difference between a *drained* and an *undrained* response. The undrained response characterizes the condition that the fluid is trapped in the porous solid frame. Such condition can be created by physically wrapping a poroelastic body with an impermeable membrane to prevent fluid from escaping; or, it characterizes the small time (instantaneous) response of any poroelastic body subject to a suddenly applied load, whatever the boundary condition is. The reason for the small time response is that fluid has a *viscosity*, which controls the velocity at which fluid can move. For a suddenly applied load, a pressure distribution is instantly generated. Once the fluid senses the pressure gradient, it moves with a velocity. However, a finite amount of time is needed for fluid to move a finite distance. Without giving such time, the fluid is trapped, and the response is undrained.

The undrained condition means there is no relative movement between the fluid and the solid frame, even though the solid phase, the fluid phases, and the frame can

instantly deform, responding to the applied load. By definition, the undrained state corresponds to the mathematical condition of

$$\zeta = 0 \tag{2.48}$$

The drained response, on the other hand, corresponds to zero pore pressure

$$p = 0 \tag{2.49}$$

This condition is established at large (infinite) time for any poroelastic body that has access to drainage surfaces, where the pressure is atmospheric (and there is no other surfaces maintained at different pressure).

In general, we may consider the undrained and drained responses as the two limiting states of a porous body subjected to a load, one at $t = 0^+$, and the other at $t \to \infty$. We shall demonstrate below that the deformation condition of these two limiting states satisfies elastic relation.

For the undrained response, we observe that the constitutive equation (2.27) becomes

$$\sigma_{ij} = \left(K_u - \frac{2G}{3}\right)\delta_{ij}e + 2Ge_{ij} \tag{2.50}$$

For the drained response, we can obtain from (2.29) the following

$$\sigma_{ij} = \left(K - \frac{2G}{3}\right)\delta_{ij}e + 2Ge_{ij} \tag{2.51}$$

Comparing (2.50) and (2.51) to (2.7), we observe that the above equations are the same as the elasticity constitutive equation, except that the dynamic variable is referred to the total stress, the kinematic variable is referred to the strain of the frame, and the material constants take different meanings. Based on such analogy, we first recognize G as the shear modulus of the frame. We next identify K_u and K as the "apparent" bulk modulus under the undrained and the drained condition, respectively. K_u is then called the *undrained bulk modulus*, and K the *drained bulk modulus*. Under drained condition, the fluid is not participating in the mechanical response; hence K is a property of the solid and solid frame only. K_u, on the other hand, is a combined property of the solid and the fluid. From (2.30), we observe that $K_u \geq K$.

Alternative expressions of constitutive equation can be given using the Poisson ratio, instead of bulk modulus, in (2.27) and (2.29). These equations are then expressed as

$$\sigma_{ij} = \frac{2G\nu_u}{1-2\nu_u}\delta_{ij}e + 2Ge_{ij} - \alpha M\delta_{ij}\zeta \tag{2.52}$$

$$\sigma_{ij} = \frac{2G\nu}{1-2\nu}\delta_{ij}e + 2Ge_{ij} - \alpha\,\delta_{ij}p \tag{2.53}$$

where we have defined the following relations

$$\nu = \frac{3K - 2G}{2(3K + G)} \tag{2.54}$$

$$\nu_u = \frac{3K_u - 2G}{2(3K_u + G)} \tag{2.55}$$

It is clear that ν_u and ν are the apparent Poisson ratios of the poroelastic material under the undrained and the drained state, respectively. Hence ν_u is called *undrained Poisson ratio*, which is a combined property of the solid and the fluid, and ν is *drained Poisson ratio*, a property of the solid only. Utilizing relations among constitutive constants as defined above, it can be shown that

$$\nu_u = \nu + \frac{9\alpha^2 GM}{2(G + 3K)(G + 3K_u)} \tag{2.56}$$

Based on the constraint on elastic material, we anticipate $0 \leq \nu \leq 1/2$; we hence conclude that $0 \leq \nu \leq \nu_u \leq 1/2$.

2.4.2 *Effective Stress Coefficient*

A large part of the development of soil mechanics is based on the concept of an *effective stress* [23, 24], which states that soil deforms elastically in response to an effective stress that is the difference between the total stress and the pore pressure. Expressed in terms of the present notations, we may write the *Terzaghi effective stress* as

$$\sigma'_{ij} = \sigma_{ij} + \delta_{ij} p \tag{2.57}$$

For poroelasticity, we can rearrange (2.29) into the following form

$$\sigma''_{ij} = \left(K - \frac{2G}{3}\right) \delta_{ij} e + 2Ge_{ij} \tag{2.58}$$

where

$$\sigma''_{ij} = \sigma_{ij} + \alpha\, \delta_{ij} p \tag{2.59}$$

As the right hand side of (2.58) is an elastic relation, σ''_{ij} is then an *effective stress*, and shall be call the *Biot effective stress*. The Biot effective stress differs from Terzaghi effective stress in the constant α multiplying the pressure. The material constant α is referred to as the *Biot effective stress coefficient*.

In Sect. 4.6.1 we shall demonstrate that α is a property of the solid and the porous frame only; in other words, it is independent of the fluid properties. Also,

in Sect. 4.8, it is shown that $0 \leq \alpha \leq 1$, which provides the justification of Terzaghi effective stress as a limiting case of Biot effective stress for materials characterized by $\alpha = 1$.

Another interpretation of the material constant α can be obtained by observing (2.28). When a poroelastic specimen is subjected to a deformation under drained condition, (2.28) becomes

$$\alpha = \frac{\zeta}{e} \tag{2.60}$$

This equation gives a meaning to α as the ratio of the fluid volume gained in a porous frame to the volume change of that frame, when the pore pressure is returned to its initial state. In the special case of incompressible solid constituent, $\alpha = 1$ (see Sect. 4.8.2), (2.60) states that the volume of fluid entering the solid frame is exactly the volume change of the frame, to compensate the increase in pore volume inside the frame.

2.4.3 *Pore Pressure Coefficient*

Skempton in 1954 [20] introduced the following pore pressure coefficients to characterize saturated soil in a triaxial compression test

$$\Delta p = B[\Delta\sigma_3 + A(\Delta\sigma_1 - \Delta\sigma_3)] \tag{2.61}$$

which can also be rearranged as

$$\Delta p = B\left[\frac{1}{3}(\Delta\sigma_1 + 2\Delta\sigma_3) + \left(A - \frac{1}{3}\right)(\Delta\sigma_1 - \Delta\sigma_3)\right] \tag{2.62}$$

In the above, Δp is the pore pressure increment in an undrained soil specimen, $\Delta\sigma_3 (= \Delta\sigma_2)$ is the increment in minor principal stress (all around chamber pressure), and $\Delta\sigma_1 > \Delta\sigma_3$ is the increment in major principal stress. (Following soil mechanics convention, compressive stress is defined as positive in this section.) The two material constants introduced are *Skempton pore pressure coefficient* A and B. Coefficient A is associated with the pore pressure generated by deviatoric stress, and is attributed to the inelastic behavior of soil. Without deviatoric stress, $\Delta\sigma_1 = \Delta\sigma_3$, (2.61) reduces to

$$\Delta p = B\Delta\sigma_3 \tag{2.63}$$

Or, without the inelastic effect, $A = 1/3$, and (2.62) becomes

$$\Delta p = B\frac{\Delta\sigma_1 + 2\Delta\sigma_3}{3} \tag{2.64}$$

Both the above equations suggest that the coefficient B represents the ratio of pore pressure generated in an undrained soil specimen to the applied all around or averaged compressive stress. For soil saturated with water, B typically takes the value of 1.

For a poroelastic body under undrained condition, we observe that (2.32) becomes [18]

$$p = -B\frac{\sigma_{kk}}{3} \tag{2.65}$$

Hence we identify the poroelastic constant B as the Skempton pore pressure coefficient B in (2.64). In Sect. 4.8 we shall demonstrate that we generally anticipate $0 \leq B \leq 1$, although $B > 1$ is possible.

2.4.4 Storage Coefficient

As fluid can be injected or extracted from a porous frame by the increase or decrease of pore pressure, it is useful to introduce the concept of a *storage coefficient* [10], which can be defined as the volume of fluid gained (or lost) in a poroelastic frame per unit volume of the frame, as a result of a unit increase (or decline) of pore pressure. It can be shown that the *Biot modulus* M, introduced by Biot [3, 4], is the inverse of one such storage coefficient.

We can examine the physical significance of M by conducting a test in which the solid frame is confined in a rigid and impermeable enclosure. We then inject a known volume of fluid into the porous frame, thus giving the variation in fluid content ζ, and observe the pressure rise p. As the solid frame is restrained from deformation, the condition $e = 0$ is established, and (2.28) reduces to

$$p = M\zeta \tag{2.66}$$

We then identify $1/M = S_e$ as a *constant strain storage coefficient* as described in Sect. 1.2.6.

We should emphasize that the storage coefficient $1/M$ is *not* the fluid compressibility. The fluid compressibility can be defined as the volume of fluid that can be accommodated into a unit volume of fluid, by a unit increment of pressure. In the current case, although the frame volume is a constant, the fluid volume is not maintained at a constant during the injection process, as the pore volume that accommodates the fluid can change, when the solid phase is compressed within the frame.

For the physical significance of C, we refer to the strain-stress relation (2.32). We observe that if we inject a volume of fluid into a wrapped poroelastic frame, while keeping the total confining stress of this frame to a constant, $\sigma_{kk} = 0$, then (2.32) reduces to

$$\zeta = Cp \tag{2.67}$$

Hence $C = S_\sigma$ is a *constant stress storage coefficient*, in contrast to $1/M$. C and M are also related through (2.45) and (2.47).

In groundwater applications, storage coefficient is an important quantity that characterizes the amount of water that can be extracted, or stored, in an elastic aquifer [28]. The definition of a *specific storativity*, S_s, is given by Hantush [11, 12] as the "*volume of water which a unit volume of the aquifer releases from storage because of expansion of water and compression of the aquifer under a unit decline in the average head within the unit volume of the aquifer*", which has the dimension of [1/L].

Following the assumptions of aquifer theory [8], the aquifer is restricted from movement in the lateral directions, such that $e_{xx} = e_{yy} = 0$, and only the vertical deformation (subsidence), e_{zz}, can exist. Also, the vertical overburden stress is assumed to remain unchanged during the injection or extraction of fluid from aquifer; hence $\sigma_{zz} = 0$ (in the incremental sense). Using these conditions in (2.28) and (2.29), we obtain

$$\sigma_{zz} = 0 = \left(K + \frac{4G}{3}\right) e_{zz} - \alpha p \tag{2.68}$$

$$p = M\left(-\alpha e_{zz} + \zeta\right) \tag{2.69}$$

Eliminating e_{zz} in the above, we obtain

$$\zeta = Sp \tag{2.70}$$

where

$$S = \frac{1}{M} + \frac{3\alpha^2}{3K + 4G} \tag{2.71}$$

is a *constant stress uniaxial strain storage coefficient*, which we shall abbreviate simply as the *storage coefficient*. If we use a different set of material constants, we can also express S as

$$S = C - \frac{4\alpha^2 G}{K(3K + 4G)} \tag{2.72}$$

The storage coefficient also plays a significant role in the definition of a *generalized consolidation coefficient* c, which will be introduced in Sect. 6.3.2, as

$$c = \frac{\kappa}{S} \tag{2.73}$$

where κ is the permeability coefficient.

We can now refer to the definition of the specific storativity used in groundwater, which is related to a unit drop of head, not pressure. The *piezometric head* of groundwater is defined as

$$h = \frac{p}{\gamma_f} + z \tag{2.74}$$

where γ_f is the specific weight of fluid, and z is the elevation. As z is a constant at a given location, it is clear that the specific storativity S_s used in groundwater applications is related to the storage coefficient S as

$$S_s = \gamma_f S \tag{2.75}$$

References

1. Bear J (1972) Dynamics of fluids in porous media. American Elsevier, New York, 764pp (also published by Dover, 1988)
2. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2):155–164
3. Biot MA (1956) General solutions of the equations of elasticity and consolidation for a porous material. J Appl Mech ASME 78:91–96
4. Biot MA, Willis DG (1957) The elastic coefficients of the theory of consolidation. J Appl Mech ASME 24:594–601
5. Boresi AP, Chong KP, Lee JD (2010) Elasticity in engineering mechanics, 3rd edn. Wiley, Hoboken, 656pp
6. Cartwright N (1983) How the laws of physics lie. Oxford University Press, New York, 232pp
7. Cauchy A-L (1828) Sur les équations qui expriment les conditions d'équilibre ou les lois du mouvement intérieur d'un corps solide, élastique, ou non émystiques (On equations that express the equilibrium conditions or the laws of motion within a solid, elastic, or inelastic). Exerc de Math 3:160–187
8. Cheng AHD (2000) Multilayered aquifer systems—fundamentals and applications. Marcel Dekker, New York/Basel, 384pp
9. Fung YC (1977) A first course in continuum mechanics, 2nd edn. Prentice-Hall, Englewood Cliffs, 351pp
10. Green DH, Wang HF (1990) Specific storage as a poroelastic coefficient. Water Resour Res 26(7):1631–1637
11. Hantush MS (1960) Modification of the theory of leaky aquifers. J Geophys Res 65(11):3713–3725
12. Hantush MS (1964) Hydraulics of wells. In: Chow VT (ed) Advances in hydroscience, vol 1. Academic, New York/London, pp 281–432
13. Hooke R (1678) De Potentia Restitutivâ, or of spring, explaining the power of springy bodies. Martyn, London, 56pp
14. Lamé G (1852) Leçons sur la théorie mathématique de l'élasticité des corps solides (Lectures on the mathematical theory of elasticity of solid bodies). Bachelier, Paris, 335pp
15. Love AEH (1892) A treatise on the mathematical theory of elasticity, vol 1. Cambridge University Press, Cambridge, 354pp

16. Navier C-L (1821) Memoire sur les lois de l'équilibre et du mouvement des corps solides élastiques (Memorandum on the laws of equilibrium and motion of elastic solids). Mem Acad Sci Inst Fr 2:375–393
17. Rappe AK, Casewit CJ, Colwell KS, Goddard WA, Skiff WM (1992) UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. J Am Chem Soc 114(25):10024–10035
18. Rice JR, Cleary MP (1976) Some basic stress diffusion solutions for fluid-saturated elastic porous media with compressible constituents. Rev Geophys 14(2):227–241
19. Saada AS (1993) Elasticity, theory and applications, 2nd edn. Kreiger, Malabar, 775pp
20. Skempton AW (1954) The pore pressure coefficients A and B. Géotechnique 4(4):143–147
21. Sokolnikoff IS (1956) Mathematical theory of elasticity, 2nd edn. McGraw-Hill, New York, 476pp
22. Terzaghi K (1923) Die Berechnung der Durchlässigkeitsziffer des Tones aus dem Verlauf der hydrodynamischen Spannungserscheinungen (A method of calculating the permeability of clay from the history of hydrodynamic stress variation). Sitzungsber. d Akad d Wiss Wien Math-Naturwiss Kl Abt IIa 132(3/4):125–138
23. Terzaghi K (1936) The shearing resistance of saturated soils and the angle between the planes of shear. In: First international conference on soil mechanics, vol 1. Harvard University, Cambridge, MA, pp 54–56
24. Terzaghi K (1943) Theoretical soil mechanics. Wiley, New York, 528pp
25. Thewlis J (ed)(1979) Concise dictionary of physics, 2nd edn. Pergamon, Oxford/New York, 370pp
26. Timoshenko SP (1983) History of strength of materials. Dover, New York, 452pp
27. Timoshenko SP, Goodier JN (1970) Theory of elasticity, 3rd edn. McGraw-Hill, New York, 567pp
28. Verruijt A (1969) Elastic storage of aquifers. In: DeWiest RJM (ed) Flow through porous media. Academic, New York, pp 331–376
29. Young T (1807) A course of lectures on natural philosophy and the mechanical arts, vol 2. Joseph Johnson, London, 738pp

Chapter 3
Micromechanics

We propose to show that a linear theory of consolidation can be established by combining the theory of elasticity with Darcy's law of flow of a fluid through a porous material. Suppose the porous skeleton considered is a continuous and homogeneous medium that satisfies the usual assumptions of the theory of elasticity. Let's ignore for the moment the liquid fills. Tension acting on a cubic unit-element side of the porous medium are similar to those that are defined in usual elasticity. The element is assumed to be large compared to the pore size.

—Maurice Biot (1935)

The bulk continuum model presented in the preceding chapter relates the *externally observable* quantities of a deforming porous specimen, such as the total stress, pore pressure, frame deformation, and fluid expelled from the frame, to each other, in order to construct constitutive relations that can be used to predict material behaviors. There are times, however, it is desirable, or even necessary, to learn what is happening *inside* a porous medium in terms of the solid and fluid phase, and the porous structure, such as change in porosity. For example, when we observe an external volume change of a porous frame, we may want to know how much of it is derived from the solid deformation, and how much is due to the pore space being taken out? When we measure a volume of fluid being expelled from a porous frame, how much of it is due to the reduction of the internal pore space, and how much is due to the expansion of fluid itself? For material constants, such as the undrained bulk modulus, how much of its apparent compliance is attributed to the compressibility of the solid constituent (which is typically small), and how much is to the pore space (which can be much larger)? Or, given an undrained bulk modulus, in what proportion does it draw its strength from the porous frame and the fluid?

Gassmann in 1951 [25] presented a model intended to partially answer these questions. In the model, Gassmann partitioned the total volume of the frame into a part occupied by the solid, and a part by the pores. In an effort to construct constitutive equations that relate the volumetric deformations to the applied stresses, he identified three micromechanical material constants, a solid, a fluid, and a pore compressibility. The Gassmann model, however, assumed that atthe grain level

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_3

(microscopic scale), the solid phase is homogeneous and isotropic, though at the macroscopic scale, the material can be heterogeneous and anisotropic. This model has been called the *ideal porous medium* model. Many porous material, particularly geomaterials, however, are not homogeneous and/or isotropic at the grain level. For example, rocks at the microscopic level are made of grains of different minerals, such as quartz, calcite, mica, and even clay minerals; hence are heterogeneous at that level. This suggests that Gassmann model is a special model.

For a general model, the *microhomogeneity* and *microisotropy* assumptions were removed by Biot and Willis in 1957 [8]. The resultant *micromechanics model* contains four independent material constants associated with volumetric deformation, one more than the ideal porous medium model. This micromechanical analysis has been widely accepted, and reformulated by many others [10, 14, 18, 46, 51, 66], in ways that are consistent with the original model. In this chapter, the Biot-Willis micromechanics model is presented.

3.1 Micromechanical Analysis

The micromechanical analysis presented in this section involves only the volumetric deformation of the porous material, and not the shear deformation. The reason is that no general analysis on shear deformation can be given. The constitutive relations for shear deformation is strongly dependent on the specific geometry of porous medium. The analysis is also difficult such that it has been performed only on regular shapes, such as spherical or ellipsoidal holes carved in solid, or regular packing of uniform size spherical particles, etc. For volumetric deformation, the constitutive equations presented below are dependent only on a single pore geometric parameter, the porosity, and not on pore size, pore shape, etc. Hence the result is general.

3.1.1 Solid and Pore Volumetric Strain

Consider a poroelastic sample of volume V, which contains a solid phase of volume V_s, and an interconnected pore space of volume V_p, such that the total volume is the sum of the two volumes

$$V = V_p + V_s \tag{3.1}$$

In the above, any isolated pores that fluid cannot infiltrate are considered as a part of the solid phase. The *porosity* is defined as

$$\phi = \frac{V_p}{V} \tag{3.2}$$

To construct constitutive equations for the volumetric deformation, we select as *kinematic variables* the pair of volumetric strains, $(\Delta V/V,\ \Delta V_p/V_p)$. For the *dynamic variables*, we select the mean total compressive stress $P = -\sigma_{kk}/3$, and the pore pressure p. These are expressed into a form that emphasizes the existence of *effective stresses*, as follows [14]:

$$\frac{\Delta V}{V} = -\frac{1}{K}(P - \alpha p) \tag{3.3}$$

$$\frac{\Delta V_p}{V_p} = -\frac{1}{K_p}(P - \beta p) \tag{3.4}$$

We note that the above relation is simply a statement of the generalized Hooke's law. A connection with the bulk continuum model in the preceding chapter can be made if we identify that $\Delta V/V = e$. Comparing (3.3) with (2.39), we can identify K and α as the drained bulk modulus and Biot effective stress coefficient, respectively. In (3.4), we define K_p as a *pore volume bulk modulus*, and β as a *pore volume effective stress coefficient*.

The coefficients appearing in (3.4) are not all independent. By invoking the *Betti-Maxwell reciprocal theorem* [52, 57] (see also Sect. 2.3.3), which states that for an energy conserving system, its internal (strain) energy is dependent only on the final state of the loading, and not on the loading path (sequence). This leads to the following constraint to the constitutive equations [10, 26]

$$\left.\frac{\partial V}{\partial p}\right|_{P=\text{const}} = -\left.\frac{\partial V_p}{\partial P}\right|_{p=\text{const}} \tag{3.5}$$

Substituting (3.3) and (3.4) into (3.5), we obtain

$$K_p = \frac{\phi}{\alpha}K \tag{3.6}$$

Hence, in micromechanics, by explicitly modeling an extra kinematic quantity, the pore volume strain $\Delta V_p/V_p$, we find one more material constant that relates volumetric deformation to the forces applied; that is, in addition to α and K defined in the bulk continuum model in Sect. 2.3, an extra constant, either β or K_p, is introduced. Also, the knowledge of porosity ϕ is needed in the above relations.

The above presentation is aimed at the interpretation of effective stresses for volumetric deformation and the associated effective stress coefficients [14]. To gain further insight into these coefficients, we can express the constitutive relations (3.3) and (3.4) in a different form, in terms of a pair of loading components (P', p), instead of (P, p), where

$$P' = P - p \tag{3.7}$$

is the difference between the total average compressive stress and the pore pressure, in the form of Terzaghi effective stress. The constitutive equations are then rewritten as

$$\frac{\Delta V}{V} = -\frac{P'}{K} - \frac{p}{K_s'} \tag{3.8}$$

$$\frac{\Delta V_p}{V_p} = -\frac{P'}{K_p} - \frac{p}{K_s''} \tag{3.9}$$

Here, following the notations introduced by Rice and Cleary [51], we have defined two bulk moduli, K_s' and K_s''. The physical meaning of K_s' and K_s'' can be elucidated if we conduct a laboratory test as described in Sect. 1.2.4 to establish the condition $P = p$, such that $P' = 0$. In that case, we find the following relations

$$K_s' = -\left.\frac{P}{\Delta V/V}\right|_{P'=0} \tag{3.10}$$

$$K_s'' = -\left.\frac{P}{\Delta V_p/V_p}\right|_{P'=0} \tag{3.11}$$

In Sect. 3.5, we demonstrate that the condition $P' = 0$ can be achieved in an *unjacketed test*, in which a test specimen is submerged in a fluid chamber and the load is applied by increasing the fluid pressure in the chamber. Hence K_s' is the *unjacketed frame bulk modulus*, and K_s'' is the *unjacketed pore volume bulk modulus* [30, 31].

As the pair of equations (3.8)–(3.9) and the pair (3.3)–(3.4) must represent the same constitutive relations, it is clear from comparison that the two effective stress coefficients are

$$\alpha = 1 - \frac{K}{K_s'} \tag{3.12}$$

$$\beta = 1 - \frac{K_p}{K_s''} \tag{3.13}$$

Since K_p is not an independent constant, we also find

$$K_p = \frac{\phi K K_s'}{K_s' - K} \tag{3.14}$$

Hence the set of independent material constants $\{K, \alpha, \beta\}$ can be replaced by the set $\{K, K_s', K_s''\}$.

Similarly, we can construct the constitutive equations for the solid strain, $\Delta V_s/V_s$, and the porosity variation, $\Delta\phi$, as linear functions of (P, p) or (P', p). Before doing so, however, we need to point out that ΔV_s and $\Delta\phi$ are not independent

kinematic variables, and they are related to ΔV and ΔV_p. To find the relation, we first take the variation of the definition of porosity (3.2), and obtain

$$\Delta\phi = \Delta\left(\frac{V_p}{V}\right) = \frac{V\Delta V_p - V_p\Delta V}{V^2} = \frac{\Delta V_p - \phi\,\Delta V}{V} \tag{3.15}$$

Based on (3.1), we then derive the following set of kinematic relations [14, 18]:

$$\frac{\Delta V}{V} = \frac{\Delta V_s}{V_s} + \frac{\Delta\phi}{1-\phi} \tag{3.16}$$

$$\frac{\Delta V_p}{V_p} = \frac{\Delta V_s}{V_s} + \frac{\Delta\phi}{\phi\,(1-\phi)} \tag{3.17}$$

Using the above equations and (3.8) and (3.9), we can assemble them into the following

$$\frac{\Delta V_s}{V_s} = -\frac{1}{(1-\phi)K_s'}P' - \frac{1}{1-\phi}\left(\frac{1}{K_s'} - \frac{\phi}{K_s''}\right)p \tag{3.18}$$

$$\Delta\phi = -\left(\frac{1-\phi}{K} - \frac{1}{K_s'}\right)P' + \phi\left(\frac{1}{K_s'} - \frac{1}{K_s''}\right)p \tag{3.19}$$

These are the constitutive equations for the solid strain and porosity variation. Or, we can rewrite (3.19) into the following:

$$\frac{\Delta\phi}{1-\phi} = -\frac{P'}{K_\varphi} + \frac{\phi}{1-\phi}\left(\frac{1}{K_s'} - \frac{1}{K_s''}\right)p \tag{3.20}$$

where

$$\frac{1}{K_\varphi} = \frac{1}{K} - \frac{1}{(1-\phi)K_s'} \tag{3.21}$$

Apparently, under drained condition, $p = 0$, the porosity variation is associated with K_φ, which may be defined as a *porosity bulk modulus*.

3.1.2 *Fluid Volumetric Strain*

We shall define V_f as the volume of fluid occupying the original pore space of a porous medium. Under saturated condition, the fluid volume is simply the pore volume

$$V_f = V_p \tag{3.22}$$

For small deformation, the constitutive equation for the fluid is simply

$$\frac{\Delta V_f}{V_f} = -\frac{p}{K_f} = -C_f p \tag{3.23}$$

where C_f is the *fluid compressibility*, and $K_f = 1/C_f$ is the *fluid bulk modulus*.

What we are more interested in, however, is a constitutive equation for the volume of fluid that the porous medium exchanged with its surrounding, that is, the volume of fluid either absorbed in, or escaped from, the porous medium frame, as it is a quantity that is more easily observed. This relation needs to be carefully constructed. First we note that although the fluid resides in the pore space, its change is not simply the pore space change, $\Delta V_f \neq \Delta V_p$, as (3.22) might suggest, because fluid can move relative to the pore space. In fact the difference between the two is exactly the volume of fluid exchanged,

$$\Delta V_f' = \Delta V_f - \Delta V_p \tag{3.24}$$

In the above, $\Delta V_f' > 0$ means fluid leaving the porous medium frame, while $\Delta V_f' < 0$ means entering.

From the definition of ζ, the variation in fluid content, as presented in Sect. 2.2.2, it is given by

$$\zeta = -\frac{\Delta V_f'}{V} \tag{3.25}$$

Here a negative sign is assigned because ζ is positive for a gain of fluid, that is, fluid entering the frame. Using (3.9), (3.14), (3.23), (3.24) and (3.25), a constitutive equation can be derived for ζ:

$$\zeta = -\left(\frac{1}{K} - \frac{1}{K_s'}\right) P' + \phi \left(\frac{1}{K_f} - \frac{1}{K_s''}\right) p \tag{3.26}$$

Equations (3.8), (3.9) and (3.26) are then the three independent constitutive equations for volumetric deformation based on micromechanics. The independent material constants involved in these equations are $\{K, K_s', K_s'', K_f, \phi\}$, or $\{K, \alpha, \beta, K_f, \phi\}$. The number of independent material constants conforms with the Biot-Willis model [8]. These constants are compared to the set $\{K, \alpha, M\}$ for the bulk continuum model presented in Chap. 2.

3.1.3 Link Among Material Constants

The link among the bulk and the micromechanical material constants can be established by comparing the constitutive equations developed in the above sections

with those in Sect. 2.3. For example, by comparing (3.26) with (2.32), we identify

$$C = \frac{\phi}{K_f} + \frac{1}{K} - \frac{1}{K_s'} - \frac{\phi}{K_s''} \tag{3.27}$$

$$B = 1 - \frac{\phi K K_s'(K_s' - K_f)}{K_f K_s''(K_s' - K) + \phi K K_s'(K_s' - K_f)} \tag{3.28}$$

Other expressions can be constructed using the various relations found in Sects. 2.3.4 and 2.3.5, and we obtain

$$M = \frac{K_f K_s'^2 K_s''}{K_f K_s''(K_s' - K) + \phi K_s'^2(K_s'' - K_f)} \tag{3.29}$$

$$K_u = K + \frac{K_f K_s''(K_s' - K)^2}{K_f K_s''(K_s' - K) + \phi K_s'^2(K_s'' - K_f)} \tag{3.30}$$

Here we may check that the relation presented by Brown and Korringa [10]

$$\frac{1}{C_u - C_s'} = \frac{1}{C_K - C_s'} + \frac{1}{\phi(C_f - C_s'')} \tag{3.31}$$

is indeed satisfied. In the above, $C_u = 1/K_u$, $C_K = 1/K$, $C_s' = 1/K_s'$, and $C_s'' = 1/K_s''$, are the corresponding compressibilities.

The above and other relations between the bulk material constants and the micromechanical constants are summarized in Sect. B.2.

3.2 Ideal Porous Medium

In this section we discuss a special porous medium model, in which the material is microscopically homogeneous and isotropic. This case appears to be first studied by Gassmann [25], and then by many others [8, 41, 46]. We shall call such material an *ideal porous medium*.

Gassmann Model Gassmann in 1951 [25] was the first to investigate the micromechanics of poroelastic material, utilizing the assumption that the solid phase at the microscopic (grain) level is homogeneous and isotropic. This idealized model, though a simplification of the Biot-Willis model presented above, brings certain physical insight into the micromechanics, and also leads to useful applications; hence is worthwhile to be examined here.

A characteristic of an ideal porous medium become apparent when it is subjected to an unjacketed test mentioned in Sect. 3.1.1, and described in more detail in Sect. 3.5. In the test, a sample is submersed in a fluid chamber, and the loading to the sample is applied by raising the fluid pressure in the chamber by an increment of

Δp. Once the pore pressure inside the sample reaches equilibrium with the chamber pressure, the loading condition is given as $P = p = \Delta p$, and $P' = P - p = 0$. Under this condition, every parts of the solid surface, external or internal (surface of the interconnected pores), are subjected to the same uniform pressure Δp. As the material is homogeneous and isotropic, the sample will deform in a geometrically similar fashion. This means that the solid and the pore space are deformed in the same proportion, leading to the conditions

$$\frac{\Delta V_s}{V_s} = \frac{\Delta V_p}{V_p} = \frac{\Delta V}{V} \tag{3.32}$$

From (3.10) and (3.11), we reach the conclusion that

$$K_s' = K_s'' = K_s \tag{3.33}$$

in which we have introduced a new notation K_s. From (3.18) with $P' = 0$, we find that

$$\frac{\Delta V_s}{V_s} = -\frac{p}{K_s} \tag{3.34}$$

As the entire surface of the solid is surrounded by the pressure p, K_s is obviously the bulk modulus of the (homogeneous and isotropic) solid. This definition clarifies the physical meaning of K_s' and K_s''. That is, under the ideal porous medium assumption, they are the bulk modulus of solid, but generally they are not, and are defined by (3.10) and (3.11). We also conclude that for ideal porous medium, there is a reduction in the number of material constants, as K_s' and K_s'' are the same. So the list of volumetric deformation related independent material constants becomes $\{K, K_s, K_f, \phi\}$.

With the reduction in micromechanical constants, we can express (3.12) into the well-known relation that interprets the Biot effective stress coefficient

$$\alpha = 1 - \frac{K}{K_s} \tag{3.35}$$

Other relations that link the bulk and the micromechanical constants, such as (3.27)–(3.30), also simplify to give the following

$$C = \frac{\phi}{K_f} + \frac{1}{K} - \frac{1+\phi}{K_s} \tag{3.36}$$

$$B = 1 - \frac{\phi K(K_s - K_f)}{K_f(K_s - K) + \phi K(K_s - K_f)} \tag{3.37}$$

$$M = \frac{K_f {K_s}^2}{K_f(K_s - K) + \phi K_s(K_s - K_f)} \tag{3.38}$$

$$K_u = K + \frac{K_f(K_s - K)^2}{K_f(K_s - K) + \phi K_s(K_s - K_f)} \tag{3.39}$$

These and other relations are summarized in Sect. B.2.

We can make some observation based on the above relations. We observe from (3.36) that C, the storage coefficient under constant stress condition, is a combination of three compressibilities: the sum of the fluid and frame compressibilities, subtracting that of the solid. Equation (3.37) shows that for geomaterials, we generally expect $K_s > K_f$, and the Skempton pore pressure coefficient B is less than unity, $B < 1$. However, for materials whose solid phase is more compliant than fluid, that is, $K_s < K_f$, B can be greater than 1. This implies that when such material is subject to a confining stress P, the undrained pore pressure rise p is greater than the applied compressive stress P.

Equation (3.39) may be referred to as the *Gassmann equation* [5, 25]. Gassmann was the first to suggest that elastic wave velocity in saturated porous medium is characterized by an undrained bulk modulus, which is a combination of the drained bulk modulus K, and a modulus related to the fluid, such that $K_u > K$.

With the ideal porous medium assumption, the various constitutive equations presented in Sect. 3.1, (3.8), (3.9), (3.18), (3.19), and (3.26), reduce to the following:

$$\frac{\Delta V}{V} = -\frac{P'}{K} - \frac{p}{K_s} \tag{3.40}$$

$$\frac{\Delta V_p}{V_p} = -\frac{1}{\phi}\left(\frac{1}{K} - \frac{1}{K_s}\right)P' - \frac{p}{K_s} \tag{3.41}$$

$$\frac{\Delta V_s}{V_s} = -\frac{1}{(1-\phi)K_s}P' - \frac{p}{K_s} \tag{3.42}$$

$$\frac{\Delta \phi}{1-\phi} = -\frac{P'}{K_\varphi} = -\left[\frac{1}{K} - \frac{1}{(1-\phi)K_s}\right]P' \tag{3.43}$$

$$\zeta = -\left(\frac{1}{K} - \frac{1}{K_s}\right)P' + \phi\left(\frac{1}{K_f} - \frac{1}{K_s}\right)p \tag{3.44}$$

Pore Collapse In the above constitutive equations, we pay particular attention to (3.43), which shows that porosity variation is a function of $P' = P - p$ only, the Terzaghi effective compressive stress. Equation (3.43) can be used to explain the experimental observation on sandstones [19, 24, 29, 68], limestone [60], and granite [54] that the ultimate strength and ductility of rocks are unique functions of the Terzaghi effective compressive stress P'. Porous materials that are granular and cemented generally have weak pore structure as compared to the solid constituent. For a material like sandstone, the first stage of compressive failure is likely to be characterized by the collapse of pore space than the yielding of solid phase. The "strength" of the pore space may be considered to consist of two parts: one part is its strength to resist a geometrically similar deformation (proportionate deformation

of solid and pore space), and the other is its strength to resist a disproportionate deformation. It seems that the integrity of porous structure is stronger resisting the former than the latter. The former does not cause a porosity change, while latter does. Hence the compressive failure of a porous medium is directly controlled by the *porosity change*. Equation (3.43) clearly indicates that for an ideal porous medium, porosity change is a unique function of P'; hence the threshold of rock failure is related to the magnitude of Terzaghi effective stress. Experimental evidences show that the fracture, tensile, and frictional strengths of rock are also controlled by the Terzaghi effective stress [11, 13, 38], though inconsistent evidences were also reported [53, 55].

3.3 Effective Modulus

There have been extensive studies conducted on composite materials for the *effective material properties*, based on a homogeneous body containing *inclusions* of different materials, or as a *mixture* of several homogeneous materials [21, 32–35]. As porous medium is an elastic body with embedded *voids*, which can be considered as a "material" with no mass and strength, it is a special case of the composite materials [12, 17, 42, 63].

There are many effective physical properties that can be examined, including elastic moduli, Darcy flow permeability, thermal, electrical, and magnetic conductivity, etc. In this section, we shall examine the *effective elastic moduli* of porous medium, particularly the (drained) bulk modulus and shear modulus. Assuming that the solid phase (the *matrix material* in composite material terminology) is homogeneous and isotropic with bulk modulus K_s and shear modulus G_s, the resultant effective bulk modulus K and shear modulus G are dependent only on the solid moduli K_s and G_s, the porosity ϕ, and the geometry of the pore space (size, shape, orientation, and their statistical distribution). In the following, we shall examine a few such simple models.

3.3.1 Mackenzie Model

Mackenzie [42] considered the effective bulk modulus and shear modulus of a homogeneous and isotropic elastic body containing spherical holes. It is assumed that the pores are far away from each other such that there is no elastic interaction among them. This assumption obviously only works for a dilute concentration of pores, implying small porosity. Based on an asymptotic analysis, the following results were obtained:

$$\frac{1}{K} = \frac{1}{(1-\phi)K_s} + \frac{3\phi}{4(1-\phi)G_s} + O(\phi^3) \tag{3.45}$$

$$\frac{G_s - G}{G_s} = \frac{5\phi\,(3K_s + 4G_s)}{9K_s + 8G_s} + O(\phi^2) \tag{3.46}$$

where K and G are respectively the effective drained bulk modulus and the effective shear modulus of the porous material. Comparing (3.45) with (3.43), we can identify the porosity bulk modulus K_φ as

$$K_\varphi = \frac{4(1-\phi)G_s}{3\phi} \tag{3.47}$$

The above equation shows that K_φ depends only on the solid shear modulus G_s, and not on K_s, implying that porosity variation is controlled by the effective pressure P' and the shear modulus of the solid G_s only.

3.3.2 Walsh Model

Based on the same dilute concentration assumption as the above, but the void geometry of a penny-shaped crack, not a sphere, Walsh [63] derived the following relation for effective bulk modulus

$$\frac{1}{K} = \frac{1}{K_s}\left(1 + \frac{16}{9}\,\frac{1-\nu_s^2}{1-2\nu_s}\,\epsilon\right) \tag{3.48}$$

where

$$\nu_s = \frac{3K_s - 2G_s}{2(3K_s + G_s)} \tag{3.49}$$

is the Poisson ratio of the matrix material, and

$$\epsilon = \frac{N\overline{c}^3}{V_b} = \frac{\overline{c}^3}{v} \tag{3.50}$$

is a crack density parameter (dimensionless), with $\overline{c}$ denoting the average crack half length (radius of penny-shape crack), N the number of cracks in the bulk sample volume V_b, and $v = V_b/N$ the average region volume (volume per crack). Equation (3.48) shows that the change in compliance (compressibility) of a porous medium is linearly dependent on the increase of crack density in the volume, which can be accomplished without porosity change if the cracks are thin and without volume.

Walsh [63] also presented the result for spherical pores as

$$\frac{1}{K} = \frac{1}{K_s}\left(1 + \frac{3}{2}\,\frac{1-\nu_s}{1-2\nu_s}\,\frac{\phi}{1-\phi}\right) \tag{3.51}$$

Using (3.49) in the above, it is easy to show that (3.51) is the same as (3.45) of the Mackenzie model. For a comparison with the crack model, we can assume that all spheres are of uniform size with radius r, thus $\overline{c} = r$. We can express the porosity as

$$\phi = \frac{4\pi}{3}\frac{\overline{c}^3}{v} = \frac{4\pi}{3}\epsilon \tag{3.52}$$

Substituting the above into (3.51), we can approximate it for small porosity to obtain

$$\frac{1}{K} \approx \frac{1}{K_s}\left(1 + 2\pi\frac{1-\nu_s}{1-2\nu_s}\epsilon\right) \tag{3.53}$$

Comparing (3.53) with (3.48), we observe that for the sphere model, the increase in material compliance with respect to the increase in crack (pore) density ϵ is roughly three times more sensitive than that for the crack model. Or, we can interpret the above result as, the effect upon compressibility of a certain concentration of narrow cracks is equal to the effect of the same number of spherical pores with a diameter roughly two-thirds that of the cracks. However, increasing the number of spherical pores means increasing porosity, while increasing the number of thin cracks may not increase porosity.

The effects of cracks on Young's modulus and Poisson ratio were also discussed by Walsh [64, 65].

3.3.3 *Budiansky and O'Connell Model*

Budiansky and O'Connell [12] investigated the case of inclusion of elliptic pores with aspect ratio a/b, where a is the semi-major and b the semi-minor axis of the ellipse, with the interaction among the inclusions using the self-consistent approach of Hill [35], and obtained the following result

$$K = K_s\left(1 - \frac{16}{9}\frac{1-\nu_s^2}{1-2\nu_s}\epsilon\right) \tag{3.54}$$

in which

$$\epsilon = \frac{2}{\pi v}\langle A^2/P\rangle \tag{3.55}$$

with A the area and P the circumference of the ellipse, and the angle brackets denoting the average. Equation (3.54) is valid for all aspect ratio a/b.

For the case of circular cracks, $a = b$, and

$$\epsilon = \frac{\langle a^3\rangle}{v} \tag{3.56}$$

In that case, the effective shear modulus was found as

$$G = G_s \left[1 - \frac{32}{45} \frac{(1-\nu_s)(5-\nu_s)}{2-\nu_s} \epsilon \right] \tag{3.57}$$

For long and narrow cracks, $b/a \to 0$, it was found

$$G = G_s \left[1 - \frac{8}{45} (10 - 7\nu_s) \epsilon \right] \tag{3.58}$$

3.3.4 Bounds on Material Constants

In the above section we observe that the effective moduli are dependent not only on the elastic properties of the matrix material, but also on the pore geometry. It is of interest to establish the bounds on these effective moduli subject to all pore geometries.

Voigt and Reuss Bound The simplest bounds that can be established are the Voigt bound [61] and the Reuss bound [49]. The Voigt average of an elastic modulus is given by

$$K = \phi_1 K_1 + \phi_2 K_2 + \phi_3 K_3 + \dots \tag{3.59}$$

where K_1, $K_2, \dots$ are the elastic modulus of material 1, 2, etc., and ϕ_1, $\phi_2, \dots$ are their volume fraction. The Reuss average is

$$\frac{1}{K} = \frac{\phi_1}{K_1} + \frac{\phi_2}{K_2} + \frac{\phi_3}{K_3} + \dots \tag{3.60}$$

The Voigt bound may be considered as stacking together layers of different homogeneous and isotropic materials and applying a uniform strain in the direction parallel to the layers. Or, it is a loading condition that all parts of a composite materials are under constant strain. It is the *upper bound* of the effective modulus for all possible configurations. The Reuss bound corresponds to the loading condition that a uniform stress is applied in the direction perpendicular to the layers. It is also a loading condition that all parts of a composite material are under uniform stress. It is the *lower bound* of the effective modulus. For a porous material, we can set $K_1 = 0$ and $K_2 = K_s$, and $\phi_1 = \phi$ and $\phi_2 = 1 - \phi$. We then obtain the following upper and lower bounds

$$K_{\text{up}} = (1-\phi) K_s \tag{3.61}$$

$$K_{\text{low}} = 0 \tag{3.62}$$

The lower bound is obvious because in the worst case scenario, such as a suspension of particles not in contact with each other, or a connected network of long and thin cracks embedded in the medium such that it can be closed without resistance, the porous matrix has no strength.

Hashin-Shtrikman Bound Hashin and Shtrikman [32] considered the bounds of effective moduli of a composite material as the mixture of two isotropic and homogeneous solid phases. The uniform stress and strain assumptions were applied only to the individual phases, and not for the whole composite material. These assumptions provide tighter bounds than the Voigt and Reuss bounds.

Consider a composite material as a mixture of a softer and a stiffer material, characterized by the moduli (K_1, G_1) and (K_2, G_2), with $K_2 > K_1$ and $G_2 > G_1$. For such a material, Hashin and Shtrikman [32] established the following upper bounds for the effective bulk modulus

$$K \le K_{\text{up}} = K_2 + \frac{\phi_1}{\frac{1}{K_1 - K_2} + \frac{3\phi_2}{3K_2 + 4G_2}} \tag{3.63}$$

and for the effective shear modulus

$$G \le G_{\text{up}} = G_2 + \frac{\phi_1}{\frac{1}{G_1 - G_2} + \frac{6\phi_2(K_2 + 2G_2)}{5G_2(3K_2 + 4G_2)}} \tag{3.64}$$

The lower bounds are

$$K \ge K_{\text{low}} = K_1 + \frac{\phi_2}{\frac{1}{K_2 - K_1} + \frac{3\phi_1}{3K_1 + 4G_1}} \tag{3.65}$$

and

$$G \ge G_{\text{low}} = G_1 + \frac{\phi_2}{\frac{1}{G_2 - G_1} + \frac{6\phi_1(K_1 + 2G_1)}{5G_1(3K_1 + 4G_1)}} \tag{3.66}$$

For a porous material, we set $K_1 = G_1 = 0$, $K_2 = K_s$, $G_2 = G_s$, $\phi_1 = \phi$, and $\phi_2 = 1 - \phi$. Equations (3.63) and (3.64) then become

$$K_{\text{up}} = K_s \left[1 - \frac{3K_s + 4G_s}{3\phi K_s + 4G_s} \phi \right] \tag{3.67}$$

$$G_{\text{up}} = G_s \left[1 - \frac{5(3K_s + 4G_s)}{3(3 + 2\phi)K_s + 4(2 + 3\phi)G_s} \phi \right] \tag{3.68}$$

It is easy to show that these upper bounds are lower than the Voigt bound given by (3.61). It is also of interest to show that K_{up} in (3.67) is the same as the K predicted by Mackenzie as (3.45). The G_{up} in (3.68) is also consistent with the G

predicted by (3.46) up to the $O(\phi)$ term. Hence the Mackenzie [42] model of non-interacting spherical pores represents the stiffest possible model of all pore geometry configurations.

For the lower bounds, we can similarly substitute $K_1 = G_1 = 0$, $K_2 = K_s$, $G_2 = G_s$, $\phi_1 = \phi$, and $\phi_2 = 1 - \phi$ into (3.65) and (3.66), and find that

$$K_{\text{low}} = G_{\text{low}} = 0 \tag{3.69}$$

which are the same as the Reuss bound (3.62).

As the shear modulus of the solid phase, G_s, is typically difficult to measure, we shall use its upper bound, $G_s \leq (3/2)K_s$, in (3.67), to obtain a more relaxed upper bound as

$$\frac{K}{K_s} \leq 1 - \frac{3\phi}{2+\phi} \tag{3.70}$$

Using (3.70) in (3.6), (3.12), and (3.13), bounds for the following quantities can also be derived:

$$\frac{K_p}{K_s} \leq \frac{2}{3} - \frac{2\phi}{3} \tag{3.71}$$

$$\alpha \geq \frac{3\phi}{2+\phi} \tag{3.72}$$

$$\beta \geq \frac{1}{3} + \frac{2\phi}{3} \tag{3.73}$$

3.4 Nonlinear Model

The definition of stresses and strains presented in Sect. 2.1, as well as the constitutive relations constructed up to this point, are based on the *small* (*infinitesimal*) *deformation* assumption. In this assumption, the deformation of an REV (Representative Elementary Volume) is small as compared to the REV itself, such that only the first order variation terms of the deformation are retained in defining the stresses and strains. In other words, we introduce a fixed spatial frame (*Eulerian frame*), and stresses and strains are considered at fixed locations as the first derivatives of the local deformation. When deformation becomes large, one must consider higher order derivatives, and also the fact that the material no longer occupies the original space, and a material frame (*Lagrangian frame*) may be necessary. These considerations lead to a finite or large deformation theory [7, 45], with *geometric nonlinearity*. The scope of this book is limited to the small (infinitesimal) deformation theory; hence the finite and large deformation theories will not be discussed.

However, even within the small deformation range, many experimental results have shown that the volumetric response of porous rocks to the change of total pressure and pore pressure is nonlinear [37, 59, 72]. Hence the linear constitutive

relations presented so far are merely approximations, applicable to small stress variations. The nonlinear behavior is generally associated with the closing/opening of crack-like pores (characterized by the small aspect ratio of the minor to major axis of the pore). Under increasing confining pressure and/or decreasing pore pressure, crack-like pores close progressively (those with the smallest aspect ratio first), and once closed they do not contribute anymore to the compressibility of the rock. The bulk compressibility of the rock thus decreases with increasing confining pressure until all the crack-like pores are closed. Pores that are approximately equi-dimensional do not close however, provided that the solid material remains elastic. But in very porous and weak rocks, such as sandstone, the solid failure can cause these pores to collapse, larger size pores first, and smaller ones later. This process, however, is not reversible; hence not elastic.

In the sections to follow, we shall investigate the *material nonlinearity* within the scope of small deformation theory.

3.4.1 Effective Stress Dependent Pore Compressibility

Investigations of the nonlinear deformation of porous rocks have been motivated by the need to quantify the effect of pore pressure decline on pore volume and the volume of the rock during depletion of an oil or gas reservoir, in order to provide more accurate information on the fluid storage, the transport property (permeability), and the mechanism of land subsidence. We shall investigate below a simple model [72] that is consistent with the micromechanics model presented in the preceding sections.

To accommodate the nonlinear deformation within the small deformation theory, the volumetric response of the porous material is now written in incremental form, for an infinitesimal transition of the loading from $\{P,\ p\}$ to $\{P+dP,\ p+dp\}$, causing an infinitesimal deformation, from $\{V,\ V_p\}$ to $\{V+dV,\ V_p+dV_p\}$, as

$$\frac{dV}{V} = -C_{bc}(P,p)\ dP + C_{bp}(P,p)\ dp \tag{3.74}$$

$$\frac{dV_p}{V_p} = -C_{pc}(P,p)\ dP + C_{pp}(P,p)\ dp \tag{3.75}$$

Comparing with (3.3) and (3.4), we have adopted here a notation in terms of compliance instead of stiffness. The *compliance coefficients* C_{bc} and C_{bp} are related to the compressibility of porous frame, and C_{pc} and C_{pp} are related to the compressibility of pore volume. These coefficients correspond to the previously defined parameters as

$$C_{bc} = \frac{1}{K}, \quad C_{bp} = \frac{\alpha}{K}, \quad C_{pc} = \frac{1}{K_p}, \quad C_{pp} = \frac{\beta}{K_p} \tag{3.76}$$

So we identify C_{bc} as the *drained frame compressibility* of the porous medium, and C_{pc} as the *pore volume compressibility*. Within the confines of elasticity theory, the deformation caused by the load $\{dP, dp\}$ should be reversible and independent of loading path, such that a relation like (3.6) should exist between two of the poroelastic constants. In the present notation it is written as

$$C_{bp} = \phi\, C_{pc} \tag{3.77}$$

The difference between the present *incremental material nonlinear theory* and the *linear theory* presented so far is that the coefficients appearing in (3.74) and (3.75) are dependent on the confining pressure P and pore pressure p, and are no longer constants.

The assumption that all coefficients are dependent on both P and p is a most general one. It is, however, rather impractical from the experiment point of view to develop such a model. To apply the theory, these coefficients need to be determined in laboratory tests covering a range of *combination* of different P and p values, which are rather costly to perform. For a more practical theory, it possible to demonstrate that, with some weak assumptions, the compliance coefficients are actually functions of the Terzaghi effective compressive stress P' only [72]. This can be established by writing (3.74) and (3.75) in a form similar to (3.8) and (3.9):

$$\frac{dV}{V} = -C_{bc}(P', p)\; dP' - C'_s(P', p)\; dp \tag{3.78}$$

$$\frac{dV_p}{V_p} = -C_{pc}(P', p)\; dP' - C''_s(P', p)\; dp \tag{3.79}$$

It is clear that

$$C'_s = C_{bc} - C_{bp} \tag{3.80}$$

$$C''_s = C_{pc} - C_{pp} \tag{3.81}$$

Consider the following assumptions:

1. The material has the characteristic of an ideal porous medium; that is, the material deforms in a geometrically similar fashion subjected to the unjacketed test ($dP' = 0$).
2. The volume variations dV and dV_p induced by the loading $\{dP, dp\}$ are reversible, and do not depend on the stress path followed. In other words, dV and dV_p are exact differentials.
3. The compressibility of the solid phase is constant (independent of stress).

Assumption 1 leads to the conditions of $dP' = 0$, and $dV/V = dV_p/V_p$; hence from (3.78) and (3.79) we conclude

$$C'_s = C''_s = C_s = \frac{1}{K_s} \tag{3.82}$$

In the above C_s is the compressibility of the solid constituent, or *solid compressibility*. With the application of assumption 2 to (3.78) and (3.79), the Euler condition for exact differential requires

$$\frac{\partial C_{bc}}{\partial p} = \frac{\partial C_s}{\partial P'} \tag{3.83}$$

$$\frac{\partial C_{pc}}{\partial p} = \frac{\partial C_s}{\partial P'} \tag{3.84}$$

Since C_s is a constant (assumption 3), we find

$$\frac{\partial C_{bc}}{\partial p} = \frac{\partial C_{pc}}{\partial p} = 0 \tag{3.85}$$

In other words, the bulk and pore compressibilities are functions of the Terzaghi effective stress only, $C_{bc} = C_{bc}(P')$ and $C_{pc} = C_{pc}(P')$. These relations are supported by experimental results [72].

In conclusion, to experimentally determine the nonlinear compressibilities, one has to determine the porosity ϕ and the solid compressibility C_s (a constant). Then it is necessary to determine one of the compressibilities, say C_{bc}, for a range of effective stress P' values. The other compressibility, C_{pc}, can then be determined as

$$C_{pc} = \frac{C_{bc} - C_s}{\phi} \tag{3.86}$$

3.4.2 Compaction Induced Permeability Change

Although the flow law (Darcy's law) and the permeability coefficient is not presented until Sect. 6.1, its dependency on the (effective) stress state is discussed here. It has been widely observed that rock permeability can rapidly decreases subject to increasing compressive stress, as a result of pore compaction. The pore space can change not only in the form of volume change, but also as shape change. Even with a small degree of compaction, the mechanisms of closing of crack-like pores, and particularly the closing of pore necks, which are the "bottlenecks" of the flow, and the collapse of larger size pores, can make the permeability highly sensitive to the applied stresses.

As discussed in the preceding sections, we anticipate that the permeability sensitivity to stress is related to the Terzaghi effective stress. Based on the mechanisms discussed above, the stress-induced permeability change can have an anisotropy effect, aligned with the principal stress directions. The anisotropic effect, however, is not discussed in this section.

The constitutive equation for porosity change is given by (3.43). Subject to the assumptions in Sect. 3.4.1, we can express it in incremental form as

$$\frac{d\phi}{1-\phi} = -C_\varphi(P')\, dP' \tag{3.87}$$

where $C_\varphi = 1/K_\varphi$ is the *porosity compressibility*. Equation (3.87) can be integrated if the function $C_\varphi(P')$ is known. Take for example the Mackenzie [42] model, as given by (3.47), expressed in the present notations

$$C_\varphi = \frac{3\phi}{4G_s(1-\phi)} \tag{3.88}$$

Integrating (3.87) from a stress-free state gives the exponential law

$$\frac{\phi}{\phi_o} = e^{-3P'/4G_s} \tag{3.89}$$

in which ϕ_o is the porosity at the unstressed state and the shear modulus of the solid constituent G_s is assumed to be a constant.

As permeability is a coefficient that expresses the ease with which a fluid flows through the void space of a porous medium (see Sect. 6.1), it is reasonable to assume that it is a function of porosity. Motivated by laboratory evidences [4], Walder and Nur [62] proposed the power law relation for the intrinsic permeability k as

$$\frac{k}{k_o} = \left(\frac{\phi}{\phi_o}\right)^a \tag{3.90}$$

where k_o is the permeability under unstressed state. According to Kozeny-Carman law, given as (6.21), the exponent a takes the value of 3. Various theoretical models as compiled in [23] give the equivalent value of a between 1 and 6. Based on experimental data compiled from various sources [16], the value of a can range from 1 for granular material such as Ottawa sand [74], to 3 for fractured rocks [40], to 11 for a cement paste [27], and to as high as 25 for high [16] and low [69] porosity sandstones.

Substituting (3.89) into (3.90), we obtain the exponential law for intrinsic permeability

$$k = k_o\, e^{-3aP'/4G_s} = k_o\, e^{-bP'} \tag{3.91}$$

where b is a composite constant, and should generally be regarded as an experimentally measured constant. Equation (3.91) shows that k is a function of the effective confining pressure P' only. This derived exponential law for stress dependent permeability is consistent with the relation proposed by Rice [50], motivated by the laboratory [9] and in situ [48] observations.

3.5 Laboratory Test

The poroelastic constitutive models developed in this and the preceding chapter are characterized by material constants. In order to apply these models, laboratory tests are need for the determination of these constants. This section discusses the laboratory setup that allows the direct or the indirect measurements of these constants.

Geertsma [26] and also Biot and Willis [8] suggested laboratory procedures known as the *jacketed* and *unjacketed* tests. In an unjacketed test, the specimen is submersed in a fluid chamber, and the loading is applied by raising the fluid pressure in the chamber. In a jacketed test, the specimen is wrapped in a flexible membrane (jacket) to prevent fluid exchange with the chamber. In fact, both of these tests can be carried out in a single apparatus schematically described as follows.

A specimen of porous material, such as a rock core, can be sealed off using a membrane. It is then placed in a pressure vessel and set between two end caps, such that a confining pressure can be applied hydraulically (see Fig. 3.1). End caps can be designed with drainage holes to enable control of the pore pressure through fluid mass exchange with the sample for the drained test. Or they can be solid and impermeable, and possibly mounted with a pressure transducer, for the undrained test. By controlling the hydraulic pressure in the chamber (P), and the pore pressure through the drainage access (p), any combination of (P, p) can be created, including the unjacketed test condition $P = p$. Variations of experimental setup that can perform the above stated tests are found in Dropek and Johnson [19], Green and Wang [28], and Zimmerman et al. [72].

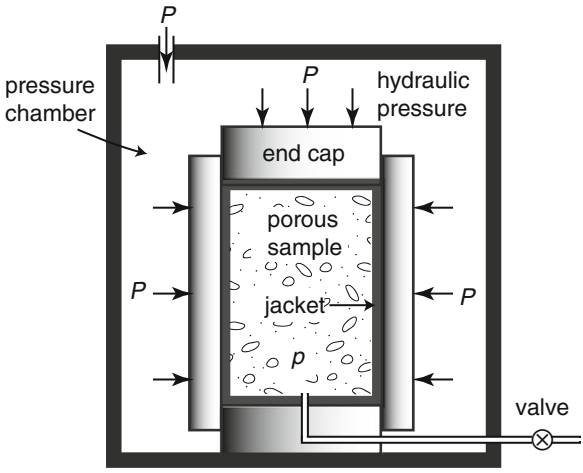


Fig. 3.1 Schematic of a jacketed test

3.5.1 Drained Test

In a drained test, the confining pressure is increased by ΔP, but the pore pressure p is maintained at its initial value. In fact, at the application of a confining pressure, an incremental pore pressure $\Delta p = B\Delta P$ is induced. To restore it to the initial value, the valve is open to allow the fluid to escape from the specimen. As the access to drainage is at the boundary, it takes time for the pore pressure to dissipate throughout the specimen. If the cylindrical sample is drained at both ends through porous caps, the drainage path is approximately half the length of the core, and the characteristic time for dissipation of pore pressure is of order L^2/c, where c is the consolidation coefficient (see Sect. 6.3.2) and L is the half length of the core. Depending on the permeability of the rock, the consolidation coefficient can be as high as 10^4 cm^2/s for a very permeable sandstone and as low as 10^{-4} cm^2/s for a low porosity shale. For a core half length of 5 cm, the time required for the pore pressure to reach equilibrium can thus vary from less than 1 s for sandstone to the order of days for shale.

Once the pore pressure is in equilibrium, two measurements can be made: the volume change of the sample, and the volume of fluid expelled from the sample. The volumetric change of the sample ΔV can be measured using strain gages mounted on the core in transverse and longitudinal directions, or from the oil volume change in the pressure chamber (Fig. 3.1). This latter method requires however calibration to account for the compressibility of the confining fluid volume and the pressure chamber itself. The expelled fluid volume, $\Delta V_f'$, measured through the control valve also needs to be calibrated to compensate for the pore pressure change. In the drained test, however, no such compensation is needed because the fluid pressure before loading and after equilibrium remains unchanged.

Given these measurements, we can define the Biot-Willis *jacketed compressibility* [8] as

$$\kappa_b = -\frac{\Delta V}{V\Delta P} \tag{3.92}$$

Based on (3.3), and $p = 0$, we then identify

$$\kappa_b = \frac{1}{K} \tag{3.93}$$

as the inverse of drained bulk modulus.

Using the information available, other bulk material constants can be determined [18]. For the fluid volume, we notice that since there is no pressure change, the fluid volume change is zero, $\Delta V_f = 0$. Based on (3.24), the expelled fluid volume $\Delta V_f'$ then represents the negative change in pore volume, $\Delta V_p = -\Delta V_f'$. From (3.4), under the drained condition $\Delta p = 0$, we obtain

$$K_p = -\frac{V_p\,\Delta P}{\Delta V_p} = \frac{\phi\, V\Delta P}{\Delta V_f'} \tag{3.94}$$

which means that with a separate measurement of porosity ϕ, we can determine K_p from the undrained test. Also, based on (3.6) and the above two relations, we find

$$\alpha = -\frac{\Delta V_f'}{\Delta V} \tag{3.95}$$

which shows that the Biot effective stress coefficient can be determined from the drained test without a porosity measurement.

3.5.2 Undrained Test

In this test, an incremental confining pressure ΔP is applied to the rock sample, without any fluid allowed to leave the sample. The volume change, ΔV, and the pore pressure rise, Δp, of the core are recorded. From these measurements, we can determine the undrained bulk modulus as

$$K_u = -\frac{V\Delta P}{\Delta V} \tag{3.96}$$

and Skempton pore pressure coefficient as

$$B = \frac{\Delta p}{\Delta P} \tag{3.97}$$

Accurate determination of B requires that the "dead" fluid volume, i.e. the volume of fluid exterior to the sample, be kept to a minimum, as the existence of this volume permits fluid to escape the core. According to Wissa [67], who conducted an experimental study of the pore measurement system, the ratio of dead fluid volume over the pore fluid volume should be less than 0.3 %. Other issues on the measurement of pore pressure coefficient are discussed in [3].

3.5.3 Unjacketed Test

As discussed above, the unjacketed test of Biot and Willis [8] can be conducted in a jacketed test configuration as shown in Fig. 3.1 by controlling the pore pressure rise Δp to be the same as the increment in confining pressure ΔP. This can be accomplished by injecting fluid into the specimen through the access tube. As in the case of the drained test, two measurements can be made: the sample volume change ΔV and the expelled fluid volume $\Delta V_f'$ (negative value, as fluid is injected). Because the pore pressure will be changing in this process, a calibration of the measuring system is needed to determine the injected fluid volume with respect to the reference pressure.

From these measurements, two parameters can be determined. The *unjacketed compressibility* is given by

$$\delta_b = -\frac{\Delta V}{V \Delta p} \tag{3.98}$$

Referring to (3.8) with $P' = 0$, we can identify δ_b as the inverse of the unjacketed frame bulk modulus

$$\delta_b = \frac{1}{K_s'} \tag{3.99}$$

The *coefficient of fluid content* is defined as

$$\gamma_b = -\frac{\Delta V_f'}{V \Delta p} = \frac{\zeta}{\Delta p} \tag{3.100}$$

From (3.26) with $P' = 0$, we obtain

$$\gamma_b = \phi \left(\frac{1}{K_f} - \frac{1}{K_s''} \right) \tag{3.101}$$

Hence with the knowledge of fluid compressibility K_f, the unjacketed pore volume bulk modulus K_s'' can be indirectly determined.

As discussed in Sect. 3.4.1, the material constants measured by the above described laboratory tests can be interpreted as the nonlinear material coefficients in the incremental sense, by recording the reference confining pressure and pore pressure at which the tests are conducted.

3.6 Table of Poroelastic Constants

In Tables 3.1 and 3.2 we compile the micromechanical as well as the bulk material constants for a number of poroelastic materials, including rocks, sand, clay, ocean sediments, man-made materials, and biomaterials. The material constants shown in Table 3.1 represent an independent set. Those in Table 3.2 are calculated based on the various formulae introduced in this and the preceding chapter, and in Sects. B.1 and B.2. The material constants k is the intrinsic permeability, μ_f is the fluid dynamic viscosity, and c is the consolidation coefficient. These coefficients are discussed in Sects. 6.1 and 6.3.2.

A note of caution about the use these tables: the material constants shown are only intended to establish some reference values for these different materials. That is, even for materials classified under the same name, their properties can differ from one sample to the next; hence those presented in the tables are just examples. We

Table 3.1 Poroelastic constants for various materials—independent constants

Porous medium	Fluid	G (N/m^2)	K (N/m^2)	K_s (N/m^2)	K_f (N/m^2)	ϕ	k (m^2)	μ_f ($N{\cdot}s/m^2$)
Ruhr sandstone	Water	1.33×10^{10}	1.31×10^{10}	3.60×10^{10}	2.25×10^{9}	0.020	2.00×10^{-16}	1.00×10^{-3}
Tennessee marble	Water	2.40×10^{10}	4.00×10^{10}	5.00×10^{10}	2.25×10^{9}	0.020	1.00×10^{-19}	1.00×10^{-3}
Charcoal granite	Water	1.87×10^{10}	3.44×10^{10}	4.54×10^{10}	2.25×10^{9}	0.020	1.00×10^{-19}	1.00×10^{-3}
Berea sandstone	Water	6.00×10^{9}	8.00×10^{9}	3.60×10^{10}	2.25×10^{9}	0.190	1.90×10^{-13}	1.00×10^{-3}
Westerly granite	Water	1.50×10^{10}	2.50×10^{10}	4.54×10^{10}	2.25×10^{9}	0.010	4.00×10^{-19}	1.00×10^{-3}
Weber sandstone	Water	1.22×10^{10}	1.34×10^{10}	3.60×10^{10}	2.25×10^{9}	0.060	1.00×10^{-15}	1.00×10^{-3}
Ohio sandstone	Water	6.80×10^{9}	8.40×10^{9}	3.10×10^{10}	2.04×10^{9}	0.191	5.60×10^{-15}	1.00×10^{-3}
Pecos sandstone	Water	5.85×10^{9}	6.64×10^{9}	3.90×10^{10}	2.25×10^{9}	0.195	8.00×10^{-16}	1.00×10^{-3}
Boise sandstone	Kerosene	4.20×10^{9}	4.60×10^{9}	3.13×10^{10}	1.38×10^{9}	0.260	8.00×10^{-13}	1.63×10^{-3}
Gulf Mexico shale	Water	7.60×10^{8}	1.10×10^{9}	3.40×10^{10}	2.25×10^{9}	0.300	1.00×10^{-19}	1.00×10^{-3}
Danian chalk	Water	2.20×10^{9}	3.30×10^{9}	1.20×10^{10}	2.25×10^{9}	0.230	1.00×10^{-17}	1.00×10^{-3}
Hard sediment	Water	2.61×10^{7}	4.36×10^{7}	3.60×10^{10}	2.25×10^{9}	0.470	1.00×10^{-10}	1.00×10^{-3}
Soft sediment	Water	2.21×10^{7}	3.69×10^{7}	3.60×10^{10}	2.25×10^{9}	0.760	1.00×10^{-11}	1.00×10^{-3}
Abyssal red clay	Salt water	7.20×10^{4}	2.00×10^{7}	5.00×10^{10}	2.00×10^{9}	0.710	3.00×10^{-16}	1.50×10^{-3}
Rock salt	Water	1.24×10^{10}	2.07×10^{10}	2.35×10^{10}	2.00×10^{9}	0.001	1.00×10^{-21}	1.00×10^{-3}
Coarse sand	Water[a]	9.79×10^{7}	1.94×10^{8}	4.86×10^{9}	6.47×10^{8}	0.480	3.62×10^{-12}	8.94×10^{-3}
Polyurethane foam	Silicon fluid[a]	3.10×10^{3}	8.14×10^{3}	1.36×10^{5}	3.68×10^{7}	0.930	9.68×10^{-10}	4.87×10^{-2}
Wool felt	Silicon fluid[a]	5.31×10^{5}	1.15×10^{6}	1.27×10^{7}	9.93×10^{7}	0.700	3.72×10^{-13}	4.87×10^{-2}
Cortical bone	Bone fluid	5.50×10^{9}	1.39×10^{10}	1.58×10^{10}	2.30×10^{9}	0.040	1.47×10^{-20}	1.00×10^{-3}
Alundum[b]	Water	2.53×10^{10}	3.14×10^{10}	7.89×10^{10}	2.04×10^{9}	0.316	2.71×10^{-12}	1.00×10^{-3}

[a] Unsaturated
[b] An artificial rock

Table 3.2 Poroelastic constants for various materials—derived constants

Porous medium	α	β	η	K_p (N/m^2)	ν	ν_u	K_u (N/m^2)	B	M (N/m^2)	c (m^2/s)
Ruhr sandstone[a]	0.637	0.989	0.275	4.10×10^8	0.120	0.299	2.87×10^{10}	0.854	3.84×10^{10}	5.10×10^{-3}
Tennessee marble[a]	0.200	0.920	0.067	4.00×10^9	0.250	0.266	4.32×10^{10}	0.371	8.01×10^{10}	7.67×10^{-6}
Charcoal granite[a]	0.242	0.937	0.076	2.85×10^9	0.270	0.292	3.87×10^{10}	0.454	7.26×10^{10}	6.77×10^{-6}
Berea sandstone[a]	0.778	0.946	0.292	1.95×10^9	0.200	0.313	1.40×10^{10}	0.551	9.92×10^9	1.37×10^0
Westerly granite[b]	0.449	0.988	0.150	5.56×10^8	0.250	0.331	3.93×10^{10}	0.810	7.08×10^{10}	2.15×10^{-5}
Weber sandstone[c]	0.629	0.965	0.259	1.27×10^9	0.150	0.272	2.27×10^{10}	0.653	2.35×10^{10}	1.79×10^{-2}
Ohio sandstone[d]	0.729	0.929	0.284	2.20×10^9	0.181	0.280	1.32×10^{10}	0.498	9.01×10^9	3.96×10^{-2}
Pecos sandstone[d]	0.830	0.960	0.336	1.56×10^9	0.159	0.309	1.33×10^{10}	0.605	9.71×10^9	5.31×10^{-3}
Boise sandstone[e]	0.853	0.955	0.351	1.40×10^9	0.150	0.279	8.11×10^9	0.507	4.82×10^9	1.76×10^0
Gulf Mexico shale[f]	0.968	0.990	0.348	3.41×10^8	0.219	0.449	7.22×10^9	0.876	6.54×10^9	1.68×10^{-7}
Danian chalk[g]	0.725	0.913	0.256	1.05×10^9	0.227	0.357	6.96×10^9	0.726	6.97×10^9	4.39×10^{-5}
Hard sediment[h]	0.999	0.999	0.333	2.05×10^7	0.250	0.497	4.51×10^9	0.992	4.47×10^9	7.72×10^0
Soft sediment[h]	0.999	0.999	0.333	2.81×10^7	0.250	0.496	2.93×10^9	0.988	2.90×10^9	6.50×10^{-1}
Abyssal red clay[i]	1.000	1.000	0.004	1.42×10^7	0.498	0.500	2.79×10^9	0.993	2.77×10^9	3.99×10^{-6}

(continued)

Table 3.2 (continued)

Porous medium	α	β	η	K_p (N/m^2)	ν	ν_u	K_u (N/m^2)	B	M (N/m^2)	c (m^2/s)
Rock salt[j]	0.119	0.993	0.040	1.74×10^8	0.250	0.274	2.33×10^{10}	0.926	1.81×10^{11}	1.69×10^{-7}
Coarse sand[k]	0.960	0.980	0.289	9.71×10^7	0.284	0.463	1.29×10^9	0.885	1.19×10^9	1.10×10^{-1}
Polyurethane foam[k]	0.940	0.941	0.237	8.05×10^3	0.331	0.500	8.93×10^6	1.06	1.01×10^7	2.76×10^{-4}
Wool felt[k]	0.910	0.931	0.260	8.82×10^5	0.299	0.493	3.63×10^7	1.06	4.25×10^7	1.63×10^{-5}
Cortical bone[l,m]	0.120	0.707	0.031	4.62×10^9	0.325	0.332	1.45×10^{10}	0.368	4.45×10^{10}	6.35×10^{-7}
Alundum[n]	0.602	0.791	0.234	1.65×10^{10}	0.182	0.200	3.37×10^{10}	0.113	6.31×10^9	1.65×10^{-1}

[a] References [18, 51]
[b] References [18, 46, 51, 73]
[c] References [18, 46, 51]
[d] References [18, 70, 71]
[e] References [18, 22]
[f] References [1, 44]
[g] References [1, 20]
[h] References [2, 58]
[i] References [36, 43]
[j] References [43, 47]
[k] References [39]
[l] Reference [15]
[m] Reference [56]
[n] Reference [70]

also note that the material constants are not always provided in the same publication as a complete set, and may have to be pulled and interpreted from different sources. As discussed in Sect. 3.4.1, poroelastic materials are generally nonlinear; hence poroelastic constants are sensitive to the stress conditions under which they are measured. Furthermore, measurement with static or dynamic techniques may yield different results [39, 70, 71]. These and other factors are not taken into consideration in the compilation of Tables 3.1 and 3.2.

References

1. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1996) Poroviscoelastic analysis of borehole and cylinder problems. Acta Mech 119(1–4):199–219
2. Badiey M, Jaya I, Cheng AHD (1994) Propagator matrix for plane wave reflection from inhomogeneous anisotropic poroelastic seafloor. J Comput Acoust 2(1):11–27
3. Berge PA, Wang HF, Bonner BP (1993) Pore pressure buildup coefficient in synthetic and natural sandstones. Int J Rock Mech Min Sci 30(7):1135–1141
4. Bernabé Y, Brace WF, Evans B (1982) Permeability, porosity and pore geometry of hot-pressed calcite. Mech Mater 1(3):173–183
5. Berryman JG (1999) Origin of Gassmann's equations. Geophysics 64(5):1627–1629
6. Biot MA (1935) El problème de la consolidation des matières argileuses sous une charge (The problem of consolidation of clay material under load). Annales de la Société Scientifique de Bruxelles, Series B 55:110–113
7. Biot MA (1973) Nonlinear and semilunar rheology of porous solids. J Geophys Res 78(23):4924–4937
8. Biot MA, Willis DG (1957) The elastic coefficients of the theory of consolidation. J Appl Mech ASME 24:594–601
9. Brace WF, Walsh JB, Frangos WT (1968) Permeability of granite under high pressure. J Geophys Res 73(6):2225–2236
10. Brown RJS, Korringa J (1975) Dependence of elastic properties of a porous rock on compressibility of pore fluid. Geophysics 40(4):608–616
11. Bruno MS, Nakagawa FM (1991) Pore pressure influence on tensile fracture propagation in sedimentary rock. Int J Rock Mech Min Sci 28(4):261–273
12. Budiansky B, O'Connell RJ (1976) Elastic moduli of a cracked solid. Int J Solids Struct 12(2):81–97
13. Byerlee JD (1975) Fracture strength and frictional strength of Weber Sandstone. Int J Rock Mech Min Sci 12(1):1–4
14. Carrell MM (1980) Mechanical response of fluid-saturated porous materials. In: Rimrott FPJ, Tabarrok B (eds) Theoretical and applied mechanics, 15th international congress on theoretical and applied mechanics, Toronto, pp 251–262
15. Cowin SC (1999) Bone poroelasticity. J Biomech 32(3):217–238
16. David C, Wong TF, Zhu WL, Zhang JX (1994) Laboratory measurement of compaction-induced permeability change in porous rocks—implications for the generation and maintenance of pore pressure excess in the crust. Pure Appl Geophys 143(1–3):425–456
17. David EC, Zimmerman RW (2011) Compressibility and shear compliance of spheroidal pores: exact derivation via the Eshelby tensor, and asymptotic expressions in limiting cases. Int J Solids Struct 48(5):680–686
18. Detournay E, Cheng AHD (1993) Fundamentals of poroelasticity. In: Fairhurst C (ed) Comprehensive rock engineering: principles, practice and projects, vol. II, Analysis and design method. Pergamon Press, Oxford/New York, pp 113–171

19. Dropek RK, Johnson JN, Walsh JB (1978) Influence of pore pressure on mechanical properties of Kayenta sandstone. J Geophys Res 83(NB6):2817–2824
20. El Rabaa AW (1989) Determination of the stress field and fracture direction in the Danian chalk. In: Maury V, Fourmaintraux D (eds) Rock at great depth. Balkema, Rotterdam/Brookfield, 1017–1024
21. Eshelby JD (1957) The determination of the elastic field of an ellipsoidal inclusion, and related problems. Proc R Soc Lond Ser A–Math Phys Sci 241(1226):376–396
22. Fatt I (1959) The Biot-Willis elastic coefficients for a sandstone. J Appl Mech ASME 26:296–297
23. Flint LE, Selker JS (2003) Use of porosity to estimate hydraulic properties of volcanic tuffs. Adv Water Resour 26(5):561–571
24. Garg SK, Nur A (1973) Effective stress laws for fluid-saturated porous rocks. J Geophys Res 78(26):5911–5921
25. Gassmann F (1951) Über die elastizität poröser medien (On elasticity of porous media). Veirteljahrsschrift der Naturforschenden Gesellschaft in Zürich 96:1–23
26. Geertsma J (1957) The effect of fluid pressure decline on volumetric changes of porous rocks. Trans Am Inst Min Metall Eng 210(12):331–340
27. Ghabezloo S, Sulem J, Saint-Marc J (2009) Evaluation of a permeability-porosity relationship in a low-permeability creeping material using a single transient test. Int J Rock Mech Min Sci 46(4):761–768
28. Green DH, Wang HF (1986) Fluid pressure response to undrained compression in saturated sedimentary rock. Geophysics 51(4):948–956
29. Handin J, Hager RV, Friedman M, Feather JN (1963) Experimental deformation of sedimentary rock under confining pressure: pore pressure effects. AAPG Bull 47:717–755
30. Hart DJ, Wang HF (1995) Laboratory measurements of a complete set of poroelastic moduli for Berea sandstone and Indiana limestone. J Geophys Res–Solid Earth 100(B9):17741–17751
31. Hart DJ, Wang HF (2010) Variation of unjacketed pore compressibility using Gassmann's equation and an overdetermined set of volumetric poroelastic measurements. Geophysics 75(1):N9–N18
32. Hashin Z, Shtrikman S (1961) Note on a variational approach to the theory of composite elastic materials. J Franklin Inst–Eng Appl Math 271(4):336–341
33. Hashin Z, Shtrikman S (1963) A variational approach to the theory of the elastic behaviour of multiphase materials. J Mech Phys Solids 11(2):127–140
34. Hill R (1952) The elastic behaviour of a crystalline aggregate. Proc Phys Soc Lond Sect A 65(389):349–355
35. Hill R (1965) A self-consistent mechanics of composite materials. J Mech Phys Solids 13(4):213–222
36. Hollister CD, Anderson DR, Heath GR (1981) Subseabed disposal of nuclear wastes. Science 213(4514):1321–1326
37. Hughes DS, Cooke CE (1953) The effect of pressure on the reduction of pore volume of consolidated sandstones. Geophysics 18:298–309
38. Jaeger JC (1963) Extension failures in rocks subject to fluid pressure. J Geophys Res 68(21):6066–6067
39. Kim YK, Kingsbury HB (1979) Dynamic characterization of poroelastic materials. Exp Mech 19(7):252–258
40. Kranz RL, Frankel AD, Engelder T, Scholz CH (1979) The permeability of whole and jointed Barre Granite. Int J Rock Mech Min Sci 16(4):225–234
41. Lubinski A (1954) The theory of elasticity for porous bodies displaying a strong pore structure. In: Proceedings of the 2nd US national congress on applied mechanics, Michigan University, pp 247–256
42. Mackenzie JK (1950) The elastic constants of a solid containing spherical holes. Proc Phys Soc Lond Sect B 63(361):2–11
43. McTigue DF (1986) Thermoelastic response of fluid-saturated porous rock. J Geophys Res–Solid Earth 91(B9):9533–9542

44. Mody FK, Hale AH (1993) Borehole-stability model to couple the mechanics and chemistry of drilling-fluid/shale interactions. J Pet Technol 45(11):1093–1101
45. Nunziato JW, Cowin SC (1979) Nonlinear theory of elastic materials with voids. Arch Ration Mech Anal 72(2):175–201
46. Nur A, Byerlee JD (1971) Exact effective stress law for elastic deformation of rock with fluids. J Geophys Res 76(26):6414–6419
47. Powers DW, Lambert SJ, Shaffer SE, Hill LR, Weart WD (eds) (1978) Geological characterization report, Waste Isolation Pilot Plant (WIPP) site, Southeastern New Mexico. Tech. Rep. SAND78-1596, Sandia Nat. Lab., Albuquerque
48. Pratt HR, Swolfs HS, Brace WF, Black AD, Handin JW (1977) Elastic and transport properties of an *in situ* jointed granite. Int J Rock Mech Min Sci 14(1):35–45
49. Reuss A (1929) Berechnung der fließgrenze von mischkristallen auf grund der plastizitätsbedingung für einkristalle (Calculation of the yield strength of mixed crystals due to the plasticity condition for single crystals). Zeitschrift für Angewandte Mathematik und Mechanik 9(1):49–58
50. Rice JR (1992) Fault stress states, pore pressure distributions, and the weakness of the San Andreas Fault. In Evans B, Wong TF (eds) Fault mechanics and transport properties of rocks. Academic, London/New York, pp 475–503
51. Rice JR, Cleary MP (1976) Some basic stress diffusion solutions for fluid-saturated elastic porous media with compressible constituents. Rev Geophys 14(2):227–241
52. Saada AS (1993) Elasticity, theory and applications, 2nd edn. Kreiger, Malabar, 775pp
53. Schmitt DR, Zoback MD (1989) Poroelastic effects in the determination of the maximum horizontal principal stress in hydraulic fracturing tests—a proposed breakdown equation employing a modified effective stress relation for tensile failure. Int J Rock Mech Min Sci 26(6):499–506
54. Schmitt DR, Zoback MD (1989) Laboratory tests of the effects of pore pressure on tensile failure. In: ISRM international symposium, Pau. Balkema
55. Schmitt DR, Zoback MD (1992) Diminished pore pressure in low-porosity crystalline rock under tensional failure—apparent strengthening by dilatancy. J Geophys Res–Solid Earth 97(B1):273–288
56. Smit TH, Huyghe JM, Cowin SC (2002) Estimation of the poroelastic parameters of cortical bone. J Biomech 35(6):829–835
57. Sokolnikoff IS (1956) Mathematical theory of elasticity, 2nd edn. McGraw-Hill, New York, 476pp
58. Stoll RD, Kan TK (1981) Reflection of acoustic waves at a water-sediment interface. J Acoust Soc Am 70(1):149–156
59. van der Knaap W (1959) Nonlinear behavior of elastic porous media. Trans Am Inst Min Metall Eng 216:179–186
60. Vincké O, Boutéca MJ, Piau JM, Fourmaintraux D (1998) Study of the effective stress at failure. In: Thimus JF, Abousleiman Y, Cheng AHD, Coussy O, Detournay E (eds) Poromechanics—a tribute to Maurice A. Biot. Balkema, Rotterdam, pp 635–640
61. Voigt W (1889) Über die beziehung zwischen den beiden elastizitätskonstanten isotroper körper (On the relationship between two elastic constants of isotropic body). Wiedemanns Ann 38:573–587
62. Walder J, Nur A (1984) Porosity reduction and crustal pore pressure development. J Geophys Res 89(NB13):1539–1548
63. Walsh JB (1965) Effect of cracks on compressibility of rock. J Geophys Res 70(2):381–389
64. Walsh JB (1965) Effect of cracks on uniaxial elastic compression of rocks. J Geophys Res 70(2):399–411
65. Walsh JB (1965) Effect of cracks in rocks on Poisson's ratio. J Geophys Res 70(20):5249–5257
66. Wang HF (2000) Theory of linear poroelasticity: with applications to geomechanics and hydrogeology. Princeton University Press, Princeton, 287pp
67. Wissa AEZ (1969) Pore pressure measurement in saturated stiff soils. J Soil Mech Found Div ASCE 95(SM4):1063–1073

68. Wong TF, David C, Zhu WL (1997) The transition from brittle faulting to cataclastic flow in porous sandstones: mechanical deformation. J Geophys Res–Solid Earth 102(B2):3009–3025
69. Yale DP (1984) Network modeling of flow, storage and deformation in porous rocks. Ph.D. thesis, Stanford University, 167pp
70. Yew CH, Jogi PN (1978) Determination of Biot's parameters for sandstones, 1. Static tests. Exp Mech 18(5):167–172
71. Yew CH, Jogi PN, Gray KE (1979) Estimation of the mechanical properties of fluid-saturated rocks using the measured wave motions. J Energy Resour Technol ASME 101(2):112–116
72. Zimmerman RW, Somerton WH, King MS (1986) Compressibility of porous rocks. J Geophys Res–Solid Earth Planets 91(B12):2765–2777
73. Zoback MD, Byerlee JD (1975) Effect of microcrack dilatancy on permeability of Westerly granite. J Geophys Res 80(5):752–755
74. Zoback MD, Byerlee JD (1976) Effect of high-pressure deformation on permeability of Ottawa sand. AAPG Bull–Am Assoc Pet Geol 60(9):1531–1542

Chapter 4
Variational Energy Formulation

In our complete study of a rock, it is not enough to know the various minerals that compose it, we must also determine their proportions; yet the solution of this issue presents some difficulties when one cannot directly solve their proportions based on the comparison of densities, when the rock contains more than two minerals. . . . If the minerals are more evenly distributed, . . . so the volume ratios of different minerals are the same as their area ratios.

—Achille Delesse (1848)

In Chap. 2, the constitutive equations for poroelasticity were constructed using the phenomenological approach. In such approach, we attempt to model a new phenomenon, such as the deformation of a saturated porous body, by drawing an analogy with a familiar phenomenon, such as the deformation of an elastic body. In the analogy, we define stresses and strains for a porous body following the elasticity concept, even though their interpretation may not be clear. (For example, see the illustration in Fig. 2.1 for the lack of a clear definition for a continuous stress field in a porous body.) We then bring in the additional force component, namely the pore pressure, and its conjugate deformation, the fluid strain (or actually the relative fluid to solid strain), to build a linear relation that is similar to that of the elasticity theory. This type of ad hoc construction of a working theory is typically motivated first by the observation, and then supported by physical insight, without formally resorting to the laws of physics.

To gain additional physical insight, in Chap. 3 we utilized the micromechanics approach to explicitly model the material phases in a porous medium, not only the solid and the fluid, but also the pore space as an additional "phase", together with their interactions. These micromechanical constitutive laws were assembled and matched up with the bulk continuum theory to provide physical insight. However, these constitutive laws were largely constructed using the "effective stress" concept; hence their theoretical basis is still phenomenological. The material constants associated with the theory, such as K, K_s', and K_s'', are still empirical constants, as their physical mechanisms are based on composite responses, and are not fully isolated to tie to the equation of the state of the phases.

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_4

There are many attempts to build porous medium constitutive models that go beyond the phenomenological model. Among the more widely pursued approaches are the *theory of mixtures* [1, 18, 19, 33, 36, 57], and the *homogenization theory* [2, 22, 56, 63]. The theory of mixtures uses the mathematical assumption that the solid and fluid phases occupies the *same* space, whose presence and influence are weighted by a *volume fraction*, in order to fulfill the requirement of a continuous mathematical function. This approach is largely mathematical, and the "material coefficients" generally do not possess physical meaning until a comparison is made with a theory that is in use, such as a phenomenological theory.

The homogenization approach, on the other hand, explicitly recognizes solid and fluid phases occupying *different* space at the microscopic level, with an assumed periodic pore geometry. The full partial differential equations, such as Navier-Stokes equation for the fluid and elasticity equation for the solid, are prescribed, together with boundary conditions. Effort is then made to simplify these equations based on the perturbation of small parameters, in order to extract mathematical terms, and physical phenomena, of the first order, second order, etc. The homogenization approach typically involves heavy mathematics. Different judgement on the mathematical treatment sometimes leads to different, and even inconsistent results. Its ultimate justification still requires the validation by physical experiments.

In this chapter we take a different approach from those above. We shall build the poroelasticity theory based on the thermodynamics principles of work and energy, first based on the reversible processes for the elastic constitutive laws, and then in the later chapters on the irreversible processes for the fluid, heat flux, and chemical diffusion laws. To cope with the highly heterogeneous nature of porous materials, the volume averaging method developed in the flow through porous medium theory [4], and the theory of heterogeneous (composite) materials [25, 59], are used to define the continuous and smooth functions needed in the construction of partial differential equations. The porous materials are considered as consisting of a heterogeneous solid phase and a homogeneous fluid phase [54]. The macroscopic stresses and strains are defined as microscopically volume or surface averaged quantities depending on whether internal energy or external work is concerned. Constitutive equations are constructed using the variational energy principle stemming from the classical physics law of minimum potential energy [52]. The resultant model consists of a set of *intrinsic material constants* [23], which are directly associated with the equations of state of the solid and fluid phase, and the fundamental deformation modes of the pore structure.

4.1 Internal and External Stress and Strain

In this section we shall discuss different types of averaging, one performed over a volume, and the other over a surface. We also recognize different volumes and surfaces. The volume averaging is associated with the internal (strain) energy stored in a system, while the surface averaging is related to the external work done to a system.

4.1.1 Porosity

Consider a *Representative Elementary Volume* (REV)[1] of porous medium that is made of a microscopically heterogeneous porous solid, filled with a homogeneous fluid. Consistent with the continuum mechanics theories, the linear dimension of the volume should be large enough to be at least one order of magnitude larger than the size of the pores (grains), yet small enough so as not to average out the spatial features of interest, such as macroscopic heterogeneity.

Figure 4.1 gives a two-dimensional illustration of an REV. The volume of the REV, Ω, consists of a solid part, Ω_s, with surface Σ_s, and a fluid part, Ω_f, with surface Σ_f. The external surface of the REV, Γ, is the sum of the solid portion Γ_s and the fluid portion Γ_f. The interface Σ_i is the intersection of the solid and the

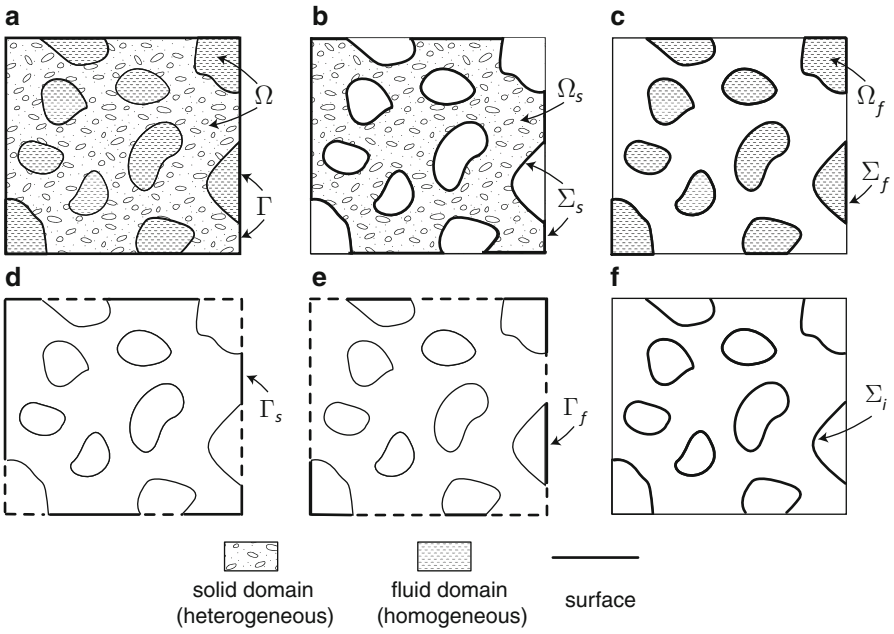


Fig. 4.1 Definitions of volumes and surfaces of solid and fluid phases in an REV. The defined volumes are denoted by Ω and shown as filled areas; defined surfaces are denoted by Γ and Σ, and shown in *thick solid lines*. (Ω: total volume; Ω_s: solid volume; Ω_f: fluid volume; Γ: total external surface; Γ_s: solid external surface; Γ_f: fluid external surface; Σ_s: solid surface; Σ_f: fluid surface; Σ_i: solid/fluid internal surface (interface))

[1]The term Representative Elementary Volume (REV) is largely used in porous medium literature [3, 4]; while in heterogeneous medium literature, it is referred to as Representative Volume Element (RVE) [44, 46].

fluid surfaces, Σ_s and Σ_f. The relations among the volumes and surfaces can be summarized as follows:

$$\begin{aligned} \Omega = \Omega_s \cup \Omega_f; \quad \Gamma = \Gamma_s \cup \Gamma_f; \\ \Sigma_i = \Sigma_s \cap \Sigma_f; \quad \Sigma_s = \Gamma_s \cup \Sigma_i; \quad \Sigma_f = \Gamma_f \cup \Sigma_i \end{aligned} \tag{4.1}$$

We should comment that in this chapter we consider only *fully saturated* (single fluid phase) poroelastic material; hence the fluid volume and fluid surface defined above are the same as pore volume and pore surface.

We should also note that there are several types of pores. First we differentiate between the *interconnected pores* and *non-interconnected pores*. The interconnected pores form a continuous space for the fluid to penetrate, and to move freely when a pressure gradient is applied. The above fluid domain and surface, Ω_f and Σ_f, refer to this type of pore space. On the other hand, the non-interconnected pores, with whatever substance residing in them, are considered as a part of the heterogeneous solid, and are volume averaged with the solid phase. For the purpose of fluid flow (see Sect. 6.1), the interconnected pores are further divided into pores that are a part of flow channel, and *dead end pores*. Both of these pores contribute to the mechanical behavior; hence no distinction is made between these two types for the purpose of this chapter.

With the above clarification, we are ready to define a *volumetric porosity*, ϕ_v, and an *areal porosity*, ϕ_a, as

$$\phi_v = \frac{\Omega_f}{\Omega} \tag{4.2}$$

$$\phi_a = \frac{\Gamma_f}{\Gamma} \tag{4.3}$$

Here we shall invoke an important theory, known as *Delesse's law*, which for the present purpose can be stated as: for a porous medium with a random arrangement of pores, the volumetric porosity and the areal porosity have the same value; hence we shall simply define a porosity ϕ, and

$$\phi_v = \phi_a = \phi \tag{4.4}$$

Delesse in 1848 [38] (see Sect. F.10 for a biography) was faced with the problem of determining the volume fraction of a certain mineral in a rock sample. To accurately determining it requires the grinding of the rock into small pieces, and even into powders, to sort out the minerals. This is sometimes an impractical practice. The area fraction of a mineral, on the other hand, is easy to determine, by polishing and observing a surface. Hence Delesse argued:

> Assume that the volume occupied by the rock is compared to a system of coordinates, and p is the area occupied by one of the mineral components in a section formed by a plane parallel to x–y, to get exactly the volume occupied by the mineral in the rock, it should be possible

> to know the successive values of p on a series of closely spaced planes, parallel to x–y, and the integral $\int p\ dz$ then gives the sought for volume. p is a function of z that can increase or decrease ... If the minerals are more evenly distributed, then p remains constant for each section ... so the volume ratios of different minerals are the same as their area ratios ...

The above statement, originally based on simple reasoning, has been confirmed through the more rigorous derivation based on geometric probability theory and integral geometry [34]. In short, we may conclude that Delesse's law holds for a porous medium containing a large number of pores, with their sizes, shapes, and orientations randomly distributed.

Up to this point, we have not explicitly stated the *frame of reference* considered for Ω, the REV—whether it is an *Eulerian frame* (fixed in space), or a *Lagrangian frame* (referred to the same group of initial mass). From thermodynamics considerations, it is important to identify the system formed by the REV as being open or closed with respect to a certain property. In a closed system, no exchange of that property with the outside environment is permitted, so the property is conserved. For an open system, the converse is true.

In the present construction of a poroelastic model, we shall require that the solid volume Ω_s be referred to the same group of solid mass. The REV Ω is a volume fixed to the solid frame (volume), but extended over to incorporates the void space enclosed therein. Hence Ω is a Lagrangian frame with the solid mass conserved within it. The fluid part of REV, Ω_f, is in fact the pore space enclosed in the solid frame, occupied by fluid. As fluid is free to enter or leave the pore space, either due to the volume change of the pore space, or due to the fluid compressibility, Ω is not a closed system for fluid mass. These are important properties that will be referred to later.

4.1.2 Volume and Surface Averaging of Elastic Material

Before constructing a theory for poroelastic material, we shall first examine the continuum mechanics theory of microscopically heterogeneous elastic materials. Continuum mechanics deals with idealized materials in which the material distribution within an infinitesimal material neighborhood is uniform. For most natural materials, however, the material at the microscopic scale is not uniform. For examples, geomaterials consist of various mineral constituents in the form of grains with different properties and shapes, which are fused together. The stress and strain field of such materials suffers large fluctuation at the microscopic level. One way to present a macroscopic continuum theory that is smooth at the microscopic level is through averaging, and particularly volume averaging [58, 59].

For any material, we can define at a material point the stress σ_{ij} and the displacement u_i. For the stress to be *statically admissible*, it must satisfy the equilibrium equation

$$\sigma_{ij,j} = 0 \tag{4.5}$$

In the above we have neglected the gravity and any other body forces. Gravity force is neglected because we consider only the incremental deformation of a body that is already in static equilibrium under its own weight. The displacement field needs to be smooth enough such that a strain field

$$e_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right) \tag{4.6}$$

is suitably defined. Such displacement field is called *kinematically admissible*.

Average Stress and Strain To arrive at a continuum theory at the macroscopic level, we perform a volume averaging over an REV. The macroscopic stress and strain are respectively defined as [59]:

$$\overline{\sigma}_{ij} = \frac{1}{\Omega}\int_{\Omega} \sigma_{ij}\, d\vec{x} \tag{4.7}$$

$$\overline{e}_{ij} = \frac{1}{\Omega}\int_{\Omega} e_{ij}\, d\vec{x} \tag{4.8}$$

in which Ω denotes the volume of the REV. These volume integrals can be converted to surface integrals with the application of the Gauss (divergence) theorem, which can be expressed in one of the following two forms:

$$\int_{\Omega} A_{i,i}\, d\vec{x} = \int_{\Gamma} A_i\, n_i\, d\vec{x} \tag{4.9}$$

where A_i is a vector, Γ is the bounding surface of Ω, and n_i is the unit outward normal vector of Γ; or for a scalar a, in the form

$$\int_{\Omega} a_{,i}\, d\vec{x} = \int_{\Gamma} a\, n_i\, d\vec{x} \tag{4.10}$$

We also note the following relation

$$(\sigma_{ik}\, x_j)_{,k} = \sigma_{ik,k}\, x_j + \sigma_{ik}\, \delta_{jk} = \sigma_{ij} \tag{4.11}$$

where x_j is the Cartesian coordinate, and we have utilized the equilibrium equation (4.5). With the above relations, it is possible to convert (4.7) and (4.8) into the following surface integrals

$$\overline{\sigma}_{ij} = \frac{1}{\Omega}\int_{\Gamma} t_i\, x_j\, d\vec{x} \tag{4.12}$$

$$\overline{e}_{ij} = \frac{1}{\Omega}\int_{\Gamma} \frac{1}{2}\left(u_i n_j + u_j n_i\right)\, d\vec{x} \tag{4.13}$$

where

$$t_i = \sigma_{ik}\, n_k \tag{4.14}$$

is the surface traction vector. Equations (4.12) and (4.13) demonstrate that the volume averaged stress and strain are completely defined by the surface traction and displacement.

Average Work Next we examine the mechanical work performed by a set of surface tractions applied to an REV, causing it to deform. This can be expressed as

$$\begin{aligned} \mathcal{W} &= \int_\Gamma t_i u_i \, d\vec{x} = \int_\Gamma \sigma_{ij} n_j u_i \, d\vec{x} \\ &= \int_\Omega (\sigma_{ij} u_i)_{,j} \, d\vec{x} = \int_\Omega \sigma_{ij} e_{ij} \, d\vec{x} \end{aligned} \tag{4.15}$$

We have again applied in the above the divergence theorem and equilibrium equation. Rather than expressing the work in the above form, we would like to express it in terms of the average stress and strain as defined in (4.12) and (4.13). By separating stress and strain into an average and a fluctuation part,

$$\sigma_{ij} = \overline{\sigma}_{ij} + \sigma'_{ij}; \quad e_{ij} = \overline{e}_{ij} + e'_{ij} \tag{4.16}$$

we can write the work per unit volume $\mathcal{W}$ as

$$\begin{aligned} \mathcal{W} &= \frac{1}{\Omega} \int_\Omega \sigma_{ij} e_{ij} \, d\vec{x} = \frac{1}{\Omega} \int_\Omega (\overline{\sigma}_{ij} + \sigma'_{ij})(\overline{e}_{ij} + e'_{ij}) \, d\vec{x} \\ &= \frac{1}{\Omega} \left(\int_\Omega \overline{\sigma}_{ij} \overline{e}_{ij} \, d\vec{x} + \int_\Omega \overline{\sigma}_{ij} e'_{ij} \, d\vec{x} + \int_\Omega \sigma'_{ij} \overline{e}_{ij} \, d\vec{x} + \int_\Omega \sigma'_{ij} e'_{ij} \, d\vec{x} \right) \\ &= \overline{\sigma}_{ij} \overline{e}_{ij} + \frac{1}{\Omega} \int_\Omega \sigma'_{ij} e'_{ij} \, d\vec{x} \\ &\approx \overline{\sigma}_{ij} \overline{e}_{ij} \end{aligned} \tag{4.17}$$

In the above, we note the operation that the average quantities are constant and can be pulled out of the integration, and the fluctuation quantities integrate to zero. The integration of the product of fluctuation terms generally does not vanish. However, under special conditions, such as uniform boundary traction, linear boundary displacement, or totally uncorrelated stress and strain distributions, which may result if the material is fine-grained and its distribution is highly random, the integration does become zero [46, 47, 59]. In general, the last relation in the above equation is an approximation.

A dry porous material is just a heterogeneous material if we consider the void space as a material with no mass and no strength. However, when the pores are filled with a fluid, the situation may be different. First, if the material is under the

drained condition, that is, the fluid does not contribute to the mechanical strength and the pores can deform without resistance from the fluid, then we may treat the fluid occupied pores just like voids. Or, if the material is undrained, that is, the fluid is trapped in the pores, then we may again consider the porous material as a heterogeneous material obeying the above averaging rules, with fluid as a material with its own bulk modulus, yet without shear modulus. Under the general condition, though, fluid can move in and out of the pores, thus the REV, and is partially contributing to the strength of the material. Then the above derived averaging relations between the volume and surface integrals, (4.7), (4.8), (4.12), and (4.13), no longer hold. The volume integral is in general not equal to the surface integral. This distinction is very important for fluid saturated porous materials. We shall demonstrate these conditions in the sections below.

4.1.3 Volume and Surface Averaging of Porous Material

To arrive at a macroscopic continuum model for poroelastic materials, we need to conduct microscopic averaging that is consistent with thermodynamic energy principles. In that process, we encounter two types of situations. First, to correctly account for the strain energy stored in the solid and fluid, volume averaging is needed. On the other hand, when referring to the work conducted on the system from external forces, surface averaging is more appropriate. These give rise to the volume and surface averaged stresses and strains. The relations among them are not necessarily the same as those given in the preceding section.

At a point inside a poroelastic material, depending on whether it is occupied by the solid phase or the fluid phase, we can define either a solid stress tensor σ_{ij}^s, or a fluid pressure p^f, with the superscripts ^{s} and ^{f} denoting the material phases. We note that these quantities are mutually exclusive at a given point, except at the interface between the two phases. This assumption is different from that of the mixture theory [18, 19, 35, 36], in which the solid and fluid coexist at the same location, but each with a partial presence weighted by a *volume fraction*. We consider the current averaging approaching more consistent with physics.

Under static equilibrium, we expect the usual equilibrium equations for the stresses and pressure within each material phase

$$\sigma_{ij,j}^s = 0; \quad \text{in } \Omega_s \tag{4.18}$$

$$p_{,i}^f = 0; \quad \text{in } \Omega_f \tag{4.19}$$

in which we again neglected the gravity and other body forces. The solid and fluid displacements in the respective material phase are expressed as u_i^s and u_i^f. Under the assumption of small deformation from a reference state, we define the solid strain tensor and the fluid volumetric strain at a point as

$$e_{ij}^s = \frac{1}{2}\left(u_{i,j}^s + u_{j,i}^s\right) \tag{4.20}$$

$$e^f = u^f_{i,i} \tag{4.21}$$

Phase Volume Averaging and Internal Stress and Strain Similar to (4.7), (4.8), (4.12) and (4.13), we can apply volume averaging to the solid stress and strain, but over the solid domain Ω_s only, and find their equivalent surface integrals as

$$\overline{s}^s_{ij} = \frac{1}{\Omega_s}\int_{\Omega_s} \sigma^s_{ij}\, d\vec{x} = \frac{1}{\Omega_s}\int_{\Sigma_s} t_i x_j\, d\vec{x} \tag{4.22}$$

$$\overline{\epsilon}^s_{ij} = \frac{1}{\Omega_s}\int_{\Omega_s} e^s_{ij}\, d\vec{x} = \frac{1}{\Omega_s}\int_{\Sigma_s} \frac{1}{2}\left(u^s_i\, n_j + u^s_j\, n_i\right)\, d\vec{x} \tag{4.23}$$

In the above, we used different symbols on the left hand sides to denote *solid internal stress*, $\overline{s}^s_{ij}$, and *solid internal strain*, $\overline{\epsilon}^s_{ij}$, and reserved $\overline{\sigma}^s_{ij}$ and $\overline{e}^s_{ij}$ for the externally averaged quantities to be introduced later. We note that $\overline{s}^s_{ij}$ and $\overline{\epsilon}^s_{ij}$ are symmetrical tensors by virtue of symmetry of σ^s_{ij} and e^s_{ij} in the volume integrals.

Similarly, we can define the fluid internal stress (pressure) and strain. For the fluid, when the REV is subject to a rapid loading, an uneven fluid pressure can develop within the REV due to the random pore structure. However, unlike solid, which can permanently sustain different stresses at different parts, the fluid pressure must equilibrate, first among the pores of different shapes and sizes within the REV, then globally, taking place at different time scales [26]. The equilibrium time at the local (REV) scale is typically very short, and can generally be neglected (except for high frequency disturbances whose wave length is smaller than the REV or pore scale). In the present formulation of a static theory, we shall adopt this assumption, which states that *the pore pressure is locally equilibrated at all times*, though it can vary at the macroscopic scale at any time. Hence the volume averaged *fluid internal stress* is given by

$$\overline{s}^f = -\frac{1}{\Omega_f}\int_{\Omega_f} p\, d\vec{x} = -\frac{1}{3\,\Omega_f}\int_{\Sigma_f} p\, x_i\, n_i\, d\vec{x} = -p \tag{4.24}$$

We note in the above that p is treated as a constant in the integration, and a negative sign is present because s^f is defined positive for tension. In the above we note the identify

$$\frac{1}{3\,\Omega_f}\int_{\Sigma_f} x_i\, n_i\, d\vec{x} = 1 \tag{4.25}$$

based on the Gauss theorem (4.9). Equation (4.25) is in fact true for any volume Ω enclosed by Γ.

Next, we define the *fluid internal strain* as

$$\overline{\epsilon}^f = \frac{1}{\Omega_f}\int_{\Omega_f} e^f\, d\vec{x} = \frac{1}{\Omega_f}\int_{\Sigma_f} u^f_i\, n_i\, d\vec{x} = e^f \tag{4.26}$$

In the above, we recognize that e^f is a constant within the REV under the conditions of locally equilibrated pressure and temperature, due to the fluid equation of state $e^f = e^f(p, T)$.

External Surface Averaging and External Stress and Strain While internal (volume) average is needed to correctly account for the average of any intensive quantities (such as internal energy density, that is, internal energy per unit volume), external surface averaging is needed for work performed by external forces. Another reason for formulating the external surface averaged variables is for the construction of boundary value problems for application purposes. For such problems, information on the external surface, known as boundary condition, is needed. For the applied boundary force (stress) and displacement, we can only know the surface averaged, and not the volume averaged stress and displacement, for use as boundary conditions.

Hence we introduce the external surface averaged *solid external stress*, $\overline{\sigma}^s_{ij}$, and *solid external strain*, $\overline{e}^s_{ij}$, as follows

$$\overline{\sigma}^s_{ij} = \frac{1}{\Gamma_s} \int_{\Gamma_s} \sigma^s_{ij} \, d\vec{x} \tag{4.27}$$

$$\overline{e}^s_{ij} = \frac{1}{\Omega_s} \int_{\Gamma_s} \frac{1}{2} \left(u^s_i \, n_j + u^s_j \, n_i \right) \, d\vec{x} \tag{4.28}$$

We note in the above that a direct conversion of (4.27) and (4.28) into volume integrals using the divergence theorem is not possible, because Γ_s is not a closed surface (see Fig. 4.1). Comparing the above with (4.22) and (4.23), we can clearly see that they are different. The need to differentiate between the volume and surface averaged quantities was also raised by Coussy [27, 28], and Berryman and Pride [8].

For the fluid, we can similarly define the external surface averaged *fluid external pressure* and *fluid external strain* as

$$\overline{p}^f = \frac{1}{\Gamma_f} \int_{\Gamma_f} p \, d\vec{x} = p \tag{4.29}$$

$$\overline{e}^f = \frac{1}{\Omega_f} \int_{\Gamma_f} u^f_i n_i \, d\vec{x} \tag{4.30}$$

We hence recognize from (4.24) and (4.29) that the internally and externally averaged pressure are in fact the same

$$\overline{s}^f = -\overline{p}^f = -p \tag{4.31}$$

4.1.4 Linkage Between Internal and External Strains

In a porous material theory, the porosity ϕ is normally considered as a material constant, not as a variable. However, we shall demonstrate below, when a material is subjected to deformation, the *variation of porosity*, $\Delta\phi$, is of the same order of magnitude as the solid and fluid strains; hence porosity needs to be modeled as a variable. In fact, the variation of porosity is exactly what will bridge the gap between the internal and external strains.

Based on the definition (4.2), we can take its variation to obtain

$$\Delta\phi = \Delta\left(\frac{\Omega_f}{\Omega}\right) = \frac{\Delta\Omega_f}{\Omega} - \frac{\Omega_f\,\Delta\Omega}{\Omega^2} = \frac{1}{\Omega}\left(\Delta\Omega_f - \phi\,\Delta\Omega\right) \tag{4.32}$$

The changes in fluid volume and total volume can be found by integrating the normal displacement over the respective surfaces, Σ_f and Γ. First we find

$$\Delta\Omega_f = \int_{\Sigma_i} u_i^f\, n_i^f\, d\vec{x} + \int_{\Gamma_f} u_i^s\, n_i^f\, d\vec{x} \tag{4.33}$$

We note in the above that we have split Σ_f into two parts, the external fluid surface Γ_f, and the interface Σ_i, following (4.1). Γ_f is in fact the extension of Γ_s to form the surface of the REV frame Γ, hence the solid displacement u_i^s, instead of the fluid displacement u_i^f is used to define its movement. The outward normal n_i^f is used to emphasize that it points away from the fluid domain. We can also find the change of volume of the REV frame by the integral

$$\phi\,\Delta\Omega = \phi\int_{\Gamma} u_i^s\, n_i\, d\vec{x} = \int_{\Gamma_f} u_i^s\, n_i^f\, d\vec{x} \tag{4.34}$$

where we note that the definition of porosity is $\phi = \Gamma_f/\Gamma$ (based on Delesse's law). Using (4.33) and (4.34) in (4.32), we obtain

$$\Delta\phi = -\frac{1}{\Omega}\int_{\Sigma_i} u_i^s\, n_i^s\, d\vec{x} \tag{4.35}$$

where we have defined $n_i^s = -n_i^f$, as the normal pointing away from solid domain, and noted the continuity condition that $\vec{u}^f = \vec{u}^s$ on Σ_i.

The solid internal volumetric strain can be obtained by contracting (4.23), which can also be separated into two parts:

$$\overline{\epsilon}^s = \overline{\epsilon}^s_{ii} = \frac{1}{\Omega_s}\int_{\Sigma_s} u^s_i\, n^s_i\, d\vec{x}$$
$$= \frac{1}{\Omega_s}\int_{\Gamma_s} u^s_i\, n^s_i\, d\vec{x} + \frac{1}{\Omega_s}\int_{\Sigma_i} u^s_i\, n^s_i\, d\vec{x} \tag{4.36}$$

in which the first part is the solid external volumetric strain $\overline{e}^s = \overline{e}^s_{ii}$, following (4.28), and the second part is related to the porosity variation (4.35). We can therefore relate the solid internal volumetric strain to the solid external volumetric strain as

$$\overline{\epsilon}^s = \overline{e}^s - \frac{\Delta\phi}{1-\phi} \tag{4.37}$$

Similarly, without going into the detailed derivation, we can demonstrate that the fluid internal and external strains are related as

$$\overline{\epsilon}^f = \overline{e}^f + \frac{\Delta\phi}{\phi} \tag{4.38}$$

The pair of kinematic equations, (4.37) and (4.38), hence provides the link between the externally observed apparent strains that are used in the bulk continuum model to the internal strains that are associated with the internal (strain) energy. The relations are rather general, dependent only on the porosity, and not on other geometric factors of the internal space. By eliminating $\Delta\phi$ between (4.37) and (4.38), we also obtain the relation

$$(1-\phi)\,(\overline{e}^s - \overline{\epsilon}^s) = -\phi\,(\overline{e}^f - \overline{\epsilon}^f) \tag{4.39}$$

which shows the interdependence of the solid and fluid internal and external strains.

The above relations involve only the volumetric strains, and not the deviatoric (shear) strains. The solid internal and external deviatoric strains are defined as

$$\langle\overline{\epsilon}^s_{ij}\rangle = \overline{\epsilon}^s_{ij} - \frac{1}{3}\,\delta_{ij}\overline{\epsilon}^s \tag{4.40}$$

$$\langle\overline{e}^s_{ij}\rangle = \overline{e}^s_{ij} - \frac{1}{3}\,\delta_{ij}\overline{e}^s \tag{4.41}$$

Unfortunately, there is no simple relation that connects the internal to the external deviatoric strains. Their relations are dependent on the geometric factors of the internal space, such as the pore or particle size, its distribution, shape, orientation, and other pore fabrics. If we symbolically denote these pore fabrics as Φ, we can then write the following relation

$$\langle\overline{\epsilon}^s_{ij}\rangle = L_s(\Phi)\,\langle\overline{e}^s_{ij}\rangle \tag{4.42}$$

where L_s is an algebraic or differential operator.

In certain conceptual models with assumed periodic structures, such as stacking of spheres, or carving of spherical, ellipsoidal, or crack-like voids in a solid, the functional relation can be theoretically investigated, as briefly discussed in Sect. 3.3 and found in a wealth of *effective modulus* or *effective medium* literatures [7, 21, 39, 45, 48, 49, 55]. Such a construction, however, is beyond the scope of the present chapter. Hence in the analysis to follow, we shall assume with impunity that $L_s = 1$, and

$$\langle \overline{\epsilon}_{ij}^s \rangle = \langle \overline{e}_{ij}^s \rangle \tag{4.43}$$

because an explicit micromechanics model will be constructed only for the volumetric deformation, and not for shear deformation. With (4.43), we can rewrite (4.37) in the following form

$$\overline{\epsilon}_{ij}^s = \overline{e}_{ij}^s - \frac{1}{3}\delta_{ij}\frac{\Delta\phi}{1-\phi} \tag{4.44}$$

4.2 Thermodynamic Principles

The derivation of constitutive laws as presented in Chaps. 2 and 3 was based on mechanical principles and phenomenological laws such as Hooke's law, without an explicit consideration of energy conservation in a thermodynamics system. In this section we shall explore the thermodynamics background of a poroelasticity theory.

First Law of Thermodynamics The *first law of thermodynamics*, which is a law of energy conservation adapted a thermodynamic system, states that in a *closed system* (a physical system that does not allow the transferring of mass), the energy balance can be described as

$$\delta \mathcal{E} = \delta \mathcal{W} + \delta Q \tag{4.45}$$

where $\mathcal{E}$ is the total energy of a closed system Ω, $\mathcal{W}$ is the external work done to the system Ω through its boundary Γ, Q is the heat energy, and the symbol δ stands for the variation of a quantity.

The total energy represented above can contain any number of components related to different physical processes, such as

$$\mathcal{E} = \mathcal{K} + \mathcal{U} + \mathcal{V} + \dots \tag{4.46}$$

where $\mathcal{K}$ is the *kinetic energy*, $\mathcal{U}$ is the *internal energy*, which can include the elastic strain energy, heat energy, and chemical energy, as appropriate, $\mathcal{V}$ is *potential energy* due to external force fields, such as gravitational, electrical, and magnetic potential. For the present consideration, we shall ignore the kinetic, potential, and other

energies, and retain only the internal energy $\mathcal{U}$. For a quasi-static, isothermal, and chemically inert poroelasticity theory considered in this chapter, $\mathcal{U}$ contains only the elastic strain energy. The heat energy will be included in the porothermoelasticity theory to be presented in Chap. 11, and the chemical energy in porochemoelasticity theory in Chap. 12. The kinetic energy will be presented in the poroelastodynamics theory in Chap. 9.

Second Law of Thermodynamics—Reversible Process The *second law of thermodynamics* introduces an extensive property called *entropy* $\mathcal{S}$, and states that the entropy of an *isolated system* (an enclose physical system that neither matter nor energy can be exchanged with its surroundings) never decreases, because isolated systems always evolve toward *thermodynamic equilibrium*, a state with the maximum entropy.

In a *reversible process* (a process that can be "reversed" by means of infinitesimal changes in some property of the system without entropy production), an increment of heat transferred, δQ, is equal to the product of the absolute temperature $\mathcal{T}$ with the increment of entropy $\delta \mathcal{S}$ in the form

$$\delta Q = \mathcal{T}\, \delta \mathcal{S} \tag{4.47}$$

Substituting the above into (4.45), we obtain

$$\delta \mathcal{E} = \delta \mathcal{W} + \mathcal{T}\, \delta \mathcal{S} \tag{4.48}$$

In the absence of kinetic and other potential energies, $\mathcal{E}$ contains only the internal energy $\mathcal{U}$, and we can express the above as

$$\delta \mathcal{U} = \delta \mathcal{W} + \mathcal{T}\, \delta \mathcal{S} \tag{4.49}$$

The mechanical work done to the system is given by

$$\delta \mathcal{W} = -p\, \delta V \tag{4.50}$$

where p is the pressure and V is the volume of the thermodynamic system. In the above, we assumed that the system is a fluid. For an elastic solid, it more appropriate to express it as

$$\delta \mathcal{W} = \sigma_{ij}\, \delta e_{ij} \tag{4.51}$$

However, for simplicity and consistency with thermodynamics literatures [37, 50], we shall utilize the form (4.50) for now, while bearing in mind its more general form (4.51). Substituting (4.50) into (4.49), we obtain

$$\delta \mathcal{U} = -p\, \delta V + \mathcal{T}\, \delta \mathcal{S} \tag{4.52}$$

Helmholtz Free Energy The *Helmholtz free energy* defined as

$$\mathcal{A} = \mathcal{U} - \mathcal{T}\,\mathcal{S} \tag{4.53}$$

is a thermodynamic potential that measures the maximum "useful work" (free energy) obtainable from a closed thermodynamic system at a constant temperature. Taking variation of the above we obtain

$$\delta\mathcal{A} = \delta\,\mathcal{U} - \mathcal{T}\delta\mathcal{S} - \mathcal{S}\,\delta\mathcal{T} \tag{4.54}$$

Local Equilibrium Assumption of Continuum Mechanics In creating a continuum model for microscopically inhomogeneous materials, such as composite or poroelastic materials, we select a small REV (representative elementary volume) and perform averaging on the physical variables, such as displacement, strain, stress, pore pressure, and temperature to produce variables that are continuous and differentiable in the macroscopic space and time (see Sect. 4.1). In so doing, we ignore the nonequilibrium thermodynamic processes taking place in the microscopic scale, such as pore pressure diffusion among the various size pores within the REV, and the heat transfer between the solid grain and the fluid surrounding it. In other words, it is assumed that the local thermodynamic equilibrium is achieved in an instant, and the locally equilibrated thermodynamic variables are given by the REV averaged ones. This requires that the external force and temperature differential be applied in small increments such that the pore pressure in the pores, and the temperature differential between the solid and fluid phases due to different thermal conductivities, are equalized within a time span that is much smaller than the rate at which the external load is applied. Or, it means that the local equilibrium time is much smaller than the time duration of the global phenomenon of interest.

The assumption of local equilibrium does not imply that the system is in global equilibrium. The REV averaged temperature and pore pressure can vary in space, and their gradients can drive heat and fluid fluxes in the global scale with the associated energy dissipation. However, ignoring the local nonequilibrium processes means ignoring the local energy dissipation. When such processes become important, as in the case of viscoelastic and dynamic phenomena, certain idealized model may be introduced to account for the local dissipation. These will be discussed in the respective chapters on poroviscoelasticity and poroelastodynamics.

An important consequence of the local equilibrium assumption is that the constitutive laws constructed based on the REV averaged variables are *reversible*. In other words, we assume that the local thermodynamic energy, such as the Helmholtz free energy, is a function of the REV averaged state variables at that location and that instant only. It is independent of the nonlocal variables, such as the temperature and pore pressure gradient, which can be determined only if the neighboring cells are referred to, and the time rate of the state variables, which requires the knowledge of its history. The Helmholtz free energy is then a *state function*. A state function implies that its differential is a *total differential*, and the thermodynamic process is *path independent*, and is *reversible* [50].

The reversible process assumption allows us to use (4.52) in (4.54), and express the Helmholtz free energy as

$$\delta \mathcal{A} = -p\,\delta V - \mathcal{S}\,\delta \mathcal{T} \tag{4.55}$$

Or, express it in terms of work,

$$\delta \mathcal{A} = \delta \mathcal{W} - \mathcal{S}\,\delta \mathcal{T} \tag{4.56}$$

In the above, the first term on the right hand side is the mechanical work, and the second term may be considered as work performed by a "thermal force".

Furthermore, consider a process that is *isothermal*, that is, the temperature $\mathcal{T}$ remains constant during the process, then (4.56) becomes

$$\delta \mathcal{A} = \delta \mathcal{W} \tag{4.57}$$

The above equation may be interpreted as, *the increase of the internal (strain) energy of a closed system is equal to the mechanical work performed to it*. This is the basic equation that we shall apply for the variational formulation of isothermal poroelasticity.

Second Law of Thermodynamics—Irreversible Process Natural processes are irreversible. A reversible process is an *idealization* of a natural process by ignoring certain mechanisms whose effects are small. For example, when a force is applied to a solid to deform it, work is performed. In the process of deformation, there may exist internal mechanisms such as the frictional sliding between grains that convert mechanical work into heat, the breaking of solid cohesive bonds to create fissures that contain surface energy, and the large time creep that is similar to the viscous fluid flow. These processes are *irreversible* in the sense that when the force is removed, the body does not return to its original undeformed state, and less useful work than originally applied is recovered. According to the second law, the entropy of the system increases.

In the theory of *elasticity*, however, we assume that the deformation is small, and the time duration is shorter than the creep time, such that all the above-mentioned energy dissipating mechanisms are negligible. We hence formulate constitutive laws assuming a reversible process.

The poroelasticity theory carries the same small deformation assumption and the constitutive laws describing the solid and fluid deformation are reversible. However, in addition to deformation, the presence of a free moving fluid allows a viscous Darcy's flow, which dissipates energy. Hence the poroelastic phenomenon is irreversible, despite the small deformation assumption.

For elastic material, even if we ignore the microscopic dissipation mechanisms such as dry friction, there can exist temperature differential at the macroscopic scale. A spontaneous heat diffusion will result, seeking equilibrium. The thermoelasticity theory models this phenomenon, which is also irreversible.

In an irreversible process, entropy always increases. The second law of thermodynamics needs to be modified from (4.47) to become the following

$$\mathcal{T}\,\delta\mathcal{S} \geq \delta Q \tag{4.58}$$

Biot's Irreversible Thermodynamics Formalism Equation (4.58) is an inequality, and does not give a predictive model. To create a model, we shall employ a formalism introduced by Biot [10, 11] (see also Nowacki [60]) for the modeling of thermoelasticity, as well as other dissipative processes [12, 13].

With the presence of dissipation, we can extend the concept of free energy by the inclusion of dissipated energy through a *dissipation energy function*, which is a measure of the local dissipation of free energy by irreversible processes. We can formally write an equation that is equivalent to (4.56), but accounting for the entropy production in the irreversible processes, as

$$\delta\mathcal{A} + \delta\mathcal{F} = \delta\mathcal{W} + \delta\mathcal{W}_{\text{irr}} \tag{4.59}$$

In the above equation, on the left hand side, $\delta\mathcal{A}$ incorporates all the useful energy of the reversible processes, $\delta\mathcal{F}$ is the dissipation energy function representing the dissipated energy in an irreversible process that cannot be recovered as useful work. On the right hand side, $\delta\mathcal{W}$ is the mechanical work, and $\delta\mathcal{W}_{\text{irr}}$ is the "work" corresponding to the generalized force and generalized displacement of the irreversible processes.

Here we need to make an important distinction between the *dissipation energy function* $\mathcal{F}$ defined here and the *dissipation function* $\mathcal{D}$, which is widely in use for modeling dynamic phenomenon to account for friction, and will be introduced for modeling poroelastodynamics in Chap. 9. The dissipation function $\mathcal{D}$, first introduced by Kelvin [65], denotes the *rate* of energy dissipation; while the dissipation energy function $\mathcal{F}$ refers to the dissipated energy.

The irreversible process of poroelasticity will be modeled in Sect. 6.1.4 to account for the porous medium flow governed by Darcy's law, as well as in later sections for porothermoelasticity and porochemoelasticity.

4.3 Variational Formulation

The energy conservation principle of an elastic body subject to a static deformation can be described as *the work done by external forces to the system is equal to the change of its internal (strain) energy* [16, 62], as given by (4.57), in which the work $\delta\mathcal{W}$ represents mechanical work only, and the Helmholtz free energy contains only the elastic strain energy. The same energy conservation principle can be applied to a poroelastic body contains two material phases, by keeping track of the internal energy and the external work done to each phase. In addition, there exist interacting forces between the two phases on the shared interface.

4.3.1 Virtual Work

We first consider the *virtual work* $\delta\mathcal{W}$, which can be decomposed into two parts, a solid part and a fluid part,

$$\delta\mathcal{W} = \delta\mathcal{W}_s + \delta\mathcal{W}_f \tag{4.60}$$

in which the solid virtual work $\delta\mathcal{W}_s$ is applied through the solid external boundary, Γ_s, and the fluid virtual work $\delta\mathcal{W}_f$ through fluid external boundary Γ_f, with $\Gamma_s + \Gamma_f = \Gamma$ the total external boundary (see Fig. 4.1). For the solid part, the boundary is under a surface traction t_i^s. Subjecting the boundary to a virtual displacements δu_i, we obtain the virtual work $\delta\mathcal{W}_s$ as

$$\begin{aligned}\delta\mathcal{W}_s &= \int_{\Gamma_s} t_i^s\, \delta u_i^s\, d\vec{x} = \int_{\Gamma_s} \sigma_{ij}^s n_j^s\, \delta u_i^s\, d\vec{x} \\ &= \int_{\Gamma_s} \overline{\sigma}_{ij}^s n_j^s\, \delta \overline{u}_i^s\, d\vec{x} + \int_{\Gamma_s} \sigma'^s_{ij} n_j^s\, \delta u'^s_i\, d\vec{x} \\ &\approx \Omega_s\, \overline{\sigma}_{ij}^s\, \delta \overline{e}_{ij}^s\end{aligned} \tag{4.61}$$

where we have utilized the definitions of external stress and strain, (4.27) and (4.28), and the symmetry of the stress tensor. In the above we note that σ'^s_{ij} and $\delta u'^s_i$ are fluctuation from the average, and we used the argument in Sect. 4.1.2 for the approximation.

For the fluid virtual work, we find that

$$\delta\mathcal{W}_f = -\int_{\Gamma_f} p\, n_i\, \delta u_i^f\, d\vec{x} = -\overline{p}^f \int_{\Gamma_f} n_i\, \delta \overline{u}_i^f\, d\vec{x} = -\Omega_f\, \overline{p}^f\, \delta \overline{e}^f \tag{4.62}$$

based on the definitions in (4.29) and (4.30). Hence the total virtual work is

$$\delta\mathcal{W} = \Omega_s\, \overline{\sigma}_{ij}^s\, \delta \overline{e}_{ij}^s - \Omega_f\, \overline{p}^f\, \delta \overline{e}^f \tag{4.63}$$

In the above, the negative sign between the two parts of work is a result of the sign convention of stress and pressure. We can divide (4.63) through by Ω to obtain

$$\delta\mathcal{W} = (1-\phi)\, \overline{\sigma}_{ij}^s\, \delta \overline{e}_{ij}^s - \phi\, \overline{p}^f\, \delta \overline{e}^f \tag{4.64}$$

where $\mathcal{W}$ is the external *work per unit volume*.

4.3.2 Internal Energy

We next examine the variation of Helmholtz free energy. We first note that $\mathcal{A}$ is a *state function*, meaning that it is a single valued function whose value is completely

determined by a set of thermodynamic parameters $(\xi_1, \xi_2, \ldots)$. By being a state function, its differential is a *total differential* with respect to these parameters, that is,

$$d\mathcal{A}(\xi_1, \xi_2, \ldots) = \frac{\partial \mathcal{A}}{\partial \xi_1} d\xi_1 + \frac{\partial \mathcal{A}}{\partial \xi_2} d\xi_2 + \ldots \tag{4.65}$$

Similar to the virtual work, we shall partition $\delta\mathcal{A}$ into a solid part $\delta\mathcal{A}_s$ and a fluid part $\delta\mathcal{A}_f$:

$$\delta\mathcal{A} = \delta\mathcal{A}_s + \delta\mathcal{A}_f = \Omega_s \, \delta\overline{\mathcal{A}}_s + \Omega_f \, \delta\overline{\mathcal{A}}_f \tag{4.66}$$

In the above, $\overline{\mathcal{A}}_s$ and $\overline{\mathcal{A}}_f$ are respectively the solid and the fluid *Helmholtz free energy density*, defined as energy per unit volume. By virtue that only strain energy is considered in the isothermal poroelasticity theory, $\overline{\mathcal{A}}_s$ and $\overline{\mathcal{A}}_f$ can also be called *strain energy density functions*. The over bar indicate volume averaging; that is

$$\overline{\mathcal{A}}_s = \frac{1}{\Omega_s} \int_{\Omega_s} \mathcal{A}_s \, d\vec{x} \tag{4.67}$$

$$\overline{\mathcal{A}}_f = \frac{1}{\Omega_f} \int_{\Omega_f} \mathcal{A}_f \, d\vec{x} \tag{4.68}$$

The fluid strain energy density function actually need not be averaged because fluid pressure and strain are locally equilibrated. In (4.66) we have ignored all other forms of energy, such as heat, chemical, etc., for the present theory.

The solid internal energy density at a point is a function of solid strain only, $\mathcal{A}_s = \mathcal{A}_s(e_{ij})$. The *averaged* solid stain energy density, however, is additionally a function of the porous medium geometry, due to the volume integration. It is further desirable to express the averaged strain energy density as function of averaged quantities, such as the averaged strain, which, in the present case, is the solid internal strain $\overline{\epsilon}_{ij}$. Hence we shall express the functional relation for $\overline{\mathcal{A}}_s$ as

$$\overline{\mathcal{A}}_s = \overline{\mathcal{A}}_s \left(\overline{\epsilon}^s_{ij}, \Phi \right) \tag{4.69}$$

where Φ represents the porous medium geometric factors. Although each unique geometry defines a unique Φ, from practical considerations it is necessary to represent Φ by some bulk geometric parameters such as porosity, specific surface, pore shape factor (aspect ratio of long and short axis), and other pore fabrics. In the simplest model, we shall introduce only porosity as a variable, whose importance has been discussed in Sect. 4.1.4. Hence we shall write (4.69) as

$$\overline{\mathcal{A}}_s = \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi) \tag{4.70}$$

For a homogeneous fluid under thermodynamic equilibrium, the fluid strain energy density is a function of fluid (mass) density ρ_f only. However, realizing the relation

$$\overline{\epsilon}^f = -\frac{\Delta \rho_f}{\rho_f} \tag{4.71}$$

we can express the fluid strain energy density function as

$$\overline{\mathcal{A}}_f = \overline{\mathcal{A}}_f(\overline{\epsilon}^f) \tag{4.72}$$

With the above dependence of variables, we can write the variation of internal energy, (4.66), as

$$\delta \mathcal{A} = \Omega_s \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{\epsilon}^s_{ij} + \Omega_s \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \phi} \delta \phi + \Omega_f \frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}^f} \delta \overline{\epsilon}^f \tag{4.73}$$

4.3.3 Porosity Equilibrium

In order to compared (4.73) to (4.63), we need to convert the internal strains, $\delta \overline{\epsilon}^s_{ij}$ and $\delta \overline{\epsilon}^f$, into the external ones, $\delta \overline{e}^s_{ij}$ and $\delta \overline{e}^f$. By utilizing (4.38) and (4.44) in (4.73), we obtain

$$\begin{aligned} \delta \mathcal{A} = {} & \Omega_s \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{e}^s_{ij} + \Omega_f \frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}_f} \delta \overline{e}^f \\ & + \left[\Omega_s \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \phi} - \frac{\Omega_s}{1-\phi} \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \overline{\epsilon}^s} + \frac{\Omega_f}{\phi} \frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}^f} \right] \delta \phi \end{aligned} \tag{4.74}$$

We notice that the conversion from internal to external volumetric strains generates additional terms associated with the porosity variation.

Based on (4.57), we are now ready to equate (4.63) and (4.74),

$$\begin{aligned} \Omega_s \, \overline{\sigma}^s_{ij} \, \delta \overline{e}^s_{ij} - \Omega_f \, \overline{p}^f \, \delta \overline{e}^f = {} & \Omega_s \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{e}^s_{ij} + \Omega_f \frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}_f} \delta \overline{e}^f \\ & + \left[\Omega_s \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \phi} - \frac{\Omega_s}{1-\phi} \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \overline{\epsilon}^s} + \frac{\Omega_f}{\phi} \frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}^f} \right] \delta \phi \end{aligned} \tag{4.75}$$

Comparing terms associated with solid and fluid strains, it is easy to see that

$$\overline{\sigma}^s_{ij}(\overline{\epsilon}^s_{ij}, \phi) = \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi)}{\partial \overline{\epsilon}^s_{ij}} \tag{4.76}$$

$$\overline{p}^f(\overline{\epsilon}^f) = -\frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}^f} \tag{4.77}$$

Equations (4.76) and (4.77) hence relate solid stresses and fluid pressure to the strain energy of porous medium.

Further examining (4.75), we observe that there is no external work to balance the porosity variation term; hence it must be zero:

$$(1-\phi)\frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij},\phi)}{\partial \phi} - \frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij},\phi)}{\partial \overline{\epsilon}^s} + \frac{\partial \overline{\mathcal{A}}_f(\overline{\epsilon}^f)}{\partial \overline{\epsilon}^f} = 0 \tag{4.78}$$

Substituting (4.76) and (4.77) into (4.78), we obtain

$$(1-\phi)\frac{\partial \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij},\phi)}{\partial \phi} + \overline{p}^s - \overline{p}^f = 0 \tag{4.79}$$

where $\overline{p}^s = -\overline{\sigma}^s_{ii}/3$ is the average solid compressive stress.

Equation (4.79) is the consequence of the explicit introduction of porosity as a state variable in the averaged internal energy density function (4.70). It describes the state that porosity must satisfy in order to minimizes the total energy of the system. The equation also shows that despite the initial assumption that porosity variation is an independent variable, it is in fact constrained by its relation with the solid and fluid internal strains, as we shall demonstrate in the following section. We shall call (4.79) the *porosity equilibrium equation*.

4.4 Constitutive Equation

Although the model developed up to this point is generally nonlinear, in this section we shall develop *linear material models*, in which the deformation of solid and fluid phases obeys linear stress-strain relationship. Even with linear materials, the resultant poroelastic model is not necessarily linear, and in fact is still nonlinear. We then *linearize* such model, which yields a poroelastic model that is consistent with the bulk continuum model [9] and micromechanics model [15].

4.4.1 Linear Material Model

It is of interest to investigate a special model in which the solid and fluid constituents are subjected to small deformation, such that their constitutive relations are linear. This assumption does not imply that the overall deformation is small, as a larger deformation can be derived from the compaction of the pore space; hence the overall behavior can be nonlinear. This model is particularly suitable for materials possessing relatively large porosity and weak pore structure, for example, granular materials.

A linear constitutive relation means that the strain energy density functional can be expressed into a *quadratic form* near the equilibrium (minimum energy) point. While the general anisotropic material case will be discussed in Chap. 5, in the following, we shall assume a *macroscopically isotropic* material. We should note that the macroscopic isotropy assumption does not imply that the material is microscopically isotropic. In fact, a porous material can be made of anisotropic grains with random orientations, such that its macroscopic behavior is isotropic.

For a macroscopically isotropic material, the quadratic form for the solid phase, in terms of the functional dependence $(\overline{\epsilon}^s_{ij}, \phi)$, is

$$\overline{\mathcal{A}}_s = \frac{1}{2}\hat{\lambda}_s\,\overline{\epsilon}^s\,\overline{\epsilon}^s + \hat{\mu}_s\,\overline{\epsilon}^s_{ij}\,\overline{\epsilon}^s_{ij} + \frac{1}{2}K_\phi\,\Delta\phi^2 + K_\psi\,\overline{\epsilon}^s\,\Delta\phi \tag{4.80}$$

in which $\hat{\lambda}_s$ and $\hat{\mu}_s$ are Lamé constants, K_ϕ and K_ψ are elastic moduli associated with porosity change. In the above we have in fact omitted terms involving the product of shear strain and porosity change by assuming that shear strain does not participate in porosity change. Certain phenomena, such as shear induced dilatancy, hence are not admitted in the above expression, yet can be modeled if desired.

Using the quadratic form (4.80) in the definition of solid stress (4.76), we find the following constitutive equation

$$\overline{\sigma}^s_{ij} = \delta_{ij}\,K_\alpha\,\overline{\epsilon}^s + 2G_s\,\langle\overline{\epsilon}^s_{ij}\rangle + \delta_{ij}\,K_\psi\,\Delta\phi \tag{4.81}$$

where

$$K_\alpha = \hat{\lambda}_s + \frac{2}{3}\hat{\mu}_s; \quad G_s = \hat{\mu}_s \tag{4.82}$$

are elastic moduli[2] associated with the volumetric and shear deformation, respectively, of the solid phase, and $\langle\overline{\epsilon}^s_{ij}\rangle$ is the solid deviatoric strain defined in (4.40). We note that (4.81) is a familiar elasticity equation, except for the last term. The reason for the presence of the last term is that $\overline{\sigma}^s_{ij}$ and $\overline{\epsilon}^s_{ij}$ are not stresses and strains at a point. They are *averaged* stresses and strains over a porous medium geometry. As a consequence, K_α, K_ϕ, K_ψ, and G_s are *effective moduli* associated with the heterogeneous solid phase and its geometric arrangement.

We recall that we do not attempt to model the micromechanics of shear deformation. Hence it is more meaningful to contract (4.81) into its volumetric form for further analysis:

$$\overline{p}^s = -K_\alpha\,\overline{\epsilon}^s - K_\psi\,\Delta\phi \tag{4.83}$$

[2]The notation K_α is used here to replace K_s used in earlier publications [23, 24, 54], and is no longer identified as the effective bulk modulus of the solid phase. K_s will be redefined and introduced in (4.88) as the effective bulk modulus of the solid phase.

where $\overline{p}^s = -\overline{\sigma}^s_{ii}/3$. We shall later identify in Sect. 4.5.2 that K_α is a bulk modulus relating solid stress to solid strain under a porosity preserving (but not pore volume preserving) deformation, and K_ψ is a bulk modulus for porosity change under a solid volume preserving deformation. These are respectively called mode I and mode III deformation.

For the fluid phase, linearity means that the fluid strain energy density function can be expressed in the quadratic form:

$$\overline{\mathcal{A}}_f = \frac{1}{2} K_f \, \overline{\epsilon}^f \, \overline{\epsilon}^f \tag{4.84}$$

where K_f is the fluid bulk modulus. Using the above in (4.77), we obtain

$$\overline{p}^f = -K_f \, \overline{\epsilon}^f \tag{4.85}$$

Next, we examine the porosity equilibrium equation (4.79). Considering ϕ as a variable, and $\phi = \phi_o + \Delta\phi$, where ϕ_o is the reference porosity, we observe that (4.79) is nonlinear in $\Delta\phi$. As discussed above, the linear material model only assumes that the deformation of solid and fluid phases is small; the deformation of the frame, however, need not be small, as a large part of the volume change can be derived from the compaction of the pores. Hence, despite the material linearity, there exists a *geometric nonlinearity*. By using (4.80) in (4.79), the porosity equilibrium equation becomes

$$\begin{aligned} \overline{p}^s - \overline{p}^f = &- (1 - \phi_o) \, K_\phi \, \Delta\phi - (1 - \phi_o) \, K_\psi \, \overline{\epsilon}^s \\ &+ K_\phi \, \Delta\phi^2 + K_\psi \, \overline{\epsilon}^s \, \Delta\phi \end{aligned} \tag{4.86}$$

Based on the above equation, we shall separately examine a linear model (Sect. 4.4.2) and a semilinear model (Sect. 4.10).

4.4.2 Linear Model

Retaining only the linear terms in (4.86), we obtain a linearized porosity equilibrium equation as:

$$\overline{p}^s - \overline{p}^f = -(1 - \phi) \, K_\phi \, \Delta\phi - (1 - \phi) \, K_\psi \, \overline{\epsilon}^s \tag{4.87}$$

As the above equation is linearized, we shall not distinguish between ϕ_o and ϕ in terms of notation. Equations (4.83), (4.85), and (4.87) form the three constitutive equations for the volumetric deformation. In these equations, we observe three constitutive constants K_α, K_ϕ and K_ψ associated with solid and porosity deformation, and one K_f associated with fluid deformation.

At this juncture, it is of interest to introduce a new constitutive constant

$$K_s = K_\alpha - \frac{K_\psi^2}{K_\phi} \tag{4.88}$$

to replace K_α in (4.83). The significance of this new constant will become clear in the section below. The three constitutive equations (4.83), (4.85), and (4.87) are in the stress-strain form. They can be solved to give the following strain-stress relations, expressed in terms of both sets of the constitutive constants,

$$\begin{aligned} \overline{\epsilon}^s &= -\frac{\left[(1-\phi)K_\phi - K_\psi\right]\overline{p}^s + K_\psi\,\overline{p}^f}{(1-\phi)(K_\alpha K_\phi - K_\psi^2)} \\ &= -\frac{\left[(1-\phi)K_\phi - K_\psi\right]\overline{p}^s + K_\psi\,\overline{p}^f}{(1-\phi)K_s K_\phi} \end{aligned} \tag{4.89}$$

$$\overline{\epsilon}^f = -\frac{\overline{p}^f}{K_f} \tag{4.90}$$

$$\begin{aligned} \Delta\phi &= -\frac{\left[K_\alpha - (1-\phi)K_\psi\right]\overline{p}^s - K_\alpha\,\overline{p}^f}{(1-\phi)(K_\alpha K_\phi - K_\psi^2)} \\ &= -\frac{[K_s K_\phi - (1-\phi)K_\phi K_\psi + K_\psi^2]\,\overline{p}^s - (K_s K_\phi + K_\psi^2)\,\overline{p}^f}{(1-\phi)K_s K_\phi^2} \end{aligned} \tag{4.91}$$

In the above equations, we note that $\Delta\phi$ is not an independent variable, and is in fact constrained by the porosity equilibrium equation (4.87), such that it is a linear combination of $\overline{\epsilon}^s$ and $\overline{\epsilon}^f$,

$$\begin{aligned} \Delta\phi &= \frac{\left[K_\alpha - (1-\phi)K_\psi\right]\overline{\epsilon}^s - K_f\,\overline{\epsilon}^f}{(1-\phi)K_\phi - K_\psi} \\ &= \frac{[K_s K_\phi - (1-\phi)K_\phi K_\psi + K_\psi^2]\,\overline{\epsilon}^s - K_\phi K_f\,\overline{\epsilon}^f}{K_\phi\left[(1-\phi)K_\phi - K_\psi\right]} \end{aligned} \tag{4.92}$$

With the above, we can also express (4.83) in terms of the internal strains only as

$$\begin{aligned} \overline{p}^s &= -\frac{(1-\phi)(K_\alpha K_\phi - K_\psi^2)\,\overline{\epsilon}^s - K_\psi K_f\,\overline{\epsilon}^f}{(1-\phi)K_\phi - K_\psi} \\ &= -\frac{(1-\phi)K_s K_\phi\,\overline{\epsilon}^s - K_\psi K_f\,\overline{\epsilon}^f}{(1-\phi)K_\phi - K_\psi} \end{aligned} \tag{4.93}$$

4.5 Intrinsic Material Constant

As commented in Chap. 2, the material constants derived from the generalized Hooke's law, such as K_u, B, α, etc., are bulk material constants, which mix properties of solid, fluid, and the porous frame without being able to tell the contribution from the individual constituents. In fact, even the micromechanical material constants presented in Chap. 3, such as K, K_s', and K_s'', failed to isolate the physical mechanisms. Using the variational energy approach, we have derived in the above sections four independent material constants associated with the volumetric deformation of porous material: K_s (or K_α), K_ϕ, K_ψ, which are properties of the solid phase and the porous structure, and K_f, which is a property of the fluid. We shall demonstrate that these constants are *intrinsic material constants*. Here we use the word "intrinsic" to signify that they are associated with isolated physical mechanisms, in contrast to the bulk constants.

4.5.1 Effective Solid Bulk Modulus

First, of the four intrinsic constants identified, the meaning of K_f is clear from (4.90), as the bulk modulus of the fluid phase (inverse of fluid compressibility), and no further elaboration is needed. We notice that up to this point we have not clearly presented an effective solid bulk modulus, which is a property of the solid phase only. The closest we get are the two constants K_s' and K_s'' from the micromechanical analysis in Sect. 3.1.1.

To seek a clear meaning of such a constant, we shall consider the unjacketed test described in Sect. 3.5.3. In such a test, an unjacketed specimen is submersed in a fluid chamber and the pressure in the chamber is raised by Δp. Time is allowed for the pore pressure to equilibrate to the chamber pressure, $\overline{p}^f = \Delta p$. Under that condition, the entire solid surface, both the external surface of the sample and the internal surface that is exposed to the pore fluid, are under the same normal surface traction of the magnitude $t_i = -n_i \Delta p$. Referring to the definitions of the solid internal stress (volume averaged) and external stress (external surface averaged), (4.22) and (4.27), we find the following

$$\overline{s}^s = \frac{\overline{s}^s_{kk}}{3} = \frac{1}{3\,\Omega_s}\int_{\Sigma_s} -\Delta p\, n_k\, x_k\, d\vec{x} = -\Delta p \tag{4.94}$$

$$-\overline{p}^s = \frac{\overline{\sigma}^s_{kk}}{3} = \frac{1}{3\,\Gamma_s}\int_{\Gamma_s} 3\,\Delta p\, d\vec{x} = -\Delta p \tag{4.95}$$

Hence we conclude that in an unjacketed test the internal and external stress and pore pressure are the same

$$\overline{p}^s = \overline{p}^f = -\overline{s}^s = -\overline{s}^f = \Delta p \tag{4.96}$$

The above conditions allow us to reduce (4.89) to the following

$$\overline{\epsilon}^s = \frac{\overline{s}^s}{K_s} \tag{4.97}$$

which relates the volume averaged strain to the volume averaged stress. Hence K_s is identified as the effective bulk modulus of the solid phase, which is free from the influence of the porosity deformation.

4.5.2 Fundamental Deformation Mode

The significance of the other intrinsic bulk moduli can be examined by analyzing the fundamental deformation modes of porous materials. In Fig. 4.2, we illustrate the three fundamental volumetric deformation modes, using a granular material as an example. Figure 4.2a shows an initially un-disturbed sample, as an assembly of spherical particles in random packing. Figure 4.2b represents the final deformed state. This deformation can be separated into three basic modes:

Mode I Figure 4.2c demonstrates the first mode, in which each solid grain is deformed in a geometrically similar fashion by the same proportion. This deformation also preserves the pore geometry, hence there is no porosity change, $\Delta\phi = 0$. As observed from (4.83), this condition shows that the solid volumetric strain is related to its average compressive stress by the following relation

$$\overline{p}^s = -K_\alpha \, \overline{\epsilon}^s \tag{4.98}$$

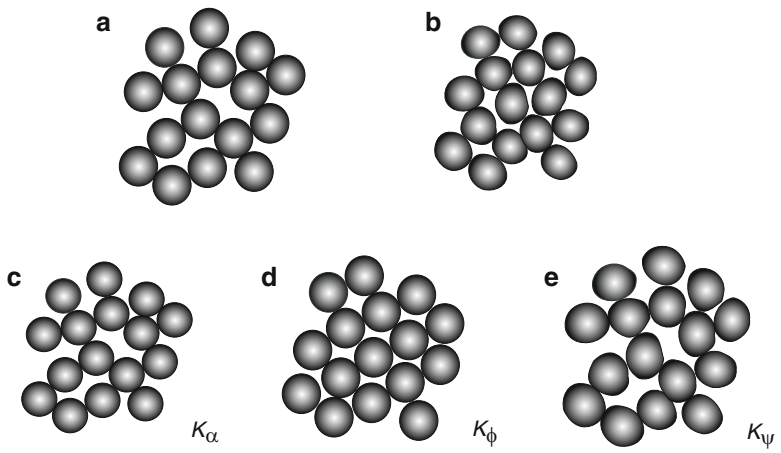


Fig. 4.2 Three fundamental volumetric deformation modes of poroelastic material and the associated bulk moduli: (**a**) Un-deformed state, (**b**) Deformed state (= Mode I + II + III), (**c**) Mode I: geometrically similar deformation, (**d**) Mode II: non-deformable solid with particle/pore rearrangement, and (**e**) Mode III: volume conserving solid shape change

We hence identify K_α as the effective bulk modulus associated with the solid geometrically similar and porosity preserving deformation.

Mode II In the second mode (Fig. 4.2d), the solid grains are assumed to be non-deformable, but they can translate and rotate to rearrange themselves into new positions, which cause the porosity to change. With $\overline{\epsilon}^s = 0$, (4.87) reduces to

$$\overline{p}^s - \overline{p}^f = -(1-\phi)\, K_\phi\, \Delta\phi \tag{4.99}$$

Hence K_ϕ is identified as the *bulk modulus of porosity*. The above equation also shows that porosity variation is associated with the difference between the solid and fluid pressure, or the Terzaghi effective stress.

Mode III Figure 4.2e depicts the third deformation mode, which is characterized by the shape change of the solid particles, without change in solid volume, $\overline{\epsilon}^s = 0$. This mode also causes the porosity to change, as (4.83) becomes

$$\overline{p}^s = -K_\psi\, \Delta\phi \tag{4.100}$$

We shall demonstrate in the section below that K_ψ is a bulk modulus associated with microinhomogeneity and microanisotropy.

With the above, we have isolated the physical mechanisms associated with these material constants, to justify the naming of these constants as *intrinsic* constants.

4.5.3 *Microisotropy and Microhomogeneity: Ideal Porous Medium*

In the volume averaging approach introduced in Sect. 4.1, we did not make any material assumption about the solid constituent at the microscopic level. That is, at the grain level, the solid is generally inhomogeneous, and can also be anisotropic. For the fluid, it is a single phase fluid, which is homogeneous and isotropic. We note that whatever the microscopic material assumption is, homogeneous or inhomogeneous, isotropic or anisotropic, once the averaging is performed, at the macroscopic level the material can be homogeneous or inhomogeneous, and isotropic or anisotropic.

To illustrated the concept of isotropy or anisotropy at the microscopic and macroscopic levels, we present these four situations in Fig. 4.3. Figure 4.3a, c show grains that are isotropic, and b and d grains that are anisotropic. In Fig. 4.3a, b, the grains are random in size, shape, and orientation; hence macroscopically, the material is isotropic, whether the grains are isotropic or not. In contrast, in Fig. 4.3c, d, the isotropic and anisotropic grains are elongated, and have the tendency to align their long axes in the same direction. The material is then macroscopically anisotropic. The above demonstration shows that macroisotropy and macroanisotropy can be a result of pore or grain structure (geometry), and are not necessarily of material origin.

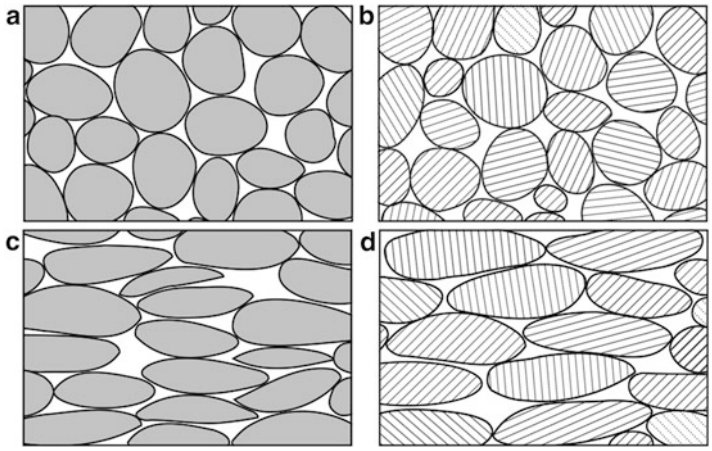


Fig. 4.3 Types of porous medium anisotropy: (**a**) microisotropy and macroisotropy; (**b**) microanisotropy and macroisotropy; (**c**) microisotropy and macroanisotropy; and (**d**) microanisotropy and macroanisotropy

We can similarly apply the above concept to micro and macro homogeneity and inhomogeneity. That is, at the grain level the material can be microhomogeneous (made of the same minerals), or microinhomogeneous (a mixture of different minerals). In either cases, the material can be macroscopically homogeneous or inhomogeneous, depending on whether the same microstructure is repeated everywhere, or it has a trend.

In Sect. 3.2 we discussed an ideal porous medium that is microscopically homogeneous and isotropic, known as the Gassmann model. Although such conditions are generally not true, for practical considerations, the ideal porous medium model is a useful simplification of a real material. It can also bring certain physical insight into the general model. Hence we shall subject the present physically based model to such assumption and observe its characteristics.

In Sect. 4.5.1 we have demonstrated that in an unjacketed test, the following loading condition exists, $\overline{p}^s = \overline{p}^f = \Delta p$. Using this condition in (4.91), we find a change in porosity

$$\Delta\phi = \frac{K_\psi}{K_s K_\phi}\Delta p \tag{4.101}$$

Following the discussion in Sect. 3.2, when an ideal porous medium is subjected to an unjacketed test, it deforms in a geometrically similar pattern, and there is no change in porosity, $\Delta\phi = 0$. From (4.101), we can easily deduce that for an ideal porous medium

$$K_\psi = 0 \tag{4.102}$$

We then realize that K_ψ is a constant that characterizes the deviation from the ideal porous medium assumption, and we may call it the *bulk modulus of microinhomogeneity and microanisotropy*. This is also consistent with the earlier characterization that K_ψ is associated with the dissimilar deformation due to volume conserving shape change.

It is of interest to examine the effect of microhomogeneity and microisotropy on the constitutive equations. With $K_\psi = 0$, we observe from (4.88) that

$$K_s = K_\alpha \tag{4.103}$$

and the constitutive equations (4.89) and (4.91) are simplified to

$$\overline{\epsilon}^s = -\frac{\overline{p}^s}{K_s} \tag{4.104}$$

$$\Delta\phi = -\frac{\overline{p}^s - \overline{p}^f}{(1-\phi)K_\phi} \tag{4.105}$$

Here we note that $\overline{p}^s - \overline{p}^f$ is related to the Terzaghi effective stress P' defined in (3.7) as

$$\overline{p}^s - \overline{p}^f = \frac{P-p}{1-\phi} = \frac{P'}{1-\phi} \tag{4.106}$$

where P is given in (4.120). Hence

$$\Delta\phi = -\frac{P'}{(1-\phi)^2 K_\phi} \tag{4.107}$$

These relations again confirm the propositions made in the preceding sections that

1. K_s is the effective bulk modulus of the solid constituent that does not require the ideal porous medium assumption. In the current case, it reduces to exactly the solid bulk modulus of the microscopically homogeneous and isotropic solid.
2. For an ideal porous medium, the Terzaghi effective stress and K_ϕ alone characterize the porosity change.

The ideal porous medium assumption, or the lack of it, will be further discussed in Sect. 4.7.

4.6 Link with Phenomenological Model

Although the constitutive equations developed in this chapter is based on the fundamental physics, it is in fact not feasible to observe these modeling parameters, such as the solid stress, solid strain, etc., in a field setting. For practical applications,

we need to rely on the observable quantities, such as the total stress and the deformation of the solid frame, as used in the phenomenological model, to construct a mathematical system for solution. Hence in the following we shall build a link between these models.

4.6.1 Link with Bulk Continuum Model

In Chap. 2, we selected the total stress σ_{ij}, rather than the solid stress, as the modeling parameter. It is of interest to discuss whether this parameter should be associated with the volume averaged or the surface averaged stress introduced in this chapter. In the solution of partial differential equations, the input data come from the boundary conditions. A boundary stress is interpreted as the force applied divided by the surface area that it is in contact with. The same is true for the interpretation of a stress loading condition in a laboratory test. Hence it is clear that we should associate σ_{ij} with the external (surface averaged) stress, and not the internal (volume averaged) stress. We then interpret the total stress as

$$\begin{aligned}\sigma_{ij} &= \frac{1}{\Gamma_s}\int_{\Gamma_s} \sigma_{ij}^s \, d\vec{x} - \frac{1}{\Gamma_f}\int_{\Gamma_f} \delta_{ij} p \, d\vec{x} \\ &= (1-\phi)\,\overline{\sigma}_{ij}^s - \delta_{ij}\,\phi\,\overline{p}^f \end{aligned} \tag{4.108}$$

based on the definition of external stress and pore pressure, (4.27) and (4.29). For the pore pressure, we simply have

$$p = \overline{p}^f \tag{4.109}$$

based on (4.29).

The "total strain" e_{ij} defined in Sect. 2.2.2 corresponds to the apparent strain of the frame. As discussed earlier, the frame is defined as the extension of the solid surface over to the pore space to incorporate the entire REV. Hence we shall extend the solid displacement u_i^s on the solid external surface Γ_s, continuously over to Γ_f, to define a continuous function u_i^{se} on Γ. We may then interpret e_{ij} as

$$\begin{aligned} e_{ij} &= \frac{1}{\Omega}\int_{\Gamma} \frac{1}{2}\left(u_i^{se} n_j + u_j^{se} n_i\right) \, d\vec{x} \\ &= \frac{1}{\Omega_s}\frac{\Omega_s}{\Omega}\frac{\Gamma}{\Gamma_s}\int_{\Gamma_s} \frac{1}{2}\left(u_i^{s} n_j + u_j^{s} n_i\right) \, d\vec{x} \\ &= \frac{1}{\Omega_s}\int_{\Gamma_s} \frac{1}{2}\left(u_i^{s} n_j + u_j^{s} n_i\right) \, d\vec{x} \end{aligned} \tag{4.110}$$

In the above we have applied Delesse's law that the volume porosity and surface porosity for a random medium are the same. Comparing with the definition of solid external strain in (4.28), we then conclude

$$e_{ij} = \overline{e}^s_{ij} \tag{4.111}$$

The existence of an apparent strain does not guarantee the existence of a corresponding displacement field. To show the existence, we can utilize the Gauss theorem (4.10) to rewrite (4.110) as the following,

$$e_{ij} = \frac{1}{\Omega}\int_\Omega \frac{1}{2}(u^{se}_{i,j} + u^{se}_{j,i})\, d\vec{x} \tag{4.112}$$

It is then possible to define a solid displacement vector corresponding to e_{ij} as

$$u_i = \frac{1}{\Omega}\int_\Omega u^{se}_i\, d\vec{x} \tag{4.113}$$

such that

$$e_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}) \tag{4.114}$$

Hence the apparent displacement of the frame in the phenomenological theory corresponds to the volume averaging of the microscopic solid displacement extended over to the pore space.

Similarly, we find the existence of the apparent fluid displacement vector U_i through the extension of the fluid external strain $\overline{e}^f$ as

$$\overline{e}^f = U_{i,i} \tag{4.115}$$

where

$$U_i = \frac{1}{\Omega}\int_\Omega u^{fe}_i\, d\vec{x} \tag{4.116}$$

with u^{fe}_i the continuous extension of the fluid displacement u^f_i over to the solid space.

The variation in fluid content ζ, defined as the amount of fluid volume entering the solid frame per unit volume of the solid frame, is the difference between the solid external strain and the fluid external strain, weighted by porosity:

$$\zeta = \phi\,(\overline{e}^s - \overline{e}^f) = \phi(u_{i,i} - U_{i,i}) \tag{4.117}$$

or, in terms of the internal strain variables,

$$\zeta = \phi\,(\overline{\epsilon}^s - \overline{\epsilon}^f) + \frac{\Delta\phi}{1-\phi} \tag{4.118}$$

Equation (4.117) also suggests that the *undrained condition*, given as $\zeta = 0$, is characterized by

$$\overline{e}^s = \overline{e}^f \tag{4.119}$$

As the micromechanical analysis is performed only on volumetric deformations, we define a total mean compressive stress as

$$P = -\frac{\sigma_{ii}}{3} = (1-\phi)\,\overline{p}^s + \phi\,\overline{p}^f \tag{4.120}$$

With the equivalency of parameters established as above, we are ready to provide a link between the phenomenological model presented in Chap. 2 and the present physically based model.

First, we need to demonstrate that the strain energy defined in the phenomenological approach, as the right hand side of (2.17), is the correct expression. Indeed, using (4.108)–(4.117), we can demonstrate that

$$\begin{aligned}\sigma_{ij}\,de_{ij} + p\,d\zeta &= [\,(1-\phi)\,\overline{\sigma}^s_{ij} - \delta_{ij}\,\phi\,\overline{p}^f\,]\,d\overline{e}^s_{ij} + \overline{p}^f\,\phi\,(d\overline{e}^s - d\overline{e}^f) \\ &= (1-\phi)\,\overline{\sigma}^s_{ij}\,d\overline{e}^s_{ij} - \phi\,\overline{p}^f\,d\overline{e}^f\end{aligned} \tag{4.121}$$

which is consistent with (4.64). The above serves to justify (2.17), presented by Biot [9] in his phenomenological approach.

Utilizing (4.37), (4.89), and (4.91), we obtain the constitutive equation for the total strain

$$\begin{aligned}e = &-\frac{K_s K_\phi + [\,(1-\phi)K_\phi - K_\psi\,]^2}{(1-\phi)^3 K_s K_\phi^2}\,P \\ &+ \frac{K_s K_\phi + [\,(1-\phi)K_\phi - K_\psi\,]\,[\phi(1-\phi)K_\phi - K_\psi\,]}{(1-\phi)^3 K_s K_\phi^2}\,p\end{aligned} \tag{4.122}$$

or, under the ideal porous medium assumption,

$$e = -\frac{K_s + (1-\phi)^2 K_\phi}{(1-\phi)^3 K_s K_\phi}\,P + \frac{K_s + \phi(1-\phi)^2 K_\phi}{(1-\phi)^3 K_s K_\phi}\,p \tag{4.123}$$

Similarly, from (4.118) and (4.89), (4.90), and (4.91), we obtain

$$\zeta = -\frac{K_s K_\phi + [(1-\phi)K_\phi - K_\psi][\phi(1-\phi)K_\phi - K_\psi]}{(1-\phi)^3 K_s K_\phi^2} P + \left\{ \frac{\phi}{K_f} + \frac{K_s K_\phi + [\phi(1-\phi)K_\phi - K_\psi]^2}{(1-\phi)^3 K_s K_\phi^2} \right\} p \tag{4.124}$$

or, for ideal porous medium

$$\zeta = -\frac{K_s + \phi(1-\phi)^2 K_\phi}{(1-\phi)^3 K_s K_\phi} P + \left[\frac{\phi}{K_f} + \frac{K_s + \phi^2(1-\phi)^2 K_\phi}{(1-\phi)^3 K_s K_\phi} \right] p \tag{4.125}$$

Equations (4.122) and (4.124) are equivalent to (2.41) and (2.32). Hence we have linked the physical based model with the phenomenological model.

Relations like (4.122) and (4.124) allows us to relate the bulk material constants to the intrinsic ones. For example, (2.39) can be expressed as

$$e = -\frac{1}{K} P + \frac{\alpha}{K} p \tag{4.126}$$

Comparing with (4.122), we can easily identify

$$K = \frac{(1-\phi)^3 K_s K_\phi^2}{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2} \tag{4.127}$$

$$\alpha = \frac{K_s K_\phi + [(1-\phi)K_\phi - K_\psi][\phi(1-\phi)K_\phi - K_\psi]}{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2} \tag{4.128}$$

Examining the above relations, we notice that the drained bulk modulus K and the Biot effective stress coefficient α are composite constants that depend on the three solid moduli, K_s, K_ϕ, and K_ψ, and are independent of the fluid modulus, K_f.

Also, comparing (2.32) written as

$$\zeta = -CBP + Cp \tag{4.129}$$

with (4.124), we find the constant stress storage coefficient and the Skempton pore pressure coefficient as

$$C = \frac{\phi}{K_f} + \frac{K_s + [\phi(1-\phi)K_\phi - K_\psi]^2}{(1-\phi)^3 K_s K_\phi^2} \tag{4.130}$$

$$B = \frac{K_f\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi][\phi(1-\phi)K_\phi - K_\psi]\}}{K_f\{K_s K_\phi + [\phi(1-\phi)K_\phi - K_\psi]^2\} + \phi(1-\phi)^3 K_s K_\phi^2} \tag{4.131}$$

Similarly, we can obtain relations for other bulk material constants, such as Biot modulus

$$M = \frac{K_f\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2\}}{(1-\phi)K_f K_\phi + \phi\,\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2\}} \tag{4.132}$$

and undrained bulk modulus

$$K_u = \frac{K_f\{K_s K_\phi + [\phi(1-\phi)K_\phi - K_\psi]^2\} + \phi(1-\phi)^3 K_s K_\phi^2}{(1-\phi)K_f K_\phi + \phi\,\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2\}} \tag{4.133}$$

We observe that C, B, M, and K_u are composite material constants depending on both solid and fluid properties. Further exposition of these relations can be found in Sect. 4.8. These and other relations are also complied in Sect. B.3.

4.6.2 Link with Micromechanics Model

To link the Rice-Cleary constants presented in Sect. 3.1.1 to the intrinsic constants, we note the following corresponding definitions of volumes and porosity

$$V = \Omega; \quad V_s = \Omega_s; \quad V_p = \Omega_f; \quad \phi = V_p/V = \Omega_f/\Omega \tag{4.134}$$

and strains,

$$\frac{\Delta V}{V} = \overline{e}^s; \quad \frac{\Delta V_s}{V_s} = \overline{\epsilon}_s; \quad \frac{\Delta V_f}{V_f} = \overline{\epsilon}_f \tag{4.135}$$

We also find

$$\frac{\Delta V_p}{V_p} = \frac{\Delta V - \Delta V_s}{\phi\, V} = \frac{1}{\phi}\,[\overline{e}^s - (1-\phi)\overline{\epsilon}^s] \tag{4.136}$$

These relations allow us to compare (4.89) with (3.18), and rearrange terms to obtain

$$K_s' = \frac{(1-\phi)K_s K_\phi}{(1-\phi)K_\phi - K_\psi} \tag{4.137}$$

as well as

$$K_s'' = \frac{\phi(1-\phi)K_s K_\phi}{\phi(1-\phi)K_\phi - K_\psi} \tag{4.138}$$

K_p is a constant dependent on K, K_s' and K_s'', and can be expressed as

$$K_p = \frac{\phi(1-\phi)^3 K_s K_\phi^2}{K_s K_\phi + [\,(1-\phi)K_\phi - K_\psi\,]\,[\phi\,(1-\phi)K_\phi - K_\psi\,]} \tag{4.139}$$

These and other relations are summarized in Sect. B.4.

We can also invert (4.127), (4.137), and (4.138) to obtain the following relations

$$K_s = \frac{(1-\phi)K_s' K_s''}{K_s'' - \phi K_s'} \tag{4.140}$$

$$K_\phi = \frac{K K_s'(K_s'' - \phi K_s')}{(1-\phi)^3[K_s'(K_s'' - \phi K_s') - K K_s'']} \tag{4.141}$$

$$K_\psi = \frac{\phi K K_s'(K_s'' - K_s')}{(1-\phi)^2[K_s'(K_s'' - \phi K_s') - K K_s'']} \tag{4.142}$$

4.7 Deviation from Ideal Porous Medium

It was demonstrated in Sect. 3.2 that an ideal porous medium is characterized by the condition of material constants $K_s' = K_s''$. In Sect. 4.5.3, we further identified the intrinsic material constant K_ψ that is directly associated with the microhomogeneity and microisotropy of the material, such that for an ideal porous medium, $K_\psi = 0$. We shall demonstrate in the sections below that this condition leads to considerable simplification of the porous medium model.

However, before such simplification is imposed, we shall first give a scrutiny of the validity of the ideal porous medium assumption for naturally occurring materials. For geomaterials, the bulk moduli of the major mineralogical components of most rocks do not differ by a large amount. Nur and Byerlee [61] reported tests on a low porosity Westerly granite ($\phi \simeq 1\%$), and indicated that the bulk modulus K_s' is identical to the grain modulus K_s. Experiments performed by Zimmerman et al. [76] on various sandstones also support this hypothesis.

Brown and Korringa [20] however suggested that the ideal porous medium condition is not satisfied by sedimentary materials, such as clayey sandstones, which contain hard minerals, such as quartz, and softer clayey materials. Also, a significant inhomogeneity can be introduced by the existence of non-interconnected pores, which is considered a part of the solid phase. Hart and Wang [42] tested a Berea sandstone and an Indiana limestone, and found that at the low effective stress level (below 5 MP), $K_s' > K_s''$ for Indiana limestone, and in fact K_s' was reported to be more than one order of magnitude larger than K_s'' for Berea sandstone [6, 41]. At high effective stress level, Hart and Wang [43] observed that for Indiana limestone, $K_s' \cong K_s''$, but for Berea sandstone, K_s' is about 2–3 times larger than K_s''. Laurent et al. [53], on the other hand, measured the compressibilities for the Vilhonneur limestone, and found that $K_s'' > K_s'$.

Utilizing (4.137) and (4.138), we can show that

$$\frac{1}{K_s'} - \frac{1}{K_s''} = \frac{K_\psi}{\phi\, K_s K_\phi} \tag{4.143}$$

It is clear that whether K_s' or K_s'' is larger is dependent on the sign of K_ψ. Based on the material stability analysis in Sect. 4.9, we find no restriction on the sign of K_ψ; hence $K_s'' > K_s'$ if $K_\psi > 0$, and $K_s' > K_s''$ if $K_\psi < 0$. Also, assuming the positiveness of K_s, we find the following bound based on (4.140)

$$K_s'' > \phi K_s' \tag{4.144}$$

The above relation suggests that given K_s', there is no upper bound for K_s'', and it can approach infinity. On the other hand, given K_s'', K_s' is bounded from the top by K_s''/ϕ; hence the laboratory measurement of K_s' being more than one order of magnitude greater than K_s'' mentioned above [6, 41] is a violation of this bound.

To gain further insight, we shall examine a special model for sedimentary rock following the argument of Brown and Korringa [20]. Consider a porous medium that is made of two isotropic and homogeneous solid constituents, a quartz, which is stiffer, and a clay, which is softer. The quartz, with a significantly larger proportion than clay, forms the porous medium skeleton with a structural strength (contributing to the strength of the frame to protect the pores). The clay, on the other hand, fills a portion of the pore space formed by the quartz minerals, and is not connected to form a continuous structure to have a structural strength. Figure 4.4 gives an idealized illustration of such a model.

If we subject such a specimen to an unjacketed test, defined by the condition $P = p$, we may examine the compressibility of the specimen based on the definition

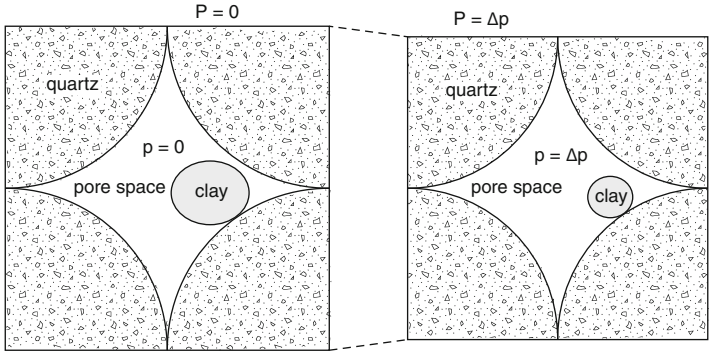


Fig. 4.4 An idealized model of sedimentary rock under unjacketed test: A quartz matrix with material and structural strength and a clay particle with only material strength. With an incremental pressure in the chamber, the clay deforms in a larger proportion than the quartz (After Hart and Wang [43])

of K_s' and K_s'' given in (3.10) and (3.11). We can begin the thought experiment by first assuming that the clay has the same bulk modulus as the quartz. In that case, we have an ideal porous medium such that the solid(s), the pore space, and the frame, all deform in the same proportion, and the porosity remains the same. Next, if we make the clay more compressible, then it will deform in a larger proportion, as illustrated in Fig. 4.4. As we assume that the clay does not contribute to the structural strength of the pore space formed by the quartz skeleton, which includes both the true pore space and the space occupied by the clay, the extra reduction in clay volume gives back a portion of the space to the true pore volume, as if the pore volume is less compressible than the frame volume, leading to the condition of $K_s'' > K_s'$. This is in fact the conclusion drawn by Brown and Korringa [20] for sedimentary rocks. Another interpretation is that under this condition, the porosity increases, $\Delta\phi > 0$, even though the pore volume decreases. According to (4.101), this implies $K_\psi > 0$, and as shown in (4.143), $K_s'' > K_s'$.

To provide a rigorous proof to the above reasoning, we shall present below a theoretical derivation. Let us define the volume occupied by the quartz as V_{s1}, by the clay as V_{s2}, and by the pore space as V_p. The solid volume is $V_s = V_{s1} + V_{s2}$, and the total volume is $V = V_{s1} + V_{s2} + V_p$. If we subject the specimen to an unjacketed test, $P' = P - p = 0$, then (3.18) becomes

$$\frac{\Delta V_s}{V_s} = -\frac{1}{1-\phi}(C_s' - \phi\, C_s'')\, p \tag{4.145}$$

where we have switched to the compressibility notation $C_s' = 1/K_s'$ and $C_s'' = K_s''$. Tracing back to the definitions of K_s' and K_s'' in (3.10) and (3.11), we observe the definition

$$\left.\frac{\Delta V}{V}\right|_{P'=0} = -C_s'\, p \tag{4.146}$$

$$\left.\frac{\Delta V_p}{V_p}\right|_{P'=0} = -C_s''\, p \tag{4.147}$$

As each of the solid constituents is microhomogeneous and microisotropic, we anticipate that the following constitutive relations under the unjacketed test,

$$\left.\frac{\Delta V_{s1}}{V_{s1}}\right|_{P'=0} = -C_{s1}\, p \tag{4.148}$$

$$\left.\frac{\Delta V_{s2}}{V_{s2}}\right|_{P'=0} = -C_{s2}\, p \tag{4.149}$$

where C_{s1} and C_{s2} are the compressibilities of the quartz and clay, respectively. The above equations can be combined to give

$$\left.\frac{\Delta V_s}{V_s}\right|_{P'=0} = -C_s\, p \tag{4.150}$$

where

$$C_s = \frac{V_{s1}}{V_{s1} + V_{s2}} C_{s1} + \frac{V_{s2}}{V_{s1} + V_{s2}} C_{s2} \tag{4.151}$$

is the volume averaged compressibility of the inhomogeneous solid (as it consists of two different solids). From the above definition, and also the assumption that $C_{s2} > C_{s1}$ (clay is more compressible than quartz), we have the condition that

$$C_{s_2} > C_s > C_{s1} \tag{4.152}$$

Substituting (4.150) into (4.145), we obtain

$$(1 - \phi) C_s + \phi\, C_s'' = C_s' \tag{4.153}$$

The above relation was given by Brown and Korringa [20]. We should emphasize, however, that (4.153) is valid only under the current stated material assumptions, and is not a general relation.

To find out whether C_s' or C_s'' is larger, we can rearrange (4.153) into the following

$$C_s' - C_s'' = \frac{1 - \phi}{\phi} (C_s - C_s') \tag{4.154}$$

Next, we recognize that the porous frame is formed by the quartz, and the clay is not contributing to the structural strength of the frame; hence under the unjacketed test condition, the geometrically similar deformation exists for the quartz frame, such that

$$\frac{\Delta V_{s1}}{V_{s1}} = \frac{\Delta V}{V} = \frac{\Delta V_p + \Delta V_{s2}}{V_p + V_{s2}} \tag{4.155}$$

Based on (4.146) and (4.148), we deduce that

$$C_s' = C_{s1} \tag{4.156}$$

Substituting the above into (4.154), we observe

$$C_s' - C_s'' = \frac{1 - \phi}{\phi} (C_s - C_{s1}) > 0 \tag{4.157}$$

that is, $C_s' > C_s''$. Or, using bulk modulus notation, $K_s'' > K_s'$, and $K_\psi > 0$.

We note that the above conclusion is the same as that made by Brown and Korringa [20] for sedimentary rock, and is also consistent with the observation of Laurent et al. [53] for Vilhonneur limestone. Hart and Wang [43], on the other hand, observed the opposite, that is, $K_s' > K_s''$, for Berea sandstone. In referring to these experiments, we notice that K_s'' is not directly measured. Particularly, in Hart and

Wang's experiments, K_s'' is evaluated using the measured values of K_s' and Skempton pore pressure coefficient B. It was also suggested, based on the observation, that such limit exists: as $B \to 1$, $K_s'' \to K_f$ [42, 43]. For this, we should comment that both K_s' and K_s'' are properties of the solid only, as indicated by their relation to the intrinsic constants, (4.137) and (4.138).

The above uncertainties on the measurement of the micromechanical constants K_s' and K_s'' show the importance of the isolation of the intrinsic constant K_ψ and its direct measurement. In Sect. 4.11 we shall discuss the laboratory measurement of K_ψ.

4.8 Limiting Material Properties

In this section we shall discuss various limiting material behaviors, such as microhomogeneity and microisotropy, incompressible or highly compressible material phases, etc., leading to simplified constitutive models.

4.8.1 Ideal Porous Medium

An ideal porous medium is characterized by the condition $K_\psi = 0$. With that condition, the various relations presented in Sect. 4.6 can be simplified. For example, the drained bulk modulus, given in (4.127), becomes

$$K = \frac{(1-\phi)^3 K_s K_\phi}{K_s + (1-\phi)^2 K_\phi} \tag{4.158}$$

Its meaning is clearer if we write it in the compliance form:

$$\frac{1}{K} = \frac{1}{(1-\phi)K_s} + \frac{1}{(1-\phi)^3 K_\phi} \tag{4.159}$$

Hence we can isolate the compliance of the solid frame into two parts, one part is due to the compliance of the solid constituent, and the other due to the porosity. This relation is similar to a series spring with its additive spring constants.

We can see this situation more clearly by examining the kinematic equation (4.37), which indicates that the total deformation, given by the solid external strain $\overline{e}^s$, is the sum of the solid deformation (solid internal strain) $\overline{\epsilon}^s$, and pore deformation $\Delta\phi/(1-\phi)$. Utilizing the constitutive equations for ideal porous medium, (4.104) and (4.105), under the drained condition $\overline{p}^f = 0$, we can express

$$\overline{e}^s = \overline{\epsilon}^s + \frac{\Delta\phi}{1-\phi} = -\frac{\overline{p}^s}{K_s} - \frac{\overline{p}^s}{(1-\phi)^2 K_\phi} = -\frac{P}{K} \tag{4.160}$$

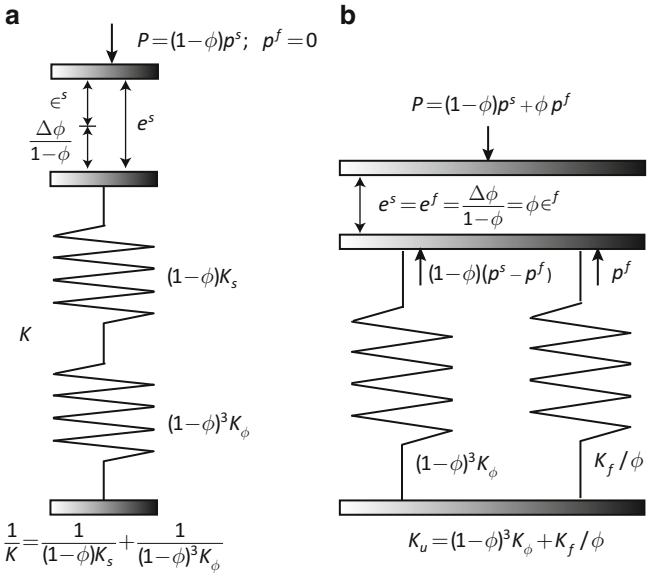


Fig. 4.5 (**a**) Series spring analogy for drained bulk modulus K, under drained condition ($p^f = 0$), for ideal porous medium ($K_\psi = 0$); (**b**) Parallel spring analogy for undrained modulus K_u, under undrained condition ($\overline{e}^s = \overline{e}^f$), for incompressible solid constituent case ($K_s \to \infty$)

where $P = (1-\phi)\,\overline{p}^s$. This series spring analogy is illustrated in Fig. 4.5a.

Or, we can obtain from (4.139) the following simplified form for the pore volume bulk modulus

$$K_p = \frac{\phi(1-\phi)^3 K_s K_\phi}{K_s + \phi\,(1-\phi)^2 K_\phi} \tag{4.161}$$

Based on the above, we can also express (4.159) as

$$\frac{1}{K} = \frac{1}{K_s} + \frac{\phi}{K_p} \tag{4.162}$$

This is another way to see the compliance of the drained frame as a series spring of the solid and pore volume compliances.

4.8.2 Granular Material

For granular materials, such as soil, its frame is formed by the packing of non-connected particles. When such a porous medium is subjected to a compression and starts to deform, we anticipate that most of the compaction is derived from the

particles rearranging their positions that takes away the pore space, and not from the actual deformation of the particles. The solid is hence much less compressible than the frame. In that case, it is reasonable to assume that $K_s \gg (K_\phi, K_\psi)$ for the intrinsic constants, or $(K_s', K_s'') \gg K$ for the Rice-Cleary constants. In the limit, we can take $(K_s/K_\phi, K_s/K_\psi) \to \infty$, and $(K_s'/K, K_s''/K) \to \infty$. Taking such limits in (3.12) and (3.13), we observe that the Biot effective stress coefficient and the pore volume effective stress coefficient take the value of unity:

$$\alpha = \beta = 1 \tag{4.163}$$

This value also suggests that under the incompressible solid assumption, the Biot effective stress is the same as the Terzaghi effective stress. This relation gives the justification of using Terzaghi effective stress in soil mechanics.

We also observe from (4.127) and (4.139) that

$$K = (1-\phi)^3 K_\phi \tag{4.164}$$

$$K_p = \phi(1-\phi)^3 K_\phi = \phi K \tag{4.165}$$

which shows that the drained frame compressibility is entirely the result of the pore compressibility.

From (4.131), (4.132), and (4.133), we obtain the following simplified relations

$$B = \frac{K_f}{\phi K + K_f} \tag{4.166}$$

$$M = \frac{K_f}{\phi} \tag{4.167}$$

$$K_u = K + \frac{K_f}{\phi} \tag{4.168}$$

The above shows that the undrained bulk modulus can be decomposed into a part contributed by the frame, and a part by the fluid. Equation (4.168) can be considered as a parallel spring model. In this case, the total applied compressive stress P is supported by two reaction forces, one part is due to the support of the fluid, and the other due to the frame. Each of the above parts is subject to the same deformation $\overline{e}^s = \overline{e}^f = \phi\,\overline{\epsilon}^f = \Delta\phi/(1-\phi)$, as a consequence of the kinematic equations (4.37) and (4.38), under the undrained condition (4.119), and the incompressible solid condition $\overline{\epsilon}^s = 0$. Utilizing the constitutive equations (4.85) and (4.87), these can be written as

$$P = (1-\phi)(\overline{p}^s - \overline{p}^f) - \overline{p}^f = -K_u\,\overline{e}^s = -(1-\phi)^3 K_\phi\,\overline{e}^s - K_f\,\overline{\epsilon}^f \tag{4.169}$$

Figure 4.5b illustrates this parallel spring analogy.

In summary, this simplified physical model is equivalent to the *elastic aquifer model* presented by Verruijt [68]. In this model, the set of independent bulk material constants is given by G, K, and K_f/ϕ $(= M)$, which simplifies the need for measuring the poroelastic material constants.

The reduction of other material constants is summarized in Appendices B.2 and B.3.

4.8.3 Soil Mechanics Model: Saturated

When a soil is saturated with water, then not only the solid, but also the fluid, can be considered as incompressible when compared to the frame compressibility. Hence it is generally true that $K_s \gg K$ and $K_f \gg K$ for saturated soil, and we can take the limits $(K_s/K, K_f/K) \to \infty$. With these limits, we deduce in addition to (4.163), (4.164), and (4.165) the following results:

$$B = 1 \tag{4.170}$$

$$C = \frac{1}{K} \tag{4.171}$$

$$M \to \infty \tag{4.172}$$

$$K_u \to \infty \tag{4.173}$$

These results also lead to

$$\nu_u = \frac{1}{2} \tag{4.174}$$

$$S = \frac{3}{3K + 4G} \tag{4.175}$$

From the values given by (4.173) and (4.174), we recognize that at the moment of a suddenly applied load, the undrained porous material is incompressible. At that moment, according to (4.170), the pore pressure rise is equal to the applied average compressive stress.

These limiting values of material constants also bring about changes in the constitutive relations. For example, by taking the limit of $M \to \infty$ in (2.28), we obtain

$$\zeta = e \tag{4.176}$$

That is, the volume of fluid exchanged with the frame is exactly the volumetric change of the frame.

Appendices B.2 and B.3 give a summary of the reduction of the above and other material constants.

4.8.4 Soil Mechanics Model: Nearly Saturated

When the pore water in the soil is unsaturated, that is, there is a presence of air in the pores, we need to deal with two fluid phases, water and air. Two-phase flow (or multiphase flow) is a complicated phenomenon. It involves the surface tension between the fluids and also with solid, the large contrast in density, compressibility, phase change at critical temperature and pressure, and instability problems. Even without considering the flow, the mechanical interaction between the multiple fluids and the solid can be complicated. Some of the constitutive models, such as that for unsaturated (water and air) porous medium [17, 28, 31], multiphase immiscible fluids [40, 69, 70], involving phase transformation of freezing [29], and drying [30, 32], have been investigated. The study of these phenomena is beyond the scope of this book; hence in the following we shall examine only a very simple case of modeling the mechanical response of two fluid phases as a single phase mixture.

For an unsaturated porous medium, when the amount of air is very small, in the form of isolated bubbles surrounded by water, called *insular air saturation* in groundwater literature [5], we can assume that its motion is in phase and its pressure is in equilibrium with the pore water. In that case, we may consider the combined water-air phase as a single fluid phase, without bothering to model the two-phase phenomenon.

Given the compressibility of the water and the air phases as C_w and C_a, respectively, we may express the apparent compressibility of the mixture as the weighted average based on Wood's equation [73]

$$C_f = S_r C_w + (1 - S_r) C_a \tag{4.177}$$

in which S_r is the *degree of water saturation*, defined as the ratio of water volume to pore volume in an REV, and $(1 - S_r)$ is then the *degree of air saturation*. In the above equation, we have neglected the gas solubility in water, or else the *Henry's coefficient of solubility* needs to be introduced [64].

A simplified relation of the above was introduced by Verruijt [68]. Based on Boyle's law, $d(V_a\, p_a) = V_a\, dp_a + p_a\, dV_a = 0$, where V_a and p_a are the volume and pressure of air, we may express the air compressibility as $C_a = 1/p_a$. Also, taking the approximation $S_r \approx 1$, Verruijt [68] expressed (4.177) as

$$C_f = C_w + \frac{1 - S_r}{P_a} \tag{4.178}$$

in which $P_a \approx p_a = p_w = p$ is intended to be taken as a constant. It should be cautioned that P_a is the absolute pressure, and not the gage pressure or incremental pressure from an initial pressure, which is typically used in the constitutive equations. Hence for an application in which the pressure change Δp is much smaller than the background absolute pressure P_a, we can treat the apparent pore water compressibility C_f as a material constant to avoid a nonlinear model.

We note that air is much more compressible than water, $C_a \gg C_w$; hence the compressibility of the mixture is dominated by C_a. For pore water containing even a small amount of air bubble, it can be highly compressible. Hence for a nearly saturated soil, although we may still consider the solid phase as incompressible, $K_s \to \infty$, for the fluid phase, the apparent fluid bulk modulus K_f is no longer infinite as for the saturated case. In fact, as demonstrated in Sect. 7.8, a mere 0.1 % of air bubble entrapped in water can increase its compressibility (or reduce K_f) by more than 10 folds.

4.8.5 Highly Compressible Solid

When the solid constituent and the frame are much more compressible than the fluid, such that $K_f \gg (K_s, K_\phi, K_\psi)$, or equivalently $K_f \gg (K_s', K_s'', K)$, we can take the limit $K_f \to \infty$, as an approximation. The reduction of material constants for this special case is summarized in Appendices B.2 and B.3. Here we pay special attention only to the Skempton pore pressure coefficient, simplified from (4.131):

$$B = 1 + \frac{\phi(1-\phi)^3 K_\phi}{K_s + \phi^2(1-\phi)^2 K_\phi} \tag{4.179}$$

in which we also used the ideal porous medium assumption. This equation shows that $B > 1$. A greater than unity Skempton pore pressure coefficient was indeed measured for polyurethane foam infiltrated by a silicone oil [51] (see Table 3.2).

4.8.6 Highly Compressible Fluid

When the fluid becomes highly compressible, for example, the fluid is a gas or a liquid containing entrapped gas bubbles, we may take the limit of $K_f \to 0$. In that case, we obtain by series expansion and retaining only the first order term of K_f the following results

$$K_u \approx K + \frac{(K_s - K)^2 K_f}{\phi K_s{}^2} \tag{4.180}$$

$$B \approx \frac{(K_s - K)K_f}{\phi K K_s} \tag{4.181}$$

$$M \approx \frac{1}{C} \approx \frac{1}{S} \approx \frac{K_f}{\phi} \tag{4.182}$$

In the limit $K_f \to 0$, we have $B \to 0$, $K_u \to K$, and $M \to 0$; in other words, the porous material behaves as an elastic material without fluid.

4.9 Material Stability and Energy Diagram

In this section, we shall examine the stability of a material subject to deformation. A material is unstable if a small perturbation in its stress or strain state leads to unbounded growth. The stability criteria can be established by examining the Helmholtz free energy. In (4.80) and (4.84) we have expressed the Helmholtz free energy density for the solid and the fluid phases, $\overline{\mathcal{A}}_s$ and $\overline{\mathcal{A}}_f$, in terms of quadratic forms of internal strains and porosity. The total Helmholtz free energy density is thus

$$\begin{aligned}\overline{\mathcal{A}} &= (1-\phi)\overline{\mathcal{A}}_s + \phi\overline{\mathcal{A}}_f \\ &= (1-\phi)\left(\frac{1}{2}K_\alpha\,\overline{\epsilon}^s\,\overline{\epsilon}^s + \frac{1}{2}K_\phi\,\Delta\phi^2 + K_\psi\,\overline{\epsilon}^s\,\Delta\phi\right) + \phi\left(\frac{1}{2}K_f\,\overline{\epsilon}^f\,\overline{\epsilon}^f\right)\end{aligned} \tag{4.183}$$

For simplicity, in the above we have ignored energy associated with shear deformation.

Based on the linearized theory, we realize the $\Delta\phi$ is not an independent quantity, and is in fact related to the solid and fluid internal strains through (4.92). Substituting (4.92) into the above, we obtain

$$\begin{aligned}\overline{\mathcal{A}} = &\frac{1}{2\left[(1-\phi)\,K_\phi - K_\psi\right]^2} \\ &\times\Big\{(1-\phi)K_s\{K_sK_\phi + [(1-\phi)K_\phi - K_\psi]^2\}\,\overline{\epsilon}^s\,\overline{\epsilon}^s - 2(1-\phi)K_sK_fK_\phi\,\overline{\epsilon}^s\,\overline{\epsilon}^f \\ &+K_f\{(1-\phi)K_fK_\phi + \phi\,[(1-\phi)K_\phi - K_\psi]^2\}\,\overline{\epsilon}^f\,\overline{\epsilon}^f\Big\}\end{aligned} \tag{4.184}$$

The above is a bivariate quadratic equation in terms of $\overline{\epsilon}^s$ and $\overline{\epsilon}^f$, and can be expressed as

$$A\,\overline{\epsilon}^s\,\overline{\epsilon}^s + B\,\overline{\epsilon}^s\,\overline{\epsilon}^f + C\,\overline{\epsilon}^f\,\overline{\epsilon}^f + F = 0 \tag{4.185}$$

Depending on the sign and magnitude of the coefficients, the above equation can represent an ellipse, a parabola, or a hyperbola, or some degenerated conic forms. The parabola and hyperbola are unbounded on the $(\overline{\epsilon}^s, \overline{\epsilon}^f)$ plane, implying unstable material; hence are not admissible. For a stable material, the quadratic form needs to represent an ellipse, which requires the conditions

$$B^2 - 4AC < 0 \tag{4.186}$$

$$AF < 0 \tag{4.187}$$

Assuming that K_s and K_f, which are directly associated with the material phases, are positive, then (4.187) shows that K_ϕ must be positive too. With these conditions, it can be shown that (4.186) is automatically satisfied regardless of the sign of K_ψ. Hence we can summarize the conditions for material stability as

$$(K_s, K_\phi, K_f) > 0 \tag{4.188}$$

with no constraint imposed on K_ψ, that is, it can be positive or negative. The consequence of $K_\psi > 0$, $K_\psi < 0$, and $K_\psi = 0$ has been discussed in Sect. 4.7. Based on the above established positiveness of bulk moduli, we can also easily show from (4.127) that the bulk modulus of the frame must be positive

$$K > 0 \tag{4.189}$$

Equation (4.184) and its variant allow us to examine the free energy of the material under different deformation and loading states [54]. The characteristics of this type of *energy diagram* may allow us to design materials that perform specific functions. For example, we can express (4.184) in terms of the volumetric strain of the solid frame, e, and the variation in fluid content, ζ, as

$$\begin{aligned}\overline{\mathcal{A}} = & \frac{1}{2\left[\phi K_s + \phi(1-\phi)^2 K_\phi + (1-\phi)K_f\right]} \\ & \times \left\{\left[\phi(1-\phi)^3 K_\phi K_s + \phi^2(1-\phi)^2 K_\phi K_f + K_s K_f\right] e^2\right. \\ & \left. -2K_f\left[\phi(1-\phi)^2 K_\phi + K_s\right] e\,\zeta + K_f\left[(1-\phi)^2 K_\phi + K_s\right]\zeta^2\right\}\end{aligned} \tag{4.190}$$

In the above, we have used the simplifying assumption $K_\psi = 0$. Equation (4.191) can be plotted on the e–ζ plane as constant strain energy density contour lines. As an example, we shall look at two materials, a Berea sandstone, and a coarse sand. For Berea sandstone, using data given in Table 3.1, $K = 8.0 \times 10^9\,\mathrm{N/m^2}$ and $K_s = 3.6 \times 10^{10}\,\mathrm{N/m^2}$, we obtain $K_\phi = 2.07 \times 10^{10}\,\mathrm{N/m^2}$. The porosity is $\phi = 0.19$, and $K_f = 2.25 \times 10^9\,\mathrm{N/m^2}$. For coarse sand, the data $K_s = 4.86 \times 10^9\,\mathrm{N/m^2}$, $K_\phi = 1.49 \times 10^9\,\mathrm{N/m^2}$, $K_f = 2.25 \times 10^9\,\mathrm{N/m^2}$, and $\phi = 0.48$, are used. In Fig. 4.6 we plot the contour corresponding to $\overline{\mathcal{A}} = 3 \times 10^4\,\mathrm{N/m^2}$ on the e–ζ plane for the two materials. First, we observe that these contours are indeed ellipses, as required for stable materials. These contours represent all the admissible solutions of (e, ζ),

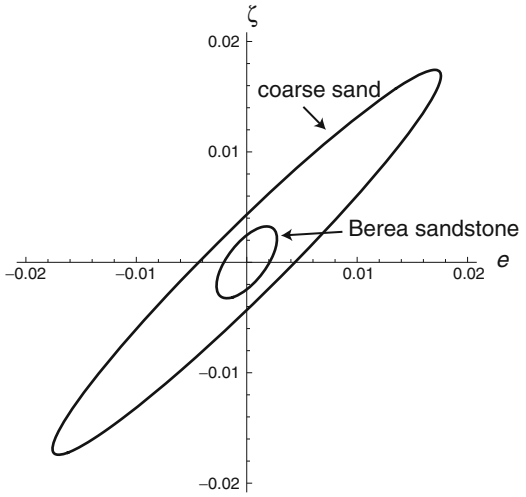


Fig. 4.6 Helmholtz energy density contour on e–ζ plane

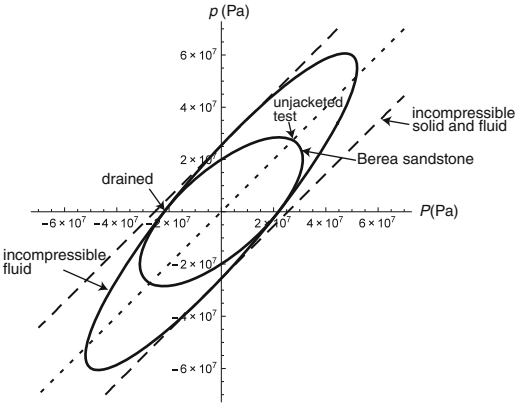


Fig. 4.7 Helmholtz energy density contour on P–p plane

given the same strain energy density. We find that the softer material (coarse sand) deforms more for the same amount of strain energy (work done). Also, to achieve the maximum deformation of the solid frame, roughly the maximum amount of fluid is expelled from the frame. The undrained state of material deformation is given by the curve's intercept on the e-axis.

Next, we present the energy as a function of the mean compressive stresses (P, p):

$$\overline{\mathcal{A}} = \frac{1}{2(1-\phi)^3 K_s K_\phi K_f} \left\{ K_f \left[K_s + (1-\phi)^2 K_\phi \right] P^2 - 2K_f \left[K_s + \phi(1-\phi)^2 K_\phi \right] P p \right.$$
$$\left. + \left[\phi^2 (1-\phi)^2 K_\phi K_f + K_s K_f + \phi(1-\phi)^3 K_\phi K_s \right] p^2 \right\} \tag{4.191}$$

The energy contour for the same Helmholtz energy density as above is plotted in Fig. 4.7 for the Berea sandstone case as the ellipse. From the contour, we can

identify the maximum P and p that can be achieved with a given energy level. Solutions located on the P axis are associated with the drained response, $p = 0$. The solution that intersects with the 45° line from the origin representing $P = p$ gives the unjacketed test result. It is of interest to further examine some special cases. Next we consider the case of an incompressible fluid, $K_f \to \infty$. Equation (4.191) then reduces to

$$\overline{\mathcal{A}} = \frac{1}{2(1-\phi)^3 K_s K_\phi} \left\{ [K_s + (1-\phi)^2 K_\phi] P^2 - 2[K_s + \phi(1-\phi)^2 K_\phi] P p + [K_s + \phi^2 (1-\phi)^2 K_\phi] p^2 \right\} \tag{4.192}$$

This equation is also plotted in Fig. 4.7 as the larger ellipse. Furthermore, we can consider the additional condition of an incompressible solid, $K_s \to \infty$. Equation (4.192) simplifies to

$$\overline{\mathcal{A}} = \frac{1}{2(1-\phi)^3 K_\phi} (P-p)^2 \tag{4.193}$$

This degenerated curve is plotted as the two parallel lines in Fig. 4.7.

4.10 Semilinear Model

Although a linear model was presented in Sect. 4.4.2, the model established in Sect. 4.3 based on variational energy principle is generally nonlinear. Nonlinear model can be constructed using the phenomenological approach as well as the physically based approach. The phenomenological approach is based on the observation of the bulk response of the material, and is largely empirical. The physically based model can be built on the physical properties of the individual material phases. Therefore it provides a rational way to model nonlinearity that can isolate the physical mechanisms into those associated with the material (solid and fluid), the porous structure, and the geometry of large deformation. We hence identify these three types of nonlinearity below:

Material nonlinearity: The constitutive laws for the solid and fluid phases are generally nonlinear. For example, the solid bulk modulus can be a function of the solid strain, $K_s = K_s(\overline{\epsilon}^s)$, and fluid bulk modulus can be a function fluid strain $K_f = K_f(\overline{\epsilon}^f)$.

Structural nonlinearity: This is related to the strength of the pore space against deformation due to mechanisms such as grain rotation and rearrangement, and closing and opening of fractures. It is characterized by the porosity modulus K_ϕ in the present model, which is likely to be dependent on the porosity change, $K_\phi = K_\phi(\Delta\phi)$.

Geometric nonlinearity: For porous materials subject to finite and large deformation, the geometric nonlinearity needs to be considered. Even if we consider

only infinitesimal deformation, in the present model, by treating porosity ϕ as a variable, the constitutive model is already nonlinear, as demonstrated in Sect. 4.4.1 and the porosity equilibrium equation (4.86).

In the following, we limit our discussion only to *semilinear models*. According to Biot [14], semilinear models refer to models that the material phases behave linearly; that is, K_s and K_f are constants, and the nonlinearities are attributed to the structural and geometric sources. The semilinear model is supported by laboratory evidences such as those reported in van der Knaap [67]. Also, for the practical consideration of the simplicity of the model, we shall only consider materials that can be approximated as ideal porous medium, that is, with microhomogeneity and microisotropy.

4.10.1 Geometric Nonlinearity

First, we examine the geometric nonlinearity caused by the porosity variation. As discussed in Sect. 4.4.1, for materials with a weak pore structure, the change of porosity is of the same order of magnitude as the volumetric strain of the solid frame. By treating porosity as a variable, the porosity equilibrium equation (4.79) is nonlinear even under the infinitesimal deformation and linear material assumptions. Taking into consideration of only the first order variation, and imposing the ideal porous medium condition, $K_\psi = 0$, (4.86) becomes

$$\overline{p}^s - \overline{p}^f = -(1-\phi_o)\, K_\phi\, \Delta\phi + K_\phi\, \Delta\phi^2 \tag{4.194}$$

Or, expressed in terms of the total compressive stress, P, and dropping the averaging notation, we obtain

$$P - p = -(1-\phi_o)^2\, K_\phi\, \Delta\phi + 2\,(1-\phi_o)\, K_\phi\, \Delta\phi^2 \tag{4.195}$$

We observe that the above geometric nonlinear model for porosity variation is defined without additional nonlinear material constant.

To examine the magnitude of nonlinearity, we can substitute numerical values into (4.195). In the linear model, $-\Delta\phi$ is predicted as $(P-p)/(1-\phi_o)^2 K_\phi$. Hence we plot $-\Delta\phi$ versus $(P-p)/(1-\phi_o)^2 K_\phi$ based on (4.195), for the case of initial porosity $\phi_o = 0.3$, in Fig. 4.8. Also plotted in Fig. 4.8 is the linear model, shown as the 45° straight line. We observe that the linear model over-predicts the porosity change. This suggests that the porosity geometric nonlinearity exhibits an apparent material strain hardening effect as porosity becomes more and more compacted.

4.10.2 Structural Nonlinearity

The strain hardening behavior in the initial stage of compaction of porous materials has been widely observed in laboratory tests [66, 71, 74]. The geometric nonlinearity examined in the above section is too weak to account for the observed behavior. This

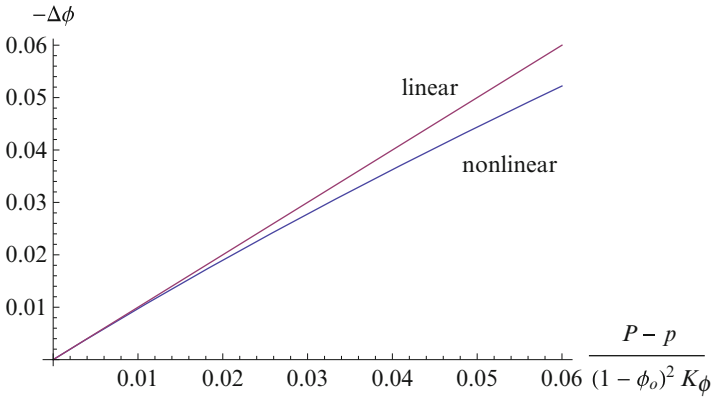


Fig. 4.8 Geometric nonlinear model of porosity variation

is an indication that the structural nonlinearity is present. As discussed above, the structural nonlinearity is characterized by the porosity modulus as a function of porosity change, $K_\phi = K_\phi(\Delta\phi)$.

To develop a nonlinear constitutive equation for the total strain e, we first express (4.123) in the following form:

$$e = -C_s p - C_K(P - p) \tag{4.196}$$

where

$$C_s = \frac{1}{K_s} \tag{4.197}$$

is the compressibility of the solid constituent, and

$$C_K = \frac{1}{K} = \frac{K_s + (1-\phi)^2 K_\phi}{(1-\phi)^3 K_s K_\phi} \tag{4.198}$$

is the drained compressibility of the solid frame. We notice that the expression of K is simplified from (4.127) by setting $K_\psi = 0$. Equation (4.196) is found both in van der Knaap [67] and Biot [14], and it was proposed that for a semilinear model, C_s is a constant and C_K is a function of $P - p$. However, based on the present *intrinsic* material constant model, K, and thus C_K, itself is composite constant involving both the solid and porosity compressibility. Hence it is necessary to isolate these mechanisms and rewrite (4.196) as

$$e = -\frac{C_s}{1-\phi}(P - \phi p) - \frac{C_\phi}{(1-\phi)^3}(P - p) \tag{4.199}$$

In the above, C_s is a constant, and

$$C_\phi = \frac{1}{K_\phi} \tag{4.200}$$

is a function of $\Delta\phi$, which in turn is a function of $P - p$. The empirical form of the function can be found in drained laboratory test similar to that in van der Knaap [67]. The typical form of such function is a power law or an exponential law. For example, for materials with a weak pore structure, $C_\phi \gg C_s$, we can ignore C_s and express (4.199) as

$$e = -\frac{C_{\phi o}}{(1-\phi_o)^3}(P-p)^a \tag{4.201}$$

where $C_{\phi o}$ is the linear porosity compressibility, ϕ_o is the initial porosity, a is an empirical constant. We anticipate $1 < a < 0$ for a strain hardening behavior. Equation (4.201) can in fact incorporate the porosity geometric nonlinearity effect as $\Delta\phi$ itself is a function of $P - p$.

We should emphasize that the above proposed semilinear model is limited to the small deformation range. It is relatively simple, but does not include many other important mechanisms, such as plasticity, grain crushing, pore collapsing, presence of crack, and flow, etc. [72, 75]. Nevertheless, we shall demonstrate through laboratory tests that the above proposed model seems to fit the data fairly well for the compression part of the loading, even for fairly large deformation.

To test the above nonlinear constitutive law, we plot in Fig. 4.9 five sets of laboratory data reported in [23] as total average compressive stress P versus the total compressive strain $-e$, under drained condition ($p = 0$). The test specimens include two rock cores provided by Ecopetrol (Colombia), a Berea sandstone, and two soft sandstone cores retrieved from the depth of 5,400 ft, and loaded to 3,000 and 5,000 psi, respectively. These data are plotted on log-log scale. The looping of data shown for the soft sandstones is caused by the unloading and reloading cycle. We observe nearly straight line behavior for all these rock cores from different origins, validating the power law shown in (4.201). The inverse of the slope of the straight line fit gives the a value. From these data, the observed a values are between 0.62 and 0.82.

4.11 Laboratory Measurement of Intrinsic Constant

As discussed in Sect. 3.5, Biot and Willis [15] proposed the laboratory measurements of the jacketed compressibility κ_b, the unjacketed compressibility δ_b, and the coefficient of fluid content γ_b. Together with an independently measured fluid compressibility $C_f = 1/K_f$, these coefficients then define the micromechanical constants K, K_s', and K_s'' through (3.93), (3.99), and (3.101).

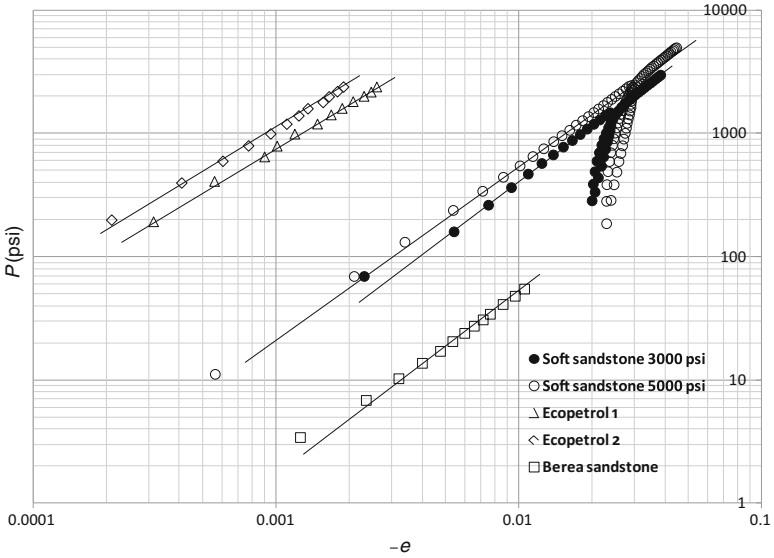


Fig. 4.9 Triaxial test data for five rock cores for testing the power law constitutive equation

For the intrinsic constants, we can utilize the relations in Sect. 4.6.2 to express them into the Biot-Willis jacketed and unjacketed coefficients as

$$K_s = \frac{1-\phi}{\gamma_b + \delta_b - \phi\, C_f} \tag{4.202}$$

$$K_\phi = \frac{\gamma_b + \delta_b - \phi\, C_f}{(1-\phi)[\kappa_b(\gamma_b + \delta_b - \phi\, C_f) - \delta_b^2]} \tag{4.203}$$

$$K_\psi = \frac{\gamma_b + \phi\, \delta_b - \phi\, C_f}{(1-\phi)[\kappa_b(\gamma_b + \delta_b - \phi\, C_f) - \delta_b^2]} \tag{4.204}$$

Although the above equations allow us to evaluate the intrinsic constants using the Biot-Willis coefficients, for better accuracy, it is more desirable to devise tests that can directly measure them. In the following, we shall propose such a procedure.

To measure the intrinsic constants, we can utilize the jacketed test as described in Sect. 3.5 by independently controlling the incremental confining compressive stress ΔP and the incremental pore pressure Δp. Following the laboratory procedure suggested by Biot and Willis [15], the initial sample volume V, the change in sample volume ΔV, and the fluid volume escaped from the sample $\Delta V_f'$ (calibrated for the pressure change), are measured. Prior to the test, the initial pore volume V_p needs to be measured, which leads to the porosity, $\phi = V_p/V$. Also needed is the compressibility of the fluid $C_f = 1/K_f$.

Using the above information and following (3.23) and (3.24), we can determine the change in pore volume ΔV_p as the difference between the change in fluid volume ΔV_f and the volume of fluid expelled from the sample $\Delta V_f'$ as

$$\Delta V_p = -\frac{\phi\, V \Delta p}{K_f} - \Delta V_f' \tag{4.205}$$

We can then evaluate

$$\Delta V_s = \Delta V - \Delta V_p \tag{4.206}$$

and

$$\Delta\phi = \frac{\Delta V_p - \phi\, \Delta V}{V} \tag{4.207}$$

As a first step of the test, we can adjust the pore pressure increment to be the same as the confining stress increment to mimic the unjacketed test; that is, $\Delta p = \Delta P$. Based on such test, we can check the ideal porous medium hypothesis by comparing ΔV_p with $\phi\, \Delta V$. If these quantities are approximately equal (within the range of experimental error), we can conclude that the tested porous material is approximately an ideal porous medium, and $K_\psi = 0$. In that case, we can obtain K_s from the unjacketed compressibility as

$$K_s = \frac{1}{\delta_b} \tag{4.208}$$

For K_ϕ, we can simplify (4.203) to the following

$$C_\phi = \frac{1}{K_\phi} = (1-\phi)^3 \kappa_b - (1-\phi)^2 \delta_b \tag{4.209}$$

That is, K_ϕ can be determined as the inverse of the difference between the Biot jacketed and unjacketed compressibilities.

If the condition $\Delta V_p \approx \phi \Delta V$ is not satisfied under the unjacketed test, then the ideal porous medium assumption does not hold, and we need to find K_ψ. As a first step, we can use the measured data to determine

$$K_s = -\frac{(1-\phi)\, V \Delta P}{\Delta V_s} \tag{4.210}$$

This set of data can also be used to evaluate the porosity change based on (4.207), which is plotted versus the pore pressure increment Δp in Fig. 4.10 and denoted as $\Delta\phi_a$. As the next step, we decrease the pore pressure while keeping ΔP constant, until the drained condition $\Delta p = 0$ is reached. Corresponding to each pore pressure change, we can evaluate $\Delta\phi$, which is also plotted it in Fig. 4.10. These data points

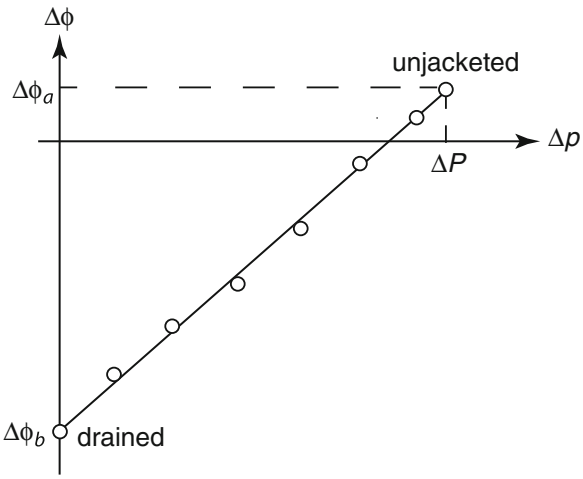


Fig. 4.10 Schematic plot of laboratory data from jacketed test

in fact are used only to check the anticipated linearity of the relation. To find the intrinsic constants, only two data points, the unjacketed test ($\Delta p = \Delta P$) and the drained test ($\Delta p = 0$), are needed, corresponding to porosity change of $\Delta\phi_a$ and $\Delta\phi_b$, respectively. Utilizing (4.91), we can find

$$K_\phi = \frac{\Delta P^2}{(1-\phi)[\Delta\phi_a - (1-\phi)\Delta\phi_b]\Delta P - K_s\,\Delta\phi_a^2} \tag{4.211}$$

$$K_\psi = \frac{K_s\,\Delta\phi_a\,\Delta P}{(1-\phi)[\Delta\phi_a - (1-\phi)\Delta\phi_b]\,\Delta P - K_s\,\Delta\phi_a^2} \tag{4.212}$$

References

1. Atkin RJ, Craine RE (1976) Continuum theories of mixtures—basic theory and historical development. Q J Mech Appl Math 29:209–244
2. Auriault JL, Sanchez-Palencia E (1977) Etude du comportement d'un milieu poreux saturé déformable (Study of macroscopic behavior of a deformable saturated porous medium). J de Méc 16(4):575–603
3. Bear J (1972) Dynamics of fluids in porous media. American Elsevier, New York, 764pp (also published by Dover, 1988)
4. Bear J, Bachmat Y (1990) Introduction to modeling phenomena of transport in porous media. Kluwer, Dordrecht/Boston, 553pp
5. Bear J, Cheng AHD (2010) Modeling groundwater flow and contaminant transport. Springer, Dordrecht/London, 834pp
6. Berge PA, Wang HF, Bonner BP (1993) Pore pressure buildup coefficient in synthetic and natural sandstones. Int J Rock Mech Mining Sci 30(7):1135–1141
7. Berryman JG (2006) Effective medium theories for multicomponent poroelastic composites. J Eng Mech ASCE 132(5):519–531

8. Berryman JG, Pride SR (1998) Volume averaging, effective stress rules, and inversion for microstructural response of multicomponent porous media. Int J Solids Struct 35(34–35):4811–4843
9. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2):155–164
10. Biot MA (1955) Variational principles in irreversible thermodynamics with application to viscoelasticity. Phys Rev 97(6):1463–1469
11. Biot MA (1956) Thermoelasticity and irreversible thermodynamics. J Appl Phys 27(3):240–253
12. Biot MA (1956) Variational and Lagrangian methods in viscoelasticity. In: Grammel R (ed) Deformation and flow of solids, IUTAM colloquium Madrid 1955. Springer, Berlin, pp 251–263
13. Biot MA (1962) Generalized theory of acoustic propagation in porous dissipative media. J Acoust Soc Am 34(9):1254–1264
14. Biot MA (1973) Nonlinear and semilinear rheology of porous solids. J Geophys Res 78(23):4924–4937
15. Biot MA, Willis DG (1957) The elastic coefficients of the theory of consolidation. J Appl Mech ASME 24:594–601
16. Boresi AP, Chong KP, Lee JD (2010) Elasticity in engineering mechanics, 3rd edn. Wiley, Hoboken, 656pp
17. Borja RI (2006) On the mechanical energy and effective stress in saturated and unsaturated porous continua. Int J Solids Struct 43(6):1764–1786
18. Bowen RM (1980) Incompressible porous media models by use of the theory of mixtures. Int J Eng Sci 18(9):1129–1148
19. Bowen RM (1982) Compressible porous media models by use of the theory of mixtures. Int J Eng Sci 20(6):697–735
20. Brown RJS, Korringa J (1975) Dependence of elastic properties of a porous rock on compressibility of pore fluid. Geophysics 40(4):608–616
21. Budiansky B, O'Connell RJ (1976) Elastic moduli of a cracked solid. Int J Solids Struct 12(2):81–97
22. Chateau X, Dormieux L (2002) Micromechanics of saturated and unsaturated porous media. Int J Numer Anal Methods Geomech 26(8):831–844
23. Cheng AHD, Abousleiman Y (2008) Intrinsic poroelasticity constants and a semilinear model. Int J Numer Anal Methods Geomech 32(7):803–831
24. Cheng AHD, Lopatnikov SL (2002) Variational formulation of fluid infiltrated porous materials and physically-based micromechanical material constants. In: Auriault JL, Geindreau C, Royer P, Bloch JF, Boutin C, Lewandowska J (eds) Poromechanics II, proceedings of the second biot conference on poromechanics. Balkema, Grenoble, pp 385–390
25. Christensen RM (1979) Mechanics of composite materials. Wiley Interscience, New York, 384pp
26. Cleary MP (1978) Elastic and dynamic response regimes of fluid-impregnated solids with diverse microstructures. Int J Solids Struct 14(10):795–819
27. Coussy O (1995) Mechanics of porous continua, 2nd edn. Wiley, Chichester/New York, 472pp
28. Coussy O (2004) Poromechanics. Wiley, Chichester, 298pp
29. Coussy O (2005) Poromechanics of freezing materials. J Mech Phys Solids 53(8):1689–1718
30. Coussy O (2006) Deformation and stress from in-pore drying-induced crystallization of salt. J Mech Phys Solids 54(8):1517–1547
31. Coussy O (2007) Revisiting the constitutive equations of unsaturated porous solids using a Lagrangian saturation concept. Int J Numer Anal Methods Geomech 31(15):1675–1694
32. Coussy O, Eymard R, Lassabatere T (1998) Constitutive modeling of unsaturated drying deformable materials. J Eng Mech ASCE 124(6):658–667
33. Crochet MJ, Naghdi PM (1966) On constitutive equations for flow of fluid through an elastic solid. Int J Eng Sci 4:383–401

34. Cruz-Orive LM, Myking AO (1981) Stereological estimation of volume ratios by systematic sections. J Microsc 122:143–157
35. de Boer R (1998) The thermodynamic structure and constitutive equations for fluid-saturated compressible and incompressible elastic porous solids. Int J Solids Struct 35(34–35):4557–4573
36. de Boer R (2000) Theory of porous media, highlights in the historical development and current state. Springer, Brelin/New York, 618pp
37. de Groot SR, Mazur P (1984) Non-equilibrium thermodynamics. Dover, New York, 526pp
38. Delesse A (1848) Procédé mécanique pour déterminer la composition des roches (Mechanical process to determine the composition of rocks). Ann des Mines Paris (Ser 4) 13:379–388
39. Eshelby JD (1957) The determination of the elastic field of an ellipsoidal inclusion, and related problems. Proc R Soc Lond Ser A Math Phys Sci 241(1226):376–396
40. Gray WG, Schrefler BA (2007) Analysis of the solid phase stress tensor in multiphase porous media. Int J Numer Anal Methods Geomech 31(4):541–581
41. Green DH, Wang HF (1986) Fluid pressure response to undrained compression in saturated sedimentary rock. Geophysics 51(4):948–956
42. Hart DJ, Wang HF (1995) Laboratory measurements of a complete set of poroelastic moduli for Berea sandstone and Indiana limestone. J Geophys Res Solid Earth 100(B9):17741–17751
43. Hart DJ, Wang HF (2010) Variation of unjacketed pore compressibility using Gassmann's equation and an overdetermined set of volumetric poroelastic measurements. Geophysics 75(1):N9–N18
44. Hashin Z (1964) Theory of mechanical behaviour of heterogeneous media. Appl Mech Rev 17:1–9
45. Hashin Z, Shtrikman S (1963) A variational approach to the theory of the elastic behaviour of multiphase materials. J Mech Phys Solids 11(2):127–140
46. Hill R (1963) Elastic properties of reinforced solids: some theoretical principles. J Mech Phys Solids 11(5):357–372
47. Hill R (1967) The essential structure of constitutive laws for metal composites and polycrystals. J Mech Phys Solids 15(2):79–95
48. Kachanov M (1987) Elastic solids with many cracks—a simple method of analysis. Int J Solids Struct 23(1):23–43
49. Kachanov M, Sevostianov I (2005) On quantitative characterization of microstructures and effective properties. Int J Solids Struct 42(2):309–336
50. Katchalsky A, Curran PF (1967) Nonequilibrium thermodynamics in biophysics. Harvard University Press, Cambridge, 248pp
51. Kim YK, Kingsbury HB (1979) Dynamic characterization of poroelastic materials. Exp Mech 19(7):252–258
52. Landau LD, Lifshitz EM (1986) Theory of elasticity, 3rd edn. Course of theoretical physics, vol 7. Butterworth-Heinemann, Oxford/Burlington, 195pp
53. Laurent J, Boutéca MJ, Sarda JP, Bary D (1993) Pore-pressure influence in the poroelastic behavior of rocks—experimental studies and results. SPE Form Eval 8(2):117–122
54. Lopatnikov SL, Cheng AHD (2002) Variational formulation of fluid infiltrated porous material in thermal and mechanical equilibrium. Mech Mater 34(11):685–704
55. Mackenzie JK (1950) The elastic constants of a solid containing spherical holes. Proc Phys Soc Lond Sect B 63(361):2–11
56. Mei CC, Auriault JL (1989) Mechanics of heterogeneous porous media with several spatial scales. Proc R Soc Lond Ser A Math Phys Eng Sci 426(1871):391–423
57. Morland LW (1972) A simple constitutive theory for a fluid-saturate porous solid. J Geophys Res 77(5):890–900
58. Mura T (1987) Micromechanics of defects in solids, 2nd edn. Springer, Dordrecht, 588pp
59. Nemat-Nasser S, Hori M (1999) Micromechanics: overall properties of heterogeneous materials, 2nd edn. North-Holland, Amsterdam/London, 810pp
60. Nowacki W (1986) Thermoelasticity, 2nd edn. Pergamon, Oxford/New York, 566pp

61. Nur A, Byerlee JD (1971) Exact effective stress law for elastic deformation of rock with fluids. J Geophys Res 76(26):6414–6419
62. Sokolnikoff IS (1956) Mathematical theory of elasticity, 2nd edn. McGraw-Hill, New York, 476pp
63. Terada K, Ito T, Kikuchi N (1998) Characterization of the mechanical behaviors of solid-fluid mixture by the homogenization method. Comput Methods Appl Mech Eng 153(3–4):223–257
64. Thomas SD (1989) A finite element model for the analysis of wave induced stresses, displacements and pore pressures in an unsaturated seabed I: theory. Comput Geotech 8(1):1–38
65. Thomson (Lord Kelvin) W (1852) On the universal tendency in nature to the dissipation of mechanical energy. Philos Mag Ser 4 4(25):304–306
66. Vajdova V, Baud P, Wong TF (2004) Compaction, dilatancy, and failure in porous carbonate rocks. J Geophys Res Solid Earth 109(B5):16
67. van der Knaap W (1959) Nonlinear behavior of elastic porous media. Trans Am Inst Mining Metall Eng 216:179–186
68. Verruijt A (1969) Elastic storage of aquifers. In: DeWiest RJM (ed) Flow through porous media. Academic, New York, pp 331–376
69. Wei C, Muraleetharan KK (2002) A continuum theory of porous media saturated by multiple immiscible fluids: I. Linear poroelasticity. Int J Eng Sci 40(16):1807–1833
70. Wei C, Muraleetharan KK (2002) A continuum theory of porous media saturated by multiple immiscible fluids: II. Lagrangian description and variational structure. Int J Eng Sci 40(16):1835–1854
71. Wong TF, Baud P (1999) Mechanical compaction of porous sandstone. Oil Gas Sci Technol Revue de l'Inst Fr du Pet 54(6):715–727
72. Wong TF, David C, Zhu WL (1997) The transition from brittle faulting to cataclastic flow in porous sandstones: mechanical deformation. J Geophys Res Solid Earth 102(B2):3009–3025
73. Wood AB (1941) Textbook of sound. Being an account of the physics of vibrations with special reference to recent theoretical and technical developments, 2nd edn. Bell & Sons, London, 578pp
74. Wu XY, Baud P, Wong TF (2000) Micromechanics of compressive failure and spatial evolution of anisotropic damage in Darley Dale sandstone. Int J Rock Mech Mining Sci 37(1–2):143–160
75. Zhang JX, Wong TF, Davis DM (1990) Micromechanics of pressure-induced grain crushing in porous rocks. J Geophys Res Solid Earth Planets 95(B1):341–352
76. Zimmerman RW, Somerton WH, King MS (1986) Compressibility of porous rocks. J Geophys Res Solid Earth Planets 91(B12):2765–2777

Chapter 5
Anisotropy

The passive strength of the materials employed in the mechanical arts depends on the cohesive and repulsive forces of their particles, and on the rigidity of their structure. The consideration of the intimate nature of these forces belongs to the discussion of the physical properties of matter; but the estimation of their magnitude, and of their relative value in various circumstances, is of undeniable importance to practical mechanics, . . . The principal effects of any force acting on a solid body may be reduced to seven denominations; extension, compression, detrusion, flexure, torsion, a1teration, and fracture.

—Thomas Young (1845)

In the preceding chapters, the constitutive law developed for poroelastic materials assumes that the materials do not exhibit directional properties at the macroscopic level, and are *isotropic*. Geomaterials however are often *anisotropic* due to the existence of bedding surfaces in sedimentary rocks, foliations in metamorphic rocks, and microcracks aligned in the direction of stresses. Biomaterials such as cortical and trabecular bones are also anisotropic, due to their growth oriented in the direction of the physiological load [17]. In this chapter we shall develop the constitutive laws for the general material anisotropy, which are also simplified to special cases.

5.1 Anisotropic Constitutive Equation

Biot in 1955 extended his isotropic theory of poroelasticity to anisotropy [3]. Carroll [5], Thompson and Willis [14], and Cheng [7] performed micromechanical analysis. In this section, we shall first construct the constitutive laws at the bulk continuum level. Micromechanics will be examined in Sect. 5.3.

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_5

5.1.1 Elasticity

Similar to the construction of the isotropic case in Sect. 2.3, we shall begin with a review of elasticity constitutive equation.

Given the stress tensor σ_{ij} and the strain tensor e_{ij}, we shall assume that they are linearly related, based on the *generalized Hooke's law*. The most general linear relation that can be constructed between these two second rank tensors is the following:

$$\sigma_{ij} = M_{ijkl}\, e_{kl} \tag{5.1}$$

where M_{ijkl} is a fourth rank tensor, representing the elastic moduli, having $3 \times 3 \times 3 \times 3 = 81$ components.

Following the elasticity analysis [4, 13], we recognize the symmetry of the stress and strain tensors, $\sigma_{ij} = \sigma_{ji}$, and $e_{ij} = e_{ji}$, such that each tensor only has six independent components. Using these conditions in (5.1), we deduce that the elastic moduli have the following symmetry

$$M_{ijkl} = M_{jikl}; \quad M_{ijkl} = M_{ijlk} \tag{5.2}$$

The number of independent moduli is reduced to $6 \times 6 = 36$. The next requirement comes from the Betti-Maxwell reciprocal relation of a reversible process. Substituting the stresses and strains into (2.23), we find that to fulfill it, it is necessary that

$$M_{ijkl} = M_{klij} \tag{5.3}$$

This condition makes the 6×6 coefficient matrix symmetric, such that there are only 21 independent material constants for the case of *general anisotropy*. This fact was first established by Green [8].

Materials, such as composite materials and geomaterials, can exhibits certain degree of symmetry. A materials that has folding symmetry with respect to one plane is called *monoclinic*. The number of independent elastic constants is reduced from 21 to 13. If the folding symmetry is extended to two orthogonal planes (it then automatically has folding symmetry with respect to the third orthogonal plane), the material is then called *orthotropic*. The number of independent elastic constants is further reduced to 9. For a material that has rotational symmetry with respect to one axis, such that it is isotropic on the plane perpendicular to that axis, it is referred to as *transversely isotropic*. In this case, the number of material constants is 5. Finally, for a material that is isotropic, there are two independent material constants, known as Lamé constants, as discussed in Sect. 2.3.1.

The elasticity constitutive equation can also be expressed in the form of strain-stress relation. In that case, the generalized Hooke's law gives

$$e_{ij} = C_{ijkl}\,\sigma_{kl} \tag{5.4}$$

where C_{ijkl} is the *elastic compliance tensor*, which can be obtained by taking the matrix inverse of a 9×9 matrix formed by the elastic moduli M_{ijkl}.

5.1.2 Poroelastic Stress-Strain Relation

Following the lead of elasticity, we can construct the stress-strain relation of poroelasticity for general anisotropy by invoking the generalized Hooke's law, which states that the stress and strain quantities are linearly related. By examining the tensor ranks, we can conclude with the following relations [7]:

$$\sigma_{ij} = M_{ijkl}\,e_{kl} - \alpha_{ij}\,p \tag{5.5}$$

$$p = M(-\alpha_{ij}\,e_{ij} + \zeta) \tag{5.6}$$

where M_{ijkl}, α_{ij} and M are 91 (81 + 9 + 1) constitutive (Hookean) constants. It is noted that on the right hand side of (5.5) the dynamic quantity p, rather than the kinematic one ζ, is used. Equation (5.5) is equivalent to (2.29) for the isotropic case. The purpose of using the form of (2.29), rather than (2.27), is to bring the effective stress concept, and the drained material response, into focus. The spirit of Hooke's law is not violated by this expression as these parameters are linearly related. We should mention that in the above relations, we already imposed the Betti-Maxwell reciprocal relation, such that the same α_{ij} appears in (5.5) and (5.6).

The symmetry of the stress and strain tensor, as well as the reciprocal relation, requires that

$$M_{ijkl} = M_{jikl} = M_{ijlk} = M_{klij} \tag{5.7}$$

The stress and strain symmetry also requires that

$$\alpha_{ij} = \alpha_{ji} \tag{5.8}$$

With the above symmetry, we find that only twenty-one M_{ij}, six α_{ij}, and one M are independent constants. The total number of constitutive coefficients for the general anisotropy of poroelasticity is therefore 28, which is compared to the 21 coefficients for anisotropic elasticity.

The physical meaning of these Hookean coefficients becomes clear when we subject the constitutive equations to special conditions. Consider first the drained condition, under which $p = 0$. We observe from (5.5) that the relation simplifies to

$$\sigma_{ij} = M_{ijkl}\, e_{kl} \tag{5.9}$$

which is identical to that of elasticity (5.1). The coefficients M_{ijkl} are then the elastic moduli of the solid skeleton, and are called the *drained elastic moduli*.

Next, we can extend the concept of effective stress to anisotropic poroelastic material by rearranging (5.5) into the following form:

$$\sigma''_{ij} = \sigma_{ij} + \alpha_{ij} p = M_{ijkl}\, e_{kl} \tag{5.10}$$

which can be compared to (2.58) and (2.59). The material constant α_{ij} is therefore the *Biot effective stress coefficient tensor*. For the isotropic case, the tensor degenerates into a scalar $\alpha_{ij} = \delta_{ij}\alpha$. It will be established in Sect. 5.3 that α_{ij} is a property of the solid constituent and pore structure only (independent of fluid modulus). For isotropic case, this fact has been established in Sect. 4.6.1.

The fact that the effective stress coefficient α_{ij} is a second rank tensor suggests that *pore pressure modifies not only effective normal stresses, but also effective shear stresses* [7]. The latter is an effect not predicted in the isotropic theory.

Finally, we turn our attention to the constant M. By comparing (5.6) with (2.28), we easily observe that M is the same as that defined in the isotropic case, that is, the inverse of constant strain storage coefficient (see Sect. 2.4.4).

We now return to the standard way of writing the generalized Hooke's law, by using the kinematic quantity ζ, instead of the dynamic one p, on the right hand side of the stress-strain relations:

$$\sigma_{ij} = M^u_{ijkl}\, e_{kl} - \alpha_{ij} M \zeta \tag{5.11}$$

Since (5.11) must be compatible with (5.5) and (5.6), it is easily shown that M^u_{ijkl} must satisfy the following relation

$$M^u_{ijkl} = M_{ijkl} + \alpha_{ij}\, \alpha_{kl}\, M \tag{5.12}$$

which can be compared with its isotropic version (2.30). The physical meaning of M^u_{ijkl} becomes clear under the undrained condition, $\zeta = 0$, and (5.11) simplifies to

$$\sigma_{ij} = M^u_{ijkl}\, e_{kl} \tag{5.13}$$

The tensor M^u_{ijkl} is then the apparent elastic moduli of the combined solid and fluid, when the fluid is trapped in the solid frame. It is referred to as the *undrained elastic modulus tensor*, in contrast to the drained elastic modulus tensor, M_{ijkl}.

We notice that (5.12) nicely separates the contribution of M^u_{ijkl} into a drained elastic part, which is a property of the solid only, and a coupling part dependent on both solid and fluid properties.

5.1.3 Poroelastic Strain-Stress Relation

The strain-stress relations are the inverse of the stress-strain relations given in (5.5) and (5.6). In shorthand, we can express them as

$$e_{ij} = C_{ijkl}\,\sigma_{kl} + \frac{1}{3}\,CB_{ij}\,p \tag{5.14}$$

$$\zeta = C\left(\frac{1}{3}\,B_{ij}\,\sigma_{ij} + p\right) \tag{5.15}$$

where C_{ijkl}, B_{ij}, and C are again Hooke's law constants. Comparing (5.15) with (2.32), we first identify C as the constant stress storage coefficient, same as the isotropy case.

Under drained condition, $p = 0$, (5.14) shows that C_{ijkl} is the *drained compliance tensor*, which is the inverse of M_{ijkl}, defined by the following relation

$$M_{ijkl}C_{klmn} = \frac{1}{2}\left(\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}\right) \tag{5.16}$$

Equivalent to the two alternative expressions, (5.5) and (5.11), the second way of writing (5.14) is

$$e_{ij} = C^u_{ijkl}\,\sigma_{kl} + \frac{1}{3}\,B_{ij}\,\zeta \tag{5.17}$$

Similarly, by setting $\zeta = 0$, we identify C^u_{ijkl} as the *undrained compliance tensor*, or the inverse of M^u_{ijkl}. Comparison of (5.14) and (5.15) with (5.17) shows that

$$C^u_{ijkl} = C_{ijkl} - \frac{1}{9}\,B_{ij}B_{kl}\,C \tag{5.18}$$

which is the counterpart of (5.12). Its isotropic version is given as (2.34).

In an undrained test, (5.15) reduces to

$$p = -\frac{1}{3}\,B_{ij}\,\sigma_{ij} \tag{5.19}$$

Comparing with (2.65), we may view B_{ij} as a generalization of the Skempton pore pressure coefficient B to a second rank tensor. For the isotropic case, it degenerates to $B_{ij} = \delta_{ij}B$. The significance of (5.19) is that *pore pressure can be generated by shear as well as normal stresses* [7]. Or, equivalently, this can be stated as *pore pressure can be generated by volumetric as well as deviatoric strains*.

This effect of pore pressure generation by deviatoric stresses is similar to that of the Skempton pore pressure coefficient A (see Sect. 2.4.3). Skempton [12] considered the effect of coefficient A as a consequence of material inelasticity. As commented by Tokunaga et al. [15], this anisotropic elastic effect can provide an explanation for the Skempton pore pressure coefficient $A \neq 1/3$, which was empirically introduced to explain undrained pore pressure responses near a shear fault [16]. Gwyther et al. [9] reported that not only areal strain but shear strains measured by a borehole tensor strain meter seemed to be associated with hydrological changes, which might be an indication of this effect too.

By manipulating the stress-strain and strain-stress relations, a number of useful formulae are obtained. For example, α_{ij} and B_{ij} are related as

$$\alpha_{ij} = \frac{CM_{ijkl}B_{kl}}{3} = \frac{M^u_{ijkl}B_{kl}}{3M} \tag{5.20}$$

$$B_{ij} = \frac{3\,C_{ijkl}\alpha_{kl}}{C} = 3MC^u_{ijkl}\alpha_{kl} \tag{5.21}$$

while the following relations are found between C and M

$$C = \frac{1}{M} + \alpha_{ij}\alpha_{kl}\,C_{ijkl} \tag{5.22}$$

$$M = \frac{1}{C} + \frac{1}{9}B_{ij}B_{kl}M^u_{ijkl} \tag{5.23}$$

which can be compared to isotropic versions, (2.45) and (2.47).

5.2 Material Symmetry

Geomaterials often exhibit certain degree of symmetry; hence the complexity of anisotropy can be reduced. Two special cases, orthotropy and transverse isotropy, are examined below.

Before such examination, it is beneficial to first switch to "engineering notations" for stresses and strains. Due to the symmetry of stress and strain tensors, it is possible to express them as vectors of six components, rather than second rank tensors, as follows:

$$\vec{\sigma} = \lfloor\, \sigma_{xx} \quad \sigma_{yy} \quad \sigma_{zz} \quad \sigma_{yz} \quad \sigma_{xz} \quad \sigma_{xy} \,\rfloor^T \tag{5.24}$$

$$\vec{e} = \lfloor\, e_{xx} \quad e_{yy} \quad e_{zz} \quad \gamma_{yz} \quad \gamma_{xz} \quad \gamma_{xy} \,\rfloor^T \tag{5.25}$$

where the superscript $\lfloor\cdots\rfloor^T$ stands for matrix transpose, and

$$\gamma_{ij} = 2\,e_{ij}; \quad i \neq j \tag{5.26}$$

is the shear strain (engineering notation). The constitutive equations (5.5), (5.6), (5.14) and (5.15) can be written in the new notations as

$$\vec{\sigma} = \boldsymbol{M}\vec{e} - \vec{\alpha}\, p \tag{5.27}$$

$$p = M\left(\zeta - \vec{\alpha}^T \vec{e}\right) \tag{5.28}$$

$$\vec{e} = \boldsymbol{C}\vec{\sigma} + \frac{C}{3}\vec{B}\, p \tag{5.29}$$

$$\zeta = C\left(p + \frac{1}{3}\vec{B}^T\vec{\sigma}\right) \tag{5.30}$$

in which the coefficients are

$$\vec{\alpha} = \lfloor\, \alpha_{xx} \quad \alpha_{yy} \quad \alpha_{zz} \quad \alpha_{yz} \quad \alpha_{xz} \quad \alpha_{xy} \,\rfloor^T \tag{5.31}$$

$$\vec{B} = \lfloor\, B_{xx} \quad B_{yy} \quad B_{zz} \quad B_{yz} \quad B_{xz} \quad B_{xy} \,\rfloor^T \tag{5.32}$$

$$\boldsymbol{M} = \left[M_{ij}\right]; \qquad i,j = 1,2,\ldots,6 \tag{5.33}$$

$$\boldsymbol{C} = \left[C_{ij}\right]; \qquad i,j = 1,2,\ldots,6 \tag{5.34}$$

In the above, we also have invoked the symmetry of material constants, α_{ij}, B_{ij}, M_{ijkl}, and C_{ijkl}, to allow the reduction of tensor ranks. In addition, the (6×6) matrices (second rank tensor), $\boldsymbol{M}$ and $\boldsymbol{C}$, are also symmetric.

The above reduction in tensor ranks allows the easy observation that there are twenty-one independent components in $\boldsymbol{M}$ (by symmetry), six components in $\vec{\alpha}$, and one M, with a total of 28 independent material constants for general anisotropy, just as stated in Sect. 5.1.2.

5.2.1 Orthotropy

For materials with three mutually orthogonal planes of elastic symmetry, known as orthotropy, there exist 13 independent material coefficients (nine $\boldsymbol{M}$, three $\vec{\alpha}$, and one M). The drained elastic moduli $\boldsymbol{M}$ can be expressed in terms of engineering constants such as Young's moduli, Poisson's ratios, etc., as [6, 10]:

$$\boldsymbol{M} = \begin{bmatrix} M_{xx} & M_{xy} & M_{xz} & 0 & 0 & 0 \\ M_{xy} & M_{yy} & M_{yz} & 0 & 0 & 0 \\ M_{xz} & M_{yz} & M_{zz} & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{yz} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{xz} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{xy} \end{bmatrix} \tag{5.35}$$

where we have denoted $M_{11} = M_{xx}, M_{12} = M_{xy}, M_{44} = G_{yz}$, etc., with the following relations:

$$M_{xx} = \frac{E_x E_y (E_z - E_y \nu_{zy}^2)}{A} \tag{5.36}$$

$$M_{yy} = \frac{E_y^2 (E_z - E_x \nu_{zx}^2)}{A} \tag{5.37}$$

$$M_{zz} = \frac{E_z^2 (E_y - E_x \nu_{yx}^2)}{A} \tag{5.38}$$

$$M_{xy} = \frac{E_x E_y (E_z \nu_{yx} + E_y \nu_{zx} \nu_{zy})}{A} \tag{5.39}$$

$$M_{xz} = \frac{E_x E_y E_z (\nu_{zx} + \nu_{yx} \nu_{zy})}{A} \tag{5.40}$$

$$M_{yz} = \frac{E_y E_z (E_x \nu_{yx} \nu_{zx} + E_y \nu_{zy})}{A} \tag{5.41}$$

in which

$$A = E_y E_z - E_x E_z \nu_{yx}^2 - E_x E_y \nu_{zx}^2 - E_y^2 \nu_{zy}^2 - 2E_x E_y \nu_{yx} \nu_{zx} \nu_{zy} \tag{5.42}$$

In the above E_x, E_y and E_z are the drained Young's moduli in directions x, y, and z, respectively; G_{xy}, G_{yz} and G_{xz} are shear moduli for coordinate planes x–y, y–z, z–x, respectively; ν_{yx}, ν_{zx} and ν_{zy} are the drained Poisson's ratios characterizing the compressive strain in the direction of the second subscript, produced by a tensile stress in the direction of the first subscript. (It is of interest to point out that Poisson's ratios are not symmetric, namely $\nu_{ij} \neq \nu_{ji}$.) There are nine such engineering constants.

Similarly, the $\boldsymbol{C}$ matrix is given by

$$\boldsymbol{C} = \begin{bmatrix} 1/E_x & -\nu_{yx}/E_y & -\nu_{zx}/E_z & 0 & 0 & 0 \\ -\nu_{yx}/E_y & 1/E_y & -\nu_{zy}/E_z & 0 & 0 & 0 \\ -\nu_{zx}/E_z & -\nu_{zy}/E_z & 1/E_z & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_{yz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_{xz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G_{xy} \end{bmatrix} \tag{5.43}$$

The other coefficients consist of

$$\vec{\alpha} = \lfloor\, \alpha_x \;\; \alpha_y \;\; \alpha_z \;\; 0 \;\; 0 \;\; 0 \,\rfloor^T \tag{5.44}$$

$$\vec{B} = \lfloor\, B_x \;\; B_y \;\; B_z \;\; 0 \;\; 0 \;\; 0 \,\rfloor^T \tag{5.45}$$

5.2.2 Transverse Isotropy

Geomaterials are often transversely isotropic, in which there exists a rotational symmetry about the axis perpendicular to the bedding plane. In this case, there exist 8 independent material constitutive constants: five $\boldsymbol{M}$, two $\vec{\alpha}$, and one M. The drained elastic moduli are

$$\boldsymbol{M} = \begin{bmatrix} M_{xx} & M_{xy} & M_{xz} & 0 & 0 & 0 \\ M_{xy} & M_{xx} & M_{xz} & 0 & 0 & 0 \\ M_{xz} & M_{xz} & M_{zz} & 0 & 0 & 0 \\ 0 & 0 & 0 & G' & 0 & 0 \\ 0 & 0 & 0 & 0 & G' & 0 \\ 0 & 0 & 0 & 0 & 0 & G \end{bmatrix} \tag{5.46}$$

in which $G = (M_{xx} - M_{xy})/2$ is not an independent parameter. In the above, the z-axis is assumed to be the axis of rotational material symmetry. By noting that $E_x = E_y = E, E_z = E'$, $\nu_{yx} = \nu_{xy} = \nu$, $\nu_{zy} = \nu_{zx} = \nu'$, and $G_{yz} = G_{xz} = G'$, (5.36), (5.37), (5.38), (5.39), (5.40), and (5.41) becomes

$$M_{xx} = \frac{E(E' - E\nu'^2)}{(1+\nu)(E' - E'\nu - 2E\nu'^2)} \tag{5.47}$$

$$M_{xy} = \frac{E(E'\nu + E\nu'^2)}{(1+\nu)(E' - E'\nu - 2E\nu'^2)} \tag{5.48}$$

$$M_{xz} = \frac{EE'\nu'}{E' - E'\nu - 2E\nu'^2} \tag{5.49}$$

$$M_{zz} = \frac{E'^2(1-\nu)}{E' - E'\nu - 2E\nu'^2} \tag{5.50}$$

$$G = \frac{M_{xx} - M_{xy}}{2} = \frac{E}{2(1+\nu)} \tag{5.51}$$

where E is the drained Young's modulus in the plane of isotropy (x–y plane), E' the drained Young's modulus perpendicular to that plane (z-direction), G the shear modulus for the plane of isotropy, G' the shear modulus normal to the plane of isotropy, ν the drained Poisson's ratio characterizing the transverse strain reduction in the plane of isotropy due to a tensile stress in the same plane, and ν' the drained Poisson's ratio corresponding to the transverse strain reduction in the plane of isotropy due to a tensile stress normal to it.

We can also express the $\boldsymbol{C}$ matrix as

$$\boldsymbol{C} = \begin{bmatrix} 1/E & -\nu/E & -\nu'/E' & 0 & 0 & 0 \\ -\nu/E & 1/E & -\nu'/E' & 0 & 0 & 0 \\ -\nu'/E' & -\nu'/E' & 1/E' & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G' & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G' & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G \end{bmatrix} \tag{5.52}$$

The other coefficients consist of

$$\vec{\alpha} = \lfloor\, \alpha_x \;\; \alpha_x \;\; \alpha_z \;\; 0 \;\; 0 \;\; 0 \,\rfloor^T \tag{5.53}$$

$$\vec{B} = \lfloor\, B_x \;\; B_x \;\; B_z \;\; 0 \;\; 0 \;\; 0 \,\rfloor^T \tag{5.54}$$

5.2.3 *Isotropy*

Finally, for isotropic material, there exist four independent poroelastic constants, as already discussed in Sects. 2.3.4 and 2.3.5. The drained elastic moduli (5.33) becomes

$$\boldsymbol{M} = \begin{bmatrix} \lambda + 2G & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2G & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2G & 0 & 0 & 0 \\ 0 & 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & 0 & G & 0 \\ 0 & 0 & 0 & 0 & 0 & G \end{bmatrix} \tag{5.55}$$

in which

$$\lambda = \frac{2G\nu}{1 - 2\nu} \tag{5.56}$$

is the drained Lamé constant. We can also express the $\boldsymbol{C}$ matrix as

$$\boldsymbol{C} = \begin{bmatrix} 1/E & -\nu/E & -\nu/E & 0 & 0 & 0 \\ -\nu/E & 1/E & -\nu/E & 0 & 0 & 0 \\ -\nu/E & -\nu/E & 1/E & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G \end{bmatrix} \tag{5.57}$$

where

$$E = 2G(1 + \nu) \tag{5.58}$$

is the drained Young's modulus. The other coefficients consist of

$$\vec{\alpha} = \lfloor\ \alpha\ \ \alpha\ \ \alpha\ \ 0\ \ 0\ \ 0\ \rfloor^T \tag{5.59}$$

$$\vec{B} = \lfloor\ B\ \ B\ \ B\ \ 0\ \ 0\ \ 0\ \rfloor^T \tag{5.60}$$

5.3 Micromechanics

The presentation of constitutive relations in the preceding sections was based on bulk parameters of total stresses and strains. In this section, the micromechanical analysis, which explicitly models the solid and fluid constituents, and the pore space, similar to that presented in Chap. 3, will be performed.

The total strain tensor e_{ij} can be decomposed into a solid part, e^s_{ij}, and a pore part, e^p_{ij}, as

$$e_{ij} = (1 - \phi)e^s_{ij} + \phi\, e^p_{ij} \tag{5.61}$$

The total stress tensor can be similarly decomposed:

$$\sigma_{ij} = (1 - \phi)\sigma^s_{ij} - \phi\, \delta_{ij} p \tag{5.62}$$

where σ^s_{ij} is the stress tenor of the solid constituent.

Based on these parameters, we present the following two constitutive relations:

$$e_{ij} = C_{ijkl}\, \sigma_{kl} + (C_{ijkk} - c_{ij})p \tag{5.63}$$

$$e^s_{ij} = C^s_{ijkl}\, \sigma^s_{kl} + \frac{\phi}{1 - \phi}(c^s_{ij} - C^s_{ijkk})p \tag{5.64}$$

We note that each equation contains a fourth order and a second order coefficient tensor, which conforms to the generalized Hooke's law. The reason that we have arranged them in the above special form will become clear later. In the above, C_{ijkl} is the drained compliance tensor as defined before. By testing the material under drained condition, $p = 0$, (5.64) shows that C^s_{ijkl} is a compliance tensor of the solid constituent. The physical significance of the other coefficients, c^s_{ij} and c_{ij}, is yet to be explored.

We now examine the fluid deformation. Similar to the relation found in (3.24), the fluid strain e_f consists of two parts: one part is the pore strain, $e^p = e^p_{ii}$, due to

the change of pore volume, and the other part is observed as the fluid expanding outside the frame, e'_f, such that

$$e'_f = e_f - e^p \tag{5.65}$$

The constitutive law for the fluid is

$$e_f = -C_f p \tag{5.66}$$

where C_f is the fluid compressibility. The variation in fluid content ζ used in the constitutive equation (5.6) is related to e'_f, and

$$\zeta = -\phi\, e'_f \tag{5.67}$$

Utilizing the definitions in (5.61) and (5.65), we can write (5.67) as the following

$$\zeta = e - (1-\phi)e^s - \phi\, e_f \tag{5.68}$$

Substitution of (5.63), (5.64) and (5.66) yields the constitutive equation

$$\zeta = (C_{iikl} - C^s_{iikl})\,\sigma_{kl} + \left[(C_{iikk} - c_{ii}) + \phi\,(C_f - c^s_{ii})\right] p \tag{5.69}$$

Knowing that (5.63) and (5.69) must represent the same constitutive laws as (5.5) and (5.6), or (5.14) and (5.15), we can draw a few conclusions by comparing these relations. First, it is revealed that the constitutive constant c_{ij} is not independent, and

$$c_{ij} = C^s_{ijkk} \tag{5.70}$$

We hence have a reduction in the compliance coefficients, leaving us with four systems: C_{ijkl}, C^s_{ijkl}, c^s_{ij} and C_f. We can therefore rewrite (5.63) and (5.69) as

$$e_{ij} = C_{ijkl}\,\sigma_{kl} + (C_{ijkk} - C^s_{ijkk})\,p \tag{5.71}$$

$$\zeta = (C_{iikl} - C^s_{iikl})\,\sigma_{kl} + \left[(C_{iikk} - C^s_{iikk}) + \phi(C_f - c^s_{ii})\right] p \tag{5.72}$$

We also find the relations among the bulk and the micromechanical constants:

$$\alpha_{ij} = \delta_{ij} - M_{ijkl}\, C^s_{klmm} \tag{5.73}$$

$$B_{ij} = \frac{3(C_{ijkk} - C^s_{ijkk})}{C} \tag{5.74}$$

$$C = (C_{iijj} - C^s_{iijj}) + \phi(C_f - c^s_{ii}) \tag{5.75}$$

$$\frac{1}{M} = M_{ijmn}\, C^s_{mnll}(C_{ijkk} - C^s_{ijkk}) + \phi(C_f - c^s_{ii}) \tag{5.76}$$

Equation (5.73) demonstrates that the Biot effective coefficient α_{ij} is a property of the porous solid only. For the other coefficients, (5.74) to (5.76), the role of fluid compressibility C_f is explicitly shown.

5.4 Ideal Porous Medium

Although the above analysis brings physical insight into the properties of poroelastic material, the resulting material coefficient system (twenty-one C_{ijkl}, twenty-one C^s_{ijkl}, six c^s_{ij}, and one C_f) are too cumbersome to be practically measured. Simplifying assumptions such as the ideal porous medium can make the model more tractable.

As discussed in Sects. 3.2 and 4.7, under the unjacketed test condition, the ideal porous medium deforms in a geometrically similar fashion, thus the following conditions are satisfied

$$e_{ij} = e^s_{ij} = e^p_{ij} \tag{5.77}$$

We also note that (5.63) and (5.64) become

$$e_{ij} = -C^s_{ijkk} P \tag{5.78}$$

$$e^s_{ij} = -C^s_{ijkk} P + \frac{\phi}{1-\phi}(c^s_{ij} - C^s_{ijkk})P \tag{5.79}$$

Judging from (5.77), we reach at the conclusion

$$c^s_{ij} = C^s_{ijkk} \tag{5.80}$$

The constitutive relation (5.64) reduces to

$$e^s_{ij} = C^s_{ijkl}\,\sigma^s_{kl} \tag{5.81}$$

The system is now defined by the compliance of the drained skeleton, C_{ijkl}, and that of the solid, C^s_{ijkl}, and the fluid compressibility, C_f. Furthermore, the microisotropic assumption shows that C^s_{ijkl} is isotropic and

$$C^s_{ijkk} = \frac{\delta_{ij} C_s}{3} \tag{5.82}$$

where C_s is the solid compressibility and

$$C_s = \frac{1}{K_s} \tag{5.83}$$

Substitution of the above into (5.73) gives

$$\alpha_{ij} = \delta_{ij} - \frac{M_{ijkk}}{3K_s} \tag{5.84}$$

Equation (5.84) is the key result of the micromechanical analysis. First, it shows that there exist only the diagonal terms $\{\alpha_{xx}, \alpha_{yy}, \alpha_{zz}\}$, and all cross terms are zero. Second, all these coefficients are defined by a single solid bulk modulus K_s if the drained frame elastic modulus tensor M_{ijkl} is known. Recall that the bulk continuum model presented in Sect. 5.1.2 suggests the existence of 28 independent material constants for general anisotropy. Under the ideal porous medium assumption, only 23 independent constants exist. In other words, the system is defined by twenty-one M_{ijkl}, one K_s, one M, and α_{ij} is defined through (5.84).

Other coefficients can be similarly simplified:

$$B_{ij} = \frac{1}{C}\left(3\,C_{ijkk} - \delta_{ij}C_s\right) \tag{5.85}$$

$$C = \left(C_K^* - C_s\right) + \phi\left(C_f - C_s\right) \tag{5.86}$$

$$\frac{1}{M} = \frac{1}{K_s}\left[\left(1 - \frac{K^*}{K_s}\right) - \phi\left(1 - \frac{K_s}{K_f}\right)\right] \tag{5.87}$$

where

$$K_f = \frac{1}{C_f} \tag{5.88}$$

is the fluid bulk modulus, and

$$C_K^* = C_{iijj} \tag{5.89}$$

$$K^* = \frac{M_{iijj}}{9} \tag{5.90}$$

can be viewed as the generalized compressibility and generalized bulk modulus for drained anisotropic porous material.

This model is very convenient for laboratory measurement purposes. The drained elastic moduli can be measured following the traditional elasticity tests, as long as we can ensure that the pore pressure is dissipated through access to drainage during the test. The solid bulk modulus K_s can be obtained by utilizing the unjacketed test described in Sect. 3.5.3. In the test, equal increments of confining stresses $\sigma_{ij} = -\delta_{ij}P$ and pore pressure $p = P$ are applied. Based on (5.78), (5.82), and (5.83), K_s is found as

$$K_s = -\frac{P}{e} \tag{5.91}$$

The material constant M, as indicated in (5.87), can be calculated from the fluid bulk modulus K_f, the porosity ϕ, and the drained elastic moduli. All the rest of the constants can be calculated using the relations provided above.

5.5 Example

There exist only a few laboratory measurements of anisotropic poroelastic coefficients [1, 2, 11, 15]. In this section, we shall use the laboratory measurement obtained by Aoki et al. [2] to illustrate the use of the various theoretical results derived in the preceding sections.

Using traditional laboratory testing facilities, Aoki et al. [2] measured the following engineering constants for a transversely isotropic Trafalgar shale: $E_x^u = 22.0\,\text{GPa}$, $E_z^u = 18.8\,\text{GPa}$, $G_{xz} = 7.23\,\text{GPa}$, $\nu_{yx}^u = 0.27$, $\nu_{zx}^u = 0.34$, $B_x = 0.51$, $B_z = 0.63$, where E_x^u and E_z^u are undrained Young's moduli, G_{xz} is shear modulus, ν_{yx}^u and ν_{zx}^u are undrained Poisson ratios, and B_x and B_z are Skempton pore pressure coefficients. The reason that the undrained, rather than drained, constants are measured is due to the low permeability of shale—it would be impractical to wait for the shale to be fully drained.

As indicated in Sect. 5.2.2, a total of eight constants are needed to define the constitutive relations of transverse isotropy, while the above measurement gives only seven; hence this data set is incomplete for modeling purposes. However, the condition of microisotropy and microhomogeneity of the ideal porous medium comes for the rescue, as only seven independent constants are needed in the model. The material is then fully defined by this set of measurements. The following conversions are demonstrated.

We note that the relations in (5.46), (5.47), (5.48) (5.49), (5.50), and (5.51) are valid if all quantities are replaced by their undrained counterparts. Substituting in the numerical values of Young's moduli and Poisson's ratios, we obtain $M_{xx}^u = 32.6\,\text{GPa}$, $M_{xy}^u = 15.3\,\text{GPa}$, $M_{xz}^u = 16.3\,\text{GPa}$, and $M_{zz}^u = 29.9\,\text{GPa}$. Next, we find from (5.20) that

$$\alpha_x = \frac{(M_{xx}^u + M_{xy}^u)B_x + M_{xz}^u B_z}{3M} \tag{5.92}$$

$$\alpha_z = \frac{2M_{xz}^u B_x + M_{zz}^u B_z}{3M} \tag{5.93}$$

Also, from (5.84) we obtain

$$\alpha_x = 1 - \frac{M_{xx} + M_{xy} + M_{xz}}{3K_s} \tag{5.94}$$

$$\alpha_z = 1 - \frac{2M_{xz} + M_{zz}}{3K_s} \tag{5.95}$$

We note from (5.12) that the drained and undrained elastic moduli are related as:

$$M_{ij} = M^u_{ij} - M\alpha_i\alpha_j \tag{5.96}$$

These are used to eliminate M_{ij} in (5.94) and (5.95). The four Eqs. (5.92), (5.93), (5.94), and (5.95), are used to solve for the four unknowns, α_x, α_z, K_s, and M. The following results are obtained: $\alpha_x = 0.733$, $\alpha_z = 0.749$, $K_s = 48.2\,\text{GPa}$, and $M = 15.8\,\text{GPa}$. Other constants are then evaluated as: $E_x = 20.6\,\text{GPa}$, $E_z = 17.3\,\text{GPa}$, $\nu_{yx} = 0.189$, $\nu_{zx} = 0.246$, $M_{xx} = 24.1\,\text{GPa}$, $M_{xy} = 6.80\,\text{GPa}$, $M_{xz} = 7.62\,\text{GPa}$, $M_{zz} = 21.0\,\text{GPa}$, and $G_{xy} = 8.66\,\text{GPa}$.

It is of interest to note that the current procedure interprets that the solid bulk modulus $K_s = 48.2\,\text{GPa}$, despite that no such measurement was made. The calculated value falls within the measured values of various rocks as shown in Table 3.1, $31\,\text{GPa} < 48.2\,\text{GPa} < 50\,\text{GPa}$; hence is reasonable.

References

1. Al-Tahini A, Abousleiman Y (2010) Pore-pressure-coefficient anisotropy measurements for intrinsic and induced anisotropy in sandstone. SPE Reserv Eval Eng 13(2):265–274
2. Aoki T, Tan CP, Bamford WE (1993) Effects of deformation and strength anisotropy on borehole failures in saturated shales. Int J Rock Mech Min Sci 30(7):1031–1034
3. Biot MA (1955) Theory of elasticity and consolidation for a porous anisotropic solid. J Appl Phys 26(2):182–185
4. Boresi AP, Chong KP, Lee JD (2010) Elasticity in engineering mechanics, 3rd edn. Wiley, Hoboken, 656pp
5. Carroll MM (1979) An effective stress law for anisotropic elastic deformation. J Geophys Res 84(B13):7510–7512
6. Chen WF, Saleeb AF (1982) Constitutive equations for engineering materials, vol I. Wiley, New York
7. Cheng AHD (1997) Material coefficients of anisotropic poroelasticity. Int J Rock Mech Min Sci 34(2):199–205
8. Green G (1839) On the laws of reflexion and refraction of light at the common surface of two non-crystallized media. Trans Camb Philos Soc 7:1–24
9. Gwyther RL, Gladwin MT, Mee GM, Hart RHG (1996) Anomalous shear strain at Parkfield during 1993–94. Geophys Res Lett 23(18):2425–2428
10. Lekhnitskii SG (1981) Theory of elasticity of an anisotropic body. Mir Publishers, Moscow, 430pp
11. Scott TE, Abousleiman Y (2005) Acoustic measurements of the anisotropy of dynamic elastic and poromechanics moduli under three stress/strain pathways. J Eng Mech ASCE 131(9):937–946
12. Skempton AW (1954) The pore pressure coefficients A and B. Géotechnique 4(4):143–147
13. Sokolnikoff IS (1956) Mathematical theory of elasticity, 2nd edn. McGraw-Hill, New York, 476pp
14. Thompson M, Willis JR (1991) A reformation of the equations of anisotropic poroelasticity. J Appl Mech ASME 58(3):612–616
15. Tokunaga T, Hart DJ, Wang HF (1998) Complete set of anisotropic poroelastic moduli for Berea sandstone. In: Thimus JF, Abousleiman Y, Cheng AHD, Coussy O, Detournay E (eds) Poromechanics—a tribute to maurice A. Biot. Balkema, Rotterdam/Brookfield, pp 629–634

16. Wang HF (1997) Effects of deviatoric stress on undrained pore pressure response to fault slip. J Geophys Res-Solid Earth 102(B8):17943–17950
17. Yoon YJ, Cowin SC (2008) An estimate of anisotropic poroelastic constants of an osteon. Biomech Model Mechanobiol 7(1):13–26
18. Young T (1845) A course of lectures on natural philosophy and the mechanical arts, a new edition, with references and notes, 2 vols. Taylor and Walton, London, 608pp

Chapter 6
Governing Equation

I now approach the account of the experiments I made in Dijon . . . to determine the laws of the water flow through sands. . . . It thus appears that for sand of comparable nature, one can conclude that output volume is proportional to the head and inversely related to the thickness of the layer traversed. . . . Thus, by calling e the thickness of the sand, s its surface area, . . . and h the height of water above this layer, . . . k is a coefficient dependent on the permeability of the layer, and q is the output volume, one has . . . $q = ks(h + e)/e$.

—Henry Darcy (1856)

In Chaps. 2 through 5 we have constructed the constitutive laws that relate the forces (stresses) applied to a porous body to its deformation (strains). In addition to these constitutive laws, there are other *physical laws* that are relevant to the deformation and motion of porous materials. These are presented in this chapter.

For the purpose of modeling, these laws are formulated in the form of mathematical equations. To reduce the size of the solution system, variables can be eliminated among the physical laws to produce *governing equations* that contain fewer variables. Given a complete set of governing equations, together with a set of well-posed boundary conditions, the mathematical system can be solved either analytically or numerically. There are times, however, that the mathematical system can be further reduced by the introduction of non-physical variables, known as *potentials*, to replace the physical ones. The physical variables are typically associated with the potentials as their spatial derivatives. We shall refer the mathematical system involving potentials as *field equations*. In this chapter we shall discuss these equations, as well as the initial and boundary conditions, leading to a complete mathematical solution system.

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_6

6.1 Darcy's Law

6.1.1 Darcy's Empirical Law

Henry Darcy (see Sect. F.9 for a biography), a French hydraulic engineer, published in 1856 a report on the water supply system that he designed for the city of Dijon, France [15]. In an appendix, he described his experiments on water flow through a column of sand filter (see Fig. 6.1), and concluded that the flow rate is directly proportional to the hydraulic head, and inversely related to the length of the column.

This observation, known as Darcy's law, can be expressed in the modern-day generalized three-dimensional and anisotropic form as

$$\vec{q} = -\mathbf{K} \cdot \nabla h \tag{6.1}$$

In the above, $\vec{q}$ is the *specific discharge vector*, defined as the volume of fluid passing through a unit area of porous medium, per unit time, in the direction normal to the area, which has the dimension of velocity, [L/T]. We also note that h is the *piezometric head* (dim. [L]), defined as

$$h = \frac{p}{\rho_f g} + z \tag{6.2}$$

where p is the pore pressure, ρ_f the fluid density, g the gravity acceleration, and z the elevation above a datum [8]. Darcy's law is an empirical (phenomenological) law. It contains an empirical coefficient $\mathbf{K}$, known as the *hydraulic conductivity*, and is given as a second rank tensor, $\mathbf{K} = \mathsf{K}_{ij}$, $i,j = 1,2,3$, for anisotropic porous medium. This coefficient expresses the ease with which a fluid flows through the

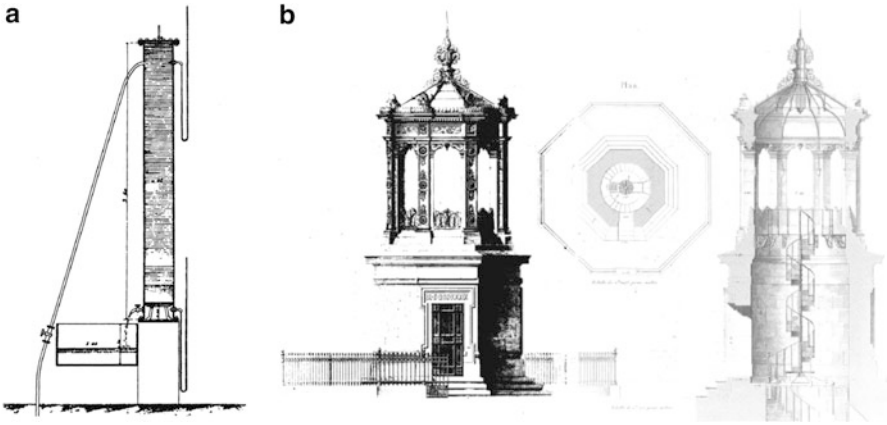


Fig. 6.1 (**a**) Sand filter of Darcy's experiment; and (**b**) the Public Fountains of the City of Dijon [15]

tortuous void space of a porous medium. It is, therefore, a coefficient that depends on both porous medium geometry and fluid properties.

The piezometric head defined in (6.2) can be viewed an extension of the concept of a *hydraulic head* in fluid mechanics [42]. The hydraulic head is the mechanical energy contained in a unit weight of incompressible fluid, which consists of three parts: a pressure energy, a (gravitational) potential energy, and a kinetic energy. It can be expressed as

$$h = \frac{p}{\rho_f g} + z + \frac{v^2}{2g} \tag{6.3}$$

in which v is the fluid velocity. In porous medium flow, it is considered that the fluid velocity is a small quantity such that the third term in the above equation can be neglected, yielding the approximation (6.2). A criterion for such determination can be based on the porous medium Reynolds number

$$\mathrm{Re} = \frac{q d}{\rho_f \mu_f} \tag{6.4}$$

where $q = \|\vec{q}\|$ is magnitude of specific discharge vector, μ_f is the fluid dynamic viscosity, and d is a characteristic length scale of the flow channel, which is typically represented by the average grain size for granular material or the average pore size for porous solid. The range of validity of Darcy's law should be confined to within $\mathrm{Re} < 1$. When such a statement is not true, then the kinetic energy needs to be considered, leading to the modification of Darcy's law. Some such non-Darcy laws include the Darcy-Forchheimer equation, Brinkman equation, etc. [8]. These non-Darcy laws are nonlinear. In the present book, only linear laws will be considered.

Equation (6.3) is based on incompressible fluid. For porous medium in deep formation subjected to large change in pressure, Hubbert [24] considered the fluid compressibility effect. For a compressible fluid under isothermal condition, the density is a function of pressure only, $\rho_f = \rho_f(p)$. In that case, the correct potential energy per unit weight is

$$h^* = \int_{p_o}^{p} \frac{dp}{\rho_f(p)\, g} + z \tag{6.5}$$

where p_o is a reference pressure, and h^* is known as *Hubbert's potential*. Darcy's law is then modified to

$$\vec{q} = -\mathbf{K} \cdot \nabla h^* \tag{6.6}$$

There have been many efforts to rationalize Darcy's empirical law using physical approaches. Among those are the volume averaging approach [6, 7] and the homogenization approach [8, 18, 23]. The homogenization theory is presented as follows.

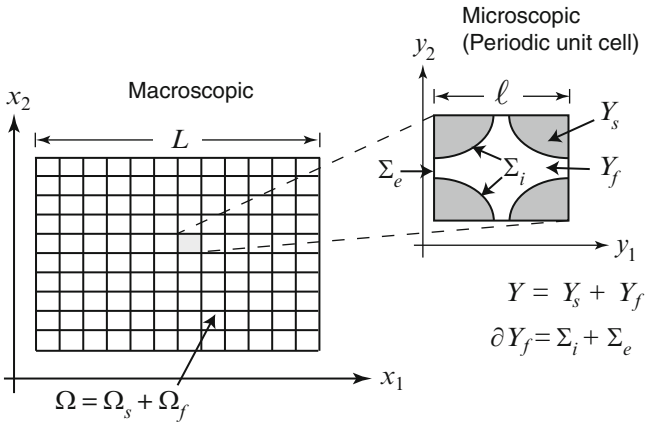


Fig. 6.2 Periodic cells of homogenization theory

6.1.2 Homogenization Theory

The homogenization approach is based on the assumption that the REVs at the microscopic level are identical periodic cells. Within the REV, the following physical assumptions are introduced for the solid and the fluid:

1. the fluid local acceleration is negligible;
2. the solid phase is non-deformable and stationary; and
3. the fluid is single-component, compressible, and Newtonian.

In other words, we consider that the fluid flow at the microscopic level is governed by the Navier-Stokes equation without the convective acceleration terms, and it exists in a domain with non-movable boundaries. Figure 6.2 gives a two-dimensional illustration of a macroscopic porous medium of size L, containing microscopic periodic cells of size ℓ, with $L \gg \ell$, and $\varepsilon = \ell/L \ll 1$, as a small parameter. The inset of Fig. 6.2 is the enlargement of one such periodic cell. In the figure, the macroscopic domain is denoted as Ω, and the microscopic domain as Y.

With such a system, our goal is to find the macroscopic governing equations that describe the relation between the *average velocity* and the *average pressure*, and the corresponding coefficients for the differential equation. To obtain a *first order* model, the mathematics is performed using asymptotic analysis by expanding the physical variables into asymptotic series in terms of the small parameter ε. Forgoing the detailed mathematics (see for example Bear and Cheng [8]), the averaging of the following steady state Navier-Stokes equation

$$\rho_f(\vec{v}\cdot\nabla)\vec{v} = -\nabla p + \mu_f\nabla^2\vec{v} + (\lambda_f + \mu_f)\nabla(\nabla\cdot\vec{v}) + \rho_f\vec{g} \tag{6.7}$$

where $\vec{v}$ is the fluid velocity, μ_f is the fluid dynamic viscosity, λ_f is the viscosity associated with fluid compressibility, and $\vec{g}$ is the gravity acceleration vector, over the microscopic fluid domain Y_f produces the macroscopic momentum equation

$$\vec{q} = -\frac{\mathbf{k}}{\mu_f} \cdot \left(\nabla p + \rho_f g \nabla z\right) \tag{6.8}$$

where $\mathbf{k}$ is a second rank tensor known as *intrinsic permeability*. With the dimension of length squared [L^2], $\mathbf{k}$ is dependent only on the pore geometry. Furthermore, $\mathbf{k}$ can be expressed as

$$\mathbf{k} = \mathbf{k}^* \ell^2 \tag{6.9}$$

where ℓ is the size of the periodic cell, and can be viewed as the characteristic length of a porous medium at microscale (such as grain size or pore size), and $\mathbf{k}^*$ is a *dimensionless intrinsic permeability*. Based on the homogenization theory, $\mathbf{k}^*$ is given by [8, 23]

$$\mathbf{k}^* = \int_{Y_f^*} \mathbf{w}^*(\vec{y}^*)\, d\vec{y}^* \tag{6.10}$$

where we note that $\vec{y}$ is the coordinate system of the microscopic cell (see Fig. 6.2), and $\vec{y}^* = \vec{y}/\ell$ is the normalized coordinate system. The REV domain Y is also normalized to a unit cube Y^*, with the fluid part as Y_f^*. The variable $\mathbf{w}^*$ is a second rank tensor satisfying the following set of governing equations

$$\nabla^2 \mathbf{w}^* = \nabla \vec{s}^* - \mathbf{I} \tag{6.11}$$

$$\nabla \cdot \mathbf{w}^* = 0 \tag{6.12}$$

where $\vec{s}^*$ is a vector, and $\mathbf{I}$ is the second rank identity tensor. Equations (6.11) and (6.12) are $9+3 = 12$ equations. The above equations are subject to the boundary conditions

$$\begin{aligned} &\mathbf{w}^* = 0, && \text{on } \vec{y}^* \in \Sigma_i^* \\ &\mathbf{w}^* \quad \text{periodic}, && \text{on } \vec{y}^* \in \Sigma_e^* \end{aligned} \tag{6.13}$$

where Σ_i^* and Σ_e^* are respectively the fluid solid interface and fluid external boundary of the unit cell Y^*, corresponding to Σ_i and Σ_e in Y (see Fig. 6.2). The above system is a well-posed boundary value problem and can be used to solve for the 12 unknowns, $\mathbf{w}^*$ and $\vec{s}^*$. Once $\mathbf{w}^*$ is known, we can obtain $\mathbf{k}^*$ based on (6.10).

Based on the above derivation, we can draw a few important conclusions:

- Equation (6.8) shows that Darcy's law is a momentum equation, as it is obtained by averaging the Navier-Stokes equation.

- As $\mathbf{w}^*$ is a second rank tensor (meaning it obeys the tensor coordinate transformation rules), the permeability $\mathbf{k}^*$ and $\mathbf{k}$ are second rank tensors too.
- Although the detail is not given, it can be proven that the governing equations (6.11) and (6.12) are *self-adjoint*, leading to the condition that the permeability tensors $\mathbf{k}^*$ and $\mathbf{k}$ are *symmetric*.
- For any symmetric second rank tensor, there exist principle directions that the non-diagonal terms vanish [2], such that $\mathbf{k}$ can be expressed as

$$\mathbf{k} = \begin{bmatrix} k_x & 0 & 0 \\ 0 & k_y & 0 \\ 0 & 0 & k_z \end{bmatrix} \tag{6.14}$$

  For an isotropic porous medium, $\mathbf{k}$ reduces to $\mathbf{k} = \delta_{ij} k$, and is a scalar.
- It is apparent from the above that $\mathbf{w}^*$, thus $\mathbf{k}^*$, are scale independent, as the length scales L and ℓ do not appear in the solution system. Its solution depends only on the geometric pattern of the pore space Y_f^* within the unit cell Y^*, for example, the grain shape (spherical, ellipsoidal, etc.), and its packing pattern (cubic, rhombohedral, etc.).
- The above also leads to the conclusion that $\mathbf{k}$ defined in (6.9) is proportional to the square of pore or grain size, for geometrically similar porous medium.
- As (6.8) is a momentum equation, we can rewrite it as

$$-\mu_f \mathbf{b} \cdot \vec{q} = \nabla p + \rho_f g \nabla z \tag{6.15}$$

  where $\mathbf{b} = \mathbf{k}^{-1}$ (dim. $[\mathrm{L}^{-2}]$) is a *resistivity tensor*. The term on the left hand side represents the viscous (dissipative) force per unit porous medium volume, from which we can define a *dissipation density function*

$$\mathcal{D} = \mu_f \mathbf{b} \cdot \vec{q} \cdot \vec{q} \tag{6.16}$$

  which is the energy dissipated per unit porous medium volume per unit time due to the porous medium flow. For isotropic porous medium, (6.16) becomes

$$\mathcal{D} = \frac{\mu_f}{k} q^2 \tag{6.17}$$

  where $q = \|\vec{q}\,\|$.

Equation (6.8) is consistent with Darcy's law for incompressible fluid, (6.1), by writing it in this form

$$\vec{q} = -\frac{\rho_f g \mathbf{k}}{\mu_f} \cdot \nabla \left(\frac{p}{\rho_f g} + z \right) \equiv -\mathbf{K} \cdot \nabla h \tag{6.18}$$

in which ρ_f is a constant. It is clear that the hydraulic conductivity, an empirical coefficient, is

$$\mathbf{K} = \frac{\rho_f g\, \mathbf{k}}{\mu_f} \tag{6.19}$$

Equation (6.8) is also consistent with Darcy's law based on Hubbert's potential, (6.6), as

$$\nabla h^* = \nabla \left(\int_{p_o}^{p} \frac{dp}{\rho_f(p)\, g} + z \right) = \frac{1}{\rho_f g} \nabla p + \nabla z \tag{6.20}$$

In the above, we have applied the chain rule $\nabla \rho_f(p) = (d\rho_f/dp)\nabla p$.

6.1.3 *Intrinsic Permeability and Mobility Coefficient*

Numerous formulae that relate intrinsic permeability to various geometric properties of the pore space, such as porosity, grain- or pore-size distribution, specific surface, aspect ratio of pores, orientation of elongated pores, and tortuosity of passages, have been developed. An often used formula is the Kozeny-Carman equation [11, 12, 19, 28]:

$$k = C_o \frac{\phi^3}{(1-\phi)^2\, \tau_t^2\, \mathcal{S}_s^2} = C_o \frac{\phi^3}{\tau_t^2\, \mathcal{S}_v^2} \tag{6.21}$$

where k is the isotropic intrinsic permeability, ϕ is porosity, $\mathcal{S}_s$ is the *specific surface* (dimension $[\mathrm{L}^{-1}]$), defined as solid surface per unit volume of the solid phase of porous medium, $\mathcal{S}_v = (1-\phi)\mathcal{S}_s$ is also a specific surface, but normalized by the volume fraction of the solid, τ_t is the *tortuosity* (dimensionless), and C_o is a coefficient for which Carman [11] suggested the value 0.2. For granular material, $\mathcal{S}_s$ is often replaced by $1/d$, where d is the mean grain size. A comprehensive review of the different models of permeability-porosity relation can be found in [34], but generally speaking, k should be a material constant directly measured for a given porous material. In Sect. 3.4.2, the nonlinear model of stress dependent (due to pore compaction) permeability is discussed. Within the scope of this chapter, permeability will be considered as a material constant.

For modeling purposes, we are typically only interested in the material response as a perturbation from its initial or at rest state, after the application of a load. A porous material at rest under gravity force is often an initial state. In that case, it is desirable to remove the *hydrostatic* stress and strain from the modeling parameters. For Darcy's law, this is accomplished by partitioning the pore pressure p into a *hydrostatic pressure* p_h, and an *excess pressure* p_e, such that

$$p = p_h + p_e \tag{6.22}$$

Based on (6.8), the hydrostatic pressure must satisfy the condition

$$\nabla p_h + \rho_f g \nabla z = 0 \tag{6.23}$$

as there is no flow under the condition. Subtracting (6.23) from (6.8), we find that the excess pressure p_e satisfies

$$\vec{q} = -\frac{\mathbf{k}}{\mu_f} \cdot \nabla p \tag{6.24}$$

In the above, for simplicity, we have dropped the subscript e for excess pressure, as from now on, only excess pressure will be modeled. Equation (6.24) hence is Darcy's law expressed in terms of the excess pressure variable.

For petroleum engineering applications, it is more convenient to define a *permeability coefficient* (a second rank tensor), also called *mobility coefficient*, as

$$\boldsymbol{\kappa} = \frac{\mathbf{k}}{\mu_f} \tag{6.25}$$

which has the dimension [L^4/FT], such that Darcy's law is expressed as

$$\vec{q} = -\boldsymbol{\kappa} \cdot \nabla p \tag{6.26}$$

For isotropic porous medium, it becomes

$$\vec{q} = -\kappa \nabla p \tag{6.27}$$

where κ is the isotropic permeability coefficient.

6.1.4 Irreversible Thermodynamics Process

In Chap. 4 we derived the constitutive equations based on a thermodynamic reversible process, assuming that the thermodynamic state variables are in equilibrium within an REV. In the global scale, poroelastic phenomena are generally not in equilibrium. According to the second law, a nonequilibrium state tends to move toward equilibrium, and the process is irreversible. This process manifests itself in Darcy's law, which is a pressure gradient driven viscous flow that dissipates kinetic energy into heat. Although we have presented Darcy's law from the momentum balance point of view in the preceding sections, it is of interest to examine the consistency of the process with the thermodynamics energy principles.

Consider the extended work-energy relation for an irreversible process presented in (4.59),

$$\delta\mathcal{A} + \delta\mathcal{F} = \delta\mathcal{W} + \delta\mathcal{W}_{\text{irr}} \tag{6.28}$$

The left hand side contains the Helmholtz free energy $\mathcal{A}$ and the dissipation energy function $\mathcal{F}$ (extensive properties), with their density functions (per unit volume) to be introduced below as $\mathcal{A}$ and $\mathcal{F}$ (intensive properties), respectively. As we are considering the energy at the macroscopic scale, we shall express $\mathcal{A}$ and $\mathcal{F}$ as functions of bulk continuum variables, rather than the micromechanical variables of solid and fluid internal and external stresses and strains.

As discussed in Sect. 4.3.2, under isothermal condition, the Helmholtz free energy density $\mathcal{A}$ represents strain energy density only; hence it is a function of solid and fluid strains. The dissipation energy density function, on the other hand, is dependent of the relative fluid and solid displacement, which induces a flow phenomenon with viscous dissipation. Hence we express these density functions as

$$\mathcal{A} = \mathcal{A}(e_{ij}, \zeta); \qquad \mathcal{F} = \mathcal{F}(w_i) \tag{6.29}$$

where w_i is the relative fluid to solid specific displacement vector to be defined below. Given a macroscopic system Ω bounded by Γ, we can express the extrinsic energy functions $\mathcal{A}$ and $\mathcal{F}$ in the integral forms

$$\begin{aligned}\delta\mathcal{A} + \delta\mathcal{F} &= \int_\Omega \delta\mathcal{A}(e_{ij}, \zeta)\, d\vec{x} + \int_\Omega \delta\mathcal{F}(w_i)\, d\vec{x} \\ &= \int_\Omega \left(\frac{\partial\mathcal{A}}{\partial e_{ij}}\delta e_{ij} + \frac{\partial\mathcal{A}}{\partial\zeta}\delta\zeta + \frac{\partial\mathcal{F}}{\partial w_i}\delta w_i \right) d\vec{x}\end{aligned} \tag{6.30}$$

As the irreversible work is performed by pore pressure on the relative fluid to solid displacement, $U_i - u_i$, we can write the right hand side of (6.28) as

$$\delta\mathcal{W} + \delta\mathcal{W}_{\text{irr}} = \int_\Gamma \sigma_{ij} n_j\, \delta u_i\, d\vec{x} - \int_\Gamma p\, n_i\, \delta w_i\, d\vec{x} \tag{6.31}$$

where the minus sign for the second term is due to the opposite sign convention for stress and pore pressure, and we have defined

$$w_i = \phi(U_i - u_i) \tag{6.32}$$

as the relative fluid to solid specific displacement vector, with the porosity factor introduced to account for the surface area that is available for flow. We also note that by virtual of this definition, $\vec{w}$ is related to the specific discharge vector as

$$\vec{q} = \frac{\partial\vec{w}}{\partial t} \tag{6.33}$$

Applying the divergence theorem to (6.31), we find

$$\delta \mathcal{W} + \delta \mathcal{W}_{\text{irr}} = \int_{\Omega} \left(\sigma_{ij} \delta e_{ij} + p \, \delta \zeta - p_{,i} \, \delta w_i \right) \, d\vec{x} \tag{6.34}$$

where we notice that based on (4.117),

$$\zeta = -\nabla \cdot \vec{w} \tag{6.35}$$

Comparing (6.34) with (6.30), it is clear that

$$\sigma_{ij} = \frac{\partial \mathcal{A}}{\partial e_{ij}} \tag{6.36}$$

$$p = \frac{\partial \mathcal{A}}{\partial \zeta} \tag{6.37}$$

$$p_{,i} = -\frac{\partial \mathcal{F}}{\partial w_i} \tag{6.38}$$

We notice that (6.36) and (6.37) are just (2.18) and (2.19) presented earlier. By introducing a quadratic form of strains for $\mathcal{A}$ as (2.21),

$$\mathcal{A} = \frac{1}{2} \lambda_u \, e^2 + \mu \, e_{ij}^2 - \alpha M e \, \zeta + \frac{1}{2} M \zeta^2 \tag{6.39}$$

where

$$\lambda_u = K_u - \frac{2G}{3} \tag{6.40}$$

is the undrained Lamé constant, we can derive the linear constitutive equations for σ_{ij} and p as in Sects. 2.3.4 and 2.3.5.

For the dissipation process, we introduce the quadratic form for $\mathcal{F}$ as

$$\mathcal{F} = \frac{1}{2\kappa} w_i \frac{\partial w_i}{\partial t} \tag{6.41}$$

where κ is the permeability coefficient introduced in Sect. 6.1.3. Differentiating the above, we obtain

$$p_{,i} = -\frac{1}{\kappa} \frac{\partial w_i}{\partial t} = -\frac{1}{\kappa} \, q_i \tag{6.42}$$

This is the Darcy's law presented as (6.27), but here it is obtained through the variational formulation of the irreversible process.

6.2 Other Physical Laws

Other physical laws needed for the modeling of poroelasticity include the mass conservation and the force equilibrium, which are examined below.

6.2.1 Mass Conservation

As demonstrated in the above section, Darcy's law is a momentum equation. To model the flow phenomenon, fluid continuity (mass conservation) is also needed. This can be accomplished by linking the specific flux vector q_i to the solid and fluid displacement vectors u_i and U_i. In the original Darcy's law, q_i is the apparent fluid flux with respect to a *nondeformable* porous frame. In the present case of a deformable frame, the definition of q_i needs to be modified to be the *relative motion* between the fluid and the solid. With such definition, we can express q_i as

$$q_i = \phi\,(\dot{U}_i - \dot{u}_i) = \frac{\phi}{\Omega}\int_{\Omega} (\dot{u}_i^{fe} - \dot{u}_i^{se})\, d\vec{x} \tag{6.43}$$

by referring to (4.113) and (4.116). If we take the divergence of q_i, we obtain

$$\nabla \cdot \vec{q} = \phi\,\frac{\partial}{\partial t}\left(\overline{e}^f - \overline{e}^s\right) \tag{6.44}$$

Based on the definition of variation in fluid content, (4.117), we obtain this continuity equation

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \vec{q} = 0 \tag{6.45}$$

This equation states that the divergence of the specific flux of an REV is equal to the (negative) rate of change of fluid content in the volume. We notice that the above statement and derivation are entirely based on the kinematic considerations, without actually introducing the mass. To provide a more rigorous basis, we shall demonstrate below that (6.45) is a linearized mass conservation equation by ignoring the spatial derivatives of fluid density.

To consider mass conservation, we first define a control volume. To be consistent with the solid and fluid constitutive equations derived in Chap. 4, this volume is selected as a porous solid frame consisting of the same group of solid mass. We then extend its exterior surface over the void area to enclose the solid as well as the pores, as described in Sect. 4.1. As this control volume contains the same group of solid mass, solid mass conservation is automatically satisfied. Fluid, on the other hand, can move in and out of the volume; hence its conservation needs to be examined.

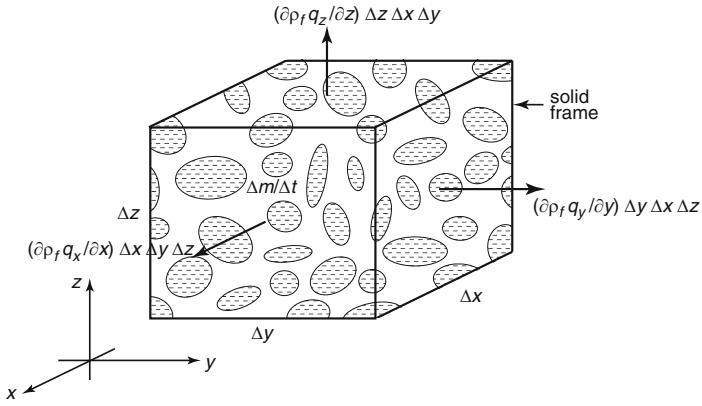


Fig. 6.3 Fluid mass conservation

In fluid mechanics, fluid mass conservation is typically balanced on an Eulerian frame. In the current case, the frame is Lagrangian and is deformable—not only the volume of the frame can change, the interior space that accommodates the fluid can also vary. Figure 6.3 plots the control volume and the net fluid mass leaving the volume through the three pairs of opposing surfaces. The sum of the mass fluxes is equal to the rate of loss of mass in the volume. We can express the statement in an equation as

$$\begin{aligned}\frac{\Delta m_f}{\Delta t} &= -\frac{\partial \rho_f q_x}{\partial x}\Delta x(\Delta y \Delta z) - \frac{\partial \rho_f q_y}{\partial y}\Delta y(\Delta x \Delta z) - \frac{\partial \rho_f q_z}{\partial z}\Delta z(\Delta x \Delta y) \\ &= -\left(\nabla \cdot \rho_f \vec{q}\right)\Delta x \Delta y \Delta z \end{aligned} \tag{6.46}$$

In the above m_f is the fluid mass contained in the control volume, and we notice that $\vec{q}$ is the *specific flux*, that is, the fluid velocity averaged over the entire surface; hence no adjustment for porosity is necessary, although only the pore surface is available for fluid flow. We can express m_f as

$$m_f = \rho_f V_p = \rho_f \phi V \tag{6.47}$$

where V_p is the pore volume and V the total volume, as defined in Sect. 3.1.1. We then find

$$\begin{aligned}\Delta m_f &= \phi V \Delta \rho_f + \rho_f V \Delta \phi + \rho_f \phi \Delta V = \rho_f V\left(-\phi\,\overline{\epsilon}^f + \Delta\phi + \phi\,\overline{\epsilon}^s\right) \\ &= \rho_f \phi V\left(-\overline{\epsilon}^f + \overline{\epsilon}^s\right) = \rho_f V \zeta \end{aligned} \tag{6.48}$$

In the above, $\overline{\epsilon}^f$ is the fluid internal strain as defined in (4.26), with its constitutive equation given by (4.71), and $\overline{e}^s = \overline{e}^s_{ii}$ is the solid external strain defined in (4.28) and (4.135). We also have utilized the internal and external strain relation (4.38) and the definition of ζ in (4.117). We observe from (6.48) that the time rate of change of fluid mass in the control volume is contributed by three parts: the change of fluid density, the change of porosity in the volume, and the change of volume itself. From the equality we notice that all these parts are properly taken into account by the variation in fluid content ζ, which is based on the relative fluid movement to the solid (control volume) through the surface area.

The divergence term on the right hand side of (6.46) can be written as

$$\nabla \cdot \rho_f \vec{q} = \rho_f \nabla \cdot \vec{q} + \vec{q} \cdot \nabla \rho_f \approx \rho_f \nabla \cdot \vec{q} \tag{6.49}$$

In the above, we have performed a *linearization* by dropping the second term on the right hand side, assuming that it is small as compared to the first term. This term is nonlinear because

$$\nabla \rho_f \sim \nabla \overline{\epsilon}^f \sim \nabla p \sim \vec{q} \tag{6.50}$$

by observing (4.71), (4.90), and Darcy's law (6.27). The omitted term is then of the order $(\vec{q} \cdot \nabla \rho_f) \sim O(|\vec{q}\,|^2)$. This approximation is consistent with the assumption of Darcy's law, in which the quadratic velocity terms was omitted in view of the small velocity involved. Utilizing (6.48) and (6.49) in (6.46), and recognizing $V = \Delta x \Delta y \Delta z$, we recover (6.45) in the limit. Equation (6.45) hence is a linearized continuity equation based on an approximation that is consistent with Darcy's law, which is also linearized.

6.2.2 *Force Equilibrium*

Next, we examine the force equilibrium under static condition, and without body force. Although at the microscopic level, equilibrium is satisfied at every single point in the solid or fluid phase, given by (4.18) and (4.19), it is not necessarily true for the averaged stresses at macroscopic level.

First, we shall show the equilibrium of the total stress $\sigma_{ij} = (1-\phi)\overline{\sigma}^s_{ij} - \phi \overline{p}^f$. Based on (4.108), we find

$$\begin{aligned} \frac{\partial \sigma_{ij}}{\partial x_j} &= \frac{1}{\Omega} \frac{\partial}{\partial x_j} \left[\int_{\Gamma_s} \sigma^s_{jk} x_i n_k \, d\vec{x} - \delta_{ij} \int_{\Gamma_f} p x_k n_k \, d\vec{x} \right] \\ &= \frac{1}{\Omega} \left[\int_{\Gamma_s} \sigma^s_{ik} n_k \, d\vec{x} - \int_{\Gamma_f} p^f n_i \, d\vec{x} \right] \end{aligned} \tag{6.51}$$

If the REV is subject to a set of self-equilibrating surface tractions, such that the resultant force and moment vanish,

$$\int_{\Gamma_s} \sigma_{ij}^s n_j \, d\vec{x} - \int_{\Gamma_f} p^f n_i \, d\vec{x} = 0 \tag{6.52}$$

$$\int_{\Gamma_s} \vartheta_{ijk} x_j \sigma_{k\ell}^s n_\ell \, d\vec{x} - \int_{\Gamma_f} \vartheta_{ijk} x_j p^f n_k \, d\vec{x} = 0 \tag{6.53}$$

where ϑ_{ijk} is the permutation symbol, then it is clear that

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \tag{6.54}$$

So the total stress tensor satisfies the equilibrium equation.

The equilibrium of the partial stresses, $(1-\phi)\overline{\sigma}_{ij}^s$ and $\phi\,\overline{p}^f$, however, can take a different form due to the presence of internal surface Σ_i, on which there exists an interaction force. Directly following (4.19), we can write

$$\frac{\partial \overline{p}^f}{\partial x_i} = 0 \tag{6.55}$$

From (6.55), it is clear that the equilibrium of fluid partial stress takes the form:

$$\frac{\partial \phi\,\overline{p}^f}{\partial x_i} - \overline{p}^f \frac{\partial \phi}{\partial x_i} = 0 \tag{6.56}$$

By the difference between (6.54) and (6.56), we find the equilibrium for the solid partial stress as

$$\frac{\partial\,(1-\phi)\,\overline{\sigma}_{ij}^s}{\partial x_j} - \overline{p}^f \frac{\partial \phi}{\partial x_i} = 0 \tag{6.57}$$

We notice the presence of the interaction term in (6.56) and (6.57), which is associated with spatial variation of porosity in the presence of macroscopic inhomogeneity. This term was recognized by Nikolaevskiy [37, 38] as equivalent to a body force. The origin of the force becomes clear if we examine Fig. 6.4. We observe that when there exists a porosity change, the force differential between the upper and lower fluid surfaces is balanced by a solid reaction force through the interface Σ_i. Such term, however, does not exist for a macroscopically homogeneous materials.

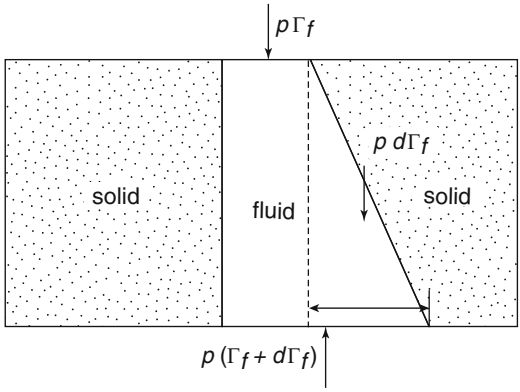


Fig. 6.4 Illustration of fluid partial stress equilibrium with spatially varying porosity

6.3 Governing Equation

Here we give a summary of the governing equations for the linear, isotropic poroelasticity:

- Definition of strain

$$e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{6.58}$$

- Constitutive equations

$$\sigma_{ij} = 2Ge_{ij} + \frac{2G\nu}{1-2\nu}\delta_{ij}e - \alpha\,\delta_{ij}p \tag{6.59}$$

$$p = M\,(\zeta - \alpha e) \tag{6.60}$$

- Equilibrium equation

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \tag{6.61}$$

- Darcy's law

$$q_i = -\kappa \frac{\partial p}{\partial x_i} \tag{6.62}$$

- Continuity equation

$$\frac{\partial \zeta}{\partial t} + \frac{\partial q_i}{\partial x_i} = 0 \tag{6.63}$$

In the above, we count the following variables: three u_i, six e_{ij} (due to symmetry), six σ_{ij}, one p, one ζ, and three q_i, a total of twenty variables. We also count the following equations: six definitions of strains, seven constitutive equations, three equilibrium equations, three equations of Darcy's law, and one continuity equation, a total of twenty equations. A set of five material constants, G, ν, α, M, and κ are needed to fully characterize these equations. As we have the same number of equations as unknowns (variables), the above forms a complete solution system.

When it comes to the mathematical or numerical solution, it is obvious that it is undesirable to solve a system with so many variables simultaneously; hence variables are often eliminated to create a more condensed solution system. This is presented in the ensuing sections.

6.3.1 Navier-Cauchy Equation

If we substitute (6.59) into (6.60), and use the definition in (6.58), we obtain the following equation in terms of the displacement vector $\vec{u}$ and pressure p,

$$G\nabla^2\vec{u} + \frac{G}{1-2\nu}\nabla(\nabla\cdot\vec{u}) = \alpha\nabla p \tag{6.64}$$

Or, when expressed in tensor notation, it becomes

$$Gu_{i,kk} + \frac{G}{1-2\nu}u_{k,ki} = \alpha p_{,i} \tag{6.65}$$

Another way to express this equation is to use ζ, instead of p, as a variable. This can be accomplished by substituting (6.60) into (6.64) to eliminate p, and we obtain

$$G\nabla^2\vec{u} + \frac{G}{1-2\nu_u}\nabla(\nabla\cdot\vec{u}) = \alpha M\nabla\zeta \tag{6.66}$$

in which we have assembled the material constants to define a new constant

$$\nu_u = \frac{2G\nu + \alpha^2 M(1-2\nu)}{2[G + \alpha^2 M(1-2\nu)]} \tag{6.67}$$

known as the undrained Poisson ratio. The physical significance of the drained and undrained materials constants has been discussed in Sects. 1.2.1 and 2.4.1.

We observe that the left-hand sides of (6.64) and (6.66) are exactly the elasticity equation known as *Navier-Cauchy equation*, or simply the *Navier equation* [43, 47]. On the right hand side, in the role of a body force, there exists the gradient of a field, either p or ζ. This is similar to the thermoelasticity equation discussed in Sect. 1.2.7, with the gradient of temperature as a body force. However, unlike a body force of gravity origin, which is known when the mass density and the gravity constants are

given, the field distributions of p and ζ are generally not known, and needs to be solved as a part of the solution process; hence an additional equation that defines the physical behavior of p or ζ is needed.

6.3.2 *Diffusion Equation*

Equations (6.62) and (6.63) can be combined to give

$$\frac{\partial \zeta}{\partial t} - \kappa \nabla^2 p = 0 \tag{6.68}$$

Using (6.60) to eliminate ζ, we obtain the following *diffusion equation* for pressure

$$\frac{\partial p}{\partial t} - \kappa M \nabla^2 p = -\alpha M \frac{\partial e}{\partial t} \tag{6.69}$$

where $e = \nabla \cdot \vec{u}$. The above equation cannot be independently solved, as its right hand side, the rate of change of volumetric strain, is not known. However, we can combine (6.64) and (6.69) to form a complete solution system, with four equations and four unknowns, $\vec{u}$ and p. In Sect. 7.3 we shall also demonstrate that under a set of assumptions of Terzaghi's one-dimensional consolidation theory, (6.69) degenerates into a one-dimensional diffusion equation without the right hand side term, thus is decoupled from (6.64).

To obtain a diffusion equation for ζ, we shall substitute the harmonic relation between p and ζ, to be derived as (6.87), into (6.68), and obtain,

$$\frac{\partial \zeta}{\partial t} - c \nabla^2 \zeta = 0 \tag{6.70}$$

where

$$c = \frac{\kappa}{S} \tag{6.71}$$

is a *generalized consolidation coefficient* [41], and S is the storage coefficient, as discussed in Sect. 2.4.4 and defined in (2.71). The reason that c is called a consolidation coefficient is because under the soil mechanics model, which is characterized by incompressible solid and fluid (see Sect. 4.8.3), c degenerates into the Terzaghi consolidation coefficient. To demonstrate this, we can write c based on an alternative expression of S in (B.14),

$$c = \frac{2\kappa G(1-\nu)(\nu_u - \nu)}{\alpha^2 (1-2\nu)^2 (1-\nu_u)} \tag{6.72}$$

Under the soil mechanics model, the following limits exist: $\alpha = 1$ and $\nu_u = 1/2$. Substituting these values into (6.72), we obtain

$$c = \frac{2\kappa G(1-\nu)}{1-2\nu} \tag{6.73}$$

This is indeed the definition of consolidation coefficient of the Terzaghi-Rendulic consolidation theory [44, 49].

Another form of diffusion equation can be obtained by using the constitutive equation (2.32), and we find [41]

$$\frac{\partial}{\partial t}\left(\sigma_{kk} + \frac{3}{B}p\right) - c\nabla^2\left(\sigma_{kk} + \frac{3}{B}p\right) = 0 \tag{6.74}$$

Equations (6.70) and (6.74) are independent diffusion equations without a coupling term. This may lead to the impression that the solution system, such as (6.66) and (6.70), is uncoupled such that we can solve (6.70) first and then substitute the result into (6.66). This is however not true for practical problems. Typical boundary value problems have boundary conditions given as tractions, displacements, pressure, and normal flux. It is not given as the variation in fluid content, because the value of ζ cannot be physically controlled at the boundary. Hence (6.70) cannot be independently solved despite the absence of coupling term in the equation. However, for certain mathematical problems, such as the solution of the free space Green's function (see Chap. 8), the domain is infinite and there is no boundary condition to satisfy. In that case, (6.70) can indeed be independently solved and the system (6.70) and (6.74) is a useful formulation.

6.3.3 *Compatibility Equation*

If the basic variable used to define the solution system is the strain tensor, e_{ij}, and not the displacement vector, u_i, then the strain field defined in (6.59) must satisfy some compatibility requirements in order to ensure the existence of a single valued continuous displacement solution u_i. These compatibility equations are identical to the St. Venant compatibility equations of elasticity [47]

$$e_{ij,kl} + e_{kl,ij} - e_{ik,jl} - e_{jl,ik} = 0 \tag{6.75}$$

Due to the symmetry of strain tensors, only 6 of the above equations are independent, which are expanded into the following:

$$\frac{\partial^2 e_{xx}}{\partial y\,\partial z} = -\frac{\partial^2 e_{yz}}{\partial x^2} + \frac{\partial^2 e_{xz}}{\partial x\,\partial y} + \frac{\partial^2 e_{xy}}{\partial x\,\partial z} \tag{6.76}$$

$$\frac{\partial^2 e_{yy}}{\partial x\,\partial z} = -\frac{\partial^2 e_{xz}}{\partial y^2} + \frac{\partial^2 e_{xy}}{\partial y\,\partial z} + \frac{\partial^2 e_{yz}}{\partial x\,\partial y} \tag{6.77}$$

$$\frac{\partial^2 e_{zz}}{\partial x\,\partial y} = -\frac{\partial^2 e_{xy}}{\partial z^2} + \frac{\partial^2 e_{yz}}{\partial x\,\partial z} + \frac{\partial^2 e_{xz}}{\partial y\,\partial z} \tag{6.78}$$

$$2\frac{\partial^2 e_{xy}}{\partial x\,\partial y} = \frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2} \tag{6.79}$$

$$2\frac{\partial^2 e_{yz}}{\partial y\,\partial z} = \frac{\partial^2 e_{yy}}{\partial z^2} + \frac{\partial^2 e_{zz}}{\partial y^2} \tag{6.80}$$

$$2\frac{\partial^2 e_{xz}}{\partial x\,\partial z} = \frac{\partial^2 e_{zz}}{\partial x^2} + \frac{\partial^2 e_{xx}}{\partial z^2} \tag{6.81}$$

Based on the above, the constitutive equations, (6.59) and (6.60), and the equilibrium equation, (6.61), we can derive the *Beltrami-Michell compatibility equation* for poroelasticity [41], analogous to that of elasticity, as follows:

$$\nabla^2\sigma_{ij} + \frac{1}{1+\nu}\sigma_{kk,ij} + 2\eta\left(\delta_{ij}\nabla^2 p + \frac{1-\nu}{1+\nu}p_{,ij}\right) = 0 \tag{6.82}$$

where η is the poroelastic stress coefficient defined in (1.56). For the elasticity version, we simply drop the terms containing p in (6.82).

Linked to these compatibility equations, there are some very useful relations that satisfy the Laplace equation. First, we can contract (6.82) to obtain:

$$\nabla^2(\sigma_{kk} + 4\eta p) = 0 \tag{6.83}$$

For plane strain, this equation reduces to

$$\nabla^2(\sigma_{kk} + 2\eta p) = 0 \tag{6.84}$$

where the summation is made for the in-plane stresses only; that is, $k = 1, 2$. And in one-dimension, it becomes

$$\frac{\partial^2}{\partial x^2}(\sigma_{xx} + 2\eta p) = 0 \tag{6.85}$$

The role of η as a poroelastic stress coefficient has been discussed in Sect. 1.2.7.

6.3.4 Harmonic Relation

In addition to (6.83), (6.84), and (6.85), there exist other harmonic relations between two poroelastic variables. These are explored as follows. First, we can take the divergence of (6.64), and obtain the following harmonic relation between p and e

$$\nabla^2\left(p-\frac{G}{\eta}e\right)=0 \tag{6.86}$$

Using (6.60) to eliminate e in (6.86), we obtain

$$\nabla^2(Sp-\zeta)=0 \tag{6.87}$$

in which we have utilized the conversion relations in (B.14) and (B.11). Finally, from (6.86) and (6.87), it is easy to see that

$$\nabla^2\left(\zeta-\frac{GS}{\eta}e\right)=0 \tag{6.88}$$

6.3.5 *Orthotropy*

For convenience of later reference, we shall construct here the governing equations for orthotropic materials. Utilizing the constitutive equations in Sect. 5.2.1, and the same equilibrium equation (6.61), we can build the following Navier equations:

$$\begin{aligned}&M_{xx}\frac{\partial^2 u_x}{\partial x^2}+G_{xy}\frac{\partial^2 u_x}{\partial y^2}+G_{xz}\frac{\partial^2 u_x}{\partial z^2}\\&\quad+(M_{xy}+G_{xy})\frac{\partial^2 u_y}{\partial x\partial y}+(M_{xz}+G_{xz})\frac{\partial^2 u_z}{\partial x\partial z}-\alpha_x\frac{\partial p}{\partial x}=0\end{aligned} \tag{6.89}$$

$$\begin{aligned}&G_{xy}\frac{\partial^2 u_y}{\partial x^2}+M_{yy}\frac{\partial^2 u_y}{\partial y^2}+G_{yz}\frac{\partial^2 u_y}{\partial z^2}\\&\quad+(M_{xy}+G_{xy})\frac{\partial^2 u_x}{\partial x\partial y}+(M_{yz}+G_{yz})\frac{\partial^2 u_z}{\partial y\partial z}-\alpha_y\frac{\partial p}{\partial y}=0\end{aligned} \tag{6.90}$$

$$\begin{aligned}&G_{xz}\frac{\partial^2 u_z}{\partial x^2}+G_{yz}\frac{\partial^2 u_z}{\partial y^2}+M_{zz}\frac{\partial^2 u_z}{\partial z^2}\\&\quad+(M_{xz}+G_{xz})\frac{\partial^2 u_x}{\partial x\partial z}+(M_{yz}+G_{yz})\frac{\partial^2 u_y}{\partial y\partial z}-\alpha_z\frac{\partial p}{\partial z}=0\end{aligned} \tag{6.91}$$

which are compared to (6.64) for isotropy case. The pore pressure diffusion equation for orthotropy, corresponding to (6.69) for isotropy, is

$$\begin{aligned}&\frac{\partial p}{\partial t}-\kappa_x M\frac{\partial^2 p}{\partial x^2}-\kappa_y M\frac{\partial^2 p}{\partial y^2}-\kappa_z M\frac{\partial^2 p}{\partial z^2}\\&\qquad=-\alpha_x M\frac{\partial e_{xx}}{\partial t}-\alpha_y M\frac{\partial e_{yy}}{\partial t}-\alpha_z M\frac{\partial e_{zz}}{\partial t}\end{aligned} \tag{6.92}$$

6.3.6 Transverse Isotropy

For transverse isotropy, the Navier equations become

$$M_{xx}\frac{\partial^2 u_x}{\partial x^2} + G\frac{\partial^2 u_x}{\partial y^2} + G'\frac{\partial^2 u_x}{\partial z^2} + (M_{xy} + G)\frac{\partial^2 u_y}{\partial x \partial y} + (M_{xz} + G')\frac{\partial^2 u_z}{\partial x \partial z} - \alpha_x \frac{\partial p}{\partial x} = 0 \tag{6.93}$$

$$G\frac{\partial^2 u_y}{\partial x^2} + M_{xx}\frac{\partial^2 u_y}{\partial y^2} + G'\frac{\partial^2 u_y}{\partial z^2} + (M_{xy} + G)\frac{\partial^2 u_x}{\partial x \partial y} + (M_{xz} + G')\frac{\partial^2 u_z}{\partial y \partial z} - \alpha_x \frac{\partial p}{\partial y} = 0 \tag{6.94}$$

$$G'\frac{\partial^2 u_z}{\partial x^2} + G'\frac{\partial^2 u_z}{\partial y^2} + M_{zz}\frac{\partial^2 u_z}{\partial z^2} + (M_{xz} + G')\frac{\partial^2 u_x}{\partial x \partial z} + (M_{xz} + G')\frac{\partial^2 u_y}{\partial y \partial z} - \alpha_z \frac{\partial p}{\partial z} = 0 \tag{6.95}$$

The pore pressure diffusion equation is

$$\frac{\partial p}{\partial t} - \kappa_x M\frac{\partial^2 p}{\partial x^2} - \kappa_x M\frac{\partial^2 p}{\partial y^2} - \kappa_z M\frac{\partial^2 p}{\partial z^2} = -\alpha_x M\frac{\partial e_{xx}}{\partial t} - \alpha_x M\frac{\partial e_{yy}}{\partial t} - \alpha_z M\frac{\partial e_{zz}}{\partial t} \tag{6.96}$$

6.4 Degenerated Governing Equation

As stated in the above section, the governing equations of poroelasticity are generally coupled. However, under special conditions, the governing equations may degenerate into some simpler forms, and may be decoupled. Some of the cases are discussed below.

6.4.1 Drained and Undrained State

When the poroelastic response is undrained ($\zeta = 0$) at the instant of a loading ($t = 0^+$), (6.64) reduces to

$$G\nabla^2\vec{u} + \frac{G}{1 - 2\nu_u}\nabla\nabla\cdot\vec{u} = 0 \tag{6.97}$$

Or, when it is drained ($p = 0$) at large time ($t \to \infty$), (6.66) becomes

$$G\nabla^2\vec{u} + \frac{G}{1-2\nu}\nabla\nabla\cdot\vec{u} = 0 \tag{6.98}$$

Both of the above equations are exactly the Navier equation of elasticity. The displacement solution is elastic, but characterized by different material constants: one by the set $\{G, \nu\}$, and the other by $\{G, \nu_u\}$. For the undrained case, once the displacement field is solved, the pore pressure can be evaluated from the constitutive law (6.60) as

$$p = -\alpha M e \tag{6.99}$$

or from (2.32) as

$$p = -\frac{B}{3}\sigma_{kk} \tag{6.100}$$

6.4.2 Soil Mechanics Model

The soil mechanics model is defined as the limiting case of incompressible solid and incompressible fluid; see Sect. 4.8.3. By taking the limits in the material constants, we obtain the following set of governing equations [31]

$$G\nabla^2\vec{u} + \frac{G}{1-2\nu}\nabla\nabla\cdot\vec{u} = \nabla p \tag{6.101}$$

$$\frac{\partial e}{\partial t} - c\nabla^2 e = 0 \tag{6.102}$$

We note that (6.102) is obtained by substituting (4.176) into (6.70). Equation (6.102) appears to be uncoupled, but again it is not. The reason is that the volumetric strain e is not available as a boundary condition. Hence, once given a physical boundary condition that is coupled, the two governing equations (6.101) and (6.102) must be solved simultaneously.

6.4.3 Irrotational Displacement Field

We consider now a special case that the displacement field is *irrotational*. According to the *Helmholtz theorem* of vector analysis [2, 43], a vector field can be decomposed into a part that is the gradient of a scalar (*lamellar* or irrotational field) and a part that is the curl of a vector (*solenoidal* or rotational field). For a displacement field that is irrotational,

$$\nabla \times \vec{u} = 0 \tag{6.103}$$

the second part does not exist and we can express the displacement as

$$\vec{u} = \nabla \Phi \tag{6.104}$$

where Φ is a scalar potential. The Navier equation (6.64) then reduces to

$$\nabla \cdot \nabla(\nabla \Phi) = \frac{\eta}{G} \nabla p \tag{6.105}$$

The above equation can be integrated to relax a gradient operator, and we find

$$\nabla \cdot \nabla \Phi = \nabla \cdot \vec{u} = e = \frac{\eta}{G} p + g(t) \tag{6.106}$$

where $g(t)$ is an unknown function of time. After insertion of (6.106) into the contracted constitutive equation (6.59), we obtain

$$\sigma_{kk} + 4\eta p = \frac{2G(1+\nu)}{1-2\nu} g(t) = 3Kg(t) \tag{6.107}$$

For plane strain, this relation becomes

$$\sigma_{kk} + 2\eta p = \frac{2G}{1-2\nu} g(t) \tag{6.108}$$

where $k = 1, 2$. For one-dimensional problem, we obtain

$$\sigma_{xx} = \frac{2G(1-\nu)}{1-2\nu} g(t) \tag{6.109}$$

As a one-dimensional displacement field $u_x(x)$ is always irrotational, (6.109) is in fact always true. The significance of (6.107), (6.108) and (6.109) is that these combinations of average normal stress and pore pressure are constant throughout the space, though it can vary with respect to time.

The pore pressure diffusion equation (6.69) can also be modified for the case of an irrotational displacement field. Utilizing (6.106) and (6.107) in (6.69), we obtain

$$\frac{\partial p}{\partial t} - c\nabla^2 p = -\frac{\eta(1-\nu)}{GS(1+\nu)} \frac{d}{dt} (\sigma_{kk} + 4\eta p) \tag{6.110}$$

For plane strain, (6.69) becomes

$$\frac{\partial p}{\partial t} - c\nabla^2 p = -\frac{\eta(1-\nu)}{GS} \frac{d}{dt} (\sigma_{kk} + 2\eta p) \tag{6.111}$$

And for one-dimensional problem, we find

$$\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial x^2} = -\frac{\eta}{GS}\frac{d\sigma_{xx}}{dt} \tag{6.112}$$

We notice that the diffusion coefficient in the above diffusion equations now becomes the consolidation coefficient, and the right hand side source terms are functions of time only.

6.4.4 Uncoupling of Diffusion Equation

In this section we examine a few special cases that the pore pressure diffusion equation (6.69) becomes uncoupled, and can be solved independently. First, for the case of irrotational displacement field, we notice that the right hand sides of (6.110), (6.111) and (6.112) are functions of time only. If in a boundary value problem we can find any part of the boundary on which such stress and pressure combination is independent of time, then the right hand side vanishes everywhere, and we have a simple diffusion equation

$$\frac{\partial p}{\partial t} - c\nabla^2 p = 0 \tag{6.113}$$

This condition can exist if a part of the domain is unbounded, such as an infinite or a semi-infinite domain. If at the far field, away from the region of disturbance, all stresses and pressure are zero, then the above condition is fulfilled and the diffusion equation is uncoupled. Another example is the one-dimensional problem. As σ_{xx} must be constant throughout the space, if the load applied at the boundary does not vary with time, an uncoupled diffusion equation also results.

It is of interest to observe that under the conditions that the function $g(t)$ vanishes, there exist one to one dependence between the various volumetric stress and strain quantities. First, from (6.106), we obtain

$$e = \frac{\eta}{G}p \tag{6.114}$$

Also, according to (6.107), we find

$$\sigma_{kk} = -4\eta p \tag{6.115}$$

Based on (6.60) and (6.114), we observe the following

$$\zeta = Sp \tag{6.116}$$

Another occasion that the decoupling is possible is when the fluid is highly compressible, such as a gas, with the condition $K_f \ll \phi K$. In the case that the flow is driven by the applied fluid pressure, not by the stress applied to the solid, we anticipate that the stress σ_{kk} and the pressure p are of the same order of magnitude. Based on the constitutive equation (2.32), and that $B \ll 1$ as given in (4.181), we can express

$$\zeta \approx Cp \tag{6.117}$$

Substituting the above into (6.68), together with conditions in (4.182), we again obtain the uncoupled diffusion equation (6.113), with the consolidation coefficient given by

$$c \approx \frac{\kappa}{\phi C_f} \tag{6.118}$$

Finally, the uncoupling of the pore pressure equation also takes place under the steady-state conditions. As the variables are not functions of time, the diffusion equation reduces to the Laplace equation

$$\nabla^2 p = 0 \tag{6.119}$$

6.5 Boundary Value Problem

For modeling purposes, we seek a mathematical system that is a complete solution system. First, we gather a set of governing equations that contain the same number of equations as the number of variables (unknowns). As these are differential equations, there generally exist an infinite number of solutions that can satisfy these equations. To model a physical phenomenon, we need to limit the solutions by imposing a set of constraints such that there exists one and one only solution that corresponds to the modeled phenomenon. These constraints are typically provided in the form of *boundary conditions*, that is, conditions prescribed on the boundary of the modeled domain. Boundary conditions need to be carefully given—when too few are prescribed, multiple solutions can exist, or, when too many are given, no solution can exist. For the mathematical solution to correspond to a physical phenomenon, the solution must exist, and must be unique. Hence we seek a *well-posed boundary value problem* that has one and one only solution. This requirement is tied to the existence and uniqueness theorem of partial differential equations.

6.5.1 Existence and Uniqueness

For *potential problems*, that is, problems governed by the Laplace equation, the *Dirichlet boundary condition* refers to the condition that the value of the potential is given on the boundary; the *Neumann boundary condition* is a condition that the derivative of potential in the direction normal to the boundary is given; and the *Robin boundary condition* is a linear combination of the above two conditions. The existence and uniqueness theorem for potential problems has been proven for boundary value problems of the Dirichlet, Neumann, Robin, and mixed types. Such proofs, however, require the bounding surface and the boundary value prescribed on it to fulfill certain smoothness conditions [13]. Particularly, the existence theorem is proven only for a *Lyapunov surface*, which is a surface in the $C^{1,a}$, $0 \leq a < 1$, continuity class [25, 26]. This type of surface does not admit a sharp edge, such as a vertex, which is found in most practical problems. Hence the existence of solution is not proven in the *strong form* for most practical problems that we encounter.

In a different approach, we may interpret the existence of solution through a *weak form*, by seeking an approximate solution using methods like the finite element method or the boundary element method. The solution is sought by minimizing the approximation error in the form of an energy norm, using schemes like the Galerkin method. In that case, it can be proven that the error norm goes to zero with finer and finer discretization. The existence theorem in this weak sense has been proven for surface in the $C^{0.1}$ class [14], known as the *Lipschitz surface*, which admits the sharp edges.

The existence and uniqueness theorem for elasticity problems is similar, as the Navier equation of elasticity is a *vector potential problem* [25]. The uniqueness theorem has been provided by Kirchhoff [27]. Similar to the potential problems, the existence proof in the strong form is limited to the Lyapunov surface; see Love [29] and Sokolnikoff [47] for a discussion. For approximate solution, however, the existence of solution is proven for the Galerkin formulation (weak form) with more general boundaries that can include edges and cracks [30].

The theory of poroelasticity is an extension of elasticity and porous medium flow (diffusion equation). The uniqueness proof was given by Deresiewicz and Skalak [16] for the quasi-static theory, and by Altay and Dökmeci [1] for the dynamic theory. The existence proof has been given in the strong form by Auriault and Sanchez-Palencia [4] and Showalter [46], and in the weak form by Augustin [3], under various conditions.

Without going into the detail of the existence and uniqueness theorem, for the presence purpose, we shall provide the following statements for a *well-posed problem of poroelasticity* (that is, the solution exists, is unique, and solvable). Sokolnikoff [47] gave a set of statements of fundamental boundary value problems of elasticity. We shall extend those statements to describe the fundamental *initial and boundary value problems* of poroelasticity:

Problem 1 Determine the distribution of stresses, displacements, pressure, and specific flux in the interior of a poroelastic body at all forward time, when the body forces are prescribed in the interior of the body, the pore pressure distribution in the body is given at the initial time, and the distribution of the forces and the specific flux acting on the surface of the body is known. The prescribed set of surface forces must satisfy the force equilibrium with the body force, and the flux must satisfy the mass conservation in the body.

Problem 2 Determine the distribution of stresses, displacements, pressure, and specific flux in the interior of a poroelastic body at all forward time, when the body forces are prescribed in the interior of the body, the pore pressure distribution in the body is given at the initial time, and the distribution of the displacements and the pore pressure on the surface of the body is known.

Problem 1 is equivalent to a Neumann problem in potential theory, and problem 2 a Dirichlet problem. In a Neumann problem, the solid body needs to be additionally constrained to prevent from rigid body motions such as translation and rotation. For the fluid part, at least one pore pressure value needs to be given to avoid non-uniqueness of pore pressure solution.

There also exist mixed boundary value problems. That is, for a given problem, it is possible that a part of the boundary has the traction ($t_i = \sigma_{ij} n_j$) boundary condition, and another part the displacement (u_i) condition. For the fluid component, a part of the boundary can have the pressure (p) condition, and another part the normal flux ($q_n = q_i n_i$) condition. Also, as displacements and tractions are vectors, the vector components can be mixed. For example, on the same part of boundary, it is possible that displacement is prescribed in some of the directions, and traction in the other.

In conclusion, for a well-posed boundary value problem, the boundary conditions must be given such that on each part of the boundary, there must exist exactly four boundary conditions for three-dimensional problems. Three of the boundary conditions are related to the solid frame, one in each mutually orthogonal spatial directions. These conditions can either be stress (traction) or displacement. The fourth boundary condition is associated with the fluid component; it can either be pressure or normal flux. For an all stress or all flux condition, additional constraints are needed as described above. Also, as the problem is time dependent, initial conditions that describe the state of the porous medium at the beginning of the solution is needed. For the solid part, an initial displacement field, and for the fluid part, an initial pressure distribution, must be known throughout the domain. Once we have a well-posed boundary value problem, then there exists a unique solution, and we are ready to find that solution either exactly through mathematical solution, or approximately by numerical solution.

6.5.2 Boundary Condition

To formulate a boundary value problem, we need to translate the physical conditions at the boundary into mathematical statements. We shall discuss a range of commonly occurring physical conditions below. For simplicity, the illustrations in Fig. 6.5 are shown in 2D.

Free surface If the surface of porous medium is not in contact with any other medium, other than air, then the surface is free from surface traction, and pore pressure is zero (atmospheric). (See Fig. 6.5a.)

Fluid contact If the contact surface is not air, but a static fluid under pressure, with $p = p_o$, then on the porous medium surface, both the normal stress and the pore pressure are the same as the fluid pressure, p_o, and there is no shear stress. (See Fig. 6.5b.)

Rigid surface smooth or rough If the surface of the porous medium is in contact with another medium which is much harder, then the contacted surface can be considered as *rigid*. On a rigid surface, the displacement normal to that surface is prohibited. Then, depending on the smoothness (roughness) of the surface, we can either assume a frictionless contact such that the porous medium surface is free to move in the direction parallel to the surface (Fig. 6.5c), or a perfect contact such that no lateral displacement is allowed (Fig. 6.5d).

Permeable and impermeable surface The contact surface can have any permeability. If the permeability is much larger than that of the porous medium, then we can assume infinite permeability, such that the pore pressure is drained on the surface (Fig. 6.5e). Or, if the permeability of the contacting surface is much smaller than the porous medium, then we can assume an impermeable surface, on which the normal flux is zero (Fig. 6.5f).

Rigid plate When a force is applied to a porous medium through a plate, the plate can be considered rigid or flexible. For a rigid plate, we know neither the normal stress nor the normal displacement directly. All we know is that the displacement on the contacting surface must conform to the rigid shape of the plate (see Fig. 6.5g). Assuming that the plate does not rotate, then the normal displacement is a constant (but unknown). Another condition is that the reaction force must equal to the applied force F, thus

$$\int_A t_y \, d\vec{x} = F \tag{6.120}$$

where A is the area of the plate. For boundary condition in the tangential direction of the plate, it can be zero displacement, or zero tangential traction, depending on the smoothness condition of the contacting surface. For the fluid condition, it can be drained, or impermeable.

Flexible plate If the plate is flexible, both the normal traction and displacement are not known (Fig.6.5h). Assuming that the plate is thin, then a two-dimensional plate equation (or a one-dimensional beam equation) needs to be solved together.

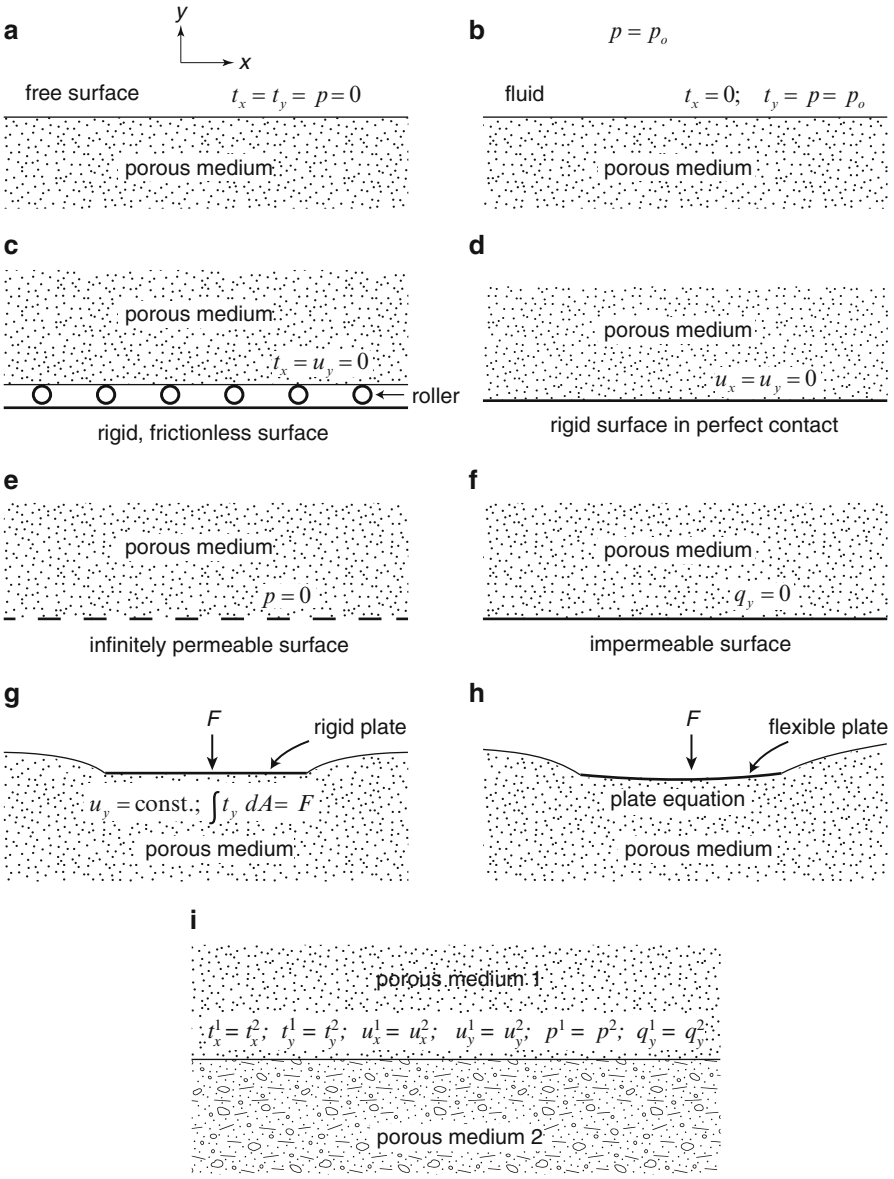


Fig. 6.5 Types of physical boundary conditions (illustrated in two dimensions): (**a**) free surface, (**b**) fluid contact, (**c**) rigid, frictionless surface, (**d**) rigid surface in perfect contact (no slippage), (**e**) infinitely permeable surface, (**f**) impermeable surface, (**g**) rigid plate, (**h**) flexible plate, and (**i**) bimaterial interface

The coupling conditions are that the deflection of the plate is equal to the vertical displacement of the porous medium, and the vertical traction of the porous medium gives a distributed load to the plate. Then depending on the smoothness of the plate surface and its permeability, boundary conditions in the horizontal direction and the fluid boundary condition can be applied.

Bimaterial interface For two porous materials in contact with each other, the conditions for each material are not directly known at the interface. However, the requirements that stress, displacement, pressure, and flux must be continuous across the interface give us eight conditions for three-dimensional problems (see Fig. 6.5i). As for each external surface, four conditions are needed, these eight conditions are sufficient for the simultaneous solution for both material zones.

6.6 Field Equation

In elasticity, it is generally difficult to directly solve the Navier equation to find analytical solution of boundary value problems. The key to effective mathematical solution is to find *potentials* that satisfy certain simpler *field equations*, such as the Laplace equation or biharmonic equation, that represent the solution of Navier equation. As the *general solutions* of the field equations are well known, for example, any function that satisfies the Laplace equation is known as harmonic function, and that satisfies biharmonic equation is known as biharmonic function, the solution to a specific elasticity problem can be constructed using a summation of a complete set of these functions to satisfy the boundary conditions [47]. Similarly, we can find the field equations for poroelasticity governing equations, and their potential representations leading to general solutions. Some of these efforts are presented below.

6.6.1 Biot Function

In elasticity, the displacement vector $\vec{u}$ can be represented by a set of Papkovitch-Neuber functions [35, 36, 39, 40], also referred to as Boussinesq-Papkovitch functions [10], in the following form

$$u_i = (\psi + x_j \Psi_j)_{,i} - 4(1-\nu)\Psi_i \tag{6.121}$$

in which the vector potential $\vec{\Psi}$ and scalar potential ψ satisfy the Laplace equations

$$\nabla^2 \Psi_i = 0 \tag{6.122}$$

$$\nabla^2 \psi = 0 \tag{6.123}$$

It is easy to show by substitution that the displacement defined in (6.121) satisfies the Navier equation of elasticity,

$$G\nabla^2\vec{u} + \frac{G}{1-2\nu}\nabla(\nabla\cdot\vec{u}) = 0 \tag{6.124}$$

Although (6.121) is a solution of the Navier equation, question may arise whether it is a *complete solution*; that is, whether there are solutions that cannot be represented by such expression. The completeness of solution of the above system has been proven by Mindlin [33]. Even if the solution is complete, we may ask whether there is redundancy. In other words, whether one or more of the above potentials can be represented by a combination of others. The answer to this question is that we cannot omit one and still retain the completeness of the solution [47]. Hence (6.121) is a general solution of the Navier equation.

Inspired by the Papkovitch-Neuber representation, Biot [9] presented the following general solution for the set of poroelasticity equations, (6.66) and (6.70),

$$u_i = (\psi + x_j\Psi_j)_{,i} - 4(1-\nu_u)\Psi_i \tag{6.125}$$

$$\zeta = \frac{GS}{\eta}\nabla^2\psi \tag{6.126}$$

We may refer ψ and Ψ_i as the *Biot functions*. In the absence of body forces and sources, the components of the vector function $\vec{\Psi}$ are harmonic

$$\nabla^2\Psi_i = 0 \tag{6.127}$$

while ψ satisfies the "biharmonic diffusion equation"

$$\frac{\partial}{\partial t}(\nabla^2\psi) - c\nabla^4\psi = 0 \tag{6.128}$$

It is easy to prove that (6.125), (6.126), (6.127) and (6.128) satisfy (6.66) and (6.70) by simple substitution. The completeness of the solution has been established by Verruijt [50].

Once u_i and ζ are given, the stresses and pressure can be obtained from the constitutive relations (2.27) and (2.28), as

$$\begin{aligned}\sigma_{ij} = 2G\big[&\psi_{,ij} - \delta_{ij}\nabla^2\psi - (1-2\nu_u)\left(\Psi_{i,j} - \Psi_{j,i}\right) + x_k\Psi_{k,ij} \\ &-2\nu_u\delta_{ij}\Psi_{k,k}\big]\end{aligned} \tag{6.129}$$

$$p = \frac{2G}{\alpha(1-2\nu)}\left[(1-\nu)\nabla^2\psi + 2(\nu_u - \nu)\Psi_{i,i}\right] \tag{6.130}$$

Of special interest to examine is the special case that the displacement field is irrotational, as discussed in Sect. 6.4.3. If we apply the curl operator to (6.125) and

based on the identity $\nabla \times \nabla\psi = 0$, we observe that for irrotational displacement, $\Psi_i = 0$. The above expressions then simplify to

$$u_i = \psi_{,i} \tag{6.131}$$

$$\sigma_{ij} = 2G\left[\psi_{,ij} - \delta_{ij}\nabla^2\psi\right] \tag{6.132}$$

$$p = \frac{2G(1-\nu)}{\alpha(1-2\nu)}\nabla^2\psi \tag{6.133}$$

6.6.2 Biot Decomposition

Biot [9] introduced a decomposition that separates the displacement into two parts, a part that satisfies the Navier equation of elasticity, and a part that is irrotational. To derive the decomposition, we shall use solid and fluid displacement, u_i and U_i, as the basic variables. First, we recall the definition of the specific flux in (6.43) as

$$q_i = \phi\frac{\partial}{\partial t}(U_i - u_i) \tag{6.134}$$

Using the above expression in Darcy's law (6.27), we obtain

$$\phi\frac{\partial}{\partial t}(U_i - u_i) = -\kappa p_{,i} \tag{6.135}$$

As the right hand side of (6.135) is the gradient of a scalar, it is possible to write $U_i - u_i$ as the gradient of a potential

$$-\phi\,(U_i - u_i) = \Phi_{,i} \tag{6.136}$$

The variation in fluid content define in (4.117) is then

$$\zeta = \phi\,(u_{i,i} - U_{i,i}) = \nabla^2\Phi \tag{6.137}$$

With the above, we can write the Navier equation (6.66) in the following form

$$Gu_{i,kk} + \frac{G}{1-2\nu_u}u_{k,ki} = \alpha M\Phi_{,kki} \tag{6.138}$$

Biot [9] defined a new variable

$$u_i^o = u_i - \frac{\alpha M(1-2\nu_u)}{2G(1-\nu_u)}\Phi_{,i} = u_i - \frac{3\alpha M}{3K_u + 4G}\Phi_{,i} \tag{6.139}$$

If we substitute (6.139) into (6.138), we can easily show that this new variable satisfies exactly the Navier equation of elasticity with undrained Poisson ratio:

$$G u^o_{i,kk} + \frac{G}{1-2\nu_u} u^o_{k,ki} = 0 \tag{6.140}$$

The pressure as defined in (2.28) can also be expressed in terms of these new variables

$$p = -\alpha M e^o - \frac{1}{S}\nabla^2 \Phi \tag{6.141}$$

where $e^o = u^o_{i,i}$. Substituting (6.136) and (6.141) into (6.135), we obtain

$$\frac{\partial \Phi_{,i}}{\partial t} - c\,\nabla^2 \Phi_{,i} = -\alpha\kappa M e^o_{,i} \tag{6.142}$$

Relaxing a gradient operator in the above, we find

$$\frac{\partial \Phi}{\partial t} - c\,\nabla^2 \Phi = -\alpha\kappa M e^o \tag{6.143}$$

where Φ is defined within a constant or an arbitrary function of time. Or, we can take the Laplacian of (6.143) and obtain

$$\frac{\partial \nabla^2 \Phi}{\partial t} - c\,\nabla^4 \Phi = 0 \tag{6.144}$$

in which we used the condition

$$\nabla^2 e^o = 0 \tag{6.145}$$

by applying the divergence operator on (6.140).

In summary, the Biot decomposition is given by

$$u_i = u^o_i + \frac{\alpha M(1-2\nu_u)}{2G(1-\nu_u)} \Phi_{,i} \tag{6.146}$$

in which u^o_i satisfies the uncoupled Navier equation (6.140), and Φ the diffusion equation, either (6.143) or (6.144). Pressure is given by (6.141), the variation in fluid content by (6.137), and the stress by

$$\sigma_{ij} = 2G e^o_{ij} + \frac{2G\nu_u}{1-2\nu_u}\delta_{ij} e^o + \frac{2\eta}{S}(\Phi_{,ij} - \delta_{ij}\nabla^2 \Phi) \tag{6.147}$$

The above decomposition may look attractive, as it implies that a solution can be found by first solving an elasticity equation, and then a diffusion equation with a source term. This statement, however, is generally not true, because in a boundary value problem, the boundary conditions are given as physical quantities such as displacement and stress. By observing the definitions in (6.146) and (6.147), clearly the boundary conditions are now coupled with the variable Φ. Hence the solution system is still coupled.

Although the Biot decomposition is not useful for solving a typical boundary value problem, it is a powerful tool in the solution of the *fundamental solutions*, also known as the *free space Green's functions*, for example, the solution of a point force or a point source located in an infinite domain. As there is no boundary condition to satisfy, the uncoupling is indeed accomplished. This property will be fully explored in Sect. 8.9.

6.6.3 McNamee-Gibson Displacement Function

McNamee and Gibson [31] presented a solution technique that expressed the poroelastic solutions in terms of two *displacement functions* for plane strain and axisymmetric problems. For three-dimensional problems, Schiffman and Fungaroli [45] provided a representation that involved three displacement functions. These representations have been applied to find analytical solutions for a range of problems, including constant loads exerted normally or horizontally over strip, circular or rectangular areas on top of semi-infinite, finite, single or multiple soil layer systems [5, 20–22, 32, 48]. The original theory was derived for the incompressible constituent (soil mechanics) model (Sect. 4.8.3). The incompressible fluid assumption was later removed by Verruijt [51]. With a slight modification, the generalized displacement functions that account for the compressible constituents of the full poroelasticity model have been derived by Detournay and Cheng [17]. The following presentation is based on the full poroelasticity model.

Plane Strain Displacement Functions It has been pointed out by Verruijt [51] that the McNamee-Gibson displacement functions [31] can be deduced from the Biot functions. For plane strain problems, defined as $u_y = 0$, $u_x = u_x(x, z, t)$ and $u_z = u_z(x, z, t)$, we define

$$\psi = -\mathcal{E}(x, z, t) \tag{6.148}$$

$$\Psi_z = \mathcal{S}(x, z, t) \tag{6.149}$$

$$\Psi_x = \Psi_y = 0 \tag{6.150}$$

where $\mathcal{E}$ and $\mathcal{S}$ are displacement functions satisfying

$$\frac{\partial}{\partial t}(\nabla^2 \mathcal{E}) - c\nabla^4 \mathcal{E} = 0 \tag{6.151}$$

$$\nabla^2 \mathcal{S} = 0 \tag{6.152}$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \tag{6.153}$$

is the two-dimensional Laplacian operator. We note that (6.151) and (6.152) are the direct consequence of (6.127) and (6.128). The resultant solution system, (6.151) and (6.152), are two equations with two unknowns, $\mathcal{E}$ and $\mathcal{S}$, which are one fewer than the original system that involves two displacement components, u_x and u_z, and a fluid component, p or ζ.

Substituting (6.148), (6.149), and (6.150) into (6.125), (6.126), (6.129), and (6.130), we obtain the displacements, stresses, etc. as[1]

$$u_x = -\frac{\partial \mathcal{E}}{\partial x} + z\frac{\partial \mathcal{S}}{\partial x} \tag{6.154}$$

$$u_z = -\frac{\partial \mathcal{E}}{\partial z} + z\frac{\partial \mathcal{S}}{\partial z} - (3 - 4\nu_u)\mathcal{S} \tag{6.155}$$

$$\zeta = -\frac{GS}{\eta}\nabla^2 \mathcal{E} \tag{6.156}$$

$$\sigma_{xx} = 2G\left[\nabla^2 \mathcal{E} - \frac{\partial^2 \mathcal{E}}{\partial x^2} + z\frac{\partial^2 \mathcal{S}}{\partial x^2} - 2\nu_u \frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.157}$$

$$\sigma_{zz} = 2G\left[\nabla^2 \mathcal{E} - \frac{\partial^2 \mathcal{E}}{\partial z^2} + z\frac{\partial^2 \mathcal{S}}{\partial z^2} - 2(1 - \nu_u)\frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.158}$$

$$\sigma_{xz} = 2G\left[-\frac{\partial^2 \mathcal{E}}{\partial x \partial z} + z\frac{\partial^2 \mathcal{S}}{\partial x \partial z} - (1 - 2\nu_u)\frac{\partial \mathcal{S}}{\partial x}\right] \tag{6.159}$$

$$p = -\frac{G}{\eta}\left[\nabla^2 \mathcal{E} - \frac{2(\nu_u - \nu)}{1 - \nu}\frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.160}$$

For the incompressible constituent model ($\nu_u = 1/2$ and $GS/\eta = 1$), the above expressions degenerate into the McNamee-Gibson case [31]

[1] A typo is found in the pore pressure equation (116 g) in Detournay and Cheng [17], in which the $\partial^2 \mathcal{S}/\partial z^2$ term should be replaced by $\partial \mathcal{S}/\partial z$.

$$u_x = -\frac{\partial \mathcal{E}}{\partial x} + z\frac{\partial \mathcal{S}}{\partial x} \tag{6.161}$$

$$u_z = -\frac{\partial \mathcal{E}}{\partial z} + z\frac{\partial \mathcal{S}}{\partial z} - \mathcal{S} \tag{6.162}$$

$$\zeta = -\nabla^2 \mathcal{E} \tag{6.163}$$

$$\sigma_{xx} = 2G\left[\nabla^2 \mathcal{E} - \frac{\partial^2 \mathcal{E}}{\partial x^2} + z\frac{\partial^2 \mathcal{S}}{\partial x^2} - \frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.164}$$

$$\sigma_{zz} = 2G\left[\nabla^2 \mathcal{E} - \frac{\partial^2 \mathcal{E}}{\partial z^2} + z\frac{\partial^2 \mathcal{S}}{\partial z^2} - \frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.165}$$

$$\sigma_{xz} = 2G\left[-\frac{\partial^2 \mathcal{E}}{\partial x \partial z} + z\frac{\partial^2 \mathcal{S}}{\partial x \partial z}\right] \tag{6.166}$$

$$p = -2G\left[\frac{1-\nu}{1-2\nu}\nabla^2 \mathcal{E} - \frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.167}$$

Axisymmetric Displacement Functions The axisymmetric case is defined by the following displacements: $u_\theta = 0$, $u_r = u_r(r, z, t)$ and $u_z = u_z(r, z, t)$. The displacement functions are obtained by assuming

$$\psi = -\mathcal{E}(r, z, t) \tag{6.168}$$

$$\Psi_z = \mathcal{S}(r, z, t) \tag{6.169}$$

$$\Psi_x = \Psi_y = 0 \tag{6.170}$$

where $\mathcal{E}$ and $\mathcal{S}$ satisfy the same equations (6.151) and (6.152), except that the Laplacian operator is in cylindrical coordinates

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2} \tag{6.171}$$

The displacement and stress expressions are

$$u_r = -\frac{\partial \mathcal{E}}{\partial r} + z\frac{\partial \mathcal{S}}{\partial r} \tag{6.172}$$

$$u_z = -\frac{\partial \mathcal{E}}{\partial z} + z\frac{\partial \mathcal{S}}{\partial z} - (3 - 4\nu_u)\mathcal{S} \tag{6.173}$$

$$\zeta = -\frac{GS}{\eta}\nabla^2 \mathcal{E} \tag{6.174}$$

$$\sigma_{rr} = 2G\left[\nabla^2 \mathcal{E} - \frac{\partial^2 \mathcal{E}}{\partial r^2} + z\frac{\partial^2 \mathcal{S}}{\partial r^2} - 2\nu_u \frac{\partial \mathcal{S}}{\partial z}\right] \tag{6.175}$$

$$\sigma_{\theta\theta} = 2G\left[\nabla^2\mathcal{E} - \frac{1}{r}\frac{\partial\mathcal{E}}{\partial r} + \frac{z}{r}\frac{\partial\mathcal{S}}{\partial r} - 2\nu_u\frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.176}$$

$$\sigma_{zz} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial z^2} + z\frac{\partial^2\mathcal{S}}{\partial z^2} - 2(1-\nu_u)\frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.177}$$

$$\sigma_{rz} = 2G\left[-\frac{\partial^2\mathcal{E}}{\partial r\partial z} + z\frac{\partial^2\mathcal{S}}{\partial r\partial z} - (1-2\nu_u)\frac{\partial\mathcal{S}}{\partial r}\right] \tag{6.178}$$

$$p = -\frac{G}{\eta}\left[\nabla^2\mathcal{E} - \frac{2(\nu_u-\nu)}{1-\nu}\frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.179}$$

The incompressible constituent case follows those in the plane strain model.

Three-Dimensional Displacement Functions The three-dimensional theory [45] is obtained by introducing three displacement functions $\mathcal{E} = \mathcal{E}(x, y, z, t)$, $\mathcal{S} = \mathcal{S}(x, y, z, t)$ and $\mathcal{Q} = \mathcal{Q}(x, y, z, t)$. While $\mathcal{E}$ and $\mathcal{S}$ satisfy (6.151) and (6.152) (in three dimensions), respectively, $\mathcal{Q}$ is also governed by the Laplace equation

$$\nabla^2\mathcal{Q} = 0 \tag{6.180}$$

As commented by Verruijt [51], while $\mathcal{Q}$ is related to the Biot functions Ψ_x and Ψ_y, there is no simple algebraic relation exists among them. Hence the function $\mathcal{Q}$ is to be regarded as an ad hoc addition to the McNamee-Gibson functions.

The displacement and stress expressions are

$$u_x = -\frac{\partial\mathcal{E}}{\partial x} + z\frac{\partial\mathcal{S}}{\partial x} + 2\frac{\partial\mathcal{Q}}{\partial y} \tag{6.181}$$

$$u_y = -\frac{\partial\mathcal{E}}{\partial y} + z\frac{\partial\mathcal{S}}{\partial y} - 2\frac{\partial\mathcal{Q}}{\partial x} \tag{6.182}$$

$$u_z = -\frac{\partial\mathcal{E}}{\partial z} + z\frac{\partial\mathcal{S}}{\partial z} - (3-4\nu_u)\mathcal{S} \tag{6.183}$$

$$\zeta = -\frac{GS}{\eta}\nabla^2\mathcal{E} \tag{6.184}$$

$$\sigma_{xx} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial x^2} + z\frac{\partial^2\mathcal{S}}{\partial x^2} - 2\nu_u\frac{\partial\mathcal{S}}{\partial z} + 2\frac{\partial^2\mathcal{Q}}{\partial x\partial y}\right] \tag{6.185}$$

$$\sigma_{yy} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial y^2} + z\frac{\partial^2\mathcal{S}}{\partial y^2} - 2\nu_u\frac{\partial\mathcal{S}}{\partial z} - 2\frac{\partial^2\mathcal{Q}}{\partial x\partial y}\right] \tag{6.186}$$

$$\sigma_{zz} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial z^2} + z\frac{\partial^2\mathcal{S}}{\partial z^2} - 2(1-\nu_u)\frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.187}$$

$$\sigma_{xy} = 2G\left[-\frac{\partial^2\mathcal{E}}{\partial x\partial y} + z\frac{\partial^2\mathcal{S}}{\partial x\partial y} - \frac{\partial^2\mathcal{Q}}{\partial x^2} + \frac{\partial^2\mathcal{Q}}{\partial y^2}\right] \tag{6.188}$$

$$\sigma_{yz} = 2G\left[-\frac{\partial^2\mathcal{E}}{\partial y\partial z} + z\frac{\partial^2\mathcal{S}}{\partial y\partial z} - (1-2\nu_u)\frac{\partial\mathcal{S}}{\partial y} - \frac{\partial^2\mathcal{Q}}{\partial x\partial z}\right] \tag{6.189}$$

$$\sigma_{xz} = 2G\left[-\frac{\partial^2\mathcal{E}}{\partial x\partial z} + z\frac{\partial^2\mathcal{S}}{\partial x\partial z} - (1-2\nu_u)\frac{\partial\mathcal{S}}{\partial x} + \frac{\partial^2\mathcal{Q}}{\partial y\partial z}\right] \tag{6.190}$$

$$p = -\frac{G}{\eta}\left[\nabla^2\mathcal{E} - \frac{2(\nu_u-\nu)}{1-\nu}\frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.191}$$

For the incompressible constituents case, some equations are reduced

$$u_z = -\frac{\partial\mathcal{E}}{\partial z} + z\frac{\partial\mathcal{S}}{\partial z} - \mathcal{S} \tag{6.192}$$

$$\zeta = -\nabla^2\mathcal{E} \tag{6.193}$$

$$\sigma_{xx} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial x^2} + z\frac{\partial^2\mathcal{S}}{\partial x^2} - \frac{\partial\mathcal{S}}{\partial z} + 2\frac{\partial^2\mathcal{Q}}{\partial x\partial y}\right] \tag{6.194}$$

$$\sigma_{yy} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial y^2} + z\frac{\partial^2\mathcal{S}}{\partial y^2} - \frac{\partial\mathcal{S}}{\partial z} - 2\frac{\partial^2\mathcal{Q}}{\partial x\partial y}\right] \tag{6.195}$$

$$\sigma_{zz} = 2G\left[\nabla^2\mathcal{E} - \frac{\partial^2\mathcal{E}}{\partial z^2} + z\frac{\partial^2\mathcal{S}}{\partial z^2} - \frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.196}$$

$$\sigma_{yz} = 2G\left[-\frac{\partial^2\mathcal{E}}{\partial y\partial z} + z\frac{\partial^2\mathcal{S}}{\partial y\partial z} - \frac{\partial^2\mathcal{Q}}{\partial x\partial z}\right] \tag{6.197}$$

$$\sigma_{xz} = 2G\left[-\frac{\partial^2\mathcal{E}}{\partial x\partial z} + z\frac{\partial^2\mathcal{S}}{\partial x\partial z} + \frac{\partial^2\mathcal{Q}}{\partial y\partial z}\right] \tag{6.198}$$

$$p = -2G\left[\frac{1-\nu}{1-2\nu}\nabla^2\mathcal{E} - \frac{\partial\mathcal{S}}{\partial z}\right] \tag{6.199}$$

while others remains the same.

The system of equations (6.151), (6.152) and (6.180) is typically solved by Laplace transform in time and Fourier or Hankel transform in space. The resultant is a set of linear algebraic equations in terms of the transformed parameters, whose solution can easily be found. The difficulty lies in the inversion of the transformations. The inversion is often done numerically

References

1. Altay GA, Dökmeci MC (1998) A uniqueness theorem in Biot's poroelasticity theory. Zeitschrift für Angewandte Mathematik und Physik 49(5):838–846
2. Aris R (1990) Vectors, tensors and the basic equations of fluid mechanics. Dover, New York, 320pp
3. Augustin MA (2015) A method of fundamental solutions in poroelasticity to model the stress field in geothermal reservoirs. Birkhäuser, Basel, 235pp

4. Auriault JL, Sanchez-Palencia E (1977) Étude du comportement d'un milieu poreux saturé déformable (Study of macroscopic behavior of a deformable saturated porous medium). Journal de Mécanique 16(4):575–603
5. Babu Shanker N, Sarma KS, Ratnam MV (1978) Consolidation due to shear loads distributed over a rectangular area. Int J Numer Anal Methods Geomech 2(4):329–342
6. Bear J (1972) Dynamics of fluids in porous media. American Elsevier (also published by Dover, 1988), New York, 764pp
7. Bear J, Bachmat Y (1990) Introduction to modeling phenomena of transport in porous media. Kluwer, Dordrecht/Boston, 553pp
8. Bear J, Cheng AHD (2010) Modeling groundwater flow and contaminant transport. Springer, Dordrecht/London, 834pp
9. Biot MA (1956) General solutions of the equations of elasticity and consolidation for a porous material. J Appl Mech ASME 78:91–96
10. Boussinesq J (1885) Applications des potentiels a l'étude de l'équilibre et du mouvement des solides élastique (Potential applications to the study of equilibrium and motion of elastic solids), Gauthiers-Villars, Paris, 721pp
11. Carman PC (1937) Fluid flow through a granular bed. Trans Inst Chem Eng 15:150–156
12. Carman PC (1956) Flow of gases through porous media. Butterworths Scientific Publications, London, 182pp
13. Cheng AHD, Cheng DT (2005) Heritage and early history of the boundary element method. Eng Anal Bound Elem 29(3):268–302
14. Costabel M (1988) Boundary integral operators on Lipschitz domains: elementary results. SIAM J Math Anal 19(3):613–626
15. Darcy H (1856) Les Fontaines Publiques de la Ville de Dijon (The public fountains of the city of Dijon). Dalmont, Paris, 647pp
16. Deresiewicz H, Skalak R (1963) On the uniqueness in dynamic poroelasticity. Bull Seismol Soc Am 53(4):783–788
17. Detournay E, Cheng AHD (1993) Fundamentals of poroelasticity. In: Fairhurst C (ed) Comprehensive rock engineering: principles, practice and projects. Analysis and design method, vol II. Pergamon Press, New York, pp 113–171
18. Ene HI, Poliševski D (1987) Thermal flow in porous media. D. Reidel, Dordrecht/Boston, 194pp
19. Epstein N (1989) On tortuosity and the tortuosity factor in flow and diffusion through porous media. Chem Eng Sci 44(3):777–779
20. Gibson RE, Knight K, Taylor PW (1963) A critical experiment to examine theories of three dimensional consolidation. In: Proceedings, European conference on soil mechanics and foundation engineering, Weisbaden, vol 1, pp 69–76
21. Gibson RE, Schiffman RL, Pu SL (1970) Plane strain and axially symmetric consolidation of a clay layer on a smooth impervious base. Q J Mech Appl Math 23:505–519
22. Harnpattanapanich T, Vardoulakis I (1987) Numerical Laplace-Fourier transform inversion technique for layered soil consolidation problems. 2. Gibson soil layer. Int J Numer Anal Methods Geomech 11(1):103–112
23. Hornung U (ed) (1997) Homogenization and porous media. Springer, New York, 279pp
24. Hubbert MK (1940) The theory of ground-water motion. J Geol 48(8):785–944
25. Jaswon MA, Symm GT (1977) Integral equation methods in potential theory and elastostatics (computational mathematics and applications). Academic, London/New York, 301pp
26. Kellogg OD (1953) Foundations of potential theory. Dover, New York, 400pp
27. Kirchhoff G (1859) Über das Gleichgewicht und die Bewegung eines unendlich dünnen elastischen Stabes (On the balance and movement of an infinitely thin elastic rod) Journal fur Mathematik (Crelle J) 56:285–313
28. Kozeny J (1927) Über kapillare leitung des wassers im boden (Aufstieg, versickerung und anwendung auf die bewässerung) (About capillary conduction of water in the soil (rise, percolation and application on irrigation)). Sitzungsber. Akad. Wiss. in Wien, Mathematisch-Naturwissenschaftliche Klasse, Abt. IIa. 136:271–309

29. Love AEH (1892) A treatise on the mathematical theory of elasticity, vol 1. Cambridge University Press, Cambridge, 354pp
30. Mazzucato AL, Nistor V (2010) Well-posedness and regularity for the elasticity equation with mixed boundary conditions on polyhedral domains and domains with cracks. Arch Ration Mech Anal 195(1):25–73
31. McNamee J, Gibson RE (1960) Displacement functions and linear transforms applied to diffusion through porous elastic media. Q J Mech Appl Math 13(1):98–111
32. McNamee J, Gibson RE (1960) Plane strain and axially symmetric problems of the consolidation of a semi-infinite stratum. Q J Mech Appl Math 13(2):210–227
33. Mindlin RD (1936) Note on the Galerkin and Papkovitch stress functions. Bull Am Math Soc 42:373–376
34. Nelson PH (1994) Permeability-porosity relationships in sedimentary rocks. Log Anal 35(3):38–62
35. Neuber H (1934) Ein neuer ansatz zur lösung der grundgleichungen der elastizitätstheorie (A new approach for the solution of the basic equations of elasticity theory). Zeitschrift für Angewandte Mathematik Und Mechanik 14:203–212
36. Neuber H (1946) Theory of notch stresses: principle for exact stress calculations (transl. from Kerbspannungslehre: Grundlagen für genaue Spannungsberechnung). J.W. Edwards Co., Ann Arbor, 181pp
37. Nikolaevskiy VN (1984) Mechanics of porous and cracked media (in Russian). Nedra, Moscow, 232pp
38. Nikolaevskiy VN, Basniev KS, Gorbunov AT, Zotov GA (1970) Mechanics of saturated porous media (in Russian). Nedra, Moscow, 339pp
39. Papkovitch PF (1932) Solution générale des équations differentielles fondamentales d'élasticité exprimée par trois functions harmoniques (General solution of the fundamental differential equations of elasticity expressed by three harmonic functions). Comptes Rendus 195:513–515
40. Papkovitch PF (1932) Expressions générales des composantes des tensions, ne refermant comme fonctions arbitraires que des fonctions harmoniques (General expressions of stress components, not closing as arbitrary functions of harmonic functions). Comptes Rendus 195:754–756
41. Rice JR, Cleary MP (1976) Some basic stress diffusion solutions for fluid-saturated elastic porous media with compressible constituents. Rev Geophys 14(2):227–241
42. Rouse H (1961) Fluid mechanics for hydraulic engineers. Dover, New York, 422pp
43. Saada AS (1993) Elasticity, theory and applications, 2nd edn. Kreiger, Malabar, 775pp
44. Schiffman RL, Chen AT-F, Jordan JC (1969) An analysis of consolidation theories. J Soil Mech Found Div ASCE 95(SM1):285–312
45. Schiffman RL, Fungaroli AA (1965) Consolidation due to tangential loads. In: Proceedings of 6th international conference on soil mechanics and foundation engineering, Montreal, pp 188–192
46. Showalter RE (2000) Diffusion in poro-elastic media. J Math Anal Appl 251(1):310–340
47. Sokolnikoff IS (1956) Mathematical theory of elasticity, 2nd edn. McGraw-Hill, New York, 476pp
48. Vardoulakis I, Harnpattanapanich T (1986) Numerical Laplace Fourier-transform inversion technique for layered-soil consolidation problems. 1. Fundamental-solutions and validation. Int J Numer Anal Methods Geomech 10(4):347–365
49. Verruijt A (1969) Elastic storage of aquifers. In: DeWiest RJM (ed) Flow through porous media. Academic, New York, pp 331–376
50. Verruijt A (1969) The completeness of Biot's solution of the coupled thermoelastic problem. Q Appl Math 26(4):485–490
51. Verruijt A (1971) Displacement functions in theory consolidation or in thermoelasticity. Zeitschrift Für Angewandte Mathematik Und Physik 22(5):891–898

Chapter 7
Analytical Solution

The stresses in any point of a section through a mass of earth can be computed from the total principal stresses n'_{I}, n'_{II}, and n'_{III} which act in this point. If the voids of the earth are filled with water under a stress n_w, the total principal stresses consist of two parts. One part, n_w, acts in the water and in the solid in every direction with equal intensity. It is called the neutral stress. The balance, $n_{\mathrm{I}} = n'_{\mathrm{I}} - n_w$, $n_{\mathrm{II}} = n'_{\mathrm{II}} - n_w$, and $n_{\mathrm{III}} = n'_{\mathrm{III}} - n_w$ represents an excess over the neutral stress n_w and it has its seat exclusively in the solid phase of the earth. This fraction of the total principal stresses will be called the effective principal stresses . . .

—Karl Terzaghi (1936)

Once we set up a boundary value problem consisting of a set of partial differential equations, together with a set of well-posed boundary and initial conditions, our goal is to find the solution of the system. It is generally desirable to find *analytical solutions*, that is, solutions explicitly expressed in terms of known mathematical functions. However, for problems arising from engineering applications, this is generally not feasible; hence *numerical solutions* are sought. Nevertheless, there are a good number of analytical solutions that can be elegantly derived. In this chapter, we shall examine some of these solutions, and their applications.

7.1 Review of Early Work

A comprehensive review of analytical solutions for geomechanics was presented by Selvadurai [144], including problems governed by poroelasticity equations. The review below extends Selvadurai's work; but is limited to the quasi-static poroelasticity theory. The poroelastodynamics theory associated with the wave propagation phenomena will be reviewed in Chap. 9.

Biot in his original paper [14] that established the theory of poroelasticity also provided the first solution by examining the problem of one-dimensional consolidation. Biot [15] and Biot and Clingan [17] continued to solve the multi-dimensional problems of settlement of a half space under a rectangular surface

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_7

load. Biot and Clingan [18] subsequently examined the plane strain problem of soil settlement subjected to a concentrated load on a flexible, pervious or impervious slab resting on the surface of a half space.

Paria [122] formulated the axisymmetric problem related to a consolidating half space and derived the axial stress. Axisymmetric, plane strain, and three-dimensional problems related to the consolidation of semi-infinite regions were presented by McNamee and Gibson [109, 110] and Gibson and McNamee [73]. Heinrich and Desoyer [80] solved similar problems, but used a different poroelasticity formulation from Biot's [14]. The consolidation due to a tangential load over a circular region of a poroelastic half space was presented by Schiffman and Fungaroli [142]. Parton [123] also considered the axisymmetric consolidation problem for a half space region. Agbezuge and Deresiewicz [10] solved the half space problem of a circular footing.

Problems related to plane strain and axisymmetric consolidation of a layer of finite thickness with a smooth impervious base were derived by Gibson et al. [74] and Booker [19]. The problem of a layer with arbitrary surface loading was investigated by Booker and Small [23, 24]. For multilayered soils, the integral transform technique, such as the Laplace, the Fourier, and the Hankel transforms, has been applied to reduce the governing partial differential equations to a set of ordinary differential equations, or algebraic equations. The technique has been utilized by many authors [25, 77, 147, 163] to solve consolidation problems due to surface and internal loads. The inversion of the integral transforms, however, is typically performed by a numerical technique; hence these solutions are semi-analytical solutions. The inclusion of the dynamic (inertial) effect in a multilayered system was studied by Bougacha et al. [27, 28] and Rajapakse and Senjuntichai [130].

Mandel [106, 107] solved the problem of a poroelastic material sandwiched between two rigid and impermeable plates, subjected to a suddenly applied force. Plan strain, cylinder, and hollow cylinder geometries were examined. Mandel's original work [107] gave only the pore pressure solution. The full solution of plane strain problem for isotropic porous material was derived by Cheng and Detournay [37], which was extended to transversely isotropic material by Abousleiman et al. [5]. The Mandel problem subject to a cyclic force has been investigated by Kameo et al. [94].

Cryer [50] investigated the problem of a poroelastic sphere with permeable boundary subjected to an external radial compressive stress. The non-monotonic pressure response observed in the Mandel [107] and the Cryer [50] problem has been called the Mandel-Cryer effect (see Sect. 1.3.2). Nowinski and Davis [118] solved the problem of a thin-walled spherical shell subjected to uniform external pressure.

Hollow circular cylinder problem was first studied by Kenyon [98]. Radially deforming solid cylinder problem was investigated by Helm [81] and Detournay and Cheng [64]. Cylinders subjected to plane deviatoric and axial stresses were examined by Abousleiman et al. [6]. Solution was extended to transversely isotropic cylinders by Abousleiman and Cui [3]. A compilation of the solid and hollow cylinder and borehole problems was given in Kanj and Abousleiman [4, 95]. The dynamic responses of solid and hollow cylinders were examined by Tajuddin [157] and Reddy and Tajuddin [131], respectively.

The problem of consolidation around pore pressure meters, which is modeled as a spherical cavity with permeable surface, and with either a rigid surface or surface free from traction, was examined by de Josselin de Jong [56]. The consolidation due to a rigid sphere embedded in a poroelastic half space and displaced in one direction was studies by de Josselin de Jong [57] using the stress function technique. Rice and Cleary [134] also looked into the spherical cavity problem. Spherical cavity subjected to shear stress on the surface, or filled with a permeable inclusion was examined by Rice et al. [136].

The problem of pore pressure distribution and effective stress around a borehole was first investigated as a plane strain poroelastic problem by Geertsma [71]. Rice and Cleary [134] solved the problem under axisymmetric loading conditions. Carter and Booker [33, 34] and Detournay and Cheng [62] extended the solution to allow non-hydrostatic far-field stress loading condition. The solution has been extended to a borehole inclined to the principle stress directions by Cui et al. [51, 52]. The solution of a borehole loaded on a finite section was given by Rajapakse [128]. Abousleiman and Cui [3] examined borehole problems in a transversely isotropic medium. The dynamic response of a borehole, that is, with the inclusion of the inertia effect, was investigated by Senjuntichai and Rajapakse [146].

For indentation problems, Nowinski [117] solved the problem of a spherical poroelastic body in contact with a spherical seat. Agbezuge and Deresiewicz [9, 59] examined the frictionless indentation of a poroelastic half space by permeable and impermeable spherical rigid indenters. Agbezuge and Deresiewicz [10] extended the analysis to include the axisymmetric indentation of a poroelastic half space by a rigid circular indenter with a flat base. The frictionless indentation problem for the isotropic poroelastic half space was also examined by Chiarella and Booker [44]. Other contact and indentation problems were investigated by Gaszyński and Szefer [69, 70, 153], and Selvadurai and Yue [145, 172].

The pore pressure and effective stress response of a porous seabed below a propagating water wave was examined by Yamamoto et al. [171] and Madsen [105]. The solution was extended to layered seabed by Yamamoto [169]. Yamamoto [170] also included the effect of inertia and Coulomb damping in the solution. The dynamic response of porous seabed was also studied by Huang and Song [86]. A review of analytical solutions of seabed response subject to water wave loading was presented in Jeng [92].

Rice and Simons [135, 148] analyzed steady state moving shear faults in a two-dimensional poroelastic space. The small time and large time behavior of a Griffith type fracture was investigated by Detournay and Cheng [63]. Asymptotic solutions of Griffith and moving semi-infinite fractures in opening and shearing mode were examined by Atkinson and Craster [13, 49].

Other analytical solutions derived include the generalized plane strain problems, such as torsion, warping, and pure bending of a beam or plate [36, 119]. Biot [16] examined the buckling of a plate subject to axial load. Taking the advantage of large aspect ratio of beams, plates, and shells, these theories have been developed for poroelasticity [104, 154, 155, 166, 173] as counterparts to elasticity theories.

When the forcing function is concentrated, that is, applied to a point or to a line, the solution becomes singular. These singular solutions are referred to as

Green's functions. When the domain of such solution is unbounded, such that there is no boundary condition to satisfy, the solution is called the *free-space Green's function* or *fundamental solution*. Nur and Booker [120] and Booker [20] were the first to derive the fundamental solution of a shear dislocation in two-dimensional space for the incompressible constituent model. Rice and Cleary [134] presented the compressible constituent fundamental solutions for a shear dislocation with permeable boundary, an opening dislocation with impermeable boundary, and a line force in two dimensions. Rudnicki [139] supplemented the shear and opening dislocation solutions by reversing the pressure and flux boundary conditions of Rice and Cleary's solution [134].

Cleary [46] provided the three-dimensional fundamental solutions of point force and fluid source to within a few evolutional integrals not in closed form. The fluid source solution of Cleary [46] was corrected by Rudnicki [137]. The closed-form point force and point source fundamental solutions in both two and three dimensions were given by Rudnicki [138] and Cheng and Predeleanu [42]. The displacement discontinuity solutions were presented by Detournay and Cheng [61]. The complete derivation of closed-form fundamental solutions in two and three dimensions for instantaneous and continuous fluid source, dipole, dilatation, and force, and solid force and displacement discontinuity has been presented by Cheng and Detournay [38]. The fundamental solutions for transversely isotropic poroelastic medium with forces aligned with the material anisotropy axes were given by Taguchi and Kurashige [156].

Two- and three-dimensional fundamental solutions were derived by Cheng [35] in the Laplace transform domain (see also [37, 39]), and by Cheng and Liu [41] in the frequency domain. Solutions of moving dislocations and fluid source are found in Cleary [47] and Rudnicki [140]. The moving force and source solutions were presented in Cheng and Liggett [40].

For problems of land subsidence due to fluid withdrawal, Geertsma [71, 72] was the first to present a solution of the steady state settlement at the surface, utilizing the elasticity solution of nuclei of strain in half space [113], and the thermal stress-pore pressure analogy (see Sect. 1.2.7). Booker and Carter [21] solved the same problem of a point sink embedded in a poroelastic half space, and provided closed form solution. We should note that the steady state poroelasticity solutions are essentially elasticity solutions with body force.

Using integral transforms applied either directly to the physical variables, or to the potentials such as McNamee and Gibson displacement functions [109], a variety of singular solutions have been obtained. These Green's functions include loading applied to the surface or interior of half plane, half space, or layered geometry, with different surface conditions. These solutions are not analytically inverted; hence numerical evaluation is needed. The solutions presented by Apirathvorakij and Karasudhi [12] and Niumpradit and Karasudhi [116] relate to uniform circular patch loads acting in the axial and horizontal directions at the interior of a poroelastic half space. Booker and Carter [22] solved the subsidence problem due to a point sink in a half space with anisotropic permeability. Puswewala and Rajapakse [126] and Rajapakse and Senjuntichai [129] obtained Green's function for the half space in

the form of point and ring force and source, respectively for the incompressible and compressible constituent cases. Kanok-Rukulchai and Chau [97] presented fluid point source solution for half space. The complete set of Green's functions in a multilayered, isotropic, and poroelastic half space that includes the total force, fluid force, fluid dilatation and source, and solid dislocation, were presented in Pan [121].

7.2 Uniaxial Strain

We first consider problems in which the solid and fluid are allowed to deform only in one spatial dimension, and the spatial variation of the deformation can exist only in the same direction, due to the constraint of geometry and loading conditions. Selecting z as that direction, the above statement can be expressed in the mathematical form as

$$u_z = u_z(z,t); \quad U_z = U_z(z,t); \quad u_x = u_y = U_x = U_y = 0 \tag{7.1}$$

The above functional dependence also leads to the following conditions for the strain quantities,

$$\begin{aligned} &e_{zz} = e_{zz}(z,t); \quad e_{xx} = e_{yy} = e_{xy} = e_{xz} = e_{yz} = 0; \\ &q_z = q_z(z,t); \quad q_x = q_y = 0; \\ &\zeta = \zeta(z,t); \quad p = p(z,t) \end{aligned} \tag{7.2}$$

We note that only the z-components exist. For stresses, however, the cross components can exist.

We shall call problems under these constraints the *uniaxial strain problems*. It will be demonstrated in the section below, the governing equations are one-dimensional.

7.2.1 Isotropy

We first examine the case of isotropic material. Based on the functional dependency in (7.2), the stress components given by the isotropic constitutive equation (2.29) simplify to the following

$$\sigma_{zz} = \frac{2G(1-\nu)}{1-2\nu} e_{zz} - \alpha p \tag{7.3}$$

$$\sigma_{xx} = \sigma_{yy} = \frac{2G\nu}{1-2\nu} e_{zz} - \alpha p \tag{7.4}$$

$$\sigma_{xy} = \sigma_{xz} = \sigma_{yz} = 0 \tag{7.5}$$

Or, expressed in terms of ζ, instead of p, (7.3) and (7.4) become

$$\sigma_{zz} = \frac{2G(1-\nu_u)}{1-2\nu_u} e_{zz} - \alpha M \zeta \tag{7.6}$$

$$\sigma_{xx} = \sigma_{yy} = \frac{2G\nu_u}{1-2\nu_u} e_{zz} - \alpha M \zeta \tag{7.7}$$

As for pressure, we have

$$p = -\alpha M e_{zz} + M\zeta \tag{7.8}$$

Based on the above simplified constitutive equations, we may examine the Skempton pore pressure effect under the uniaxial strain condition. At the instant of an applied vertical load σ_{zz}, the undrained condition $\zeta = 0$ exists. Equations (7.6) and (7.8) show the following relations between the applied vertical stress and the responding pore pressure

$$p = -\frac{\alpha M(1-2\nu_u)}{2G(1-\nu_u)} \sigma_{zz} = -\frac{B(1+\nu_u)}{3(1-\nu_u)} \sigma_{zz} = -\frac{\eta}{GS} \sigma_{zz} \tag{7.9}$$

In the above, the various equivalent expressions are obtained through the conversion relations in Sect. B.1.

As all quantities are functions of (z, t) only, it is easy to show that the equilibrium equation (6.61) is trivial, except for the z component, which becomes

$$\frac{\partial \sigma_{zz}}{\partial z} = 0 \tag{7.10}$$

From the above, we immediately conclude that σ_{zz} is independent of the z-coordinate. We then observe that the functional dependency of the stress components are

$$\sigma_{zz} = \sigma_{zz}(t); \quad \sigma_{xx} = \sigma_{xx}(z,t); \quad \sigma_{yy} = \sigma_{yy}(z,t) \tag{7.11}$$

The differential equation for u_z is deduced from the Navier equation (6.64)

$$\frac{2G(1-\nu)}{1-2\nu} \frac{\partial^2 u_z}{\partial z^2} - \alpha \frac{\partial p}{\partial z} = 0 \tag{7.12}$$

The diffusion equation can be obtained from (6.69) as

$$\frac{\partial p}{\partial t} - \kappa M \frac{\partial^2 p}{\partial z^2} = -\alpha M \frac{\partial e_{zz}}{\partial t} \tag{7.13}$$

By expressing e_{zz} in terms of p and σ_{zz} using (7.3), the above equation becomes

$$\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial z^2} = -\frac{\eta}{GS}\frac{d\sigma_{zz}}{dt} \tag{7.14}$$

Here we note that the right hand side of (7.14) is changed into an ordinary derivative, due to the functional dependency of σ_{zz}. This equation was earlier derived as (6.112) as a consequence of an irrotational displacement field.

7.2.2 Transverse Isotropy

As many geological materials are transversely isotropic due to the geological bedding process, it is of interest to investigate such special case here. Assuming that the direction of the deformation coincides with the axis of material symmetry (see Sect. 5.2.2), the constitutive equations are simplified from (5.27), (5.28), (5.46), and (5.53), based on the uniaxial strain condition, as

$$\sigma_{zz} = M_{zz}\, e_{zz} - \alpha_z p \tag{7.15}$$

$$\sigma_{xx} = \sigma_{yy} = M_{xz}\, e_{zz} - \alpha p \tag{7.16}$$

$$p = -\alpha_z M e_{zz} + M\zeta \tag{7.17}$$

Under undrained condition, $\zeta = 0$, the Skempton pore pressure effect can be expressed as

$$p = -\frac{\alpha_z M}{M_{zz} + \alpha_z^2 M}\,\sigma_{zz} \tag{7.18}$$

The Navier equation, equivalent to (7.12), is given by

$$M_{zz}\frac{\partial^2 u_z}{\partial z^2} - \alpha_z \frac{\partial p}{\partial z} = 0 \tag{7.19}$$

From Darcy's law (6.26), continuity equation (6.45), and constitutive equation (7.17), we obtain the diffusion equation

$$\frac{\partial p}{\partial t} - \kappa_z M\frac{\partial^2 p}{\partial z^2} = -\alpha_z M\frac{\partial e_{zz}}{\partial t} \tag{7.20}$$

where we assume that the principle axes of permeability coefficient coincide with the material symmetry axes, and κ_z is the vertical permeability coefficient. Using (7.15) to eliminate e_{zz} on the right hand side, we obtain the equation that is equivalent to (7.14) as

$$\frac{\partial p}{\partial t} - c_z\frac{\partial^2 p}{\partial z^2} = -\frac{\alpha_z M}{M_{zz} + \alpha_z^2 M}\frac{d\sigma_{zz}}{dt} \tag{7.21}$$

where

$$c_z = \frac{\kappa_z M M_{zz}}{M_{zz} + \alpha_z^2 M} \tag{7.22}$$

is the consolidation coefficient in the vertical direction.

We notice that the two governing equations for the transversely isotropic case, (7.19) and (7.21), are the same as the isotropic case, (7.12) and (7.14), and differ only in the coefficients of the differential equations. Hence the mathematical solutions for these two material cases are the same, requiring only the proper substitution of the material constants.

7.3 One-Dimensional Consolidation Problem

7.3.1 Terzaghi's Consolidation Problem

As a first example of analytical solution, we shall examine the one-dimensional consolidation problem investigated by Terzaghi [158], but based on the poroelasticity theory. Consider an initially undisturbed soil layer of thickness L, resting on a rigid impermeable base (see Fig. 7.1a). A uniformly distributed downward surcharge is instantly applied to the top of the layer and is maintained at that magnitude. The top surface of the layer is drained, meaning that the pressure is kept at atmospheric. These physical descriptions can be translated into the mathematical conditions as follows:

$$\begin{aligned} \sigma_{zz} = -P_o\,\mathrm{H}(t-0), \quad p = 0; \qquad & \text{at } z = 0; \qquad t \geq 0 \\ u_z = 0, \quad \frac{\partial p}{\partial z} = 0; \qquad & \text{at } z = L; \qquad t \geq 0 \\ p = 0; \qquad & \text{at } t = 0^-; \quad 0 \leq z \leq L \end{aligned} \tag{7.23}$$

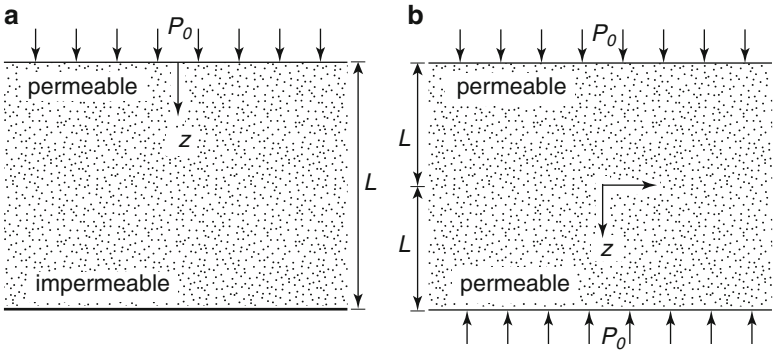


Fig. 7.1 One-dimensional consolidation problems: (**a**) Terzaghi problem; and (**b**) compression of a finite thickness permeable layer

In the above, P_o is the magnitude of the vertical stress, and $\mathrm{H}(t-0)$ is the Heaviside unit step function.

Before moving on to solve this problem, we shall observe a problem with similar geometry, as depicted in Fig. 7.1b. In this case, a permeable layer of thickness $2L$ is compressed by a uniform stress P_o at the top and bottom. The mathematical solution for this problem is the same as that for case a, if the origin of the coordinate is shifted to the center of the layer, and the solution is valid between $-L \leq z \leq L$. We hence shall not repeat this solution.

Pore Pressure Response To solve the above one-dimensional consolidation problem, we first substitute the stress condition in (7.23) into the right hand side of (7.14), and obtain

$$\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial z^2} = \frac{P_o \eta}{GS}\delta(t-0) \tag{7.24}$$

where $\delta(t-0)$ is the Dirac delta function. The Dirac delta function is the result of differentiation with respect to time the Heaviside unit step function. The above equation can be solved with the pair of pressure boundary conditions and the initial condition in (7.23). To solve it, we first need to express the pressure solution into a homogeneous and a particular part,

$$p = p_h + p_p \tag{7.25}$$

With a particular solution of this form

$$p_p = \frac{P_o \eta}{GS} H(t-0) \tag{7.26}$$

we can remove the right hand side of (7.24). The homogeneous governing equation is then solved with a set of modified boundary and initial conditions.

As an alternative way to solve the system, we can modify the mathematical problem by placing the time origin at $t = 0^+$, instead of 0^-. In that case, the boundary condition for the vertical stress is modified to $\sigma_{zz} = -P_o$ for all $t > 0$. As σ_{zz} now is a constant in time, (7.14) becomes the homogeneous diffusion equation

$$\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial z^2} = 0 \tag{7.27}$$

From the physical reasoning of Skempton pore pressure effect, at the instant of the applied load, there is a pore pressure rise of the magnitude

$$p(z, 0^+) = \frac{P_o \eta}{GS} \tag{7.28}$$

given by (7.9). Equation (7.28) then gives the initial condition.

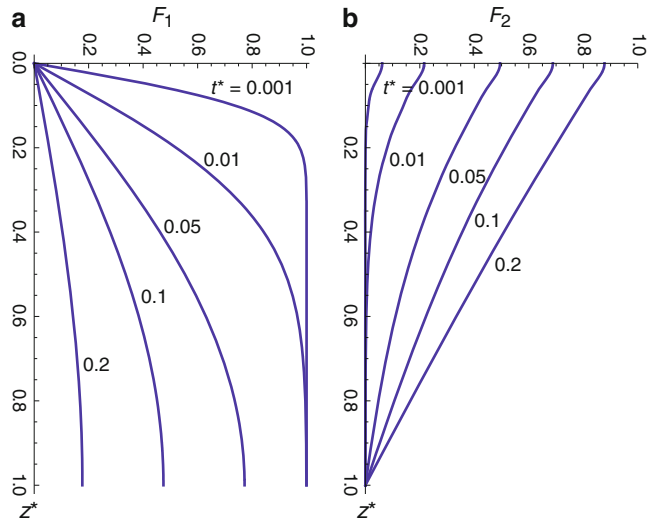


Fig. 7.2 Plot of two functions: (**a**) F_1, and (**b**) F_2, versus z^* for various t^* values

Either way we consider the problem, the diffusion equation, (7.24) or (7.27), can be solved with the proper initial and boundary conditions to give the following solution [14]

$$p(z,t) = \frac{P_o\,\eta}{GS} F_1(z^*, t^*) \tag{7.29}$$

where

$$F_1(z^*, t^*) = \sum_{m=1,3,\cdots}^{\infty} \frac{4}{m\pi} \sin\left(\frac{m\pi z^*}{2}\right) e^{-m^2\pi^2 t^*} \tag{7.30}$$

with $z^* = z/L$ and $t^* = ct/4L^2$ are dimensionless distance and time, respectively. The function F_1 is plotted versus z^* in Fig. 7.2a for various t^* values, representing the normalized pressure profiles in the depth direction at various times. It can be shown that F_1 has such asymptotic behaviors

$$F_1(z^*, 0) = 1; \quad F_1(z^*, \infty) = 0 \tag{7.31}$$

We hence observe an initial pore pressure given by (7.28), which is then monotonically dissipated to zero as time approaches infinity.

Settlement The displacement u_z is found by substituting the pressure solution into (7.12), integrating with respect to z twice, subject to the proper boundary conditions. We then obtain

$$u_z(z,t) = \frac{P_o L(1-2\nu_u)}{2G(1-\nu_u)}(1-z^*) + \frac{P_o L(\nu_u-\nu)}{2G(1-\nu_u)(1-\nu)} F_2(z^*,t^*) \tag{7.32}$$

where

$$F_2(z^*,t^*) = \sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2\pi^2} \cos\left(\frac{m\pi z^*}{2}\right)\left(1-e^{-m^2\pi^2 t^*}\right) \tag{7.33}$$

In Fig. 7.2b, F_2 is plotted versus z^* for various t^* values. It can be shown that F_2 has these asymptotic behaviors

$$F_2(z^*,0) = 0; \quad F_2(z^*,\infty) = (1-z^*) \tag{7.34}$$

We hence obtain the following displacement profiles at $t = 0$ and ∞

$$u_z(z,0) = \frac{P_o L(1-2\nu_u)}{2G(1-\nu_u)}(1-z^*) \tag{7.35}$$

$$u_z(z,\infty) = \frac{P_o L(1-2\nu)}{2G(1-\nu)}(1-z^*) \tag{7.36}$$

These displacements vary linearly with depth, and are elasticity solutions. They are identical expressions, except for the Poisson ratio: at $t = 0$, it takes the undrained value, and at $t \to \infty$, the drained value. At the top of the layer, the settlement is given by

$$\begin{aligned} u(0,t) &= \frac{P_o L(1-2\nu_u)}{2G(1-\nu_u)} \\ &\quad + \frac{P_o L(\nu_u-\nu)}{2G(1-\nu_u)(1-\nu)} \sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2\pi^2}\left(1-e^{-m^2\pi^2 t^*}\right) \\ &= \frac{P_o L(1-2\nu)}{2G(1-\nu)} - \frac{P_o L(\nu_u-\nu)}{2G(1-\nu)(1-\nu_u)} \sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2\pi^2} e^{-m^2\pi^2 t^*} \end{aligned} \tag{7.37}$$

In the above, we have utilized the identity

$$\sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2\pi^2} = 1 \tag{7.38}$$

to derive the second equality of the equation. From the above, we can obtain the settlement at the top of the layer at $t = 0^+$ and ∞ as

$$u(0,0^+) = \frac{P_o L(1-2\nu_u)}{2G(1-\nu_u)} \tag{7.39}$$

$$u(0,\infty) = \frac{P_o L(1-2\nu)}{2G(1-\nu)} \tag{7.40}$$

with the progressive settlement between $t = \infty$ and 0^+ as

$$u(0,\infty) - u(0,0^+) = \frac{P_o L(\nu_u - \nu)}{2G(1-\nu)(1-\nu_u)} \tag{7.41}$$

Terzaghi Consolidation Problem We are now ready to compare the poroelastic consolidation problem with that of the classical Terzaghi problem. Based on the poroelasticity theory, the diffusion equation for pore pressure is coupled with the solid deformation through a term of rate of volumetric strain on the right hand side, as demonstrated in (7.13). Only under the special loading condition that the surcharge is a constant with time the diffusion equation decouples to become the same as the Terzaghi case, given as (7.27). The pore pressure solution hence is the same, given as (7.29) and (7.30).

The material constants, however, need to be reinterpreted. The material considered in Terzaghi's case is soil (clay), which is saturated with water. It corresponds to the incompressible constituent case of poroelasticity (see Sect. 4.8.3), which is characterized by the limiting material values of $\alpha = 1$ and $\nu_u = 1/2$. The consolidation coefficient c is differently defined for the two cases, as (6.72) and (6.73). The pore pressure response hence is given by taking the material limits in (7.29), which leads to $\eta/GS = 1$, and

$$p(z,t) = P_o F_1(z^*, t^*) \tag{7.42}$$

The initial pore pressure rise is then

$$p(z,0^+) = P_o \tag{7.43}$$

as compared to (7.28). The displacement is obtained from (7.32) as

$$u_z(z,t) = \frac{P_o L(1-2\nu)}{2G(1-\nu)} F_2(z^*, t^*) \tag{7.44}$$

According to (7.34), this shows that the soil stratum is initially incompressible, and deforms only after the pore water has time to be expelled. The soil settlement at the top is

$$u_z(0,t) = \frac{P_o L(1-2\nu)}{2G(1-\nu)} \left(1 - \sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2\pi^2} e^{-m^2\pi^2 t^*}\right) \tag{7.45}$$

Energy Dissipation As discussed in Sect. 6.1.4, when seepage flow takes place, there is energy dissipation. To find the energy dissipated by the consolidation process, we first find the specific flux as

$$q_z(z,t) = -\frac{2P_o\,\eta\,\kappa}{GSL}\,F_3(z^*,t^*) \tag{7.46}$$

where

$$F_3(z^*,t^*) = \sum_{m=1,3,\cdots}^{\infty} \cos\left(\frac{m\pi z^*}{2}\right) e^{-m^2\pi^2 t^*} \tag{7.47}$$

The energy dissipation rate, based on (6.17), is

$$\mathcal{D}(z,t) = \frac{1}{\kappa}\,q_z^2 = \frac{4P_o^2\,\eta^2\kappa}{G^2S^2L^2}\,F_4(z^*,t^*) \tag{7.48}$$

where

$$F_4(z^*,t^*) = \sum_{m=1,3,\cdots}^{\infty}\sum_{n=1,3,\cdots}^{\infty} \cos\left(\frac{m\pi z^*}{2}\right)\cos\left(\frac{n\pi z^*}{2}\right) e^{-(m^2+n^2)\pi^2 t^*} \tag{7.49}$$

The total energy dissipated for the entire depth and the whole time duration, per unit area, is obtained as

$$E_D = \int_0^t \int_0^L \mathcal{D}(z,t)\,dz\,dt = \frac{P_o^2\,\eta^2 L}{2G^2S} = \frac{P_o^2 L(\nu_u-\nu)}{4G(1-\nu)(1-\nu_u)} \tag{7.50}$$

In the above integration, we have utilized the orthogonal property of the cosine function. Or, comparing the above with (7.41), we observe

$$E_D = \frac{1}{2}P_o\left[u_z(0,\infty) - u_z(0,0^+)\right] \tag{7.51}$$

which can be interpreted as the work performed to the soil layer.

We can visualize this effect by observing the work energy diagram during a loading and unloading cycle, as Fig. 7.3. When a load σ_{zz} is instantly applied, the material response is elastic, hence there is a linear relation between σ_{zz} and $u_z(0,t)$. From $t = 0^+$ to ∞, the layer settle from $u_z(0,0^+)$ to $u_z(0,\infty)$ under the constant load. When the load is removed, the layer will instantly rebound with a magnitude the same as the initial deformation. Given time, the layer will return to the initial undisturbed state, that is, the origin of the diagram. In Fig. 7.3, the area contained in the parallelogram is the energy dissipated, and half of it is attributed to the loading, and the other half the unloading, process. The result is consistent with that obtained by integrating the dissipation density function, given as (7.50) and (7.51).

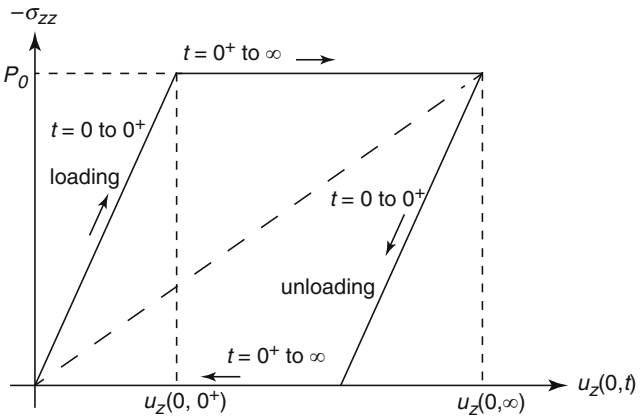


Fig. 7.3 Work done during loading and unloading of a soil stratum

7.3.2 *Loading by Fluid Pressure*

We consider next a problem of the same geometry as that in Fig. 7.1a, except that the upper surface of the soil stratum is in contact with a fluid. At $t = 0^+$, a sudden pressure increment of the magnitude p_o is applied. In this case, we have the same boundary and initial value problem as in (7.23), except that the pressure boundary condition at the top, $z = 0$, needs to be modified to $p = p_o \mathrm{H}(t - 0)$.

Solution of this problem can be obtained by the superposition of two loading modes:

- mode 1: $\sigma_{zz}(0, t) = -p_o \mathrm{H}(t - 0), \quad p(0, t) = 0$
- mode 2: $\sigma_{zz}(0, t) = 0, \quad p(0, t) = p_o \mathrm{H}(t - 0)$

with the rest of the boundary conditions remaining the same. We observe that the mode 1 problem is exactly that investigated in Sect. 7.3; hence only the mode 2 problem needs to be solved.

The pressure solution of the second mode is given by

$$p^{(2)}(z, t) = p_o[1 - F_1(z^*, t^*)] \tag{7.52}$$

which has the following limits: $p^{(2)}(z, 0^+) = 0$ and $p^{(2)}(z, \infty) = p_o$. The displacement field is again found by integrating (7.12), and we find

$$u_z^{(2)}(z, t) = -\frac{p_o L \eta}{G} F_2(z^*, t^*) \tag{7.53}$$

The surface displacement is given by

$$u_z^{(2)}(0, t) = -\frac{p_o L \eta}{G} \left(1 - \sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2 \pi^2} e^{-m^2 \pi^2 t^*}\right) \tag{7.54}$$

The above solution shows that the surface displacement is zero at $t = 0^+$, and then gradually rebounds (u_z negative) to its large time value of $u^{(2)}(0, \infty) = -p_o L\eta/G$. The rebound is a consequence of the dilation of the porous solid due to the increase of pore pressure. The analogy of this dilation to thermal expansion is discussed in Sect. 1.2.7.

The complete solution of this problem is obtained by the superposition of the two modes. Adding (7.29) to (7.52), the pressure is given by

$$p(z,t) = p_o\left[1 + \left(\frac{\eta}{GS} - 1\right) F_1(z^*, t^*)\right] \tag{7.55}$$

It varies from $p(z, 0^+) = p_o\eta/GS$ to $p(z, \infty) = p_o$. The displacement becomes

$$u_z(z,t) = \frac{p_o L(1-2\nu_u)}{2G(1-\nu_u)}(1-z^*) - \frac{p_o L[\alpha(1-\nu_u)(1-2\nu) - (\nu_u - \nu)]}{2G(1-\nu_u)(1-\nu)} F_2(z^*, t^*) \tag{7.56}$$

The surface settlement is then

$$u_z(0,t) = \frac{p_o L(1-2\nu_u)}{2G(1-\nu_u)} - \frac{p_o L[\alpha(1-\nu_u)(1-2\nu) - (\nu_u - \nu)]}{2G(1-\nu_u)(1-\nu)} \times \sum_{m=1,3,\cdots}^{\infty} \frac{8}{m^2\pi^2}\left(1 - e^{-m^2\pi^2 t^*}\right) \tag{7.57}$$

which has the following small time and large time limits:

$$u_z(0,0^+) = \frac{p_o L(1-2\nu_u)}{2G(1-\nu_u)} \tag{7.58}$$

$$u_z(0,\infty) = \frac{p_o L(1-\alpha)(1-2\nu)}{2G(1-\nu)} \tag{7.59}$$

In Fig. 7.4 the settlement as a function of dimensionless time t^* as predicted by (7.57), normalized by its initial value (7.58), is expressed as that due to mode 1, mode 2, and combined loading. The material is assumed to be Ruhr sandstone (Table 3.2). The mode 1 solution, which is the same as (7.37), shows an initial settlement, and continues to settle. The mode 2 solution starts from zero, and rebounds. The combined effect is an initial settlement given by (7.58) and a gradual rebound to the final value of (7.59), which is still a net settlement.

It is of interest to discuss the case of incompressible constituents, characterized by the material constants $\alpha = 1$ and $\nu_u = 0.5$. As the constituents are incompressible, the soil initially does not deform. Based on the Skempton effect, an instant pore pressure rise of p_o will result from the loading. As this pore pressure rise is the same as the applied pressure at the top surface, there is no further pore pressure diffusion, and the stratum remains stationary. These effects are consistent with that predicted by (7.58) and (7.59), which show that $u_z(0, 0^+) = 0$ and $u_z(0, \infty) = 0$.

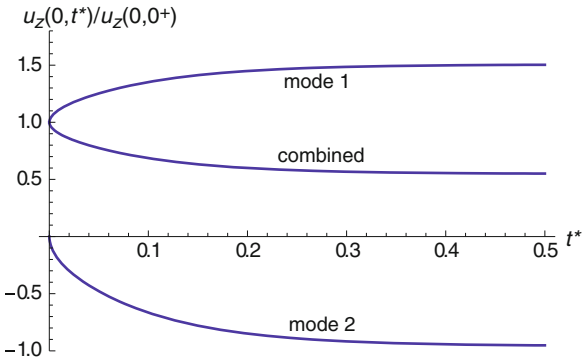


Fig. 7.4 Surface settlement due to fluid pressure loading, normalized by initial settlement, expressed as mode 1 ($\sigma_{zz} = -p_o,\ p = 0$), mode 2 ($\sigma_{zz} = 0,\ p = p_o$), and combined ($\sigma_{zz} = -p_o,\ p = p_o$) loading. Material is Ruhr sandstone with $\alpha = 0.637$, $\nu = 0.12$, and $\nu_u = 0.299$; see Table 3.2

7.3.3 *Variable Rete Loading*

The solutions presented in Sects. 7.3.1 and 7.3.2 are related to the loading condition of a Heaviside unit step function; that is, the load is suddenly applied at $t = 0^+$, and maintained at constant value for $t > 0$. These solutions, and other analytical solutions of similar type, can be utilized to construct solutions for problems whose loading is variable with time. The idea is based on the principle of superposition of a linear system.

A solution, whether it is a stress, pore pressure, or displacement, obtained based on a Heaviside step loading $\mathrm{H}(t-0)$, can be considered as an *influence function* $U(\vec{x}, t)$. If the boundary condition is replaced by a function of time $A(t)$, the solution for this new problem can be obtained through the application of the *Duhamel principle of superposition*:

$$V(\vec{x}, t) = A(0^+)\, U(\vec{x}, t) + \int_0^t U(\vec{x}, \tau) \frac{\partial A(t-\tau)}{\partial t}\, d\tau \tag{7.60}$$

where the first term on the right hand side is needed if $A(0) \neq 0$.

As an example, we can examine the pressure solution obtained as (7.29), expressed as an influence function due to a loading of $\sigma_{zz} = -\mathrm{H}(t-0)$,

$$p_1(z, t) = \frac{\eta}{GS} F_1(z^*, t^*) \tag{7.61}$$

If the loading is changed into one that increases linearly with time, that is, the stress boundary condition in (7.23) is replaced by

$$\sigma_{zz} = -A\, t, \quad \text{at } z = 0 \tag{7.62}$$

then the solution to this problem is obtained as

$$p_2(z,t) = \frac{A\eta}{GS}\int_0^t F_1(z^*,t^*)\,dt$$
$$= \frac{A\eta L^2}{GSc}\sum_{m=1,3,\cdots}^{\infty}\frac{16}{m^3\pi^3}\sin\left(\frac{m\pi z^*}{2}\right)\left(1-e^{-m^2\pi^2 t^*}\right) \tag{7.63}$$

In a laboratory setup, it is often difficult, or undesirable, to apply the load as a step, and a ramp type loading as shown in Fig. 7.5a is used. Solution to this problem can be obtained by the superposition of two linear loadings at different time and opposite magnitude, as illustrated in Fig. 7.5a, as

$$p_3(z,t) = \mathrm{H}(t-0)\,p_2(z,t) - \mathrm{H}(t-t_o)\,p_2(z,t-t_o) \tag{7.64}$$

Similarly, as demonstrated in Fig.7.5b, the solution for a box type loading can be obtained as follows

$$p_4(z,t) = P_o\left[\mathrm{H}(t-0)\,p_1(z,t) - \mathrm{H}(t-t_o)\,p_1(z,t-t_o)\right] \tag{7.65}$$

This box solution, in analytical form, can be designed for a small time interval Δt and shifted with time to approximate the solution of an arbitrary loading in time.

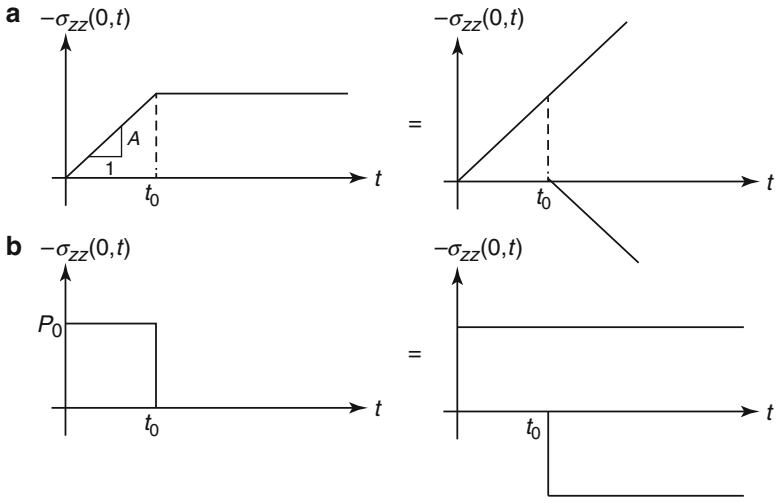


Fig. 7.5 Solutions obtained through superposition: (**a**) Ramp loading, and (**b**) box loading

7.3.4 Harmonic Excitation

In this problem the top of the stratum is in contact with a fluid, which is subjected to a pulsating pressure of $p_o \cos \omega t$, where ω is the excitation frequency (in radian). As the loading is applied through a fluid, we shall again examine this problem as a mode 1 and a mode 2 problem, and the final solution is the summation of the two:

- mode 1: $\sigma_{zz}(0,t) = -p_o \cos \omega t, p(0,t) = 0$
- mode 2: $\sigma_{zz}(0,t) = 0, p(0,t) = p_o \cos \omega t$

Semi-infinite Layer Before examining the problem of a layer with finite thickness, we first solve the problem of a semi-infinite layer. As a convenient technique for solving the problem, we shall utilize the complex variable representation. First we recognizing that

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t); \quad \Re(e^{i\omega t}) = \cos(\omega t) \tag{7.66}$$

where $i = \sqrt{-1}$ is the imaginary unit, and $\Re$ is the real part of a complex variable. In the mathematical solution, we extend the variables to include both the real and imaginary parts of the complex variable, and in the end, only the real part is taken to represent the physical solution.

Consider first the mode 1 loading. The top boundary conditions are:

$$\sigma_{zz}(0,t) = -p_o\, e^{i\omega t}; \quad p(0,t) = 0 \tag{7.67}$$

Because of the special form of the boundary conditions, it is possible to isolate the time dependent part of the solution and express the variables in the following form

$$p(z,t) = \tilde{p}(z,\omega)\, e^{i\,\omega t}; \quad u_z(z,t) = \tilde{u}_z(z,\omega)\, e^{i\,\omega t} \tag{7.68}$$

where $\tilde{p}$ and $\tilde{u}_z$ are the complex amplitude of the periodic responses. We can substitute the above pressure expression into the coupled diffusion equation (7.14), and divide it through by $e^{i\,\omega t}$, to obtain

$$c\,\frac{d^2\tilde{p}}{dz^2} - i\,\omega\tilde{p} = -\frac{i\,\omega p_o \eta}{GS} \tag{7.69}$$

which is an ordinary differential equation. Solution of this equation with the stated mode 1 boundary conditions is easily obtained as

$$\tilde{p}^{(1)}(z,\omega) = \frac{p_o \eta}{GS}\left(1 - e^{-\sqrt{i\omega z^2/c}}\right) \tag{7.70}$$

In Fig. 7.6a we plot the amplitude of the pressure response, obtained as the absolute value of the complex number $\tilde{p}^{(1)}$, normalized by $p_o\eta/GS$, versus the dimensionless depth $\sqrt{\omega/c}\,z$; and in Fig. 7.6b we plot the phase shift, given as the argument of

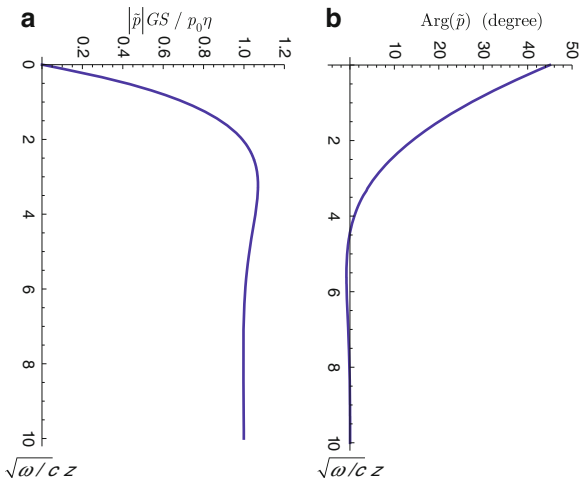


Fig. 7.6 Pressure response for mode 1 harmonic loading for the semi-infinite stratum case. (**a**) Normalized pressure amplitude, and (**b**) phase shift

$\tilde{p}^{(1)}$, in degree unit, as the function of the dimensionless depth. We note that when the solution is normalized this way, it is independent of any material properties.

We first observe in Fig. 7.6a that the pore pressure amplitude is zero at the soil surface, which is controlled by the mode 1 boundary condition. It then approaches the large depth amplitude ($p_o\eta/GS$) in a distance about $z \approx 3\sqrt{c/\omega}$. We also notice that the pore pressure amplitude profile is non-monotonic.

We may consider the length $\delta_\ell = 3\sqrt{c/\omega}$ as a boundary layer thickness [98, 99], within which the pore pressure is affected by the boundary condition (drained) at the top. Outside the boundary layer, the stratum largely behaves as undrained with a pressure amplitude predicted by the Skempton effect. To get some idea about the boundary layer thickness, we assume a loading frequency of $f(= \omega/2\pi) = 0.1$ Hz. The thickness δ_ℓ can be calculated based on the consolidation coefficient values reported in Table 3.2, which give values ranging from 10 m for hard sediment, 3 m for soft sediment, 27 cm for Ruhr sandstone, to 2 cm for Westerly granite.

Figure 7.6b shows that near the surface, there is a time lag of pore pressure response to the applied σ_{zz} loading. At the surface, the phase shift angle is $\pi/4$ (45°). Somewhat beyond the boundary layer, the pore pressure response becomes in phase with the surface loading.

Of particular interest to mention here is the effect of the coupling term in the pore pressure diffusion equation (7.14), which give rise to the right hand side of (7.69). The uncoupled diffusion equation based on Terzaghi theory does not contain this term. The pressure response due to a harmonic excitation, after the initial transient has died out, predicted by the uncoupled model is then zero. This gives a drastically different prediction as the poroelastic case.

Now let us examine the mode 2 problem. The pressure solution is simply

$$\tilde{p}^{(2)}(z,\omega) = p_o\, e^{-\sqrt{i\omega z^2/c}} \tag{7.71}$$

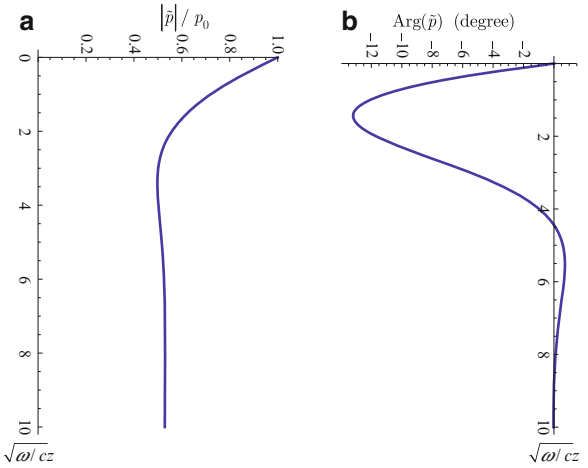


Fig. 7.7 Pore pressure response due to harmonic fluid pressure loading (combined mode 1 and 2) for one-dimensional half-space problem. (**a**) Normalized pressure amplitude, and (**b**) phase shift. Material is Ruhr sandstone with $GS/\eta = 2.38$

whose amplitude changes from p_o at the surface and diminishes to zero as depth increases. The penetration of pore pressure into the layer is also limited to within a boundary layer of thickness about $3\sqrt{c/\omega}$ as described above. The total pressure response due to a harmonic fluid pressure excitation is the summation of (7.70) and (7.71). In Fig. 7.7 we plot the combined solution as pore pressure amplitude normalized by the loading magnitude p_o and phase shift versus depth. Based on the normalization, the solution is dependent on the combination of material constant $GS/\eta = \alpha(1-2\nu)(1-\nu)/(\nu_u - \nu)$ only. The result presented in Fig. 7.7 is based on Ruhr sandstone with $\alpha = 0.637$, $\nu = 0.12$, and $\nu_u = 0.299$ (see Table 3.2).

The displacement solution can also be found, but is not reported here because for the half-space case, it is infinite at the surface. It will be presented in the finite thickness layer case.

Finite Thickness Layer We can examine the same harmonic fluid excitation problem for a layer with finite thickness L (see Fig. 7.1a). At the bottom of the layer, $z = L$, the same no displacement and no flux condition as those in (7.23) apply. The mode 1 solution for pressure is given by

$$\tilde{p}^{(1)}(z^*, \omega^*) = \frac{p_o \eta}{GS}\left[1 + \tanh\sqrt{i\omega^*}\,\sinh(\sqrt{i\omega^*}\,z^*) - \cosh(\sqrt{i\omega^*}\,z^*)\right] \tag{7.72}$$

where we have defined the dimensionless distance $z^* = z/L$ and dimensionless frequency $\omega^* = \omega L^2/c$.

The displacement can be obtained by integrating the above, and is found as

$$\tilde{u}_z^{(1)}(z^*, \omega^*) = \frac{p_o L(1-2\nu_u)}{2G(1-\nu_u)}(1-z^*) + \frac{p_o L(\nu_u - \nu)}{2G(1-\nu)(1-\nu_u)\sqrt{i\omega^*}}\,\frac{\sinh[(1-z^*)\sqrt{i\omega^*}]}{\cosh\sqrt{i\omega^*}} \tag{7.73}$$

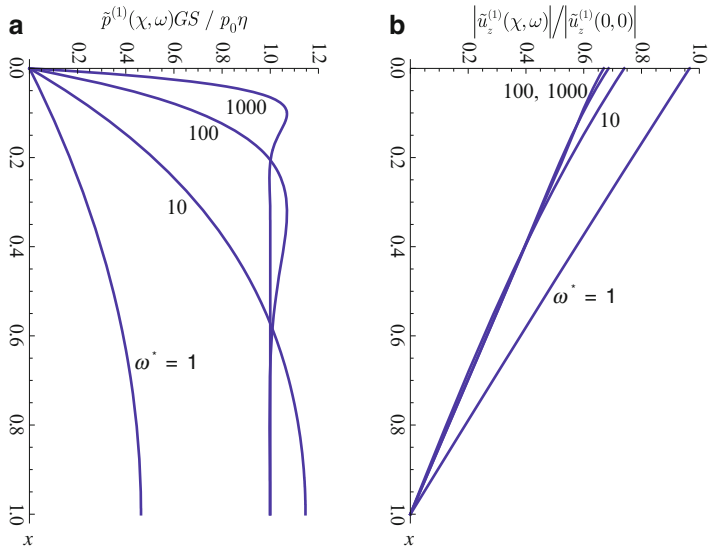


Fig. 7.8 Response of harmonic mode 1 loading for a finite thickness layer versus dimensionless depth z^*, at various dimensionless frequencies ω^*, for Ruhr sandstone ($\nu = 0.12$ and $\nu_u = 0.299$): (**a**) normalized pore pressure amplitude, and (**b**) normalized displacement amplitude

In Fig. 7.8 we plot the normalized pore pressure and displacement amplitude as function of dimensionless depth z^* at various dimensionless frequencies ω^*. The normalized pore pressure profile is independent of material constants, while the normalized displace profile is dependent on the Poisson ratios ν and ν_u only.

The settlement at $z = 0$ is

$$\tilde{u}_z^{(1)}(0,\omega^*) = \frac{p_o L(1-2\nu_u)}{2G(1-\nu_u)} + \frac{p_o L(\nu_u-\nu)}{2G(1-\nu)(1-\nu_u)\sqrt{i\omega^*}}\tanh\sqrt{i\omega^*} \tag{7.74}$$

In the above, we observe the following limits:

$$\left|\frac{\tanh\sqrt{i\omega^*}}{\sqrt{i\omega^*}}\right| \to 1 \text{ as } \omega^* \to 0; \quad \left|\frac{\tanh\sqrt{i\omega^*}}{\sqrt{i\omega^*}}\right| \to 0 \text{ as } \omega^* \to \infty \tag{7.75}$$

From these limits, we can show that

$$\left|\tilde{u}_z^{(1)}(0,0)\right| = \frac{p_o L(1-2\nu)}{2G(1-\nu)}; \quad \left|\tilde{u}_z^{(1)}(0,\infty)\right| = \frac{p_o L(1-2\nu_u)}{2G(1-\nu_u)} \tag{7.76}$$

In other words, when the loading frequency is low, the layer responds as a drained material, and when the frequency is high, it behaves as undrained. It should be remarked that when the frequency becomes high, the dynamic (inertia) effect needs

to be considered, which is not present in the quasi-static theory. The dynamic effect will be investigated in Chap. 9.

Next, we examine the mode 2 response for the harmonic excitation. Solving the governing equations together with the boundary conditions, we obtain the following: the pressure

$$\tilde{p}^{(2)}(z^*, \omega^*) = p_o \left[\cosh(\sqrt{i\omega^*}\, z^*) - \tanh \sqrt{i\omega^*}\ \sinh(\sqrt{i\omega^*}\, z^*) \right] \tag{7.77}$$

the displacement

$$\tilde{u}_z^{(2)}(z^*, \omega^*) = -\frac{p_o L \alpha (1-2\nu)}{2G(1-\nu)\sqrt{i\omega^*}} \ \frac{\sinh[(1-z^*)\sqrt{i\omega^*}]}{\cosh \sqrt{i\omega^*}} \tag{7.78}$$

and the settlement at the top

$$\tilde{u}_z^{(2)}(0, \omega^*) = -\frac{p_o L \alpha (1-2\nu)}{2G(1-\nu)\sqrt{i\omega^*}} \tanh \sqrt{i\omega^*} \tag{7.79}$$

7.4 Plane Strain

For plane strain problems, the displacement is allowed only in a plane geometry (chosen as x–y plane here), and prohibited in the direction perpendicular to it (chosen as z-axis). These problems can be characterized by the following functional dependence of displacement components:

$$\begin{aligned} &u_x = u_x(x, y, t); \quad u_y = u_y(x, y, t); \quad u_z = 0 \\ &U_x = U_x(x, y, t); \quad U_y = U_y(x, y, t); \quad U_z = 0 \end{aligned} \tag{7.80}$$

The above relations also suggest the following

$$\begin{aligned} &e_{xx} = e_{xx}(x, y, t); \quad e_{yy} = e_{yy}(x, y, t); \quad e_{xy} = e_{xy}(x, y, t); \\ &e_{zz} = e_{xz} = e_{yz} = 0; \\ &q_x = q_x(x, y, t); \quad q_y = q_y(x, y, t); \quad q_z = 0; \\ &\zeta = \zeta(x, y, t); \quad p = p(x, y, t) \end{aligned} \tag{7.81}$$

Of the strain compatibility relations (6.76)–(6.81), only one remain non-trivial:

$$2\frac{\partial^2 e_{xy}}{\partial x\, \partial y} = \frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2} \tag{7.82}$$

For stresses, it is easy to conclude from the constitutive equations and the conditions in (7.81) that

$$\sigma_{xz} = \sigma_{yz} = 0 \tag{7.83}$$

We also find the reduction of equilibrium equations

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0 \tag{7.84}$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0 \tag{7.85}$$

$$\frac{\partial \sigma_{zz}}{\partial z} = 0 \tag{7.86}$$

7.4.1 Orthotropy

For orthotropic material with its material axes coinciding with the plane strain coordinate system, we observe the following stress-strain relations

$$\sigma_{xx} = M_{xx} e_{xx} + M_{xy} e_{yy} - \alpha_x p \tag{7.87}$$

$$\sigma_{yy} = M_{xy} e_{xx} + M_{yy} e_{yy} - \alpha_y p \tag{7.88}$$

$$\sigma_{zz} = M_{xz} e_{xx} + M_{yz} e_{yy} - \alpha_z p \tag{7.89}$$

$$\sigma_{xy} = 2 G_{xy} e_{xy} \tag{7.90}$$

$$p = M(\zeta - \alpha_x e_{xx} - \alpha_y e_{yy}) \tag{7.91}$$

Or, (7.87) and (7.88) can be inverted to give the strain-stress relations:

$$e_{xx} = \frac{M_{yy}\sigma_{xx} - M_{xy}\sigma_{yy} + \alpha_x M_{yy} p - \alpha_y M_{xy} p}{M_{xx}M_{yy} - M_{xy}^2} \tag{7.92}$$

$$e_{yy} = \frac{M_{xx}\sigma_{yy} - M_{xy}\sigma_{xx} + \alpha_y M_{xx} p - \alpha_x M_{xy} p}{M_{xx}M_{yy} - M_{xy}^2} \tag{7.93}$$

The governing equations are obtained by simplifying (6.89)–(6.92), and we find the following Navier equations

$$M_{xx}\frac{\partial^2 u_x}{\partial x^2} + (M_{xy} + G_{xy})\frac{\partial^2 u_y}{\partial x \partial y} + G_{xy}\frac{\partial^2 u_x}{\partial y^2} - \alpha_x \frac{\partial p}{\partial x} = 0 \tag{7.94}$$

$$G_{xy}\frac{\partial^2 u_y}{\partial x^2} + (M_{xy} + G_{xy})\frac{\partial^2 u_x}{\partial x \partial y} + M_{yy}\frac{\partial^2 u_y}{\partial y^2} - \alpha_y \frac{\partial p}{\partial y} = 0 \tag{7.95}$$

and the diffusion equation

$$\frac{\partial p}{\partial t} - M\left(\kappa_x \frac{\partial^2 p}{\partial x^2} + \kappa_y \frac{\partial^2 p}{\partial y^2}\right) = -M\left(\alpha_x \frac{\partial^2 u_x}{\partial t \partial x} + \alpha_y \frac{\partial^2 u_y}{\partial t \partial y}\right) \tag{7.96}$$

These three coupled equations are used to solve for the three unknowns, u_x, u_y, and p.

Or, if we choose the set of primary variables as u_x, u_y, and ζ, the above set of Eqs. (7.94), (7.95), and (7.96), becomes

$$M^u_{xx}\frac{\partial^2 u_x}{\partial x^2} + (M^u_{xy} + G_{xy})\frac{\partial^2 u_y}{\partial x \partial y} + G_{xy}\frac{\partial^2 u_x}{\partial y^2} - \alpha_x M\frac{\partial \zeta}{\partial x} = 0 \tag{7.97}$$

$$G_{xy}\frac{\partial^2 u_y}{\partial x^2} + (M^u_{xy} + G_{xy})\frac{\partial^2 u_x}{\partial x \partial y} + M^u_{yy}\frac{\partial^2 u_y}{\partial y^2} - \alpha_y M\frac{\partial \zeta}{\partial y} = 0 \tag{7.98}$$

$$\frac{\partial \zeta}{\partial t} = c_x \frac{\partial^2 \zeta}{\partial x^2} + c_y \frac{\partial^2 \zeta}{\partial y^2} \tag{7.99}$$

where M^u_{ij} are undrained coefficients, and c_x and c_y are consolidation coefficients in the respective directions.

7.4.2 Isotropy

For isotropy, the stress-strain relation are

$$\sigma_{xx} = \frac{2G\nu}{1-2\nu} e + 2G e_{xx} - \alpha p \tag{7.100}$$

$$\sigma_{yy} = \frac{2G\nu}{1-2\nu} e + 2G e_{yy} - \alpha p \tag{7.101}$$

$$\sigma_{zz} = \frac{2G\nu}{1-2\nu} e - \alpha p \tag{7.102}$$

$$\sigma_{xy} = 2G e_{xy} \tag{7.103}$$

$$p = M(\zeta - \alpha e) \tag{7.104}$$

where

$$e = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \tag{7.105}$$

Or, we can also obtain σ_{zz} from the other two components of normal stresses and pressure as

$$\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy}) - \alpha(1-2\nu)p \tag{7.106}$$

The governing equations are

$$G\left(\frac{\partial^2 u_x}{\partial x^2}+\frac{\partial^2 u_x}{\partial y^2}\right)+\frac{G}{1-2\nu}\frac{\partial e}{\partial x}=\alpha\frac{\partial p}{\partial x} \tag{7.107}$$

$$G\left(\frac{\partial^2 u_y}{\partial x^2}+\frac{\partial^2 u_y}{\partial y^2}\right)+\frac{G}{1-2\nu}\frac{\partial e}{\partial y}=\alpha\frac{\partial p}{\partial y} \tag{7.108}$$

$$\frac{\partial p}{\partial t}-\kappa M\left(\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}\right)=-\alpha M\frac{\partial e}{\partial t} \tag{7.109}$$

We can also express the above in terms of (u_x, u_y, ζ) variables as

$$G\left(\frac{\partial^2 u_x}{\partial x^2}+\frac{\partial^2 u_x}{\partial y^2}\right)+\frac{G}{1-2\nu_u}\frac{\partial e}{\partial x}=\alpha M\frac{\partial \zeta}{\partial x} \tag{7.110}$$

$$G\left(\frac{\partial^2 u_y}{\partial x^2}+\frac{\partial^2 u_y}{\partial y^2}\right)+\frac{G}{1-2\nu_u}\frac{\partial e}{\partial y}=\alpha M\frac{\partial \zeta}{\partial y} \tag{7.111}$$

$$\frac{\partial \zeta}{\partial t}-c\left(\frac{\partial^2 \zeta}{\partial x^2}+\frac{\partial^2 \zeta}{\partial y^2}\right)=0 \tag{7.112}$$

7.4.3 Volumetric Strain and Rotation Formulation

In the above governing equations, the primary variables are the displacement components and the pore pressure p or variation in fluid content ζ. In the following we shall present an alternative formulation in variables that are governed by lower order partial differential equations. We first refer to the Navier equation in vector notation as (6.64) or (6.66), and then introduce the vector identity of

$$\nabla^2\vec{A}=\nabla(\nabla\cdot\vec{A})-\nabla\times(\nabla\times\vec{A}) \tag{7.113}$$

We can use the above to transform (6.64) into the following

$$\frac{2G(1-\nu)}{1-2\nu}\nabla(\nabla\cdot\vec{u})-G\nabla\times(\nabla\times\vec{u})=\alpha\nabla p \tag{7.114}$$

By recognizing that $\nabla\cdot\vec{u}=e$ and $\nabla\times\vec{u}=2\vec{\Omega}$, where $\vec{\Omega}$ is the rotation vector (see Appendix C), we can express (6.64) in terms of these variables. However, for such a system in three-dimension, there are more variables than equations; hence it is not suitable for a solution system.

In plane strain, on the other hand, we observe that only the z-component rotation Ω_z exists, and (7.114) can be expressed as

$$\frac{2G(1-\nu)}{1-2\nu}\frac{\partial e}{\partial x} - G\frac{\partial \Omega_z}{\partial y} = \alpha\frac{\partial p}{\partial x} \tag{7.115}$$

$$\frac{2G(1-\nu)}{1-2\nu}\frac{\partial e}{\partial y} + G\frac{\partial \Omega_z}{\partial x} = \alpha\frac{\partial p}{\partial y} \tag{7.116}$$

These equations, combined with (7.109), can be used to solve for (e, Ω_z, p). Similarly, we can write for the set of variables (e, Ω_z, ζ)

$$\frac{2G(1-\nu_u)}{1-2\nu_u}\frac{\partial e}{\partial x} - G\frac{\partial \Omega_z}{\partial y} = \alpha M\frac{\partial \zeta}{\partial x} \tag{7.117}$$

$$\frac{2G(1-\nu_u)}{1-2\nu_u}\frac{\partial e}{\partial y} + G\frac{\partial \Omega_z}{\partial x} = \alpha M\frac{\partial \zeta}{\partial y} \tag{7.118}$$

7.5 Generalized Plane Strain

Many problems in mining, petroleum and geotechnical engineering, such as tunnel, borehole, dike, and retaining wall, are characterized by a cross-sectional geometry that is invariant along a longitudinal direction. In these problems, the dimension may be considered as infinite in that direction. If the material properties and boundary conditions also do not change, the three-dimensional stresses and strains are invariant in that direction as well, meaning that they are functions of two spatial dimensions only. These conditions lead to two-dimensional governing equations. However, unlike the plane strain problem, the displacement can have three-dimensional components, and these components can have three-dimensional variations. This added degree of freedom allows the deformations such as axial strain, out-of-plane shear, warping, bending, and torsion, which are not present in plane strain problems, to be modeled in the two-dimensional solution system. These problems are called *generalized plane strain* problems, which incorporated plane strain as a special case.

Generalized plane strain has been investigated in the context of elasticity [11, 103], and found applications in petroleum engineering and rock mechanics [30, 31, 168]. For poroelasticity, generalized plane strain problems of torsion and pure bending was first examined by Nowinski and Davis [119]. A complete exploration of all possible cases was presented in Cheng [36], which constitutes the present section.

7.5.1 Definition of Generalized Plane Strain

Figure 7.9 shows two cylindrical surfaces generated in the z-direction. Together with the x-y planes, they can bound an interior, an exterior, or a simply-connected region

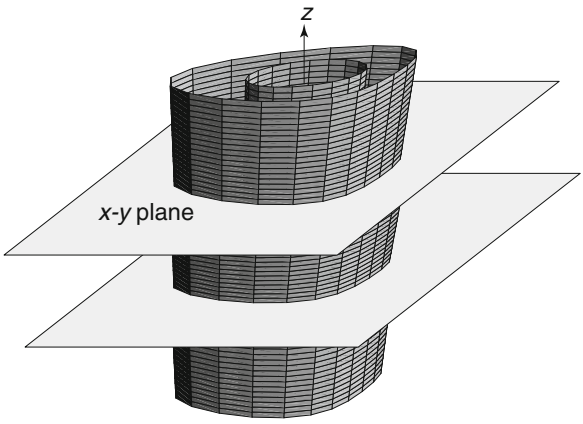


Fig. 7.9 Generalized plane strain geometry

of finite or infinite length in the z-direction. Material properties can be homogeneous or heterogeneous. In the case of heterogeneous material, properties are allowed to vary in the x-y plane, but not in the z-direction. Boundary conditions in the form of surface tractions, pore pressure, displacements, and normal flux must not change in the longitudinal (z) direction. For a finite-length cylinders, boundary conditions at the end surfaces must belong to a special class that allows the generalized plane strain solution to exist.

When these conditions are met, every plane perpendicular to the z-axis is a plane of symmetry. The resultant stresses and strains may have three-dimensional components, but they are functions of two spatial dimensions (x, y) and time t only. Under these conditions, we anticipate to find two-dimensional governing equations that allows the full solutions to these problems.

The above physical statements suggest the following functional dependence of the stress and strain quantities:

$$\sigma_{ij} = \sigma_{ij}(x, y, t); \quad p = p(x, y, t); \quad e_{ij} = e_{ij}(x, y, t); \quad \zeta = \zeta(x, y, t) \tag{7.119}$$

The strains, however, cannot be arbitrary function of (x, y), as they must satisfy compatibility conditions. Taking into consideration the above functional dependence, the six St. Venant compatibility equations (6.76)–(6.81) are reduced to the following:

$$\frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2} = 2\,\frac{\partial^2 e_{xy}}{\partial x\,\partial y} \tag{7.120}$$

$$\frac{\partial^2 e_{xz}}{\partial x\,\partial y} - \frac{\partial^2 e_{yz}}{\partial x^2} = \frac{\partial^2 e_{yz}}{\partial x\,\partial y} - \frac{\partial^2 e_{xz}}{\partial y^2} = 0 \tag{7.121}$$

$$\frac{\partial^2 e_{zz}}{\partial x^2} = \frac{\partial^2 e_{zz}}{\partial y^2} = \frac{\partial^2 e_{zz}}{\partial x\,\partial y} = 0 \tag{7.122}$$

Equation (7.122) shows that e_{zz} must be restricted to this form:

$$e_{zz} = A(t)\,x + B(t)\,y + C(t) \tag{7.123}$$

By the kinematic equation

$$e_{zz} = \frac{\partial u_z}{\partial z} \tag{7.124}$$

it is easily shown that the z-component displacement is given by:

$$u_z(x, y, z, t) = [A(t)\,x + B(t)\,y + C(t)]\,z + h(x, y, t) \tag{7.125}$$

Utilizing the rest of the kinematic equations and the functional dependence shown in (7.119), we observe that

$$u_x(x, y, z, t) = -\frac{A(t)}{2}\,z^2 - D(t)\,yz + F(t)\,z + f(x, y, t) \tag{7.126}$$

$$u_y(x, y, z, t) = -\frac{B(t)}{2}\,z^2 + D(t)\,xz + H(t)\,z + g(x, y, t) \tag{7.127}$$

In the above, A, B, C, D, F, H are arbitrary functions of time, and f, g, h are arbitrary functions of time and space. Equations (7.125), (7.126), and (7.127) are the most general forms of displacement field that satisfy the generalized plane strain condition. We shall demonstrate below that A and B are associated with bending deformation, C with axial strain, D with torsion, F and H with shear deformation, h with warping, and f and g are related to the classical plane strain deformation.

The compatibility requirements (7.120), (7.121), and (7.122) are based on kinematic relations among displacements and strains, which are the same for poroelasticity and elasticity. Hence the discussion below is valid for both elasticity and poroelasticity.

There is, in fact, no universally agreed upon definition of generalized plane strain. Cases considered by Wu and Li [168] and also Rencis and Huang [133] correspond to the case of $D = F = H = h = 0$. The *complete plane strain* studied by Brady and Bray [30, 31] is equivalent to setting $A = B = D = F = H = 0$. Definition in elasticity textbook [141] is equivalent to the case of $A = B = D = F = H = h = 0$. Result obtained by Lekhnitskii [103] is the closest to the present case, yet it still contains the simplification of assigning $F = H = 0$. Hence the definitions in (7.125), (7.126), and (7.127) give the most general case of generalized plane strain for both elasticity and poroelasticity.

Equations (7.125), (7.126), and (7.127) contain two parts, a part given by coefficients A, B, C, D, F, H, which are particular solutions tied to certain elementary deformation modes, and a part defined by the functions f, g, h, which must be solved as boundary value problems. Since the system discussed here is

linear, these problems can be separately solved, and then superposed. We shall discuss these elementary and non-elementary solutions in the following.

7.5.2 Pure Shear

The simple displacement field

$$u_x = F(t)\, z; \quad u_y = u_z = 0 \tag{7.128}$$

is associated with a pure shear in the x-z plane, because the stresses obtained from constitutive equations (5.27) and (5.35), for the orthotropic material, contain only one non-vanishing component

$$\sigma_{xz} = F(t)\, G_{xz}; \quad \sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = \sigma_{yz} = 0 \tag{7.129}$$

A second pure shear solution is defined by

$$u_y = H(t)\, z; \quad u_x = u_z = 0 \tag{7.130}$$

which leads to

$$\sigma_{yz} = H(t)\, G_{yz}; \quad \sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = \sigma_{xz} = 0 \tag{7.131}$$

As these are pure shear, there is no volumetric deformation, and no pore pressure is generated. So the poroelastic solution is the same as the elastic solution.

7.5.3 Warping

Warping is associated with the function h in (7.125):

$$u_z = h(x, y, t); \quad u_x = u_y = 0 \tag{7.132}$$

It is easily shown that

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = 0 \tag{7.133}$$

The only non-zero components are the shear stresses σ_{yz} and σ_{xz}. Again, there is no pore pressure generation, and the poroelastic solution is the same as the elastic one. This type of deformation, which has no in-plane movement, but warps out of plane, belongs to a class called *anti-plane* problems, which has been extensively studied in elasticity [85, 112].

As a consequence of (7.133), the x- and y-component equilibrium equations are automatically satisfied, and the z-component equation takes the form

$$\frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} = 0 \tag{7.134}$$

The constitutive equations for the surviving stresses are

$$\sigma_{yz} = G_{yz} \frac{\partial u_z}{\partial y} \tag{7.135}$$

$$\sigma_{xz} = G_{xz} \frac{\partial u_z}{\partial x} \tag{7.136}$$

Combining (7.135) and (7.136) with (7.134), we obtain the following elliptic equation in two dimensions

$$G_{xz} \frac{\partial^2 u_z}{\partial x^2} + G_{yz} \frac{\partial^2 u_z}{\partial y^2} = 0 \tag{7.137}$$

which is a single governing equation, solving for a single unknown.

Typical boundary conditions for (7.137) are either u_z given, or t_z given on Γ, where Γ is the contour of the cylinder cross-section on the x-y plane, and t_z is the z-component boundary traction defined as

$$t_z = \sigma_{xz} n_x + \sigma_{yz} n_y \tag{7.138}$$

with n_x and n_y the x- and y-component of the unit outward normal $\vec{n}$ on Γ, respectively. Equation (7.138) is then expressed in terms of the displacement variable as

$$G_{xz} n_x \frac{\partial u_z}{\partial x} + G_{yz} n_y \frac{\partial u_z}{\partial y} = t_z \quad \text{on } \Gamma \tag{7.139}$$

where the values of t_z is given. Equation (7.137) together with proper boundary conditions has been solved for warping of orthotropic cylinders in elliptic and rectangular cross-sections [79].

Also of interest is the special case of transverse isotropy with x-y plane as the plane of rotational symmetry. In that case, $G_{xz} = G_{yz} = G'$, where G' is the shear modulus normal to the plane of isotropy. Equation (7.137) reduces to the Laplace equation

$$\frac{\partial^2 u_z}{\partial x^2} + \frac{\partial^2 u_z}{\partial y^2} = 0 \tag{7.140}$$

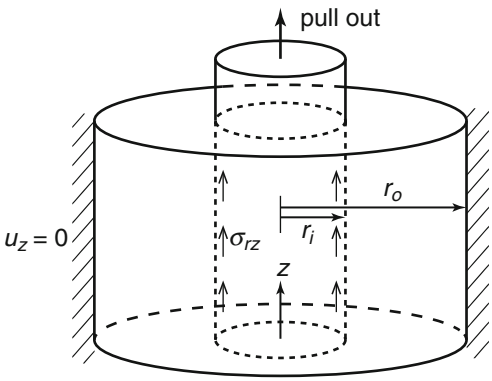


Fig. 7.10 Pull out of a rigid rod from a cylindrical porous medium

and the boundary condition (7.139) becomes

$$\frac{\partial u_z}{\partial n} = \frac{t_z}{G'} \quad \text{on } \Gamma \tag{7.141}$$

Warping can exist for problems such as excavation of boreholes and tunnels. If a borehole is drilled in a direction that is inclined to the principal axes of far-field stresses, far-field deviatoric stress components are present. In a standard solution procedure far-field stresses are removed and then added back. The subtraction of deviatoric stresses generates shear stresses on the borehole wall which becomes a cause for warping (see Sect. 7.14.2).

To gain an idea of a simple warping, we can examine a pull-out problem. Figure 7.10 shows an long, circular, saturated porous cylinder with a rigid rod embedded in it. When the rod is pulled in the axial (z) direction, it exerts a constant shear $\sigma_{rz} = \tau_o$ on the inner wall, $r = r_i$. The outer wall of the porous medium, at $r = r_o$, is fixed to a rigid surface and no displacement is allowed, $u_z = 0$. Under axial symmetry, (7.140) can be expressed as

$$\frac{1}{r}\left[\frac{\partial}{\partial r}\left(\frac{\partial u_z}{\partial r}\right)\right] = 0 \tag{7.142}$$

Solution of the above equation with the two boundary conditions at $r = r_i$ and r_o can be easily found as

$$u_z = \frac{\tau_o r_i}{G'} \ln\left(\frac{r}{r_o}\right) \tag{7.143}$$

This elastic solution is admitted as a poroelastic solution as no pore pressure is generated by this deformation.

Another source of warping come from the torsion of a cylinder, which is discussed next.

7.5.4 Torsion

The displacement field

$$u_x = -D(t)\, yz; \quad u_y = D(t)\, xz; \quad u_z = 0 \tag{7.144}$$

gives a *pure torsion* about the z-axis with $D(t)$ as the twist per unit length. This displacement field can generate shear stresses σ_{yz} and σ_{xz}. For a cylinder subjected to a pure torsion, shear stress is found on its surface. For the surface to remain stress free, the x-y plane must warp to compensate for it to restore the surface to zero stress condition. Hence torsion problems need to be solved together with warping. This requires the modification of the displacement field to

$$u_x = -D(t)\, yz; \quad u_y = D(t)\, xz; \quad u_z = h(x, y, t) \tag{7.145}$$

From the constitutive equations, the following stresses vanish

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = 0 \tag{7.146}$$

As there is no normal stress, again no pore pressure is generated. So the elasticity solution is admitted as a poroelastic solution.

The system can be solved as follows. Based on the constitutive equations, the non-vanishing shear stresses are

$$\sigma_{yz} = G_{yz} D(t)\, x + G_{yz} \frac{\partial u_z}{\partial y} \tag{7.147}$$

$$\sigma_{xz} = -G_{xz} D(t)\, y + G_{xz} \frac{\partial u_z}{\partial x} \tag{7.148}$$

Substitution into the equilibrium equation (7.134) produces the identical governing equation as warping problems, i.e., (7.137) for orthotropic material, and (7.140) for transversely isotropic material. The traction free boundary condition, following (7.138), (7.139), (7.147), and (7.148), is

$$G_{xz} n_x \frac{\partial u_z}{\partial x} + G_{yz} n_y \frac{\partial u_z}{\partial y} = D(t) \left(G_{yz} n_x\, y - G_{yz} n_y\, x \right) \quad \text{on } \Gamma \tag{7.149}$$

The final solution consists of solving u_z from (7.137), with (7.149) as boundary condition, and u_x and u_y are given as (7.145). Stresses can then be evaluated from (7.147) and (7.148).

For the special case of isotropic material with circular cross-section, (7.149) becomes

$$\frac{\partial u_z}{\partial n} = 0 \quad \text{on } \Gamma \tag{7.150}$$

The solution of (7.140) is then trivial, $u_z = 0$. Hence a circular cylinder subject to torsion does not warp.

7.5.5 Plane Strain

The classical plane strain problem is given by the following displacement field

$$u_x = f(x, y, t); \quad u_y = g(x, y, t); \quad u_z = 0 \tag{7.151}$$

which is the same as (7.80). The constitutive equations and governing equations for orthotropic and isotropic cases are already presented in Sect. 7.4, and are not repeated here.

7.5.6 Axial Strain

Pure axial strain is given by

$$u_z = C(t)\, z; \quad u_x = u_y = 0 \tag{7.152}$$

Given a cylinder, the above conditions requires the cylinder surface to be constrained from displacement on the x-y plane. When such constraint does not exist, axial strain can generate displacements in the x-y plane. Hence depending on the boundary condition on the surface, the axial strain problem generally needs to be solved together with the plane strain problem to fulfill the boundary conditions. The basic solution form is therefore modified to

$$u_x = f(x, y, t); \quad u_y = g(x, y, t); \quad u_z = C(t)\, z \tag{7.153}$$

With the above displacement components, we find that the Navier equations (7.94) and (7.95) remain unchanged. The diffusion equation (7.96) however is modified to

$$\begin{aligned} &\frac{\partial p}{\partial t} - M\left(\kappa_x \frac{\partial^2 p}{\partial x^2} + \kappa_y \frac{\partial^2 p}{\partial y^2}\right) = \\ &\qquad -M\left(\alpha_x \frac{\partial^2 u_x}{\partial t\, \partial x} + \alpha_y \frac{\partial^2 u_y}{\partial t\, \partial y}\right) - M\alpha_z \frac{dC(t)}{dt} \end{aligned} \tag{7.154}$$

Hence the solution system, (7.94), (7.95) and (7.154), is the same as plane strain problem subject to the same boundary conditions on the cylinder side surface, except that there exist an extra pore pressure generation term as the last term on the right hand side of (7.154), which is proportional to the rate of axial strain de_{zz}/dt, and is

a function of time only. This extra term can be removed by finding its particular solution, and then solving for a homogeneous solution with modified boundary conditions.

If there exists a poroelastic solver for plane strain problem, such as a computer program using a certain numerical method, then the same solver can be used to solve the combined plane strain and axial strain problem. This can be achieved by defining a set of new variables:

$$\sigma_{xx}^{\dagger} = \sigma_{xx} - M_{xz}^{u} C(t) \tag{7.155}$$

$$\sigma_{yy}^{\dagger} = \sigma_{yy} - M_{yz}^{u} C(t) \tag{7.156}$$

$$\sigma_{zz}^{\dagger} = \sigma_{zz} - M_{zz}^{u} C(t) \tag{7.157}$$

$$p^{\dagger} = p + M\alpha_z C(t) \tag{7.158}$$

where M_{ij}^{u} are undrained elastic moduli, defined in (5.12), and written here as

$$M_{ij}^{u} = M_{ij} + \alpha_i \alpha_j M; \qquad \text{for} \quad i,j = 1,2,3 \tag{7.159}$$

Other variables, u_x, u_y, σ_{xy}, q_x, q_y, and ζ, remain the same. This new set of variables satisfies all the plane strain equations in Sect. 7.4. The given boundary values in terms of tractions and pore pressure used as input to the solver, however, need to be modified to

$$t_x^{\dagger} = t_x - n_x M_{xz}^{u} C(t) \tag{7.160}$$

$$t_y^{\dagger} = t_y - n_y M_{yz}^{u} C(t) \tag{7.161}$$

$$p^{\dagger} = p + M\alpha_z C(t) \tag{7.162}$$

Boundary conditions in terms of displacements and normal flux are unchanged.

7.5.7 Pure Bending

Bending of a prismatic bar is associated with coefficients $A(t)$ and $B(t)$ in (7.125). Similar to the axial strain problem, bending is generally accompanied by plane strain deformation in order to maintain a traction free side surface. For bending about y-axis, such as the cantilever beam shown in Fig. 7.11, displacements are characterized by

$$u_x = f(x,y,t); \quad u_y = -\frac{B(t)}{2} z^2 + g(x,y,t); \quad u_z = B(t)\, yz \tag{7.163}$$

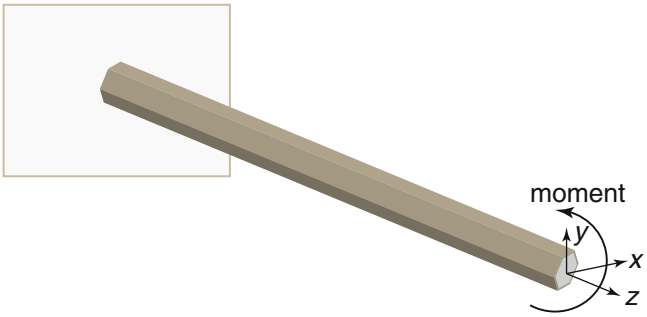


Fig. 7.11 Bending of a prismatic bar

It is easily shown that these shear stresses vanish

$$\sigma_{yz} = \sigma_{xz} = 0 \tag{7.164}$$

which is the same as the plane strain and axial strain cases. We note that for this problem the only loading allowed is a moment characterized by the magnitude $B(t)$, and no concentrated or distributed forces can be admitted; otherwise the generalized plane strain condition does not exist, and the solution is three-dimensional.

Utilizing the constitutive and equilibrium equations, we find these Navier and diffusion equations that form a two-dimensional solution system:

$$M_{xx}\frac{\partial^2 u_x}{\partial x^2} + (M_{xy} + G_{xy})\frac{\partial^2 u_y}{\partial x\,\partial y} + G_{xy}\frac{\partial^2 u_x}{\partial y^2} - \alpha_x\frac{\partial p}{\partial x} = 0 \tag{7.165}$$

$$G_{xy}\frac{\partial^2 u_y}{\partial x^2} + (M_{xy} + G_{xy})\frac{\partial^2 u_x}{\partial x\,\partial y} + M_{yy}\frac{\partial^2 u_y}{\partial y^2} - \alpha_y\frac{\partial p}{\partial y} = -M_{yz}\,B(t) \tag{7.166}$$

$$\begin{aligned}&\frac{\partial p}{\partial t} - M\left(\kappa_x\frac{\partial^2 p}{\partial x^2} + \kappa_y\frac{\partial^2 p}{\partial y^2}\right)\\ &\quad = -M\left(\alpha_x\frac{\partial^2 u_x}{\partial t\,\partial x} + \alpha_y\frac{\partial^2 u_y}{\partial t\,\partial y}\right) - M\alpha_z\,y\,\frac{dB(t)}{dt}\end{aligned} \tag{7.167}$$

Comparing to the plane strain case, (7.94), (7.95), and (7.96), we find a term equivalent to a body force in the y-direction in (7.166), and a y-dependent pore pressure production term in (7.167). These extra terms can be removed as particular solutions. Or, we can define a set of new variables:

$$u_y' = u_y + \frac{M_{yz}^u B(t)\,y^2}{2M_{yy}^u} \tag{7.168}$$

$$\sigma'_{xx} = \sigma_{xx} + \frac{B(t)\,y}{M^u_{yy}}\left(M^u_{yz}M^u_{xy} - M^u_{xz}M^u_{yy}\right) \tag{7.169}$$

$$\sigma'_{zz} = \sigma_{zz} + \frac{B(t)\,y}{M^u_{yy}}\left(M^u_{yz}M^u_{yz} - M^u_{yy}M^u_{zz}\right) \tag{7.170}$$

$$p' = p + \frac{MB(t)\,y}{M^u_{yy}}\left(M^u_{yy}\alpha_z - M^u_{yz}\alpha_y\right) \tag{7.171}$$

$$q'_y = q_y - \kappa_y \frac{MB(t)}{M^u_{yy}}\left(M^u_{yy}\alpha_z - M^u_{yz}\alpha_y\right) \tag{7.172}$$

Other variables, u_x, σ_{yy}, σ_{xy}, q_x and ζ, remain unchanged. These new variables satisfy the identical equations for plane strain in Sect. 7.4. Hence the same solver for plane strain problems can be used, except that the traction and pore pressure free boundary conditions

$$t_x = t_y = p = 0 \quad \text{on } \Gamma \tag{7.173}$$

should be modified to

$$t'_x = \frac{M^u_{yz}M^u_{xy} - M^u_{xz}M^u_{yy}}{M^u_{yy}}\,B(t)\,n_x\,y \tag{7.174}$$

$$p' = \frac{M(M^u_{yy}\alpha_z - M^u_{yz}\alpha_y)}{M^u_{yy}}\,B(t)\,y \tag{7.175}$$

and t'_y remains the same.

7.6 Pure Bending of Plate

As an example of the pure bending problem introduced in Sect. 7.5.7, we shall assume that the bar cross section as shown in Fig. 7.11 extends to infinity in the x-direction, and the problem becomes the pure bending of a plate. This gives the plane strain condition in the x-direction and the displacement field in (7.163) becomes

$$u_x = 0; \quad u_y = -\frac{B(t)}{2}\,z^2 + g(y,t); \quad u_z = B(t)\,yz \tag{7.176}$$

The governing equations presented in Sect. 7.5.7 are reduced to partial differential equations in one spatial dimension. The solution of the problem is to determine the functions $B(t)$ and $g(y,t)$ under certain loading conditions.

It is easy to show from (7.176) that the following strains and stresses vanish

$$e_{xx} = e_{yz} = e_{xz} = e_{xy} = 0 \tag{7.177}$$

and

$$\sigma_{yz} = \sigma_{xz} = \sigma_{xy} = 0 \tag{7.178}$$

We also find

$$e_{yy} = \frac{\partial g}{\partial y} \tag{7.179}$$

$$e_{zz} = B(t)\, y \tag{7.180}$$

The stress-strain relations for an orthotropic material are simplified to the following

$$\sigma_{xx} = M_{xy}\, e_{yy} + M_{xz}\, e_{zz} - \alpha_x p \tag{7.181}$$

$$\sigma_{yy} = M_{yy}\, e_{yy} + M_{yz}\, e_{zz} - \alpha_y p \tag{7.182}$$

$$\sigma_{zz} = M_{yz}\, e_{yy} + M_{zz}\, e_{zz} - \alpha_z p \tag{7.183}$$

Using the conditions of (7.178) in the y-component equilibrium equation, we conclude that

$$\frac{\partial \sigma_{yy}}{\partial y} = 0 \tag{7.184}$$

The diffusion equation (7.167) becomes

$$\frac{\partial p}{\partial t} - M\kappa_y \frac{\partial^2 p}{\partial y^2} = -M\alpha_y \frac{\partial e_{yy}}{\partial t} - M\alpha_z y \frac{dB(t)}{dt} \tag{7.185}$$

7.6.1 Bending of Cantilever Plate

We shall first examine the problem of bending of a cantilever plate of thickness h and width b (in the x-direction), subject to a suddenly applied end moment,

$$\mathcal{M} = \mathcal{M}_o\, \mathrm{H}(t - 0) \tag{7.186}$$

where $\mathcal{M}_o$ is the magnitude of the moment, as illustrated in Fig. 7.12a. We assume that at the top and bottom of the plate, $y = \pm h/2$, the surfaces are traction free

$$\sigma_{yy}(\pm h/2, z, t) = 0; \tag{7.187}$$

Integrating (7.184) together with the boundary condition (7.187), we conclude that

$$\sigma_{yy} = 0 \tag{7.188}$$

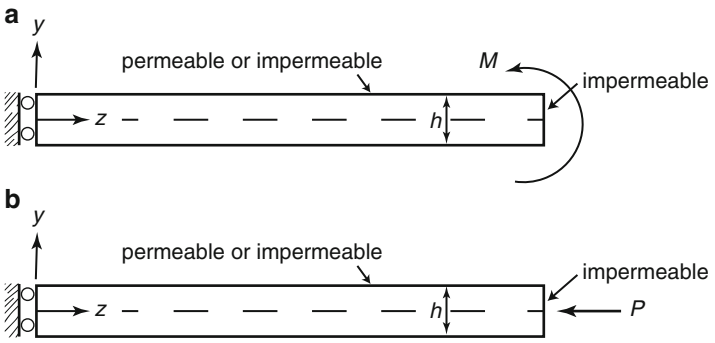


Fig. 7.12 Bending of a plate: (**a**) Cantilever plate subject to an end moment; and (**b**) buckling under axial stress

everywhere. Applying the above to the constitutive equation (7.182), and substituting in (7.180), we obtain

$$e_{yy} = -\frac{M_{yz}}{M_{yy}} B(t)\, y + \frac{\alpha_y}{M_{yy}} p \tag{7.189}$$

Using the above in (7.185), we obtain the following diffusion equation

$$\frac{\partial p}{\partial t} - c_y \frac{\partial^2 p}{\partial y^2} = -\frac{M\left(\alpha_z M_{yy} - \alpha_y M_{yz}\right)}{M^u_{yy}} \frac{dB(t)}{dt}\, y \tag{7.190}$$

where

$$c_y = \frac{\kappa_y M M_{yy}}{M^u_{yy}} \tag{7.191}$$

is the y-component consolidation coefficient.

The solution of (7.190) is dependent on the pressure boundary conditions on the surfaces. We shall examine two cases below: permeable and impermeable surface.

Permeable Surface When the top and bottom surfaces are permeable, the pore pressure is atmospheric, and

$$p(\pm h/2, z, t) = 0 \tag{7.192}$$

To solve (7.190), we shall utilized the Laplace transform (see Appendix D), and it becomes

$$s\tilde{p} - c_y \frac{\partial^2 \tilde{p}}{\partial y^2} = -\frac{s\tilde{B} M(\alpha_z M_{yy} - \alpha_y M_{yz})}{M^u_{yy}}\, y \tag{7.193}$$

Solution of the above subject to the boundary condition (7.192) is

$$\tilde{p} = \frac{\tilde{B}M(\alpha_z M_{yy} - \alpha_y M_{yz})}{M^u_{yy}} \left[\frac{h \sinh \sqrt{s/c_y}\, y}{2 \sinh \sqrt{h^2 s/4c_y}} - y \right] \tag{7.194}$$

Using (7.182) and (7.188) to eliminate e_{yy} in (7.183), we obtain the axial stress as

$$\begin{aligned} \tilde{\sigma}_{zz} &= \frac{M_{yy}M_{zz} - M_{yz}M_{yz}}{M_{yy}} \tilde{e}_{zz} - \frac{\alpha_z M_{yy} - \alpha_y M_{yz}}{M_{yy}} \tilde{p} \\ &= \frac{\tilde{B}(M^u_{yy}M^u_{zz} - M^u_{yz}M^u_{yz})}{M^u_{yy}} y \\ &\quad - \frac{h\tilde{B}M(\alpha_z M_{yy} - \alpha_y M_{yz})^2}{2M_{yy}M^u_{yy}} \frac{\sinh \sqrt{s/c_y}\, y}{\sinh \sqrt{h^2 s/4c_y}} \end{aligned} \tag{7.195}$$

The bending moment for a plate of width b is obtained by integrating the axial stress σ_{zz} as follows

$$\begin{aligned} \tilde{\mathcal{M}} &= -b \int_{-h/2}^{h/2} \tilde{\sigma}_{zz}\, y\, dy \\ &= -\frac{b\, h^3 \tilde{B}(M^u_{yy}M^u_{zz} - M^u_{yz}M^u_{yz})}{12M^u_{yy}} - \frac{b\, h^3 \tilde{B}M(\alpha_z M_{yy} - \alpha_y M_{yz})^2}{4M_{yy}M^u_{yy}} \\ &\quad \times \left(\frac{4c_y}{h^2 s} - \sqrt{\frac{4c_y}{h^2 s}} \coth \sqrt{\frac{h^2 s}{4c_y}} \right) \end{aligned} \tag{7.196}$$

Based on (7.186), we also obtain

$$\tilde{\mathcal{M}} = \frac{\mathcal{M}_o}{s} \tag{7.197}$$

Equating the above, we can solve for the unknown function $\tilde{B}(s)$ as

$$\begin{aligned} \tilde{B}(s) = -\frac{4\mathcal{M}_o}{bh^3 s} \Bigg[&\frac{M^u_{yy}M^u_{zz} - M^u_{yz}M^u_{yz}}{3M^u_{yy}} + \frac{M(\alpha_z M_{yy} - \alpha_y M_{yz})^2}{M_{yy}M^u_{yy}} \\ &\times \left(\frac{4c_y}{h^2 s} - \sqrt{\frac{4c_y}{h^2 s}} \coth \sqrt{\frac{h^2 s}{4c_y}} \right) \Bigg]^{-1} \end{aligned} \tag{7.198}$$

By equating (7.179) and (7.189), and integrate, we obtain

$$\tilde{g}(y,s) = -\frac{\tilde{B}M^u_{yz}}{2M^u_{yy}}\,y^2 - \frac{h^2\tilde{B}\alpha_y M(\alpha_z M_{yy} - \alpha_y M_{yz})}{4M_{yy}M^u_{yy}} \times \frac{1-\cosh\sqrt{s/c_y}\,y}{\sqrt{h^2 s/4c_y}\,\sinh\sqrt{h^2 s/4c_y}} \tag{7.199}$$

With the expressions (7.198) and (7.199), we have fully defined the displacement fields in (7.176). What is of most interest is the *deflection* of the beam; that is, u_y along the neutral axis $y = 0$. This is given by

$$u_y(0,z,t) = -\frac{B(t)}{2}\,z^2 \tag{7.200}$$

with $B(t)$ provided in its Laplace transform form in (7.198).

For an isotropic material,

$$\begin{aligned} M_{yy} &= M_{zz} = \lambda + 2G; \qquad M_{yz} = \lambda \\ M^u_{yy} &= M^u_{zz} = \lambda_u + 2G; \quad M^u_{yz} = \lambda_u \end{aligned} \tag{7.201}$$

where λ and λ_u are respectively the drained and undrained Lamé constant. Hence for isotropy, the deflection can be expressed as

$$\tilde{u}_y(0,z,s) = \frac{6\mathcal{M}_o}{bh^3 s\,\mathbb{E}_1(s)}\,z^2 \tag{7.202}$$

where

$$\mathbb{E}_1(s) = \frac{4G(3K_u+G)}{3K_u+4G} + \frac{108\,G^2\alpha^2 M}{(3K+4G)(3K_u+4G)}\left(\frac{4c}{h^2 s} - \sqrt{\frac{4c}{h^2 s}}\coth\sqrt{\frac{h^2 s}{4c}}\right) \tag{7.203}$$

is an evolutional function analogous to a Young's modulus.

Equation (7.202) is written in such form in order to compare with the elastic beam theory. For an elastic cantilever beam of rectangular cross section subject to an end moment $\mathcal{M}_o$, the beam deflection is

$$u_y = \frac{\mathcal{M}_o}{2EI}\,z^2 = \frac{6\mathcal{M}_o}{bh^3 E}\,z^2 \tag{7.204}$$

where $I = bh^3/12$ is the moment of inertia, and E is Young's modulus. Equation (7.204) however is for a beam under *plane stress* condition in the x-direction. Under the plane strain condition, such as the current problem, the deflection formula is modified to

$$u_y = \frac{6\mathcal{M}_o}{bh^3E'} z^2 \tag{7.205}$$

where E' is the plane strain equivalent of Young's modulus [152], and

$$E' = \frac{E}{1-\nu^2} = \frac{4G(3K+G)}{3K+4G} \tag{7.206}$$

Equation (7.202) is then analogous to (7.205). We further observe the following asymptotic behaviors for the term contained in $\mathbb{E}_1(s)$,

$$\frac{4c}{h^2 s} - \sqrt{\frac{4c}{h^2 s}} \coth \sqrt{\frac{h^2 s}{4c}} \to 0, \quad \text{as } s \to \infty \tag{7.207}$$

$$\frac{4c}{h^2 s} - \sqrt{\frac{4c}{h^2 s}} \coth \sqrt{\frac{h^2 s}{4c}} \to -\frac{1}{3}, \quad \text{as } s \to 0 \tag{7.208}$$

We then conclude that at large time, $t \to \infty$ and $s \to 0$,

$$u_y(0,z,\infty) = \frac{3\,\mathcal{M}_o(3K+4G)}{2bh^3\,G(3K+G)} z^2 = \frac{6\mathcal{M}_o}{bh^3E'} z^2 \tag{7.209}$$

where E' is defined in (7.206). On the other hand, at the instant of the loading, $t \to 0$ and $s \to \infty$, the deflection is

$$u_y(0,z,0^+) = \frac{3\,\mathcal{M}_o(3K_u+4G)}{2\,b\,h^3\,G(3K_u+G)} z^2 = \frac{6\,\mathcal{M}_o}{b\,h^3E'_u} z^2 \tag{7.210}$$

where

$$E'_u = \frac{E_u}{1-\nu_u^2} = \frac{4G(3K_u+G)}{3K_u+4G} \tag{7.211}$$

with E_u the undrained Young's modulus. The deflection then creeps from the small time value given by (7.210) to the large time value given by (7.209).

For numerical result, we assume that the porous material is a Westerly granite saturated with water (see Tables 3.1 and 3.2), with $G = 1.5 \times 10^{10}$ Pa, $K = 2.5 \times 10^{10}$ Pa, $K_u = 3.93 \times 10^{10}$ Pa, $M = 7.08 \times 10^{10}$ Pa, $\alpha = 0.449$, and $c = 2.15 \times 10^{-5}$ m^2/s. A cantilever plate of thickness $h = 2$ m and length $L = 10$ m is suddenly

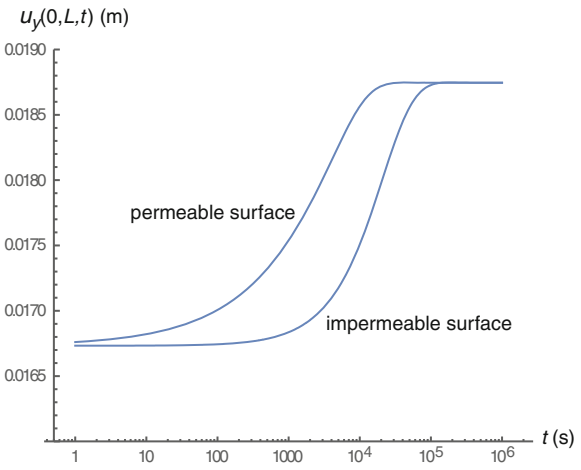


Fig. 7.13 Deflection of a cantilever beam subject to an end moment, with permeable or impermeable surface

subjected to an end moment of $\mathcal{M}_o = 10 \times 10^7\,\mathrm{N}\cdot\mathrm{m}$ per meter of width in the x-direction ($b = 1\,\mathrm{m}$). The deflection as function of time is evaluated by performing numerical inversion on (7.202). Particularly, the Stehfest method of approximate Laplace inversion as described in Sect. D.3 is used. In Fig. 7.13 the deflection at the end of the plate is plotted versus time, as the permeable surface case.

Impermeable Surface In the second case, we assume that the top and bottom surfaces of the plate are sealed off from contact with atmosphere. We hence modify the pore pressure boundary condition to

$$\frac{\partial p}{\partial y} = 0, \quad \text{at } y = \pm h/2 \tag{7.212}$$

Solution of (7.193) then becomes

$$\tilde{p} = \frac{\tilde{B}M(\alpha_z M_{yy} - \alpha_y M_{yz})}{M^u_{yy}} \left[\frac{\sinh\sqrt{s/c_y}\,y}{\sqrt{s/c_y}\cosh\sqrt{h^2 s/4c_y}} - y \right] \tag{7.213}$$

Similar to (7.195), σ_{zz} is obtained as

$$\begin{aligned} \tilde{\sigma}_{zz} = {} & \frac{\tilde{B}(M^u_{yy}M^u_{zz} - M^u_{yz}M^u_{yz})}{M^u_{yy}}\, y \\ & - \frac{\tilde{B}M(\alpha_z M_{yy} - \alpha_y M_{yz})^2}{M_{yy}M^u_{yy}} \frac{\sinh\sqrt{s/c_y}\,y}{\sqrt{s/c_y}\cosh\sqrt{h^2 s/4c_y}} \end{aligned} \tag{7.214}$$

The moment is found as

$$\tilde{\mathcal{M}} = -\frac{bh^3\tilde{B}(M^u_{yy}M^u_{zz} - M^u_{yz}M^u_{yz})}{12M^u_{yy}} + \frac{bh^3\tilde{B}M(\alpha_z M_{yy} - \alpha_y M_{yz})^2}{4M_{yy}M^u_{yy}} \times \left[\frac{4c_y}{h^2 s} - \left(\frac{4c_y}{h^2 s}\right)^{3/2} \tanh\sqrt{\frac{h^2 s}{4c_y}}\right] \tag{7.215}$$

Based on the above, we can solve for the function $\tilde{B}(s)$

$$\tilde{B}(s) = -\frac{4\mathcal{M}_o}{bh^3 s}\left\{\frac{M^u_{yy}M^u_{zz} - M^u_{yz}M^u_{yz}}{3M^u_{yy}} - \frac{M(\alpha_z M_{yy} - \alpha_y M_{yz})^2}{M_{yy}M^u_{yy}} \times \left[\frac{4c_y}{h^2 s} - \left(\frac{4c_y}{h^2 s}\right)^{3/2} \tanh\sqrt{\frac{h^2 s}{4c_y}}\right]\right\}^{-1} \tag{7.216}$$

By equating (7.179) to (7.189), and integrate, we obtain

$$\tilde{g}(y,s) = -\frac{\tilde{B}M^u_{yz}}{2M^u_{yy}}y^2 - \frac{h^2\tilde{B}\alpha_y M(\alpha_z M_{yy} - \alpha_y M_{yz})}{4M_{yy}M^u_{yy}} \times \frac{1 - \cosh\sqrt{s/c_y}\,y}{(h^2 s/4c_y)\cosh\sqrt{h^2 s/4c_y}} \tag{7.217}$$

The functions $\tilde{B}$ and $\tilde{g}$ in (7.216) and (7.217) then fully define the solutions.

For an isotropic material, the deflection at the neutral axis is given by

$$\tilde{u}_y(0,z,s) = \frac{6\mathcal{M}_o z^2}{bh^3 s\mathbb{E}_2(s)} \tag{7.218}$$

where the evolutional function $\mathbb{E}_2(s)$ is defined as

$$\mathbb{E}_2(s) = \frac{4G(3K_u + G)}{3K_u + 4G} - \frac{108\,G^2\alpha^2 M}{(3K + 4G)(3K_u + 4G)} \times \left[\frac{4c}{h^2 s} - \left(\frac{4c}{h^2 s}\right)^{3/2} \tanh\sqrt{\frac{h^2 s}{4c}}\right] \tag{7.219}$$

With the same porous material and plate dimension, the deflection at the end of the plate is plotted versus time in Fig. 7.13 for the impermeable surface case. We observe that the diffusion time for the impermeable case is much longer than the permeable case due to the lack of access to external drainage. The small and large time asymptotic displacement values are the same.

7.6.2 Buckling of Axially Loaded Plate

Next we examine the problem of buckling of a porous plate subject to an axial stress, as illustrated in Fig. 7.12b. This problem was first investigated by Biot [16] for a plate with impermeable surface. We shall examine both the permeable and impermeable surface cases below.

Permeable Surface We shall begin with the equation already derived for the bending moment, as (7.196), but simplified for the isotropic case,

$$\tilde{\mathcal{M}} = -\frac{bh^3\tilde{B}G(3K_u+G)}{3(3K_u+4G)} - \frac{9bh^3\tilde{B}G^2M\alpha^2}{(3K+4G)(3K_u+4G)} \times\left(\frac{4c}{h^2s} - \sqrt{\frac{4c}{h^2s}}\coth\sqrt{\frac{h^2s}{4c}}\right) \tag{7.220}$$

The loading condition for the plate is a uniform compressive axial stress

$$\sigma_{zz} = -P_o\mathrm{H}(t-0) \tag{7.221}$$

applied on the cross section of the plate. Although there is no external moment applied to the plate, we shall examine the stability problem by subjecting the cross section of the plate to a small deflection $u_y(0,z,t)$, thus producing a moment

$$\mathcal{M} = -P_o bhu_y(0,z,t) \tag{7.222}$$

Equating the above with (7.220) allows us to solve for $\tilde{B}$ as

$$\tilde{B}(s) = \frac{12P_o}{h^2\mathbb{E}_1(s)}\tilde{u}_y(0,z,s) \tag{7.223}$$

where $\mathbb{E}_1(s)$ is the analogous Young's modulus defined in (7.203).

Differentiating the deflection $u_y(0,z,t)$ given in (7.176) with respect to z twice allows us to identify that $B(t)$ is the curvature of the plate

$$-\frac{\partial^2 u_y(0,z,t)}{\partial z^2} = B(t) \tag{7.224}$$

In the above we have utilized the condition $g(0,z,t) = 0$. Equating the Laplace transform of (7.224) with (7.223), we obtain the following differential equation

$$\frac{d^2\tilde{u}_y(0,z,s)}{dz^2} + \frac{12P_o}{h^2\mathbb{E}_1(s)}\tilde{u}_y(0,z,s) = 0 \tag{7.225}$$

The above equation is the well-known Euler equation for the buckling of a column with the moment of inertia $I = bh^3/12$. The solution of (7.225) that satisfies the pin-pin end support condition

$$u_y(0, 0, t) = u_y(0, L, t) = 0 \tag{7.226}$$

is given by

$$\tilde{u}_y(0, z, s) = A \sin \sqrt{\frac{12P_o}{h^2 \mathbb{E}_1(s)}}\, z \tag{7.227}$$

where A is an arbitrary constant, and

$$\sqrt{\frac{12P_o}{h^2 \mathbb{E}_1(s)}} = \frac{n\pi}{L}; \quad n = 0, 1, 2, \ldots \tag{7.228}$$

The above equation allows us to solve for the critical buckling stress. For the first mode, $n = 1$, it is

$$\tilde{P}_{\text{cr}}(s) = \frac{\pi^2 h^2 \mathbb{E}_1(s)}{12sL^2} \tag{7.229}$$

We then observe that the critical buckling stress is a function of the slenderness ratio L/h and the material property $\mathbb{E}_1$, which is a function of time. Based on the discussion of the asymptotic behaviors of $\mathbb{E}_1(s)$ in Sect. 7.6.1, we find the limiting values of $P_{\text{cr}}(t)$ as

$$P_{\text{cr}}(0^+) = \frac{\pi^2 h^2 E_u'}{12L^2} \tag{7.230}$$

$$P_{\text{cr}}(\infty) = \frac{\pi^2 h^2 E'}{12L^2} \tag{7.231}$$

The above relation suggests that $P_{\text{cr}}(0^+) > P_{\text{cr}}(\infty)$. This means that it takes a larger axial stress to buckle a plate at small time than at large time. Hence it is possible that when a stress P that is between the two critical values, $P_{\text{cr}}(0^+) > P > P_{\text{cr}}(\infty)$, is applied, the plate does not buckle right away, and buckles at a later time when the critical stress required to buckle a plate is reduced with time.

Impermeable Surface The impermeable surface case is treated the same way as the permeable surface case. The only difference between these two solutions is the definition of the evolutional function that defines the analogous Young's modulus. Hence we conclude that the critical buckling stress for the impermeable surface case is given by

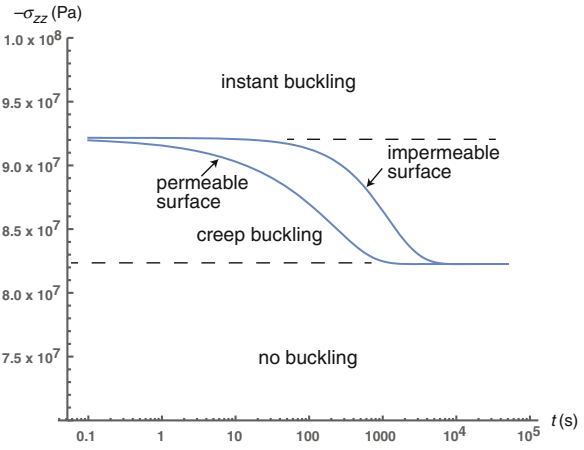


Fig. 7.14 Critical buckling stress of a plate subject to axial stress with pin-pin end condition

$$\tilde{P}_{\text{cr}}(s) = \frac{\pi^2 h^2 \mathbb{E}_2(s)}{12L^2} \tag{7.232}$$

where $\mathbb{E}_2(s)$ is defined in (7.219).

Assuming the same material of Westerly granite as in the preceding section, but changing the plate dimension to $h = 0.5\,\text{m}$ and $L = 10\,\text{m}$ to give a slenderness ratio of 20, we plot in Fig. 7.14 the critical buckling stress as function of time for the permeable and impermeable surface cases.

7.7 Mandel Problem

As discussed in Sect. 1.3.2, the Mandel problem [107] is one of the first solutions that demonstrate a non-monotonic pressure behavior, characteristic of a poroelastic response. Mandel's original solution was for the incompressible constituent model and only pore pressure solution was derived. Here we extend Mandel's work by not only providing the full solution that include the displacement, stress, etc., but also solving for the compressible constituents and anisotropic material cases [5, 37].

Figure 7.15 depicts the geometry of the extended Mandel problem, which shows a transversely isotropic poroelastic specimen, such as shale, sandwiched between two rigid, frictionless plates. Two cases are examined, one with loading applied perpendicular to the bedding plane (plane of isotropy), and the other parallel to it. We assume that the specimen is infinitely long in the y-direction, such that the plane strain condition ($u_y = 0$ and $q_y = 0$) exists in that direction. In addition, in the x-z plane the deformation fulfills a generalized plane strain condition. As a consequence, even though the geometry of the problem has a two-dimensional appearance, with two-dimensional displacement components u_x and u_z, the solution system is one-dimensional. This is demonstrated below.

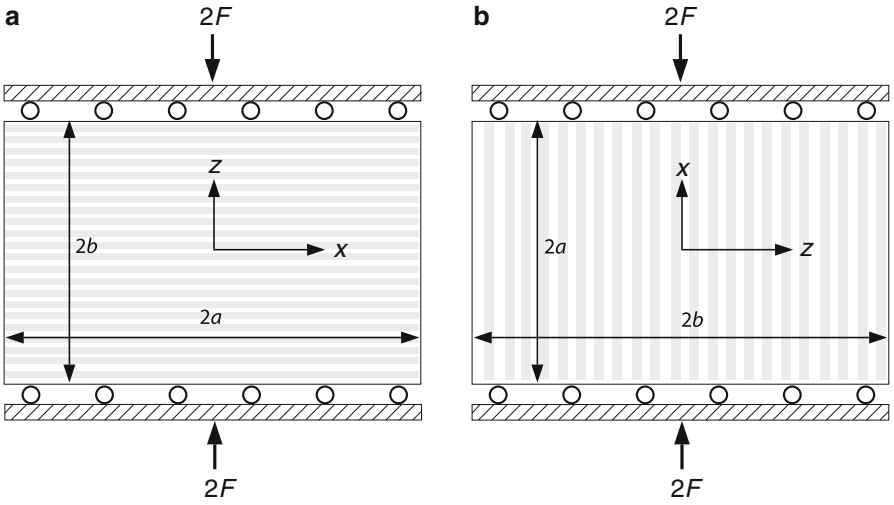


Fig. 7.15 Mandel's problem with transverse isotropic material: (**a**) loading in the direction of axis of rotational symmetry, and (**b**) in the direction parallel to it

Applied Force The boundary conditions for the problem illustrated in Fig. 7.15a are the following:

$$\begin{aligned} \sigma_{xx} = \sigma_{xz} = p = 0, \quad &\text{at} \quad x = \pm a \\ \sigma_{xz} = q_z = 0, \quad &\text{at} \quad z = \pm b \\ u_z = u_z(t), \quad &\text{at} \quad z = \pm b \\ \int_{-a}^{a} \sigma_{zz}\, dx = -2F, \quad &\text{at} \quad z = \pm b \end{aligned} \tag{7.233}$$

This set of boundary conditions is that of the original Mandel problem. We shall also examine a problem of applied displacement later in this section. Of particular interest to this problem is the implicit boundary conditions at $z = \pm b$. The third line of the above equations shows that at the top and bottom surfaces of the specimen, we know only that the vertical displacement u_z is uniform at any given time, imposed by the rigid plate condition; but we do not know its magnitude. The fourth line shows that although we know the magnitude of the vertical force applied, we do not know the distribution of σ_{zz} along the surfaces.

Restricted by the above-mentioned geometry and loading conditions, it is realized that every horizontal plane is a plane of folding symmetry. In other words, a horizontal plane will move in the vertical direction without distortion and remain a horizontal plane. It is also recognized that the flux must be parallel to the confining impermeable plates. More precisely, this is a case of generalized plane strain with an axial strain in the z-direction. Based on (7.153), and the additional condition of plan strain in the y-direction, we conclude

$$u_x = f(x,t); \quad u_y = 0; \quad u_z = C(t)\, z \tag{7.234}$$

With the above, we may express the strain and flux component as follows

$$e_{zz} = e_{zz}(t); \quad q_z = 0 \tag{7.235}$$

Utilizing the constitutive and other governing equations, the following conditions can be further established:

$$\begin{aligned} &e_{xx} = e_{xx}(x,t); \quad q_x = q_x(x,t); \quad \zeta = \zeta(x,t); \\ &\sigma_{zz} = \sigma_{zz}(x,t); \quad p = p(x,t); \quad \sigma_{xx} = \sigma_{xz} = 0 \end{aligned} \tag{7.236}$$

These conditions were introduced by Mandel [107].

The governing equations can be obtained by utilizing the plane strain orthotropic material case presented in Sect. 7.4.1. To adjust to the convention of designating z-axis as the axis of symmetry for transverse isotropy, and the plane strain condition in the x-z plane, rather than the x-y plane, the roles of y and z coordinates are swapped in all equations in Sect. 7.4.1. Based on the conditions in (7.234), (7.235), and (7.236), the Navier equation (7.94) is simplified to

$$M_{xx}\frac{\partial^2 u_x}{\partial x^2} - \alpha_x \frac{\partial p}{\partial x} = 0 \tag{7.237}$$

Substituting the anisotropic Darcy's law (6.26) into the continuity equation (6.63), we also obtain the diffusion equation

$$\frac{\partial \zeta}{\partial t} - \kappa_x \frac{\partial^2 p}{\partial x^2} = 0 \tag{7.238}$$

We note that under the plane strain condition, the anisotropic material properties in the y direction does not come into play. So the solution presented below is valid for both the orthotropic and transversely isotropic cases.

Equation (7.237) can be differentiated with respect to x to give

$$M_{xx}\frac{\partial^2 e_{xx}}{\partial x^2} - \alpha_x \frac{\partial^2 p}{\partial x^2} = 0 \tag{7.239}$$

We can also differentiate the constitutive equation (7.91) to obtain

$$\frac{\partial^2 p}{\partial x^2} = M\frac{\partial^2 \zeta}{\partial x^2} - \alpha_x M \frac{\partial^2 e_{xx}}{\partial x^2} \tag{7.240}$$

Between (7.239) and (7.240) we can eliminate the pressure term in (7.238) to obtain the diffusion equation

$$\frac{\partial \zeta}{\partial t} - c_x \frac{\partial^2 \zeta}{\partial x^2} = 0 \tag{7.241}$$

where

$$c_x = \frac{\kappa_x M M_{xx}}{M_{xx} + \alpha_x^2 M} = \frac{\kappa_x M M_{xx}}{M_{xx}^u} \tag{7.242}$$

is the x-component consolidation coefficient.

The constitutive equations (7.92) and (7.93) become

$$e_{xx} = -\frac{M_{xz}}{M_{xx}M_{zz} - M_{xz}^2}\sigma_{zz} + \frac{\alpha_x M_{zz} - \alpha_z M_{xz}}{M_{xx}M_{zz} - M_{xz}^2} p \tag{7.243}$$

$$e_{zz} = \frac{M_{xx}}{M_{xx}M_{zz} - M_{xz}^2}\sigma_{zz} + \frac{\alpha_z M_{xx} - \alpha_x M_{xz}}{M_{xx}M_{zz} - M_{xz}^2} p \tag{7.244}$$

Substituting (7.243) and (7.244) into (7.91), we find:

$$\zeta = \frac{\alpha_z M_{xx} - \alpha_x M_{xz}}{M_{xx}M_{zz} - M_{xz}^2}(\sigma_{zz} + A_1 p) \tag{7.245}$$

where

$$A_1 = \frac{\alpha_x^2 M_{zz} - 2\alpha_x\alpha_z M_{xz} + \alpha_z^2 M_{xx}}{\alpha_z M_{xx} - \alpha_x M_{xz}} + \frac{M_{xx}M_{zz} - M_{xz}^2}{M(\alpha_z M_{xx} - \alpha_x M_{xz})} \tag{7.246}$$

Using (7.245) in (7.241), the diffusion equation for such a combination of stress and pore pressure exists:

$$\left(\frac{\partial}{\partial t} - c_x \frac{\partial^2}{\partial x^2}\right)(\sigma_{zz} + A_1 p) = 0 \tag{7.247}$$

This equation can be compared to (6.74) for three-dimensional isotropic case.

Another useful equation is the compatibility relation. Under the current simplified strain conditions, (7.82) becomes (y is replaced by z):

$$\frac{\partial^2 e_{zz}}{\partial x^2} = 0 \tag{7.248}$$

Substituting in (7.244), we obtain the orthotropy equivalent of Beltrami-Michell compatibility relation (6.83) as

$$\frac{\partial^2}{\partial x^2}(\sigma_{zz} + A_2 p) = 0 \tag{7.249}$$

with

$$A_2 = \frac{\alpha_z M_{xx} - \alpha_x M_{xz}}{M_{xx}} \tag{7.250}$$

The pair of Equations (7.247) and (7.249) forms a sufficient solution system for the solution of the two variables σ_{zz} and p, and we are ready for their solution.

First, (7.249) can be integrated to give

$$\sigma_{zz} + A_2 p = C_1(t) + C_2(t)\, x \tag{7.251}$$

Due to the symmetry of the solution with respect to the centerline (z-axis), based on the applied loading, it is clear that $C_2(t) = 0$. In fact, if we retain C_2, yet discard C_1, then the solution corresponds to the problem of bending of a specimen between two rigid and impermeable plates. This problem has already been solved as the generalized plane strain bending problem in Sect. 7.6. For the present problem, we drop C_2 to obtain

$$\sigma_{zz} + A_2 p = C_1(t) \tag{7.252}$$

Using separation of variables, the solution of (7.247) can be expressed in terms of the series:

$$\sigma_{zz} + A_1 p = -\frac{F}{a} - \sum_{i=1}^{\infty} D_i \cos\left(\frac{\beta_i x}{a}\right) \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.253}$$

In the above, we have extracted a constant term in front of the series based on the following argument: as $t \to \infty$, $p \to 0$, the material behaves as a drained elastic material. According to elasticity solution, $\sigma_{zz}(x) = -F/a$. We note the condition that as $t \to \infty$, the series goes to zero. Hence there must exist such a constant term in the above expression.

Eliminating σ_{zz} between (7.252) and (7.253), and expanding the time function $C_1(t)$ into a similar time series, we obtain the expression for pore pressure:

$$p = \frac{1}{A_2 - A_1} \sum_{i=1}^{\infty} D_i \left[\cos\left(\frac{\beta_i x}{a}\right) - \cos\beta_i\right] \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.254}$$

We note that as $t \to \infty$, the series and p goes to zero. In the above expression we have incorporated the boundary condition $p = 0$ at $x = \pm a$ through the introduction of the $\cos\beta_i$ term. The stress can be solved from (7.253) in the following form:

$$\begin{aligned}\sigma_{zz} = &-\frac{F}{a} - \frac{A_2}{A_2 - A_1} \sum_{i=1}^{\infty} D_i \cos\left(\frac{\beta_i x}{a}\right) \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \\ &+ \frac{A_1}{A_2 - A_1} \sum_{i=1}^{\infty} D_i \cos\beta_i \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right)\end{aligned} \tag{7.255}$$

To satisfy the loading condition, we substitute (7.255) into the last boundary condition in (7.233) and integrate. From the result, we find that in order for the total force to equal to F, the left hand side of the expression below must cancel with that on the right:

$$\frac{A_2}{A_2 - A_1} \sum_{i=1}^{\infty} \frac{2D_i\, a}{\beta_i} \sin\beta_i \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) = \frac{A_1}{A_2 - A_1} \sum_{i=1}^{\infty} 2D_i\, a \cos\beta_i \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.256}$$

This is possible only if β_i satisfies the characteristic equation:

$$\frac{\tan\beta_i}{\beta_i} = \frac{A_1}{A_2} \tag{7.257}$$

where β_i, $i = 1, 2, 3, \ldots$ are roots of the above transcendental equation.

The constants D_i are determined from the initial condition. At the instant of the loading, the system is undrained, i.e., $\zeta = 0$. The stress distribution is again elastic, $\sigma_{zz}(x, 0) = -F/a$. From the constitutive relations (7.245), the pore pressure generated from the Skempton effect is found as:

$$p(x,0) = -\frac{\sigma_{zz}(x,0)}{A_1} = \frac{F}{aA_1} = \frac{1}{A_2 - A_1} \sum_{i=1}^{\infty} D_i \left[\cos\left(\frac{\beta_i x}{a}\right) - \cos\beta_i\right] \tag{7.258}$$

Utilizing the orthogonal property of the cosine function, we can multiply (7.258) by $\cos(\beta_j x/a)$ and integrate from $-a$ to $+a$ to find the coefficients:

$$D_i = \frac{2F(A_2 - A_1)}{aA_1} \frac{\sin\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \tag{7.259}$$

Hence the solution in terms of pore pressure and vertical stress is complete. The displacement can be found by integrating according to the constitutive equation.

In the following, the complete solution is presented:

$$u_x = \left[\frac{F}{a} \frac{M_{xz}}{M_{xx}M_{zz} - M_{xz}^2} - \frac{2F}{a} \frac{\alpha_x\alpha_z M + M_{xz}}{A_1 M(\alpha_z M_{xx} - \alpha_x M_{xz})} \times \sum_{i=1}^{\infty} \frac{\sin\beta_i \cos\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right)\right] x + \frac{2F\alpha_x}{A_2 M_{xx}} \sum_{i=1}^{\infty} \frac{\cos\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \sin\left(\frac{\beta_i x}{a}\right) \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.260}$$

$$u_z = -\frac{F}{a}\frac{M_{xx}}{M_{xx}M_{zz} - M_{xz}^2}\left[1 + 2\left(\frac{A_2}{A_1} - 1\right)\right.$$
$$\left. \times \sum_{i=1}^{\infty} \frac{\sin\beta_i \cos\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right)\right] z \tag{7.261}$$

$$\sigma_{xx} = 0 \tag{7.262}$$

$$\sigma_{zz} = -\frac{F}{a} - \frac{2F}{a}\frac{A_2}{A_1}\sum_{i=1}^{\infty} \frac{\sin\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \cos\left(\frac{\beta_i x}{a}\right) \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right)$$
$$+ \frac{2F}{a}\sum_{i=1}^{\infty} \frac{\sin\beta_i \cos\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.263}$$

$$\sigma_{xy} = 0 \tag{7.264}$$

$$p = \frac{2F}{aA_1}\sum_{i=1}^{\infty} \frac{\sin\beta_i}{\beta_i - \sin\beta_i \cos\beta_i}\left[\cos\left(\frac{\beta_i x}{a}\right) - \cos\beta_i\right] \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.265}$$

$$q_x = \frac{2F\kappa_x}{a^2 A_1}\sum_{i=1}^{\infty} \frac{\beta_i \sin\beta_i}{\beta_i - \sin\beta_i \cos\beta_i} \sin\left(\frac{\beta_i x}{a}\right) \exp\left(-\frac{\beta_i^2 c_x t}{a^2}\right) \tag{7.266}$$

$$q_z = 0 \tag{7.267}$$

As mentioned above, the above solution is valid for both the transverse isotropy and orthotropy cases. The only difference between the two cases is in the stress σ_{yy} induced by the plane strain condition. We find that for orthotropy

$$\sigma_{yy} = M_{xy}e_{xx} + M_{yz}e_{zz} - \alpha_y p \tag{7.268}$$

and for transverse isotropy

$$\sigma_{yy} = M_{xy}e_{xx} + M_{xz}e_{zz} - \alpha_x p \tag{7.269}$$

due to symmetry.

The above solution corresponds to the material orientation as shown in Fig. 7.15a. For a laboratory test that the force is applied in the direction parallel to the bedding plane of a transversely isotropic material, such as that shown in Fig. 7.15b, the solution is obtained by swapping between the x and z coordinates, as well as the specimen dimensions a and b, on all expressions. For example, the consolidation coefficient c_x is replaced by c_z defined as follows

$$c_z = \frac{\kappa_z M M_{zz}}{M_{zz}^u} \tag{7.270}$$

which is compared with (7.242).

To gain an idea of the behavior of the derived solution and the physical characteristics of Mandel's problem, we shall calculate the solution using the material properties of a transversely isotropic Trafalgar shale as analyzed in Sect. 5.5. The material properties needed are summarized here: $M_{xx} = 24.1\,\text{GPa}$, $M_{xy} = 6.80\,\text{GPa}$, $M_{xz} = 7.62\,\text{GPa}$, $M_{zz} = 21.0\,\text{GPa}$, $\alpha_x = 0.733$, $\alpha_z = 0.749$, and $M = 15.8\,\text{GPa}$. In addition to these mechanical constants, the hydraulic properties are also needed. These data are estimated as follows: $k_x = 1\times10^{-7}$ darcy (1 darcy = 0.987×10^{-12} m^2), $k_z = 2\times10^{-8}$ darcy, μ (for water) = 0.001 Pa·s. Utilizing (7.242) and (7.270), the consolidation coefficients are calculated as $c_x = 1.15\times10^{-6}$ m^2/s and $c_z = 2.19\times10^{-7}$ m^2/s.

Using the above data set, we examine a specimen with a cross-section of $a = b = 10$ cm. The force on the specimen is given as $F = 1\times10^6$ N/m, which suggests an initial stress level of $\sigma_{zz} = -1\times10^7$ Pa. For a specimen with orientation as shown in Fig. 7.15a, Fig. 7.16a illustrates the evolution of pore pressure along the x-axis (at any elevation z). At $t = 0^+$, a uniform pressure of magnitude given by (7.258) arises due to the Skempton effect. As time progresses we observe that the pore pressure in the center region rises above the initial value, which is shaded in a light tone for easy visualization, before it dissipates. This phenomenon is known as the Mandel-Cryer effect, whose physical mechanism is discussed in Sect. 1.3.2.

We next inspect the normal compressive stress $-\sigma_{zz}$ along the same horizontal section. Figure 7.16b shows that at $t = 0^+$, a uniform stress $\sigma_{zz}(x, 0) = -F/a$ exists. Due to the pore pressure dissipation near the edges and the associated load transfer mechanism as discussed in Sect. 1.3.2, the stress near the center rises above the initial value, which is again marked in a light shade, while that near the edges drops below. As $t \to \infty$, the same uniform stress must return, namely $\sigma_{zz}(x, \infty) = -F/a$.

Next, we turn the specimen sideways, as illustrated in Fig. 7.15b. In Fig. 7.17 we plot the pore pressure history at the center line, $p(0, t)$, as the case B. In the same

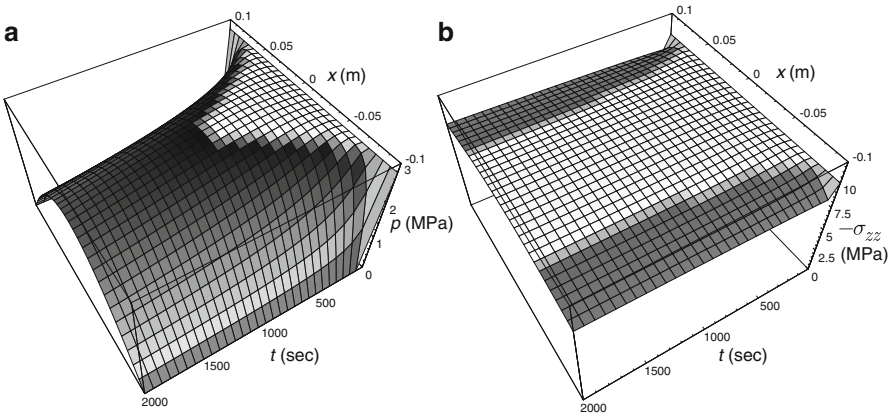


Fig. 7.16 Solution of Mandel's problem with transversely isotropic material: (**a**) pore pressure and (**b**) normal stress along the x-axis as function of time

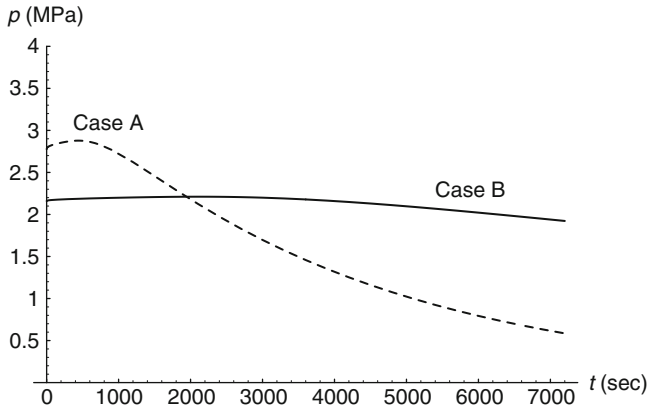


Fig. 7.17 Pore pressure history at the centerline of specimen. *Case A*: force perpendicular to bedding plane; *Case B*: force parallel to bedding plane

figure we also plotted the pore pressure for the case illustrated in Fig. 7.15a as case A for comparison. We note that as the material is stiffer in the x than in the z-direction, the pore pressure rise is smaller for case B. Also, the consolidation coefficient c_z is much smaller than c_x; hence the diffusion time for case B is much longer.

Applied Displacement Also of interest is the problem in which the load is applied as a given vertical displacement, rather than a given force. For a step change of displacement, this boundary condition is applied

$$u_z = \mp u_o \mathrm{H}(t-0), \quad \text{at} \quad z = \pm b \tag{7.271}$$

instead of those in (7.233). As the boundary condition is explicit, this is an easier problem to solve than the original Mandel problem and the solution has been given in Cheng [36].

To solve the problem we begin with the diffusion equation (7.154) simplified for the present conditions

$$\frac{\partial p}{\partial t} - M\kappa_x \frac{\partial^2 p}{\partial x^2} = -M\left(\alpha_x \frac{\partial e_{xx}}{\partial t} + \alpha_z \frac{\partial e_{zz}}{\partial t}\right) \tag{7.272}$$

Utilizing the constitutive equation (7.87), and setting $\sigma_{xx} = 0$ based on (7.236), we find

$$e_{xx} = -\frac{M_{xz}}{M_{xx}} e_{zz} + \frac{\alpha_x}{M_{xx}} p \tag{7.273}$$

The above can be used to eliminate e_{xx} in (7.272) to obtain

$$\frac{\partial p}{\partial t} - c_x \frac{\partial^2 p}{\partial x^2} = -\frac{M(\alpha_z M_{xx} - \alpha_x M_{xz})}{M^u_{xx}} \frac{\partial e_{zz}}{\partial t} \tag{7.274}$$

Based on (7.235) and the boundary condition (7.271) we can easily solve for

$$u_z = -\frac{u_o}{b} z \tag{7.275}$$

and

$$e_{zz} = -\frac{u_o}{b} \tag{7.276}$$

In the above equations we have neglected a Heaviside unit step function on the right hand side for solution at $t > 0$. Equation (7.276) allows us to solve (7.274) as a homogeneous diffusion equation

$$\frac{\partial p}{\partial t} - c_x \frac{\partial^2 p}{\partial x^2} = 0 \tag{7.277}$$

with the initial condition

$$p(x, 0^+) = p_o = \frac{M(\alpha_z M_{xx} - \alpha_x M_{xz})}{M^u_{xx}} \frac{u_o}{b} \tag{7.278}$$

due to the Skempton effect. Solution of (7.277) is then

$$p = \frac{4p_o}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} \exp\left[-\frac{(2n-1)^2 \pi^2 c_x t}{a^2}\right] \cos\left[\frac{(2n-1)\pi x}{2a}\right] \tag{7.279}$$

We notice that (7.279) is the same solution as that of heat diffusion of a finite length bar initially at a constant temperature, and suddenly cooled from both ends to zero temperature [45]. Due to the nature of diffusion equation, the pore pressure given by (7.279) drops monotonically to zero; hence the well-known Mandel-Cryer effect is not observed in this problem.

The x-component displacement is obtained by integrating (7.273), and we find

$$u_x = \frac{u_o M_{xz}}{b M_{xx}} x + \frac{u_o \alpha_x M(\alpha_z M_{xx} - \alpha_x M_{xz})}{b M_{xx} M^u_{xx}} \times \frac{8a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} \exp\left[-\frac{(2n-1)^2 \pi^2 c_x t}{a^2}\right] \cos\left[\frac{(2n-1)\pi x}{2a}\right] \tag{7.280}$$

We note that the infinite sum shown as the second line in the above converges to x and 0 respectively as $t \to 0$ and ∞. We hence observe the small time and large time limits

$$u_x(x, 0^+) = \frac{u_o M^u_{xz}}{b M^u_{xx}} x \tag{7.281}$$

$$u_x(x, \infty) = \frac{u_o M_{xz}}{b M_{xx}} x \tag{7.282}$$

The vertical stress is

$$\sigma_{zz} = -\frac{M_{xx}M_{zz} - M^2_{xz}}{M_{xx}} \frac{u_o}{b} - \frac{\alpha_z M_{xx} - \alpha_x M_{xz}}{M_{xx}} p \tag{7.283}$$

Finally, the force acting on the top and bottom of rigid plates is

$$F = \frac{M_{xx}M_{zz} - M^2_{xz}}{M_{xx}} \frac{a u_o}{b} + \frac{M(\alpha_z M_{xx} - \alpha_x M_{xz})^2}{M_{xx} M^u_{xx}} \frac{a u_o}{b} \times \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \exp\left[-\frac{(2n-1)^2 \pi^2 c_x t}{a^2}\right] \tag{7.284}$$

7.8 Water Wave Over Seabed

In Sect. 1.3.9 we discussed the porous seabed response to a propagating water wave. In this section we shall examine the analytical solution of a semi-infinite seabed loaded on top by a propagating pressure wave, as the result of a propagating water wave on the ocean surface. The poroelastic response in terms of pore pressure, effective stress, and displacement, are derived.

Figure 7.18 gives a definition sketch of the problem investigated. The seabed is considered as semi-infinite in the y-direction. On top of it, a water wave propagates in the x-direction. Based on the linear water wave theory [55], a dynamic pressure is exerted at the bottom of the sea (top of seabed), with the magnitude

$$p = \frac{\gamma_w H}{2\cosh(\ell d)} \cos(\ell x - \omega t) \tag{7.285}$$

where H is the wave height, γ_w the specific weight of water, d the mean water depth, ℓ the *wave number*, and ω the *wave frequency*. The two quantities ℓ and ω are related to *wavelength* L and *wave period* $\mathcal{T}$ in the following fashion

$$\ell = \frac{2\pi}{L}; \quad \omega = \frac{2\pi}{\mathcal{T}} \tag{7.286}$$

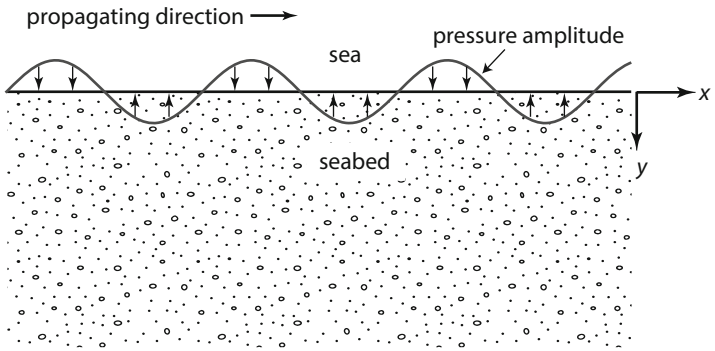


Fig. 7.18 Definition sketch of a semi-infinite poroelastic seabed loaded by a propagating pressure wave

Based on linear water wave theory, there exists such dispersion relation

$$\omega = \sqrt{g\ell \tanh(\ell d)} \tag{7.287}$$

where g is the gravity acceleration.

This water pressure exerts both a vertical total stress and a pore pressure on the seabed. As water is considered inviscid in the linear water wave theory, there is no shear stress exerted to the seabed. These statements are expressed in terms of the following boundary conditions:

$$\begin{aligned} \sigma_{yy}(x,0,t) &= -p_o \cos(\ell x - \omega t) \\ \sigma_{xy}(x,0,t) &= 0 \\ p(x,0,t) &= p_o \cos(\ell x - \omega t) \end{aligned} \tag{7.288}$$

where p_o is given by

$$p_o = \frac{\gamma_w H}{2\cosh(\ell d)} \tag{7.289}$$

In the far field, as $y \to \infty$, all stresses, pressure, displacements, and flux must vanish

$$\sigma_{xx} = \sigma_{yy} = \sigma_{xy} = p = u_x = u_y = q_x = q_y = 0 \tag{7.290}$$

Here it is important to mention that in the mathematical problem defined above and below, we model only the "dynamic" response due to the water wave dynamic loading. In the background, there is a hydrostatic water pressure on top of the seabed, as well as the hydrostatic stress state of the saturated seabed itself. These solutions are simple and are not modeled. However, when it comes to the evaluation

of the true stress state of the seabed, particularly leading to the failure and stability analysis, these background stresses must be added as a part of the solution.

Considering the seabed as a isotropic poroelastic material, the governing equations for this problem are the set of plane strain equations (7.107), (7.108), and (7.109). In the present problem, the special periodic boundary condition in (7.288) allows a "steady state" solution, which is also periodic, after an initial transient has died down. Utilizing the complex variable representation, we shall express the solution variables as

$$u_x(x, y, t) = \tilde{u}_x(y)\, e^{i(\ell x - \omega t)} \tag{7.291}$$

$$u_y(x, y, t) = \tilde{u}_y(y)\, e^{i(\ell x - \omega t)} \tag{7.292}$$

$$p(x, y, t) = \tilde{p}(y)\, e^{i(\ell x - \omega t)} \tag{7.293}$$

Substituting the above into (7.107), (7.108), and (7.109), and dividing the equations through by $e^{i(\ell x - \omega t)}$, we obtain the following set of second order ordinary differential equations

$$G\frac{d^2\tilde{u}_x}{dy^2} + \frac{i\ell G}{1-2\nu}\frac{d\tilde{u}_y}{dy} - \frac{2\ell^2 G(1-\nu)}{1-2\nu}\tilde{u}_x = i\ell\alpha\tilde{p} \tag{7.294}$$

$$\frac{2G(1-\nu)}{1-2\nu}\frac{d^2\tilde{u}_y}{dy^2} + \frac{i\ell G}{1-2\nu}\frac{d\tilde{u}_x}{dy} - \ell^2 G\tilde{u}_y = \alpha\frac{d\tilde{p}}{dy} \tag{7.295}$$

$$\kappa M\frac{d^2\tilde{p}}{dy^2} + \left(i\omega - \ell^2\kappa M\right)\tilde{p} = -i\omega\alpha M\frac{d\tilde{u}_y}{dy} + \ell\omega\alpha M\tilde{u}_x \tag{7.296}$$

In the following, we shall derive the analytical solution following Yamamoto et al. [171], but generalized for the compressible constituents case.

Equations (7.294), (7.295), and (7.296) can be differentiated to eliminate the variables among themselves, to arrive at a 6th order ordinary differential equation in the following form:

$$\left(\frac{d^2}{dy^2} - \ell^2\right)^2 \left(\frac{d^2}{dy^2} - \ell'^2\right) = 0 \tag{7.297}$$

where

$$\ell' = \sqrt{\ell^2 - \frac{i\omega}{c}} \tag{7.298}$$

in which c is the consolidation coefficient.

With the above characteristic equation, the solutions of $\tilde{u}_x$, $\tilde{u}_y$, and $\tilde{p}$ can be expressed in these general forms

$$\tilde{u}_x = a_1\, e^{-\ell y} + a_2\, \ell y\, e^{-\ell y} + a_3\, e^{-\ell' y} + a_4\, e^{\ell y} + a_5\, \ell y\, e^{\ell y} + a_6\, e^{\ell' y} \tag{7.299}$$

$$\tilde{u}_y = b_1\, e^{-\ell y} + b_2\, \ell y\, e^{-\ell y} + b_3\, e^{-\ell' y} + b_4\, e^{\ell y} + b_5\, \ell y\, e^{\ell y} + b_6\, e^{\ell' y} \tag{7.300}$$

$$\tilde{p} = c_1\, e^{-\ell y} + c_2\, \ell y\, e^{-\ell y} + c_3\, e^{-\ell' y} + c_4\, e^{\ell y} + c_5\, \ell y\, e^{\ell y} + c_6\, e^{\ell' y} \tag{7.301}$$

However, if we consider the boundary condition at $y \to \infty$, as given in (7.290), it is clear that all terms with positive argument of the exponential function must vanish, or the solution becomes unbounded at infinity. (For problem of finite thickness seabed, however, all terms remains.) So for the present infinite depth problem, (7.299), (7.300), and (7.301) simplify to

$$\tilde{u}_x(y) = a_1\, e^{-\ell y} + a_2\, \ell y\, e^{-\ell y} + a_3\, e^{-\ell' y} \tag{7.302}$$

$$\tilde{u}_y(y) = b_1\, e^{-\ell y} + b_2\, \ell y\, e^{-\ell y} + b_3\, e^{-\ell' y} \tag{7.303}$$

$$\tilde{p}(y) = c_1\, e^{-\ell y} + c_2\, \ell y\, e^{-\ell y} + c_3\, e^{-\ell' y} \tag{7.304}$$

We therefore observe that the seabed response in terms of displacement and pressure is characterized by the exponential decay with increasing depth. The decay has two length scales, associated with the inverse of the wave numbers ℓ and ℓ'. According to (7.286) and (7.298), ℓ is a function of the wavelength L only, while ℓ' is in addition dependent on the material properties. The significance of these two length scales will be explored further in the special cases below.

These nine coefficients, $a_1,\ a_2,\ \ldots,\ c_3$, are not all independent. In fact, the above solutions need to satisfy two of the three governing equations (7.294), (7.295), and (7.296). Substituting (7.302), (7.303), and (7.304) into two of the governing equations, and equating terms of $e^{-\ell y}$, $\ell y e^{-\ell y}$, and $e^{-\ell' y}$, we obtain six equations that allow six of the coefficients to be solved as functions of the other three. If we resolve $b_1,\ b_2,\ \ldots,\ c_3$, in terms of a_1, a_2, and a_3, we obtain

$$b_1 = i a_1 + \frac{i[1 + m(3 - 4\nu)]}{1 + m} a_2 \tag{7.305}$$

$$b_2 = i a_2 \tag{7.306}$$

$$b_3 = \frac{i\ell'}{\ell} a_3 \tag{7.307}$$

$$c_1 = \frac{2iG\ell}{\alpha(1 + m)} a_2 \tag{7.308}$$

$$c_2 = 0 \tag{7.309}$$

$$c_3 = -\frac{2G(1 - \nu)\omega}{\alpha c \ell (1 - 2\nu)} a_3 \tag{7.310}$$

where we have utilized the shorthand notation

$$m = \frac{G}{\alpha^2 M(1-2\nu)} = \frac{1-2\nu_u}{2(\nu_u-\nu)} \tag{7.311}$$

We seek to solve the three remaining coefficients a_1, a_2, and a_3 from the boundary conditions in (7.288), which can be expressed as follows:

$$\frac{2i\ell G\nu}{1-2\nu}\tilde{u}_x(0) + \frac{2G(1-\nu)}{1-2\nu}\frac{d\tilde{u}_y(0)}{dy} - \alpha\tilde{p}(0) = -p_o \tag{7.312}$$

$$\tilde{p}(0) = p_o \tag{7.313}$$

$$\frac{d\tilde{u}_x(0)}{dy} + i\ell\tilde{u}_y(0) = 0 \tag{7.314}$$

From the above, these coefficients are solved as

$$a_1 = -\frac{ip_o}{2G}\frac{(1+\ell'')[1-\alpha(1+m)] - m(1-2\nu)(\alpha\ell'' + i\omega'')}{\ell[\ell'' + i(1+m)\omega'']} \tag{7.315}$$

$$a_2 = -\frac{ip_o}{2G}\frac{(1+m)(\alpha\ell'' + i\omega'')}{\ell[\ell'' + i(1+m)\omega'']} \tag{7.316}$$

$$a_3 = \frac{ip_o}{2G}\frac{1-\alpha(1+m)}{\ell[\ell'' + i(1+m)\omega'']} \tag{7.317}$$

By back substitution into (7.305)–(7.310), the other coefficients are found:

$$b_1 = \frac{p_o}{2G}\frac{(1+\ell'')[1-\alpha(1+m)] + [1+2m(1-\nu)](\alpha\ell'' + i\omega'')}{\ell[\ell'' + i(1+m)\omega'']} \tag{7.318}$$

$$b_2 = ia_2 \tag{7.319}$$

$$b_3 = -\frac{p_o}{2G}\frac{(1+\ell'')[1-\alpha(1+m)]}{\ell[\ell'' + i(1+m)\omega'']} \tag{7.320}$$

$$c_1 = p_o\frac{\alpha\ell'' + i\omega''}{\alpha[\ell'' + i(1+m)\omega'']} \tag{7.321}$$

$$c_2 = 0 \tag{7.322}$$

$$c_3 = -ip_o\frac{[1-\alpha(1+m)]\omega''}{\alpha[\ell'' + i(1+m)\omega'']} \tag{7.323}$$

where

$$\ell'' = \frac{\ell' - \ell}{\ell} \tag{7.324}$$

$$\omega'' = \frac{\omega(1-\nu)}{\ell^2 c(1-2\nu)} \tag{7.325}$$

We now have solved the problem with displacements and pressure given by (7.291)–(7.293), (7.302)–(7.304), and (7.315)–(7.323).

The stresses can be obtained from the constitutive equations, and they are expressed as

$$\tilde{\sigma}_{xx}(y) = a_1' e^{-\ell y} + a_2' \ell y e^{-\ell y} + a_3' e^{-\ell' y} \tag{7.326}$$

$$\tilde{\sigma}_{yy}(y) = b_1' e^{-\ell y} + b_2' \ell y e^{-\ell y} + b_3' e^{-\ell' y} \tag{7.327}$$

$$\tilde{\sigma}_{xy}(y) = c_1' e^{-\ell y} + c_2' \ell y e^{-\ell y} + c_3' e^{-\ell' y} \tag{7.328}$$

with

$$a_1' = -p_o \left[1 - \frac{(1+2\ell'')[1-\alpha(1+m)]}{\ell'' + i(1+m)\,\omega''} \right] \tag{7.329}$$

$$a_2' = p_o \frac{(1+m)(\alpha\ell'' + i\,\omega'')}{\ell'' + i(1+m)\,\omega''} \tag{7.330}$$

$$a_3' = p_o \frac{[1-\alpha(1+m)]\,[(1-\nu) - i(1-2\nu)\,\omega'']}{(1-\nu)[\ell'' + i(1+m)\,\omega'']} \tag{7.331}$$

$$b_1' = -p_o \left[1 + \frac{1-\alpha(1+m)}{\ell'' + i(1+m)\,\omega''} \right] \tag{7.332}$$

$$b_2' = -a_2' \tag{7.333}$$

$$b_3' = p_o \frac{1-\alpha(1+m)}{\ell'' + i(1+m)\,\omega''} \tag{7.334}$$

$$c_1' = i p_o \frac{(1+\ell'')[1-\alpha(1+m)]}{\ell'' + i(1+m)\,\omega''} \tag{7.335}$$

$$c_2' = i a_2' \tag{7.336}$$

$$c_3' = -c_1' \tag{7.337}$$

The above solutions, (7.302)–(7.304), (7.315)–(7.323), and (7.326)–(7.337), when properly normalized, that is, stresses and pore pressure by p_o, and displacements by $p_o/2G\ell$, are functions of the dimensionless depth ℓy, and four dimensionless parameters: α, m, ℓ'', and ω'', only. Or, in terms of physical parameters, these four dimensionless constants can be replaced by α, ν, M/G, and $\omega/\ell^2 c$.

To gain physical insights into the above analytical solution, we shall examine some limiting cases.

Saturated Seabed—Incompressible Pore Water Consider the soil mechanics model with saturated pore water (see Sect. 4.8.3). The material properties are characterized by $\alpha = 1$ and $\nu_u = 1/2$, which further lead to $m = 0$, as defined in (7.311). We find that the solutions significantly simplify, and

$$\frac{2G\ell\tilde{u}_x}{p_o} = -i\ell y\, e^{-\ell y} \tag{7.338}$$

$$\frac{2G\ell\tilde{u}_y}{p_o} = (1+\ell y)e^{-\ell y} \tag{7.339}$$

$$\frac{\tilde{p}}{p_o} = e^{-\ell y} \tag{7.340}$$

$$\frac{\tilde{\sigma}_{xx}}{p_o} = -(1-\ell y)\, e^{-\ell y} \tag{7.341}$$

$$\frac{\tilde{\sigma}_{yy}}{p_o} = -(1+\ell y)\, e^{-\ell y} \tag{7.342}$$

$$\frac{\tilde{\sigma}_{xy}}{p_o} = i\,\ell y\, e^{-\ell y} \tag{7.343}$$

It is of interest to observe that these solutions are free from the $e^{-\ell' y}$ term. Hence for saturated seabed, the penetration depth of water wave pressure into the seabed is entirely characterized by the wavelength L of the water wave, and is independent of the poroelastic properties of the seabed. In fact, the pore pressure solution (7.340) is exactly the same as that obtained by Putnam [127] for studying water wave damping due to energy dissipated by seepage flow into the seabed. Putnam's model assumes Darcy's flow of an incompressible fluid in a rigid, nondeformable porous medium. The governing equation is Laplace equation without any material properties in its coefficients; hence the pore pressure is independent of any material properties, even the permeability coefficient. The porous medium flow theory however does not give any prediction of seabed deformation and stresses. The poroelastic theory gives the complete response of the seabed, including displacements, stresses, and pore pressure, and in the general case, these responses are dependent on the seabed material properties.

The solutions presented as (7.338)–(7.343) can be normalized and plotted as their complex amplitudes versus dimensionless depth ℓy in Fig. 7.19. There are only three curves shown because we recognize the following relations: $2G\ell|\tilde{u}_x|/p_o = |\tilde{\sigma}'_{xx}|/p_o = |\tilde{\sigma}'_{yy}|/p_o = |\tilde{\sigma}_{xy}|/p_o$, where $\sigma'_{xx} = \sigma_{xx} + p$ and $\sigma'_{yy} = \sigma_{yy} + p$ are effective stresses.

In another limit, we consider a highly permeable seabed, and in the limit, $\kappa \to \infty$. This condition leads to $c \to \infty$, and $\ell' \to \ell$. In this case, the penetration depth is also independent of ℓ'. By combining terms in (7.304), with c_1 and c_3 given by (7.321) and (7.323), we find that $c_1 + c_3 = p_o$; hence the pressure reduces to

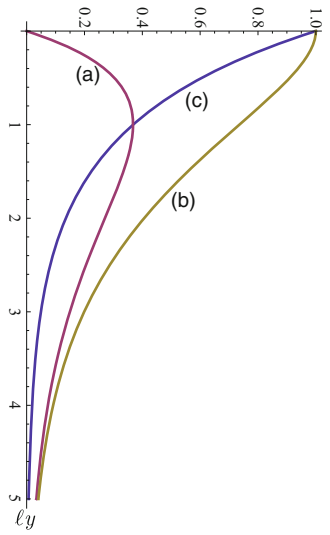


Fig. 7.19 Normalized displacement, stress, and pressure amplitude versus dimensionless depth for a poroelastic seabed subjected a traveling water wave loading: (*a*) $2G\ell|\tilde{u}_x|/p_o = |\tilde{\sigma}'_{xx}|/p_o = |\tilde{\sigma}'_{yy}|/p_o = |\tilde{\sigma}_{xy}|/p_o$, (*b*) $2G\ell|\tilde{u}_y|/p_o$, and (*c*) $\tilde{p}/p_o$, versus ℓy

the same expression as (7.340). However, the displacements and the stresses do not reduce to those expressed in (7.338)–(7.343).

Unsaturated Seabed—Compressible Pore Water Although the incompressible solid constituent assumption used in the above, given as $\alpha = 1$, is a good approximation for a typical seabed, because the seabed material is largely loosely packed sand and clay, it has been observed that the pore water in sea bottom sediment is not always saturated. In fact, a small amount of gas bubble may be present due to the decomposing organic matters in sediments (see discussion in Sect. 1.3.9). Hence the apparent compressibility of the air-water mixture C_f defined in (4.178) is large, rather than small as stated in the incompressible pore water assumption. Based on the highly compressible fluid approximation (4.182), we can express the consolidation coefficient as

$$c \approx \frac{\kappa K_f}{\phi} \tag{7.344}$$

Taking the limit of small consolidation coefficient, ℓ' is approximately

$$\Re(\ell') \approx \sqrt{\frac{\omega}{2c}} \tag{7.345}$$

Hence in addition to wavelength L, there exists a second length scale that limits the pore pressure penetration. Based on (7.344), we observe that both increasing pore water compressibility and decreasing the permeability have the same effect in limiting the penetration depth of pore pressure.

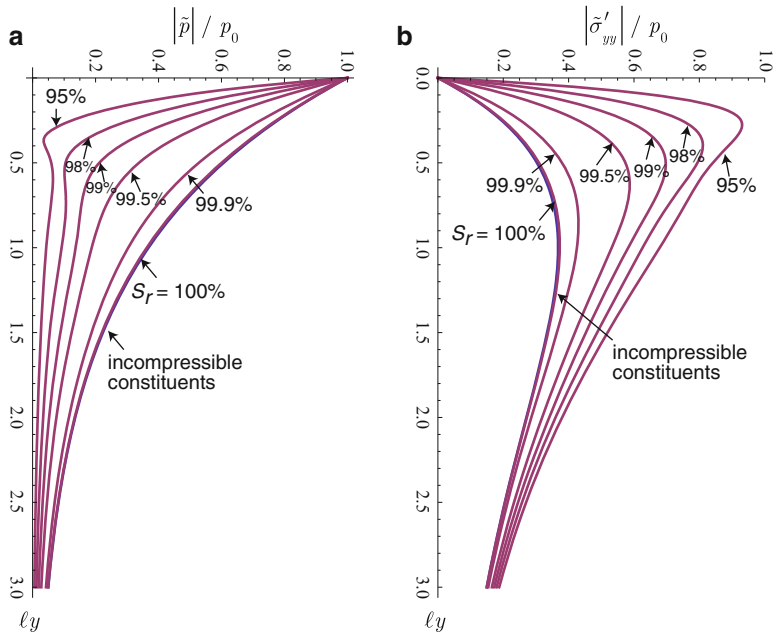


Fig. 7.20 Normalized pressure and effective stress amplitude versus dimensionless depth, for various saturation degrees: (**a**) $|\tilde{p}|/p_o$, and (**b**) $|\tilde{\sigma}'_{yy}|/p_o$

To fix ideas, we shall examine some numerical results. Take a water wave with water depth $d = 7\,\text{m}$, wavelength $L = 63\,\text{m}$, wave period $\mathcal{T} = 8\,\text{s}$, and specific weight of water $\gamma_w = 1 \times 10^4\,\text{N/m}^3$ [105]. For seabed material, we assume the material properties of a hard sediment with saturated pore water based on Tables 3.1 and 3.2, with $\alpha = 1$, $\nu = 0.25$, $G = 2.6 \times 10^7\,\text{N/m}^2$, $M = 4.5 \times 10^9\,\text{N/m}^2$, $\kappa = 1 \times 10^{-7}\,\text{m}^4/\text{N·s}$, $\phi = 0.47$, and $c = 7.7\,\text{m}^2/\text{s}$. The pore pressure amplitude $|\tilde{p}|/p_o$ and the effective stress amplitude $|\tilde{\sigma}'_{yy}|/p_o$ are respectively plotted in Fig. 7.20a, b. These are marked as the 100 % saturation curves. Also plotted is the case of incompressible constituents, as predicted by (7.339) and (7.340). We observe that the responses of these two cases are largely indistinguishable.

Next, we assume that the pore water is partially saturated, with degree of saturation $S_r = \{99.9\,\%,\ 99.5\,\%,\ 99\,\%,\ 98\,\%,\ 95\,\%\}$. Based on (4.177), and $P_a = 1.7 \times 10^5\,\text{N/m}^2$ (7 m of hydrostatic pressure of sea water plus atmospheric pressure), we calculate the apparent bulk modulus of pore water as $K_f = \{1.6 \times 10^8,\ 3.4 \times 10^7,\ 1.7 \times 10^7,\ 8.5 \times 10^6,\ 3.4 \times 10^6\}\,\text{N/m}^2$, which are compared to $K_f = 2.25 \times 10^9\,\text{N/m}^2$ for the saturated case. Hence based on (4.177), even as little as 0.1 % air content can cause drastic increase in apparent pore water compressibility.

In Fig. 7.20 we also present these solutions. For the pore pressure, we observe a significant reduction in its penetration within a boundary layer thickness of $\delta_\ell = 3/\ell' \approx 3\sqrt{2c/\omega}$, or the dimensionless thickness $\ell\delta_\ell \approx 3\sqrt{2c\,\ell^2/\omega}$. For the unsaturated case, we calculate $c = \{7.7,\ 6.3,\ 3.7,\ 2.5,\ 1.5,\ 0.67\}\,\text{m}^2/\text{s}$

corresponding to the various saturation levels shown above. From these values, we find $\ell\delta_\ell \approx \{1.3,\ 1.2,\ 0.92,\ 0.75,\ 0.58,\ 0.39\}$. For the more compressible pore water cases, we clearly observe the presence of such feature.

In Fig. 7.20b we observe that the magnitude of vertical effective stress significantly increases as pore water becomes more compressible. This is a direct consequence of the reduction of pore pressure penetration. The effective stress peaks around the dimensionless boundary layer thickness $\ell\delta_\ell$, with values given above. As the wave loading is sinusoidal, the wave generated effective stress is added to and subtracted from the background effective stress, as the wave crest and trough pass by. Hence the larger the generated effective stress is, the less stable the seabed becomes.

Flux Exchange The fluctuating pressure at sea bottom not only induces a seepage flux in the seabed, but also pumps fluid in and out of it. The volume of fluid exchanged per wave cycle is indicative of the rate of contaminant mass released from the sediment into the ocean [93, 114]. Hence we shall examine the seepage exchange due to the wave and seabed interaction.

The total volume of water exchanged per wave length and per wave period (volume entering and leaving the seabed are all counted as positive) can be obtained by integrating the normal flux at top of seabed with respect to x and t, and

$$V_w = \frac{2L\mathcal{T}}{\pi} |\tilde{q}_y(0)| \tag{7.346}$$

where V_w is the volume per unit length of seabed perpendicular to the water wave propagation direction. We may obtain $\tilde{q}_y$ by differentiating pressure according to Darcy's law, and

$$\tilde{q}_y(0) = -p_o \kappa \ell \left[1 - \frac{i\ell'' \omega'' [1 - \alpha(1+m)]}{\alpha[\ell'' + i(1+m)\,\omega'']} \right] \tag{7.347}$$

As this expression is rather complicated, we can examine two special cases.

When the seabed is saturated, we can assume incompressible constituents, and take $\alpha = 1$ and $m = 0$ in the above equation we obtain

$$V_w = 4 p_o \kappa \mathcal{T} \tag{7.348}$$

Half of the above volume is pumped in and half pumped out. Hence the volume of water pumped in or out of seabed is proportional to the magnitude of dynamic pressure at sea bottom, the permeability, and the wave period, but is independent of the wavelength.

When the seabed is partially saturated, and the pore water is highly compressible, we may take the limit of $K_f/G = \epsilon \to 0$, and then $M \sim O(\epsilon)$, $S \sim O(\epsilon^{-1})$, $c \sim O(\epsilon)$, $\ell'' \sim O(\epsilon^{-1/2})$, $\omega'' \sim O(\epsilon^{-1})$, and $m \sim O(\epsilon^{-1})$. Based on (7.347), we obtain

$$\tilde{q}_y(0) \approx p_o \kappa \ell \ell'' \tag{7.349}$$

and

$$V_w \approx 2p_o L\sqrt{\frac{\phi\kappa\mathcal{T}}{\pi K_f}} \tag{7.350}$$

where we have used the relations $\ell'' \approx \sqrt{\omega/2c}/\ell$, and $c \approx \kappa K_f/\phi$. We observe that the prediction provided by (7.350) is very different from that of (7.348). Particularly, the compressibility of pore water plays an important role. The more compressible the pore water is, the larger the exchanged volume.

To get an idea of the difference, we can calculate the volume of water exchanged per meter of seabed perpendicular to the wave propagation direction based on (7.348) for the incompressible constituents case using the wave and material parameters mentioned above. For a 2 m wave height, we calculate from (7.285) a bottom pressure $p_o = 8\times10^3$ N/m^2. From (7.348), we obtain $V_w = 2.6\times10^{-2}$ m^3/m. For unsaturated pore water, assuming $S_r = 99\,\%$, we obtain $V_w = 8.2\times10^{-2}$ m^3/m, which is about 3 times larger than the incompressible constituents case.

Energy Dissipation When a water wave propagates over a seabed, there are many mechanisms that its energy can be dissipated, including the viscosity of water, the deformation of a viscoelastic sea bottom mud, and a seepage flow in the seabed driven by the wave induced pressure fluctuation [150]. For wave damping over a permeable seabed, the seabed has been modeled as a rigid porous medium with Darcy's flow [127, 132], and also as a poroelastic medium [54, 90, 91]. In the following, we shall have a simple examination of the energy dissipated due to the seepage flow.

As discussed in Sect. 6.1.2, the energy dissipated in porous medium is proportional to the square of the magnitude of the specific flux. First, we obtain

$$\tilde{q}_x(y) = -i\ell\kappa\tilde{p}(y) \tag{7.351}$$

$$\tilde{q}_y(y) = -\kappa\frac{d\tilde{p}(y)}{dy} \tag{7.352}$$

where $\tilde{p}(y)$ is given by (7.304). The energy dissipated per wave length, per wave period, and per unit length in the z direction, for the entire depth, can be obtained by integrating over x, y, and t, and is given by

$$E_D = \frac{2L\mathcal{T}}{\pi\kappa}\int_0^\infty \left(|\tilde{q}_x|^2 + |\tilde{q}_y|^2\right)\, dy \tag{7.353}$$

For the incompressible constituents case, with $\tilde{p}$ given as (7.340), we obtain the following simple result

$$E_D = 4\kappa\mathcal{T}p_o^2 \tag{7.354}$$

of which half of the energy dissipated is due to the vertical flux, and the other half the horizontal flux. Equation (7.354) shows that energy dissipated is proportional to the permeability coefficient and the wave period.

7.9 Spherical Symmetry

Definitions of the poroelastic variables and governing equations in the spherical coordinate system (R, θ, φ) are presented in Sect. C.3. In this section, we examine the special case of problems possessing spherical symmetry, that is, all variables are functions of R and t only. This condition places restriction on displacement such that only the radial component can exist

$$u_R = u_R(R, t); \quad u_\theta = u_\varphi = 0 \tag{7.355}$$

For stresses and strains, all normal components exist, while all shear components vanish.

As observed in Sect. C.3, the equilibrium equations (C.55) and (C.56) become

$$\frac{\partial \sigma_{RR}}{\partial R} + \frac{2\sigma_{RR} - \sigma_{\theta\theta} - \sigma_{\varphi\varphi}}{R} = 0 \tag{7.356}$$

$$\sigma_{\theta\theta} = \sigma_{\varphi\varphi} \tag{7.357}$$

The stress-strain relations are

$$\sigma_{RR} = \frac{2G\nu}{1-2\nu} e + 2G e_{RR} - \alpha p \tag{7.358}$$

$$\sigma_{\varphi\varphi} = \frac{2G\nu}{1-2\nu} e + 2G e_{\varphi\varphi} - \alpha p \tag{7.359}$$

with

$$e_{RR} = \frac{\partial u_R}{\partial R} \tag{7.360}$$

$$e_{\varphi\varphi} = e_{\theta\theta} = \frac{u_R}{R} \tag{7.361}$$

$$e = \frac{\partial u_R}{\partial R} + 2\frac{u_R}{R} = \frac{1}{R^2}\frac{\partial R^2 u_R}{\partial R} \tag{7.362}$$

Substituting the above into the equilibrium equation (7.356), we obtain the Navier equation

$$\frac{2G(1-\nu)}{1-2\nu}\left(\frac{\partial^2 u_R}{\partial R^2} + \frac{2}{R}\frac{\partial u_R}{\partial R} - \frac{2u_R}{R^2}\right) - \alpha\frac{\partial p}{\partial R} = 0 \tag{7.363}$$

We can rearrange the above into

$$\frac{\partial e}{\partial R} = \frac{\partial}{\partial R}\left(\frac{1}{R^2}\frac{\partial R^2 u_R}{\partial R}\right) = \frac{\eta}{G}\frac{\partial p}{\partial R} \tag{7.364}$$

Integrating once, we obtain

$$e = \frac{\eta}{G}p + 3A_1(t) \tag{7.365}$$

where A_1 is an integration constant (function of time). Integrating another time we find

$$u_R(R,t) = \frac{\eta}{G}\frac{1}{R^2}\int R^2 p(R,t)\, dR + A_1(t)R + \frac{A_2(t)}{R^2} \tag{7.366}$$

where $A_2(t)$ is another arbitrary functions of time. These coefficients are to be determined by the boundary conditions of a specific problem.

Substituting the above into (7.358) and (7.359), we obtain the solution for the stresses

$$\sigma_{RR} = -4\eta\frac{1}{R^3}\int R^2 p(R,t)\, dR + \frac{2G(1+\nu)}{1-2\nu}A_1(t) - \frac{4G}{R^3}A_2(t) \tag{7.367}$$

$$\sigma_{\varphi\varphi} = 2\eta\frac{1}{R^3}\int R^2 p(R,t)\, dR - 2\eta p + \frac{2G(1+\nu)}{1-2\nu}A_1(t) + \frac{2G}{R^3}A_2(t) \tag{7.368}$$

For pore pressure equation, we notice that the displacement field with spherical symmetry is irrotational, that is

$$\nabla \times \vec{u} = \nabla \times (u_R(R,t)\,\hat{e}_R) = 0 \tag{7.369}$$

based on (C.41). We can therefore utilize the earlier result in Sect. 6.4.3 to express the pressure diffusion equation as (6.110), which is repeated here as

$$\frac{\partial p}{\partial t} - c\nabla^2 p = -\frac{\eta(1-\nu)}{GS(1+\nu)}\frac{d}{dt}(\sigma_{kk} + 4\eta p) \tag{7.370}$$

We notice that the right hand side term $(\sigma_{kk} + 4\eta p)$ has been proven to be a function of time only. This is confirmed from (7.367) and (7.368), and we find

$$\sigma_{kk} + 4\eta p = \sigma_{RR} + 2\sigma_{\varphi\varphi} + 4\eta p = \frac{6G(1+\nu)}{1-2\nu}A_1(t) \tag{7.371}$$

We hence can write (7.370) as the following

$$\frac{\partial p}{\partial t} - c\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial p}{\partial R}\right) = -\frac{3\alpha}{S}\frac{dA_1(t)}{dt} \tag{7.372}$$

7.10 Cryer Problem

Cryer [50] investigated the problem of a saturated porous sphere subject to a suddenly applied constant radial compressive stress. An initial uniform pore pressure rise will result due to the Skempton pore pressure effect. The surface of the sphere is permeable and fluid can freely escape. As discussed in Sect. 1.3.2, the non-monotonic pressure diffusion observed has been referred to as a Mandel-Cryer effect.

Cryer considered the porous material as soil saturated with water, and used the incompressible constituent model. The exact solution was only provided for the pore pressure at the center, and the radial displacement on the surface. The full solution for the incompressible constituent model has been derived by Mason et al. [108]. In the following, we shall derive the full solution for the general poroelastic model (compressible constituents).

The initial condition is an undisturbed specimen, and then at $t = 0^+$ a radial stress is suddenly applied. The boundary conditions are

$$\begin{aligned} &\sigma_{RR} = -P_o \mathrm{H}(t-0), \quad p = 0, \quad \text{at } R = R_o \\ &u_R = 0, \quad \frac{\partial p}{\partial R} = 0, \quad \text{at } R = 0 \end{aligned} \tag{7.373}$$

where R_o is the radius of the sphere.

Laplace Transform Solution To solve the problem, we begin with (7.372) and perform Laplace transform over it to remove the time variable and derivative (see Appendix D), such that it is transformed from a partial differential equation to an ordinary differential equation

$$\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2 \frac{\partial \tilde{p}}{\partial R}\right) - \frac{s}{c}\tilde{p} = \frac{3\alpha s}{\kappa}\tilde{A}_1(s) \tag{7.374}$$

where the $\tilde{}$ overbar represents the Laplace transformed quantities, and s is the transform parameter. In the above we have used the initial condition $p(R, 0^-) = 0$.

Equation (7.374) is a second order ordinary differential equation, and can be solved to give

$$\tilde{p} = C_1 \frac{\sinh\sqrt{R^2 s/c}}{R} + C_2 \frac{\cosh\sqrt{R^2 s/c}}{R} - \frac{3\alpha}{S}\tilde{A}_1(s) \tag{7.375}$$

For a solid sphere, the solution needs to be bounded at $R = 0$; hence it is necessary that $C_2 = 0$. For a hollow sphere or a cavity, however, such term should be retained. We shall not present the hollow sphere solution here (see Nowinski and Davis [118] for hollow sphere solution for incompressible constituents case). Its solution, however, follows the same procedure below, with only lengthier algebra.

Applying the boundary condition of $p = 0$ at $R = R_o$, we obtain for pressure

$$\tilde{p} = -\frac{3\alpha}{S}\tilde{A}_1(s)\left(1 - \frac{R_o \sinh\sqrt{R^2 s/c}}{R \sinh\sqrt{R_o^2 s/c}}\right) \tag{7.376}$$

We are however not yet at the final solution, as we need to find what the term $\tilde{A}_1(s)$ represents. To find $\tilde{A}_1(s)$, we can utilize the stress solution (7.367). First, we recognize that for the stress to be bounded at $R = 0$, we must have $A_2(t) = 0$. We then take Laplace transform of (7.367) and perform the integration to obtain

$$\begin{aligned}\tilde{\sigma}_{RR}(R,s) &= -4\eta\frac{1}{R^3}\int R^2\tilde{p}(R,s)\,dR + \frac{2G(1+\nu)}{1-2\nu}\tilde{A}_1(s)\\ &= \left[-\frac{24\eta^2(1-\nu)}{S(1-2\nu)}\,\frac{R^*\sqrt{s^*}\,\cosh(R^*\sqrt{s^*}) - \sinh(R^*\sqrt{s^*})}{R^{*3}s^*\sinh\sqrt{s^*}}\right.\\ &\quad\left. + \frac{8\eta^2(1-\nu) + 2GS(1+\nu)}{S(1-2\nu)}\right]\tilde{A}_1(s)\end{aligned} \tag{7.377}$$

where we have defined the dimensionless parameters

$$s^* = \frac{R_o^2 s}{c}; \quad R^* = \frac{R}{R_o} \tag{7.378}$$

Utilizing the boundary condition in (7.373), $\sigma_{RR} = -P_o$ at $R = R_o$, we can solve the above equation for $\tilde{A}(s)$ as

$$s\tilde{A}_1(s) = \frac{P_o S(1-2\nu)(\nu_u - \nu)\,s^*\sinh\sqrt{s^*}}{4\eta^2(1-\nu)D(s^*)} \tag{7.379}$$

where we have defined

$$D(s^*) = 6(\nu_u - \nu)\sqrt{s^*}\cosh\sqrt{s^*} - [6(\nu_u - \nu) + (1-\nu)(1+\nu_u)s^*]\sinh\sqrt{s^*} \tag{7.380}$$

for shorthand.

Substituting (7.379) into (7.376), we obtain the Laplace transform solution for pressure

$$\frac{s\tilde{p}}{P_o} = \frac{3(\nu_u - \nu)s^*}{2\eta D(s^*)R^*}\left[\sinh(R^*\sqrt{s^*}) - R^*\sinh\sqrt{s^*}\right] \tag{7.381}$$

To compare with the Cryer solution, we can take the incompressible constituent limits, in which we note (see Sect. B.1)

$$\nu_u = \frac{1}{2}; \quad \eta = \frac{1-2\nu}{2(1-\nu)} \tag{7.382}$$

and (7.381) becomes

$$\frac{s\tilde{p}}{P_o} = \frac{s^*[\sinh(R^*\sqrt{s^*}) - R^*\sinh\sqrt{s^*}]}{R^*[4\eta\sqrt{s^*}\cosh\sqrt{s^*} - (s^* + 4\eta)\sinh\sqrt{s^*}]} \tag{7.383}$$

At the center of the sphere, $R^* \to 0$, the above reduces to

$$\frac{s\tilde{p}}{P_o} = \frac{s^*\left(\sqrt{s^*} - \sinh\sqrt{s^*}\right)}{4\eta\sqrt{s^*}\cosh\sqrt{s^*} - (s^* + 4\eta)\sinh\sqrt{s^*}} \tag{7.384}$$

which is Cryer's result [50].

For stresses, we can obtain from (7.377) and (7.379) the following expression for σ_{RR}

$$\begin{aligned}\frac{s\tilde{\sigma}_{RR}}{P_o} = -1 - \frac{6(\nu_u - \nu)}{D(s^*)R^{*3}}\Big[&R^{*3}(\sinh\sqrt{s^*} - \sqrt{s^*}\cosh\sqrt{s^*}) \\ &+R^*\sqrt{s^*}\cosh(R^*\sqrt{s^*}) - \sinh(R^*\sqrt{s^*})\Big]\end{aligned} \tag{7.385}$$

And similarly, we find for $\tilde{\sigma}_{\varphi\varphi}$

$$\begin{aligned}\frac{s\tilde{\sigma}_{\varphi\varphi}}{P_o} = -1 - \frac{3(\nu_u - \nu)}{D(s^*)R^{*3}}\Big[&2R^{*3}(\sinh\sqrt{s^*} - \sqrt{s^*}\cosh\sqrt{s^*}) \\ &+(1 + R^{*2}s^*)\sinh(R^*\sqrt{s^*}) - R^*\sqrt{s^*}\cosh(R^*\sqrt{s^*})\Big]\end{aligned} \tag{7.386}$$

Based on (7.366), we also obtain the Laplace transform of u_R as

$$\begin{aligned}\frac{s\tilde{u}_R}{u_R(R_o,\infty)} = R^* - \frac{3(\nu_u - \nu)}{(1-2\nu)D(s^*)R^{*2}}\Big\{&2(1-2\nu)R^{*3}\sqrt{s^*}\cosh\sqrt{s^*} \\ &-[2(1-2\nu) + (1-\nu)s^*]R^{*3}\sinh\sqrt{s^*} \\ &+(1+\nu)\left[R^*\sqrt{s^*}\cosh(R^*\sqrt{s^*}) - \sinh(R^*\sqrt{s^*})\right]\Big\}\end{aligned} \tag{7.387}$$

where

$$u_R(R_o,\infty) = -\frac{P_o R_o(1-2\nu)}{2G(1+\nu)} \tag{7.388}$$

is the steady state ($t \to \infty$) radial displacement at the surface of the sphere.

Solution in Time Domain To obtain the solution in the time domain, we can perform the inverse Laplace transform on the complex s plane as the line integral

$$p(R,t) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} \tilde{p}(R,s)\, e^{st}\, ds \tag{7.389}$$

However, as shown in Appendix D, the integration is actually performed on a closed semi-circular contour as shown in Fig. D.1, and the result is obtained by summing the *residue* of the *poles* according to the *Cauchy residue theorem* (D.11).

Furthermore, as observed above, the Laplace transform solutions, such as the pore pressure solution in (7.381), are expressed as the quotient of two analytic functions. The poles are located at the zeros of the denominator of the solution, in other words, the root of the expression $D(s^*)$ defined in (7.380). Denoting the infinite number of roots as $s = s_n$, $n = 1, 2, \ldots, \infty$, they are to be found from the following algebraic equation obtained by equating $D(s^*) = 0$:

$$\tanh\sqrt{s_n} = \frac{6(\nu_u - \nu)\sqrt{s_n}}{6(\nu_u - \nu) + (1-\nu)(1+\nu_u)s_n} \tag{7.390}$$

Based on the Cauchy integral theorem, and particularly (D.13), the inverse Laplace transform (7.389) can be performed by summing the residues at the poles; hence

$$\frac{p(R,t)}{P_o} = \sum_{n=1}^{\infty} \frac{18(\nu_u - \nu)^2}{\eta E(s_n)} \left[\frac{\sinh(R^*\sqrt{s_n})}{R^*\sinh\sqrt{s_n}} - 1\right] e^{s_n t^*} \tag{7.391}$$

where

$$E(s_n) = -(1-\nu)^2(1+\nu_u)^2 s_n - 18(1+\nu)(\nu_u - \nu)(1-\nu_u) \tag{7.392}$$

and

$$t^* = \frac{c\,t}{R_o^2} \tag{7.393}$$

is the dimensionless time. In (7.391), the denominator is obtained by differentiating $D(s^*)$, and we also used (7.390) to eliminate $\cosh\sqrt{s_n}$ for shorter expression.

For the solution (7.391) to be bounded at $t \to \infty$, it is necessary that $s_n \le 0$. This condition produces imaginary argument for the hyperbolic functions in (7.391). By recognizing that

$$\sinh ix = i\sin x; \quad \cosh ix = \cos x \tag{7.394}$$

we can rewrite the above expressions by replacing s_n with $-x_n$, and obtain

$$\frac{p(R,t)}{P_o} = \sum_{n=1}^{\infty} \frac{18(\nu_u - \nu)^2}{\eta E(x_n)} \left[\frac{\sin(R^*\sqrt{x_n})}{R^*\sin\sqrt{x_n}} - 1\right] e^{-x_n t^*} \tag{7.395}$$

where

$$E(x_n) = (1-\nu)^2(1+\nu_u)^2 x_n - 18(1+\nu)(\nu_u - \nu)(1-\nu_u) \tag{7.396}$$

and x_n are the positive roots of the algebraic equation

$$\tan\sqrt{x_n} = \frac{6(\nu_u - \nu)\sqrt{x_n}}{6(\nu_u - \nu) - (1-\nu)(1+\nu_u)x_n}; \quad x_n > 0 \tag{7.397}$$

In the above, we have skipped the root $x_n = 0$ as the residue associated with the term is zero.

The roots defined in (7.397) can be found using a root finder discussed in a numerical analysis book [125]. For large n values, these roots approaches

$$x_n \approx (n\pi)^2 \tag{7.398}$$

which means we can search for roots by starting from these locations. For the incompressible constituents case, (7.395) and (7.397) reduce to

$$\frac{p(R,t)}{P_o} = \sum_{n=1}^{\infty} \frac{4(1-\nu)(1-2\nu)\left[\sin(R^*\sqrt{x_n}) - R^*\sin\sqrt{x_n}\right]}{[(1-\nu)^2 x_n - 2(1+\nu)(1-2\nu)]R^*\sin\sqrt{x_n}} e^{-x_n t^*} \tag{7.399}$$

and

$$\tan\sqrt{x_n} = \frac{2(1-2\nu)\sqrt{x_n}}{2(1-2\nu) - (1-\nu)x_n} \tag{7.400}$$

Equations (7.399) and (7.400) are the same as that derived in Mason et al. [108]. We observe from (7.399) that the non-dimensionalized pressure solution is a function of a single material property only: the drained Poisson ratio ν. The general solution given by (7.395), however, is dependent on three dimensionless poroelastic parameters $\{\nu, \nu_u, \eta\}$, or $\{\nu, \nu_u, \alpha\}$. At the center of the sphere, $R^* = 0$, Eq. (7.399) is further reduced to

$$\frac{p(R,t)}{P_o} = \sum_{n=1}^{\infty} \frac{8\eta(\sqrt{x_n} - \sin\sqrt{x_n})}{(x_n - 12\eta + 16\eta^2)\sin\sqrt{x_n}} e^{-x_n t^*} \tag{7.401}$$

where η is defined by (7.382). The above result was obtained by Cryer [50].

In Fig. 7.21 we plot the normalized pressure response p/P_o as a function of the dimensionless radius R^* at various dimensionless times t^*, for an incompressible constituent model with $\nu = 0.25$. For comparison, we present the first few values of x_n for this case as determined from (7.400) as

$$x_n = \{6.57119,\ 36.7075,\ 86.1158,\ 155.223,\ 244.058,\ 352.628,\ 480.936,\ \ldots\}$$

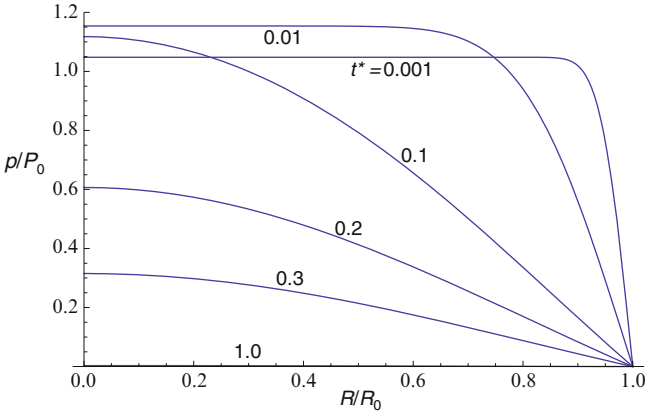


Fig. 7.21 Normalized pressure distribution for the Cryer problem at various times, for an incompressible constituent model with $\nu = 0.25$

Based on the Skempton effect, we anticipate a uniform pore pressure rise of $p = BP_o$ at the instant of the application of the load $\sigma_{RR} = -P_o$. (For the incompressible constituents case, $B = 1$.) Then the pore pressure will dissipate due to the drained boundary condition $p = 0$ at $R = R_o$. We observe in Fig. 7.21 that at small times, $t^* = 0.001, 0.01, 0.1$, the pressure rises above its initial value $p/P_o = 1$. As commented earlier, this is the well-known Mandel-Cryer effect.

The Mandel-Cryer effect can be more clearly observed from the analytical solution if we express it into a power series of small time. This can be accomplished by taking Taylor series expansion of (7.381) into small values of $1/s$ (large values of s). In doing so, we notice that as $s^* \to \infty$, the terms $\{\sinh\sqrt{s^*}, \cosh\sqrt{s^*}\} \to \infty$, but $\coth\sqrt{s^*} \to 1$; and $\{\sinh(R^*\sqrt{s^*}), \cosh(R^*\sqrt{s^*})\}/\sinh\sqrt{s^*} \to 0$ for $R^* < 1$. We hence approximate (7.381) as

$$\begin{aligned}\frac{s\tilde{p}}{P_o} &\approx \frac{3(\nu_u-\nu)s^*}{2\eta[6(\nu_u-\nu)(\sqrt{s^*}-1)-(1-\nu)(1+\nu_u)s^*]}\\ &\approx \frac{3(\nu_u-\nu)}{2\eta(1-\nu)(1+\nu_u)}+\frac{9(\nu_u-\nu)^2}{\eta(1-\nu)^2(1+\nu_u)^2\sqrt{s^*}}\\ &\quad +O\left(s^{*-3/2}\right); \quad \text{for } R^*<1\end{aligned} \tag{7.402}$$

The above expansion can be inverted term by term. By keeping only the first two terms, and using the relations in Sect. B.1 to convert the poroelastic constants to bring out the role of Skempton pore pressure coefficient B, we obtain

$$\frac{p(R,t)}{P_o} = B + \frac{8B^2\eta}{\sqrt{\pi}}\sqrt{t^*} + O(t^*); \quad \text{for } R^* < 1 \tag{7.403}$$

The second term on the right hand side shows that at small time the pressure increases rather than decreases.

The displacement solution can be found by performing the Laplace inversion on (7.387), and we obtain

$$\frac{u_R(R,t)}{u_R(R_o,\infty)} = R^* - \sum_{n=1}^{\infty} \frac{12(1+\nu)(\nu_u-\nu)}{(1-2\nu)E(x_n)R^{*2}x_n \sin\sqrt{x_n}} \times\{3(\nu_u-\nu)[\sin(R^*\sqrt{x_n}) - R^*\sqrt{x_n}\cos(R^*\sqrt{x_n})] +(1-\nu)(1-2\nu)R^{*3}x_n \sin\sqrt{x_n}\}\, e^{-x_n t^*} \tag{7.404}$$

For incompressible constituents, it reduces to

$$\frac{u_R(R,t)}{u_R(R_o,\infty)} = R^* - \sum_{n=1}^{\infty} \frac{4(1+\nu)(1-2\nu)}{[(1-\nu)^2 x_n - 2(1+\nu)(1-2\nu)]R^{*2}x_n \sin\sqrt{x_n}} \times[\sin(R^*\sqrt{x_n}) - R^*\sqrt{x_n}\cos(R^*\sqrt{x_n})]\, e^{-x_n t^*} \tag{7.405}$$

In (7.404) and (7.405), we notice that the leading terms on the right hand side represent the steady-state solution at $t \to \infty$, which is purely elastic.

In Fig. 7.22 we plot the normalized radial displacement, $u_R(R,t)/u_R(R_o,\infty)$ as function of R^*, at various t^*, for the incompressible constituents case, with $\nu = 0.25$. We observe that as $t \to 0$, the displacement vanishes, which is a direct consequence of the incompressible constituents assumption. This is however not

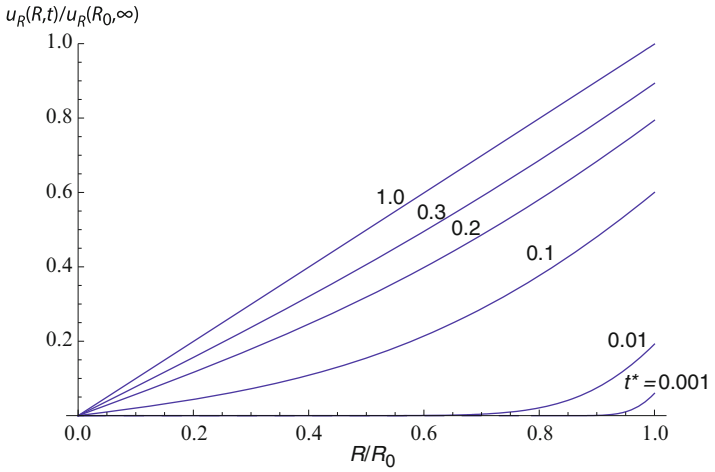


Fig. 7.22 Normalized radial displacement distribution for the Cryer problem at various times, for an incompressible constituent model with $\nu = 0.25$

true for the general poroelastic model. To observe the small time response, we can take the large s expansion of (7.387) and obtain

$$\frac{u_R(R,t)}{u_R(R_o,0^+)} = R^* + \frac{12(\nu_u-\nu)}{\sqrt{\pi}(1-\nu)(1+\nu_u)} R^* \sqrt{t^*} + O\left(t^*\right) \tag{7.406}$$

where

$$u_R(R_o,0^+) = -\frac{P_o R_o(1-2\nu_u)}{2G(1+\nu_u)} \tag{7.407}$$

is the undrained elastic solution at $R = R_o$.

For the stresses, we obtain from (7.385) and (7.386) the following solutions in time

$$\begin{aligned}\frac{\sigma_{RR}(R,t)}{P_o} &= -1 - \sum_{n=1}^{\infty} \frac{12(\nu_u-\nu)}{E(x_n){R^*}^3 x_n \sin\sqrt{x_n}} \\ &\times\{6(\nu_u-\nu)[\sin(R^*\sqrt{x_n}) - R^*\sqrt{x_n}\cos(R^*\sqrt{x_n})] \\ &-(1-\nu)(1+\nu_u){R^*}^3 \sin\sqrt{x_n}\}\, e^{-x_n t^*}\end{aligned} \tag{7.408}$$

$$\begin{aligned}\frac{\sigma_{\varphi\varphi}(R,t)}{P_o} &= -1 - \sum_{n=1}^{\infty} \frac{12(\nu_u-\nu)}{E(x_n){R^*}^3 x_n \sin\sqrt{x_n}} \\ &\times\{3(\nu_u-\nu)[({R^*}^2 x_n - 1)\sin(R^*\sqrt{x_n}) + R^*\sqrt{x_n}\cos(R^*\sqrt{x_n})] \\ &-(1-\nu)(1+\nu_u){R^*}^3 x_n \sin\sqrt{x_n}\}\, e^{-x_n t^*}\end{aligned} \tag{7.409}$$

And for incompressible constituents, we have

$$\begin{aligned}\frac{\sigma_{RR}(R,t)}{P_o} &= -1 - \sum_{n=1}^{\infty} \frac{4(1-2\nu)}{[(1-\nu)^2 x_n - 2(1-2\nu)(1+\nu)]{R^*}^3 x_n \sin\sqrt{x_n}} \\ &\times\{2(1-2\nu)\left[\sin(R^*\sqrt{x_n}) - R^*\sqrt{x_n}\cos(R^*\sqrt{x_n})\right] \\ &-(1-\nu){R^*}^3 x_n \sin\sqrt{x_n}\}\, e^{-x_n t^*}\end{aligned} \tag{7.410}$$

$$\begin{aligned}\frac{\sigma_{\varphi\varphi}(R,t)}{P_o} &= -1 - \sum_{n=1}^{\infty} \frac{4(1-2\nu)}{[(1-\nu)^2 x_n - 2(1-2\nu)(1+\nu)]{R^*}^3 x_n \sin\sqrt{x_n}} \\ &\times\{(1-2\nu)[({R^*}^2 x_n - 1)\sin(R^*\sqrt{x_n}) + R^*\sqrt{x_n}\cos(R^*\sqrt{x_n})] \\ &-(1-\nu){R^*}^3 x_n \sin\sqrt{x_n}\}\, e^{-x_n t^*}\end{aligned} \tag{7.411}$$

In Fig. 7.23 we plot the normalized radial stress in compression, $-\sigma_{RR}/P_o$, versus R/R_o at various times, for the incompressible constituents case. We realize that both

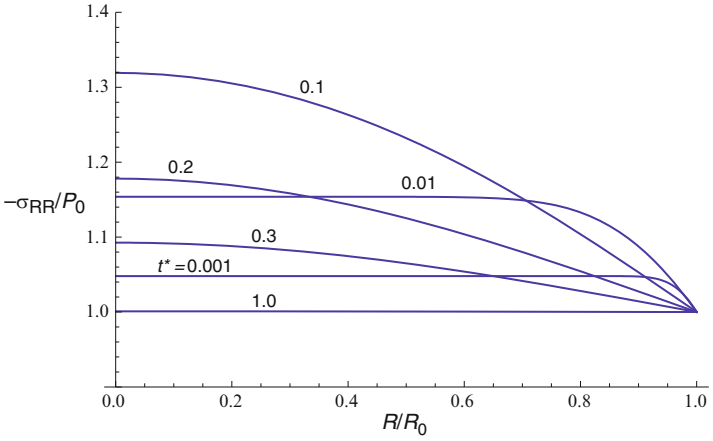


Fig. 7.23 Radial stress distribution for the Cryer problem at various times, for an incompressible constituent model with $\nu = 0.25$

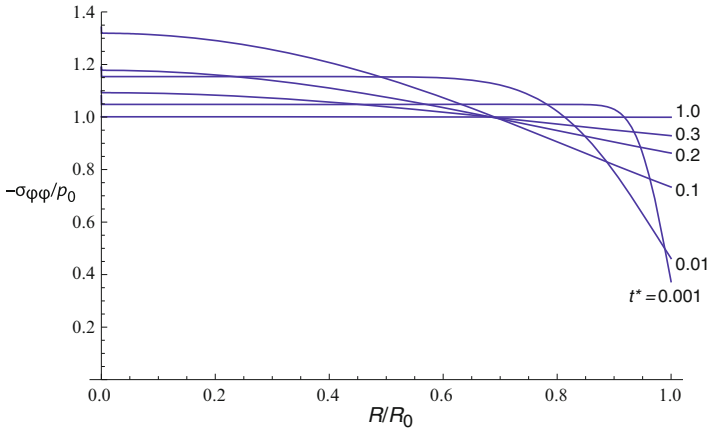


Fig. 7.24 Circumferential stress distribution for the Cryer problem at various times, for an incompressible constituent model with $\nu = 0.25$

at $t = 0^+$ and $t = \infty$, the stress distribution must be elastic; that is, $\sigma_{RR} = -P_o$. Between these times, the stress becomes more and more compressive in the inner region, but must eventually return to the original stress level.

In Fig. 7.24 we presents the evolution of the circumferential stress with time, $-\sigma_{\varphi\varphi}/P_o$. Similar to radial stress, it must begin as the uniform stress $-P_o$ and end at the same value at large time.

7.11 Spherical Cavity

Next, we examine the problem of a spherical cavity. Similar to the sphere problem in the preceding section, governing equations are those given in Sect. 7.9, except that the porous material now occupies the space $R_o \leq R < \infty$. The general solutions for displacement and stress relating to pore pressure are given by (7.366), (7.367), and (7.368). Observing the terms in these solutions, it is obvious that for the displacement and stresses to be bounded at $R \to \infty$, the time function A_1 must vanish. With this condition, the pore pressure diffusion equation (7.372) becomes uncoupled:

$$\frac{\partial p}{\partial t} - c\,\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial p}{\partial R}\right) = 0 \tag{7.412}$$

Once pressure is solved, displacement and stresses can be found as

$$u_R = \frac{\eta}{G}\frac{1}{R^2}\int R^2 p(R,t)\,dR + \frac{A_2(t)}{R^2} \tag{7.413}$$

$$\sigma_{RR} = -4\eta\frac{1}{R^3}\int R^2 p(R,t)\,dR - \frac{4G}{R^3}A_2(t) \tag{7.414}$$

$$\sigma_{\varphi\varphi} = 2\eta\frac{1}{R^3}\int R^2 p(R,t)\,dR - 2\eta p + \frac{2G}{R^3}A_2(t) \tag{7.415}$$

We shall examine a few problems involving the spherical cavity geometry.

7.11.1 Pressurized Cavity

Consider a spherical cavity initially filled with a fluid that is in equilibrium with the formation fluid. At $t = 0^+$, the pressure in the cavity is instantly raised by the amount p_o. This pressure increment has two effects: one is a pressure boundary condition that drives a pressure diffusion, and the other is a radial stress pushing the cavity outward. For the convenience of solving not only the pressurized cavity problem, but also other problems, we shall decompose these effects into two solution modes, with boundary conditions given as below:

- mode 1: $\sigma_{RR}(R_o,t) = -P_o\mathrm{H}(t-0),\quad p(R,t) = 0$
- mode 2: $\sigma_{RR}(R_o,t) = 0,\quad p(R_o,t) = p_o\mathrm{H}(t-0)$

where R_o is the radius of the cavity. We note that if the loading is applied by the pressurization of a fluid filling the cavity, the boundary condition, and thus the solution, is the summation of that of mode 1 and 2.

Mode 1 As the pore pressure diffusion equation (7.412) is uncoupled from the solid deformation, and the boundary condition is zero, its solution is trivial

$$p = 0 \tag{7.416}$$

Utilizing σ_{RR} in (7.414) and its boundary condition, we can solve for

$$A_2(t) = \frac{P_o R_o^3}{4G} \tag{7.417}$$

With the substitution, the solutions based on (7.413) and (7.415) become

$$\frac{G\,u_R}{P_o R_o} = \frac{1}{4}\left(\frac{R_o}{R}\right)^2 \tag{7.418}$$

$$\frac{\sigma_{RR}}{P_o} = -\left(\frac{R_o}{R}\right)^3 \tag{7.419}$$

$$\frac{\sigma_{\varphi\varphi}}{P_o} = \frac{1}{2}\left(\frac{R_o}{R}\right)^3 \tag{7.420}$$

The above are simply the elastic solutions.

As discuss before, at the instant and the infinite time of the loading, the solution is elastic. Equation (7.418) shows that with the sudden application of the radial stress at the cavity surface, the volumetric strain e as defined in (7.362) is in fact identically zero. According to Skempton effect, there is no pore pressure generated. Hence the solution remains in the elastic state at all times.

Mode 2 For the mode 2 problem, we solve the diffusion equation (7.412) with the stated boundary conditions. Such solution is easily found [32]

$$\frac{p}{p_o} = \frac{1}{R^*}\,\mathrm{erfc}\left(\frac{R^* - 1}{2\sqrt{t^*}}\right) \tag{7.421}$$

where erfc is the complimentary error function, and R^* and t^* are defined in (7.378) and (7.393). Given the above, we can perform the integration

$$\int R^2 p(R,T)\,dt = p_o R_o \left[-(R+R_o)\sqrt{\frac{c\,t}{\pi}}\,e^{-(R-R_o)^2/4ct} + \frac{1}{2}\left(R^2 - R_o^2 - 2c\,t\right)\,\mathrm{erfc}\left(\frac{R-R_o}{2\sqrt{c\,t}}\right)\right] \tag{7.422}$$

Using the above for σ_{RR} in (7.414) and the null boundary condition, we obtain

$$A_2(t) = \frac{\eta p_o R_o}{G}\left(c\,t + 2R_o\sqrt{\frac{c\,t}{\pi}}\right) \tag{7.423}$$

Hence we can obtain the stress and displacement solutions based on (7.366), (7.367), and (7.368) as

$$\frac{Gu_R}{p_o R_o} = \frac{\eta}{R^{*2}}\left[2\sqrt{\frac{t^*}{\pi}} + t^* - (R^*+1)\sqrt{\frac{t^*}{\pi}}\, e^{-(R^*-1)^2/4t^*} + \frac{1}{2}(R^{*2}-1-2t^*)\,\mathrm{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}}\right)\right] \tag{7.424}$$

$$\frac{\sigma_{RR}}{p_o} = -\frac{4\,\eta}{R^{*3}}\left[2\sqrt{\frac{t^*}{\pi}} + t^* - (R^*+1)\sqrt{\frac{t^*}{\pi}}\, e^{-(R^*-1)^2/4t^*} + \frac{1}{2}(R^{*2}-1-2t^*)\,\mathrm{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}}\right)\right] \tag{7.425}$$

$$\frac{\sigma_{\varphi\varphi}}{p_o} = \frac{2\,\eta}{R^{*3}}\left[2\sqrt{\frac{t^*}{\pi}} + t^* - (R^*+1)\sqrt{\frac{t^*}{\pi}}\, e^{-(R^*-1)^2/4t^*} - \frac{1}{2}(R^{*2}+1+2t^*)\,\mathrm{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}}\right)\right] \tag{7.426}$$

As an illustration, in Fig. 7.25 we plot the normalized circumferential stress $-\sigma_{\varphi\varphi}/p_o$ (in compression) versus the dimensionless radial distance R/R_o, at various dimensionless times, for the mode 2 problem, given by (7.426). The material constant selected is $\eta = 0.333$ (or $\alpha = 1$ and $\nu = 0.25$). We observe that at the instant of pore pressure penetration, $-\sigma_{\varphi\varphi}$ at $R = R_o$ immediately rises up to $2\eta p_o$, and remains constant throughout. As time increases, the pore pressure penetrates deeper, and the compressive region grows with it. At large distance, however, the $\sigma_{\varphi\varphi}$ is in slight tension.

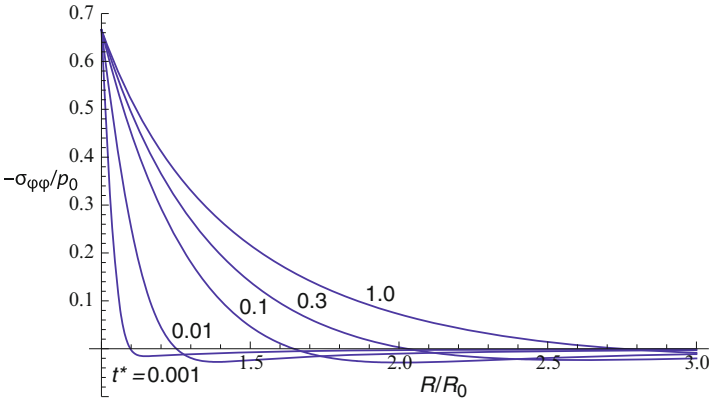


Fig. 7.25 Normalized compressive circumferential stress $-\sigma_{\varphi\varphi}/p_o$ versus radial distance at various times, for a spherical cavity subject to mode 2 loading, for material constant $\eta = 0.333$

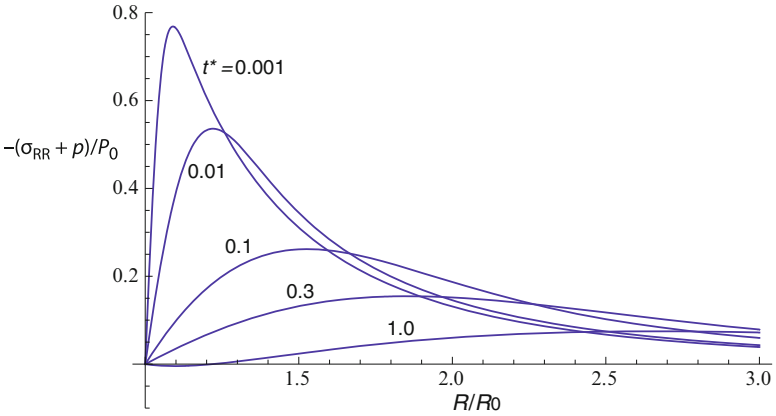


Fig. 7.26 Terzaghi effective radial stress $-(\sigma_{RR} + p)/P_o$ distribution at various times, for a spherical cavity subject to a fluid pressurization (mode 1 + mode 2), with material constant $\eta = 0.333$

As discussed earlier, if the physical loading is applied through a fluid pressure increment in the cavity, the solution is obtained by summing the mode 1 and 2 solutions with $P_o = p_o$, and

$$p = p^{(2)}, \quad \sigma_{RR} = \sigma_{RR}^{(1)} + \sigma_{RR}^{(2)}, \quad \sigma_{\varphi\varphi} = \sigma_{\varphi\varphi}^{(1)} + \sigma_{\varphi\varphi}^{(2)}, \quad u_R = u_R^{(1)} + u_R^{(2)} \tag{7.427}$$

In the above, the superscripts $^{(1)}$ and $^{(2)}$ represent the mode 1 and 2 solutions, respectively. In Fig. 7.26 we present such solution for the normalized Terzaghi effective radial stress $-(\sigma_{RR} + p)/P_o$ versus radial distance at various times.

7.11.2 Excavated Cavity

In this problem we assume an infinite porous medium under a uniform compressive stress and pore pressure

$$\sigma_{RR} = \sigma_{\theta\theta} = \sigma_{\varphi\varphi} = -P_o, \quad p = p_o \tag{7.428}$$

At $t = 0^+$, a spherical cavity is instantly created, with the surface of the cavity, $R = R_o$, subjected to the boundary conditions

$$\sigma_{RR} = 0, \quad p = 0 \tag{7.429}$$

The conditions at far field, $R \to \infty$, remain the same as that given by (7.428). To solve this problem, we define a new set of variables, $\hat{\sigma}_{RR}$, $\hat{p}$, etc., which are

the difference between the true value and the far field value. The boundary value problem for these new variables is then defined by the boundary conditions

$$\hat{\sigma}_{RR} = P_o, \quad \hat{p} = -p_o \tag{7.430}$$

Solution of the above is just the negative of the mode 1 and 2 solutions presented in the preceding section. Adding back the far field values, we obtain the solution of the instantly excavated spherical cavity problem as

$$p = -p^{(2)} + p_o \tag{7.431}$$

$$\sigma_{RR} = -\sigma_{RR}^{(1)} - \sigma_{RR}^{(2)} - P_o \tag{7.432}$$

$$\sigma_{\varphi\varphi} = -\sigma_{\varphi\varphi}^{(1)} - \sigma_{\varphi\varphi}^{(2)} - P_o \tag{7.433}$$

$$u_R = -u_R^{(1)} - u_R^{(2)} \tag{7.434}$$

7.11.3 Pore Pressure Meter Problem

De Josselin de Jong in 1953 [56] formulated the pore pressure meter (piezometer) problem as a consolidation problem, and solved it as one of the earliest analytical solutions of poroelasticity. A piezometer is a hollow sphere inserted into the soil. Its surface is in contact with the pore water pressure, and is immediately in equilibrium with it. The magnitude of the pressure is sensed by a diaphragm and the signal is transmitted to the ground surface.

The problem was simplified to two boundary values problems, one involving a rigid and permeable sphere, and the other an open spherical cavity. The original problem was based on the incompressible constituent model, and only the solution of transient pressure in the piezometer was provided. Here we shall derive the full solution based on the general model.

Rigid Type Consider a hollow sphere of radius R_o with rigid but permeable surface, embedded in an infinite porous medium (Fig. 7.27a). The cavity is filled with the same fluid as the porous medium, and its pressure p_m is in equilibrium with the pore water pressure at the sphere surface, $p_m = p(R_o, t)$. These are summarized to give the following boundary conditions

$$u_R = 0, \quad p = p_m; \quad \text{at } R = R_o \tag{7.435}$$

At $t = 0^+$, the porous medium is subjected to a uniform far field stress of $\sigma_{RR} = \sigma_{\theta\theta} = \sigma_{\varphi\varphi} = -P_o$. Upon the application of the uniform compressive far field stress P_o, an unsupported spherical cavity will be pushed inward, with a radial displacement predicted by the excavated cavity problem in Sect. 7.11.2. The presence of a rigid sphere, however, counteracts it with a radial stress in order to

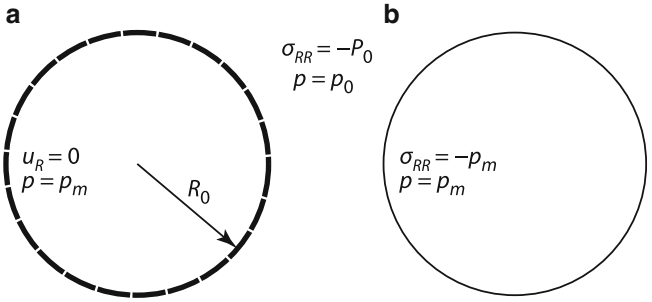


Fig. 7.27 Far field and boundary conditions for the pore pressure meter problem: (**a**) rigid type, and (**b**) cavernous type

maintain the no displacement condition. The reaction force, according to the mode 1 solution (7.418), is exactly P_o. These combined loadings lead to a uniform stress state of

$$\sigma_{RR} = \sigma_{\theta\theta} = \sigma_{\varphi\varphi} = -P_o; \quad \text{at } t = 0^+, \text{ for } R > R_o \tag{7.436}$$

By the Skempton pore pressure effect, an initial uniform pore pressure is generated

$$p = p_o = BP_o; \quad \text{at } t = 0^+, \text{ for } R > R_o \tag{7.437}$$

The pressure p_m in the piezometer, however, is not fixed. Its value is controlled by the volume fluid entering the piezometer and the constitutive law of the fluid. Denoting the volume of fluid entering the cavity as ΔV, the cavity volume $V = 4\pi R_o^3/3$, and the compressibility of fluid C_f, the constitutive law gives

$$p_m = \frac{1}{C_f}\frac{\Delta V}{V} \tag{7.438}$$

The rate of fluid entering the cavity can be found by integrating the specific flux over the surface of the sphere

$$\frac{dV}{dt} = -q_R(R_o, t) \times 4\pi R_o^2 \tag{7.439}$$

According to Darcy's law

$$q_R = -\kappa \frac{\partial p}{\partial R} \tag{7.440}$$

Putting the above conditions together, we can rewrite the pore pressure boundary condition in (7.435) into the following

$$\frac{C_f R_o}{3}\frac{\partial p}{\partial t} = \kappa \frac{\partial p}{\partial R}, \quad \text{at} \quad R = R_o \tag{7.441}$$

The pore pressure diffusion equation (7.412) can then be solved with the initial and boundary conditions (7.437) and (7.441).

Similar to the Cryer problem in Sect. 7.10, we shall solve this problem in Laplace transform domain. After Laplace transform, (7.412) becomes

$$\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial \tilde{p}}{\partial R}\right) - \frac{s}{c}\tilde{p} + \frac{1}{c}p_o = 0 \tag{7.442}$$

Solution of the above is

$$\tilde{p} = \frac{p_o}{s} + C_1\frac{e^{-R\sqrt{s/c}}}{R} \tag{7.443}$$

in which we have dropped the exponential term with positive power for the solution to be bounded at infinity. Substituting the above into the boundary condition (7.441) to solve for the constant C_1, we obtain the solution

$$\frac{s\tilde{p}}{p_o} = 1 - \frac{C_f R_o^3\, s}{C_f R_o^2\, s + 3\kappa R_o\sqrt{s/c} + 3\kappa}\,\frac{e^{-(R-R_o)\sqrt{s/c}}}{R} \tag{7.444}$$

In dimensionless form, it can be expressed as

$$\frac{s\tilde{p}}{p_o} = 1 - \frac{\mu_1 s^*\, e^{-(R^*-1)\sqrt{s^*}}}{R^*\sqrt{1-4\mu_1}}\left(\frac{1}{\sqrt{s^*}+D_2} - \frac{1}{\sqrt{s^*}+D_1}\right) \tag{7.445}$$

where s^* and R^* are dimensionless parameters defined in (7.378), and

$$D_1 = \frac{1+\sqrt{1-4\mu_1}}{2\mu_1} \tag{7.446}$$

$$D_2 = \frac{1-\sqrt{1-4\mu_1}}{2\mu_1} \tag{7.447}$$

with μ_1 as the ratio of fluid compressibility to the storage coefficient

$$\mu_1 = \frac{c\,C_f}{3\kappa} = \frac{C_f}{3S} \tag{7.448}$$

Utilizing Table D.1, we can analytically invert the above to obtain the solution in time as

$$\frac{p}{p_o} = 1 - \frac{\mu_1}{R^*\sqrt{1-4\mu_1}}\left[D_1\, e^{D_1(R^*-1)+D_1^2 t^*}\,\mathrm{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}} + D_1\sqrt{t^*}\right)\right.$$
$$\left. - D_2\, e^{D_2(R^*-1)+D_2^2 t^*}\,\mathrm{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}} + D_2\sqrt{t^*}\right)\right] \tag{7.449}$$

in which t^* is the dimensionless time defined in (7.393). It is of interest to note that the above solution, expressed in dimensionless form, is dependent on only one dimensionless material parameter, μ_1.

Once the pressure solution is obtained, the displacement and stresses can be found through (7.413), (7.414), and (7.415), by utilizing the displacement boundary condition in (7.435). As the main interest is in the pressure p_m of the pore pressure meter, which is the same as the pressure at the boundary, we can set $R = R_o$ in (7.449) to obtain

$$\frac{p_m}{p_o} = 1 - \frac{\mu_1}{\sqrt{1-4\mu_1}}\left[D_1\, e^{D_1^2 t^*}\,\text{erfc}\left(D_1\sqrt{t^*}\right) - D_2\, e^{D_2^2 t^*}\,\text{erfc}\left(D_2\sqrt{t^*}\right)\right] \tag{7.450}$$

Particularly, for the soil mechanics case (incompressible constituents), we set $B = 1$ in (7.437), thus $p_o = P_o$, and S can be reduced based on (B.24), to give

$$\mu_1 = \frac{2GC_f(1-\nu)}{3(1-2\nu)} \tag{7.451}$$

The above is the result obtained by de Josselin de Jong [56].

Cavernous Type In this case, the cavity wall is free to move (Fig. 7.27b). The porous medium is subject to the following physical conditions

$$\begin{aligned} \sigma_{RR} = -p_m, \quad p = p_m, \quad \text{at} \quad R = R_o \\ p = p_o, \quad \text{at} \quad t = 0^+ \end{aligned} \tag{7.452}$$

First, we examine the initial pore pressure response. Upon the application of a far field uniform compressive stress P_o, the cavity will deform inward. The compressed fluid in the cavity then generates a reaction force of the magnitude p_m, which exerts a mode 1 loading on the cavity. In this case, the resultant stress field is no longer uniform. However, the average stress $\sigma_{kk}/3$ is still uniform and is equal to $-P_o$. Again, an initial uniform pore pressure $p_o = BP_o$ is generated. (A key reason for this result is that mode 1 solution generates neither an average stress, nor a pore pressure.)

The initial cavity pressure p_{mo} can be found through the following reasoning. Based on the mode 1 solution (7.418), the combined loading of the far field stress $-P_o$ and the cavity pressure p_{mo} creates a radial displacement at the cavity surface as

$$u_R(R_o, 0^+) = -\frac{(P_o - p_{mo})R_o}{4G} \tag{7.453}$$

The reduction in cavity volume due to this loads is then

$$\Delta V = \frac{(P_o - p_{mo})R_o}{4G} \times 4\pi R_o^2 \tag{7.454}$$

which can be equated to the volume change predicted by the fluid constitutive equation

$$\Delta V = C_f\, p_{mo} \times \frac{4}{3}\pi R_o^3 \tag{7.455}$$

From the above we can solve for

$$p_{mo} = \frac{P_o}{1 + \frac{4}{3}GC_f} \tag{7.456}$$

The boundary condition (7.441) for the rigid sphere needs to be modified for the open cavity case. On the right hand side of (7.441), we need to account for two parts of fluid volume change: one part is due to the fluid flux q_R into the cavity, and the second is caused by the deformation of the cavity wall, u_R. Hence we modify (7.441) to include the second part, as the following

$$\frac{C_f R_o}{3}\frac{\partial p}{\partial t} = \kappa \frac{\partial p}{\partial R} - \frac{\partial u_R}{\partial t}, \quad \text{at} \quad R = R_o \tag{7.457}$$

The above boundary condition requires the coupling of the solution for pressure with that for displacement.

Upon the substitution of pressure and displacement solutions, (7.443) and (7.413), into the boundary condition (7.457), and that for stress in (7.452), we can solve for the two constants C_1 and A_2. The solution for pressure, presented in dimensionless form, is

$$\frac{s\tilde{p}}{p_o} = 1 - \frac{[\mu_1 - (1-B)\mu_2]s^*\, e^{-(R^*-1)\sqrt{s^*}}}{BR^*\sqrt{1-4\mu_2}}\left(\frac{1}{\sqrt{s^*}+D_4} - \frac{1}{\sqrt{s^*}+D_3}\right) \tag{7.458}$$

where

$$D_3 = \frac{1+\sqrt{1-4\mu_2}}{2\mu_2} \tag{7.459}$$

$$D_4 = \frac{1-\sqrt{1-4\mu_2}}{2\mu_2} \tag{7.460}$$

and

$$\mu_2 = \mu_1 + \frac{1}{GS} \tag{7.461}$$

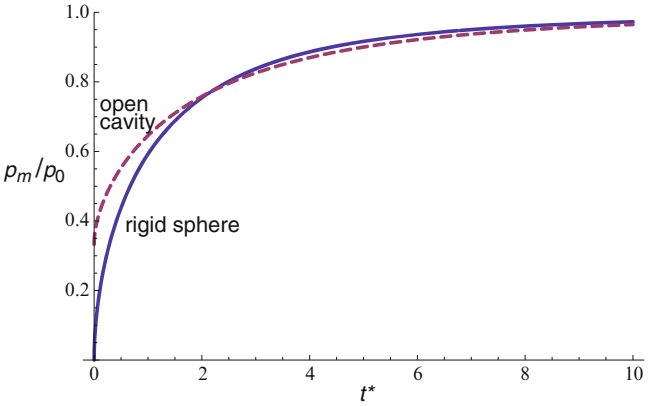


Fig. 7.28 Pressure history in pore pressure meter. *Solid line*: rigid sphere, *dashed line*: open cavity

The solution in time is then

$$\frac{p}{p_o} = 1 - \frac{\mu_1 - (1-B)\mu_2}{BR^*\sqrt{1-4\mu_2}} \left[D_3\, e^{D_3(R^*-1)+D_3^2 t^*} \operatorname{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}} + D_3\sqrt{t^*} \right) \right.$$
$$\left. - D_4\, e^{D_4(R^*-1)+D_4^2 t^*} \operatorname{erfc}\left(\frac{R^*-1}{2\sqrt{t^*}} + D_4\sqrt{t^*} \right) \right] \tag{7.462}$$

We note that the above solution is dependent on three dimensionless material parameters, C_f/S, GS, and B.

For the soil mechanics case, the pressure in the cavity is obtained as

$$\frac{p_m}{p_o} = 1 - \frac{\mu_1}{\sqrt{1-4\mu_2}} \left[D_3\, e^{D_3^2 t^*} \operatorname{erfc}\left(D_3\sqrt{t^*} \right) - D_4\, e^{D_4^2 t^*} \operatorname{erfc}\left(D_4\sqrt{t^*} \right) \right] \tag{7.463}$$

where

$$\mu_2 = \mu_1 + \frac{1-\nu}{2(1-2\nu)} \tag{7.464}$$

with μ_1 defined in (7.448). The above result was obtained by de Josselin de Jong [56]. We should mention, however, that there is a contradiction in the physical assumption of the soil mechanics case, which assumes that the water is incompressible in the porous medium; while the water in the cavity is compressible.

In Fig. 7.28 we plot these two solutions, (7.450) and (7.463), for the parameter values $\mu_1 = 2$ and $\mu_2 = 3$, for the soil mechanics case.

7.12 Axial Symmetry

In this section we present governing equations in cylindrical coordinate system $(r,\ \theta,\ z)$ (see Sect. C.2) with axial symmetry. Particularly, we restrict our attention to plane strain problems with the following functional dependency for displacement and pressure

$$u_r = u_r(r,t); \quad p = p(r,t); \quad \zeta = \zeta(r,t); \quad u_\theta = u_z = 0 \tag{7.465}$$

Apparently, all stress and strain components depend on (r,t) only, and these components vanish

$$e_{zz} = e_{r\theta} = e_{rz} = e_{\theta z} = \sigma_{r\theta} = \sigma_{rz} = \sigma_{\theta z} = 0 \tag{7.466}$$

It is easy to show that the displacement field define in (7.465) is irrotational:

$$\nabla \times \vec{u} = \nabla \times (u_r(r,t)\,\hat{e}_r) = 0 \tag{7.467}$$

based on (C.22), and the solution possesses the properties described in Sect. 6.4.3.

In the following, we shall present two solution strategies, one is more suitable for interior domain problems, and the other exterior domain problems.

Pore Pressure Representation With axial symmetry, the equilibrium equation (C.36) reduces to

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0 \tag{7.468}$$

The stress-strain relations are

$$\sigma_{rr} = \frac{2G\nu}{1-2\nu} e + 2G e_{rr} - \alpha p \tag{7.469}$$

$$\sigma_{\theta\theta} = \frac{2G\nu}{1-2\nu} e + 2G e_{\theta\theta} - \alpha p \tag{7.470}$$

$$\sigma_{zz} = \frac{2G\nu}{1-2\nu} e - \alpha p \tag{7.471}$$

with

$$e_{rr} = \frac{\partial u_r}{\partial r} \tag{7.472}$$

$$e_{\theta\theta} = \frac{u_r}{r} \tag{7.473}$$

$$e = \frac{\partial u_r}{\partial r} + \frac{u_r}{r} = \frac{1}{r}\frac{\partial r u_r}{\partial r} \tag{7.474}$$

Substituting the above into the equilibrium equation (7.468), we obtain the Navier equation

$$\frac{2G(1-\nu)}{1-2\nu}\left(\frac{\partial^2 u_r}{\partial r^2}+\frac{1}{r}\frac{\partial u_r}{\partial r}-\frac{u_r}{r^2}\right)-\alpha\frac{\partial p}{\partial r}=0 \tag{7.475}$$

which can be rearranged into

$$\frac{\partial e}{\partial r}=\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\, r u_r}{\partial r}\right)=\frac{\eta}{G}\frac{\partial p}{\partial r} \tag{7.476}$$

Integrating the above equation once with respect to r, we obtain

$$e=\frac{\eta}{G}p+2A_1(t) \tag{7.477}$$

and another time, it becomes

$$u_r=\frac{\eta}{G}\frac{1}{r}\int rp(r,t)\,dr+A_1(t)\,r+\frac{A_2(t)}{r} \tag{7.478}$$

Utilizing the constitutive equations (7.469), (7.470), and (7.471), we arrive at the following

$$\sigma_{rr}=-2\eta\frac{1}{r^2}\int rp(r,t)\,dr+\frac{2G}{1-2\nu}A_1(t)-\frac{2G}{r^2}A_2(t) \tag{7.479}$$

$$\sigma_{\theta\theta}=2\eta\frac{1}{r^2}\int rp(r,t)\,dr-2\eta p+\frac{2G}{1-2\nu}A_1(t)+\frac{2G}{r^2}A_2(t) \tag{7.480}$$

$$\sigma_{zz}=-2\eta p+\frac{4G\nu}{1-2\nu}A_1(t) \tag{7.481}$$

We notice from (7.479) and (7.480) that

$$\sigma_{rr}+\sigma_{\theta\theta}+2\eta p=\frac{4G}{1-2\nu}A_1(t) \tag{7.482}$$

That is, the above combination of stresses and pressure is a function of time only, and not a function of space, as predicted by (6.108) for a plane strain irrotational displacement field. The irrotational condition also shows that the pressure satisfies the diffusion equation (6.111), which becomes

$$\frac{\partial p}{\partial t}-c\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right)=-\frac{2\alpha}{S}\frac{dA_1(t)}{dt} \tag{7.483}$$

We shall demonstrate in Sect. 7.14 that the above representation is more convenient for the solution of a borehole problem, which involves an exterior domain that extends to infinity.

Variation in Fluid Content Representation For the cylinder problem, involving an interior domain, we seek a slightly different solution strategy, utilizing the variable ζ, rather than p, for its representation.

Based on these stress-strain relations

$$\sigma_{rr} = \frac{2G\nu_u}{1-2\nu_u} e + 2Ge_{rr} - \alpha M\zeta \tag{7.484}$$

$$\sigma_{\theta\theta} = \frac{2G\nu_u}{1-2\nu_u} e + 2Ge_{\theta\theta} - \alpha M\zeta \tag{7.485}$$

$$\sigma_{zz} = \frac{2G\nu_u}{1-2\nu_u} e - \alpha M\zeta \tag{7.486}$$

we obtain the Navier equation

$$\frac{2G(1-\nu_u)}{1-2\nu_u}\left(\frac{\partial^2 u_r}{\partial r^2} + \frac{1}{r}\frac{\partial u_r}{\partial r} - \frac{u_r}{r^2}\right) - \alpha M\frac{\partial \zeta}{\partial r} = 0 \tag{7.487}$$

which can be rearranged into

$$\frac{\partial e}{\partial r} = \frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\, r u_r}{\partial r}\right) = \frac{\eta}{GS}\frac{\partial \zeta}{\partial r} \tag{7.488}$$

Integrating the above equation once, we obtain

$$e = \frac{\eta}{GS}\zeta + 2B_1(t) \tag{7.489}$$

and twice, it yields

$$u_r = \frac{\eta}{GS}\frac{1}{r}\int r\,\zeta(r,t)\,dr + B_1(t)\,r + \frac{B_2(t)}{r} \tag{7.490}$$

Substituting the above into the stress-strain relation, we obtain the solutions for the stresses and pressure

$$\sigma_{rr} = -\frac{2\eta}{S}\frac{1}{r^2}\int r\,\zeta(r,t)\,dr + \frac{2G}{1-2\nu_u}B_1(t) - \frac{2G}{r^2}B_2(t) \tag{7.491}$$

$$\sigma_{\theta\theta} = \frac{2\eta}{S}\frac{1}{r^2}\int r\,\zeta(r,t)\,dr - \frac{2\eta}{S}\zeta + \frac{2G}{1-2\nu_u}B_1(t) + \frac{2G}{r^2}B_2(t) \tag{7.492}$$

$$\sigma_{zz} = -\frac{2\eta}{S}\zeta + \frac{4G\nu_u}{1-2\nu_u}B_1(t) \tag{7.493}$$

$$p = \frac{\zeta}{S} - 2\alpha M B_1(t) \tag{7.494}$$

We notice from the above solutions that

$$\sigma_{rr} + \sigma_{\theta\theta} + 2\eta p = \frac{4G(1-\nu_u)}{(1-2\nu_u)(1-\nu)} B_1(t) \tag{7.495}$$

Comparing (7.477) and (7.489), and utilizing the constitutive equation, we can show that

$$B_1 = \frac{(1-\nu)(1-2\nu_u)}{(1-\nu_u)(1-2\nu)} A_1 \tag{7.496}$$

$$B_2 = A_2 \tag{7.497}$$

As shown in (6.70), ζ satisfies the diffusion equation, which is written in the axial symmetry case as

$$\frac{\partial \zeta}{\partial t} - c \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \zeta}{\partial r} \right) = 0 \tag{7.498}$$

7.13 Cylinder Problem

The poroelastic cylinder problem was first investigated by de Leeuw [58]. However, only the solid cylinder mode 1 problems was solved for the incompressible constituent case. The complete solution is given as follows.

7.13.1 Solid Cylinder

Consider a circular cylinder of radius r_o of finite or infinite length. We shall examine the analytical solutions of four different loading conditions (see Fig. 7.29):

Mode 1 The surface of the cylinder is subject to a uniform radial stress, $\sigma_{rr} = -P_o$, and the pore pressure is drained, $p = 0$.

Mode 2 The surface is subject to a uniform pore pressure $p = p_o$, and is otherwise stress free, $\sigma_{rr} = 0$.

Mode 3 The surface is subject to the equivalent of a far-field deviatoric stress of the magnitude S_o, with drained pore pressure.

Mode 4 The cylinder is subject to an uniform axial strain e_{zz}, which is a function of time only, and the surface is stress and pore pressure free.

The above decomposition of loading into different modes allows problems with different physical boundary conditions be solved by superposition. For example, for a loading applied through fluid pressurization on the surface of the cylinder, the solution is obtained by adding mode 2 to mode 1. For the problem of a retrieved

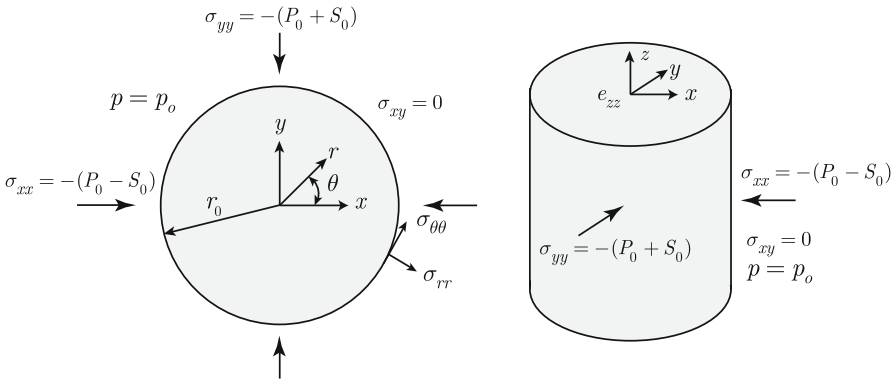


Fig. 7.29 Loading modes for cylinder problem

cylindrical core from a deep formation, in which the surface is suddenly relieved of a pre-existing non-hydrostatic stress and pore pressure, the relaxation of the stress and strain can be obtained by summing or subtracting the four modes described above.

Mode 1: Radial Stress The boundary conditions are given as

$$\sigma_{rr} = -P_o; \quad \sigma_{r\theta} = 0; \quad p = 0; \quad \text{at } r = r_o \tag{7.499}$$

To solve the problem, we begin with the diffusion equation (7.498). Performing Laplace transform, we obtain

$$\frac{d^2\tilde{\zeta}}{dr^2} + \frac{1}{r}\frac{d\tilde{\zeta}}{dr} - \frac{s}{c}\tilde{\zeta} = 0 \tag{7.500}$$

The general solution of the above is

$$\tilde{\zeta} = D_1 \, \mathrm{I}_0 \left(r\sqrt{\frac{s}{c}} \right) \tag{7.501}$$

where I_0 is the modified Bessel function of the first kind of order 0, and D_1 is a constant (functions of s) to be determined. We should note that in (7.501) we have omitted a term containing K_0, the modified Bessel function of the second kind of order 0, in order for the solution to be bounded at $r = 0$ for the current problem.

Take the Laplace transform of (7.491) and (7.494), we obtain

$$\tilde{\sigma}_{rr} = -\frac{2\eta}{S}\frac{1}{r^2}\int r\,\tilde{\zeta}(r,s)\,dr + \frac{2G}{1-2\nu_u}\tilde{B}_1(s) \tag{7.502}$$

$$\tilde{p} = \frac{\tilde{\zeta}}{S} - 2\alpha M\tilde{B}_1(s) \tag{7.503}$$

In the above, for the solution to be bounded at $r = 0$, we have drop the term associated with B_2. Substituting (7.501) into (7.502) and (7.503), and utilizing the two boundary conditions in (7.499), we are able to determine the two constants, $\tilde{B}_1$ and D_1. The solutions in the Laplace transform domain are then obtained as

$$\frac{s\tilde{p}^{(1)}}{P_o} = \frac{(\nu_u - \nu)\sqrt{s^*}}{\eta D(s^*)}\left[\mathrm{I}_0(\sqrt{s^*}) - \mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.504}$$

$$\frac{s\tilde{\sigma}_{rr}^{(1)}}{P_o} = -1 - \frac{2(\nu_u - \nu)}{r^* D(s^*)}\left[r^*\mathrm{I}_1(\sqrt{s^*}) - \mathrm{I}_1(r^*\sqrt{s^*})\right] \tag{7.505}$$

$$\frac{s\tilde{\sigma}_{\theta\theta}^{(1)}}{P_o} = -1 - \frac{2(\nu_u - \nu)}{r^* D(s^*)}\left[r^*\mathrm{I}_1(\sqrt{s^*}) + \mathrm{I}_1(r^*\sqrt{s^*}) - r^*\sqrt{s^*}\,\mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.506}$$

$$\frac{s\tilde{\sigma}_{zz}^{(1)}}{P_o} = -2\nu - \frac{2(\nu_u - \nu)}{D(s^*)}\left[(1-\nu)\sqrt{s^*}\,\mathrm{I}_0(\sqrt{s^*}) + 2\nu\,\mathrm{I}_1(\sqrt{s^*}) - \sqrt{s^*}\,\mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.507}$$

$$\frac{s\tilde{u}_r^{(1)}}{u_r(r_o,\infty)} = r^* - \frac{2(\nu_u - \nu)}{(1-2\nu)D(s^*)}\left[(1-\nu)r^*\sqrt{s^*}\,\mathrm{I}_0(\sqrt{s^*}) - (1-2\nu)r^*\mathrm{I}_1(\sqrt{s^*}) - \mathrm{I}_1(r^*\sqrt{s^*})\right] \tag{7.508}$$

where

$$r^* = \frac{r}{r_o}; \quad s^* = \frac{r_o^2 s}{c} \tag{7.509}$$

$$D(s^*) = \sqrt{s^*}(1-\nu)\,\mathrm{I}_0(\sqrt{s^*}) - 2(\nu_u - \nu)\,\mathrm{I}_1(\sqrt{s^*}) \tag{7.510}$$

and

$$u_r(r_o,\infty) = -\frac{P_o r_o(1-2\nu)}{2G} \tag{7.511}$$

is the final ($t \to \infty$) radial displacement at the surface of the cylinder. In the above we have used the superscript $^{(1)}$ to denote the 'mode 1' solution.

To obtain the solution in time, we apply the inverse Laplace transform by utilizing the contour integral and Cauchy residue theorem as described in Appendix D, and a procedure similar to that in Sect. 7.10. We find

$$\frac{p^{(1)}}{P_o} = \frac{4(\nu_u - \nu)^2}{\eta}\sum_{n=1}^{\infty}\frac{1}{E(x_n)}\left[\frac{\mathrm{J}_0(r^*\sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - 1\right]e^{-x_n t^*} \tag{7.512}$$

where

$$t^* = \frac{c\,t}{r_o^2} \tag{7.513}$$

is the dimensionless time, J_0 is the Bessel function of the first kind of order 0, $E(x_n)$ is the shorthand for

$$E(x_n) = (1-\nu)^2 x_n - 4(\nu_u - \nu)(1-\nu_u) \tag{7.514}$$

and x_n are the n-th positive real roots of the equation

$$(1-\nu)\sqrt{x_n}\,\mathrm{J}_0(\sqrt{x_n}) = 2(\nu_u - \nu)\,\mathrm{J}_1(\sqrt{x_n}); \quad \text{for } x_n > 0 \tag{7.515}$$

The values of x_n can be found using a numerical root finder. By the asymptotic behavior of Bessel function with large argument, we can show that

$$x_n \to \left[\left(n - \frac{1}{4}\right)\pi\right]^2 \quad \text{as } x_n \to \infty \tag{7.516}$$

The above equation can be used as the starting search position for a root finder subroutine. For example, for a material with incompressible constituents, with $\nu_u = 0.5$ and $\nu = 0.25$, we list the first few roots as follows:

$$x_n = \{4.28241,\ 29.114,\ 73.5443,\ 137.702,\ 221.596,\ 325.228\} \tag{7.517}$$

In Fig. 7.30 we plot the pore pressure history for such material. We notice the similarity of the non-monotonic pressure response as the Cryer problem, shown in Fig. 7.21.

Similarly, we obtain the solution in time for the stresses and displacement as

$$\frac{\sigma_{rr}^{(1)}}{P_o} = -1 - 4(1-\nu)(\nu_u - \nu) \times \sum_{n=1}^{\infty} \frac{1}{E(x_n)} \left[\frac{\mathrm{J}_1(r^*\sqrt{x_n})}{r^*\,\mathrm{J}_1(\sqrt{x_n})} - 1\right] e^{-x_n t^*} \tag{7.518}$$

$$\frac{\sigma_{\theta\theta}^{(1)}}{P_o} = -1 - 4(1-\nu)(\nu_u - \nu) \sum_{n=1}^{\infty} \frac{1}{E(x_n)} \times \left[\frac{r^*\sqrt{x_n}\,\mathrm{J}_0(r^*\sqrt{x_n}) - \mathrm{J}_1(r^*\sqrt{x_n})}{r^*\,\mathrm{J}_1(\sqrt{x_n})} - 1\right] e^{-x_n t^*} \tag{7.519}$$

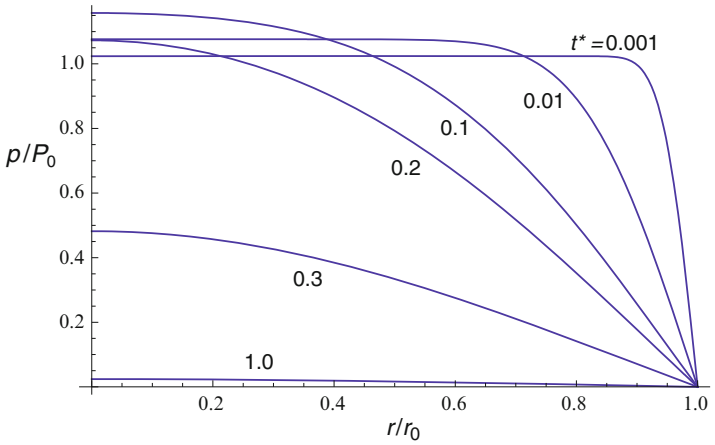


Fig. 7.30 Pressure history for a circular cylinder under mode 1 loading, with $\alpha = 1$, $\nu_u = 0.5$ and $\nu = 0.25$

$$\frac{\sigma_{zz}^{(1)}}{P_o} = -2\nu - 8(\nu_u - \nu)\sum_{n=1}^{\infty}\frac{1}{E(x_n)} \times\left[\frac{(\nu_u - \nu)\,\mathrm{J}_0(r^*\sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - \nu_u(1-\nu)\right] e^{-x_n t^*} \tag{7.520}$$

$$\frac{u_r^{(1)}}{u_r(r_o,\infty)} = r^* - \frac{4(1-\nu)(\nu_u-\nu)}{1-2\nu}\sum_{n=1}^{\infty}\frac{1}{E(x_n)} \times\left[(1-2\nu_u)\,r^* + \frac{\mathrm{J}_1(r^*\sqrt{x_n})}{\mathrm{J}_1(\sqrt{x_n})}\right] e^{-x_n t^*} \tag{7.521}$$

in which $u_r(r_o,\infty)$ is defined in (7.511). We also note that all shear stresses vanish

$$\sigma_{r\theta}^{(1)} = \sigma_{rz}^{(1)} = \sigma_{\theta z}^{(1)} = 0 \tag{7.522}$$

In Fig. 7.31 we plot the distribution of vertical stress σ_{zz} as function of r at various dimensionless times for a material with $\nu_u = 0.5$ and $\nu = 0.25$. By the asymptotic behavior, at $t = 0^+$ the cylinder behaves as an undrained elastic material, characterized by the undrained Poisson ratio ν_u; hence a uniform stress of $\sigma_{zz}(r,0^+) = -2\nu_u P_o = -P_o$ exists. As observed in Fig. 7.31, at the surface of the cylinder, the stress immediately reduces due to pore pressure dropping to zero caused by the boundary condition. The vertical stress at the surface immediately takes the value

$$\sigma_{zz}(r_o,0^+) = -\frac{2\nu(1-\nu_u)}{1-\nu}P_o \tag{7.523}$$

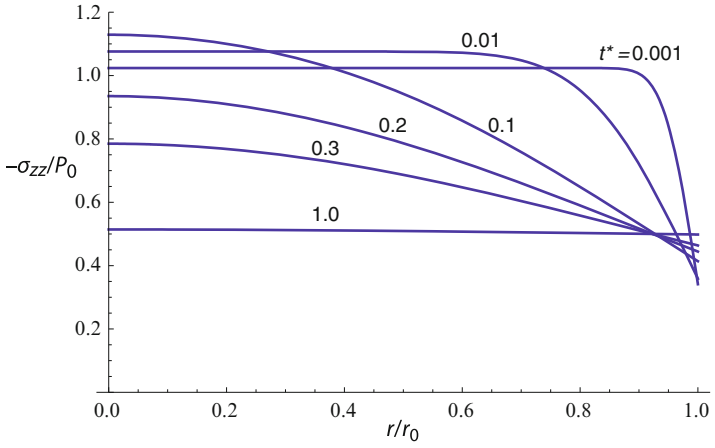


Fig. 7.31 Vertical stress distribution for mode 1 cylinder problem, at various dimensionless times for $\nu_u = 0.5$ and $\nu = 0.25$

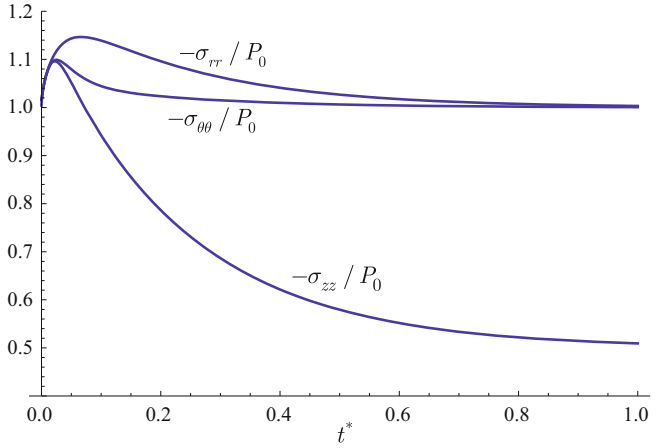


Fig. 7.32 Time history of radial, circumferential, and vertical stresses for mode 1 cylinder problem, at $r = r_o/2$, for $\nu_u = 0.5$ and $\nu = 0.25$

At large time, the cylinder behaves as drained elastic material, and the vertical stress becomes uniform again, $\sigma_{zz}(r, \infty) = -2\nu P_o$, as observed in Fig. 7.31.

To observe the evolution of stresses with time, we plot in Fig. 7.32 the three stresses, radial, circumferential, and vertical, at a location half way inside the cylinder ($r = r_o/2$) as functions of dimensionless time t^*. The poroelasticity theory predicts that the material response is elastic at both the small time (undrained) and the large time (drained). The stresses σ_{rr} and $\sigma_{\theta\theta}$ hence start as $-P_o$ at $t = 0^+$, and return to $-P_o$ at $t \to \infty$, as observed in Fig. 7.32. The small time and large time asymptotic behavior of the vertical stress σ_{zz} has already been discussed above.

Mode 2: Fluid Pressure Next we examine the mode 2 problem, which is defined by the following boundary conditions:

$$\sigma_{rr} = 0; \quad \sigma_{r\theta} = 0; \quad p = p_o; \quad \text{at } r = r_o \tag{7.524}$$

Following the same procedure as the mode 1 problem, the solutions in the Laplace transform domain are obtained as

$$\frac{s\tilde{p}^{(2)}}{p_o} = 1 - \frac{(1-\nu)\sqrt{s^*}}{D(s^*)}\left[\mathrm{I}_0(\sqrt{s^*}) - \mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.525}$$

$$\frac{s\tilde{\sigma}_{rr}^{(2)}}{p_o} = \frac{2\eta(1-\nu)}{r^* D(s^*)}\left[r^*\mathrm{I}_1(\sqrt{s^*}) - \mathrm{I}_1(r^*\sqrt{s^*})\right] \tag{7.526}$$

$$\frac{s\tilde{\sigma}_{\theta\theta}^{(2)}}{p_o} = \frac{2\eta(1-\nu)}{r^* D(s^*)}\left[r^*\mathrm{I}_1(\sqrt{s^*}) + \mathrm{I}_1(r^*\sqrt{s^*}) - r^*\sqrt{s^*}\,\mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.527}$$

$$\frac{s\tilde{\sigma}_{zz}^{(2)}}{p_o} = -\frac{2\eta}{1-\nu} + \frac{2\eta(1-\nu)}{D(s^*)}\left[(1-\nu)\sqrt{s^*}\,\mathrm{I}_0(\sqrt{s^*}) + 2\nu\,\mathrm{I}_1(\sqrt{s^*}) - \sqrt{s^*}\,\mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.528}$$

$$\frac{s\tilde{u}_r^{(2)}}{u_r(r_o,\infty)} = r^* - \frac{1}{D(s^*)}\left[(1-\nu)\,r^*\sqrt{s^*}\,\mathrm{I}_0(\sqrt{s^*}) - (1-2\nu)\,r^*\mathrm{I}_1(\sqrt{s^*}) - \mathrm{I}_1(r^*\sqrt{s^*})\right] \tag{7.529}$$

where

$$u_r(r_o,\infty) = \frac{p_o r_o\,\eta(1-\nu)}{G} \tag{7.530}$$

is the displacement at the surface of cylinder at large time. In the above, the superscript $^{(2)}$ denotes the mode 2 solution.

Applying inverse Laplace transform to the above solutions, we obtain

$$\frac{p^{(2)}}{p_o} = 1 - 4(1-\nu)(\nu_u - \nu)\sum_{n=1}^{\infty}\frac{1}{E(x_n)} \times\left[\frac{\mathrm{J}_0(r^*\sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - 1\right] e^{-x_n t^*} \tag{7.531}$$

$$\frac{\sigma_{rr}^{(2)}}{p_o} = 4\eta(1-\nu)^2\sum_{n=1}^{\infty}\frac{1}{E(x_n)}\left[\frac{\mathrm{J}_1(r^*\sqrt{x_n})}{r^*\mathrm{J}_1(\sqrt{x_n})} - 1\right] e^{-x_n t^*} \tag{7.532}$$

$$\frac{\sigma_{\theta\theta}^{(2)}}{p_o} = 4\eta(1-\nu)^2 \sum_{n=1}^{\infty} \frac{1}{E(x_n)} \times \left[\frac{r^* \sqrt{x_n}\, \mathrm{J}_0(r^* \sqrt{x_n}) - \mathrm{J}_1(r^* \sqrt{x_n})}{r^* \mathrm{J}_1(\sqrt{x_n})} - 1 \right] e^{-x_n t^*} \tag{7.533}$$

$$\frac{\sigma_{zz}^{(2)}}{p_o} = -\frac{2\eta}{1-\nu} + 8\eta(1-\nu) \sum_{n=1}^{\infty} \frac{1}{E(x_n)} \times \left[\frac{(\nu_u - \nu)\, \mathrm{J}_0(r^* \sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - \nu_u(1-\nu) \right] e^{-x_n t^*} \tag{7.534}$$

$$\frac{u_r^{(2)}}{u_r(r_o, \infty)} = r^* - 2(1-\nu) \sum_{n=1}^{\infty} \frac{1}{E(x_n)} \times \left[(1-2\nu_u) r^* + \frac{\mathrm{J}_1(r^* \sqrt{x_n})}{\mathrm{J}_1(\sqrt{x_n})} \right] e^{-x_n t^*} \tag{7.535}$$

where $E(x_n)$ and x_n are the same as that defined in (7.514) and (7.515). It is of interest to observe from the above expressions that as $t \to \infty$, the radial and circumferential stresses vanish, while $\sigma_{zz}(r, \infty) = -2\eta p_o/(1-\nu)$, due to the presence of pore pressure and Poisson effect.

In Fig. 7.33 we plot the pore pressure history for a porous material with $\alpha = 1$, $\nu_u = 0.5$, and $\nu = 0.25$. We observe that at smaller times, the pore pressure penetrates the outer shell of the cylinder. The swelling of the outer shell causes the pore pressure in the inner core to become negative. At large time, the entire cylinder must be under constant pressure p_o.

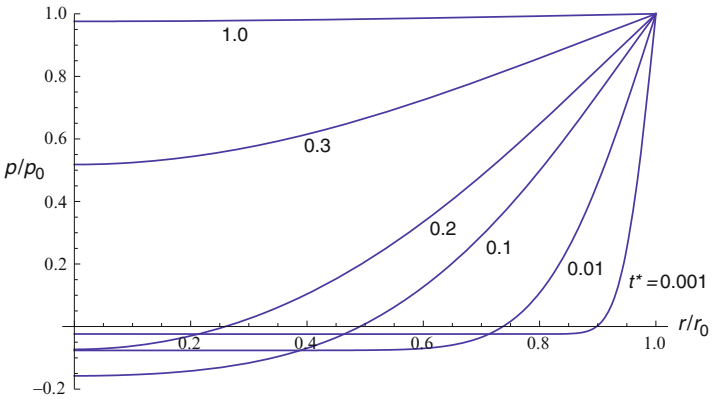


Fig. 7.33 Pressure history for a circular cylinder under mode 2 loading, with $\alpha = 1$, $\nu_u = 0.5$ and $\nu = 0.25$

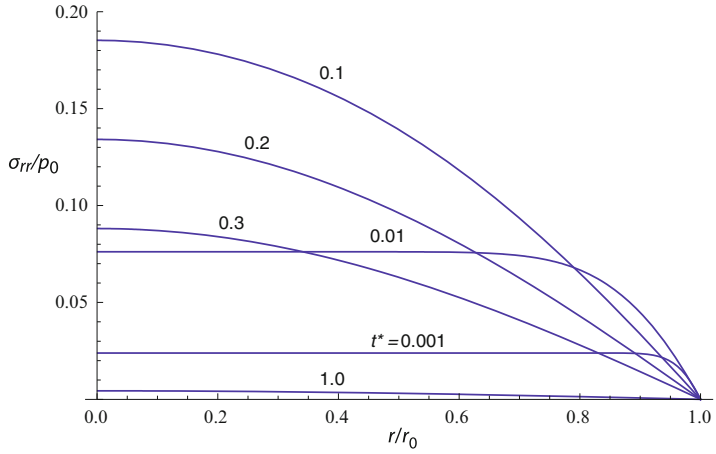


Fig. 7.34 Radial stress history for a circular cylinder under mode 2 loading, with $\alpha = 1$, $\nu_u = 0.5$ and $\nu = 0.25$

The evolution of radial stress with time is plotted as Fig. 7.34. We observe that the stress distribution starts from zero, and becomes more and more tensile as pore pressure penetrates the outer shell. At large time, however, the stress must return to zero.

Mode 3: Deviatoric Stress Boundary conditions for the mode 3 problem are

$$\sigma_{rr} = S_o \cos 2\theta; \quad \sigma_{r\theta} = -S_o \sin 2\theta; \quad p = 0; \quad \text{at } r = r_o \tag{7.536}$$

Or, in Cartesian coordinates, the above are written as

$$\sigma_{xx} = S_o; \quad \sigma_{yy} = -S_o; \quad p = 0; \quad \text{at } r = r_o \tag{7.537}$$

This set of boundary conditions generally does not arise from a direct loading on the cylinder surface. It can, however, be an *unloading* condition, such as that resulting from the retrieval of a cylindrical core in a non-hydrostatic underground stress environment, which will be discussed in Sect. 7.15.1.

These boundary conditions no longer define an axisymmetric problem. Their specific dependence on the sine and cosine functions, however, allows a simple solution to be sought. In fact, the solution is just the pure shear condition of elasticity:

$$\frac{\sigma_{rr}^{(3)}}{S_o} = \cos 2\theta \tag{7.538}$$

$$\frac{\sigma_{\theta\theta}^{(3)}}{S_o} = -\cos 2\theta \tag{7.539}$$

$$\frac{\sigma_{r\theta}^{(3)}}{S_o} = -\sin 2\theta \tag{7.540}$$

$$\frac{p^{(3)}}{S_o} = 0 \tag{7.541}$$

$$\frac{2Gu_r^{(3)}}{S_o r_o} = r^* \cos 2\theta \tag{7.542}$$

$$\frac{2Gu_\theta^{(3)}}{S_o r_o} = -r^* \sin 2\theta \tag{7.543}$$

where the superscript $^{(3)}$ denotes the mode 3 solution. As $\sigma_{rr} + \sigma_{\theta\theta} = 0$, there is no volumetric strain resulting by this load. For a cylinder initially without pore pressure, there is no new pore pressure generated; hence the solution remains elastic, and is independent of time.

Mode 4: Axial Strain Boundary conditions for the mode 4 problem are as follows

$$\sigma_{rr} = 0; \quad \sigma_{r\theta} = 0; \quad p = 0; \quad \text{at } r = r_o \tag{7.544}$$

and

$$e_{zz} = C(t) \tag{7.545}$$

where $C(t)$ is an arbitrary function of time. The above set of conditions defines the case of axial strain in the generalized plane strain problem, as discussed in Sect. 7.5.6. This problem can be solved as a plane strain problem by utilizing the new variables defined in (7.155)–(7.158), expressed for the present isotropic and axisymmetric case, as

$$\sigma_{rr}^\dagger = \sigma_{rr} - \frac{2G\nu_u}{1-2\nu_u} C(t) \tag{7.546}$$

$$\sigma_{\theta\theta}^\dagger = \sigma_{\theta\theta} - \frac{2G\nu_u}{1-2\nu_u} C(t) \tag{7.547}$$

$$\sigma_{zz}^\dagger = \sigma_{zz} - \frac{2G(1-\nu_u)}{1-2\nu_u} C(t) \tag{7.548}$$

$$p^\dagger = p + M\alpha\, C(t) \tag{7.549}$$

Other variables remain the same. In these new variables, the boundary conditions in (7.544) become

$$\sigma_{rr}^\dagger = -\frac{2G\nu_u}{1-2\nu_u} C(t); \quad \sigma_{r\theta}^\dagger = 0; \quad p^\dagger = M\alpha\, C(t); \quad \text{at } r = r_o \tag{7.550}$$

As these new variables satisfy the plane strain governing equations, the solutions can be obtained by summing up the mode 1 and mode 2 solutions, with appropriate multiplication constants, as given in (7.550).

Although we can express the solution in the time domain directly, in the following, we shall first present the solution in the Laplace transform domain, for the benefit of the application to poroviscoelastic materials, which will be discussed in Chap. 10. For pore pressure, we can sum (7.504) and (7.525) to obtain

$$\tilde{p}^{\dagger} = -\frac{G(\nu_u - \nu)\sqrt{s^*}\,\tilde{C}(s)}{\eta D(s^*)}\left[\mathrm{I}_0(\sqrt{s^*}) - \mathrm{I}_0(r^*\sqrt{s^*})\right] + M\alpha\tilde{C}(s) \tag{7.551}$$

where $\tilde{C}(s)$ is the Laplace transform of $C(t)$. Restoring to the original variable, we find

$$\tilde{p} = -\frac{G(\nu_u - \nu)\sqrt{s^*}\,\tilde{C}(s)}{\eta D(s^*)}\left[\mathrm{I}_0(\sqrt{s^*}) - \mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.552}$$

If we consider that the vertical strain is applied as a step increment in compression, expressed as

$$C(t) = -C_o\,\mathrm{H}(t-0) \tag{7.553}$$

then (7.552) becomes

$$\frac{s\tilde{p}^{(4)}}{C_o G} = \frac{(\nu_u - \nu)\sqrt{s^*}}{\eta D(s^*)}\left[\mathrm{I}_0(\sqrt{s^*}) - \mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.554}$$

where the superscript $^{(4)}$ denotes the mode 4 solution. Similarly, we can obtain the rest of the solution for the step increment of vertical strain as

$$\frac{s\tilde{\sigma}_{rr}^{(4)}}{C_o G} = -\frac{2(\nu_u - \nu)}{r^* D(s^*)}\left[r^*\,\mathrm{I}_1(\sqrt{s^*}) - \mathrm{I}_1(r^*\sqrt{s^*})\right] \tag{7.555}$$

$$\frac{s\tilde{\sigma}_{\theta\theta}^{(4)}}{C_o G} = -\frac{2(\nu_u - \nu)}{r^* D(s^*)}\left[r^*\,\mathrm{I}_1(\sqrt{s^*}) + \mathrm{I}_1(r^*\sqrt{s^*}) - r^*\sqrt{s^*}\,\mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.556}$$

$$\frac{s\tilde{\sigma}_{zz}^{(4)}}{C_o G} = -\frac{2}{1+\nu} - \frac{2(\nu_u - \nu)}{D(s^*)}\left[(1-\nu)\sqrt{s^*}\,\mathrm{I}_0(\sqrt{s^*}) + 2\nu\,\mathrm{I}_1(\sqrt{s^*}) - \sqrt{s^*}\,\mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{7.557}$$

$$\frac{s\tilde{u}_r^{(4)}}{u_r(r_o,\infty)} = r^* + \frac{\nu_u - \nu}{\nu D(s^*)}\Big[(1-\nu)r^*\sqrt{s^*}\,\mathrm{I}_0(\sqrt{s^*}) - (1-2\nu)r^*\,\mathrm{I}_1(\sqrt{s^*}) - \mathrm{I}_1(r^*\sqrt{s^*})\Big] \tag{7.558}$$

where

$$u_r(r_o,\infty) = C_o r_o \nu \tag{7.559}$$

Applying inverse Laplace transform to the above solutions, we obtain the solutions in time

$$\frac{p^{(4)}}{C_o G} = \frac{4(\nu_u-\nu)^2}{\eta}\sum_{n=1}^{\infty}\frac{1}{E(x_n)}\left[\frac{\mathrm{J}_0(r^*\sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - 1\right]e^{-x_n t^*} \tag{7.560}$$

$$\frac{\sigma_{rr}^{(4)}}{C_o G} = -4(1-\nu)(\nu_u-\nu)\sum_{n=1}^{\infty}\frac{1}{E(x_n)}\left[\frac{\mathrm{J}_1(r^*\sqrt{x_n})}{r^*\mathrm{J}_1(\sqrt{x_n})} - 1\right]e^{-x_n t^*} \tag{7.561}$$

$$\frac{\sigma_{\theta\theta}^{(4)}}{C_o G} = -4(1-\nu)(\nu_u-\nu)\sum_{n=1}^{\infty}\frac{1}{E(x_n)} \times\left[\frac{r^*\sqrt{x_n}\,\mathrm{J}_0(r^*\sqrt{x_n}) - \mathrm{J}_1(r^*\sqrt{x_n})}{r^*\mathrm{J}_1(\sqrt{x_n})} - 1\right]e^{-x_n t^*} \tag{7.562}$$

$$\frac{\sigma_{zz}^{(4)}}{C_o G} = -2(1+\nu) - 8(\nu_u-\nu)\sum_{n=1}^{\infty}\frac{1}{E(x_n)} \times\left[\frac{(\nu_u-\nu)\,\mathrm{J}_0(r^*\sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - \nu_u(1-\nu)\right]e^{-x_n t^*} \tag{7.563}$$

$$\frac{u_r^{(4)}}{u_r(r_o,\infty)} = r^* + \frac{2(1-\nu)(\nu_u-\nu)}{\nu}\sum_{n=1}^{\infty}\frac{1}{E(x_n)} \times\left[(1-2\nu_u)r^* + \frac{\mathrm{J}_1(r^*\sqrt{x_n})}{\mathrm{J}_1(\sqrt{x_n})}\right]e^{-x_n t^*} \tag{7.564}$$

In Fig. 7.35 we plot the radial displacement at various times. We note the following limiting behaviors: $u_r(r,0^+) = C_o\,\nu_u\,r$ and $u_r(r,\infty) = C_o\,\nu\,r$.

We note that the solutions given in (7.560)–(7.564) refer to a loading of a constant compressive strain as a step function $e_{zz} = -C_o\,\mathrm{H}(t-0)$. For the loading of an axial strain which is an arbitrary function of time, $e_{zz} = C(t)$, we can find its solution by utilizing the *Duhamel principle of superposition*. To do so, we first find the *influence functions*, which are the material response due to a unit step loading in time, $e_{zz} = \mathrm{H}(t-0)$. For example, the pressure influence function p^* can be obtained from (7.560) as

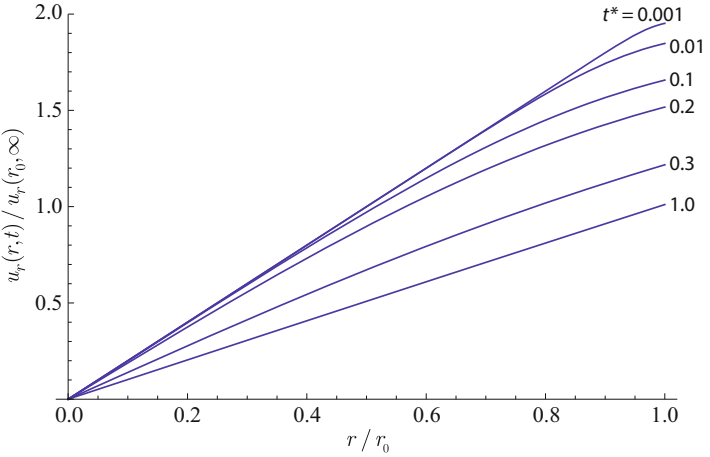


Fig. 7.35 Radial displacement for cylinder problem at various times due to a step increment of axial strain (mode 4), for $\nu_u = 0.5$ and $\nu = 0.25$

$$p^*(r,t) = -\frac{4G(\nu_u - \nu)^2}{\eta} \sum_{n=1}^{\infty} \frac{1}{E(x_n)} \left[\frac{\mathrm{J}_0(r^* \sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - 1 \right] e^{-x_n t^*} \tag{7.565}$$

The pressure history due to the loading $e_{zz} = C(t)$ is then

$$p(r,t) = C(0^+)\, p^*(r,t) + \int_0^t C'(t-\tau)\, p^*(r,\tau)\, d\tau \tag{7.566}$$

where C' denotes the derivative of C. For the case of a linearly increasing compressive strain

$$e_{zz} = -At \tag{7.567}$$

equation (7.566) can be integrated to obtain the pressure response as

$$\frac{cp}{GAr_o^2} = \frac{4(\nu_u - \nu)^2}{\eta} \sum_{n=1}^{\infty} \frac{1}{x_n E(x_n)} \left[\frac{\mathrm{J}_0(r^* \sqrt{x_n})}{\mathrm{J}_0(\sqrt{x_n})} - 1 \right] \left(1 - e^{-x_n t^*} \right) \tag{7.568}$$

7.13.2 Hollow Cylinder

In this section we consider the problem of a saturated hollow circular cylinder deforming under plane strain condition, with its inner surface subject to a fluid

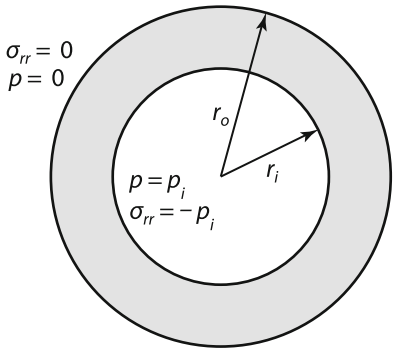


Fig. 7.36 Boundary condition for hollow cylinder problem

pressure rise, and outer surface maintained as traction and pore pressure free. More detailed analysis of hollow cylinder solution can be found in Kanj et al. [96].

The boundary conditions of this problem are given as follows (see Fig. 7.36):

$$\sigma_{rr}(r_i, t) = -p_i(t); \quad p(r_i, t) = p_i(t); \quad \sigma_{rr}(r_o, t) = 0; \quad p(r_o, t) = 0 \tag{7.569}$$

and the initial condition is

$$p(r, 0^-) = 0; \quad \text{for } r_i \le r \le r_o \tag{7.570}$$

where r_i and r_o are respectively the inner and outer radius of the hollow cylinder, and $p_i(t)$ is the pressurization history of the inner surface.

To solve this boundary value problem, the beginning point is the general solutions derived in Sect. 7.12, and particularly, Eqs. (7.490)–(7.494), and the governing equation (7.498). Again, we shall apply Laplace transform to remove the time variable. We hence obtain from (7.491) and (7.494) the following

$$\tilde{\sigma}_{rr} = -\frac{2\eta}{S}\frac{1}{r^2}\int r\,\tilde{\zeta}(r,s)\,dr + \frac{2G}{1-2\nu_u}\tilde{B}_1(s) - \frac{2G}{r^2}\tilde{B}_2(s) \tag{7.571}$$

$$\tilde{p} = \frac{\tilde{\zeta}}{S} - 2\alpha M\tilde{B}_1(s) \tag{7.572}$$

Comparing with (7.502) and (7.503) of the solid cylinder problem, we observe that we have an additional term involving $\tilde{B}_2(s)$.

The general solution of the governing equation (7.498) in the Laplace transform domain, given as (7.500), is

$$\tilde{\zeta} = D_1 \mathrm{I}_0\left(r\sqrt{\frac{s}{c}}\right) + D_2 \mathrm{K}_0\left(r\sqrt{\frac{s}{c}}\right) \tag{7.573}$$

Comparing to (7.501), we observe that we need to retain both the Bessel function terms for the present problem. We can now substitute (7.573) into (7.571)

and (7.572), and utilize the four boundary conditions in (7.569), to solve for the four unknowns, $\tilde{B}_1$, $\tilde{B}_2$, D_1 and D_2, as

$$\begin{aligned}\tilde{B}_1 = &-\frac{(1-2\nu_u)\tilde{p}_i}{2D(s)}\left\{\mathrm{K}_0(\xi_o)\left[-\xi_i^2\,\mathrm{I}_0(\xi_i)+2\eta\,(\xi_i\,\mathrm{I}_1(\xi_i)-\xi_o\,\mathrm{I}_1(\xi_o))\right]\right.\\ &\left.+\mathrm{I}_0(\xi_o)\left[\xi_i^2\,\mathrm{K}_0(\xi_i)+2\eta\,(\xi_i\,\mathrm{K}_1(\xi_i)-\xi_o\,\mathrm{K}_1(\xi_o))\right]\right\} \qquad (7.574)\\ \tilde{B}_2 = &\frac{c\,\tilde{p}_i\,\xi_i\,\xi_o}{2s(1-\nu)D(s)}\{4\eta(\nu_u-\nu)\,[\,\mathrm{I}_1(\xi_i)\,\mathrm{K}_1(\xi_o)-\mathrm{I}_1(\xi_o)\,\mathrm{K}_1(\xi_i)\,]\\ &+2(\nu_u-\nu)\,\xi_i\,\mathrm{I}_1(\xi_o)\,[\,\mathrm{K}_0(\xi_o)-\mathrm{K}_0(\xi_i)\,]\\ &+2(\nu_u-\nu)\,\xi_i\,\mathrm{K}_1(\xi_o)\,[\,\mathrm{I}_0(\xi_o)-\mathrm{I}_0(\xi_i)\,]\\ &+2\eta(1-\nu)\,\mathrm{K}_0(\xi_o)\,[\,\xi_i\,\mathrm{I}_1(\xi_o)-\xi_o\,\mathrm{I}_1(\xi_i)\,]\\ &+2\eta(1-\nu)\,\mathrm{I}_0(\xi_o)\,[\,\xi_i\,\mathrm{K}_1(\xi_o)-\xi_o\,\mathrm{K}_1(\xi_i)\,]\\ &+(1-\nu)\,\xi_i\,\xi_o\,[\,\mathrm{I}_0(\xi_i)\mathrm{K}_0(\xi_o)-\mathrm{I}_0(\xi_o)\mathrm{K}_0(\xi_i)\,]\,\} \qquad (7.575)\\ D_1 = &\frac{2\eta(1-\nu_u)\tilde{p}_i}{(\nu_u-\nu)D(s)}\,\{\eta(1-\nu)(\xi_o^2-\xi_i^2)\,\mathrm{K}_0(\xi_o)\\ &+(\nu_u-\nu)\,\xi_i^2\,[\,\mathrm{K}_0(\xi_o)-\mathrm{K}_0(\xi_i)\,]\\ &+2\eta(\nu_u-\nu)\,[\,\xi_o\,\mathrm{K}_1(\xi_o)-\xi_i\,\mathrm{K}_1(\xi_i)\,]\,\} \qquad (7.576)\\ D_2 = &-\frac{2\eta(1-\nu_u)\tilde{p}_i}{(\nu_u-\nu)D(s)}\,\{\eta(1-\nu)(\xi_o^2-\xi_i^2)\,\mathrm{I}_0(\xi_o)\\ &+(\nu_u-\nu)\,\xi_i^2\,[\,\mathrm{I}_0(\xi_o)-\mathrm{I}_0(\xi_i)\,]\\ &-2\eta(\nu_u-\nu)\,[\,\xi_o\,\mathrm{I}_1(\xi_o)-\xi_i\,\mathrm{I}_1(\xi_i)\,]\,\} \qquad (7.577)\end{aligned}$$

where

$$\begin{aligned}D(s) = &\frac{2G(\nu_u-\nu)}{1-\nu}\,\{\,[\,\xi_o\,\mathrm{I}_1(\xi_o)-\xi_i\,\mathrm{I}_1(\xi_i)\,]\,[\mathrm{K}_0(\xi_i)-\mathrm{K}_0(\xi_o)\,]\\ &+\,[\,\xi_o\,\mathrm{K}_1(\xi_o)-\xi_i\,\mathrm{K}_1(\xi_i)\,]\,[\,\mathrm{I}_0(\xi_i)-\mathrm{I}_0(\xi_o)\,]\,\}\\ &+G(\xi_o^2-\xi_i^2)\,[\,\mathrm{I}_0(\xi_i)\,\mathrm{K}_0(\xi_o)-\mathrm{I}_0(\xi_o)\,\mathrm{K}_0(\xi_i)\,] \qquad (7.578)\end{aligned}$$

and

$$\xi_i = r_i\sqrt{\frac{s}{c}};\quad \xi_o = r_o\sqrt{\frac{s}{c}} \qquad (7.579)$$

We hence find the solution in the Laplace transform domain as

$$\tilde{u}_r = \frac{\eta}{GS\sqrt{s/c}}\left[D_1 \mathrm{I}_1(\xi) - D_2 \mathrm{K}_1(\xi)\right] + \tilde{B}_1 r + \frac{\tilde{B}_2}{r} \tag{7.580}$$

$$\tilde{\sigma}_{rr} = -\frac{2\eta}{S\xi}\left[D_1 \mathrm{I}_1(\xi) - D_2 \mathrm{K}_1(\xi)\right] + \frac{2G}{1-2\nu_u}\tilde{B}_1 - \frac{2G}{r^2}\tilde{B}_2 \tag{7.581}$$

$$\tilde{\sigma}_{\theta\theta} = \frac{2\eta}{S\xi}\left[D_1 \mathrm{I}_1(\xi) - D_2 \mathrm{K}_1(\xi)\right] - \frac{2\eta}{S}\left[D_1 \mathrm{I}_0(\xi) + D_2 \mathrm{K}_0(\xi)\right] + \frac{2G}{1-2\nu_u}\tilde{B}_1 + \frac{2G}{r^2}\tilde{B}_2 \tag{7.582}$$

$$\tilde{\sigma}_{zz} = -\frac{2\eta}{S}\left[D_1 \mathrm{I}_0(\xi) + D_2 \mathrm{K}_0(\xi)\right] + \frac{2G}{1-2\nu_u}\tilde{B}_1 \tag{7.583}$$

$$\tilde{p} = \frac{1}{S}\left[D_1 \mathrm{I}_0(\xi) + D_2 \mathrm{K}_0(\xi)\right] - 2\alpha M \tilde{B}_1 \tag{7.584}$$

where

$$\xi = r\sqrt{\frac{s}{c}} \tag{7.585}$$

Following the inverse Laplace transform procedure described in Appendix D, it is possible to obtain analytical expressions in the time domain for the above expressions. However, the results are too lengthy to be of interest to present here. Rather, in the following, we shall rely on numerical inverse Laplace transform to obtain results in the time domain. Particularly, we shall use the Stehfest method as described in Sect. D.3 to perform the inversion.

First, we consider a case of a thick-walled cylinder, say, with an r_i/r_o ratio of 0.3, such as the ones used in the laboratory burst tests of hollow rock cylinders for testing tensile strength of rocks [67]. These tests can be used for the prediction of *breakdown pressure* in an in situ pressurization of a borehole, related to hydraulic fracturing or in situ stress measurement applications. Assuming the material properties of a Berea sandstone saturated with water with $\nu_u = 0.313$, $\nu = 0.2$, and $\eta = 0.292$ ($\alpha = 0.778$) (Table 3.2), and a step rise in fluid pressure on the inner wall $p_i(t) = p_o \mathrm{H}(t-0)$ we plot the pore pressure history in the cylinder with respect to dimensionless time $t^* = ct/r_o^2$ as Fig. 7.37. We notice that at small times, the pore pressure is negative in a portion of the cylinder. The reason is that the wall is in tension by the internal pressurization, creating a pore pressure drop due to the Skempton effect. At large time, the solution approaches the steady state solution

$$\frac{p(r,\infty)}{p_o} = \frac{\log(r/r_o)}{\log(r_i/r_o)} \tag{7.586}$$

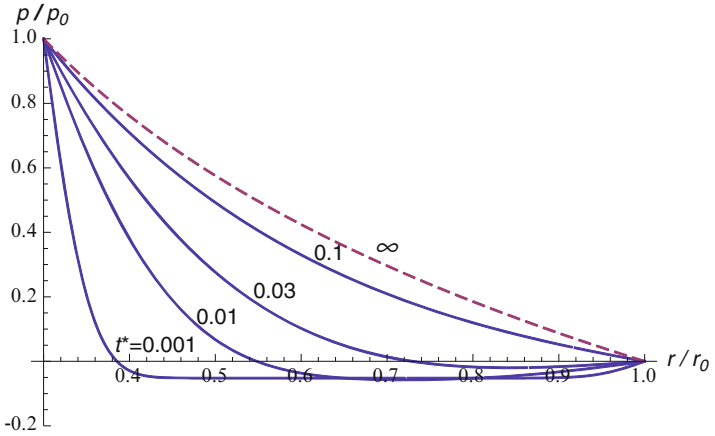


Fig. 7.37 Pressure history in a thick-walled hollow cylinder with $r_i/r_o = 0.3$, and $\nu_u = 0.313$, $\nu = 0.2$, and $\eta = 0.292$

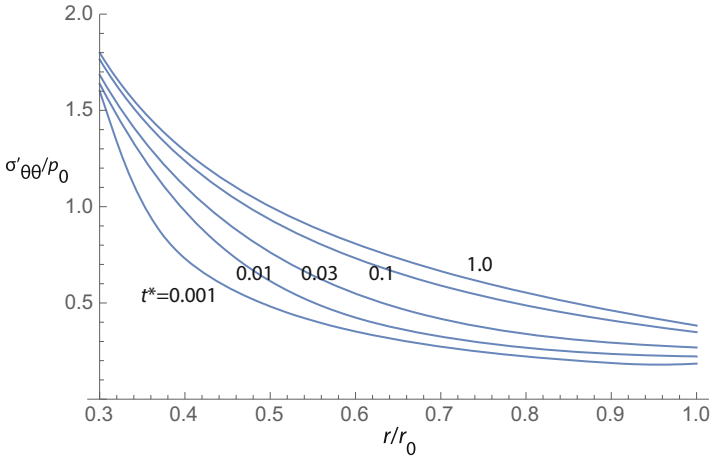


Fig. 7.38 Terzaghi effective hoop stress history in the wall of a hollow cylinder with $r_i/r_o = 0.3$, and $\nu_u = 0.313$, $\nu = 0.2$, and $\eta = 0.292$

Also of interest to examine is the Terzaghi effective circumferential (hoop) stress $\sigma'_{\theta\theta} = \sigma_{\theta\theta} + p$. In Fig. 7.38 we plot $\sigma'_{\theta\theta}$ versus r/r_o at various dimensionless times. We observe that the effective hoop stress is tensile and is largest at the inner wall. It also increases with time, due to the pore pressure penetration, making it effectively more tensile.

Next, we examine a case of thin-walled tube, such as a blood vessel [43, 98]. Assume incompressible constituents, $\nu_u = 0.5$ and $\alpha = 1$, and $\nu = 0.1$ ($\eta = 0.444$), with a wall thickness of $0.1\, r_o$. The loading is a step drop of pressure in the tube, $p_i(t) = -p_o \mathrm{H}(t - 0)$ (in contrast to the step rise in the thick-walled case).

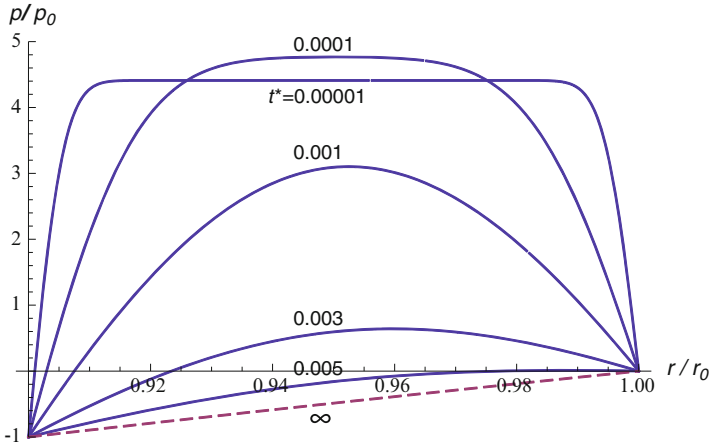


Fig. 7.39 Pressure history in a thin-walled tube with $r_i/r_o = 0.9$, and $\nu_u = 0.5$, $\nu = 0.1$, and $\eta = 0.333$

Figure 7.39 presents the pore pressure history in the tube wall as a function of dimensionless time. We observe that when pressure drops in the tube, the vessel contracts, and a rise in pore pressure is found in the tube wall. In contrast with the thick-walled cylinder case, the magnitude of the pore pressure rise is rather large; and in fact, the thinner the wall is, the larger the pore pressure rise. This large pressure rise in a thin wall creates a large pressure gradient, which can drive a large flux toward the inner and outer surface of the wall. Such a flux is speculated to have the physiological consequences of atherosclerosis [100].

7.14 Borehole Problem

The stress concentration near an excavated borehole is of critical interest to the oil and gas industry. Stress levels exceeding material limits can cause costly wellbore damage. For vertical borehole orientation, the linear elastic analysis based on the plane strain assumption is well known [87] and has been extensively applied. However, in petroleum engineering applications, the formation surrounding the borehole is often permeated with fluid. The pore pressure field near the borehole is perturbed by the excavation. Fluid diffusion leads to a time-dependent redistribution of total and effective stresses. These coupled effects should be examined within the framework of the Biot theory of poroelasticity.

The borehole problem refers to a long circular shaft excavated from a geological formation. We assume that the in situ stress is uniform, but can be *non-hydrostatic*. In general, if the stress in a geological formation is induced by its self-weight, then the stress is *hydrostatic*, that is, the compressive stress is the same in all directions,

and linearly increases with depth. However, tectonic movement can alter the stress field, such that the three principal stresses can differ in magnitude.

If the geometry, material properties, and loading conditions do not change in the borehole axis direction, then a generalized plane strain condition exists. Furthermore, if the axis of the borehole is aligned with one of the principal stress directions, the problem reduces to plane strain.

The analytical solutions of such problems have many applications, such as borehole stability after excavation, mud weight control during drilling operation, in situ stress determination, hydraulic fracture initiation, hydrocarbon production, well stimulation, etc. Some of the these applications were discussed in Chap. 1. As demonstrated in the solution strategy presented in the earlier sections, these problems need not be separately solved. They can be studied as a few fundamental loading modes. Then solutions of different problems can be assembled by using the principle of superposition for linear systems. We shall take this approach to study these problems first as plane strain problems for simplicity, and then extend the analysis to generalized plane strain problems.

7.14.1 Plane Strain Borehole Problem

Figure 7.40 shows the x-y plane of a non-hydrostatic far-field stress field with the minimum and the maximum principal compressive stresses aligned with the x- and y-axis, respectively, together with a virgin pore pressure p_o:

$$\sigma_{xx} = -P_{\min}, \quad \sigma_{yy} = -P_{\max}, \quad \sigma_{xy} = 0, \quad p = p_o \tag{7.587}$$

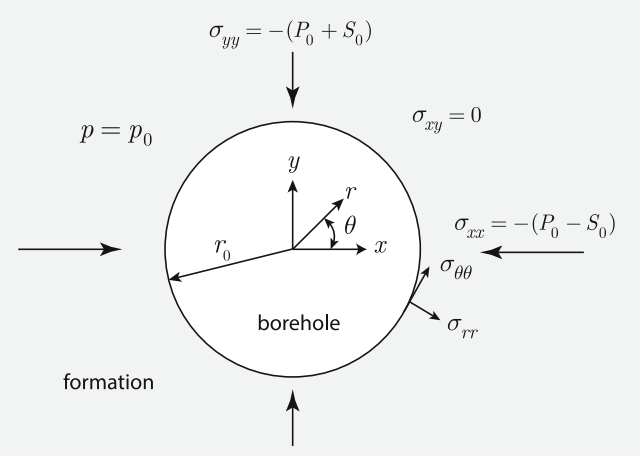


Fig. 7.40 A borehole is a non-hydrostatic stress field

These normal stresses can also be expressed as

$$\sigma_{xx} = -(P_o - S_o), \quad \sigma_{yy} = -(P_o + S_o) \tag{7.588}$$

with

$$P_o = \frac{P_{\max} + P_{\min}}{2}, \quad S_o = \frac{P_{\max} - P_{\min}}{2} \tag{7.589}$$

Or, we can express the above in polar coordinate system as

$$\sigma_{rr} = -P_o + S_o \cos 2\theta \tag{7.590}$$

$$\sigma_{\theta\theta} = -P_o - S_o \cos 2\theta \tag{7.591}$$

$$\sigma_{r\theta} = -S_o \sin 2\theta \tag{7.592}$$

A borehole can be excavated from the formation, leaving the wall both stress and pore pressure free. A stress concentration will instantly develop on the wall, and then evolves with time, as the pore fluid is drained. Or, we can have the problem of an existing borehole filled with drilling mud. The mud weight can be adjusted to push back the formation to prevent the borehole from collapsing due to concentrated compressive or shear stress on the wall. When the mud becomes too heavy, it can create tensile hoop stress and fracture the wall. In a different situation, the fluid pressure can be lowered to below the formation pore pressure to extract the formation fluid (water or hydrocarbon), and the borehole is a production well. As mentioned above, all these problems can be studies as the superposition of the fundamental loading modes; hence we shall first seek the solutions in three loading modes. The solution presented below largely follows that of Detournay and Cheng [62].

Axial Symmetry The mode 1 and 2 loading conditions satisfy axial symmetry. The governing equations and general solutions are those presented in Sect. 7.12. As compared to the cylinder problem solved in Sect. 7.13, the present problem is an exterior domain problem with its domain extends to infinity. For any loading applied at the near field, all the resultant stresses, pore pressure, and displacement must vanish as $r \to \infty$. Hence we may conclude that the time functions $A_1(t)$ and $B_1(t)$ found in (7.477)–(7.481), and (7.489)–(7.494) must vanish. We can then express (7.478)–(7.481) as follows

$$u_r = \frac{\eta}{G}\frac{1}{r}\int rp(r,t)\, dr + \frac{A_2(t)}{r} \tag{7.593}$$

$$\sigma_{rr} = -2\eta \frac{1}{r^2}\int rp(r,t)\, dr - \frac{2G}{r^2}A_2(t) \tag{7.594}$$

$$\sigma_{\theta\theta} = 2\eta \frac{1}{r^2} \int rp(r,t)\, dr - 2\eta p + \frac{2G}{r^2} A_2(t) \tag{7.595}$$

$$\sigma_{zz} = -2\eta p \tag{7.596}$$

Particularly, we observe that the diffusion equation for pore pressure (7.483) no longer has the right hand side, and is decoupled:

$$\frac{\partial p}{\partial t} - c \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) = 0 \tag{7.597}$$

The above equation can be solved independent of the solid response.

Mode 1: Radial Stress For the mode 1 problem, the external domain of the borehole is initially at equilibrium, with zero stresses and pore pressure. At $t = 0^+$, the borehole wall is subjected to a stress P_o pushing radially outward, with the pore pressure maintained at zero. The boundary conditions are expressed as follows:

$$\sigma_{rr} = -P_o; \quad \sigma_{r\theta} = 0; \quad p = 0; \quad \text{at } r = r_o \tag{7.598}$$

As the pore pressure diffusion equation (7.597) is decoupled, we can solve it first. From the boundary condition in (7.598), it is clear that the solution is trivial

$$p^{(1)} = 0 \tag{7.599}$$

In the above and below we shall use the superscript $^{(1)}$ to denote the 'mode 1' solution for ease of future reference. The rest of the solution is simply the classical Lamé solution of elasticity [161]:

$$\frac{2Gu_r^{(1)}}{P_o r_o} = \frac{1}{r^*} \tag{7.600}$$

$$\frac{\sigma_{rr}^{(1)}}{P_o} = -\frac{1}{r^{*2}} \tag{7.601}$$

$$\frac{\sigma_{\theta\theta}^{(1)}}{P_o} = \frac{1}{r^{*2}} \tag{7.602}$$

where

$$r^* = \frac{r}{r_o} \tag{7.603}$$

We observe from the above and (7.596) that $\sigma_{rr} + \sigma_{\theta\theta} + \sigma_{zz} = 0$; hence there is no volumetric strain, and no pore pressure is generated by the Skempton effect.

The above solution is for a hole that is pressurized inside. For the problem that the hole is free from stress, while the load is applied by stretching a plate (Kirsch

solution [101]), or from a formation far-field compressive stress [111], the solution can be obtained by superposition. For borehole excavated in a formation with a preexisting uniform compressive stress $\sigma_{rr} = \sigma_{\theta\theta} = -P_o$, with the borehole wall freed from radial stress, the solution becomes

$$\sigma_{rr} = P_0 \left(\frac{r_o^2}{r^2} - 1 \right) \tag{7.604}$$

$$\sigma_{\theta\theta} = -P_o \left(\frac{r_o^2}{r^2} + 1 \right) \tag{7.605}$$

Particularly, we find that at the borehole wall, $r = r_o$, a circumferential stress

$$\sigma_{\theta\theta}(r_o, t) = -2P_o \tag{7.606}$$

is developed, which is twice the original stress level of the formation. This means that the excavation creates a stress concentration, which may cause a stability problem for the integrity of the hole.

Mode 2: Fluid Pressure The boundary conditions are

$$\sigma_{rr} = 0; \quad \sigma_{r\theta} = 0; \quad p = p_o; \quad \text{at } r = r_o \tag{7.607}$$

The pore pressure field is solved by taking the Laplace transform of (7.597). Solving this ordinary differential equation with the boundary condition (7.607), we obtain

$$\frac{s\tilde{p}^{(2)}}{p_o} = \frac{\mathrm{K}_0(r^*\sqrt{s^*})}{\mathrm{K}_0(\sqrt{s^*})} \tag{7.608}$$

where

$$s^* = \frac{r_o^2 s}{c} \tag{7.609}$$

In the above and below, the superscript $^{(2)}$ is used to denote the mode 2 solution.

The radial stress σ_{rr} is obtained by integrating the pore pressure according to (7.594). By utilizing the boundary condition (7.607), the constant $A_2(t)$ can be solved. It then follows from (7.593), (7.594), and (7.595) that the displacement, stress, and specific flux solutions are

$$\frac{2Gs\tilde{u}_r^{(2)}}{p_o r_o} = -2\eta \left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_0(\sqrt{s^*})} - \frac{1}{r^*} \frac{\mathrm{K}_1(\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_0(\sqrt{s^*})} \right] \tag{7.610}$$

$$\frac{s\tilde{\sigma}_{rr}^{(2)}}{p_o} = 2\eta \left[\frac{1}{r^*} \frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_0(\sqrt{s^*})} - \frac{1}{r^{*2}} \frac{\mathrm{K}_1(\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_0(\sqrt{s^*})} \right] \tag{7.611}$$

$$\frac{s\tilde{\sigma}_{\theta\theta}^{(2)}}{p_o} = -2\eta\left[\frac{1}{r^*}\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_0(\sqrt{s^*})} - \frac{1}{r^{*2}}\frac{\mathrm{K}_1(\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_0(\sqrt{s^*})} + \frac{\mathrm{K}_0(r^*\sqrt{s^*})}{\mathrm{K}_0(\sqrt{s^*})}\right] \tag{7.612}$$

$$\frac{sr_o\tilde{q}_r^{(2)}}{\kappa p_o} = \frac{\sqrt{s^*}\,\mathrm{K}_1(r^*\sqrt{s^*})}{\mathrm{K}_0(\sqrt{s^*})} \tag{7.613}$$

It is possible to perform analytical inverse Laplace transform to the above solutions. For example, the pore pressure solution (7.608) can be inverted to [32]

$$\frac{p^{(2)}}{p_o} = 1 + \frac{2}{\pi}\int_0^\infty e^{-cu^2t}\,\frac{\mathrm{J}_0(ur)\,\mathrm{Y}_0(ur_o) - \mathrm{Y}_0(ur)\,\mathrm{J}_0(ur_o)}{u\left[\mathrm{J}_0^2(ur_o) + \mathrm{Y}_0^2(ur_o)\right]}\,du \tag{7.614}$$

where J_0 and Y_0 are respectively the Bessel function of the first and the second kind of order zero. Numerical evaluation of expressions like (7.614), however, takes some effort. We hence shall keep these Laplace transform expressions, and rely on the approximate inverse Laplace transform as described in Sect. D.3 to find the solution in time domain.

In Fig. 7.41 the normalized pore pressure p/p_o of the mode 2 solution is plotted versus the dimensionless radial distance r/r_o at various dimensionless times $t^* = ct/r_o^2$. We notice that the normalized solution is independent of the material constants. We observe that the pore pressure progressively increases with time, and as $t \to \infty, p \to p_o$.

In Fig. 7.42 we present the normalized circumferential stress $-\sigma_{\theta\theta}/\eta p_o$ as function of radial distance at various dimensionless times. We observe that at the borehole wall, a compressive stress of the magnitude $\sigma_{\theta\theta}(r_o, t) = -2\eta p_o$ instantly

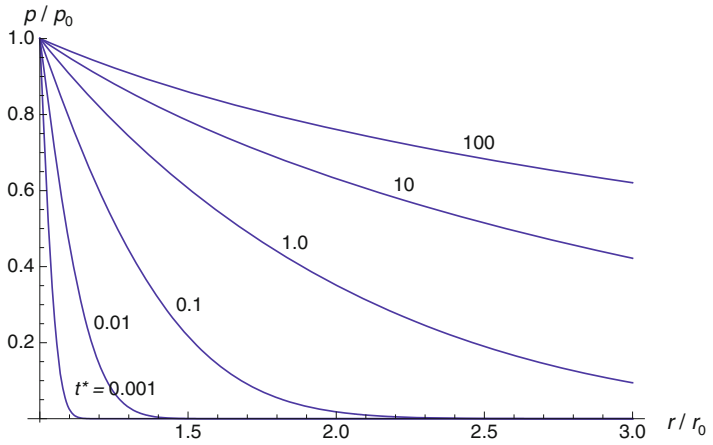


Fig. 7.41 Pressure history of mode 2 borehole solution

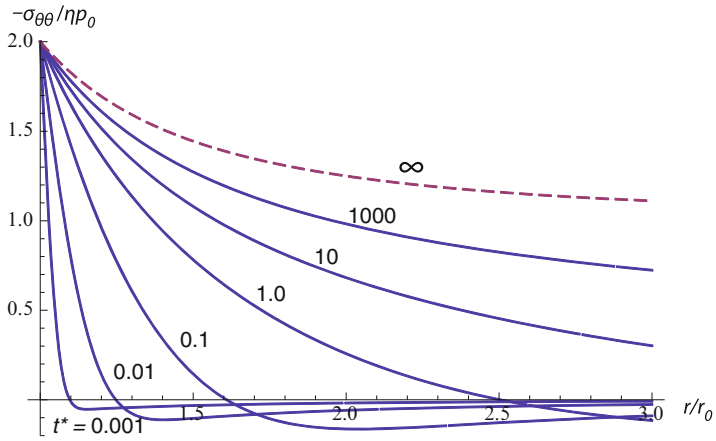


Fig. 7.42 Circumferential stress history of mode 2 borehole solution

develops, and remains so all the time, due to the penetration of pore pressure. We also find that at smaller times, there is a region of small tensile stress propagating ahead of the compression region. In fact, at any distance inside the formation, the stress history always starts as tensile before it becomes compressive. As $t \to \infty$, a steady state is reached with

$$\sigma_{\theta\theta}(r, \infty) = -\eta p_o \left(1 + \frac{r_o^2}{r^2}\right) \tag{7.615}$$

For future reference, we also note the asymptotic value for the radial stress

$$\sigma_{rr}(r, \infty) = -\eta p_o \left(1 - \frac{r_o^2}{r^2}\right) \tag{7.616}$$

An important observation from (7.615) is that although the total circumferential stress at the borehole wall is compressive, the Terzaghi effective stress (associated with material failure) is in fact tensile, $\sigma'_{\theta\theta}(r_o, t) = (1 - 2\eta)p_o$, with $0 \le \eta \le 0.5$.

Next we examine the radial displacement u_r. Figure 7.43 presents the normalized radial displacement $Gu_r/\eta r_o p_o$ versus dimensionless distance at various dimensionless times. We first notice that at the wall, there is no displacement. Inside the wall, the displacement moves radially outward. At steady state, we find

$$u_r(r, \infty) = \frac{\eta r_o p_o}{2G} \left(\frac{r}{r_o} - \frac{r_o}{r}\right) \tag{7.617}$$

Mode 3: Deviatoric Stress The boundary conditions for the mode 3 problem are

$$\sigma_{rr} = S_o \cos 2\theta; \quad \sigma_{r\theta} = -S_o \sin 2\theta; \quad p = 0; \quad \text{at } r = r_o \tag{7.618}$$

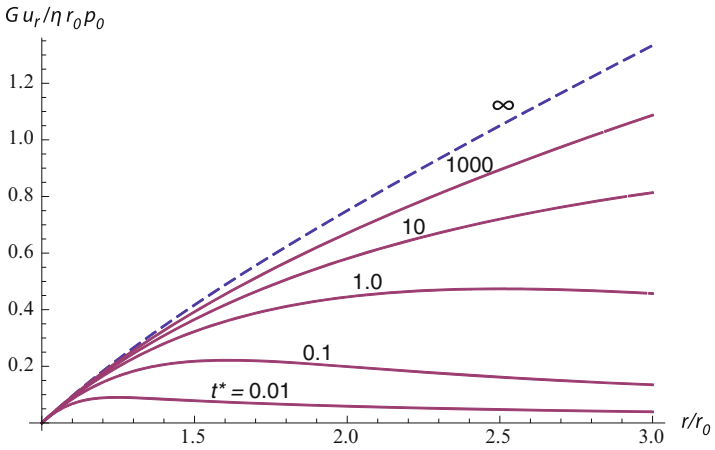


Fig. 7.43 Radial displacement history of mode 2 borehole solution

For elasticity problems, the above loading conditions leads to the Kirsch solution [101, 111]. It is however not so for poroelasticity.

Mode 3 is obviously not axisymmetric; hence the full plane strain equations in Sect. 7.4 should be used. Particularly, we shall use the volumetric strain and rotation formulation developed in Sect. 7.4.3. For the present problem, we perform Laplace transform to (7.112), (7.117), and (7.118), and express them in polar coordinates, as the following

$$\frac{1-\nu_u}{1-2\nu_u}\frac{\partial \tilde{e}}{\partial r} - \frac{1}{r}\frac{\partial \tilde{\Omega}_z}{\partial \theta} - \frac{B(1+\nu_u)}{3(1-2\nu_u)}\frac{\partial \tilde{\zeta}}{\partial r} = 0 \tag{7.619}$$

$$\frac{1-\nu_u}{1-2\nu_u}\frac{1}{r}\frac{\partial \tilde{e}}{\partial \theta} + \frac{\partial \tilde{\Omega}_z}{\partial r} - \frac{B(1+\nu_u)}{3(1-2\nu_u)}\frac{1}{r}\frac{\partial \tilde{\zeta}}{\partial \theta} = 0 \tag{7.620}$$

$$\frac{\partial^2 \tilde{\zeta}}{\partial r^2} + \frac{1}{r}\frac{\partial \tilde{\zeta}}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \tilde{\zeta}}{\partial \theta^2} - \frac{s}{c}\tilde{\zeta} = 0 \tag{7.621}$$

where

$$\tilde{e} = \frac{\partial \tilde{u}_r}{\partial r} + \frac{1}{r}\frac{\partial \tilde{u}_\theta}{\partial \theta} + \frac{\tilde{u}_r}{r} \tag{7.622}$$

is the volumetric strain, and

$$\tilde{\Omega}_z = \frac{1}{2r}\left(\frac{\partial\, r\tilde{u}_\theta}{\partial r} - \frac{\partial \tilde{u}_r}{\partial \theta}\right) \tag{7.623}$$

denotes the rotation of the displacement field.

Using symmetry considerations, it can be argued that the dependence of the displacement and stress solution upon the polar angle θ is in the following form

$$(\tilde{\zeta},\, \tilde{e},\, \tilde{u}_r,\, \tilde{\sigma}_{rr},\, \tilde{\sigma}_{\theta\theta},\, \tilde{p}) = (\tilde{\mathbb{Z}},\, \tilde{\mathbb{E}},\, \tilde{\mathbb{U}}_r,\, \tilde{\mathbb{S}}_{rr},\, \tilde{\mathbb{S}}_{\theta\theta},\, \tilde{\mathbb{P}}) \times \cos 2\theta$$
$$(\tilde{\Omega}_z,\, \tilde{u}_\theta,\, \tilde{\sigma}_{r\theta}) = (\tilde{\mathbb{W}}_z,\, \tilde{\mathbb{U}}_\theta,\, \tilde{\mathbb{S}}_{r\theta}) \times \sin 2\theta \tag{7.624}$$

in which $\tilde{\mathbb{Z}}$, $\tilde{\mathbb{E}}$, $\tilde{\mathbb{W}}_z$, $\tilde{\mathbb{U}}_r$, $\tilde{\mathbb{U}}_\theta$, $\tilde{\mathbb{S}}_{rr}$, $\tilde{\mathbb{S}}_{r\theta}$, $\tilde{\mathbb{S}}_{\theta\theta}$, and $\tilde{\mathbb{P}}$ are functions of r and s only. Substituting the above expressions into Eqs. (7.619), (7.620), and (7.621) produces the set of ordinary differential equations

$$\frac{1-\nu_u}{1-2\nu_u}\frac{d\tilde{\mathbb{E}}}{dr} - 2\frac{\tilde{\mathbb{W}}_z}{r} - \frac{B(1+\nu_u)}{3(1-2\nu_u)}\frac{d\tilde{\mathbb{Z}}}{dr} = 0 \tag{7.625}$$

$$\frac{1-\nu_u}{1-2\nu_u}\frac{\tilde{\mathbb{E}}}{r} - \frac{1}{2}\frac{d\tilde{\mathbb{W}}_z}{dr} - \frac{B(1+\nu_u)}{3(1-2\nu_u)}\frac{\tilde{\mathbb{Z}}}{r} = 0 \tag{7.626}$$

$$r^2\frac{d^2\tilde{\mathbb{Z}}}{dr^2} + r\frac{d\tilde{\mathbb{Z}}}{dr} - \left(\frac{s}{c}r^2 + 4\right)\tilde{\mathbb{Z}} = 0 \tag{7.627}$$

Equation (7.627) is a modified Bessel equation and can be solved for $\tilde{\mathbb{Z}}$

$$s\tilde{\mathbb{Z}} = C_1' \mathrm{K}_2(r^*\sqrt{s^*}) \tag{7.628}$$

where we have drop the term associated with I_2, which is unbounded at $r \to \infty$. Substituting (7.628) into (7.625) and (7.626), and after some manipulation, we can solve for

$$s\tilde{\mathbb{W}}_z = \frac{C_2'}{r^{*2}} \tag{7.629}$$

$$s\tilde{\mathbb{E}} = C_1'\frac{B(1+\nu_u)}{3(1-\nu_u)}\mathrm{K}_2(r^*\sqrt{s^*}) - \frac{1-2\nu_u}{1-\nu_u}\frac{C_2'}{r^{*2}} \tag{7.630}$$

In (7.629) and (7.630) we have dropped the unbounded terms as the solution must vanish at infinity. The displacement components $\tilde{\mathbb{U}}_r$ and $\tilde{\mathbb{U}}_\theta$ are found by solving (7.629) and (7.630), and

$$\frac{2Gs\tilde{\mathbb{U}}_r^{(3)}}{S_o r_o} = C_1\left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\sqrt{s^*}} + \frac{2\,\mathrm{K}_2(r^*\sqrt{s^*})}{r^*s^*}\right] + \frac{C_2}{r^*} + \frac{C_3}{r^{*3}} \tag{7.631}$$

$$\frac{2Gs\tilde{\mathbb{U}}_\theta^{(3)}}{S_o r_o} = 2C_1\frac{\mathrm{K}_2(r^*\sqrt{s^*})}{r^*s^*} - \frac{1-2\nu_u}{2(1-\nu_u)}\frac{C_2}{r^*} + \frac{C_3}{r^{*3}} \tag{7.632}$$

in which we have redefined the constants into C_1, C_2 and C_3. In the above and below, the superscript $^{(3)}$ denotes the mode 3 solution.

The stress and pore pressure can now be deduced from (7.631) and (7.632) as

$$\frac{s\tilde{\mathbb{P}}^{(3)}}{S_o} = -\frac{B(1-\nu)(1+\nu_u)}{3(\nu_u-\nu)} C_1 \mathrm{K}_2(r^*\sqrt{s^*}) + \frac{B(1+\nu_u)}{3(1-\nu_u)} \frac{C_2}{r^{*2}} \tag{7.633}$$

$$\frac{s\tilde{\mathbb{S}}^{(3)}_{rr}}{S_o} = -C_1 \left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{r^*\sqrt{s^*}} + \frac{6\,\mathrm{K}_2(r^*\sqrt{s^*})}{r^{*2}s^*} \right] - \frac{1}{1-\nu_u} \frac{C_2}{r^{*2}} - \frac{3C_3}{r^{*4}} \tag{7.634}$$

$$\frac{s\tilde{\mathbb{S}}^{(3)}_{\theta\theta}}{S_o} = C_1 \left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{r^*\sqrt{s^*}} + \left(1 + \frac{6}{r^{*2}s^*}\right) \mathrm{K}_2(r^*\sqrt{s^*}) \right] + \frac{3C_3}{r^{*4}} \tag{7.635}$$

$$\frac{s\tilde{\mathbb{S}}^{(3)}_{r\theta}}{S_o} = -2C_1 \left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{r^*\sqrt{s^*}} + \frac{3\,\mathrm{K}_2(r^*\sqrt{s^*})}{r^{*2}s^*} \right] - \frac{1}{2(1-\nu_u)} \frac{C_2}{r^{*2}} - \frac{3C_3}{r^{*4}} \tag{7.636}$$

The three constants, C_1, C_2 and C_3, are obtained from the boundary conditions (7.618). They are

$$C_1 = -\frac{4(\nu_u-\nu)s^*}{(1-\nu)s^*\mathrm{K}_2(\sqrt{s^*}) - 2(\nu_u-\nu)\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} \tag{7.637}$$

$$C_2 = -\frac{4(1-\nu_u)(1-\nu)s^*\mathrm{K}_2(\sqrt{s^*})}{(1-\nu)s^*\mathrm{K}_2(\sqrt{s^*}) - 2(\nu_u-\nu)\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} \tag{7.638}$$

$$C_3 = \frac{2(\nu_u-\nu)\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*}) + [8(\nu_u-\nu) + (1-\nu)s^*]\,\mathrm{K}_2(\sqrt{s^*})}{(1-\nu)s^*\mathrm{K}_2(\sqrt{s^*}) - 2(\nu_u-\nu)\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} \tag{7.639}$$

In Fig. 7.44 we plot the normalized pore pressure history versus the radial distance in the $\theta = 90°$ or $270°$ direction, for the parameters $\nu = 0.2$, $\nu_u = 0.4$, and $B = 0.8$. The solution's dependence on $\cos 2\theta$ is as shown in (7.624); that is, at $\theta = 0°$ or $180°$, the pressure carries the opposite sign as $\theta = 90°$ or $270°$. We observe that at the instant of the deviatoric stress loading, the undrained response gives an initial pore pressure as

$$p(r,\theta,0) = -\frac{4}{3} S_o B(1+\nu_u) \frac{r_o^2}{r^2} \cos 2\theta \tag{7.640}$$

The boundary condition $p = 0$ then causes a steep pore pressure gradient near the wall and the drainage of pore fluid. Due to the pore pressure dependence on the polar angle θ, a tangential flow also develops.

Next, we examine the interesting behavior of the tangential (circumferential) stress $\sigma_{\theta\theta}$. As discussed before, poroelastic solutions approach elastic solutions at the initial and final times, characterized by the undrained and drained material

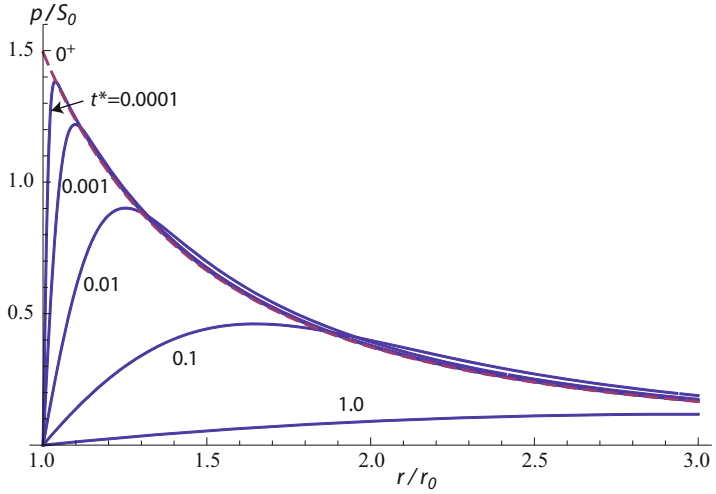


Fig. 7.44 Normalized pore pressure p/S_o versus dimensionless radial distance r/r_o in the $\theta = 90°$ direction, at various times, for the mode 3 borehole problem, with $\nu = 0.2$, $\nu_u = 0.4$, and $B = 0.8$

constants, respectively. The elastic solution of tangential stress for the current problem, however, turns out to be independent of material constants. Hence the tangential stress solutions for poroelasticity at $t = 0$ (undrained) and $t \to \infty$ (drained) are the same:

$$\sigma_{\theta\theta}(r, \theta, 0) = \sigma_{\theta\theta}(r, \theta, \infty) = 3\,\frac{r_o^4}{r^4}\,S_o \cos 2\theta \tag{7.641}$$

In between these time limits, the tangential stress can either increase or decrease, but it must return to its initial value at large time. Similarly, we find for the radial stress the following asymptotic values:

$$\sigma_{rr}(r, \theta, 0) = \sigma_{rr}(r, \theta, \infty) = \left(4\,\frac{r_o^2}{r^2} - 3\,\frac{r_o^4}{r^4}\right) S_o \cos 2\theta \tag{7.642}$$

In Fig. 7.45 we present the normalized tangential stress versus radial distance along the $\theta = 90°$ direction at various times. The dashed line represents both the small time and large time asymptotic solution. First we observe that at $t = 0^+$, a compressive hoop stress is immediately developed, with the magnitude of $\sigma_{\theta\theta} = -3S_o$ at the borehole wall, and the material is undrained. As time progresses, the generated pore pressure starts to dissipate due to the drained boundary condition. Near the borehole wall, the material effectively softens, and the comparatively harder material inside tends to shield the near borehole region, and reducesits stress

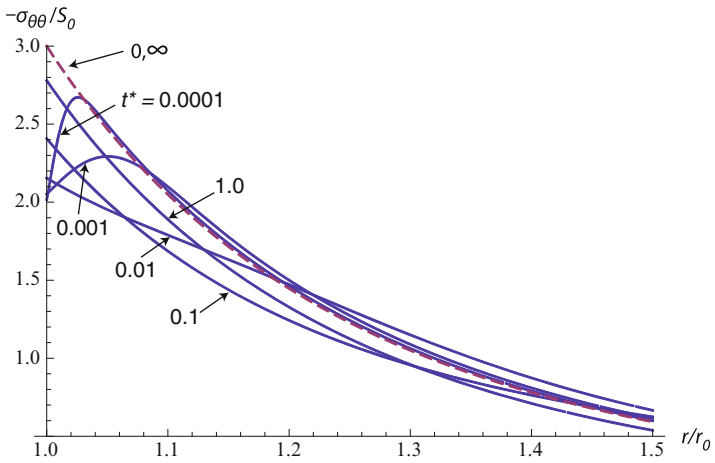


Fig. 7.45 Normalized compressive tangential stress $-\sigma_{\theta\theta}/S_o$ versus dimensionless radial distance r/r_o in the $\theta = 90°$ direction, at various times, for the mode 3 borehole problem, with $\nu = 0.2$, $\nu_u = 0.4$, and $B = 0.8$

concentration. In fact, the stress at the borehole wall, $r = r_o$, is instantaneously reduced to

$$\sigma_{\theta\theta}(r_o, \theta, 0^+) = -\frac{3 + \nu - 4\nu_u}{1 - \nu} S_o \cos 2\theta \tag{7.643}$$

As a result, at very small times ($t^* \approx 10^{-3}$ and smaller), the peak of tangential stress is located inside the rock and not at the wall of the borehole, as predicted by the elastic analysis. At larger times, as the pore pressure is more uniformly drained, the tangential stress monotonically decreases toward the inside of the formation, but increases with time. Finally, the stress is restored to its original elastic value given by (7.641).

It is of interest to observe that the above described phenomenon takes place in a small region near the borehole, as the horizontal scale of Fig. 7.45 is only half of the borehole radius, and the region with most stress changes is within a distance of $0.1\, r_o$ from the wall. In contrast to $\sigma_{\theta\theta}$, the radial stress component σ_{rr} (not plotted) experiences little variation. The poroelastic mechanism is thus limited to a skin region. As the failure of an excavated borehole typically initiates from the near wall region, this stress evolution is critical in the time-dependent failure mechanism of a borehole, as will be discussed in the next section.

Finally, the history of the normalized radial displacement, $2Gu_r/S_o r_o$, at the borehole wall $r = r_o$, in the $\theta = 90°$ direction, is plotted as Fig. 7.46. The radial displacement varies from small time to large time with these limits

$$u_r(r_o, 90°, 0^+) = \frac{(3 - 4\nu_u) r_o S_o}{2G}, \quad u_r(r_o, 90°, \infty) = \frac{(3 - 4\nu) r_o S_o}{2G} \tag{7.644}$$

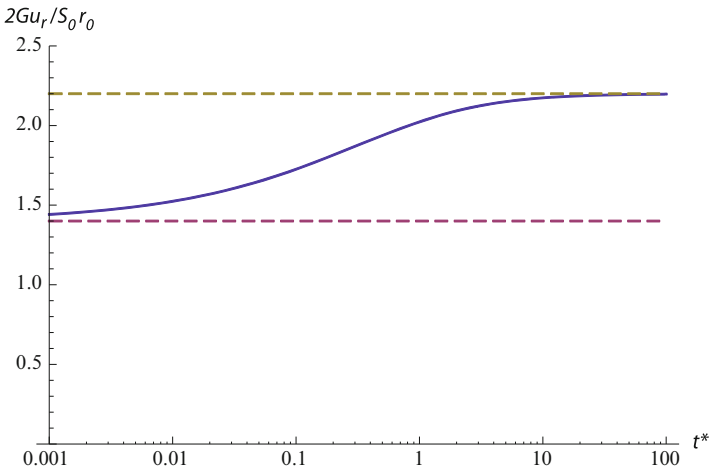


Fig. 7.46 Normalized radial displacement at borehole wall ($r = r_o$), along the $\theta = 90°$ direction, versus dimensionless time, for the mode 3 borehole problem, with $\nu = 0.2$, $\nu_u = 0.4$, and $B = 0.8$. *Dashed lines*: asymptotic solution at $t = 0$ and $t \to \infty$

These are marked in dashed lines in Fig. 7.46. Since the radial displacement at the boundary is proportional to $\cos 2\theta$, the deformed borehole is elliptic in shape. Indeed, the measured *borehole ellipticity* is a way to interpret the directions of the in situ principal stresses and the magnitude of the deviatoric stress. The poroelastic effect makes the ellipticity a time-dependent phenomenon. In the case of the evacuation of a borehole, the existence of a far-field stress deviator produces a time-dependent opening in the direction of the minimum compressive stress and a progressive closure in the perpendicular direction. The phenomenon of progressive opening was noted by Carter and Booker [33], and called it a 'reverse consolidation'.

Mode 4: Constant Flux As an alternative to the mode 2 problem of constant pressure, we can examine the problem of constant flux at borehole wall, with the boundary conditions

$$\sigma_{rr} = 0; \quad \sigma_{r\theta} = 0; \quad q_r = -\kappa \frac{\partial p}{\partial r} = q_o; \quad \text{at } r = r_o \tag{7.645}$$

It is easy to show that the solution of (7.597) in the Laplace transform domain, subject to the last of the above boundary conditions, is

$$\frac{\kappa s \tilde{p}^{(4)}}{q_o r_o} = \frac{\mathrm{K}_0(r^* \sqrt{s^*})}{\sqrt{s^*}\, \mathrm{K}_1(\sqrt{s^*})} \tag{7.646}$$

The above can be inverted to the time domain to give [32]

$$\frac{\kappa p^{(4)}}{q_o r_o} = -\frac{2}{\pi}\int_0^\infty \left(1 - e^{-cu^2 t}\right) \frac{\mathrm{J}_0(ur)\mathrm{Y}_1(ur_o) - \mathrm{Y}_0(ur)\mathrm{J}_1(ur_o)}{u^2\left[\mathrm{J}_1^2(ur_o) + \mathrm{Y}_1^2(ur_o)\right]}\, du \tag{7.647}$$

The remainder of the solutions are found from (7.593), (7.594), and (7.595) as

$$\frac{2\kappa G s\tilde{u}_r^{(4)}}{q_o r_o^2} = -\frac{2\eta}{s^*}\left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\mathrm{K}_1(\sqrt{s^*})} - \frac{1}{r^*}\right] \tag{7.648}$$

$$\frac{\kappa s\tilde{\sigma}_{rr}^{(4)}}{q_o r_0} = \frac{2\eta}{r^* s^*}\left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\mathrm{K}_1(\sqrt{s^*})} - \frac{1}{r^*}\right] \tag{7.649}$$

$$\frac{\kappa s\tilde{\sigma}_{\theta\theta}^{(4)}}{q_o r_0} = -2\eta\left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{r^* s^*\,\mathrm{K}_1(\sqrt{s^*})} + \frac{\mathrm{K}_0(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} - \frac{1}{r^{*2} s^*}\right] \tag{7.650}$$

$$\frac{s\tilde{q}_r^{(4)}}{q_o} = \frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\mathrm{K}_1(\sqrt{s^*})} \tag{7.651}$$

Numerical inverse Laplace transform (Sect. D.3) can be used to evaluate the above expressions.

In Fig. 7.47 the normalized pore pressure $\kappa p/q_o r_o$ is plotted versus the dimensionless radial distance r/r_o at various dimensionless times $t^* = ct/r_o^2$. We observe that the pore pressure progressively increases with time. At borehole wall, $r = r_o$, the pressure increases at the rate

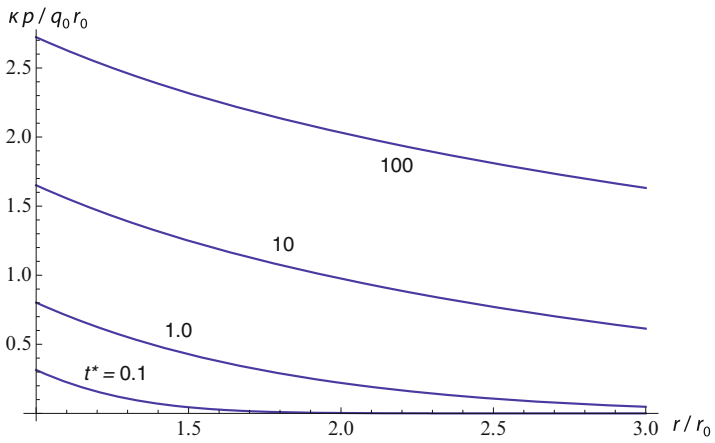


Fig. 7.47 Normalized pore pressure versus radial distance for mode 4 borehole problem

$$p(r_o, t) \approx \frac{2\sqrt{c}\, q_o}{\sqrt{\pi}\kappa} \sqrt{t} - \frac{c\, q_o}{2\kappa r_o} t \tag{7.652}$$

at small time, $t \to 0$. At large time, $t \to \infty$, the pressure increases without bound

$$p(r_o, t) \approx \frac{q_o r_o}{2\kappa} \left(\ln 4 - \gamma + \ln \frac{c\, t}{r_o^2} \right) \tag{7.653}$$

where $\gamma = 0.5772\ldots$ is the Euler constant. The pressure at the borehole wall is the pressure that needs to be maintained in the borehole for an injection at that rate, or the "drawdown" needed in a well when fluid is extracted at that rate.

7.14.2 Inclined Borehole Problem

Deep geological formations largely have one of its principal stress axes aligned with the vertical direction (direction of gravity), though not necessarily so. As discussed in the preceding section, when a vertical borehole is drilled in such a formation, the borehole axis coincides with a principal stress axis, and the problem can be modeled as a plane strain problem. However, when a borehole is drilled in an angle inclined to the vertical direction, or the formation principal stress axes are misaligned with the gravity direction, we have an *inclined borehole problem*. In fact, with the present-day drilling technology, *directional drilling* is widely employed to exploit the often horizontal pay formation that contains oil, gas, coal, or other mineral resources. In the inclined borehole situation, when the borehole has a long straight section (say, 100 times the borehole diameter), and the material properties and the loading conditions do not vary in the direction of borehole axis, then the problem can be modeled as a generalized plane strain problem, as presented in Sect. 7.5. The solution of an inclined borehole in an elastic formation has been investigated by Bradley [31]. The extension of the solution to poroelastic medium was given by Cui et al. [51], which is presented below.

Figure 7.48 gives a definition sketch of the inclined borehole problem. In the figure, the Cartesian coordinate system $x'y'z'$ is chosen to coincide with the principal axes of the in situ compressive stresses, respectively designated as $S_{x'}$, $S_{y'}$ and $S_{z'}$. The virgin pore pressure is given by p_o. The borehole is defined by a local coordinate system xyz with the borehole axis coinciding with the z-axis. The local system is formed by a rotation of an azimuth angle $\varphi_{z'}$ about the z'-axis, and then by an inclination of a zenith angle φ_y toward the x-axis. See Fig. 7.48a for illustration.

For the convenience of solution presentation, the local (borehole) coordinate system xyz is chosen to display the solution domain, as shown in Fig. 7.48b. We note that the cube in the figure is drawn only for the easy presentation of far-field stresses, as the domain actually extends to infinity. The in situ stresses can be converted to the local coordinate system via the transformation [68]

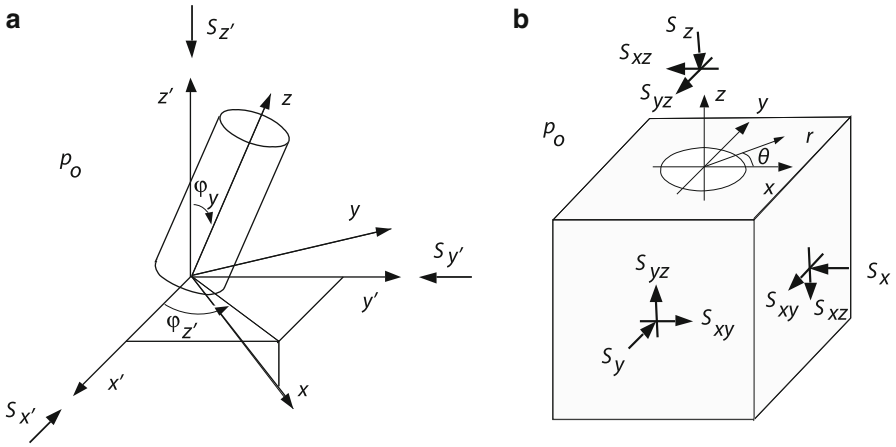


Fig. 7.48 Definition sketch of an inclined borehole problem

$$\begin{Bmatrix} S_x \\ S_y \\ S_z \\ S_{xy} \\ S_{yz} \\ S_{xz} \end{Bmatrix} = \begin{bmatrix} \ell^2_{xx'} & \ell^2_{xy'} & \ell^2_{xz'} \\ \ell^2_{yx'} & \ell^2_{yy'} & \ell^2_{yz'} \\ \ell^2_{zx'} & \ell^2_{zy'} & \ell^2_{zz'} \\ \ell_{xx'}\ell_{yx'} & \ell_{xy'}\ell_{yy'} & \ell_{xz'}\ell_{yz'} \\ \ell_{yx'}\ell_{zx'} & \ell_{yy'}\ell_{zy'} & \ell_{yz'}\ell_{zz'} \\ \ell_{zx'}\ell_{xx'} & \ell_{zy'}\ell_{xy'} & \ell_{zz'}\ell_{xz'} \end{bmatrix} \begin{Bmatrix} S_{x'} \\ S_{y'} \\ S_{z'} \end{Bmatrix} \tag{7.654}$$

where

$$\begin{bmatrix} \ell_{xx'} & \ell_{xy'} & \ell_{xz'} \\ \ell_{yx'} & \ell_{yy'} & \ell_{yz'} \\ \ell_{zx'} & \ell_{zy'} & \ell_{zz'} \end{bmatrix} = \begin{bmatrix} \cos\varphi_{z'}\cos\varphi_y & \sin\varphi_{z'}\cos\varphi_y & -\sin\varphi_y \\ -\sin\varphi_{z'} & \cos\varphi_{z'} & 0 \\ \cos\varphi_{z'}\sin\varphi_y & \sin\varphi_{z'}\sin\varphi_y & \cos\varphi_y \end{bmatrix} \tag{7.655}$$

The conditions at the far field, $r \to \infty$, become

$$\begin{aligned} &\sigma_{xx} = -S_x, \quad \sigma_{yy} = -S_y, \quad \sigma_{zz} = -S_z, \\ &\sigma_{xy} = -S_{xy}, \quad \sigma_{yz} = -S_{yz}, \quad \sigma_{xz} = -S_{xz}, \quad p = p_o \end{aligned} \tag{7.656}$$

At the borehole wall, $r = r_o$, surface tractions and pore pressure are assumed to vanish at the instant of excavation

$$\begin{aligned} &\sigma_{rr} = -S_r\,\mathrm{H}(-t), \quad \sigma_{r\theta} = -S_{r\theta}\,\mathrm{H}(-t), \\ &\sigma_{rz} = -S_{rz}\,\mathrm{H}(-t), \quad p = p_o\,\mathrm{H}(-t) \end{aligned} \tag{7.657}$$

In the above, H denotes the Heaviside unit step function, and $\mathrm{H}(-t)$ is defined as $\mathrm{H}(-t) = 1 - \mathrm{H}(t)$ such that $\mathrm{H}(-t) = 1$ for $t \leq 0$, and $\mathrm{H}(-t) = 0$ for $t > 0$. We also note that S_r, $S_{r\theta}$, and S_{rz} are the far-field compressive stress components in cylindrical coordinates:

$$S_r = \frac{S_x + S_y}{2} + \frac{S_x - S_y}{2} \cos 2\theta + S_{xy} \sin 2\theta \tag{7.658}$$

$$S_{r\theta} = S_{xy} \cos 2\theta - \frac{S_x - S_y}{2} \sin 2\theta \tag{7.659}$$

$$S_{rz} = S_{xz} \cos \theta + S_{yz} \sin \theta \tag{7.660}$$

Similar to the plane strain borehole and the cylinder problem, the best way to solve the above problem is to decompose the boundary and far-field conditions (7.656) and (7.657) into different modes that can be superposed to create solutions of different problems. In the following, we shall define three problems (see Fig. 7.49 for illustration):

Problem I: In the far-field ($r \to \infty$)

$$\begin{aligned} &\sigma_{xx} = -S_x, \quad \sigma_{yy} = -S_y, \\ &\sigma_{zz} = -\nu(S_x + S_y) - \alpha(1 - 2\nu)p_o = -S_z^{(1)}, \\ &\sigma_{xy} = -S_{xy}, \quad \sigma_{yz} = \sigma_{xz} = 0, \quad p = p_o \end{aligned} \tag{7.661}$$

At the borehole wall ($r = r_o$)

$$\begin{aligned} &\sigma_{rr} = -S_r \mathrm{H}(-t), \quad \sigma_{r\theta} = -S_{r\theta} \mathrm{H}(-t), \\ &\sigma_{rz} = 0, \quad p = p_o \mathrm{H}(-t) \end{aligned} \tag{7.662}$$

Problem II: In the far-field ($r \to \infty$)

$$\begin{aligned} &\sigma_{zz} = -S_z + \left[\nu(S_x + S_y) + \alpha(1 - 2\nu)p_o\right] = -S_z^{(2)}, \\ &\sigma_{xx} = \sigma_{yy} = \sigma_{xy} = \sigma_{yz} = \sigma_{xz} = p = 0 \end{aligned} \tag{7.663}$$

At the borehole wall ($r = r_o$)

$$\sigma_{rr} = \sigma_{r\theta} = \sigma_{rz} = p = 0 \tag{7.664}$$

Problem III: In the far-field ($r \to \infty$)

$$\begin{aligned} &\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = p = 0, \\ &\sigma_{yz} = -S_{yz}, \quad \sigma_{xz} = -S_{xz} \end{aligned} \tag{7.665}$$

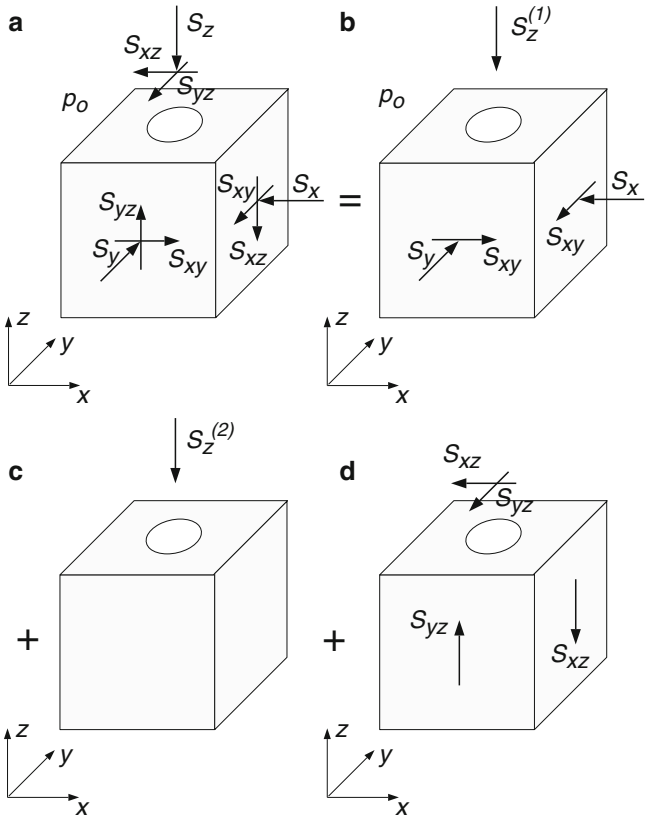


Fig. 7.49 Decomposition of inclined borehole problem, where $S_z = S_z^{(1)} + S_z^{(2)}$

At the borehole wall ($r = r_o$)

$$\sigma_{rr} = \sigma_{r\theta} = p = 0, \quad \sigma_{rz} = -S_{rz}\mathrm{H}(-t) \tag{7.666}$$

We note in the above the decomposition of $S_z^{(1)}$ and $S_z^{(2)}$ such that

$$S_z = S_z^{(1)} + S_z^{(2)} \tag{7.667}$$

By summing the above far-field and boundary conditions (7.661)–(7.666), we find that (7.656) and (7.657) are reproduced.

Problem I: Poroelastic Plane-Strain Problem Problem I is designed to be a poroelastic plane strain problem, namely that the out-of-plane displacement u_z and flux q_z vanish. We note that in a plane strain problem, the far-field conditions of σ_{zz}, σ_{yz}, and σ_{xz} in (7.661), and the boundary condition of σ_{rz} in (7.662) are exactly

the results of a plane strain condition, as given in (7.83) and (7.106); hence are redundant, but admissible.

As the plane strain borehole problem has been solved in Sect. 7.14.1, we shall utilize these solutions by converting the present problem to the standard form of mode 1 through 3 as presented in Sect. 7.14.1. To match up the solutions, we first perform a rotation of coordinates about the z-axis such that the x''–axis coincides with the minimum horizontal principal stress and y'' with the maximum. The angle of rotation needed, measured between the x and the x'' axes, is given by

$$\theta_r = \frac{1}{2}\tan^{-1}\frac{2S_{xy}}{S_x - S_y} \tag{7.668}$$

The polar angle under the new coordinate system is then

$$\theta'' = \theta - \theta_r \tag{7.669}$$

The far-field conditions become: at $r \to \infty$

$$\sigma_{x''x''} = -(P_o - S_o), \quad \sigma_{y''y''} = -(P_o + S_o), \quad \sigma_{x''y''} = 0, \quad p = p_o \tag{7.670}$$

where P_o is the mean compressive stress and S_o the deviatoric stress given by

$$P_o = \frac{S_x + S_y}{2} \tag{7.671}$$

$$S_o = \sqrt{\left(\frac{S_x - S_y}{2}\right)^2 + S_{xy}^2} \tag{7.672}$$

At the borehole wall, $r = r_o$, all surface tractions and the pore pressure vanish.

We observe in the above that we have transformed the in situ stress data from the global (formation) coordinate system $x'y'z'$ to the local (borehole) coordinate system xyz, and then aligned it with the local planar principal axes system $x''y''$. Equation (7.670) is now of the same form as (7.587) and (7.588) in the plane strain problem. Solutions of Problem I can then be obtained as the mode 1, 2, and 3 solutions presented in Sect. 7.14.1.

With the above decomposition and coordinate transformation, the solution of the first problem can be obtained as:

$$u_r^{(\mathrm{I})} = -u_r^{(1)} - u_r^{(2)} - u_r^{(3)} \tag{7.673}$$

$$u_\theta^{(\mathrm{I})} = -u_\theta^{(3)} \tag{7.674}$$

$$p^{(\mathrm{I})} = p_o - p^{(2)} - p^{(3)} \tag{7.675}$$

$$\sigma_{rr}^{(\mathrm{I})} = -P_o + S_o \cos 2(\theta - \theta_r) - \sigma_{rr}^{(1)} - \sigma_{rr}^{(2)} - \sigma_{rr}^{(3)} \tag{7.676}$$

$$\sigma_{\theta\theta}^{(\mathrm{I})} = -P_o - S_o \cos 2(\theta - \theta_r) - \sigma_{\theta\theta}^{(1)} - \sigma_{\theta\theta}^{(2)} - \sigma_{\theta\theta}^{(3)} \tag{7.677}$$

$$\sigma_{r\theta}^{(\mathrm{I})} = -S_o \sin 2(\theta - \theta_r) - \sigma_{r\theta}^{(3)} \tag{7.678}$$

$$\sigma_{rz}^{(\mathrm{I})} = \sigma_{\theta z}^{(\mathrm{I})} = 0 \tag{7.679}$$

$$\sigma_{zz}^{(\mathrm{I})} = \nu \left(\sigma_{rr}^{(\mathrm{I})} + \sigma_{\theta\theta}^{(\mathrm{I})} \right) - \alpha(1 - 2\nu) p^{(\mathrm{I})} \tag{7.680}$$

where the superscript $^{(\mathrm{I})}$ denotes the solution for problem I, and the superscripts $^{(1)}$, $^{(2)}$, and $^{(3)}$ represent the solution of the three fundamental loading modes of a plane strain borehole problem, as presented in Sect. 7.14.1. Particularly, the mode 1 solution is given by (7.599)–(7.602), and mode 2 by (7.608)–(7.613). The mode 3 solution needs to be aligned with the principle stress axes in the x–y plane, and is give as

$$\tilde{u}_r^{(3)} = \tilde{\mathbb{U}}_r^{(3)} \cos 2(\theta - \theta_r) \tag{7.681}$$

$$\tilde{u}_\theta^{(3)} = \tilde{\mathbb{U}}_\theta^{(3)} \sin 2(\theta - \theta_r) \tag{7.682}$$

$$\tilde{p}^{(3)} = \tilde{\mathbb{P}}^{(3)} \cos 2(\theta - \theta_r) \tag{7.683}$$

$$\tilde{\sigma}_{rr}^{(3)} = \tilde{\mathbb{S}}_{rr}^{(3)} \cos 2(\theta - \theta_r) \tag{7.684}$$

$$\tilde{\sigma}_{\theta\theta}^{(3)} = \tilde{\mathbb{S}}_{\theta\theta}^{(3)} \cos 2(\theta - \theta_r) \tag{7.685}$$

$$\tilde{\sigma}_{r\theta}^{(3)} = \tilde{\mathbb{S}}_{r\theta}^{(3)} \sin 2(\theta - \theta_r) \tag{7.686}$$

with $\tilde{\mathbb{U}}_r^{(3)}$, $\tilde{\mathbb{U}}_\theta^{(3)}$, etc., given in (7.631)–(7.636).

Problem II: Elastic Uniaxial Stress Problem For the second problem, we observe from (7.663) and (7.664) that the stress condition is uniaxial in the borehole axis direction. We also notice that σ_{zz} has a part that is aimed at compensating for the σ_{zz} generated by the plane strain solution in Problem I, given as (7.661). The set of stresses and pore pressure defined in (7.663) satisfies the equilibrium equation as well as the stress compatibility equation (6.82); hence is an admissible poroelastic solution. The initial stress state is simply

$$\sigma_{zz}^{(\mathrm{II})} = -S_z + \left[\nu(S_x + S_y) + \alpha(1 - 2\nu) p_o \right] \tag{7.687}$$

$$\sigma_{xx}^{(\mathrm{II})} = \sigma_{yy}^{(\mathrm{II})} = \sigma_{xy}^{(\mathrm{II})} = \sigma_{yz}^{(\mathrm{II})} = \sigma_{xz}^{(\mathrm{II})} = p^{(\mathrm{II})} = 0 \tag{7.688}$$

everywhere. The excavation of the borehole then introduces the boundary conditions as given by (7.664). These conditions are identical with the initial stress state given as (7.688); hence no perturbation is generated. The solution of Problem II is then (7.687) and (7.688), which is independent of time.

Problem III: Elastic Anti-plane Shear Problem For the third problem, we notice that a disturbance is introduced by the boundary condition (7.666) as a sudden change of shear stress σ_{rz} from the value $-S_{rz}$ to zero at the borehole wall (as defined by the Heaviside function $\mathrm{H}(-t)$). This anti-plane shear disturbance does not create any mean normal stress, hence no pore pressure will be generated by the Skempton effect. The solution is the same as the elastic one. Following Bradley [29], the solution for Problem III can be obtained as

$$\sigma_{rz}^{(\mathrm{III})} = -(S_{xz}\cos\theta + S_{yz}\sin\theta)\left[1 - \mathrm{H}(t)\frac{r_o^2}{r^2}\right] \tag{7.689}$$

$$\sigma_{\theta z}^{(\mathrm{III})} = (S_{xz}\sin\theta - S_{yz}\cos\theta)\left[1 + \mathrm{H}(t)\frac{r_o^2}{r^2}\right] \tag{7.690}$$

$$\sigma_{rr}^{(\mathrm{III})} = \sigma_{\theta\theta}^{(\mathrm{III})} = \sigma_{zz}^{(\mathrm{III})} = \sigma_{r\theta}^{(\mathrm{III})} = p^{(\mathrm{III})} = 0 \tag{7.691}$$

To obtain the full solution of the inclined borehole problem, the solutions developed in Problems I, II and III can be added together with the proper weight by the loading conditions and the background (far field) stresses. These will be demonstrated in Sect. 7.15.5 for application problems.

7.15 Borehole and Cylinder Application Problems

In this section we shall assemble the basic solutions derived in Sects. 7.13 and 7.14, presented in terms of different loading modes, to obtain solutions for various applications involving cylinder and borehole problems.

7.15.1 Retrieval of Cylindrical Core

Consider a formation with a non-hydrostatic in situ stress field, shown as the background stress in Fig. 7.40, and expressed as (7.587)–(7.592). A long, circular cylindrical core is instantly extracted from the formation, perpendicular to the plane. This initial and boundary value problem can be described as follows: at $t < 0$ and within $r_o \geq r \geq 0$, the following stress and pore pressure conditions exist

$$\sigma_{rr} = -P_o + S_o\cos 2\theta \tag{7.692}$$

$$\sigma_{\theta\theta} = -P_o - S_o\cos 2\theta \tag{7.693}$$

$$\sigma_{r\theta} = -S_o\sin 2\theta \tag{7.694}$$

$$p = p_o \tag{7.695}$$

And at $t \geq 0$ and $r = r_o$, the following boundary condition is imposed:

$$\sigma_{rr} = \sigma_{r\theta} = p = 0 \tag{7.696}$$

For the convenience of solution, rather than directly solving the above problem, we shall define a new set of stresses and pore pressure, such that they are the difference between the true values and the initial values given as (7.692)–(7.695). With this set of new variables, marked with a † superscript, the initial and boundary value problem is modified to the following: at $t < 0$ and $r_o \geq r \geq 0$,

$$\sigma^\dagger_{rr} = \sigma^\dagger_{\theta\theta} = \sigma^\dagger_{r\theta} = p^\dagger = 0 \tag{7.697}$$

and at $t \geq 0$ and $r = r_o$,

$$\sigma^\dagger_{rr} = P_o - S_o \cos 2\theta \tag{7.698}$$

$$\sigma^\dagger_{r\theta} = S_o \sin 2\theta \tag{7.699}$$

$$p^\dagger = -p_o \tag{7.700}$$

We recognize that the solution of the above problem can be obtained as the negative of the summation of the mode 1, 2, and 3 solutions as derived in Sect. 7.13.1. To restore to the original variables, we can add back the in situ stresses and pore pressure; hence the final solution is obtained as

$$\sigma_{rr} = -\sigma^{(1)}_{rr} - \sigma^{(2)}_{rr} - P_o \tag{7.701}$$

$$\sigma_{\theta\theta} = -\sigma^{(1)}_{\theta\theta} - \sigma^{(2)}_{\theta\theta} - P_o \tag{7.702}$$

$$\sigma_{r\theta} = 0 \tag{7.703}$$

$$p = -p^{(1)} - p^{(2)} + p_o \tag{7.704}$$

$$u_r = -u^{(1)}_r - u^{(2)}_r - u^{(3)}_r \tag{7.705}$$

$$u_\theta = -u^{(3)}_\theta \tag{7.706}$$

In the above, the superscripts $^{(1)}$, $^{(2)}$, and $^{(3)}$ represent the mode 1, 2, and 3 solutions, respectively. Terms that are identically zero are omitted, and the mode 3 stress terms cancel with the in situ deviatoric stress terms. It is of interest to note that as the deviatoric stress is an elastic solution (does not generate pore pressure), it vanishes the instant the surface loading is removed. So the stress inside the cylinder after its retrieval, given as (7.701), (7.702), and (7.703), is free from the influence of the formation deviatoric stress.

In Fig. 7.50 we plot the deformed cylinder surface with the radial displacement u_r exaggerated, magnified by a factor $G/P_o r_o$, for the case of $\nu = 0.25$, $\nu_u = 0.5$, $\eta = 0.33$, $p_o/P_o = 0.4$, and $S_o/P_o = 0.2$. The result is shown only in the first quadrant because of the symmetry with respect to the x- and y-axis. We observe

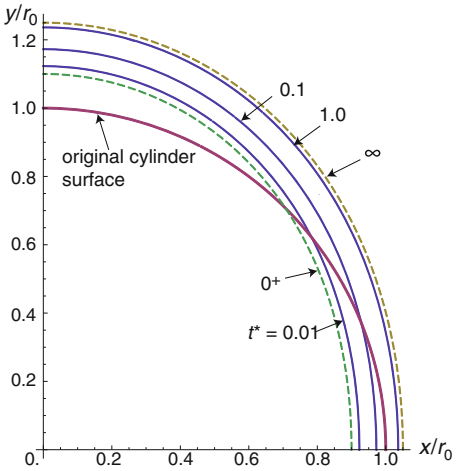


Fig. 7.50 Deformed cylinder surface at various dimensionless times, with exaggerated radial displacement

that at the instant of the removal of the surface stress, the cylinder contracts in the minimum in situ stress direction, and expands in the other, to become elliptic in shape. The pore pressure at that instant is, according to (7.704), $p = -BP_o + p_o$, where $B = 1$. As $p_o/P_o = 0.4$, the pressure inside the cylinder is negative. As time progresses, the pressure becomes equilibrated with the boundary pressure $p = 0$, and the cylinder surface rebounds radially.

7.15.2 Excavated Borehole

Consider a section of borehole instantly drilled in a non-hydrostatic stress field, as shown in Fig. 7.40. If the hole is kept empty without fluid, the boundary conditions at $r = r_o$ are

$$\sigma_{rr} = \sigma_{r\theta} = p = 0 \tag{7.707}$$

for $t \geq 0$. Similar to the retrieved cylinder problem presented in the preceding section, this problem can be solved using the proper superposition of fundamental solution modes derived in Sect. 7.14.1. Particularly, the solution of stresses and displacements are

$$\sigma_{rr} = -\sigma_{rr}^{(1)} - \sigma_{rr}^{(2)} - \sigma_{rr}^{(3)} - P_o + S_o \cos 2\theta \tag{7.708}$$

$$\sigma_{\theta\theta} = -\sigma_{\theta\theta}^{(1)} - \sigma_{\theta\theta}^{(2)} - \sigma_{\theta\theta}^{(3)} - P_o - S_o \cos 2\theta \tag{7.709}$$

$$\sigma_{r\theta} = -\sigma_{r\theta}^{(3)} - S_o \sin 2\theta \tag{7.710}$$

$$p = -p^{(2)} - p^{(3)} + p_o \tag{7.711}$$

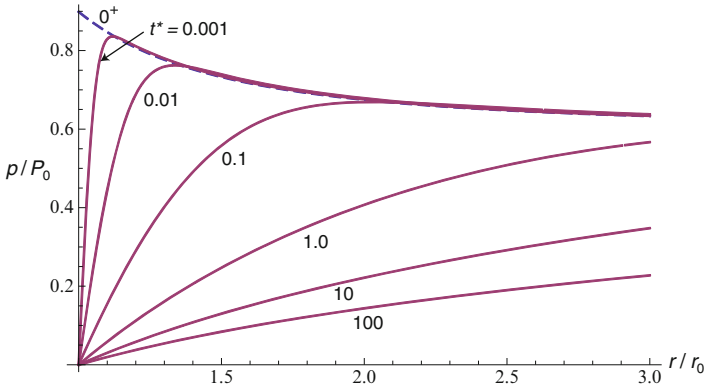


Fig. 7.51 Formation pore pressure history due to an instantly excavated borehole, along $\theta = 0°$

$$u_r = -u_r^{(1)} - u_r^{(2)} - u_r^{(3)} \tag{7.712}$$

$$u_\theta = -u_\theta^{(3)} \tag{7.713}$$

In the above, the superscripts $^{(1)}$, $^{(2)}$, and $^{(3)}$ represent the mode 1, 2, and 3 solutions, respectively; and again, we have omitted terms that are identically zero.

In the following examples, we shall use the set of material constants and in situ stress conditions for the numerical results: $B = 0.8$, $\nu = 0.2$, $\nu_u = 0.4$, which also lead to $\alpha = 0.893$ and $\eta = 0.335$, a stress deviator $S_o = 0.2P_o$, and a virgin pore pressure $p_o = 0.6P_o$.

Pressure Response Figure 7.51 presents the pore pressure history along the $\theta = 0°$ direction. At $t = 0^+$, the instantaneous pore pressure response is

$$p(r, 0°, 0^+) = \frac{4B(1+\nu_u)}{3}\frac{r_o^2}{r^2}S_o + p_o \tag{7.714}$$

as shown in dashed line in Fig. 7.51. It then progressively dissipates due to the boundary condition $p = 0$ at $r = r_o$.

Effective Radial Stress and Tensile Failure For radial stress σ_{rr}, we find the following small time and large time asymptotic stresses in the 0° direction

$$\sigma_{rr}(r, 0°, 0^+) = -P_o\left(1 - \frac{r_o^2}{r^2}\right) + S_o\left(1 - 4\frac{r_o^2}{r^2} + 3\frac{r_o^4}{r^4}\right) \tag{7.715}$$

$$\sigma_{rr}(r, 0°, \infty) = -P_o\left(1 - \frac{r_o^2}{r^2}\right) + S_o\left(1 - 4\frac{r_o^2}{r^2} + 3\frac{r_o^4}{r^4}\right) + \eta p_o\left(1 - \frac{r_o^2}{r^2}\right) \tag{7.716}$$

It is easy to see that these are compressive stresses (negative), but must drop to zero at the borehole wall, $r = r_o$.

It is, however, more important to look at the effective stress, particularly the Terzaghi effective stress, $\sigma'_{rr} = \sigma_{rr} + p$, than the stress itself, as the former is associated with the material failure. From (7.715) we observe that at a small distance, $\Delta r^* = (r - r_o)/r_o$, inside the wall, the radial stress at small time is approximately

$$\sigma_{rr}(r, 0^\circ, t) \approx -(2P_o + 4S_o)\Delta r^* \tag{7.717}$$

which is compressive. However, the pore pressure generated at $t = 0^+$, as given by (7.714), hardly has time to dissipate

$$p(r, 0^\circ, t) \approx \frac{4B(1 + \nu_u)}{3} S_o + p_o \tag{7.718}$$

for small time and small Δr^*. Hence, before the pore pressure in the near wall region, $r_o < r < 2r_o$, is significantly dissipated, that region is generally under tension, in terms of effective stress.

In Fig. 7.52 we plot the Terzaghi effective radial stress σ'_{rr} along the $\theta = 0^\circ$ direction as function of radial distance, at various times. We clearly observe a tensile stress region within one borehole radius at small times. As rocks generally have much small tensile strength than compressive strength, such region is prone to tensile failure.

The tendency for tensile failure in the near wall region can in fact be enhanced in an industry technique called coal cavitation, as discussed in Sect. 1.3.7. To enhance the pore pressure, a borehole section can be sealed off and pressurized by air. Time

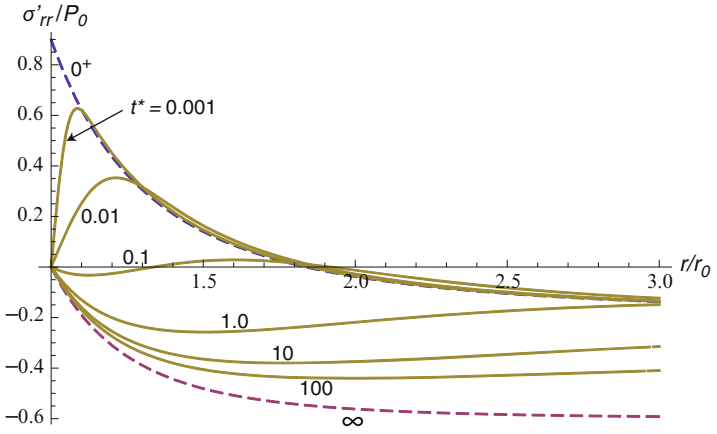


Fig. 7.52 Terzaghi effective radial stress in the near wall region, along $\theta = 0^\circ$ direction, due to an instantly excavated borehole

is allowed for the pressure to permeate the region around the borehole. Then the air pressure is suddenly vented and the radial stress on the wall is instantly reduced to zero. The effect is equivalent to the evacuation of a borehole. The artificially created large pore pressures works in favor of creating a large tensile stress by the effective stress principle. The coal then sloughs off the wall to create a large underground cavity for in situ coal gasification.

Shear Failure Next we examine the potential for shear failure for an instantly excavated borehole. While there exist a number of sophisticated failure models, here we shall employ a simple one, the Mohr-Coulomb failure criterion, which can be expressed as

$$S_{\max} = P_{\text{ave}} \sin\phi' + c' \cos\phi' \tag{7.719}$$

where $S_{\max}$ is the in-plane maximum shear stress,

$$S_{\max} = \frac{1}{2}\sqrt{(\sigma_{rr} - \sigma_{\theta\theta})^2 + 4\,\sigma_{r\theta}^2} \tag{7.720}$$

P_{ave} is the average in-plane compressive stress

$$P_{\text{ave}} = -\frac{1}{2}(\sigma_{rr} + \sigma_{\theta\theta}) \tag{7.721}$$

c' is *cohesion*, and ϕ' is the *angle of internal friction*. For poroelastic material, the Terzaghi effective average compressive stress

$$P'_{\text{ave}} = P_{\text{ave}} - p \tag{7.722}$$

should be used in place of P_{ave} in (7.719).

In Fig. 7.53 we present two sets of parametric plots of $S_{\max}$ versus P'_{ave} along $\theta = 0°$ direction. In the first set, the family of solid lines represents the parametric plot of $S_{\max}$ versus P'_{ave} as function of r^*, at various t^*, plotted for the range $1 \le r^* \le 100$. We also present in short dashed lines the limiting curves corresponding to $t = 0^+$ and ∞. For the second set of curves, we plot in long dashed lines the parametric plot of $S_{\max}$ versus P'_{ave} as function of t^*, at various r^* values, for $0.0001 \le t^* \le 10$. By interpolating between these two sets of curves, $S_{\max}$ and P'_{ave} can be determined for any r^* and t^* values.

Before analyzing these curves, it is of interest to examine the elastic solution. For an instantly excavated borehole in a non-hydrostatic stress field as shown in Fig. 7.40, the elastic solution is given by

$$\sigma_{rr} = -\left(1 - \frac{r_o^2}{r^{*2}}\right) P_o + \left(1 - 4\frac{r_o^2}{r^{*2}} + 3\frac{r_o^4}{r^{*4}}\right) S_o \cos 2\theta \tag{7.723}$$

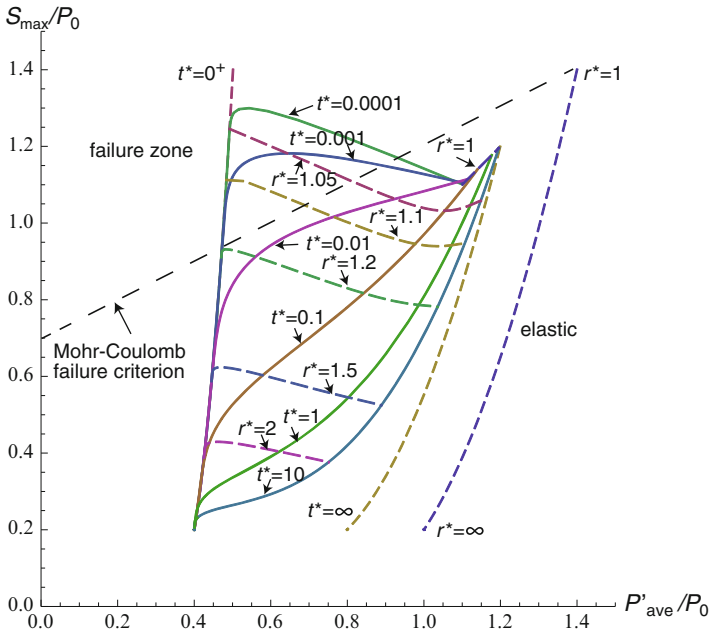


Fig. 7.53 Parametric plot of maximum shear stress $S_{\max}$ versus Terzaghi effective average compressive stress P'_{ave}, as function of r^* (*solid lines*), for $1 \leq r^* \leq 100$, and as function of t^* (*long dashed lines*), for $0.0001 < t^* < 10$, along the $\theta = 0°$ direction, at various times, together with the Mohr-Coulomb failure criterion and elastic solution

$$\sigma_{\theta\theta} = -\left(1 + \frac{r_o^2}{r^{*2}}\right) P_o - \left(1 + 3\frac{r_o^4}{r^{*4}}\right) S_o \cos 2\theta \tag{7.724}$$

$$\sigma_{r\theta} = -\left(1 + 2\frac{r_o^2}{r^{*2}} - 3\frac{r_o^4}{r^{*4}}\right) S_o \sin 2\theta \tag{7.725}$$

The parametric plot of $S_{\max}$ versus P_{ave} for the elasticity solution is also presented in Fig. 7.53. We observe that the elastic solution is different from the large time poroelastic solution ($t \to \infty$), although when the pore pressure is dissipated, $p = 0$, the poroelastic solution reduces to an elastic solution. The reason is that the draining of a uniform pore pressure $p = p_o$ in an infinite domain relieves the formation of a poroelastic stress of the magnitude $P_{\text{ave}} = \eta p_o$ (similar to the thermoelastic stress, see Sect. 1.2.7 and (1.57)); hence the curve is shifted by that amount from the purely elastic solution. The elastic solution shows that at the borehole wall, $r = r_o$, the ratio $S_{\max}/P_{\text{ave}}$ attains its maximum value of 1; hence the borehole is most prone to shear failure at the wall.

The poroelastic solution, however, is more complex. As observe in Fig. 7.53, the relation between $S_{\max}$ and P'_{ave} is not only time dependent, but also non-monotonic. To examine its consequence on the Mohr-Coulomb shear failure, we

shall assume a material with an angle of internal friction $\phi' = 30°$ (which gives a 1:2 slope in Fig. 7.53). We also assume that the combination of in situ stresses and the rock cohesion is such that based on the elastic solution, the borehole is at impending failure at the wall, but is just safe. This Mohr-Coulomb criterion is plotted in Fig. 7.53 as the dashed line intersecting the top part of elastic solution. Any pair of $(P'_{\text{ave}}, S_{\max})$ values that is located above the line indicates failure. From the figure we observe that the elastic solution is non-conservation and the poroelastic solution indicates failure during small times $t^* < 0.01$ within a skin layer of the wall, $1 < r/r_o < 1.2$. As pointed out by Detournay and Cheng [62], this poroelastic failure mechanism provides a potential explanation for the time-dependent borehole breakouts with finite size chips, a pervasive feature of deep wells [75, 124, 174].

7.15.3 Fluid Extraction and Injection

In the production of hydrocarbon or water from a borehole, fluid is pumped out at a given rate. For an injection operation, fluid is pumped in. Given a constant rate injection, the boundary conditions can be expressed as

$$\sigma_{rr} = -p_w; \quad \sigma_{r\theta} = 0; \quad q_r = q_o; \quad \text{at } r = r_o \tag{7.726}$$

where p_w is the transient pressure history in the wellbore that needs to be maintained in order to drive a constant flux q_o into the formation. As the pore pressure at the borehole wall must be in equilibrium with the wellbore pressure, we have

$$p_w = p(r_o, t) \tag{7.727}$$

Since this pressure is not known a priori, the boundary condition is implicit. However, utilizing the load decomposition solution technique that we have been practicing, we shall demonstrate below that this problem can be solved explicitly.

First we observe that if we set the radial stress condition in (7.726) to zero, the solution is given by the mode 4 solution in (7.646)–(7.651). The pore pressure is a part of the solution and its value at the wall $p(r_o, t)$ is found by setting $r = r_o$ in (7.646). We hence identify the wellbore pressure as

$$\tilde{p}_w = \frac{q_o r_o}{\kappa s} \frac{\mathrm{K}_0(\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} \tag{7.728}$$

Using the above to replace P_o in the radial stress boundary condition in (7.598) for mode 1 problem, we can obtain the mode 1 solution (7.599)–(7.602) in the Laplace transform domain. We note that the mode 1 solution is now function of time. As observed in (7.599)–(7.602), this mode generates neither pressure nor flux; hence

there is no interference among the boundary conditions of the two modes, and the solutions can be superposed. As a result, we obtain the solution as

$$\tilde{u}_r = \tilde{u}_r^{(1)} + \tilde{u}_r^{(4)} \tag{7.729}$$

$$\tilde{\sigma}_{rr} = \tilde{\sigma}_{rr}^{(1)} + \tilde{\sigma}_{rr}^{(4)} \tag{7.730}$$

$$\tilde{\sigma}_{\theta\theta} = \tilde{\sigma}_{\theta\theta}^{(1)} + \tilde{\sigma}_{\theta\theta}^{(4)} \tag{7.731}$$

$$\tilde{p} = \tilde{p}^{(4)} \tag{7.732}$$

$$\tilde{q}_r = \tilde{q}_r^{(4)} \tag{7.733}$$

Here we are reminded that for the mode 1 solution, p_w should replace P_o.

As an illustration, we express the radial and circumferential Terzaghi effective stress solution as the following

$$\frac{\kappa s\tilde{\sigma}'_{rr}}{q_o r_0} = \frac{2\eta}{r^* s^*}\left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{\mathrm{K}_1(\sqrt{s^*})} - \frac{1}{r^*}\right] - \frac{\mathrm{K}_0(\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})}\frac{1}{r^{*2}} + \frac{\mathrm{K}_0(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} \tag{7.734}$$

$$\frac{\kappa s\tilde{\sigma}'_{\theta\theta}}{q_o r_0} = -2\eta\left[\frac{\mathrm{K}_1(r^*\sqrt{s^*})}{r^* s^*\,\mathrm{K}_1(\sqrt{s^*})} + \frac{\mathrm{K}_0(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} - \frac{1}{r^{*2}s^*}\right] + \frac{\mathrm{K}_0(\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})}\frac{1}{r^{*2}} + \frac{\mathrm{K}_0(r^*\sqrt{s^*})}{\sqrt{s^*}\,\mathrm{K}_1(\sqrt{s^*})} \tag{7.735}$$

In Fig. 7.54 we plot the normalized Terzaghi effective radial stress versus radial distance at various times, for the case of material constant $\eta = 0.25$. We observe a non-monotonic trend, and particularly at large time σ'_{rr} becomes tensile due to the penetration of pore pressure.

At borehole wall, $r = r_o$, we find

$$\sigma'_{rr} = 0 \tag{7.736}$$

and

$$\tilde{\sigma}'_{\theta\theta} = 2(1-\eta)\tilde{p}_w \tag{7.737}$$

Hence at the borehole wall there is an effective tensile stress in the circumferential direction that is proportional to the pressure in the borehole. It increases with time as injection progresses with asymptotic behaviors at small and large times similar to that of pore pressure, given as (7.652) and (7.653).

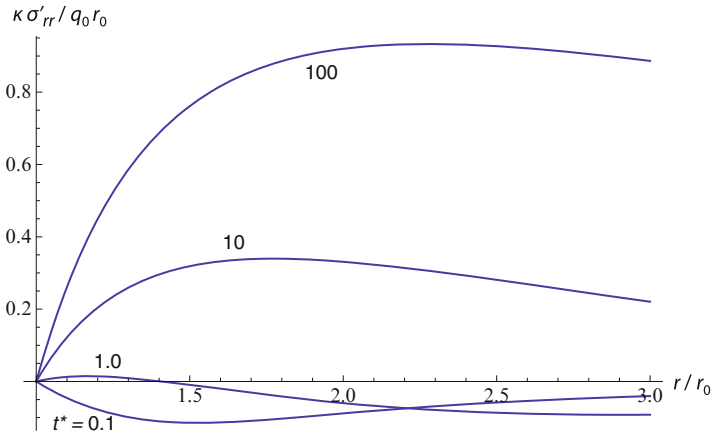


Fig. 7.54 Terzaghi effective radial stress versus radial distance for the constant rate injection problem

The above solution refer to constant rate injection. For production, that is, extraction of fluid at constant rate, the signs of the above solutions are reversed. We also note that the above solution does not take into consideration of the existence of the far-field stress and formation virgin pore pressure. The effect of these stresses can be easily superposed based on the mode 1 through 3 solutions developed in Sect. 7.14.1.

7.15.4 Borehole Breakdown Pressure

This problem has its root in the hydraulic fracturing technology used by the petroleum industry for stimulating wells. In this technique a section of a borehole is sealed-off, and fluid is gradually injected to raise the pressure in the section. When the fluid pressure in the borehole is large enough to initiate a fracture in the hydrocarbon containing formation, it causes a *breakdown*, and a sudden drop of borehole pressure due to the loss of fluid into the propagating fracture.

A rigorous study of the breakdown process should be carried out from the perspective of fracture mechanics, by analyzing the requirements for the propagation of a pre-existing flaw at the borehole [2, 26, 88, 143]. However, it is often postulated that breakdown takes place when the Terzaghi effective tensile stress at the borehole wall is equal to the tensile strength of the rock, T, i.e.

$$\sigma'_{\theta\theta} = \sigma_{\theta\theta} + p = T \tag{7.738}$$

The stress concentration induced by pressurization of the borehole, $p = p_w$, in the presence of in situ stresses, can be deduced by superposing the three modes of solutions analyzed earlier.

Let us assume that the borehole has been excavated, and is in equilibrium with the formation by filling it with a fluid that is at the same pressure as the formation virgin pore pressure p_o. To fracture the formation, the borehole pressure p_w is continuously raised, until the breakdown takes place. We shall examine the hoop stress $\sigma_{\theta\theta}$ at the borehole wall in several parts.

First, the existence of a stress-free borehole wall in the in situ stress field P_o and S_o creates a hoop stress concentration of $\sigma_{\theta\theta} = -2P_o + 4S_o$ at the borehole wall, in the $\theta = 90°$ direction, according to the mode 1 and large time mode 3 solution, (7.602) and (7.641), or the elasticity solution (7.724). Here we choose $\theta = 90°$ because it is where the minimum compressive stress exists. The initial borehole pressure $p = p_o$, however, imposes a mode 1 loading of $\sigma_{rr} = -p_o$, resulting in an increment of $\sigma_{\theta\theta}$ in the amount of p_o. As this pressure is the same as the formation pore pressure, there is no mode 2 loading. Then the raising of borehole pressure in the amount of $p_w - p_o$ generates a mode 1 as well as a mode 2 loading. Mode 1 loading gives an increment of $p_w - p_o$ to $\sigma_{\theta\theta}$, and mode 2 loading creates a (compressive) poroelastic stress of $-2\eta(p_w - p_o)$ at the wall, as predicted by (7.615). It is of interest to note that the large time mode 2 solution is instantly reached at the borehole wall where it is in contact with the borehole pressure, so this poroelastic stress is independent of the pressurization rate in the borehole. Finally, as we are interested in the Terzaghi effective stress, the pore pressure at the wall p_w needs to be added to the hoop stress. In summary, we obtain

$$\sigma'_{\theta\theta}(r_o, 90°, t) = -2P_o + 4S_o + p_o + (p_w - p_o) - 2\eta(p_w - p_o) + p_w \tag{7.739}$$

At the breakdown, $\sigma'_{\theta\theta}$ is given by (7.738), and we designate p_w as p_{bl}, indicating lower bound. Hence (7.739) can be expressed as

$$p_{bl} = \frac{T + 2P_o - 4S_o - 2\eta p_o}{2(1-\eta)} \tag{7.740}$$

This expression, first derived by Haimson and Fairhurst [76], represents the lower bound for the breakdown pressure.

Another case of interest is when the borehole fluid cannot penetrate the rock, due to the existence of a filter cake on the wall, or the practice of sleeve fracturing. In that case, through appropriate combination, the predicted breakdown pressure, p_{bu}, is

$$p_{bu} = T + 2P_o - 4S_o - 2\eta p_o - (1 - 2\eta)p_i \tag{7.741}$$

where p_i is the pore pressure at the borehole wall, which takes the value of p_o in the case of sleeve fracturing. This case gives the upper bound of breakdown pressure.

The above formulae, (7.740) and (7.741), can be used to determine the minimum formation in situ stress, if the rock tensile strength T is known, or negligible comparing to the in situ stress.

7.15.5 Borehole Stability Analysis

In this section, we shall carry out a stability analysis for an inclined borehole using a more sophisticated failure criterion than the Mohr-Coulomb criterion used in Sect. 7.15.2.

Excavated Borehole First, we shall examine a borehole drilled in a direction inclined to the principal stresses. As analyzed in Sect. 7.14.2, the solution to such problem can be obtained as the combination of solutions of three problems:

$$u_r = u_r^{(\mathrm{I})} \tag{7.742}$$

$$u_\theta = u_\theta^{(\mathrm{I})} \tag{7.743}$$

$$\sigma_{rr} = \sigma_{rr}^{(\mathrm{I})} \tag{7.744}$$

$$\sigma_{\theta\theta} = \sigma_{\theta\theta}^{(\mathrm{I})} \tag{7.745}$$

$$\sigma_{zz} = \sigma_{zz}^{(\mathrm{I})} + \sigma_{zz}^{(\mathrm{II})} \tag{7.746}$$

$$\sigma_{r\theta} = \sigma_{r\theta}^{(\mathrm{I})} \tag{7.747}$$

$$\sigma_{rz} = \sigma_{rz}^{(\mathrm{III})} \tag{7.748}$$

$$\sigma_{\theta z} = \sigma_{\theta z}^{(\mathrm{III})} \tag{7.749}$$

$$p = p^{(\mathrm{I})} \tag{7.750}$$

in which we have omitted components that are identically zero. In the above, the superscripts $^{(\mathrm{I}),\,(\mathrm{II}),\,(\mathrm{III})}$ denote solutions of problem I, II and III, respectively, defined in Sect. 7.14.2. Problem I is the plane strain problem of an excavated borehole, whose solution is further obtained by the superposition of mode 1, 2, and 3 solutions, as in Sect. 7.15.2.

A number of different criteria have been used to describe rock failure [60, 89], such as Mohr-Coulomb, Drucker-Prager [65], Hoek-Brown [83, 84], and Lade [102]. Although not considered as one of the most accurate [48], here we shall use the classical *Drucker-Prager criterion* for the failure analysis. The criterion is based on a mean shear stress $\sqrt{J_2}$ defined by

$$J_2 = \frac{1}{6}\left[(\sigma_{rr} - \sigma_{\theta\theta})^2 + (\sigma_{\theta\theta} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{rr})^2\right] + \sigma_{r\theta}^2 + \sigma_{rz}^2 + \sigma_{\theta z}^2 \tag{7.751}$$

and a Terzaghi effective mean compressive stress S_p given as

$$S_p = -\frac{\sigma_{rr} + \sigma_{\theta\theta} + \sigma_{zz}}{3} - p \tag{7.752}$$

The Drucker-Prager criterion can be expressed as

$$\sqrt{J_2} = 3AS_p + D \tag{7.753}$$

where A and D are positive material constants. It is known as the failure envelope in the $\sqrt{J_2}$ versus S_p space.

For easy visualization of the failure potential, we shall use the "stress cloud" idea proposed by Bradley [29]. From the solution (7.744)–(7.750), for each set of fixed r and t value, a family of the pairs (S_p, $\sqrt{J_2}$) can be calculated for θ varying from 0 to 360°. Plotting these pairs in a graph of $\sqrt{J_2}$ versus S_p, a closed curve is formed, which is referred as the stress cloud. It delineates the range of stress state for potential failure.

To fix ideas, we shall examine a case with the following far-field stress condition [167]:

$$\begin{aligned} S_{z'} &= 25 \text{ MPa}, \quad S_{x'} = 29 \text{ MPa}, \\ S_{y'} &= 20 \text{ MPa}, \quad p_o = 9.8 \text{ MPa} \end{aligned} \tag{7.754}$$

The borehole is drilled along the direction of azimuth angle $\varphi_{z'} = 30°$ and zenith angle $\varphi_y = 60°$ (see Fig. 7.48). The radius of borehole is $r_o = 0.1$ m. The material properties for a Gulf of Mexico shale (Tables 3.1 and 3.2) are used.

In Fig. 7.55a we plot the stress cloud at borehole wall, $r = r_o$, at various times, $t = 0.001$, 0.1, and 10 days, based on poroelastic solution, in solid lines. For comparison, the stress cloud based on Bradley's work [29] is presented as the loops in dashed lines in Fig. 7.55a. First, if we ignore the existence of pore pressure, the stress cloud evaluated for $r = r_o$ based on the elastic solution is represented by the loop on the right. Realizing the existence of pore pressure, a few earlier studies [1, 29, 167] have included this effect by assuming the pore pressure to be a constant given by the formation pore pressure p_o. The modified stress cloud based on the Terzaghi effective stress is shown as dashed loop on the left in Fig. 7.55a. We observe that the effect of pore pressure is to reduce the effective mean compressive stress S_p, while leaving the mean shear stress $\sqrt{J_2}$ unchanged. The net result is to shift the stress cloud to the left, thus to increase the failure potential. Clearly, the use of a constant pore pressure is inappropriate as it is time and space dependent. Moreover, the pore pressure near the well bore, where according to the elasticity solution the failure potential is the greatest, is closer to zero than to p_o. To more accurately describe the phenomenon, the poroelastic solution, shown as the solid loops, is needed.

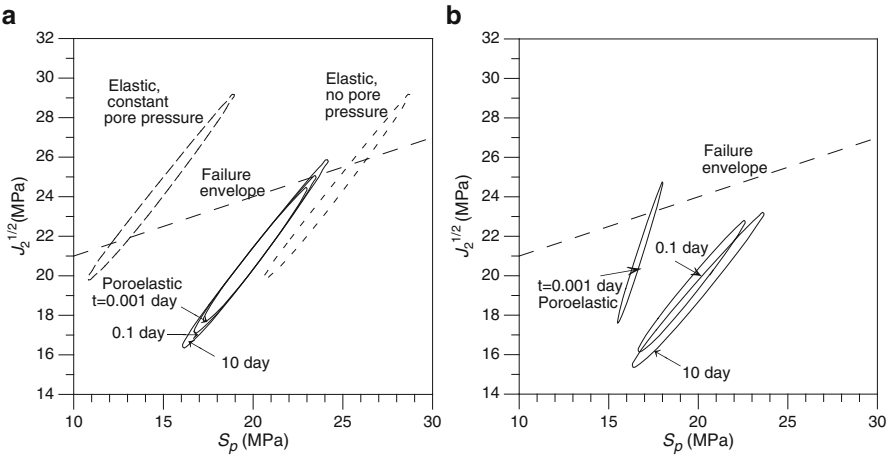


Fig. 7.55 Stress cloud at (**a**) $r = r_o$ and (**b**) $r = 1.05\, r_o$, at various times

For the illustration of potential failure, we shall choose the following values for the constants in the Drucker-Prager criterion (7.751): $A = 0.1$ and $D = 18$ MPa. This envelope is shown in Fig. 7.55 as the dashed straight line. If a stress cloud is entirely below the failure envelope, the borehole is stable. If a part of it is above the envelope, failure takes place in the corresponding region, and time.

We observe that the poroelastic solutions are shifted down and to the left as compared to the elastic, no pore pressure solution. The poroelastic cloud also expands with time along its long axis. At $t = 0.1$ day the borehole is at the threshold of failure. At $t = 10$ days the cloud is practically at steady state and a portion of the borehole wall has failed. This trend suggests that borehole failure potential increases with time. It is possible that a borehole does not fail at the time of excavation, but at a later time. Indeed, delayed failures have been observed in the field. This interpretation was first suggested by Detournay and Cheng [62].

As already discussed in Sect. 7.15.2, at steady state, when pore pressure has vanished around the borehole, the poroelastic solution does not approach the elastic one, due to the poroelastic stress effect caused by the draining of pore pressure. This draining of pore pressure is equivalent to the imposition of a uniform tensile stress $\sigma_{xx} + \sigma_{yy} + \sigma_{zz} = 4\eta p_o$, or, a reduction in S_p by $4\eta p_o/3$. Hence the poroelastic stress cloud at large time is shifted to the left by that amount as compared to the elastic solution. This magnitude is nearly exactly reproduced by the $t = 10$ days stress cloud in Fig.7.55a. Similar poroelastic effect exists for the mean shear stress which shifts the poroelastic cloud down. From the relative positions of the clouds, it seems that the elastic solutions, with or without the consideration of pore pressure, overestimate the failure potential at the wall. Hence a careful borehole design by poroelastic theory can allow a wider range of operation condition.

Next, we turn our attention to the failure potential at a small distance inside the wall, rather than at the wall. Figure 7.55b presents the poroelastic clouds at the radial

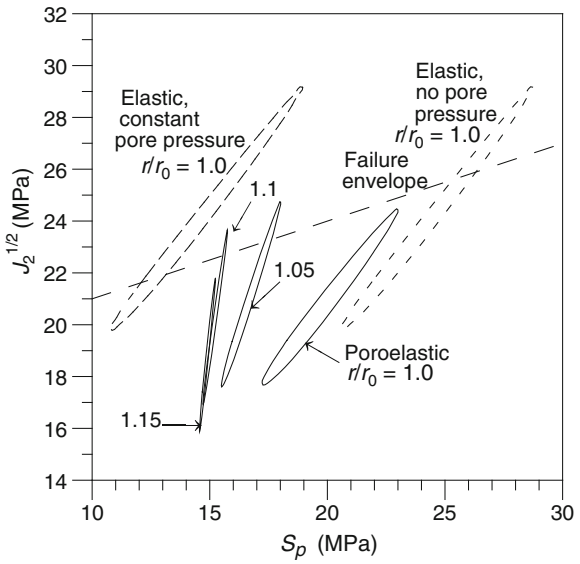


Fig. 7.56 Stress cloud at $t = 0.001$ day, at various radial distances

distance $r = 1.05\, r_o$, at various times. The transient location of the cloud has a trend that is opposite to that shown in Fig. 7.55a. We observe that the formation is more susceptible to failure at small times, rather than at larger times. We also recognize that for poroelastic solution, the potential for failure is larger at a small distance inside the wall than right at the wall.

To further expound this phenomenon, Fig. 7.56 shows the stress clouds at different radial distances from the wall at $t = 0.001$ day. We once again confirm that the stress state is more critical at $r = 1.05\, r_o$, as well as at $1.1\, r_o$, than at the wall. However, away from a thin layer near the wall, the failure potential decreases, as shown by the cloud corresponding to $r = 1.15\, r_o$.

As discussed in Sect. 7.15.2, a rapidly excavated borehole is also prone to tensile failure. This tensile regime, which exists only at small times, can also be demonstrated through the present solution. In Fig. 7.57, the negative radial Terzaghi effective stress $-\sigma'_{rr}$ is plotted as a function of the borehole azimuthal angle θ for various r/r_o values, at $t = 0.001$ day. The upper part of the figure shows compression and lower part tension. Assuming a tensile strength of $T = 5$ MPa for shale [66], this failure criterion is shown as the dashed line in Fig. 7.57. It predicts a region of failure at $r = 1.1\, r_o$ between $20° < \theta < 110°$ (and the symmetrical part across the borehole). At $r = 1.05\, r_0$, the failure zone is restricted to $40° < \theta < 90°$. At $r = 1.2\, r_o$, the formation is safe from tensile failure.

Pressurized Borehole To prevent a deep borehole from collapsing after drilling, the hole is typically filled with a drilling mud to create a *mud weight*, which can be controlled through mud density, to push back the formation to reduce the stress concentration created by the excavation. When the mud weight is not sufficient, the borehole can fail by a combination of compressive and shear stresses. However,

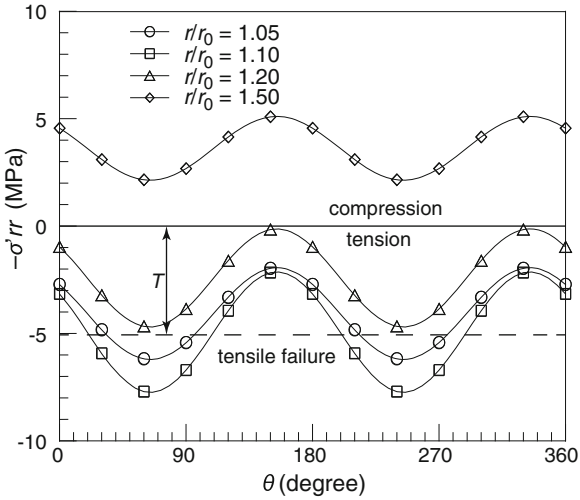


Fig. 7.57 Radial Terzaghi effective stress as a function of borehole azimuthal angle θ at $t = 0.001$ day and various r/r_o values

when the mud weight becomes too much, it can create a tensile hoop stress and fracture the formation, as described in Sect. 7.15.4. Hence the borehole stability can only be maintained within a certain range of the mud density. As the borehole integrity may need to be maintained at different depths with different rocks and strengths, the design and management of the recirculating mud during a drilling operation is of high importance to a successful drilling of a deep borehole. Computer software (PBORE-3D) [7, 52, 53] has been designed based on these analytical solutions, and broadly applied for industry applications [82, 115, 162].

7.16 Moving Load on Half Plane

For some applications, the applied load is moving at a certain speed. This could be a propagating hydraulic fracture driven by injected fluid, a borehole drilled by an advancing bit, or the weight exerted by the wheels of a moving car on a roadway. When the speed is constant, the response of the porous medium becomes steady-state as viewed by an observer moving at the same speed as the applied load. This type of problem is best solved on a moving coordinate system.

Moving Coordinate System Mathematically, we can modify the governing equations as follows. Assuming that the load is moving at a speed of V in the positive x-direction, we can define a moving coordinate system (x', y, z), where $x' = x - Vt$. All the dependent variables are expressed in terms of the new coordinate system and become independent of time: for example, $u_i(x, y, z, t)$ becomes $u_i(x', y, z)$. For the governing equations, (6.64) and (6.69), we note that they remain the same with the substitution of x by x', except that the time derivative terms in (6.69) need to be modified. Applying chain rule,

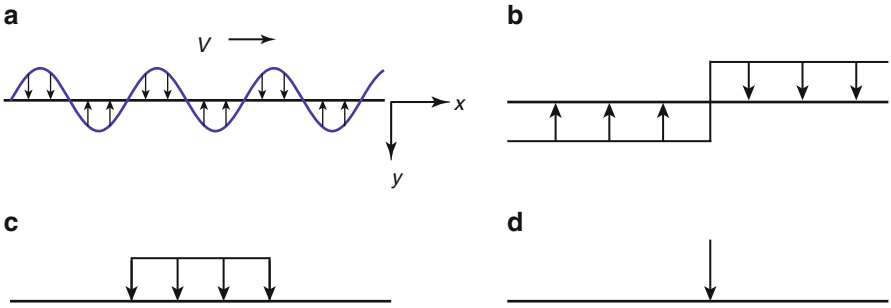


Fig. 7.58 Various moving loads on a half plane

$$\frac{\partial}{\partial t} = \frac{\partial x'}{\partial t}\frac{\partial}{\partial x'} = -V\frac{\partial}{\partial x'} \tag{7.755}$$

we obtain

$$\kappa M\nabla^2 p + V\frac{\partial p}{\partial x'} + \alpha MV\frac{\partial e}{\partial x'} = 0 \tag{7.756}$$

In the following, we shall investigate the problem of a moving load applied to the surface of a half plane, under plane strain conditions.

Sinusoidal Load Referring to Fig. 7.58a, we first examine the case of a traveling sinusoidal load given by the following boundary conditions

$$\begin{aligned} \sigma_{yy}(x,0,t) &= -P_o \sin(\ell x - \omega t) \\ \sigma_{xy}(x,0,t) &= 0 \\ p(x,0,t) &= 0 \end{aligned} \tag{7.757}$$

We observe that this problem is in fact the same as the propagating water wave problem presented in Sect. 7.8, except that the boundary conditions in (7.288) have been changed such that the pore pressure is set to zero, and the vertical stress is given by the sine function, instead of cosine function. The reason that the sine function is used will become clear below.

Recognizing that the velocity of a traveling wave is

$$V = \frac{\omega}{\ell} \tag{7.758}$$

we can express the first equation of (7.757) as

$$\sigma_{yy}(x',0) = -P_o \sin(\ell x') \tag{7.759}$$

Similar to (7.291), (7.292), and (7.293), we can utilize the complex variable expressions to simplify the solution process. In the moving coordinate system, we express

$$u_x(x', y) = \tilde{u}_x(y)\, e^{i\ell x'} \tag{7.760}$$

$$u_y(x', y) = \tilde{u}_y(y)\, e^{i\ell x'} \tag{7.761}$$

$$p(x', y) = \tilde{p}(y)\, e^{i\ell x'} \tag{7.762}$$

and similarly for the stresses. Due to the fact that the applied loading is a sine instead of cosine function, the physical solutions are obtained by taking the imaginary part, instead of real part, of the above expressions.

Following the same procedure as in Sect. 7.8, we obtain the general solutions as the following

$$\tilde{u}_x(y) = a_1\, e^{-\ell y} + a_2\, \ell y\, e^{-\ell y} + a_3\, e^{-\ell' y} \tag{7.763}$$

$$\tilde{u}_y(y) = b_1\, e^{-\ell y} + b_2\, \ell y\, e^{-\ell y} + b_3\, e^{-\ell' y} \tag{7.764}$$

$$\tilde{p}(y) = c_1\, e^{-\ell y} + c_2\, \ell y\, e^{-\ell y} + c_3\, e^{-\ell' y} \tag{7.765}$$

$$\tilde{\sigma}_{xx}(y) = a_1'\, e^{-\ell y} + a_2'\, \ell y\, e^{-\ell y} + a_3'\, e^{-\ell' y} \tag{7.766}$$

$$\tilde{\sigma}_{yy}(y) = b_1'\, e^{-\ell y} + b_2'\, \ell y\, e^{-\ell y} + b_3'\, e^{-\ell' y} \tag{7.767}$$

$$\tilde{\sigma}_{xy}(y) = c_1'\, e^{-\ell y} + c_2'\, \ell y\, e^{-\ell y} + c_3'\, e^{-\ell' y} \tag{7.768}$$

in which

$$\ell' = \sqrt{\ell^2 - \frac{i\ell V}{c}} \tag{7.769}$$

The three coefficients a_1, a_2, and a_3 can be solved from the boundary conditions in (7.757), which are expressed as

$$\frac{2i\ell G\nu}{1-2\nu}\,\tilde{u}_x(0) + \frac{2G(1-\nu)}{1-2\nu}\,\frac{d\tilde{u}_y(0)}{dy} - \alpha\,\tilde{p}(0) = -P_o \tag{7.770}$$

$$\tilde{p}(0) = 0 \tag{7.771}$$

$$\frac{d\tilde{u}_x(0)}{dy} + i\ell\,\tilde{u}_y(0) = 0 \tag{7.772}$$

We then obtain

$$a_1 = -\frac{P_o}{2G}\,\frac{i(1+\ell'') + m(1-2\nu)\,\omega''}{\ell\,[\ell'' + i(1+m)\,\omega'']} \tag{7.773}$$

$$a_2 = \frac{P_o}{2G} \frac{(1+m)\,\omega''}{\ell'' + i(1+m)\,\omega''} \tag{7.774}$$

$$a_3 = \frac{iP_o}{2G} \frac{1}{\ell\,[\ell'' + i\,(1+m)\,\omega'']} \tag{7.775}$$

where m, ℓ'' and ω'' are the dimensionless quantities defined as

$$m = \frac{G}{\alpha^2 M(1-2\nu)} \tag{7.776}$$

$$\ell'' = \frac{\ell'}{\ell} - 1 \tag{7.777}$$

$$\omega'' = \frac{V(1-\nu)}{\ell c\,(1-2\nu)} \tag{7.778}$$

The other coefficients are found to be

$$b_1 = \frac{P_o}{2G} \frac{(1+\ell'') + i\omega''[1+2m(1-\nu)]}{\ell\,[\ell'' + i\,(1+m)\,\omega'']} \tag{7.779}$$

$$b_2 = i\,a_2 \tag{7.780}$$

$$b_3 = -\frac{P_o}{2G} \frac{(1+\ell'')}{\ell\,[\ell'' + i\,(1+m)\,\omega'']} \tag{7.781}$$

$$c_1 = iP_o \frac{\omega''}{\alpha[\ell'' + i(1+m)\,\omega'']} \tag{7.782}$$

$$c_2 = 0 \tag{7.783}$$

$$c_3 = -c_1 \tag{7.784}$$

From the constitutive equations, we then obtain the coefficients for the stresses

$$a_1' = P_o \frac{1+\ell'' - i(1+m)\,\omega''}{\ell'' + i(1+m)\,\omega''} \tag{7.785}$$

$$a_2' = iP_o \frac{\ell\,(1+m)\,\omega''}{\ell'' + i(1+m)\,\omega''} \tag{7.786}$$

$$a_3' = P_o \frac{\ell''(2+\ell'')\nu - (1-2\nu)(1-i\omega'')}{(1-2\nu)[\ell'' + i(1+m)\,\omega'']} \tag{7.787}$$

$$b_1' = -P_o \frac{1+\ell'' + i(1+m)\,\omega''}{\ell'' + i(1+m)\,\omega''} \tag{7.788}$$

$$b_2' = -a_2' \tag{7.789}$$

$$b_3' = P_o \frac{(1-\nu)\ell''(2+\ell'') + (1-2\nu)(1+i\omega'')}{(1-2\nu)[\ell'' + i(1+m)\omega'']} \tag{7.790}$$

$$c_1' = iP_o \frac{1+\ell''}{\ell'' + i(1+m)\omega''} \tag{7.791}$$

$$c_2' = ia_2' \tag{7.792}$$

$$c_3' = -c_1' \tag{7.793}$$

Incompressible Constituent Model For incompressible constituents, we set $\alpha = 1$, and $M \to \infty$, which leads to $m = 0$. We find

$$a_1 = -\frac{P_o}{2G} \frac{i(1+\ell'')}{\ell(\ell'' + i\omega'')} \tag{7.794}$$

$$a_2 = \frac{P_o}{2G} \frac{\omega''}{\ell'' + i\omega''} \tag{7.795}$$

$$a_3 = \frac{iP_o}{2G} \frac{1}{\ell(\ell'' + i\omega'')} \tag{7.796}$$

and other coefficients are similarly simplified.

Step Load The sinusoidal load given as (7.759) can be transformed into a step load of magnitude P_o as shown in Fig. 7.58b by observing the following integral

$$\frac{1}{\pi}\int_0^\infty \frac{\Im\{e^{i\ell x'}\}}{\ell}\, d\ell = \frac{1}{\pi}\int_0^\infty \frac{\sin(\ell x')}{\ell}\, d\ell = \mathrm{H}(x'-0) - \frac{1}{2} \tag{7.797}$$

where H is the Heaviside unit step function, and $\Im\{\}$ denotes the imaginary part. We observe that the right hand side of the above equation takes the value of $-1/2$ for $x' < 0$, and $1/2$ for $x' > 0$. Here we note that the integration of the real part of $e^{i\ell x'}$ in the above equation diverges and should be discarded.

The above relation suggests that the solution of the step load problem can be obtained by integrating the solution of the sinusoidal load as

$$p(x', y) = \frac{1}{\pi}\int_0^\infty \frac{\Im\{\tilde{p}(y)e^{i\ell x'}\}}{\ell}\, d\ell \tag{7.798}$$

$$\sigma_{xx}(x', y) = \frac{1}{\pi}\int_0^\infty \frac{\Im\{\tilde{\sigma}_{xx}(y)e^{i\ell x'}\}}{\ell}\, d\ell \tag{7.799}$$

and similarly for σ_{xy} and σ_{yy}. In the above we did not write the solution for the displacement u_x and u_y because the integration diverges. According to Verruijt [164], for solution of half plane problems with a non-zero resultant load, the displacement solution based on integral transform will not converge in the inverse transform. This appears to be the case for the current step load problem. For pore pressure

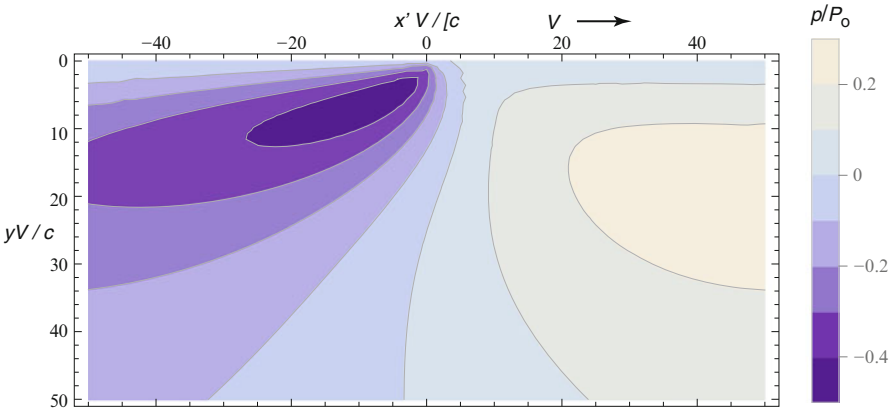


Fig. 7.59 Contour plot of normalized pore pressure p/P_o for a moving step load, for $\nu = 0.25$

and stresses, the above integrations are well defined. However, these integrations cannot be obtained in closed form; hence numerical integration is needed for their evaluation.

In Fig. 7.59 we present the contour plot of the normalized pressure p/P_o versus the dimensionless coordinates $(x'V/c, yV/c)$ assuming an incompressible constituent material ($\alpha = 1$ and $M \to \infty$), which gives the maximum pore pressure response. When solution is presented in these parameters, it is a function of the Poisson ratio only. For the present plot, $\nu = 0.25$ is selected. We observe that the pore pressure at certain depth is generally positive ahead of the step load, and negative behind it. On the surface, it must be zero to satisfy the boundary condition.

Strip Load For practical applications, it is of interest to obtain the solution of a strip load of magnitude P_o between $-L \geq x' \geq L$, as shown in Fig. 7.58c. This load pattern can be obtained by the superposition of two step loads

$$\begin{aligned} \mathrm{H}(x'+L) - \mathrm{H}(x'-L) &= 1 \quad \text{for} \quad |x'| \leq L \\ &= 0 \quad \text{for} \quad |x'| > L \end{aligned} \tag{7.800}$$

Hence the solution of the current problem can be obtained from the step load solution through the following superposition

$$p(x', y) = \frac{1}{\pi} \int_0^\infty \frac{\Im\{\tilde{p}_x(y)[e^{i\ell(x'+L)} - e^{i\ell(x'-L)}]\}}{\ell} \, d\ell \tag{7.801}$$

and similarly for $\sigma_{xx}(x', y)$, etc.

In Fig. 7.60 we present the contour plot of the normalized pore pressure versus the dimensionless spatial coordinates, for two strip load sizes, $VL/c = 10$ and $VL/c = 100$. Assuming a dimensional strip load size of $L = 0.1$ m and a velocity

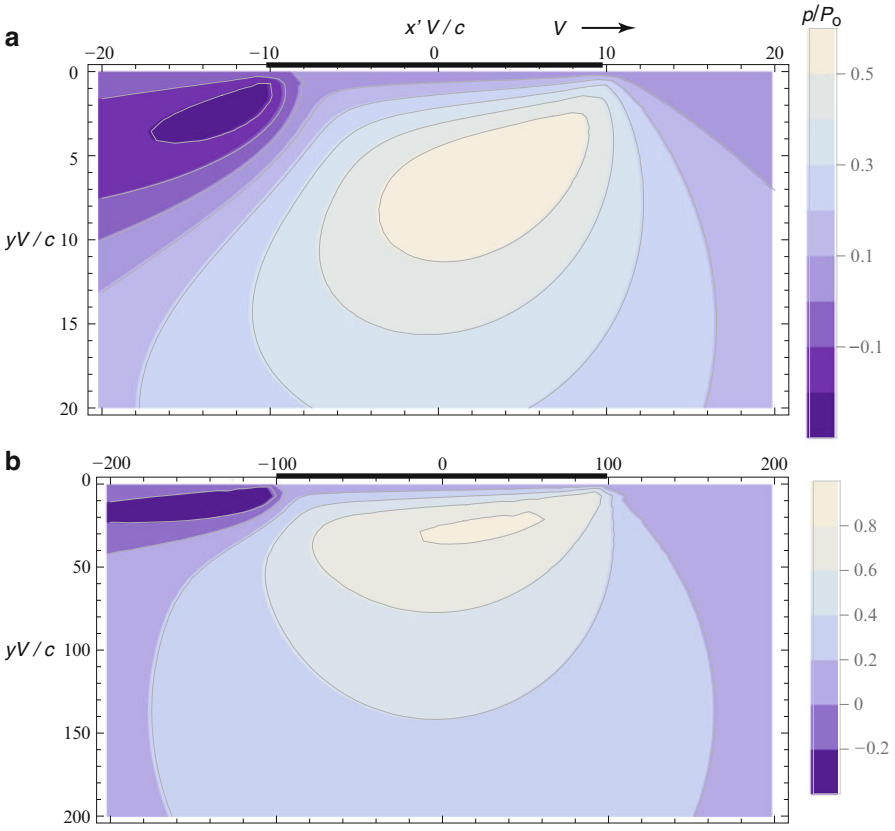


Fig. 7.60 Contour plot of normalized pore pressure p/P_o for a moving strip load, for $\nu = 0.25$. Strip size (**a**) $VL/c = 10$, and (**b**) $VL/c = 100$

of $V = 1$ m/s, the above corresponds to materials with consolidation coefficient of $c = 10^{-2}$ and 10^{-3} m^2/s, respectively. In the contour plots, we observe a positive pore pressure below and ahead of the moving strip load, and a negative pressure trailing the load. The pore pressure magnitude is larger for the larger load size. As VL/c increases, which corresponds to increasing strip load size or velocity, or decreasing consolidation coefficient, the pore pressure distribution becomes more symmetrical. In fact, in the limit, as the velocity approaches infinity, or the consolidation coefficient approaches zero, there is no time for pore pressure to dissipate, and the material behaves as undrained. The solution then approaches elastic solution, which is symmetrical to the applied load, and the velocity has no effect.

In Fig. 7.61 the stress contour plots are presented. Figure 7.61a gives the normalized vertical stress σ_{yy}/P_o for the incompressible constituent material with $\nu = 0.25$, and the strip load size $VL/c = 10$. We observe that the stress contours

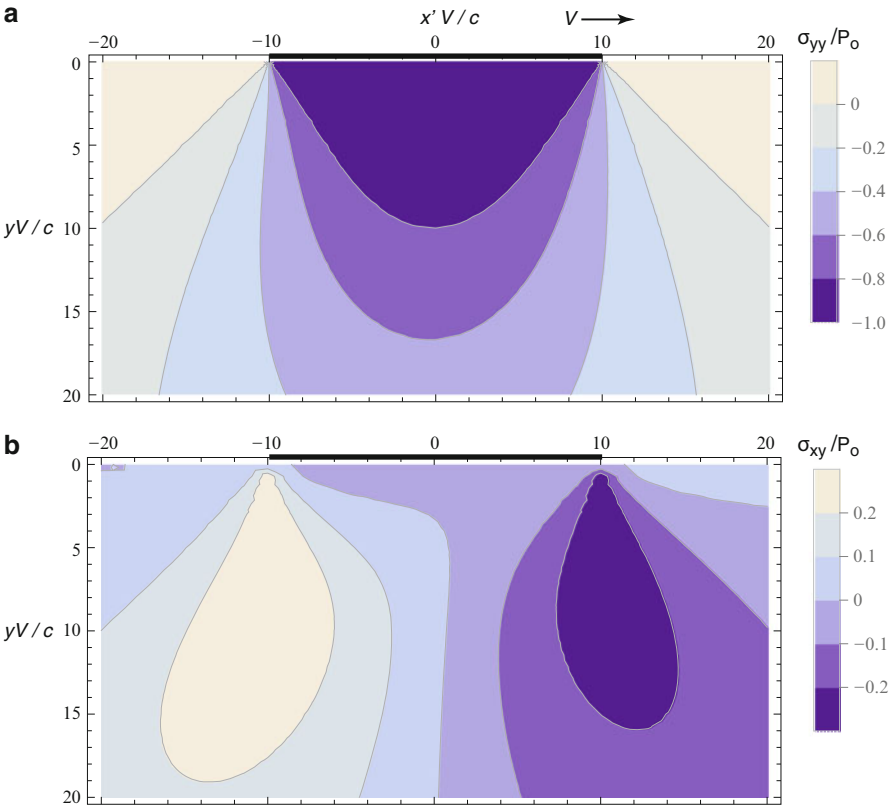


Fig. 7.61 Contour plot of (**a**) vertical stress σ_{yy}/P_o, and (**b**) shear stress σ_{xy}/P_o, for a moving strip load with $\nu = 0.25$ and strip size $VL/c = 10$

are nearly symmetrical with the x'-axis, meaning that the velocity has little effect on the stress distribution, and it is largely elastic. The effective stress, however, will be strongly influenced by the pore pressure response as shown in Fig. 7.60. Figure 7.61b plots the shear stress. Similarly, the velocity has only a small effect, as the stress distribution is only slightly off from the antisymmetry expected from an elastic solution.

Concentrated Load As the last example, we consider the case of concentrated load of unit magnitude, as shown in Fig. 7.58d. The solution can be obtained by a simple differentiation with respect to x' of the step load solution. This is evident by differentiating the step load on the surface given in (7.797), which yields

$$\frac{\partial \mathrm{H}(x'-0)}{\partial x'} = \delta(x'-0) \tag{7.802}$$

Hence we obtain from (7.798) and (7.799) the following solution for concentrated load

$$p(x', y) = \frac{1}{\pi} \int_0^\infty \frac{\Im\{i\ell \tilde{p}(y) e^{i\ell x'}\}}{\ell} \, d\ell \tag{7.803}$$

$$\sigma_{xx}(x', y) = \frac{1}{\pi} \int_0^\infty \frac{\Im\{i\ell \tilde{\sigma}_{xx}(y) e^{i\ell x'}\}}{\ell} \, d\ell \tag{7.804}$$

7.17 Plane Strain Half Space and Layered Problem

For plane strain problems with horizontally layered geometry (see Fig. 7.62), and the special case of half space (a single layer that is infinitely thick), the McNamee-Gibson displacement functions introduced in Sect. 6.6.3 are often utilized for solution.

The displacement functions $\mathcal{E}(x, z, t)$ and $\mathcal{S}(x, z, t)$ satisfy the following set of partial differential equations

$$\frac{\partial}{\partial t}(\nabla^2 \mathcal{E}) - c\nabla^4 \mathcal{E} = 0 \tag{7.805}$$

$$\nabla^2 \mathcal{S} = 0 \tag{7.806}$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \tag{7.807}$$

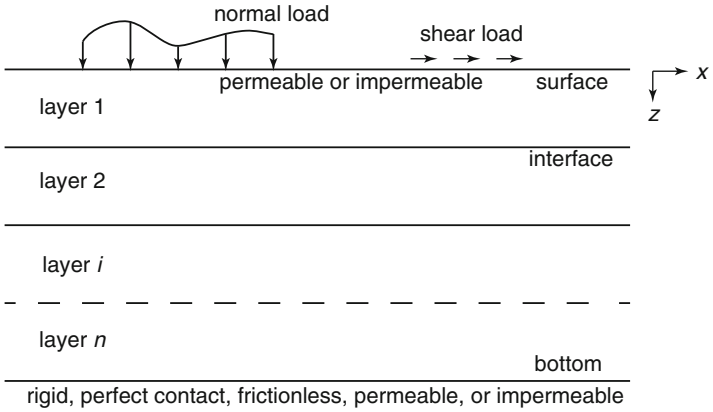


Fig. 7.62 Horizontally layered formation with surface loading

The layered geometry investigated here allows the application of integral transform in the x-direction to the above set of equations to remove the dependence on the x-coordinate. When Laplace transform is further applied to the time variable, the governing equations (7.805) and (7.806) reduce to ordinary differential equations whose solutions are readily found.

7.17.1 General Solution for Layered Problem

For a domain made of multiple layers, with each layer containing a homogeneous poroelastic material, we define a set of dependent variables, displacement, stress, pore pressure, etc., for each layer. Using the displacement function representation, these reduce to the set of functions $\mathcal{E}_i$ and $\mathcal{S}_i$, with $i = 1, 2, \ldots, n$, for an n-layer system.

The Fourier transform and its inverse are defined as follows [151]

$$\bar{f}(\xi) = \frac{2}{\pi} \int_{-\infty}^{\infty} f(x)\, e^{-ix\xi}\, dx \tag{7.808}$$

$$f(x) = \int_{-\infty}^{\infty} \bar{f}(\xi)\, e^{ix\xi}\, dx \tag{7.809}$$

where the overbar denotes Fourier transform, and ξ is the Fourier transform parameter. To simplify the governing equations (7.805) and (7.806), we apply simultaneously the Laplace and the Fourier transform, and obtain

$$\left(\frac{d^2}{dz^2} - \xi^2\right)\left(\frac{d^2}{dz^2} - \xi^2 - \frac{s}{c}\right)\tilde{\bar{\mathcal{E}}}_i(z) = 0 \tag{7.810}$$

$$\left(\frac{d^2}{dz^2} - \xi^2\right)\tilde{\bar{\mathcal{S}}}_i(z) = 0 \tag{7.811}$$

where the tilde denotes Laplace transform, and s is the Laplace transform parameter. The above are ordinary differential equations, whose solutions are readily found

$$\tilde{\bar{\mathcal{E}}}_i(z) = A_1^i\, e^{-z\xi} + A_2^i\, e^{-z\sqrt{\xi^2 + s/c}} + A_3^i\, e^{z\xi} + A_4^i\, e^{z\sqrt{\xi^2 + s/c}} \tag{7.812}$$

$$\tilde{\bar{\mathcal{S}}}_i(z) = B_1^i\, e^{-z\xi} + B_2^i\, e^{z\xi} \tag{7.813}$$

In the above, the subscripts and superscripts i indicate the layer number, and the constants (actually functions of ξ and s) $A_1^i, \ldots, A_4^i, B_1^i, B_2^i$ are to be determined from the boundary and the interface matching conditions. For an n-layer system, there are $6\,n$ such constants to determine. On the top and bottom surfaces, we must define three boundary conditions each, one in the x direction, such as u_x or σ_{xz}, one in the z-direction, such as u_z or σ_{zz}, and one condition related to the fluid, such as pore pressure p or the normal flux q_z. One each interface, we must match six displacement and stress continuity conditions,

$$u_x^i = u_x^{i+1}; \quad u_z^i = u_z^{i+1}; \quad q_z^i = q_z^{i+1};$$
$$\sigma_{xz}^i = \sigma_{xz}^{i+1}; \quad \sigma_{zz}^i = \sigma_{zz}^{i+1}; \quad p^i = p^{i+1} \tag{7.814}$$

where the superscripts indicate the layer number. As there exist $n-1$ interfaces, plus the top and bottom surfaces, we count a total of $6n$ boundary and interface conditions. These conditions are used to set up a linear algebraic system involving a $6n \times 6n$ banded matrix, for the determination of the $6n$ constants.

What remains for the solution step is the inverse transforms to bring the solutions to the time t and the spatial x domain. The analytical inversion of Laplace and Fourier transform, however, is generally not feasible. Hence the solutions need to be evaluated by numerical quadrature. In that case, the coefficient matrix needs to be inverted numerous times, each time with different ξ and s values; hence special algorithms, such as the propagator matrix method [78, 160], has been employed. There are a number of solutions of this type [25, 77, 147, 149, 163, 165]. Due to their numerical nature, these solutions are not presented here. In the following, we shall examine the simplest of these problems—a single, semi-infinite layer.

7.17.2 Plane Strain Half Space Problem

The following solution follows that of Verruijt [164]. Referring to Fig. 7.62, we assume that there exists a single homogeneous layer occupying the half space $0 \leq z < \infty$. On the surface, $z = 0$, a strip load of normal stress of constant magnitude P_o is applied, and the surface is otherwise stress and pore pressure free,

$$\sigma_{zz} = -P_o, \quad \text{for } |x| \leq L; \qquad \sigma_{zz} = 0, \quad \text{for } |x| > L;$$
$$\sigma_{xz} = 0; \qquad p = 0 \tag{7.815}$$

As the applied normal load is symmetrical with respect to z-axis, the displacement field is also symmetrical,

$$u_x(x,z,t) = u_x(-x,z,t); \qquad u_z(x,z,t) = u_z(-x,z,t) \tag{7.816}$$

Due to this condition, we shall apply the Fourier cosine transform, instead of the Fourier transform, to the governing equations. The Fourier cosine transform and its inverse are defined as [151]

$$\bar{f}(\xi) = \frac{2}{\pi} \int_0^\infty f(x) \cos(x\xi)\, dx \tag{7.817}$$

$$f(x) = \int_0^\infty \bar{f}(\xi) \cos(x\xi)\, d\xi \tag{7.818}$$

The advantage of using the Fourier cosine transform is that, when numerical inversion is performed, the integration is needed only between $0 \leq \xi < \infty$, instead of $-\infty < \xi < \infty$, and the variables are real, instead of complex, as compared to the Fourier transform.

The general solution of the problems is still given by (7.812) and (7.813). However, the condition at $z \to \infty$ for the half space problem requires that $A_3 = A_4 = B_2 = 0$, for the solution to remain bounded. Hence the solution simplifies to

$$\bar{\tilde{\mathcal{E}}}(z) = A_1 e^{-z\xi} + A_2 e^{-z\sqrt{\xi^2+s/c}} \tag{7.819}$$

$$\bar{\tilde{\mathcal{S}}}(z) = B_1 e^{-z\xi} \tag{7.820}$$

Performing Fourier inverse transform of the above, we find

$$\tilde{\mathcal{E}}(x,z) = \int_0^\infty \left(A_1 e^{-z\xi} + A_2 e^{-z\sqrt{\xi^2+s/c}}\right) \cos(x\xi)\, d\xi \tag{7.821}$$

$$\tilde{\mathcal{S}}(x,z) = \int_0^\infty B_1 e^{-z\xi} \cos(x\xi)\, d\xi \tag{7.822}$$

For the convenience of solution, we shall present the following expressions [164] by differentiating the above,

$$\frac{\partial^2 \tilde{\mathcal{E}}}{\partial x^2} = -\int_0^\infty \left(A_1 \xi^2 e^{-z\xi} + A_2 \xi^2 e^{-z\sqrt{\xi^2+s/c}}\right) \cos(x\xi)\, d\xi \tag{7.823}$$

$$\frac{\partial^2 \tilde{\mathcal{E}}}{\partial z^2} = \int_0^\infty \left[A_1 \xi^2 e^{-z\xi} + A_2 \left(\xi^2 + \frac{s}{c}\right) e^{-z\sqrt{\xi^2+s/c}}\right] \cos(x\xi)\, d\xi \tag{7.824}$$

$$\nabla^2 \tilde{\mathcal{E}} = \int_0^\infty A_2 \frac{s}{c} e^{-z\sqrt{\xi^2+s/c}} \cos(x\xi)\, d\xi \tag{7.825}$$

$$\frac{\partial^2 \tilde{\mathcal{E}}}{\partial x \partial z} = \int_0^\infty \left[A_1 \xi^2 e^{-z\xi} + A_2 \xi \sqrt{\xi^2 + s/c}\, e^{-z\sqrt{\xi^2+s/c}}\right] \sin(x\xi)\, d\xi \tag{7.826}$$

$$\frac{\partial \tilde{\mathcal{S}}}{\partial x} = -\int_0^\infty B_1 \xi\, e^{-z\xi} \sin(x\xi)\, d\xi \tag{7.827}$$

$$\frac{\partial \tilde{\mathcal{S}}}{\partial z} = -\int_0^\infty B_1 \xi\, e^{-z\xi} \cos(x\xi)\, d\xi \tag{7.828}$$

$$\frac{\partial^2 \tilde{\mathcal{S}}}{\partial z^2} = \int_0^\infty B_1 \xi^2 e^{-z\xi} \cos(x\xi)\, d\xi \tag{7.829}$$

$$\frac{\partial^2 \tilde{\mathcal{S}}}{\partial x \partial z} = \int_0^\infty B_1 \xi^2 e^{-z\xi} \sin(x\xi)\, d\xi \tag{7.830}$$

Based on the pore pressure expression (6.160) and the zero pore pressure condition in (7.815), we obtain

$$\begin{aligned}\tilde{p}(x,0) &= -\frac{G}{\eta}\left[\int_0^\infty A_2\frac{s}{c}\cos(x\xi)\,d\xi + \frac{2(\nu_u-\nu)}{1-\nu}\int_0^\infty B_1\xi\cos(x\xi)\,d\xi\right]\\ &= 0\end{aligned} \tag{7.831}$$

It is clear that

$$A_2\frac{s}{c} + \frac{2(\nu_u-\nu)}{1-\nu}B_1\xi = 0 \tag{7.832}$$

Similarly, based on (6.159), we obtain from the boundary condition for σ_{xz} the following

$$A_1\xi + A_2\sqrt{\xi^2+s/c} - (1-2\nu_u)B_1 = 0 \tag{7.833}$$

For the strip load boundary condition of σ_{zz}, we can refer to (7.797) and (7.800), and express it as

$$\begin{aligned}\tilde{\sigma}_{zz}(x,0) &= -\frac{P_o}{\pi s}\int_0^\infty \frac{\sin[\xi(x+L)]-\sin[\xi(x-L)]}{\xi}\,d\xi\\ &= -\frac{2P_o}{\pi s}\int_0^\infty \frac{\sin(L\xi)}{\xi}\cos(x\xi)\,d\xi\end{aligned} \tag{7.834}$$

From (6.158) and the above, we therefore find

$$A_1\xi^2 + A_2\xi^2 - 2(1-\nu_u)B_1\xi = \frac{P_o\sin(L\xi)}{\pi G\xi s} \tag{7.835}$$

The three Eqs. (7.832), (7.833), and (7.835), allow us to solve for

$$A_1 = -\frac{P_o\left[s(1-\nu)(1-2\nu_u)+2c(\nu_u-\nu)\xi\sqrt{\xi^2+s/c}\right]\sin(L\xi)}{\pi G\xi^3 s\left[s(1-\nu)+2c\xi(\nu_u-\nu)\left(\xi-\sqrt{\xi^2+s/c}\right)\right]} \tag{7.836}$$

$$A_2 = \frac{2P_o c(\nu_u-\nu)\sin(L\xi)}{\pi G\xi s\left[s(1-\nu)+2c\xi(\nu_u-\nu)\left(\xi-\sqrt{\xi^2+s/c}\right)\right]} \tag{7.837}$$

$$B_1 = -\frac{P_o(1-\nu)\sin(L\xi)}{\pi G\xi^2\left[s(1-\nu)+2c\xi(\nu_u-\nu)\left(\xi-\sqrt{\xi^2+s/c}\right)\right]} \tag{7.838}$$

We hence in principle have defined all solutions.

Pore Pressure With the substitution of the constant (7.836)–(7.838) into (6.160) and (7.823)–(7.830), we find the pore pressure solution as

$$\tilde{p}(x,z) = \frac{2P_o(\nu_u-\nu)}{\pi\eta} \times \int_0^\infty \frac{\left(e^{-z\xi}-e^{-z\sqrt{\xi^2+s/c}}\right)\sin(L\xi)}{\xi\left[s(1-\nu)+2c\xi(\nu_u-\nu)\left(\xi-\sqrt{\xi^2+s/c}\right)\right]}\cos(x\xi)\,d\xi \tag{7.839}$$

The Laplace inverse transform of the above has been carried out by Verruijt [164] as

$$p(x,z,t) = \frac{P_o(\nu_u-\nu)}{\pi\eta(1-\nu_u)}\int_0^\infty \frac{1}{\xi}\left[e^{-z\xi}F_2(\xi,t)-F_1(\xi,z,t)\right]\sin(L\xi)\cos(x\xi)\,d\xi \tag{7.840}$$

where

$$F_1(\xi,z,t) = \exp(-z\xi)\left[1+\operatorname{erf}\left(\xi\sqrt{ct}-\frac{z}{2\sqrt{ct}}\right)\right] + \frac{1+\nu-2\nu_u}{1-\nu}\exp\left[-\frac{4(\nu_u-\nu)(1-\nu_u)}{(1-\nu)^2}\xi^2 ct+\frac{1+\nu-2\nu_u}{1-\nu}z\xi\right] \times \operatorname{erfc}\left(\frac{1+\nu-2\nu_u}{1-\nu}\xi\sqrt{ct}+\frac{z}{2\sqrt{ct}}\right) \tag{7.841}$$

$$F_2(\xi,t) = 1+\operatorname{erf}\left(\xi\sqrt{ct}\right)+\frac{1+\nu-2\nu_u}{1-\nu}\exp\left[-\frac{4(\nu_u-\nu)(1-\nu_u)}{(1-\nu)^2}\xi^2 ct\right] \times \operatorname{erfc}\left(\frac{1+\nu-2\nu_u}{1-\nu}\xi\sqrt{ct}\right) \tag{7.842}$$

The analytical inversion of the Fourier cosine transform is not possible; hence the final step of the solution needs to be carried out numerically by performing quadrature on (7.840).

In Fig. 7.63 we present the contour plot of normalized pore pressure p/P_o for an incompressible constituent material ($\nu_u = 0.5$, $\alpha = 1$), with $\nu = 0$, versus dimensionless coordinates $x^* = x/L$ and $z^* = z/L$, for two dimensionless times $t^* = ct/L^2 = 0.01$ and 1.0. We note that the normalized pore pressure solution is independent of the shear modulus. These plots show that at $t^* = 0.01$, the maximum pore pressure observed is $p/P_o = 0.85$, and at $t^* = 1$, it is reduced to $p/P_o = 0.27$.

The displacement for the plane strain problem is unbounded, and cannot be obtained by the integral transform technique. The stresses can be obtained from (6.157), (6.158), and (6.159). Analytical Laplace inverse transform for these expressions, however, is not possible. Hence numerical inversion is needed for both the Laplace and the Fourier transform. Details of these solutions are found in Verruijt [164].

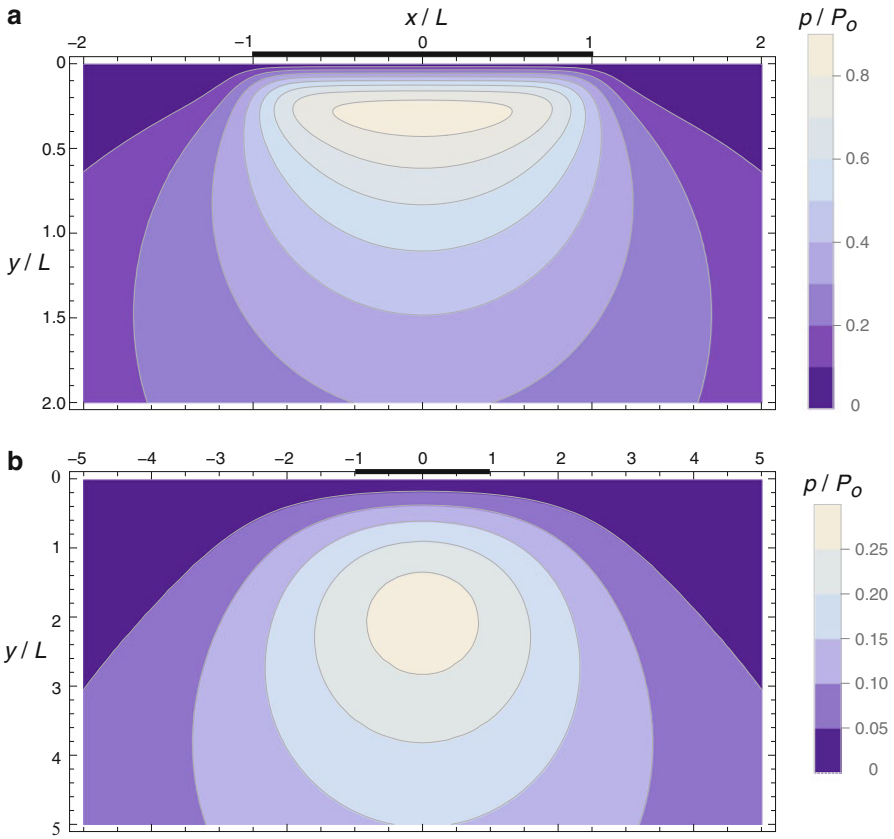


Fig. 7.63 Contour plot of normalized pore pressure p/P_o at (**a**) $t^* = 0.01$, and (**b**) $t^* = 1.0$, for incompressible constituent material with $\nu = 0$

7.18 Axial Symmetry Half Space Problem

For layered formation as shown in Fig. 7.62, with a surface loading that possesses axial symmetry with respect to the z-axis, all variables are independent of the θ coordinate, and are functions of (r, z, t) only. In this case, we shall utilize the axisymmetric displacement functions $\mathcal{E}(r, z, t)$ and $\mathcal{S}(r, z, t)$ defined in Sect. 6.6.3. These two displacement functions satisfy the same governing equations (7.805) and (7.806), except that the Laplacian operator is given by

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{\partial^2}{\partial z^2} \tag{7.843}$$

To simplify the governing equations, we shall utilize the zeroth order Hankel transform and its inverse, defined as [151]

$$\bar{f}(\xi) = \int_0^\infty f(r)\, r\, \mathrm{J}_0(r\xi)\, dr \tag{7.844}$$

$$f(r) = \int_0^\infty \bar{f}(\xi)\, \xi\, \mathrm{J}_0(r\xi)\, d\xi \tag{7.845}$$

where the overbar denotes Hankel transform, and ξ is the Hankel transform parameter.

Applying both the Laplace and the Hankel transform to the governing equations (7.805) and (7.806), we obtain the same ordinary differential equations (7.810) and (7.811). The general solutions also take the form as (7.812) and (7.813). While such system can generally allow us to solve multilayered problem by some matrix and numerical integration techniques, in the following we shall only examine the half space problem, following the solution of Verruijt [164].

For the axisymmetric problem, we assume a vertical load of constant magnitude P_o, distributed over a disk area on the surface, with the boundary conditions,

$$\begin{aligned} \sigma_{zz} = -P_o, \quad \text{for } r \le L; \qquad \sigma_{zz} = 0, \quad \text{for } r > L; \\ \sigma_{rz} = 0; \qquad p = 0 \end{aligned} \tag{7.846}$$

Due to the semi-infinite space, the general solutions (7.812) and (7.813) are reduced to (7.819) and (7.820). Performing inverse Hankel transform to these solutions, we find

$$\tilde{\mathcal{E}}(r,z) = \int_0^\infty \left(A_1\, e^{-z\xi} + A_2\, e^{-z\sqrt{\xi^2+s/c}} \right) \xi\, \mathrm{J}_0(r\xi)\, d\xi \tag{7.847}$$

$$\tilde{\mathcal{S}}(r,z) = \int_0^\infty B_1\, e^{-z\xi}\, \xi\, \mathrm{J}_0(r\xi)\, d\xi \tag{7.848}$$

For the convenience of solution, we shall present the following expressions [164]

$$\frac{\partial \tilde{\mathcal{E}}}{\partial r} = -\int_0^\infty \left(A_1 e^{-z\xi} + A_2 e^{-z\sqrt{\xi^2+s/c}} \right) \xi^2 \mathrm{J}_1(r\xi)\, d\xi \tag{7.849}$$

$$\frac{\partial \tilde{\mathcal{E}}}{\partial z} = -\int_0^\infty \left(A_1 \xi\, e^{-z\xi} + A_2 \sqrt{\xi^2 + s/c}\, e^{-z\sqrt{\xi^2+s/c}} \right) \xi\, \mathrm{J}_0(r\xi)\, d\xi \tag{7.850}$$

$$\frac{\partial^2 \tilde{\mathcal{E}}}{\partial r^2} = -\int_0^\infty \left(A_1 e^{-z\xi} + A_2 e^{-z\sqrt{\xi^2+s/c}} \right) \xi^3 \left[\mathrm{J}_0(r\xi) - \frac{\mathrm{J}_1(r\xi)}{r\xi} \right] d\xi \tag{7.851}$$

$$\frac{\partial^2 \tilde{\mathcal{E}}}{\partial z^2} = \int_0^\infty \left[A_1 \xi^3 e^{-z\xi} + A_2 \xi \left(\xi^2 + \frac{s}{c} \right) e^{-z\sqrt{\xi^2+s/c}} \right] \mathrm{J}_0(r\xi)\, d\xi \tag{7.852}$$

$$\nabla^2 \tilde{\mathcal{E}} = \int_0^\infty A_2 \frac{s}{c}\, e^{-z\sqrt{\xi^2+s/c}}\, \xi\, \mathrm{J}_0(r\xi)\, d\xi \tag{7.853}$$

$$\frac{\partial^2 \tilde{\mathcal{E}}}{\partial r \partial z} = \int_0^\infty \left[A_1 \xi^3 e^{-z\xi} + A_2 \xi^2 \sqrt{\xi^2 + s/c}\, e^{-z\sqrt{\xi^2+s/c}} \right] \mathrm{J}_1(r\xi)\, d\xi \tag{7.854}$$

$$\frac{\partial \tilde{\mathcal{S}}}{\partial r} = -\int_0^\infty B_1 \xi^2 e^{-z\xi} \mathrm{J}_1(r\xi)\, d\xi \tag{7.855}$$

$$\frac{\partial \tilde{\mathcal{S}}}{\partial z} = -\int_0^\infty B_1 \xi^2 e^{-z\xi} \mathrm{J}_0(r\xi)\, d\xi \tag{7.856}$$

$$\frac{\partial^2 \tilde{\mathcal{S}}}{\partial z^2} = \int_0^\infty B_1 \xi^3 e^{-z\xi} \mathrm{J}_0(r\xi)\, d\xi \tag{7.857}$$

$$\frac{\partial^2 \tilde{\mathcal{S}}}{\partial r \partial z} = \int_0^\infty B_1 \xi^3 e^{-z\xi} \mathrm{J}_1(r\xi)\, d\xi \tag{7.858}$$

Based on the pore pressure and shear stress expressions, (6.178) and (6.179), and the null boundary conditions in (7.846), we obtain

$$A_2 \frac{s}{c} + \frac{2(\nu_u - \nu)}{1-\nu} B_1 \xi = 0 \tag{7.859}$$

$$A_1 \xi + A_2 \sqrt{\xi^2 + s/c} - (1 - 2\nu_u) B_1 = 0 \tag{7.860}$$

For the disk load boundary condition of σ_{zz} in (7.846), we observe this Hankel transform

$$\int_0^\infty [1 - \mathrm{H}(r-L)] r \mathrm{J}_0(r\xi)\, dr = \int_0^L r \mathrm{J}_0(r\xi)\, dr = \frac{L}{\xi} \mathrm{J}_1(L\xi) \tag{7.861}$$

We therefore find from (6.158) and the above

$$A_1 \xi^3 + A_2 \xi^3 - 2(1-\nu_u) B_1 \xi^2 = \frac{P_o L}{2Gs} \mathrm{J}_1(L\xi) \tag{7.862}$$

From (7.859), (7.860), and (7.862), we can solve for

$$A_1 = -\frac{P_o L \left[s(1-\nu)(1-2\nu_u) + 2c(\nu_u - \nu)\xi \sqrt{\xi^2 + s/c} \right] \mathrm{J}_1(L\xi)}{2G\xi^3 s \left[s(1-\nu) + 2c\xi(\nu_u - \nu) \left(\xi - \sqrt{\xi^2 + s/c} \right) \right]} \tag{7.863}$$

$$A_2 = \frac{P_o L c (\nu_u - \nu) \mathrm{J}_1(L\xi)}{G\xi s \left[s(1-\nu) + 2c\xi(\nu_u - \nu) \left(\xi - \sqrt{\xi^2 + s/c} \right) \right]} \tag{7.864}$$

$$B_1 = -\frac{P_o L (1-\nu) \mathrm{J}_1(L\xi)}{2G\xi^2 \left[s(1-\nu) + 2c\xi(\nu_u - \nu) \left(\xi - \sqrt{\xi^2 + s/c} \right) \right]} \tag{7.865}$$

These constants formally define the solution.

Vertical Displacement at Surface Unlike the strip load case, the displacement due to a disk load is bounded; hence it can be obtained by the present integral transform technique. Based on (6.173) and the above constants, the vertical displacement on the surface, $z = 0$, is given by

$$\tilde{u}_z(r,0) = \frac{P_o L(1-\nu)(1-\nu_u)}{G} \times \int_0^\infty \frac{\mathrm{J}_1(L\xi)\,\mathrm{J}_0(r\xi)}{\xi\left[s(1-\nu) + 2c\xi(\nu_u-\nu)\left(\xi - \sqrt{\xi^2 + s/c}\right)\right]}\, d\xi \tag{7.866}$$

The Laplace inverse transform of the above was given by Verruijt [164] as

$$u_z(r,0,t) = \frac{P_o L(1-\nu)}{2G} \int_0^\infty \frac{1}{\xi} F_2(\xi,t)\,\mathrm{J}_1(L\xi)\,\mathrm{J}_0(r\xi)\, d\xi \tag{7.867}$$

where $F_2(\xi,t)$ is defined in (7.842). The Hankel inverse transform has to be numerically evaluated.

In Fig. 7.64 we plot the normalized vertical displacement at the surface of the half space, $Gu_z(r,0,t)/P_oL$, for various dimensionless times, $t^* = ct/L^2$, for an incompressible constituent material with $\nu = 0$. In the figure, we also present in dashed lines the elastic solutions [161] based on undrained and drained Poisson ratios, which corresponds to the poroelastic solutions at $t = 0$ and $t \to \infty$, respectively. In other words, we anticipate the poroelastic solution to have the following limits

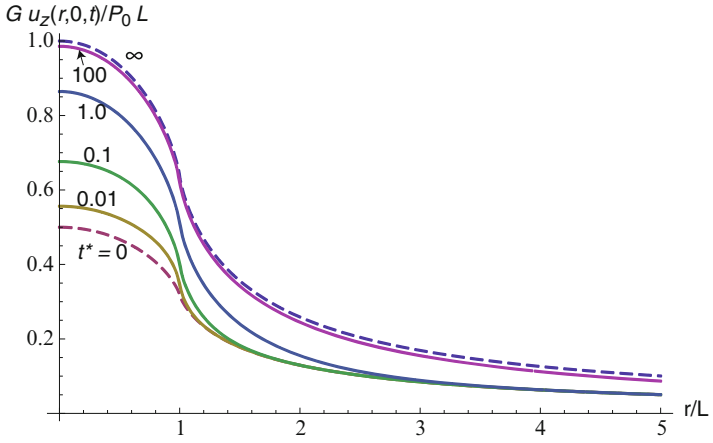


Fig. 7.64 Normalized vertical displacement Gu_z/P_oL at the surface of half space due to a disk load, at various dimensionless times, for an incompressible constituent material with $\nu = 0$

$$u(r,0,0) = \frac{2P_o L(1-\nu_u)}{\pi G}\,\mathrm{E}\left(\frac{r^2}{L^2}\right), \qquad r \le L$$
$$= \frac{r}{L}\left[\mathrm{E}\left(\frac{L^2}{r^2}\right) - \left(1-\frac{L^2}{r^2}\right)\mathrm{K}\left(\frac{L^2}{r^2}\right)\right], \quad r > L \tag{7.868}$$

where K and E are respectively the complete elliptic integral of the first kind and second kind [8]. For $u(r,0,\infty)$, we replace ν_u by ν in the above. The maximum displacement is located at the center and is given by

$$u(0,0,0) = \frac{P_o L(1-\nu_u)}{G} \tag{7.869}$$

$$u(0,0,\infty) = \frac{P_o L(1-\nu)}{G} \tag{7.870}$$

Pore Pressure Pore pressure is obtained as

$$\tilde{p}(r,z) = \frac{P_o L(\nu_u-\nu)}{\eta} \times \int_0^\infty \frac{\left(e^{-z\xi} - e^{-z\sqrt{\xi^2+s/c}}\right)\mathrm{J}_1(L\xi)\,\mathrm{J}_0(r\xi)}{s(1-\nu)+2c\xi(\nu_u-\nu)\left(\xi-\sqrt{\xi^2+s/c}\right)}\,d\xi \tag{7.871}$$

Performing Laplace inverse transform, we find

$$p(r,z,t) = \frac{P_o(\nu_u-\nu)}{2\eta(1-\nu_u)}\int_0^\infty \left[e^{-z\xi}F_2(\xi,t) - F_1(\xi,z,t)\right]\mathrm{J}_1(L\xi)\,\mathrm{J}_0(r\xi)\,d\xi \tag{7.872}$$

with F_1 and F_2 defined in (7.841) and (7.842). Other solutions, such as the stresses, require numerical evaluation of both the Laplace and Hankel transform, and are not shown here. The detail can be found in Verruijt [164].

References

1. Aadnoy BS, Chenevert ME (1987) Stability of highly inclined boreholes. SPE Drill Eng 2(4):364–374
2. Abou-Sayed AS, Brechtel CE, Clifton RJ (1978) In situ stress determination by hydrofracturing: a fracture mechanics approach. J Geophys Res 83(NB6):2851–2862
3. Abousleiman Y, Cui L (1998) Poroelastic solutions in transversely isotropic media for wellbore and cylinder. Int J Solids Struct 35(34–35):4905–4929
4. Abousleiman YN, Kanj MY (2004) The generalized Lamé problem–part II: applications in poromechanics. J Appl Mech ASME 71(2):180–189
5. Abousleiman Y, Cheng AHD, Cui L, Detournay E, Roegiers JC (1996) Mandel's problem revisited. Géotechnique 46(2):187–195

6. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1996) Poroviscoelastic analysis of borehole and cylinder problems. Acta Mechanica 119(1–4):199–219
7. Abousleiman Y, Cui L, Ekbote S, Zaman M, Roegiers JC, Cheng AHD (1997) Applications of time-dependent pseudo-3D stress analysis in evaluating wellbore stability. Int J Rock Mech Min Sci 34(3–4):paper no. 1
8. Abramowitz M, Stegun IA (1965) Handbook of mathematical functions: with formulas, graphs, and mathematical tables. Dover, New York, 1046pp
9. Agbezuge LK, Deresiewicz H (1974) Indentation of a consolidating half-space. Isr J Technol 12(5–6):322–338
10. Agbezuge LK, Deresiewicz H (1975) Consolidation settlement of a circular footing. Isr J Technol 13(4):264–269
11. Amadei B (1983) Rock anisotropy and the theory of stress measurements. Springer, Berlin/Heidelberg, 478pp
12. Apirathvorakij V, Karasudhi P (1980) Quasi-static bending of a cylindrical elastic bar partially embedded in a saturated elastic half-space. Int J Solids Struct 16(7):625–644
13. Atkinson C, Craster RV (1991) Plane strain fracture in poroelastic media. Proc R Soc Lond Ser A Math Phys Eng Sci 434:605–633
14. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2): 155–164
15. Biot MA (1941) Consolidation settlement under a rectangular load distribution. J Appl Phys 12(5):426–430
16. Biot MA (1964) Theory of buckling of a porous slab and its thermoelastic analogy. J Appl Mech 31(2):194–198
17. Biot MA, Clingan FM (1941) Consolidation settlement of a soil with an impervious top surface. J Appl Phys 12(7):578–581
18. Biot MA, Clingan FM (1942) Bending settlement of a slab resting on a consolidating foundation. J Appl Phys 13(1):35–40
19. Booker JR (1974) The consolidation of a finite layer subject to surface loading. Int J Solids Struct 10:1053–1065
20. Booker JR (1974) Time dependent strain following faulting of a porous medium. J Geophys Res 79(14):2037–2044
21. Booker JR, Carter JP (1986) Analysis of a point sink embedded in a porous elastic half space. Int J Numer Anal Methods Geomech 10(2):137–150
22. Booker JR, Carter JP (1987) Elastic consolidation around a point sink embedded in a half-space with anisotropic permeability. Int J Numer Anal Methods Geomech 11(1):61–77
23. Booker JR, Small JC (1982) Finite layer analysis of consolidation. I. Int J Numer Anal Methods Geomech 6(2):151–171
24. Booker JR, Small JC (1982) Finite layer analysis of consolidation. II. Int J Numer Anal Methods Geomech 6(2):173–194
25. Booker JR, Small JC (1987) A method of computing the consolidation behavior of layered soils using direct numerical inversion of Laplace transforms. Int J Numer Anal Methods Geomech 11(4):363–380
26. Boone TJ, Wawrzynek PA, Ingraffea AR (1986) Simulation of the fracture process in rock with application to hydrofracturing. Int J Rock Mech Min Sci 23(3):255–265
27. Bougacha S, Roesset JM, Tassoulas JL (1993) Dynamic stiffness of foundations on fluid-filled poroelastic stratum. J Eng Mech ASCE 119(8):1649–1662
28. Bougacha S, Tassoulas JL, Roesset JM (1993) Analysis of foundations on fluid-filled poroelastic stratum. J Eng Mech ASCE 119(8):1632–1648
29. Bradley WB (1979) Failure of inclined boreholes. J Energy Resour Technol ASME 101(4):232–239
30. Brady BHG (1979) Direct formulation of the boundary element method of stress analysis for complete plane strain. Int J Rock Mech Min Sci 16(4):235–244

31. Brady BHG, Bray JW (1978) Boundary element method for determining stresses and displacements around long openings in a triaxial stress field. Int J Rock Mech Min Sci 15(1):21–28
32. Carslaw HS, Jaeger JC (1959) Conduction of heat in solids. Oxford University Press, Oxford, 510pp
33. Carter JP, Booker JR (1982) Elastic consolidation around a deep circular tunnel. Int J Solids Struct 18(12):1059–1074
34. Carter JP, Booker JR (1983) Creep and consolidation around circular openings in infinite media. Int J Solids Struct 19(8):663–675
35. Cheng AHD (1981) Boundary integral equation formulation for porous-elasticity with applications in soil mechanics and geophysics. Doctoral thesis, Cornell University, 155pp
36. Cheng AHD (1998) On generalized plane strain poroelasticity. Int J Rock Mech Min Sci 35(2):183–193
37. Cheng AHD, Detournay E (1988) A direct boundary element method for plane-strain poroelasticity. Int J Numer Anal Methods Geomech 12(5):551–572
38. Cheng AHD, Detournay E (1998) On singular integral equations and fundamental solutions of poroelasticity. Int J Solids Struct 35(34–35):4521–4555
39. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to soil consolidation. Int J Numer Methods Eng 20(2):255–278
40. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to fracture propagation. Int J Numer Methods Eng 20(2):279–296
41. Cheng AHD, Liu PLF (1986) Seepage force on a pipeline buried in a poroelastic seabed under wave loadings. Appl Ocean Res 8(1):22–32
42. Cheng AHD, Predeleanu M (1987) Transient boundary element formulation for linear poroelasticity. Appl Math Model 11(4):285–290
43. Cheng AHD, Skalak R, Weinbaum S, Ni JCY (1984) Consolidation of poroelastic arterial tissue. In: Boresi AP, Chong KP (eds) Proceedings of the 5th ASCE/EMD specialty conference on engineering mechanics in civil engineering, University of Wyoming, Laramie, pp 891–894
44. Chiarella C, Booker JR (1975) The time-settlement behavior of a rigid die resting on a deep clay layer. Q J Mech Appl Math 28:317–328
45. Churchill RV (1972) Operational mathematics, 3rd edn. McGraw-Hill, New York, 481pp
46. Cleary MP (1977) Fundamental solutions for a fluid-saturated porous solid. Int J Solids Struct 13(9):785–806
47. Cleary MP (1978) Moving singularities in elasto-diffusive solids with applications to fracture propagation. Int J Solids Struct 14(2):81–97
48. Colmenares LB, Zoback MD (2002) A statistical evaluation of intact rock failure criteria constrained by polyaxial test data for five different rocks. Int J Rock Mech Min Sci 39(6): 695–729
49. Craster RV, Atkinson C (1992) Shear cracks in thermoplastic and poroelastic media. J Mech Phys Solids 40(4):887–924
50. Cryer CW (1963) A comparison of the three-dimensional consolidation theories of Biot and Terzaghi. Q J Mech Appl Math 16(4):401–412
51. Cui L, Cheng AHD, Abousleiman Y (1997) Poroelastic solution for an inclined borehole. J Appl Mech ASME 64(1):32–38
52. Cui L, Abousleiman Y, Cheng AHD, Roegiers JC (1999) Time-dependent failure analysis of inclined boreholes in fluid-saturated formations. J Energy Resour Technol ASME 121(1): 31–39
53. Cui L, Abousleiman Y, Ekbote S, Roegiers JC, Zaman M (1999) A software for poroelastic analyses of borehole stability. In: Latin American and Caribbean petroleum engineering conference, SPE 54013, Caracas
54. Dalrymple RA, Liu PLF (1982) Gravity wave over a poro-elastic seabed. In: Proceedings of Ocean structural dynamics symposium, ASCE, Covallis, pp 181–195

55. Dean RG, Dalrymple RA (1991) Water wave mechanics for engineers and scientists. World Scientific, Singapore/Teaneck, 353pp
56. de Josselin de Jong G (1953) Consolidation around pore pressure meters. J Appl Phys 24: 922–928
57. de Josselin de Jong G (1957) Application of stress functions to consolidation problems. In: Proceedings of the 4th international conference on soil mechanics and foundation engineering, London, pp 320–323
58. de Leeuw EH (1965) The theory of three-dimensional consolidation applied to cylindrical bodies. In: Proceedings of 6th international conference on soil mechanics and foundation engineering, Montreal, vol 1, pp 287–290
59. Deresiewicz H (1977) Indentation of a consolidating half-space. 2. Effect of Poisson's ratio. Isr J Technol 15(3):89–97
60. Desai CS, Siriwardane HJ (1984) Constitutive laws for engineering materials, with emphasis on geologic materials. Prentice-Hall, Englewood Cliffs, 468pp
61. Detournay E, Cheng AHD (1987) Poroelastic solution of a plane-strain point displacement discontinuity. J Appl Mech ASME 54(4):783–787
62. Detournay E, Cheng AHD (1988) Poroelastic response of a borehole in a non-hydrostatic stress field. Int J Rock Mech Min Sci 25(3):171–182
63. Detournay E, Cheng AHD (1991) Plane strain analysis of a stationary hydraulic fracture in a poroelastic medium. Int J Solids Struct 27(13):1645–1662
64. Detournay E, Cheng AHD (1993) Fundamentals of poroelasticity. In: Fairhurst C (ed) Comprehensive rock engineering: principles, practice and projects. Analysis and design method, vol II. Pergamon Press, Oxford/New York, pp 113–171
65. Drucker DC, Prager W (1952) Soil mechanics and plastic analysis or limit design. Q Appl Math 10(2):157–165
66. Eseme E, Urai JL, Krooss BM, Littke R (2007) Review of mechanical properties of oil shales: implications for exploitation and basin modelling. Oil Shale 24(2):159–174
67. Ewy RT, Cook NGW (1990) Deformation and fracture around cylindrical openings in rock–I. Observations and analysis of deformations. Int J Rock Mech Min Sci 27(5):387–407
68. Fjær E, Holt RM, Horsrud P, Raaen AM, Risnes R (2002) Petroleum related rock mechanics, 2nd edn. Elsevier, Amsterdam/Boston, 514pp
69. Gaszyński J (1976) Certain solution of dual integral-equations and its application to contact problems of consolidation. Arch Mech 28(1):75–88
70. Gaszyński J, Szefer G (1978) Axisymmetric problem of punch for consolidating semi-space with mixed boundary permeability conditions. Arch Mech 30(1):17–26
71. Geertsma J (1966) Problems of rock mechanics in petroleum production engineering. In: Proceedings of 1st congress of international society of rock mechanics, Lisbon, pp 585–594
72. Geertsma J (1973) Land subsidence above compacting oil and gas reservoirs. J Pet Technol 25(6):734–744
73. Gibson RE, McNamee J (1963) A three-dimensional problem of consolidation of a semi-infinite clay stratum. Q J Mech Appl Math 16(1):115–127
74. Gibson RE, Schiffman RL, Pu SL (1970) Plane strain and axially symmetric consolidation of a clay layer on a smooth impervious base. Q J Mech Appl Math 23:505–519
75. Gough DI, Bell JS (1982) Stress orientations from borehole wall fractures with examples from Colorado, East Texas, and Northern Canada. Can J Earth Sci 19(7):1358–1370
76. Haimson B, Fairhurst C (1967) Initiation and extension of hydraulic fractures in rocks. Soc Pet Eng J 7:310–318
77. Harnpattanapanich T, Vardoulakis I (1987) Numerical Laplace-Fourier transform inversion technique for layered soil consolidation problems. 2. Gibson soil layer. Int J Numer Anal Methods Geomech 11(1):103–112
78. Haskell NA (1953) The dispersion of surface waves on multilayered media. Bull Seismol Soc Am 43(1):17–34
79. Hearmon RFS (1961) An introduction to applied anisotropic elasticity. Oxford University Press, London, 135pp

80. Heinrich G, Desoyer K (1961) Theorie of dreidimensionaler setzungsvorgänge in tonschichten (Theory of three-dimensional settlement processes in clay). Ing-Arch 30(4): 225–253
81. Helm DC (1987) Three-dimensional consolidation theory in terms of the velocity of solids. Géotechnique 37(3):369–392
82. Hoang S, Abousleiman Y, Al-Tahini AM (2010) Multilaterals drilling and sustainable openhole production from theory to field-case studies. SPE J 15(3):878–892
83. Hoek E, Brown ET (1980) Empirical strength criterion for rock masses. J Geotech Eng Div ASCE 106(9):1013–1035
84. Hoek E, Brown ET (1997) Practical estimates of rock mass strength. Int J Rock Mech Min Sci 34(8):1165–1186
85. Horgan CO, Miller KL (1994) Antiplane shear deformations for homogeneous and inhomogeneous anisotropic linearly elastic solids. J Appl Mech ASME 61(1):23–29
86. Huang LH, Song CH (1993) Dynamic response of poroelastic bed to water waves. J Hydraul Eng ASCE 119(9):1003–1020
87. Hubbert MK, Willis DG (1957) Mechanics of hydraulic fracturing. Trans Am Inst Min Eng 210:153–166
88. Ishijima Y, Roegiers JC (1983) Fracture initiation and breakdown pressure—are they similar? In: 24th US symposium rock mechanics, College Station, pp 761–772
89. Jaeger JC, Cook NGW, Zimmerman C (2007) Fundamentals of rock mechanics, 4th edn. Wiley-Blackwell, Malden, 488pp
90. Jeng DS (2000) On calculating the length of a short-crested wave over a porous seabed. Appl Ocean Res 22(2):63–73
91. Jeng DS (2001) Wave dispersion equation in a porous seabed. Ocean Eng 28(12):1585–1599
92. Jeng DS (2001) Mechanism of the wave-induced seabed instability in the vicinity of a breakwater: a review. Ocean Eng 28(5):537–570
93. Jeng DS, Barry DA, Li L (2001) Water wave-driven seepage in marine sediments. Adv Water Res 24(1):1–10
94. Kameo Y, Adachi T, Hojo M (2008) Transient response of fluid pressure in a poroelastic material under uniaxial cyclic loading. J Mech Phys Solids 56(5):1794–1805
95. Kanj MY, Abousleiman YN (2004) The generalized Lamé problem–part I: coupled poromechanical solutions. J Appl Mech ASME 71(2):168–179
96. Kanj M, Abousleiman Y, Ghanem R (2003) Poromechanics of anisotropic hollow cylinders. J Eng Mech ASCE 129(11):1277–1287
97. Kanok-Rukulchai W, Chau KT (1990) Point sink fundamental solutions for subsidence prediction. J Eng Mech ASCE 116(5):1176–1181
98. Kenyon DE (1976) Transient filtration in a porous elastic cylinder. J Appl Mech ASME 43(4):594–598
99. Kenyon DE (1979) Consolidation in transversely isotropic solids. J Appl Mech ASME 46(1):65–70
100. Kenyon DE (1979) Mathematical model of water flux through aortic tissue. Bull Math Biol 41(1):79–90
101. Kirsch EG (1898) Die Theorie der Elastizität und die Bedürfnisse der Festigkeitslehre (The theory of elasticity and its application to the strength of materials). Zeitschrift des Vereines deutscher Ingenieure 42:797–807
102. Lade PV (1977) Elasto-plastic stress-strain theory for cohesionless soil with curved yield surfaces. Int J Solids Struct 13(11):1019–1035
103. Lekhnitskii SG (1981) Theory of elasticity of an anisotropic body. MIR Publishers, Moscow, 430pp
104. Li LP, Schulgasser K, Cederbaum G (1995) Theory of poroelastic beams with axial diffusion. J Mech Phys Solids 43(12):2023–2042
105. Madsen OS (1978) Wave-induced pore pressures and effective stresses in a porous bed. Géotechnique 28(4):377–393

106. Mandel J (1950) Etude mathematique de la consolidation des sols (Mathematical study of consolidation of soils). Actes du Colloque International de Mecanique de Poitiers 4(9):9–19
107. Mandel J (1953) Consolidation des sols (etude mathematique) (Consolidation of soils (mathematical study)). Géotechnique 3:287–299
108. Mason DP, Solomon A, Nicolaysen LO (1991) Evolution of stress and strain during the consolidation of a fluid-saturated porous elastic sphere. J Appl Phys 70(9):4724–4740
109. McNamee J, Gibson RE (1960) Displacement functions and linear transforms applied to diffusion through porous elastic media. Q J Mech Appl Math 13(1):98–111
110. McNamee J, Gibson RE (1960) Plane strain and axially symmetric problems of the consolidation of a semi-infinite stratum. Q J Mech Appl Math 13(2):210–227
111. Miles AJ, Topping AD (1949) Stresses around a deep well. Trans AIME 179:186–191
112. Milne-Thomson LM (1962) Antiplane elastic systems. Academic, New York, 265pp
113. Mindlin RD, Cheng DH (1950) Nuclei of strain in the semi-infinite solid. J Appl Phys 21(9):926–930
114. Mu YK, Cheng AHD, Badiey M, Bennett R (1999) Water wave driven seepage in sediment and parameter inversion based on pore pressure data. Int J Numer Anal Methods Geomech 23(13):1655–1674
115. Nguyen VX, Abousleiman YN, Hoang S (2009) Analyses of wellbore instability in drilling through chemically active fractured-rock formations. SPE J 14(2):283–301
116. Niumpradit B, Karasudhi P (1981) Load transfer from an elastic pile to a saturated porous elastic soil. Int J Numer Anal Methods Geomech 5(2):115–138
117. Nowinski JL (1971) Bone articulations as systems of poroelastic bodies in contact. AIAA J 9(1):62–67
118. Nowinski JL, Davis CF (1970) A model of the human skull as a poroelastic spherical shell subjected to a quasistatic load. Math Biosci 8(3–4):397–416
119. Nowinski JL, Davis CF (1972) Flexure and torsion of bones viewed as anisotropic poroelastic bodies. Int J Eng Sci 10(12):1063–1079
120. Nur A, Booker JR (1972) Aftershocks caused by pore fluid flow? Science 175(4024):885–887
121. Pan E (1999) Green's functions in layered poroelastic half-spaces. Int J Numer Anal Methods Geomech 23(13):1631–1653
122. Paria G (1958) Axisymmetric consolidation for a porous elastic material containing a fluid. J Math Phys 36:338–346
123. Parton VZ (1965) An axially symmetrical problem in the theory of consolidation of compressible porous media saturated by a liquid. Sov Phys Dokl 10:127–130 (Transl. from Doklady Akademii Nauk SSSR 160(4):785–788)
124. Plumb RA, Cox JW (1987) Stress directions in eastern North America determined to 4.5 km from borehole elongation measurements. J Geophys Res-Solid Earth Planets 92(B6):4805–4816
125. Press WH, Teukolsky SA, Vetterling WT, Flannery BP (1992) Numerical recipes in Fortran, the art of scientific computing, 2nd edn. Cambridge University Press, Cambridge/New York
126. Puswewala UGA, Rajapakse RKND (1988) Axisymmetric fundamental solutions for a completely saturated porous elastic solid. Int J Eng Sci 26(5):419–436
127. Putnam JA (1949) Loss of wave energy due to percolation in a permeable sea bottom. Trans Am Geophys Union 30:349–356
128. Rajapakse RKND (1993) Stress-analysis of borehole in poroelastic medium. J Eng Mech ASCE 119(6), 1205–1227
129. Rajapakse RKND, Senjuntichai T (1993) Fundamental solutions for a poroelastic half-space with compressible constituents. J Appl Mech ASME 60(4):847–856
130. Rajapakse RKND, Senjuntichai T (1995) Dynamic response of a multilayered poroelastic medium. Earthq Eng Struct Dyn 24(5):703–722
131. Reddy PM, Tajuddin M (2000) Exact analysis of the plane-strain vibrations of thick-walled hollow poroelastic cylinders. Int J Solids Struct 37(25):3439–3456
132. Reid RO, Kajuira K (1957) On the damping of gravity waves over a permeable seabed. Trans Am Geophys Union 30:662–666

133. Rencis JJ, Huang Q (1992) Boundary element formulation for generalized plane strain. Eng Anal Bound Elem 9(3):263–271
134. Rice JR, Cleary MP (1976) Some basic stress diffusion solutions for fluid-saturated elastic porous media with compressible constituents. Rev Geophys 14(2):227–241
135. Rice JR, Simons DA (1976) Stabilization of spreading shear faults by coupled deformation-diffusion effects in fluid-infiltrated porous materials. J Geophys Res 81(29):5322–5334
136. Rice JR, Rudnicki JW, Simons DA (1978) Deformation of spherical cavities and inclusions in fluid-infiltrated elastic materials. Int J Solids Struct 14(4):289–303
137. Rudnicki JW (1981) On 'Fundamental solutions for a fluid-saturated porous solid' by M.P. Cleary. Int J Solids Struct 17(9):855–857
138. Rudnicki JW (1986) Fluid mass sources and point forces in linear elastic diffusive solids. Mech Mater 5(4):383–393
139. Rudnicki JW (1987) Plane strain dislocations in linear elastic diffusive solids. J Appl Mech ASME 54(3):545–552
140. Rudnicki JW (1996) Moving and stationary dislocations in poroelastic solids and applications to aseismic slip in the Earth's crust. In: Selvadurai APS (ed) Mechanics of poroelastic media. Kluwer Academic, Dordrecht/Boston, pp 3–22
141. Saada AS (1993) Elasticity, theory and applications, 2nd edn. Kreiger, Malabar, 775pp
142. Schiffman RL, Fungaroli AA (1965) Consolidation due to tangential loads. In: Proceedings 6th international conference on soil mechanics and foundation engineering, Montreal, pp 188–192
143. Schmitt DR, Zoback MD (1989) Poroelastic effects in the determination of the maximum horizontal principal stress in hydraulic fracturing tests—a proposed breakdown equation employing a modified effective stress relation for tensile failure. Int J Rock Mech Min Sci 26(6):499–506
144. Selvadurai APS (2007) The analytical method in geomechanics. Appl Mech Rev ASME 60(1–6):87–106
145. Selvadurai APS, Yue ZQ (1994) On the indentation of a poroelastic layer. Int J Numer Anal Methods Geomech 18(3):161–175
146. Senjuntichai T, Rajapakse RKND (1993) Transient response of a circular cavity in a poroelastic medium. Int J Numer Anal Methods Geomech 17(6):357–383
147. Senjuntichai T, Rajapakse RKND (1995) Exact stiffness method for quasi-statics of a multilayered poroelastic medium. Int J Solids Struct 32(11):1535–1553
148. Simons DA (1977) Boundary-layer analysis of propagating mode II cracks in porous elastic media. J Mech Phys Solids 25(2):99–115
149. Singh S, Rani S (2006) Plane strain deformation of a multi-layered poroelastic half-space by surface loads. J Earth Syst Sci 115(6):685–694
150. Sleath JFA (1984) Sea bed mechanics. Wiley, New York, 335pp
151. Sneddon IN (1995) Fourier transforms. Dover, New York, 542pp
152. Sokolnikoff IS (1956) Mathematical theory of elasticity, 2nd edn. McGraw-Hill, New York, 476pp
153. Szefer G, Gaszyński J (1975) Axisymmetric punch problem under condition of consolidation. Arch Mech 27(3):497–515
154. Taber LA (1992) A theory for transverse deflection of poroelastic plates. J Appl Mech ASME 59(3):628–634
155. Taber LA (1992) Axisymmetric deformation of poroelastic shells of revolution. Int J Solids Struct 29(24):3125–3143
156. Taguchi I, Kurashige M (2002) Fundamental solutions for a fluid-saturated, transversely isotropic, poroelastic solid. Int J Numer Anal Methods Geomech 26(3):299–321
157. Tajuddin M (1978) Vibrations in fluid-saturated porous elastic cylinders. Revue Roumaine des Sciences Techniques–Série de Mécanique Appliquée 23:371–379
158. Terzaghi K (1923) Die Berechnung der Durchlässigkeitsziffer des Tones aus dem Verlauf der hydrodynamischen Spannungserscheinungen (A method of calculating the permeability of clay from the history of hydrodynamic stress variation). Sitzungsber. d. Akad. d. Wiss., Wien Math.-Naturwiss. Kl., Abt. IIa 132(3/4):125–138

159. Terzaghi K (1936) The shearing resistance of saturated soils and the angle between the planes of shear. In: First international conference on soil mechanics, vol 1. Harvard University, Cambridge, pp 54–56
160. Thomson WT (1950) Transmission of elastic waves through a stratified solid medium. J Appl Phys 21(2):89–93
161. Timoshenko SP, Goodier JN (1970) Theory of elasticity, 3rd edn. McGraw-Hill, New York, 567pp
162. Tran MH, Abousleiman Y, Nguyen VX (2011) The effects of filter-cake buildup and time-dependent properties on the stability of inclined wellbores. SPE J 16(4):1010–1028
163. Vardoulakis I, Harnpattanapanich T (1986) Numerical Laplace Fourier-transform inversion technique for layered-soil consolidation problems. 1. Fundamental-solutions and validation. Int J Numer Anal Methods Geomech 10(4):347–365
164. Verruijt A (2014) Theory and problems of poroelasticity, 263pp. http://geo.verruijt.net
165. Wang J, Fang S (2003) State space solution of non-axisymmetric Biot consolidation problem for multilayered porous media. Int J Eng Sci 41(15):1799–1813
166. Wang Z-H, Prévost JH, Coussy O (2009) Bending of fluid-saturated linear poroelastic beams with compressible constituents. Int J Numer Anal Methods Geomech 33(4):425–447
167. Woodland DC (1990) Borehole instability in the Western Canadian overthrust belt. SPE Drill Eng 5:27–33
168. Wu ZY, Li SP (1990) The generalized plane strain problem and its application in three-dimensional stress measurement. Int J Rock Mech Min Sci 27(1):43–49
169. Yamamoto T (1981) Wave-induced pore pressures and effective stresses in inhomogeneous seabed foundations. Ocean Eng 8(1):1–16
170. Yamamoto T (1983) On the response of a Coulomb-damped poroelastic bed to water waves. Mar Geotechnol 5(2):93–130
171. Yamamoto T, Koning HL, Sellmeijer H, Van Hijum E (1978) On the response of a poro-elastic bed to water waves. J Fluid Mech 87:193–206
172. Yue ZQ, Selvadurai APS (1994) On the asymmetric indentation of a consolidating poroelastic half space. Appl Math Model 18(4):170–185
173. Zhang DJ, Cowin SC (1994) Oscillatory bending of a poroelastic beam. J Mech Phys Solids 42(10):1575–1599
174. Zoback MD, Moos D, Mastin L, Anderson RN (1985) Well bore breakouts and in situ stress. J Geophys Res 90(B7):5523–5530

Chapter 8
Fundamental Solution and Integral Equation

We will now show, that if any two functions whatever are taken, satisfying these conditions, it will always be in our power to assign one, and only one value of ρ, which will produce them for corresponding potential functions. For this we may remark, that the equation (3) art. 3 being applied to the space within the body, becomes, by making $U = \frac{1}{r}$

$$\int \frac{d\sigma}{r}\left(\frac{\overline{dV}}{dw}\right) = \int d\sigma\,\overline{V}\left(\frac{d\frac{1}{r}}{dw}\right) - 4\pi V;$$

—George Green (1828)

In the mathematical terminology, a *fundamental solution* is a singular solution of a linear partial differential equation that is not required to satisfy boundary conditions. A *Green's function*, on the other hand, is the singular solution tied to a certain domain geometry and boundary condition. For this reason, a fundamental solution is also called a *free space Green's function*.

Using the symbolism of a generalized function, known as the *Dirac delta function*, δ, a fundamental solution is a solution of linear partial differential equation with the Dirac delta as its right hand side. From physical considerations, Dirac delta can be used to approximate a forcing applied to a small region. For example, a fluid mass injected through an injection well, whose radius is small compared to the formation that it is flooding, can be considered as a *point source*. Similarly, a force acting on a small area can be considered as a *point force*, and a defect in crystal structure (dislocation) or a local slippage of a geological fault can be approximated as a *displacement discontinuity*, etc. Hence fundamental solution is not just an abstract mathematical construct; it has its root in physics and can be used to model physical phenomena.

The use of fundamental solution to simulate a physical phenomenon and to solve mathematical problems can be traced to George Green [30] (see Sect. F.7 for a biography). Green utilized the fundamental solution $1/r$ of the Laplace equation to model the electrical and magnetic *potential* created by concentrated electrostatic and magnetic charges. The potential is a mathematical construct, whose derivative gives the force associated with the field. The use of fundamental solution to solve mathematical problems is closely tied to the integral equations known as *Green's*

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_8

identities. Particularly, the third identify, shown in its original form in the prologue of the chapter, provides the general solution to the boundary value problem known as the *Dirichlet problem*.

In this chapter, we extend the classical work on fundamental solutions and integral equations for potential (satisfying Laplace equation) and elasticity problems to poroelasticity. We shall demonstrate that the several varieties of integral equations derived from different origins and the many different fundamental solutions are intricately related. The presentation begins with the integral equations for general anisotropic poroelastic materials. The fundamental solutions are then derived, but only for the case of isotropy. The presentation follows the work of Cheng and Detournay [16].

8.1 Reciprocal Theorem

8.1.1 Green's Second Identity

Green's second identity gives the reciprocal relation between any two sufficiently smooth functions ϕ and ψ

$$\int_\Gamma \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) \, d\vec{x} = \int_\Omega \left(\phi \nabla^2 \psi - \psi \nabla^2 \phi \right) \, d\vec{x} \tag{8.1}$$

where Γ is a closed, smooth surface [14], Ω is the region enclosed by it, and n is the unit outward normal of Γ. If ϕ and ψ satisfy the Poisson equations

$$\nabla^2 \phi = \rho_\phi \tag{8.2}$$

$$\nabla^2 \psi = \rho_\psi \tag{8.3}$$

then (8.1) becomes

$$\int_\Gamma \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) \, d\vec{x} = \int_\Omega \left(\phi \, \rho_\psi - \psi \rho_\phi \right) \, d\vec{x} \tag{8.4}$$

We can in fact generalize the above relation from the Laplacian operator to any linear partial differential operators [43]

$$\int_\Gamma \left(\phi \, \mathfrak{B}^* \{\psi\} - \psi \, \mathfrak{B}\{\phi\} \right) \, d\vec{x} = \int_\Omega \left(\phi \, \mathfrak{L}^* \{\psi\} - \psi \, \mathfrak{L}\{\phi\} \right) \, d\vec{x} \tag{8.5}$$

In the above $\mathfrak{L}$ is a linear partial differential operator, $\mathfrak{L}^*$ is its *adjoint operator*, $\mathfrak{B}$ is the generalized boundary normal derivative, and $\mathfrak{B}^*$ is its adjoint operator. The left hand of (8.5) is the consequence of integration by parts on the right hand side operators. Equation (8.5) is then the *generalized Green's reciprocal theorem*.

As an example, we consider the general second order linear partial differential equation in two dimensions

$$\mathfrak{L}\{\phi\} = A\frac{\partial^2\phi}{\partial x^2} + 2B\frac{\partial^2\phi}{\partial x\partial y} + C\frac{\partial^2\phi}{\partial y^2} + D\frac{\partial\phi}{\partial x} + E\frac{\partial\phi}{\partial y} + F\phi \tag{8.6}$$

where the coefficients A, B, ..., and F are functions of x and y. The generalized Green's second identity in the form of (8.6) exists with the definition of the operators [32]

$$\mathfrak{L}^*\{\psi\} = \frac{\partial^2 A\psi}{\partial x^2} + 2\frac{\partial^2 B\psi}{\partial x\partial y} + \frac{\partial^2 C\psi}{\partial y^2} - \frac{\partial D\psi}{\partial x} - \frac{\partial E\psi}{\partial y} + F\psi \tag{8.7}$$

$$\mathfrak{B}\{\phi\} = \left(A\frac{\partial\phi}{\partial x} + 2B\frac{\partial\phi}{\partial y}\right)n_x + \left(C\frac{\partial\phi}{\partial y} + E\phi\right)n_y \tag{8.8}$$

$$\mathfrak{B}^*\{\psi\} = \left(\frac{\partial A\psi}{\partial x} - D\psi\right)n_x + \left(2\frac{\partial B\psi}{\partial x} + \frac{\partial C\psi}{\partial y}\right)n_y \tag{8.9}$$

where n_x and n_y are vector components of n.

8.1.2 Betti-Maxwell Reciprocal Theorem

The construction of integral equations for elasticity was based on the *Betti-Maxwell reciprocal theorem*. James Clerk Maxwell [41] was the first to formulate the theory for trusses, and stated: "*The extension in BC (a truss member) due to the tension along DE is always equal to the extension in DE due to the tension in BC*".

Enrico Betti [3] generalized Green's second identity (8.1) for the Laplacian operator to the theory of elasticity. It can be stated as follows: Given two independent elastic states $\{\vec{u}^{(1)}, \vec{t}^{(1)}, \vec{F}^{(1)}\}$ and $\{\vec{u}^{(2)}, \vec{t}^{(2)}, \vec{F}^{(2)}\}$ in static equilibrium, where $\vec{u}$ is the displacement, $\vec{t}$ is the boundary traction, and $\vec{F}$ is the body force, with Γ the surface enclosing an elastic body, and Ω the region enclosed by Γ, they satisfy the following reciprocal relation

$$\int_\Gamma [\vec{t}^{(1)} \cdot \vec{u}^{(2)} - \vec{t}^{(2)} \cdot \vec{u}^{(1)}]\, d\vec{x} = \int_\Omega [\vec{F}^{(2)} \cdot \vec{u}^{(1)} - \vec{F}^{(1)} \cdot \vec{u}^{(2)}]\, d\vec{x} \tag{8.10}$$

In the above, we have substituted the Navier operator of elasticity by its right hand side, the body force term, to arrive at the elasticity counterpart of (8.4). Rayleigh [48] further generalized the above theorem to elastodynamics in the frequency domain, and extended the forces and displacements to generalized forces and generalized displacements for electric circuit theory.

The physical background of *Betti's reciprocal theorem* (8.10) is based on the energy conservation principle for a *reversible* (*non-dissipative*) thermodynamic

process, or the *principle of least action*. In a reversible process, when two forces are applied to a body, making it to deform, transforming the work done into internal (strain) energy, it does not matter which force is applied first, and the end state is the same. This process applies to unloading as well—when the forces are removed, it does not matter the sequence of removal, and the body will restore to its original unloaded and undeformed state. If a material obeys this physical law, then the constitutive equations developed for it must contain this property.

8.1.3 Reciprocal Theorem of Poroelasticity

As the poroelasticity constitutive relations developed in Chaps. 3 through 5 were constructed based on the assumption of a reversible (elastic) process, the reciprocal relation similar to (8.10) is a built-in property of the theory. We shall give a brief proof of such relation.

We begin by summarizing the anisotropic constitutive equations developed in Chap. 5 as follows:

$$\sigma_{ij} = M^u_{ijkl} e_{kl} - \alpha_{ij} M \zeta \tag{8.11}$$

$$p = M\zeta - \alpha_{ij} M e_{ij} \tag{8.12}$$

We note the symmetry of the following material coefficients:

$$M^u_{ijkl} = M^u_{jikl} = M^u_{ijlk} = M^u_{klij} \tag{8.13}$$

$$\alpha_{ij} = \alpha_{ji} \tag{8.14}$$

Assuming that $\{\sigma^{(1)}_{ij}, p^{(1)}, e^{(1)}_{ij}, \zeta^{(1)}\}$ and $\{\sigma^{(2)}_{ij}, p^{(2)}, e^{(2)}_{ij}, \zeta^{(2)}\}$ are two independent states of stresses and strains of the same material, we can show that

$$\begin{aligned} &\sigma^{(1)}_{ij} e^{(2)}_{ij} + p^{(1)} \zeta^{(2)} \\ &\quad = M^u_{ijkl} e^{(1)}_{kl} e^{(2)}_{ij} - \alpha_{ij} M \zeta^{(1)} e^{(2)}_{ij} - \alpha_{ij} M e^{(1)}_{ij} \zeta^{(2)} + M \zeta^{(1)} \zeta^{(2)} \end{aligned} \tag{8.15}$$

In the above we have substituted $\sigma^{(1)}_{ij}$ and $p^{(1)}$ by their constitutive relation (8.11) and (8.12). Similarly, we can write

$$\begin{aligned} &\sigma^{(2)}_{ij} e^{(1)}_{ij} + p^{(2)} \zeta^{(1)} \\ &\quad = M^u_{ijkl} e^{(2)}_{kl} e^{(1)}_{ij} - \alpha_{ij} M \zeta^{(2)} e^{(1)}_{ij} - \alpha_{ij} M e^{(2)}_{ij} \zeta^{(1)} + M \zeta^{(2)} \zeta^{(1)} \end{aligned} \tag{8.16}$$

By noting that

$$M^u_{ijkl}\, e^{(1)}_{kl}\, e^{(2)}_{ij} = M^u_{klij}\, e^{(1)}_{ij}\, e^{(2)}_{kl} = M^u_{ijkl}\, e^{(1)}_{ij}\, e^{(2)}_{kl} \tag{8.17}$$

in which we have utilized the symmetry property of the material coefficient (8.13), it is clear that such reciprocal relation exists

$$\sigma^{(1)}_{ij} e^{(2)}_{ij} + p^{(1)}\zeta^{(2)} = \sigma^{(2)}_{ij} e^{(1)}_{ij} + p^{(2)}\zeta^{(1)} \tag{8.18}$$

which is the reciprocal relation (2.23) introduced earlier for isotropy.

To derive the poroelasticity counterpart of Betti's reciprocal theorem (8.10), we shall reintroduce the physical laws presented in Sects. 6.1 and 6.2 by explicitly considering the body force and source terms

- Equilibrium equation

$$\sigma_{ij,j} = -F_i \tag{8.19}$$

- Darcy's law

$$q_i = -\kappa_{ij}(p_{,j} - f_j) \tag{8.20}$$

- Continuity equation

$$\frac{\partial \zeta}{\partial t} + q_{i,i} = \gamma \tag{8.21}$$

In the above, F_i is the total body force (force per unit volume), f_i the fluid body force, and γ the fluid source (volume of injected fluid per unit volume of porous medium per unit time). These body force terms were previously neglected because when solving for boundary value problems, we typically consider only the perturbed solution from its initial at rest stress and strain state due to gravity (body) force. Here we introduce these body forces not for the purpose of modeling the gravity effect, but for introducing singular forcing functions, such as a concentrated force acting at a single point, based on the mathematical concept of a Dirac delta function. Similar concept is applied to the fluid source term in (8.21).

To further extend the above concept of singular forcing term to accommodate a *displacement discontinuity*, we shall introduce the definition of a total strain e^*_{ij} as the sum of an elastic strain e_{ij} and an *eigenstrain* E_{ij}, as

$$e^*_{ij} = e_{ij} + E_{ij} \tag{8.22}$$

The term eigenstrain is a generic name given to such nonelastic strains as thermal expansion, initial strains, plastic strains, and misfit strains by Mura [44]. Here we shall use it to introduce a singularity similar to the *nucleus of strain* concept of Love

[40]. The singularity, however, differs from that of Love's, and corresponds more closely to that introduced by Nedelec [45] and Becache et al. [2]. Replacing the elastic strain in (8.18) by the total strain using the relation (8.22), and then drop the asterisk superscript from the total strain notation for the simplicity, we obtain

$$\sigma_{ij}^{(1)} e_{ij}^{(2)} + p^{(1)} \zeta^{(2)} - \sigma_{ij}^{(1)} E_{ij}^{(2)} = \sigma_{ij}^{(2)} e_{ij}^{(1)} + p^{(2)} \zeta^{(1)} - \sigma_{ij}^{(2)} E_{ij}^{(1)} \tag{8.23}$$

We can interpret (8.23) as follows. When $E_{ij} = 0$, the stresses and strains are regular, and (8.23) is exactly (8.18). When $E_{ij} \neq 0$, e_{ij} is actually e_{ij}^* and the stresses and strains are singular solutions corresponding to the introduction of the singular forcing function.

We notice that the derivations leading to (8.18) and (8.23) are pure mathematical manipulation, and the two systems can be assigned to different space and time without changing the validity of these equations. We shall set the first system to be at one location and time, $(\vec{\chi}, \tau)$, and the second system at another, $(\vec{\chi} - \vec{x}, t - \tau)$. Equation (8.23) can be integrated over a domain Ω, $\vec{\chi} \in \Omega$, and time τ to obtain

$$\begin{aligned} \int_0^t \int_\Omega & \left[\sigma_{ij}^{(1)}(\vec{\chi}, \tau)\, e_{ij}^{(2)}(\vec{\chi} - \vec{x}, t - \tau) - \sigma_{ij}^{(2)}(\vec{\chi} - \vec{x}, t - \tau)\, e_{ij}^{(1)}(\vec{\chi}, \tau) \right] \\ & + \left[p^{(1)}(\vec{\chi}, \tau)\, \zeta^{(2)}(\vec{\chi} - \vec{x}, t - \tau) - p^{(2)}(\vec{\chi} - \vec{x}, t - \tau)\, \zeta^{(1)}(\vec{\chi}, \tau) \right] \\ & + \left[E_{ij}^{(1)}(\vec{\chi}, \tau) \sigma_{ij}^{(2)}(\vec{\chi} - \vec{x}, t - \tau) - E_{ij}^{(2)}(\vec{\chi} - \vec{x}, t - \tau) \sigma_{ij}^{(1)}(\vec{\chi}, \tau) \right] d\vec{\chi}\, d\tau = 0 \end{aligned} \tag{8.24}$$

We notice the following relation

$$\left(\sigma_{ij}^{(1)} u_i^{(2)} \right)_{,j} = \sigma_{ij}^{(1)} u_{i,j}^{(2)} + \sigma_{ij,j}^{(1)} u_i^{(2)} = \sigma_{ij}^{(1)} e_{ij}^{(2)} - F_i^{(1)} u_i^{(2)} \tag{8.25}$$

where we have utilized the equilibrium equation (8.19), which brings in the body force term. The symmetry of σ_{ij} and the summation of indices i and j allow the conversion of $u_{i,j}$ to e_{ij}. Integrating the continuity equation (8.21) with respect to time, we obtain

$$\zeta = -w_{i,i} + Q \tag{8.26}$$

in which we assumed that all initial values are zero, and

$$w_i = \int_0^t q_i \, dt \tag{8.27}$$

is the specific relative fluid to solid displacement vector

$$w_i = \phi \, (U_i - u_i) \tag{8.28}$$

and

$$Q = \int_0^t \gamma \, dt \tag{8.29}$$

is the volume of injected fluid due to the source γ. It can then be shown that

$$\begin{aligned} p^{(1)}\zeta^{(2)} &= -\left(p^{(1)}w_i^{(2)}\right)_{,i} + p_{,i}^{(1)}w_i^{(2)} + p^{(1)}Q^{(2)} \\ &= -\left(p^{(1)}w_i^{(2)}\right)_{,i} - b_{ij}\, q_i^{(1)}w_j^{(2)} + w_i^{(2)}f_i^{(1)} + p^{(1)}Q^{(2)} \end{aligned} \tag{8.30}$$

where we have utilized Darcy's law (8.20), and defined the *resistivity tensor* as the matrix inverse of permeability tensor

$$[b_{ij}] = [\kappa_{ij}]^{-1} \tag{8.31}$$

which is symmetric. We note that the results presented in (8.25) and (8.30) also apply to their conjugate parts by switching the system index (1) and (2). Substituting the above and their conjugate parts into (8.24), we obtain

$$\begin{aligned} &\int_0^t \int_\Omega \left[\left(\sigma_{ij}^{(1)}u_i^{(2)}\right)_{,j} - \left(\sigma_{ij}^{(2)}u_i^{(1)}\right)_{,j}\right] - \left[\left(p^{(1)}w_i^{(2)}\right)_{,i} - \left(p^{(2)}w_i^{(1)}\right)_{,i}\right] \\ &\quad + \left[F_i^{(1)}u_i^{(2)} - F_i^{(2)}u_i^{(1)}\right] + \left[f_i^{(1)}w_i^{(2)} - f_i^{(2)}w_i^{(1)}\right] - \left[Q^{(1)}p^{(2)} - Q^{(2)}p^{(1)}\right] \\ &\quad + \left[E_{ij}^{(1)}\sigma_{ij}^{(2)} - E_{ij}^{(2)}\sigma_{ij}^{(1)}\right] - \left[b_{ij}\, q_i^{(1)}w_j^{(2)} - b_{ij}\, q_i^{(2)}w_j^{(1)}\right] d\vec{\chi}\, d\tau = 0 \end{aligned} \tag{8.32}$$

We note that the last group of the integrand can be transformed as follows:

$$\begin{aligned} b_{ij}\left(q_i^{(1)}w_j^{(2)} - q_i^{(2)}w_j^{(1)}\right) &= b_{ij}\left(q_i^{(1)}w_j^{(2)} - q_j^{(2)}w_i^{(1)}\right) \\ &= b_{ij}\left[\frac{\partial w_i^{(1)}(\vec{\chi},\tau)}{\partial\tau}\, w_j^{(2)}(\vec{\chi}-\vec{x},t-\tau) - \frac{\partial w_j^{(2)}(\vec{\chi}-\vec{x},t-\tau)}{\partial(t-\tau)}\, w_i^{(1)}(\vec{\chi},\tau)\right] \\ &= b_{ij}\,\frac{\partial w_i^{(1)}(\vec{\chi},\tau)\, w_j^{(2)}(\vec{\chi}-\vec{x},t-\tau)}{\partial\tau} \end{aligned} \tag{8.33}$$

Integrating the above we find:

$$\begin{aligned} &\int_0^t b_{ij}\,\frac{\partial w_i^{(1)}(\vec{\chi},\tau)\, w_j^{(2)}(\vec{\chi}-\vec{x},t-\tau)}{\partial\tau}\, d\tau \\ &\quad = b_{ij}\, w_i^{(1)}(\vec{\chi},\tau)\, w_j^{(2)}(\vec{\chi}-\vec{x},t-\tau)\Big|_{\tau=0}^{\tau=t} = 0 \end{aligned} \tag{8.34}$$

where we note that $w_i(0) = 0$, consistent with the assumption in (8.26). Hence the last part of the integrand of (8.32) vanishes. Finally, for the divergence terms contained in (8.25) and (8.30), we can apply the divergence theorem to convert volume integral to surface integral

$$\int_\Omega \left(\sigma_{ij}^{(1)} u_i^{(2)}\right)_{,j} d\vec{\chi} = \int_\Gamma \sigma_{ij}^{(1)} n_j u_i^{(2)} \, d\vec{\chi} \tag{8.35}$$

$$\int_\Omega \left(p^{(1)} w_i^{(2)}\right)_{,i} d\vec{\chi} = \int_\Gamma p^{(1)} n_i w_i^{(2)} \, d\vec{\chi} \tag{8.36}$$

in which n_i is the component of the unit outward normal of Γ. Utilizing the above results, we can transform (8.32) to the following

$$\begin{aligned}\int_0^t \int_\Gamma \left[\left(\sigma_{ij}^{(1)} n_j u_i^{(2)} - \sigma_{ij}^{(2)} n_j u_i^{(1)}\right) - \left(p^{(1)} w_i^{(2)} n_i - p^{(2)} w_i^{(1)} n_i\right)\right] d\vec{\chi}\, d\tau \\ + \int_0^t \int_\Omega \left[\left(F_i^{(1)} u_i^{(2)} - F_i^{(2)} u_i^{(1)}\right) + \left(f_i^{(1)} w_i^{(2)} - f_i^{(2)} w_i^{(1)}\right)\right. \\ \left. - \left(Q^{(1)} p^{(2)} - Q^{(2)} p^{(1)}\right) + \left(E_{ij}^{(1)} \sigma_{ij}^{(2)} - E_{ij}^{(2)} \sigma_{ij}^{(1)}\right)\right] d\vec{\chi}\, d\tau = 0\end{aligned} \tag{8.37}$$

The above equation is the *reciprocal theorem of poroelasticity*, which was derived in its present form by Cheng and Detournay [16], and in less general forms by Predeleanu [47], as well as by Cleary [20], and Cheng and Predeleanu [19].

8.2 Somigliana Integral Equation

8.2.1 Green's Third Identity

If we replace the function ψ in (8.4) by the fundamental solution of Laplace equation G, which satisfies

$$\nabla^2 G(\vec{\chi} - \vec{x}) = \delta(\vec{\chi} - \vec{x}) \tag{8.38}$$

where $\{\vec{x}, \vec{\chi}\} \in \Omega$, and $\delta(\vec{\chi} - \vec{x})$ is the Dirac delta function located at $\vec{x}$, and integrate with respect to $\vec{\chi}$, we obtain from (8.4) *Green's third identity*

$$\phi(\vec{x}) = \int_\Gamma \left[\phi(\chi) \frac{\partial G(\vec{\chi} - \vec{x})}{\partial n(\vec{\chi})} - G(\vec{\chi} - \vec{x}) \frac{\partial \phi(\vec{\chi})}{\partial n(\vec{\chi})}\right] d\vec{\chi} \tag{8.39}$$

where we have assumed $\rho_\phi = 0$. Equation (8.39) formally gives the solution of ϕ of a boundary value problem.

8.2.2 Elasticity

For elasticity, utilizing the singular solution known as the *hexapole*[1] [40]

$$\vec{u}^{\,q} = \frac{1-2\nu}{8\pi G(1-\nu)} \nabla \left(\frac{1}{r}\right) \tag{8.40}$$

as one of the elastic states in Betti's reciprocal theorem (8.10), Betti [3, 4] obtained the following integral equation

$$e = \int_{\Gamma} (\vec{t} \cdot \vec{u}^{\,q} - \vec{t}^{\,q} \cdot \vec{u}\,)\, d\vec{\chi} \tag{8.41}$$

where we have neglected the body force term. The above equation, however, is not very useful for the solution of boundary value problem, because typically it is the displacement, not the volumetric strain that is sought after as the solution.

Carlos Somigliana in 1885 [50] utilized Kelvin solution [53] of point force in infinite space

$$u_{ij}^{F} = \frac{1}{16\pi G(1-\nu)} \frac{1}{r} \left[\frac{x_i x_j}{r^2} + (3-3\nu)\delta_{ij}\right] \tag{8.42}$$

to derive the *Somigliana integral equation*

$$u_j = \int_{\Gamma} \left(t_i\, u_{ij}^{F} - t_{ij}^{F}\, u_i\right)\, d\vec{\chi} \tag{8.43}$$

This equation formally gives the solution of displacement at any point within the domain Ω, based on the boundary values of displacement and traction, and is equivalent to Green's third identity (8.39) extended to elasticity.

In the following, we shall apply the similar idea to construct Somigliana integral equations for poroelasticity.

8.2.3 Poroelasticity

For poroelasticity, we shall construct integral equation representations for not only the solid and fluid displacement, but also the stress and pore pressure. To obtain these singular integral equations, we shall substitute the body force, source, and the eigenstrain terms of the second system in (8.37) by the following delta functions

[1] The hexapole is related to the center of dilatation by a factor $-(1-2\nu)/2G(1-\nu)$; see Sect. 8.8.2.

$$F_{ik}^{(2)} = \delta_{ik}\,\delta(\vec{\chi} - \vec{x})\,\delta(t - \tau) \tag{8.44}$$

$$f_{ik}^{(2)} = \delta_{ik}\,\delta(\vec{\chi} - \vec{x})\,\delta(t - \tau) \tag{8.45}$$

$$Q^{(2)} = \delta(\vec{\chi} - \vec{x})\,\delta(t - \tau) \tag{8.46}$$

$$E_{ijkl}^{(2)} = -\frac{1}{2}\left(\delta_{ik}\delta_{jl} + \delta_{jk}\delta_{il}\right)\,\delta(\vec{\chi} - \vec{x})\,\delta(t - \tau) \tag{8.47}$$

where δ_{ik} is the Kronecker delta, and $\delta(\vec{\chi} - \vec{x})$ and $\delta(t - \tau)$ the Dirac delta functions with singularity located at point $\vec{x}$ and time τ, respectively. The above substitutions correspond to instantaneous point (total) force, instantaneous point fluid force, instantaneous fluid volume dilatation, and instantaneous displacement discontinuity, respectively. We notice that the indices for the forces are extended from a vector to a second rank tensor, and for the displacement discontinuity from a second rank to fourth rank tensor, for the following reason. As force is a vector, we can make three separate substitutions, a force in the x_1, x_2, and x_3 directions. As each of the forces is a vector, F_{ik} becomes a second order tensor, with i denoting the vector component, and k representing substitution of force in the x_k direction, for $i, k = 1, 2$ for 2D problems, and $1, 2, 3$ for 3D. The same situation exists for displacement discontinuity; hence the second rank tensor is expanded to fourth rank.

With the introduction of each of the non-zero forcing terms into the set of governing equations, (8.19)–(8.22), we seek its solution. Particularly, we seek the solution in the infinite space, that is, without boundary and without boundary condition to satisfy. These solutions are *free space Green's functions*, or *fundamental solutions*. For each substitution, we mark the associated solution with a descriptive superscript. For example, for the substitution of an instantaneous point force given in (8.44), solution in terms of displacement, stress, pore pressure, etc., are denoted as u_{ik}^{Fi}, σ_{ijk}^{Fi}, p_k^{Fi}, etc. Here we again notice the extended indices due to that of the forcing function. In Table 8.1 the notation system is summarized, in which the first letter is used to mark the type of the forcing function, and the second letter the time duration of the forcing function, i for instantaneous (Dirac delta function), and c for continuous (Heaviside unit step function).

In Fig. 8.1 we show the sign convention for the various singularities. In particular, the figure shows that a source is defined as the injection of fluid mass into a domain, a dipole is defined as a source (positive sign) and a sink (negative sign) brought together in the orientation shown, and the displacement discontinuity is similar to the stress—it is a tensor. We shall discuss these in more detail in the derivation of the fundamental solutions in Sect. 8.9.

We first look at the substitution of $F_{ik}^{(2)}$ as given in (8.44) into the reciprocal relation (8.37), with the rest of the forcing terms set to zero, $f_{ik}^{(2)} = Q^{(2)} = E_{ijkl}^{(2)} = 0$. We note the following properties of Dirac delta function

$$\begin{aligned} \delta(\vec{x} - \vec{\chi}) &= \infty, \quad \text{for} \quad \vec{x} = \vec{\chi} \\ &= 0, \quad \text{for} \quad \vec{x} \neq \vec{\chi} \end{aligned}$$

Table 8.1 Convention for naming singular solutions

Singularity type	Forcing term	Index notation	
		Instantaneous	Continuous
Total force	F_i	*Fi*	*Fc*
Double force		*bi*	*bc*
Quadrupole and hexapole		*qi*	*qc*
Solid dilatation	E_{ij}	*ci*	*cc*
Displacement discontinuity	E_{ijkl}	*di*	*dc*
Edge dislocation		*ei*	*ec*
Fluid force	f_i	*fi*	*fc*
Fluid source	γ	*si*	*sc*
Fluid dipole	γ_i	*pi*	*pc*
Fluid dilatation	Q	*li*	*lc*
Fluid dodecapole		*oi*	*oc*

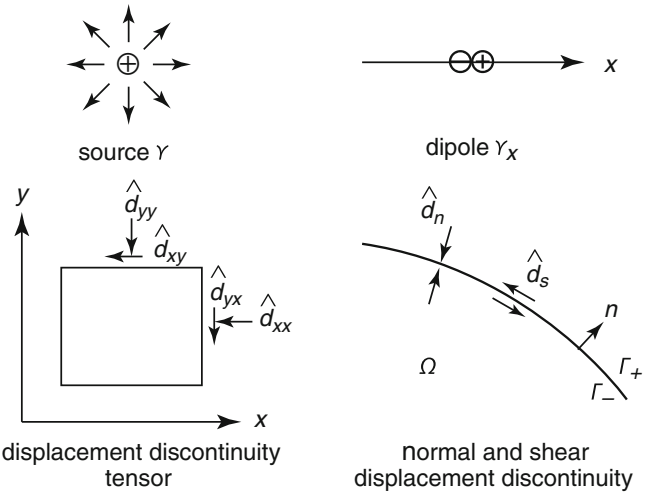


Fig. 8.1 Sign convention for the various singularities

$$\int_{\Omega} \delta(\vec{x} - \vec{\chi})\, d\vec{x} = 1, \qquad \text{for} \quad \vec{\chi} \in \Omega$$
$$= 0, \qquad \text{for} \quad \vec{\chi} \notin \Omega \tag{8.48}$$

The same applies to $\delta(t - \tau)$, except that it is one-dimensional in time. In the substitution, we observe that

$$\int_0^{t^+} \int_{\Omega} \delta_{ik}\, \delta(\vec{\chi} - \vec{x})\, \delta(t - \tau) u_i^{(1)}(\vec{\chi}, \tau)\, d\vec{\chi}\, d\tau = u_k^{(1)}(\vec{x}, t) \tag{8.49}$$

for $\vec{x} \in \Omega$. In the above we have emphasized that the integration for τ is from 0 to t^+ so it will pick up the value of 1. In the following, we will drop the superscript $^{(1)}$, as the first system is the solution of the boundary value problem sought, and system 2 is replaced by the known fundamental solutions. With the above substitutions, we obtain

$$\begin{aligned}\overline{\theta}\, u_k(\vec{x},t) = \int_0^t \int_\Gamma &\left[u_{ik}^{Fi}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_{ik}^{Fi}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[w_k^{Fi}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p_k^{Fi}(\vec{\chi}-\vec{x},t-\tau)\, w(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.50}$$

$$\begin{aligned}\overline{\theta}\, w_k(\vec{x},t) = \int_0^t \int_\Gamma &\left[u_{ik}^{fi}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_{ik}^{fi}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[w_k^{fi}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p_k^{fi}(\vec{\chi}-\vec{x},t-\tau)\, w(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.51}$$

$$\begin{aligned}-\overline{\theta}\, p(\vec{x},t) = \int_0^t \int_\Gamma &\left[u_i^{li}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_i^{li}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[w^{li}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p^{li}(\vec{\chi}-\vec{x},t-\tau)\, w(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.52}$$

$$\begin{aligned}-\overline{\theta}\, \sigma_{kl}(\vec{x},t) = \int_0^t \int_\Gamma &\left[u_{ikl}^{di}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_{ikl}^{di}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[w_{kl}^{di}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p_{kl}^{di}(\vec{\chi}-\vec{x},t-\tau)\, w(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.53}$$

In the above equations, $\overline{\theta} = 1$ for $\vec{x} \in \Omega$ and $\vec{x} \notin \Gamma$, and we have utilized these symbols for shorthand

$$\begin{aligned} t_i &= \sigma_{ij} n_j, \quad w = w_i n_i, \\ t_{ik}^{Fi} &= \sigma_{ijk}^{Fi} n_j, \quad w_k^{Fi} = w_{ik}^{Fi} n_i, \\ t_{ik}^{fi} &= \sigma_{ijk}^{fi} n_j, \quad w_k^{fi} = w_{ik}^{Fi} n_i, \\ t_i^{li} &= \sigma_{ij}^{li} n_j, \quad w^{li} = w_i^{li} n_i, \\ t_{ikl}^{di} &= \sigma_{ijkl}^{di} n_j, \quad w_{kl}^{di} = w_{ikl}^{di} n_i \end{aligned} \tag{8.54}$$

where t_i is the traditional definition of boundary traction, and w is the specific relative fluid to solid displacement normal to the boundary, and similar definitions are utilized for the fundamental solutions, whose indices are contracted.

In (8.50)–(8.53) we have assumed that the solid and fluid body (gravity) forces are separately considered, there is no physical fluid source and solid displacement discontinuity within the domain, and the initial conditions are zero. In doing so, all the volume integrals in (8.37) vanish, and (8.50)–(8.53) are *boundary integral equations*, as all the information needed to obtain the solutions on the left hand sides is contained on the boundary. (For the full formula including the body forces and initial conditions, see Cheng and Detournay [16].)

Although (8.50)–(8.53) formally give the solution of u_i, w_i, p and σ_{ij} at any location $\vec{x} \in \Omega$ and time t, in a typical boundary value problem, not all the information on the right hand side are readily available. Hence before applying these formulae, a boundary value problem needs to be solved. The methodology for solving boundary value problems based on these integral equations is called *boundary integral equation method*, or *boundary element method* (BEM) [15, 17, 18]. The strategy for such methodology is to place the field point right on the boundary, $\vec{x} \in \Gamma$. In that case, the integration of the Dirac delta function $\delta(\vec{\chi} - \vec{x})$ needs to be carefully handled, as the integration path for $\vec{\chi}$ crosses $\vec{x}$. In that case, the integration is interpreted as the *Cauchy principal value integration*, and a small hemisphere (3D) or semicircle (2D) is used to exclude the singularity from the domain Ω, as shown in Fig. 8.2, and the limit is taken to shrink the radius to zero. On a surface/line that is smooth (having continuous first derivative), the Cauchy principal value gives $\overline{\theta} = 1/2$. On a "corner", where the first derivative has a jump, $\overline{\theta}$ is given by the interior angle divided by 2π (2D) or the interior solid angle divided by 4π (3D). By doing so, (8.50)–(8.53) are entirely defined on the boundary, and can be discretized in a numerical procedure to solve a boundary value problem. In another formulation, $\vec{x}$ is moved outside the domain, $\vec{x} \notin \Omega$, and $\overline{\theta} = 0$, leading to the *null-field integral equations* [14, 57, 58].

Equations (8.50)–(8.53) are in fact not yet in a form suitable for a BEM implementation, because in a boundary value problem the boundary conditions are hardly given in terms of the normal relative fluid to solid displacement w, as needed on the right hand sides. As discussed in Sect. 6.5.2, the physical boundary conditions are typically given as stress σ_{ij}, solid displacement u_i, pore pressure p, or fluid normal specific discharge q. Hence an integration by parts with respect to time is

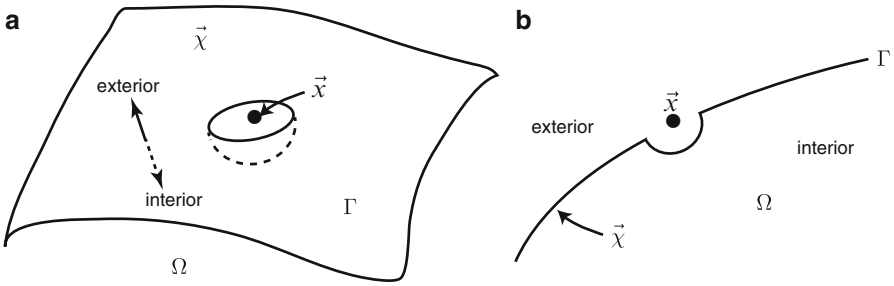


Fig. 8.2 Cauchy principal value integration: (**a**) 3D and (**b**) 2D

performed to convert w on the right hand sides into q, and the following integral equations are obtained

$$\begin{aligned}\overline{\theta}\, u_k(\vec{x},t) = &\int_0^t \int_\Gamma \left[u_{ik}^{Fi}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_{ik}^{Fi}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[q_k^{Fc}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p_k^{Fc}(\vec{\chi}-\vec{x},t-\tau)\, q(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.55}$$

$$\begin{aligned}\overline{\theta}\, w_k(\vec{x},t) = &\int_0^t \int_\Gamma \left[u_{ik}^{fi}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_{ik}^{fi}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[q_k^{fc}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p_k^{fc}(\vec{\chi}-\vec{x},t-\tau)\, q(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.56}$$

$$\begin{aligned}-\overline{\theta}\, p(\vec{x},t) = &\int_0^t \int_\Gamma \left[u_i^{li}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_i^{li}(\vec{\chi}-\vec{x},t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[q^{lc}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p^{lc}(\vec{\chi}-\vec{x},t-\tau)\, q(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.57}$$

$$\begin{aligned}-\overline{\theta}\, \sigma_{kl}(\vec{x},t) = &\int_0^t \int_\Gamma \left[u_{ikl}^{di}(\vec{\chi}-\vec{x},t-\tau)\, t_i(\vec{\chi},\tau) - t_{ikl}^{di}(\vec{\chi}-\vec{x},,t-\tau)\, u_i(\vec{\chi},\tau)\right] \\ &- \left[q_{kl}^{dc}(\vec{\chi}-\vec{x},t-\tau)\, p(\vec{\chi},\tau) - p_{kl}^{dc}(\vec{\chi}-\vec{x},t-\tau)\, q(\vec{\chi},\tau)\right] d\vec{\chi}\, d\tau\end{aligned} \tag{8.58}$$

In the above we have utilized the following relation, according to (8.27),

$$q = q_i\, n_i = \dot{w}_i\, n_i = \dot{w} \tag{8.59}$$

where the over dot indicates time derivative. Also, we note the relations among the fundamental solutions

$$\begin{aligned} q_k^{Fc} &= w_k^{Fi},\ \dot{p}_k^{Fc} = p_k^{Fi}, \\ q_k^{fc} &= w_k^{fi},\ \dot{p}_k^{fc} = p_k^{fi}, \\ q^{lc} &= w^{li},\ \dot{p}^{lc} = p^{li}, \\ q_{kl}^{dc} &= w_{kl}^{di},\ \dot{p}_{kl}^{dc} = p_{kl}^{di} \end{aligned} \tag{8.60}$$

in which we recognize that the time derivative of a continuous (in time) fundamental solution (marked by the superscript ^{c}) gives the instantaneous fundamental solution (marked by ^{i}). The former is associated with a Heaviside unit step function input, and the latter a Dirac delta function input. Although there are four integral

equations presented as (8.55)–(8.58), only (8.55) and (8.57) are needed in a BEM implementation for the solution of boundary value problem [15–17].

Similar to the technique utilized for analytical solutions in Chap. 7, to avoid the handling of time, Laplace transform can be applied to the above integral equations and fundamental solutions. Numerical inverse Laplace transform is then applied to obtain solution in time. To do so, we apply Laplace transform to (8.55) and (8.57), and utilize the convolutional theorem (D.9), to obtain the following [15, 17]

$$\begin{aligned}\overline{\theta}\,\tilde{u}_k(\vec{x},s) = &\int_\Gamma \left[\tilde{u}_{ik}^{Fi}(\vec{\chi}-\vec{x},s)\,\tilde{t}_i(\vec{\chi},s) - \tilde{t}_{ik}^{Fi}(\vec{\chi}-\vec{x},s)\,\tilde{u}_i(\vec{\chi},s)\right] \\ &-\frac{1}{s}\left[\tilde{q}_k^{Fi}(\vec{\chi}-\vec{x},s)\,\tilde{p}(\vec{\chi},s) - \tilde{p}_k^{Fi}(\vec{\chi}-\vec{x},s)\,\tilde{q}(\vec{\chi},s)\right] d\vec{\chi} \qquad (8.61)\\ -\overline{\theta}\,\tilde{p}(\vec{x},s) = &\int_\Gamma s\left[\tilde{u}_i^{si}(\vec{\chi}-\vec{x},s)\,\tilde{t}_i(\vec{\chi},s) - \tilde{t}_i^{si}(\vec{\chi}-\vec{x},s)\,\tilde{u}_i(\vec{\chi},s)\right] \\ &-\left[\tilde{q}^{si}(\vec{\chi}-\vec{x},s)\,\tilde{p}(\vec{\chi},s) - \tilde{p}^{si}(\vec{\chi}-\vec{x},s)\,\tilde{q}(\vec{\chi},s)\right] d\vec{\chi} \qquad (8.62)\end{aligned}$$

In the above we have converted the fundamental solutions to unify the expressions such that only two types of singularities appear: an instantaneous point force (*Fi*), and an instantaneous point source (*si*), based on time differentiation and the Laplace transform formula (D.3), such that

$$\begin{aligned}\tilde{q}_k^{Fi} &= s\,\tilde{q}_k^{Fc},\ \ \tilde{p}_k^{Fi} = s\,\tilde{p}_k^{Fc},\\ \tilde{u}_i^{li} &= s\,\tilde{u}_i^{si},\ \ \tilde{t}_i^{li} = s\,\tilde{t}_i^{si},\\ \tilde{q}^{lc} &= \tilde{q}^{si},\ \ \tilde{p}^{lc} = \tilde{p}^{si}\end{aligned} \qquad (8.63)$$

In the above, we have utilized the fact that fluid source γ is the time derivative of fluid dilatation Q, as indicated in (8.29); therefore the superscripts ^{lc} and ^{si} are interchangeable for all fundamental solutions, in time and in Laplace transform domain.

8.3 Fredholm Integral Equation

8.3.1 Potential Problem

Fredholm in 1903 [27] proved the existence and uniqueness of solution of the linear integral equation

$$\mu(x) - \lambda \int_a^b K(x,\chi)\,\mu(\chi)\,d\chi = f(x); \quad a \le x \le b \qquad (8.64)$$

where λ is a constant, $f(x)$ and $K(x, \chi)$ are given continuous functions, and $\mu(x)$ is the sought for solution. Equation (8.64) is known as the *Fredholm integral equation of the second kind*. By virtue of the Fredholm theorem, we can solve a Dirichlet problem by the following formula [38, 52]

$$\overline{\theta}\,\phi(\vec{x}) = \int_{\Gamma} K(\vec{\chi} - \vec{x})\,\mu(\vec{\chi})\,d\vec{\chi} \tag{8.65}$$

In the above, ϕ is the solution of Laplace equation subject to Dirichlet boundary condition, the kernel

$$K(\vec{\chi} - \vec{x}) = \frac{\partial(1/r)}{\partial n(\vec{\chi})} \tag{8.66}$$

with $r = |\vec{\chi} - \vec{x}|$, is known as a *dipole*, or *double layer potential*, for its physical meaning of representing the potential field of bringing a positive and a negative electrical charge close together, μ is a distribution density, Γ is a closed *Lyapunov surface*, and $\overline{\theta}$ takes the value of 0, 1/2, and 1, respectively for $\vec{x} \notin \Omega$, $\vec{x} \in \Gamma$, and $\vec{x} \in \Omega$. The Fredholm theorem guarantees the existence and uniqueness of μ.

An alternative equation to (8.65) is based on the distribution of a *source*, or *single layer potential*, given as a *Fredholm integral equation of the first kind*

$$\overline{\theta}\,\phi(\vec{x}) = \int_{\Gamma} \frac{1}{r}\,\sigma(\vec{\chi})\,d\vec{\chi} \tag{8.67}$$

where σ is a distribution density. Equations (8.65) and (8.67), together with their normal derivatives at the boundary Γ,

$$\overline{\theta}\,\frac{\partial\phi(\vec{x})}{\partial n(\vec{x})} = \int_{\Gamma} \frac{\partial^2(1/r)}{\partial n(\vec{x})\,\partial n(\vec{\chi})}\,\mu(\vec{\chi})\,d\vec{\chi} \tag{8.68}$$

$$\overline{\theta}\,\frac{\partial\phi(\vec{x})}{\partial n(\vec{x})} = \int_{\Gamma} \frac{\partial(1/r)}{\partial n(\vec{x})}\,\sigma(\vec{\chi})\,d\vec{\chi} \tag{8.69}$$

can be discretized in a numerical procedure to solve mixed boundary value problems [38]. These methods are called *single layer method* and *double layer method* in the BEM literature [14]. They are also called the *indirect method* in contrast to the Green's third identity (8.39) based method, which is called the *direct method*. The difference between (8.39), and (8.65) and (8.67) is that the 'distribution densities' in (8.39) are physical quantities given as ϕ and $\partial\phi/\partial n$, while those for (8.65) and (8.67) are 'fictitious densities'. The equivalence of Green's third identity and the Fredholm integral equations has been established by recognizing that the fictitious densities are in fact the jumps of potentials and its normal derivatives on the boundary, between an interior domain and an exterior domain problem [8, 38].

It is of interest to point out that, when the field point $\vec{x}$ is located on the boundary Γ, the integration in (8.67) contains a *weak* (*integrable*) *singularity*, while (8.65)

and (8.69) contain a *strong* (*non-integrable*) *singularity*. The integrals in (8.65) and (8.69) need to be interpreted in the *Cauchy principal value* sense, an idea introduced by Cauchy [10]. The integral in (8.68) contains a *hypersingularity*. Its integration is performed in the *Hadamard finite part* sense, a concept introduced by Hadamard [35]. Its numerical implementation needs to be carefully handled [36].

8.3.2 Elasticity

Volterra [56] in 1907 presented the dislocation solution of elasticity, as well as other singular solutions such as the force double and the disclination, generally known as the nuclei of strain [40]. For a point dislocation in unbounded three-dimensional space, the resultant displacement field is

$$u^d_{ijk} = \frac{1}{4\pi(1-\nu)}\frac{1}{r^2}\left[(1-2\nu)\left(\delta_{kj}x_i - \delta_{ij}x_k - \delta_{ik}x_j\right) - \frac{2}{r^2}x_i x_j x_k\right] \tag{8.70}$$

This singular solution can be distributed over the boundary Γ to give the *Volterra integral equation of the first kind* [55]:

$$u_k(\vec{x}) = \int_\Gamma u^d_{kji}(\vec{x}-\vec{\chi})\, n_j(\vec{\chi})\, \mu_i(\vec{\chi})\, d\vec{\chi} \tag{8.71}$$

where the distribution density μ_i is a vector, also known as the *displacement discontinuity*. Equation (8.71) is equivalent to (8.65) of the potential problem. In the numerical implementation, it is called the *double layer method*, or the *displacement discontinuity method* [22].

The counterpart of (8.67) is given by the *Somigliana integral equation* [51]

$$u_j(\vec{x}) = \int_\Gamma u^F_{ij}(\vec{x}-\vec{\chi})\, \sigma_i(\vec{\chi})\, d\vec{\chi} \tag{8.72}$$

where the distribution density σ_i is the *single layer potential*, or the *stress discontinuity*.

8.3.3 Poroelasticity

In the following, we shall utilize the poroelasticity equivalent of Green's third identify, (8.55)–(8.58), to construct the equivalent of the Fredholm integral equations, in the form of single layer (stress discontinuity) and double layer (displacement discontinuity) methods.

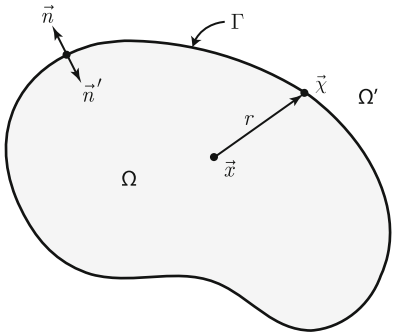


Fig. 8.3 Complementary regions Ω and Ω' bounded by a common boundary Γ

Consider an interior region Ω bounded by Γ. Its complementary (exterior) region is denoted as Ω', as depicted in Fig. 8.3. The unit outward normal $\vec{n}$ is associated with Ω, and $\vec{n}'$ with Ω', where we note the relation $\vec{n} = -\vec{n}'$. We can write (8.55)–(8.58) for the exterior domain Ω', instead of the interior domain Ω, while keeping the field point $\vec{x}$ in Ω, and obtain

$$
\begin{aligned}
0 = \int_0^t \int_\Gamma & \left[u_{ik}^{Fi}(\vec{\chi} - \vec{x}, t-\tau)\, t_i'(\vec{\chi}, \tau) + t_{ik}^{Fi}(\vec{\chi} - \vec{x}, t-\tau)\, u_i'(\vec{\chi}, \tau) \right] \\
& + \left[q_k^{Fc}(\vec{\chi} - \vec{x}, t-\tau)\, p'(\vec{\chi}, \tau) + p_k^{Fc}(\vec{\chi} - \vec{x}, t-\tau)\, q'(\vec{\chi}, \tau) \right] d\vec{\chi}\, d\tau
\end{aligned} \tag{8.73}
$$

$$
\begin{aligned}
0 = \int_0^t \int_\Gamma & \left[u_{ik}^{fi}(\vec{\chi} - \vec{x}, t-\tau)\, t_i'(\vec{\chi}, \tau) + t_{ik}^{fi}(\vec{\chi} - \vec{x}, t-\tau)\, u_i'(\vec{\chi}, \tau) \right] \\
& + \left[q_k^{fc}(\vec{\chi} - \vec{x}, t-\tau)\, p'(\vec{\chi}, \tau) + p_k^{fc}(\vec{\chi} - \vec{x}, t-\tau)\, q'(\vec{\chi}, \tau) \right] d\vec{\chi}\, d\tau
\end{aligned} \tag{8.74}
$$

$$
\begin{aligned}
0 = \int_0^t \int_\Gamma & \left[u_i^{li}(\vec{\chi} - \vec{x}, t-\tau)\, t_i'(\vec{\chi}, \tau) + t_i^{li}(\vec{\chi} - \vec{x}, t-\tau)\, u_i'(\vec{\chi}, \tau) \right] \\
& + \left[q^{lc}(\vec{\chi} - \vec{x}, t-\tau)\, p'(\vec{\chi}, \tau) + p^{lc}(\vec{\chi} - \vec{x}, t-\tau)\, q'(\vec{\chi}, \tau) \right] d\vec{\chi}\, d\tau
\end{aligned} \tag{8.75}
$$

$$
\begin{aligned}
0 = \int_0^t \int_\Gamma & \left[u_{ikl}^{di}(\vec{\chi} - \vec{x}, t-\tau)\, t_i'(\vec{\chi}, \tau) + t_{ikl}^{di}(\vec{\chi} - \vec{x}, t-\tau)\, u_i'(\vec{\chi}, \tau) \right] \\
& + \left[q_{kl}^{dc}(\vec{\chi} - \vec{x}, t-\tau)\, p'(\vec{\chi}, \tau) + p_{kl}^{dc}(\vec{\chi} - \vec{x}, t-\tau)\, q'(\vec{\chi}, \tau) \right] d\vec{\chi}\, d\tau
\end{aligned} \tag{8.76}
$$

where the prime is used to denote quantities associated with Ω', and

$$
t_i' = \sigma_{ij}' n_j', \quad q' = q_i' n_i' \tag{8.77}
$$

The definitions of the fundamental solutions remain the same as those in (8.54); that is, they are referred to n instead of n', thus causing the sign changes in (8.73)–(8.76). The left hand sides of the above equations become zero because $\vec{x} \notin \Omega'$.

Adding (8.55)–(8.58) to (8.73)–(8.76), the following expressions are obtained

$$\overline{\theta}\, u_k = \int_0^t \int_\Gamma \left[u_{ik}^{Fi}(\sigma_{ij} - \sigma'_{ij})n_j - t_{ik}^{Fi}(u_i - u'_i)\right] - \left[q_k^{Fc}(p - p') - p_k^{Fc}(q_i - q'_i)n_i\right] d\vec{\chi}\, d\tau \tag{8.78}$$

$$\overline{\theta}\, w_k = \int_0^t \int_\Gamma \left[u_{ik}^{fi}(\sigma_{ij} - \sigma'_{ij})n_j - t_{ik}^{fi}(u_i - u'_i)\right] - \left[q_k^{fc}(p - p') - p_k^{fc}(q_i - q'_i)n_i\right] d\vec{\chi}\, d\tau \tag{8.79}$$

$$-\overline{\theta}\, p = \int_0^t \int_\Gamma \left[u_i^{li}(\sigma_{ij} - \sigma'_{ij})n_j - t_i^{li}(u_i - u'_i)\right] - \left[q^{lc}(p - p') - p^{lc}(q_i - q'_i)n_i\right] d\vec{\chi}\, d\tau \tag{8.80}$$

$$-\overline{\theta}\, \sigma_{kl} = \int_0^t \int_\Gamma \left[u_{ikl}^{di}(\sigma_{ij} - \sigma'_{ij})n_j - t_{ikl}^{di}(u_i - u'_i)\right] - \left[q_{kl}^{dc}(p - p') - p_{kl}^{dc}(q_i - q'_i)n_i\right] d\vec{\chi}\, d\tau \tag{8.81}$$

These expressions form the basis of the stress and displacement discontinuity methods presented in the sections below.

8.4 Stress Discontinuity Method

For a boundary value problem defined in an interior domain Ω, we can artificially create a complementary problem in the exterior domain Ω' whose solid displacement and fluid flux along the boundary Γ are identical to that of the primary problem,

$$u_i = u'_i, \qquad q_i = q'_i \tag{8.82}$$

Substituting the above into (8.78)–(8.81), we obtain

$$\overline{\theta}\, u_k(\vec{x}, t) = \int_0^t \int_\Gamma \left[u_{ki}^{Fi}(\vec{x} - \vec{\chi}, t - \tau)\, \hat{s}_i(\vec{\chi}, \tau) + q_k^{Fc}(\vec{x} - \vec{\chi}, t - \tau)\, \hat{s}(\vec{\chi}, \tau)\right] d\vec{\chi}\, d\tau \tag{8.83}$$

$$\overline{\theta}\, w_k(\vec{x}, t) = \int_0^t \int_\Gamma \left[u_{ki}^{fi}(\vec{x} - \vec{\chi}, t - \tau)\, \hat{s}_i(\vec{\chi}, \tau) + q_k^{fc}(\vec{x} - \vec{\chi}, t - \tau)\, \hat{s}(\vec{\chi}, \tau)\right] d\vec{\chi}\, d\tau \tag{8.84}$$

$$\overline{\theta}\, p(\vec{x},t) = \int_0^t \int_\Gamma \left[u_i^{li}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_i(\vec{\chi},\tau) + q^{lc}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}(\vec{\chi},\tau) \right] d\vec{\chi}\, d\tau \tag{8.85}$$

$$\overline{\theta}\, \sigma_{kl}(\vec{x},t) = \int_0^t \int_\Gamma \left[u_{ikl}^{di}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_i(\vec{\chi},\tau) + q_{kl}^{dc}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}(\vec{\chi},\tau) \right] d\vec{\chi}\, d\tau \tag{8.86}$$

where $\hat{s}_{ij}$, $\hat{s}_i$, and $\hat{s}$ represent the stress discontinuity, the traction jump, and the pressure jumps across the boundary Γ, respectively, with the following definitions

$$\hat{s}_{ij} = (\sigma_{ij} - \sigma'_{ij}) \tag{8.87}$$

$$\hat{s}_i = \hat{s}_{ij}\, n_j \tag{8.88}$$

$$\hat{s} = -(p - p') \tag{8.89}$$

We may argue that although $\hat{s}_i$ and $\hat{s}$ have their mathematical roots, and are not arbitrary functions, they do not have a physical meaning as the complementary problem is fictitious. Hence these quantities may be referred to as 'fictitious densities'.

We note that in (8.83)–(8.86) we have switched the role of $\vec{x}$ and $\vec{\chi}$ in the fundamental solutions. The difference is that the singularities (delta function) are now located at $\vec{\chi}$, instead of $\vec{x}$, as in the case of the Somigliana integral equations shown in (8.55)–(8.58). The switch is performed in the spirit of a physical interpretation, and is not a mathematical requirement—we now distribute the fundamental solutions along the boundary Γ with certain density, and integrate to create the needed solution, rather than fixing it at $\vec{x}$ and integrate its *influence* at $\vec{\chi}$ on the boundary. As the fundamental solutions are not necessarily symmetrical with respect to $\vec{x}$ and $\vec{\chi}$, sign adjustments were made in (8.83)–(8.86). Particularly, based on the derived fundamental solutions in Appendix E, we observe the following relations

$$\begin{aligned}
u_{ik}^{Fi}(\vec{\chi}-\vec{x},t-\tau) &= u_{ki}^{Fi}(\vec{x}-\vec{\chi},t-\tau), \quad q_k^{Fc}(\vec{\chi}-\vec{x},t-\tau) = q_k^{Fc}(\vec{x}-\vec{\chi},t-\tau) \\
u_{ik}^{fi}(\vec{\chi}-\vec{x},t-\tau) &= u_{ki}^{fi}(\vec{x}-\vec{\chi},t-\tau), \quad q_k^{fc}(\vec{\chi}-\vec{x},t-\tau) = q_k^{fc}(\vec{x}-\vec{\chi},t-\tau) \\
u_i^{li}(\vec{\chi}-\vec{x},t-\tau) &= -u_i^{li}(\vec{x}-\vec{\chi},t-\tau), \quad q^{lc}(\vec{\chi}-\vec{x},t-\tau) = -q^{lc}(\vec{x}-\vec{\chi},t-\tau) \\
u_{ikl}^{di}(\vec{\chi}-\vec{x},t-\tau) &= -u_{ikl}^{di}(\vec{x}-\vec{\chi},t-\tau), \quad q_{kl}^{dc}(\vec{\chi}-\vec{x},t-\tau) = -q_{kl}^{dc}(\vec{x}-\vec{\chi},t-\tau)
\end{aligned} \tag{8.90}$$

In the above, these rules were followed: if a fundamental solution is a function of r only, $\vec{x}$ and $\vec{\chi}$ can be freely exchanged; and if there exists a differentiation with respect to spatial coordinates, switching will cause a sign change. In (8.83)

and (8.84), we also switched the subscript indices of u_{ik}^{Fi} and u_{ik}^{fi}, such that the vector component of the fundamental solution matches that of the displacement solution on the left hand side.

Although (8.83)–(8.86) are mathematically correct, they are not in a physically appealing form, typically constructed through physical intuition and not mathematical rigor. We observe that the right hand side fundamental solutions contain a variety of variables. Based on physical intuition, we should distribute displacement fundamental solutions in order to obtain displacement solution, pressure fundamental solutions to obtain pressure solution, etc. The connection between these two approaches can be accomplished by exploring the various intricate and interesting relations among the fundamental solutions. These relations can be proven using the reciprocal theorem of poroelasticity (8.37), as demonstrated in Sect. 8.20. In the following we only list those used for the present purpose:

$$\begin{aligned} u_{ki}^{fi} &= w_{ki}^{Fi}, \qquad u_i^{li} = p_i^{Fi}, \qquad u_{ikl}^{di} = \sigma_{kli}^{Fi}, \\ q_{ki}^{fc} &= w_{ki}^{fi}, \qquad q_i^{lc} = p_i^{fi}, \qquad q_{ikl}^{dc} = \sigma_{kli}^{fi}, \\ q_{ki}^{Fc} &= u_{ki}^{fi} \end{aligned} \tag{8.91}$$

These relations conveniently transform (8.83), (8.84), (8.85) and (8.86) into the following

$$\begin{aligned} \overline{\theta}\, u_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[& u_{ki}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_i(\vec{\chi},\tau) \\ & + u_{ki}^{fi}(\vec{x}-\vec{\chi},t-\tau)\, n_i(\vec{\chi})\, \hat{s}(\vec{\chi},\tau) \Big]\, d\vec{\chi}\, d\tau \end{aligned} \tag{8.92}$$

$$\begin{aligned} \overline{\theta}\, w_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[& w_{ki}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_i(\vec{\chi},\tau) \\ & + w_{ki}^{fi}(\vec{x}-\vec{\chi},t-\tau)\, n_i(\vec{\chi})\, \hat{s}(\vec{\chi},\tau) \Big]\, d\vec{\chi}\, d\tau \end{aligned} \tag{8.93}$$

$$\begin{aligned} \overline{\theta}\, p(\vec{x},t) = \int_0^t \int_\Gamma \Big[& p_i^{Fi}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_i(\vec{\chi},\tau) \\ & + p_i^{fi}(\vec{x}-\vec{\chi},t-\tau)\, n_i(\vec{\chi})\, \hat{s}(\vec{\chi},\tau) \Big]\, d\vec{\chi}\, d\tau \end{aligned} \tag{8.94}$$

$$\begin{aligned} \overline{\theta}\, \sigma_{kl}(\vec{x},t) = \int_0^t \int_\Gamma \Big[& \sigma_{kli}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_i(\vec{\chi},\tau) \\ & + \sigma_{kli}^{fi}(\vec{x}-\vec{\chi},t-\tau)\, n_i(\vec{\chi})\, \hat{s}(\vec{\chi},\tau) \Big]\, d\vec{\chi}\, d\tau \end{aligned} \tag{8.95}$$

We note that the right hand side fundamental solutions are now consistent with the solution sought on the left hand side.

As a first step to solve boundary value problems, we need to solve for the yet unknown distribution densities, $\hat{s}$ and $\hat{s}_i$. This is accomplished by placing the filed point $\vec{x}$ on the boundary Γ, where the boundary conditions are available. In a mixed boundary value problem, boundary conditions are given as u_i or t_i, and p or q. Hence we need the following additional integral equations,

$$\overline{\theta}\, t_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[\sigma_{kji}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\,\hat{s}_i(\vec{\chi},\tau) + \sigma_{kji}^{fi}(\vec{x}-\vec{\chi},t-\tau)\,n_i(\vec{\chi})\,\hat{s}(\vec{\chi},\tau)\Big]\, n_j(\vec{x})\; d\vec{\chi}\; d\tau \tag{8.96}$$

$$\overline{\theta}\, q(\vec{x},t) = \int_0^t \int_\Gamma \Big[q_{ij}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\hat{s}_j(\vec{\chi},\tau) + q_{ij}^{fi}(\vec{x}-\vec{\chi},t-\tau)\,n_j(\vec{\chi})\,\hat{s}(\vec{\chi},\tau)\Big]\, n_i(\vec{x})\; d\vec{\chi}\; d\tau \tag{8.97}$$

which are obtained by differentiating and assembling the integral equations according to the constitutive equations. The four equations (8.92), (8.94), (8.96), and (8.97) can now be exploited in a collocation procedure to determine the unknown distribution densities $\hat{s}_i$ and $\hat{s}$. This method can be referred to as a poroelastic *stress discontinuity method* [29].

Similar to the discussion in Sect. 8.3.1, the two (8.92) and (8.94) are weakly singular (containing $\ln r$ singularity for 2D, and $1/r$ for 3D), and are equivalent to the single layer method of the potential theory. Equations (8.96) and (8.97), on the other hand, are Cauchy singular, and Cauchy principal value is used to interpret the integration. Once the distribution densities are solved (in discrete form), these integral equations can be used to obtain the solution at any point in the domain by placing $\vec{x}$ at the location of interest.

8.5 Displacement Discontinuity Method

In contrast to the stress discontinuity method, we now consider the case where the boundary stresses and pore pressure for the interior and exterior domain problems are set equal

$$\sigma_{ij} = \sigma'_{ij}, \qquad p = p' \tag{8.98}$$

Substituting the above into (8.78)–(8.81), we obtain

$$\overline{\theta}\, u_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[\sigma_{ijk}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\,n_j(\vec{\chi})\,\hat{d}_i(\vec{\chi},\tau) + p_k^{Fc}(\vec{x}-\vec{\chi},t-\tau)\,\hat{d}(\vec{\chi},\tau)\Big]\; d\vec{\chi}\; d\tau \tag{8.99}$$

$$\overline{\theta}\, w_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[\sigma^{fi}_{ijk}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + p^{fc}_k(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.100}$$

$$\overline{\theta}\, p(\vec{x},t) = \int_0^t \int_\Gamma \Big[\sigma^{li}_{ij}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + p^{lc}(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.101}$$

$$\overline{\theta}\, \sigma_{kl}(\vec{x},t) = \int_0^t \int_\Gamma \Big[\sigma^{di}_{klij}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + p^{dc}_{kl}(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.102}$$

where

$$\hat{d}_i = u_i - u_i' = \hat{d}_{ij} n_j \tag{8.103}$$

$$\hat{d} = -(q_i - q_i')\, n_i \tag{8.104}$$

In the above, $\hat{d}$ is the fluid flux discontinuity normal to the surface, and $\hat{d}_i$ is the surface displacement discontinuity vector, reduced from the general displacement discontinuity tensor $\hat{d}_{ij}$. We note that relations similar to (8.90) have been used for the adjustment of signs in (8.99)–(8.102).

The integral equations can again be cast into a physically more appealing form, in which influence functions of the same type as the physical quantity on the left hand side are distributed:

$$\overline{\theta}\, u_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[u^{di}_{kji}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + u^{si}_k(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.105}$$

$$\overline{\theta}\, w_k(\vec{x},t) = \int_0^t \int_\Gamma \Big[w^{di}_{kji}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + w^{si}_k(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.106}$$

$$\overline{\theta}\, p(\vec{x},t) = \int_0^t \int_\Gamma \Big[p^{di}_{ji}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + p^{si}(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.107}$$

$$\overline{\theta}\, \sigma_{kl}(\vec{x},t) = \int_0^t \int_\Gamma \Big[\sigma^{di}_{klij}(\vec{x}-\vec{\chi},t-\tau)\, n_j(\vec{\chi})\, \hat{d}_i(\vec{\chi},\tau) + \sigma^{si}_{kl}(\vec{x}-\vec{\chi},t-\tau)\, \hat{d}(\vec{\chi},\tau)\Big]\, d\vec{\chi}\, d\tau \tag{8.108}$$

In the above we have utilized (8.91) as well as these relations below

$$\begin{aligned} p_k^{Fc} &= u_k^{si}, \;\; \sigma_{ijk}^{fi} = w_{kji}^{di}, \\ p_k^{fc} &= w_k^{si}, \;\; \sigma_{ij}^{li} = p_{ji}^{di}, \\ p^{lc} &= p^{si}, \;\; p_{kl}^{dc} = \sigma_{kl}^{si} \end{aligned} \tag{8.109}$$

The above relations are proven in Sect. 8.20.

Equations (8.105) and (8.107) hence define another indirect method in which solid displacement discontinuity and fluid source singularities are distributed at fictitious densities to obtain the same physical quantities on the left hand side. These equations are Cauchy singular. This indirect method may be viewed as equivalent to the double layer method of potential theory.

In order to solve mixed boundary value problems, (8.106) and (8.108) are utilized to obtain the following hypersingular equations:

$$\begin{aligned} \overline{\theta}\, t_k(\vec{x}, t) = \int_0^t \int_\Gamma \Big[& \sigma_{kjil}^{di}(\vec{x} - \vec{\chi}, t - \tau)\, n_i(\vec{\chi})\, \hat{d}_l(\vec{\chi}, \tau) \\ & + \sigma_{kj}^{si}(\vec{x} - \vec{\chi}, t - \tau)\, \hat{d}(\vec{\chi}, \tau) \Big] n_j(\vec{x})\, d\vec{\chi}\, d\tau \end{aligned} \tag{8.110}$$

$$\begin{aligned} \overline{\theta}\, q(\vec{x}, t) = \int_0^t \int_\Gamma \Big[& q_{ijk}^{di}(\vec{x} - \vec{\chi}, t - \tau)\, n_j(\vec{\chi})\, \hat{d}_k(\vec{\chi}, \tau) \\ & + q_i^{si}(\vec{x} - \vec{\chi}, t - \tau)\, \hat{d}(\vec{\chi}, \tau) \Big] n_i(\vec{x})\, d\vec{\chi}\, d\tau \end{aligned} \tag{8.111}$$

By this derivation we can loosely interpret that the constant $\overline{\theta}$ takes the same value as that in (8.106) and (8.108). More rigorously, however, the Hadamard finite-part argument needs to be invoked to obtain its limiting value. In the numerical implementation, certain regularization process is needed to evaluate this kind of hypersingular equations [39]. Equations (8.105), (8.107), (8.110) and (8.111) now form the bases of a poroelastic *displacement discontinuity method* [23, 54].

8.6 Dislocation Method

In elasticity, the *dislocation method* has been used to model linear fracture problems. Here the method is extended to poroelasticity. The dislocation method is a special case of the displacement discontinuity method, in which the 'fictitious densities' become actual physical quantities. For a fracture, the surface Γ in (8.107) and (8.108) now represents the fracture (dual) surface, with the density $\hat{d}_i$ as the actual displacement jump due to the opening and sliding of the fracture walls, and $\hat{d}$ the actual normal fluid flux jump due to the injection or extraction of fluid through the fracture, as defined in (8.103) and (8.104).

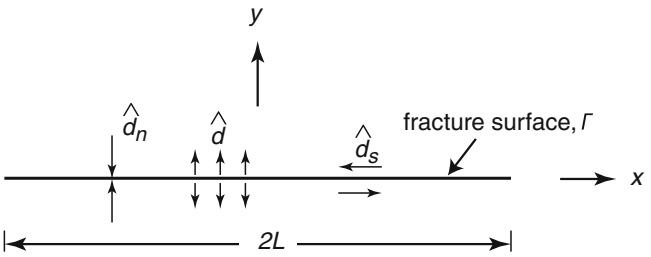


Fig. 8.4 A straight line fracture in a porous medium

Figure 8.4 gives an illustration of a two-dimensional straight line fracture of length $2L$. The width of the opening aperture is denoted as $\hat{d}_n$, the normal displacement discontinuity, and the sliding of the two surfaces is denoted as $\hat{d}_s$, the shear displacement discontinuity. Based on the above, (8.107) and (8.108) can be written as

$$\sigma_n(x,t) = \int_0^t \int_{-L}^{L} \Big[\sigma^{di}_{2222}(x-\chi, t-\tau)\,\hat{d}_n(\chi,\tau) + \sigma^{di}_{2221}(x-\chi, t-\tau)\,\hat{d}_s(\chi,\tau) + \sigma^{si}_{22}(x-\chi, t-\tau)\,\hat{d}(\chi,\tau)\Big]\, d\chi\, d\tau \tag{8.112}$$

$$\sigma_s(x,t) = \int_0^t \int_{-L}^{L} \Big[\sigma^{di}_{1222}(x-\chi, t-\tau)\,\hat{d}_n(\chi,\tau) + \sigma^{di}_{1221}(x-\chi, t-\tau)\,\hat{d}_s(\chi,\tau) + \sigma^{si}_{12}(x-\chi, t-\tau)\,\hat{d}(\chi,\tau)\Big]\, d\chi\, d\tau \tag{8.113}$$

$$p(x,t) = \int_0^t \int_{-L}^{L} \Big[p^{di}_{22}(x-\chi, t-\tau)\,\hat{d}_n(\chi,\tau) + p^{di}_{21}(x-\chi, t-\tau)\,\hat{d}_s(\chi,\tau) + p^{si}(x-\chi, t-\tau)\,\hat{d}(\chi,\tau)\Big]\, d\chi\, d\tau \tag{8.114}$$

in which σ_n and σ_s are respectively the normal and shear stress on the fracture surface, p is the pore pressure, $\hat{d}_n$ and $\hat{d}_s$ are the normal (mode 1) and shear (mode 2) displacement jumps across the fracture, and $\hat{d}$ the fluid injection rate. These equations have been utilized for the numerical solution of hydraulic fracture embedded in porous formation, opened by fluid pressurization [23, 49, 54].

Similar to the *edge dislocation method* in elasticity [5], its poroelastic version can also be devised. Performing an integration by parts with respect to χ on (8.112) produces the edge dislocation equation

$$\sigma_n(x,t) = \int_0^t \int_{-L}^{L} \Big[\sigma^{ei}_{2222}(x-\chi, t-\tau)\,\hat{d}'_n(\chi,\tau) + \sigma^{ei}_{2221}(x-\chi, t-\tau)\,\hat{d}'_s(\chi,\tau) + \sigma^{si}_{22}(x-\chi, t-\tau)\hat{d}(\chi,\tau)\Big]\, d\chi\, d\tau \tag{8.115}$$

In the above, $\hat{d}_n'$ and $\hat{d}_s'$ are the derivatives of $\hat{d}_n$ and $\hat{d}_s$ with respect to χ. The influence functions denoted by the superscript ^{e} are edge dislocation solutions obtained as

$$\sigma_{2222}^{ei}(x-\chi, t-\tau) = \int_{-\infty}^{\chi} \sigma_{2222}^{di}(x-\chi', t-\tau)\, d\chi' \tag{8.116}$$

and so forth. Equation (8.115) has a $1/r$ type singularity as compared to the $1/r^2$ type in (8.112) ($r = |x-\chi|$), which is of some numerical advantage. In addition, the stress intensity factor can directly be solved in the edge dislocation method, which should be more accurate than that determined from the slope of displacement in the dislocation method. The poroelastic edge dislocation method has been numerically implemented [13, 24].

For curved fracture, an orthonormal coordinate system can be used. A two-dimensional representation, in which Γ is a curve with a $+$ and a $-$ side, is illustrated in Fig. 8.1, where u_i^- corresponds to u_i, and u_i^+ to u_i'. The normal and shear component of displacement discontinuity, $\hat{d}_n$ and $\hat{d}_s$, associated with the surface Γ, are related to the displacement discontinuity tensor as

$$\hat{d}_n = \hat{d}_{ij} n_i n_j \tag{8.117}$$

$$\hat{d}_s = \hat{d}_{jk} n_i n_k \varepsilon_{ji} \tag{8.118}$$

where ε_{ij} is the two-dimensional Levi-Cevita permutation symbol.

8.7 Galerkin Integral Equation

A shortcoming common to the boundary integral equation methods presented so far is that the coefficient matrix of the discretized, linear solution system is non-symmetric, in contrast to the symmetric coefficient matrices of finite element methods. Since there are considerable computational advantages to be gained from a symmetric coefficient matrix, symmetric "Galerkin formulations", mimicking the Galerkin weighted residual formulations of finite elements, have been proposed for boundary integral equations. The symmetric Galerkin BEM formulation for poroelasticity was presented by Pan and Maier [46]. Based on the Galerkin formulation viewpoint, the Green's functions in the integral equations are simply weighing functions, among many other possible weighing functions. In this section, we investigate the theoretical foundation of the symmetric Galerkin integral equation system and unify it with the present family of integral equations.

For the present purpose, we shall rewrite the single and double layer integral equations for displacement, (8.92) and (8.105), into the following

$$\overline{\theta}\, u_i(\vec{x},t) = \int_0^t \int_\Gamma \Big[u_{ik}^{Fi}(\vec{x}-\vec{\chi},t-\tau)\, \hat{s}_k(\vec{\chi},\tau) + u_{ik}^{fi}(\vec{x}-\vec{\chi},t-\tau)\, n_k(\vec{\chi})\, \hat{s}(\vec{\chi},\tau) \Big] d\vec{\chi}\, d\tau \tag{8.119}$$

$$\overline{\theta}\, u_i(\vec{x},t) = \int_0^t \int_\Gamma \Big[u_{ikl}^{di}(\vec{x}-\vec{\chi},t-\tau)\, n_k(\vec{\chi})\, \hat{d}_l(\vec{\chi},\tau) + u_i^{li}(\vec{x}-\vec{\chi},t-\tau)\, \hat{D}(\vec{\chi},\tau) \Big]\, d\vec{\chi}\, d\tau \tag{8.120}$$

In the above, (8.119) is repeated from (8.92), but with rearranged indices, while (8.120) is modified from (8.105) by performing integration by parts on the second term in the integrand, in which

$$u_k^{li} = \frac{\partial u_k^{si}}{\partial t} \tag{8.121}$$

and based on (8.104),

$$\hat{D} = \int_0^t \hat{d}\, dt = -(w_i - w_i')\, n_i \tag{8.122}$$

is the fluid relative displacement discontinuity.

Equations (8.119) and (8.120) can be added together to give a mixed-type integral equation

$$2\overline{\theta}\, u_i = \int_0^t \int_\Gamma [u_{ik}^{Fi}\hat{s}_k + u_{ik}^{fi} n_k \hat{s} + u_{ikl}^{di} n_k \hat{d}_l + u_i^{li}\hat{D}]\, d\vec{\chi}\, d\tau \tag{8.123}$$

Following the same strategy, and utilizing the single and double layer integral equations of Sects. 8.4 and 8.5, three more mixed type integral equations can be constructed

$$2\overline{\theta}\, t_i = \int_0^t \int_\Gamma [\sigma_{ijk}^{Fi}\hat{s}_k + \sigma_{ijk}^{fi} n_k \hat{s} + \sigma_{ijkl}^{di} n_k \hat{d}_l + \sigma_{ij}^{li}\hat{D}] n_j\, d\vec{\chi}\, d\tau \tag{8.124}$$

$$2\overline{\theta}\, w = \int_0^t \int_\Gamma [w_{ik}^{Fi}\hat{s}_k + w_{ik}^{fi} n_k \hat{s} + w_{ikl}^{di} n_k \hat{d}_l + w_i^{li}\hat{D}] n_i\, d\vec{\chi}\, d\tau \tag{8.125}$$

$$2\overline{\theta}\, p = \int_0^t \int_\Gamma [p_k^{Fi}\hat{s}_k + p_k^{fi} n_k \hat{s} + p_{kl}^{di} n_k \hat{d}_l + p^{li}\hat{D}]\, d\vec{\chi}\, d\tau \tag{8.126}$$

In the above, $w = v_i n_i$ is the normal fluid relative displacement. These four equations (8.123)–(8.126) can be put into a matrix form

$$\int_0^t \int_\Gamma \begin{bmatrix} \sigma_{ijkl}^{di} n_k n_j & \sigma_{ijk}^{Fi} n_j & \sigma_{ijk}^{fi} n_k n_j & \sigma_{ij}^{li} n_j \\ u_{ikl}^{di} n_k & u_{ik}^{Fi} & u_{ik}^{fi} n_k & u_i^{li} \\ w_{ikl}^{di} n_k n_i & w_{ik}^{Fi} n_i & w_{ik}^{fi} n_k n_i & w_i^{li} n_i \\ p_{kl}^{di} n_k & p_k^{Fi} & p_k^{fi} n_k & p^{li} \end{bmatrix} \begin{Bmatrix} \hat{d}_l \\ \hat{s}_k \\ \hat{s} \\ \hat{D} \end{Bmatrix} d\vec{\chi}\, d\tau = 2\overline{\theta} \begin{Bmatrix} t_i \\ u_i \\ w \\ p \end{Bmatrix} \tag{8.127}$$

We notice that the coefficient matrix is symmetric because of the following relations $\sigma_{ijk}^{Fi} = u_{kji}^{di}$, $\sigma_{ijk}^{fi} = w_{kji}^{di}$, $\sigma_{ij}^{li} = p_{ji}^{di}$, $u_{ij}^{fi} = w_{ji}^{Fi}$, $u_i^{li} = p_i^{Fi}$, and $w_i^{li} = p_i^{fi}$, which have been shown in (8.91) and (8.109). Similar to the stress and the displacement discontinuity methods, only half of the above equations are needed at each collocation node, depending on the type of boundary conditions (traction vs. displacement, and pressure vs. flux).

8.8 Fundamental Solution

Before deriving the poroelastic fundamental solutions, we shall first review a few elementary and elasticity fundamental solutions, as the poroelastic ones will be constructed based on these solutions.

8.8.1 Elementary Fundamental Solution

In the historical development, the first fundamental solution $1/r$ was presented by Joseph-Louis Lagrange as a potential whose derivative represents the gravitational force satisfying Newton's law of gravitation [14]. The partial differential equation satisfying such solution was found later by Pierre-Simon Laplace in his study of celestial mechanics, which is known as the *Laplace equation*.

Laplace Equation The solution of Laplace equation with a Dirac delta function as its right hand side

$$\nabla^2 \phi = \delta(\vec{x} - \vec{\chi}) \tag{8.128}$$

is given by

$$\begin{aligned} \phi &= \frac{1}{2\pi} \ln r, \quad \text{(2D)} \\ &= -\frac{1}{4\pi} \frac{1}{r}, \quad \text{(3D)} \end{aligned} \tag{8.129}$$

where $\vec{x}$ and $\vec{\chi}$ are spatial points in 2D or 3D, with $\vec{x}$ as the *field point*, and $\vec{\chi}$ the *source point*, that is, where the singularity is located, and $r(\vec{x} - \vec{\chi}) = |\vec{x} - \vec{\chi}|$ is the Euclidean distance between these two points.

Biharmonic Equation For the biharmonic equation,

$$\nabla^4 \phi = \delta(\vec{x} - \vec{\chi}) \tag{8.130}$$

the fundamental solutions are [12]

$$\begin{aligned} \phi &= \frac{r^2}{8\pi} (\ln r - 1), \quad \text{(2D)} \\ &= -\frac{r}{8\pi}, \quad \text{(3D)} \end{aligned} \tag{8.131}$$

Diffusion Equation For *diffusion equation*, we first present that due to a *continuous* source

$$\nabla^2 \phi - \frac{1}{c} \frac{\partial \phi}{\partial t} = \delta(\vec{x} - \vec{\chi})\, \mathrm{H}(t - \tau) \tag{8.132}$$

The fundamental solutions are [9, 21]

$$\begin{aligned} \phi &= -\frac{1}{4\pi} \mathrm{E}_1(\xi^2), \quad \text{(2D)} \\ &= -\frac{1}{4\pi} \frac{\mathrm{erfc}(\xi)}{r}, \quad \text{(3D)} \end{aligned} \tag{8.133}$$

where E_1 is the elliptic integral [1], and

$$\xi = \frac{r}{\sqrt{4c\,(t - \tau)}} \tag{8.134}$$

When the source is *instantaneous*, we replace the Heaviside unit step function $\mathrm{H}(t - \tau)$ in (8.132) by the Dirac delta function $\delta(t - \tau)$. As $\delta(t - \tau) = \partial \mathrm{H}(t - \tau)/\partial t$, the fundamental solutions for the instantaneous source are obtained by differentiating (8.133) with respect to time [43], and

$$\phi = -\frac{c}{(\sqrt{\pi}\, r)^n} \xi^n e^{-\xi^2} \tag{8.135}$$

in which $n = 2$ for two-dimensional and $n = 3$ for three-dimensional problems.

Helmholtz Equation When Laplace transform is applied to the diffusion equation (8.132), it becomes the *modified Helmholtz equation*

$$\nabla^2 \phi - k^2 \phi = \delta(\vec{x} - \vec{\chi}) \tag{8.136}$$

The fundamental solutions are

$$\begin{aligned} \phi &= -\frac{1}{2\pi} \mathrm{K}_0(kr), \qquad \text{(2D)} \\ &= -\frac{1}{4\pi} \frac{e^{-kr}}{r}, \qquad \text{(3D)} \end{aligned} \tag{8.137}$$

Or, for the *Helmholtz equation*,

$$\nabla^2 \phi + k^2 \phi = \delta(\vec{x} - \vec{\chi}) \tag{8.138}$$

The fundamental solutions are

$$\begin{aligned} \phi &= \frac{1}{4} \mathrm{Y}_0(kr), \qquad \text{(2D)} \\ &= -\frac{1}{4\pi} \frac{\cos(kr)}{r}, \qquad \text{(3D)} \end{aligned} \tag{8.139}$$

Bi-Helmholtz Equation For the product of modified Helmholtz operators,

$$(\nabla^2 - k_1^2)(\nabla^2 - k_2^2)\phi = \nabla^4 \phi - (k_1^2 + k_2^2)\nabla^2 \phi + k_1^2 k_2^2 \phi = \delta(\vec{x} - \vec{\chi}) \tag{8.140}$$

fundamental solutions are given by [12]

$$\begin{aligned} \phi &= -\frac{1}{2\pi} \left[\frac{\mathrm{K}_0(k_1 r)}{k_1^2 - k_2^2} + \frac{\mathrm{K}_0(k_2 r)}{k_2^2 - k_1^2} \right], \qquad \text{(2D)} \\ &= -\frac{1}{4\pi r} \left[\frac{e^{-k_1 r}}{k_1^2 - k_2^2} + \frac{e^{-k_2 r}}{k_2^2 - k_1^2} \right], \qquad \text{(3D)} \end{aligned} \tag{8.141}$$

8.8.2 Elasticity Fundamental Solution

Elasticity fundamental solutions are referred to the solution of the Navier equation of elasticity with various singularities introduced as concentrated force, moment, strain, displacement, etc.

Point Force The Navier equation can be expressed in the following form

$$G u_{i,jj} + \frac{G}{1 - 2\nu} u_{j,ji} = -F_i \tag{8.142}$$

where we have introduced a *body force* term on the right hand side, which normally represents the gravitational body force when the gravity effect is to be considered.

Here, we utilize this term to introduce a *concentrated point force* of unit magnitude acting in the x_k direction, and express the terms as

$$F_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi}) \tag{8.143}$$

The above notation shows that we are actually making three separate substitutions, $k = 1, 2, 3$, representing forces acting in these respective directions. This requires us to expand the tensor notation from that of a first rank tensor (vector) to a second rank tensor. We then express (8.142) into the following

$$G u^F_{ik,jj} + \frac{G}{1-2\nu} u^F_{jk,ji} = -\delta_{ik}\delta(\vec{x} - \vec{\chi}) \tag{8.144}$$

where the superscript ^{F} denotes the point force fundamental solution, the first index of u^F_{ik} represents the vector component, and the second index the direction of the concentrated force. But we shall see below that $u^F_{ik} = u^F_{ki}$.

This solution of concentrated force in three-dimensional space was derived by Kelvin [53] (see Sect. F.7 for Kelvin's involvement in Green's function). We shall give a quick demonstration how this solution can be derived using variable decomposition and by keeping track of the forcing function in the decomposition. Similar technique will be used to derive the poroelastic fundamental solutions in the subsequent sections.

In Sect. 6.6.1 we presented the well-known Papkovitch-Neuber functions for elasticity as (6.121), (6.122) and (6.123). Such relations, however, satisfies only the homogeneous Navier equation. Mindlin in 1953 [42] carefully tracked the body force term to define the following functions. First, we repeat (6.121) here:

$$u_i = (\psi + x_j\Psi_j)_{,i} - 4(1-\nu)\Psi_i \tag{8.145}$$

These functions now satisfy

$$\nabla^2\Psi_i = \frac{1}{4G(1-\nu)} F_i \tag{8.146}$$

$$\nabla^2\psi = -\frac{1}{4G(1-\nu)} x_i F_i \tag{8.147}$$

The above is the complete solution of (8.142). We can substitute (8.143) into (8.146) and (8.147), and obtain

$$\nabla^2\Psi^F_{ik} = \frac{1}{4G(1-\nu)} \delta_{ik}\delta(\vec{x} - 0) \tag{8.148}$$

$$\nabla^2\psi_k = 0 \tag{8.149}$$

In the above, for convenience, we have located the force at the origin, $\vec{\chi} = 0$, and notice that $x_i \delta(\vec{x} - 0) = 0$. Based on (8.129), we easily find

$$\begin{aligned} \Psi_{ik}^F &= \frac{1}{8\pi G(1-\nu)} \delta_{ik} \ln r, \qquad \text{(2D)} \\ &= -\frac{1}{16\pi G(1-\nu)} \delta_{ik} \frac{1}{r}, \qquad \text{(3D)} \end{aligned} \tag{8.150}$$

and

$$\psi_k = 0 \tag{8.151}$$

Substituting the above into (8.145), we find the point force solution in terms of displacement in two- and three-dimensional space as

$$\begin{aligned} u_{ik}^F &= \frac{1}{8\pi G(1-\nu)} \left[\frac{x_i x_k}{r^2} - (3-4\nu)\delta_{ik} \ln r \right], \qquad \text{(2D)} \\ &= \frac{1}{16\pi G(1-\nu)} \frac{1}{r} \left[\frac{x_i x_k}{r^2} + (3-4\nu)\delta_{ik} \right], \qquad \text{(3D)} \end{aligned} \tag{8.152}$$

Double Force Here we define a double force as a pair of forces of equal magnitude acting along an axis, but opposing each other. The forces are brought together with intensity inversely proportional to the distance separating them, until they meet at a point. This limiting process is equivalent to taking the derivative of the point force solution along the axis of its application [40]. Hence the double force is obtained as

$$u_{ik}^b = \frac{\partial u_{ik}^F}{\partial x_k} \tag{8.153}$$

where the superscript ^{b} denotes double force, and the second subscript of displacement indicates the direction of the force.

Quadrupole and Hexapole If we sum two and three pairs of mutually orthogonal double forces at a point, we obtain respectively a *quadrupole* and a *hexapole* for 2D and 3D problems. In terms of forcing function, we introduce

$$F_{ik,k} = \delta_{ik}\delta_{,k} = \delta_{,i} \tag{8.154}$$

to the Navier equation; and it becomes

$$G u_{i,jj}^q + \frac{G}{1-2\nu} u_{j,ji}^q = -\delta_{,i} \tag{8.155}$$

where the superscript ^{q} denotes both the quadrupole and hexapole solution. It is easy to show that

$$u_i^q = u_{ik,k}^F \tag{8.156}$$

We hence obtain

$$\begin{aligned} u_i^q &= -\frac{1-2\nu}{4\pi G(1-\nu)}\frac{x_i}{r^2}, \quad \text{(2D)} \\ &= -\frac{1-2\nu}{8\pi G(1-\nu)}\frac{x_i}{r^3}, \quad \text{(3D)} \end{aligned} \tag{8.157}$$

Center of Dilatation Another important elasticity fundamental solution is the *center of dilatation*. The solution represents a unit volume expansion centered at the origin. The solution of such problem was given by Dougall in 1897 [25]. Here we shall give it a quick derivation.

Utilizing the idea of eigenstrain in (8.22), the solution corresponds to the introduction of

$$\begin{aligned} E_{ij} &= \left(\frac{\ln r}{2\pi}\right)_{,ij}, \quad \text{(2D)} \\ &= \left(-\frac{1}{4\pi r}\right)_{,ij}, \quad \text{(3D)} \end{aligned} \tag{8.158}$$

to the Navier equation. It is easy to show that the above definitions lead to the volumetric strain of

$$E = E_{ii} = \delta \tag{8.159}$$

hence fulfilling the declaration that it is a singularity representing a unit volume expansion at the origin.

Substituting the total strain e_{ij}^* into the Navier equation, and realizing that the elastic strain e_{ij} satisfies the Navier equation exactly, we obtain

$$\begin{aligned} G u_{i,jj}^c + \frac{G}{1-2\nu} u_{j,ji}^c &= 2GE_{ij,j} + \frac{2G\nu}{1-2\nu} E_{jj,i} \\ &= \frac{2G(1-\nu)}{1-2\nu}\delta_{,i} \end{aligned} \tag{8.160}$$

In the above, we have replaced the asterisk by the superscript ^{c} to denote the center of dilatation fundamental solution. Comparing the above with (8.155), it is clear that the center of dilatation solution is

$$u_i^c = -\frac{2G(1-\nu)}{1-2\nu} u_i^q \tag{8.161}$$

We hence obtain this well-known but curious result—adding two (three) pairs of mutually orthogonal double forces together creates a circular (spherical) dilation at the center (see Huynen and Detournay [37] for a physical argument and illustration). Although by this equivalence, we already have derived the solution, in the following, we shall give it a different derivation to gain additional insight.

By the axial symmetry of a center of dilation, we realize that the displacement is a function of the radial distance r only. This also means that the displacement satisfies the irrotational field condition, and can be expressed as the gradient of a scalar function,

$$u_i^c = \Phi^c_{,i} \tag{8.162}$$

Substituting the above into (8.160), and relaxing a gradient operator, we find

$$\nabla^2 \Phi^c = \delta(\vec{x} - \vec{\chi}) \tag{8.163}$$

We then easily find Φ^c from (8.129), and the displacement becomes

$$\begin{aligned} u_i^c &= \frac{1}{2\pi} \frac{x_i}{r^2}, \quad \text{(2D)} \\ &= \frac{1}{4\pi} \frac{x_i}{r^3}, \quad \text{(3D)} \end{aligned} \tag{8.164}$$

or, in polar and spherical coordinates, respectively for the 2D and 3D cases,

$$\begin{aligned} u_r^c &= \frac{1}{2\pi} \frac{1}{r}, \quad \text{(2D)} \\ &= \frac{1}{4\pi} \frac{1}{r^2}, \quad \text{(3D)} \end{aligned} \tag{8.165}$$

If we take a circle (2D) and a sphere (3D) of any radius centered at the origin, the volume change due to the surface movement is obtained by multiplying the surface area $2\pi r$ and $4\pi r^2$ by the radial displacement u_r, and we obtain

$$\Delta V = 1 \tag{8.166}$$

The above volume change is constant for any circle and sphere, even for the one with $r = 0$. In fact, from (8.163), we can show that

$$e^c = \delta(\vec{x} - \vec{\chi}) \tag{8.167}$$

So once again we have shown that the solution is a center of dilatation.

8.9 Poroelasticity Fundamental Solution

As mentioned in the preamble of this chapter, fundamental solution is a particular solution that satisfies a set of linear governing equations in an unbounded domain, except at a certain isolated geometric object of lower dimension, such as a point, a line, or a surface. On these objects, certain singularity or function discontinuity exists. The mathematical symbolism of a Dirac delta function δ, which is a generalized function, is often used to represent the singularity. The solution in the domain needs to satisfy the singular or discontinuous behavior in a limiting sense when the object is approached from the domain. As a result, solution singularity and/or discontinuity exist in the domain that is inclusive of these objects. Fundamental solution is a particular solution and is not unique in the sense that it can incorporate any general solution without affecting the conditions that it needs to satisfy. However, as a common practice, the general solutions are excluded from the expression such that the fundamental solution becomes unique.

In the preceding sections we have demonstrated the use of fundamental solutions for the solution of poroelastic boundary value problems through integral equation representation. In addition to being a useful mathematical construct, fundamental solution can have many important physical applications. For example, the point source (sink) solution has long been used in groundwater applications to simulate injection and pumping wells in various types of aquifers [11]. In Sect. 1.3.6 we discussed the use of a point displacement discontinuity to represent the pore pressure field generated by a slippage of geological fault. In elasticity, various fundamental solutions, known as *nuclei of strain*, have been used to simulate material defects, initial stress, thermal stress, effective properties of heterogeneous materials, among many other applications [44]. In geomechanics and geophysics, applications have been found in volcanology for simulating collapsing of magna chamber [33, 34], and land subsidence due to fluid extraction [7, 28].

In Sect. 7.1, a literature review of the available poroelastic fundamental solutions and Green's functions has been given. Some of these solutions were derived in closed forms, while others in semi-analytical forms (such as under integral transform), which require numerical evaluation. In this section, we shall only present and derive these solutions in closed form. This limits the presentation below to only the fundamental solutions of isotropic poroelasticity. As demonstrated by Cheng and Detournay [16] that using a heuristic solution methodology based on a variable decomposition scheme of Biot [6], the entire list of physically based fundamental solutions, such as concentrated forces, sources, displacement discontinuity, etc., can be derived in closed form. We should emphasize that, although only the isotropic fundamental solutions are provided here, the reciprocal relations and integral equations presented in Sects. 8.1, 8.2, 8.3, 8.4, 8.5 and 8.6 are valid for the general anisotropy.

As discussed in Sect. 8.1.3, to derive the fundamental solutions, the governing equations for poroelasticity need to be modified to include the appropriate singular

forcing terms, such as those shown in Table 8.1. We hence repeat here the governing equations (8.19), (8.20) and (8.21), simplified for isotropy, as

$$\sigma_{ij,j} = -F_i \tag{8.168}$$

$$q_i = -\kappa(p_{,i} - f_i) \tag{8.169}$$

$$\frac{\partial \zeta}{\partial t} + q_{i,i} = \gamma \tag{8.170}$$

The isotropic constitutive equations are also modified to accommodate the presence of a nucleus of strain, as shown in (8.22), to become the following:

$$\sigma_{ij} = 2Ge_{ij} + \frac{2G\nu}{1-2\nu}\delta_{ij}e - \alpha\delta_{ij}p - 2GE_{ij} - \frac{2G\nu}{1-2\nu}\delta_{ij}E_{kk} \tag{8.171}$$

$$p = M(\zeta - \alpha e) + M\alpha E_{ii} \tag{8.172}$$

Here for convenience we have dropped the asterisk superscripts for the strain.

The above equations can be assembled to give the field equations in the form of a Navier equation and a coupled diffusion equation in terms of the variables (u_i, p) as

$$Gu_{i,jj} + \frac{G}{(1-2\nu)}u_{j,ji} - \alpha p_{,i} = -F_i + 2GE_{ij,j} + \frac{2G\nu}{(1-2\nu)}E_{jj,i} \tag{8.173}$$

$$\frac{\partial p}{\partial t} - \kappa M\nabla^2 p + \alpha M\frac{\partial e}{\partial t} = -\kappa M f_{i,i} + M\gamma + \alpha M\frac{\partial E_{ii}}{\partial t} \tag{8.174}$$

Or, these equations can be expressed in terms of the variables (u_i, ζ), and we obtain

$$Gu_{i,jj} + \frac{G}{(1-2\nu_u)}u_{j,ji} - \alpha M\zeta_{,i} = -F_i + 2GE_{ij,j} + \frac{2G\nu_u}{(1-2\nu_u)}E_{jj,i} \tag{8.175}$$

$$\frac{\partial \zeta}{\partial t} - c\nabla^2\zeta = \frac{\eta c}{G}F_{i,i} - \kappa f_{i,i} + \gamma + 2\eta c\left(\nabla^2 E_{ii} - E_{ij,ij}\right) \tag{8.176}$$

Biot Decomposition Here we introduce the *Biot decomposition* discussed in Sect. 6.6.2, and particularly Eq. (6.146), repeated below in terms of shorthand material parameters

$$u_i = u_i^o + \frac{\eta}{GS}\Phi_{,i} \tag{8.177}$$

If we require u_i^o to satisfy the undrained Navier equation of elasticity incorporating all the forcing term on the right hand side,

$$Gu_{i,jj}^o + \frac{G}{1-2\nu_u}u_{j,ji}^o = -F_i + 2GE_{ij,j} + \frac{2G\nu_u}{(1-2\nu_u)}E_{jj,i} \tag{8.178}$$

then the substitution of (8.177) and (8.178) into (8.175) shows that

$$\zeta = \nabla^2 \Phi \tag{8.179}$$

Using the above in the diffusion equation (8.176) leads to

$$\frac{\partial(\nabla^2 \Phi)}{\partial t} - c\nabla^4 \Phi = \frac{\eta c}{G} F_{i,i} - \kappa f_{i,i} + \gamma + 2\eta c \left(\nabla^2 E_{ii} - E_{ij,ij}\right) \tag{8.180}$$

At this point we shall drop the terms involving the eigenstrain E_{ij} to simplify the presentation. These terms will be restored in the sections where the center of dilatation and displacement discontinuity solutions are derived.

Relaxing a Laplacian operator, we find

$$\frac{\partial \Phi}{\partial t} - c\nabla^2 \Phi = \frac{\eta c}{G} g_1 - \kappa g_2 + g_3 \tag{8.181}$$

in which the functions g_1, g_2, and g_3 satisfy the following equations

$$\nabla^2 g_1 = \nabla \cdot \vec{F} \tag{8.182}$$

$$\nabla^2 g_2 = \nabla \cdot \vec{f} \tag{8.183}$$

$$\nabla^2 g_3 = \gamma \tag{8.184}$$

We note that by the above action, Φ is determined to only within an arbitrary harmonic function. This is not a concern as fundamental solution is not unique, and we are seeking only the singular part of the solution.

Once Φ is solved, ζ can be determined from (8.179). However, taking Laplacian of a singular quantity needs to be carefully handled—sometimes a Dirac delta function can be generated from it, which may not be obvious in the differentiation process. An alternative way to determine ζ can be based on (8.181),

$$\zeta = \frac{1}{c}\frac{\partial \Phi}{\partial t} - \frac{\eta}{G} g_1 + S g_2 - \frac{1}{c} g_3 \tag{8.185}$$

in which the time derivative, instead of spatial derivative, is taken, and the terms g_1, g_2, and g_3 can be easily determined.

Using (8.179) and (8.184) in the continuity equation (8.170), we find

$$\nabla^2 \dot{\Phi} + \nabla \cdot \vec{q} = \nabla^2 g_3 \tag{8.186}$$

which leads to

$$q_i = \left(-\dot{\Phi} + g_3\right)_{,i} \tag{8.187}$$

Integrating with respect to time we find

$$w_i = \left(-\Phi + \int_0^t g_3 \, dt \right)_{,i} \tag{8.188}$$

Similarly, the use of (8.183) and (8.187) in Darcy's law (8.169), and relaxing a gradient operator, lead to

$$p = \frac{1}{\kappa} \dot{\Phi} + g_2 - \frac{1}{\kappa} g_3 \tag{8.189}$$

The above relations mean that once the close form expression of Φ is obtained in the solution process, ζ, q_i, w_i, and p can be found without the need of u_i^o. Substituting (8.189) into (8.181), we also obtain this useful formula

$$p = \frac{1}{S} \nabla^2 \Phi + \frac{\eta}{GS} g_1 \tag{8.190}$$

which allows the pore pressure to be obtained by differentiating with respect to spatial coordinates, which is an easier process, instead of integration with respect to time, as in (8.189).

From (8.179) and (8.190), we also observe

$$\zeta = Sp - \frac{\eta}{G} g_1 \tag{8.191}$$

The stress can be obtained as

$$\sigma_{ij} = \sigma_{ij}^o + \frac{2\eta}{S} (\Phi_{,ij} - \delta_{ij} \nabla^2 \Phi) \tag{8.192}$$

where σ_{ij}^o is defined by the elasticity relation

$$\sigma_{ij}^o = 2G \, e_{ij}^o + \frac{2G\nu_u}{1 - 2\nu_u} \delta_{ij} e^o \tag{8.193}$$

These relations will be used for the derivation of the fundamental solutions to be presented below.

Before deriving the fundamental solutions, we shall offer some comment about the decomposition presented by Biot in 1956 [6]. On the surface, the decomposition looks highly powerful as it leads to the uncoupling of governing equations, as (8.178) and (8.181), in terms of u_i^o and Φ. Further exploration gives simple relations for the variables, such as (8.179) and (8.187)–(8.190). However, we should be aware that solutions obtained thisway are only *particular solutions*, and are not

complete solutions, as they generally cannot satisfy a set of given physical boundary conditions. Whenever a boundary is present, and a physical boundary condition is prescribed, such as displacement, traction, pressure, etc., we find that the current variables become coupled through these physical boundary conditions; hence the problem cannot be solved in a decoupled fashion after all. This is the reason that the Biot decomposition has not been used to solve any boundary value problem, including the problem of finding Green's function of point force, source, etc., in half space, where a boundary is present. Our present problem, however, is to find a particular singular solution in free space (infinite domain), and there is no boundary condition to satisfy. Hence in this case, the Biot decomposition is truly a powerful tool for deriving fundamental solutions.

Irrotational Fluid Fundamental Solutions Based on the decomposition presented in the preceding section, we notice that for all singular solutions that do not involve a total force, that is, $F_i = 0$, Eq. (8.178) immediately leads to the result

$$u_i^o = 0 \tag{8.194}$$

The displacement is now the gradient of a potential

$$u_i = \frac{\eta}{GS} \Phi_{,i} \tag{8.195}$$

and is irrotational, as discussed in Sect. 6.4.3. With $g_1 = 0$, (8.191) gives the following relation

$$\zeta_{...}^{*\dagger} = Sp_{...}^{*\dagger} \tag{8.196}$$

where the * superscript denotes the type of fundamental solutions, which includes all fluid related ones, such as source ^{s}, dipole ^{p}, dilatation ^{l}, and fluid force ^{f}, and the † superscript indicates instantaneous ^{i} or continuous ^{c} in time solutions, as summarized in Table 8.1. The subscript is for whatever tensor index needed in its definition. This relation will be used to obtain p from ζ, and vice versa, in the sections to follow.

8.10 Fluid Source

Fluid source refers to the emanation of fluid mass from a single point into a domain at a certain time rate. In order to have a finite volume of fluid produced at a point, the magnitude of flux at the point must be infinite; hence there exists a singularity. In the following we shall derive a continuous (in time) and an instantaneous source solution.

8.10.1 Continuous Source

For a continuous fluid source, we introduce for the source term a Dirac delta function is space and a Heaviside unit step function in time, as

$$\gamma = \delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.197}$$

Substituting the above into the right hand side of (8.180), we obtain

$$\frac{\partial(\nabla^2 \Phi^{sc})}{\partial t} - c\,\nabla^4 \Phi^{sc} = \delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.198}$$

in which we have introduced the superscript ^{sc} to denote continuous source, following the convention in Table 8.1. Equation (8.198) can also be expressed in the form of (8.181), with the relaxation of a Laplacian operator,

$$\frac{\partial \Phi^{sc}}{\partial t} - c\nabla^2 \Phi^{sc} = g_3 \tag{8.199}$$

Based on (8.184) and the elementary fundamental solution (8.129), we obtain

$$\begin{aligned} g_3 &= \frac{\ln r}{2\pi}\,\mathrm{H}(t - \tau), \qquad \text{(2D)} \\ &= -\frac{1}{4\pi r}\,\mathrm{H}(t - \tau), \quad \text{(3D)} \end{aligned} \tag{8.200}$$

where $r = |\vec{x} - \vec{\chi}|$.

There are multiple ways to solve the above equations. If we choose to start from (8.198), then based on the fundamental solution of diffusion equation (8.133), we find

$$\begin{aligned} \nabla^2 \Phi^{sc} \;=\; \zeta^{sc} &= \frac{1}{4\pi c}\,\mathrm{E}_1(\xi^2), \qquad \text{(2D)} \\ &= \frac{1}{4\pi c}\,\frac{\mathrm{erfc}(\xi)}{r}, \quad \text{(3D)} \end{aligned} \tag{8.201}$$

where

$$\xi = \frac{r}{\sqrt{4c(t - \tau)}} \tag{8.202}$$

Using (8.190) with $g_1 = 0$, we obtain the continuous source fundamental solution in terms of pore pressure as

$$p^{sc} = \frac{1}{4\pi\kappa}\mathrm{E}_1(\xi^2), \quad \text{(2D)}$$
$$= \frac{1}{4\pi\kappa}\frac{\mathrm{erfc}(\xi)}{r}, \quad \text{(3D)} \tag{8.203}$$

Equation (8.189) with $g_2 = 0$ gives

$$\dot{\Phi}^{sc} = \kappa p^{sc} + g_3 \tag{8.204}$$

where g_3 is given in (8.200). We hence have two ways to find Φ^{sc}, either by integrating (8.201) twice with respect to r, or by integrating (8.204) once with respect to t, and we obtain

$$\Phi^{sc} = \frac{r^2}{16\pi c}\left[\frac{1+\xi^2}{\xi^2}\mathrm{E}_1(\xi^2) + \frac{2}{\xi^2}\ln r - \frac{e^{-\xi^2}}{\xi^2}\right], \quad \text{(2D)}$$
$$= \frac{r}{16\pi c}\left[\frac{1+2\xi^2}{\xi^2}\mathrm{erfc}(\xi) - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi} - \frac{1}{\xi^2}\right], \quad \text{(3D)} \tag{8.205}$$

We are now ready to derive the rest of the fundamental solutions. Displacement is obtained from (8.195), and

$$u_i^{sc} = \frac{\eta}{8\pi G\kappa} r r_{,i}\left[\mathrm{E}_1(\xi^2) + \frac{1-e^{-\xi^2}}{\xi^2}\right], \quad \text{(2D)}$$
$$= \frac{\eta}{16\pi G\kappa} r_{,i}\left[2\,\mathrm{erfc}(\xi) + \frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right], \quad \text{(3D)} \tag{8.206}$$

where

$$r_{,i} = \frac{\partial r}{\partial x_i} = \frac{x_i - \chi_i}{r} \tag{8.207}$$

Stress is obtained from (8.192) (with $\sigma_{ij}^o = 0$), and

$$\sigma_{ij}^{sc} = -\frac{\eta}{4\pi\kappa}\left[(2r_{,i}r_{,j} - \delta_{ij})\frac{1-e^{-\xi^2}}{\xi^2} + \delta_{ij}\mathrm{E}_1(\xi^2)\right], \quad \text{(2D)}$$
$$= -\frac{\eta}{8\pi\kappa}\frac{1}{r}\left\{(3r_{,i}r_{,j} - \delta_{ij})\left[\frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right]\right.$$
$$\left. + 2(\delta_{ij} + r_{,i}r_{,j})\,\mathrm{erfc}(\xi)\right\}, \quad \text{(3D)} \tag{8.208}$$

The specific discharge is obtained by differentiating p^{sc}, and

$$q_i^{sc} = \frac{1}{2\pi}\frac{r_{,i}}{r}e^{-\xi^2}, \qquad \text{(2D)}$$
$$= \frac{1}{4\pi}\frac{r_{,i}}{r^2}\left[\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2} + \operatorname{erfc}(\xi)\right], \qquad \text{(3D)} \tag{8.209}$$

The complete list of the continuous source fundamental solutions is summarized in Sect. E.1.

As the continuous source fundamental solution can represent the physical problem of injection and extraction of fluid into and from a fluid saturated poroelastic formation, through a well of small or infinitesimal diameter, it is of interest to examine the physical characteristics of these solutions. First, we observe that there is a 2D and a 3D solution. The 3D solution corresponds to an injection well that can be approximated as a point at large distance, such as a short, screened section of a well. The 2D solution can represent a fully penetrating (and screened) well into a layered formation confined from top and bottom by impermeable layers.

Due to the nature of the singularity, the solutions presented above possess axial and spherical symmetry respectively for 2D and 3D problems. For example, we can express q_r^{sc} as

$$q_r^{sc} = \frac{1}{2\pi r}e^{-\xi^2}, \qquad \text{(2D)}$$
$$= \frac{1}{4\pi r^2}\left[\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2} + \operatorname{erfc}(\xi)\right], \qquad \text{(3D)} \tag{8.210}$$

Here for convenience, we used the same symbol r for r in the cylindrical coordinate system and R in the spherical coordinate system (Appendix C). We observe that q_r^{sc} is a function of r only. Particularly, as $r \to 0$, we find

$$\lim_{r\to 0} q_r^{sc} = \frac{1}{2\pi r}, \qquad \text{(2D)}$$
$$= \frac{1}{4\pi r^2}, \qquad \text{(3D)} \tag{8.211}$$

When the above is integrated over a circle or a spherical surface of radius r, we confirm that the volume injection rate is unity at the origin.

Based on axial and spherical symmetry, we can also express the displacement and stresses in (8.206) and (8.208) in these coordinates as functions of r only,

$$u_r^{sc} = \frac{\eta}{8\pi G\kappa} r\left[\mathrm{E}_1(\xi^2) + \frac{1-e^{-\xi^2}}{\xi^2}\right], \qquad \text{(2D)}$$
$$= \frac{\eta}{16\pi G\kappa}\left[2\operatorname{erfc}(\xi) + \frac{\operatorname{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right], \qquad \text{(3D)} \tag{8.212}$$

$$\sigma_{rr}^{sc} = -\frac{\eta}{4\pi\kappa}\left[\mathrm{E}_1(\xi^2) + \frac{1-\mathrm{e}^{-\xi^2}}{\xi^2}\right], \qquad \text{(2D)}$$

$$= -\frac{\eta}{4\pi\kappa}\frac{1}{r}\left[2\,\mathrm{erfc}(\xi) + \frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right], \qquad \text{(3D)} \qquad (8.213)$$

$$\sigma_{\theta\theta}^{sc} = -\frac{\eta}{4\pi\kappa}\left[\mathrm{E}_1(\xi^2) - \frac{1-e^{-\xi^2}}{\xi^2}\right], \qquad \text{(2D)}$$

$$= -\frac{\eta}{8\pi\kappa}\frac{1}{r}\left[2\,\mathrm{erfc}(\xi) - \frac{\mathrm{erf}(\xi)}{\xi^2} + \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right], \qquad \text{(3D)} \qquad (8.214)$$

We also note that $u_\theta^{sc} = \sigma_{r\theta}^{sc} = 0$. In the above, θ represents both (θ, φ) in the spherical coordinate system.

In Fig. 8.5 we present p^{sc}, u_r^{sc}, σ_{rr}^{sc} and $\sigma_{\theta\theta}^{sc}$ versus radial distance for the two-dimensional case. For convenience, in these plots we have assumed the following material constants: $c = 1$, $\kappa = 1$, $G = 1$, and $\eta = 0.25$. In Fig. 8.5a, the pore pressure at various times is presented. We observe that with the fluid mass injection, the pressure rises with time. Particularly, the pore pressure is infinite at the origin.

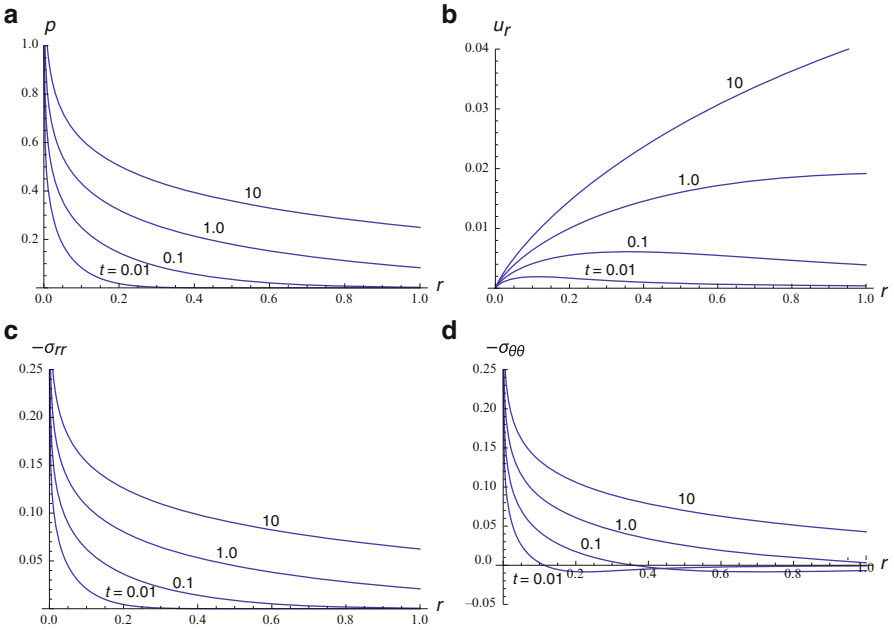


Fig. 8.5 Pore pressure, radial displacement, radial and tangential stresses of a two-dimensional continuous source at various times. (**a**) p^{sc}, (**b**) u_r^{sc}, (**c**) σ_{rr}^{sc}, and (**d**) $\sigma_{\theta\theta}^{sc}$

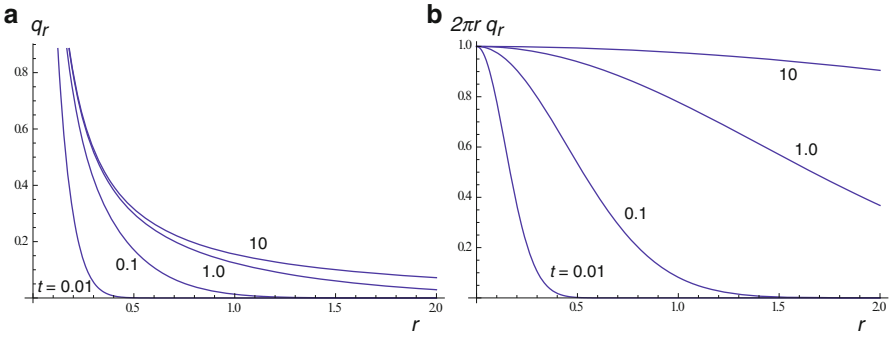


Fig. 8.6 Radial specific flux of a two-dimensional continuous source at various times: (**a**) q_r^{sc} vs. r, and (**b**) $2\pi r q_r^{sc}$ vs. r

Radial displacement is shown in Fig. 8.5b. We observe that u_r^{sc} is zero at the origin, and it has a maximum value propagating with time into the formation. Radial and tangential stresses are plotted as $-\sigma_{rr}^{sc}$ and $-\sigma_{\theta\theta}^{sc}$ in Fig. 8.5c, d. The injection of fluid causes a compressional stress in the near field, except for $\sigma_{\theta\theta}^{sc}$, which has a small tensile zone propagating with time into the formation with diminishing magnitude.

In Fig. 8.6 we plot the radial specific flux and the discharge for the 2D continuous source, (8.209), versus radial distance r at various times. Again, $c = 1$ is assumed. Figure 8.6a plots q_r^{sc} versus r. We observe that the flux diminishes rapidly at large distance. In Fig. 8.6b, we plot the total discharge crossing a circle of radius r, calculated as $2\pi r q_r^{sc}$, at a distance r. We note that if the fluid and the formation were incompressible, then the volume discharge crossing the circle should be unity (the discharge of the source) at any distance. The solution shows that at small times, the far field is hardly affected, and the injected volume is largely stored in the near field by the compressibility effect. A constant discharge develops only at large time.

Observing from the above, it is of interest to examine the asymptotic behaviors of the solution at small and large times, and near and far away from the source origin. First, we take the Taylor series expansion for small r, and obtain the near source behavior

$$\begin{aligned} p^{sc} &\approx \frac{1}{4\pi\kappa}\left[-\gamma - \ln\left(\frac{r^2}{4ct}\right)\right], \qquad \text{(2D)} \\ &\approx \frac{1}{4\pi\kappa}\left(\frac{1}{r} - \frac{1}{\sqrt{\pi ct}}\right), \qquad \text{(3D)} \end{aligned} \tag{8.215}$$

where $\gamma = 0.5772157\ldots$ is the Euler number. We observe that the pore pressure is singular at the origin. At small time, $t \to 0$, the expansion gives

$$p^{sc} \approx \frac{1}{\pi\kappa}\frac{ct}{r^2}e^{-r^2/4ct}, \qquad \text{(2D)}$$

$$\approx \frac{1}{2\pi^{3/2}\kappa}\sqrt{\frac{ct}{r^2}}\,e^{-r^2/4ct}, \qquad \text{(3D)} \tag{8.216}$$

Hence pore pressure is zero everywhere as $t \to 0$, except at the origin, where it is infinite. For solution at large time, the expansion produces the same result as (8.215). This shows that the as $t \to \infty$, the pore pressure grows unbounded for the 2D case, while it is bounded by $1/4\pi\kappa r$ for the 3D case. The large distance expansion gives the same result as (8.216); so at any finite time, the pore pressure decays exponentially toward large distance.

The stresses have the similar asymptotic behavior near the origin as the pore pressure, and are given by

$$\sigma_{rr}^{sc} \approx \frac{\eta}{4\pi\kappa}\left[-1+\gamma+\ln\left(\frac{r^2}{4ct}\right)\right], \qquad \text{(2D)}$$

$$\approx -\frac{\eta}{2\pi\kappa}\left(\frac{1}{r}-\frac{2\eta}{3\sqrt{\pi ct}}\right), \qquad \text{(3D)} \tag{8.217}$$

$$\sigma_{\theta\theta}^{sc} \approx \frac{\eta}{4\pi\kappa}\left[1+\gamma+\ln\left(\frac{r^2}{4ct}\right)\right], \qquad \text{(2D)}$$

$$\approx -\frac{\eta}{4\pi\kappa}\left(\frac{1}{r}-\frac{4\eta}{3\sqrt{\pi ct}}\right), \qquad \text{(3D)} \tag{8.218}$$

8.10.2 Instantaneous Source

A continuous source is a step rise of injection rate from 0 at $t = 0^-$ to unity at $t = 0^+$, and maintained at that rate at all times. The total injected volume rises linearly with time. An instantaneous source is an injection of a unit volume of fluid within an infinitesimal time duration. Hence the instantaneous source is represented by the following

$$\gamma = \delta(\vec{x}-\vec{\chi})\,\delta(t-\tau) \tag{8.219}$$

which is compared to (8.197) for the continuous source case. As there exists such relation

$$\frac{d\,\mathrm{H}(t-\tau)}{dt} = \delta(t-\tau) \tag{8.220}$$

we can easily deduce that the instantaneous source fundamental solutions, denoted by the ^{si} superscript, are the time derivatives of the continuous source solutions

$$\square^{si}_{\dots} = \frac{\partial \square^{sc}_{\dots}}{\partial t} \tag{8.221}$$

where the $\square$ symbol is replaced by u_i, p, σ_{ij}, etc., and the $\dots$ subscript represents the necessary tensor indices. The above argument in fact applies to all continuous and instantaneous fundamental solution, including dipole, total force, displacement discontinuity, etc. Hence we can introduce the following general formula

$$\square^{*i}_{\dots} = \frac{\partial \square^{*c}_{\dots}}{\partial t} \tag{8.222}$$

where the asterisk superscript can be replaced by ^{F,f,d}, etc., for the singularities defined in Table 8.1.

Following (8.221), we can find the pore pressure, displacement, and stresses by differentiating (8.203), and (8.212), (8.213) and (8.214), and obtain

$$\begin{aligned}
p^{si} &= \frac{1}{\pi S}\frac{1}{r^2}\xi^2 e^{-\xi^2}, && \text{(2D)} \\
&= \frac{1}{\pi^{3/2} S}\frac{1}{r^3}\xi^3 e^{-\xi^2}, && \text{(3D)} && (8.223) \\
u^{si}_r &= \frac{\eta}{2\pi G S}\frac{1}{r}\left(1 - e^{-\xi^2}\right), && \text{(2D)} \\
&= \frac{\eta}{4\pi G S}\frac{1}{r^2}\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right], && \text{(3D)} && (8.224) \\
\sigma^{si}_{rr} &= -\frac{\eta}{\pi S}\frac{1}{r^2}(1 - e^{-\xi^2}), && \text{(2D)} \\
&= -\frac{\eta}{\pi S}\frac{1}{r^3}\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right], && \text{(3D)} && (8.225) \\
\sigma^{si}_{\theta\theta} &= \frac{\eta}{\pi S}\frac{1}{r^2}\left[1 - (1 + 2\xi^2)e^{-\xi^2}\right], && \text{(2D)} \\
&= \frac{\eta}{2\pi S}\frac{1}{r^3}\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\xi(1 + 2\xi^2)e^{-\xi^2}\right], && \text{(3D)} && (8.226)
\end{aligned}$$

The rest of the instantaneous source solutions are listed in Sect. E.2.

In Fig. 8.7 we plot p^{si}, u^{si}_r, σ^{si}_{rr} and $\sigma^{si}_{\theta\theta}$ versus radial distance for the two-dimensional case, using the same set of material constants as the continuous source case. Comparing with Fig. 8.5, we observe that there is no singularity at the origin (except at $t = 0$), and all pore pressure and stresses are of finite value which decays with time.

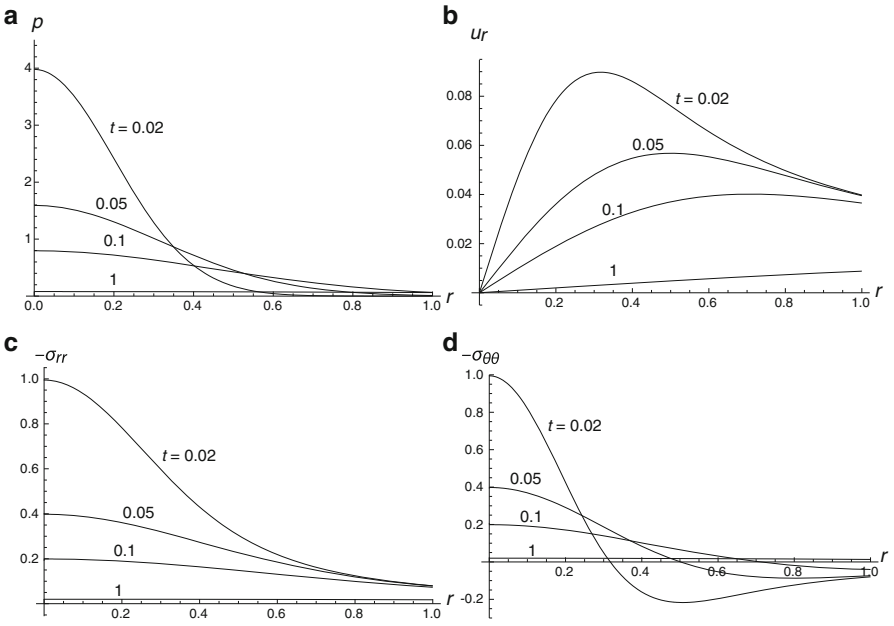


Fig. 8.7 Pore pressure, radial displacement, radial and tangential stresses of a two-dimensional instantaneous source at various times. (**a**) p^{si}, (**b**) u_r^{si}, (**c**) σ_{rr}^{si}, and (**d**) $\sigma_{\theta\theta}^{si}$

For an asymptotic analysis, we take the limit of $r \to 0$ and find

$$\begin{aligned} p^{si} &\approx \frac{1}{4\pi Sc}\left(\frac{1}{t} - \frac{r^2}{4ct^2}\right), && \text{(2D)} \\ &\approx \frac{1}{8\pi^{3/2}Sc^{3/2}}\left(\frac{1}{t^{3/2}} - \frac{r^2}{4ct^{5/2}}\right), && \text{(3D)} \end{aligned} \tag{8.227}$$

Comparing to the continuous source solution, we find that the pore pressure is no longer infinite at the origin, and in fact is given by

$$\begin{aligned} p^{si}(0,t) &= \frac{1}{4\pi Sc}\, t^{-1}, && \text{(2D)} \\ &= \frac{1}{8\pi^{3/2}Sc^{3/2}}\, t^{-3/2}, && \text{(3D)} \end{aligned} \tag{8.228}$$

Hence except at $t = 0$, the pore pressure is of a finite value that decays with time in the fashion shown in (8.228). We note that the expansion (8.227) is also valid for large time; hence the pore pressure at a given distance decays with time following (8.227).

The full solution (8.223) on the other hand shows that the pore pressure decays exponentially with distance at large distance, but increases with time at small time. In fact, the pore pressure at a given location is initially zero, and then increases with time, before it eventually dissipates to zero. The arrival time of the maximum pore pressure at a location r is

$$\begin{aligned} t_{\max} &= \frac{r^2}{4c}, \qquad \text{(2D)} \\ &= \frac{r^2}{6c}, \qquad \text{(3D)} \end{aligned} \tag{8.229}$$

The maximum pore pressure at the location is

$$\begin{aligned} p_{\max} &= \frac{1}{\pi e S}\frac{1}{r^2}, \qquad \text{(2D)} \\ &= \frac{1}{\pi^{3/2} e S}\frac{1}{r^3}, \qquad \text{(3D)} \end{aligned} \tag{8.230}$$

The peak pore pressure moves outward at a velocity of

$$\begin{aligned} v &= \sqrt{c/t}, \qquad \text{(2D)} \\ &= \sqrt{3c/2t}, \qquad \text{(3D)} \end{aligned} \tag{8.231}$$

Similarly, we can find that the stresses at the origin are finite, and decay with time as

$$\begin{aligned} \sigma_{rr}^{si}(0,t) \;=\; \sigma_{\theta\theta}^{si}(0,t) &= -\frac{\eta}{4\pi S c}\, t^{-1}, \qquad \text{(2D)} \\ &= -\frac{\eta}{6\pi^{3/2} S c^{3/2}}\, t^{-3/2}, \qquad \text{(3D)} \end{aligned} \tag{8.232}$$

while displacement is zero at the origin.

8.11 Fluid Dipole

In electrostatics, a *dipole* is a pair of electric charges of equal magnitude but opposite sign brought next to each other. In fluid mechanics, we may conceptually bring a source and a sink of equal (but opposite) intensity closer together along a line, with their intensities increasing proportional to the inverse of the distance separating them. In the limit, we obtain a dipole. It can be shown that the resultant potential field of this limiting process is equivalent to taking the spatial derivative of the source potential in the direction of the line.

8.11.1 Continuous Dipole

A poroelastic continuous dipole can be created by introducing a forcing function that is the spatial derivative of the continuous source. In other words, we can replace the source term γ by the following

$$\gamma_k = -\frac{\partial \delta(\vec{x} - \vec{\chi})}{\partial x_k} \mathrm{H}(t - \tau) \tag{8.233}$$

In the above we have added a subscript k to the source term to make it a vector because of the existence of three dipoles, one in each direction of differentiation. Although the action of bringing a source and a sink together does not sound like a physical process, we shall find the presence of dipole in the point force and displacement discontinuity solutions, to be presented in Sects. 8.15 and 8.18. Hence dipole does have a physical meaning and application.

Substituting (8.233) into the right hand side of (8.180), we obtain

$$\frac{\partial(\nabla^2 \Phi_k^{pc})}{\partial t} - c\nabla^4 \Phi_k^{pc} = -\frac{\partial \delta(\vec{x} - \vec{\chi})}{\partial x_k} \mathrm{H}(t - \tau) \tag{8.234}$$

in which we have introduced the superscript ^{pc} to denote continuous dipole, following the naming convention in Table 8.1. Comparing the above with (8.198), it is clear that we can obtain Φ^{pc} by differentiating Φ^{sc} given in (8.205), and

$$\begin{aligned} \Phi^{pc} &= -\frac{1}{8\pi c} r r_{,i} \left[\mathrm{E}_1(\xi^2) - \frac{e^{-\xi^2}}{\xi^2} \right], && \text{(2D)} \\ &= -\frac{1}{16\pi c} r_{,i} \left[\frac{1}{\xi^2} - \frac{2}{\sqrt{\pi}} \frac{e^{-\xi^2}}{\xi} - \frac{1 - 2\xi^2}{\xi^2} \operatorname{erfc}(\xi) \right], && \text{(3D)} \end{aligned} \tag{8.235}$$

In fact all dipole fundamental solutions can be obtained by differentiating the source fundamental solutions as follows

$$\Box^{p\dagger}_{\cdots k} = -\frac{\partial \Box^{s\dagger}_{\cdots}}{\partial x_k} \tag{8.236}$$

where the superscript ^{p} denotes dipole, and ^{s} is for source, $\Box$ is replaced by physical parameters such as u_i, p, σ_{ij}, etc., and the superscript † is replaced either by ^{i} for instantaneous, or by ^{c} for continuous solutions. As the right hand side is differentiated with respect to x_k, it prompts the addition of a tensor index k to the dipole solution, as shown on the left hand side. We also note that the negative sign is assigned to corresponds to the orientation of the source-sink pair as illustrated in Fig. 8.1.

As a demonstration, we obtain the following pore pressure solution based on (8.203)

$$p_k^{pc} = -\frac{\partial p^{sc}}{\partial x_k} = \frac{1}{2\pi\kappa}\frac{r_{,k}}{r}e^{-\xi^2}, \qquad \text{(2D)}$$

$$= \frac{1}{4\pi\kappa}\frac{r_{,k}}{r^2}\left[\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2} + \operatorname{erfc}(\xi)\right], \qquad \text{(3D)} \qquad (8.237)$$

By realizing that q_i^{sc} in (8.209) is obtained by differentiating p^{sc} with respect to x_i, we quickly conclude that

$$p_k^{p\dagger} = \frac{1}{\kappa} q_k^{s\dagger} \qquad (8.238)$$

where the † superscript is replaced by ^{c} for continuous and by ^{i} for instantaneous solutions. The displacement and stress solutions are

$$u_{ik}^{pc} = \frac{\eta}{8\pi G\kappa}\left[(2r_{,i}r_{,k} - \delta_{ik})\frac{1-e^{-\xi^2}}{\xi^2} - \delta_{ik}\mathrm{E}_1(\xi^2)\right], \qquad \text{(2D)}$$

$$= \frac{\eta}{16\pi G\kappa}\frac{1}{r}\left\{(3r_{,i}r_{,k} - \delta_{ik})\left[\frac{\operatorname{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right]\right.$$
$$\left. + 2(r_{,i}r_{,k} - \delta_{ik})\operatorname{erfc}(\xi)\right\}, \qquad \text{(3D)} \qquad (8.239)$$

$$\sigma_{ijk}^{pc} = \frac{\eta}{2\pi\kappa}\frac{1}{r}\left[(\delta_{ik}r_{,j} + \delta_{jk}r_{,i} + \delta_{ij}r_{,k} - 4r_{,i}r_{,j}r_{,k})\frac{1-e^{-\xi^2}}{\xi^2}\right.$$
$$\left. + 2(r_{,i}r_{,j}r_{,k} - \delta_{ij}r_{,k})e^{-\xi^2}\right], \qquad \text{(2D)}$$

$$= \frac{\eta}{8\pi\kappa}\frac{1}{r^2}\left\{(\delta_{ij}r_{,k} + \delta_{ik}r_{,j} + \delta_{jk}r_{,i} - 5r_{,i}r_{,j}r_{,k})\right.$$
$$\times\left[\frac{3\operatorname{erf}(\xi)}{\xi^2} - \frac{6}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi} + 2\operatorname{erfc}(\xi)\right]$$
$$\left. + 4(r_{,i}r_{,j}r_{,k} - \delta_{ij}r_{,k})\left[\operatorname{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right]\right\}, \text{(3D)} \qquad (8.240)$$

The rest of the continuous dipole solutions are listed in Sect. E.3.

In Fig. 8.8 we plot the pore pressure, specific flux, and stress components for a continuous dipole in the x direction, at $t = 1$. Figure 8.8a plots the pore pressure. We notice that for a dipole in the x direction, the source is on the positive side of x-axis, and the sink is on the negative side. We also observe that pore pressure is zero along the y-axis. Figure 8.8b is a vector plot of the flow magnitude and direction.

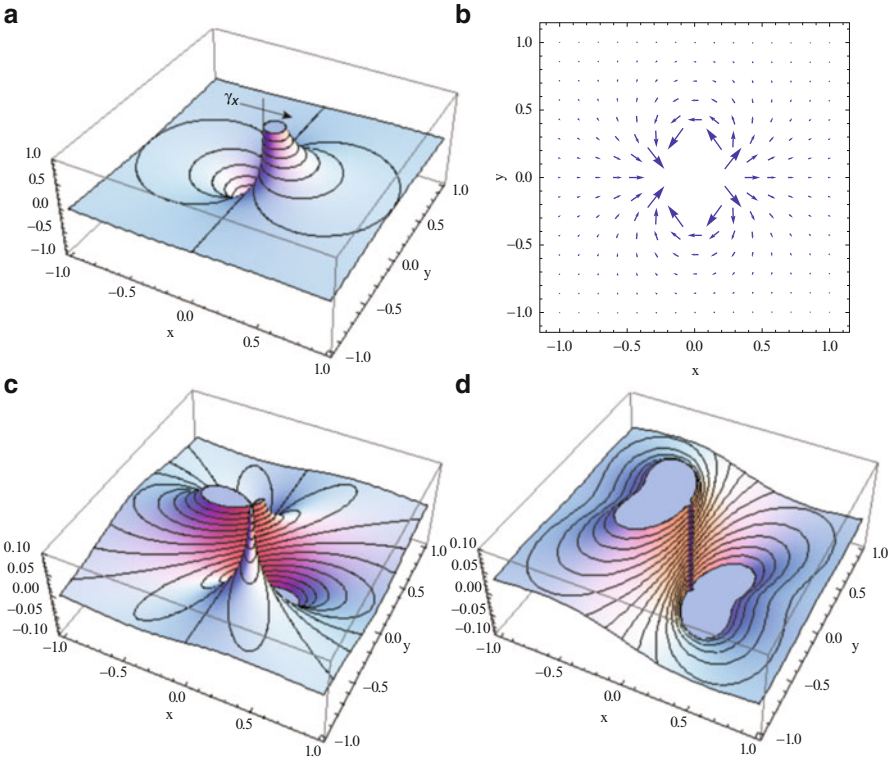


Fig. 8.8 Pore pressure, specific flux, and stresses of a two-dimensional continuous dipole in the x direction at $t = 1$. (**a**) p_x^{pc}, (**b**) $\vec{q}_x^{pc}$, (**c**) σ_{xxx}^{pc}, and (**d**) σ_{yyx}^{pc}

It shows that the x-axis is equivalent to an impermeable surface, as there is no flux across it. The induced normal stresses σ_{xxx}^{pc} and σ_{yyx}^{pc} by the fluid dipole are shown in Fig. 8.8c, d respectively.

For asymptotic behaviors, we find for small r the following

$$\begin{aligned} p_k^{pc} &\approx \frac{r_{,k}}{2\pi\kappa}\left(\frac{1}{r} - \frac{r}{4ct}\right), && \text{(2D)} \\ &\approx \frac{r_{,k}}{4\pi\kappa}\left(\frac{1}{r^2} - \frac{r}{6\pi^{1/2}c^{3/2}t^{3/2}}\right), && \text{(3D)} \end{aligned} \tag{8.241}$$

Hence there exist a $1/r$ and $1/r^2$ singularity, respectively, at the origin. The stresses have the same order of singularity as pore pressure.

8.11.2 Instantaneous Dipole

The instantaneous dipole solution corresponds to the introduction of the forcing term

$$\gamma_k = -\frac{\partial \delta(\vec{x}-\vec{\chi})}{\partial x_k}\,\delta(t-\tau) \tag{8.242}$$

It can be obtained by either differentiating the instantaneous source in the spatial direction, or differentiating the continuous dipole with respect to time. For example, the pore pressure solution can be obtained as

$$\begin{aligned} p_k^{pi} &= -\frac{\partial p^{si}}{\partial x_k} = \frac{\partial p^{pc}}{\partial t} = \frac{2}{\pi S}\frac{r_{,k}}{r^3}\xi^4 e^{-\xi^2}, \qquad \text{(2D)} \\ &= \frac{2}{\pi^{3/2} S}\frac{r_{,k}}{r^4}\xi^5 e^{-\xi^2}, \qquad \text{(3D)} \end{aligned} \tag{8.243}$$

The displacement and stress solutions are

$$\begin{aligned} u_{ik}^{pi} &= \frac{\eta}{2\pi GS}\frac{1}{r^2}\left[(2r_{,i}r_{,k}-\delta_{ik})(1-e^{-\xi^2}) - 2r_{,i}r_{,k}\xi^2 e^{-\xi^2}\right], \quad \text{(2D)} \\ &= \frac{\eta}{4\pi GS}\frac{1}{r^3}\left\{(3r_{,i}r_{,k}-\delta_{ik})\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right]\right. \\ &\qquad \left. -\frac{4}{\sqrt{\pi}}r_{,i}r_{,k}\xi^3 e^{-\xi^2}\right\}, \qquad \text{(3D)} \end{aligned} \tag{8.244}$$

$$\begin{aligned} \sigma_{ijk}^{pi} &= \frac{2\eta}{\pi S}\frac{1}{r^3}\left[(\delta_{ik}r_{,j}+\delta_{jk}r_{,i}+\delta_{ij}r_{,k}-4r_{,i}r_{,j}r_{,k})(1-e^{-\xi^2}-\xi^2 e^{-\xi^2})\right. \\ &\qquad \left. +2(r_{,i}r_{,j}r_{,k}-\delta_{ij}r_{,k})\xi^4 e^{-\xi^2}\right], \qquad \text{(2D)} \\ &= \frac{\eta}{2\pi S}\frac{1}{r^4}\left\{\frac{8}{\sqrt{\pi}}(r_{,i}r_{,j}r_{,k}-\delta_{ij}r_{,k})\xi^5 e^{-\xi^2}\right. \\ &\qquad +(\delta_{ij}r_{,k}+\delta_{ik}r_{,j}+\delta_{jk}r_{,i}-5r_{,i}r_{,j}r_{,k}) \\ &\qquad \left.\times\left[3\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}(3+2\xi^2)\xi e^{-\xi^2}\right]\right\}, \quad \text{(3D)} \end{aligned} \tag{8.245}$$

The rest of the instantaneous dipole solutions are listed in Sect. E.4.

In Fig. 8.9 we plot the displacement, pore pressure and stress components for a two-dimensional instantaneous dipole in the x direction, at $t = 0.05$, for comparison with the continuous dipole solution in Fig. 8.8. We note that these solutions are no longer singular at the origin. They are either zero, or of finite value and decay with time.

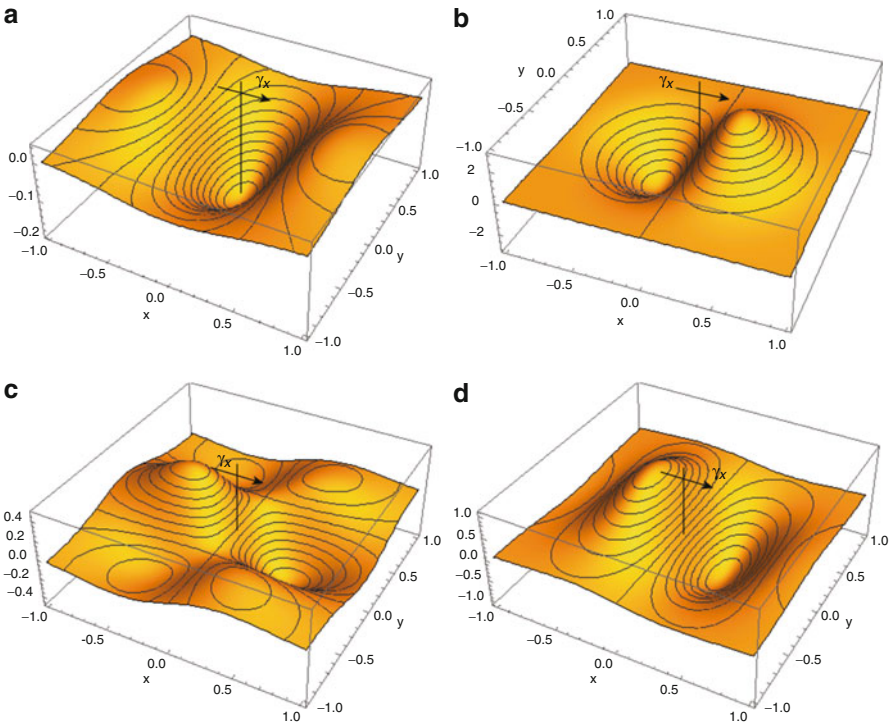


Fig. 8.9 Pore pressure and stress of a two-dimensional instantaneous dipole in the x direction at $t = 0.05$. (**a**) u^{pi}_{xx}, (**b**) p^{pi}_{x}, (**c**) σ^{pi}_{xxx}, and (**d**) σ^{pi}_{yyx}

For asymptotic behaviors, we find for small r the following

$$\begin{aligned} p^{pi}_k &\approx \frac{r_{,k}\, r}{8\pi S c^2 t^2}, \qquad \text{(2D)} \\ &\approx \frac{r_{,k}\, r}{16\pi^{3/2} S c^{5/2} t^{5/2}}, \qquad \text{(3D)} \end{aligned} \tag{8.246}$$

It is clear that the pore pressure is not only finite at the origin, it is zero. As observe in Fig. 8.9a, the pore pressure has a peak (trough) along the x-axis. The arrival time for the peak at a given r is

$$\begin{aligned} t_{\max} &= \frac{r^2}{8c}, \qquad \text{(2D)} \\ &= \frac{r^2}{10c}, \qquad \text{(3D)} \end{aligned} \tag{8.247}$$

The maximum pore pressure at the location is

$$\begin{aligned} p_{\max} &= \frac{8}{\pi e^2 S}\frac{1}{r^3}, && \text{(2D)} \\ &= \frac{25\sqrt{5}}{2\sqrt{2}\,\pi^{3/2} e^{5/2} S}\frac{1}{r^4}, && \text{(3D)} \end{aligned} \tag{8.248}$$

The peak pore pressure moves at a velocity of

$$\begin{aligned} v &= \sqrt{2c/t}, && \text{(2D)} \\ &= \sqrt{5c/2t}, && \text{(3D)} \end{aligned} \tag{8.249}$$

We also note that the displacement in the direction of the dipole at the origin is

$$\begin{aligned} u_{ik}^{pi}(0,t) &= -\frac{\eta}{8\pi GSc}\delta_{ik} t^{-1}, && \text{(2D)} \\ &= -\frac{\eta}{24\pi^{3/2} GSc^{3/2}}\delta_{ik} t^{-3/2}, && \text{(3D)} \end{aligned} \tag{8.250}$$

8.12 Fluid Dilatation

The fluid dilatation fundamental solution appeared in the various integral equations presented earlier. The fluid dilatation term Q is defined as the volume of injected fluid, which is obtained by integrating the source term γ with respect to time, as shown in (8.29). We shall discuss a continuous case, that is, a volume expansion is introduced at $t = 0^+$ and stays there all the time, and an instantaneous case, in which a spike (Dirac delta function in time) expansion of unit volume takes place at the origin, and disappears instantly.

8.12.1 Continuous Fluid Dilatation

A continuous fluid dilatation is represented by the following

$$Q = \delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.251}$$

By the definition of Q and γ, we clearly see that the above is equivalent to the introduction of an instantaneous source

$$\gamma = \delta(\vec{x} - \vec{\chi})\,\delta(t - \tau) \tag{8.252}$$

In other words, an instantaneous source injects a unit volume of fluid within an infinitesimal amount of time, and then ceases to exist. The action is the same as the sudden appearance of a unit fluid volume dilatation at a point that stays constant with time. We hence recognize the following relation

$$\square^{lc}_{\cdots} = \square^{si}_{\cdots} \tag{8.253}$$

where the superscript ^{l} denotes fluid dilatation. Also, as q_i is the time derivative of w_i, as given in (8.187) and (8.188), we have the following relation

$$w_i^{l\dagger} = q_i^{s\dagger} \tag{8.254}$$

where † is replaced by ^{c} for continuous and by ^{i} for instantaneous solutions.

As we have already derived the instantaneous source solution, the entire continuous fluid dilatation solution is available, as listed in Sect. E.5.

8.12.2 Instantaneous Fluid Dilatation

For an instantaneous fluid dilatation

$$Q = \delta(\vec{x} - \vec{\chi})\,\delta(t - \tau) \tag{8.255}$$

it is equivalent to

$$\gamma = \delta(\vec{x} - \vec{\chi})\,\frac{\partial \delta(t-\tau)}{\partial t} \tag{8.256}$$

Hence we obtain

$$\square^{li}_{\cdots} = \frac{\partial \square^{si}_{\cdots}}{\partial t} \tag{8.257}$$

For example, we find from (8.223) that

$$\begin{aligned} p^{li} \;=\; \frac{\partial p^{si}}{\partial t} &= \frac{4c}{\pi S}\frac{1}{r^4}(\xi^2 - 1)\xi^4 e^{-\xi^2}, && \text{(2D)} \\ &= \frac{2c}{\pi^{3/2} S}\frac{1}{r^5}(2\xi^2 - 3)\xi^5 e^{-\xi^2}, && \text{(3D)} \end{aligned} \tag{8.258}$$

We also find that

$$u_r^{li} = \frac{\eta}{2\pi GS}\frac{1}{r}\left[\delta(t-\tau) - 4c\frac{1}{r^2}\xi^4 e^{-\xi^2}\right], \qquad \text{(2D)}$$

$$= \frac{\eta}{4\pi GS}\frac{1}{r^2}\left[\delta(t-\tau) - \frac{8c}{\sqrt{\pi}}\frac{1}{r^2}\xi^5 e^{-\xi^2}\right], \qquad \text{(3D)} \qquad (8.259)$$

$$\sigma_{rr}^{li} = \frac{\eta}{\pi S}\frac{1}{r^2}\left[-\delta(t-\tau) + 4c\frac{1}{r^2}\xi^4 e^{-\xi^2}\right], \qquad \text{(2D)}$$

$$= \frac{\eta}{\pi S}\frac{1}{r^3}\left[-\delta(t-\tau) + \frac{8c}{\sqrt{\pi}}\frac{1}{r^2}\xi^5 e^{-\xi^2}\right], \qquad \text{(3D)} \qquad (8.260)$$

$$\sigma_{\theta\theta}^{li} = \frac{\eta}{\pi S}\frac{1}{r^2}\left[\delta(t-\tau) + 4c\frac{1}{r^2}(1-2\xi^2)\xi^4 e^{-\xi^2}\right], \qquad \text{(2D)}$$

$$= \frac{\eta}{2\pi S}\frac{1}{r^3}\left[\delta(t-\tau) + \frac{16c}{\sqrt{\pi}}\frac{1}{r^2}(1-\xi^2)\xi^5 e^{-\xi^2}\right], \qquad \text{(3D)} \qquad (8.261)$$

In the above we notice the presence of a Dirac delta function in time, otherwise these quantities are finite at the origin. The complete list of fundamental solutions is shown in Sect. E.6.

In Fig. 8.10 we present p^{li}, u_r^{li}, σ_{rr}^{li} and $\sigma_{\theta\theta}^{li}$ versus radial distance for the two-dimensional case, using the same materials constants as the cases before. In Fig. 8.10a, we observe that a large negative pore pressure p^{li} is generated near the origin (infinite at $r = 0$ and $t = 0^+$), which dissipates with time. Figure 8.10b shows that the fluid dilatation causes an radial displacement u_r^{li} moving inward towards the origin. At the origin, the displacement is zero as no displacement discontinuity is allowed. At large time, the fluid effect diminishes and the displacement returns to zero everywhere. The inward movement of radial displacement causes a tensile radial stress σ_{rr}^{li} as demonstrated in Fig. 8.10c. The tangential stress $\sigma_{\theta\theta}^{li}$ as plotted in Fig. 8.10d shows both a tensile and a compressive region.

For limiting behaviors, we find at origin and for $t > 0$, the following

$$p^{li}(0,t) = -\frac{1}{4\pi Sc}\,t^{-2}, \qquad \text{(2D)}$$

$$= -\frac{3}{16\pi^{3/2}Sc^{3/2}}\,t^{-5/2}, \qquad \text{(3D)} \qquad (8.262)$$

and

$$\sigma_{rr}^{li}(0,t) = \sigma_{\theta\theta}^{li}(0,t) = \frac{\eta}{4\pi Sc}\,t^{-2}, \qquad \text{(2D)}$$

$$= \frac{\eta}{4\pi^{3/2}Sc^{3/2}}\,t^{-5/2}, \qquad \text{(3D)} \qquad (8.263)$$

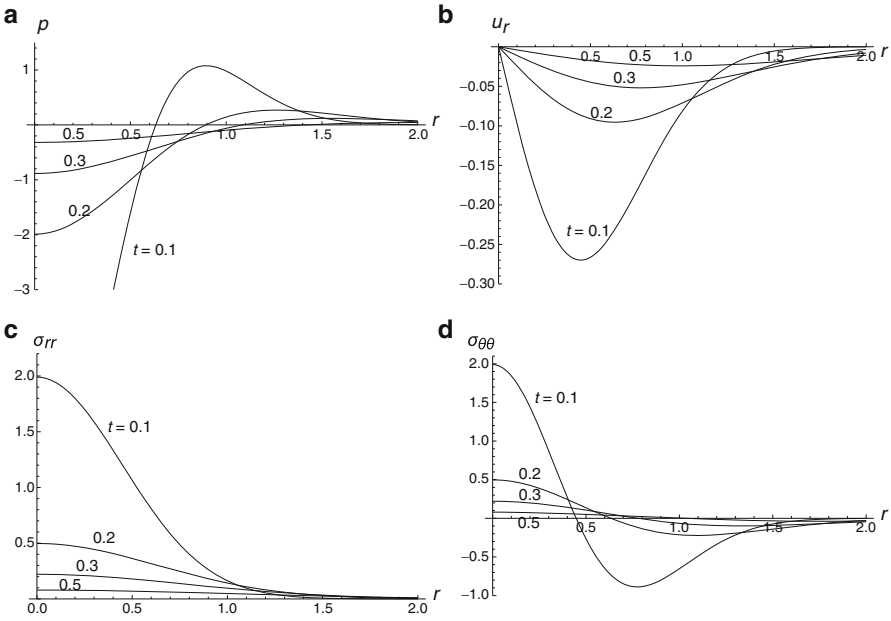


Fig. 8.10 Pore pressure, radial displacement, radial and tangential stresses of a two-dimensional instantaneous fluid dilatation at various times. (**a**) p^{li}, (**b**) u_r^{li}, (**c**) σ_{rr}^{li}, and (**d**) $\sigma_{\theta\theta}^{li}$

8.13 Fluid Force

A fluid force is similar to a fluid dipole, but is not a fluid dipole, as we shall demonstrate below.

8.13.1 Continuous Fluid Force

A continuous fluid force in the x_k direction is given by

$$f_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.264}$$

We observe from the above expressions that the substitution of (8.264) into the right hand side of (8.176) produces

$$f_{ik,i} = \frac{\partial\delta(\vec{x} - \vec{\chi})}{\partial x_k}\,\mathrm{H}(t - \tau) \tag{8.265}$$

which is equivalent to the introduction of a continuous fluid dipole defined by (8.233), but adjusted for a multiplication factor κ. We hence conclude that if we solve (8.175) and (8.176) for u_i and ζ, we have the following equivalency

$$\Box^{f\dagger}_{\dots} = \kappa \, \Box^{p\dagger}_{\dots} \tag{8.266}$$

in which $\Box$ represents u_i and ζ, as well as variables derived from them, such as σ_{ij} and p, and the superscript † is replaced by either ^{c} or ^{i}.

The above relation, however, does not extend to q_i and w_i. As observed from (8.187) and (8.188), the source and fluid body force equivalence does not exist. Rather, from (8.188) and (8.195), we find that in the absence of F_i ($g_1 = 0$) and γ ($g_3 = 0$), u_i and w_i are related as

$$w^{f\dagger}_{ik} = -\frac{GS}{\eta} u^{f\dagger}_{ik} \tag{8.267}$$

Differentiating the above equation with respect to time, we obtain

$$q^{fc}_{ik} = -\frac{GS}{\eta} u^{fi}_{ik} = -\frac{GS\kappa}{\eta} u^{pi}_{ik} \tag{8.268}$$

These fundamental solutions are summarized in Sect. E.7.

8.13.2 Instantaneous Fluid Force

An instantaneous fluid force is

$$f_{ik} = \delta_{ik} \delta(\vec{x} - \vec{\chi}) \, \delta(t - \tau) \tag{8.269}$$

The fundamental solutions can be obtained by differentiating with respect to time the continuous fluid force solutions, and are summarized in Sect. E.8.

The concept of a double force, created by differentiating a point force, and the assemblage of mutually orthogonal pairs of double force to created quadrupole in 2D (plane strain) and hexapole in 3D, was discussed in Sect. 8.8.2, for elasticity. Here we shall construct their poroelastic counterparts.

8.14 Fluid Dodecapole

A fluid *dodecapole* is formed by adding two and three pairs of opposing and approaching dipoles, on two and three mutually orthogonal axes, respectively for 2D and 3D problems, in the limit converging on a single point [37]. In other words,

it is the negative of divergence of a dipole, or the Laplacian of a source. Such fluid singularity is created in a solid quadrupole and sexapole, which will be discussed in Sect. 8.16. The forcing function of a continuous dodecapole is given by

$$\gamma = \nabla^2 \delta(\vec{x} - \vec{\chi})\, \mathrm{H}(t - \tau) \tag{8.270}$$

This means that we can obtain the dodecapole solutions as

$$u_i^{oc} = -u_{ij,j}^{pc} = u_{i,jj}^{sc} \tag{8.271}$$

where the superscript ^{o} denotes dodecapole. But we shall directly seek its solution.

The problem is governed by

$$\frac{\partial(\nabla^2 \Phi^{oc})}{\partial t} - c\nabla^4 \Phi^{oc} = \mathrm{H}(t - \tau)\nabla^2 \delta(\vec{x} - \vec{\chi}) \tag{8.272}$$

Relaxing a Laplacian in the above produces

$$\frac{\partial \Phi^{oc}}{\partial t} - c\nabla^2 \Phi^{oc} = \delta(\vec{x} - \vec{\chi})\, \mathrm{H}(t - \tau) \tag{8.273}$$

Referring to (8.181), the above expression also suggests

$$g_3 = \delta(\vec{x} - \vec{\chi})\, \mathrm{H}(t - \tau) \tag{8.274}$$

It is obvious from (8.132) and (8.133) that

$$\begin{aligned} \Phi^{oc} &= \frac{1}{4\pi c}\, \mathrm{E}_1(\xi^2), \qquad \text{(2D)} \\ &= \frac{1}{4\pi c}\, \frac{\mathrm{erfc}(\xi)}{r}, \qquad \text{(3D)} \end{aligned} \tag{8.275}$$

Based on (8.195), we obtain the displacement

$$\begin{aligned} u_r^{oc} &= -\frac{\eta}{2\pi G\kappa}\, \frac{1}{r}\, e^{-\xi^2}, \qquad \text{(2D)} \\ &= -\frac{\eta}{4\pi G\kappa}\, \frac{1}{r^2} \left[\mathrm{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\, \xi e^{-\xi^2} \right], \qquad \text{(3D)} \end{aligned} \tag{8.276}$$

The limiting behavior of the displacement is observed as follows. First, it is clear that as $t \to 0$, we find

$$u_r^{oc}(r, 0) = 0 \tag{8.277}$$

At large time, $t \to \infty$, the displacement evolves into

$$\begin{aligned} u_r^{oc}(r,\infty) &= -\frac{\eta}{2\pi G\kappa}\frac{1}{r}, && \text{(2D)} \\ &= -\frac{\eta}{4\pi G\kappa}\frac{1}{r^2}, && \text{(3D)} \end{aligned} \tag{8.278}$$

Comparing the above with (8.165), we observe that this is a *center of compression* (or a negative *center of dilatation*). Or, we can present (8.276) in terms of the volumetric strain, which gives

$$\begin{aligned} e^{oc} &= -\frac{\eta}{G\kappa}\left[\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau) - \frac{1}{\pi}\frac{1}{r^2}\xi^2 e^{-\xi^2}\right], && \text{(2D)} \\ &= -\frac{\eta}{G\kappa}\left[\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau) - \frac{1}{\pi^{3/2}}\frac{1}{r^3}\xi^3 e^{-\xi^2}\right], && \text{(3D)} \end{aligned} \tag{8.279}$$

The above shows that a center of compression is instantly generated at the origin. We shall discuss its relation with the solid quadrupole and hexapole in Sect. 8.16.

Pore pressure can be found from (8.189) with g_3 given as (8.274)

$$\begin{aligned} p^{oc} &= -\frac{1}{\pi\kappa}\frac{1}{r^2}\xi^2 e^{-\xi^2} - \frac{1}{\kappa}\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau), && \text{(2D)} \\ &= -\frac{1}{\pi^{3/2}\kappa}\frac{1}{r^3}\xi^3 e^{-\xi^2} - \frac{1}{\kappa}\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau), && \text{(3D)} \end{aligned} \tag{8.280}$$

and ζ^{oc} from (8.191) as

$$\zeta^{oc} = Sp^{oc} \tag{8.281}$$

These and other solutions are listed in Sect. E.9. The instantaneous solutions are simply the time derivatives of the above.

8.15 Total Force

The fundamental solutions presented in the preceding sections are of fluid origin, which possesses the property of an irrotational field. When a force is applied to the solid portion, such statement is no longer true, and a different treatment is needed.

8.15.1 Continuous Total Force

The solution of a continuous point force in the x_k-direction corresponds to the substitution of

$$F_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau) \tag{8.282}$$

into the right hand sides of (8.175) and (8.176), and we obtain

$$G u^{Fc}_{ik,jj} + \frac{G}{(1-2\nu_u)}\, u^{Fc}_{jk,ji} - \alpha M \zeta^{Fc}_{k,i} = -\delta_{ik}\,\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau) \tag{8.283}$$

$$\frac{\partial \zeta^{Fc}_k}{\partial t} - c\,\nabla^2 \zeta^{Fc}_k = \frac{\eta\, c}{G}\,\frac{\partial \delta(\vec{x} - \vec{\chi})}{\partial x_k}\,\mathrm{H}(t-\tau) \tag{8.284}$$

in which the superscript ^{Fc} denotes the continuous total force solution. According to (8.177), the displacement solution can be decomposed into two parts, one part satisfies the undrained Navier equation (8.178), and the other part the diffusion equation (8.180), as follows

$$G u^{Fo}_{ik,jj} + \frac{G}{1-2\nu_u}\, u^{Fo}_{jk,ji} = -\delta_{ik}\,\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau) \tag{8.285}$$

$$\frac{\partial(\nabla^2 \Phi^{Fc})}{\partial t} - c\,\nabla^4 \Phi^{Fc} = \frac{\eta\, c}{G}\,\frac{\partial \delta(\vec{x} - \vec{\chi})}{\partial x_k}\,\mathrm{H}(t-\tau) \tag{8.286}$$

Equation (8.285) leads to the Kelvin solution (8.152), except that the undrained Poisson ratio ν_u replaces the drained one ν,

$$\begin{aligned} u^{Fo}_{ik} &= \frac{1}{8\pi G(1-\nu_u)}\,[r_{,i} r_{,k} - (3-4\nu_u)\delta_{ik}\ln r]\,\mathrm{H}(t-\tau), \quad \text{(2D)} \\ &= \frac{1}{16\pi G(1-\nu_u)}\,\frac{1}{r}\,[r_{,i} r_{,k} + (3-4\nu_u)\,\delta_{ik}]\,\mathrm{H}(t-\tau), \quad \text{(3D)} \end{aligned} \tag{8.287}$$

For (8.286), we can relax a Laplacian to obtain

$$\frac{\partial \Phi^{Fc}}{\partial t} - c\,\nabla^2 \Phi^{Fc} = \frac{\eta\, c}{G}\, g_1 \tag{8.288}$$

Based on (8.182), g_1 satisfies

$$\nabla^2 g_1 = \frac{\partial \delta(\vec{x} - \vec{\chi})}{\partial x_k}\,\mathrm{H}(t-\tau) \tag{8.289}$$

hence

$$g_1 = \frac{1}{2\pi}\frac{r_{,i}}{r}\mathrm{H}(t-\tau), \qquad \text{(2D)}$$
$$= \frac{1}{4\pi}\frac{r_{,i}}{r^2}\mathrm{H}(t-\tau), \qquad \text{(3D)} \tag{8.290}$$

Rather than solving (8.288) directly, we can compare (8.286) with (8.234) of the dipole solution, and quickly conclude that

$$\Phi_k^{Fc} = -\frac{\eta c}{G}\Phi_k^{pc} \tag{8.291}$$

Hence the application of a point force to poroelastic medium generates not only an elastic field, but also a continuous fluid dipole at the origin, which gives the transient nature of the solution. We then deduce from the above that

$$u_{ik}^{Fc} = u_{ij}^{Fo} - \frac{\eta c}{G}u_{ik}^{pc} \tag{8.292}$$

We can assemble these together to get

$$u_{ik}^{Fc} = \frac{1}{8\pi G(1-\nu_u)}\left[r_{,i}r_{,k} - (3-4\nu_u)\delta_{ik}\ln r\right]\mathrm{H}(t-\tau)$$
$$+\frac{\eta^2}{8\pi G^2 S}\left[(\delta_{ik}-2r_{,i}r_{,k})\frac{1-e^{-\xi^2}}{\xi^2} + \delta_{ik}\mathrm{E}_1(\xi^2)\right], \qquad \text{(2D)} \tag{8.293}$$
$$= \frac{1}{16\pi G(1-\nu_u)}\frac{1}{r}\left[r_{,i}r_{,k} + (3-4\nu_u)\delta_{ik}\right]\mathrm{H}(t-\tau)$$
$$+\frac{\eta^2}{16\pi G^2 S}\frac{1}{r}\left\{(\delta_{ik}-3r_{,i}r_{,k})\left[\frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right]\right.$$
$$\left. + 2(\delta_{ik}-r_{,i}r_{,k})\,\mathrm{erfc}(\xi)\right\}, \qquad \text{(3D)} \tag{8.294}$$

From (8.192), we find the stress solution

$$\sigma_{ijk}^{Fc} = \sigma_{ijk}^{Fo} - \frac{\eta c}{G}\sigma_{ijk}^{pc} \tag{8.295}$$

where

$$\sigma_{ijk}^{Fo} = \frac{1}{4\pi(1-\nu_u)}\frac{1}{r}\left[(1-2\nu_u)\left(\delta_{ij}r_{,k} - \delta_{jk}r_{,i} - \delta_{ik}r_{,j}\right)\right.$$
$$\left. -2\,r_{,i}r_{,j}r_{,k}\right]\mathrm{H}(t-\tau), \qquad \text{(2D)}$$

$$= \frac{1}{8\pi(1-\nu_u)}\frac{1}{r^2}\left[(1-2\nu_u)\left(\delta_{ij}r_{,k}-\delta_{jk}r_{,i}-\delta_{ik}r_{,j}\right)\right.$$
$$\left.-3\,r_{,i}r_{,j}r_{,k}\right]\mathrm{H}(t-\tau), \qquad \text{(3D)} \qquad (8.296)$$

And (8.179) gives the relation

$$\zeta_k^{Fc} = -\frac{\eta c}{G}\zeta_k^{pc} \qquad (8.297)$$

The above simple relation however does not extend to p_k^{Fc}, w_{ik}^{Fc}, and q_{ik}^{Fc}. This is because the pressure due to a dipole is zero everywhere at $t = 0^+$, except at the singularity. For the point force loading, there is an instantaneous pressure response everywhere due to the Skempton effect. This effect is introduced through the g_1 term in (8.191); hence we obtain

$$p_k^{Fc} = -\frac{\eta c}{G}p_k^{pc} + \frac{\eta}{GS}g_1 \qquad (8.298)$$

Combining (8.237) and (8.290), we find

$$p_k^{Fc} = \frac{\eta}{2\pi GS}\frac{r_{,k}}{r}\left(1-e^{-\xi^2}\right)\mathrm{H}(t-\tau), \qquad \text{(2D)}$$
$$= \frac{\eta}{4\pi GS}\frac{r_{,k}}{r^2}\left[\mathrm{erf}(\xi)-\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right]\mathrm{H}(t-\tau), \qquad \text{(3D)} \qquad (8.299)$$

These fundamental solutions are summarized in Sect. E.10.

To visualize the fundamental solution, we plot in Fig. 8.11 the stress field at $t = 0^+$ due to a continuous point force in x-direction for the 2D problem.

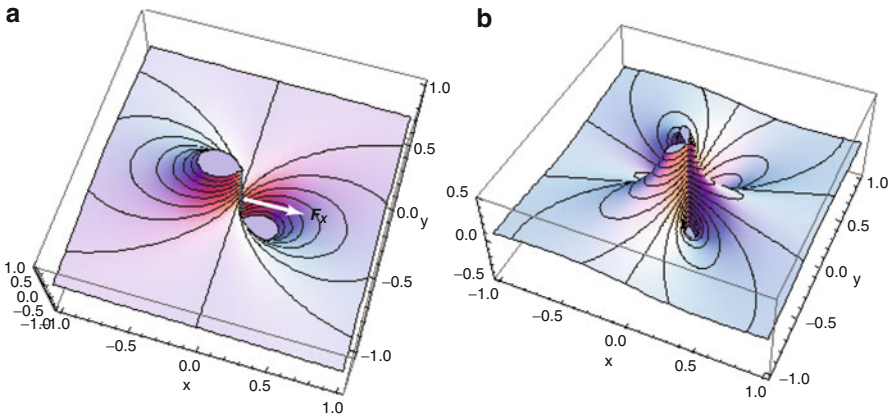


Fig. 8.11 Stress field of continuous point force in the x-direction at $t = 0^+$: (**a**) σ_{xxx}^{Fo}, and (**b**) σ_{yyx}^{Fo}

Figure 8.11a, b respectively show σ^{Fo}_{xxx} and σ^{Fo}_{yyx}, by assuming the parameter value $\nu_u = 0.4$. This response is in fact elastic but with undrained Poisson ratio replacing the drained on. As discussed above, the evolutional part of the displacement and stress solution is of the same pattern as a negative dipole, which can be observed in Fig. 8.8; hence is not reproduced here. It is of interest to note that due to the drained and undrained characteristics of poroelastic material, the step rise of stress response will go through an evolution; but as $t \to \infty$, it will return to the same pattern as shown in Fig. 8.11, except that its magnitude is modulated by the drained, instead of the undrained, Poisson ratio.

8.15.2 *Instantaneous Total Force*

The instantaneous force in the x_k-direction corresponds to the substitution of

$$F_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\,\delta(t - \tau) \tag{8.300}$$

The fundamental solutions can be obtained by taking the time derivative of the continuous total force solutions presented in the preceding section. Particularly we note from (8.292) and (8.295) that

$$u^{Fi}_{ik} = u^{Fo}_{ik}\,\delta(t - \tau) - \frac{\eta\, c}{G}\, u^{pi}_{ik} \tag{8.301}$$

$$\sigma^{Fi}_{ijk} = \sigma^{Fo}_{ijk}\,\delta(t - \tau) - \frac{\eta\, c}{G}\, \sigma^{pi}_{ijk} \tag{8.302}$$

where u^{Fo}_{ik} and σ^{Fo}_{ijk} are the undrained elastic solutions defined in (8.287) and (8.296), and u^{pi}_{ik} and σ^{pi}_{ijk} are the instantaneous fluid dipole solutions given in Sect. 8.11.2. The full solutions are summarized in Sect. E.11.

We observe from (8.301) and (8.302) that with the application of an instantaneous force, an infinitely large elastic field is generated at $t = 0^+$, which then instantly disappears. For an elastic material, the system returns to the original undisturbed state. For poroelastic material, however, the effect lingers. This is because an instantaneous fluid dipole is generated by the force. As observed from the solutions in Sect. 8.11.2 and Fig. 8.9, a pore pressure and stress peak is generated at the origin, which propagates outward, with its magnitude decaying with time. For practical consideration, we realize that an instantaneous force is not possible create. However, any force applied for a short period of time and then removed can be approximated as an instantaneous force a little while after the force is removed. The field then continuous to show the effect of an instantaneous dipole.

8.16 Solid Quadrupole and Hexapole

The concept of the creation of a double force, and its assemblage into mutually orthogonal pairs to from quadrupole in 2D (plane strain) and hexapole in 3D, has been discussed in Sect. 8.8.2 for elasticity. Their counterpart in poroelasticity has been derived and analyzed by Huynen and Detournay [37]. Here and in the section below we give a slightly different presentation.

Based on (8.282), the forcing function of a continuous quadrupole/hexapole is given by

$$F_{ik,k} = \delta_{ik}\delta_{,k}\,\mathrm{H}(t-\tau) = \delta_{,i}\,\mathrm{H}(t-\tau) \tag{8.303}$$

This means that we can obtain the quadrupole/hexapole solutions as

$$u_i^{qc} = u_{ik,k}^{Fc} \tag{8.304}$$

where the superscript ^{q} is used to denote both the quadrupole and the hexapole. However, rather than carrying out the lengthy differentiation and simplification, which can sometimes miss the proper singularity, we shall derive them directly in the following.

Utilizing the Biot decomposition (8.177), and the uncoupled Eqs. (8.178) and (8.180), we obtain these two equations corresponding to the current forcing function

$$G\,u_{i,jj}^{qo} + \frac{G}{1-2\nu_u}\,u_{j,ji}^{qo} = -\delta_{,i}\,\mathrm{H}(t-\tau) \tag{8.305}$$

$$\frac{\partial(\nabla^2\Phi^{qc})}{\partial t} - c\,\nabla^4\Phi^{qc} = \frac{\eta\,c}{G}\,\mathrm{H}(t-\tau)\,\nabla^2\delta \tag{8.306}$$

The above relation also suggests that

$$g_1 = \delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau) \tag{8.307}$$

Comparing (8.305) with (8.155), we quickly obtain from (8.157) that

$$\begin{aligned} u_r^{qo} &= -\frac{1-2\nu_u}{4\pi G(1-\nu_u)}\,\frac{1}{r}\,\mathrm{H}(t-\tau), \quad \text{(2D)} \\ &= -\frac{1-2\nu_u}{8\pi G(1-\nu_u)}\,\frac{1}{r^2}\,\mathrm{H}(t-\tau), \quad \text{(3D)} \end{aligned} \tag{8.308}$$

For (8.306), comparison with (8.272) shows that

$$\Phi^{qc} = \frac{\eta\,c}{G}\,\Phi^{oc} \tag{8.309}$$

Hence the application of a quadrupole/hexapole creates a fluid dodecapole. Assembling the terms, we find

$$u_r^{qc} = -\frac{1-2\nu_u}{4\pi G(1-\nu_u)}\frac{1}{r}\mathrm{H}(t-\tau) - \frac{\eta^2}{2\pi G^2 S}\frac{1}{r}e^{-\xi^2}, \quad \text{(2D)}$$

$$= -\frac{1-2\nu_u}{8\pi G(1-\nu_u)}\frac{1}{r^2}\mathrm{H}(t-\tau) - \frac{\eta^2}{4\pi G^2 S}\frac{1}{r^2}\left[\mathrm{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right], \quad \text{(3D)} \tag{8.310}$$

Based on the asymptotic behavior of dodecapole at small and large times, (8.277) and (8.278), we can show that the displacement for quadrupole and hexapole satisfies these elastic relations with undrained and drained material constants

$$u_r^{qc}(r,0) = -\frac{1-2\nu_u}{4\pi G(1-\nu_u)}\frac{1}{r}, \quad \text{(2D)}$$

$$= -\frac{1-2\nu_u}{8\pi G(1-\nu_u)}\frac{1}{r^2}, \quad \text{(3D)} \tag{8.311}$$

and

$$u_r^{qc}(r,\infty) = -\frac{1-2\nu}{4\pi G(1-\nu)}\frac{1}{r}, \quad \text{(2D)}$$

$$= -\frac{1-2\nu}{8\pi G(1-\nu)}\frac{1}{r^2}, \quad \text{(3D)} \tag{8.312}$$

The singularity at the origin needs to be more carefully analyzed. Equation (8.311) implies that a center of compression related to the undrained material constant is instantly generated. From (8.279), however, we observe that the dodecapole also generates an instant center of compression. The volumetric strain is actually given by

$$e^{qc} = -\frac{1-\nu}{2G(1-\nu)}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) + \frac{\eta^2}{\pi G^2 S}\frac{1}{r^2}\xi^2 e^{-\xi^2}, \quad \text{(2D)}$$

$$= -\frac{1-\nu}{2G(1-\nu)}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) + \frac{\eta^2}{\pi^{3/2}G^2 S}\frac{1}{r^3}\xi^3 e^{-\xi^2}, \quad \text{(3D)} \tag{8.313}$$

Hence there exists a center of compression characterized by the drained material properties, plus an evolutional part caused by the fluid dodecapole.

Pore pressure can be obtained as

$$p^{qc} = \frac{\eta c}{G}p^{oc} + \frac{\eta}{GS}g_1 \tag{8.314}$$

and we find

$$p^{qc} = -\frac{\eta}{\pi GS}\frac{1}{r^2}\xi^2 e^{-\xi^2}, \qquad \text{(2D)}$$
$$= -\frac{\eta}{\pi^{3/2}GS}\frac{1}{r^3}\xi^3 e^{-\xi^2}, \qquad \text{(3D)} \tag{8.315}$$

The above solution shows that the pore pressure induced by a quadrupole/hexapole is equivalent to an instantaneous source

$$p^{qc} = -\frac{\eta}{G}p^{si} \tag{8.316}$$

Eq. (8.315) indicates that there is a negative pore pressure at the origin,

$$p^{qc}(0,t) = -\frac{\eta}{4\pi GSc}t^{-1}, \qquad \text{(2D)}$$
$$= -\frac{\eta}{8\pi^{3/2}GSc^{3/2}}t^{-3/2}, \qquad \text{(3D)} \tag{8.317}$$

which is infinite at $t = 0$. From (8.191) we obtain

$$\zeta^{qc} = -\frac{\eta}{\pi G}\frac{1}{r^2}\xi^2 e^{-\xi^2} - \frac{\eta}{G}\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau), \qquad \text{(2D)}$$
$$= -\frac{\eta}{\pi^{3/2}G}\frac{1}{r^3}\xi^3 e^{-\xi^2} - \frac{\eta}{G}\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau), \qquad \text{(3D)} \tag{8.318}$$

The above shows that not only there exists a solid center of dilatation, there is a center of variation in fluid content, that is, the relative fluid to solid volumetric strain. Other quantities are found in Sect. E.12.

8.17 Solid Center of Dilatation

Similar to the center of dilatation for elastic material discussed in Sect. 8.8, we can define its counterpart for poroelasticity. For elasticity, it has been proven that the center of dilation is equivalent to the quadrupole/hexapole, and they differ only by a sign and a constant factor. This is however not so for poroelasticity. We shall demonstrated below.

For continuous solid dilatation, we introduce the eigenstrain

$$E_{ij} = \left(\frac{\ln r}{2\pi}\right)_{,ij}\mathrm{H}(t-\tau), \qquad \text{(2D)}$$
$$= \left(-\frac{1}{4\pi r}\right)_{,ij}\mathrm{H}(t-\tau), \qquad \text{(3D)} \tag{8.319}$$

which corresponds to

$$E = E_{ii} = \delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.320}$$

Substituting the above into the Biot decomposition equations (8.178) and (8.180), we obtain these two equations

$$G\,u^{co}_{i,jj} + \frac{G}{1-2\nu_u}\,u^{co}_{j,ji} = \frac{2G(1-\nu_u)}{1-2\nu_u}\delta_{,i}\,\mathrm{H}(t-\tau) \tag{8.321}$$

$$\frac{\partial(\nabla^2\Phi^{cc})}{\partial t} - c\,\nabla^4\Phi^{cc} = 0 \tag{8.322}$$

where we used the superscript ^{c} to denote center of dilatation. Comparing the above with (8.305) and (8.306) for quadrupole and hexapole, we observe that there is no fluid dodecapole in this case. Solution of the above is simply elastic

$$\begin{aligned} u^{cc}_r &= \frac{1}{2\pi}\frac{1}{r}, \quad \text{(2D)} \\ &= \frac{1}{4\pi}\frac{1}{r^2}, \quad \text{(3D)} \end{aligned} \tag{8.323}$$

and

$$p^{cc} = \zeta^{cc} = w^{cc}_i = q^{cc}_i = 0 \tag{8.324}$$

This is not surprising as the above solutions are the same as the mode 1 borehole and spherical cavity problems, discussed in Sects. 7.11.1 and 7.14.1. We already know that these solutions are elastic and no pore pressure is generated. It becomes clear if the volumetric strain is presented

$$e^{cc} = \delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.325}$$

The above shows that there is no volumetric strain everywhere in the domain, but there is a unit dilatation at the center.

As both the quadrupole/hexapole and the center of dilatation are acceptable poroelastic solutions, it may be of interest to discuss which of these is the more relevant solution for application purpose. This is dependent on the physical condition taking place near the center. The quadruple/hexapole solution shows that there is not only a solid movement, but also a net relative fluid to solid movement at the origin, which requires the instant injection or extraction of a volume of fluid. At the origin, the relative fluid to solid displacement is infinite ($w^{qc}_r \sim 1/r$). While for the center of dilatation solution, there is no relative fluid to solid movement. At the center, a cavity that is devoid of solid and fluid is created.

8.18 Displacement Discontinuity

Based on the work in elasticity [2, 45], a continuous displacement discontinuity tensor can be generated by the introduction through eigenstrain the following Dirac delta function

$$E_{ijkl} = -\frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau) \tag{8.326}$$

Substituting the above into (8.173) and (8.174), and setting $F_i = \gamma = 0$, we obtain

$$\begin{aligned} & G\,u^{dc}_{ikl,jj} + \frac{G}{(1-2\nu)}\,u^{dc}_{jkl,ji} - \alpha p^{dc}_{kl,i} \\ & \quad = -G\left[\delta_{ik}\,\delta_{,l}(\vec{x} - \vec{\chi}) + \delta_{il}\,\delta_{,k}(\vec{x} - \vec{\chi})\right] H(t - \tau) \\ & \qquad - \frac{2G\nu}{(1-2\nu)}\delta_{kl}\,\delta_{,i}(\vec{x} - \vec{\chi}) H(t - \tau) \end{aligned} \tag{8.327}$$

$$\frac{\partial p^{dc}_{kl}}{\partial t} - \kappa M \nabla^2 p^{dc}_{kl} + \alpha M \frac{\partial e^{dc}_{kl}}{\partial t} = -\alpha M \delta_{kl}\,\delta(\vec{x} - \vec{\chi})\,\delta(t - \tau) \tag{8.328}$$

In the above, the ^{dc} superscript denotes continuous displacement discontinuity, $\delta_{,i}$ means the differentiation of the Dirac delta function with respect to x_i, and the tensor ranks for displacement and pore pressure are increased by two because the introduced forcing function is a second rank tensor.

The solution to the above equations can be constructed as follows. We observe in Sect. 8.15 that the introduction of a forcing term $F_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau)$ creates a solution of u^{Fc}_{ik}. This means that if we introduce the spatial derivative of a Dirac delta function $\delta_{ik}\delta_{,l}(\vec{x} - \vec{\chi})\,\mathrm{H}(t - \tau)$ to the right hand side of (8.327), a solution $u^{Fc}_{ik,l}$ results. Similar situation exists in (8.328), as we observe that the right hand side forcing term is equivalent to the introduction of an instantaneous source through the term γ, such that a u^{si}_i solution is produced. Adding these contributions together, we can obtain the displacement and pore pressure part of the displacement discontinuity fundamental solution as follows

$$u^{dc}_{ikl} = G\left(u^{Fc}_{ik,l} + u^{Fc}_{il,k}\right) + \frac{2G\nu}{1-2\nu}\delta_{kl}u^{Fc}_{mi,m} - \alpha\delta_{kl}u^{si}_i \tag{8.329}$$

$$p^{dc}_{kl} = G\left(p^{Fc}_{k,l} + p^{Fc}_{l,k}\right) + \frac{2G\nu}{1-2\nu}\delta_{kl}\,p^{Fc}_{m,m} - \alpha\delta_{kl}\,p^{si} \tag{8.330}$$

without formally solving for (8.327) and (8.328).

Rather than differentiating the various expressions in (8.329) and (8.330), we shall explore the following interesting relations to relate thefundamental solutions

to the already derived results. Based on the relation (8.350), to be presented in Sect. 8.20, and (8.253), we observe

$$u_i^{si} = u_i^{lc} = p_i^{Fc} \tag{8.331}$$

Making the substitution in (8.329), we obtain

$$u_{ikl}^{dc} = G\left(u_{ki,l}^{Fc} + u_{li,k}^{Fc}\right) + \frac{2G\nu}{1-2\nu}\delta_{kl}\, u_{mi,m}^{Fc} - \alpha\delta_{kl} p_i^{Fc} \tag{8.332}$$

where all the right hand side fundamental solutions are related to continuous force. We observe that the right hand side is exactly the constitutive relation for stress, with some adjustment of indices. We hence conclude the relation

$$u_{ikl}^{dc} = \sigma_{lki}^{Fc} \tag{8.333}$$

where the right hand side solution is already obtained in Sect. 8.15. In the above, we note that the permutation of indices is important as symmetry does not exist for all permutations.

Also, based on the same substitution of the relation in (8.331), we can rewrite (8.330) as

$$p_{kl}^{dc} = G\left(u_{k,l}^{si} + u_{l,k}^{si}\right) + \frac{2G\nu}{1-2\nu}\delta_{kl}\, u_{m,m}^{si} - \alpha\delta_{kl} p^{si} \tag{8.334}$$

It is then obvious that

$$p_{kl}^{dc} = \sigma_{kl}^{si} \tag{8.335}$$

Other solutions, such as σ_{ijkl}^{dc}, can be obtained by differentiation, or by utilizing the various relations among the fundamental solutions. These solutions, as well as the instantaneous displacement discontinuity solutions, are listed in Sects. E.14 and E.15.

In Fig. 8.12 we plot the pore pressure field generated by a continuous displacement discontinuity at the origin, with parameter values $c = 1, \kappa = 1$, and $\eta = 0.25$, at $t = 1$. Figure 8.12a is the result of a normal displacement discontinuity d_{xx}, which is the closing of two points at the origin in the x direction, as indicated in the figure. (See Fig. 8.1 for sign convention.) We observe that the pore pressure is negative along the x-axis due to the positive solid strain dilatation caused by the movement of the two points. Along the y-axis at a distance from the origin, we observe peaks of positive pore pressure. Figure 8.12b is the result of a sliding displacement discontinuity d_{xy} with orientation as shown in the figure. We observe the anti-symmetric pressure pattern developed in the four quadrants. This pattern can be compared to the field observation of perturbed piezometric head in water wells after an earthquake induced by fault slippage, as shown in Fig. 1.7.

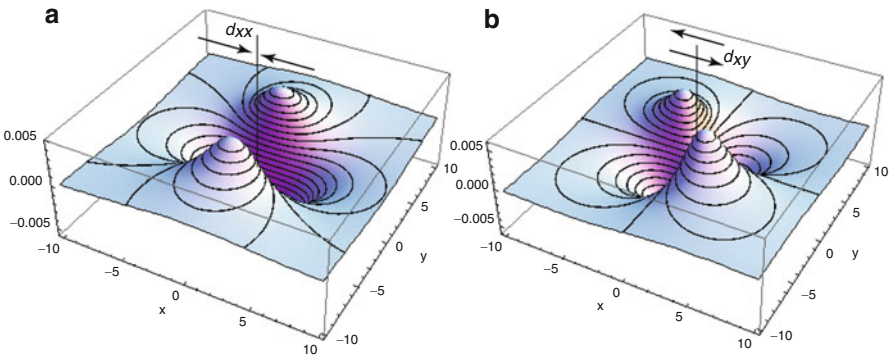


Fig. 8.12 Pressure field due to 2D continuous displacement discontinuity. (**a**) normal displacement discontinuity p^{dc}_{xx}, and (**b**) sliding displacement discontinuity p^{dc}_{xy}

8.19 Edge Dislocation

Edge dislocation represents a constant closing (normal mode) or sliding (slip mode) along a cut in continuous medium that creates two surfaces. Figure 8.13 illustrated the plane strain case with a cut along the positive x-axis. As the displacement discontinuity represents a jump in displacement characterized by a Dirac delta function, as in (8.326), the edge dislocation can be obtained by integrating the displacement discontinuity in the spatial dimension. For a cut in the orientation shown in Fig. 8.13, we can obtain the fundamental solutions of edge dislocation by the following integration

$$\Box^{e\dagger}_{\ldots j}(\vec{x}, t-\tau) = \int_0^\infty \Box^{d\dagger}_{\ldots 1l}(\vec{x}-\vec{\chi}, t-\tau)\, d\chi_1 \tag{8.336}$$

where the $\Box$ represent the displacement, stress, etc., the superscript ^{e} denotes edge dislocation, the † superscript is either ^{i} or ^{c} for instantaneous and continuous solutions, and the tensor index $l = 1$ represents the slip mode, and $l = 2$ for normal mode. We note that the next to last tensor index for displacement discontinuity solutions is set to 1, corresponding to the cut orientation shown in Fig. 8.13. As a result, the edge dislocation has the reduction of tensor index by one. Although the edge dislocation solutions can be constructed through (8.336), the integration is not necessarily easy. Hence in the following we shall seek to directly derive the solutions, following Detournay and Cheng [23].

First, it is recognized that, at the instant of loading, the material response is elastic, characterized by the undrained material constants. Hence the instantaneous part of the solution, represented as u^o_i in (8.177), can be obtain from the classical elasticity theory as [26]

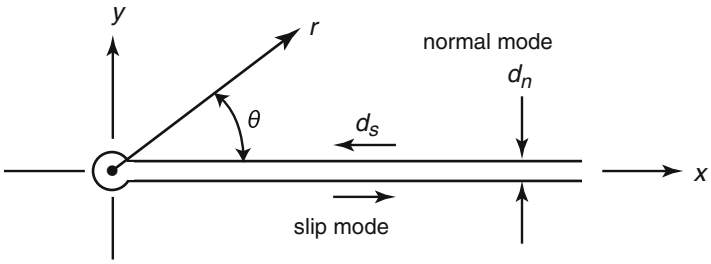


Fig. 8.13 An edge dislocation

$$(u_{11}^{ec})^o = \frac{1}{4\pi(1-\nu_u)} \mathrm{H}(t-\tau)\,[2(1-\nu_u)\theta + r_{,1}r_{,2}] \tag{8.337}$$

$$(u_{21}^{ec})^o = -\frac{1}{4\pi(1-\nu_u)} \mathrm{H}(t-\tau)\,[(1-2\nu_u)\ln r + r_{,1}r_{,1}] \tag{8.338}$$

$$(u_{12}^{ec})^o = \frac{1}{4\pi(1-\nu_u)} \mathrm{H}(t-\tau)\,[(1-2\nu_u)\ln r + r_{,2}r_{,2}] \tag{8.339}$$

$$(u_{22}^{ec})^o = \frac{1}{4\pi(1-\nu_u)} \mathrm{H}(t-\tau)\,[2(1-\nu_u)\theta - r_{,1}r_{,2}] \tag{8.340}$$

In the above, θ is the polar angle, which has a branch cut along the positive x-axis ($0 \le \theta \le 2\pi$), see Fig. 8.13. Note that the first subscript of displacement denotes the displacement component, while the second subscript is reserved for the dislocation mode: 1 for slip, and 2 for normal mode.

By substituting (8.337), (8.338), (8.339) and (8.340) into the left hand side of (8.178), and taking divergence of the whole equation, we find the equivalent body force term $F_{ik,i}$ on the right hand side. Transferring this term to the right hand side of (8.176), and set $f_i = \gamma = E_{ij} = 0$, we obtain

$$\frac{\partial \zeta_k^{ec}}{\partial t} - c\nabla^2 \zeta_k^{ec} = 2\eta c\,\mathrm{H}(t-\tau)\,\epsilon_{ki}\delta_{,i}(\vec{x}-\vec{\chi}) \tag{8.341}$$

where the subscript k for ζ denotes the dislocation mode, and ϵ_{ki} is the two-dimensional permutation symbol (i.e. $\epsilon_{11} = \epsilon_{22} = 0$ and $\epsilon_{12} = -\epsilon_{21} = 1$). We hence observe that the introduction of edge dislocation creates a dipole in the $-y$ direction for the slip mode, and a dipole in the $+x$ direction for the normal mode. As observed in Fig. 8.8a, b, these dipole orientations create a no-flux boundary condition across the dislocation line for the normal mode, and a zero pressure condition for the slip mode.

As dipole solution has been presented, (8.341) is easily solved for

$$\zeta_k^{ec} = -2\eta c\,\epsilon_{ki}\zeta_i^{pc} = -\frac{\eta}{\pi}\epsilon_{ki}\frac{r_{,i}}{r}e^{-\xi^2} \tag{8.342}$$

From (8.179), we can solve for

$$\Phi_k^{ec} = \frac{\eta}{4\pi}\epsilon_{ki} r r_{,i}\left[\mathrm{E}_1(\xi^2) + \xi^{-2}(1 - e^{-\xi^2})\right] \tag{8.343}$$

From (8.177) the time-dependent part of displacement is then given by:

$$\Delta u_{ik}^{ec} = \frac{\eta^2}{4\pi GS}\epsilon_{kj}\left[(\delta_{ij} - 2r_{,i}r_{,j})\xi^{-2}(1 - e^{-\xi^2}) + \delta_{ij}\mathrm{E}_1(\xi^2)\right] \tag{8.344}$$

And the complete displacement solution is

$$u_{ik}^{ec} = (u_{ik}^{ec})^o + \Delta u_{ik}^{ec} \tag{8.345}$$

The fundamental solution for stress, σ_{ijk}^{ec}, can also be separated into a time independent part and dependent part

$$\sigma_{ijk}^{ec} = (\sigma_{ijk}^{ec})^o + \Delta\sigma_{ijk}^{ec} \tag{8.346}$$

Based on (8.192) and constitutive equation for stress, we find

$$(\sigma_{ijk}^{ec})^o = \frac{G}{2\pi(1-\nu_u)}\mathrm{H}(t-\tau)\epsilon_{k\ell}\frac{1}{r}\left(\delta_{i\ell}r_{,j} + \delta_{j\ell}r_{,i} - \delta_{ij}r_{,\ell} - 2r_{,i}r_{,j}r_{,\ell}\right) \tag{8.347}$$

$$\begin{aligned}\Delta\sigma_{ijk}^{ec} = -\frac{\eta^2}{\pi S}\epsilon_{k\ell}\frac{1}{r}\Big[&(\delta_{i\ell}r_{,j} + \delta_{j\ell}r_{,i} + \delta_{ij}r_{,\ell} - 4r_{,i}r_{,j}r_{,\ell})\,\xi^{-2}(1 - e^{-\xi^2}) \\ &+ 2(r_{,i}r_{,j}r_{,\ell} - \delta_{ij}r_{,\ell})e^{-\xi^2}\Big]\end{aligned} \tag{8.348}$$

We note that although displacement has a jump across the positive x-axis, the stress is in fact continuous. Based on (8.191), pore pressure is obtained as

$$p_k^{ec} = \frac{\eta}{\pi S}\epsilon_{ki}\frac{r_{,i}}{r}\left(1 - e^{-\xi^2}\right) \tag{8.349}$$

8.20 Fundamental Solution Relation Based on Reciprocity

The reciprocal relations of poroelasticity (8.37) allows us to explore some interesting relations among the fundamental solutions of poroelasticity that are otherwise not obvious. Particularly, we shall demonstrated below these relations:

$$\begin{aligned} u_i^{li} &= p_i^{Fi}, \quad w_{ij}^{Fi} = u_{ij}^{fi}, \\ w_i^{li} &= p_i^{fi}, \quad u_{ijk}^{di} = \sigma_{kji}^{Fi}, \\ w_{ijk}^{di} &= \sigma_{kji}^{fi}, \quad p_{ij}^{di} = \sigma_{ji}^{li} \end{aligned} \tag{8.350}$$

The above relations are valid for instantaneous as well as continuous solutions; thus we can replace the superscript ^{i} in the above by ^{c}. They are also valid for isotropic as well as general anisotropic materials, although the closed form solutions of general anisotropy are not presented.

Referring to the reciprocal theorem of poroelasticity (8.37), we can consider Ω as a domain enclosed by a circle (2D) or a spherical surface (3D) Γ with radius R. We shall take the limit of $R \to \infty$. We note that the first two integrals are performed over the surface Γ. If the products in the integrand, $\sigma_{ij}^{(1)} u_i^{(2)}$, $p^{(1)} w_i^{(2)}$, etc., decay fast enough, say, of order R^{-n}, where $n > 1$ for 2D and $n > 2$ for 3D cases, then the surface integrals vanish in the limit. As we plan to introduce fundamental solutions, which have the above stated property, so we shall eliminate the surface integrals in (8.37), which becomes the following

$$\int_0^t \int_{\Omega_\infty} \Big[\Big(F_i^{(1)} u_i^{(2)} - F_i^{(2)} u_i^{(1)} \Big) + \Big(f_i^{(1)} w_i^{(2)} - f_i^{(2)} w_i^{(1)} \Big) - \big(Q^{(1)} p^{(2)} - Q^{(2)} p^{(1)} \big) + \Big(E_{ij}^{(1)} \sigma_{ij}^{(2)} - E_{ij}^{(2)} \sigma_{ij}^{(1)} \Big) \Big] \, d\vec{\chi} \, d\tau = 0 \tag{8.351}$$

To prove the first relation in (8.350), we substitute in the following singularities and their corresponding influence functions:

$$F_{ij}^{(1)} = \delta_{ij} \, \delta(\vec{\chi} - \vec{x}_1) \, \delta(t - \tau), \quad Q^{(2)} = \delta(\vec{\chi} - \vec{x}_2) \, \delta(t - \tau) \tag{8.352}$$

with the rest of the forcing functions set to zero. In the above, $\vec{x}_1$ and $\vec{x}_2$ are two arbitrary points where the singularities are applied. The introduction of these singularities means that the first system of solution is associated with the fundamental solution ^{Fi}, and the second system with ^{li}. Integrating with respect to space and time over the delta functions, (8.351) reduces to

$$u_i^{li}(\vec{x}_1 - \vec{x}_2, t) + p_i^{Fi}(\vec{x}_2 - \vec{x}_1, t) = 0 \tag{8.353}$$

As these solutions are directional, involving the differentiation of r with respect to the first spatial variable in the argument, switching $\vec{x}_1$ and $\vec{x}_2$ causes a change of sign; hence we have proven

$$u_i^{li} = p_i^{Fi} \tag{8.354}$$

Relation (8.350) is thus proven.

A different substitution can be made as

$$F_{ij}^{(1)} = \delta_{ij} \, \delta(\vec{\chi} - \vec{x}_1) \, \delta(t - \tau), \quad f_{ij}^{(2)} = \delta_{ij} \, \delta(\vec{\chi} - \vec{x}_2) \, \delta(t - \tau) \tag{8.355}$$

from which we deduce that

$$u_{ij}^{fi} = w_{ij}^{Fi} \tag{8.356}$$

By alternating the pairs, $\{f_i^{(1)}, Q^{(2)}\}$, $\{E_{ij}^{(1)}, F_i^{(2)}\}$, $\{E_{ij}^{(1)}, f_i^{(2)}\}$, and $\{E_{ij}^{(1)}, Q^{(2)}\}$, the remaining relations in (8.350) can be proven.

References

1. Abramowitz M, Stegun IA (1965) Handbook of mathematical functions: with formulas, graphs, and mathematical tables. Dover, New York, 1046pp
2. Becache E, Nedelec JC, Nishimura N (1993) Regularization in 3-D for anisotropic elastodynamic crack and obstacle problems. J Elast 31:25–46
3. Betti E (1872) Teoria della elasticità (Theory of elasticity). Nuovo Cimento 2:6–10
4. Betti E (1874) Sopra l'equazioni di equilibrio dei corpi solidi elastici (On the equations of equilibrium of elastic solids). Annali delle Università Toscane 10:143–158
5. Bilby BA, Eshelby JD (1968) Dislocations and the theory of fracture. In: Liebowitz H (ed) Fracture, an advanced treatise, vol 1. Academic, New York, pp 99–182
6. Biot MA (1956) General solutions of the equations of elasticity and consolidation for a porous material. J Appl Mech ASME 78:91–96
7. Booker JR, Carter JP (1986) Long term subsidence due to fluid extraction from a saturated, anisotropic, elastic soil mass. Q J Mech Appl Math 39:85–97
8. Brebbia CA, Butterfield R (1978) Formal equivalence of direct and indirect boundary element methods. Appl Math Model 2(2):132–134
9. Carslaw HS, Jaeger JC (1959) Conduction of heat in solids. Oxford University Press, Oxford, 510pp
10. Cauchy A-L (1827) Mémoire sur les intégrales définies (Memoir on definite integrals). Mémoires des divers savants 2(1):601–799
11. Cheng AHD (2000) Multilayered aquifer systems—fundamentals and applications. Marcel Dekker, New York, Basel, 384pp
12. Cheng AHD, Antes H, Ortner N (1994) Fundamental solutions of products of Helmholtz and polyharmonic operator. Eng Anal Bound Elem 14(2):187–191
13. Cheng AH-D, Badmus T, Detournay E (1988) Singular integral equation method for cracks embedded in poroelastic space. In: Brebbia CA (ed) Boundary elements X, vol 3 Stress analysis. Springer, Berlin, pp 265–278
14. Cheng AHD, Cheng DT (2005) Heritage and early history of the boundary element method. Eng Anal Bound Elem 29(3):268–302
15. Cheng AHD, Detournay E (1988) A direct boundary element method for plane-strain poroelasticity. Int J Numer Anal Methods Geomech 12(5):551–572
16. Cheng AHD, Detournay E (1998) On singular integral equations and fundamental solutions of poroelasticity. Int J Solids Struct 35(34–35):4521–4555
17. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to soil consolidation. Int J Numer Methods Eng 20(2):255–278
18. Cheng AHD, Liggett JA (1984) Boundary integral equation method for linear porous-elasticity with applications to fracture propagation. Int J Numer Methods Eng 20(2):279–296
19. Cheng AHD, Predeleanu M (1987) Transient boundary element formulation for linear poroelasticity. Appl Math Model 11(4):285–290
20. Cleary MP (1977) Fundamental solutions for a fluid-saturated porous solid. Int J Solids Struct 13(9):785–806

21. Crank J (1980) The mathematics of diffusion, 2nd edn. Oxford University Press, Oxford, 424pp
22. Crouch SL, Starfield AM (1983) Boundary element methods in solid mechanics. Allen and Unwin, London/Boston, 322pp
23. Detournay E, Cheng AHD (1987) Poroelastic solution of a plane-strain point displacement discontinuity. J Appl Mech ASME 54(4):783–787
24. Detournay E, Cheng AHD (1991) Plane strain analysis of a stationary hydraulic fracture in a poroelastic medium. Int J Solids Struct 27(13):1645–1662
25. Dougall J (1897) A general method of solving the equations of elasticity. Proc Edinb Math Soc 16(1):82–98
26. Dundurs J (1969) Elastic interaction of dislocations with inhomogeneities. In: Mura T (ed) Mathematical theory of dislocation. ASME, New York, pp 70–115
27. Fredholm I (1903) Sur une classe d'équations fonctionelles (On a class of functional equations). Acta Math 27(1):365–390
28. Geertsma J (1973) Land subsidence above compacting oil and gas reservoirs. J Pet Technol 25(6):734–744
29. Ghassemi A, Cheng AHD, Diek A, Roegiers JC (2001) A complete plane strain fictitious stress boundary element method for poroelastic media. Eng Anal Bound Elem 25(1):41–48
30. Green G (1828) An essay on the application of mathematical analysis to the theories of electricity and magnetism, printed for the author, by T. Wheelhouse, Nottingham
31. Green G (1871) Mathematical papers of George Green. N.M. Ferrers, reprinted by Chelsea Publ. Co., New York, 1970
32. Greenberg MD (1971) Application of green's function in science and engineering. Prentice-Hall, Englewood Cliffs, 141pp
33. Gudmundsson A (1998) Magma chambers modeled as cavities explain the formation of rift zone central volcanoes and their eruption and intrusion statistics. J Geophys Res–Solid Earth 103(B4):7401–7412
34. Gudmundsson A (1998) Formation and development of normal-fault calderas and the initiation of large explosive eruptions. Bull Volcanol 60(3):160–170
35. Hadamard J (1908) Theorie des équations aux dérivées partielles linéaires hyperboliques et du probléme de Cauchy (Theories of equations with partial linear hyperbolic derivatives and the problem of Cauchy). Acta Math 31(4):333–380
36. Hong HK, Chen JT (1988) Derivations of integral equations of elasticity. J Eng Mech ASCE 114(6):1028–1044
37. Huynen A, Detournay E (2014) A remark on the poroelastic center of dilation. J Elast 116(2):189–206
38. Jaswon MA, Symm GT (1977) Integral equation methods in potential theory and elastostatics (Computational mathematics and applications). Academic, London/New York, 301pp
39. Krishnasamy G, Rizzo FJ, Rudolphi TJ (1991) Hypersingular boundary integral equations: their occurrence, interpretation, regularization, and computation, Chap. 7. In: Banerjee PK, Kobayashi S (eds) Developments in boundary element methods. Elsevier, London/New York, pp 207–252
40. Love AEH (1927) A treatise on the mathematical theory of elasticity, 4th edn. Cambridge University Press, Cambridge, 643pp
41. Maxwell JC (1864) On the calculation of the equilibrium and stiffness of frames. Philos Mag 27:294–299
42. Mindlin RD (1953) Force at a point in the interior of a semi-infinite solid. In: Proceedings of the first midwestern conference on solid mechanics, Urbana, pp 56–59
43. Morse P, Feshbach H (1953) Methods of theoretical physics, part 1. McGraw-Hill, New York, 997pp
44. Mura T (1987) Micromechanics of defects in solids, 2nd edn. Springer, Dordrecht, 588pp
45. Nedelec JC (1986) The double layer potential for periodic elastic waves in R^3. In: Du QH (ed) Boundary elements. Pergamon, Oxford/New York, pp 439–448
46. Pan E, Maier G (1997) A symmetric boundary integral approach to transient poroelastic analysis. Comput Mech 19(3):169–178

47. Predeleanu M (1968) O teoremă de reciprocitate în teoria consolidării mediilor poroase (A reciprocal theorem in the consolidation theory of the porous media). Analele Universității București, Seria Științele Naturii, Matematică-Mecanică, Extras 17(2):69–74
48. Rayleigh JWS (1877) The theory of sound, vols 1 and 2. Macmillan, London, (Also published by Dover, 2nd edn., 1945)
49. Renshaw CE, Harvey CF (1994) Propagation velocity of a natural hydraulic fracture in a poroelastic medium. J Geophys Res–Solid Earth 99(B11):21667–21677
50. Somigliana C (1885) Sopra l'equilibrio di un corpo elastico isotropo (On the equilibrium of an isotropic elastic body). Nuovo Cimento 3:17–20
51. Somigliana C (1914/1915) Sulla teoria delle distorsioni elastiche (on the theory of elastic distortions). Note I e II. Atti della Accademia Nazionale dei Lincei Rendiconti Classe di Scienze Fisiche Matematiche e Naturali. 23:463–472 and 24:655–666
52. Sternberg WJ, Smith TL (1944) The theory of potential and spherical harmonics, 2nd edn. University of Toronto Press, Toronto, 324pp
53. Thomson (Lord Kelvin) W (1848) Note on the integrations of the equations of equilibrium of an elastic solid. Camb Dublin Math J 3:76–99
54. Vandamme L, Detournay E, Cheng AHD (1989) A two-dimensional poroelastic displacement discontinuity method for hydraulic fracture simulation. Int J Numer Anal Methods Geomech 13(2):215–224
55. Volterra V (1896) Sulla inversione degli integrali multipli (On the inversion of multiple integrals). Atti della Accademia Nazionale dei Lincei Rendiconti Classe di Scienze Fisiche Matematiche e Naturali 5:289–300
56. Volterra V (1907) Sur l'équilibre des carps élastiques multiplement connexes (On the equilibrium of multiply connected elastic body). Annales Scientifiques de l'École Normale Superieure, Paris 24:401–517
57. Waterman PC (1965) Matrix formulation for electromagnetic scattering. Proc Inst Electr Electron Eng 53(8):805–812
58. Waterman PC (1969) New formulation of acoustic scattering. J Acoust Soc Am 45(6):1417–1429

Chapter 9
Poroelastodynamics

> *We shall not write down the expressions for its roots and shall only remark that . . . one of them corresponds to waves with a very small damping, and the other to waves with a very large damping. The waves of the second kind are thus really non-existent.*
>
> —Yacov Frenkel (1944)

In the preceding chapters we have been dealing with poroelastic theories and problems under the assumptions similar to *elastostatics*; that is, at any instant of a loading, the poroelastic body is at a state of *static equilibrium*. In other words, for a body of any size, finite or infinitesimal, the summation of all forces, including surface and body forces, must equal to zero, $\sum \vec{F} = 0$, such that there is no acceleration created by the imbalance of forces. This, however, does not mean that there is no motion. One of the characteristics of poroelastic body is that its deformation is time-dependent, giving the impression of a creeping-like motion, even if the applied load is constant in time. This transient behavior is the consequence of a fluid phase. Fluid has no shear strength to resist shear deformation, but has a viscosity that resists the rate of shear deformation. Hence the force equilibrium of a fluid can be accompanied by motion. So even without considering the acceleration caused by force imbalance, the poroelastic body is not exactly static, and the poroelastic theory presented in the preceding chapters can be called a *quasi-static* theory.

When a force is rapidly applied, such as by an explosion in the air, by the impact of a solid body, or due to the slippage of a fault, the *inertial effect*, that is, the right hand side of Newton's second law of motion $\vec{F} = m\vec{a}$, cannot be neglected. A *dynamic theory* should be introduced.

When the inertial effect is considered in a continuum body that is compressible, a *wave* phenomenon results. Particularly, the stress caused by the applied force is not instantly felt throughout the body—it has a finite speed of propagation. Sound propagation in the air as a wave phenomenon was recognized by philosophers and scientists as early as Aristotle (384–322 BC), and then by Galileo Galilei (1564–1642) [64]. In fact, Aristotle already recognized that sound is a *longitudinal wave* when he wrote "(the air) *is set in motion . . . by contraction or expansion or compression*" [13]. This is indicative of the definition of a longitudinal wave, in

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_9

which the particle motion is parallel to the wave propagation direction. In a solid, elastic medium, there exist two types of waves: in addition to the longitudinal, or *compressional wave*, there also exists a *transverse wave*, also called a *shear wave*, in which the particle motion is perpendicular to the wave propagation direction. These two waves propagate at different wave speed. The historical development of elastic wave theory, or *elastodynamics*, is well summarized in Love [83].

In this chapter we are interested in the wave propagation in porous medium, or the theory of *poroelastodynamics*. In such medium, a third wave, called the *second compressional wave*, is observed, due to the existence of two phases, a solid and a fluid. The reasoning and theoretical demonstration of such waves were first presented by Yackov Frenkel [54] (see Sect. F.13 for a biography). Frenkel's work was motivated by the field observation of Ivanov [66, 67], who discovered the so-called *seismoelectric effect of the second kind* (E-effect) generated by underground explosion—when a seismic wave is generated by an explosion, electric potential differences can be observed between electrodes situated at different distances from the source of the waves. Based on the continuum mechanics theory, Frenkel demonstrated that in a fluid infiltrated isotropic porous medium, in addition to a longitudinal and a shear wave, there existed a second longitudinal wave characterized by the out-of-phase movement between solid and fluid. He then showed that in the presence of electrolytes in liquids, electric current was generated due to the relative movement between the phases. The alternating directions of the electrical current in turn generate an electromagnetic wave. However, as quoted in the prologue of the chapter, after the proclamation of the discovery of a second wave, Frenkel did not further pursue its characteristics.

Twelve years later, citing Frenkel's original contribution, Biot [19, 20] re-derived the theory of wave propagation in porous medium. Biot not only demonstrated the existence of the waves, but also presented the wave speeds. Particularly, it was shown that the second compressional wave is highly dissipative, and propagates at a much lower speed than the first compressional wave; hence they are respectively called the *slow wave* and the *fast wave*. In the higher frequency range, Biot also introduced a physical model of capillary flow in parallel plates or tubes, to account for the viscous-inertial attenuation. He also discovered a characteristic frequency at which the attenuation reaches its maximum. Biot's model became immensely popular and the second wave has largely been referred to as the Biot second wave.

9.1 Dynamic Equilibrium Equation

With the effect of acceleration, or the inclusion of *inertial force*, the equilibrium equation for elastodynamics is given by

$$\sigma_{ij,j} = \rho \ddot{u}_i - F_i \tag{9.1}$$

where the double dot denotes the second derivative with respect to time.

The dynamic force balance for porous medium is more complicated. Not only it contains two material phases that can be differently accelerated, but also it is highly heterogeneous at the microscopic scale. The macroscopic continuum mechanics theory is developed by performing averaging over a microscopic representative elementary volume (REV). This averaging can be conducted using the homogenization theory [7, 80, 104], the mixture theory [27, 28, 38], or the volume averaging theory [36, 82, 93, 94]. For the scope of the book, in the following, we shall give a limited derivation by applying the *variational Lagrangian approach* to the volume averaging theory in order to examine the inertia effect for poroelastodynamics. A more general derivation can be found in Lopatnikov and Cheng [82].

Variational Lagrangian Formulation In the classical Lagrangian mechanics, the kinetic and potential energies are expressed in terms of the coordinates q_i and the velocities $\dot{q}_i$ of each particle i. The *Lagrangian* $\mathcal{L} = \mathcal{L}(q_i, \dot{q}_i, t)$ is the difference between the total kinetic energy $\mathcal{K}$ and the total potential energy $\mathcal{V}$ of the system

$$\mathcal{L} = \mathcal{K} - \mathcal{V} \tag{9.2}$$

Based on Hamilton's *principle of least action*, the motion of the particles from time t_1 to t_2 follows a path that makes the *action*

$$A = \int_{t_1}^{t_2} \mathcal{L}\, dt \tag{9.3}$$

a stationary value; in other words

$$\delta A = 0 \tag{9.4}$$

This leads to the *Lagrange's equation of motion* for a conservative system

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = 0 \tag{9.5}$$

When the system is nonconservative, such as there exists a friction, a *Rayleigh dissipation function* $\mathcal{D}(\dot{q})$ can be introduced, and the Lagrange's equation is modified to

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} + \frac{\partial \mathcal{D}}{\partial \dot{q}_i} = 0 \tag{9.6}$$

The above system can be generalized to view q_i are the *generalized coordinates*, and $\dot{q}_i$ the *generalized velocities*. We can then identify in (9.6) that $\partial \mathcal{L}/\partial q_i$ are the generalized (conservative) force components, $\partial \mathcal{L}/\partial \dot{q}_i$ the generalized momentum, and $\partial \mathcal{D}/\partial \dot{q}_i$ the generalized dissipative forces.

For a continuum, we shall define a Lagrangian density $\mathcal{L}$, and a dissipation density function $\mathcal{D}$, as energy per unit volume, such that

$$\mathcal{L} = \int_\Omega \mathcal{L}\, d\vec{x}\,; \qquad \mathcal{D} = \int_\Omega \mathcal{D}\, d\vec{x} \tag{9.7}$$

where Ω is the domain of interest, or the Representative Elementary Volume (REV). The principle of least action is then expressed as

$$\begin{aligned}\delta A &= \int_{t_1}^{t_2}\int_\Omega \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta q_i \, d\vec{x}\, dt - \int_{t_1}^{t_2}\int_\Omega \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i \, d\vec{x}\, dt + \int_{t_1}^{t_2}\int_\Omega \frac{\partial \mathcal{D}}{\partial \dot{q}_i}\delta q_i \, d\vec{x}\, dt \\ &= 0\end{aligned} \tag{9.8}$$

Kinetic Energy and Apparent Mass Consider a deforming porous medium in motion. The averaged Lagrangian density in an REV of volume Ω consists of the following energy components

$$\mathcal{L} = \mathcal{K} + \mathcal{W} - \mathcal{U} - \mathcal{V} \tag{9.9}$$

where $\mathcal{K}$ is the kinetic energy density, $\mathcal{W}$ the work energy density, $\mathcal{U}$ the elastic strain energy density, and $\mathcal{V}$ the potential (gravitational) energy density.

As the porous medium contains a solid and a fluid phase, respectively occupying the volume Ω_s and Ω_f, the kinetic energy density can be expressed as the weighted sum of the energy density of the solid phase $\mathcal{K}_s$ and the fluid phase $\mathcal{K}_f$:

$$\begin{aligned}\mathcal{K} &= (1-\phi)\mathcal{K}_s + \phi\mathcal{K}_f \\ &= \frac{1-\phi}{\Omega_s}\int_{\Omega_s} \frac{\rho_s v_i^s v_i^s}{2}\, d\vec{x} + \frac{\phi}{\Omega_f}\int_{\Omega_f} \frac{\rho_f v_i^f v_i^f}{2}\, d\vec{x}\end{aligned} \tag{9.10}$$

where ρ_s and ρ_f are the solid and fluid density, and $v_i^s = \dot{u}_i^s$ and $v_i^f = \dot{u}_i^f$ are the velocity components of solid and fluid, with u_i^s and u_i^f the solid and fluid displacement vector, respectively. These microscopic densities and velocities can be expresses as the sum of an average and a fluctuation:

$$\rho_\alpha = \overline{\rho}_\alpha + \overline{\overline{\rho}}_\alpha; \qquad v_i^\alpha = \overline{v}_i^\alpha + \overline{\overline{v}}_i^\alpha \tag{9.11}$$

where $\alpha = s, f$ denotes the material phase, the overbar indicates the volumetric average, and the double overbar denotes fluctuation from the average. Substituting (9.11) into (9.10), and assuming that velocity and density fluctuations are *uncorrelated*, we obtain

$$\mathcal{K}_s = \overline{\rho}_s\left(\frac{\overline{v}_i^s\,\overline{v}_i^s}{2} + k_s\right) \tag{9.12}$$

$$\mathcal{K}_f = \overline{\rho}_f \left(\frac{\overline{v}_i^f \overline{v}_i^f}{2} + k_f \right) \tag{9.13}$$

where

$$k_\alpha = \frac{1}{\Omega_\alpha} \int_{\Omega_\alpha} \frac{\overline{\overline{v}}_i^\alpha \overline{\overline{v}}_i^\alpha}{2} \, d\vec{x}, \qquad \alpha = s, f \tag{9.14}$$

is the kinetic energy density (per unit mass) due to the local fluctuation of velocities for phase α.

To construct a model at the macroscopic level, it is necessary to express k_α as functions of macroscopic parameters. In a more general form, it is suggested that they are functions of the following variables [82]

$$k_s = k_s(\overline{v}_i^-, \phi, \dot{\phi}, \overline{\epsilon}_{ij}^s); \qquad k_f = k_f(\overline{v}_i^-, \dot{\phi}) \tag{9.15}$$

where $\overline{\epsilon}_{ij}^s$ is the solid internal strain as defined in Sect. 4.1, and ϕ is porosity. In the above we also assumed, to the first order approximation, that the microscopic velocity fluctuations are caused by the macroscopic asynchronous motion of solid and fluid phases, hence only the relative fluid to solid velocity

$$\overline{v}_i^- = \overline{v}_i^f - \overline{v}_i^s \tag{9.16}$$

appears in the arguments of k_s and k_f.

For a simplified model, and particularly to make a connection with the Biot model [19, 20], we shall retain in the above functions only the most important effect, the nonuniform velocity in the fluid phase, and set

$$k_s = 0; \qquad k_f = k_f(\overline{v}_i^-) \tag{9.17}$$

In making that judgement, we recognize that at the microscale, the fluid velocity variation within the pores is much larger than the solid velocity variation within and between the grains, and also that the time rate change terms are of secondary effects, which are important only in very high frequency range. These secondary effects have been discussed [82], but are beyond the present scope to examine. For a linear theory, we can express k_f as a quadratic function of $\overline{v}_i^-$

$$k_f = \frac{1}{2} a_{ij}^f \overline{v}_i^- \overline{v}_j^- \tag{9.18}$$

where a_{ij}^f is an *apparent mass* constitutive constant (a second rank tensor), which is dependent only on the pore geometry. For isotropic material, we can express the coefficient as $a_{ij}^f = \delta_{ij} a^f$.

Taking the variation of the kinetic energy part of the *action density* (action per unit volume) with respect to the solid and fluid displacements, with the following correspondences of variables

$$q_i = \{\overline{u}_i^s, \overline{u}_i^f\}; \qquad \dot{q}_i = \{\dot{\overline{u}}_i^s, \dot{\overline{u}}_i^f\} = \{\overline{v}_i^s, \overline{v}_i^f\} \tag{9.19}$$

we obtain

$$\begin{aligned}
\delta \int_{t_1}^{t_2} \mathcal{K}\, dt &= \int_{t_1}^{t_2} \left(\frac{d}{dt} \frac{\partial \mathcal{K}}{\partial \overline{v}_i^s} \delta \overline{u}_i^s + \frac{d}{dt} \frac{\partial \mathcal{K}}{\partial \overline{v}_i^f} \delta \overline{u}_i^f \right) dt \\
&= \int_{t_1}^{t_2} \left\{ \frac{d}{dt} \left[(1-\phi)\, \overline{\rho}_s \overline{v}_i^s - \phi\, \overline{\rho}_f\, a_{ij}^f (\overline{v}_j^f - \overline{v}_j^s) \right] \delta \overline{u}_i^s \right. \\
&\qquad \left. + \frac{d}{dt} \left[\phi\, \overline{\rho}_s \overline{v}_i^s + \phi\, \overline{\rho}_f\, a_{ij}^f (\overline{v}_j^f - \overline{v}_j^s) \right] \delta \overline{u}_i^f \right\} dt \\
&= \int_{t_1}^{t_2} \left\{ \left[(1-\phi) \overline{\rho}_s \ddot{\overline{u}}_i^s - \rho_{ij}^a (\ddot{\overline{u}}_j^f - \ddot{\overline{u}}_j^s) \right] \delta \overline{u}_i^s \right. \\
&\qquad \left. + \left[\phi\, \overline{\rho}_f \ddot{\overline{u}}_i^f + \rho_{ij}^a (\ddot{\overline{u}}_j^f - \ddot{\overline{u}}_j^s) \right] \delta \overline{u}_i^f \right\} dt
\end{aligned} \tag{9.20}$$

In the above, we have defined an *apparent mass density*

$$\rho_{ij}^a = \phi\, \overline{\rho}_f\, a_{ij}^f \tag{9.21}$$

We note that the concept of (isotropic) apparent mass density was first introduced by Biot [19] to "*take into account the fact that the relative fluid flow through the pores is not uniform*".

The apparent mass is also called an *added mass* in fluid dynamics. Friedrich Bessel suggested this effect in 1826 [16] to describe the motion of a pendulum swinging in a fluid. As the pendulum, or any solid body, accelerates or decelerates in the fluid, a volume of surrounding fluid is forced to move with it. The inertia of the fluid then exerts a net force on the body. Based on d'Alembert's principle, the fluid force may be considered as an "added mass", and Newton's second law can be expressed as

$$F = (m + m_a)\, a = (\rho_s + \rho_a) V_s\, a \tag{9.22}$$

where m is the mass of the solid body, m_a is the added mass, V_s is the volume of the solid body, ρ_a is an *added mass density*, and

$$\rho_a = C_a \rho_f \tag{9.23}$$

with C_a as an *added mass coefficient*, whose value is dependent only on the shape of the body (sphere, ellipsoid, etc.). For a sphere accelerating in an infinite body of inviscid fluid, the added mass coefficient has been theoretically determined to be $C_a = 1/2$ [77]. In other words, the fluid volume dragged by the solid body is equivalent to half of the solid volume. For a non-spherical and non-rotating body, the added mass coefficient can be expressed as a symmetric second rank tensor C^a_{ij}. It is of interest to emphasize that this added mass force exists only if there is an acceleration; and it does not exist for a body moving at constant velocity in an inviscid fluid.

When viscosity exists, the boundary condition on the solid surface changes from free slippage of the ideal fluid to no slippage between solid and fluid. As a consequence, a layer of fluid, known as a *boundary layer*, is dragged with the solid, introducing another added mass. The added mass force of viscous origin, known as the *Basset force* [14], is proportional to the velocity gradient on the surface, and is independent of the acceleration. We shall defer its discussion to a later part in this section on dynamic permeability.

For porous medium, the apparent mass effect has been examined based on the homogenization theory [53, 62, 123]. Or, as suggested by Berryman [15], we may draw a simple analogy with fluid dynamics and express the added mass density in (9.21) as

$$\rho^a_{ij} = (1-\phi) C^a_{ij} \rho_f \tag{9.24}$$

where C^a_{ij} is the porous medium added mass coefficient. In the above equation, the $(1-\phi)$ factor is introduced because C^a_{ij} is normalized with respect to the porous medium volume and not the solid volume.

For a porous medium of high porosity (low solid content) made of isolated spherical particles, we can use $C_a = 1/2$ as if there were an infinite body of fluid surrounding each sphere. In the limit, when $\phi \to 1$, we observe from (9.24) that $\rho_a \to 0$, because the volume of fluid goes to zero as porosity approaches 1. In addition to the modulation by porosity, in a porous medium of low porosity, there is not sufficient amount of fluid to be dragged as added mass; hence the value of C_a is dependent on porosity and must be reduced. In the limit, as $\phi \to 0$, there is no fluid; hence $C_a \to 0$ and $\rho_a \to 0$.

By considering the hypothetical experiment of pulling a porous solid frame through a superfluid (frictionless), Johnson [68] argued that the apparent drag (added mass) is related to the tortuosity factor of the porous medium. The added mass density is then expressed as [70]

$$\rho_a = (\tau_t - 1)\, \phi\, \rho_f \tag{9.25}$$

where $\tau_t > 1$ is the *tortuosity*. Comparing with (9.21), apparently $\tau_t = a_f + 1$. Also, comparing with (9.24), we find

$$C_a = \frac{\phi}{1-\phi}(\tau_t - 1) \tag{9.26}$$

Based on a self-similar model, Sen et al. [109], established that the tortuosity is a function of the porosity

$$\tau_t = \phi^{-\beta} \tag{9.27}$$

Using fused glass beads, Johnson et al. [71] measured $\beta = 1/2$, which is the same as the theoretical value for the random packing of spheres. For a random array of needles, the theoretical value of $\beta = 3/2$ is also consistent with experiment [72]. Adopting the value of $\beta = 1/2$, we can express (9.26) as

$$C_a = \frac{\phi}{1-\phi}\left(\frac{1}{\sqrt{\phi}} - 1\right) \tag{9.28}$$

In Fig. 9.1a we plot the added mass coefficient C_a versus the porosity. We observe the limits of $C_a = \{0,\ 1/2\}$ in the limits of $\phi = \{0,\ 1\}$, as discussed in the above. Figure 9.1b present ρ_a/ρ_f as a function of ϕ. We also observe the anticipated value of $\rho_a = 0$ at $\phi = \{0,\ 1\}$. Particularly, at $\phi = 0.25$ the added mass density attains a maximum value of $\rho_a = 0.25\,\rho_f$. Equation (9.28) can be used as a guideline for selecting the value of added mass coefficient when no experimental measurement is available.

Work and Strain Energy For the strain energy density, we can express

$$\overline{\mathcal{U}} = (1-\phi)\,\overline{\mathcal{U}}_s + \phi\,\overline{\mathcal{U}}_f \tag{9.29}$$

where $\overline{\mathcal{U}}_s$ and $\overline{\mathcal{U}}_f$ are the REV averaged strain energy density functions of the solid and fluid phase, respectively. These functions are the Helmholtz free energy

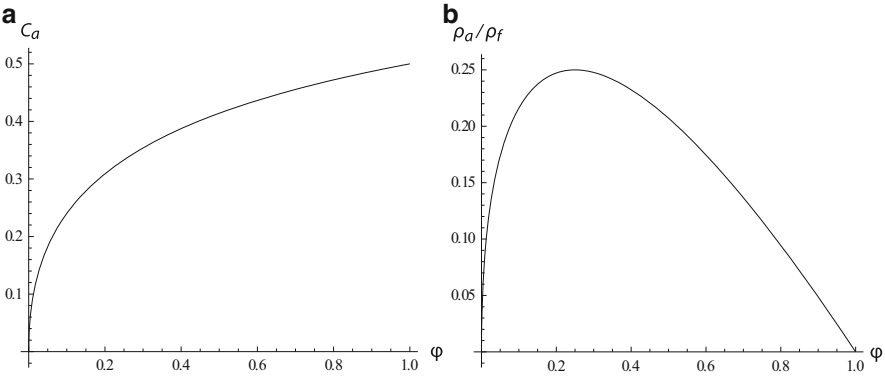


Fig. 9.1 Added mass as function of porosity: (**a**) C_a vs. ϕ, and (**b**) ρ_a/ρ_f vs. ϕ

density $\overline{\mathcal{A}}_s$ and $\overline{\mathcal{A}}_f$ under isothermal condition, as defined in (4.67) and (4.68). The functional dependence of $\overline{\mathcal{U}}_s$ and $\overline{\mathcal{U}}_f$ on macroscopic parameters has been determined in (4.70) and (4.72), which is repeated here

$$\overline{\mathcal{U}}_s = \overline{\mathcal{U}}_s(\overline{\epsilon}^s_{ij}, \phi); \qquad \overline{\mathcal{U}}_f = \overline{\mathcal{U}}_f(\overline{\epsilon}^f) \tag{9.30}$$

Taking variation of the action density, we obtain

$$\begin{aligned}
\delta \int_{t_1}^{t_2} \overline{\mathcal{U}}\, dt &= \int_{t_1}^{t_2} \left[(1-\phi) \frac{\partial\, \overline{\mathcal{U}}_s}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{\epsilon}^s_{ij} + (1-\phi) \frac{\partial\, \overline{\mathcal{U}}_s}{\partial \phi} \delta\phi + \phi \frac{\partial\, \overline{\mathcal{U}}_f}{\partial \overline{\epsilon}^f} \delta \overline{\epsilon}^f \right] dt \\
&= \int_{t_1}^{t_2} \left[(1-\phi) \frac{\partial\, \overline{\mathcal{U}}_s}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{e}^s_{ij} + \phi \frac{\partial\, \overline{\mathcal{U}}_f}{\partial \overline{\epsilon}^f} \delta \overline{e}^f \right] dt \\
&= \int_{t_1}^{t_2} \left[(1-\phi) \overline{\sigma}^s_{ij} \delta \overline{e}^s_{ij} - \phi \overline{p}^f \delta \overline{e}^f \right] dt
\end{aligned} \tag{9.31}$$

In the above derivation, we have utilized (4.73)–(4.78) to convert from internal strains $\{\overline{\epsilon}^s_{ij}, \overline{\epsilon}^f\}$ to external strains $\{\overline{e}^s_{ij}, \overline{e}^f\}$, and have substituted in the definitions of stresses and pore pressure.

For the work energy density, we can examine its variation as

$$\begin{aligned}
\delta \mathcal{W} &= \frac{1}{\Omega} \int_{\Gamma_s} \sigma^s_{ij} n_j \delta u^s_i \, d\vec{x} - \frac{1}{\Omega} \int_{\Gamma_f} p^f n_i \delta u^f_i \, d\vec{x} \\
&= \frac{1}{\Omega} \int_{\Omega_s} (\sigma^s_{ij,j} \delta u^s_i + \sigma^s_{ij} \delta e^s_{ij}) \, d\vec{x} - \frac{1}{\Omega} \int_{\Omega_f} (p^f_{,i} \delta u^f_i + p^f \delta e^f) \, d\vec{x} \\
&= (1-\phi)(\overline{\sigma}^s_{ij,j} \delta \overline{u}^s_i + \overline{\sigma}^s_{ij} \delta \overline{e}^s_{ij}) - \phi (\overline{p}^f_{,i} \delta \overline{u}^f_i + \overline{p}^f \delta \overline{e}^f)
\end{aligned} \tag{9.32}$$

In the above we have applied the divergence theorem. Combining (9.31) and (9.32), we obtain

$$\delta \int_{t_1}^{t_2} (\mathcal{W} - \mathcal{U})\, dt = \int_{t_1}^{t_2} \left[(1-\phi)\, \overline{\sigma}^s_{ij,j} \delta \overline{u}^s_i - \phi\, \overline{p}^f_{,i} \delta \overline{u}^f_i \right] dt \tag{9.33}$$

Potential Energy Consider gravity potential only, we can express the potential energy density as

$$\mathcal{V} = (1-\phi)\, g \int_{\Omega_s} \rho_s z \, d\vec{x} + \phi\, g \int_{\Omega_f} \rho_f z \, d\vec{x} \tag{9.34}$$

in which g is the gravity acceleration, and z is the Cartesian coordinate with its axis pointing to the opposite direction of gravity. The variation its action is then

$$\delta \int_{t_1}^{t_2} \mathcal{V}\, dt = (1-\phi)\, g \int_{t_1}^{t_2} \overline{\rho}_s \frac{\partial z}{\partial x_i} \delta \overline{u}_i^s \, d\vec{x} + \phi\, g \int_{t_1}^{t_2} \overline{\rho}_f \frac{\partial z}{\partial x_i} \delta \overline{u}_i^f \, d\vec{x} \tag{9.35}$$

In the above we note that δx_i is replaced by $\{\delta \overline{u}_i^s,\ \delta \overline{u}_i^f\}$.

Dissipation Function The dissipation function for continuum mechanics was introduced by Biot [17, 18] as a formalism to model irreversible thermodynamic processes based on the minimum production of entropy of a thermodynamic system. For porous media, we can consider two types of dissipative processes. The first type includes processes associated with the relaxation of temperature, mass diffusion, etc., that do not accompany physical movement of porous media. The second type is the irreversible processes related to the movement of material. In this chapter, we consider only dissipations of the second type.

For porous medium, we define the dissipation density function as the weighted sum of that in the solid and the fluid phase

$$\mathcal{D} = (1-\phi)\mathcal{D}_s + \phi\, \mathcal{D}_f \tag{9.36}$$

As discussed in Lopatnikov and Cheng [82], the dissipation density functions can be expressed as functions of these macroscopic variables

$$\mathcal{D}_s = \mathcal{D}_s(\overline{v}_i^{-},\ \dot{\phi},\ \dot{\overline{\epsilon}}_{ij}^{\,s}); \qquad \mathcal{D}_f = \mathcal{D}_f(\overline{v}^{-},\ \dot{\phi},\ \dot{\rho}_f) \tag{9.37}$$

For the present purpose, however, we shall ignore all the dissipations taking place in the solid phase, thus excluding mechanisms such as viscoelasticity and dry friction, etc., and consider only the viscous dissipation in the fluid phase; that is,

$$\mathcal{D}_s = 0; \qquad \mathcal{D}_f = \mathcal{D}_f(\overline{v}^{-}) \tag{9.38}$$

which is consistent with the Darcy's law, as adopted in Biot theory. For a linear theory, the dissipation density function is expressed in a quadratic form

$$\mathcal{D} = \frac{1}{2}\beta_{ij}\, \overline{v}_i^{-}\, \overline{v}_j^{-} \tag{9.39}$$

where β_{ij} is a resistance coefficient (tensor). Taking variation of (9.39) with respect to solid and fluid displacements, based on the principle of least action (9.8), we obtain

$$\begin{aligned} &\int_{t_1}^{t_2} \left(\frac{\partial \mathcal{D}}{\partial \overline{v}_i^s} \delta \overline{u}_i^s + \frac{\partial \mathcal{D}}{\partial \overline{v}_i^f} \delta \overline{u}_i^f \right) dt \\ &\qquad = \int_{t_1}^{t_2} \left[-\beta_{ij} (\dot{\overline{u}}_j^f - \dot{\overline{u}}_j^s) \delta \overline{u}_i^s + \beta_{ij} (\dot{\overline{u}}_j^f - \dot{\overline{u}}_j^s) \delta \overline{u}_i^f \right] dt \end{aligned} \tag{9.40}$$

These may be viewed as the Darcy dissipative force in an equilibrium equation.

Equilibrium Equation To construct the dynamic equilibrium equations, we collect the terms of the same variation quantities, $\delta\overline{u}_i^s$ and $\delta\overline{u}_i^f$, to obtain the following two equations, respectively for the solid and fluid phase

$$(1-\phi)\rho_s\ddot{u}_i^s - \rho_{ij}^a(\ddot{u}_j^f - \ddot{u}_j^s) - (1-\phi)\frac{\partial\sigma_{ij}^s}{\partial x_j} + (1-\phi)\rho_s g\frac{\partial z}{\partial x_i} - \beta_{ij}(\dot{u}_j^f - \dot{u}_j^s) = 0 \tag{9.41}$$

$$\phi\rho_f\ddot{u}_i^f + \rho_{ij}^a(\ddot{u}_j^f - \ddot{u}_j^s) + \phi\frac{\partial p^f}{\partial x_i} + \phi\rho_f g\frac{\partial z}{\partial x_i} + \beta_{ij}(\dot{u}_j^f - \dot{u}_j^s) = 0 \tag{9.42}$$

In the above we have dropped the overbar notation for simplicity, as from this point on, only the averaged, macroscopic quantities will be referred.

Equations (9.41) and (9.42) can be summed to obtain the total equilibrium equation for the porous medium as

$$(1-\phi)\rho_s\ddot{u}_i^s + \phi\rho_f\ddot{u}_i^f - \frac{\partial\sigma_{ij}}{\partial x_j} + \rho g\frac{\partial z}{\partial x_i} = 0 \tag{9.43}$$

where

$$\sigma_{ij} = (1-\phi)\sigma_{ij}^s - \phi\,\delta_{ij}p^f \tag{9.44}$$

is the total stress, and

$$\rho = (1-\phi)\rho^s + \phi\rho_f \tag{9.45}$$

is the total density. As the body force effect is not important for the wave propagation phenomenon, which is a result of a dynamic fluctuation from the hydrostatic state, we can drop such term in (9.43) and express it as

$$\sigma_{ij,j} = \rho\ddot{u}_i^s + \phi\rho_f(\ddot{u}_i^f - \ddot{u}_i^s) \tag{9.46}$$

which is just the previously derived equilibrium equation (6.54) modified to include the inertial effect.

We can also examine (9.42) without the body force term. By recognizing the Darcy specific flux as

$$q_i = \phi(\dot{u}_i^f - \dot{u}_i^s) \tag{9.47}$$

we can express (9.46) as the following

$$-\mu_f b_{ij}q_j = p_{,i} + \rho_f\ddot{u}_i^f + \frac{\rho_{ij}^a}{\phi}(\ddot{u}_j^f - \ddot{u}_j^s) \tag{9.48}$$

where we have dropped the superscript of p^f, and

$$b_{ij} = \frac{\beta_{ij}}{\phi^2 \mu_f} = [k_{ij}]^{-1} \tag{9.49}$$

is the resistivity coefficient defined in (6.15). Equation (9.48) is Darcy's law modified for the inertial effect.

9.2 Dynamic Permeability

In this section we shall examine the resistivity coefficient β_{ij} derived in the preceding section in some detail. For simplicity, we shall discuss only the isotropic case. In that case, we can express (9.48) as

$$-\frac{\phi \mu_f}{k}(\dot{u}_i^f - \dot{u}_i^s) = p^f_{,i} + \rho_f \ddot{u}_i^f + \frac{\rho_a}{\phi}(\ddot{u}_i^f - \ddot{u}_i^s) \tag{9.50}$$

in which we have substituted q_i by its definition (9.47). We note that by writing the fluid equilibrium equation (9.42) into the above, we have interpreted the constitutive coefficient β as $\beta = \phi^2 \mu_f / k$, based on the quasi-static Darcy's law derived in Sect. 6.1. Darcy's law, however, ignores the local acceleration as seen in the steady state Navier-Stokes equation (6.7) used in the homogenization theory (see Sect. 6.1.2); hence it needs to be modified for the dynamic effect.

Boundary Layer Consider the conceptual model of porous medium flow as a group of straight, parallel capillary tubes of uniform diameter. Under the steady state assumption, the flow inside the tubes is the Hagen-Poiseuille flow with discharge

$$Q = \frac{\pi a^4}{8\mu_f} \frac{\partial p}{\partial z} \tag{9.51}$$

where a is the tube radius, and the velocity profile is a paraboloid (see Fig. 9.2a). The resistance to the flow is due to the shear stress at the wall, which is proportional to the viscosity of the fluid and the velocity gradient at the wall. For the fully developed Hagen-Poiseuille flow case, we can interpret a permeability coefficient (inverse of resistivity) as

$$\kappa = \frac{\phi a^2}{8\mu_f} \tag{9.52}$$

When the flow is accelerating, and not at a steady state, the velocity profile and its gradient at the wall will be different. We can observe this effect by examining the oscillating plate solution of Stokes [108]. Figure 9.2b shows the velocity profile

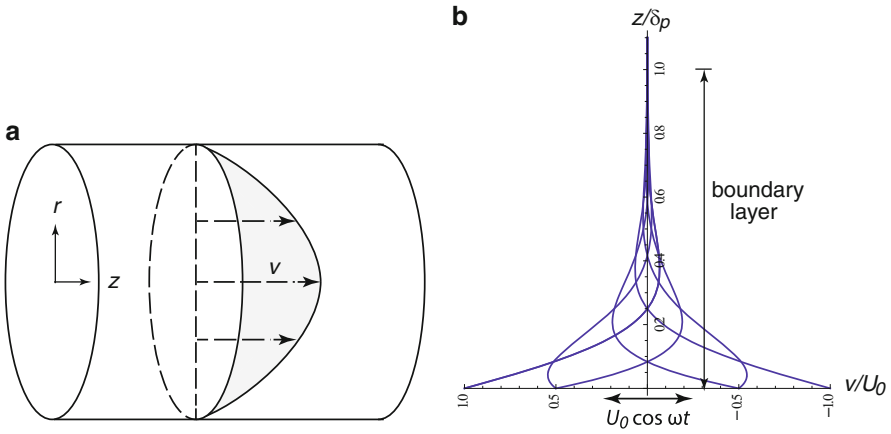


Fig. 9.2 Conceptual permeability model: (**a**) Fully developed velocity profile of Hagen-Poiseuille flow, and (**b**) Velocity profiles at different times of an oscillating plate

at various times in a semi-infinite fluid body, induced by a plate oscillating at the velocity $U_o \cos(\omega t)$. The flow profile exhibits a *boundary layer* structure, that is, significant flow exists only within a certain distance from the plate, denoted as the boundary layer thickness δ_ℓ, with the magnitude

$$\delta_\ell = 2\pi \sqrt{\frac{2\mu_f}{\rho_f \omega}} \tag{9.53}$$

At a distance δ_ℓ from the plate, the velocity amplitude is reduced to $e^{-2\pi} \approx 0.002\, U_o$. We observe that as frequency increases, the boundary layer thickness decreases, leading to a larger velocity gradient and larger viscous resistance. Hence it is necessary to modify the "static" permeability to a dynamic one to properly take into account the acceleration effect.

Conceptual Model To account for the boundary layer effect of fluid motion in flow channels, Biot [20] proposed a frequency dependent viscosity correction factor F. To accommodate for such model, we first perform Fourier transform, defined as

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt \tag{9.54}$$

to the fluid equilibrium equation (9.50). Or, in an equivalent operation, we assume that the time dependence of all variables in (9.50) can be expressed in terms of a sinusoidal function in complex variable form, $e^{-i\omega t}$, and a function can be expressed as $f(t) = \tilde{f}(\omega) e^{-i\omega t}$. Either way, (9.50) can be transformed into the following

$$\frac{i\omega\phi\mu_a}{k}(\tilde{u}_i^f - \tilde{u}_i^s) = \tilde{p}_{,i} - \rho_f \omega^2 \tilde{u}_i^f - \frac{\rho_a \omega^2}{\phi}(\tilde{u}_i^f - \tilde{u}_i^s) \tag{9.55}$$

In the above, we have replaced the fluid viscosity μ_f by an *apparent viscosity* μ_a with the definition

$$\mu_a = \mu_f F(\kappa_\mu) \tag{9.56}$$

in which $F(\kappa_\mu)$ is a *viscosity correction factor* with the argument

$$\kappa_\mu = a\sqrt{\frac{\rho_f \omega}{\mu_f}} \tag{9.57}$$

where a is a pore size factor, corresponding to the radius of a circular tube, or the half gap width of parallel plates, for the *conceptual model* of choice.

As an idealized case of flow in fractures, or slit-like pores, Biot [20] constructed the *parallel plates* flow model, and found the following viscosity correction factor[1]

$$F_1(\kappa_\mu) = -\frac{\sqrt{i}\,\kappa_\mu \tan(\sqrt{i}\,\kappa_\mu)}{3[1 - \tan(\sqrt{i}\,\kappa_\mu)/(\sqrt{i}\,\kappa_\mu)]} \tag{9.58}$$

We note that the above expression is modified from Biot's result, which assumed an $e^{i\omega t}$ time factor in flow motion, while here we have used an $e^{-i\omega t}$ factor. The consequence of the above change is that the current function is the complex conjugate of Biot's.

Assuming a circular tube pore geometry, Biot also constructed a *capillary tube* model, with the following viscosity correction factor

$$F_2(\kappa_\mu) = \frac{i\kappa_\mu T(\kappa_\mu)}{4[1 + 2T(\kappa_\mu)/\kappa_\mu]} \tag{9.59}$$

where

$$T(\kappa_\mu) = \frac{\mathrm{ber}'(i\kappa_\mu) + i\,\mathrm{bei}'(i\kappa_\mu)}{\mathrm{ber}\,(i\kappa_\mu) + i\,\mathrm{bei}\,(i\kappa_\mu)} = -\frac{\mathrm{J}_1(\sqrt{i}\,\kappa_\mu)}{\sqrt{i}\,\mathrm{J}_0(\sqrt{i}\,\kappa_\mu)} \tag{9.60}$$

In the above, ber and bei are, respectively, the real and imaginary parts of the Kelvin function of the first kind of order zero, and ber$'$ and bei$'$ are their derivatives. The last relation in (9.60) is obtained by referring to the table of mathematical functions [3]. It is obvious that the function F is a complex quantity. In Fig. 9.3 we plot the real and the imaginary part of F versus its real argument κ_μ, respectively for the parallel plate[2] and capillary tube case. Particularly we note that the imaginary part of the function takes the negative value.

[1] Equation (9.58) corrects a typographical error of Biot's equation (2.17) [20], in which an i factor is replaced by $\sqrt{i}$.

[2] The vertical scale of Fig. 9.3a corrects an error in Biot's Fig. 2 [20]; see footnote in [22].

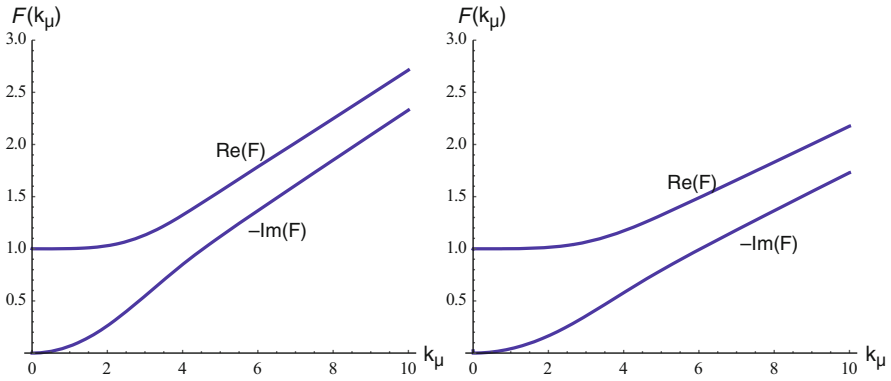


Fig. 9.3 Real and imaginary part of viscosity correction factor: (**a**) Parallel plate model, and (**b**) Capillary tube model

To understand these curves, we first examine the parameter κ_μ. Comparing with the boundary layer thickness defined in (9.53), we realize that κ_μ is a ratio of pore size to the boundary layer thickness:

$$\kappa_\mu = 2\sqrt{2}\pi \frac{a}{\delta_\ell} \approx 9\frac{a}{\delta_\ell} \tag{9.61}$$

At low frequency, that is, small κ_μ, the boundary layer thickness is larger than the pore size; hence the flow is fully developed. In this case, little or no modification to viscosity is needed. This is observed by expanding F into a series of small κ_μ, and we find, as $\omega \to 0$,

$$F_1(\kappa_\mu) \to 1 - i\frac{\kappa_\mu^2}{15} \tag{9.62}$$

$$F_2(\kappa_\mu) \to 1 - i\frac{\kappa_\mu^2}{24} \tag{9.63}$$

Particularly, at zero frequency, $F(0) = 1$.

At high frequency, $\delta_\ell \ll a$, a larger velocity gradient is developed near the wall, than that of the fully developed flow. The increase in resistance can be modeled as if the fluid has a larger viscosity, μ_a; hence F increases with κ_μ. As $\kappa_\mu \to \infty$ ($\omega \to \infty$), the following asymptotic behaviors exists:

$$F_1(\kappa_\mu) \to \frac{\sqrt{2}}{6}(1-i)\kappa_u \tag{9.64}$$

$$F_2(\kappa_\mu) \to \frac{\sqrt{2}}{8}(1-i)\kappa_u \tag{9.65}$$

As the above conceptual models are somewhat arbitrary, and they do not exactly represent the random porous medium geometry, Johnson et al. [69] proposed the following simple viscosity correction factor

$$F_3(\kappa_\mu) = \sqrt{1 - i\frac{\kappa_\mu^2}{16}} \tag{9.66}$$

to mimic F_2 in (9.59). F_3 has the same asymptotic behavior as F_2 as $\kappa_\mu \to \infty$. However, as $\kappa_\mu \to 0$,

$$F_3(\kappa_\mu) \to 1 - i\frac{\kappa_\mu^2}{32} \tag{9.67}$$

Although (9.66) is not based on a conceptual model, it is easier to calculate than expressions like (9.59). Also, it can be adjusted to fit experimental data at high frequency by manipulating the constant factor within the radical. Hence it can be a useful alternative to Biot's parallel plate and circular tube viscosity correction factors.

In order to apply the apparent viscosity model, values of material constants need to be estimated, particularly the pore size factor a. For the conceptual models, Biot [20] suggested

$$a = \delta_p \sqrt{\frac{k}{\phi}} \tag{9.68}$$

where k is the intrinsic permeability, and δ_p is a factor dependent only on the pore geometry, given as

$$\delta_p = \sqrt{\frac{16}{3}\tau_t}, \quad \sqrt{8\tau_t} \tag{9.69}$$

respectively for the parallel plates and circular tube model. In the above, τ_t is tortuosity.

Laboratory measurements gave $\delta_p = 6.3$ for crenel slits [8], and 3.2 for an assembly of spheres [24]. For oceans sediments, Stoll [114] suggested that the following empirical equation

$$a = \frac{1}{7} d_{\text{mean}} \tag{9.70}$$

can be used, where d_{mean} is the mean grain size. Other studies of *dynamic permeability* can be found in Sheng and Zhou [111] and Charlaix et al. [32].

As indicated by Biot [20], the two functions $F_1(\kappa_\mu)$ and $F_2(\kappa_\mu)$ are similar in shape. In fact, if we replace the argument κ_μ in F_2 by $4\kappa_\mu/3$, then

$$F_1(\kappa_\mu) \approx F_2(4\kappa_\mu/3) \tag{9.71}$$

So it does not matter which conceptual model is chosen, as long as a is properly determined, for example, by fitting experimental data at high frequency.

Basset Force and Viscous Added Mass For a partial differential equation classified as a *wave equation*, such as (9.50), the second time derivative term together with the second spatial derivative terms are responsible for the wave propagation, while the first time derivative term is associated with wave attenuation. When the wave equation is transformed to a frequency domain equation, such as (9.55), the first time derivative creates an imaginary coefficient, as the term on the left hand side of (9.55).

In the above discussed conceptual models of oscillatory flow, we find that the apparent viscosity μ_a in (9.55) is a complex quantity, as a consequence of the viscosity correction factor $F(\kappa_\mu)$, which is complex and can be expressed into a real and an imaginary part:

$$F = F_r + iF_i \tag{9.72}$$

When the above is substituted into (9.55), the real and imaginary terms can be separated and rearranged into the following

$$\frac{i\omega\phi F_r\mu_f}{k}(\tilde{u}_i^f - \tilde{u}_i^s) = \tilde{p}_{,i} - \rho_f\omega^2\tilde{u}_i^f - (\rho_a + \rho_a')\frac{\omega^2}{\phi}(\tilde{u}_i^f - \tilde{u}_i^s) \tag{9.73}$$

in which

$$\rho_a' = -\frac{\phi^2 F_i \mu_f}{\omega k} \tag{9.74}$$

As F_i is negative, the quantity ρ_a' is positive. We hence observe that the fluid viscosity has two effects: one is that energy is dissipated in a shear flow, and the other is that a thin layer of fluid near the solid surface (a boundary layer) is dragged to moves with the solid, as if there is an *added mass*. The apparent force due to this effect is known as the *Basset force* [14, 79, 105]. We may call ρ_a' a *Basset apparent mass density*.

To compare ρ_a with ρ_a', we notice that ρ_a' is dependent on the viscosity μ_f and the excitation frequency ω, while ρ_a is independent of those and is dependent on the geometry of the pore space only. In the limit of high frequency, as well as low viscosity, we find the Basset added mass density goes to zero

$$\rho_a' \to 0, \quad \text{as} \quad \omega \to \infty \quad \text{or} \quad \mu_f \to 0 \tag{9.75}$$

At low frequency, it approaches a constant

$$\rho_a' \to C_d \, \phi \, \rho_f, \quad \text{as} \quad \omega \to 0 \tag{9.76}$$

In deriving the above result we have utilized (9.57), (9.65), (9.68), and (9.69), and defined C_d as a constant dependent on the pore geometry. It is of interest to observe that the low frequency limit of ρ_a' is independent of fluid viscosity. For the capillary tube model,

$$C_d = \frac{1}{3} \tau_t \tag{9.77}$$

in which τ_t can be estimated using (9.27). For parallel plates model,

$$C_d = \frac{16}{45} \tau_t \approx \frac{1}{3} \tau_t \tag{9.78}$$

hence the low frequency limit is not sensitive to the pore geometry and we can simply use (9.77). Again comparing ρ_a' with ρ_a, we observe that ρ_a dominates in the high frequency range, while ρ_a' and ρ_a are of comparable importance at low frequency. To gain some idea about the magnitude of the added mass densities, we can assume a random packing of spheres with a porosity of $\phi = 0.4$. Using the above formulae, we calculate $\rho_a = 0.23 \, \rho_f$, and $\rho_a' = 0.21 \, \rho_f$ as $\omega \to 0$.

Characteristic Frequency and Attenuation Biot [19, 20] suggested that wave damping per wave cycle goes to zero in the static limit, that is, when the frequency approaches zero. At high frequency, as observed from (9.64) or (9.65), the real part of the viscosity correction factor F_r increases with the square root of frequency. However, when it is multiplied by $1/\omega$, the energy dissipated per wave cycle also approaches zero as $\omega \to \infty$. In between these two limits, a maximum attenuation exists. Biot suggested that this occurs around a *characteristic frequency*

$$\omega_c = \frac{\phi \, \mu_f}{k \, \rho_f} = \frac{\delta_p^2 \, \mu_f}{a^2 \rho_f} \tag{9.79}$$

We observe that when $\omega = \omega_c$,

$$a = \delta_p \sqrt{\frac{\mu_f}{\rho_f \, \omega}} \tag{9.80}$$

Comparing with (9.53), the above equation suggests that at the characteristic frequency, the boundary layer thickness δ_ℓ is of the order of pore size a. Based on the definitions (9.57) and (9.79) , we also find the correspondence

$$\kappa_\mu = \delta_p \sqrt{\frac{\omega}{\omega_c}} \tag{9.81}$$

In the case of water at 15°C, with pore size $a = 0.1$ mm, the critical frequency is approximately $f_c \approx 100$ Hz ($f_c = \omega_c/2\pi$).

9.3 Governing Equation

Governing equations for poroelastodynamics for isotropic material are similar to those for the quasi-static theory, shown as (6.58)–(6.63), except that the inertial terms need to be introduced, and the permeability coefficient is replaced by the dynamic permeability. In the following, we shall present the whole set of poroelastodynamics governing equations with a slight change of notation.

First, the equilibrium equation (9.43) is expressed in the following form

$$\sigma_{ij,j} = \rho \ddot{u}_i + \rho_f \ddot{w}_i - F_i \tag{9.82}$$

in which ρ is the total density defined in (9.45), F_i replaces the last term in (9.43) and is the body force. We shall henceforth use u_i and U_i to denote solid and fluid displacements, which also leads to the definition

$$w_i = \phi(U_i - u_i) \tag{9.83}$$

as the *specific relative fluid to solid displacement*. It is related to the *specific flux* as

$$q_i = \dot{w}_i \tag{9.84}$$

Darcy's law with inertia effect is given by (9.50), and expressed as follows

$$\dot{w}_i = -\kappa \left(p_{,i} + \rho_f \ddot{u}_i + \rho' \ddot{w}_i - f_i \right) \tag{9.85}$$

in which f_i is the fluid body force, and κ is the permeability coefficient. Here the permeability coefficient is interpreted as $\kappa = k/\mu_f$ at low frequency, and needs to be modified to

$$\kappa = \frac{k}{\mu_a} \tag{9.86}$$

at high frequency, with μ_a given by (9.56). In the above, we also used the shorthand notation

$$\rho' = \frac{\rho_f}{\phi} + \frac{\rho_a}{\phi^2} \tag{9.87}$$

The continuity equation with a source term (8.21) is expressed as

$$\dot{\zeta} + \dot{w}_{i,i} = \gamma \tag{9.88}$$

The above can be integrated with respect to time to give the definition of ζ,

$$\zeta = -w_{i,i} + Q \tag{9.89}$$

where Q is the volume of injected fluid, and

$$\gamma = \dot{Q} \tag{9.90}$$

Finally, we need to introduce the constitutive equations, (6.59) and (6.60),

$$\sigma_{ij} = 2Ge_{ij} + \lambda\,\delta_{ij}e - \alpha\delta_{ij}p \tag{9.91}$$

$$p = M\,(\zeta - \alpha e) \tag{9.92}$$

The above set of equations completely defines the poroelastodynamic problem.

Time Domain Equations The above set of equations contains more variables than desirable for a mathematical solution. We hence seek to eliminate some of them. It is possible to select the set of basic variables as u_i and w_i, and condense the governing equations into the following

$$G\nabla^2\boldsymbol{u} + (\lambda_u + G)\nabla(\nabla\cdot\boldsymbol{u}) + \alpha M\nabla(\nabla\cdot\boldsymbol{w}) = \rho\ddot{\boldsymbol{u}} + \rho_f\ddot{\boldsymbol{w}} - \boldsymbol{F} + \alpha M\nabla Q \tag{9.93}$$

$$\alpha M\nabla(\nabla\cdot\boldsymbol{u}) + M\nabla(\nabla\cdot\boldsymbol{w}) - \frac{1}{\kappa}\dot{\boldsymbol{w}} = \rho_f\ddot{\boldsymbol{u}} + \rho'\ddot{\boldsymbol{w}} - \boldsymbol{f} + M\nabla Q \tag{9.94}$$

In the above we have switched the notation by using boldface letters to denote vectors, and

$$\lambda_u = \lambda + \alpha^2 M \tag{9.95}$$

is the undrained Lamé constant. Equations (9.93) and (9.94) are six equations with six unknowns; hence is a sufficient solution system.

Frequency Domain Equations To simplify the differential equations, we can apply Fourier transform as defined in (9.54) to (9.93) and (9.94) to obtain

$$G\nabla^2\tilde{\boldsymbol{u}} + (\lambda_u + G)\nabla(\nabla\cdot\tilde{\boldsymbol{u}}) + \alpha M\nabla(\nabla\cdot\tilde{\boldsymbol{w}}) + \omega^2\rho\tilde{\boldsymbol{u}} + \omega^2\rho_f\tilde{\boldsymbol{w}} = -\tilde{\boldsymbol{F}} + \alpha M\nabla\tilde{Q} \tag{9.96}$$

$$\alpha M\nabla(\nabla\cdot\tilde{\boldsymbol{u}}) + M\nabla(\nabla\cdot\tilde{\boldsymbol{w}}) + \omega^2\rho_f\tilde{\boldsymbol{u}} + \omega^2\rho''\tilde{\boldsymbol{w}} = -\tilde{\boldsymbol{f}} + M\nabla\tilde{Q} \tag{9.97}$$

where

$$\rho'' = \rho' + \frac{i}{\omega\kappa} \tag{9.98}$$

with ρ' defined in (9.87).

Displacement-Pressure Formulation It was demonstrated by Cheng et al. [33] that in the frequency domain, the governing equations can be simplified from the 6-variable (u_i, w_i) formulation to a 4-variable (u_i, p) formulation. This is accomplished by expressing Darcy's law (9.85) in frequency domain in the form

$$\omega^2\rho''\tilde{\boldsymbol{w}} = \nabla\tilde{p} - \omega^2\rho_f\,\tilde{\boldsymbol{u}} - \tilde{\boldsymbol{f}} \tag{9.99}$$

We can also take divergence of the above equation to obtain

$$\omega^2\rho''\nabla\cdot\tilde{\boldsymbol{w}} = \nabla^2\tilde{p} - \omega^2\rho_f\nabla\cdot\tilde{\boldsymbol{u}} - \nabla\cdot\tilde{\boldsymbol{f}} \tag{9.100}$$

Combining (9.89) and (9.92) gives

$$\nabla\cdot\tilde{\boldsymbol{w}} = -\frac{\tilde{p}}{M} - \alpha\,\nabla\cdot\tilde{\boldsymbol{u}} + \tilde{Q} \tag{9.101}$$

Equations (9.99) and (9.101) can be used to eliminate $\tilde{\boldsymbol{w}}$ and $\nabla\cdot\tilde{\boldsymbol{w}}$ in (9.96) and (9.100) to obtain the pair of equations

$$G\nabla^2\tilde{\boldsymbol{u}} + (\lambda+G)\nabla(\nabla\cdot\tilde{\boldsymbol{u}}) + \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''}\tilde{\boldsymbol{u}} - \frac{\alpha\rho'' - \rho_f}{\rho''}\nabla\tilde{p} = -\tilde{\boldsymbol{F}} + \frac{\rho_f}{\rho''}\tilde{\boldsymbol{f}} \tag{9.102}$$

$$\nabla^2\tilde{p} + \frac{\omega^2\rho''}{M}\tilde{p} + \omega^2(\alpha\rho'' - \rho_f)\nabla\cdot\tilde{\boldsymbol{u}} = \nabla\cdot\tilde{\boldsymbol{f}} + i\omega\rho''\tilde{\gamma} \tag{9.103}$$

in which we have replaced the volume of injected fluid per unit porous medium volume $\tilde{Q}$ by its rate $\tilde{\gamma} = -i\omega\tilde{Q}$. Equations (9.102) and (9.103) are now four equations with four variables $\tilde{\boldsymbol{u}}$ and $\tilde{p}$. It is more desirable to solve the 4-variable system in a boundary value problem not only because of its reduction in degree-of-freedom as compared to the 6-variable system, but also due to its relevance associated with the physical boundary conditions, which will be explained below.

We should note that the elimination of the variable $\boldsymbol{w}$ in the time-domain equations (9.93) and (9.94) to arrive at a $(\boldsymbol{u}, p)$ formulation is also possible. The process however leads to higher order partial differential equations.

Boundary Conditions To solve the above governing equations for a given physical problem, initial and boundary conditions are needed. The initial conditions are: at $t = 0$, values of $\{\boldsymbol{u}, \dot{\boldsymbol{u}}, \boldsymbol{w}, \dot{\boldsymbol{w}}\}$ for the 6-variable formulation, or $\{\boldsymbol{u}, \dot{\boldsymbol{u}}, p\}$ for the 4-variable formulation, must be given for all $\boldsymbol{x} \in \Omega$. Section 6.5.2 discussed the boundary conditions corresponding to the various physical situations, for the

4-variable case. For the 6-variable case, additional conditions are needed. These conditions, however, are generally not known, or not clear. We shall examine some of these conditions below.

Take, for example, the free surface boundary illustrated in Fig. 6.5a. For simplicity, we consider only the two-dimensional case. For the two solid variables, u_x and u_y, the traction free conditions, $t_x = t_y = 0$, provide the needed boundary conditions. For the fluid variables, w_x and w_y, two conditions are also needed. However, from physical reasoning, we find only one

$$p = 0 \tag{9.104}$$

which can be expressed in terms of displacement variables as

$$\alpha \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \right) + \left(\frac{\partial w_x}{\partial x} + \frac{\partial w_y}{\partial y} \right) = 0 \tag{9.105}$$

following the constitutive equation (9.92). We still need a fluid related boundary condition in either the x or the y direction. The question is, whether a condition can be independently prescribed, and if so, what is its physical meaning? Assuming that an independent condition can indeed be prescribed, we may choose the simplest one

$$w_x = 0 \tag{9.106}$$

which also leads to $\dot{w}_x = \ddot{w}_x = 0$. The pore pressure condition (9.104) on a flat boundary also gives $\partial p/\partial x = 0$. An inspection on the dynamic Darcy's law (9.85) shows that

$$\ddot{u}_x = 0 \tag{9.107}$$

This means that the imposed boundary condition (9.106) prohibits wave propagation along the free surface. This is not a physically attainable condition. So it remains unclear what is a physically meaningful boundary condition on the free surface.

In another example, we examine a rigid, impermeable surface in perfect contact with porous medium, shown in Fig. 6.5d, f. For the solid boundary conditions, it is clear that $u_x = u_y = 0$. For the fluid part, we can deduce from the no-flux condition that

$$\dot{w}_y = 0 \tag{9.108}$$

For the condition is the x-direction, we may be tempted to impose a 'no-slip' condition of fluid dynamics for viscous flow

$$\dot{w}_x = 0 \tag{9.109}$$

on the solid surface. This however may not be compatible with Darcy's flow. Although the underlying physical mechanism of porous medium flow is a viscous flow, the specific discharge $\boldsymbol{q} = \dot{\boldsymbol{w}}$ appearing in Darcy's law is an REV averaged macroscopic flux, not the microscopic flow in the pore space, where the no-slip condition must be enforced on the channel walls and any solid surface. As an illustration, we consider a porous medium flow bounded by two parallel impermeable surfaces. The flow profile is uniform across the depth, and not a parabolic one. Or, we can also examine the dynamic Darcy's law (9.85), and find

$$\frac{\partial p}{\partial x} = 0 \tag{9.110}$$

meaning that pore pressure is not allowed to vary along the impermeable surface, which is also an unattainable physical condition.

As an ultimate case, we consider a forcing function that is sinusoidal in time. After the initial transient dies out, the solution becomes sinusoidal and satisfies the frequency domain governing equations, either (9.96) and (9.97), or (9.102) and (9.103). A boundary value problem can be solved using the 4-variable formulation (9.102) and (9.103) without specifying a boundary condition on w_x, for the two cases mentioned above. The solutions gives $\{\tilde{\boldsymbol{u}}, \tilde{p}\}$ for all $\boldsymbol{x} \in \Omega$. The dynamic Darcy's law (9.85) in the frequency domain allows the evaluation of $\tilde{\boldsymbol{w}}$ for all $\boldsymbol{x} \in \Omega$. It is obvious that an extra boundary condition on w_x cannot be prescribed. Hence we conclude that when the 6-variable formulation is used to solve boundary value problems, one needs to be extra careful not to specify boundary conditions that violate the physical nature of the problem.

9.4 Wave Propagation

In this section we shall demonstrate that the governing equations presented in Sect. 9.3 are *wave equations*. We then examine the propagation of harmonic waves in homogeneous medium to show the existence of three distinctive waves. Due to the dissipative nature of the viscous fluid, the propagation velocities are complex numbers. The physical meaning of the complex velocities, in terms of phase velocity and attenuation, are explored in Sect. 9.5.

Before deriving the poroelastic waves, however, we shall first present the classical elastic wave theory to get familiarized with the subject.

9.4.1 Elastic Wave

The governing equation for elastodynamics is

$$G\nabla^2\tilde{\boldsymbol{u}} + (\lambda + G)\nabla(\nabla \cdot \tilde{\boldsymbol{u}}) + \omega^2\rho\tilde{\boldsymbol{u}} = 0 \tag{9.111}$$

In the above we have already performed Fourier transform. Taking divergence of the above equation, with $\nabla \cdot \boldsymbol{u} = e$, we can transform it to the standard form of wave equation

$$v_{pe}^2 \nabla^2 \tilde{e} + \omega^2 \tilde{e} = 0 \tag{9.112}$$

where

$$v_{pe} = \sqrt{\frac{E_d}{\rho}} \tag{9.113}$$

is the *dilatational wave velocity* of elastic medium. In the above, we have defined a *dynamic elastic modulus*

$$E_d = \lambda + 2G \tag{9.114}$$

As dilatation is the opposite of compression, such wave is also called a *compressional wave*. Consider a uniaxial strain problem that has strain only in one spatial direction, say x-direction, such that $e = e_{xx} = e_{xx}(x,t)$, and $e_{yy} = e_{zz} = 0$. We then conclude that only $u_x = u_x(x,t)$ exists, and $u_y = u_z = 0$. This suggests that the movement of the particle is in the same direction of wave propagation; hence dilatational wave is also called a *longitudinal wave*.

We can also apply the curl operator to (9.111), and denote $\nabla \times \boldsymbol{u} = \boldsymbol{\Omega}$. Realizing the vector identity $\nabla \cdot (\nabla \times \boldsymbol{A}) = 0$, where $\boldsymbol{A}$ is any vector, we obtain

$$v_{se}^2 \nabla^2 \tilde{\boldsymbol{\Omega}} + \omega^2 \tilde{\boldsymbol{\Omega}} = 0 \tag{9.115}$$

where

$$v_{se} = \sqrt{\frac{G}{\rho}} \tag{9.116}$$

is the *rotational wave velocity*. As rotation is made of shear components (see Appendix C), such wave is also called a *shear wave*. For a wave in the x-y plane propagating in the x-direction, only the term $\Omega_z = (1/2)\,\partial u_y(x,t)/\partial x$ exists; hence the motion of the particle is perpendicular to the wave propagation direction. Such wave is also called a *transverse wave*.

The above simple analysis shows that for an elastic medium, there exist two propagating waves, one compressional and one shear. The fact that these are independent waves, and there is no other wave in existence in free space, can be established by invoking the *Helmholtz theorem* in vector calculus [4], also known as *Helmholtz decomposition*, which states that any sufficiently smooth, rapidly decaying vector field in three dimensions can be resolved into the sum of an irrotational (curl-free) vector field and a solenoidal (divergence-free) vector field.

The elastic wave velocities are functions of material properties only, and are independent of wave frequency. These waves are also called *non-dispersive waves*. We shall demonstrate below that the poroelastic wave velocities are dependent on frequency; hence the waves are *dispersive*.

9.4.2 Poroelastic Wave

For poroelastic waves, we can apply the similar technique used in the preceding section to isolate the dilatational and rotational waves. We shall first examine the rotational wave.

Shear Wave For shear wave (S-wave), we apply the curl operator to (9.96) and (9.97), and obtain the pair of equations

$$G\nabla^2\tilde{\boldsymbol{\Omega}}_s + \omega^2\rho\,\tilde{\boldsymbol{\Omega}}_s + \omega^2\rho_f\,\tilde{\boldsymbol{\Omega}}_r = 0 \tag{9.117}$$

$$\omega^2\rho_f\,\tilde{\boldsymbol{\Omega}}_s + \omega^2\rho''\,\tilde{\boldsymbol{\Omega}}_r = 0 \tag{9.118}$$

in which we have neglected the body force and source terms. In the above, we have defined

$$\boldsymbol{\Omega}_s = \nabla \times \boldsymbol{u} \tag{9.119}$$

$$\boldsymbol{\Omega}_r = \nabla \times \boldsymbol{w} \tag{9.120}$$

From (9.118) we can solve for

$$\tilde{\boldsymbol{\Omega}}_r = -\frac{\rho_f}{\rho''}\,\tilde{\boldsymbol{\Omega}}_s \tag{9.121}$$

Before making substitution into (9.117), we observe from the definition (9.83) that

$$\boldsymbol{\Omega}_r = \phi\,(\boldsymbol{\Omega}_f - \boldsymbol{\Omega}_s) \tag{9.122}$$

with

$$\boldsymbol{\Omega}_f = \nabla \times \boldsymbol{U} \tag{9.123}$$

Hence we can write (9.121) as

$$\boldsymbol{\Omega}_f = \frac{\rho_a + (i\phi^2/\omega\kappa)}{\phi\rho_f + \rho_a + (i\phi^2/\omega\kappa)}\boldsymbol{\Omega}_s \tag{9.124}$$

The above relation shows that the rotation of solid causes a partial rotation of fluid through inertia coupling. As the imaginary part is associated with fluid viscosity and wave damping, we observe that the entrainment of fluid acts as an added mass effect involving ρ_a as well as ρ_a' (defined within κ).

Using (9.122) to eliminate $\boldsymbol{\Omega}_r$ in (9.117), we obtain the following wave equation

$$v_s^2 \nabla^2 \tilde{\boldsymbol{\Omega}}_s + \omega^2 \tilde{\boldsymbol{\Omega}}_s = 0 \tag{9.125}$$

with the shear wave velocity

$$v_s = \sqrt{\frac{G}{\rho - (\rho_f^2/\rho'')}} \tag{9.126}$$

Based on (9.124), it is easy to conclude that $\boldsymbol{\Omega}_f$ propagates at the same velocity. Comparing (9.126) with the elastic shear wave velocity (9.116), we observe that the poroelastic shear wave velocity is a complex quantity, with its imaginary part related to wave damping. We also observe that both the real and imaginary parts of (9.126) are frequency dependent, and are no longer constant.

Compressional Waves For compressional wave, we take divergence of (9.96) and (9.97), leading to the following

$$(\lambda_u + 2G)\nabla^2 \tilde{e} - \alpha M \nabla^2 \tilde{\zeta} + \omega^2 \rho\, \tilde{e} - \omega^2 \rho_f\, \tilde{\zeta} = 0 \tag{9.127}$$

$$\alpha M \nabla^2 \tilde{e} - M \nabla^2 \tilde{\zeta} + \omega^2 \rho_f \tilde{e} - \omega^2 \rho'' \tilde{\zeta} = 0 \tag{9.128}$$

which can be expressed in the matrix form

$$\begin{bmatrix} (\lambda_u + 2G)\nabla^2 + \omega^2 \rho & -(\alpha M \nabla^2 + \omega^2 \rho_f) \\ \alpha M \nabla^2 + \omega^2 \rho_f & -(M \nabla^2 + \omega^2 \rho'') \end{bmatrix} \begin{Bmatrix} \tilde{e} \\ \tilde{\zeta} \end{Bmatrix} = 0 \tag{9.129}$$

Our goal is to eliminate one variable from the above equations. As a linear partial differential operator follows the algebraic rule of addition and multiplication [63], (9.129) can be treated as a linear algebraic system. Using elimination, we obtain

$$\left\{ \left[(\lambda_u + 2G)\nabla^2 + \omega^2 \rho \right] (M \nabla^2 + \omega^2 \rho'') - (\alpha M \nabla^2 + \omega^2 \rho_f)^2 \right\} \tilde{e} = 0 \tag{9.130}$$

The above can be expanded into the following

$$A \nabla^4 \tilde{e} - B\omega^2 \nabla^2 \tilde{e} + C\omega^4 \tilde{e} = 0 \tag{9.131}$$

where

$$A = M E_d \tag{9.132}$$

$$B = \rho'' E_{du} + \rho M - 2\rho_f \alpha M \tag{9.133}$$

$$C = \rho\rho'' - \rho_f^2 \tag{9.134}$$

with E_d the *drained dynamic modulus* as defined in (9.114), and

$$E_{du} = \lambda_u + 2G = E_d + \alpha^2 M \tag{9.135}$$

the *undrained dynamic modulus*. In turn, (9.131) can be factored into

$$\left(v_{p1}^2 \nabla^2 \tilde{e} + \omega^2 \tilde{e}\right)\left(v_{p2}^2 \nabla^2 \tilde{e} + \omega^2 \tilde{e}\right) = 0 \tag{9.136}$$

in which

$$v_{p1} = \sqrt{\frac{2A}{B - \sqrt{B^2 - 4AC}}} \tag{9.137}$$

$$v_{p2} = \sqrt{\frac{2A}{B + \sqrt{B^2 - 4AC}}} \tag{9.138}$$

Equation (9.136) is satisfied by either of the following wave equations

$$v_{p1}^2 \nabla^2 \tilde{e}_{p1} + \omega^2 \tilde{e}_{p1} = 0 \tag{9.139}$$

$$v_{p2}^2 \nabla^2 \tilde{e}_{p2} + \omega^2 \tilde{e}_{p2} = 0 \tag{9.140}$$

in which we have separately denoted the strain as e_{p1} and e_{p2}. Hence the system of Eqs. (9.127) and (9.128) can sustain two waves with velocity v_{p1} and v_{p2}.

We can apply the same procedure to (9.127) and (9.128) to eliminate $\tilde{e}$, and obtain an equation for $\tilde{\zeta}$. The characteristic equation, however, has the same coefficients as (9.131); hence no additional wave equation is generated.

Fast Wave and Slow Wave The above result leads us to conclude that for poroelasticity, there are two compressional waves. We shall call them the *first compressional wave* (or compressional wave of the first kind, or P1-wave), and the *second compressional wave* (or compressional wave of the second kind, or P2-wave), corresponding to the propagation velocity v_{p1} and v_{p2} in (9.137) and (9.138), respectively. We shall show that the first compressional wave has the faster velocity, and is called the *fast wave*, and the second wave is called the *slow wave*.

First, we shall show that the real parts of the three constants A, B and C, defined by (9.132), (9.133), and (9.134), are all positive. It is obvious that A is real and positive, as the elastic moduli E_d and M are real and positive. As the phase velocities (to be discussed in Sect. 9.5) are largely associated with the real part of the complex velocities, in the following we shall ignore the imaginary part of ρ'', defined in (9.98), such that $\rho'' = \rho'$. The imaginary part, $i/\omega\kappa$, is largely associated with the wave attenuation, although it does contribute a

small part in wave dispersion (frequency dependent wave velocity). With the above condition, we can express C as

$$C = \frac{\rho\rho_a}{\phi^2} + \frac{(1-\phi)\rho_s\rho_f}{\phi} > 0 \tag{9.141}$$

given the physical conditions $\{\rho, \rho_s, \rho_f, \rho_a\} > 0$. For the real part of B, we find

$$\begin{aligned} B &= \rho' E_{du} + \rho M - 2\rho_f \alpha M > \rho' E_{du} + \rho M - 2\sqrt{\rho\rho' M E_{du}} \\ &= \left(\sqrt{\rho' E_{du}} - \sqrt{\rho M}\right)^2 > 0 \end{aligned} \tag{9.142}$$

In the above, we have substituted the condition $E_{du} > \alpha^2 M$ by virtue of (9.135), and $\rho\rho' > \rho_f^2$, because C is positive.

Furthermore, we shall demonstrate that $B^2 - 4AC > 0$. For brevity, we switch to a set of Biot [19] material constants, P, Q and R, which are all positive, with definitions given in Sect. B.6. We can demonstrate that

$$A = (PR - Q^2)/\phi^2 \tag{9.143}$$

$$B = (\rho_{11}R + \rho_{22}P + 2\rho_a Q)/\phi^2 \tag{9.144}$$

$$C = (\rho_{11}\rho_{22} - \rho_a^2)/\phi^2 \tag{9.145}$$

where ρ_{11} and ρ_{22} are defined following Biot's convention

$$\rho_{11} = (1-\phi)\rho_s + \rho_a; \qquad \rho_{22} = \phi\rho_f + \rho_a \tag{9.146}$$

We can show that [39]

$$\begin{aligned} \phi^4(B^2 - 4AC) &= (\rho_{11}R + \rho_{22}P + 2\rho_a Q)^2 - 4(PR - Q^2)(\rho_{11}\rho_{22} - \rho_a^2) \\ &= (\rho_{22}P - \rho_{11}R)^2 + 4(\rho_{11}Q + \rho_a P)(\rho_{22}Q + \rho_a R) > 0 \end{aligned} \tag{9.147}$$

We are now ready to demonstrate that

$$\frac{v_{p1}^2}{v_{p2}^2} = \frac{B + \sqrt{B^2 - 4AC}}{B - \sqrt{B^2 - 4AC}} > 1 \tag{9.148}$$

Hence v_{p1} is the fast wave velocity, and v_{p2} is the slow wave velocity. It is also of interest to make a comparison with the shear wave,

$$\frac{v_{p1}^2}{v_s^2} = \frac{2AC}{\rho' G\left(B - \sqrt{B^2 - 4AC}\right)} > \frac{B}{2\rho' G} > \frac{\phi^2 B}{\rho_{22} P} > 1 \tag{9.149}$$

Hence the fast compressional wave is always faster than the shear wave. The slow compressional wave, however, can be faster or slower than the shear wave.

Dynamic Compatibility Condition Biot [19] pointed out that under certain material condition, it is possible that there is no relative fluid to solid movement, and the two compressional waves degenerate into one. To obtain such condition, we set $\zeta = 0$ in (9.127) and (9.128). To avoid contradiction between these two equations, we must have

$$\frac{E_{du}}{\alpha M} = \frac{\rho}{\rho_f} \tag{9.150}$$

The above is called the *dynamic compatibility condition*. Under such condition, the compressional wave velocity becomes

$$v_p = \sqrt{\frac{E_{du}}{\rho}} \tag{9.151}$$

9.5 Phase Velocity and Attenuation

The complex wave velocities obtained in the preceding section are mathematical artifacts resulting from a complex variable representation of the differential equations. In this section we shall explore the physical quantities associated with them, such as the phase velocity and wave attenuation. These quantities are related to the real and imaginary parts of the complex velocities. Due to the complicated expressions, it is impractical to analytically separate the velocities into a real and an imaginary part. Hence, in this section we shall numerically examine the phase velocity and attenuation based on different material constants, and observe their trend. In addition, we shall also perform asymptotic analysis in the high and low frequency ranges. In these limits, simpler mathematical relations emerges, which can provide valuable physical insight.

High and Low Frequency As a first step, it is necessary to define what is considered a high frequency, and what is a low frequency. As indicated by Biot [19], the characteristic frequency (9.79), written here in the ordinary frequency form

$$f_c = \frac{\phi}{2\pi\kappa\rho_f} \tag{9.152}$$

serves such role. High frequency is defined as $f \gg f_c$, and low frequency as $f \ll f_c$.

We observe in (9.152) that for a given fluid, f_c is dominated by the permeability coefficient κ. As shown in Table 3.1, values of κ can span 10 orders of magnitude for different porous materials; so the characteristic frequency also has a wide range of variation. In Table 9.1 we present f_c for a range of materials assuming water as the

Table 9.1 Critical frequency and applicable frequency range assuming a pore size of 0.1 mm for various porous media

Porous medium	Characteristic frequency (Hz)	Applicable frequency range (Hz)
Hard sediment	7.5×10^2	$<2 \times 10^7$
Polyurethane foam	7.5×10^3	$<1 \times 10^6$
Berea sandstone	1.6×10^5	$<3 \times 10^7$
Pecos sandstone	3.9×10^7	$<3 \times 10^7$
Westerly granite	4.0×10^9	$<5 \times 10^7$
Gulf Mexico shale	4.8×10^{11}	$<2 \times 10^7$

fluid. We observe that a low frequency for one material can be a very high frequency for another, in terms of their frequency dependent behavior.

Range of Validity of Continuum Theory Another frequency of interest is the frequency range in which the continuum mechanics theory remains valid. The continuum theory is based on the averaging of an inhomogeneous material over an REV of size at least one order of magnitude larger than the pore or grain size. As the wave frequency increases, the wave length decreases. When the wave length becomes shorter than the pore or grain size, it is not clear how the solid borne waves can transmit from one grain to the other, and how the multiple reflection, transmission, and scattering of waves at the fluid and grain boundaries affect the propagation of the bulk waves. In other words, the REV averaged mechanical properties may no longer be the relevant quantities that characterize the propagation and dissipation of the very high frequency (short wave length) waves. Bourbié et al. [25] suggested that the minimum REV should be at least three times the pore diameter. Given a wave velocity V, and minimum wave length comparable to the pore size a, we may find that frequency limit as $f = V/a$. In Table 9.1, such frequency is evaluated using compressional wave velocity based on properties given in Tables 3.1 and 3.2, with an assumed pore size of 0.1 mm. The above result shows that the Biot theory probably should not be applied to the high megahertz, or ultrasound, range.

9.5.1 Phase Velocity

The wave velocities derived in the preceding section are complex quantities. It can be expressed in the form of a quotient of wave number ℓ and frequency ω as

$$v = v_r + i v_i = \frac{\omega}{\ell} \tag{9.153}$$

In the above, ω is a real number, and ℓ a complex number, with

$$\ell = \ell_r + i \ell_i \tag{9.154}$$

The *phase velocity* v_{ph} is the velocity at which a wave form propagates, and is given as

$$v_{ph} = \frac{\omega}{|\ell_r|} = \frac{|v|^2}{|v_r|} \approx |v_r| \tag{9.155}$$

where the last approximation is obtained by assuming $v_r \gg v_i$, which is true for the shear and first compressional waves, but not necessarily true for the second compressional wave. Before presenting the phase velocity, we shall first examine the real and imaginary parts of the complex velocities as follows.

Real and Imaginary Part of Wave Velocity To obtain numerical results, we select two different porous materials, a hard sediment and a Berea sandstone, saturated with water. The material constants are found in Tables 3.1 and 3.2, with additional properties needed in the model given by $(\rho,\, \rho_a) = (1875,\, 215)$ kg/m^3 for hard sediment, and $(\rho,\, \rho_a) = (2458,\, 246)$ kg/m^3 for Berea sandstone. We also used a $\delta_p = 3.2$ defined in (9.68) for the evaluation of the viscosity correction factor $F_2(\kappa_\mu)$ in (9.59). The real parts of the shear and compressional wave velocities are plotted in Fig. 9.4a, b, and the imaginary parts in Fig. 9.4c, d, respectively for the two materials.

We observe in Fig. 9.4a, b that the real part of v_{p1} is much larger than that of v_{p2}. We also find that although the real parts of v_{p1} and v_s are frequency dependent, they only slightly increase in magnitude from low to high frequency. The real part of v_{p2}, on the other hand, is strongly frequency dependent. Figure 9.4c, d show that the imaginary parts of the velocities are highly frequency dependent and go to zero at both frequency ends, $f \to 0$ and $f \to \infty$. They peak around a frequency that can be identified as the characteristic frequency ω_c defined in (9.79). We shall further explore the imaginary part when we discuss wave attenuation.

Shear Wave Phase Velocity In Fig. 9.5a, b we plot the phase velocity of the shear wave in solid lines as function of frequency f. As commented above, we observe that the shear wave phase velocity slightly increases with frequency. Based on (9.126) and the asymptotic behavior of $F(\kappa_\mu)$, it can be shown that as $f \to 0$,

$$v_{ph} \approx \sqrt{\frac{G}{\rho}}\left[1 + \frac{\phi^2}{8\rho^2}(4\rho\rho' - \rho_f^2)\left(\frac{f}{f_c}\right)^2\right] \to \sqrt{\frac{G}{\rho}} \tag{9.156}$$

where f_c is the characteristic frequency given by (9.152), and as $f \to \infty$,

$$v_{ph} \to \sqrt{\frac{G}{\rho - (\rho_f^2/\rho')}} \tag{9.157}$$

The above behavior can be explained by the fluid rotational entrainment as observed in (9.124). At low frequency, the viscous effect dominates, and there exists full entrainment, $\boldsymbol{\Omega}_f = \boldsymbol{\Omega}_s$; hence the density within the radical in (9.156) is the total

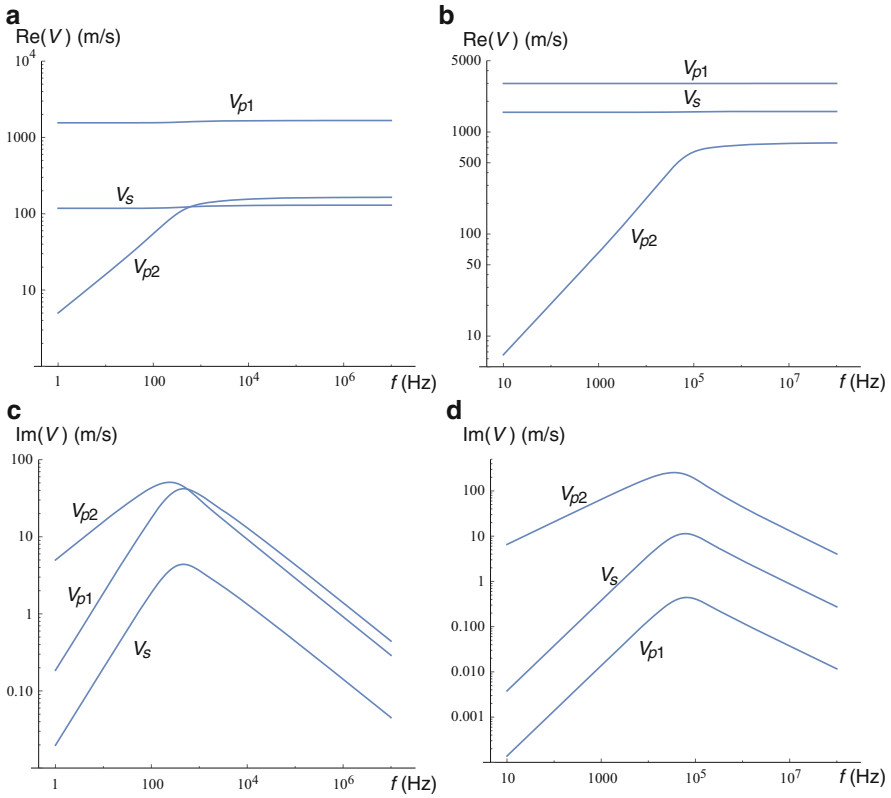


Fig. 9.4 Real and imaginary parts of velocity as functions of frequency: (**a**) Hard sediment, real part; (**b**) Berea sandstone, real part; (**c**) Hard sediment, imaginary part; and (**d**) Berea sandstone, imaginary part

density ρ. At high frequency, the viscous added mass effect ρ_a' diminishes and the added mass ρ_a dominates. There exists only partial fluid entrainment, given by the proportion $1 - [\phi\rho_f/(\phi\,\rho_f + \rho_a)]$. The smaller effective density leads to higher velocity at higher frequency.

The frequency dependent wave velocity means that the wave is dispersive. In other words, for the propagation of a chromatic wave (wave that contains many frequency components), the higher frequency components travel faster and will get ahead of the lower frequency components, so the wave is dispersed and the original wave form is distorted.

Compressional Wave Phase Velocity These phase velocities are presented in Fig. 9.5c through f in solid lines for the first and the second waves. It is clear that the first wave is the *fast wave*, and the second the *slow wave*. For asymptotic behavior, we find that as $f \to 0$,

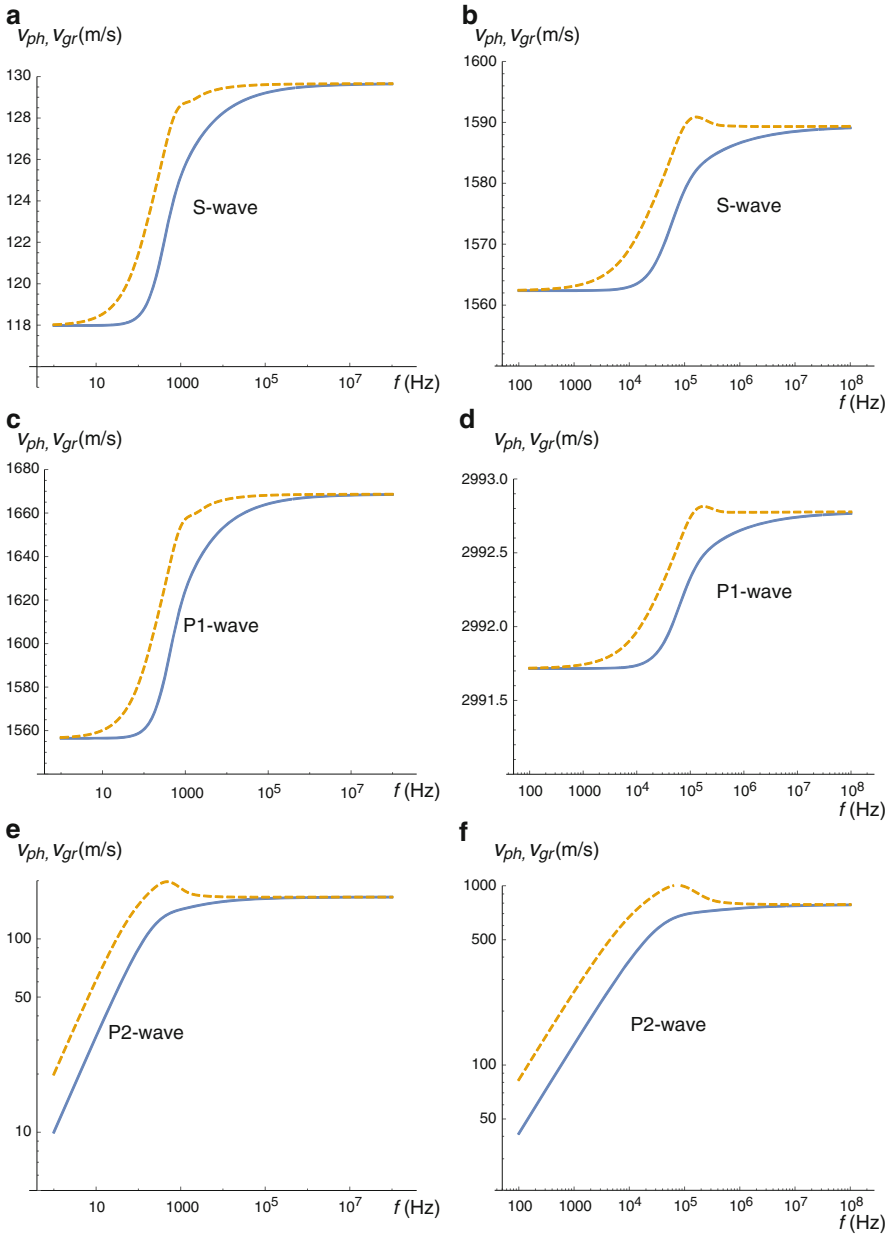


Fig. 9.5 Phase velocity (*solid line*) and group velocity (*dashed line*) as function of frequency for: shear wave (**a**) Hard sediment, (**b**) Berea sandstone; first compressional wave (**c**) Hard sediment, (**d**) Berea sandstone; and second compressional wave (**e**) Hard sediment, (**f**) Berea sandstone

$$v_{ph} \to \sqrt{\frac{E_{du}}{\rho}} \tag{9.158}$$

for the first compressional wave. That is, the first compressional wave velocity approaches that of the elastic wave velocity of the undrained material. This is the same result as obtained by Gassmann [55]. For the second wave we find, as $f \to 0$,

$$v_{ph} \approx \left[\frac{2\phi M E_d}{\rho_f E_{du}}\right]^{1/2} \left(\frac{f}{f_c}\right)^{1/2} \tag{9.159}$$

From Fig. 9.5c, d, we observe that (9.158) is a good approximation for frequency up to 100 Hz for hard sediment, and up to 10 kHz for Berea sandstone. In the other limit, $f \to \infty$, there is no simplification for the velocity expressions, except that the quantity ρ'' in (9.133) and (9.134) is replaced by ρ'.

Group Velocity *Group velocity* is the velocity at which the wave energy is transmitted. When the wave velocity is constant, group velocity is the same as the phase velocity. When the wave velocity is frequency dependent, which is the case for the poroelastodynamic waves, the energy is transmitted at a different rate as the phase velocity.

By examining the linear superposition of two progressive waves of the same amplitude, with slightly different frequency, we find that waves travel in groups, with the envelope modulating them traveling at the group velocity. Such group velocity v_{gr} can be obtained through the following formula [81]

$$v_{gr} = \frac{\partial \omega}{\partial |\ell_r|} \tag{9.160}$$

Utilizing the equation for phase velocity (9.155), we can convert the above into the following

$$\frac{1}{v_{gr}} = \frac{\partial(\omega / v_{ph})}{\partial \omega} \tag{9.161}$$

As the phase velocities for the shear and compressional waves have been defined, the above equation allows us to find the group velocities. In Fig. 9.5 we plot the group velocities in dashed lines together with the phase velocities. We observe that the group velocities are larger than the phase velocities. In the limit, such as $f \to 0$ for S and P1-waves, and $f \to \infty$ for S, P1 and P2-wave, the phase velocities approach constants, hence the group velocities becomes the same as the phase velocities.

Observation of the Second Compressional Wave A distinctive feature of the poroelastodynamics theory is the existence of a second compressional wave, also known as the slow wave. Although the slow wave has been theoretically predicted [19, 20, 54], its observation is difficult due to its highly dissipative nature, particularly in the low frequency range. The first convincing observation of the

P2-wave is attributed to Plona [100], who conducted laboratory tests in water saturated sintered glass beads in the ultrasonic range, and found two compressional waves: a fast wave with speed comparable to the compressional wave speed of the constituent glass beads, and a slow compressional wave with speed less than the speed of sound in water. These wave speeds were analyzed by Berryman [15] to confirm their consistency with Biot's theory. Further experiments were conducted in the superfluid ^{4}He with nearly zero viscosity, and demonstrated that the second wave existed as a consequence of the added mass and tortuosity effects [68, 70, 71]. Other experimental observations were conducted in air and water saturated sand [119, 120], river sediment [29], and water saturated sandstone [74].

9.5.2 Attenuation

The most commonly used measure of acoustic energy dissipation is the *attenuation coefficient* α_l, which is used to characterize the exponential decay of wave amplitude in the form of $e^{-\alpha_l x}$, where x is the propagating distance. We can express α_l as

$$\alpha_l = |\ell_i| = 2\pi f \frac{v_i}{|v|^2} \tag{9.162}$$

The above shows that α_l has the dimension of $[\mathrm{L}^{-1}]$.

Another often used measure is the *attenuation coefficient per wave cycle*, which is expressed as the *inverse of quality factor* Q^{-1} as

$$Q^{-1} = 2\left|\frac{\ell_i}{\ell_r}\right| = 2\left|\frac{v_i}{v_r}\right| \tag{9.163}$$

where Q^{-1} is dimensionless. We can also see that these two attenuation coefficients are related to each other through the relation

$$Q^{-1} = \frac{\alpha_l\, v_{ph}}{\pi f} \tag{9.164}$$

In Fig. 9.6 we plot the inverse of quality factor as a function of frequency for hard sediment and Berea sandstone, with the same data set presented in the preceding section. We observe that for the shear and first compressional wave, attenuation goes to zero for both $f \to 0$ and $f \to \infty$. In between, there exists a frequency where the attenuation is maximum. As pointed out by Biot [19, 20], that frequency is near the characteristic frequency f_c (see Table 9.1 for f_c values). We also notice that shear wave attenuates faster than the first compressional wave. For the second compressional wave, even larger attenuation is observed. Particularly, at low frequency, $f \ll f_c$, the inverse of quality factor $Q^{-1} \approx 2$, which means that the wave is dampened within one wave cycle; hence the wave behaves like a diffusion wave, and does not propagate.

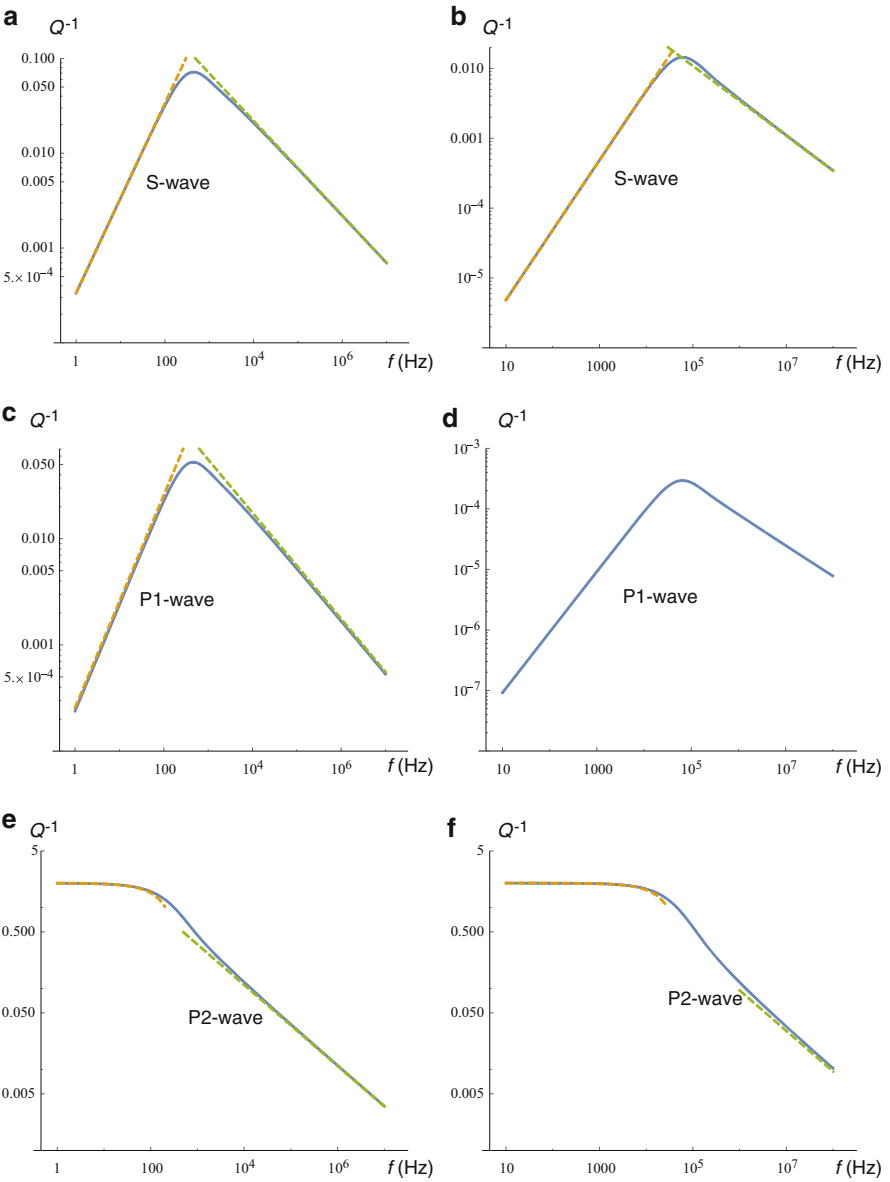


Fig. 9.6 Inverse quality factor (Q^{-1}) versus frequency for: shear wave (**a**) Hard sediment, (**b**) Berea sandstone; first compressional wave (**c**) Hard sediment, (**d**) Berea sandstone; and second compressional wave (**e**) Hard sediment, (**f**) Berea sandstone. *Dashed lines* are based on asymptotic formulae

Shear Wave For the attenuation of shear waves, the asymptotic formulae of (9.126) have been provided by Biot [20]. At low frequencies, it is given as

$$Q_s^{-1} \approx \frac{\phi \rho_f}{\rho} \frac{f}{f_c} \tag{9.165}$$

and at high frequencies,

$$Q_s^{-1} \approx \frac{\sqrt{2}\,\delta_p}{8} \frac{\rho_f^3}{\phi \rho' (\rho \rho' - \rho_f^2)} \left(\frac{f_c}{f}\right)^{1/2} \tag{9.166}$$

The above asymptotic equations are plotted as the dashed lines in Figs. 9.6a, b.

According to Biot [19, 20], the maximum attenuation takes place roughly around the characteristic frequency, which are evaluated to give $f_c = 7.5 \times 10^2$ Hz for hard sediment, and 1.6×10^5 Hz for Berea sandstone. Substituting the above into the shear velocity equation (9.126), we obtain $Q^{-1} = 6.5 \times 10^{-2}$ and 5.4×10^{-3}, respectively. The actual maximum attenuation occurs at $f = 4.5 \times 10^2$ Hz for hard sediment, and 6.0×10^4 Hz for Berea sandstone, with $Q^{-1} = 7.2 \times 10^{-2}$ and 8.8×10^{-3}, respectively. So although not highly accurate, the Biot criterion can be used to locate the maximum attenuation frequency and to evaluate the attenuation.

Compressional Waves Biot [20] also investigated the asymptotic behavior of the compressional waves, and concluded that for the first wave

$$Q_{p1}^{-1} \sim \frac{f}{f_c} \qquad \text{as} \quad f \to 0 \tag{9.167}$$

$$Q_{p1}^{-1} \sim \left(\frac{f_c}{f}\right)^{1/2} \qquad \text{as} \quad f \to \infty \tag{9.168}$$

and for the second wave

$$Q_{p1}^{-1} \approx 2 \qquad \text{as} \quad f \to 0 \tag{9.169}$$

$$Q_{p1}^{-1} \sim \left(\frac{f_c}{f}\right)^{1/2} \qquad \text{as} \quad f \to \infty \tag{9.170}$$

However, the above formulae give only the general trend with the frequency. In terms of magnitude, they can be off by several orders of magnitude.

Using the assumption $K_s \gg K_f \gg K$, which is a good assumption for granular materials, such as soil and sand, better estimates have been obtained [10, 118, 126]. For the first compressional wave at low frequencies, a better asymptotic formula is

$$Q_{p1}^{-1} \approx \frac{\phi (\rho - \rho_f)^2}{\rho \rho_f} \frac{f}{f_c} \tag{9.171}$$

and at high frequencies, the estimate is modified to

$$Q_{p1}^{-1} \approx \frac{\sqrt{2}\,\delta_p}{8} \frac{\rho_f(\rho-\rho_f)^2}{\phi(\rho\rho'-\rho_f^2)(\rho+\rho'-2\rho_f)} \left(\frac{f_c}{f}\right)^{1/2} \tag{9.172}$$

These asymptotic formulae are plotted as dashed lines in Fig. 9.6c for the hard sediment case, and we observe excellent agreement. For the Berea sandstone case, for which these formulae are not intended, the estimates are still off; hence they are not plotted.

Similarly, we obtained for the second compressional wave these better estimates: at low frequencies,

$$Q_{p2}^{-1} \approx 2 - \left[\frac{2\phi(\rho+\rho'-2\rho_f)}{\rho_f} + \frac{\delta_p^2}{12}\right] \frac{f}{f_c} \tag{9.173}$$

and at high frequencies,

$$Q_{p2}^{-1} \approx \frac{\sqrt{2}\,\delta_p}{8} \frac{\rho_f}{\phi(\rho+\rho'-2\rho_f)} \left(\frac{f_c}{f}\right)^{1/2} \tag{9.174}$$

These results are plotted in dashed line in Fig. 9.6e, f. Good agreement is observed for both the hard sediment and the Berea sandstone case.

9.5.3 Extended Biot Models

Biot's theory can provide accurate prediction of wave speed and attenuation for synthetic porous media such as packed and sintered glass beads, and natural granular materials such as sand. However, in a number of applications, such as seismic wave propagation in partially saturated sedimentary rocks [56, 57, 124], acoustic wave propagation in ocean sediment [112, 114], and acoustics in sound absorbing materials [5, 6, 31], discrepancies in the predicted frequency dispersion relations were found. Biot theory has been modified to suit these and other applications.

Partially Saturated Sedimentary Rocks and Squirt Flow Partial saturation of porous rock by two or more different fluids can occur in a multitude of geological settings. For instance, gas, oil and brine commonly share the available pore space in the upper part of gas capped reservoirs. Extending the Biot theory to model wave propagation in porous media saturated by two or more immiscible fluids is not a trivial matter [117]. However, when two immiscible fluids are distributed on a relatively fine scale, they can be regarded as a single composite fluid whose compressibility is given by the average of its constituent compressibilities, based on Wood's equation [125]

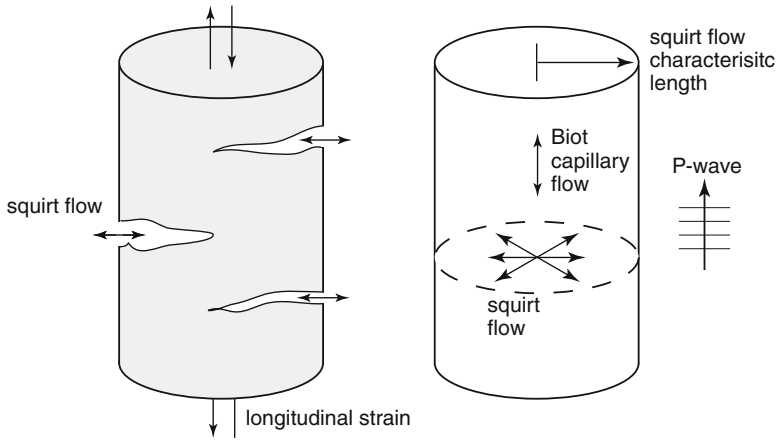


Fig. 9.7 Squirt flow phenomenon: longitudinal strain can squeeze fluid in and out of cracks oriented perpendicular to the wave propagation direction, causing flow in the transverse directions

$$C_f = S_1 C_{f1} + S_2 C_{f2} \tag{9.175}$$

where C_{f1} and C_{f2} are respectively the compressibility of fluid 1 and 2, C_f is the apparent compressibility of the composite fluid, and S_1 and S_2 are the respective degree of saturation of the two fluids. With this simplification, the Biot model of a single fluid phase can apply.

White [122] was the first to show theoretically that partial fluid saturation can cause significant attenuation and phase velocity dispersion. Experimental studies [45, 58] also indicated that partially saturated porous rocks exhibited different phase velocity behavior from fully saturated ones. A number of mechanisms have been suggested to account for the discrepancy. Largely the *grain-scale local flow* effects [88] are considered to be the most important.

For a unidirectional wave, the Biot REV averaged continuum theory allows the macroscopic fluid movement only in the direction of the wave propagation. Mavko and Nur [90] suggested that with the passage of a longitudinal wave, fluid can be squeezed in and out of microcracks, or between the imperfect contact of grains. This so-called *squirt flow* [89] can occur in all directions, and particularly perpendicular to the wave propagating direction (see Fig. 9.7), giving rise to a viscoelastic like phenomenon [92, 97]. This additional dissipation mechanism is not accounted for in the Biot capillary tube model discussed in Sect. 9.2.

Mavko and Nur [90] have shown that the squirt flow mechanism results in much higher attenuation values than those predicted by Biot's theory. This mechanism has been considered to be responsible for the measured seismic energy losses and velocity dispersion of both the P- and S-waves [92, 121]. The squirt flow mechanism has been traditionally investigated by examining fluid flow in an individual crack [90, 91] or at a grain contact point [98]. Attenuation and velocities are calculated by considering viscous energy losses through a complex modulus [92]. By so doing,

the squirt flow mechanism has been treated independent of Biot's theory. Yet, it is clear that these two modes of solid-fluid interaction should be intimately connected [49–51, 102]. A model that combines the squirt flow mechanism with the Biot model is the BISQ (Biot-squirt) model [50, 51]. In the following, we shall give a brief presentation.

The BISQ model is built on the Biot model. The compressional wave velocities are defined in (9.137) and (9.138), with coefficients A, B and C given by (9.132), (9.133), and (9.134). The only difference is that the Biot modulus M contained in (9.132) and (9.133) is replaced by a complex modulus M_{sq}, in order to accommodate the apparent viscoelastic behavior introduced by the squirt flow. In the above, M_{sq} is defined as

$$M_{sq} = M\left[1 - \frac{2\,\mathrm{J}_1(\lambda_{sq}R_{sq})}{\lambda_{sq}R_{sq}\,\mathrm{J}_0(\lambda_{sq}R_{sq})}\right] \tag{9.176}$$

where

$$\lambda_{sq} = \sqrt{\frac{i\mu_f\,\omega}{kM}} \tag{9.177}$$

and R_{sq} is a squirt flow characteristic length (see Fig. 9.7). The BISQ model hence contains an extra material parameter R_{sq}, which introduces an extra frequency dependency as compared to the Biot model. This parameter is normally obtained through data fitting of the observed dispersion relation.

It is of interest to observe that in the low frequency limit, $\omega \to 0$, the second term in the square brackets of (9.176) approaches 1, and $M_{sq} \to 0$. By observing the low frequency limit of phase velocity given by (9.158), and the modified definition of (9.135)

$$E_{du} = E_d + \alpha^2 M_{sq} \tag{9.178}$$

in the BISQ case, we find

$$v_{ph} \to \sqrt{\frac{E_d}{\rho}} \tag{9.179}$$

Hence, as $\omega \to 0$, the BISQ model predicts a phase velocity approaching that of the drained porous material, in contrast to that of the undrained material, as predicted by the Biot [19, 20] and the Gassmann [55] model, shown as (9.158). This phenomenon can be interpreted as follows: at low frequencies, the fluid is slowly and easily squeezed perpendicular to the direction of wave propagation and offers no strength against the seismic excitation. At high frequency, $\omega \to \infty$, it is easily shown that in (9.176) $M_{sq} \to M$, and the BISQ model approaches the Biot model.

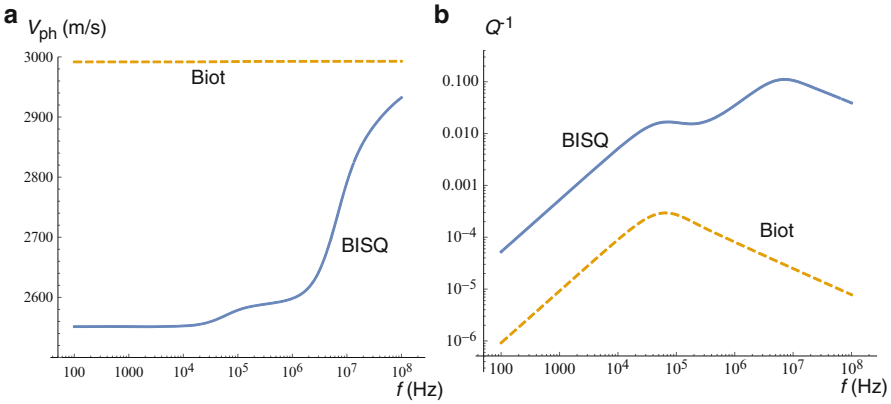


Fig. 9.8 Comparison of first compressional wave based on BISQ and Biot model for Berea sandstone: (**a**) Phase velocity versus frequency; and (**b**) Inverse quality factor versus frequency

Using Berea sandstone saturated with water as an example, and selecting a squirt flow characteristic length of $R_{sq} = 0.5$ mm, we plot in Fig. 9.8 phase velocity and inverse quality factor versus frequency for the first compressional wave. We observe that for the BISQ model, the velocity dispersion covers a fairly wide range, from the drained to the undrained elastic velocity, for about 440 m/s; while for the Biot model, the velocity dispersion is less than 1 m/s from low to high frequency (see Fig. 9.5d). From Fig. 9.8b we also observe that the BISQ model predicts a much larger wave attenuation, which can be several orders of magnitude larger than the Biot case.

Ocean Sediment and Coulomb Friction For wave propagating in porous medium, whether it is dry, saturated, or partially saturated with fluid, there exist a number of energy dissipation mechanisms. For granular materials, such as sand, loosely consolidated soil, and ocean sediment, a significant energy loss can be caused by the *Coulomb friction* resulting from the intergranular solid to solid contact or the anelasticity of cementing material between grains [112, 113]. Other energy dissipation mechanisms can include the solid viscoelastic and thermoelastic dissipation [21, 22]. In this part, we shall consider only a simple model of intergranular friction that can be easily incorporated into the Biot model.

Based on the ample amount of field and laboratory data on acoustic waves in ocean sediments in the 1 Hz to 100 kHz range, Stoll [112, 113, 115] concluded that energy loss is due to two main mechanisms: one due to the viscous losses of water as it moves through the skeletal frame, which is frequency dependent and can be accurately predicted by the Biot theory, and the other to the intergranular friction that is nearly independent of frequency. This Coulomb friction is considered to be the dominant damping mechanism for granular sediments such as sand and silt.

A simple way to model this effect is to replace the real-valued elastic moduli by complex quantities, based on the *Kryloff-Bogoliuboff equivalent linearization* [75]

theory. Here we shall assume that the friction is caused by two independent modes of frame deformation, a volumetric and a shear deformation. Hence we introduce two complex moduli, G and K, such that

$$G = G_o(1 - i\delta') \tag{9.180}$$

$$K = K_o(1 - i\delta'') \tag{9.181}$$

where G_o and K_o are respectively the shear modulus and drained bulk modulus of the frame, and δ' and δ'' are respectively specific losses associated with shear and volumetric deformation of the frame.

Values of the shear damping coefficient δ' have been reported to range from 0.003 to 0.015 for sand, 0.02 to 0.03 for silt, and 0.003 to 0.006 for sedimentary rocks [114]. It is suggested that δ' should be dependent on porosity, effective overburden stress, and frequency. For the volumetric damping coefficient δ'', such empirical relation

$$\delta'' = 0.75\delta' \tag{9.182}$$

is suggested [61]. By using the hard sediment as an example, in which we assume the Coulomb damping coefficients as $\delta' = 0.02$ and $\delta'' = 0.015$, we can evaluate the wave attenuation expressed in terms of the inverse quality factor. In Fig. 9.9 we present the inverse quality factor versus frequency for the S-wave as the solid line. For comparison, we also present the case without such damping in dashed line. We observe the extra damping from the limits

$$Q_s^{-1} \to \delta', \quad \text{as} \quad f \to 0, \infty \tag{9.183}$$

For the P-waves, we present in Fig. 9.10 the attenuation as function of frequency. Based on the asymptotic analysis [10], we obtain the following limiting values

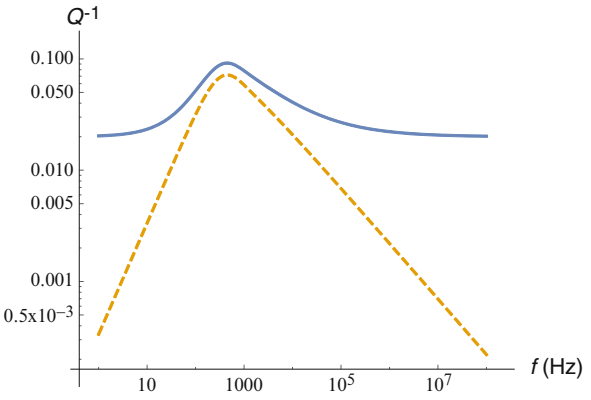


Fig. 9.9 Inverse quality factor versus frequency for S-wave for the hard sediment case. *Solid line*: with Coulomb damping; *Dashed line*: without Coulomb damping

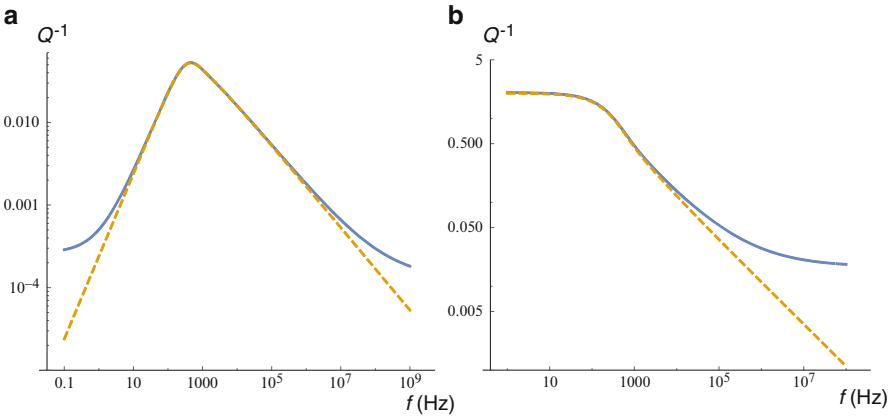


Fig. 9.10 Inverse quality factor versus frequency for (**a**) P1-wave, and (**b**) P2-wave, for the hard sediment case. *Solid line*: with Coulomb damping; *Dashed line*: without Coulomb damping

$$Q_{p1}^{-1} \approx \frac{\phi E_{do}}{K_f}\delta_E, \quad \text{as} \quad f \to 0 \tag{9.184}$$

$$Q_{p1}^{-1} \approx \left[\frac{(1-\phi)\rho_f + \rho_a}{\phi\rho + (1-2\phi)\rho_f + \rho_a}\right]^2 \frac{\phi E_{do}}{K_f}\delta_E, \quad \text{as} \quad f \to \infty \tag{9.185}$$

$$Q_{p2}^{-1} \approx \delta_E, \quad \text{as} \quad f \to \infty \tag{9.186}$$

where for convenience we have extended the definition of the drained dynamic modulus to its complex form

$$E_d = \frac{4}{3}G + K = E_{do}(1 - i\delta_E) \tag{9.187}$$

9.6 One-Dimensional Wave Problem

To examine analytical solutions, we start with the simplest case: a one-dimensional plane wave in a homogeneous medium. This leads to the condition that only one displacement component for the solid and the fluid, exist. Choosing y as the direction, these two displacements are expressed as $u_y = u_y(y,t)$ and $w_y = w_y(y,t)$. Or, in the $(\boldsymbol{u}, p)$ formulation, the second condition is replaced by $p = p(y,t)$.

The governing equations are simplified from (9.102) and (9.103) to the following set of coupled second order ordinary differential equations

$$E_d\frac{\partial^2\tilde{u}_y}{\partial y^2} + \omega^2(\rho - \alpha_\rho\rho_f)\tilde{u}_y - (\alpha - \alpha_\rho)\frac{\partial\tilde{p}}{\partial y} = 0 \tag{9.188}$$

$$M\frac{\partial^2\tilde{p}}{\partial y^2}+\frac{\omega^2\rho_f}{\alpha_\rho}\tilde{p}+\frac{\omega^2\rho_f M}{\alpha_\rho}(\alpha-\alpha_\rho)\frac{\partial\tilde{u}_y}{\partial y}=0 \tag{9.189}$$

where we have used the shorthand notation of

$$\alpha_\rho=\frac{\rho_f}{\rho''} \tag{9.190}$$

and E_d is the drained dynamic modulus as defined in (9.114).

The system (9.188) and (9.189) can be solved by the following exponential ansatz

$$\tilde{u}_y(y)=Ue^{i\ell y} \tag{9.191}$$

$$\tilde{p}(y)=Pe^{i\ell y} \tag{9.192}$$

where U and P are constants to be determined by boundary conditions of a specific problem, and

$$\ell=\frac{\omega}{v} \tag{9.193}$$

is the complex wave number, with V the complex wave velocity. Substituting (9.191) and (9.192) into (9.188) and (9.189), and divide the equations through by $e^{i\ell y}$, we obtain the following matrix system

$$\begin{bmatrix} -E_d\,\ell^2+\omega^2(\rho-\alpha_\rho\rho_f) & -i(\alpha-\alpha_\rho)\ell \\ i\omega^2\rho_f M(\alpha-\alpha_\rho)\ell/\alpha_\rho & -M\ell^2+(\omega^2\rho_f/\alpha_\rho) \end{bmatrix}\begin{Bmatrix} U \\ P \end{Bmatrix}=0 \tag{9.194}$$

As the right hand side of (9.194) is zero, for the above system to have a solution, the determinant of the matrix must also be zero, leading to the following characteristic equation,

$$A\left(\frac{\ell}{\omega}\right)^4-B\left(\frac{\ell}{\omega}\right)^2+C=0 \tag{9.195}$$

where the coefficients A, B and C are the same as those given in (9.132), (9.133), and (9.134). The above equation can be solved for the eigenvalues as the four complex roots

$$\ell_1=-\ell_3=\frac{\omega}{v_{p1}} \tag{9.196}$$

$$\ell_2=-\ell_4=\frac{\omega}{v_{p2}} \tag{9.197}$$

where v_{p1} and v_{p2} are the two compressional wave velocities defined in (9.137) and (9.138). It is clear from the above that due to the one-dimensional nature of the

problem, the shear wave does not exist, and we find only two compressional waves. We also note that ℓ_1 and ℓ_2 have positive real part, and are associated with waves travelling in the positive y-direction, while ℓ_3 and ℓ_4 represent waves in the negative y-direction.

The full solution of (9.188) and (9.189) can then be expressed as

$$\tilde{u}_y(y) = \sum_{i=1}^{4} U_i \, e^{i\ell_i y} \tag{9.198}$$

$$\tilde{p}(y) = \sum_{i=1}^{4} P_i \, e^{i\ell_i y} \tag{9.199}$$

These eight unknown constants U_i and P_i, $i = 1, \ldots, 4$, are not all independent. Solving for the eigenvectors of the system, we find

$$P_i = d_i \, U_i \tag{9.200}$$

where

$$d_i = \frac{-E_d \, \ell_i^2 + \omega^2(\rho - \alpha_\rho \rho_f)}{i(\alpha - \alpha_\rho)\ell_i} \tag{9.201}$$

with $d_1 = -d_3$ and $d_2 = -d_4$. The remaining 4 constants U_i then need four boundary conditions for their determination.

Once these constants are found, the stress can be determined from the constitutive equation as

$$\begin{aligned} \tilde{\sigma}_{yy} &= E_d \frac{\partial \tilde{u}_y}{\partial y} - \alpha \tilde{p} \\ &= \sum_{i=1}^{4} (iE_d \, \ell_i - \alpha d_i) \, U_i \, e^{i\ell_i y} \end{aligned} \tag{9.202}$$

and specific flux from the dynamic Darcy's law

$$\begin{aligned} \tilde{q}_y &= -\frac{i\alpha_\rho}{\omega \rho_f} \left(\frac{\partial \tilde{p}}{\partial y} - \omega^2 \rho_f \tilde{u}_y \right) \\ &= \frac{\alpha_\rho}{\omega \rho_f} \sum_{i=1}^{4} \left(d_i \ell_i + i\omega^2 \rho_f \right) U_i \, e^{i\ell_i y} \end{aligned} \tag{9.203}$$

In the following, we shall examine a few problems with different boundary conditions.

9.6.1 Half Space

Consider a half space uniformly loaded on top by a sinusoidal forcing of either a stress or a pressure. As before, we shall solve it as a mode 1 and a mode 2 problem, which can be superimposed as appropriate.

Mode 1: Stress Loading Given a half space occupying $y \geq 0$, the surface is excited by a vertical stress $\sigma_{yy} = -P_o \cos \omega t$, with pressure maintained at zero. This lead to the following boundary conditions: at $y = 0$,

$$\tilde{\sigma}_{yy} = -P_o, \qquad \tilde{p} = 0 \tag{9.204}$$

Based on the solution (9.198), the half space geometry dictates that only the outgoing waves, that is, waves travelling in the positive y-direction, can exist. This condition requires that the two coefficients U_3 and U_4 vanish. By utilizing the two boundary conditions in (9.204), we can then solve for U_1 and U_2, and obtain the following solution

$$\tilde{u}_y = \frac{-iP_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left[d_2\, e^{i\ell_1 y} - d_1\, e^{i\ell_2 y}\right] \tag{9.205}$$

$$\tilde{p} = \frac{-iP_o d_1 d_2}{E_d(d_1\ell_2 - d_2\ell_1)}\left[e^{i\ell_1 y} - e^{i\ell_2 y}\right] \tag{9.206}$$

$$\tilde{\sigma}_{yy} = \frac{-iP_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left[(iE_d\ell_1 - \alpha d_1)\, d_2\, e^{i\ell_1 y} - (iE_d\ell_2 - \alpha d_2)\, d_1 e^{i\ell_2 y}\right] \tag{9.207}$$

Given an excitation magnitude of $P_o = 100$ kPa, we plot in Fig. 9.11 the amplitude of displacement at the top of the semi-infinite layer versus frequency. We observe that as $f \to 0$, the displacement $|\tilde{u}_y(0)|$ approaches infinity. This is anticipated, as the displacement is unbounded for the static case, due to the infinite layer thickness. Under the dynamic condition, however, the displacement is bounded, and is roughly proportional to the wave length, or inversely proportional to the wave frequency.

In the same figure, we also plot the elastic solution based on the undrained elastic modulus

$$\tilde{u}_y = \frac{iP_o}{\sqrt{E_{du}\rho}\,\omega}\, e^{i\sqrt{E_{du}/\rho}\, y} \tag{9.208}$$

For the hard sediment case, we observe that the undrained elastic assumption much underpredicts the deformation. For the Berea sandstone, the two solutions are nearly indistinguishable.

Mode 2: Pressure Loading For a pressure loading $p = p_o \cos \omega t$ at $y = 0$, we find the following boundary conditions:

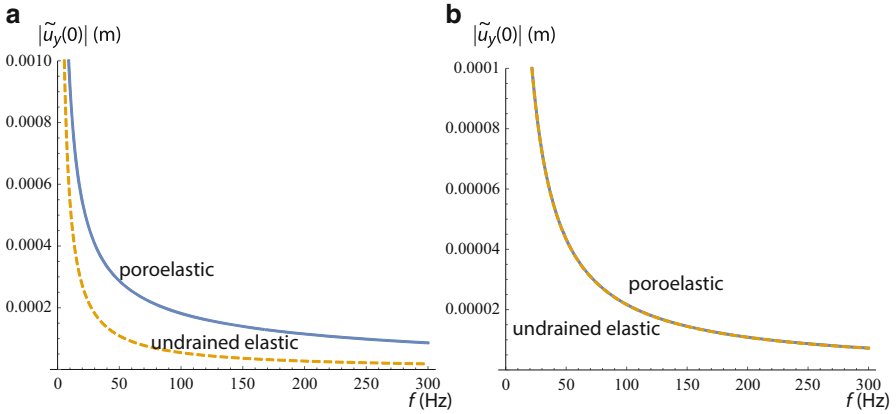


Fig. 9.11 Displacement amplitude at the top of half space ($y = 0$) for mode 1 problem: (**a**) hard sediment, and (**b**) Berea sandstone

$$\tilde{\sigma}_{yy} = 0, \qquad \tilde{p} = p_o \tag{9.209}$$

The solution is obtained as

$$\tilde{u}_y = \frac{-ip_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left[(iE_d\ell_2 - \alpha d_2)\, e^{i\ell_1 y} - (iE_d\ell_1 - \alpha d_1)\, e^{i\ell_2 y}\right] \tag{9.210}$$

$$\tilde{p} = \frac{-ip_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left[d_1(iE_d\ell_2 - \alpha d_2)\, e^{i\ell_1 y} - d_2(iE_d\ell_1 - \alpha d_1)\, e^{i\ell_2 y}\right] \tag{9.211}$$

$$\tilde{\sigma}_{yy} = \frac{-ip_o(iE_d\ell_1 - \alpha d_1)(iE_d\ell_2 - \alpha d_2)}{E_d(d_1\ell_2 - d_2\ell_1)}\left(e^{i\ell_1 y} - e^{i\ell_2 y}\right) \tag{9.212}$$

9.6.2 Finite Thickness Layer

Consider a horizontal layer of constant thickness L, with its bottom supported by a rigid and impermeable foundation, and top a permeable surface; see Fig. 9.12.

Mode 1: Stress Excitation at the Top For the first problem, we excite the top of the layer with a vertical stress $\sigma_{yy} = -P_o \cos \omega t$, and keep the pressure zero. The boundary conditions are

$$\begin{aligned} &\tilde{u}_y = 0; \qquad \tilde{q}_y = 0; \qquad \text{at} \quad y = 0 \\ &\tilde{\sigma}_{yy} = -P_o; \quad \tilde{p} = 0; \qquad \text{at} \quad y = L \end{aligned} \tag{9.213}$$

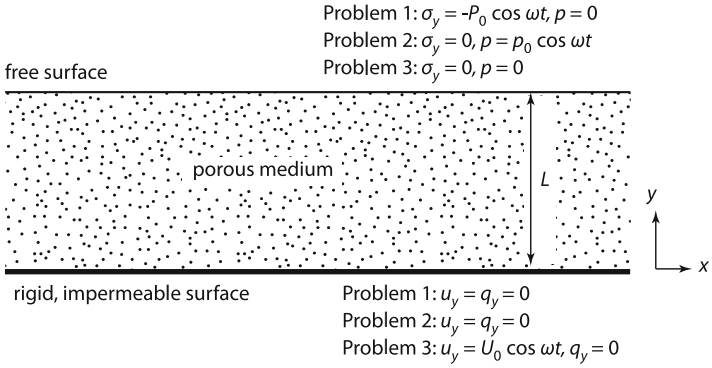


Fig. 9.12 Boundary conditions for one-dimensional poroelastodynamic wave propagation problems

Using the first two boundary conditions, it is easy to show that $U_1 = -U_3$ and $U_2 = -U_4$. Utilizing the remainder of the boundary conditions, we obtain the following solutions [33, 107]

$$\tilde{u}_y = \frac{P_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left(d_2\frac{\sin \ell_1 y}{\cos \ell_1 L} - d_1\frac{\sin \ell_2 y}{\cos \ell_2 L}\right) \tag{9.214}$$

$$\tilde{p} = \frac{i\, d_1 d_2 P_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left(\frac{\cos \ell_1 y}{\cos \ell_1 L} - \frac{\cos \ell_2 y}{\cos \ell_2 L}\right) \tag{9.215}$$

$$\tilde{\sigma}_{yy} = \frac{iP_o}{E_d(d_1\ell_2 - d_2\ell_1)}\left[(iE_d\ell_1 - \alpha d_1)d_2\frac{\cos \ell_1 y}{\cos \ell_1 L} - (iE_d\ell_2 - \alpha d_2)\, d_1\frac{\cos \ell_2 y}{\cos \ell_2 L}\right] \tag{9.216}$$

Given a excitation magnitude of $P_o = 100$ kPa and a layer thickness of $L = 10$ m, we can evaluate the above solutions. In Figs. 9.13a, b we plot the displacement amplitude at the top of the layer, $|\tilde{u}_y(L)|$, versus frequency, in solid lines, for hard sediment and Berea sandstone, respectively. For comparison, we also plot the elastic solution in dashed line. The elastic solution is based on the undrained poroelastic properties, given as

$$u_y = -\frac{P_o \sin \ell_u y}{E_{du}\, \ell_u \cos \ell_u L} \tag{9.217}$$

where E_{du} is the undrained dynamic modulus as defined in (9.135), and

$$\ell_u = \frac{\omega}{v_{pu}} \tag{9.218}$$

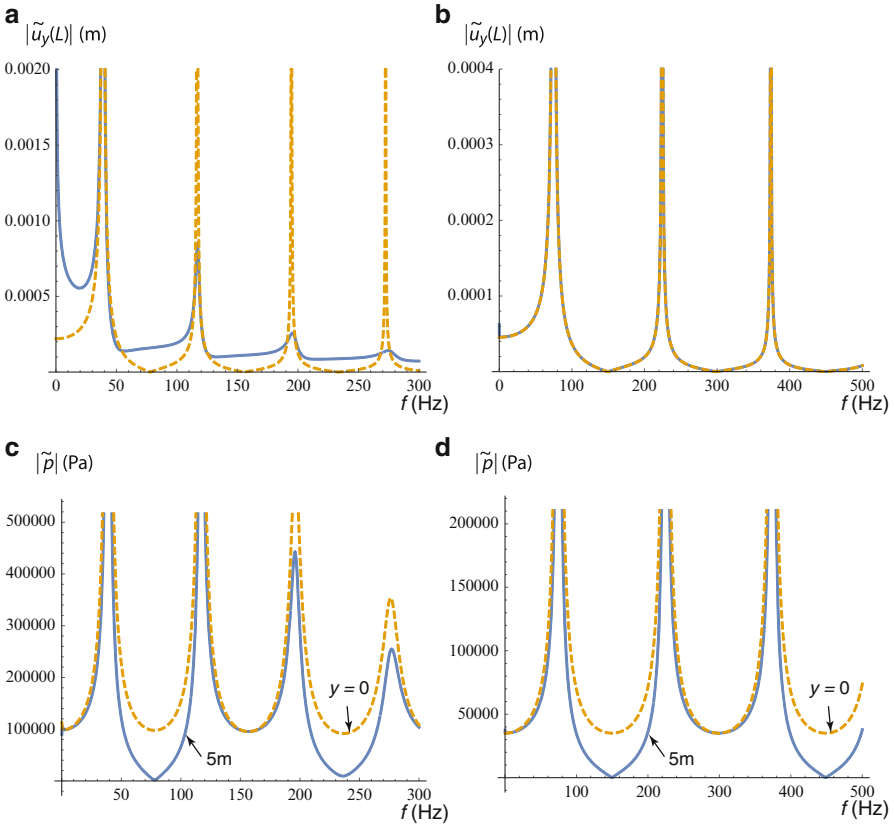


Fig. 9.13 One-dimensional wave propagation problem with force excitation at the top of finite thickness layer. (**a**) and (**b**): Displacement amplitude at the top $|\tilde{u}_y(L)|$, with (**a**) hard sediment, (**b**) Berea sandstone, *solid line* the poroelastic solution, and *dashed line* the elastic solution. (**c**) and (**d**): Pressure amplitude $|\tilde{p}(y)|$ at the bottom ($y = 0$) and mid-depth ($y = 5\,\text{m}$) for (**c**) hard sediment and (**d**) Berea sandstone

is the wave number, with

$$v_{pu} = \sqrt{\frac{E_{du}}{\rho}} \tag{9.219}$$

the elastic compressional wave velocity of the undrained material.

It is of interest to observe that although the poroelastic solution approaches the drained elastic solution as $\omega \to 0$, this behavior exists only at very low frequency, say $f \ll 1\,\text{Hz}$. At other frequencies, the overall behavior is closer to the undrained, rather than drained, elastic solution. This is particularly evident in terms of the resonance frequency. For the undrained elastic solution, the first cutoff frequency takes place at

$$\omega_1 = \frac{\pi v_{pu}}{2L} \tag{9.220}$$

and the subsequent ones at $3\,\omega_1,\ 5\,\omega_1, \ldots$ Figure 9.13a, b show that these resonance peaks are closely matched by the poroelastic ones. Figure 9.13b further shows that for Berea sandstone, the entire poroelastic solution is closely matched by the undrained elastic solution, due to its low permeability, thus justifying the approximate modeling of wave propagation in saturated rocks as elastic waves, if the undrained poroelastic modulus is used. Figure 9.13a, on the other hand, shows significant damping around resonance peaks for the case of hard sediment, due to the larger permeability allowing the relative movement of viscous fluid. It also shows that away from the resonance frequencies, the layer top displacement can be in fact much larger than the elastic response.

In Figs. 9.13c, d we plot the pressure response as function of frequency at two different depths, $y = 0$ and $0.5L$, for the hard sediment and Berea sandstone cases. We notice that the pressure behavior at the different depths for these two materials are fairly similar, except that there is larger damping around the peaks for the hard sediment.

Mode 2: Pressure Excitation at the Top When the top of the layer is in contact with a fluid, and the excitation is caused by the pressure fluctuation $p = p_o \cos \omega t$, then in addition to a total stress loading $\tilde{\sigma}_{yy} = -p_o$, there is a pressure loading $\tilde{p} = p_o$. These effects can be separately investigated and superposed as a mode 1 and a mode 2 solution, similar to the quasi-static case in Sect. 7.3.2. The mode 1 solution has been presented in the preceding section. Solution to the mode 2 problem is sought here.

The boundary conditions for the mode 2 problem is as follows:

$$\begin{aligned} &\tilde{u}_y = 0; \qquad \tilde{q}_y = 0; \qquad \text{at} \quad y = 0 \\ &\tilde{\sigma}_{yy} = 0; \qquad \tilde{p} = p_o; \qquad \text{at} \quad y = L \end{aligned} \tag{9.221}$$

Using the same methodology as the mode 1 problem, we obtain the following solutions

$$\tilde{u}_y = \frac{p_o}{E_d(d_1\ell_2 - d_2\ell_1)} \left[(iE_d\ell_2 - \alpha d_2) \frac{\sin \ell_1 y}{\cos \ell_1 L} + (iE_d\ell_1 - \alpha d_1) \frac{\sin \ell_2 y}{\cos \ell_2 L} \right] \tag{9.222}$$

$$\tilde{p} = \frac{-ip_o}{E_d(d_1\ell_2 - d_2\ell_1)} \left[d_1(iE_d\ell_2 - \alpha d_2) \frac{\cos \ell_1 y}{\cos \ell_1 L} - d_2(iE_d\ell_1 - \alpha d_1) \frac{\cos \ell_2 y}{\cos \ell_2 L} \right] \tag{9.223}$$

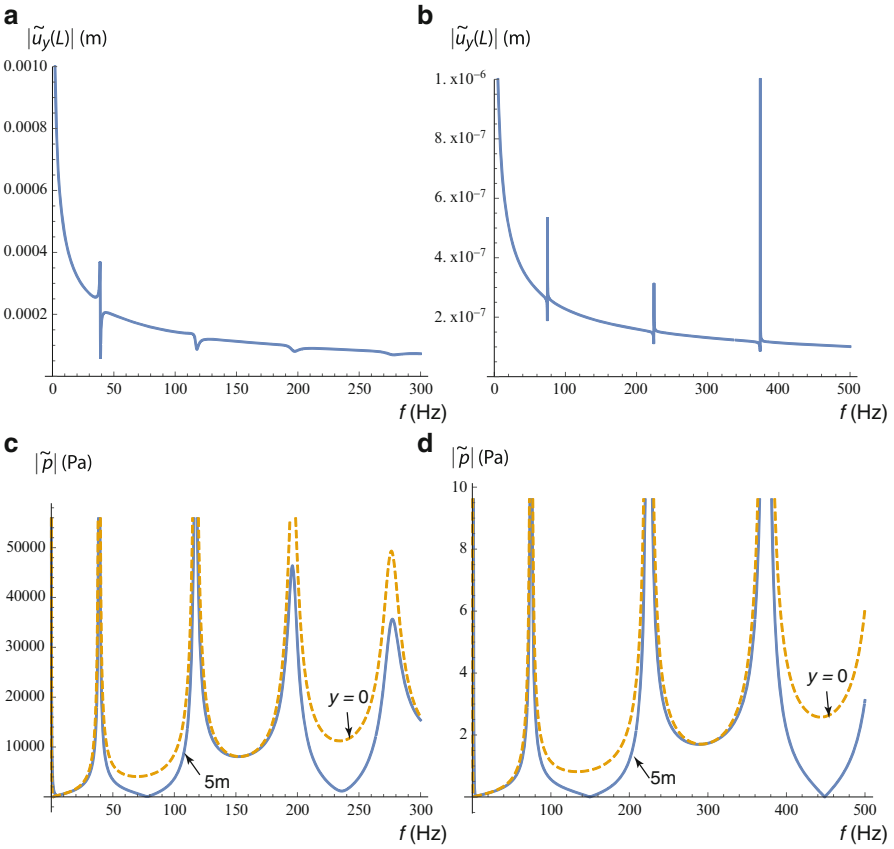


Fig. 9.14 One-dimensional wave propagation problem with pressure excitation at the top of finite thickness layer. (**a**) and (**b**): Displacement amplitude at the *top* $|\tilde{u}_y(L)|$, with (**a**) hard sediment, (**b**) Berea sandstone. (**c**) and (**d**): Pressure amplitude $|\tilde{p}(y)|$ at the *bottom* ($y = 0$) and mid-depth ($y = 5$ m) for (**c**) hard sediment and (**d**) Berea sandstone

$$\tilde{\sigma}_y = \frac{-ip_o}{E_d(d_1\ell_2 - d_2\ell_1)}(iE_d\ell_1 - \alpha d_1)(iE_d\ell_2 - \alpha d_2) \times \left(\frac{\cos \ell_1 y}{\cos \ell_1 L} - \frac{\cos \ell_2 y}{\cos \ell_2 L}\right) \tag{9.224}$$

Given a loading of $p_o = 100$ kPa, we plot in Fig. 9.14 the displacement amplitude at the top of the layer, and the pore pressure response at two different depths, as functions of excitation frequency for the two materials, hard sediment and Berea sandstone. We observe in Fig. 9.14a, b a much smaller displacement response, particularly for the Berea sandstone case. The response peak for the hard sediment case is much dampened. As frequency increases, the displacement decreases. In Fig. 9.14c, d we observe similar pressure response pattern as in Fig. 9.13a, b.

Their magnitudes, however, are much smaller. The reason is that the pore pressure response in the mode 1 problem is generated by the stress and the Skempton effect; while in the mode 2 problem, it relies on pore pressure diffusion, and is strongly dependent on permeability.

Mode 3: Displacement Excitation at the Bottom To simulate an earthquake phenomenon, we examine a problem that the bottom of the layer is excited by a sinusoidal displacement of $u_y(0) = U_o \cos \omega t$. The boundary conditions are

$$\begin{aligned} &\tilde{u}_y = U_o; \qquad \tilde{q}_y = 0; \qquad \text{at} \quad y = 0 \\ &\tilde{\sigma}_{yy} = 0; \qquad \tilde{p} = 0; \qquad \text{at} \quad y = L \end{aligned} \tag{9.225}$$

For this problem, we obtain the solutions as

$$\tilde{u}_y = -\frac{U_o}{d_1\ell_1 - d_2\ell_2}\left[(d_2\ell_2 + i\rho_f\omega^2)\frac{\cos \ell_1(L-y)}{\cos \ell_1 L} - (d_1\ell_1 + i\rho_f\omega^2)\frac{\cos \ell_2(L-y)}{\cos \ell_2 L}\right] \tag{9.226}$$

$$\tilde{p} = \frac{iU_o}{d_1\ell_1 - d_2\ell_2}\left[d_1(d_2\ell_2 + i\rho_f\omega^2)\frac{\sin \ell_1(L-y)}{\cos \ell_1 L} - d_2(d_1\ell_1 + i\rho_f\omega^2)\frac{\sin \ell_2(L-y)}{\cos \ell_2 L}\right] \tag{9.227}$$

$$\tilde{\sigma}_y = \frac{iU_o}{d_1\ell_1 - d_2\ell_2}\left[(d_2\ell_2 + i\rho_f\omega^2)(iE_d\ell_1 - \alpha d_1)\frac{\sin \ell_1(L-y)}{\cos \ell_1 L} - (d_1\ell_1 + i\rho_f\omega^2)(iE_d\ell_2 - \alpha d_2)\frac{\sin \ell_2(L-y)}{\cos \ell_2 L}\right] \tag{9.228}$$

For comparison, we also present the elastic solution as

$$\tilde{u}_y = U_o \frac{\cos \ell_u(L-y)}{\cos \ell_u L} \tag{9.229}$$

For a bottom displacement excitation magnitude of $U_o = 1\,\text{cm}$, we present the results of displacement at the top, and pore pressure at the bottom and mid-depth, in Fig. 9.15. We observe similar responses as the two previous problems. Of particular interest is the liquefaction potential due to the reduction in effective stress. In Fig. 9.16 the Terzaghi effective stress versus depth due to the excitation at two different frequencies is presented for the hard sediment case. In the same

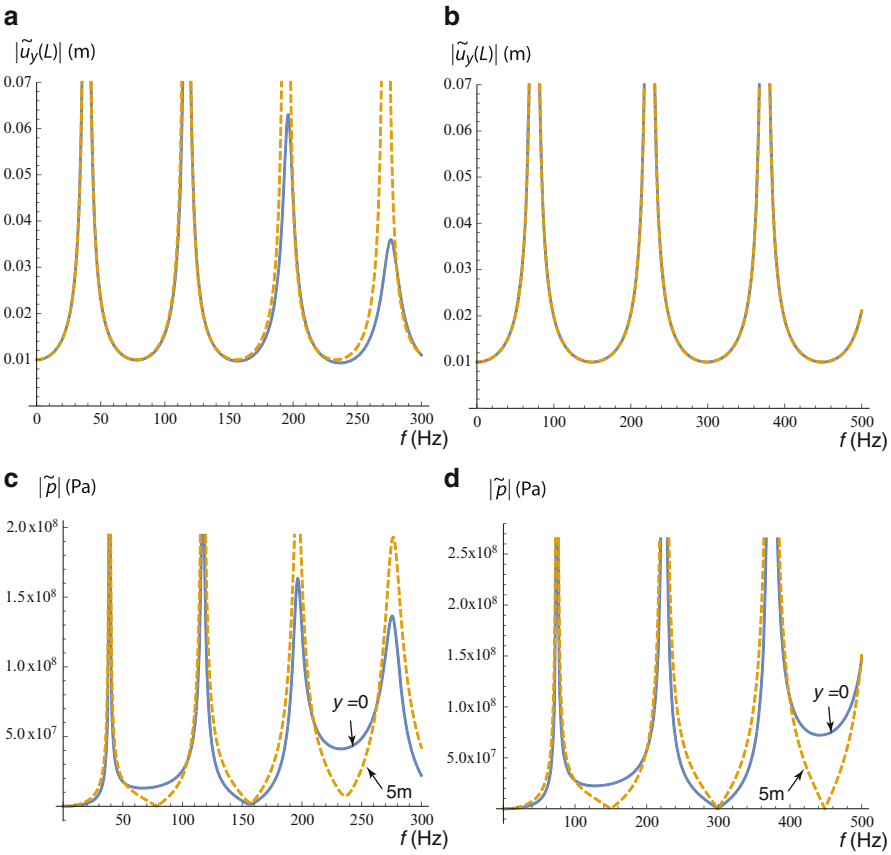


Fig. 9.15 One-dimensional wave propagation problem with displacement excitation at the bottom of finite thickness layer. (**a**) and (**b**): Displacement amplitude at the *top* $|\widetilde{u}_y(L)|$, with (**a**) hard sediment, (**b**) Berea sandstone, *solid line* the poroelastic solution, and *dashed line* the elastic solution. (**c**) and (**d**): Pressure amplitude $|\widetilde{p}(y)|$ at the *bottom* ($y = 0$) and mid-depth ($y = 5$ m) for (**c**) hard sediment and (**d**) Berea sandstone

figure the Terzaghi effective stress due to the hydrostatic effect (self-weight), $|\sigma'_{yy}| = (\rho - \rho_f)g\,(L-y)$ is plotted in dashed line. As the excitation is sinusoidal, that is, changing between positive and negative values, a stress larger than hydrostatic stress means that tension can develop during the cycle. Figure 9.16 shows that the layer is stable at $f = 20$ Hz, and can liquefy at 25 Hz.

9.7 Thermoelasticity Analogy

The analogy between thermoelasticity and poroelasticity for the quasi-static case has been presented in Sect. 1.2.7. Here we shall examine the dynamic case. The governing equations for *thermoelastodynamics* are as follows [96]

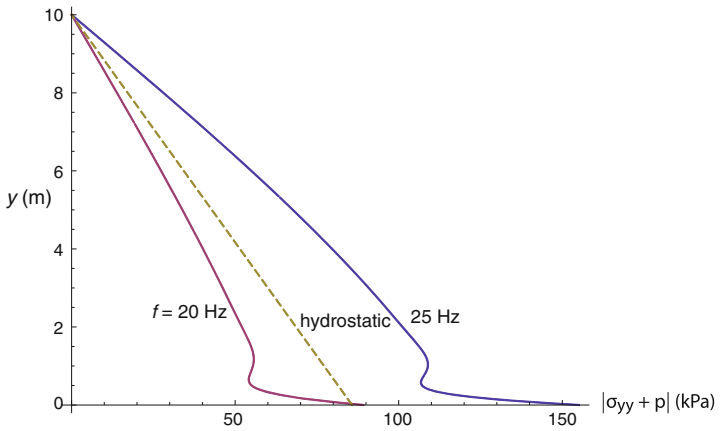


Fig. 9.16 Hard sediment layer with bottom displacement excitation at two different frequencies. Plot of Terzaghi effective stress versus depth. *Solid line*: effective stress due to excitation. *Dashed line*: hydrostatic effective stress

$$\sigma_{ij} = 2G\, e_{ij} + \lambda \delta_{ij} e - \alpha_T\, \delta_{ij} T \tag{9.230}$$

$$s = \alpha_T\, e + \frac{c_v}{\mathcal{T}_o} T \tag{9.231}$$

$$\sigma_{ij,j} = \rho \ddot{u}_i - X_i \tag{9.232}$$

$$q_i = -k_T T_{,i} \tag{9.233}$$

$$\mathcal{T}_o\, \dot{s} + q_{i,i} = \gamma_T \tag{9.234}$$

In the above, T is temperature increment from a reference temperature $\mathcal{T}_o$, s is entropy density, q_i is heat flux, c_v is specific heat at constant strain, $\alpha_T = 3K\beta_l$ is a thermoelastic constitutive constant, with K the bulk modulus of elasticity and β_l the coefficient of linear thermal expansion, k_T is the thermal conductivity, X_i is the solid body force, and γ_T is the heat source.

Comparing the above governing equations with the poroelastic ones given in Sect. 9.3, we find that the mathematical correspondence is incomplete between the poroelastic and thermoelastic systems, due to the extra presence of fluid density, fluid body force, and the added mass density terms of poroelasticity. Nevertheless, a partial correspondence can be built by equating variables and parameters as shown in Table 9.2. This table demonstrates that the poroelasticity mathematical system is the more general one that incorporates thermoelasticity, and any solution found for the former can be applied to the latter by simplification, but not vice versa.

It is of interest to demonstrate that the thermoelasticity system can be reduced to the following coupled second order partial differential equations involving four independent variables (u_i, T),

Table 9.2 Thermoelasticity/poroelasticity analogy in time domain. (Identical variables and parameters are not shown)

Poroelasticity	Thermoelasticity
p	T
ζ	s
q_i	$q_i/\mathcal{T}_o$
F_i	X_i
f_i	–
γ	$\gamma_T/\mathcal{T}_o$
α	α_T
κ	$k_T/\mathcal{T}_o$
M	$\mathcal{T}_o/c_v$
ρ_f	–
ρ_a	–

$$G\nabla^2\boldsymbol{u} + (\lambda + G)\nabla(\nabla\cdot\boldsymbol{u}) - \alpha_T\nabla T = \rho\ddot{\boldsymbol{u}} - \boldsymbol{X} \tag{9.235}$$

$$k_T\nabla^2 T - c_v\dot{T} - \alpha_T\mathcal{T}_o\,\nabla\cdot\dot{\boldsymbol{u}} = -\gamma_T \tag{9.236}$$

As commented earlier, the reduction into a four-variable (u_i, p) system similar to the one above is not possible for poroelastodynamics, unless the order of the differential equations is raised to fourth order.

In frequency domain, (9.235) and (9.236) can be transformed into

$$G\nabla^2\tilde{\boldsymbol{u}} + (\lambda + G)\nabla(\nabla\cdot\tilde{\boldsymbol{u}}) + \omega^2\rho\,\tilde{\boldsymbol{u}} - \alpha_T\nabla\tilde{T} = -\tilde{\boldsymbol{X}} \tag{9.237}$$

$$\nabla^2\tilde{T} + \frac{i\omega}{\kappa_T}\tilde{T} + i\omega\eta_t\nabla\cdot\tilde{\boldsymbol{u}} = -\frac{\tilde{Q}_T}{\kappa_T} \tag{9.238}$$

where

$$\kappa_T = \frac{k_T}{c_v};\quad \eta_t = \frac{\alpha_T\mathcal{T}_o}{k_T};\quad Q_T = \frac{\gamma_T}{c_v} \tag{9.239}$$

Comparing (9.237) and (9.238) with (9.102) and (9.103), we find a complete analogy between poroelastodynamics and thermoelastodynamics based on the correspondences shown in Table 9.3, as demonstrated by Cheng et al. [33]. We also note that if we set $(\rho_f, \rho_a, f_i) = 0$, such that $\rho'' = i/\omega\kappa$, then Tables 9.2 and 9.3 are consistent.

9.8 Poroelastodynamics Fundamental Solution

As discussed in Sect. 8.9, fundamental solutions are solutions resulting from singular forcing functions, such as point force and source, in infinite space. They are also called free-space Green's functions. In this section we shall examine the fundamental solutions of poroelastodynamics.

Table 9.3 Thermoelasticity/poroelasticity analogy in frequency domain. (Identical variables and parameters are not shown)

Poroelasticity	Thermoelasticity
$\widetilde{p}$	$\widetilde{T}$
$\widetilde{\sigma}_{ij} + (\rho_f/\rho'')\delta_{ij}\widetilde{p}$	$\widetilde{\sigma}_{ij}$
$\widetilde{\zeta} - (\rho_f/\rho'')\widetilde{e}$	$\widetilde{s}$
$\widetilde{q}_i - (i\omega\rho_f/\rho'')\widetilde{u}_i - (i/\omega\rho'')\widetilde{f}_i$	$\widetilde{q}_i/\mathcal{T}_o$
$\widetilde{F}_i - (\rho_f/\rho'')\widetilde{f}_i$	$\widetilde{X}_i$
$M\widetilde{\gamma} - (iM/\omega\rho'')\widetilde{f}_{i,i}$	$\widetilde{Q}_T$
$\alpha - (\rho_f/\rho'')$	α_T
$iM/\omega\rho''$	κ_T
$-i\omega\rho''[\alpha - (\rho_f/\rho'')]$	η_t
$\rho - (\rho_f^2/\rho'')$	ρ

Burridge and Vargas [30] was the first to derive the fundamental solution of 3D point force for poroelastodynamics. Based on the six-variable formulation, involving solid and fluid displacements (u_i, w_i), solution was presented in the Laplace transform domain. Utilizing perturbation, solution in time domain was found as large-time asymptotic solution. Norris [95] derived the 3D fundamental solution of point force, as well as fluid force, in the frequency domain, based on the (u_i, w_i) formulation. Manolis and Beskos [86, 87] presented a 3D point force fundamental solution in the Laplace transform domain, also using the (u_i, w_i) formulation.

Using a thermoelasticity analogy [76] and the four-variable (u_i, p) formulation, Bonnet [23] presented the 2D and 3D fundamental solutions of point force and source in frequency domain. The force solution was corrected by Dominguez [46], and the source solution was pointed out as incorrect [26, 47]. Also utilizing a thermoelasticity analogy [96] and the four-variable (u_i, p) formulation, Cheng et al. [33] derived 2D and 3D fundamental solutions of point force and fluid source.

For fundamental solutions in the time domain, Chen [34, 35] presented point force and fluid source solutions in 2D and 3D, using the (u_i, p) formulation. These solutions were obtained by inverse Laplace transform of Bonnet's solution [23]. However, part of the solution remained in integral form, which requires numerically integration.

Other attempts at deriving poroelastodynamic fundamental solutions include Kaynia and Banerjee [73], Zimmerman and Stern [127], Philippacopoulos [99], and Sahay [103]. Green's functions are also available for half plane and half space geometry [59, 110]. Reviews of these efforts can be found in Schanz [106] and Ding et al. [44].

In the following, we shall derive the 2D and 3D point force and fluid source fundamental solutions in the frequency domain, based on the four-variable (u_i, p) formulation, following the methodology of Ding et al. [44].

9.8.1 Elastodynamics Fundamental Solution

Before deriving the poroelastodynamic fundamental solutions, it is of interest to first present the three-dimensional elastodynamic fundamental solution due to an instantaneous point force in the frequency domain. This classical result is given as [83]

$$\tilde{u}_{ij}^{F} = \frac{1}{\rho\,\omega^2}\left[\left(\frac{e^{i\ell_{se}r}}{4\pi r} - \frac{e^{i\ell_{pe}r}}{4\pi r}\right)_{,ij} + \delta_{ij}\ell_{se}^2\,\frac{e^{i\ell_{se}r}}{4\pi r}\right] \tag{9.240}$$

where

$$\ell_{se} = \frac{\omega}{v_{se}}; \qquad \ell_{pe} = \frac{\omega}{v_{pe}} \tag{9.241}$$

are wave numbers, and v_{pe} and v_{se} are elastic compressional and shear wave velocities respectively define in (9.113) and (9.116).

The above solution is actually the combination of a shear and a compressional wave, as a point force can generate both types of waves. By decomposing the Dirac delta function representing a point force into a rotational and irrotational part [2], (9.240) can be separated into a shear wave and a compressive wave, as follows

$$\tilde{u}_{ij}^{s} = \frac{1}{\rho\,\omega^2}\left[\left(\frac{e^{i\ell_{se}r}}{4\pi r} - \frac{1}{4\pi r}\right)_{,ij} + \delta_{ij}\ell_{se}^2\,\frac{e^{i\ell_{se}r}}{4\pi r}\right] \tag{9.242}$$

$$\tilde{u}_{ij}^{p} = -\frac{1}{\rho\,\omega^2}\left(\frac{e^{i\ell_{pe}r}}{4\pi r} - \frac{1}{4\pi r}\right)_{,ij} \tag{9.243}$$

We note the presence of the $(1/4\pi r)_{,ij}$ term, which is not evident in the combined solutions.

9.8.2 Helmholtz Decomposition

As shown in Sect. 9.4, there are three types of poroelastic waves, one shear and two compressional waves. We shall demonstrate that a point force generates all three types of waves, while a fluid source generates only compressional waves. In order to isolate these wave mechanisms, we shall apply the *Helmholtz decomposition* to separate the governing equations, as well as the forcing functions, into components responsible for the shear and the compressional waves.

Given as a fundamental theorem of vector calculus [4], the Helmholtz theorem states that any twice differentiable vector function can be decomposed into a

curl-free (*irrotational*) and a divergence-free (*solenoidal*) component. Hence it is possible to express the displacement vectors $\boldsymbol{u}$ and $\boldsymbol{w}$ into the following

$$\boldsymbol{u} = \boldsymbol{u}^p + \boldsymbol{u}^s; \qquad \boldsymbol{w} = \boldsymbol{w}^p + \boldsymbol{w}^s \tag{9.244}$$

In the above, the superscript ^{p} denotes the irrotational (*dilatational*) components, which are associated with the compressional (P) waves, and superscript ^{s} denotes the rotational (*distortional*) components, which are associated with the shear (S) waves. We shall examine these components separately.

Rotational Field Considering the rotational (divergence-free) field only, we can substitute the $(\boldsymbol{u}^s, \boldsymbol{w}^s)$ parts of (9.244) into (9.96) and (9.97), to obtain the following

$$G\nabla^2\tilde{\boldsymbol{u}}^s + \omega^2\rho\tilde{\boldsymbol{u}}^s + \omega^2\rho_f\tilde{\boldsymbol{w}}^s = -\tilde{\boldsymbol{F}}^s \tag{9.245}$$

$$\omega^2\rho_f\tilde{\boldsymbol{u}}^s + \omega^2\rho''\tilde{\boldsymbol{w}}^s = -\tilde{\boldsymbol{f}}^s \tag{9.246}$$

in which the conditions $\nabla\cdot\tilde{\boldsymbol{u}}^s = \nabla\cdot\tilde{\boldsymbol{w}}^s = 0$ have been applied. In the above, we also decomposed the forcing terms into rotational and irrotational parts,

$$\tilde{\boldsymbol{F}} = \tilde{\boldsymbol{F}}^p + \tilde{\boldsymbol{F}}^s; \qquad \tilde{\boldsymbol{f}} = \tilde{\boldsymbol{f}}^p + \tilde{\boldsymbol{f}}^s \tag{9.247}$$

The fluid source term $\nabla\tilde{Q}$ does not appear because it does not contain rotational part, by the virtue of the vector identity $\nabla\times\nabla\tilde{Q} = 0$.

Equation (9.246) can be solved for $\tilde{\boldsymbol{w}}^s$,

$$\tilde{\boldsymbol{w}}^s = -\frac{1}{\omega^2\rho''}(\omega^2\rho_f\tilde{\boldsymbol{u}}^s + \tilde{\boldsymbol{f}}^s) \tag{9.248}$$

which in turn can be used to eliminate the same quantity in (9.245), to obtain

$$G\nabla^2\tilde{\boldsymbol{u}}^s + \omega^2\left(\rho - \frac{\rho_f^2}{\rho''}\right)\tilde{\boldsymbol{u}}^s = -\tilde{\boldsymbol{F}}^s + \frac{\rho_f}{\rho''}\tilde{\boldsymbol{f}}^s \tag{9.249}$$

We can rewrite the above into a standard wave equation

$$\left(\rho - \frac{\rho_f^2}{\rho''}\right)\left(v_s^2\nabla^2\tilde{\boldsymbol{u}}^s + \omega^2\tilde{\boldsymbol{u}}^s\right) = -\tilde{\boldsymbol{F}}^s + \frac{\rho_f}{\rho''}\tilde{\boldsymbol{f}}^s \tag{9.250}$$

where v_s is the shear wave velocity as defined in (9.126). Furthermore, as $\tilde{\boldsymbol{u}}^s$ is divergence-free, we can express it as the curl of a vector potential $\tilde{\boldsymbol{\psi}}^s$

$$\tilde{\boldsymbol{u}}^s = \nabla\times\tilde{\boldsymbol{\psi}}^s \tag{9.251}$$

Substituting (9.251) into (9.250), and relaxing a curl operator, we obtain

$$\left(\rho - \frac{\rho_f^2}{\rho''}\right)\left(v_s^2 \nabla^2 \tilde{\boldsymbol{\psi}}^s + \omega^2 \tilde{\boldsymbol{\psi}}^s\right) = -\tilde{\boldsymbol{\theta}}^s + \frac{\rho_f}{\rho''}\tilde{\boldsymbol{\eta}}^s \tag{9.252}$$

where we have defined

$$\tilde{\boldsymbol{F}}^s = \nabla \times \tilde{\boldsymbol{\theta}}^s; \qquad \tilde{\boldsymbol{f}}^s = \nabla \times \tilde{\boldsymbol{\eta}}^s \tag{9.253}$$

Irrotational Field We can utilize the following vector identity

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A} \tag{9.254}$$

to transform (9.96) into the following

$$\begin{aligned}(\lambda_u + 2G)\nabla(\nabla \cdot \tilde{\boldsymbol{u}}) + G\nabla \times (\nabla \times \tilde{\boldsymbol{u}}) \\ + \alpha M \nabla(\nabla \cdot \tilde{\boldsymbol{w}}) + \omega^2 \rho \tilde{\boldsymbol{u}} + \omega^2 \rho_f \tilde{\boldsymbol{w}} = -\tilde{\boldsymbol{F}} + \alpha M \nabla \tilde{Q}\end{aligned} \tag{9.255}$$

Since $\tilde{\boldsymbol{u}}$ and $\tilde{\boldsymbol{w}}$ are irrotational, we can express (9.255) and (9.97) into the following

$$(\lambda_u + 2G)\nabla(\nabla \cdot \tilde{\boldsymbol{u}}^p) + \alpha M \nabla(\nabla \cdot \tilde{\boldsymbol{w}}^p) + \omega^2 \rho \tilde{\boldsymbol{u}}^p + \omega^2 \rho_f \tilde{\boldsymbol{w}}^p = -\tilde{\boldsymbol{F}}^p + \alpha M \nabla \tilde{Q} \tag{9.256}$$

$$\alpha M \nabla(\nabla \cdot \tilde{\boldsymbol{u}}^p) + M \nabla(\nabla \cdot \tilde{\boldsymbol{w}}^p) + \omega^2 \rho_f \tilde{\boldsymbol{u}}^p + \omega^2 \rho'' \tilde{\boldsymbol{w}}^p = -\tilde{\boldsymbol{f}}^p + M \nabla \tilde{Q} \tag{9.257}$$

As we already know that there exist two compressional waves, we can further decompose $\tilde{\boldsymbol{u}}^p$ into two parts, corresponding to the first and the second wave

$$\tilde{\boldsymbol{u}}^p = \tilde{\boldsymbol{u}}^{p1} + \tilde{\boldsymbol{u}}^{p2} \tag{9.258}$$

We should note that the above partition is not arbitrary. The variable needs to be separated into the right proportions such that (9.256) and (9.257) can be the isolated into two wave equations with different wave speeds. Such condition is sought below.

To solve (9.256) and (9.257), we notice that they are linear partial differential equations. According to the linear partial differential operator theory [63], the differential operators, ∇^2, ∇, $\partial^2/\partial x^2$, $\partial/\partial x$, etc., can be manipulated as algebraic quantities; that is, they can be added, subtracted, multiplied, and divided following algebraic rules. These equations can then be solved as linear algebraic equations. As these are vector equations, (9.256) and (9.257) are six equations containing 6 unknowns, $\tilde{u}_i^p$ and $\tilde{w}_i^p$.

A solution strategy is to solve (9.257) for $\tilde{w}_i^p$ as linear combinations of $\tilde{u}_i^p$, and then use these relations to eliminate $\tilde{w}_i^p$ in (9.256). So in principle we can write

$$\tilde{\boldsymbol{w}}^p = \boldsymbol{\xi}\, \tilde{\boldsymbol{u}}^p \tag{9.259}$$

where $\boldsymbol{\xi}$ is a 3×3 matrix with its elements containing not only the material constants, but also the partial differential operators. As $\tilde{\boldsymbol{u}}^p$ and $\tilde{\boldsymbol{w}}^p$ are irrotational, they can be expressed as the gradient of a scalar potential

$$\tilde{\boldsymbol{u}}^p = \nabla\tilde{\phi}^p; \qquad \tilde{\boldsymbol{w}}^p = \nabla\tilde{\psi}^p \tag{9.260}$$

Substituting the above into (9.256) and (9.257), we observe that each of the vector equations can be expressed into component equations such that the x-component equation involves only $(\partial\phi/\partial x, \partial\psi/\partial x)$, and y-component only $(\partial\phi/\partial y, \partial\psi/\partial y)$, etc. In other words, the x-component of $\tilde{\boldsymbol{u}}^p$ is dependent only on the x-component of $\tilde{\boldsymbol{w}}^p$, etc. This leads to the conclusion that $\boldsymbol{\xi}$ is diagonal. In addition, each of the component equations is governed by the same partial differential operators with the same coefficients; hence we further conclude that the diagonal terms of identical, and we can modify (9.259) to the following

$$\tilde{\boldsymbol{w}}^p = \xi\,\tilde{\boldsymbol{u}}^p \tag{9.261}$$

where ξ is a scalar.

As we have introduced the partition (9.258), we further express

$$\tilde{\boldsymbol{w}}^p = \xi_1\,\tilde{\boldsymbol{u}}^{p1} + \xi_2\,\tilde{\boldsymbol{u}}^{p2} \tag{9.262}$$

We note that in the above we did not set $\xi_1 = \xi_2 = \xi$. In fact, only the right partition will allow us to isolate the wave equations, as commented above. In addition, we express $\tilde{\boldsymbol{u}}^{p1}$ and $\tilde{\boldsymbol{u}}^{p2}$ as the gradient of scalar function

$$\tilde{\boldsymbol{u}}^{p1} = \nabla\tilde{\phi}^{p1}; \qquad \tilde{\boldsymbol{u}}^{p2} = \nabla\tilde{\phi}^{p2} \tag{9.263}$$

Substituting (9.258), (9.262) and (9.263) into (9.256) and (9.257), and relax a gradient operator, we obtain

$$\begin{aligned}&(\lambda_u + 2G + \alpha M\xi_1)\nabla^2\tilde{\phi}^{p1} + (\lambda_u + 2G + \alpha M\xi_2)\nabla^2\tilde{\phi}^{p2}\\ &\quad + \omega^2(\rho + \rho_f\xi_1)\tilde{\phi}^{p1} + \omega^2(\rho + \rho_f\xi_2)\tilde{\phi}^{p2} = -\tilde{\theta}^p + \alpha M\tilde{Q} \qquad (9.264)\\ &M(\alpha + \xi_1)\nabla^2\tilde{\phi}^{p1} + M(\alpha + \xi_2)\nabla^2\tilde{\phi}^{p2}\\ &\quad + \omega^2(\rho_f + \rho''\xi_1)\tilde{\phi}^{p1} + \omega^2(\rho_f + \rho''\xi_2)\tilde{\phi}^{p2} = -\tilde{\eta}^p + M\tilde{Q} \qquad (9.265)\end{aligned}$$

in which we have defined

$$\tilde{\boldsymbol{F}}^p = \nabla\tilde{\theta}^p; \qquad \tilde{\boldsymbol{f}}^p = \nabla\tilde{\eta}^p \tag{9.266}$$

The above is permissible because the forcing functions are also irrotational.

Equations (9.264) and (9.265) can be rearranged into

$$(\rho+\rho_f\xi_1)(v_{p1}^2\nabla^2\tilde{\phi}^{p1}+\omega^2\tilde{\phi}^{p1}) +(\rho+\rho_f\xi_2)(v_{p2}^2\nabla^2\tilde{\phi}^{p2}+\omega^2\tilde{\phi}^{p2})=-\tilde{\theta}^p+\alpha M\tilde{Q} \tag{9.267}$$

$$(\rho_f+\rho''\xi_1)(v_{p3}^2\nabla^2\tilde{\phi}^{p1}+\omega^2\tilde{\phi}^{p1}) +(\rho_f+\rho''\xi_2)(v_{p4}^2\nabla^2\tilde{\phi}^{p2}+\omega^2\tilde{\phi}^{p2})=-\tilde{\eta}^p+M\tilde{Q} \tag{9.268}$$

in which

$$v_{p1}^2=(\lambda_u+2G+\alpha M\xi_1)/(\rho+\rho_f\xi_1) \tag{9.269}$$

$$v_{p2}^2=(\lambda_u+2G+\alpha M\xi_2)/(\rho+\rho_f\xi_2) \tag{9.270}$$

$$v_{p3}^2=M(\alpha+\xi_1)/(\rho_f+\rho''\xi_1) \tag{9.271}$$

$$v_{p4}^2=M(\alpha+\xi_2)/(\rho_f+\rho''\xi_2) \tag{9.272}$$

For (9.267) and (9.268) to represent the same two wave equations, one for $\tilde{\phi}^{p1}$ and one for $\tilde{\phi}^{p2}$, we must have the conditions

$$v_{p1}^2=v_{p3}^2; \qquad v_{p2}^2=v_{p4}^2 \tag{9.273}$$

With the above, we can solve (9.267) and (9.268) through elimination to obtain the following two independent wave equations

$$v_{p1}^2\nabla^2\tilde{\phi}^{p1}+\omega^2\tilde{\phi}^{p1}=\frac{1}{(\rho\rho''-\rho_f^2)(\xi_1-\xi_2)}\Big\{(\rho_f+\rho''\xi_2)\tilde{\theta}^p -(\rho+\rho_f\xi_2)\tilde{\eta}^p+M[(\rho+\rho_f\xi_2)-\alpha(\rho_f+\rho''\xi_2)]\tilde{Q}\Big\} \tag{9.274}$$

$$v_{p2}^2\nabla^2\tilde{\phi}^{p2}+\omega^2\tilde{\phi}^{p2}=\frac{1}{(\rho\rho''-\rho_f^2)(\xi_2-\xi_1)}\Big\{(\rho_f+\rho''\xi_1)\tilde{\theta}^p -(\rho+\rho_f\xi_1)\tilde{\eta}^p+M[(\rho+\rho_f\xi_1)-\alpha(\rho_f+\rho''\xi_1)]\tilde{Q}\Big\} \tag{9.275}$$

one for P1-wave and one for P2-wave, each with its own wave velocity.

The conditions that allows such separation, (9.273), leads to

$$\xi_i=\frac{\lambda_u+2G-\rho v_{pi}^2}{\rho_f v_{pi}^2-\alpha M}=\frac{\rho_f v_{pi}^2-\alpha M}{M-\rho'' v_{pi}^2}; \qquad i=1,2 \tag{9.276}$$

The above equation not only defines ξ_1 and ξ_2, but also gives the characteristic equation for the wave velocities. This is done by cross multiplying the numerators and denominators of the two right hand side expressions, which yields

$$A\frac{1}{v_{pi}^4} - B\frac{1}{v_{pi}^2} + C = 0 \tag{9.277}$$

where A, B and C are the same of those defined in (9.132), (9.133), and (9.134). The compressional wave velocities v_{p1} and v_{p2} solved from (9.277) are therefore the same as those defined in (9.137) and (9.138), just as expected. It is of interest to observe that ξ_1 and ξ_2 as solved in (9.276) turn out to be pure algebraic expressions, and do not contain partial differential operators. This is a direct consequence of the derivation and no prior assumption was made.

Utilizing the wave equations presented in this section, we can derive the fundamental solutions of force and fluid source by introducing Dirac delta function to properly replace the forcing functions.

9.8.3 Three-Dimensional Point Force Solution

We may express an instantaneous concentrated force of unit magnitude in the x_j-direction, acting at the time and space origin, as $\boldsymbol{F}_j = \delta(\boldsymbol{x})\delta(t)\boldsymbol{e}_j$, where $\boldsymbol{e}_j$ is a unit vector in the x_j-direction. The Fourier transform of it is

$$\tilde{\boldsymbol{F}}_j = \delta(\boldsymbol{x})\boldsymbol{e}_j \tag{9.278}$$

For the point force solution, we also set $\tilde{\boldsymbol{f}} = \tilde{Q} = 0$.

In order to isolate the solution into a shear and a compressional wave, the forcing function needs to be decomposed into a divergence-free part $\tilde{\boldsymbol{F}}_j^s$ and a curl-free part $\tilde{\boldsymbol{F}}_j^p$. The decomposition of Dirac delta function can be accomplished by utilizing the fundamental solution of Laplace equation (8.129), and the vector identity (9.254), such that [1]

$$\begin{aligned}
\tilde{\boldsymbol{F}}_j = \delta(\boldsymbol{x})\boldsymbol{e}_j &= \nabla^2\left(-\frac{1}{4\pi r}\boldsymbol{e}_j\right) \\
&= \nabla\times\left(\nabla\times\frac{1}{4\pi r}\boldsymbol{e}_j\right) - \nabla\left(\nabla\cdot\frac{1}{4\pi r}\boldsymbol{e}_j\right) \\
&= \tilde{\boldsymbol{F}}_j^s + \tilde{\boldsymbol{F}}_j^p
\end{aligned} \tag{9.279}$$

Based on the definitions in (9.253) and (9.266), we find

$$\tilde{\boldsymbol{\theta}}_j^s = \nabla \times \frac{1}{4\pi r}\boldsymbol{e}_j \tag{9.280}$$

$$\tilde{\theta}_j^p = \nabla \cdot \frac{1}{4\pi r}\boldsymbol{e}_j \tag{9.281}$$

Shear Wave First we examine the shear wave component. Based on the wave equation (9.252) and the forcing function in (9.280), we obtain

$$\left(\rho - \frac{\rho_f^2}{\rho''}\right)\left(v_s^2\nabla^2\tilde{\boldsymbol{\psi}}_j^s + \omega^2\tilde{\boldsymbol{\psi}}_j^s\right) = -\nabla \times \left(\frac{1}{4\pi r}\boldsymbol{e}_j\right) \tag{9.282}$$

where the subscript j for $\tilde{\boldsymbol{\psi}}_j^s$ is used to mark the solution corresponding to the force in the x_j-direction.

To solve the above, we further express

$$\tilde{\boldsymbol{\psi}}_j^s = \nabla \times \tilde{\boldsymbol{\Psi}}_j^s \tag{9.283}$$

Substituting the above into (9.282) and again relaxing a curl operator, we find

$$\left(\rho - \frac{\rho_f^2}{\rho''}\right)\left(v_s^2\nabla^2\tilde{\boldsymbol{\Psi}}_j^s + \omega^2\tilde{\boldsymbol{\Psi}}_j^s\right) = -\frac{1}{4\pi r}\boldsymbol{e}_j \tag{9.284}$$

The solution of the above equation can be obtained by observing the fundamental solution of the Laplace equation (8.129), and that of the Helmholtz equation (8.137), which lead to the following

$$\tilde{\boldsymbol{\Psi}}_j^s = \frac{\rho''}{\omega^2(\rho\rho'' - \rho_f^2)}\left(\frac{e^{i\ell_s r}}{4\pi r} - \frac{1}{4\pi r}\right)\boldsymbol{e}_j \tag{9.285}$$

where

$$\ell_s = \omega/v_s \tag{9.286}$$

is the wave number. And finally, we can find the displacement of the fundamental solution as

$$\begin{aligned}\tilde{u}_{ij}^s &= \nabla \times \nabla \times \tilde{\Psi}_j^s \\ &= \frac{\rho''}{\omega^2(\rho\rho'' - \rho_f^2)}\left[\delta_{ij}\ell_s^2\frac{e^{i\ell_s r}}{4\pi r} + \left(\frac{e^{i\ell_s r}}{4\pi r} - \frac{1}{4\pi r}\right)_{,ij}\right]\end{aligned} \tag{9.287}$$

where the subscript i marks the ith vector component of the displacement vector $\tilde{\boldsymbol{u}}_j^s$, and the subscript j indicates the solution due to the force in the x_j-direction. We note the similarity between the above solution and the elastic solution (9.242).

For the complete solution, we also need the pore pressure. It is easy to show that no pore pressure is generated by the shear wave

$$\tilde{p}_j^s = 0 \tag{9.288}$$

Compressional Waves Next we examine the compressional waves. Based on the two wave equations (9.274) and (9.275), and the right hand side given by (9.281), we find

$$v_{p1}^2 \nabla^2 \tilde{\phi}_j^{p1} + \omega^2 \tilde{\phi}_j^{p1} = -\frac{\rho_f + \rho'' \xi_2}{(\rho\rho'' - \rho_f^2)(\xi_1 - \xi_2)} \left(\frac{1}{4\pi r} \right)_{,j} \tag{9.289}$$

$$v_{p2}^2 \nabla^2 \tilde{\phi}_j^{p2} + \omega^2 \tilde{\phi}_j^{p2} = -\frac{\rho_f + \rho'' \xi_1}{(\rho\rho'' - \rho_f^2)(\xi_2 - \xi_1)} \left(\frac{1}{4\pi r} \right)_{,j} \tag{9.290}$$

Solutions of the above are

$$\tilde{\phi}_j^{p1} = \frac{\rho_f + \rho'' \xi_2}{\omega^2(\rho\rho'' - \rho_f^2)(\xi_1 - \xi_2)} \left(\frac{e^{i\ell_{p1} r}}{4\pi r} - \frac{1}{4\pi r} \right)_{,j} \tag{9.291}$$

$$\tilde{\phi}_j^{p2} = \frac{\rho_f + \rho'' \xi_1}{\omega^2(\rho\rho'' - \rho_f^2)(\xi_2 - \xi_1)} \left(\frac{e^{i\ell_{p2} r}}{4\pi r} - \frac{1}{4\pi r} \right)_{,j} \tag{9.292}$$

where

$$\ell_{pi} = \omega / v_{pi}; \qquad i = 1, 2 \tag{9.293}$$

are the wave numbers. Finally, the displacement solutions for the first and the second compressional waves are

$$\tilde{u}_{ij}^{p1} = -\frac{\ell_{p2}^2(\ell_{p1}^2 M - \rho''\omega^2)}{\omega^4(\ell_{p1}^2 - \ell_{p2}^2)(\rho\rho'' - \rho_f^2)} \left(\frac{e^{i\ell_{p1} r}}{4\pi r} - \frac{1}{4\pi r} \right)_{,ij} \tag{9.294}$$

$$\tilde{u}_{ij}^{p2} = -\frac{\ell_{p1}^2(\ell_{p2}^2 M - \rho''\omega^2)}{\omega^4(\ell_{p2}^2 - \ell_{p1}^2)(\rho\rho'' - \rho_f^2)} \left(\frac{e^{i\ell_{p2} r}}{4\pi r} - \frac{1}{4\pi r} \right)_{,ij} \tag{9.295}$$

In the above, we have used (9.276) to eliminate ξ_i.

The pore pressure can be found through the constitutive equations (9.89) and (9.92), with $\tilde{\boldsymbol{w}}^p$ given by (9.262), such that

$$\tilde{p}_j^{p1} = -M(\alpha + \xi_1)\tilde{u}_{ij,i}^{p1}; \qquad \tilde{p}_j^{p2} = -M(\alpha + \xi_2)\tilde{u}_{ij,i}^{p2} \tag{9.296}$$

Using the above, we obtain

$$\tilde{p}_j^{p1} = \frac{\omega^2(\alpha\rho'' - \rho_f)}{(\lambda + 2G)(\ell_{p1}^2 - \ell_{p2}^2)} \left(\frac{e^{i\ell_{p1}r}}{4\pi r}\right)_{,j} \tag{9.297}$$

$$\tilde{p}_j^{p2} = \frac{\omega^2(\alpha\rho'' - \rho_f)}{(\lambda + 2G)(\ell_{p2}^2 - \ell_{p1}^2)} \left(\frac{e^{i\ell_{p2}r}}{4\pi r}\right)_{,j} \tag{9.298}$$

The detail of the above derivation can be found in Ding et al. [44].

Complete Wave Solution In the above, we have isolated the three wave components in order to see more clearly the contribution from different wave mechanisms. These wave components, (9.287), (9.294) and (9.295), can be summed to give the complete wave solution of a point force as

$$\tilde{u}_{ij}^{Fi} = \frac{\rho''}{\omega^2(\rho\rho'' - \rho_f^2)} \left[\left(\frac{e^{i\ell_s r}}{4\pi r} - A_1 \frac{e^{i\ell_{p1}r}}{4\pi r} - A_2 \frac{e^{i\ell_{p2}r}}{4\pi r} \right)_{,ij} + \delta_{ij}\ell_s^2 \frac{e^{i\ell_s r}}{4\pi r} \right] \tag{9.299}$$

where

$$A_1 = \frac{\ell_{p2}^2(\ell_{p1}^2 M - \rho''\omega^2)}{\rho''\omega^2(\ell_{p1}^2 - \ell_{p2}^2)}; \qquad A_2 = \frac{\ell_{p1}^2(\ell_{p2}^2 M - \rho''\omega^2)}{\rho''\omega^2(\ell_{p2}^2 - \ell_{p1}^2)} \tag{9.300}$$

In doing so, we observe that the $(1/4\pi r)_{,ij}$ terms present in (9.287), (9.294) and (9.295) all cancel out. The above solution can be compared to the elastic solution (9.240).

The pressure solution is found by summing (9.288), (9.297) and (9.298),

$$\tilde{p}_j^{Fi} = \frac{\omega^2(\alpha\rho'' - \rho_f)}{(\lambda + 2G)(\ell_{p1}^2 - \ell_{p2}^2)} \left(\frac{e^{i\ell_{p1}r}}{4\pi r} - \frac{e^{i\ell_{p2}r}}{4\pi r} \right)_{,j} \tag{9.301}$$

The fundamental solutions (9.299) and (9.301), formally derived here, are identical to that presented in Cheng et al. [33], based on the analogy with thermoelasticity.

Numerical Result To visualize the fundamental solutions, we shall evaluate the above expressions using real material properties. The material chosen is Pecos sandstone, with properties given in Tables 3.1 and 3.2. In addition, the following material constants are needed: $\rho = 2.27 \times 10^3$ kg/m^3, $\rho_f = 1.0 \times 10^3$ kg/m^3, and $\rho_a = 1.17 \times 10^2$ kg/m^3. As the solution is a function of frequency, we select $f = 10$ Hz, where $f = \omega/2\pi$. From these data, we can calculate the complex wave velocity as $v_{p1} = 3049 - 3.31 \times 10^{-6}\, i$ m/s, $v_{p2} = 0.408 - 0.408\, i$ m/s, and $v_s = 1604 - 1.77 \times 10^{-5}\, i$ m/s.

First, we examine the wave field generated by an instantaneous point force of 1 MN acting at the origin, pointing in the x-direction. We plot in Fig. 9.17 the real and imaginary part of the x-component complex displacement $\tilde{u}_{11}^s$, separately for the

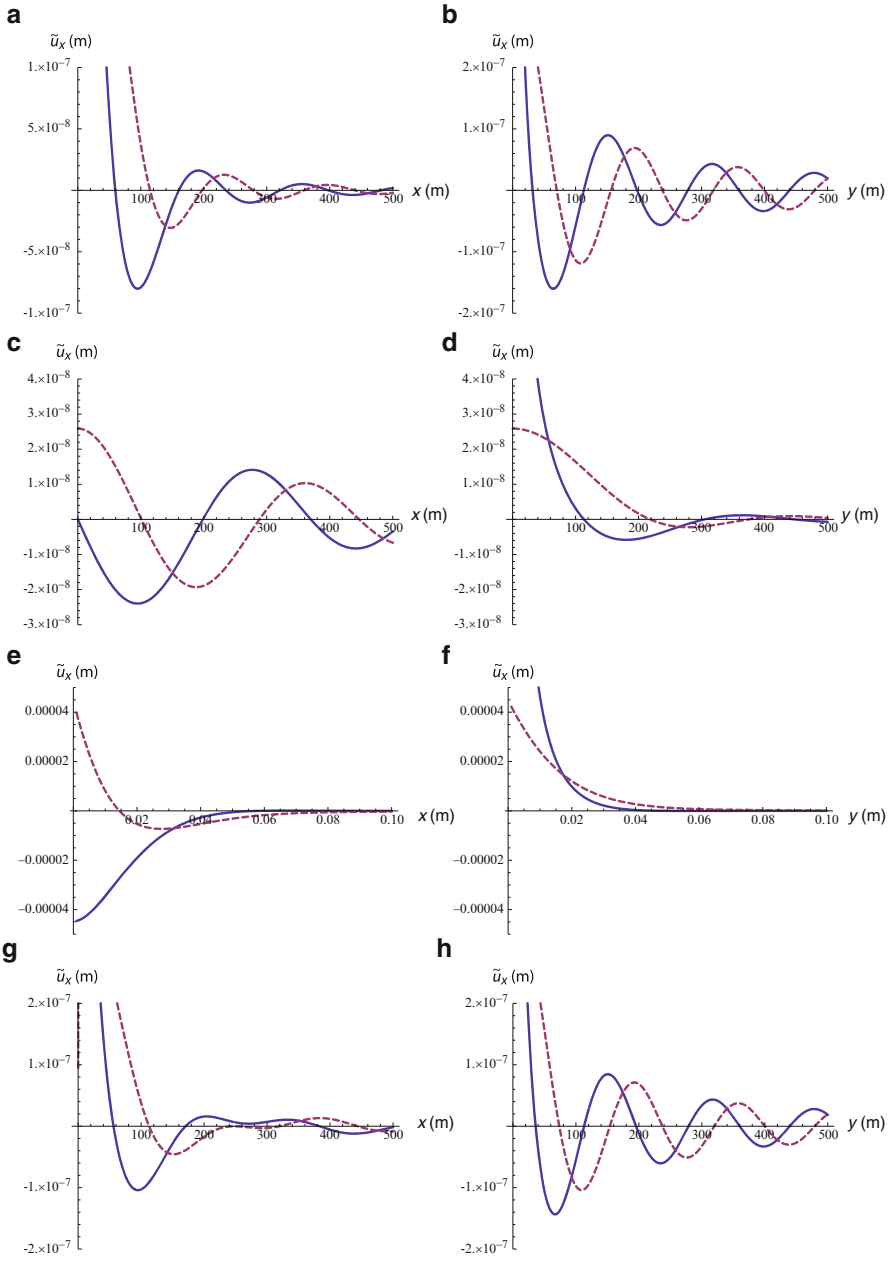


Fig. 9.17 Real (*solid line*) and imaginary (*dashed line*) part of x-component displacement due to an instantaneous point force in the x-direction $\tilde{u}_{xx}$, at $f = 10$ Hz. (**a**) S-wave along the x-axis, and (**b**) along the y-axis; (**c**) P1-wave along the x-axis, and (**d**) along the y-axis; (**e**) P2-wave along the x-axis, and (**f**) along the y-axis; (**g**) Combined wave along the x-axis, and (**h**) along the y-axis

S, P1 and P2-waves, and then combined. The displacement $\tilde{u}_x$ is plotted along the x-axis (in the direction of the force) and the y-axis (perpendicular to the force).

We observe in Fig. 9.17 that some of the displacement profiles are singular at the origin, and some are not. Hence it is of interest to examine the singularity of the displacement solution. First we can expand the exponential term into a series around $r = 0$,

$$\frac{e^{i\ell r}}{4\pi r} = \frac{1}{4\pi r} + \frac{i\ell}{4\pi} - \frac{\ell^2 r}{8\pi} - \frac{i\ell^3 r^2}{24\pi} + \cdots \tag{9.302}$$

It is interesting to observe that the leading term is independent of the wave number. The above relation shows the importance of the term $(1/4\pi r)$ in (9.287), (9.294) and (9.295), for the S, P1 and P2-waves, which is not present in the combined wave solution (9.299). In fact, the singularity of these isolated waves is of the order

$$\left(\frac{e^{i\ell r}}{4\pi r} - \frac{1}{4\pi r}\right)_{,ij} \sim O(r_{,ij}) = O\left(\frac{\delta_{ij}}{r} - \frac{x_i x_j}{r^3}\right) \tag{9.303}$$

We hence observe that some of the displacement profiles are singular, of the order $O(r^{-1})$.

We also notice in Fig. 9.17 that the P2-wave is highly dissipative for this geomaterial and does not propagate. While the S and the P1-waves are plotted for the first 500 m from the origin, the P2-wave is plotted only for the first 10 cm. As a result, the off-phase motion ceases to exist after a very short distance, and the solid and fluid motions are in phase with each other.

The total pore pressure field due to the point force given as (9.301) is presented in Fig. 9.18. Figure 9.18a plots the real and imaginary part of pore pressure along the x-axis for the first 500 m. The pore pressure field is in fact non-singular at the origin, although it contain very large peaks near it, which rapidly decreases. Figure 9.18b plots the real part of pore pressure response on the x-y plane for a 20 cm × 20 cm region.

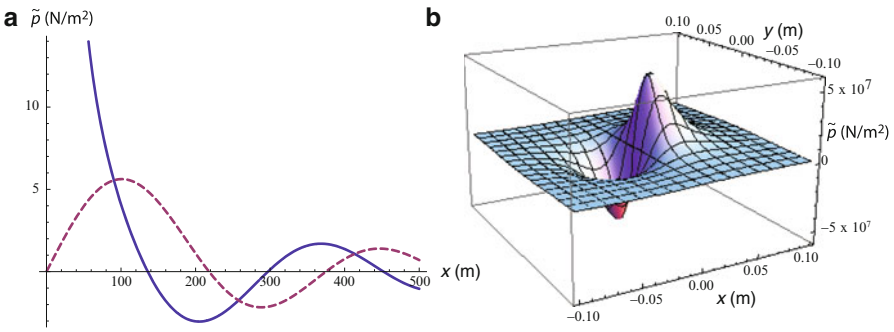


Fig. 9.18 Pore pressure due to an instantaneous point force in the x-direction, $\tilde{p}_x$, at $f = 10$ Hz. (**a**) Real (*solid line*) and imaginary (*dashed line*) part of pressure along the x-axis, and (**b**) Real part of pressure on x-y plane

9.8.4 Three-Dimensional Fluid Source Solution

We have used the symbol γ to represent a fluid source, that is, the rate of injected fluid volume per unit volume of porous medium. We also have used the symbol $\dot{Q} = \gamma$, where Q is the injected fluid volume per unit volume of porous medium. These definitions are found in (9.88), (9.89), and (9.90).

An instantaneous fluid source of unit magnitude at the time and space origin is given by $\gamma = \delta(\boldsymbol{x})\delta(t)$, from which we obtain the Fourier transform of Q as

$$\tilde{Q} = \frac{i\delta(\boldsymbol{x})}{\omega} \tag{9.304}$$

For the source solution, we further require that $\tilde{\boldsymbol{F}} = \tilde{\boldsymbol{f}} = 0$.

Shear Wave First we examine the shear wave. Referring to the wave equation (9.250), we observe that the right hand side is identically zero; hence a fluid source does not generate shear wave.

Compressional Waves Next we examine the compressional waves. Substituting the forcing term in (9.304) into the pair of wave equations (9.274) and (9.275), we obtain

$$v_{p1}^2 \nabla^2 \tilde{\phi}^{p1} + \omega^2 \tilde{\phi}^{p1} = \frac{iM[(\rho + \rho_f \xi_2) - \alpha(\rho_f + \rho'' \xi_2)]}{\omega(\rho\rho'' - \rho_f^2)(\xi_1 - \xi_2)} \delta(\boldsymbol{x}) \tag{9.305}$$

$$v_{p2}^2 \nabla^2 \tilde{\phi}^{p2} + \omega^2 \tilde{\phi}^{p2} = \frac{iM[(\rho + \rho_f \xi_1) - \alpha(\rho_f + \rho'' \xi_1)]}{\omega(\rho\rho'' - \rho_f^2)(\xi_2 - \xi_1)} \delta(\boldsymbol{x}) \tag{9.306}$$

The above equations are easily solved for $\tilde{\phi}^{p1}$ and $\tilde{\phi}^{p2}$. After taking the gradient and simplifying the coefficients [44], we obtain

$$\tilde{u}_i^{p1} = \frac{i\omega(\alpha\rho'' - \rho_f)}{(\lambda + 2G)(\ell_{p1}^2 - \ell_{p2}^2)} \left(\frac{e^{i\ell_{p1} r}}{4\pi r} \right)_{,i} \tag{9.307}$$

$$\tilde{u}_i^{p2} = \frac{i\omega(\alpha\rho'' - \rho_f)}{(\lambda + 2G)(\ell_{p2}^2 - \ell_{p1}^2)} \left(\frac{e^{i\ell_{p2} r}}{4\pi r} \right)_{,i} \tag{9.308}$$

The complete solution is the sum of the above, and

$$\tilde{u}_i^{si} = \frac{i\omega(\alpha\rho'' - \rho_f)}{(\lambda + 2G)(\ell_{p1}^2 - \ell_{p2}^2)} \left(\frac{e^{i\ell_{p1} r}}{4\pi r} - \frac{e^{i\ell_{p2} r}}{4\pi r} \right)_{,i} \tag{9.309}$$

By comparing with (9.301), we notice the similarity between the point force pressure solution and the fluid source displacement solution. In fact the following relation exists

$$\tilde{p}_i^{Fi} = -i\omega\tilde{u}_i^{si} \tag{9.310}$$

Such relation can be proven using the Betti-Maxwell reciprocal relation, as demonstrated by Nowacki [96] for thermoelasticity. This relation will also be proven in Sect. 9.9, where the integral equation representation is introduced.

The pressure solution can be obtained from the constitutive equation (9.296), and

$$\tilde{p}^{p1} = \frac{i\rho''\omega}{\ell_{p2}^2 - \ell_{p1}^2}\left[\ell_{p1}^2 - \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''(\lambda + 2G)}\right]\frac{e^{i\ell_{p1}r}}{4\pi r} \tag{9.311}$$

$$\tilde{p}^{p2} = \frac{i\rho''\omega}{\ell_{p1}^2 - \ell_{p2}^2}\left[\ell_{p2}^2 - \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''(\lambda + 2G)}\right]\frac{e^{i\ell_{p2}r}}{4\pi r} \tag{9.312}$$

The complete solution is

$$\tilde{p}^{si} = \frac{i\rho''\omega}{\ell_{p2}^2 - \ell_{p1}^2}\left\{\left[\ell_{p1}^2 - \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''(\lambda + 2G)}\right]\frac{e^{i\ell_{p1}r}}{4\pi r} - \left[\ell_{p2}^2 - \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''(\lambda + 2G)}\right]\frac{e^{i\ell_{p2}r}}{4\pi r}\right\} \tag{9.313}$$

Numerical Result For the same material, Pecos sandstone, and frequency of $f = 10\,\text{Hz}$, we introduce an instantaneous injection of a fluid volume of 1 liter at the origin. The pattern of displacement is similar to the pressure response of instantaneous point force, but with real and imaginary part reversed, as indicated by (9.310); hence it is not presented. In Fig. 9.19 we plot the pressure field along the radial distance r, as the solution has axial symmetry, for the P1 and P2-waves. We observe that the P2-wave response is very large near the origin, but it is dissipated within a small distance.

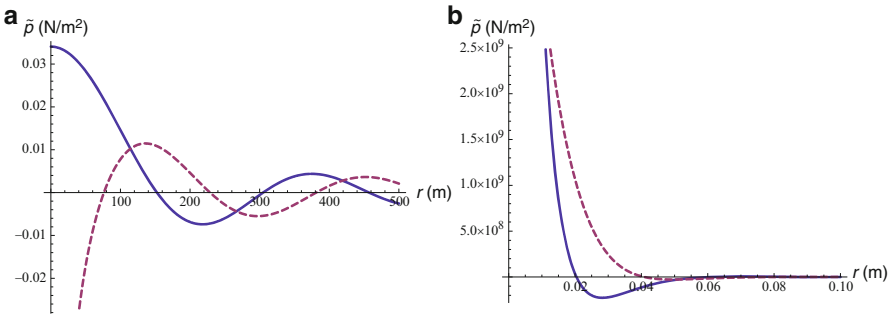


Fig. 9.19 Real (*solid line*) and imaginary (*dashed line*) part of pressure due to an instantaneous source, at $f = 10\,\text{Hz}$. (**a**) P1-wave along radial distance r, and (**b**) P2-wave along r

9.8.5 Two-Dimensional Fundamental Solution

The two-dimensional fundamental solutions can be similarly constructed. The results are shown below.

Point Force Solution For a two-dimensional instantaneous point force in the x_j-direction, the shear wave component of displacement is

$$\tilde{u}_{ij}^{s} = \frac{\rho''}{\omega^2(\rho\rho'' - \rho_f^2)} \left\{ \delta_{ij}\ell_s^2 \frac{i\mathrm{H}_0^{(1)}(\ell_s r)}{4} + \left[\frac{i\mathrm{H}_0^{(1)}(\ell_s r)}{4} + \frac{\ln r}{2\pi} \right]_{,ij} \right\} \tag{9.314}$$

where $\mathrm{H}_0^{(1)}$ is the Hankel function of the first kind of order zero. And for compressional waves:

$$\tilde{u}_{ij}^{p1} = \frac{\ell_{p2}^2(\ell_{p1}^2 M - \rho''\omega^2)}{\omega^4(\ell_{p2}^2 - \ell_{p1}^2)(\rho\rho'' - \rho_f^2)} \left[\frac{i\mathrm{H}_0^{(1)}(\ell_{p1} r)}{4} + \frac{\ln r}{2\pi} \right]_{,ij} \tag{9.315}$$

$$\tilde{u}_{ij}^{p2} = \frac{\ell_{p1}^2(\ell_{p2}^2 M - \rho''\omega^2)}{\omega^4(\ell_{p1}^2 - \ell_{p2}^2)(\rho\rho'' - \rho_f^2)} \left[\frac{i\mathrm{H}_0^{(1)}(\ell_{p2} r)}{4} + \frac{\ln r}{2\pi} \right]_{,ij} \tag{9.316}$$

The above leads to the complete wave solution

$$\tilde{u}_{ij}^{Fi} = \frac{i\rho''}{4\,\omega^2(\rho\rho'' - \rho_f^2)} \left\{ \left[A_1 \mathrm{H}_0^{(1)}(\ell_{p1} r) + A_2 \mathrm{H}_0^{(1)}(\ell_{p2} r) + \mathrm{H}_0^{(1)}(\ell_s r) \right]_{,ij} + \delta_{ij}\ell_s^2 \mathrm{H}_0^{(1)}(\ell_s r) \right\} \tag{9.317}$$

with A_1 and A_2 defined in (9.300). The complete pressure solution is given by

$$\tilde{p}_j^{Fi} = \frac{i\omega^2(\alpha\rho'' - \rho_f)}{4(\lambda + 2G)(\ell_{p1}^2 - \ell_{p2}^2)} \left[\mathrm{H}_0^{(1)}(\ell_{p1} r) - \mathrm{H}_0^{(1)}(\ell_{p2} r) \right]_{,j} \tag{9.318}$$

Fluid Source Solution The two-dimensional instantaneous fluid source fundamental solutions are

$$\tilde{u}_i^{si} = -\frac{\omega(\alpha\rho'' - \rho_f)}{4(\lambda + 2G)(\ell_{p1}^2 - \ell_{p2}^2)} \left[\mathrm{H}_0^{(1)}(\ell_{p1} r) - \mathrm{H}_0^{(1)}(\ell_{p2} r) \right]_{,i} \tag{9.319}$$

$$\tilde{p}^{si} = -\frac{\omega\rho''}{4(\ell_{p2}^2 - \ell_{p1}^2)} \left\{ \left[\ell_{p1}^2 - \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''(\lambda + 2G)} \right] \mathrm{H}_0^{(1)}(\ell_{p1} r) - \left[\ell_{p2}^2 - \frac{\omega^2(\rho\rho'' - \rho_f^2)}{\rho''(\lambda + 2G)} \right] \mathrm{H}_0^{(1)}(\ell_{p2} r) \right\} \tag{9.320}$$

9.9 Integral Equation Representation

Similar to the integral equation representation for quasi-static poroelasticity, as developed in Chap. 8, the solution of poroelastodynamics can also be presented in the integral form. These integral equations have been derived for poroelastodynamics [33, 46, 86, 101], as well as for thermoelastodynamics [65, 96]. The basis of existence of such integral equation is the reciprocity relation of poroelasticity given as (8.18). Rather than giving a detailed derivation, the resultant integral equation is given as follows [33]

$$\begin{aligned}
&\int_0^t \int_\Gamma \left[\left(\sigma_{ij}^{(1)} n_j u_i^{(2)} - \sigma_{ij}^{(2)} n_j u_i^{(1)} \right) - \left(p^{(1)} w_i^{(2)} n_i - p^{(2)} w_i^{(1)} n_i \right) \right] d\chi \, d\tau \\
&\quad + \int_0^t \int_\Omega \left[\left(F_i^{(1)} u_i^{(2)} - F_i^{(2)} u_i^{(1)} \right) + \left(f_i^{(1)} w_i^{(2)} - f_i^{(2)} w_i^{(1)} \right) \right. \\
&\quad \left. - \left(Q^{(1)} p^{(2)} - Q^{(2)} p^{(1)} \right) \right] d\chi \, d\tau = 0
\end{aligned} \tag{9.321}$$

The frequency domain equation can be obtained by performing Fourier transform to the above equation, and we obtain

$$\begin{aligned}
&\int_\Gamma \left[\left(\tilde{t}_i^{(1)} \tilde{u}_i^{(2)} - \tilde{t}_i^{(2)} \tilde{u}_i^{(1)} \right) - \frac{i}{\omega} \left(\tilde{p}^{(1)} \tilde{q}^{(2)} - \tilde{p}^{(2)} \tilde{q}^{(1)} \right) \right] d\chi \\
&\quad + \int_\Omega \left[\left(\tilde{F}_i^{(1)} \tilde{u}_i^{(2)} - \tilde{F}_i^{(2)} \tilde{u}_i^{(1)} \right) + \frac{i}{\omega} \left(\tilde{f}_i^{(1)} \tilde{q}_i^{(2)} - \tilde{f}_i^{(2)} \tilde{q}_i^{(1)} \right) \right. \\
&\quad \left. - \frac{i}{\omega} \left(\tilde{\gamma}^{(1)} \tilde{p}^{(2)} - \tilde{\gamma}^{(2)} \tilde{p}^{(1)} \right) \right] d\chi = 0
\end{aligned} \tag{9.322}$$

In the above, $t_i = \sigma_{ij} n_j$ is the boundary traction, and $q = q_i n_i$ with $\tilde{q}_i = i\omega \tilde{w}_i$ is the boundary normal specific flux.

The Somigliana-type integral equation representation in the frequency domain can be obtained by substituting in the fundamental solutions as derived in the preceding sections, and they are

$$\begin{aligned}
\overline{\theta}\, \tilde{u}_k(\boldsymbol{x}, \omega) = &\int_\Gamma \left[\tilde{u}_{ik}^{Fi}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{t}_i(\boldsymbol{\chi}, \omega) - \tilde{t}_{ik}^{Fi}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{u}_i(\boldsymbol{\chi}, \omega) \right] \\
&- \frac{i}{\omega} \left[\tilde{q}_k^{Fi}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{p}(\boldsymbol{\chi}, \omega) - \tilde{p}_k^{Fi}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{q}(\boldsymbol{\chi}, \omega) \right] d\chi
\end{aligned} \tag{9.323}$$

$$\begin{aligned}
\overline{\theta}\, \tilde{p}(\boldsymbol{x}, \omega) = i\omega &\int_\Gamma \left[\tilde{u}_i^{si}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{t}_i(\boldsymbol{\chi}, \omega) - \tilde{t}_i^{si}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{u}_i(\boldsymbol{\chi}, \omega) \right] \\
&+ \left[\tilde{q}^{si}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{p}(\boldsymbol{\chi}, \omega) - \tilde{p}^{si}(\boldsymbol{\chi} - \boldsymbol{x}, \omega)\, \tilde{q}(\boldsymbol{\chi}, \omega) \right] d\chi
\end{aligned} \tag{9.324}$$

where $\overline{\theta}$ is the Cauchy principal value integration coefficient defined in Sect. 8.2.3. The above integral equations have been utilized to numerically solve poroelastodynamic problems in the frequency domain, including the Mohr-Coulomb shear failure due to vibrating loading on soil stratum [33], dynamic compliance of a shallow circular foundation [37], the dam-reservoir-foundation system under earthquake excitation [9, 48, 84], and the dynamic impedance of pile groups [85].

Although we shall not discuss the numerical implementation of the above integral equations, we shall utilize these equations to demonstrate some interesting relations among the fundamental solutions, similar to those presented in Sect. 8.20. Referring to (9.321), we can choose the domain Ω as a sphere with radius R, and take the limit of $R \to \infty$. If the integrands decay faster than $1/r^2$ as $r \to \infty$, then the surface integrals in (9.321) vanish, and the equation reduces to

$$\int_0^t \int_{\Omega_\infty} \Big[\Big(F_i^{(1)} u_i^{(2)} - F_i^{(2)} u_i^{(1)} \Big) + \Big(f_i^{(1)} w_i^{(2)} - f_i^{(2)} w_i^{(1)} \Big) - \big(Q^{(1)} p^{(2)} - Q^{(2)} p^{(1)} \big) \Big] \, d\boldsymbol{\chi} \, d\tau = 0 \tag{9.325}$$

We shall substituting in two different sets of fundamental solutions for the above. For the first system, we introduce the point force fundamental solution, corresponding to the substitution of $\boldsymbol{F}_j^{(1)} = \delta(\boldsymbol{x}) \delta(t) \boldsymbol{e}_j$, and $\boldsymbol{f}^{(1)} = Q^{(1)} = 0$, and for the second system, we use the fluid dilatation fundamental solution, corresponding to $Q^{(2)} = \delta(\boldsymbol{x}) \delta(t)$, and $\boldsymbol{F}^{(2)} = \boldsymbol{f}^{(2)} = 0$. These fundamental solutions decay fast enough as $r \to \infty$ to satisfy the above stated condition. For the volume integrals in (9.325), they are either zero, or they integrate on a Dirac delta function, to produce

$$u_j^{li} + p_j^{Fi} = 0 \tag{9.326}$$

As the fluid source fundamental solution is obtained by the substitution $\gamma = \dot{Q} = \delta(\boldsymbol{x}) \delta(t)$, so it is clear that

$$u_j^{si} = \dot{u}_j^{li} \tag{9.327}$$

Performing Fourier transform to the above equations, we can show that

$$\tilde{p}_j^{Fi} = -i \omega \tilde{u}_j^{si} \tag{9.328}$$

which has been observed as (9.310).

9.10 Plane Wave Reflection and Refraction

It is well known that when a propagating wave encounters a boundary, it can bounce back to the same medium, known as *reflection*, and it can penetrate into the next medium with a change in direction of travel, known as *refraction*. A boundary can be an *interface* between two materials, or a *free surface* with no material on the other side. In reality, however, a free surface means that the material on the other side is air, with negligible mass density and rigidity.

As already discussed, an elastic medium can sustain two types of waves, a P-wave and an S-wave. For each type of waves impinging on a boundary, two types of waves can be generated, as reflected and refracted waves. For porous medium, there are three types of wave. For wave refraction, there can exist a *critical incident angle* beyond which the wave cannot penetrated the other medium. In addition to knowing the directions of the waves, it is of interest to know the amplitudes, and particularly the partition of energy, among the reflected and refracted waves.

There are also many other interesting phenomena associated with wave interaction with boundaries. For example, waves can be trapped within a layer by repeatedly reflected up and down by the two bounding surfaces, a phenomenon known as *Love wave* [83]. Or, waves can travel along a free surface, known as *Rayleigh wave*, or on an interface between a fluid and a solid, known as *Stoneley wave* [52, 78, 83].

These and other problems involving the effects of boundaries on the propagation of poroelastic waves have been extensively studied by Deresiewicz and coworkers in a sequence of papers [39–43]. In the following, we shall examine three cases: the reflection of poroelastic plane wave on a free surface and a clamped impermeable surface, and the reflection and refraction of a plane wave on a fluid and porous medium interface.

9.10.1 Plane Strain Wave Solution

As shown in Sect. 9.8.2, the poroelastic wave field is governed by the three wave equations, (9.252), (9.274) and (9.275). Equation (9.252) in fact is a vector equation with a vector potential $\tilde{\boldsymbol{\psi}}^s$. Under the plane strain condition, $u_x = u_x(x, z)$, $u_z = u_z(x, z)$, and $u_y = 0$, only the y-component vector potential $\tilde{\psi}^s_y(x, z)$ exists[3]. Hence we can rewrite these equations for plane strain, in the absence of the body force and source terms, as

$$v_s^2 \nabla^2 \tilde{\psi}_s + \omega^2 \tilde{\psi}_s = 0 \tag{9.329}$$

[3]The shear wave under plane strain condition is called an SV wave. Under a generalized plane strain condition $u_x = u_z = 0$ and $u_y = u_y(x, z)$, or $\tilde{\psi}^s_x = \tilde{\psi}^s_x(x, z)$, $\tilde{\psi}^s_z = \tilde{\psi}^s_z(x, z)$ and $\tilde{\psi}^s_y = 0$, an SH wave also exists.

$$v_{p1}^2 \nabla^2 \tilde{\phi}_{p1} + \omega^2 \tilde{\phi}_{p1} = 0 \tag{9.330}$$

$$v_{p2}^2 \nabla^2 \tilde{\phi}_{p2} + \omega^2 \tilde{\phi}_{p2} = 0 \tag{9.331}$$

In the above, we have moved the superscripts of $\tilde{\psi}$ and $\tilde{\phi}$ to subscripts, for convenience of notation. The general solutions of (9.329), (9.330), and (9.331) can be expressed as

$$\tilde{\psi}_s = A_s \exp(i \boldsymbol{\ell}_s \cdot \boldsymbol{x}) \tag{9.332}$$

$$\tilde{\phi}_{p1} = A_{p1} \exp(i \boldsymbol{\ell}_{p1} \cdot \boldsymbol{x}) \tag{9.333}$$

$$\tilde{\phi}_{p2} = A_{p2} \exp(i \boldsymbol{\ell}_{p2} \cdot \boldsymbol{x}) \tag{9.334}$$

where A_s, A_{p1}, A_{p2} are wave amplitudes, $\boldsymbol{x} = x\hat{i} + z\hat{k}$ is a position vector, and $\boldsymbol{\ell}_s$, $\boldsymbol{\ell}_{p1}$, $\boldsymbol{\ell}_{p2}$ are wave number vectors given by

$$\boldsymbol{\ell}_i \cdot \boldsymbol{\ell}_i = \ell_i^2, \quad i = s,\ p1,\ p2 \tag{9.335}$$

in which ℓ_s, ℓ_{p1} and ℓ_{p2} are scalar wave numbers defined in (9.286) and (9.293). We note that both $\boldsymbol{\ell}$ and ℓ in general are complex numbers.

Once the potentials are solved, the displacement is found from the definitions (9.251) and (9.260), and

$$\tilde{u}_x = \frac{\partial \tilde{\phi}_p}{\partial x} - \frac{\partial \tilde{\psi}_s}{\partial z} \tag{9.336}$$

$$\tilde{u}_z = \frac{\partial \tilde{\phi}_p}{\partial z} + \frac{\partial \tilde{\psi}_s}{\partial x} \tag{9.337}$$

where $\tilde{\phi}_p = \tilde{\phi}_{p1} + \tilde{\phi}_{p2}$. Based on plane strain constitutive equations, we obtain

$$\tilde{\sigma}_{xz} = G\left(2\frac{\partial^2 \tilde{\phi}_p}{\partial x\, \partial z} + \frac{\partial^2 \tilde{\psi}_s}{\partial x^2} - \frac{\partial^2 \tilde{\psi}_s}{\partial z^2}\right) \tag{9.338}$$

Following (9.248) and (9.262), the specific relative fluid to solid displacement is related to the solid displacement as

$$\tilde{\boldsymbol{w}} = \xi_1 \tilde{\boldsymbol{u}}^{p1} + \xi_2 \tilde{\boldsymbol{u}}^{p2} - \frac{\rho_f}{\rho''} \tilde{\boldsymbol{u}}^s \tag{9.339}$$

where ξ_1 and ξ_2 are defined in (9.276). Substituting in (9.336) and (9.337), we obtain

$$\tilde{w}_x = \xi_1 \frac{\partial \tilde{\phi}_{p1}}{\partial x} + \xi_2 \frac{\partial \tilde{\phi}_{p2}}{\partial x} - \frac{\rho_f}{\rho''} \frac{\partial \tilde{\psi}_s}{\partial z} \tag{9.340}$$

$$\tilde{w}_z = \xi_1 \frac{\partial \tilde{\phi}_{p1}}{\partial z} + \xi_2 \frac{\partial \tilde{\phi}_{p2}}{\partial z} + \frac{\rho_f}{\rho''} \frac{\partial \tilde{\psi}_s}{\partial x} \tag{9.341}$$

Pore pressure can be obtained from (9.296), which gives

$$\tilde{p} = -M(\alpha + \xi_1)\nabla^2 \tilde{\phi}_{p1} - M(\alpha + \xi_2)\nabla^2 \tilde{\phi}_{p2} \tag{9.342}$$

where $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial z^2$. For the normal stress, we obtain

$$\tilde{\sigma}_{zz} = \lambda \frac{\partial^2 \tilde{\phi}_p}{\partial x^2} + (\lambda + 2G)\frac{\partial^2 \tilde{\phi}_p}{\partial z^2} + 2G\frac{\partial^2 \tilde{\psi}_s}{\partial x \partial z} - \alpha \tilde{p} \tag{9.343}$$

The above quantities are needed for describing the boundary conditions to be presented in the cases below.

9.10.2 *Reflection on Free Surface—Non-Dissipative Medium*

Given a poroelastic half space bounded by a horizontal plane (see Fig. 9.20), we choose a Cartesian coordinate system such that $z = 0$ represents the boundary, with the z-axis directed into the interior of the medium. Consider a plane wave in the x-z plane incident on the free surface at an incident angle θ. The incident wave can be one of the three types, P1, P2, and S. Generally speaking, even with one type of incident wave, three types of waves can be generated at the free surface, as shown in Fig. 9.20.

On the free surface, all surface tractions and pore pressure must vanish, leading to the following boundary conditions

$$\tilde{\sigma}_{zz} = \tilde{\sigma}_{xz} = \tilde{p} = 0, \quad \text{at} \quad z = 0 \tag{9.344}$$

Non-dissipative Medium Before investigating the problem for the real (dissipative) porous medium, we shall first examine the idealized case of a non-dissipative porous

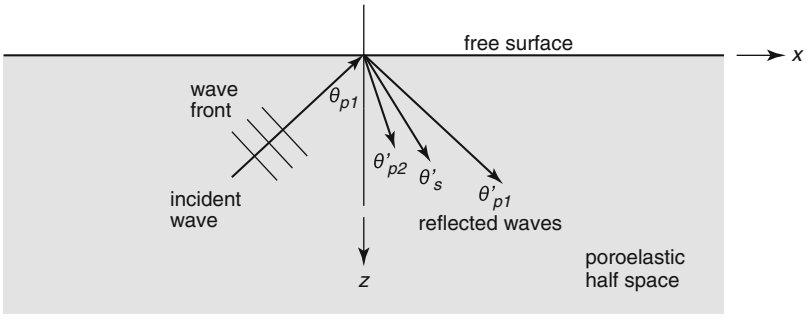


Fig. 9.20 Incident and reflection angles for an incident P1-wave

medium [19, 39], for its simplicity and to gain physical insight. Non-dissipative porous medium means that there is no loss of energy due to viscous dissipation. This can be accomplished by setting the fluid viscosity μ_f to zero. A direct consequence of this assumption is that the quantity ρ'' defined in (9.98) is reduced to

$$\rho'' = \rho' = \frac{\rho_f}{\phi} + \frac{\rho_a}{\phi^2} \tag{9.345}$$

which is real valued. Otherwise all previously derived results remain the same. As ρ'' was the only quantity that introduced an imaginary part to all results, the wave velocities v_s, v_{p1} and v_{p2} defined in (9.126), (9.137) and (9.138) now become real valued. In turn, the wave numbers ℓ_s, ℓ_{p1} and ℓ_{p2} are all real valued.

Incident P1-Wave Consider first the case of an incident P1-wave arriving at the free surface with the incident angle θ_{p1}. This wave can generate three reflected waves, S, P1 and P2, respectively with the reflection angles θ'_s, θ'_{p1}, θ'_{p2}, as shown in Fig. 9.20. By decomposing the wave number vector into its vector components $\boldsymbol{\ell} = \ell_x \hat{i} + \ell_z \hat{k}$, we can express the wave potential of an incident wave as

$$\tilde{\phi}_{p1} = A_{p1} \exp\left[i\,\ell_{p1}(x \sin\theta_{p1} - z\cos\theta_{p1})\right] \tag{9.346}$$

In the above, $\ell_{p1} = |\boldsymbol{\ell}_{p1}|$, with $\ell_{p1} = \omega/v_{p1}$ the scalar wave number. Similarly, the three reflected waves are expressed as

$$\tilde{\psi}'_s = A'_s \exp\left[i\,\ell_s(x \sin\theta'_s + z\cos\theta'_s)\right] \tag{9.347}$$

$$\tilde{\phi}'_{p1} = A'_{p1} \exp\left[i\,\ell_{p1}(x \sin\theta'_{p1} + z\cos\theta'_{p1})\right] \tag{9.348}$$

$$\tilde{\phi}'_{p2} = A'_{p2} \exp\left[i\,\ell_{p2}(x \sin\theta'_{p2} + z\cos\theta'_{p2})\right] \tag{9.349}$$

where we have used the prime superscript to denote all reflected wave components. We should note that in (9.346)–(9.349) we have assumed that all the reflected waves are of the same frequency as the incident wave. This condition is obvious when the boundary conditions are enforced. The complete wave field is then the sum of the above wave fields.

We can substitute the wave potentials (9.346)–(9.349) into the stress and pore pressure expressions (9.338)–(9.343), and enforce the boundary condition (9.344). However, before writing out the full expressions, we shall make the following observation: we find that in each of the terms there exists a factor $\exp[i\,\ell(x\sin\theta + z\cos\theta)]$. On the free surface, $z = 0$, the expression reduces to $\exp(i\,\ell\,x\sin\theta)$, which is a function of x. For the null boundary conditions to be matched at all x values, it is necessary that all the exponential terms have the same argument. In other words, we must have the condition

$$\ell_{p1}\sin\theta_{p1} = \ell_s \sin\theta'_s = \ell_{p1}\sin\theta'_{p1} = \ell_{p2}\sin\theta'_{p2} \tag{9.350}$$

which is known as *Snell's law*. From the above, we can immediately conclude that

$$\theta'_{p1} = \theta_{p1} \tag{9.351}$$

In other words, for the same type of wave (P1), the reflection angle is equal to the incident angle. We can also easily obtain the reflection angle for the other two waves

$$\theta'_s = \sin^{-1}\left(\frac{v_s}{v_{p1}}\sin\theta_{p1}\right) \tag{9.352}$$

$$\theta'_{p2} = \sin^{-1}\left(\frac{v_{p2}}{v_{p1}}\sin\theta_{p1}\right) \tag{9.353}$$

As we have demonstrated in Sect. 9.4.2 that the P1-wave is the fastest among the poroelastic waves, $v_{p1} > v_s$ and $v_{p1} > v_{p2}$, it can be easily shown that for any incident angle $0° < \theta_{p1} < 90°$, θ'_s and θ'_{p2} exist. Hence the incident P1-wave always generates three reflected waves, with the exception of *normal incidence*, $\theta_{p1} = 0°$, and the special case of *complete mode conversion*, in which the reflected P1-wave vanishes. These will be discussed below.

The constitutive equations (9.338), (9.342), (9.343), and the condition (9.350) allows us to set up the stress and pore pressure free boundary conditions (9.344) as

$$\ell^2_{p1}\sin 2\theta_{p1}R_{p1} + \ell^2_{p2}\sin 2\theta'_{p2}R_{p2} - \ell^2_s\cos 2\theta'_s R_s = \ell^2_{p1}\sin 2\theta_{p1} \tag{9.354}$$

$$\ell^2_{p1}(\alpha+\xi_1)R_{p1} + \ell^2_{p2}(\alpha+\xi_2)R_{p2} = -\ell^2_{p1}(\alpha+\xi_1) \tag{9.355}$$

$$\begin{aligned}&\ell^2_{p1}(\lambda+2G\cos^2\theta_{p1})R_{p1} + \ell^2_{p2}(\lambda+2G\cos^2\theta'_{p2})R_{p2}\\ &\quad+\ell^2_s G\sin 2\theta'_s R_s = -\ell^2_{p1}(\lambda+2G\cos^2\theta_{p1})\end{aligned} \tag{9.356}$$

where

$$R_{p1} = A'_{p1}/A_{p1}, \quad R_{p2} = A'_{p2}/A_{p1}, \quad R_s = A'_s/A_{p1} \tag{9.357}$$

are the amplitude ratios of the reflected waves as compared to the incident wave, also called the *reflection coefficient*. The above equations can be solved for

$$\begin{aligned}R_{p1} = -\big\{&G(\alpha+\xi_1)\cos[2(\theta'_{p2}-\theta'_s)] - G(\alpha+\xi_2)\cos[2(\theta_{p1}+\theta'_s)]\\ &+(\lambda+G)(\xi_1-\xi_2)\cos(2\theta'_s)\big\}\big/D\end{aligned} \tag{9.358}$$

$$R_{p2} = 2G(v^2_{p2}/v^2_{p1})(\alpha+\xi_1)\sin(2\theta_{p1})\sin(2\theta'_s)\big/D \tag{9.359}$$

$$\begin{aligned}R_s = -\big\{&2(v^2_s/v^2_{p1})[G(\alpha+\xi_1)\cos(2\theta'_{p2}) - G(\alpha+\xi_2)\cos(2\theta_{p1})\\ &+(\lambda+G)(\xi_1-\xi_2)]\sin(2\theta_{p1})\big\}\big/D\end{aligned} \tag{9.360}$$

where

$$D = G(\alpha + \xi_1)\cos[2(\theta'_{p2} - \theta'_s)] - G(\alpha + \xi_2)\cos[2(\theta_{p1} - \theta'_s)] + (\lambda + G)(\xi_1 - \xi_2)\cos(2\theta'_s) \tag{9.361}$$

In the following, we shall make a few observations based on the well-known results of elastic waves [1].

First, similar to the elasticity case, we observe that for non-dissipative medium, the reflection angle and amplitude are independent of the wave frequency (which is not the case for dissipative medium). Next we examine the case of a normal incident P1-wave. With $\theta_{p1} = 0°$, we clearly see from (9.352) and (9.353) that $\theta'_s = \theta'_{p2} = 0$. Substituting into (9.358)–(9.361), we obtain

$$R_{p1} = -1, \qquad R_{p2} = R_s = 0 \tag{9.362}$$

This means that the P1-wave reflects on itself and no other wave is generated. Based on (9.346) and (9.348), we obtain

$$\begin{aligned} \Re\{\tilde{\phi}_{p1} + \tilde{\phi}'_{p1}\} &= \Re\left\{A_{p1}\left[\exp(-i\ell_{p1}z) - \exp(i\ell_{p1}z)\right]\exp(-i\omega t)\right\} \\ &= -2\sin(\ell_{p1}z)\sin(\omega t) \end{aligned} \tag{9.363}$$

This further indicates that the wave becomes a *standing wave*.

We also note that a complete mode conversion, that is, the incident P1-wave vanishes and the other two types of waves are generated, is possible if the numerator of (9.358) is identically zero

$$G(\alpha + \xi_1)\cos[2(\theta'_{p2} - \theta'_s)] - G(\alpha + \xi_2)\cos[2(\theta_{p1} + \theta'_s)] + (\lambda + G)(\xi_1 - \xi_2)\cos(2\theta'_s) = 0 \tag{9.364}$$

As θ'_s and θ'_{p2} are functions of θ_{p1}, the above equation can be used to find roots of θ_{p1} for a given set of material constants.

In Fig. 9.21a we plot the reflection angles θ'_s and θ'_{p2} versus the incident angle θ_{p1} using material properties of Pecos sandstone. In Fig. 9.21b, the amplitude ratios as defined in (9.358), (9.359), and (9.360) are presented versus incident angle. We observe that for this material, the three reflected waves always exist within $0 < \theta_{p1} < 90°$, and the complete mode conversion of P1-wave does not occur. For elastic material, complete mode conversion can exist only if $\nu < 0.263$. For poroelastic material, the P1-wave velocity is dominated by the undrained Poisson ratio ν_u. Table 3.2 shows that for poroelastic materials saturated with water, ν_u is generally greater than 0.263; hence complete mode conversion does not occur.

Incident P2-Wave For an incident P2-wave, the incident potential is

$$\tilde{\phi}_{p2} = A_{p2}\exp\left[i\,\ell_{p2}(x\sin\theta_{p2} - z\cos\theta_{p2})\right] \tag{9.365}$$

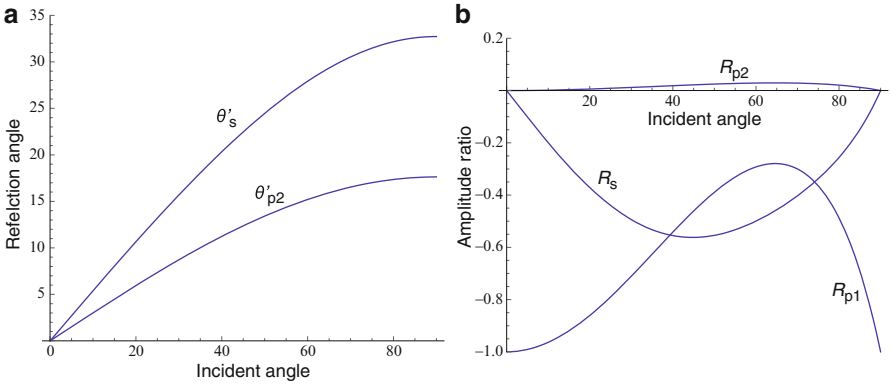


Fig. 9.21 P1 incident wave on free surface of a Pecos sandstone. (**a**) Reflection angle for P2 and S-waves, and (**b**) reflection amplitude ratio $R_{p1},\ R_{p2},\ R_s$, as functions of incident angle θ_{p1}

Based on Snell's law, the reflection angles are

$$\theta'_{p2} = \theta_{p2} \tag{9.366}$$

and

$$\theta'_s = \sin^{-1}\left(\frac{v_s}{v_{p2}}\sin\theta_{p2}\right) \tag{9.367}$$

$$\theta'_{p1} = \sin^{-1}\left(\frac{v_{p1}}{v_{p2}}\sin\theta_{p2}\right) \tag{9.368}$$

As $v_{p2} < v_{p1}$, there exists a critical incident angle θ^{cp}_{p2} such that P1-wave in not reflected when $\theta_{p2} > \theta^{cp}_{p2}$. This angle is easily found by setting θ'_{p1} to 90° in (9.368) and solving for θ_{p2}; we thus obtain

$$\theta^{cp}_{p2} = \sin^{-1}\left(\frac{v_{p2}}{v_{p1}}\right) \tag{9.369}$$

In addition, if $v_{p2} < v_s$, as it is generally the case, there exists a second critical angle

$$\theta^{cs}_{p2} = \sin^{-1}\left(\frac{v_{p2}}{v_s}\right) \tag{9.370}$$

such that when $\theta_{p2} > \theta^{cs}_{p2}$, only the P2-wave is reflected.

In Fig. 9.22a we plot the reflection angles θ'_s and θ'_{p1} versus the incident angle θ_{p2}, again using Pecos sandstone as an example. We observe that the critical angle for the P1-wave is 17.6°, and for S-wave is 34.1°.

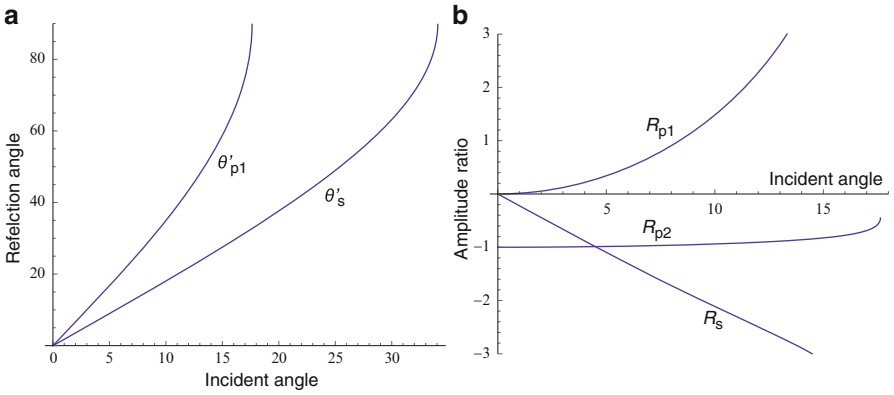


Fig. 9.22 P2 incident wave on free surface of a Pecos sandstone. (**a**) Reflection angle for P1 and S-waves, and (**b**) reflection amplitude ratio $R_{p1},\ R_{p2},\ R_s$, as functions of incident angle θ_{p2}

The reflected wave potentials are the same as (9.347), (9.348), and (9.349). Matching the same boundary conditions leading to the same equations as (9.354), (9.355), and (9.356), except that the right hand sides are replaced by

$$\ell_{p2}^2 \sin 2\theta_{p2}, \quad -\ell_{p2}^2(\alpha+\xi_2), \quad -\ell_{p2}^2(\lambda+2G\cos^2\theta_{p2}) \tag{9.371}$$

respectively. The amplitude ratios can then be solved as

$$R_{p1} = -\,2G(v_{p1}^2/v_{p2}^2)(\alpha+\xi_2)\sin(2\theta_{p2})\sin(2\theta_s')\big/ D \tag{9.372}$$

$$\begin{aligned} R_{p2} = -\big\{&G(\alpha+\xi_1)\cos[2(\theta_{p2}+\theta_s')] - G(\alpha+\xi_2)\cos[2(\theta_{p1}'-\theta_s')] \\ &+(\lambda+G)(\xi_1-\xi_2)\cos(2\theta_s')\big\}\big/ D \end{aligned} \tag{9.373}$$

$$\begin{aligned} R_s = -\big\{&2(v_s^2/v_{p2}^2)[G(\alpha+\xi_1)\cos(2\theta_{p2}) - G(\alpha+\xi_2)\cos(2\theta_{p1}') \\ &+(\lambda+G)(\xi_1-\xi_2)]\sin(2\theta_{p2})\big\}\big/ D \end{aligned} \tag{9.374}$$

where the reflected amplitudes are normalized by the incident amplitude A_2. Similar to the P1-wave case, at normal incidence, only the P2-wave is reflected. Also, complete mode conversion can occur if the numerator of (9.373) becomes zero.

In Fig. 9.22b, the amplitude ratios are plotted versus the incident angle. We observe the amplitude ratios no longer exist after the smaller of the critical angle, $\theta_{p2}^{cp} = 17.6°$, because they becomes complex quantities, and a *homogeneous wave* no longer exist. We shall discuss the *inhomogeneous waves* in Sect. 9.10.3

Incident S-Wave For an incident S-wave, the incident potential is

$$\tilde{\psi}_s = A_s \exp\left[i\,\ell_s(x\sin\theta_s - z\cos\theta_s)\right] \tag{9.375}$$

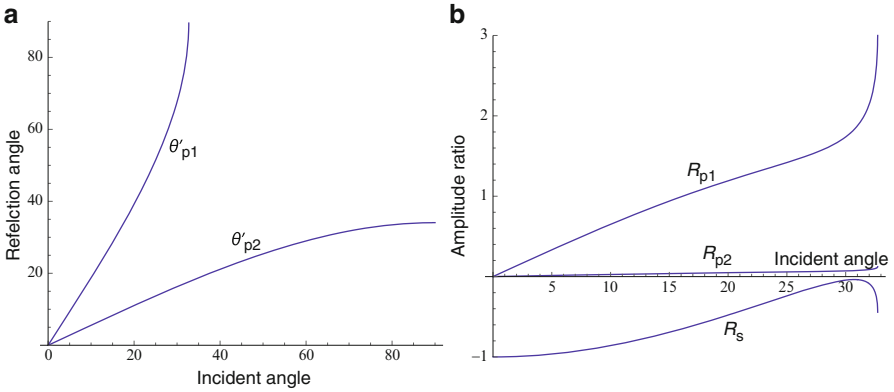


Fig. 9.23 S incident wave on free surface of a Pecos sandstone. (**a**) Reflection angle for P1 and P2-waves, and (**b**) reflection amplitude ratio $R_{p1},\ R_{p2},\ R_s$, as functions of incident angle θ_s

Snell's law gives

$$\theta_s' = \theta_s \tag{9.376}$$

and

$$\theta_{p1}' = \sin^{-1}\left(\frac{v_{p1}}{v_s}\sin\theta_s\right) \tag{9.377}$$

$$\theta_{p2}' = \sin^{-1}\left(\frac{v_{p2}}{v_s}\sin\theta_s\right) \tag{9.378}$$

Critical incident angle beyond which P1-wave is not reflected is

$$\theta_s^c = \sin^{-1}\left(\frac{v_s}{v_{p1}}\right) \tag{9.379}$$

In Fig. 9.23a we plot the reflection angles θ_{p1}' and θ_{p2}' versus the incident angle θ_s. We observe a critical angle for the P1-wave at 32.7°.

For fulfilling the boundary conditions, (9.354), (9.355), and (9.356) still hold, except that the right hand sides are replaced by

$$\ell_s^2\cos 2\theta_s, \quad 0, \quad \ell_s^2 G\sin 2\theta_s \tag{9.380}$$

The amplitude ratios, normalized by A_s, are solved as

$$R_{p1} = -\,G(v_{p1}^2/v_s^2)(\alpha+\xi_2)\sin(4\theta_s)\big/\,D \tag{9.381}$$

$$R_{p2} = \ G(v_{p2}^2/v_s^2)(\alpha+\xi_1)\sin(4\theta_s)\big/\,D \tag{9.382}$$

$$R_s = -\big\{G(\alpha+\xi_1)\cos[2(\theta'_{p2}+\theta_s)] - G(\alpha+\xi_2)\cos[2(\theta'_{p1}+\theta_s)] + (\lambda+G)(\xi_1-\xi_2)\cos(2\theta_s)\big\}/D \tag{9.383}$$

Again, at normal incidence, only the S-wave is reflected. Complete mode conversion can occur if the numerator of (9.383) becomes zero. Same as elasticity, a *zero mode conversion* can occur at incident angle $\theta_s = 45°$, based on (9.381) and (9.382). At that angle, both P1 and P2-waves vanish, and only the S-wave is reflected. The amplitude ratios are plotted in Fig. 9.23b. Similar to elasticity, homogeneous wave no longer exists after the critical angle. For elasticity, it can be shown that the amplitude ratio $R_s = -1$ at critical angle [1]. This is however not the case for the poroelastic wave. For Pecos sandstone, we find $R_s = -0.451$.

9.10.3 Reflection on Free Surface—Dissipative Medium

For dissipative medium, we can follow the same approach as in the preceding sections, except that the wave number vectors in (9.332), (9.333), and (9.334) are complex numbers. We can write a wave number vector in terms of its real and imaginary components

$$\boldsymbol{\ell} = \boldsymbol{\ell}_r + i\boldsymbol{\ell}_i \tag{9.384}$$

We note that $\boldsymbol{\ell}_r$ and $\boldsymbol{\ell}_i$ are each a real vector, and they are not necessarily in the same direction. The general solution of the wave equation can then be expressed as

$$\tilde{\phi} = A\exp(i\boldsymbol{\ell}\cdot\boldsymbol{x}) = A\exp(i\boldsymbol{\ell}_r\cdot\boldsymbol{x})\exp(-\boldsymbol{\ell}_i\cdot\boldsymbol{x}) \tag{9.385}$$

We observe that $\boldsymbol{\ell}_r$ is in the direction normal to the planes of equal phase, and $\boldsymbol{\ell}_i$ is in the direction normal to the planes of equal amplitude. We hence can define two angles, of which

$$\theta_{ph} = \arg(\boldsymbol{\ell}_r) \tag{9.386}$$

is the angle of propagation of plane of equal phase, and

$$\theta_a = \arg(\boldsymbol{\ell}_i) \tag{9.387}$$

is the angle of propagation of equal amplitude (the direction of maximum wave attenuation). We may also call

$$\ell_r = |\boldsymbol{\ell}_r|, \qquad \ell_i = |\boldsymbol{\ell}_i| \tag{9.388}$$

respectively the equal phase and equal amplitude wave numbers. When $\boldsymbol{\ell}_r$ and $\boldsymbol{\ell}_i$ are in the same direction, that is, $\theta_{ph} = \theta_a$, the wave is called a *homogeneous wave*; or else it is an *inhomogeneous wave*.

We assume that the incident wave, either a P1, P2, or S-wave, is homogeneous. This allows the incident potential to be expressed as

$$\tilde{\phi} = A \exp\left[i\,\ell(x\sin\theta - z\cos\theta)\right] \tag{9.389}$$

where ℓ is a scalar, no longer a vector. In the case of an S-wave, ϕ is replaced by ψ. The above expression of incident potential is the same as the non-dissipative medium case, such as (9.346), except that the wave number is complex, and no longer real.

The reflected waves are generally inhomogeneous, and we can write the complex wave number vector as

$$\boldsymbol{\ell} = \ell\,\boldsymbol{n} = \ell\,(n_x\hat{i} + n_z\hat{k}) \tag{9.390}$$

where $\boldsymbol{n}$ is a unit vector in the direction of $\boldsymbol{\ell}$, with $n_x^2 + n_z^2 = 1$, and $\ell = |\boldsymbol{\ell}|$. These two vector components n_x and n_z are complex quantities. The phase angle defined in (9.386) for the reflected wave is then given by the real parts of the vector components as

$$\theta' = \tan^{-1}\frac{\Re(n_z)}{\Re(n_x)} \tag{9.391}$$

As before, we shall use the prime to denote quantities associated with the reflected waves. The phase velocity, that is, the velocity in the direction perpendicular to the equal phase planes, is found as

$$v_{ph} = \frac{\omega}{|\ell_r|} = \frac{\omega}{|\ell|\sqrt{\Re(n_x)^2 + \Re(n_z)^2}} \tag{9.392}$$

So the phase velocity is dependent not only on material properties and frequency, but also on the reflection angle. For real materials, however, the dependence is negligible, as the imaginary part of the complex velocity is much smaller than the real part, such that $v_{ph} \approx \omega/|\ell|$.

For convenience, we shall express the vector components of $\boldsymbol{n}$ as

$$n_x = \cos\alpha', \qquad n_z = \sin\alpha' \tag{9.393}$$

This allows us to write the reflected wave potentials as

$$\tilde{\phi}' = A' \exp\left[i\,\ell\,(x\cos\alpha' + z\sin\alpha')\right] \tag{9.394}$$

Comparing the above with the reflected wave potentials for the non-dissipative case, (9.347), (9.348), and (9.349), we see a formal correspondence between the potentials if θ' is replaced by α'. The only difference is that, as n_x and n_z are complex numbers, $\cos\alpha'$ and $\sin\alpha'$ must also be complex numbers. This means that α' is complex, not real.

The purpose of the above exercise is to reduce the mathematical form of the dissipative case to that of the non-dissipative case. Once we build such correspondence, all the solutions presented in Sect. 9.10.2 remain valid. The only difference is that we need to use ρ'' instead of ρ' to calculate the complex velocities v_{p1}, v_{p2} and v_s, instead of real velocities. These lead to complex wave numbers instead of real wave numbers. In the end, we obtain the complex angle α' and the complex amplitude ratio R. From the complex angle α' we then extract the angle of equal phase θ' based on (9.391) and (9.393).

Reflection Angle For a given incident wave, P1, P2 or S, the reflection angles are obtained from Snell's law. Take, for example, a P1 incident, we find

$$\ell_{p1}\sin\theta_{p1} = \ell_s\sin\alpha'_s = \ell_{p1}\sin\alpha'_{p1} = \ell_{p2}\sin\alpha'_{p2} \tag{9.395}$$

From the above, we immediately conclude that

$$\alpha'_{p1} = \theta_{p1} \tag{9.396}$$

As θ_{p1} is real, the reflection angle for P1-wave is then the same as the incident angle,

$$\theta'_{p1} = \theta_{p1} \tag{9.397}$$

Also, the reflected P1-wave remains homogeneous. The above conclusion can be extended to the other two types of incident waves, P2 and S; that is, waves of the same type reflect at the same angle and remains homogeneous.

The above is not true for reflected waves of different types. First, we can obtain α' for the other two waves from these relations

$$\alpha'_s = \sin^{-1}\left(\frac{v_s}{v_{p1}}\sin\theta_{p1}\right) \tag{9.398}$$

$$\alpha'_{p2} = \sin^{-1}\left(\frac{v_{p2}}{v_{p1}}\sin\theta_{p1}\right) \tag{9.399}$$

As the velocities are complex, it is obvious that α'_s and α'_{p2} are also complex. In addition, they are frequency dependent. Based on (9.391), we can calculate the reflection angles θ'_{p2} and θ'_s as function of incident angle θ_{p1} for the material Pecos sandstone, at different frequencies, as shown in Fig. 9.24a, b.

Figure 9.24a presents the reflected P2-wave at frequencies 10 Hz, 1 kHz, and 100 kHz. We observe that in these frequency ranges, the P2 wave is nearly reflected perpendicular to the free surface. The reason is that Pecos sandstone has very low

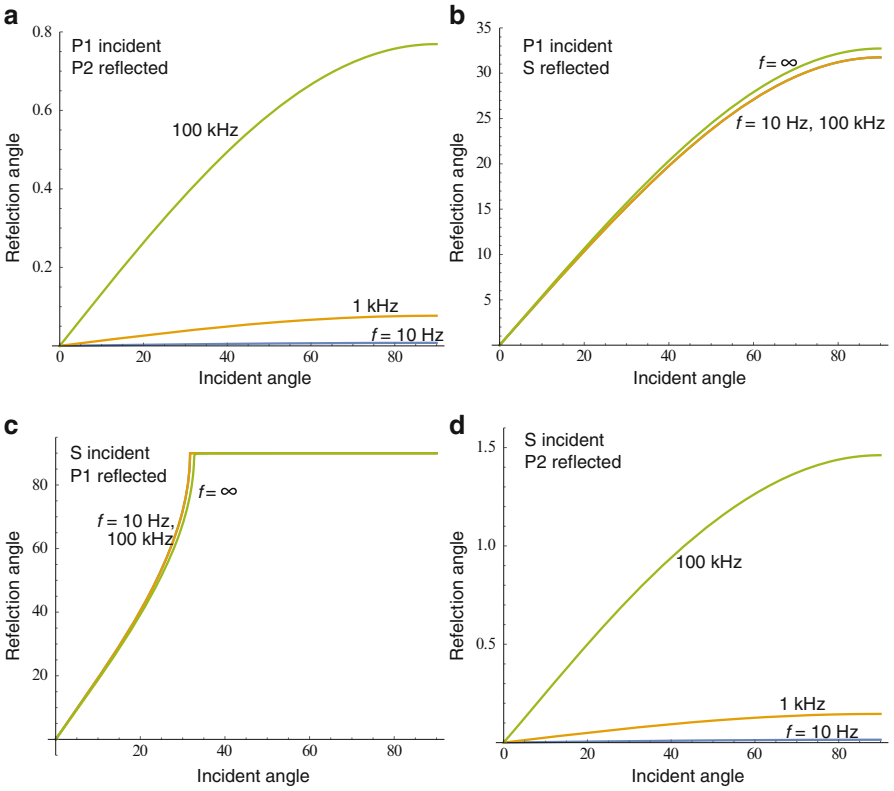


Fig. 9.24 Reflection versus incident angle for Pecos sandstone at various frequencies. (**a**) P1 incident, P2 reflected; (**b**) P1 incident, S reflected; (**c**) S incident, P1 reflected; and (**d**) S incident, P2 reflected

permeability and the calculated velocity at $f = 10\,\text{Hz}$ is $v_{p2} = 0.408 - 0.408\,i$ m/s. This is compared to $v_{p2} = 924$ m for the non-dissipative (infinite permeability) case.

Figure 9.24b shows the reflected S-wave at these frequencies. The S-wave reflection angle is only slightly dependent on frequency. The curves for frequency range from 10 Hz to 100 kHz pretty much fall on each other. In the same figure, we also show the curve for the non-dissipative case, as presented in Fig. 9.21a. This curve corresponds to the asymptotic case of $f \to \infty$.

The case of an S incident wave is presented as Fig. 9.24c, d. We observe in Fig. 9.24c that the reflected P1 wave is only slightly frequency dependent within the range of frequency of 10 Hz to 100 kHz. In the figure we also plot the asymptotic case of $f \to \infty$ which converges to the non-dissipative medium case. Of particular interest to observe is that a critical angle does not exist for the P1 wave, although beyond the critical angle of the non-dissipative case, the reflection angle is nearly 90°. Figure 9.24d shows the reflected P2-wave. Similar to the P1 incident case, the P2 wave is reflected nearly perpendicular to the free surface.

Amplitude Ratio The solutions of amplitude ratio of the reflected over incident waves for dissipative medium are the same as those obtained for the non-dissipative medium, as defined in Sect. 9.10.2, except that they are no longer real valued.

In Fig. 9.25 we plot the moduli of amplitude ratios $|R_{p1}|$, $|R_{p2}|$, and $|R_s|$, which give a comparison of the reflected versus incident wave magnitude, due to P1 and S incident waves. Figure 9.25a–c refer to a P1 incident wave. We observe that the effect of dissipation on the amplitude of the reflected P1 and S-waves is relatively small as compared to the limiting case $f \to \infty$, which is the same as the non-dissipative case, $\kappa \to \infty$. The curves corresponding to frequency between 10 Hz to 100 kHz are indistinguishable from each other. The dissipation effect on the reflected P2-wave, however, is strong. The reflected P2-wave is so small that it is almost non-existent, due to the low permeability of Pecos sandstone. Figure 9.25b is plotted in log-scale to highlight the small values. In the limit, $f \to \infty$, the maximum magnitude of reflected P2-wave is about 3 % of the incident P1-wave.

Figure 9.25d–f show the results of an S incident wave. The response is more complex due to the existence of critical angle and the angle of zero mode conversion. For the reflected P1 and P2-waves, we observe the peak amplitude response at the critical angle, which can be compared to the non-dissipative case, Fig. 9.23b. For the reflected S-wave, the minimum amplitude takes place before the critical angle. At zero mode conversion angle, $\theta_s = 45°$, both R_{p1} and R_{p2} become zero, while R_s takes the value of 1.

9.10.4 Impermeable Surface

Next we examine wave reflection on a clamped and impermeable surface. On a clamped surface, both displacement components vanish, and the z-component relative fluid to solid displacement also vanish. The boundary conditions are

$$\tilde{u}_x = \tilde{u}_z = \tilde{w}_z = 0, \quad \text{at} \quad z = 0 \tag{9.400}$$

Incident P1-Wave The relations among the incident and reflection angles are unaffected by the boundary conditions, and are given by (9.351), (9.352), and (9.353). For fulfilling the boundary conditions, we utilize the constitutive equations (9.336), (9.337) and (9.341), and obtain

$$\ell_{p1} \sin\theta_{p1} R_{p1} + \ell_{p2} \sin\theta'_{p2} R_{p2} - \ell_s \cos\theta'_s R_s = -\ell_{p1} \sin\theta_{p1} \tag{9.401}$$

$$\ell_{p1} \cos\theta_{p1} R_{p1} + \ell_{p2} \cos\theta'_{p2} R_{p2} + \ell_s \sin\theta'_s R_s = \ell_{p1} \cos\theta_{p1} \tag{9.402}$$

$$\ell_{p1}\xi_1 \cos\theta_{p1} R_{p1} + \ell_{p2}\xi_2 \cos\theta'_{p2} R_{p2} + \ell_s \frac{\rho_f}{\rho''} \sin\theta'_s R_s = \ell_{p1}\xi_1 \cos\theta_{p1} \tag{9.403}$$

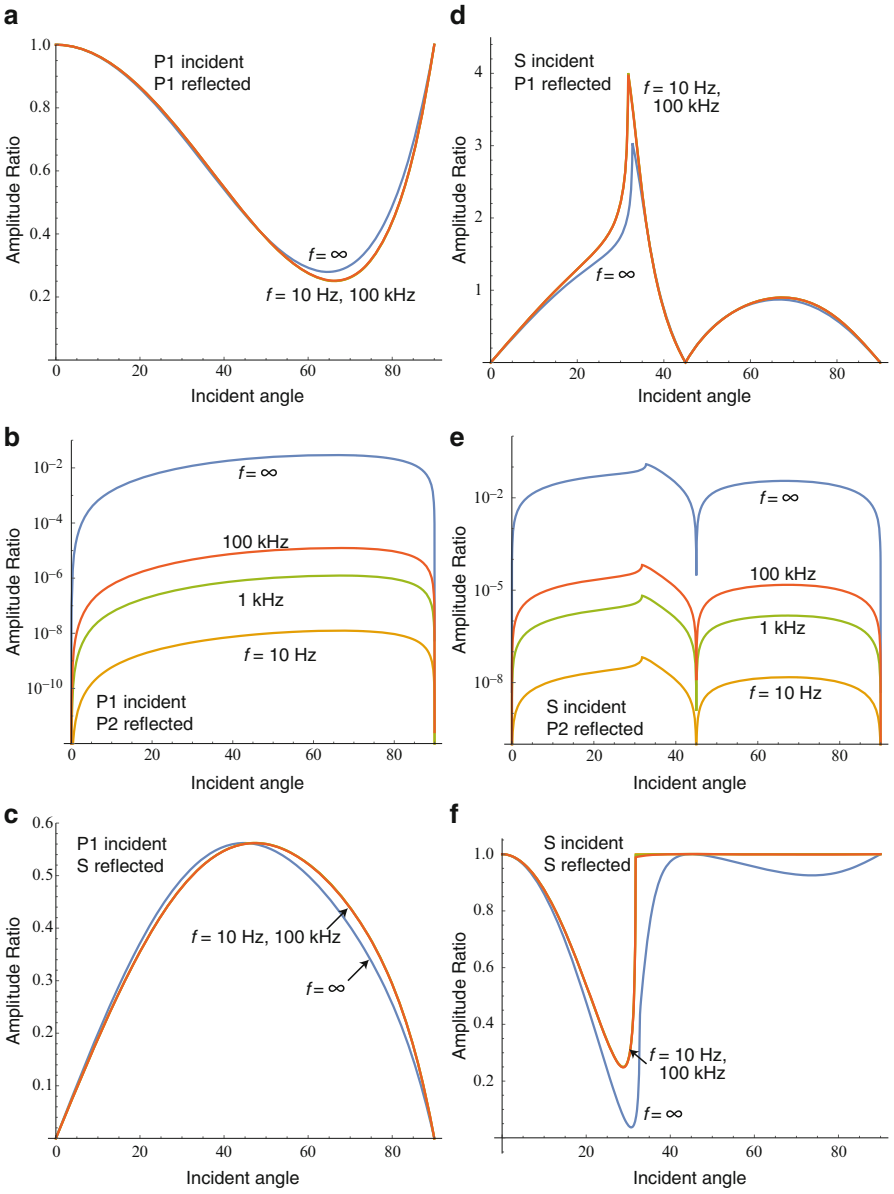


Fig. 9.25 Reflection on free surface: modulus of reflected amplitude ratio, $|R_{p1}|$, $|R_{p2}|$, and $|R_s|$, for Pecos sandstone, at various frequencies. (**a**) P1 incident, P1 reflected; (**b**) P1 incident, P2 reflected; (**c**) P1 incident, S reflected; (**d**) S incident, P1 reflected; (**e**) S incident, P2 reflected; and (**f**) S incident, S reflected

The above can be solved for

$$R_{p1} = \big[(\rho_f - \rho''\xi_1)\cos\theta_{p1}\sin\theta'_{p2}\sin\theta'_s + (\rho_f - \rho''\xi_2)\sin\theta_{p1}\cos\theta'_{p2}\sin\theta'_s - \rho''(\xi_1 - \xi_2)\cos\theta_{p1}\cos\theta'_{p2}\cos\theta'_s\big]\big/ D \tag{9.404}$$

$$R_{p2} = -(v_{p2}/v_{p1})(\rho_f - \rho''\xi_1)\sin 2\theta_{p1}\sin\theta'_s\big/ D \tag{9.405}$$

$$R_s = -(v_s/v_{p1})(\xi_1 - \xi_2)\sin 2\theta_{p1}\cos\theta'_{p2}\big/ D \tag{9.406}$$

where

$$D = (\rho_f - \rho''\xi_1)\cos\theta_{p1}\sin\theta'_{p2}\sin\theta'_s - (\rho_f - \rho''\xi_2)\sin\theta_{p1}\cos\theta'_{p2}\sin\theta'_s - \rho''(\xi_1 - \xi_2)\cos\theta_{p1}\cos\theta'_{p2}\cos\theta'_s \tag{9.407}$$

Incident S-Wave For an incident S-wave, we replace the right hand sides of (9.401), (9.402), and (9.403) by

$$-\ell_s\cos\theta_s, \quad -\ell_s\sin\theta_s, \quad -\frac{\rho_f}{\rho''}\ell_s\sin\theta_s \tag{9.408}$$

Amplitude ratios are obtained as

$$R_{p1} = \big[(\rho_f - \rho''\xi_1)\cos\theta'_{p1}\sin\theta'_{p2}\sin\theta_s + (\rho_f - \rho''\xi_2)\sin\theta'_{p1}\cos\theta'_{p2}\sin\theta_s - \rho''(\xi_1 - \xi_2)\cos\theta'_{p1}\cos\theta'_{p2}\cos\theta_s\big]\big/ D \tag{9.409}$$

$$R_{p2} = -(v_{p2}/v_{p1})(\rho_f - \rho''\xi_1)\sin 2\theta'_{p1}\sin\theta_s\big/ D \tag{9.410}$$

$$R_s = -(v_s/v_{p1})(\xi_1 - \xi_2)\sin 2\theta'_{p1}\cos\theta'_{p2}\big/ D \tag{9.411}$$

Moduli of these amplitude ratios are presented in Fig. 9.26.

9.10.5 Fluid and Porous Medium Interface

The problem of acoustic wave reflection and refraction on fluid and porous medium interface is of importance to the study of sound propagation in shallow ocean, where the interaction with the sea bottom plays a role, as well as to the geoacoustic exploration of continental shelves, where ocean water lays on top [12, 60]. In this section we shall examine the reflection and refraction of an ocean acoustic wave incident on the ocean-sediment interface [11, 116]. Figure 9.27 gives an illustration of such situation.

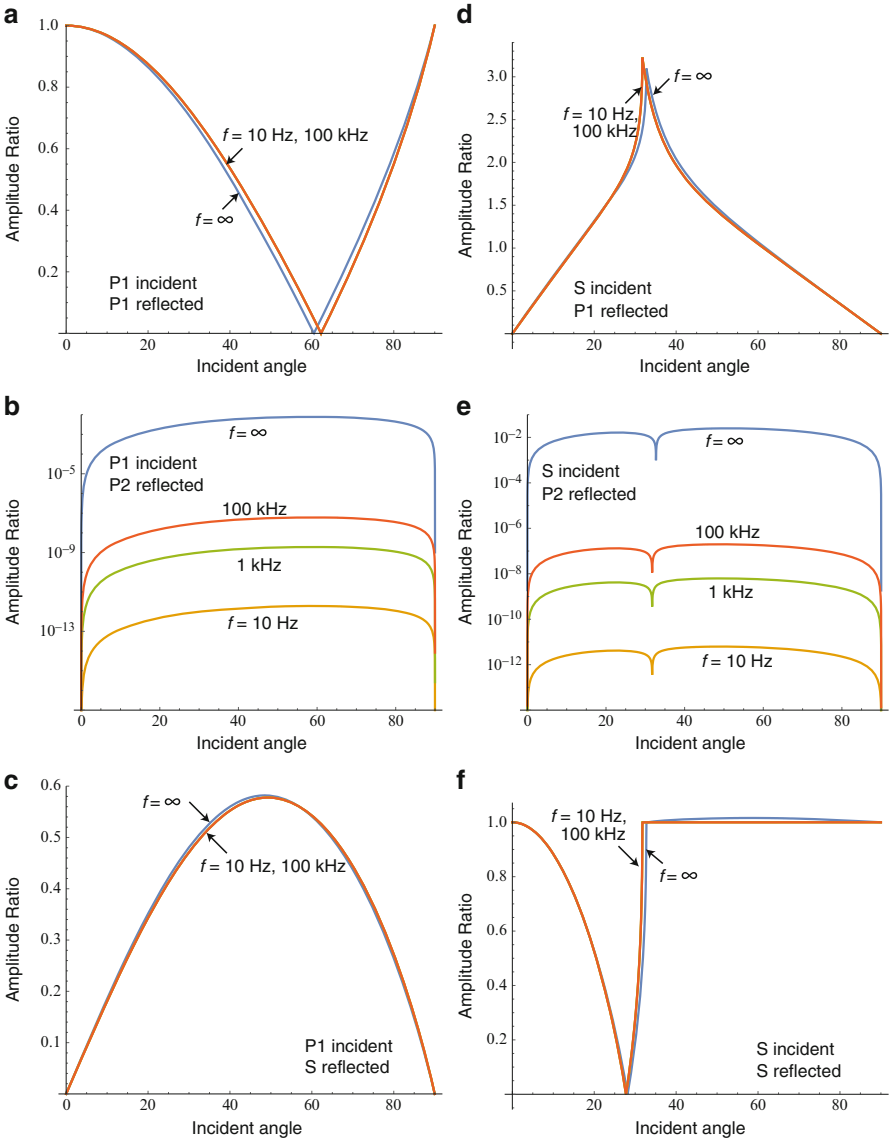


Fig. 9.26 Reflection on clamped impermeable surface: modulus of reflected amplitude ratio, $|R_{p1}|$, $|R_{p2}|$, and $|R_s|$, for Pecos sandstone, at various frequencies. (**a**) P1 incident, P1 reflected; (**b**) P1 incident, P2 reflected; (**c**) P1 incident, S reflected; (**d**) S incident, P1 reflected; (**e**) S incident, P2 reflected; and (**f**) S incident, S reflected

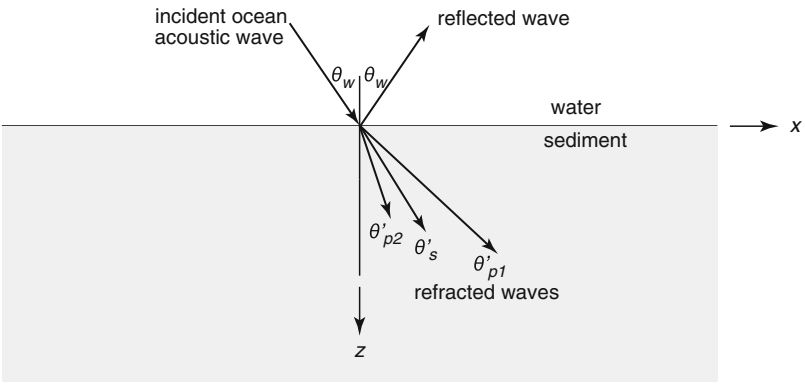


Fig. 9.27 The reflection and refraction of an ocean acoustic wave incident on sea bottom

For the fluid region, we define a fluid displacement $\boldsymbol{u}^w$, which can be expressed as the gradient of a potential

$$\boldsymbol{u}^w = \nabla \phi_w \tag{9.412}$$

The fluid pressure is given by

$$p_w = -K_f \nabla \cdot \boldsymbol{u}^w = -K_f \nabla^2 \phi_w \tag{9.413}$$

Similar to the derivation in Sects. 9.10.2 and 9.10.3, we define an incident displacement potential for acoustic wave in ocean as

$$\tilde{\phi}_w = A_w \exp\left[i\,\ell_w (x \sin\theta_w + z \cos\theta_w)\right] \tag{9.414}$$

where $\ell_w = \omega / v_w$ is the wave number of the ocean acoustic wave, and

$$v_w = \sqrt{\frac{K_f}{\rho_f}} \tag{9.415}$$

is the acoustic wave velocity in ocean. Here v_w and ℓ_w are real numbers. The reflected wave potential in ocean is given by

$$\tilde{\phi}'_w = A'_w \exp\left[i\,\ell_w (x \sin\theta_w - z \cos\theta_w)\right] \tag{9.416}$$

In the above we have already imposed the condition that the reflection angle is equal to the incident angle. The refracted wave potentials in the sediment are the same as those defined in (9.347), (9.348), and (9.349), except that the real valued angles θ' should be replaced by the complex valued α'. The total wave potential in each medium is the sum of these incident, reflected, and refracted waves within the respective medium.

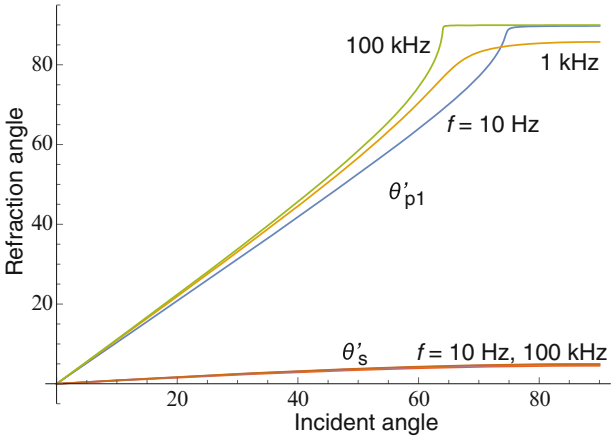


Fig. 9.28 Refraction angles for P1 and S-wave in porous medium for incident ocean acoustic wave on hard sediment surface

Before introducing the boundary conditions, we can first study the refraction angles in porous medium. Again we can apply Snell's law, which gives

$$\alpha'_{p1} = \sin^{-1}\left(\frac{v_{p1}}{v_w}\sin\theta_w\right) \tag{9.417}$$

$$\alpha'_{p2} = \sin^{-1}\left(\frac{v_{p2}}{v_w}\sin\theta_w\right) \tag{9.418}$$

$$\alpha'_{s} = \sin^{-1}\left(\frac{v_{s}}{v_w}\sin\theta_w\right) \tag{9.419}$$

From the above we can obtain the angle of refraction of equal phase, θ'_{p1}, θ'_{p2} and θ'_s. In Fig. 9.28 we plot the refraction angles for the P1 and S-wave using the data set of a hard sediment as reported in Tables 3.1 and 3.2, and Sect. 9.5.1. First, we observe that the refraction angle for the P1-wave is sensitive to the frequency, as a consequence of the relatively high permeability. The S-wave, on the other hand, is refracted nearly vertical to the surface due to the small shear modulus and shear wave velocity of the sediment.

The reflection and refraction coefficients can be found by matching the boundary conditions. At the water-sediment interface, the three stress continuity boundary conditions need to be satisfied:

$$\tilde{\sigma}_{zz} = -\tilde{p}_w, \quad \tilde{\sigma}_{xz} = 0, \quad \tilde{p} = \tilde{p}_w, \quad \text{at} \quad z = 0 \tag{9.420}$$

where p_w is the fluid pressure in the ocean. In addition, a fourth boundary condition on the continuity of vertical displacement at the interface is needed

$$\tilde{u}^w_z = (1-\phi)\tilde{u}^s_z + \phi\,\tilde{u}^f_z = \tilde{u}_z + \phi\,\tilde{w}_z, \quad \text{at} \quad z = 0 \tag{9.421}$$

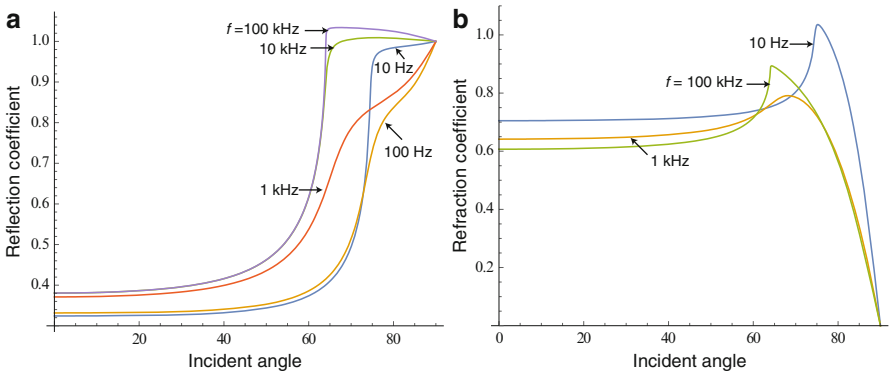


Fig. 9.29 Reflection and refraction coefficients for ocean acoustic wave incident on hard sediment surface at various frequencies. (**a**) Reflection coefficient of ocean acoustic wave; and (**b**) refraction coefficient for porous medium P1-wave

Based on these conditions, we can construct the following set of linear equations in terms of amplitude ratios:

$$\ell_{p1}^2 \sin 2\theta'_{p1} R_{p1} + \ell_{p2}^2 \sin 2\theta'_{p2} R_{p2} - \ell_s^2 \cos 2\theta'_s R_s = 0 \tag{9.422}$$

$$\ell_{p1}^2 M(\alpha + \xi_1) R_{p1} + \ell_{p2}^2 M(\alpha + \xi_2) R_{p2} - \ell_w^2 K_f R_w = \ell_w^2 K_f \tag{9.423}$$

$$\ell_{p1}^2 (\lambda + 2G\cos^2\theta_{p1}) R_{p1} + \ell_{p2}^2 (\lambda + 2G\cos^2\theta'_{p2}) R_{p2} + \ell_s^2 G \sin 2\theta'_s R_s + \ell_w^2 (1-\alpha) K_f R_w = -\ell_w^2 (1-\alpha) K_f \tag{9.424}$$

$$\ell_{p1}(1 + \phi\xi_1)\cos\theta_{p1} R_{p1} + \ell_{p2}(1 + \phi\xi_2)\cos\theta'_{p2} R_{p2} + \ell_s \left(1 + \frac{\phi\rho_f}{\rho''}\right) \sin\theta'_s R_s + \ell_w \cos\theta_w R_w = \ell_w \cos\theta_w \tag{9.425}$$

The above equations can be solved for the amplitude ratios R_{p1}, R_{p2}, R_s and R_w, using a computer algebra program. The expressions however are somewhat lengthy; hence are not listed here.

Figures 9.29a, b respectively plot the reflection coefficient $|R_w|$ of the ocean acoustic wave, and the refraction coefficient $|R_{p1}|$ of the porous medium P1-wave. We observe that the reflection of the ocean acoustic wave is strongly dependent on the frequency. We should remark that if the ocean bottom is modeled as an elastic medium, then the reflection coefficient becomes independent of the wave frequency. This shows the importance of correctly modeling ocean bottom as a poroelastic material. Figure 9.29b also shows the strong frequency dependence of the refracted P1-wave.

References

1. Achenbach JD (1973) Wave propagation in elastic solids. North-Holland, Amsterdam, 439pp
2. Achenbach JD (2003) Reciprocity in elastodynamics. Cambridge University Press, Cambridge/New York, 255pp
3. Abramowitz M, Stegun IA (1965) Handbook of mathematical functions: with formulas, graphs, and mathematical tables. Dover, New York, 1046pp
4. Aris R (1990) Vectors, tensors and the basic equations of fluid mechanics. Dover, New York, 320pp
5. Attenborough K (1982) Acoustical characteristics of porous materials. Phys Rep—Rev Sect Phys Lett 82(3):179–227
6. Attenborough K (1987) On the acoustic slow wave in air-filled granular media. J Acoust Soc Am 81(1):93–102
7. Auriault JL (1980) Dynamic behavior of a porous medium saturated by a Newtonian fluid. Int J Eng Sci 18(6):775–785
8. Auriault JL, Borne L, Chambon R (1985) Dynamics of porous saturated media, checking of the generalized law of Darcy. J Acoust Soc Am 77(5):1641–1650
9. Aznárez JJ, Maeso O, Domínguez J (2006) BE analysis of bottom sediments in dynamic fluid-structure interaction problems. Eng Anal Bound Elem 30(2):124–136
10. Badiey M, Cheng AHD, Mu YK (1998) From geology to geoacoustics—evaluation of Biot-Stoll sound speed and attenuation for shallow water acoustics. J Acoust Soc Am 103(1):309–320
11. Badiey M, Jaya I, Cheng AHD (1994) Propagator matrix for plane wave reflection from inhomogeneous anisotropic poroelastic seafloor. J Comput Acoust 2(1):11–27
12. Badiey M, Jaya I, Cheng AHD (1994) A shallow water acoustic/geoacoustic experiment near the New Jersey Atlantic Generating Station site. J Acoust Soc Am 96(6):3593–3604
13. Barnes J (ed) (1984) The complete works of Aristotle. Revised Oxford translation, vol 1. Princeton University Press, Princeton, 1256pp
14. Basset AB (1888) Treatise on hydrodynamics, vol 2. Deighton, Bell, London, 328pp
15. Berryman JG (1980) Confirmation of Biot's theory. Appl Phys Lett 37(4):382–384
16. Bessel FW (1826) Untersuchungen über die Länge des einfachen Sekundenpendels (Studies on the length of the simple seconds pendulum). In: Abhandlungen der Königliche Akademie der Wissenschaften zu Berlin (Proceedings of the Royal Academy of Sciences in Berlin), Berlin
17. Biot MA (1955) Variational principles in irreversible thermodynamics with application to viscoelasticity. Phys Rev 97(6):1463–1469
18. Biot MA (1956) Thermoelasticity and irreversible thermodynamics. J Appl Phys 27(3):240–253
19. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid. 1. Low-frequency range. J Acoust Soc Am 28(2):168–178
20. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid. 2. Higher frequency range. J Acoust Soc Am 28(2):179–191
21. Biot MA (1962) Mechanics of deformation and acoustic propagation in porous media. J Appl Phys 33(4):1482–1498
22. Biot MA (1962) Generalized theory of acoustic propagation in porous dissipative media. J Acoust Soc Am 34(9):1254–1264
23. Bonnet G (1987) Basic singular solutions for a poroelastic medium in the dynamic range. J Acoust Soc Am 82(5):1758–1762
24. Bonnet G, Auriault JL (1985) Dynamics of saturated and deformable porous media: homogenization theory and determination of the solid-liquid coupling coefficients. In: Boccara N, Daoud ZM (eds) Physics of finely divided matter, Proc. Winter School. Springer, Les Houches, pp 306–316

25. Bourbié T, Coussy O, Zinszner B (1987) Acoustics of porous media. Editions Technip, Paris, 334pp
26. Boutin C, Bonnet G, Bard PY (1987) Green functions and associated sources in infinite and stratified poroelastic media. Geophys J R Astron Soc 90(3):521–550
27. Bowen RM (1982) Compressible porous media models by use of the theory of mixtures. Int J Eng Sci 20(6):697–735
28. Bowen RM, Lockett RR (1983) Inertial effects in poroelasticity. J Appl Mech, ASME 50(2):334–342
29. Boyle FA, Chotiros NP (1992) Experimental detection of a slow acoustic wave in sediment at shallow grazing angles. J Acoust Soc Am 91(5):2615–2619
30. Burridge R, Vargas CA (1979) Fundamental solution in dynamic poroelasticity. Geophys J R Astron Soc 58(1):61–90
31. Champoux Y, Allard JF (1991) Dynamic tortuosity and bulk modulus in air-saturated porous media. J Appl Phys 70(4):1975–1979
32. Charlaix E, Kushnick AP, Stokes JP (1988) Experimental study of dynamic permeability in porous media. Phys Rev Lett 61(14):1595–1598
33. Cheng AHD, Badmus T, Beskos DE (1991) Integral equation for dynamic poroelasticity in frequency domain with BEM solution. J Eng Mech ASCE 117(5):1136–1157
34. Chen J (1994) Time domain fundamental solution to Biot's complete equations of dynamic poroelasticity. Part I: two-dimensional solution. Int J Solids Struct 31(10):1447–1490
35. Chen J (1994) Time domain fundamental solution to Biot's complete equations of dynamic poroelasticity. Part II: three-dimensional solution. Int J Solids Struct 31(2):169–202
36. Coussy O (2010) Mechanics and physics of porous solids. Wiley, Chichester/Hoboken, 296pp
37. Dargush G, Chopra M (1996) Dynamic analysis of axisymmetric foundations on poroelastic media. J Eng Mech ASCE 122(7):623–632
38. de Boer R, Ehlers W, Liu ZF (1993) One-dimensional transient wave propagation in fluid-saturated incompressible porous media. Arch Appl Mech 63(1):59–72
39. Deresiewicz H (1960) The effect of boundaries on wave propagation in a liquid-filled porous solid: I. Reflection of plane waves at a free plane boundary (non-dissipative case). Bull Seismol Soc Am 50(4):599–607
40. Deresiewicz H (1961) The effect of boundaries on wave propagation in a liquid-filled porous solid: II. Love waves in a porous layer. Bull Seismol Soc Am 51(1):51–59
41. Deresiewicz H (1962) The effect of boundaries on wave propagation in a liquid-filled porous solid: IV. Surface waves in a half-space. Bull Seismol Soc Am 52(3):627–638
42. Deresiewicz H, Rice JT (1962) The effect of boundaries on wave propagation in a liquid-filled porous solid: III. Reflection of plane waves at a free plane boundary (general case). Bull Seismol Soc Am 52(3):595–625
43. Deresiewicz H, Rice JT (1964) The effect of boundaries on wave propagation in a liquid-filled porous solid: V. Transmission across a plane interface. Bull Seismol Soc Am 54(1):409–416
44. Ding BY, Cheng AHD, Chen ZL (2013) Fundamental solutions of poroelastodynamics in frequency domain based on wave decomposition. J Appl Mech ASME 80(6):061201, 12 p
45. Domenico S (1976) Effect of brine-gas mixture on velocity in an unconsolidated sand reservoir. Geophysics 41(5):882–894
46. Dominguez J (1991) An integral formulation for dynamic poroelasticity. J Appl Mech ASME 58(2):588–591
47. Dominguez J (1992) Boundary element approach for dynamic poroelastic problems. Int J Numer Methods Eng 35(2):307–324
48. Domínguez J, Gallego R, Japón BR (1997) Effects of porous sediments on seismic response of concrete gravity dams. J Eng Mech ASCE 123(4):302–311
49. Dvorkin J, Mavko G, Nur A (1995) Squirt flow in fully saturated rocks. Geophysics 60(1):97–107
50. Dvorkin J, Nolen-Hoeksema R, Nur A (1994) The squirt-flow mechanism: macroscopic description. Geophysics 59(3):428–438

51. Dvorkin J, Nur A (1993) Dynamic poroelasticity—a unified model with the squirt and the Biot mechanisms. Geophysics 58(4):524–533
52. Ewing WM, Jardetsky WS, Press F (1957) Elastic waves in layered media. McGraw-Hill, New York
53. Ferrin JL, Mikelić A (2003) Homogenizing the acoustic properties of a porous matrix containing an incompressible inviscid fluid. Math Methods Appl Sci 26(10):831–859
54. Frenkel J (1944) On the theory of seismic and seismoelectric phenomena in moist soil. J Phys USSR 13(4):230–241. Republished (2005) J Eng Mech ASCE 131(9):879–887
55. Gassmann F (1951) Über die elastizität poröser medien (On elasticity of porous media) Veirteljahrsschrift der Naturforschenden Gesellschaft in Zürich, Zürich, vol 96, pp 1–23
56. Gist GA (1994) Interpreting laboratory velocity measurements in partially gas-saturated rocks. Geophysics 59(7):1100–1109
57. Gist GA (1994) Fluid effects on velocity and attenuation in sandstones. J Acoust Soc Am 96(2):1158–1173
58. Gregory A (1976) Fluid saturation effects on dynamic elastic properties of sedimentary rocks. Geophysics 41(5):895–921
59. Halpern MR, Christiano P (1986) Response of poroelastic halfspace to steady-state harmonic surface tractions. Int J Numer Anal Methods Geomech 10(6):609–632
60. Hamilton EL (1980) Geoacoustic modeling of the sea floor. J Acoust Soc Am 68(5):1313–1340
61. Hardin BO (1965) The nature of damping in sands. J Soil Mech Found Div ASCE 91:63–97
62. Hill RJ, Koch DL, Ladd AJC (2001) The first effects of fluid inertia on flows in ordered and random arrays of spheres. J Fluid Mech 448:213–241
63. Hörmander L (1969) Linear partial differential operators. Springer, Berlin, 285pp
64. Imelda SC, Subramaniam R (2007) From Pythagoras to Sauveur: tracing the history of ideas about the nature of sound. Phys Educ 42(2):173–179
65. Ionescu-Cazimir V (1964) Problem of linear coupled thermoelasticity. Theorems of reciprocity for the dynamic problem of coupled thermoelasticity. I and II. Bulletin de l'Académie Polonaise des Sciences, Série des Sciences Techniques 12:473–480, 481–488
66. Ivanov AG (1939) Effect of electrization of earth layers by elastic waves passing through them. Doklady Akademii Nauk SSSR 24(1):42–45
67. Ivanov AG (1940) The electroseismic effect of the second kind. Izvestiya Akademii Nauk SSSR, Ser. Geogr. Geofiz 5:699–727
68. Johnson DL (1980) Equivalence between 4th sound in liquid He II at low temperatures and the Biot slow wave in consolidated porous media. Appl Phys Lett 37(12):1065–1067
69. Johnson DL, Koplik J, Dashen R (1987) Theory of dynamic permeability and tortuosity in fluid-saturated porous media. J Fluid Mech 176:379–402
70. Johnson DL, Plona TJ (1982) Acoustic slow waves and the consolidation transition. J Acoust Soc Am 72(2):556–565
71. Johnson DL, Plona TJ, Scala C, Pasierb F, Kojima H (1982) Tortuosity and acoustic slow waves. Phys Rev Lett 49(25):1840–1844
72. Johnson DL, Sen PN (1981) Multiple scattering of acoustic waves with application to the index of refraction of 4th sound. Phys Rev B 24(5):2486–2496
73. Kaynia AM, Banerjee PK (1993) Fundamental solutions of Biot's equations of dynamic poroelasticity. Int J Eng Sci 31(5):817–830
74. Kelder O, Smeulders DMJ (1997) Observation of the Biot slow wave in water-saturated Nivelsteiner sandstone. Geophysics 62(6):1794–1796
75. Kryloff N, Bogoliuboff N (1949) Introduction to non-linear mechanics. Princeton University Press, Princeton, 114pp
76. Kupradze VD, Gezelia TG, Basheleishvili MO, Burchuladze TV (1979) Three-dimensional problems of the mathematical theory of elasticity and thermoelasticity. North-Holland, Amsterdam/New York, 929pp
77. Lamb H (1948) Hydrodynamics, 6th edn. Dover, New York, 738pp

78. Landau LD, Lifshitz EM (1986) Theory of elasticity, vol 7 in course of theoretical physics, 3rd edn. Butterworth-Heinemann, Oxford/New York, 195pp
79. Landau LD, Lifshitz EM (1987) Fluid mechanics, vol 6 in course of theoretical physics, 2nd edn. Butterworth-Heinemann, Oxford/New York, 552pp
80. Levy T (1979) Propagation of waves in a fluid-saturated porous elastic solid. Int J Eng Sci 17:1005–1014
81. Lighthill J (2001) Waves in fluids, 2nd edn. Cambridge University Press, Cambridge/New York, 524pp
82. Lopatnikov SL, Cheng AHD (2004) Macroscopic Lagrangian formulation of poroelasticity with porosity dynamics. J Mech Phys Solids 52(12):2801–2839
83. Love AEH (1892) A treatise on the mathematical theory of elasticity, vol 1. Cambridge University Press, Cambridge, 354pp
84. Maeso O, Aznárez JJ, Domínguez J (2004) Three-dimensional models of reservoir sediment and effects on the seismic response of arch dams. Earthq Eng Struct Dyn 33(10):1103–1123
85. Maeso O, Aznárez JJ, García F (2005) Dynamic impedances of piles and groups of piles in saturated soils. Comput Struct 83(10–11):769–782
86. Manolis GD, Beskos DE (1989) Integral formulation and fundamental solutions of dynamic poroelasticity and thermoelasticity. Acta Mech 76(1–2):89–104
87. Manolis GD, Beskos DE (1990) Errata in 'Integral formulation and fundamental solutions of dynamic poroelasticity and thermoelasticity'. Acta Mech 83(3–4):223–226
88. Mavko G, Jizba D (1991) Estimating grain-scale fluid effects on velocity dispersion in rocks. Geophysics 56(12):1940–1949
89. Mavko G, Nur A (1975) Melt squirt in the asthenosphere. J Geophys Res 80(11):1444–1448
90. Mavko G, Nur A (1979) Wave attenuation in partially saturated rocks. Geophysics 44:161–178
91. Miksis MJ (1988) Effect of contact line movement on the dissipation of waves in partially saturated rocks. J Geophys Res 93:6624–6634
92. Murphy W, Winkler K, Kleinberg R (1986) Acoustic relaxation in sedimentary rocks: dependence on grain contacts and fluid saturation. Geophysics 51(3):757–766
93. Nigmatulin RI (1990) Dynamics of multiphase systems, vol 1 & 2. Hemisphere, New York, 878pp
94. Nikolaevskiy VN (1984) Mechanics of porous and cracked media (in Russian). Nedra, Moscow, 232pp
95. Norris AN (1985) Radiation from a point source and scattering theory in a fluid-saturated porous solid. J Acoust Soc Am 77(6):2012–2022
96. Nowacki W (1975) Dynamics problems of thermoelasticity. Noordhoff, Leyden, 456pp
97. O'Connell RJ, Budiansky B (1977) Viscoelastic properties of fluid-saturated cracked solids. J Geophys Res 82(36):5719–5735
98. Palmer I, Traviolia M (1980) Attenuation by squirt flow in undersaturated gas sands. Geophysics 45(12):1780–1792
99. Philippacopoulos AJ (1998) Spectral Green's dyadic for point sources in poroelastic media. J Eng Mech ASCE 124(1):24–31
100. Plona TJ (1980) Observation of a second bulk compressional wave in a porous medium at ultrasonic frequencies. Appl Phys Lett 36(4):259–261
101. Predeleanu M (1984) Development of boundary element method to dynamic problems for porous media. Appl Math Model 8(6):378–382
102. Pride SR, Berryman JG, Harris JM (2004) Seismic attenuation due to wave-induced flow. J Geophys Res B: Solid Earth 109(1):B01201 01201–01219
103. Sahay PN (2001) Dynamic green's function for homogeneous and isotropic porous media. Geophys J Int 147(3):622–629
104. Sanchez-Palencia E (1980) Non-homogeneous media and vibration theory. Springer, Berlin/New York, 398pp

105. Sangani AS, Zhang DZ, Prosperetti A (1991) The added mass, Basset, and viscous drag coefficients in nondilute bubbly liquids undergoing small-amplitude oscillatory motion. Phys Fluids A-Fluid Dyn 3(12):2955–2970
106. Schanz M (2009) Poroelastodynamics: linear models, analytical solutions, and numerical methods. Appl Mech Rev 62(3):1–15
107. Schanz M, Cheng AHD (2000) Transient wave propagation in a one-dimensional poroelastic column. Acta Mech 145(1–4):1–18
108. Schlichting H, Gersten K (2000) Boundary-layer theory, 8th edn. Springer, Berlin/New York, 826pp
109. Sen PN, Scala C, Cohen MH (1981) A self-similar model for sedimentary rocks with application to the dielectric constant of fused glass beads. Geophysics 46(5):781–795
110. Senjuntichai T, Rajapakse RKND (1994) Dynamic green's functions of homogeneous poroelastic half-plane. J Eng Mech ASCE 120(11):2381–2464
111. Sheng P, Zhou MY (1988) Dynamic permeability in porous media. Phys Rev Lett 61(14):1591–1594
112. Stoll RD (1974) Acoustic waves in saturated sediments. In: Hampton L (ed) Physics of sound in marine sediment. Plenum, New York, pp 19–39
113. Stoll RD (1977) Acoustic waves in ocean sediments. Geophysics 42(4):715–725
114. Stoll RD (1989) Sediment acoustics. Springer, Berlin/New York, 155pp
115. Stoll RD, Bryan GM (1970) Wave attenuation in saturated sediments. J Acoust Soc Am 47(5):1440–1447
116. Stoll RD, Kan TK (1981) Reflection of acoustic waves at a water-sediment interface. J Acoust Soc Am 70(1):149–156
117. Toms J, Müller TM, Ciz R, Gurevich B (2006) Comparative review of theoretical models for elastic wave attenuation and dispersion in partially saturated rocks. Soil Dyn Earthq Eng 26(6–7):548–565
118. Turgut A, Yamamoto T (1988) Synthetic seismograms for marine sediments and determination of porosity and permeability. Geophysics 53(8):1056–1067
119. van der Grinten JGM, van Dongen MEH, van der Kogel H (1985) A shock-tube technique for studying pore-pressure propagation in a dry and water-saturated porous medium. J Appl Phys 58(8):2937–2942
120. van der Grinten JGM, van Dongen MEH, van der Kogel H (1987) Strain and pore pressure propagation in a water-saturated porous medium. J Appl Phys 62(12):4682–4687
121. Wang Z, Nur A (1990) Dispersion analysis of acoustic velocities in rocks. J Acoust Soc Am 87:2384–2395
122. White JE (1975) Computed seismic speeds and attenuation in rocks with partial gas saturation. Geophysics 40(2):224–232
123. Wilmanski K (2005) Tortuosity and objective relative accelerations in the theory of porous materials. Proc R Soc Lond, Ser A-Math Phys Sci 461(2057):1533–1561
124. Winkler KW (1985) Dispersion analysis of velocity and attenuation in Berea sandstone. J Geophys Res: Solid Earth 90(B8):6793–6800
125. Wood AB (1941) Textbook of sound. Being an account of the physics of vibrations with special reference to recent theoretical and technical developments, 2nd edn. Bell & Sons, London, 578pp
126. Yamamoto T, Turgut A (1988) Acoustic wave propagation through porous media with arbitrary pore size distributions. J Acoust Soc Am 83(5):1744–1751
127. Zimmerman C, Stern M (1993) Boundary element solution of 3-D wave scatter problems in a poroelastic medium. Eng Anal Bound Elem 12(4):223–240

Chapter 10
Poroviscoelasticity

We will call pressure in general (and sometimes tension, traction, or more generally, elastic strength), on both sides of a small flat face imagined out from inside of a body, or the limit of separation of two bodies, the resultant of all the actions of molecules located on that side, the molecules on the opposite side, and whose directions across the face, by regarding it as positive when attractive and negative when repulsive.

—Claude-Louis Navier (1821)

The poroelasticity theory presented so far has explored the coupling between pore pressure diffusion and elastic matrix deformation in fluid impregnated porous solid, giving rise to a variety of time-dependent phenomena that are important to engineering applications. However, in addition to the macroscopic Darcy flow induced effects, there exist many other time-dependent physical mechanisms. For example, the rock mass itself without fluid can exhibit creep behavior [16]. This and other behaviors can result from the intergranular frictional sliding, the intrinsic solid grain creep deformation, and even the solid-fluid interactions in fissures at the sub-granular scale [6, 11]. These time-dependent phenomena can have a time scale that is much smaller than Darcy flow, for example, the squirt flow due to fluid pressure equilibrium at the grain scale, or a much larger time scale, such as the secondary consolidation of clay, or the creep of rock under high confining stresses. These phenomena can be modeled as apparent *viscoelastic* mechanisms at the macroscopic level.

The coexistence of poroelastic and viscoelastic phenomena has been observed in the laboratory and in the field for a range of rock types [13, 14, 28]. The simultaneous modeling of these mechanisms should help to interpret those observations. In field applications, one of the major problems encountered is borehole stability. Drilling difficulties are associated mainly with shales and tufaceous sediments. Typical problems encountered in shales are drill-tool sticking, sloughing, and creep or swelling in the borehole wall. The cause of these problems can be of chemical and/or mechanical nature. However, even in oil-base mud, which is free from chemical effects, borehole instability remains aproblem. Part of

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_10

the difficulty can be attributed to mechanical effects of poroelastic and viscoelastic origin [22, 24]. A poroviscoelastic solution can help elucidate these phenomena and the associated creep events.

Poroviscoelasticity was first investigated by Biot [4]. As pointed out by Biot [6], in addition to the Darcy flow, there are a number of physical, chemical, mechanical, thermal, and electrical mechanisms that can introduce time-dependent properties to a fluid infiltrated porous solid. For example:

- The pressure equilibrium and fluid exchange between the macroscopic flow channels and the micropores introduces a 'sponginess effect' to the material at small time, that is, before the macroscopic fluid migration takes place.
- The same fluid viscosity effect is expected if the solid is made of deformable grains in point contact, or grains containing small cracks, such that fluid has to be squeezed from these narrow gaps before the solid can deform. In particular, if the solid has flexible membrane, such as biological cells, a lubrication type squeeze film may develop between the cells and a spring-dashpot phenomenon is observed.
- The initial temperature differential arising in solid and fluid when stress is applied tends to even out in time because of the thermal conductivity, thus giving rise to a thermoelastic relaxation spectrum.
- The interfacial surface tension for clayey material, as a result of physical-chemical-electrical equilibrium of ions and molecules, plays an important role in the overall rigidity of the material. This effect, which has a relatively large relaxation time, may be the cause for the secondary creep event long after the pore pressure reaches equilibrium.
- For crystal in equilibrium with its solution, or solid grains with adsorbed substances, the solubility of the species may be sensitive to solid stress and fluid pressure. A relaxation effect is observed as the substances enter into the solution or precipitate on the solid surface, which tends to modified the solid and the fluid properties.
- The porous solid itself without fluid may exhibit time-dependent behavior, resulting from intergranular frictional sliding or the intrinsic solid creep deformation.

These and other time-dependent mechanisms [11] can be explicitly and individually modeled, if they are important enough. An example is the squirt flow model in elastodynamics (Sect. 9.5.3). Or, they can be implicitly modeled as a series of viscoelastic relaxation phenomena, based on a set of empirical parameters obtained by fitting data to the theory.

In this chapter, we shall present the viscoelastic models and incorporate them into poroelasticity. As a first step, we shall give a quick review of the basic viscoelasticity theory without porous medium flow [9, 15].

10.1 Viscoelasticity

The viscoelasticity models were inspired by the *creep* phenomenon observed in solids. The linear elasticity model based on Hooke's law states that the strain of a material is proportional to the stress applied. So if we fix the stress, the observed strain should be uniquely related on the stress, and should be independent of the time. However, this is not so for certain materials. This prompted the introduction of the rate of stress (time derivative of stress) and rate of strain into the constitutive modeling. For easy visualization of the process, an analogy is typically drawn between the observed phenomenon and the simple mechanical devices of *spring* and *dashpot*.

10.1.1 Spring and Dashpot Model

When a force F is applied to a helical spring (Fig. 10.1a), the length of the spring increases by an amount u that is proportional to the force applied. This relation is similar to the elasticity constitutive relation

$$\sigma = Ke \tag{10.1}$$

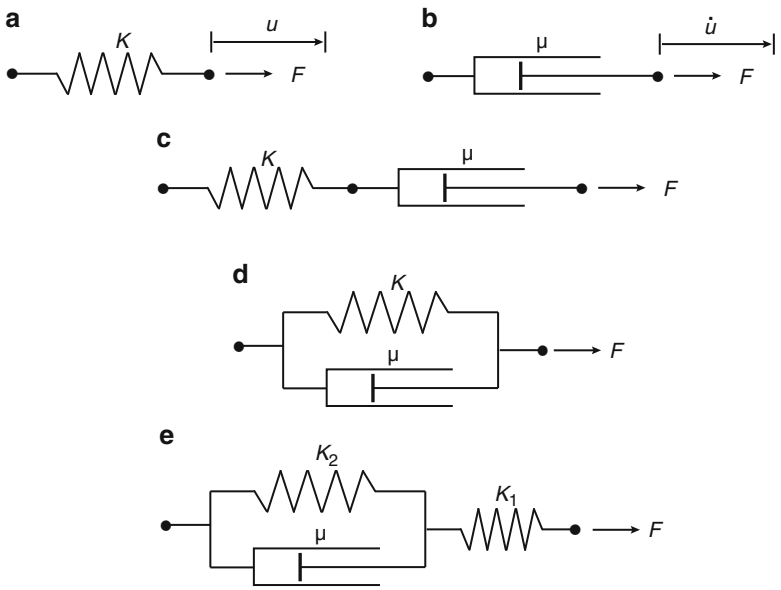


Fig. 10.1 Spring-dashpot models: (**a**) spring, (**b**) dashpot, (**c**) Maxwell model, (**d**) Kelvin-Voigt model, and (**e**) standard linear solid model

where σ represents some stress, volumetric or shear, and e represents strain, volumetric or shear, and K is the proportionality constant, which is a material property. Hence we shall use a spring to conceptualize such a linear elastic relation for a solid.

Fluid is altogether a different material, particularly its behavior in shear. According to Newton's law of viscosity, when a shear stress is applied, the fluid continuously to deform without an end. However, the time rate of the shear deformation $\dot{e}$, where the over dot denotes time derivative, is proportional to the shear stress σ applied, which can be written as the following law

$$\sigma = \mu \dot{e} \tag{10.2}$$

where μ is known as a *viscosity*. Solid can display this type of behavior too. In the mechanical analogy, this relation is represented by a *dashpot* (Fig. 10.1b), which is a piston pulled or pushed in a container of viscous fluid.

To model the solid behavior of creep, the *Maxwell model*, which is a serial assembly of a spring and a dashpot (Fig. 10.1c), can be used. In terms of a constitutive relation, we can sum the time derivative of (10.1) and (10.2) to obtain

$$\frac{\dot{\sigma}}{K} + \frac{\sigma}{\mu} = \dot{e} \tag{10.3}$$

where $\dot{e}$ is the sum of the two parts of the strain.

A deficiency of using Maxwell model to represent solid is that, like a fluid, it creeps without an end. A different model, the *Kelvin-Voigt model* (Fig. 10.1d), can be used. In this case, the stress is the sum of the two parts, and we obtain

$$\sigma = \mu \dot{e} + Ke \tag{10.4}$$

This model will reach an elastic equilibrium, in which the entire force is sustained by the elastic spring, when the fluid motion ceases at large time. It, however, in turn is deficient in that there is no instantaneous deformation of an elastic body subject to a force.

So the next model, the *standard linear solid model*, or the *Zener model* [30], as shown in Fig. 10.1e, is the minimally acceptable model for a typical viscoelastic solid. This mechanical assembly will give an initial deformation, followed by a creep, and then settle into a final, and larger, deformation. The constitutive equation is given by

$$K_1 K_2 e + K_1 \mu \dot{e} = (K_1 + K_2)\sigma + \mu \dot{\sigma} \tag{10.5}$$

where K_1 and K_2 are spring constants for the two springs. Solving the above differential equation with an initially applied stress σ_o, we find the evolution of deformation as

$$e(t) = \frac{\sigma_o}{K_1}\left(1 + \frac{K_1}{K_2} - \frac{K_1}{K_2} e^{-K_2 t/\mu}\right) \tag{10.6}$$

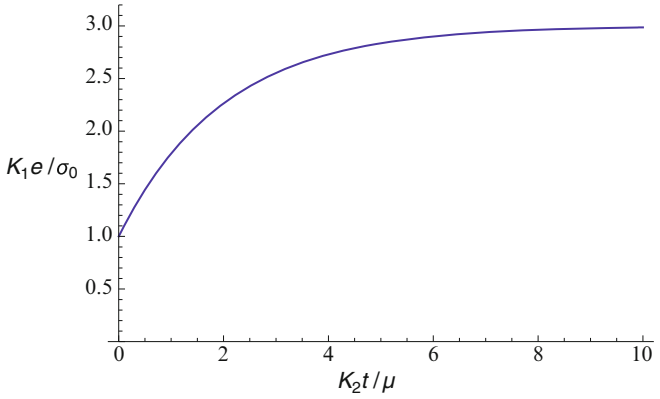


Fig. 10.2 Time evolution of strain based on standard linear solid model, for $K_1/K_2 = 2$

Figure 10.2 plots this relation for the selection of parameter $K_1/K_2 = 2$.

We observe that the behavior in Fig. 10.2 resembles the consolidation behavior of a clay layer due to pore pressure diffusion. Hence we may be tempted to model consolidation as a viscoelastic phenomenon. Physically, however, these two are totally different processes, as observed from the consolidation time. The consolidation time of pore pressure diffusion is a function of both the material property (consolidation coefficient) and the problem geometry (distance to drainage surface). The larger the physical size, the longer the consolidation time (proportion to length squared). The viscoelastic creep time scale, however, is entirely a material property. The creep time is always the same, whatever the sample size is.

By assembling the springs of different elastic constants, and dashpots of different viscosities, in parallel and in series, more complicated viscoelastic models can be constructed. These models are necessary if different physical processes with different time scales are simultaneously present. As commented earlier, these models are largely empirical models, driven by observations.

10.1.2 Correspondence Principle

As observed from the several spring-dashpot models presented in the preceding section, as well as the more complicated ones that can be found in textbooks of viscoelasticity [9, 15], the differential equations representing the models can in general be expressed in the following form

$$\sum_{k=0}^{m} p_k \frac{d^k \sigma}{dt^k} = \sum_{k=0}^{n} q_k \frac{d^k e}{dt^k} \tag{10.7}$$

where p_k and q_k are the proper combination of the constitutive constants. The more complex the spring-dashpot model becomes, the higher the time derivative order is found in the above constitutive equation.

We can subject (10.7) to Laplace transform, and obtain

$$\sum_{k=0}^{m} p_k s^k \tilde{\sigma} = \sum_{k=0}^{n} q_k s^k \tilde{e} \tag{10.8}$$

where s is the Laplace transform parameter. The above equation can be rearranged to obtain

$$\tilde{\sigma} = \overline{K}\,\tilde{e} \tag{10.9}$$

where

$$\overline{K} = \frac{\sum_{k=0}^{n} q_k s^k}{\sum_{k=0}^{m} p_k s^k} \tag{10.10}$$

We hence observe that the constitutive equations for viscoelasticity in the Laplace transform space is formally the same as that of elasticity, except that the constitutive constant is a function of the Laplace transform parameter s and many sub-constitutive constants. With the constitutive equations, and the other governing equations, formally identical, this means that *all solutions developed for elasticity in the Laplace transform domain are the same for viscoelasticity, as long as the proper material constants are referred to*. This is called the *correspondence principle*. The difference between the viscoelastic and the elastic effects can be observed in the time domain via the inverse Laplace transform.

The constitutive equation (10.9) is expressed in the Laplace transform domain. It is of interest to develop it in the time domain. This can be done by rewriting (10.9) into the following form

$$\tilde{\sigma} = s\tilde{K}\,\tilde{e} \tag{10.11}$$

where

$$\tilde{K} = \frac{\overline{K}}{s} \tag{10.12}$$

Performing Laplace inverse to (10.11), and utilizing the convolutional theorem (Appendix D), we obtain

$$\sigma(t) = K(t)\,e(0) + \int_0^t K(t-t')\frac{de(t')}{dt'}\,dt' \tag{10.13}$$

where

$$K(t) = \mathscr{L}^{-1}\{\tilde{K}(s)\} \tag{10.14}$$

Equation (10.13) can be viewed as a Duhamel principle of superposition, and in the present form it is called the *hereditary integral* in viscoelasticity [15].

It may be of interest to note that among the above notations, $\overline{K}(s)$ and $K(t)$ are of the same physical dimension as the modulus of elasticity K. The quantity $\tilde{K}(s)$, on the other hand, is not of the same dimension. This is because that the Laplace transform and inversion do not preserve physical dimension. Rather, the *Laplace-Carson transform*

$$\bar{f}(s) = \mathscr{C}\{f(t)\} = \int_0^\infty sf(t)\, e^{-st}\, dt \tag{10.15}$$

preserves the physical dimension. By this relation, we see that $\overline{K}(s) = s\tilde{K}$ is actually the Laplace-Carson transform of $K(t)$.

We observe from (10.10) and (10.12) that $\tilde{K}$ is the quotient of two polynomials in s. This means that if we can factor the denominator of (10.10) into

$$\sum_{k=0}^{m} p_k s^k = p_m(s-\beta_1)(s-\beta_2)\cdots(s-\beta_m) \tag{10.16}$$

then we can find the Laplace inversion of $\tilde{K}(s)$ as (see Table D.1)

$$K(t) = -\frac{1}{p_m}\sum_{n=1}^{m} \frac{q(\beta_n)}{\beta_n\, p'(\beta_n)}\left(1 - e^{\beta_n t}\right) \tag{10.17}$$

For a physically acceptable model, we must have $\beta_1, \beta_2, \ldots, \beta_m < 0$.

We can now return to (10.13), and perform integration by parts, to obtain

$$\sigma(t) = \int_{-\infty}^{+\infty} K(t-t')\, de(t') = K(t) \otimes e(t) \tag{10.18}$$

where we have extended the lower integration limit to $-\infty$ such that the initial condition is absorbed into the integral. We also used the symbol $\otimes$ to denote such integral operation. Equation (10.18) is known as a *Riemann-Stieltjes integral*.

The above procedure suggests that we can construct the constitutive equations of viscoelasticity for each stress and strain components as

$$\sigma_{ij}(t) = M_{ijkl}(t) \otimes e_{kl}(t) \tag{10.19}$$

and for isotropic material,

$$\sigma_{ij}(t) = 2G(t) \otimes e_{ij}(t) + \delta_{ij}\lambda(t) \otimes e(t) \tag{10.20}$$

These equations mean, if we interpret the multiplication operator in the elasticity constitutive equation as the Riemann-Stieltjes operator $\otimes$, we then have the same constitutive equation, which leads to the correspondence principle in the time domain.

10.2 Poroviscoelasticity

Biot modeled viscoelasticity [3, 5] and poroviscoelasticity [4, 6] based on the variational Lagrangian thermodynamic considerations. Without going into its theoretical foundation, here we shall directly resort to the phenomenological approach based on generalized Hooke's law.

10.2.1 Correspondence Principle

Following the lead in the preceding section on viscoelasticity, we can construct the poroviscoelasticity constitutive equations based on the correspondence principle [2, 20].

Time Domain In the time domain, we can construct

$$\sigma_{ij}(t) = 2G(t) \otimes e_{ij}(t) + \delta_{ij}\lambda(t) \otimes e(t) - \delta_{ij}\,\alpha(t) \otimes p(t) \tag{10.21}$$

$$p(t) = M(t) \otimes [\zeta(t) - \alpha(t) \otimes e(t)] \tag{10.22}$$

where $\lambda(t)$, $G(t)$, $M(t)$, and $\alpha(t)$ are the four time dependent viscoelastic moduli, based on their elastic counterparts. In the above we note that the operator $\otimes$ has commutative, associative, and distributive properties [17].

Laplace Transform Domain It is more convenient to formulate the theory and to solve the boundary value problems in the Laplace transform space. Hence we apply Laplace transform to (10.21) and (10.22) and obtain

$$\tilde{\sigma}_{ij} = 2\overline{G}\,\tilde{e}_{ij} + \delta_{ij}\overline{\lambda}\,\tilde{e} - \delta_{ij}\overline{\alpha}\,\tilde{p} \tag{10.23}$$

$$\tilde{p} = \overline{M}(\tilde{\zeta} - \overline{\alpha}\,\tilde{e}) \tag{10.24}$$

where $\overline{\lambda}$, $\overline{G}$, $\overline{M}$ and $\overline{\alpha}$ are the Laplace-Carson transform of the time-dependent material constants.

Applying Laplace transform to the equilibrium equation, Darcy's law, and continuity equation, gives the following

$$\tilde{\sigma}_{ij,j} = 0 \tag{10.25}$$

$$\tilde{q}_i = -\kappa\,\tilde{p}_{,i} \tag{10.26}$$

$$s\tilde{\zeta} + \tilde{q}_{i,i} = 0 \tag{10.27}$$

For simplicity, the permeability coefficient is assumed to be free from viscoelastic effects. Based on the above set of equations, the differential equation solution

system is identical for poroelasticity and poroviscoelasticity. Hence all the solutions found in the Laplace transform space for the former can be utilized for the latter. The same principle can in fact be applied to extend the poroelastodynamic theory to poroviscoelastodynamics. In that case, Fourier transform, instead of Laplace transform, should be used.

It should be cautioned that not all boundary value problems can be solved in the Laplace transform space. First of all, Laplace transform does not work for nonlinear problems, either nonlinear governing equations, or nonlinear boundary conditions. Also, even for linear problems, if the boundary condition changes its type with time, such as changing from a displacement condition to a stress condition as time progresses, or if the boundary geometry is not fixed, and moves with time, then Laplace transform cannot be applied for the solution. But as already commented, when the Laplace transform solution is applicable to a poroelastic problem, no extra effort is needed in solving the poroviscoelastic problem.

10.2.2 Intrinsic Models

Under the correspondence principle, the viscoelastic model is contained in the material constants, $\overline{\lambda}$, $\overline{G}$, $\overline{M}$ and $\overline{\alpha}$, in which the appropriate spring-dashpot model can be assigned. However, before we do so, we need to ask the question: is it reasonable to directly construct a model for $\overline{\lambda}$, realizing that it is a composite constant of the bulk modulus $\overline{K}$ and shear modulus $\overline{G}$? And what about $\overline{M}$ and $\overline{\alpha}$?

The answer is, the viscoelastic models should be based on the physical processes, such as volumetric and shear deformation. In addition, they should be constructed on isolated material constituents, such as solid, fluid, and the void space structure, and not on an arbitrary Hooke's law constant. In other words, we need to build the model based on the micromechanical processes as described in Chap. 3, and the intrinsic material constants discussed in Chap. 4 [1].

Based on the above argument, $\overline{\lambda}$ should be constructed as

$$\overline{\lambda} = \overline{K} - \frac{2}{3}\overline{G} \tag{10.28}$$

which separates the volumetric and shear deformation processes, as they can each have their own time scale and magnitude of creep. Also, the Biot effective stress coefficient should be expressed in terms of the micromechanical material constants

$$\overline{\alpha} = 1 - \frac{\overline{K}}{\overline{K}_s} \tag{10.29}$$

It is more likely that the porous frame will creep due to grain sliding and rearranging its orientation, than the solid constituent itself. So different model needs to be applied to $\overline{K}$ and $\overline{K}_s$. Furthermore, we realize that $\overline{K}$ is still a composite constant, and can be express in terms of the intrinsic material constants

$$\frac{1}{\overline{\overline{K}}} = \frac{1}{(1-\phi)\overline{K}_s} + \frac{1}{(1-\phi)^3\overline{K}_\phi} \tag{10.30}$$

based on (4.159); therefore $\overline{\alpha}$ becomes

$$\overline{\alpha} = 1 - \frac{(1-\phi)^3\overline{K}_\phi}{\overline{K}_s + (1-\phi)^2\overline{K}_\phi} \tag{10.31}$$

Based on the same argument, the Biot modulus $\overline{M}$ should be constructed as

$$\frac{1}{\overline{\overline{M}}} = \frac{\phi}{K_f} + \frac{1-\phi}{\overline{K}_s + (1-\phi)^2\overline{K}_\phi} \tag{10.32}$$

where the fluid bulk modulus K_f should be introduced as a constant.

In the following, we shall examine some of the viscoelastic processes.

Shear Flow Generally speaking, materials have a higher potential to creep under shear than volumetric deformation. For materials like clay, it can creep for so long that in the short term, it behaves like a very viscous fluid, so the Maxwell fluid model can be adopted

$$\mu\dot{\sigma}_{ij} + G_o\sigma_{ij} = 2G_o\mu\,\dot{e}_{ij}, \quad i \neq j \tag{10.33}$$

where G_o is the instantaneous elastic shear modulus and μ a viscosity. The equivalent shear modulus under the correspondence principle is

$$\overline{G} = \frac{sG_o\mu}{s\mu + G_o} \tag{10.34}$$

Or, as proposed by Biot [6], a more flexible relation that covers both a short-range fast creep and the long-range steady-state secondary flow can be given as follows

$$\frac{1}{\overline{\overline{G}}} = \frac{s\mu_1 + G_o}{sG_o\mu_1} + \frac{1}{\mu_2 s^b}, \tag{10.35}$$

in which $0 < b < 1$, and μ_1, μ_2 and b are parameters determined from experimental fit using short time and long time data. Hence for a shear stress applied as a step load $\tau_o\mathrm{H}(t-0)$, with its Laplace transform as τ_o/s, the time evolution of the engineering shear strain γ is

$$\gamma = \frac{\tau_o}{G_o}\left[1 + \frac{G_o t}{\mu_1} + \frac{G_o t^b}{\mu_2\Gamma(1+b)}\right] \tag{10.36}$$

where Γ is the Gamma function.

Secondary Consolidation Consider the volumetric deformation of clay. It has been observed that the consolidation history consists of two stages [10, 27], of which only the first is associated with the exudation of fluid from the pores. The second stage, generally known as *secondary consolidation*, can be modeled in a phenomenological approach by assuming that the solid skeleton against volumetric deformation behaves as a standard linear solid model [8, 23].

Examining the relation (10.30), and making the assumption that the solid grain itself is not viscoelastic, that is, $\overline{K}_s = K_s$ and is time independent, we conclude that modeling either $\overline{K}_\phi$ or $\overline{K}$ as a standard linear solid model leads to the same creep behavior. Hence we shall directly construct the viscoelastic model on $\overline{K}$ as

$$\overline{K} = \frac{K_1 K_2 + s K_1 \mu_K}{K_1 + K_2 + s \mu_K} \tag{10.37}$$

The time dependent volumetric stain due to the application of a step load of compressional stress is given by (10.6).

As earlier commented, in poroviscoelasticity there are two different physical processes that create the same macroscopic observation of a creep—one is due to pore pressure diffusion, and the other the material viscosity. When the time scales are comparable, not only the responses are mixed, but also a non-monotonic behavior may be observed, due to the counteractions of the processes, which will be demonstrated in Sect. 10.4. In a laboratory experiment, it is necessary to separate the composite response to isolate the contribution due to each process in order to obtain the viscoelastic parameters. This can be accomplished by manipulating the time scale of response. Although the material response time cannot be changed, the diffusion time scale can be easily adjusted due to its relation with the square of sample size. Hence it may be possible to design the sample size and geometric configuration such that the response times are separated by at least one order of magnitude. Different parts of the curve can then be used for curve fitting.

Grain Viscoelasticity Although the solid itself is less likely to creep than the pore structure, there is a phenomenon associated with the grain-scale local flow, when fluid is squeeze out, or absorbed into, the microcracks within the grain, or the imperfect contact space between grain boundaries. This is already described as the squirt flow phenomenon in Sect. 9.5.3. As this effect is associated with the local grain geometry, and not the global pore pressure diffusion, it can be treated as a material property and modeled as a viscoelastic phenomenon related to K_s.

As discussed in Sect. 3.5.3, K_s can be measured in the laboratory as the inverse of unjacketed compressibility by observing the volume change of the sample in an unjacketed test, which requires the pore pressure throughout the sample to equilibrate with the incremental fluid pressure in the chamber. If the solid has long term creep property, it can be observed by measuring the continuous change of sample volume while keeping the chamber fluid pressure constant. The time scale of the local flow, however, is typically much smaller than the global diffusion time scale, unless the sample size is very small, and the permeability is large. So in

general the unjacketed test is not suitable for the determination of the short-term viscoelastic effect of the solid grains. Dynamic testing, in which acoustic waves with period comparable to the time scale of the local flow, can be used to explore the viscoelasticity of the grain.

Biot Effective Stress Coefficient As demonstrated in (10.29) and (10.31), $\overline{\alpha}$ can be controlled by two time scales, one due to pore volume viscoelastic deformation, and the other due to the grain viscoelasticity. Although for most geomaterials, the creep of $\overline{K}_s$ should be negligible compared to $\overline{K}_\phi$, it may not be so for biomaterials. Also, in the acoustic range, the short-term local flow effect can introduce viscoelastic effect to the grain. Hence in the following we shall investigate the variability of $\overline{\alpha}$ over a range of both viscoelastic effects.

When the viscoelastic effect of $\overline{K}_s$ is significant, and is of a time scale comparable to that of the frame viscoelasticity, then according to (10.30), $\overline{K}$ is a composite material constant involving two processes. Hence a viscoelastic model should be constructed on the intrinsic material constant $\overline{K}_\phi$, instead of $\overline{K}$. Although K_ϕ cannot be directly measured, as shown in (4.209), it can be obtained as the inverse of the difference between jacketed and unjacketed compressibilities, both of which are function of time. It is then feasible to build a spring-dashpot model for $\overline{K}_\phi$.

We assume the standard linear solid model for both $\overline{K}_\phi$ and $\overline{K}_s$, such that

$$\overline{K}_\phi = \frac{K_{\phi 1}K_{\phi 2} + sK_{\phi 1}\mu_\phi}{K_{\phi 1} + K_{\phi 2} + s\mu_\phi} \tag{10.38}$$

$$\overline{K}_s = \frac{K_{s1}K_{s2} + sK_{s1}\mu_s}{K_{s1} + K_{s2} + s\mu_s} \tag{10.39}$$

We can substitute the above into (10.31), and perform Laplace-Carson inverse transform to obtain the analytical expression of $\alpha(t)$. Such expression, however, is too lengthy to be of interest to be presented here. We hence only present the numerical result below.

In Fig. 10.3 we plot $\alpha(t)$ versus the dimensionless time $K_{\phi 2}t/\mu_\phi$. In all the cases presented, we have selected $\phi = 0.2$ and $K_{\phi 1}/K_{s1} = 0.52$, such that the initial values of α is 0.8. We then vary the value of initial and final values of K_ϕ and K_s by controlling the ratios $K_{\phi 1}/K_{\phi_2}$ and K_{s1}/K_{s2}. The relative relaxation time is controlled by the ratio $K_{s2}\,\mu_\phi/K_{\phi 2}\,\mu_s$.

We observe in Fig. 10.3 that the initial and final value of $\alpha(t)$ are given by

$$\alpha(0) = 1 - \frac{(1-\phi)^3 C_{s1}}{C_{\phi 1} + (1-\phi)^2 C_{s1}} \tag{10.40}$$

$$\alpha(\infty) = 1 - \frac{(1-\phi)^3 (C_{s1} + C_{s2})}{C_{\phi 1} + C_{\phi 2} + (1-\phi)^2 (C_{s1} + C_{s2})} \tag{10.41}$$

In the above, we have denoted $C_{\phi 1} = 1/K_{\phi 1}$, $C_{\phi 2} = 1/K_{\phi 2}, \ldots$, to shorten the mathematical expression. We observe that the final value can be greater or smaller

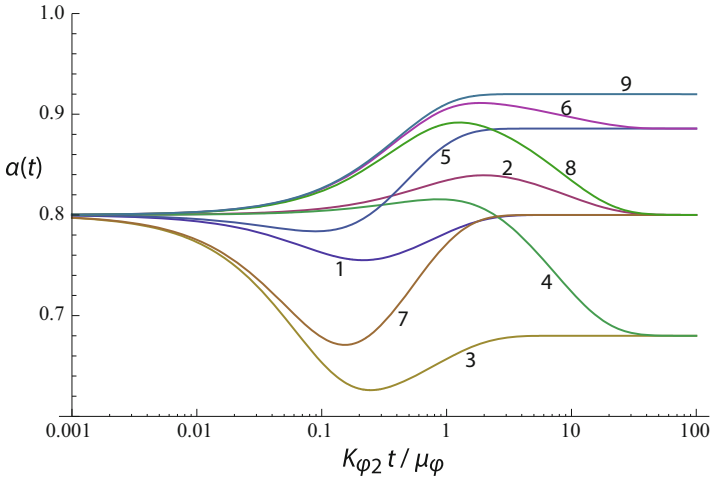


Fig. 10.3 Viscoelastic Biot effective stress coefficient $\alpha(t)$ versus dimensionless time $K_{\phi 2} t/\mu_{\phi}$, for $\phi = 0.2$, $K_{\phi 1}/K_{s1} = 0.52$, and a range of dimensionless parameters $(K_{\phi 1}/K_{\phi 2},\ K_{s1}/K_{s2},\ K_{s2}\,\mu_{\phi}/K_{\phi 2}\,\mu_{s})$ given as: Case 1: $(0.5, 0.5, 10)$; 2: $(0.5, 0.5, 0.1)$; 3: $(0.5, 2, 10)$; 4: $(0.5, 2, 0.1)$; 5: $(2, 0.5, 10)$; 6: $(2, 0.5, 0.1)$; 7: $(2, 2, 10)$; 8: $(2, 2, 0.1)$; 9: $(2, -, 0)$

than the initial value. In fact, it can be shown that $\alpha(\infty) > \alpha(0)$ if $(K_{\phi 1}/K_{\phi 2}) > (K_{s1}/K_{s2})$, and vice versa. The time scale of the evolution is controlled by the factors $\mu_{\phi}/K_{\phi 2}$ and μ_{s}/K_{s2}. The larger the value, the longer it takes to creep. In the cases presented, we have made the time scale for one process to be 10 times greater, or 10 times smaller, than the other. We observe that the interplay of these factors can create non-monotonic time evolution of the α value.

The above phenomenon also indicates that if we conduct a jacketed drained test, as described in Sect. 3.5.1, and observe the volume of fluid expelled from the sample, as a ratio of the change of sample volume, we may then observe the time evolution of $\alpha(t)$ based on (3.95).

10.3 Borehole Problem

As mentioned earlier, based on the correspondence principle, all analytical solutions derived for the poroelastic problems in the Laplace transform domain, such as those presented in Chap. 7, directly give the solutions of poroviscoelastic problems, provided that the proper substitution of the material constants by their viscoelastic models is made. Using this principle, poroviscoelastic analysis has been performed for problems with borehole, cylinder [2, 19, 21], and Mandel problem geometries [18]. In this and the following section, we shall perform the poroviscoelastic analysis for the borehole and cylinder problems.

The analytical solutions for plane strain borehole problems were derived in Sect. 7.14.1, both in the time domain and the Laplace transform domain. The solutions are decomposed into mode 1, 2, and 3, and we shall separately examine them below.

Mode 1 The mode 1 problem involves the axisymmetric radial stress loading on the borehole wall. The solution given as (7.599)–(7.602) shows that the porous medium deforms in deviatoric strain only; hence no pore pressure is generated. The radial and tangential stresses are independent of material properties, and remain constant at all times. The radial displacement, u_r, however, is dependent on the shear modulus. Its creep is entirely controlled by the time-dependent behavior of $G(t)$. The trend is straightforward; hence is not illustrated here.

Mode 2 Mode 2 problem is associated with pore pressure loading on the wall. The solution is given as (7.608)–(7.613). For the pore pressure response, the only material coefficient that enters into the solution is the consolidation coefficient, which can be expressed in its micromechanics definition as:

$$\bar{c} = \frac{\kappa K_f K_s^2 (4\overline{G} + 3\overline{K})}{K_f (K_s - \overline{K})(4\overline{G} + 3K_s) + \phi K_s (K_s - K_f)(4\overline{G} + 3\overline{K})} \tag{10.42}$$

In the above, we have assumed that K_f and K_s are non-viscoelastic. We observe that the viscoelastic effect of $\overline{K}$ and $\overline{G}$ can change the consolidation coefficient, which in turn changes the time scale of pore pressure diffusion.

To illustrate the interaction between the poroelastic and viscoelastic effects, we shall examine this problem, and problems below, using three rocks: a Berea sandstone, a Danian chalk, and a deep water Gulf of Mexico shale, with their poroelastic properties given in Tables 3.1 and 3.2. The selection of these materials is based on the wide range of permeabilities of these rocks, spanning six orders of magnitude.

For viscoelastic properties, we lack direct data for these rocks; hence we shall make assumptions. First, we choose the standard linear solid model for $\overline{K}$ as (10.37), and the same for $\overline{G}$ as

$$\overline{G} = \frac{G_1 G_2 + s G_1 \mu_G}{G_1 + G_2 + s \mu_G} \tag{10.43}$$

The initial values G_1 and K_1 are given by the G and K values in Table 3.1. We shall assume $G_2 = G_1$ and $K_2 = K_1$ such that the final moduli are 1/2 of their initial values. Loosely based on the observation of some retrieved cores, we choose here the μ_G and μ_K values such that the time scale μ_G/G_2 is 6 h, and μ_K/K_2 is 2 days for all three rocks.

For a borehole with radius $r_o = 0.1$ m, we plot in Fig. 10.4a the normalized pressure versus time at the location $r = 1$ m for the three rocks in solid lines. For comparison, the poroelastic solutions are shown by dashed lines. We observe that

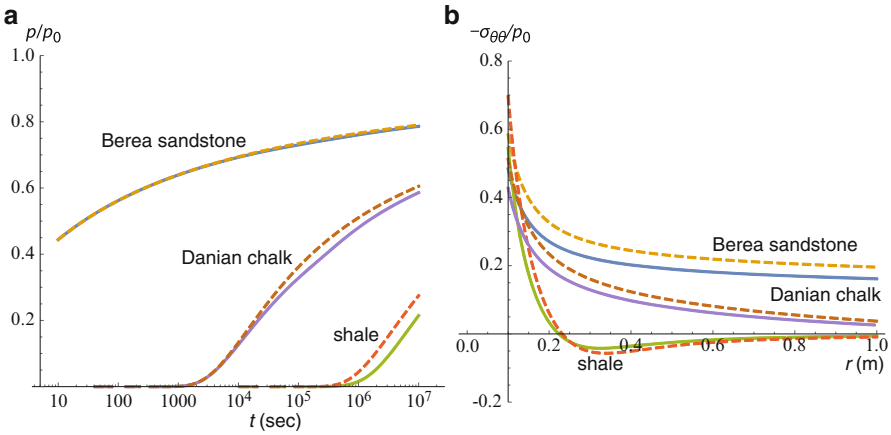


Fig. 10.4 Borehole problem mode 2. *Solid line*: poroviscoelastic solution; *dashed line*: poroelastic solution. (**a**) Pore pressure at $r = 1$ m versus time; and (**b**) tangential stress at 12 h versus r

for Berea sandstone, which has the largest permeability, the viscoelastic effect is too slow to interfere with the pore pressure diffusion. For the other two cases with small permeability, the viscoelastic effect acts to delay the pore pressure diffusion.

For the stresses, we observe from (7.611) and (7.612) that the magnitude of the stresses is directly affected by

$$\overline{\eta} = \frac{3\overline{G}(K_s - \overline{K})}{(4\overline{G} + 3\overline{K})K_s} \tag{10.44}$$

Figure 10.4b shows the normalized tangential compressive stress $-\sigma_{\theta\theta}/p_o$ as a function of radial distance r, at $t = 12$ h, for the mode 2 problem. Again, the solid lines are poroviscoelastic, and dashed line are poroelastic solutions. We observe that for the Berea sandstone and Danian chalk cases, the viscoelastic effect reduces the tangential compressive stress. For the shale, the tangential stress is in the tensile range, because $t = 12$ h is in the very small time range for pore pressure diffusion.

Mode 3 Mode 3 involves the deviatoric stress loading, with solution given as (7.631)–(7.639). In these expressions, we need to interpret $\overline{\nu}$ and $\overline{\nu}_u$ as

$$\overline{\nu} = \frac{3\overline{K} - 2\overline{G}}{6\overline{K} + 2\overline{G}} \tag{10.45}$$

$$\overline{\nu}_u = \frac{3\overline{K}_u - 2\overline{G}}{6\overline{K}_u + 2\overline{G}} \tag{10.46}$$

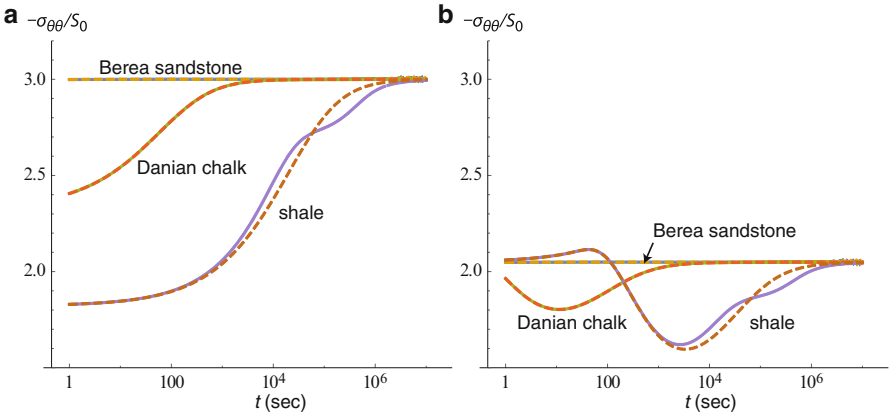


Fig. 10.5 Borehole problem mode 3. *Solid line*: poroviscoelastic solution; *dashed line*: poroelastic solution. (**a**) Tangential stress at borehole wall versus time; and (**b**) tangential stress at 1 cm into the wall

with

$$\overline{K}_u = \overline{K} + \frac{K_f(K_s - \overline{K})^2}{K_f(K_s - \overline{K}) + \phi K_s(K_s - K_f)} \tag{10.47}$$

in which we again assumed that K_s and K_f are constants.

Figure 10.5 depicts the tangential compressive stress $-\sigma_{\theta\theta}/S_o$ in the $\theta = 90°$ direction for the mode III problem, at two locations, $r = r_o = 10\,\text{cm}$, and $r = 11\,\text{cm}$, that is, 1 cm into the wall. We observe that the viscoelastic effect makes a difference only for the shale. At large time, the asymptotic value of $\sigma_{\theta\theta} = -3S_o$ at the wall is reached for all cases. Comparing Fig. 10.5a, b, we observe that very large stress variation happens within a short distance from the wall and within a short time. This effect has been discussed in Sect. 7.15.2, and we observe here that it is largely due to the poroelastic effect, as the viscoelastic effect manifests itself at larger time.

10.4 Cylinder Problem

The cylinder problem was investigated as mode 1 through 4 in Sect. 7.13. For the numerical results, we shall refer to the same three rocks investigated in the preceding section, and assume the same viscoelastic properties for these rocks. We also assume that the size of the cylinder is $r_o = 10\,\text{cm}$.

Mode 1 Figure 10.6a presents the radial displacement at the cylinder surface, $u_r(r_o, t)$, normalized by its initial value, $u_r(r_o, 0)$, versus time, where

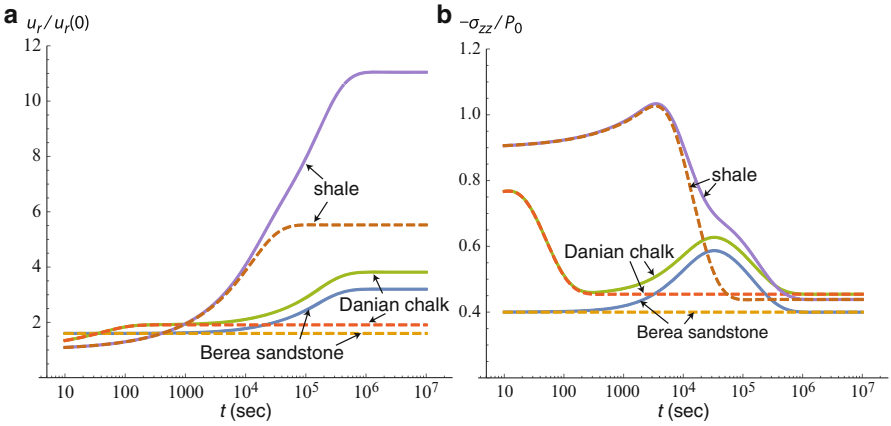


Fig. 10.6 Cylinder problem mode 1. *Solid line*: poroviscoelastic solution; *dashed line*: poroelastic solution. (**a**) Radial displacement at cylinder surface, $r = r_o$, versus time; and (**b**) vertical stress at cylinder center, $r = 0$

$$u_r(r_o, 0) = -\frac{P_o r_o(1 - 2\nu_{u1})}{2G_1} \tag{10.48}$$

with ν_{u1} and G_1 the initial values of these material constants. For the poroelastic solutions, shown in dashed lines, we anticipate the cylinder to consolidate as time progresses, with the large to small time displacement ratio given by

$$\frac{u_r(r_o, \infty)}{u_r(r_o, 0)} = \frac{1 - 2\nu_1}{1 - 2\nu_{u1}} \tag{10.49}$$

For the Berea sandstone, the poroelastic process takes place within a fraction of a second, due to its large permeability and small sample size; hence the dashed line shows no variation in the plotted time range. For the combined poroviscoelastic response, the Berea sandstone exhibits only one stage of consolidation, due to the viscoelastic effect. For Danian chalk, we find two consolidation stages, with the first one due to pore pressure diffusion, and the second attributed to viscoelastic creep. For the shale, the time scale for these two processes are close to each other; hence the response curve is mixed, and it is difficult to separate the stages.

Figure 10.6b shows the axial stress σ_{zz} at the center of the cylinder, $r = 0$. The stress is created by the Poisson effect, as the cylinder is prevented from expanding in the z-direction. For the poroelastic effect, the stress changes from the small time to large time value as

$$\sigma_{zz}(0, 0) = -2\nu_{u1} P_o; \qquad \sigma_{zz}(0, \infty) = -2\nu_1 P_o \tag{10.50}$$

For Berea sandstone, we again observe that the sample is completely drained in the time range of the plot, and the variation is entirely due to viscoelasticity. For the viscoelastic effect, the evolution of the Poisson ratio is controlled by the relaxation time of $G(t)$ and $K(t)$, which are selected to be different. As the shear modulus decreases first, the drained Poisson ratio increases with time, which leads to the increase in vertical compressive stress. At larger time, the bulk modulus starts to decrease. In the end, both shear modulus and bulk modulus are reduced by the same proportion, and the drained Poisson ratio return to its original values. These behaviors explain the rising and falling of vertical compressive stress for the Berea sandstone case.

For Danian chalk, we clearly observe that the poroelastic and viscoelastic effects are separated. In the first stage, the stress drops due to poroelastic effect; and then the interaction between the shear and bulk modulus causes the stress to rise and then fall. For the shale, these effects are mixed, as demonstrated in Fig. 10.6b.

Mode 2 The mode 2 problem subject the cylinder surface to a pore pressure increment, yet without changing the total stress. In Fig. 10.7a the radial displacement at the outer surface, $u_r(r_o, t)$, normalized by its final poroelastic value, which is

$$u_r(r_o, \infty) = \frac{p_o r_o \eta_1 (1 - \nu_1)}{G_1} \tag{10.51}$$

is plotted for Danian chalk and shale. Here we note that the initial radial displacement $u_r(r_o, 0)$ is zero. The Berea sandstone case is not presented because it is purely viscoelastic in the time range of the plot. For Danian chalk, we again observe two clear stages of creep—the earlier one due to poroelastic effect, and the later one due to viscoelastic effect. For the shale, these two stages are mixed and hard to distinguish from each other. Figure 10.7b present the pressure response at the center

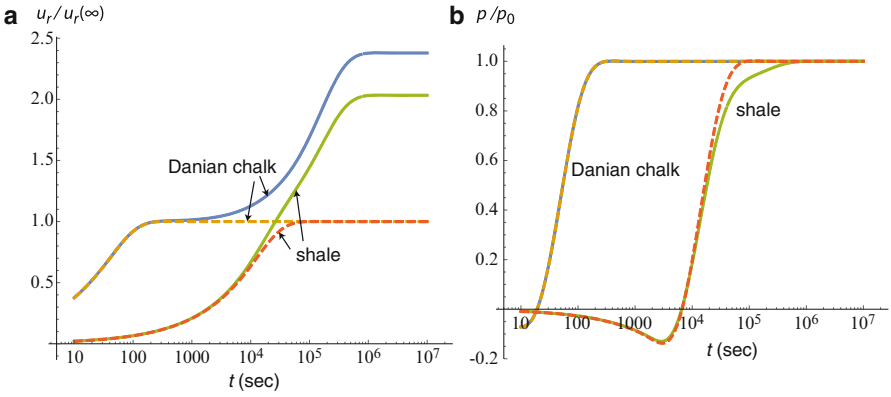


Fig. 10.7 Cylinder problem mode 2. *Solid line*: poroviscoelastic solution; *dashed line*: poroelastic solution. (**a**) Radial displacement at cylinder surface, $r = r_o$, versus time; and (**b**) pressure at center of cylinder, $r = 0$

of the cylinder ($r = 0$), for these two rocks. We observe that the viscoelastic effect somewhat delays the pore pressure response for the shale case.

Mode 3 Mode 3 solution is given as (7.538)–(7.543), which consists of only shear strain, and is without pore pressure. Hence only the viscoelastic effect exists, and is not elaborated.

Mode 4 Mode 4 problem is caused by the loading of an axial strain $e_{zz}(t)$. Here we examine the case of a compressive strain that increases linearly with time

$$e_{zz} = -At \tag{10.52}$$

The solution in Laplace transform domain due to a Heaviside step loading has been given as (7.554)–(7.558). For the loading given as above, the solution for pressure becomes

$$s\tilde{p} = \frac{G(\nu_u - \nu)Ar_o^2}{c\,\eta\sqrt{s^*}D(s^*)}\left[\mathrm{I}_0(\sqrt{s^*}) - \mathrm{I}_0(r^*\sqrt{s^*})\right] \tag{10.53}$$

Figure 10.8 depicts the pore pressure response at the center of the cylinder, normalized by its steady state poroelastic solution

$$p(0,\infty) = \frac{AG_1(\nu_{u1} - \nu_1)r_o^2}{4c_1\eta_1(1 - \nu_{u1})} \tag{10.54}$$

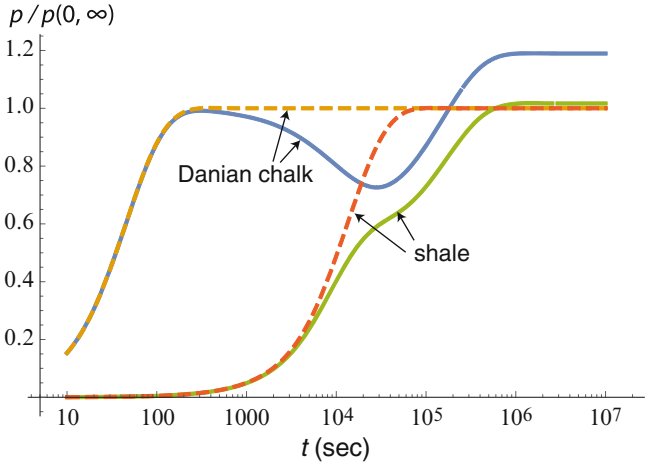


Fig. 10.8 Pore pressure response at the center of the cylinder $p(0, t)$ due to mode 4 loading. *Solid line*: poroviscoelastic solution; *dashed line*: poroelastic solution

We again observe that Danian chalk exhibits a poroelastic and a viscoelastic stage, due to the different relaxation time of these two responses. We also see that the viscoelastic effect on the pore pressure is non-monotonic, due to the time evolution of Poisson ratio. For the shale case, these two responses are mixed, and the pore pressure rise is monotonic.

As the strain continuously to increase with time, the stresses behave the same way. The pore pressure, however, reaches a steady state at large time. This is because pore pressure is dissipated by the diffusion from the center of the core to the surface. When the rate of diffusion becomes equal to the rate of generation by Skempton effect, a steady state is reached.

Anelastic Strain Recovery As discussed in Sect. 1.3.12, rock cores drilled and retrieved from great depth experience a sudden relief of confining stresses and pore pressure on the cylinder surface, and can be observed to creep for a few hours to a few days. The creep can be measured and used as a way to determine not only the direction, but also the magnitude, of the in situ stress, a method known as the anelastic strain recovery (ASR). According to the viscoelasticity theory, once the confining stress is removed, the core will expand with time. However, the actual core behavior is more complex. Some cores were observed to contract and then expand [7, 12, 14, 29]. This can be attributed to the poroviscoelastic effect.

The poroelastic solution of a retrieved core has been presented in Sect. 7.15.1 as the superposition of mode 1 and mode 2 problems (assuming mode 3 is not present), based on (7.701)–(7.706). For the ASR, we are interested in the radial displacement at the cylinder surface. The poroviscoelastic solution can be obtained as the proper superposition of the results in Figs. 10.6a and 10.7a.

Assume that a cylindrical core is initially under a uniform compressive stress P_o and a virgin pore pressure $p_o = 0.8P_o$. At $t = 0^+$, the normal stress and pore pressure on the cylinder surface are suddenly reduced to zero. The solution of radial displacement at the cylinder surface, $u_r(r_o, t)$, is presented in Fig. 10.9a. For Danian chalk, we observe that the creep takes place in two stages: first a contraction due to the poroelastic effect, and then an expansion due to the viscoelastic effect. For the shale, these two stages are mixed and only an expansion is observed. This means that if we interpret this expansion as a viscoelastic only effect, it will lead to erroneous result.

To gain further insight into the non-monotonic behavior of Danian chalk as shown in Fig. 10.9a, we can separately plot the contributions from the mode 1 and mode 2 effect, which can be further isolated into the poroelastic and viscoelastic effect, as shown in Fig. 10.9b. For mode 1 poroelastic effect (dashed line), the removal of surface radial stress is equivalent to pulling the cylinder surface outward, causing it to instantly expand. By Skempton effect, the pore pressure drops to negative. The zero pore pressure condition at the surface then causes a diffusion of pore pressure into the cylinder, which has the effect of increasingthe effective tensile

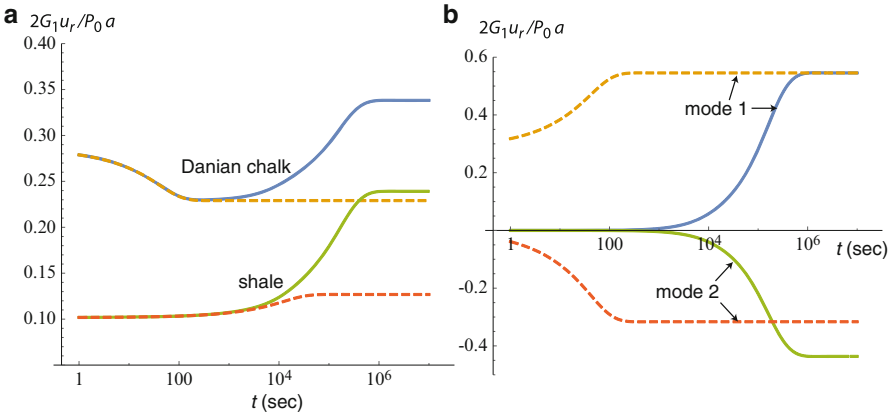


Fig. 10.9 (**a**) Radial displacement of retrieved cylindrical core, $u_r(r_o, t)$. *Solid line*: poroviscoelastic solution; *dashed line*: poroelastic solution. (**b**) The Danian chalk solution decomposed into mode 1 and mode 2 contributions. *Solid line*: viscoelastic contribution; *dashed line*: poroelastic contribution

stress in the core. Hence the core continues to expand. The additional viscoelastic effect (difference between poroviscoelastic and poroelastic solution) due to the removal of the stress adds to the time-dependent expansion.

For mode 2, a reduction of pore pressure on the surface of the core induces a pore pressure diffusion, causing fluid to flow outward. The pore pressure reduction has the effect of making the cylinder in effective compression; hence the cylinder contracts with time. The viscoelastic effect further adds on to this trend, but manifests itself at larger time. These four effects combines together to give the non-monotonic behavior shown in Fig. 10.9a.

Time Evolution of Parameters For the all the above poroviscoelastic responses for borehole and cylinder, we use two viscoelastic models, one for shear modulus G, and one for drained bulk modulus K, both are based on the standard linear solid model. For both G and K, we assume that their final values are reduced to one half of their initial. The viscoelastic relaxation times for these parameters are chosen to be μ_G/G_2 for 6 h, and μ_K/K_2 for 2 days. To gain insight into what has caused the stress, pressure, and displacement behaviors shown, it may be of interest to examine the time evolution of the various bulk material constants, such as drained and undrained Poisson ratio, effective stress coefficient, Skempton pore pressure coefficient, etc., that are responsible for the variety of physical phenomena discussed in Sect. 1.2.

In Fig. 10.10 we plot the time evolution of these coefficients normalized by their initial values. We observe that most of their behaviors are more complex than the standard linear solid model; hence it is important to implement the viscoelastic model at the micromechanical and intrinsic material constant level.

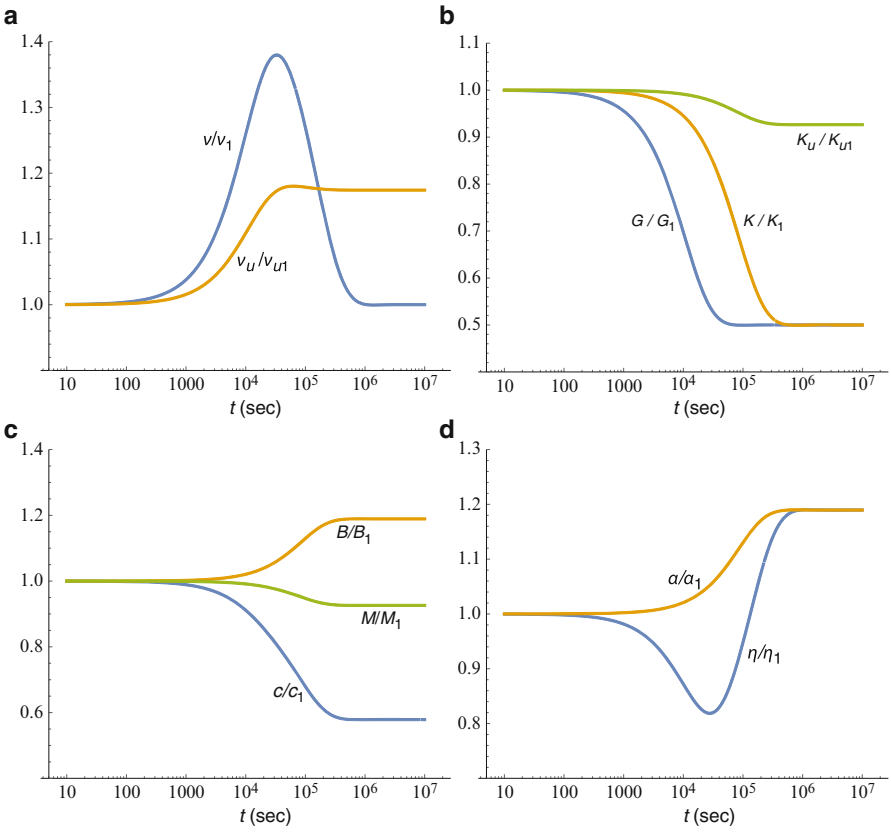


Fig. 10.10 The time evolution of material constants normalized by their initial values. (**a**) Drained and undrained Poisson ratios; (**b**) shear, and drained and undrained bulk moduli; (**c**) consolidation and Skempton pore pressure coefficients, and Biot modulus; and (**d**) Biot effective stress and poroelastic coefficients

10.5 Poroviscoelastodynamics

The above established viscoelasticity models can be extended to dynamic poroelasticity [26]. For dynamics, the models should be constructed in the Fourier transform, rather than Laplace transform, domain. In fact, Fourier transform and Laplace transform are formally equivalent if we replace the Laplace transform parameter s by $-i\omega$; and then all the earlier derived correspondence principles remain valid.

For example, the standard linear solid model for drained bulk modulus expressed in the frequency domain is

$$\overline{K} = \frac{K_1 K_2 - i\omega K_1 \mu_K}{K_1 + K_2 - i\omega \mu_K} = \overline{K}' + i\overline{K}'' \tag{10.55}$$

where the real part

$$\overline{K}'(\omega) = \frac{K_1(K_1K_2 + K_2^2 + \mu_K^2\omega^2)}{(K_1 + K_2)^2 + \mu_K^2\omega^2} \tag{10.56}$$

is called the *storage modulus*, and the imaginary part

$$\overline{K}''(\omega) = -\frac{K_1^2\mu_K\omega}{(K_1 + K_2)^2 + \mu_K^2\omega^2} \tag{10.57}$$

is known as the *loss modulus*. The squirt (BISQ) model described in Sect. 9.5.3 is already a poroviscoelastodynamic model, in which the Biot modulus M is replaced by a complex modulus M_{sq}, given as (9.176).

These models can be used in the poroelastodynamic problems with known analytical solution in the frequency domain, such as the one-dimensional wave problems derived in Sect. 9.6. In Fig. 10.11 we present the solution for the problem of stress excitation at the top of a finite thickness layer, given as (9.214) and (9.216). The material is assumed to be Berea sandstone. In the figure, the dashed line corresponds to the poroelastic solution already presented in Fig. 9.13. Here we further introduce the poroviscoelastic solution by using the same standard linear solid model parameters adopted in the preceding section, and also the BISQ model using the squirt flow characteristic length $R_{sq} = 0.5$ mm. In Fig. 10.11a the solution is shown for the amplitude of displacement at the top of the layer. We observe that resonance peaks are shifted because both viscoelastic models can change the wave propagation velocity. Figure 10.11b presents the pore pressure response. We note that results are shown only for the poroelastic and the standard linear solid cases

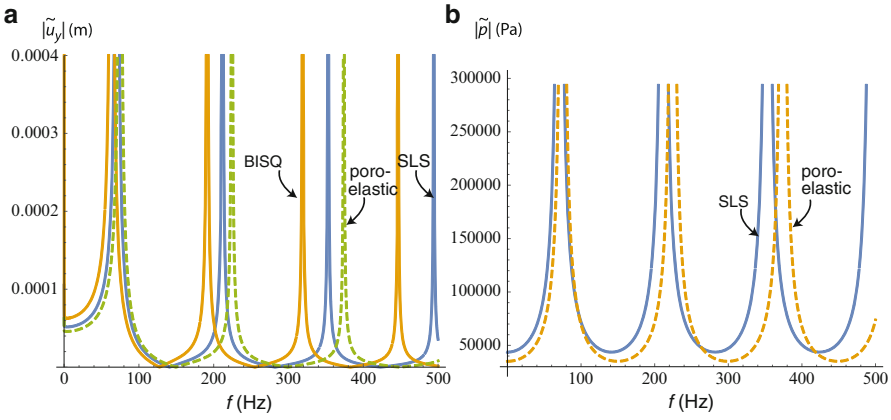


Fig. 10.11 Force excitation at the top of finite thickness layer for Berea sandstone. Responses due to poroelastic, standard linear solid (SLS) and squirt (BISQ) model. (**a**) Displacement amplitude at the top of layer $|\widetilde{u}_y(L)|$, and (**b**) pressure amplitude at the bottom $|\widetilde{p}(0)|$

because the BISQ model predicts nearly zero pore pressure response. This is due to the limit of $M_{sq} \to 0$ as $f \to 0$, an inherent assumption of the model. This suggests that the BISQ model may be suitable for the prediction of acoustic wave speed, but may not be suitable for the investigation of geotechnical problems such as pore pressure response and the associated effective stress.

References

1. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1993) A micromechanically consistent poroviscoelasticity theory for rock mechanics applications. Int J Rock Mech Min Sci 30(7):1177–1180
2. Abousleiman Y, Cheng AHD, Jiang C, Roegiers JC (1996) Poroviscoelastic analysis of borehole and cylinder problems. Acta Mechanica 119(1–4):199–219
3. Biot MA (1955) Variational principles in irreversible thermodynamics with application to viscoelasticity. Phys Rev 97(6):1463–1469
4. Biot MA (1956) Theory of deformation of a porous viscoelastic anisotropic solid. J Appl Phys 27(5):459–467
5. Biot MA (1956) Variational and Lagrangian methods in viscoelasticity. In: Grammel R (ed) Deformation and flow of solids, IUTAM colloquium Madrid, 1955. Springer, Berlin, pp 251–263
6. Biot MA (1962) Mechanics of deformation and acoustic propagation in porous media. J Appl Phys 33(4):1482–1498
7. Brereton NR, Chroston PN, Evans CJ, Hudson JA, Whitmarsh RB (1992) Anelastic strain recovery and elastic properties of oceanic basaltic rocks. In: Gradstein FM, Ludden JN et al (eds) Proceedings of the ocean drilling program, scientific results, College Station, vol 123, pp 469–491
8. Brutsaert W, Corapcioglu MY (1976) Pumping of aquifer with viscoelastic properties. J Hydraul Div ASCE 102(11):1663–1675
9. Christensen RM (1982) Theory of viscoelasticity, an introduction, 2nd edn. Academic, New York, 364pp
10. Christie IF (1964) A re-appraisal of Merchant's contribution to the theory of consolidation. Géotechnique 14(4):309–320
11. Cleary MP (1978) Elastic and dynamic response regimes of fluid-impregnated solids with diverse microstructures. Int J Solids Struct 14(10):795–819
12. Dey TN, Kranz RL (1988) State of stress and relationship of mechanical properties to hydrothermal alteration at valles caldera core hole 1, New Mexico. J Geophys Res–Solid Earth Planets 93(B6):6108–6112
13. El Rabaa AW (1989) Determination of the stress field and fracture direction in the Danian chalk. In: Maury V, Fourmaintraux D (eds) Rock at great depth. Balkema, Rotterdam/Brookfield, pp 1017–1024
14. El Rabaa AW, Meadows DL (1986) Laboratory and field application of the strain relaxation method. SPE 15072, SPE 56th California regional meeting, Oakland, pp 259–272
15. Flügge W (1975) Viscoelasticity, 2nd edn. Springer, Berlin/New York, 194pp
16. Griggs D (1939) Creep of rocks. J Geol 47(3):225–251
17. Hirschman II, Widder DV (2005) The convolution transform. Dover, New York, 288pp
18. Hoang SK, Abousleiman YN (2009) Poroviscoelastic two-dimensional anisotropic solution with application to articular cartilage testing. J Eng Mech ASCE 135(5):367–374
19. Hoang SK, Abousleiman YN (2010) Poroviscoelasticity of transversely isotropic cylinders under laboratory loading conditions. Mech Res Commun 37(3):298–306

20. Hoang SK, Abousleiman YN (2012) Correspondence principle between anisotropic poro-viscoelasticity and poroelasticity using micromechanics and application to compression of orthotropic rectangular strips. J Appl Phys 112:044907
21. Mehrabian A, Abousleiman Y (2011) General solutions to poroviscoelastic model of hydrocephalic human brain tissue. J Theor Biol 291:105–118
22. Mody FK, Hale AH (1993) Borehole-stability model to couple the mechanics and chemistry of drilling-fluid/shale interactions. J Pet Technol 45(11):1093–1101
23. Murad MA, Guerreiro JN, Loula AFD (2001) Micromechanical computational modeling of secondary consolidation and hereditary creep in soils. Comput Methods Appl Mech Eng 190(15–17):1985–2016
24. Nakken SJ, Christensen TL, Marsden R, Holt RM (1989) Mechanical behaviour of clays at high stress levels for well bore stability applications. In: Maury V, Fourmaintraux D (eds) Rock at great depth, ISRM international symposium. Balkema, Pau, pp 141–148
25. Navier C-L (1821) Memoire sur les lois de l'équilibre et du mouvement des corps solides élastiques (Memorandum on the laws of equilibrium and motion of elastic solids). Mem Acad Sci Inst Fr 2:375–393
26. Schanz M, Cheng AHD (2001) Dynamic analysis of a one-dimensional poroviscoelastic column. J Appl Mech 68(2):192–198
27. Taylor DW, Merchant W (1940) A theory of clay consolidation accounting for secondary compression. J Math Phys 19:167–185
28. Warpinski NR, Teufel LW (1989) A viscoelastic constitutive model for determining in-situ stress magnitudes from anelastic strain recovery of core. SPE Prod Eng 4(3):274–279
29. Warpinski NR, Teufel LW (1989) Author's reply to discussion of "a viscoelastic constitutive model for determining in-situ stress magnitudes from anelastic strain recovery of core". SPE Prod Eng 4(3):287–289
30. Zener C (1948) Elasticity and anelasticity of metals. University of Chicago Press, Chicago, 170pp

Chapter 11
Porothermoelasticity

If two corpuscles of a body lie infinitely close and have different temperature, the warmer corpuscle transmits a certain amount of its heat to the other one; and this heat—given from the warmer corpuscle to the colder one at a given time and during a given moment—is proportional to the temperature difference, if the difference has a small value.

—Joseph Fourier (1822)

The poroelasticity theory presented so far assumes an *isothermal* condition, that is, temperature remains unchanged during the deformation and diffusion process. In practice, however, temperature of a porous medium can change. Not only it can change if it is in contact with a body of different temperature, and heat is transferred by conduction, but also the deformation itself can generate internal heat. In addition, heat can be transported in and out of the porous medium by a fluid flow.

It is a common knowledge that a change of temperature in solid body induces a stress or a strain, depending on the boundary constraint. This phenomenon has been modeled in solid mechanics as the *theory of thermal stress* [6]. It is however less well known that without an external heat input, the deformation of a body can cause a temperature change. This reciprocal effect is ignored in the theory of thermal stress because its magnitude is small for most materials. The theory that considers both of these effects is known as the *coupled theory of thermoelasticity* [53]. For porous medium, the heat effect is applied to two material phases, a solid and a fluid, and even more complicated physical phenomena can result from their interaction. In this chapter, we shall formulate the coupled thermal-hydraulic-mechanical theory of *porothermoelasticity*.

Heat plays an important role in the mechanical responses of porous materials in many natural phenomena and man-made processes. For example, heat can be extracted from geothermal reservoir by circulating colder water in the formation and recovering it at a higher temperature [28, 29]; in a fault zone, the friction generated heat can raise the pore water pressure to cause a reduction of effective stress, and trigger earthquakes [40, 43, 44, 71]; in a nuclear waste repository, the heat generated by radioactive decay can damage the integrity of the containment structure and the surrounding saturated and unsaturated rocks, and also induce a circulation of fluid

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_11

flow by thermal convection [7, 8, 56, 66]; drilling fluid of a different temperature in a borehole can cause spalling failure on the wall [1, 11, 48, 72]; the burning of a low permeability and saturated cement can cause it to burst due to the differential in solid and fluid thermal expansion coefficients [26, 51]; a rapidly deforming shear band in the soil can generate heat that causes the softening of clay and activates fluid pressurization, leading to large scale landslide [75, 76]; and the cycling freezing and thawing of cement and rock causes damage in their strength [16, 17, 52, 58].

To extend the theory of poromechanics to include the thermal effects, Biot in 1977 [5] presented a finite strain, non-isothermal theory, which includes the thermomolecular diffusion and the dynamic forces, using the variational Lagrangian thermodynamics approach. Schiffman [61] and Booker and Savvidou [7], on the other hand, provided a simple extension of Biot's quasi-static poroelasticity theory by adding a thermal stress term to the constitutive equations. Palciauskas and Domenico [56] and McTigue [47] performed micromechanical analysis by explicitly modeling the constitutive equations of solid and fluid phases. These theories [7, 47, 56, 61], however, are similar to the theory of thermal stress, which uncouples the thermal diffusion process from the solid and fluid deformation. The fully coupled porothermoelasticity theory was derived by Derski and Kowalski [23], as well as others [2, 39, 65].

In the following section, we shall derive the coupled porothermoelasticity theory using the variational energy approach as introduced in Chap. 4.

11.1 Constitutive Equation

11.1.1 Thermodynamic Formulation

Following the derivation in thermoelasticity [4, 53], we shall introduce the Helmholtz free energy $\mathcal{A}$ as defined in (4.53), with its variation for a reversible process given by (4.56), which is repeated here

$$\delta \mathcal{A} = \delta \mathcal{W} - \mathcal{S} \delta \mathcal{T} \tag{11.1}$$

Defining $\overline{\mathcal{A}}_s$ and $\overline{\mathcal{A}}_f$ as the volume averaged Helmholtz free energy density, respectively for the solid and the fluid phase, we can express

$$\mathcal{A} = \Omega_s \overline{\mathcal{A}}_s + \Omega_f \overline{\mathcal{A}}_f \tag{11.2}$$

Following the development in Sect. 4.3.2, we can express the solid and fluid energy density as functions of these state variables

$$\overline{\mathcal{A}}_s = \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi, \overline{\mathcal{T}}_s) \tag{11.3}$$

$$\overline{\mathcal{A}}_f = \overline{\mathcal{A}}_f \left(\overline{\epsilon}_f, \overline{\mathcal{T}}_f \right) \tag{11.4}$$

where $\overline{\mathcal{T}}_s$ and $\overline{\mathcal{T}}_f$ are respectively the solid and fluid phase volume averaged absolute temperature. Comparing to (4.70) and (4.72) for the isothermal case, we observe the additional temperature dependence in the free energy density.

With the above state variables, we can write the variation of Helmholtz free energy as

$$\begin{aligned}
\delta \mathcal{A} &= \Omega_s \left[\frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{\epsilon}^s_{ij} + \frac{\partial \overline{\mathcal{A}}_s}{\partial \phi} \delta \phi + \frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\mathcal{T}}_s} \delta \overline{\mathcal{T}}_s \right] + \Omega_f \left[\frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\epsilon}^f} \delta \overline{\epsilon}^f + \frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\mathcal{T}}_f} \delta \overline{\mathcal{T}}_f \right] \\
&= \Omega_s \frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s_{ij}} \delta \overline{e}^s_{ij} + \Omega_f \frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\epsilon}^f} \delta \overline{e}^f + \Omega_s \frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\mathcal{T}}_s} \delta \overline{\mathcal{T}}_s + \Omega_f \frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\mathcal{T}}_f} \delta \overline{\mathcal{T}}_f \\
&\quad + \left[\Omega_s \frac{\partial \overline{\mathcal{A}}_s}{\partial \phi} - \frac{\Omega_s}{1-\phi} \frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s} + \frac{\Omega_f}{\phi} \frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\epsilon}^f} \right] \delta \phi
\end{aligned} \tag{11.5}$$

In the above we have converted from the internal to external strains.

Substituting in (4.63), we can write the right hand side of (11.1) as

$$\delta \mathcal{W} - \mathcal{S} \delta \mathcal{T} = \Omega_s \overline{\sigma}^s_{ij} \delta \overline{e}^s_{ij} - \Omega_f \overline{p}^f \delta \overline{e}^f - \Omega_s \overline{s}_s \delta \overline{\mathcal{T}}_s - \Omega_f \overline{s}_f \delta \overline{\mathcal{T}}_f \tag{11.6}$$

In the above $\overline{s}_s$ and $\overline{s}_f$ are respectively the *entropy density* (entropy per unit volume) of the solid and fluid phase. Equating (11.5) and (11.6), we obtain the following definitions for stresses, pore pressure, and entropy

$$\overline{\sigma}^s_{ij} = \frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s_{ij}} \tag{11.7}$$

$$\overline{p}^f = -\frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\epsilon}_f} \tag{11.8}$$

$$\overline{s}_s = -\frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\mathcal{T}}_s} \tag{11.9}$$

$$\overline{s}_f = -\frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\mathcal{T}}_f} \tag{11.10}$$

Based on the variation of $\delta\phi$, we also obtain the porosity equilibrium equation

$$(1-\phi) \frac{\partial \overline{\mathcal{A}}_s}{\partial \phi} + \overline{p}^s - \overline{p}^f = 0 \tag{11.11}$$

11.1.2 Linear Model and Intrinsic Material Constant

For small deformation and small change in temperature, we can obtain linear constitutive equations by expanding the free energies with respect to their state variables around an equilibrium point to obtain the quadratic forms

$$\overline{\mathcal{A}}_s = \frac{1}{2}\hat{\lambda}_s \overline{\epsilon}^s \overline{\epsilon}^s + \hat{\mu}_s \overline{\epsilon}^s_{ij} \overline{\epsilon}^s_{ij} + \frac{1}{2} K_\phi \, \Delta\phi^2 + K_\psi \, \overline{\epsilon}^s \, \Delta\phi - B_s \overline{\epsilon}^s \, \overline{T}_s$$
$$- B_\psi \, \overline{T}_s \, \Delta\phi - \frac{1}{2} m_\alpha \overline{T}_s^2 \tag{11.12}$$

$$\overline{\mathcal{A}}_f = \frac{1}{2} K_f \overline{\epsilon}^f \overline{\epsilon}^f - B_f \overline{\epsilon}^f \, \overline{T}_f - \frac{1}{2} m_\beta \overline{T}_f^2 \tag{11.13}$$

where

$$\overline{T}_s = \overline{\mathcal{T}}_s - \mathcal{T}_o; \qquad \overline{T}_f = \overline{\mathcal{T}}_f - \mathcal{T}_o \tag{11.14}$$

are temperature changes from a reference state $\mathcal{T}_o$. Similar to the infinitesimal strain assumption of linear elasticity, here we assume that $\overline{T}_s/\mathcal{T}_o \ll 1$ and $\overline{T}_f/\mathcal{T}_o \ll 1$. In the above, the constitutive constants $\hat{\lambda}_s$, $\hat{\mu}_s$, K_ϕ, K_ψ, and K_f have been earlier introduced for the isothermal case in (4.80). The thermal coupling requires the additional constitutive constants B_s, B_ψ, B_f, m_α, and m_β. We note that in the above quadratic forms we have made the assumption that shear strain does not generate temperature; hence such interaction terms are omitted.

Stress-Strain Relations Substituting the quadratic forms (11.12) and (11.13) into (11.7) and (11.8), we obtain the following constitutive equations

$$\overline{\sigma}^s_{ij} = \delta_{ij} \left(K_\alpha \, \overline{\epsilon}^s + K_\psi \, \Delta\phi - B_s \overline{T}_s \right) + 2G_s \langle \overline{\epsilon}^s_{ij} \rangle \tag{11.15}$$

$$\overline{p}^f = -K_f \overline{\epsilon}^f + B_f \overline{T}_f \tag{11.16}$$

where K_α and G_s are defined in (4.82). Equation (11.15) can be contracted to give the solid volumetric response as

$$\overline{p}^s = -K_\alpha \, \overline{\epsilon}^s - K_\psi \, \Delta\phi + B_s \overline{T}_s \tag{11.17}$$

From the porosity equilibrium equation (11.11), we obtain the linearized constitutive equation for porosity change

$$\overline{p}^s - \overline{p}^f = -(1-\phi) K_\phi \, \Delta\phi - (1-\phi) K_\psi \, \overline{\epsilon}^s + (1-\phi) B_\psi \, \overline{T}_s \tag{11.18}$$

which, as discussed in Sect. 4.4.2, is not an independent equation, but it relates the porosity change $\Delta\phi$ to the solid stress and strain variables.

Strain-Stress Relations The three Eqs. (11.16), (11.17), and (11.18), can be used to eliminate variables to obtain the following two strain-stress relations

$$\overline{\epsilon}^s = -\frac{\left[(1-\phi)K_\phi - K_\psi\right]\overline{p}^s + K_\psi\,\overline{p}^f}{(1-\phi)(K_\alpha K_\phi - K_\psi^2)} + \frac{(K_\phi B_s - K_\psi B_\psi)\overline{T}_s}{K_\alpha K_\phi - K_\psi^2} \tag{11.19}$$

$$\overline{\epsilon}^f = \frac{-\overline{p}^f + B_f\,\overline{T}_f}{K_f} \tag{11.20}$$

We can also express the change of porosity as

$$\Delta\phi = -\frac{\left[K_\alpha - (1-\phi)K_\psi\right]\overline{p}^s - K_\alpha\,\overline{p}^f}{(1-\phi)(K_\alpha K_\phi - K_\psi^2)} + \frac{(K_\alpha B_\psi - K_\psi B_s)\overline{T}_s}{K_\alpha K_\phi - K_\psi^2} \tag{11.21}$$

The above equations can be compared to the isothermal case, (4.89), (4.90), and (4.91).

As in the poroelasticity case, it is of interest to replace some of the constitutive constants appearing in the quadratic form by ones that have more direct physical meaning. For this, we shall replace K_α by K_s using (4.88), and introduce the following three thermal constants

$$\beta_s = \frac{K_\phi B_s - K_\psi B_\psi}{K_\alpha K_\phi - K_\psi^2} = \frac{K_\phi B_s - K_\psi B_\psi}{K_s K_\phi} \tag{11.22}$$

$$\beta_f = \frac{B_f}{K_f} \tag{11.23}$$

$$\beta_\psi = \frac{K_\alpha B_\psi - K_\psi B_s}{K_\alpha K_\phi - K_\psi^2} = \frac{(K_s K_\phi + K_\psi^2)B_\psi - K_\phi K_\psi B_s}{K_s K_\phi^2} \tag{11.24}$$

or, their inverse relations

$$B_s = \frac{(K_s K_\phi + K_\psi^2)\beta_s + K_\phi K_\psi \beta_\psi}{K_\phi} \tag{11.25}$$

$$B_f = K_f \beta_f \tag{11.26}$$

$$B_\psi = K_\psi \beta_s + K_\phi \beta_\psi \tag{11.27}$$

With the substitution, we can rewrite (11.19), (11.20), and (11.21) into the following

$$\overline{\epsilon}^s = -\frac{\left[(1-\phi)K_\phi - K_\psi\right]\overline{p}^s + K_\psi\,\overline{p}^f}{(1-\phi)K_s K_\phi} + \beta_s\,\overline{T}_s \tag{11.28}$$

$$\overline{\epsilon}^f = -\frac{\overline{p}^f}{K_f} + \beta_f\,\overline{T}_f \tag{11.29}$$

$$\Delta\phi = -\frac{[K_s K_\phi - (1-\phi)K_\phi K_\psi + K_\psi^2]\,\overline{p}^s - (K_s K_\phi + K_\psi^2)\,\overline{p}^f}{(1-\phi)K_s K_\phi^2} + \beta_\psi\,\overline{T}_s \tag{11.30}$$

Interpretation of Material Constants Based on the above relations, we observe that when the temperature is raised for a porous medium that is free to expand without solid and fluid stresses, $\overline{p}^s = \overline{p}^f = 0$, these constitution equations reduce to

$$\overline{\epsilon}^s = \beta_s\,\overline{T}_s \tag{11.31}$$

$$\overline{\epsilon}^f = \beta_f\,\overline{T}_f \tag{11.32}$$

$$\Delta\phi = \beta_\psi\,\overline{T}_s \tag{11.33}$$

Hence we can identify β_s, β_f, and β_ψ as the coefficient of volumetric thermal expansion (three times the coefficient of linear thermal expansion) for solid, fluid, and porosity. These three coefficients are independent constants and will replace B_s, B_f, and B_ψ.

Entropy Constitutive Equation Using (11.12) and (11.13) in (11.9) and (11.10), we find the constitutive equations for the solid and fluid entropy density as

$$\overline{s}_s = m_\alpha\,\overline{T}_s + B_s\,\overline{\epsilon}^s + B_\psi\,\Delta\phi \tag{11.34}$$

$$\overline{s}_f = m_\beta\,\overline{T}_f + B_f\,\overline{\epsilon}^f \tag{11.35}$$

where we have set the zero entropy state at the reference absolute temperature $\mathcal{T}_o$ and zero strain. The above relations can also be written in terms of stresses and temperature as

$$\overline{s}_s = -\frac{\big[(1-\phi)\beta_s + \beta_\psi\big]\,\overline{p}^s - \beta_\psi\,\overline{p}^f}{1-\phi} + m_s\,\overline{T}_s \tag{11.36}$$

$$\overline{s}_f = -\beta_f\,\overline{p}^f + m_f\,\overline{T}_f \tag{11.37}$$

In the above we have introduced two constitutive constants m_s and m_f to replace m_α and m_β for their direct physical meaning, with the relations

$$m_s = m_\alpha + \frac{K_s K_\phi \beta_s^2 + (K_\psi \beta_s + K_\phi \beta_\psi)^2}{K_\phi} \tag{11.38}$$

$$m_f = m_\beta + K_f \beta_f^2 \tag{11.39}$$

We observe that under the condition of free expansion for solid and fluid, $\overline{p}^s = \overline{p}^f = 0$, (11.36) and (11.37) reduce to

$$\bar{s}_s = m_s \overline{T}_s \tag{11.40}$$

$$\bar{s}_f = m_f \overline{T}_f \tag{11.41}$$

We can further express

$$m_s = \frac{c_s}{\mathcal{T}_o}, \qquad m_f = \frac{c_f}{\mathcal{T}_o} \tag{11.42}$$

Hence c_s and c_f are respectively the specific heat of solid and fluid phase under the condition of constant stress and pore pressure.

11.1.3 Ideal Porous Medium

In poroelasticity, an ideal porous medium model was constructed by assuming that the solid phase is homogeneous and isotropic at the microscopic level. Although this assumption is not always true, it leads to considerable simplification of the micromechanics model, and also sheds important physical insight into some of the micromechanical relations, as demonstrated in Sects. 3.2 and 4.7. Hence it is important to also discuss the implication of this assumption in the thermal model.

Consider that the microscopic material is not only mechanically, but also thermally homogeneous and isotropic. When such a material is subjected to a uniform heating, and allowed to freely expand without solid stress and pore pressure, the solid will expand in a geometrically self-similar fashion. In other words, the solid, the pore space, and the frame volume will expand in the same proportion, such that there is no porosity change. Equation (11.33) clearly indicate that under such condition, we must have

$$\beta_\psi = 0 \tag{11.43}$$

Combining with the condition $K_\psi = 0$, we find the following simplification of the material constant relations

$$B_s = K_s \beta_s \tag{11.44}$$

$$B_\psi = 0 \tag{11.45}$$

$$m_s = m_\alpha + K_s \beta_s^2 \tag{11.46}$$

as well as the constitutive equations

$$\overline{\epsilon}^s = -\frac{\overline{p}^s}{K_s} + \beta_s \overline{T}_s \tag{11.47}$$

$$\Delta\phi = -\frac{\overline{p}^s - \overline{p}^f}{(1-\phi)K_\phi} \tag{11.48}$$

$$\overline{s}_s = -\beta_s \overline{p}^s + m_s \overline{T}_s \tag{11.49}$$

11.1.4 Link with Bulk Continuum Model

The above analysis gives the physical origin of the intrinsic material constants. For practical applications, these constitutive equations need to be assembled into the measurable bulk quantities such as total stress, frame deformation, etc., as presented in Sect. 4.6.1.

Up to this point, we have considered two distinct temperatures, $\overline{T}_s$ and $\overline{T}_f$, respectively for the solid and fluid phase. On each of the temperatures, local equilibrium, that is, at the grain scale, the local variation of temperature reaches equilibrium at a time scale much smaller than the global time scale, is assumed. So far we have not made an assumption on the temperature equilibrium between the two phases. In general, $\overline{T}_s$ and $\overline{T}_f$ are two distinct variables as heat can be separately conducted in the solid and fluid phases with different thermal conductivity coefficients. In addition, heat can be transported in the fluid phase. But there also exists a heat transfer between the solid and fluid phases through the surface contact. The heat transfer can be difficult to model as it is dependent on the microstructure of the pore space. A conceptual model needs to be introduced.

As a first step in modeling, we shall introduce a simplified model. We shall make the additional assumption that due to the close contact between the solid and fluid phases at the grain scale, a *local thermal equilibrium* is established between the two phases much faster than the large scale heat conduction, such that only one temperature

$$\overline{T}_s = \overline{T}_f = T \tag{11.50}$$

exists at a given location and time. This assumption is the best if the solid and the fluid are well mixed, have small grain or pore size, and large specific surface. It also requires the condition that the fluid movement is minimal. In other words, the fluid movement is largely caused by the frame and solid deformation induced changes in fluid content, and not by a large scale sustained pressure differential driven porous medium flow. When such a flow exists, we must consider *advective heat transport*, that is, heat transported by the migration of fluid, which can be much faster than heat conduction. In that case, there may not be time enough for the sold and fluid temperatures to equilibrate, and a heat transfer law needs to be established between the two phases. This phenomenon is not considered in the present porothermoelasticity model.

Volumetric Strain To construct the bulk continuum model, we note the relations between the internal and external strains, (4.37), (4.38), and (4.117), and the total and partial stresses, (4.120). Following a similar procedure as in Sect. 4.6.1, and assembling terms, we obtain

$$e = -\frac{1}{K}(P - \alpha p) + \beta_d T \tag{11.51}$$

where K and α are related to the intrinsic material constants as (4.127) and (4.128). We observe that if we set $T = 0$ in (11.51), it reduces to the poroelastic case; hence K and α are the *isothermal* drained bulk modulus and the Biot effective stress coefficient, respectively. We also define a new material constant

$$\beta_d = \beta_s + \frac{\beta_\psi}{1 - \phi} \tag{11.52}$$

If we conduct a thermal expansion test that the porous medium is free from stress and pore pressure, $P = p = 0$, (11.51) reduces to

$$e = \beta_d T \tag{11.53}$$

Hence β_d is the *drained coefficient of volumetric thermal expansion for porous medium frame*. We observe that β_d is associated with the thermal properties of the solid and the porous structure only, and not the fluid. Particularly, under the ideal porous medium assumption, (11.52) reduces to

$$\beta_d = \beta_s \tag{11.54}$$

and its physical meaning is clearer.

To promote the effective stress concept, we can also express (11.51) as

$$e = -\frac{1}{K}(P - \alpha p - \alpha_d T) \tag{11.55}$$

where

$$\alpha_d = K\beta_d \tag{11.56}$$

is a drained thermoelastic effective stress coefficient, which plays a role similar to α in defining the effective stress relation.

For the variation in fluid content, it can be assembled into a form similar to (2.32), as

$$\zeta = -\frac{\alpha}{K}\left(P - \frac{p}{B}\right) - \beta_v T \tag{11.57}$$

where

$$\beta_v = \phi\,(\beta_f - \beta_s) - \frac{\beta_\psi}{1-\phi} \tag{11.58}$$

is a coefficient of volumetric thermal expansion for variation in fluid content. Its meaning is clear by conducting a thermal expansion test in which the sample is free to expand ($P = p = 0$). We then measure the fluid volume expelled from or absorbed into the frame, and find β_v from this relation

$$\zeta = -\beta_v T \tag{11.59}$$

We also observe that for ideal porous medium, (11.58) reduces to

$$\beta_v = \phi(\beta_f - \beta_s) \tag{11.60}$$

So β_v is the differential expansion coefficient between fluid and solid. For geomaterials saturated with a liquid, we generally anticipate $\beta_f > \beta_s$; hence β_v is generally positive.

Alternatively, we can express the constitutive equation (11.51) from a strain-stress form to a stress-strain form by replacing p on the right hand side by ζ, using (11.57), and obtain

$$P = K_u(-e + B\zeta + \beta_u T) \tag{11.61}$$

where

$$\beta_u = \beta_d + B\beta_v \tag{11.62}$$

By setting $\zeta = 0$ (undrained) and $P = 0$ (free expansion) in (11.61), we observe

$$e = \beta_u T \tag{11.63}$$

It is clear that β_u is the undrained coefficient of volumetric thermal expansion, in contrast to the drained coefficient defined in (11.51) and (11.52). For ideal porous medium, we can express β_u as

$$\beta_u = (1 - \phi B)\beta_s + \phi B \beta_f \tag{11.64}$$

based on (11.54) and (11.60). For the special case of incompressible solid and fluid, $B \to 1$, the above becomes

$$\beta_u = (1 - \phi)\beta_s + \phi\beta_f \tag{11.65}$$

hence β_u is the volume fraction weighted sum of solid and fluid thermal expansion coefficient. For highly compressible fluid, $B \to 0$, β_u reduces to the solid thermal expansion coefficient β_s.

We can also express (11.57) into a stress-strain form by utilizing (11.51):

$$p = M(-\alpha e + \zeta + \beta_e T) \tag{11.66}$$

where

$$\beta_e = \alpha\beta_d + \beta_v \tag{11.67}$$

is a coefficient of volumetric thermal expansion for variation in fluid content at constant frame volume. The significance of β_e is found by setting $e = p = 0$ and raising the temperature in a thermal expansion test. By observing the fluid expelled from the fixed frame volume, we can then determine β_e as

$$\zeta = -\beta_e T \tag{11.68}$$

Entropy To complete the constitutive equations, we define the total entropy density as

$$s = (1-\phi)\bar{s}_s + \phi\,\bar{s}_f \tag{11.69}$$

With the substitution of (11.36) and (11.37), and converting to the bulk variables, we can express the entropy as

$$s = -\beta_d P - \beta_v p + mT \tag{11.70}$$

where

$$m = (1-\phi)m_s + \phi\, m_f \tag{11.71}$$

We can express the constitutive constant m as

$$m = \frac{c_p}{\mathcal{T}_o} \tag{11.72}$$

By setting $P = p = 0$ in (11.70), we can identify c_p as the specific heat of the porous medium (solid and fluid) at constant stress.

Or, it is more common to define a specific heat at constant strain (referring to e) than at constant stress. We can in fact define two, one under drained and the other undrained condition. The drained one is defined in this relation

$$s = K\beta_d e - \beta_e p + m_d T \tag{11.73}$$

where

$$m_d = m - K\beta_d^2 \tag{11.74}$$

If we express the constitutive constant m_d as

$$m_d = \frac{c_d}{\mathcal{T}_o} \tag{11.75}$$

and set $e = p = 0$ in (11.73), we identify c_d as the drained specific heat at constant strain.

Alternatively, we can express s as a function of (e, ζ, T),

$$s = K_u\beta_u e - M\beta_e\zeta + m_u T \tag{11.76}$$

where

$$m_u = m_d - M\beta_e^2 \tag{11.77}$$

We can also write m_u as

$$m_u = \frac{c_u}{\mathcal{T}_o} \tag{11.78}$$

with c_u as the undrained specific heat at constant strain.

Reciprocal Work Theorem In the above, we have constructed the bulk continuum model by carefully assembling the thermodynamically consistent microscopic constitutive equations. As a consequence, the macroscopic model should also be thermodynamically consistent. This may be checked by summarizing the three constitutive equations, (11.51), (11.57), and (11.70), into the following form:

$$e = -\frac{1}{K}P + \frac{\alpha}{K}p + \beta_d T \tag{11.79}$$

$$\zeta = -\frac{\alpha}{K}P + \frac{\alpha}{BK}p - \beta_v T \tag{11.80}$$

$$s = -\beta_d P - \beta_v p + mT \tag{11.81}$$

We notice the symmetry of the constitutive constants in the above equations. It is easy to prove that these equations satisfy the reciprocal work theorem in the following form (see also Sect. 2.3.3)

$$-P^{(1)}e^{(2)} + p^{(1)}\zeta^{(2)} + T^{(1)}s^{(2)} = -P^{(2)}e^{(1)} + p^{(2)}\zeta^{(1)} + T^{(2)}s^{(1)} \tag{11.82}$$

Based on the above, we can back construct a Helmholtz free energy density using the bulk continuum variables as

$$\delta \mathcal{A} = \sigma_{ij} \delta e_{ij} + p \delta \zeta - s \delta T \tag{11.83}$$

We may then seek a quadratic form for $\mathcal{A}$ and derive a seemingly thermodynamics consistent model without going through the micromechanical analysis. In doing so, however, we should be aware that there is no particular reason that the Helmholtz free energy constructed as (11.83) is correct. The parameters used in the construction are apparent stress and strain quantities; for example, e is the frame deformation, and ζ is fluid escaping from the frame. These are not directly associated with the individual material phase and energy. The reason (11.83) is correct only because it is supported by the micromechanical analysis, as demonstrated in the above derivation. In fact, the micromechanical analysis leading to the above result contains a linearization of the porosity equilibrium equation, as discussed in Sects. 4.3.3 and 4.4.2.

Having stated the above caution, we can express the quadratic form of the Helmholtz free energy density as

$$\mathcal{A} = \frac{1}{2}\lambda_u e^2 + \mu e_{ij}^2 - \alpha M e \zeta - K_u \beta_u e T + \beta_e M \zeta T + \frac{1}{2} M \zeta^2 - \frac{1}{2} m_u T^2 \tag{11.84}$$

which can be compared with the isothermal case (6.39). In the above, we identify 7 independent constitutive constants, λ_u, μ (G), α, M, β_u, β_e, and m_u.

11.1.5 Bulk Continuum Constitutive Equation

For application purposes, the bulk continuum model is used in the construction of partial differential equations for the solution of boundary value problems. Here we summarize the constitutive equations based on the bulk continuum variables.

Stress-Strain Relation Combining with the constitution equation for shear deformation, which does not participate in the pore pressure and thermal processes, (11.61) becomes

$$\sigma_{ij} = \left(K_u - \frac{2G}{3} \right) \delta_{ij} e + 2G e_{ij} - \alpha M \delta_{ij} \zeta - \alpha_u \delta_{ij} T \tag{11.85}$$

where

$$\alpha_u = K_u \beta_u \tag{11.86}$$

is an undrained thermoelastic effective stress coefficient. In the above, we note the conversion of constants based on the relation among $\{\alpha, M, K_u, B\}$ as defined in (2.42). For pore pressure, we repeat (11.66) here

$$p = M(-\alpha e + \zeta + \beta_e T) \tag{11.87}$$

For the thermal constitutive equation, we rewrite (11.76) into the following form

$$s = \alpha_u e - M\beta_e \zeta + m_u T \tag{11.88}$$

The three Eqs. (11.85), (11.87), and (11.88), form a complete set of constitutive equations for porothermoelasticity that relates the variables $(\sigma_{ij},\ p,\ s)$ to $(e_{ij},\ \zeta,\ T)$, defined within the seven independent material constants $\{K_u, G, \alpha, M, \alpha_u, \beta_e, m_u\}$. We also observe the symmetry of material constants in the above equations.

It is sometimes preferable to use p, instead of ζ, on the right hand side of the above constitutive equations. In that case, we can express (11.85) as

$$\sigma_{ij} = \left(K - \frac{2G}{3}\right)\delta_{ij} e + 2Ge_{ij} - \alpha\delta_{ij} p - \alpha_d \delta_{ij} T \tag{11.89}$$

Equation (11.87) is expressed as

$$\zeta = \alpha e + \frac{p}{M} - \beta_e T \tag{11.90}$$

and (11.73) becomes

$$s = \alpha_d e - \beta_e p + m_d T \tag{11.91}$$

Equations (11.89), (11.90), and (11.91) then relate $(\sigma_{ij},\ \zeta,\ s)$ to $(e_{ij},\ p,\ T)$, defined by the set of material constants $\{K, G, \alpha, M, \alpha_d, \beta_e, m_d\}$.

For the above constitutive equations, it is possible to replace the temperature on the right hand side by entropy. For (11.85), we obtain

$$\sigma_{ij} = \left(K_b - \frac{2G}{3}\right)\delta_{ij} e + 2Ge_{ij} - \alpha_b M\delta_{ij}\zeta - \frac{\alpha_u}{m_u}\delta_{ij} s \tag{11.92}$$

where

$$K_b = K_u + \frac{\alpha_u^2}{m_u} \tag{11.93}$$

$$\alpha_b = \alpha + \frac{\alpha_u \beta_e}{m_u} \tag{11.94}$$

are the adiabatic undrained bulk modulus and the adiabatic undrained Biot effective stress coefficient, respectively. For pore pressure, (11.87) becomes

$$p = M\left(-\alpha_b e + \frac{m_d}{m_u}\zeta + \frac{\beta_e}{m_u}s\right) \tag{11.95}$$

The thermal constitutive equation is given by (11.76)

$$T = \frac{1}{m_u}(-\alpha_u e + M\beta_e \zeta + s) \tag{11.96}$$

The above system defines $(\sigma_{ij},\ p,\ T)$ as function of $(e_{ij},\ \zeta,\ s)$.

Similarly, we can express $(\sigma_{ij},\ \zeta,\ T)$ as function of $(e_{ij},\ p,\ s)$. From (11.89) we obtain

$$\sigma_{ij} = \left(K_a - \frac{2G}{3}\right)\delta_{ij}e + 2Ge_{ij} - \alpha_a\delta_{ij}p - \frac{\alpha_d}{m_d}\delta_{ij}s \tag{11.97}$$

where

$$K_a = K + \frac{\alpha_d^2}{m_d} \tag{11.98}$$

$$\alpha_a = \alpha + \frac{\alpha_d\beta_e}{m_d} \tag{11.99}$$

are the adiabatic drained bulk modulus and the adiabatic drained Biot effective stress coefficient, respectively. The fluid and thermal constitutive equations are

$$\zeta = \alpha_a e + \left(\frac{1}{M} + \frac{\beta_e^2}{m_d}\right)p - \frac{\beta_e}{m_d}s \tag{11.100}$$

and

$$T = \frac{1}{m_d}(-\alpha_d e + \beta_e p + s) \tag{11.101}$$

Strain-Stress Relation The stress-strain relations can be inverted to become strain-stress relations. Based on (11.51), we can write

$$e_{ij} = \left(\frac{1}{9K} - \frac{1}{6G}\right)\delta_{ij}\sigma_{kk} + \frac{1}{2G}\sigma_{ij} + \frac{\alpha}{3K}\delta_{ij}p + \frac{\beta_d}{3}\delta_{ij}T \tag{11.102}$$

The variation in fluid content is given by

$$\zeta = \frac{\alpha}{K}\left(\frac{\sigma_{kk}}{3} + \frac{p}{B}\right) - \beta_v T \tag{11.103}$$

and entropy is

$$s = \beta_d \frac{\sigma_{kk}}{3} - \beta_v p + mT \tag{11.104}$$

The above system expresses $(e_{ij},\ \zeta,\ s)$ as functions of $(\sigma_{ij},\ p,\ T)$. Alternatively, the system can be written as

$$e_{ij} = \left(\frac{1}{9K_u} - \frac{1}{6G}\right)\delta_{ij}\sigma_{kk} + \frac{1}{2G}\sigma_{ij} + \frac{B}{3}\delta_{ij}\zeta + \frac{\beta_u}{3}\delta_{ij}T \tag{11.105}$$

$$p = B\left(-\frac{\sigma_{kk}}{3} + \frac{K}{\alpha}\zeta + \frac{K\beta_v}{\alpha}T\right) \tag{11.106}$$

$$s = \frac{\beta_u}{3}\sigma_{kk} - \frac{BK\beta_v}{\alpha}\zeta + (m_u + K_u\beta_u^2)T \tag{11.107}$$

for $(e_{ij},\ p,\ s)$ as functions of $(\sigma_{ij},\ \zeta,\ T)$. Relations that utilize the adiabatic material constants can be similarly constructed.

11.1.6 Porothermoelastic Physical Mechanism

Armed with the constitutive equations, we can discuss some basic physical mechanisms of porothermoelastic responses. First, consider an isotropic specimen that is free from external stress, $P = 0$. The specimen can be sealed, or have a small permeability such that heat conduction takes place much faster than pressure diffusion, to be under the undrained condition, $\zeta = 0$. When heat is added to raise the temperature by an increment ΔT, we observe from (11.61) that a volume expansion results

$$e = \beta_u \Delta T \tag{11.108}$$

From (11.57), we find a pore pressure increment of

$$p = \frac{BK\beta_v}{\alpha}\Delta T \tag{11.109}$$

We observe from (11.60) that β_v can be positive or negative, depending on the relative magnitude of the solid and fluid expansion coefficient; hence the pressure change can be positive or negative. For geomaterials, we generally find $\beta_f > \beta_s$, and there is a positive pressure rise. As there is no external stress, the specimen is then in tension, in terms of Terzaghi effective stress. Hence heating a poroelastic material, such as saturated concrete, can cause it to burst due to the internal tension generated by fluid expansion.

Under the same condition of free expansion, $P = 0$, but allowing the fluid to drain, $p = 0$, the volumetric strain given by (11.51) becomes

$$e = \beta_d \Delta T \tag{11.110}$$

For a material with $\beta_f > \beta_s$, it is easy to show that $\beta_u > \beta_d$. Hence a heated specimen will initially expand under undrained condition, and then contract (to a smaller expansion) when pore pressure is dissipated.

Not only raising the temperature can cause porous material to fail by tension, lowering the temperature can have the same effect. For an elastic material restrained from deformation, the lowering of temperature causes the material to contract, and the material is in tension. This thermoelastic effect has been utilized in the field to enhance formation permeability. It is accomplished by injecting fluid at lower than formation temperature into borehole or fracture in order to induce tensile cracks on the wall. In the following, the porothermoelastic mechanism is examined.

Consider a specimen that is restrained from deformation, $e = 0$, and is undrained, $\zeta = 0$. From (11.61) and (11.87), we find

$$P = K_u \beta_u \Delta T \tag{11.111}$$

$$p = M \beta_e \Delta T \tag{11.112}$$

We observe from (11.111) that if the temperature is lowered, P is negative; hence the specimen is in tension. This effect however is mitigated by the presence of pore fluid, as the Terzaghi effective stress is

$$P' = P - p = [(K_u - \alpha M)\beta_d - M(1-\alpha)\beta_v]\Delta T \tag{11.113}$$

where we have utilized the definitions of β_u and β_e in (11.62) and (11.67). For ideal porous medium, the above is further expressed as

$$P' = \frac{\phi(1-\phi)^3 K_\phi}{(1-\phi)K_f + \phi[K_s + (1-\phi)^2 K_\phi]} (K_s\beta_s - K_f\beta_f)\Delta T \tag{11.114}$$

Although there is no guarantee that $K_s\beta_s > K_f\beta_f$, for geomaterials saturated with water, K_s is at least one order of magnitude larger than K_f. Hence for geomaterials we can safely assume $K_s\beta_s > K_f\beta_f$, and the cooling of rock in a confined underground environment causes a Terzaghi effective stress initially in tension. As time increases, and the pore pressure is dissipated, $p = 0$, we find from (11.51) that

$$P' = K\beta_d \Delta T \tag{11.115}$$

With a negative ΔT, the drained material is definitely in tension. Comparing with (11.113), it shows that the tension increases with time in the amount

$$P'_{\text{drained}} - P'_{\text{undrained}} = (1-\alpha)M\beta_e \Delta T \tag{11.116}$$

11.2 Balance Law

To complete the theory of porothermoelasticity, we need to introduce several other fundamental physical laws, such as force, mass, and energy balance. These are presented below.

Force Equilibrium The force equilibrium remains unchanged with the thermal effect. In the absence of body force and the inertial effect, the static equilibrium is given by (6.61), which is shown here as

$$\sigma_{ij,j} = 0 \tag{11.117}$$

Fluid Mass Balance The fluid mass balance has been derived in Sect. 6.2.1, expressed in terms of the continuity equation (6.45), which is repeated here

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \vec{q} = 0 \tag{11.118}$$

Thermal Energy Balance In a closed thermodynamic system (no mass transfer), the time rate of change of heat (power) within the volume Ω is balanced by the heat flux entering and leaving the system through its surface Γ, as

$$\frac{dQ}{dt} = -\int_{\Gamma} \vec{q} \cdot \vec{n} \, d\vec{x} \tag{11.119}$$

where Q is the heat energy, $\vec{q}$ is the heat flux vector, and $\vec{n}$ is the outward normal of the surface Γ. The left hand side can be further expressed as

$$\frac{dQ}{dt} = \frac{d}{dt}\int_{\Omega} \hbar \, d\vec{x} = \int_{\Omega} \frac{\partial \hbar}{\partial t} \, d\vec{x} = \int_{\Omega} \mathcal{T} \frac{\partial s}{\partial t} \, d\vec{x} \tag{11.120}$$

where $\hbar$ is the REV averaged *heat density* (per unit volume), and s is entropy density as defined before. In the above, we have utilized the second law of thermodynamics (4.47) applied at the REV level in the following form

$$\delta \hbar = \mathcal{T} \delta s \tag{11.121}$$

Applying the divergence theorem to the right hand side of (11.119), we find

$$\int_{\Gamma} \vec{q} \cdot \vec{n} \, d\vec{x} = \int_{\Omega} \nabla \cdot \vec{q} \, d\vec{x} \tag{11.122}$$

Assembling (11.119), (11.120), and (11.122) together, we obtain

$$\mathcal{T} \frac{\partial s}{\partial t} + \nabla \cdot \vec{q} = 0 \tag{11.123}$$

which is a thermal energy conservation equation for a closed system.

The REV of the porous medium considered here, however, is not a closed system. It is selected to enclose the same solid mass, but the fluid mass can move in and out, and transport heat with it. We hence need to consider not only the heat flux through conduction, but also that due to *advection*. For this, we shall modify (11.123) to become

$$\mathcal{T}\frac{\partial s}{\partial t} + \nabla \cdot \vec{q} + \nabla \cdot (\vec{q}\, \hbar_f) = 0 \tag{11.124}$$

where $\hbar_f$ is the heat density of the fluid phase. We can express $\hbar_f$ as

$$\hbar_f = \mathcal{T} s_f = \mathcal{T}\left[(m_f + K_f \beta_f^2) T - \beta_f p\right] \tag{11.125}$$

In the above, we have utilized the constitutive equation for fluid entropy $\bar{s}_f$ in (11.35), and for fluid internal strain $\overline{\epsilon}^f$ in (11.20). With the above, (11.124) can be expressed in terms of bulk continuum variables, consistent with the rest of the continuum model constructed.

Equation (11.124), however, is nonlinear. To build a linear theory, we shall examine the advective transport term, which can be expanded into

$$\nabla \cdot (\vec{q}\, \hbar_f) = \vec{q} \cdot \nabla(\mathcal{T} s_f) - \mathcal{T} s_f \frac{\partial \zeta}{\partial t} \tag{11.126}$$

where we have utilized the fluid continuity equation (11.118). The second term on the right hand side represents the entropy (heat) gained or lost by the REV due to change in the variation in fluid content. Comparing to the first term in (11.124), this term should be negligible as $s_f < s$, and $\zeta \ll 1$. The first term on the right hand side of (11.126) is the entropy brought into (or out of) the REV by fluid flow if there exists a temperature gradient. This term may not be neglected if the fluid flow is significant. In fact, for applications like geothermal energy extraction, the whole idea is to circulate water in the formation to bring out heat by advection; the advective terms are definitely not to be dropped.

In the present chapter, however, we are interested in the thermoelastic deformation and stress problems, resulting from stress and temperature loadings, and not a sustained porous medium flow problem. In an effort of linearization, we shall simply drop the third term in (11.124), due to the minimal fluid flux involved in the problems to be examined below. It is important to bear in mind that this simplification will limit the type of physical problems that we investigate.

Finally, we present the linearized version of (11.124) as

$$\mathcal{T}_o \frac{\partial s}{\partial t} + \nabla \cdot \vec{q} = 0 \tag{11.127}$$

where we have replaced $\mathcal{T}$ by $\mathcal{T}_o$, based on the assumption $|\mathcal{T} - \mathcal{T}_o|/\mathcal{T}_o \ll 1$.

11.3 Nonequilibrium Thermodynamics and Transport Law

The classical thermodynamics laws describe systems in mechanical, thermal, and chemical equilibria. According to these principles, the entropy of an isolated system attains a maximum value while its free energy reaches a minimum at equilibrium. All physical processes performed under the thermodynamic equilibrium conditions are reversible, such as the elastic and thermoelastic deformation processes described by the constitutive equations developed in Sects. 4.4 and 11.1. However, many real world processes occur under nonequilibrium conditions, are irreversible, and are not closed systems. For such systems, the classical laws provide only a set of inequalities describing the direction of changes, and not equalities that give quantitative information. Many phenomenological laws, such as Darcy's law for porous medium flow, Fourier's law for heat flow, Fick's law of solute mass diffusion, and Ohm's law for electric current, were created to fill the application gaps. The nonequilibrium thermodynamics is an attempt to provide theoretical foundation to these observed transport phenomena [21].

11.3.1 Onsager's Reciprocal Relation

When two or more irreversible transport processes, such as heat conduction, electrical conduction and mass diffusion take place simultaneously in a thermodynamic system, the processes may interfere with each other. For example, as early as 1801, Rouss carried out experiments on the electrical and osmotic behavior of porous media, and found that the application of an electromotive force can produce not only an electrical current, but also a fluid flow [37]. If the processes are not too far from equilibrium, and the rate of reaching equilibrium is not too fast, it may be possible to expressed these coupled relations in linear laws [54, 55], such as

$$\vec{J}_T = L_{TT}\vec{X}_T + L_{Tp}\vec{X}_p + L_{Tc}\vec{X}_c + L_{Te}\vec{X}_e \tag{11.128}$$

$$\vec{J}_p = L_{pT}\vec{X}_T + L_{pp}\vec{X}_p + L_{pc}\vec{X}_c + L_{pe}\vec{X}_e \tag{11.129}$$

$$\vec{J}_c = L_{cT}\vec{X}_T + L_{cp}\vec{X}_p + L_{cc}\vec{X}_c + L_{ce}\vec{X}_e \tag{11.130}$$

$$\vec{J}_e = L_{eT}\vec{X}_T + L_{ep}\vec{X}_p + L_{ec}\vec{X}_c + L_{ee}\vec{X}_e \tag{11.131}$$

In the above, we have selected four fluxes that are of interest to geophysical and geochemical phenomena, the heat flux $\vec{J}_T$, the fluid volume flux $\vec{J}_p$, the solute mass flux $\vec{J}_c$, and the electrical current $\vec{J}_e$, and related them to the respective *driving forces*, as the temperature gradient $\vec{X}_T$, pore pressure gradient $\vec{X}_p$, solute mass concentration

Table 11.1 Physical phenomena associated with coupled transport laws (Adopted from [67, 81])

	Force $\vec{X}$			
Flux $\vec{J}$	Temperature	Hydraulic	Chemical	Electrical
Heat	Thermal conduction	Mechano-caloric	Dufour effect	Peltier effect
Fluid	Thermo-osmosis	Seepage	Osmosis	Electro-osmosis
Solute	Soret effect	Hyperfiltration	Diffusion	Electrophoresis
Current	Seebeck effect	Rouss effect	Diffusion current	Electric conduction

(chemical potential) gradient $\vec{X}_c$, and the electrical potential gradient $\vec{X}_e$. The coefficients L_{ij}, where the subscripts $\{i,j\} = \{T,p,c,e\}$ are phenomenological constants obtained experimentally. In Table 11.1 we present the various physical phenomena associated with these coupled flux laws [20].

According to Onsager's reciprocal theorem [54, 55], the following symmetry exists in the coefficient matrix

$$L_{ij} = L_{ji} \tag{11.132}$$

The above relation not only reduces the number of independent coefficients, but also reveals the correlations among the flow phenomena. Onsager's proof on such relation had been based on the statistical mechanics principals that held for processes near their equilibrium states. An impressive amount of experimental observations had been collected that supported this theorem [49, 50]. For porous media flow, evidences had also been collected for the simultaneous flow of water and salt [42], and of water and electricity [32]. We should note, however, although the Onsager reciprocal relation has been widely accepted, there also exist evidences that such relations are violated for certain systems [18, 45].

The rate of entropy production in a thermodynamic nonequilibrium system can be written as [19, 37]

$$\sigma = \sum_{i=T,p,c,e} \vec{J}_i \cdot \vec{X}_i > 0 \tag{11.133}$$

In the above, the greater than sign is a consequence of the second law of thermodynamics. Substituting (11.128)–(11.133), we can demonstrate the conditions for the coefficients:

$$L_{ii} \geq 0; \qquad L_{ii}L_{jj} \geq L_{ij}^2 \tag{11.134}$$

In the above, summation is not performed on repeated indices. The above conditions also suggest that the off-diagonal coefficients L_{ij}, $i \neq j$, can be positive or negative.

11.3.2 Thermal-Hydraulic Transport Phenomena

In porothermoelasticity, our interest is focused on thermal-hydraulic effects and their coupling; hence we shall ignore the chemical and electrical effects here. These phenomena will be investigated in Chap. 12. With such assumption, and the proper selection of the flux and thermodynamic force pairs, (11.128)–(11.131) reduce to

$$\vec{J}_p = -L_{pp}\nabla p - L_{pT}\left(\frac{\nabla \mathcal{T}}{\mathcal{T}}\right) \tag{11.135}$$

$$\vec{J}_T = -L_{Tp}\nabla p - L_{TT}\left(\frac{\nabla \mathcal{T}}{\mathcal{T}}\right) \tag{11.136}$$

Linearizing, we find

$$\vec{J}_p = -L_{pp}\nabla p - \frac{L_{pT}}{\mathcal{T}_o}\nabla \mathcal{T} \tag{11.137}$$

$$\vec{J}_T = -L_{Tp}\nabla p - \frac{L_{TT}}{\mathcal{T}_o}\nabla \mathcal{T} \tag{11.138}$$

For porous medium applications, we select to model the fluid specific flux $\vec{q}$ and the heat flux $\vec{q}$, with reference to the cross-sectional area of the porous medium frame, including solid and pore space, and establish the following gradient laws

$$\vec{q} = -\kappa\nabla p - k_{pT}\nabla T \tag{11.139}$$

$$\vec{q} = -k_{Tp}\nabla p - k_T\nabla T \tag{11.140}$$

where κ is the permeability coefficient, k_T is the *effective thermal conductivity* (see Sect. 11.3.4), k_{pT} is the *thermo-osmosis coefficient*, and k_{Tp} is the *mechano-caloric coefficient*. In the above, we note that the fluid flow and pressure gradient take place in the pore space, while the heat flow and temperature gradient are referred to the average of both the solid and the fluid; hence there is no particular reason that the Onsager's reciprocal relation (11.132) exists. This means that k_{pT} and k_{Tp} are two independent coefficients.

When the coupling effect is weak, k_{pT}, $k_{Tp} \to 0$, and (11.139) and (11.140) respectively reduce to Darcy's law and Fourier's law:

$$\vec{q} = -\kappa\,\nabla p \tag{11.141}$$

$$\vec{q} = -k_T\nabla T \tag{11.142}$$

For the convenience of later use, we can invert (11.139) and (11.140) to obtain this form

$$\nabla p = -b_{pp}\,\vec{q} - b_{pT}\,\vec{q} \tag{11.143}$$

$$\nabla T = -b_{Tp}\,\vec{q} - b_{TT}\,\vec{q} \tag{11.144}$$

where

$$\begin{bmatrix} b_{pp} & b_{pT} \\ b_{Tp} & b_{TT} \end{bmatrix} = \begin{bmatrix} \kappa & k_{pT} \\ k_{Tp} & k_T \end{bmatrix}^{-1} \tag{11.145}$$

Thermo-Osmosis Effect Thermo-osmosis is the phenomenon of a porous medium flow driven by a temperature gradient, independent of a pressure gradient. Such phenomenon was observed by Derjaguin and Sidorenkov for porous glasses [22], Rastogi et al. [59, 60] for cellophane membrane, Dirksen [24] in clay, and Trémosa et al. [74] in Liassic argillite. For very low permeability geomaterials such as clay and shale, this effect can be quite significant. Based on the experimental results of Srivastava and Avasthi [69], Carnahan [9] estimated that the thermo-osmotic flow through kaolinite can be 800 times larger than Darcy's flow. Ghassemi and Diek [27] analyzed the mud filtrate invasion due to temperature gradient and found that the thermal-osmotic flow can be several times larger than hydraulic flow.

Derjaguin et al. [22] proposed that thermo-osmosis results from the alteration of water properties close to the solid surface. The importance of these effects depends on: (i) the surface hydrophilicity, (ii) the temperature, as the structure of boundary layers can be destroyed by increasing temperature, and (iii) the pore size, as thermo-osmosis conductivity is larger in narrow pores.

Mechano-Caloric Effect Experimental observations show that an isothermal fluid flow in the clay pores, driven by a pressure gradient, produces an excess heat flux in the flow direction. Derjaguin et al. [22] used nonequilibrium thermodynamics concepts to explain this phenomenon, known as the *mechano-caloric effect*. At the pore level, the interaction between a solid and the electrolytes carried by the fluid at the contact surface yields a substantial alteration of the water structure in terms of hydrogen bond network. This causes an alteration of the specific enthalpy of the solution in the pore space of a clay. An isothermal fluid flow across the porous medium would thus produce an excess heat flow and a related temperature gradient. Further theoretical insight was given by Gonçalvès and Trémosa [31].

Based on estimated parameters for shale, Ghassemi et al. [30] examined the impact of mud temperature and salinity on water flux, stress, and pore pressure around the wellbore, and concluded that while the thermo-osmosis effect can be significant, the mechano-caloric effect is negligible.

11.3.3 Irreversible Thermodynamics Process

Similar to the presentation in Sect. 6.1.4, it is of interest to demonstrate the consistency between the thermal-hydraulic transport process presented in the preceding section and the thermodynamic principles of irreversible process. Comparing with poroelasticity, the porothermoelasticity has two dissipative processes—in addition

to the one caused by the porous medium flow, there is one resulting from heat diffusion.

We shall once again refer to the generalized work-energy relation for irreversible process presented in (4.59). To model the equivalent thermal dissipative work, we shall follow the lead of Biot [4] (see also Nowacki [53]) to introduce an entropy displacement vector $\vec{H}$ based on the relation

$$s = -\nabla \cdot \vec{H} \tag{11.146}$$

By virtue of (11.127), we find that the entropy displacement is related to the heat flux $\vec{q}$ as

$$\vec{q} = \mathcal{T}_o \frac{\partial \vec{H}}{\partial t} \tag{11.147}$$

In an analogy, we can compare the above equations with (6.35) for the variation in fluid content ζ and its displacement vector $\vec{w}$, and with (6.33) for fluid specific flux and relative fluid to solid displacement.

The dissipation energy density function $\mathcal{F}$ is now function of both displacements, as

$$\mathcal{F} = \mathcal{F}(w_i, H_i) \tag{11.148}$$

The Helmholtz free energy is now also function of heat energy. Adopting the notation in Sect. 11.1.1, the Helmholtz free energy density function is given as

$$\mathcal{A} = \mathcal{A}(e_{ij}, \zeta, s) \tag{11.149}$$

We note that not only we have expressed the energy density function in terms of the bulk continuum variables, but also we have selected entropy density s, instead of temperature T, as the state variable.

With the above, we can then express the energy functions $\mathcal{A}$ and $\mathcal{F}$ as the integration of the density functions over a macroscopic system Ω bounded by Γ, as

$$\begin{aligned} \delta\mathcal{A} + \delta\mathcal{F} &= \int_\Omega \delta\mathcal{A}(e_{ij}, \zeta, s)\, d\vec{x} + \int_\Omega \delta\mathcal{F}(w_i, H_i)\, d\vec{x} \\ &= \int_\Omega \left(\frac{\partial \mathcal{A}}{\partial e_{ij}} \delta e_{ij} + \frac{\partial \mathcal{A}}{\partial \zeta} \delta\zeta + \frac{\partial \mathcal{A}}{\partial s} \delta s + \frac{\partial \mathcal{F}}{\partial w_i} \delta w_i + \frac{\partial \mathcal{F}}{\partial H_i} \delta H_i \right) d\vec{x} \end{aligned} \tag{11.150}$$

For the reversible and irreversible work, we find

$$\delta\mathcal{W} + \delta\mathcal{W}_{\text{irr}} = \int_\Gamma \sigma_{ij} n_j \delta u_i \, d\vec{x} - \int_\Gamma p\, n_i\, \delta w_i \, d\vec{x} - \int_\Gamma T n_i\, \delta H_i \, d\vec{x} \tag{11.151}$$

Applying the divergence theorem to the above, we find

$$\delta\mathcal{W} + \delta\mathcal{W}_{\text{irr}} = \int_\Omega \left(\sigma_{ij}\delta e_{ij} + p\,\delta\zeta + T\delta s - p_{,i}\,\delta w_i - T_{,i}\,\delta H_i\right)\, d\vec{x} \tag{11.152}$$

Comparing (11.152) with (11.150), it is clear that

$$\sigma_{ij} = \frac{\partial \mathcal{A}}{\partial e_{ij}} \tag{11.153}$$

$$p = \frac{\partial \mathcal{A}}{\partial \zeta} \tag{11.154}$$

$$T = \frac{\partial \mathcal{A}}{\partial s} \tag{11.155}$$

$$p_{,i} = -\frac{\partial \mathcal{F}}{\partial w_i} \tag{11.156}$$

$$T_{,i} = -\frac{\partial \mathcal{F}}{\partial H_i} \tag{11.157}$$

A quadratic form can be introduced for $\mathcal{A}$ as the following

$$\mathcal{A} = \frac{1}{2}\lambda_b e^2 + \mu e_{ij}^2 - \alpha_b M e\,\zeta - \frac{\alpha_u}{m_u} e\,s + \frac{\beta_e M}{m_u}\zeta\, s + \frac{1}{2}\frac{M m_d}{m_u}\zeta^2 + \frac{1}{2}\frac{1}{m_u}s^2 \tag{11.158}$$

where

$$\lambda_b = K_b - \frac{2G}{3} \tag{11.159}$$

is the adiabatic undrained Lamé constant. The quadratic form (11.158) can be compared with (11.84). In (11.158), we also identify seven independent material constants, λ_b, μ (G), α_b, M, α_u, β_e, and m_d, by noting that m_u is defined in (11.77). It is easy to check that (11.153)–(11.157) produce the same constitutive equations as derived in Sect. 11.1.5.

Presenting $\mathcal{F}$ in the quadratic form

$$\mathcal{F} = \frac{1}{2}b_{pp}w_i\frac{\partial w_i}{\partial t} + b_{pT}\mathcal{T}_o w_i\frac{\partial H_i}{\partial t} + b_{Tp}H_i\frac{\partial w_i}{\partial t} + \frac{1}{2}b_{TT}\mathcal{T}_o H_i\frac{\partial H_i}{\partial t} \tag{11.160}$$

we can differentiate the above with respect to w_i and H_i, and obtain

$$\frac{\partial \mathcal{F}}{\partial w_i} = b_{pp}\vec{q} + b_{pT}\vec{q} = -\nabla p \tag{11.161}$$

$$\frac{\partial \mathcal{F}}{\partial H_i} = b_{Tp}\vec{q} + b_{TT}\vec{q} = -\nabla T \tag{11.162}$$

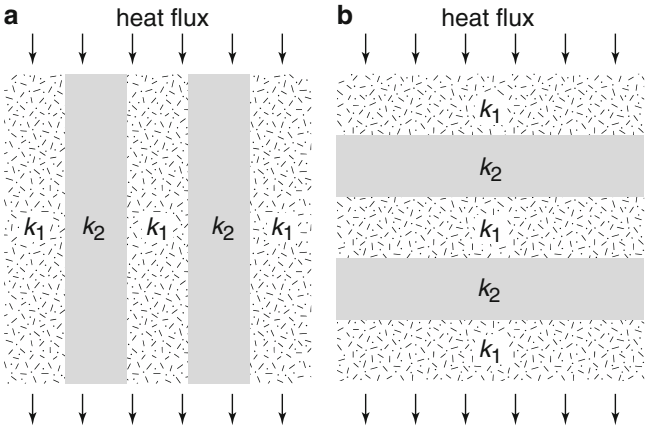


Fig. 11.1 Effective thermal conductivity models: (**a**) parallel model, and (**b**) series model

which are the same as (11.143) and (11.144). We hence have shown that the flux laws can be derived based on the variational principle applied to irreversible thermodynamics.

11.3.4 Effective Thermal Conductivity

As discussed in Sect. 11.1.4, to construct the bulk continuum model, we have made the simplifying assumption that the solid and fluid phases are in intimate contact with each other at the grain scale, such that they are at a local thermal equilibrium at all times. Hence the Fourier law (11.142) for the bulk material is written with a single temperature T and an *effective thermal conductivity* k_T.

The effective thermal conductivity obviously is dependent on the thermal conductivities of the solid and fluid phases, k_T^s and k_T^f, and the geometry of the porous medium. Of all the porous medium geometry, it is of interest to examine the bounds of effective thermal conductivity based on the conceptual models.

Given two materials, one with larger thermal conductivity k_1 and the other with smaller conductivity k_2, it is easy to show that the most efficient way that heat can be conducted is to arrange the materials in layers as shown in Fig. 11.1a, and the heat is conducted in the direction parallel to them. This is because heat will seek the path of least resistance and conduct through the larger conductivity layers. On the other hand, if the heat flow is in the direction perpendicular to these layers, such as shown in Fig. 11.1b, then the less conductive layers become the bottleneck, and the heat flow is minimized. These situations are in fact analogous to the *series circuit* and the *parallel circuit* of electric circuit theory. From the well-known results, the equivalent thermal conductivity is obtained as follows: for the parallel model

Table 11.2 Thermal conductivity for various materials (After Coussy [15])

Material	k_T (W/m°K)
Concrete	1.5–2.1
Rock	3
Sand	0.4–0.8
Clay	0.8–2
Granite	2.5–3.8
Sandstone	1.5–4.3
Water (0 °C)	0.598
Air	0.026

$$k_u = (1-\phi)k_T^s + \phi k_T^f \tag{11.163}$$

and for the series model

$$k_l = \frac{1}{(1-\phi)/k_T^s + \phi/k_T^f} \tag{11.164}$$

where we have assumed the two materials are a solid and a fluid, and ϕ is porosity. It is easy to show that k_u is the upper bound, and k_l is the lower bound. We may refer these bounds as the Wiener bounds [78].

To gain some idea about these bounds, we show in Table 11.2 the typical values of thermal conductivity for a few materials. Assuming a porous rock saturated with water, the thermal conductivity ratio of these two phases is $k_T^f/k_T^s = 1/5$. In Fig. 11.2 we plot the effective thermal conductivity normalized by that of the solid phase, k_T/k_T^s, versus porosity ϕ, based on the upper bound and lower bound formulas, (11.163) and (11.164). For a given porosity, the effective thermal conductivity of an actual porous medium must fall between these two bounds.

Observing from Fig. 11.1, we realize that these two models, series and parallel, are actually anisotropic. We are interested in the bounds for isotropic medium. Hashin and Shtrikman [33] derived effective conductivity bounds that were the best (i.e. narrowest) possible bounds for macroscopically homogeneous, isotropic, two-phase materials, as

$$k_u = k_T^s \frac{2k_T^s + k_T^f - 2(k_T^s - k_T^f)\phi}{2k_T^s + k_T^f + (k_T^s - k_T^f)\phi} \tag{11.165}$$

$$k_l = k_T^f \frac{2k_T^f + k_T^s - 2(k_T^f - k_T^s)(1-\phi)}{2k_T^f + k_T^s + (k_T^f - k_T^s)(1-\phi)} \tag{11.166}$$

These two bounds are also plotted in Fig. 11.2. The Hashin-Shtrikman bounds always lie within the Wiener bounds.

The Hashin-Shtrikman bounds, although derived based on different lines of reasoning, are equivalent to the Maxwell-Eucken model [46]. The Maxwell-Eucken

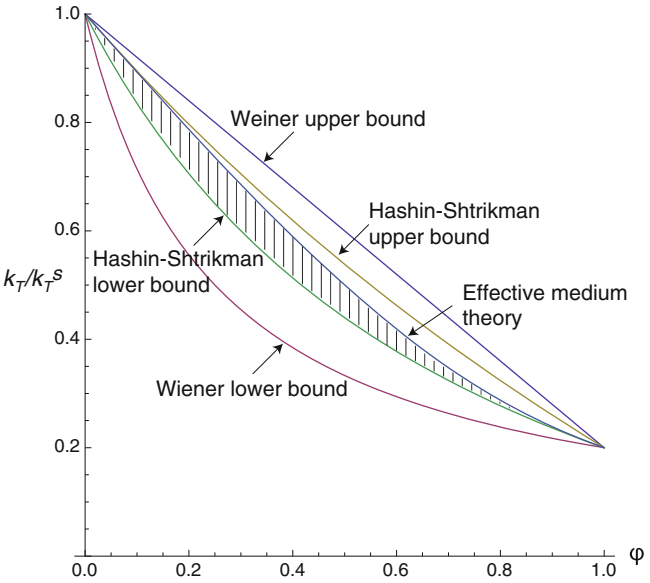


Fig. 11.2 Upper and lower bounds for normalized effective thermal conductivity versus porosity, for porous medium with $k_T^f/k_T^s = 1/5$

upper bound assumes that the solid phase is continuous, and the fluid phase is isolated in cavities. The lower bound, on the other hand, is associated with a mixture of solid particles suspended in fluid. These models have a "bias" either toward the solid or the fluid. An unbiased medium created by the random distribution of solid and fluid phases, not requiring a priori either phase to be connected or isolated, can be modeled by the *effective medium theory*, which leads to the following theoretical result [38, 41]

$$k_T = \frac{1}{4}\left\{ (3\phi - 1)k_T^f + (2 - 3\phi)k_T^s + \sqrt{\left[(3\phi - 1)k_T^f + (2 - 3\phi)k_T^s\right]^2 + 8k_T^s k_T^f}\right\} \tag{11.167}$$

This equation is also plotted in Fig. 11.2.

Based on the observation of a few geomaterials, construction sand [73], unconsolidated sands [79], consolidated rocks [80], and sandstone [70], permeated with air, Carson et al. [10] found that the effective thermal conductivities for these specimens all fall between the effective medium theory and the lower Hashin-Shtrikman bound, shown as the hatched area in Fig. 11.2. Hence (11.166) and (11.167) can be used to estimate the effective thermal conductivity of geomaterials. For example, given a sandstone with $k_T^s = 3\,\mathrm{W/m^\circ K}$, and porosity $\phi = 0.2$,

saturated with water with $k_T^f = 0.6\,\mathrm{W/m^\circ K}$, we can determine that the effective thermal conductivity is between 2.1 and 2.4 $\mathrm{W/m^\circ K}$, which is reasonably narrow.

11.4 Governing Equation

For the purpose of mathematical solution, variables can be eliminated among the various constitutive relations, equilibrium and continuity equations, and the flux laws introduced in Sects. 11.1, 11.2, and 11.3 to form a system of partial differential equations for the solution of initial and boundary value problems. In the following we shall present three models:

Complete Model This model uses the full set of equations developed in the above sections. Particularly it uses the coupled constitutive equations derived in Sect. 11.1.4, and the coupled flux laws in Sect. 11.3.2; hence it is the complete model.

Thermally Coupled Model In the thermally coupled model, we use the uncoupled flux laws, Darcy's law and Fourier law, given as (11.141) and (11.142), because very little is known about the coupling coefficients in (11.139) and (11.140). The resulting model, however, is still coupled; that is, all the thermal, hydraulic, and mechanical variables need to be simultaneously solved. To emphasize it, it is called the thermally coupled model.

Thermally Uncoupled Model This model further ignores the coupling term in constitutive equation that deformation can generated heat. As a consequence, the heat diffusion equation is decoupled from the elastic deformation; hence can be independently solved. This model is most commonly adopted for application purposes, and most existing analytical solutions are provided for this model.

In the presentation below we shall first present the thermally coupled model, as it is the more general model than the thermally uncoupled model. All the analytical solutions will be derived for the thermally coupled model, and then simplified for the uncoupled model. We then demonstrate that the complete model, although containing more physical parameters, can be arranged in a way that is mathematically equivalent to the thermally coupled model. Hence all the derived analytically solutions apply to the complete model by a proper substitution of material coefficients.

11.4.1 Thermally Coupled Model

Similar to those constructed in Sect. 6.3, the governing equations can be assembled into a number of different forms.

Navier Equation The Navier-type equation for porothermoelasticity is obtained by substituting the constitutive equation (11.89) into the static equilibrium equation (11.117), which yields

$$G\nabla^2\vec{u} + \left(K + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha\nabla p - \alpha_d\nabla T = 0 \tag{11.168}$$

Or, we can switch to ζ as variable instead of p, and obtain from (11.85)

$$G\nabla^2\vec{u} + \left(K_u + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha M\nabla\zeta - \alpha_u\nabla T = 0 \tag{11.169}$$

Similar to switching between p and ζ as variable, we can replace T by s in the above equations. Based on (11.92) and (11.97), we obtain the following

$$G\nabla^2\vec{u} + \left(K_a + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha_a\nabla p - \frac{\alpha_d}{m_d}\nabla s = 0 \tag{11.170}$$

$$G\nabla^2\vec{u} + \left(K_b + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha_b M\nabla\zeta - \frac{\alpha_u}{m_u}\nabla s = 0 \tag{11.171}$$

Compatibility and Harmonic Equations Similar to relations presented in Sects. 6.3.3 and 6.3.4 we can obtain a few harmonic relations among the variables.

First, following the same steps that led to the Beltrami-Michell compatibility equation for poroelasticity (6.82), we obtain the corresponding equation for porothermoelasticity:

$$\begin{aligned}\nabla^2\sigma_{ij} + \frac{1}{1+\nu}\sigma_{kk,ij} + 2\eta\left(\delta_{ij}\nabla^2 p + \frac{1-\nu}{1+\nu}p_{,ij}\right)\\ +2\eta_d\left(\delta_{ij}\nabla^2 T + \frac{1-\nu}{1+\nu}T_{,ij}\right) = 0\end{aligned} \tag{11.172}$$

where

$$\eta_d = \frac{\alpha_d(1-2\nu)}{2(1-\nu)} \tag{11.173}$$

is a *thermoelastic stress coefficient*, defined parallel to the poroelastic stress coefficient η in (1.56). Equation (11.172) can be contracted to yield a Laplace equation

$$\nabla^2(\sigma_{kk} + 4\eta p + 4\eta_d T) = 0 \tag{11.174}$$

Its two-dimensional and one-dimensional forms are similar to (6.84) and (6.85).

Other harmonic relations can be obtained by taking divergence of (11.168) and (11.169), and we find

$$\nabla^2\left[\left(K+\frac{4G}{3}\right)e-\alpha p-\alpha_d T\right]=0 \tag{11.175}$$

$$\nabla^2\left[\left(K_u+\frac{4G}{3}\right)e-\alpha M\zeta-\alpha_u T\right]=0 \tag{11.176}$$

Similarly, (11.170) and (11.171) yield

$$\nabla^2\left[\left(K_a+\frac{4G}{3}\right)e-\alpha_a p-\frac{\alpha_d}{m_d}s\right]=0 \tag{11.177}$$

$$\nabla^2\left[\left(K_b+\frac{4G}{3}\right)e-\alpha_b M\zeta-\frac{\alpha_u}{m_u}s\right]=0 \tag{11.178}$$

Using (11.66) to eliminate e in (11.175), we obtain

$$\nabla^2(\zeta-Sp+\beta_c T)=0 \tag{11.179}$$

where we have utilized the definition of a storage coefficient (see (B.14))

$$S=\frac{3K_u+4G}{M(3K+4G)} \tag{11.180}$$

and defined a new variable

$$\beta_c=\beta_e-\frac{\eta\alpha_d}{G}=\beta_v+\frac{4}{3}\eta\beta_d \tag{11.181}$$

This and other definitions of porothermoelastic constants are summarized in Sect. B.7.

Similarly, using the constitutive equation (11.91) to eliminate e in (11.177), we find

$$\nabla^2(s-S_a T+\beta_c p)=0 \tag{11.182}$$

where, by analogy, we have defined an adiabatic drained thermal storage coefficient as

$$S_a=\frac{m_d(3K_a+4G)}{3K+4G} \tag{11.183}$$

As demonstrated below, (11.179) and (11.182) are useful in the derivation of the diffusion equations for the variation in fluid content ζ and entropy density s.

Fluid Diffusion Equation To form a mathematical system for solution, the mass conservation equation derived in Sect. 11.2 can be combined with the flux gradient law in Sect. 11.3.2 to eliminate the flux variable, and to form a diffusion-type equation. For the current model, rather than using the coupled flux law (11.139), we shall use the simple Darcy's law (11.141). Combining (11.118) with (11.141), we obtain

$$\frac{\partial \zeta}{\partial t} - \kappa \nabla^2 p = 0 \tag{11.184}$$

A pore pressure diffusion equation can be created by substituting (11.66) into the above to eliminate ζ, and we find

$$\frac{\partial p}{\partial t} - \kappa M \nabla^2 p = -\alpha M \frac{\partial e}{\partial t} + \beta_e M \frac{\partial T}{\partial t} \tag{11.185}$$

The right hand side shows that not only the rate of change of strain, but also the rate of change of temperature, serves as a production term for the process of pressure diffusion.

An alternative diffusion equation can be obtained by utilizing the harmonic relation (11.179) to eliminate p in (11.184), and it produces

$$\frac{\partial \zeta}{\partial t} - c \nabla^2 \zeta = c \beta_c \nabla^2 T \tag{11.186}$$

Comparing it with (6.70) for poroelasticity, we observe that the right hand side of (11.186) introduces the thermal effect. Or, we can replace the temperature variable on the right hand side by entropy, to give

$$\frac{\partial \zeta}{\partial t} - c_c \nabla^2 \zeta = \frac{c_c \beta_g}{m_d} \nabla^2 s \tag{11.187}$$

where we have used the shorthand notations

$$c_c = \frac{\kappa}{S_c} \tag{11.188}$$

$$S_c = \frac{m_u (3K_b + 4G)}{m_d M (3K_d + 4G)} \tag{11.189}$$

$$K_d = K_b - \frac{m_u M \alpha_b^2}{m_d} \tag{11.190}$$

$$\beta_g = \beta_e - \frac{3\alpha_b (m_d \alpha_u - M m_u \alpha_b \beta_e)}{m_d (3K_d + 4G)} \tag{11.191}$$

Thermal Diffusion Equation Combining (11.127) and (11.142), we obtain

$$\frac{\partial s}{\partial t} - \frac{k_T}{\mathcal{T}_o}\nabla^2 T = 0 \tag{11.192}$$

To obtain diffusion equation for temperature, we substitute (11.91) into the above to eliminate s, and find

$$\frac{\partial T}{\partial t} - \kappa_T \nabla^2 T = -\frac{\alpha_d}{m_d}\frac{\partial e}{\partial t} + \frac{\beta_e}{m_d}\frac{\partial p}{\partial t} \tag{11.193}$$

where

$$\kappa_T = \frac{k_T}{m_d \mathcal{T}_o} = \frac{k_T}{c_d} \tag{11.194}$$

is a heat diffusivity coefficient. We notice from the right hand side of (11.193) that the rate of strain and pressure contributes to the heat production. Or, we can replace the pore pressure term on the right hand side by the variation in fluid content to obtain

$$\frac{\partial T}{\partial t} - \frac{m_d \kappa_T}{m_u}\nabla^2 T = -\frac{\alpha_u}{m_u}\frac{\partial e}{\partial t} + \frac{M\beta_e}{m_u}\frac{\partial \zeta}{\partial t} \tag{11.195}$$

Next, we examine the diffusion equation for entropy. Substituting (11.182) into (11.192) to eliminate T, we find the diffusion equation for entropy

$$\frac{\partial s}{\partial t} - c_a \nabla^2 s = c_a \beta_c \nabla^2 p \tag{11.196}$$

where

$$c_a = \frac{k_T}{S_a \mathcal{T}_o} = \frac{m_d \kappa_T}{S_a} \tag{11.197}$$

is an entropy diffusivity coefficient analogous to the consolidation coefficient c. Or, if we replace the pore pressure term on right hand side of (11.196) by variation in fluid content, we obtain

$$\frac{\partial s}{\partial t} - c_b \nabla^2 s = M c_b \beta_h \nabla^2 \zeta \tag{11.198}$$

where

$$c_b = \frac{m_d \kappa_T}{S_b} \tag{11.199}$$

$$S_b = \frac{m_u(3K_b + 4G)}{3K_u + 4G} \tag{11.200}$$

$$\beta_h = \beta_e + \frac{3\alpha_u(\alpha_u\beta_e - \alpha_b m_u)}{m_u(3K_u + 4G)} \tag{11.201}$$

It is of interest to note that the pair of diffusion equations, (11.187) and (11.198), are defined within the two variables ζ and s, and are coupled to each other. But they are otherwise uncoupled from the solid deformation.

Complete Solution System The governing equations (11.168), (11.185), and (11.193), together with appropriate boundary conditions, form a complete solution system to solve for the variables $(\vec{u},\ p,\ T)$. The system is defined on the following set of 9 independent material constants $\{G, K, M, \alpha, \alpha_d, \beta_e, m_d, \kappa, \kappa_T\}$.

Boundary Conditions The boundary conditions required to satisfy the poroelastic part of variables have been discussed in Sect. 6.5.2. With the addition of temperature as a variable, an additional boundary condition is needed. This can be a Dirichlet boundary condition, as prescribed temperature

$$T(\vec{x}, t) = f_1(\vec{x}, t) \tag{11.202}$$

where $\vec{x} \in \Gamma$, Γ is the boundary of the solution domain, and the right hand side is a given function. Or it can be a Neumann boundary condition, as known heat flux

$$-k_T \frac{\partial T}{\partial n} = f_2(\vec{x}, t) \tag{11.203}$$

when n is the boundary outward normal. Particularly, if $f_2 = 0$, the boundary is insulated. Or, we can have a heat transfer condition

$$-k_T \frac{\partial T}{\partial n} = h(T_\infty - T) \tag{11.204}$$

where h is the heat transfer coefficient, and T_∞ is the far-field temperature of the medium in contact.

11.4.2 Porothermoelastic Constant

To apply the porothermoelastic model, measurement of a minimum set of material constant is needed. In Table 11.3 we present these porothermoelastic constants for several geomaterials compiled from the literature.[1] Their poroelastic properties have

[1] McTigue [47] used the notations $(\alpha_s',\ \alpha_s'',\ \alpha_f)$ for coefficient of thermal expansion. The correspondences with the current notations are $\alpha_s' = \beta_d$ and $\phi(\alpha_f - \alpha_s'') = \beta_v$. In the table, McTigue further assumed $\alpha_s' = \alpha_s'' = \alpha_s$, which corresponds to the ideal porous medium model of $K_\psi = \beta_\psi = 0$. These assumptions lead to the consistent result of $\beta_d = \beta_s$ and $\beta_v = \phi(\beta_f - \beta_s)$ in (11.54)

Table 11.3 Thermal constants for poroelastic material saturated with water at 20 °C ($\mathcal{T}_o$ = 293 °K), for deep sea clay [34, 47], rock salt [47, 57], Berea sandstone [68, 82], and Westerly granite [48, 64]. Upper table: independent constants; lower tables: derived constants

Porous medium	k_T W/m °K	c_d J/m^3 °K	$\beta_s = \beta_d$ °K^{-1}	β_f °K^{-1}
Abyssal red clay	1.02	3.92×10^6	3.4×10^{-5}	3.0×10^{-4}
Rock salt	6.6	1.89×10^6	1.2×10^{-4}	3.0×10^{-4}
Berea sandstone	2.24	1.76×10^6	4.5×10^{-6}	2.1×10^{-4}
Westerly granite	2.5	2.30×10^6	2.4×10^{-5}	2.1×10^{-4}
Porous medium	β_v °K^{-1}	β_e °K^{-1}	β_u °K^{-1}	β_c °K^{-1}
Abyssal red clay	1.89×10^{-4}	2.23×10^{-4}	2.22×10^{-4}	1.89×10^{-4}
Rock salt	1.80×10^{-7}	1.45×10^{-5}	1.20×10^{-4}	6.53×10^{-6}
Berea sandstone	3.90×10^{-5}	4.25×10^{-5}	2.60×10^{-5}	4.08×10^{-5}
Westerly granite	1.86×10^{-6}	1.26×10^{-5}	2.55×10^{-5}	6.65×10^{-6}
Porous medium	α_d N/m^2 °K	α_u N/m^2 °K	α_e N/m^2 °K	η_d N/m^2 °K
Abyssal red clay	6.80×10^2	6.18×10^5	3.77×10^3	2.44×10^0
Rock salt	2.48×10^6	2.80×10^6	1.10×10^6	8.27×10^5
Berea sandstone	3.60×10^4	3.64×10^5	2.94×10^5	1.35×10^4
Westerly granite	6.00×10^5	1.00×10^6	3.58×10^5	2.00×10^5
Porous medium	m_d J/m^3	m_u J/m^3	κ_T m^2/s	c_a m^2/s
Abyssal red clay	1.34×10^4	1.32×10^4	2.60×10^{-7}	2.60×10^{-7}
Rock salt	6.45×10^3	6.41×10^3	3.49×10^{-6}	3.40×10^{-6}
Berea sandstone	6.01×10^3	5.99×10^3	1.27×10^{-6}	1.27×10^{-6}
Westerly granite	7.85×10^3	7.84×10^3	1.09×10^{-6}	1.09×10^{-6}

been reported in Tables 3.1 and 3.2. In Table 11.3, the upper table shows a set of four independent material constants, k_T, c_d, β_s, and β_f. Applying the ideal porous medium assumption discussed in Sect. 11.1.3, we can evaluate β_d and β_v from β_s and β_f, based on (11.54) and (11.60). The rest of the material constants in the lower tables are evaluated from the various definitions summarized in Sect. B.7. We also note that to obtain m_d, m_u, and c_a, the value of $\mathcal{T}_o$, the reference temperature, is needed, which is selected as $\mathcal{T}_o = 20\,°\text{C}$ (293 °K).

and (11.60) in the current model. Also, the value of α_e ($= b'$ of McTigue) for rock salt has been corrected in the table.

11.4.3 Irrotational Displacement Field

Under certain deformation conditions, such as problems of uniaxial strain, $u_x = u_x(x)$ and $u_y = u_z = 0$, of axial symmetry, $u_r = u_r(r)$ and $u_\theta = u_z = 0$, and of spherical symmetry, $u_R = u_R(R)$ and $u_\theta = u_\varphi = 0$, we can prove that the displacement field is irrotational:

$$\nabla \times \vec{u} = 0 \tag{11.205}$$

As demonstrated in Sect. 6.4.3, the displacement vector can be expressed as the gradient of a scalar potential, as (6.104). The Navier equation (11.168) then becomes

$$\nabla^2 \vec{u} = \frac{\eta}{G} \nabla p + \frac{\eta_d}{G} \nabla T \tag{11.206}$$

The above can be integrated to yield

$$e = \frac{\eta}{G} p + \frac{\eta_d}{G} T + g(t) \tag{11.207}$$

where $g(t)$ is an arbitrary function of t. From the above, and the constitutive equation (11.89), we find that the following expression is a function of time only

$$\sigma_{kk} + 4\eta p + 4\eta_d T = \frac{2G(1+\nu)}{1-2\nu} g(t) \tag{11.208}$$

And under plane strain condition, we have

$$\sigma_{kk} + 2\eta p + 2\eta_d T = \frac{2G}{1-2\nu} g(t) \tag{11.209}$$

Using (11.207) and (11.208) in the pore pressure diffusion equation (11.185), it can be transformed to

$$\frac{\partial p}{\partial t} - c \nabla^2 p = \frac{\beta_c}{S} \frac{\partial T}{\partial t} - \frac{\eta(1-\nu)}{GS(1+\nu)} \frac{d}{dt} (\sigma_{kk} + 4\eta p + 4\eta_d T) \tag{11.210}$$

Similarly, from the heat diffusion equation (11.193), we find

$$\frac{\partial T}{\partial t} - c_a \nabla^2 T = \frac{\beta_c}{S_a} \frac{\partial p}{\partial t} - \frac{\eta_d(1-\nu)}{GS_a(1+\nu)} \frac{d}{dt} (\sigma_{kk} + 4\eta p + 4\eta_d T) \tag{11.211}$$

For plane strain problem, in which $\sigma_{kk} = \sigma_{xx} + \sigma_{yy}$, the above equations can be express as

$$\frac{\partial p}{\partial t} - c \nabla^2 p = \frac{\beta_c}{S} \frac{\partial T}{\partial t} - \frac{\eta(1-\nu)}{GS} \frac{d}{dt} (\sigma_{kk} + 2\eta p + 2\eta_d T) \tag{11.212}$$

$$\frac{\partial T}{\partial t} - c_a \nabla^2 T = \frac{\beta_c}{S_a}\frac{\partial p}{\partial t} - \frac{\eta_d(1-\nu)}{GS_a}\frac{d}{dt}(\sigma_{kk} + 2\eta p + 2\eta_d T) \tag{11.213}$$

The advantage of the above diffusion equations is that, if the value of $\sigma_{kk} + 4\eta p + 4\eta_d T$ is known at a given location, it is known at all locations, as it is a function of t only. Particularly, for exterior domain problems, the boundary condition requires that the stress and pressure vanish at infinity; hence the above diffusion equations simplify to

$$\frac{\partial p}{\partial t} - c\nabla^2 p = \frac{\beta_c}{S}\frac{\partial T}{\partial t} \tag{11.214}$$

$$\frac{\partial T}{\partial t} - c_a \nabla^2 T = \frac{\beta_c}{S_a}\frac{\partial p}{\partial t} \tag{11.215}$$

Comparing to the original diffusion equations (11.185) and (11.193), we find that (11.214) and (11.215) are uncoupled from solid displacement; hence these equations can be solved independent of the Navier equation (11.168). The pore pressure and temperature, however, are still coupled and (11.214) and (11.215) need to be solved simultaneously.

11.4.4 Thermally Uncoupled Model

The porothermoelasticity model constructed so far contains the full coupling among the solid, the fluid, and the thermal effects. In practice, however, depending on the material constants involved, the physical strengths of the various coupling may not be the same, and certain coupling may be ignored. Particularly, the thermal coupling is known to be weak (with the exception in thermoelastodynamics, in which the coupling can be important). The thermoelasticity theory [53] is generally uncoupled to reduce to the theory of thermal stress [6].

Coupling means two-way effects. For poroelasticity, the presence of pore pressure affects the solid deformation, and solid deformation generates pore pressure (Skempton effect). Equivalently, for coupled thermoelasticity, not only heat causes solid to deform, but also solid deformation generates heat. As demonstrated by Zimmerman [82], the poroelastic coupling can be ignored if the *poroelastic coupling parameter* is much less than one,

$$\alpha B \ll 1 \tag{11.216}$$

and the thermoelastic coupling can be ignored if the *thermoelastic coupling parameter* is much less than one,

$$\frac{K\beta_d^2 \mathcal{T}_o}{c_d} = \frac{K\beta_d^2}{m_d} \ll 1 \tag{11.217}$$

where $\mathcal{T}_o$ is a reference absolute temperature in °K. Based on the poroelastic material constants listed in Table 3.2, the condition (11.216) is obviously not satisfied. For condition (11.217), Zimmerman [82] estimated that the parameter is around 4×10^{-5} for Berea sandstone. Based on data in Tables 3.2 and 11.3, we can evaluate the thermoelastic coupling parameter as 1.7×10^{-6}, 4.6×10^{-2}, 2.7×10^{-5}, and 1.8×10^{-3}, respectively for deep sea clay, rock salt, Berea sandstone and Westerly granite; hence confirm that for most geomaterials, the thermoelastic coupling is weak and can be ignored.

In terms of governing equations, the thermal uncoupling is achieved by taking the limit of

$$S_a \to \infty \tag{11.218}$$

in (11.213) or (11.215), which is equivalent to taking

$$m_d \to \infty \tag{11.219}$$

This condition also leads to

$$c_a \to \kappa_T \tag{11.220}$$

with other parameter limiting conditions shown in Sect. B.7. With the above, the diffusion equation for temperature (11.213) reduces to

$$\frac{\partial T}{\partial t} - \kappa_T \nabla^2 T = 0 \tag{11.221}$$

Hence the heat diffusion is uncoupled from the poroelastic process.

For the other diffusion equations, while the equation for pressure (11.185) remains unchanged, the equation for the variation in fluid content (11.186) can be expressed as

$$\frac{\partial \zeta}{\partial t} - c\nabla^2 \zeta = \frac{c\,\beta_c}{\kappa_T}\frac{\partial T}{\partial t} \tag{11.222}$$

In the solution of a porothermoelasticity problem, (11.221) can be independently solved based on the temperature boundary conditions such as (11.202), (11.203), and (11.204). In turn, (11.222) can be solved with the known temperature information. However, as ζ is typically not given as a boundary condition, it is more often that we transfer the temperature information to the other two governing equations, (11.168) and (11.185), which must be solved simultaneously. We also notice that the material constant m_d is no longer needed in the uncoupled model, and the total number of material constants in reduced to 8.

11.4.5 Complete Model

The complete model differs from the thermally coupled model in the fluid and heat flux laws, (11.139) and (11.140) as compared to (11.141) and (11.142). Combining the continuity equation (11.118) with (11.139) we obtain

$$\frac{\partial \zeta}{\partial t} - \kappa \nabla^2 p - k_{pT} \nabla^2 T = 0 \tag{11.223}$$

From (11.127) and (11.140), we find

$$\frac{\partial s}{\partial t} - \frac{k_T}{\mathcal{T}_o} \nabla^2 T - \frac{k_{Tp}}{\mathcal{T}_o} \nabla^2 p = 0 \tag{11.224}$$

A pore pressure diffusion equation can be obtained by substituting (11.66) into (11.223) to eliminate ζ, and we find

$$\frac{\partial p}{\partial t} - \kappa M \nabla^2 p = -\alpha M \frac{\partial e}{\partial t} + \beta_e M \frac{\partial T}{\partial t} + k_{pT} M \nabla^2 T \tag{11.225}$$

Similarly, to obtain an diffusion equation for temperature, we substitute (11.91) into the above to eliminate s, and find

$$\frac{\partial T}{\partial t} - \kappa_T \nabla^2 T = -\frac{\alpha_d}{m_d} \frac{\partial e}{\partial t} + \frac{\beta_e}{m_d} \frac{\partial p}{\partial t} + \kappa_{Tp} \nabla^2 p \tag{11.226}$$

where

$$\kappa_{Tp} = \frac{k_{Tp}}{m_d \mathcal{T}_o} \tag{11.227}$$

Equations (11.225) and (11.226) can be compare to (11.185) and (11.193), and we observe the extra right hand side term.

In the following sections we shall present analytical solutions for the thermally coupled model by solving (11.185) and (11.193), together with (11.168). For the complete model, rather than attempting to solve (11.225) and (11.226), we can demonstrate that mathematically they are equivalent to (11.185) and (11.193). This can be accomplished by using (11.226) to eliminate the $\nabla^2 T$ term in (11.225), and conversely, using (11.225) to eliminate the $\nabla^2 p$ term in (11.226). We hence obtain

$$\begin{aligned} &\frac{\partial p}{\partial t} - \frac{M m_d (\kappa \kappa_T - k_{pT} \kappa_{Tp})}{M \beta_e k_{pT} + m_d \kappa_T} \nabla^2 p \\ &\quad = -\frac{M(\alpha m_d \kappa_T - \alpha_d k_{pT})}{M \beta_e k_{pT} + m_d \kappa_T} \frac{\partial e}{\partial t} + \frac{M m_d (\kappa_T + \beta_e \kappa_T)}{M \beta_e k_{pT} + m_d \kappa_T} \frac{\partial T}{\partial t} \end{aligned} \tag{11.228}$$

$$\frac{\partial T}{\partial t} - \frac{\kappa\kappa_T - k_{pT}\kappa_{Tp}}{\kappa + \beta_e\kappa_{Tp}}\nabla^2 T = -\frac{\alpha_d\kappa - \alpha\, m_d\kappa_{Tp}}{m_d(\kappa + \beta_e\kappa_{Tp})}\frac{\partial e}{\partial t} + \frac{M\beta_e\kappa + m_d\kappa_{Tp}}{Mm_d(\kappa + \beta_e\kappa_{Tp})}\frac{\partial p}{\partial t} \tag{11.229}$$

These two equations are the same as (11.185) and (11.193), except for the coefficients. Hence we can obtain the solutions for the complete model by the proper replacement of the coefficients of the thermally coupled model solutions.

11.5 Uniaxial Strain

In the following sections we shall explore a few analytical solutions involving simple geometries. The simplest one is the uniaxial strain problem, characterized by the set of assumptions

$$u_z = u_z(z,t);\quad p = p(z,t);\quad T = T(z,t);\quad u_x = u_y = 0 \tag{11.230}$$

which also lead to

$$e_{zz} = e_{zz}(z,t);\quad \zeta = \zeta(z,t);\quad e_{xx} = e_{yy} = e_{xy} = e_{xz} = e_{yz} = 0 \tag{11.231}$$

Based on the above, the Navier equation (11.168) reduces to

$$\frac{2G(1-\nu)}{1-2\nu}\frac{\partial^2 u_z}{\partial z^2} - \alpha\frac{\partial p}{\partial z} - \alpha_d\frac{\partial T}{\partial z} = 0 \tag{11.232}$$

The diffusion equations are reduced from (11.185) and (11.193) and become

$$\frac{\partial p}{\partial t} - \kappa M\frac{\partial^2 p}{\partial z^2} = -\alpha M\frac{\partial e_{zz}}{\partial t} + \beta_e M\frac{\partial T}{\partial t} \tag{11.233}$$

and

$$\frac{\partial T}{\partial t} - \kappa_T\frac{\partial^2 T}{\partial z^2} = -\frac{\alpha_d}{m_d}\frac{\partial e_{zz}}{\partial t} + \frac{\beta_e}{m_d}\frac{\partial p}{\partial t} \tag{11.234}$$

We can also express the constitutive equations as

$$\sigma_{zz} = \frac{2G(1-\nu)}{1-2\nu}e_{zz} - \alpha p - \alpha_d T \tag{11.235}$$

$$\sigma_{xx} = \sigma_{yy} = \frac{2G\nu}{1-2\nu}e_{zz} - \alpha p - \alpha_d T \tag{11.236}$$

$$\sigma_{xy} = \sigma_{xz} = \sigma_{yz} = 0 \tag{11.237}$$

$$p = -\alpha M e_{zz} + M\zeta + M\beta_e T \tag{11.238}$$

From the above, and the equilibrium equation

$$\frac{\partial \sigma_{zz}}{\partial z} = 0 \tag{11.239}$$

we obtain the functional dependency of the stresses

$$\sigma_{zz} = \sigma_{zz}(t); \quad \sigma_{xx} = \sigma_{xx}(z,t); \quad \sigma_{yy} = \sigma_{yy}(z,t) \tag{11.240}$$

We particularly note that σ_{zz} is a function of t only. By eliminating e_{zz} in (11.233) and (11.234) using (11.235), we obtain

$$\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial z^2} = -\frac{\eta}{GS}\frac{d\sigma_{zz}}{dt} + \alpha_e \frac{\partial T}{\partial t} \tag{11.241}$$

$$\frac{\partial T}{\partial t} - c_a\frac{\partial^2 T}{\partial z^2} = -\frac{\eta_d}{GS_a}\frac{d\sigma_{zz}}{dt} + \alpha_p \frac{\partial p}{\partial t} \tag{11.242}$$

where we have adopted the shorthand notations

$$\alpha_e = \frac{\beta_c}{S} \tag{11.243}$$

$$\alpha_p = \frac{\beta_c}{S_a} \tag{11.244}$$

Equations (11.241) and (11.242) are in fact the direct consequence of the irrotational displacement field, as discussed in Sect. 11.4.3.

11.6 Heating of a Half Space

To illustrate the porothermoelastic effect, we shall first examine the problem of heating of a half space from its surface. The solutions were presented by McTigue [47] for the thermally uncoupled case. In this section we shall solve both the thermally coupled and the uncoupled cases.

Consider a half space occupying $z \geq 0$, and is initially in equilibrium in temperature and pressure, or conveniently set as $T = 0$ and $p = 0$. The surface at $z = 0$ is kept stress free at all times. For the fluid, two types of boundary conditions are applied, zero pressure and zero flux. For the temperature part, the surface can be subjected to a step rise of temperature, or a constant heat flux. Table 11.4 summarizes the boundary condition at the surface for four different cases. At the far field, all stresses, pressure and temperature vanish. The initial condition is zero (constant) temperature everywhere.

Table 11.4 Boundary conditions at $z = 0$ for heating of half space problems, for four cases

Case	σ_{zz}	p	$\partial p/\partial z$	T	$\partial T/\partial z$
1	0	0	–	$T_1 \mathrm{H}(t-0)$	–
2	0	–	0	$T_1 \mathrm{H}(t-0)$	–
3	0	0	–	–	$-q_0/k_T$
4	0	–	0	–	$-q_0/k_T$

11.6.1 Constant Temperature, Permeable Boundary

The boundary conditions are given as Case 1 in Table 11.4. We shall separately solve the coupled and uncoupled cases.

Thermally Coupled Solution Based on the boundary condition $\sigma_{zz} = 0$, (11.241) and (11.242) simplify to the following

$$\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial z^2} = \alpha_e \frac{\partial T}{\partial t} \tag{11.245}$$

$$\frac{\partial T}{\partial t} - c_a\frac{\partial^2 T}{\partial z^2} = \alpha_p \frac{\partial p}{\partial t} \tag{11.246}$$

Performing Laplace transform, we find

$$s\tilde{p} - c\frac{d^2\tilde{p}}{dz^2} = s\alpha_e \tilde{T} \tag{11.247}$$

$$s\tilde{T} - c_a\frac{d^2\tilde{T}}{dz^2} = s\alpha_p \tilde{p} \tag{11.248}$$

Variables can be eliminated between these two equations and we obtain higher order differential equations:

$$\tilde{T}''''(z) - \frac{s(c+c_a)}{c\,c_a}\tilde{T}''(z) + \frac{s^2(1-\alpha_p\alpha_e)}{c\,c_a}\tilde{T}(z) = 0 \tag{11.249}$$

and the same for $\tilde{p}(z)$. The general solution of the above is

$$\tilde{T} = C_1 e^{-\lambda_1\sqrt{s}z} + C_2 e^{-\lambda_2\sqrt{s}z} \tag{11.250}$$

where we have dropped terms that diverge at $z \to \infty$, and

$$\lambda_{1,2} = \sqrt{\frac{(c+c_a) \pm C_Q}{2c\,c_a}} \tag{11.251}$$

in which

$$C_Q = \sqrt{(c - c_a)^2 + 4c c_a \alpha_p \alpha_e} \tag{11.252}$$

Substituting the above into (11.248), we can obtain $\tilde{p}(z)$. Utilizing the two boundary conditions at $z = 0$ as shown in Table 11.4, the two constants C_1 and C_2 can be solved and we find

$$\frac{s\tilde{T}}{T_1} = \frac{c - c_a + C_Q}{2C_Q} e^{-\lambda_1 \sqrt{s} z} - \frac{c - c_a - C_Q}{2C_Q} e^{-\lambda_2 \sqrt{s} z} \tag{11.253}$$

$$\frac{s\tilde{p}}{\alpha_e T_1} = -\frac{c_a}{C_Q} \left(e^{-\lambda_1 \sqrt{s} z} - e^{-\lambda_2 \sqrt{s} z} \right) \tag{11.254}$$

Performing inverse Laplace transform using Table D.1, the solution in time domain is obtained

$$\begin{aligned} \frac{T(z,t)}{T_1} = & \frac{c - c_a + C_Q}{2C_Q} \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}} \right) \\ & - \frac{c - c_a - C_Q}{2C_Q} \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}} \right) \end{aligned} \tag{11.255}$$

$$\frac{p(z,t)}{\alpha_e T_1} = -\frac{c_a}{C_Q} \left[\operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}} \right) - \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}} \right) \right] \tag{11.256}$$

From (11.235) with the condition $\sigma_{zz} = 0$, we obtain

$$e_{zz} = \frac{1 - 2\nu}{2G(1 - \nu)} (\alpha p + \alpha_d T) \tag{11.257}$$

The displacement u_z can be found by integrating the above, and

$$\begin{aligned} u_z(z,t) = \frac{T_1(1 - 2\nu)}{4GC_Q(1 - \nu)} \Bigg\{ & \left[\alpha_d (c - c_a + C_Q) - 2c_a \alpha \alpha_e \right] \\ & \times \left[z \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}} \right) - \frac{2}{\lambda_1} \sqrt{\frac{t}{\pi}} e^{-\lambda_1^2 z^2 / 4t} \right] \\ & - \left[\alpha_d (c - c_a - C_Q) - 2c_a \alpha \alpha_e \right] \times \\ & \left[z \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}} \right) - \frac{2}{\lambda_2} \sqrt{\frac{t}{\pi}} e^{-\lambda_2^2 z^2 / 4t} \right] \Bigg\} \end{aligned} \tag{11.258}$$

Utilizing (11.257) in (11.236), we can find the horizontal stress as

$$\sigma_{xx} = -\frac{1-2\nu}{1-\nu}(\alpha p + \alpha_d T) \tag{11.259}$$

Thermally Uncoupled Solution The solution for the thermally uncoupled case was obtained by McTigue [47]. Here we shall derive it by simplifying it from the thermally coupled case. First, we notice that the difference between the coupled and uncoupled case is that the heat diffusion equation is uncoupled for the latter, that is, (11.246) becomes

$$\frac{\partial T}{\partial t} - \kappa_T \frac{\partial^2 T}{\partial z^2} = 0 \tag{11.260}$$

As discussed in Sect. 11.4.4, this is a consequence of the condition (11.219), which further leads to

$$\begin{aligned} &\alpha_p \to 0; \quad C_Q \to c - c_a; \quad c_a \to \kappa_T; \\ &\lambda_1 \to \sqrt{1/\kappa_T}; \quad \lambda_2 \to \sqrt{1/c} \end{aligned} \tag{11.261}$$

Using the above in (11.255) and (11.256), we obtain

$$\frac{T(\xi)}{T_1} = \operatorname{erfc}(R_q \xi) \tag{11.262}$$

$$\frac{p(\xi)}{\alpha_e T_1} = \frac{1}{1-R_q^2}\left[\operatorname{erfc}(R_q \xi) - \operatorname{erfc}(\xi)\right] \tag{11.263}$$

where

$$R_q = \sqrt{\frac{c}{\kappa_T}} \tag{11.264}$$

is the square root of the ratio of consolidation coefficient and the heat diffusivity coefficient, and

$$\xi = \frac{z}{2\sqrt{ct}} \tag{11.265}$$

For the special case $R_q = 1$, the limit can be taken from (11.263) and it becomes

$$\frac{p(\xi)}{\alpha_e T_1} = \frac{1}{\sqrt{\pi}} \xi e^{-\xi^2} \tag{11.266}$$

The above solutions are *self-similar* in the sense that the space and time variables are combined into one parameter ξ. Hence a single curve can represent the solution at all space and time.

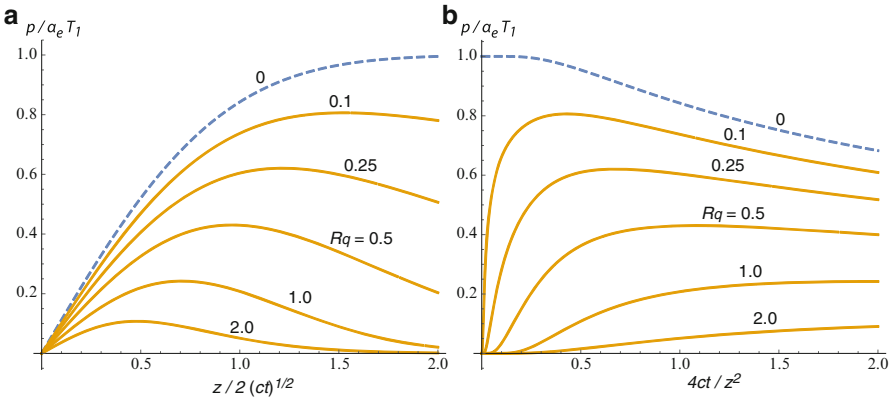


Fig. 11.3 Pressure solution for case 1, for different diffusivity ratios $R_q = \sqrt{c/\kappa_T}$. (**a**) $p/\alpha_e T_1$ versus $\xi = z/2\sqrt{ct}$, and (**b**) versus $\xi^{-2} = 4ct/z^2$

In Fig. 11.3a we plot the pressure solution versus the parameter ξ for different R_q values. For a given time, these curves represent the pressure profile versus depth z. As observe in the figure, when $R_q > 1$, the pressure diffusion is faster than the heat conduction; hence the pressure response is smaller than the case of $R_q < 1$. By taking derivative of (11.263) we can show that the maximum pressure takes place at

$$\xi = \sqrt{-\frac{\ln R_q}{1 - R_q^2}} \tag{11.267}$$

We also find that as $R_q \to \infty$, $p_{\max} \to 0$, and as $R_q \to 0$, $p_{\max}/\alpha_e T_1 \to 1$. Figure 11.3a also presents the limiting curve of $R_q \to 0$, corresponding to the case of $\kappa_T \gg c$, and (11.263) reduces to

$$\frac{p(\xi)}{\alpha_e T_1} = \text{erf}(\xi) \tag{11.268}$$

In Fig. 11.3b we plot the same solution (11.263), but versus $\xi^{-2} = 4ct/z^2$. At a fixed depth z, these curves show the time history of pore pressure response. We observe that the pore pressure will rise to a maximum at the time defined by (11.267). At large time, the pressure decay to zero at the rate of $1/\sqrt{t}$.

The displacement solution can be obtained by integrating (11.235) as

$$u_z(z,t) = \frac{1-2\nu}{2G(1-\nu)} \int_{-\infty}^{z} (\alpha_d T + \alpha p)\, dz \tag{11.269}$$

where we have utilized the condition $\sigma_{zz} = 0$. Substituting in (11.262) and (11.263), we obtain

$$u_z(z,t) = -\frac{(1-2\nu)T_1\sqrt{ct}}{G(1-\nu)}\left\{\alpha_d\left[\frac{e^{-R_q^2\xi^2}}{\sqrt{\pi}R_q} - \xi\,\mathrm{erfc}(R_q\xi)\right]\right.$$
$$\left. + \frac{\alpha\alpha_e}{1-R_q^2}\left[\xi\,\mathrm{erfc}(\xi) - \xi\,\mathrm{erfc}(R_q\xi) + \frac{e^{-R_q^2\xi^2}}{\sqrt{\pi}R_q} - \frac{e^{-\xi^2}}{\sqrt{\pi}}\right]\right\} \tag{11.270}$$

We note that the above expression has two parts: the first part is due to the heat induced thermal expansion of the drained material, characterized by the drained thermoelastic effective stress coefficient α_d, and the second part is due to the thermally induced pore pressure, which causes a tensile stress, thus additional expansion, by the effective stress effect. On the surface, $z = 0$, (11.270) reduces to

$$u_z(0,t) = -\frac{(1-2\nu)T_1\sqrt{ct}}{\sqrt{\pi}G(1-\nu)R_q}\left[\alpha_d + \frac{\alpha\alpha_e}{1+R_q}\right] \tag{11.271}$$

So with the heating of the surface, the surface rebounds at the rate of $\sqrt{t}$. Of the two contributions, that due to thermal effect and the pore pressure effect, we observe the ratio

$$\frac{u_{\text{pressure}}}{u_{\text{thermal}}} = \frac{\alpha\alpha_e}{(1+R_q)\alpha_d} \tag{11.272}$$

For the special case that the pore pressure diffusivity is very small, $c \to 0$, (11.271) reduces to

$$u_z(0,t) = -\frac{2(1+\nu_u)\beta_u T_1}{3(1-\nu_u)}\sqrt{\frac{\kappa_T t}{\pi}} \tag{11.273}$$

which is the same as the thermoelastic solution, except that the Poisson ratio and the thermal expansion coefficient take the undrained value.

To obtain numerical values of the above solutions, we shall assume that the temperature on the surface of the half space is suddenly raised by $T_1 = 10\,^\circ\mathrm{C}$. We investigate the responses for two porous media, a abyssal red clay, and a rock salt, with their material constants summarized in Tables 3.1, 3.2, and 11.3. As commented in Sect. 11.4.4, the thermoelastic coupling for these materials is weak; hence only the thermally uncoupled case is evaluated in the following.

In Fig. 11.4 we plot the negative vertical displacement $-u_z$ (as the layer expands) versus the layer depth z at $t = 30$ days for the two materials. We also plot the partial components contributed by the thermal and pore pressure effect, which show that one effect can be larger than the other as dictated by (11.272).

Another result of interest is the stress distribution. We note that although the vertical stress is zero, the horizontal stresses are not. Particularly, we find the horizontal Biot effective stress $\sigma''_{xx} = \sigma_{xx} + \alpha p$ as

$$\sigma''_{xx}(z,t) = \frac{\nu}{1-\nu}\alpha p - \frac{1-2\nu}{1-\nu}\alpha_d T \tag{11.274}$$

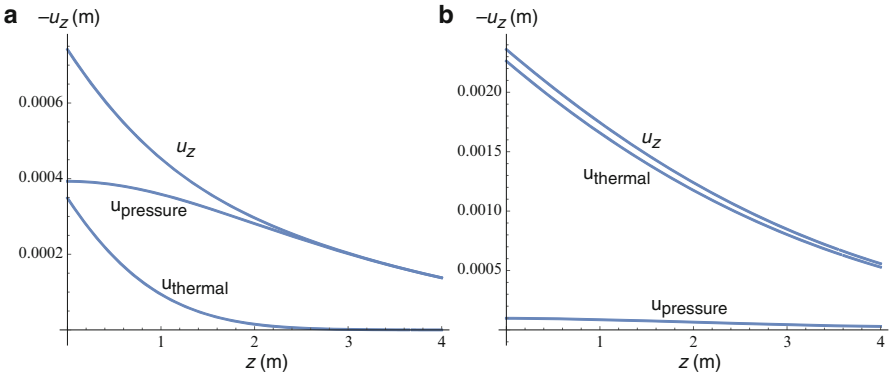


Fig. 11.4 Displacement solution for case 1: $-u_z$ versus depth z for (**a**) abyssal red clay, and (**b**) rock salt, at $t = 30$ days

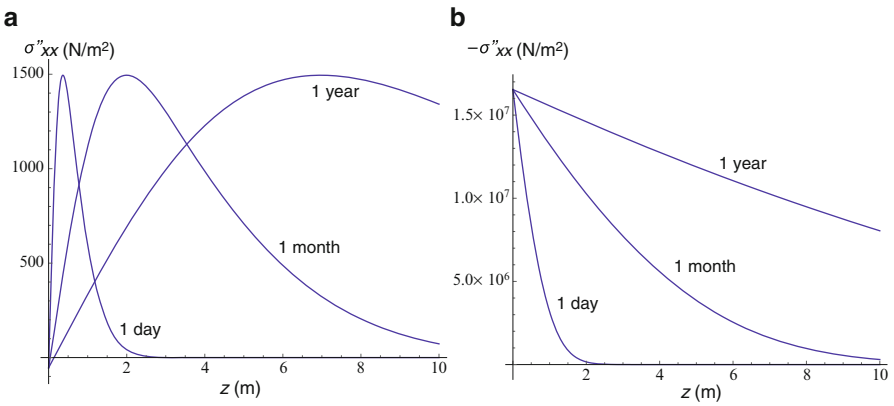


Fig. 11.5 Horizontal Biot effective stress versus depth at various time for case 1: (**a**) abyssal red clay, and (**b**) rock salt

Figure 11.5 plots the effective stress versus depth at various times for the two materials. We notice that at the surface, there is no pore pressure and the thermal effect dominates. The effective stress is

$$\sigma''_{xx}(0,t) = -\frac{1-2\nu}{1-\nu}\alpha_d T_1 \tag{11.275}$$

which is a constant and is always in compression. At a depth, however, the behavior is much different. Figure 11.5a shows that for red clay, pore pressure has a large effect and the effective stress is in tension except very near the surface. Figure 11.5b plots the negative effective stress for rock salt, and it shows that the thermal effect dominates and the layer is entirely in compression.

11.6.2 Constant Temperature, Impermeable Boundary

This problem corresponds to Case 2 in Table 11.4. Again, we shall examine a coupled and an uncoupled solution.

Thermally Coupled Solution Following the same procedure as in Sect. 11.6.1, but with different boundary conditions, we obtain the solution in Laplace transform domain

$$\frac{s\tilde{T}}{T_1} = \frac{\lambda_2(c - c_a + C_Q)e^{-\lambda_1\sqrt{s}z} - \lambda_1(c - c_a - C_Q)e^{-\lambda_2\sqrt{s}z}}{\lambda_2(c - c_a + C_Q) - \lambda_1(c - c_a - C_Q)} \tag{11.276}$$

$$\frac{s\tilde{p}}{\alpha_e T_1} = \frac{2c_a(\lambda_2 e^{-\lambda_1\sqrt{s}z} - \lambda_1 e^{-\lambda_2\sqrt{s}z})}{\lambda_1(c - c_a + C_Q) - \lambda_2(c - c_a - C_Q)} \tag{11.277}$$

And the solution in time is

$$\begin{aligned}\frac{T(z,t)}{T_1} &= \frac{\lambda_2(c - c_a + C_Q)}{\lambda_2(c - c_a + C_Q) - \lambda_1(c - c_a - C_Q)}\operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right) \\ &\quad - \frac{\lambda_1(c - c_a - C_Q)}{\lambda_2(c - c_a + C_Q) - \lambda_1(c - c_a - C_Q)}\operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\end{aligned} \tag{11.278}$$

$$\begin{aligned}\frac{p(z,t)}{\alpha_e T_1} &= \frac{2c_a}{\lambda_1(c - c_a + C_Q) - \lambda_2(c - c_a - C_Q)} \\ &\quad \times \left[\lambda_2 \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right) - \lambda_1 \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\right]\end{aligned} \tag{11.279}$$

Thermally Uncoupled Solution Taking the limits following (11.261), we can obtain the solution of the uncoupled case. The temperature is given as (11.262), and the pore pressure as

$$\frac{p(\xi)}{\alpha_e T_1} = \frac{1}{1 - R_q^2}\left[\operatorname{erfc}(R_q\xi) - R_q \operatorname{erfc}(\xi)\right] \tag{11.280}$$

For $R_q = 1$, it becomes

$$\frac{p(\xi)}{\alpha_e T_1} = \frac{1}{2}\operatorname{erfc}(\xi) - \frac{1}{\sqrt{\pi}}\xi e^{-\xi^2} \tag{11.281}$$

At $z = 0$, the pressure is independent of time:

$$\frac{p(0,t)}{\alpha_e T_1} = \frac{1}{1 + R_q} \tag{11.282}$$

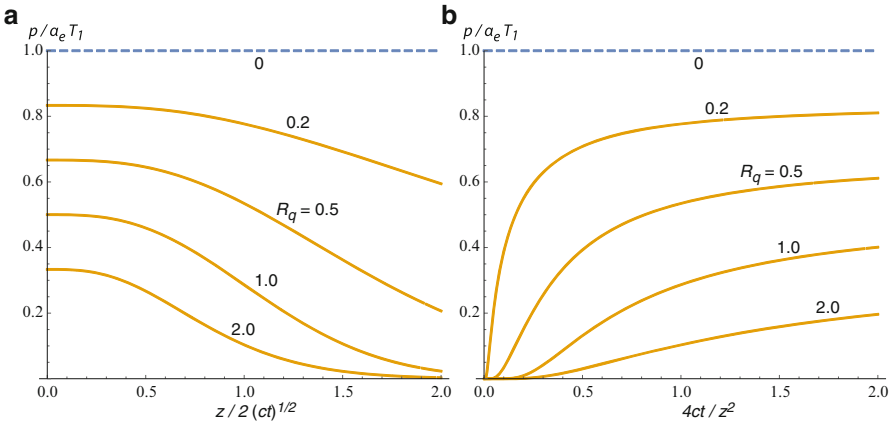


Fig. 11.6 Pressure solution for case 2 with different R_q values: (**a**) $p/\alpha_e T_1$ versus $\xi = z/2\sqrt{ct}$, and (**b**) versus $\xi^{-2} = 4ct/z^2$

In Fig. 11.6a we plot the pressure solution versus ξ for different R_q values. These curves give the pressure profile versus depth at fixed time. In Fig. 11.6b, the solution is plotted versus ξ^{-2}, which gives the pressure evolution in time at fixed depth. The displacement solution is

$$u_z(z,t) = -\frac{(1-2\nu)T_1\sqrt{ct}}{G(1-\nu)}\left\{\alpha_d\left[\frac{\xi^2 e^{-R_q^2}}{\sqrt{\pi}R_q} - \xi\,\mathrm{erfc}(R_q\xi)\right] + \frac{\alpha\,\alpha_e}{1-R_q^2}\right.$$
$$\left.\times\left[\xi\,\mathrm{erfc}(R_q\xi) - R_q\xi\,\mathrm{erfc}(\xi) - \frac{e^{-R_q^2\xi^2}}{\sqrt{\pi}R_q} + \frac{R_q e^{-\xi^2}}{\sqrt{\pi}}\right]\right\} \tag{11.283}$$

11.6.3 Constant Heat Flux, Permeable Boundary

The constant heat flux and permeable boundary case is referred to as Case 3 with boundary conditions given in Table 11.4.

Thermally Coupled Solution The Laplace transform temperature and pore pressure solutions are

$$\tilde{T} = \frac{q_o}{k_T s^{3/2}}\frac{(c-c_a+C_Q)e^{-\lambda_1\sqrt{s}z} - (c-c_a-C_Q)e^{-\lambda_2\sqrt{s}z}}{\lambda_1(c-c_a+C_Q) - \lambda_2(c-c_a-C_Q)} \tag{11.284}$$

$$\tilde{p} = \frac{2c_a\alpha_e q_o(e^{-\lambda_1\sqrt{s}z} - e^{-\lambda_2\sqrt{s}z})}{k_T s^{3/2}[\lambda_2(c-c_a+C_Q) - \lambda_1(c-c_a-C_Q)]} \tag{11.285}$$

And the solution in time is

$$T(z,t) = \frac{q_o}{k_T[\lambda_1(c-c_a+C_Q)-\lambda_2(c-c_a-C_Q)]} \times \left\{ (c-c_a+C_Q)\left[2\sqrt{\frac{t}{\pi}}e^{-\lambda_1^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right)\right] -(c-c_a-C_Q)\left[2\sqrt{\frac{t}{\pi}}e^{-\lambda_2^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\right]\right\} \tag{11.286}$$

$$p(z,t) = \frac{2c_a\alpha_e q_o}{k_T[\lambda_2(c-c_a+C_Q)-\lambda_1(c-c_a-C_Q)]} \times \left[2\sqrt{\frac{t}{\pi}}e^{-\lambda_1^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right) -2\sqrt{\frac{t}{\pi}}e^{-\lambda_2^2 z^2/4t} + \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\right] \tag{11.287}$$

Thermally Uncoupled Solution The temperature solution is given by

$$T(z,t) = \frac{2q_o\sqrt{\kappa_T t}}{k_T}\left[\frac{e^{-R_q^2\xi^2}}{\sqrt{\pi}} - R_q\xi\operatorname{erfc}\left(R_q\xi\right)\right] \tag{11.288}$$

The pressure is

$$p(z,t) = \frac{2\alpha_e q_o\sqrt{\kappa_T t}}{k_T(1-R_q^2)}\left[\frac{e^{-R_q^2\xi^2}}{\sqrt{\pi}} - R_q\xi\operatorname{erfc}(R_q\xi) - \frac{e^{-\xi^2}}{\sqrt{\pi}} + \xi\operatorname{erfc}(\xi)\right] \tag{11.289}$$

For $R_q = 1$, the above becomes

$$p(z,t) = \frac{\alpha_e q_o\sqrt{\kappa_T t}}{k_T}\xi\operatorname{erfc}(\xi) \tag{11.290}$$

The fluid flux is obtained from Darcy's law (6.27) and is

$$q_z(z,t) = -\frac{\kappa\,\alpha_e\,q_o}{k_T(1-R_q^2)}\left[\operatorname{erfc}(\xi) - R_q\operatorname{erfc}(R_q\xi)\right] \tag{11.291}$$

It is interesting to observe that at the boundary $z = 0$, the fluid flux is a constant

$$q_z(0,t) = -\frac{\kappa\,\alpha_e\,q_o}{k_T R_q(1+R_q)} \tag{11.292}$$

Displacement and horizontal stress can be obtained from (11.269) and (11.274).

11.6.4 Constant Heat Flux, Impermeable Boundary

The boundary conditions for Case 4 are found in Table 11.4.

Thermally Coupled Solution The Laplace transform temperature and pore pressure solutions are

$$\tilde{T} = \frac{q_o}{2k_T C_Q \lambda_1 \lambda_2 s^{3/2}} \left[\lambda_2 (c - c_a + C_Q) e^{-\lambda_1 \sqrt{s} z} \right. \\ \left. - \lambda_1 (c - c_a - C_Q) e^{-\lambda_2 \sqrt{s} z} \right] \tag{11.293}$$

$$\tilde{p} = \frac{c_a \alpha_e q_o (\lambda_1 e^{-\lambda_2 \sqrt{s} z} - \lambda_2 e^{-\lambda_1 \sqrt{s} z})}{k_T C_Q \lambda_1 \lambda_2 s^{3/2}} \tag{11.294}$$

And the solution in time is

$$T(z,t) = \frac{q_o}{2k_T C_Q \lambda_1 \lambda_2} \\ \times \left\{ \lambda_2 (c - c_a + C_Q) \left[2\sqrt{\frac{t}{\pi}} e^{-\lambda_1^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}} \right) \right] \right. \\ \left. - \lambda_1 (c - c_a - C_Q) \left[2\sqrt{\frac{t}{\pi}} e^{-\lambda_2^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}} \right) \right] \right\} \tag{11.295}$$

$$p(z,t) = \frac{c_a \alpha_e q_o}{k_T C_Q \lambda_1 \lambda_2} \left\{ \lambda_1 \left[2\sqrt{\frac{t}{\pi}} e^{-\lambda_2^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}} \right) \right] \right. \\ \left. - \lambda_2 \left[2\sqrt{\frac{t}{\pi}} e^{-\lambda_1^2 z^2/4t} - \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}} \right) \right] \right\} \tag{11.296}$$

Thermally Uncoupled Solution The temperature solution is the same as (11.288). The pressure solution is

$$p(z,t) = \frac{2\alpha_e q_o \sqrt{\kappa_T t}}{k_T (1 - R_q^2)} \left[\frac{e^{-R_q^2 \xi^2}}{\sqrt{\pi}} - \frac{R_q e^{-\xi^2}}{\sqrt{\pi}} - R_q \xi \operatorname{erfc}(R_q \xi) + R_q \xi \operatorname{erfc}(\xi) \right] \tag{11.297}$$

When $R_q = 1$, the above becomes

$$p(z,t) = \frac{\alpha_e q_o \sqrt{\kappa_T t}}{k_T} \frac{e^{-\xi^2}}{\sqrt{\pi}} \tag{11.298}$$

At the surface, $z = 0$, the pressure increases with the square root of time

$$p(z,t) = \frac{2\alpha_e q_o \sqrt{\kappa_T t}}{\sqrt{\pi}\, k_T (1 + R_q)} \tag{11.299}$$

11.7 Axial Symmetry

In this section we shall examine solutions that possess axial symmetry, and are under the plan strain condition. The dependent variables satisfying the above conditions have the following functional dependence on the spatial coordinates

$$u_r = u_r(r,t); \quad u_\theta = u_z = 0; \quad p = p(r,t); \quad T = T(r,t); \quad \zeta = \zeta(r,t) \tag{11.300}$$

Similar to the solutions derived in Sect. 7.12, we shall present two formulations, one utilizing pore pressure as one of the primary variable, and the other utilizing the variation in fluid content. Depending on the problems solved, such as interior domain or exterior domain problems, one formulation may be easier than the other to implement.

Pore Pressure Representation Based on the stress-strain relations

$$\sigma_{rr} = \frac{2G\nu}{1-2\nu} e + 2G e_{rr} - \alpha p - \alpha_d T \tag{11.301}$$

$$\sigma_{\theta\theta} = \frac{2G\nu}{1-2\nu} e + 2G e_{\theta\theta} - \alpha p - \alpha_d T \tag{11.302}$$

$$\sigma_{zz} = \frac{2G\nu}{1-2\nu} e - \alpha p - \alpha_d T \tag{11.303}$$

and the equilibrium equation, we obtain the Navier equation

$$\frac{2G(1-\nu)}{1-2\nu}\left(\frac{\partial^2 u_r}{\partial r^2} + \frac{1}{r}\frac{\partial u_r}{\partial r} - \frac{u_r}{r^2}\right) - \alpha \frac{\partial p}{\partial r} - \alpha_d \frac{\partial T}{\partial r} = 0 \tag{11.304}$$

The above equation can be integrated once to obtain

$$e = \frac{\eta}{G} p + \frac{\eta_d}{G} T + 2A_1(t) \tag{11.305}$$

Integrating a second time, we can find the radial displacement as

$$u_r = \frac{\eta}{G}\frac{1}{r}\int r p(r,t)\, dr + \frac{\eta_d}{G}\frac{1}{r}\int r T(r,t)\, dr + A_1(t)\, r + \frac{A_2(t)}{r} \tag{11.306}$$

Based on the constitutive equations (11.301), (11.302), and (11.303), we arrive at the following

$$\sigma_{rr} = -2\eta\frac{1}{r^2}\int rp(r,t)\,dr - 2\eta_d\frac{1}{r^2}\int rT(r,t)\,dr + \frac{2G}{1-2\nu}A_1(t) - \frac{2G}{r^2}A_2(t) \tag{11.307}$$

$$\sigma_{\theta\theta} = 2\eta\frac{1}{r^2}\int rp(r,t)\,dr + 2\eta_d\frac{1}{r^2}\int rT(r,t)\,dr - 2\eta p - 2\eta_d T + \frac{2G}{1-2\nu}A_1(t) + \frac{2G}{r^2}A_2(t) \tag{11.308}$$

$$\sigma_{zz} = -2\eta p - 2\eta_d T + \frac{4G\nu}{1-2\nu}A_1(t) \tag{11.309}$$

We notice from (11.307) and (11.308) that

$$\sigma_{rr} + \sigma_{\theta\theta} + 2\eta p + 2\eta_d T = \frac{4G}{1-2\nu}A_1(t) \tag{11.310}$$

That is, the above combination of stresses, pore pressure, and temperature is a function of time only, as predicted by (11.209) for a plane strain irrotational displacement field.

The above solutions are presented in terms of temperature and pore pressure. Before we can utilize such formulae, the temperature and pore pressure solutions must first be obtained. The diffusion equations for the irrotational field are given by (11.212) and (11.213), written here as

$$\frac{\partial T}{\partial t} - c_a\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) = \alpha_p\frac{\partial p}{\partial t} - \frac{2\alpha_d}{S_a}\frac{dA_1(t)}{dt} \tag{11.311}$$

and

$$\frac{\partial p}{\partial t} - c\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \alpha_e\frac{\partial T}{\partial t} - \frac{2\alpha}{S}\frac{dA_1(t)}{dt} \tag{11.312}$$

where α_e and α_p are defined in (11.243) and (11.244). We note that although these two equations are coupled, they are uncoupled from the displacement field; hence can be solved independently from the Navier equation for displacement. Furthermore, for the thermally uncoupled theory, $(\alpha_p, \alpha_d/S_a) \to 0$, (11.311) reduces to

$$\frac{\partial T}{\partial t} - \kappa_T\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) = 0 \tag{11.313}$$

The temperature field can then be independently solved.

Variation in Fluid Content Representation In this formulation, we utilize the Navier equation in the form of (11.169), with axial symmetry, to obtain

$$e = \frac{\eta}{GS}\zeta + \frac{\eta\beta_u}{GSB}T + 2B_1(t) \tag{11.314}$$

Integrating once more, we find

$$u_r = \frac{\eta}{GS}\frac{1}{r}\int r\zeta(r,t)\,dr + \frac{\eta\beta_u}{GSB}\frac{1}{r}\int rT(r,t)\,dr + B_1(t)r + \frac{B_2(t)}{r} \tag{11.315}$$

From the constitutive equations, we obtain the following

$$\sigma_{rr} = -\frac{2\eta}{S}\frac{1}{r^2}\int r\zeta(r,t)\,dr - \frac{2\eta\beta_u}{SB}\frac{1}{r^2}\int rT(r,t)\,dr + \frac{2G}{1-2\nu_u}B_1(t) - \frac{2G}{r^2}B_2(t) \tag{11.316}$$

$$\sigma_{\theta\theta} = \frac{2\eta}{S}\left[\frac{1}{r^2}\int r\zeta(r,t)\,dr - \zeta\right] + \frac{2\eta\beta_u}{SB}\left[\frac{1}{r^2}\int rT(r,t)\,dr - T\right] + \frac{2G}{1-2\nu_u}B_1(t) + \frac{2G}{r^2}B_2(t) \tag{11.317}$$

$$\sigma_{zz} = -\frac{2\eta}{S}\zeta - \frac{2\eta\beta_u}{SB}T + \frac{4G\nu_u}{1-2\nu_u}B_1(t) \tag{11.318}$$

$$p = \frac{\zeta}{S} + \alpha_e T - 2\alpha M B_1(t) \tag{11.319}$$

The above representation requires the solutions of T and ζ.

Substituting (11.314) into (11.195), we can decouple the temperature diffusion equation from the volumetric strain in the following form

$$\frac{\partial T}{\partial t} - c_b\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) = M\alpha_t\frac{\partial\zeta}{\partial t} - \frac{2\alpha_u}{S_b}B_1(t) \tag{11.320}$$

where c_b and S_b are given by (11.199) and (11.200), and we have defined the following parameters

$$\alpha_t = \frac{\beta_b}{S_b} \tag{11.321}$$

$$\beta_b = \beta_e - \frac{\alpha_u\eta}{MGS} \tag{11.322}$$

The diffusion equation for variation in fluid content (11.186) is already decoupled from volumetric strain, and is written here as

$$\frac{\partial \zeta}{\partial t} - c\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \zeta}{\partial r}\right) = c\beta_c \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) \tag{11.323}$$

For the thermally uncoupled case, the temperature is governed by (11.313), which also allows the above equation to be expressed as

$$\frac{\partial \zeta}{\partial t} - c\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \zeta}{\partial r}\right) = \frac{c\beta_c}{\kappa_T}\frac{\partial T}{\partial t} \tag{11.324}$$

11.8 Borehole Problem

Consider an infinitely long circular borehole of radius r_o in an unbounded domain, which is initially at temperature and pressure equilibrium. The problem has been solved for various stress and pressure loading conditions, as demonstrated in Sect. 7.14.1. We recognize that while the solutions in Sect. 7.14.1 are for the isothermal case, some of the solutions remain valid for the nonisothermal case. This is true if the deformation does not generate volumetric strain, such as the mode 1 solution corresponding to axial stress loading on the borehole wall. In this section we shall examine problems with temperature as well as pore pressure loading. We shall solve these problems based on both the coupled and uncoupled porothermoelastic theories.

The borehole problem involves an exterior domain that extends to infinity. For the solution to be bounded at $r \to \infty$, it is necessary that the constant $A_1(t) = 0$ in (11.306). The set of equations (11.305)–(11.309) then reduce to

$$e = \frac{\eta}{G}p + \frac{\eta_d}{G}T \tag{11.325}$$

$$u_r = \frac{\eta}{G}\frac{1}{r}\int rp(r,t)\,dr + \frac{\eta_d}{G}\frac{1}{r}\int rT(r,t)\,dr + \frac{A_2(t)}{r} \tag{11.326}$$

$$\sigma_{rr} = -2\eta\frac{1}{r^2}\int rp(r,t)\,dr - 2\eta_d\frac{1}{r^2}\int rT(r,t)\,dr - \frac{2G}{r^2}A_2(t) \tag{11.327}$$

$$\sigma_{\theta\theta} = 2\eta\frac{1}{r^2}\int rp(r,t)\,dr + 2\eta_d\frac{1}{r^2}\int rT(r,t)\,dr - 2\eta p - 2\eta_d T + \frac{2G}{r^2}A_2(t) \tag{11.328}$$

$$\sigma_{zz} = -2\eta p - 2\eta_d T \tag{11.329}$$

The diffusion equations become

$$\frac{\partial p}{\partial t} - c\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \alpha_e\frac{\partial T}{\partial t} \tag{11.330}$$

and

$$\frac{\partial T}{\partial t} - c_a \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) = \alpha_p \frac{\partial p}{\partial t} \tag{11.331}$$

11.8.1 Constant Temperature and Pore Pressure Problem

The constant temperature problem is defined as follows. At $t = 0$, the porous medium domain is at zero temperature and pore pressure everywhere. At $t = 0^+$, the temperature on the borehole wall is raised by T_1, and the borehole is otherwise stress and pore pressure free. The boundary conditions can be expressed as

$$\sigma_{rr} = 0; \quad p = 0; \quad T = T_1; \quad \text{at } r = r_o \tag{11.332}$$

Thermally Coupled Solution Performing Laplace transform to (11.330) and (11.331), we obtain the pair of modified Helmholtz equations

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \tilde{T}}{\partial r}\right) - \frac{s}{c_a}\tilde{T} = -\frac{s\alpha_p}{c_a}\tilde{p} \tag{11.333}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \tilde{p}}{\partial r}\right) - \frac{s}{c}\tilde{p} = -\frac{s\alpha_e}{c}\tilde{T} \tag{11.334}$$

General solutions of such coupled systems have been explored for the multilayered aquifer system [13, 14], as well as for the multiple porosity media [12]. Hence we can express the temperature as

$$\tilde{T} = C_1 \mathrm{K}_0(\lambda_1\sqrt{s}r) + C_2 \mathrm{K}_0(\lambda_2\sqrt{s}r) \tag{11.335}$$

where λ_1 and λ_2 are solved from the same eigenvalue system as the uniaxial strain problem, and are given as (11.251). The pore pressure is obtained from (11.333) as

$$\tilde{p} = \frac{1}{2c\alpha_p}\left[(c - c_a - C_Q)C_1 \mathrm{K}_0(\lambda_1\sqrt{s}r) + (c - c_a + C_Q)C_2 \mathrm{K}_0(\lambda_2\sqrt{s}r)\right] \tag{11.336}$$

Utilizing the boundary conditions for temperature and pore pressure in (11.332), we can solve for C_1 and C_2 from (11.335) and (11.336). Substituting these constants, the temperature and pore pressure solutions are

$$\tilde{T} = \frac{T_1}{2C_Q s}\left[\frac{(c - c_a + C_Q)\mathrm{K}_0(\lambda_1\sqrt{s}r)}{\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{(c - c_a - C_Q)\mathrm{K}_0(\lambda_2\sqrt{s}r)}{\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{11.337}$$

$$\tilde{p} = -\frac{T_1 c_a \alpha_e}{C_Q s}\left[\frac{\mathrm{K}_0(\lambda_1\sqrt{s}r)}{\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{\mathrm{K}_0(\lambda_2\sqrt{s}r)}{\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{11.338}$$

These Laplace transform solutions can be inverted to the time domain, similar to the inversion between (7.608) and (7.614). However, for the simplicity of numerical evaluation, we shall retain the Laplace transform expressions and utilize the numerical Laplace inversion for their evaluation.

To obtain the displacement and stress solutions, we carry out these two integrals

$$\int r\tilde{T}(r,s)\,dr = -\frac{T_1 r}{2C_Q s^{3/2}}\left[\frac{(c-c_a+C_Q)\mathrm{K}_1(\lambda_1\sqrt{s}r)}{\lambda_1\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{(c-c_a-C_Q)\mathrm{K}_1(\lambda_2\sqrt{s}r)}{\lambda_2\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{11.339}$$

$$\int r\tilde{p}(r,s)\,dr = \frac{T_1 c_a \alpha_e r}{C_Q s^{3/2}}\left[\frac{\mathrm{K}_1(\lambda_1\sqrt{s}r)}{\lambda_1\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{\mathrm{K}_1(\lambda_2\sqrt{s}r)}{\lambda_2\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{11.340}$$

Substituting the above into (11.327) for the radial stress, and enforcing the zero stress boundary condition at $r = r_o$, we can solve for the coefficients $\tilde{A}_2$ as

$$\tilde{A}_2 = \frac{T_1 r_o}{2C_Q G s^{3/2}}\left\{\frac{[(c-c_a+C_Q)\eta_d - 2\eta\alpha_e c_a]\mathrm{K}_1(\lambda_1\sqrt{s}r_o)}{\lambda_1\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{[(c-c_a-C_Q)\eta_d - 2\eta\alpha_e c_a]\mathrm{K}_1(\lambda_2\sqrt{s}r_o)}{\lambda_2\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right\} \tag{11.341}$$

The above expressions, (11.337)–(11.341), fully define the solutions (11.326)–(11.329) in the Laplace transform domain.

Next we examine the problem of constant pressure loading, with the boundary conditions

$$\sigma_{rr} = 0; \quad p = p_o; \quad T = 0; \quad \text{at } r = r_o \tag{11.342}$$

Before solving this problem, we observe that there exists a symmetry between the temperature and pore pressure variables in the governing equations (11.311) and (11.312) if we substitute $p \leftrightarrow T$, $c \leftrightarrow c_a$, and $\alpha_p \leftrightarrow \alpha_e$. Hence solution for p and T for this problem can be obtained by making these substitutions in (11.337) and (11.338) without repeating the solution process.

Thermally Uncoupled Solution Solution of this problem has been examined by Wang and Papamichos [77], and used for borehole stability analysis [1, 11]. The uncoupled solution can be obtained by taking the limit based on the relations in (11.261), or by directly solving the uncoupled system, which we shall attempt below.

The solution of temperature can be obtained from the uncoupled heat diffusion equation (11.313) in Laplace transform domain

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \tilde{T}}{\partial r}\right) - \frac{s}{c_a}\tilde{T} = 0 \tag{11.343}$$

Given the constant temperature boundary condition, we find

$$\frac{s\tilde{T}}{T_1} = \frac{\mathrm{K}_0(r\sqrt{s/\kappa_T})}{\mathrm{K}_0(r_o\sqrt{s/\kappa_T})} \tag{11.344}$$

Substituting the above into (11.334), we obtain

$$s\tilde{p} - c\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \tilde{p}}{\partial r}\right) = \alpha_e T_1 \frac{\mathrm{K}_0(r\sqrt{s/\kappa_T})}{\mathrm{K}_0(r_o\sqrt{s/\kappa_T})} \tag{11.345}$$

The solution of the above, subject to the pressure boundary condition, is

$$\tilde{p} = \frac{\alpha_e T_1}{(1-R_q^2)s}\left[\frac{\mathrm{K}_0(r\sqrt{s/\kappa_T})}{\mathrm{K}_0(r_o\sqrt{s/\kappa_T})} - \frac{\mathrm{K}_0(r\sqrt{s/c})}{\mathrm{K}_0(r_o\sqrt{s/c})}\right] \tag{11.346}$$

where $R_q = \sqrt{c/\kappa_T}$.

To obtain the rest of the solutions, we carry out these two integrals

$$\int r\tilde{T}(r,s)\,dr = -\frac{T_1 r\,\mathrm{K}_1(r\sqrt{s/\kappa_T})}{s\sqrt{s/\kappa_T}\,\mathrm{K}_0(r_o\sqrt{s/\kappa_T})} \tag{11.347}$$

$$\int r\tilde{p}(r,s)\,dr = -\frac{\alpha_e T_1 r}{(1-R_q^2)s}\left[\frac{\mathrm{K}_1(r\sqrt{s/\kappa_T})}{\sqrt{s/\kappa_T}\,\mathrm{K}_0(r_o\sqrt{s/\kappa_T})} - \frac{\mathrm{K}_1(r\sqrt{s/c})}{\sqrt{s/c}\,\mathrm{K}_0(r_o\sqrt{s/c})}\right] \tag{11.348}$$

Substituting the above into (11.327) for the radial stress, and enforcing the zero stress boundary condition at $r = r_o$, we can solve for the coefficients $\tilde{A}_2$ as

$$\tilde{A}_2 = \frac{T_1 r_o}{(1-R_q^2)Gs}\left\{\frac{[\eta\alpha_e + \eta_d(1-R_q^2)]\,\mathrm{K}_1(r_o\sqrt{s/\kappa_T})}{\sqrt{s/\kappa_T}\,\mathrm{K}_0(r_o\sqrt{s/\kappa_T})} - \frac{\eta\alpha_e\,\mathrm{K}_1(r_o\sqrt{s/c})}{\sqrt{s/c}\,\mathrm{K}_0(r_o\sqrt{s/c})}\right\} \tag{11.349}$$

The above expressions, (11.344), and (11.346)–(11.349), fully define the solutions (11.326)–(11.329) in the Laplace transform domain.

For numerical result, we again use the case of a rock salt. Assume a borehole with radius $r_o = 10\,\text{cm}$, and an temperature increment of $100\,^\circ\text{K}$. In Fig. 11.7 we plot the responses in terms of temperature, pressure, and stresses. First, Fig. 11.7a presents the temperature profile versus radial distance for two times borehole radius into the formation at various times. We observe that although as $t \to \infty$ the entire formation should be at the boundary temperature, that temperature will take a very long time to establish.

Next, the pore pressure response in presented in Fig. 11.7b. With the temperature rising near the wall, a large pore pressure is developed, which however must be drained to zero at the wall. Hence we observe a sharp peak that spread out with time with its maximum value migrate into the formation. As $t \to \infty$, however, all pore pressure must dissipate to zero.

Figure 11.7c plots the negative radial stress $-\sigma_{rr}$ versus radial distance. We observe that the radial stress is in compression, with its value increases with time as the temperature rises. At the boundary the stress must return to zero due to the boundary condition. As $t \to \infty$, it has an asymptotic profile of

$$\sigma_{rr}(r, \infty) = -\eta_d T_1 \left(1 - \frac{r_o^2}{r^2}\right) \tag{11.350}$$

which reaches a constant value of $\sigma_{rr} = -\eta_d T_1$ as $r \to \infty$.

The negative circumferential stress $-\sigma_{\theta\theta}$ is plotted as Fig. 11.9d. The heating at the wall causes a constant compressive hoop stress at the wall as

$$\sigma_{\theta\theta}(r_o, t) = -2\eta_d T_1 \tag{11.351}$$

A distance ahead, however, the compression drops to tension. As time progresses, the compression region grows. As $t \to \infty$, it develops into the asymptotic profile

$$\sigma_{\theta\theta}(r, \infty) = -\eta_d T_1 \left(1 + \frac{r_o^2}{r^2}\right) \tag{11.352}$$

Finally, we examine the Terzaghi effective stresses. In Fig. 11.7e, f the radial and circumferential effective stress, σ'_{rr} and $\sigma'_{\theta\theta}$, are evaluated and plotted. For the effective radial stress, we observe that at small times it is dominated by the near wellbore pore pressure response, such that there exists a tensile stress region with a sharp peak. As the pore pressure magnitude diminishes toward the inside of the formation, the tensile stress also decreases and in fact becomes compression at larger distance. As time progresses, the tensile region diminishes, and at large time the entire region is in compression. We can also examine the negative effective circumferential stress $-\sigma'_{\theta\theta}$ in Fig. 11.7f. In this case, as the magnitude of the circumferential stress is significantly larger than the pore pressure in the near wellbore region, its behavior is similar to that of $-\sigma_{\theta\theta}$.

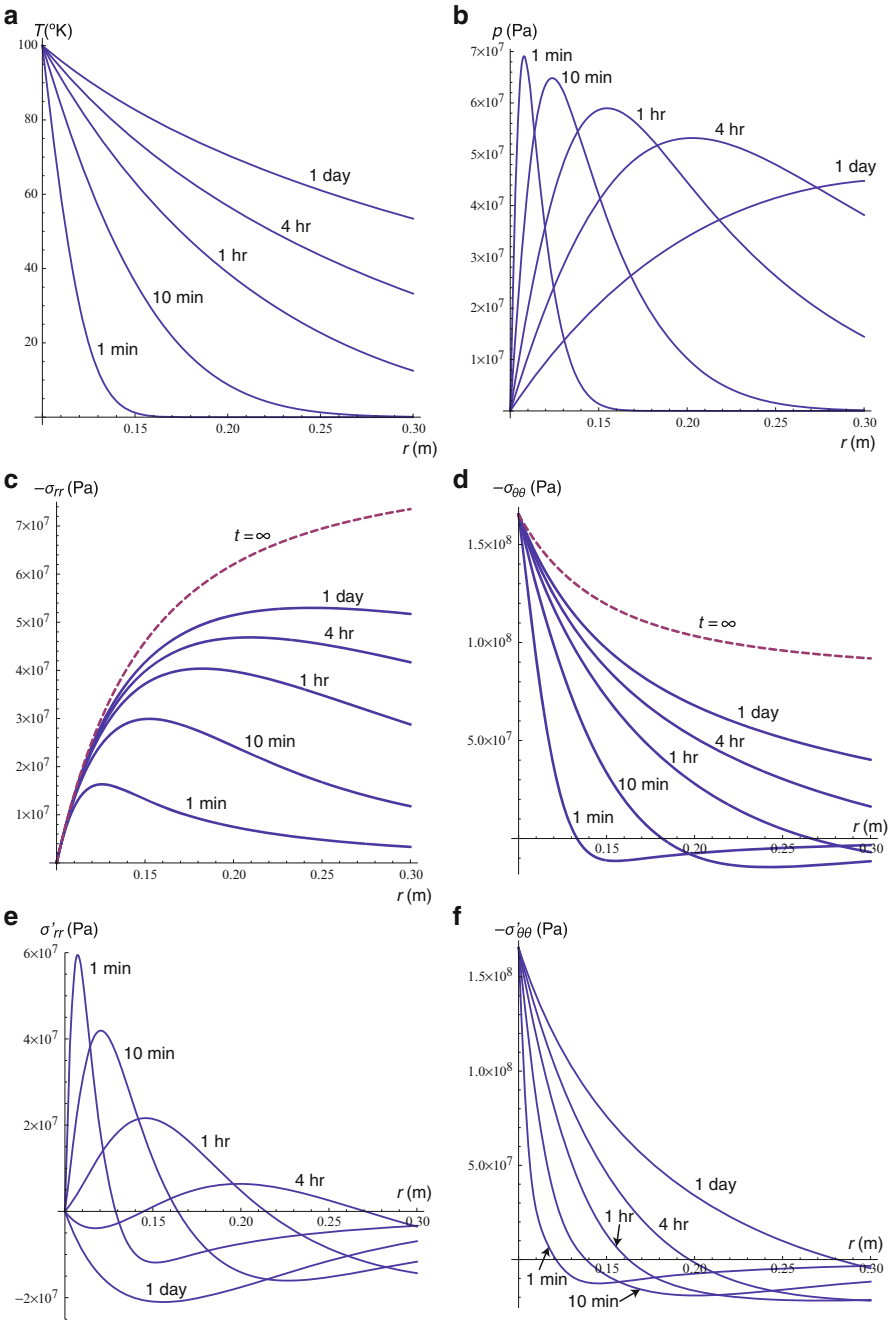


Fig. 11.7 Porothermoelastic solution of borehole problem, step rise in temperature: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial stress $-\sigma_{rr}$, (**d**) circumferential stress $-\sigma_{\theta\theta}$, (**e**) Terzaghi effective radial stress σ'_{rr}, and (**f**) Terzaghi effective circumferential stress $-\sigma'_{\theta\theta}$

11.8.2 Constant Heat and Fluid Flux Problem

The boundary conditions for the constant heat flux problem are, at $t = 0^+$,

$$\sigma_{rr} = 0; \quad p = 0; \quad \frac{\partial T}{\partial r} = -\frac{q_o}{k_T}; \quad \text{at } r = r_o \tag{11.353}$$

Thermally Coupled Solution The general solutions of temperature and pore pressure have been obtained as (11.335) and (11.336). Utilizing the boundary condition (11.353), we can solve for C_1 and C_2. Substituting these constants, the temperature and pore pressure solutions are

$$\begin{aligned}\tilde{T} = \frac{\sqrt{2c\,c_a}q_o}{k_T s^{3/2} D}\big[&(c - c_a + C_Q)\,\mathrm{K}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{K}_0(\lambda_1\sqrt{s}r) \\ &-(c - c_a - C_Q)\,\mathrm{K}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{K}_0(\lambda_2\sqrt{s}r)\big]\end{aligned} \tag{11.354}$$

$$\begin{aligned}\tilde{p} = -\frac{2^{3/2}c^{1/2}c_a^{3/2}q_o\alpha_e}{k_T s^{3/2} D}\big[&\mathrm{K}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{K}_0(\lambda_1\sqrt{s}r) \\ &-\mathrm{K}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{K}_0(\lambda_2\sqrt{s}r)\big]\end{aligned} \tag{11.355}$$

where

$$\begin{aligned}D = &(c - c_a + C_Q)(c + c_a + C_Q)^{1/2}\,\mathrm{K}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{K}_1(\lambda_1\sqrt{s}r_o) \\ &-(c - c_a - C_Q)(c + c_a - C_Q)^{1/2}\,\mathrm{K}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{K}_1(\lambda_2\sqrt{s}r_o)\end{aligned} \tag{11.356}$$

The integrals are

$$\begin{aligned}\int r\tilde{T}(r,s)\,dr = \frac{\sqrt{2c\,c_a}q_o r}{k_T\lambda_1\lambda_2 s^2 D}\big[&(c - c_a - C_Q)\lambda_1\,\mathrm{K}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{K}_1(\lambda_2\sqrt{s}r) \\ &-(c - c_a + C_Q)\lambda_2\,\mathrm{K}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{K}_1(\lambda_1\sqrt{s}r)\big]\end{aligned} \tag{11.357}$$

$$\begin{aligned}\int r\tilde{p}(r,s)\,dr = \frac{2^{3/2}c^{1/2}c_a^{3/2}q_o\alpha_e r}{k_T\lambda_1\lambda_2 s^2 D}\big[&\lambda_2\,\mathrm{K}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{K}_1(\lambda_1\sqrt{s}r) \\ &-\lambda_1\,\mathrm{K}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{K}_1(\lambda_2\sqrt{s}r)\big]\end{aligned} \tag{11.358}$$

The constant $\tilde{A}_2$ can be solved from the radial stress boundary condition as

$$\begin{aligned}\tilde{A}_2 = &\frac{\sqrt{2c\,c_a}\,q_o r_o}{k_T G \lambda_1 \lambda_2 s^2 D} \\ &\times \left\{[(c - c_a + C_Q)\eta_d - 2\eta\alpha_e c_a]\lambda_2 \mathrm{K}_0(\lambda_2\sqrt{s}\,r_o)\,\mathrm{K}_1(\lambda_1\sqrt{s}\,r_o)\right. \\ &\left. -[(c - c_a - C_Q)\eta_d - 2\eta\alpha_e c_a]\lambda_1 \mathrm{K}_0(\lambda_1\sqrt{s}\,r_o)\,\mathrm{K}_1(\lambda_2\sqrt{s}\,r_o)\right\}\end{aligned} \tag{11.359}$$

The above expressions fully define the solutions in the Laplace transform domain.

The problem of constant fluid flux can be solved by invoking the symmetry condition between T and p in the solution system.

Thermally Uncoupled Solution Solution of constant heat flux on a borehole wall based on the uncoupled theory has been provided by McTigue [48]. Here we derive the solution following the similar procedure as the constant temperature problem presented above.

The solution of temperature from (11.343) with the current boundary condition is

$$\frac{s k_T \tilde{T}}{q_o r_o} = \frac{\mathrm{K}_0(r\sqrt{s/\kappa_T})}{r_o\sqrt{s/\kappa_T}\,\mathrm{K}_1(r_o\sqrt{s/\kappa_T})} \tag{11.360}$$

Introducing the above in (11.334), and solve it with the pore pressure boundary condition, we find

$$\tilde{p} = \frac{q_o \alpha_e \kappa_T^{1/2}}{(1 - R_q^2)s^{3/2}k_T}\left[\frac{\mathrm{K}_0(r\sqrt{s/\kappa_T})}{\mathrm{K}_1(r_o\sqrt{s/\kappa_T})} - \frac{\mathrm{K}_0(r_o\sqrt{s/\kappa_T})\,\mathrm{K}_0(r\sqrt{s/c})}{\mathrm{K}_1(r_o\sqrt{s/\kappa_T})\,\mathrm{K}_0(r_o\sqrt{s/c})}\right] \tag{11.361}$$

To obtain the rest of the solutions, we carry out these two integrations

$$\int r\tilde{T}(r,s)\,dr = -\frac{q_o r \kappa_T \mathrm{K}_1(r\sqrt{s/\kappa_T})}{s^2 k_T \mathrm{K}_1(r_o\sqrt{s/\kappa_T})} \tag{11.362}$$

$$\begin{aligned}\int r\tilde{p}(r,s)\,dr = &\frac{q_o \alpha_e \kappa_T^{1/2} r}{(1 - R_q^2)s^{3/2}k_T}\left[\frac{\mathrm{K}_1(r\sqrt{s/\kappa_T})}{\sqrt{s/\kappa_T}\,\mathrm{K}_1(r_o\sqrt{s/\kappa_T})}\right. \\ &\left. - \frac{\mathrm{K}_0(r_o\sqrt{s/\kappa_T})\,\mathrm{K}_1(r\sqrt{s/c})}{\sqrt{s/c}\,\mathrm{K}_1(r_o\sqrt{s/\kappa_T})\,\mathrm{K}_0(r_o\sqrt{s/c})}\right]\end{aligned} \tag{11.363}$$

Using the radial stress boundary condition, we can solve for the coefficients $\tilde{A}_2$ as

$$\tilde{A}_2 = \frac{q_o \kappa_T^{1/2} r_o}{(1-R_q^2) G s^{3/2} k_T} \left[\frac{\eta \alpha_e + \eta_d (1-R_q^2)}{\sqrt{s/\kappa_T}} - \frac{\eta \alpha_e \mathrm{K}_1(r_o\sqrt{s/c})\,\mathrm{K}_0(r_o\sqrt{s/\kappa_T})}{\sqrt{s/c}\,\mathrm{K}_0(r_o\sqrt{s/c})\,\mathrm{K}_1(r_o\sqrt{s/\kappa_T})} \right] \tag{11.364}$$

The above expressions then fully define the solutions (11.326)–(11.329) in the Laplace transform domain.

For numerical result, we use the case of a rock salt in Table 11.3. Assuming a borehole with radius $r_o = 10\,\mathrm{cm}$, and a heat flux of $1{,}000\,\mathrm{W/m^2}$, in Fig. 11.8 we plot the responses in terms of temperature, pressure, and stresses, similar to those in Fig. 11.7 for the constant temperature case. We notice that for constant flux, the temperature, as well as pressure and stress responses, are small at small time. At large time, the temperature and other responses are unbounded.

11.9 Cylinder Problem

The cylinder problem is an interior domain problem and has been solved for the isothermal case in Sect. 7.13. Here we shall examine the nonisothermal case, for both the thermally coupled and uncoupled problems. The initial condition is given as $T = 0$ and $p = 0$. The boundary conditions for the problem of a step rise in temperature on the cylinder surface are as follows:

$$\sigma_{rr} = 0; \quad p = 0; \quad T = T_1; \quad \text{at } r = r_o \tag{11.365}$$

Thermally Coupled Solution By utilizing the pore pressure formulation, it is necessary to set the constant $A_2(t) = 0$ to avoid unbounded solution at $r = 0$. The solutions (11.306), (11.307), and (11.308) then become

$$u_r = \frac{\eta}{G}\frac{1}{r}\int rp(r,t)\,dr + \frac{\eta_d}{G}\frac{1}{r}\int rT(r,t)\,dr + A_1(t)\,r \tag{11.366}$$

$$\sigma_{rr} = -2\eta\frac{1}{r^2}\int rp(r,t)\,dr - 2\eta_d\frac{1}{r^2}\int rT(r,t)\,dr + \frac{2G}{1-2\nu}A_1(t) \tag{11.367}$$

$$\sigma_{\theta\theta} = 2\eta\frac{1}{r^2}\int rp(r,t)\,dr + 2\eta_d\frac{1}{r^2}\int rT(r,t)\,dr - 2\eta p - 2\eta_d T + \frac{2G}{1-2\nu}A_1(t) \tag{11.368}$$

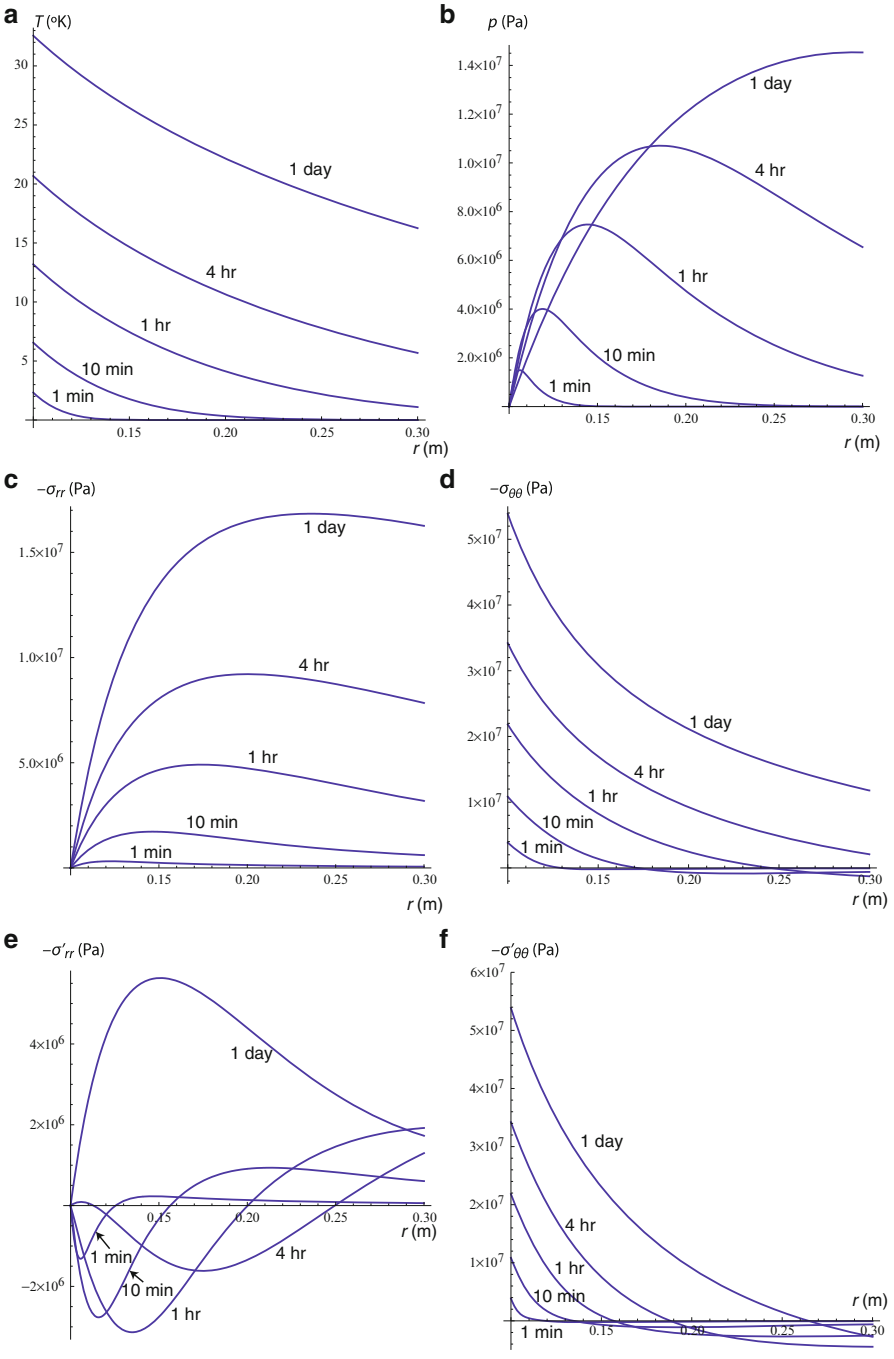


Fig. 11.8 Porothermoelastic solution of borehole problem, constant heat flux: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial stress $-\sigma_{rr}$, (**d**) circumferential stress $-\sigma_{\theta\theta}$, (**e**) Terzaghi effective radial stress $-\sigma'_{rr}$, and (**f**) Terzaghi effective circumferential stress $-\sigma'_{\theta\theta}$

while (11.305) and (11.309) remain the same. The two diffusion equations (11.311) and (11.312) are also unchanged. Performing Laplace transform, they become

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \tilde{T}}{\partial r}\right) - \frac{s}{c_a}\tilde{T} = -\frac{s\alpha_p}{c_a}\tilde{p} + \frac{2s\alpha_d}{c_a S_a}\tilde{A}_1(s) \tag{11.369}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \tilde{p}}{\partial r}\right) - \frac{s}{c}\tilde{p} = -\frac{s\alpha_e}{c}\tilde{T} + \frac{2s\alpha}{cS}\tilde{A}_1(s) \tag{11.370}$$

Solutions of the above pair of equations are

$$\tilde{T} = C_1 \mathrm{I}_0(\lambda_1\sqrt{s}r) + C_2 \mathrm{I}_0(\lambda_2\sqrt{s}r) - \frac{2(\alpha_d S + \alpha\alpha_p S_a)}{SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1(s) \tag{11.371}$$

$$\begin{aligned}\tilde{p} = {} & \frac{C_1(c-c_a-C_Q)\mathrm{I}_0(\lambda_1\sqrt{s}r)}{2c\alpha_p} + \frac{C_2(c-c_a+C_Q)\mathrm{I}_0(\lambda_2\sqrt{s}r)}{2c\alpha_p} \\ & - \frac{2(\alpha S_a + \alpha_d\alpha_e S)}{SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1(s)\end{aligned} \tag{11.372}$$

Again, λ_1 and λ_2 are eigenvalues defined in (11.251), with parameter C_Q given in (11.252).

Utilizing the temperature and pore pressure boundary condition in (11.365), we can solve for C_1 and C_2 as

$$\begin{aligned}C_1 = {} & \frac{1}{2sC_Q SS_a(1-\alpha_p\alpha_e)\mathrm{I}_0(\lambda_1\sqrt{s}r_o)} \\ & \times\{2s[(c-c_a+C_Q)(S\alpha_d + S_a\alpha\alpha_p) - 2c\alpha_p(S_a\alpha + S\alpha_d\alpha_p)]\tilde{A}_1(s) \\ & + SS_a T_1(c-c_a+C_Q)(1-\alpha_p\alpha_e)\}\end{aligned} \tag{11.373}$$

$$\begin{aligned}C_2 = {} & -\frac{1}{2sC_Q SS_a(1-\alpha_p\alpha_e)\mathrm{I}_0(\lambda_2\sqrt{s}r_o)} \\ & \times\{2s[(c-c_a-C_Q)(S\alpha_d + S_a\alpha\alpha_p) - 2c\alpha_p(S_a\alpha + S\alpha_d\alpha_e)]\tilde{A}_1(s) \\ & + SS_a T_1(c-c_a-C_Q)(1-\alpha_p\alpha_e)\}\end{aligned} \tag{11.374}$$

The integrals are

$$\begin{aligned}\int r\tilde{T}(r,s)\,dr = {} & \frac{C_1 r \mathrm{I}_1(\lambda_1\sqrt{s}r)}{\lambda_1\sqrt{s}} + \frac{C_2 r \mathrm{I}_1(\lambda_2\sqrt{s}r)}{\lambda_2\sqrt{s}} \\ & - \frac{r^2(\alpha_d S + \alpha\alpha_p S_a)}{SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1(s)\end{aligned} \tag{11.375}$$

$$\int r\tilde{p}(r,s)\,dr = \frac{(c-c_a-C_Q)C_1 r\mathrm{I}_1(\lambda_1\sqrt{s}r)}{2c\alpha_p\lambda_1\sqrt{s}} + \frac{(c-c_a+C_Q)C_2 r\mathrm{I}_1(\lambda_2\sqrt{s}r)}{2c\alpha_p\lambda_2\sqrt{s}} - \frac{r^2(\alpha S_a+\alpha_d\alpha_e S)}{SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1(s) \tag{11.376}$$

For the above solutions, we still need to resolve the constant $\tilde{A}_1(s)$. This is accomplished by utilizing the radial stress boundary condition, and we obtain

$$\begin{aligned}\tilde{A}_1(s) = {} & \frac{SS_aT_1(1-2\nu)(1-\alpha_p\alpha_e)}{D} \\ & \times\{\lambda_1(c-c_a-C_Q)\left[\eta(c-c_a+C_Q)+2c\alpha_p\eta_d\right] \\ & \times\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\mathrm{I}_1(\lambda_2\sqrt{s}r_o) \\ & -\lambda_2(c-c_a+C_Q)\left[\eta(c-c_a-C_Q)+2c\alpha_p\eta_d\right] \\ & \times\mathrm{I}_0(\lambda_2\sqrt{s}r_o)\mathrm{I}_1(\lambda_1\sqrt{s}r_o)\}\end{aligned} \tag{11.377}$$

where

$$\begin{aligned}D = {} & 2s(1-2\nu)\left\{\lambda_2\left[(c-c_a+C_Q)(S\alpha_d+S_a\alpha\alpha_p)-2c\alpha_p(S_a\alpha+S\alpha_d\alpha_e)\right]\right. \\ & \times\left[\eta(c-c_a-C_Q)+2c\alpha_p\eta_d\right]\mathrm{I}_0(\lambda_2\sqrt{s}r_o)\mathrm{I}_1(\lambda_1\sqrt{s}r_o) \\ & -\lambda_1\left[(c-c_a-C_Q)(S\alpha_d+S_a\alpha\alpha_p)-2c\alpha_p(S_a\alpha+S\alpha_d\alpha_e)\right] \\ & \left.\times\left[\eta(c-c_a+C_Q)+2c\alpha_p\eta_d\right]\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\mathrm{I}_1(\lambda_2\sqrt{s}r_o)\right\} \\ & -4cC_Q\alpha_p r_o\lambda_1\lambda_2 s^{3/2}\left\{GSS_a(1-\alpha_p\alpha_e)\right. \\ & \left.+(1-2\nu)\left[S\alpha_d(\alpha_e\eta+\eta_d)+S_a\alpha(\eta+\alpha_p\eta_d)\right]\right\} \\ & \times\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\mathrm{I}_0(\lambda_2\sqrt{s}r_o)\end{aligned} \tag{11.378}$$

With the back substitution, the entire solutions are defined.

Thermally Uncoupled Solution The problem of an anisotropic hollow cylinders subjected to a temperature rise has been solved by Kanj and Abousleiman [36], based on the thermally uncoupled formulation. Here we shall examine the problem of an isotropic, solid cylinder subject to a step temperature rise on the surface. The solution of such problems can be obtained by taking the limit of the coupled solution presented above. Here, however, we shall directly solve it using the variation in fluid content representation (11.314)–(11.319).

For interior domain problems, for the solution to be bounded at $r = 0$, it is necessary that we set $B_2(t) = 0$ in those representations, which reduce to

$$u_r = \frac{\eta}{GS}\frac{1}{r}\int r\zeta(r,t)\,dr + \frac{\eta\beta_u}{GSB}\frac{1}{r}\int rT(r,t)\,dr + B_1(t)\,r \tag{11.379}$$

$$\sigma_{rr} = -\frac{2\eta}{S}\frac{1}{r^2}\int r\zeta(r,t)\,dr - \frac{2\eta\beta_u}{SB}\frac{1}{r^2}\int rT(r,t)\,dr + \frac{2G}{1-2\nu_u}B_1(t) \tag{11.380}$$

$$\sigma_{\theta\theta} = \frac{2\eta}{S}\left[\frac{1}{r^2}\int r\zeta(r,t)\,dr - \zeta\right] + \frac{2\eta\beta_u}{SB}\left[\frac{1}{r^2}\int rT(r,t)\,dr - T\right] + \frac{2G}{1-2\nu_u}B_1(t) \tag{11.381}$$

The other expressions, (11.314), (11.318), and (11.319), remain the same.

The temperature is solved from the uncoupled heat diffusion equation (11.343), and the variation in fluid content from (11.324), which expressed in Laplace transform domain is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\tilde{\zeta}}{\partial r}\right) - \frac{s}{c}\tilde{\zeta} = -\frac{\beta_c s}{\kappa_T}\tilde{T} \tag{11.382}$$

With the boundary condition in (11.365), the temperature is easily solved

$$\frac{s\tilde{T}}{T_1} = \frac{\mathrm{I}_0(r\sqrt{s/\kappa_T})}{\mathrm{I}_0(r_o\sqrt{s/\kappa_T})} \tag{11.383}$$

Substituting the above to the right hand side of (11.382), we find

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\tilde{\zeta}}{\partial r}\right) - \frac{s}{c}\tilde{\zeta} = -\frac{\beta_c T_1}{\kappa_T}\frac{\mathrm{I}_0(r\sqrt{s/\kappa_T})}{\mathrm{I}_0(r_o\sqrt{s/\kappa_T})} \tag{11.384}$$

The general and particular solution of the above is

$$\tilde{\zeta} = \frac{R_q^2\beta_c T_1}{(1-R_q^2)s}\frac{\mathrm{I}_0(r\sqrt{s/\kappa_T})}{\mathrm{I}_0(r_o\sqrt{s/\kappa_T})} + D_1\,\mathrm{I}_0(r\sqrt{s/c}) \tag{11.385}$$

where $R_q = \sqrt{c/\kappa_T}$ is the square root of the ratio of poroelastic to thermoelastic diffusivity, as defined in (11.264).

To obtain the rest of the solutions, we first carry out these two integrations

$$\int r\tilde{T}(r,s)\,dr = \frac{T_1 r \mathrm{I}_1(r\sqrt{s/\kappa_T})}{s\sqrt{s/\kappa_T}\,\mathrm{I}_0(r_o\sqrt{s/\kappa_T})} \tag{11.386}$$

$$\int r\tilde{p}(r,s)\,dr = \frac{T_1 R_q^2 \beta_c r \mathrm{I}_1(r\sqrt{s/\kappa_T})}{(1-R_q^2)s\sqrt{s/\kappa_T}\,\mathrm{I}_0(r_o\sqrt{s/\kappa_T})} + D_1\frac{r\mathrm{I}_1(r\sqrt{s/c})}{\sqrt{s/c}} \tag{11.387}$$

Substituting the above into the radial stress and pressure solution, (11.316) and (11.319), and utilizing the boundary conditions in (11.448), we can solve for the two coefficients D_1 and $\tilde{B}_1$ as

$$D_1 = \frac{T_1}{R_q(1-R_q^2)D}\left\{-3\beta_c r_o\sqrt{s/\kappa_T}\,\mathrm{I}_0(r_o\sqrt{s/\kappa_T}) + 4\eta(1+\nu_u)[BR_q^2\beta_c + (1-R_q^2)\beta_u]\mathrm{I}_1(r_o\sqrt{s/\kappa_T})\right\} \tag{11.388}$$

$$\tilde{B}_1 = \frac{T_1(1+\nu_u)(1-2\nu_u)}{(1-\nu_u)R_q(1-R_q^2)D}\left\{-B\beta_c\sqrt{s/\kappa_T}\,\mathrm{I}_1(r_o\sqrt{s/c})\,\mathrm{I}_0(r_o\sqrt{s/\kappa_T}) + [BR_q^2\beta_c + (1-R_q^2)\beta_u]\mathrm{I}_0(r_o\sqrt{s/c})\,\mathrm{I}_1(r_o\sqrt{s/\kappa_T})\right\} \tag{11.389}$$

where

$$D = s\mathrm{I}_0(r_o\sqrt{s/\kappa_T})\left[3r_o\sqrt{s/c}\,\mathrm{I}_0(r_o\sqrt{s/c}) - 4\eta B(1+\nu_u)\mathrm{I}_1(r_o\sqrt{s/c})\right] \tag{11.390}$$

The above expressions, (11.383), and (11.385)–(11.390), fully define the solutions.

For the numerical result, we shall examine the case of a rock salt. The poroelastic and thermoelastic material constants are given in Tables 3.1, 3.2, and 11.3. Assume a cylinder with radius of $r_o = 10$ cm. At $t = 0^+$, its surface temperature is instantly raise by 100 °K. In Fig. 11.9a, we plot the temperature profile along the radial distance at various times based on (11.383). The temperature is initially zero, and increases with time as heat penetrates from the surface. At $t = 30$ min, the entire cylinder is almost at a constant temperature of 100 °K.

In Fig. 11.9b we plot the pore pressure profile at various times. At small times, the temperature penetrates a surface layer and the fluid is heated to create a positive pore pressure. At the surface, the pore pressure must return to zero due to the boundary condition; hence a peak profile is developed. The pore pressure decreases toward the inner core, and in fact becomes negative, which is due to the elastic effect—as the solid in the outer region expands, it stretches the inner part into tension, thus creating a negative pore pressure. As time progresses, the pore pressure peak declines due to fluid diffusion, and the negative pore pressure increases at the center. At even larger time, the inner core becomes heated and the pore pressure rises to positive value and

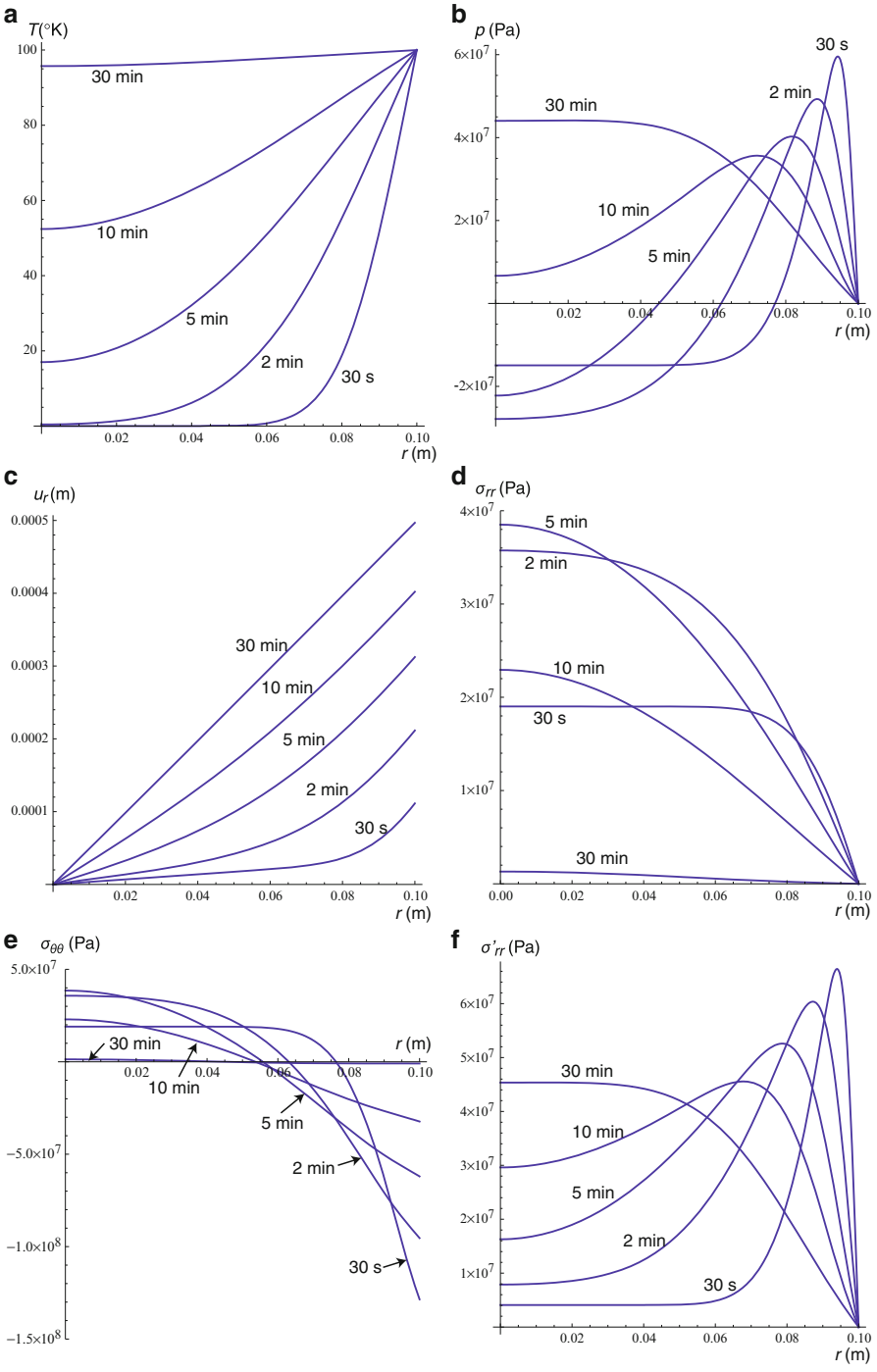


Fig. 11.9 Porothermoelastic solution for cylinder problem: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial displacement u_r, (**d**) radial stress σ_{rr}, (**e**) circumferential stress $\sigma_{\theta\theta}$, and (**f**) Terzaghi effective radial stress σ'_{rr}

the peak is located at the center. The solution is presented only until $t = 30$ min, at which time the entire cylinder is heated based on the relatively large heat diffusivity, but the pore pressure is not yet dissipated due to the relatively small fluid diffusivity, as the diffusivity ratio for rock salt is given as $R_q^2 = c/\kappa_T = 0.048$. As time further increases, the entire pore pressure must be dissipated and return to zero everywhere. Hence at the center of the cylinder, the pore pressure starts from the initial zero to become negative, and then positive, and eventually return to zero again.

Figure 11.9c presents the time history of radial displacement, which increases with time. At large time, the cylinder is in uniform expansion and we observe the asymptotic solution

$$u_r(r, \infty) = \frac{(1+\nu)\beta_d T_1}{3} r \tag{11.391}$$

as expected.

The radial stress σ_{rr} is presented in Fig. 11.9d. We observe that as the outer layer is heated and expands, the inner core is stretched to become tensile. The tensile stress increases with time, and then decreases. It must return to zero at large time. In the present case, it has nearly disappeared at 30 min.

The circumferential stress $\sigma_{\theta\theta}$ is plotted as Fig. 11.9e. The initial heating causes it to be in compression near the surface layer, and tension inside. As time increases, both the compression and tension decreases, and eventually it must become stress free at large time.

Finally, we examine the Terzaghi effective stress which is related to the failure of the material. In Fig. 11.9f the effective stress σ'_{rr} is evaluated and plotted. We find that the entire cylinder is in effective tension, with large peak inside the cylinder. If the tensile stress exceeds the tensile strength of the material, a whole portion can fail. Indeed, the rapid heating of saturated cylindrical cores can cause it to burst.

11.10 Spherical Symmetry

Similar to the axial symmetry problems, we can construct the general solution for problems possessing spherical symmetry, which are function of (R, t) only. We shall present below two formulations.

Pore Pressure Representation Following similar procedure in Sect. 7.9, we obtain the solution for volumetric strain as

$$e = \frac{\eta}{G} p + \frac{\eta_d}{G} T + 3A_1(t) \tag{11.392}$$

and the radial displacement becomes

$$u_R = \frac{\eta}{G}\frac{1}{R^2}\int R^2 p(R,t)\, dR + \frac{\eta_d}{G}\frac{1}{R^2}\int R^2 T(R,t)\, dR + A_1(t)R + \frac{A_2(t)}{R^2} \tag{11.393}$$

The stresses are

$$\sigma_{RR} = -4\eta\frac{1}{R^3}\int R^2 p(R,t)\,dR - 4\eta_d\frac{1}{R^3}\int R^2 T(R,t)\,dR + \frac{2G(1+\nu)}{1-2\nu}A_1(t) - \frac{4G}{R^3}A_2(t) \tag{11.394}$$

$$\sigma_{\varphi\varphi} = 2\eta\frac{1}{R^3}\int R^2 p(R,t)\,dR + 2\eta_d\frac{1}{R^3}\int R^2 T(R,t)\,dR - 2\eta p - 2\eta_d T + \frac{2G(1+\nu)}{1-2\nu}A_1(t) + \frac{2G}{R^3}A_2(t) \tag{11.395}$$

From the above, it is clear that

$$\sigma_{RR} + 2\sigma_{\varphi\varphi} + 4\eta p + 4\eta_d T = \frac{6G(1+\nu)}{1-2\nu}A_1(t) \tag{11.396}$$

The temperature and pore pressure can be solved from these diffusion equations

$$\frac{\partial T}{\partial t} - c_a\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial T}{\partial R}\right) = \alpha_p\frac{\partial p}{\partial t} - \frac{3\alpha_d}{S_a}\frac{dA_1(t)}{dt} \tag{11.397}$$

and

$$\frac{\partial p}{\partial t} - c\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial p}{\partial R}\right) = \alpha_e\frac{\partial T}{\partial t} - \frac{3\alpha}{S}\frac{dA_1(t)}{dt} \tag{11.398}$$

For the thermally uncoupled theory, the temperature Eq. (11.397) simplifies to

$$\frac{\partial T}{\partial t} - \kappa_T\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial T}{\partial R}\right) = 0 \tag{11.399}$$

and the pore pressure diffusion equation (11.398) remains the same.

Variation in Fluid Content Representation In this formulation, the volumetric strain is given by

$$e = \frac{\eta}{GS}\zeta + \frac{\eta\beta_u}{GSB}T + 3B_1(t) \tag{11.400}$$

an radial displacement by

$$u_R = \frac{\eta}{GS}\frac{1}{R^2}\int R^2\zeta(R,t)\,dR + \frac{\eta\beta_u}{GSB}\frac{1}{R^2}\int R^2 T(R,t)\,dR + B_1(t)R + \frac{B_2(t)}{R^2} \tag{11.401}$$

The stresses and pressure are

$$\sigma_{RR} = -\frac{4\eta}{S}\frac{1}{R^3}\int R^2\zeta(R,t)\,dR - \frac{4\eta\beta_u}{SB}\frac{1}{R^3}\int R^2 T(R,t)\,dR + \frac{2G(1+\nu_u)}{1-2\nu_u}B_1(t) - \frac{4G}{R^3}B_2(t) \tag{11.402}$$

$$\sigma_{\varphi\varphi} = \frac{2\eta}{S}\frac{1}{R^3}\int R^2\zeta(R,t)\,dR + \frac{2\eta\beta_u}{SB}\frac{1}{R^3}\int R^2 T(R,t)\,dR - \frac{2\eta}{S}\zeta - \frac{2\eta\beta_u}{SB}T + \frac{2G(1+\nu_u)}{1-2\nu_u}B_1(t) + \frac{2G}{R^3}B_2(t) \tag{11.403}$$

$$p = \frac{\zeta}{S} + \alpha_e T - 2\alpha M B_1(t) \tag{11.404}$$

Temperature can be solved from

$$\frac{\partial T}{\partial t} - c_b\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial T}{\partial R}\right) = M\alpha_t\frac{\partial\zeta}{\partial t} - \frac{2\alpha_u}{S_b}B_1(t) \tag{11.405}$$

and the variation in fluid content from

$$\frac{\partial\zeta}{\partial t} - c\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial\zeta}{\partial R}\right) = c\beta_c\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial T}{\partial R}\right) \tag{11.406}$$

For the thermally uncoupled case, the temperature is governed by (11.399), which also allows (11.406) to be expressed as

$$\frac{\partial\zeta}{\partial t} - c\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial\zeta}{\partial R}\right) = \frac{c\beta_c}{\kappa_T}\frac{\partial T}{\partial t} \tag{11.407}$$

11.11 Cavity Problem

As the cavity problem is an exterior domain problem, the set of representations (11.392), (11.393), (11.394), and (11.395) are simplified by setting $A_1(t) = 0$, and we obtain

$$e = \frac{\eta}{G}p + \frac{\eta_d}{G}T \tag{11.408}$$

$$u_R = \frac{\eta}{G}\frac{1}{R^2}\int R^2 p(R,t)\,dR + \frac{\eta_d}{G}\frac{1}{R^2}\int R^2 T(R,t)\,dR + \frac{A_2(t)}{R^2} \tag{11.409}$$

$$\sigma_{RR} = -4\eta\frac{1}{R^3}\int R^2 p(R,t)\,dR - 4\eta_d\frac{1}{R^3}\int R^2 T(R,t)\,dR - \frac{4G}{R^3}A_2(t) \tag{11.410}$$

$$\sigma_{\varphi\varphi} = 2\eta \frac{1}{R^3} \int R^2 p(R,t)\, dR + 2\eta_d \frac{1}{R^3} \int R^2 T(R,t)\, dR$$
$$-2\eta p - 2\eta_d T + \frac{2G}{R^3} A_2(t) \tag{11.411}$$

11.11.1 Constant Temperature Problem

We shall solve the problem of a step rise in temperature on the cavity wall, given by the following boundary conditions:

$$\sigma_{RR} = 0; \quad p = 0; \quad T = T_1; \quad \text{at } R = R_o \tag{11.412}$$

As commented in Sect. 11.8.1, the problem of step rise in pore pressure can be obtained by invoking the symmetry between the variables T and p; hence is not explicitly solved here. In the following, we shall seek the solution for both the thermally coupled and uncoupled problems.

Thermally Coupled Solution The temperature and pore pressure are to be solved from the set of coupled diffusion equations

$$\frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial \tilde{T}}{\partial R} \right) - \frac{s}{c_a} \tilde{T} = -\frac{s\alpha_p}{c_a} \tilde{p} \tag{11.413}$$

and

$$\frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial \tilde{p}}{\partial R} \right) - \frac{s}{c} \tilde{p} = -\frac{s\alpha_e}{c} \tilde{T} \tag{11.414}$$

The general solutions of the above are

$$\tilde{T} = C_1 \frac{e^{-\lambda_1 \sqrt{s} R}}{R} + C_2 \frac{e^{-\lambda_2 \sqrt{s} R}}{R} \tag{11.415}$$

$$\tilde{p} = \frac{C_1 (c - c_a - C_Q)}{2c\alpha_p} \frac{e^{-\lambda_1 \sqrt{s} R}}{R} + \frac{C_2 (c - c_a + C_Q)}{2c\alpha_p} \frac{e^{-\lambda_2 \sqrt{s} R}}{R} \tag{11.416}$$

in which the definitions of λ_1 and λ_2 are found in (11.251). Using the boundary conditions to solve for C_1 and C_2, we find

$$\tilde{T} = \frac{T_1 R_o}{2C_Q s} \left[(c - c_a + C_Q) \frac{e^{-\lambda_1 \sqrt{s} (R - R_o)}}{R} - (c - c_a - C_Q) \frac{e^{-\lambda_2 \sqrt{s} (R - R_o)}}{R} \right] \tag{11.417}$$

$$\tilde{p} = \frac{c_a T_1 \alpha_e R_o}{C_Q s} \left[-\frac{e^{-\lambda_1 \sqrt{s}(R-R_o)}}{R} + \frac{e^{-\lambda_2 \sqrt{s}(R-R_o)}}{R} \right] \tag{11.418}$$

Performing Laplace inverse transform (see Table D.1), the above becomes

$$T(R,t) = \frac{T_1 R_o}{2C_Q R} \left\{ (c - c_a + C_Q) \operatorname{erfc}\left[\frac{\lambda_1 (R-R_o)}{2\sqrt{t}} \right] - (c - c_a - C_Q) \operatorname{erfc}\left[\frac{\lambda_2 (R-R_o)}{2\sqrt{t}} \right] \right\} \tag{11.419}$$

$$p(R,T) = \frac{c_a T_1 \alpha_e R_o}{C_Q R} \left\{ -\operatorname{erfc}\left[\frac{\lambda_1 (R-R_o)}{2\sqrt{t}} \right] + \operatorname{erfc}\left[\frac{\lambda_2 (R-R_o)}{2\sqrt{t}} \right] \right\} \tag{11.420}$$

The integrals are

$$\begin{aligned} \int R^2 T(R,t)\, dR = {} & \frac{T_1 R_o}{2C_Q} \\ & \times \left\{ (c - c_a + C_Q) \left(-\frac{R+R_o}{\lambda_1} \sqrt{\frac{t}{\pi}}\, e^{-\lambda_1^2 (R-R_o)^2/4t} \right.\right. \\ & \left. + \left[\frac{R^2 - R_o^2}{2} - \frac{t}{\lambda_1^2} \right] \operatorname{erfc}\left[\frac{\lambda_1 (R-R_o)}{2\sqrt{t}} \right] \right) \\ & - (c - c_a - C_Q) \left(-\frac{R+R_o}{\lambda_2} \sqrt{\frac{t}{\pi}}\, e^{-\lambda_2^2 (R-R_o)^2/4t} \right. \\ & \left.\left. + \left[\frac{R^2 - R_o^2}{2} - \frac{t}{\lambda_2^2} \right] \operatorname{erfc}\left[\frac{\lambda_2 (R-R_o)}{2\sqrt{t}} \right] \right) \right\} \end{aligned} \tag{11.421}$$

$$\begin{aligned} \int R^2 p(R,t)\, dR = {} & \frac{c_a T_1 \alpha_e R_o}{C_Q} \times \left\{ \frac{R+R_o}{\lambda_1} \sqrt{\frac{t}{\pi}}\, e^{-\lambda_1^2 (R-R_o)^2/4t} \right. \\ & - \left[\frac{R^2 - R_o^2}{2} - \frac{t}{\lambda_1^2} \right] \operatorname{erfc}\left[\frac{\lambda_1 (R-R_o)}{2\sqrt{t}} \right] \\ & - \frac{R+R_o}{\lambda_2} \sqrt{\frac{t}{\pi}}\, e^{-\lambda_2^2 (R-R_o)^2/4t} \\ & \left. + \left[\frac{R^2 - R_o^2}{2} - \frac{t}{\lambda_2^2} \right] \operatorname{erfc}\left[\frac{\lambda_2 (R-R_o)}{2\sqrt{t}} \right] \right\} \end{aligned} \tag{11.422}$$

The function $A_2(t)$ can be found from the radial stress boundary condition, as

$$A_2(t) = \frac{R_o T_1}{2GC_Q\lambda_1^2\lambda_2^2}\sqrt{\frac{t}{\pi}}\left\{(\lambda_1-\lambda_2)\left[\eta_d(c_a-c)+2c_a\alpha_e\eta\right]\right.$$
$$\times\left[2R_o\lambda_1\lambda_2+\sqrt{\pi t}(\lambda_1+\lambda_2)\right]$$
$$\left.+C_Q\eta_d\left[2R_o\lambda_1\lambda_2(\lambda_1+\lambda_2)+\sqrt{\pi t}(\lambda_1^2+\lambda_2^2)\right]\right\} \tag{11.423}$$

These expressions then fully define the displacement and stress quantities in (11.408)–(11.411).

Thermally Uncoupled Solution The spherical cavity problem has been solved based on the thermally uncoupled model [7, 39, 63] for the investigation of the heat induced consolidation around an underground storage of nuclear waste. Here we shall present the solution as follows.

First, the temperature can be solved from the uncoupled heat diffusion equation (11.399) and the boundary condition (11.412) as

$$T = \frac{T_1 R_o}{R}\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right) \tag{11.424}$$

Pore pressure can be solved from (11.398) with $A_1(t) = 0$, and we obtain

$$p = \frac{\kappa_T\alpha_e T_1 R_o}{(c-\kappa_T)R}\left[\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{ct}}\right)-\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)\right] \tag{11.425}$$

The integrals are

$$\int R^2T(R,t)\,dR = T_1R_o\left[-(R+R_o)\sqrt{\frac{\kappa_T t}{\pi}}e^{-(R-R_o)^2/4\kappa_T t}\right.$$
$$\left.+\left(\frac{R^2-R_o^2}{2}-\kappa_T t\right)\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)\right] \tag{11.426}$$

$$\int R^2p(R,t)\,dR = \frac{\kappa_T T_1\alpha_e R_o}{c-\kappa_T}\left[(R+R_o)\sqrt{\frac{\kappa_T t}{\pi}}e^{-(R-R_o)^2/4\kappa_T t}\right.$$
$$-\left(\frac{R^2-R_o^2}{2}-\kappa_T t\right)\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)$$
$$-(R+R_o)\sqrt{\frac{ct}{\pi}}e^{-(R-R_o)^2/4ct}$$
$$\left.+\left(\frac{R^2-R_o^2}{2}-ct\right)\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{ct}}\right)\right] \tag{11.427}$$

And $A_2(t)$ is solved as

$$A_2(t) = \frac{R_o T_1}{G(\sqrt{c} + \sqrt{\kappa_T})} \sqrt{\frac{t}{\pi}} \left[2R_o \eta_d \sqrt{\kappa_T c} + \kappa_T(\alpha_e \eta + \eta_d)(2R_o + \sqrt{\pi c t} + \sqrt{\pi \kappa_T t}) \right] \tag{11.428}$$

These expressions fully define the displacement and stress solutions.

In Fig. 11.10 we plot the temperature, pore pressure, stresses and effective stresses for the case of a cavity of radius $R_o = 10\,\text{m}$ in rock salt, subjected to a temperature increment of 100 °K. These results can be compared with the borehole case in Fig. 11.7.

11.11.2 Constant Heat Flux Problem

The boundary condition for the constant heat flux problem is given as follows:

$$\sigma_{RR} = 0; \quad p = 0; \quad \frac{\partial T}{\partial R} = -\frac{q_o}{k_T}; \quad \text{at } R = R_o \tag{11.429}$$

Thermally Coupled Solution The general solution of temperature and pore pressure are given by (11.415) and (11.416). With the boundary condition (11.429), C_1 and C_2 can be solved. With the substitution, we obtain

$$\tilde{T} = \frac{q_o R_o^2}{k_T s D} \left[(c - c_a + C_Q) \frac{e^{-\lambda_1 \sqrt{s}(R - R_o)}}{R} - (c - c_a - C_Q) \frac{e^{-\lambda_2 \sqrt{s}(R - R_o)}}{R} \right] \tag{11.430}$$

$$\tilde{p} = \frac{2 c_a q_o \alpha_e R_o^2}{k_T s D} \left[-\frac{e^{-\lambda_1 \sqrt{s}(R - R_o)}}{R} + \frac{e^{-\lambda_2 \sqrt{s}(R - R_o)}}{R} \right] \tag{11.431}$$

where

$$D = R_o \sqrt{s} \left[(c - c_a + C_Q)\lambda_1 - (c - c_a - C_Q)\lambda_2 \right] + 2C_Q \tag{11.432}$$

Applying the inverse Laplace transform (see Table D.1), we obtain

$$T(R,t) = \frac{q_o R_o^2}{2 C_Q k_T R} \left\{ (c - c_a + C_Q) \times \left[-e^{b(a_1 + bt)} \operatorname{erfc}\left(b\sqrt{t} + \frac{a_1}{2\sqrt{t}} \right) + \operatorname{erfc}\left(\frac{a_1}{2\sqrt{t}} \right) \right] + (c - c_a - C_Q) \times \left[-e^{b(a_2 + bt)} \operatorname{erfc}\left(b\sqrt{t} + \frac{a_2}{2\sqrt{t}} \right) + \operatorname{erfc}\left(\frac{a_2}{2\sqrt{t}} \right) \right] \right\} \tag{11.433}$$

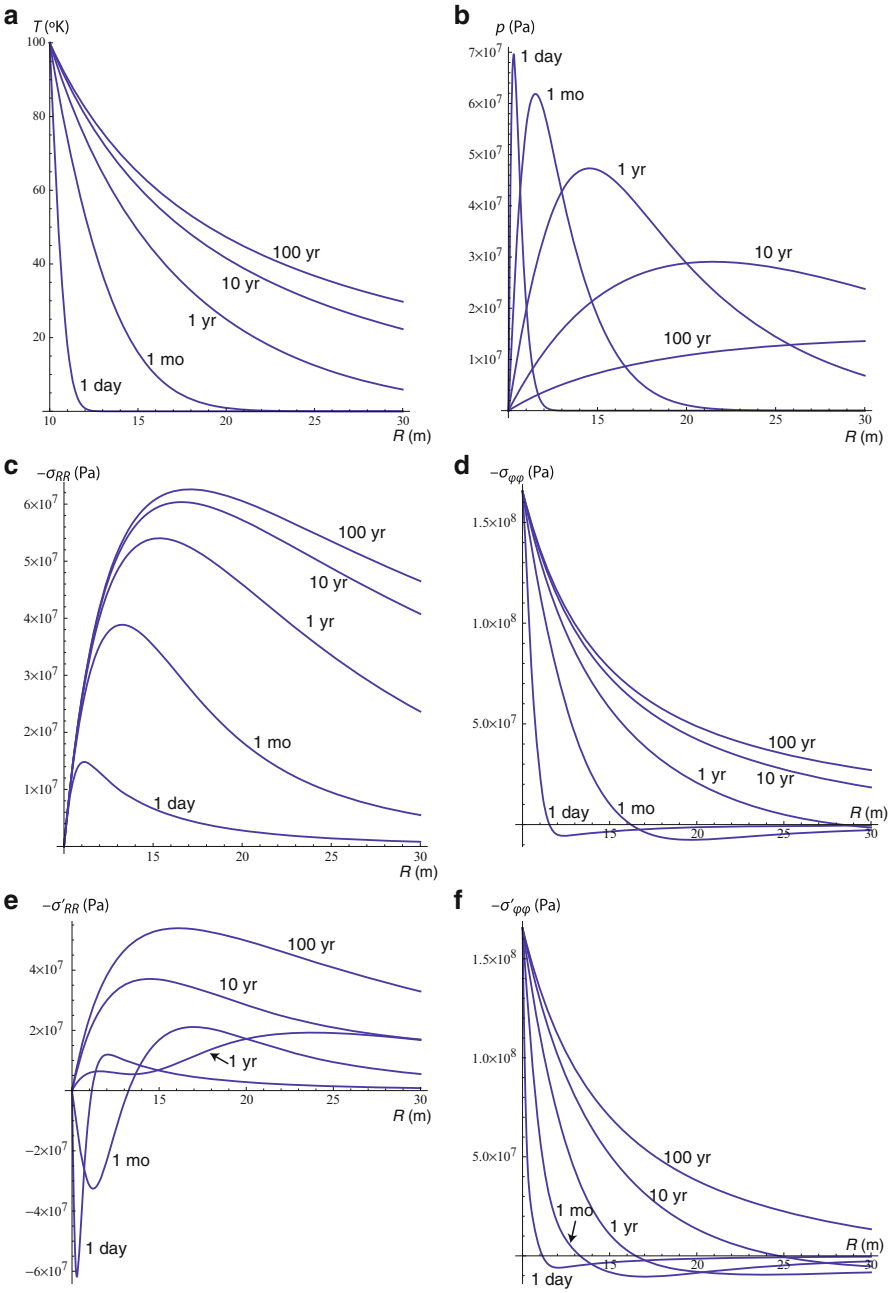


Fig. 11.10 Porothermoelastic solution of cavity problem, step rise in temperature: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial stress $-\sigma_{RR}$, (**d**) circumferential stress $-\sigma_{\varphi\varphi}$, (**e**) Terzaghi effective radial stress $-\sigma'_{RR}$, and (**f**) Terzaghi effective circumferential stress $-\sigma'_{\varphi\varphi}$

$$p(R,t) = \frac{c_a q_o \alpha_e R_o^2}{C_Q k_T R}\left[e^{b(a_1+bt)}\operatorname{erfc}\left(b\sqrt{t}+\frac{a_1}{2\sqrt{t}}\right)-\operatorname{erfc}\left(\frac{a_1}{2\sqrt{t}}\right)\right.$$
$$\left.-e^{b(a_2+bt)}\operatorname{erfc}\left(b\sqrt{t}+\frac{a_2}{2\sqrt{t}}\right)+\operatorname{erfc}\left(\frac{a_2}{2\sqrt{t}}\right)\right] \tag{11.434}$$

where

$$a_1 = \lambda_1(R-R_o) \tag{11.435}$$
$$a_2 = \lambda_2(R-R_o) \tag{11.436}$$
$$b = \frac{2C_Q}{R_o[(c-c_a+C_Q)\lambda_1-(c-c_a-C_Q)\lambda_2]} \tag{11.437}$$

The integrals are found as

$$\int R^2 T(R,T)\,dR = \frac{q_o R_o^2}{2C_Q k_T}\times\left\{(c-c_a+C_Q)\left[\frac{2-b\lambda_1(R+R_o)}{b\lambda_1^2}\sqrt{\frac{t}{\pi}}\,e^{-a_1/4t}\right.\right.$$
$$+\frac{1-b\lambda_1 R}{b^2\lambda_1^2}e^{b(a_1+bt)}\operatorname{erfc}\left(\frac{a_1+2bt}{2\sqrt{t}}\right)$$
$$\left.-\frac{2(1-b\lambda_1 R+b^2t)-b^2\lambda_1^2(R^2-R_o^2)}{2b^2\lambda_1^2}\operatorname{erfc}\left(\frac{a_1}{2\sqrt{t}}\right)\right]$$
$$-(c-c_a-C_Q)\left[\frac{2-b\lambda_2(R+R_o)}{b\lambda_2^2}\sqrt{\frac{t}{\pi}}\,e^{-a_2/4t}\right.$$
$$+\frac{1-b\lambda_2 R}{b^2\lambda_2^2}e^{b(a_2+bt)}\operatorname{erfc}\left(\frac{a_2+2bt}{2\sqrt{t}}\right)$$
$$\left.\left.-\frac{2(1-b\lambda_2 R+b^2t)-b^2\lambda_2^2(R^2-R_o^2)}{2b^2\lambda_2^2}\operatorname{erfc}\left(\frac{a_2}{2\sqrt{t}}\right)\right]\right\} \tag{11.438}$$

$$\int R^2 p(R,T)\,dR = \frac{c_a q_o \alpha_e R_o^2}{C_Q k_T}\left[-\frac{2-b\lambda_1(R+R_o)}{b\lambda_1^2}\sqrt{\frac{t}{\pi}}\,e^{-a_1/4t}\right.$$
$$-\frac{1-b\lambda_1 R}{b^2\lambda_1^2}e^{b(a_1+bt)}\operatorname{erfc}\left(\frac{a_1+2bt}{2\sqrt{t}}\right)$$
$$+\frac{2(1-b\lambda_1 R+b^2t)-b^2\lambda_1^2(R^2-R_o^2)}{2b^2\lambda_1^2}\operatorname{erfc}\left(\frac{a_1}{2\sqrt{t}}\right)$$
$$\left.+\frac{2-b\lambda_2(R+R_o)}{b\lambda_2^2}\sqrt{\frac{t}{\pi}}\,e^{-a_2/4t}+\frac{1-b\lambda_2 R}{b^2\lambda_2^2}e^{b(a_2+bt)}\operatorname{erfc}\left(\frac{a_2+2bt}{2\sqrt{t}}\right)\right.$$

$$-\frac{2(1-b\lambda_2 R+b^2 t)-b^2\lambda_2^2(R^2-R_o^2)}{2b^2\lambda_2^2}\operatorname{erfc}\left(\frac{a_2}{2\sqrt{t}}\right)\Bigg] \tag{11.439}$$

The constant $A_2(t)$ is solved from the radial stress boundary condition as

$$\begin{aligned}A_2(t) = -\frac{q_o R_o^2}{2\sqrt{\pi}\,b^2 C_Q G k_T \lambda_1^2\lambda_2^2}\Big\{ & b^2\sqrt{\pi}\,t\big[\eta_d\lambda_1^2(c-c_a-C_Q)\\ & -\eta_d\lambda_2^2(c-c_a+C_Q)-2\alpha_e\eta c_a(\lambda_1^2-\lambda_2^2)\big]\\ & -\big[\eta_d\lambda_1^2(c-c_a-C_Q)(1-\lambda_2 bR_o)\\ & -\eta_d\lambda_2^2(c-c_a+C_Q)(1-\lambda_1 bR_o)\\ & +2\alpha_e\eta c_a(\lambda_1-\lambda_2)(\lambda_1+\lambda_2-bR_o\lambda_1\lambda_2)\big]\\ & \times\left[-\sqrt{\pi}+2b\sqrt{t}+\sqrt{\pi}\,e^{b^2 t}\operatorname{erfc}\left(b\sqrt{t}\right)\right]\Big\}\end{aligned} \tag{11.440}$$

These expressions completely define the solution.

Thermally Uncoupled Solution The solutions for temperature and pore pressure are

$$\tilde{T} = \frac{q_o R_o^2\sqrt{\kappa_T}}{k_T s(R_o\sqrt{s}+\sqrt{\kappa_T})}\frac{e^{-\sqrt{s/\kappa_T}(R-R_o)}}{R} \tag{11.441}$$

$$\begin{aligned}\tilde{p} = & \frac{q_o R_o^2\alpha_e\kappa_T^{3/2}}{k_T s(c-\kappa_T)(R_o\sqrt{s}+\sqrt{\kappa_T})}\\ & \times\left[\frac{e^{-\sqrt{s/c}(R-R_o)}}{R}-\frac{e^{-\sqrt{s/\kappa_T}(R-R_o)}}{R}\right]\end{aligned} \tag{11.442}$$

Applying the inverse Laplace transform, we obtain

$$\begin{aligned}T(R,t) = & \frac{q_o R_o^2}{k_T R}\left[-e^{(R-R_o)/R_o+\kappa_T t/R_o^2}\operatorname{erfc}\left(\frac{\sqrt{\kappa_T t}}{R_o}+\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)\right.\\ & \left.+\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)\right]\end{aligned} \tag{11.443}$$

$$\begin{aligned}p(R,t) = & \frac{q_o R_o^2\alpha_e\kappa_T}{k_T R(c-\kappa_T)}\left[e^{(R-R_o)/R_o+\kappa_T t/R_o^2}\times\operatorname{erfc}\left(\frac{\sqrt{\kappa_T t}}{R_o}+\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)\right.\\ & -\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)-e^{\sqrt{\kappa_T/c}(R-R_o)/R_o+\kappa_T t/R_o^2}\operatorname{erfc}\left(\frac{\sqrt{\kappa_T t}}{R_o}+\frac{R-R_o}{2\sqrt{c t}}\right)\\ & \left.+\operatorname{erfc}\left(\frac{R-R_o}{2\sqrt{c t}}\right)\right]\end{aligned} \tag{11.444}$$

The integrals are

$$\int R^2 T(R,T)\, dR = \frac{q_o R_o^2}{k_T}\left[-(R-R_o)\sqrt{\frac{\kappa_T t}{\pi}}\, e^{-(R-R_o)^2/4\kappa_T t}\right.$$
$$+\frac{1}{2}(R^2 - R_o^2 - 2\kappa_T t)\,\mathrm{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)$$
$$\left.-R_o(R-R_o)\, e^{(R-R_o)/R_o+\kappa_T t/R_o^2}\,\mathrm{erfc}\left(\frac{\sqrt{\kappa_T t}}{R_o}+\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)\right] \tag{11.445}$$

$$\int R^2 p(R,T)\, dR = \frac{q_o R_o^2 \alpha_e \kappa_T}{k_T(c-\kappa_T)}\left\{(R-R_o)\sqrt{\frac{\kappa_T t}{\pi}}\, e^{-(R-R_o)^2/4\kappa_T t}\right.$$
$$-\frac{1}{2}(R^2 - R_o^2 - 2\kappa_T t)\,\mathrm{erfc}\left(\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)$$
$$+R_o(R-R_o)\, e^{(R-R_o)/R_o+\kappa_T t/R_o^2}\,\mathrm{erfc}\left(\frac{\sqrt{\kappa_T t}}{R_o}+\frac{R-R_o}{2\sqrt{\kappa_T t}}\right)$$
$$-\left[(R+R_o)-2R_o\sqrt{\frac{c}{\kappa_T}}\right]\sqrt{\frac{ct}{\pi}}\, e^{-(R-R_o)^2/4ct}$$
$$+\frac{1}{2}\left[R^2 - R_o^2 - 2ct + 2R_o^2\left(\sqrt{\frac{c}{\kappa_T}}-\frac{c}{\kappa_T}\right)\right]\mathrm{erfc}\left(\frac{R-R_o}{2\sqrt{ct}}\right)$$
$$-R_o\left(R\sqrt{\frac{c}{\kappa_T}}-R_o\frac{c}{\kappa_T}\right)e^{\sqrt{\kappa_T}(R-R_o)/R_o\sqrt{c}+\kappa_T t/R_o^2}$$
$$\left.\times\,\mathrm{erfc}\left(\frac{\sqrt{\kappa_T t}}{R_o}+\frac{R-R_o}{2\sqrt{ct}}\right)\right\} \tag{11.446}$$

And the constant $A_2(t)$ is obtained from the radial stress boundary condition as

$$A_2(t) = -\frac{q_o R_o^2}{Gk_T(\sqrt{c}+\sqrt{\kappa_T})}\left\{-\kappa_T t(\alpha_e\eta+\eta_d)(\sqrt{c}+\sqrt{\kappa_T})\right.$$
$$\left.+R_o\alpha_e\eta\sqrt{c}\left[2\sqrt{\frac{\kappa_T t}{\pi}}-R_o+R_o e^{\kappa_T t/R_o^2}\,\mathrm{erfc}\left(\sqrt{\frac{\kappa_T t}{R_o^2}}\right)\right]\right\} \tag{11.447}$$

Based on the above solution, we plot the temperature, pore pressure, stresses and effective stresses for the case of a cavity with radius $R_o = 10\,\mathrm{m}$, and a constant heat flux of $100\,\mathrm{W/m^2}$ on the wall, in Fig. 11.11

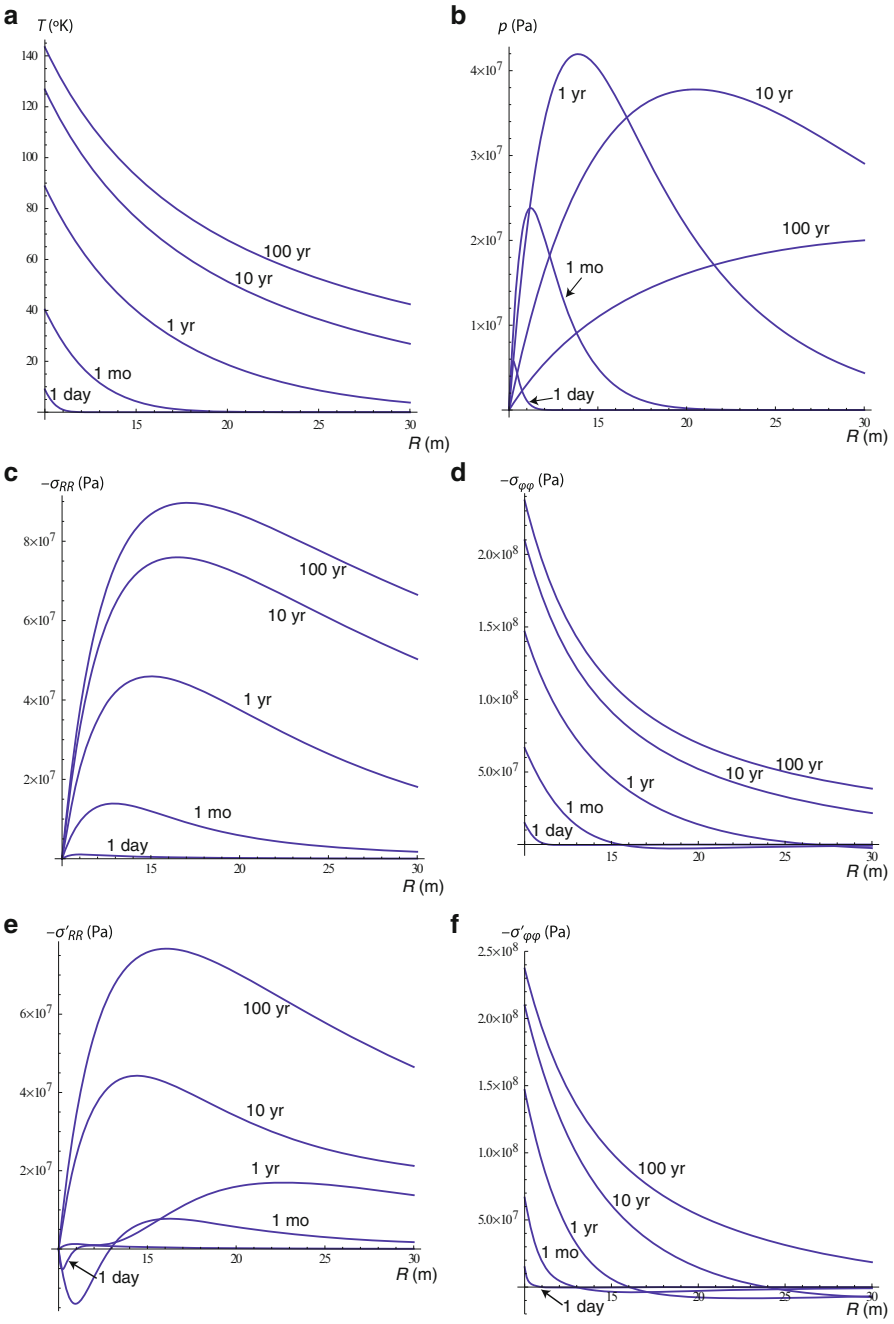


Fig. 11.11 Porothermoelastic solution of cavity problem, constant heat flux: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial stress $-\sigma_{RR}$, (**d**) circumferential stress $-\sigma_{\varphi\varphi}$, (**e**) Terzaghi effective radial stress $-\sigma'_{RR}$, and (**f**) Terzaghi effective circumferential stress $-\sigma'_{\varphi\varphi}$

11.12 Sphere Problem

Consider a sphere of radius R_o with its surface temperature suddenly raised by T_1, and the surface is otherwise stress and pressure free:

$$\sigma_{RR} = 0; \quad p = 0; \quad T = T_1 \quad \text{at } R = R_o \tag{11.448}$$

Thermally Coupled Solution The thermally coupled problem was investigated by Belotserkovets and Prevost [3]. Here we solve it by utilizing the pore pressure representation given by (11.392), (11.393), (11.394), and (11.395). Setting $A_2(t) = 0$ to avoid unbounded solution for interior domain problem, we obtain

$$u_R = \frac{\eta}{G}\frac{1}{R^2}\int R^2 p(R,t)\, dR + \frac{\eta_d}{G}\frac{1}{R^2}\int R^2 T(R,t)\, dR + A_1(t)R \tag{11.449}$$

$$\sigma_{RR} = -4\eta\frac{1}{R^3}\int R^2 p(R,t)\, dR - 4\eta_d\frac{1}{R^3}\int R^2 T(R,t)\, dR + \frac{2G(1+\nu)}{1-2\nu}A_1(t) \tag{11.450}$$

$$\sigma_{\varphi\varphi} = 2\eta\frac{1}{R^3}\int R^2 p(R,t)\, dR + 2\eta_d\frac{1}{R^3}\int R^2 T(R,t)\, dR - 2\eta p - 2\eta_d T + \frac{2G(1+\nu)}{1-2\nu}A_1(t) \tag{11.451}$$

The temperature and pore pressure can be solved from the following coupled diffusion equations in Laplace transform domain

$$\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial \tilde{T}}{\partial R}\right) - \frac{s}{c_a}\tilde{T} = -\frac{s\alpha_p}{c_a}\tilde{p} - \frac{2s\alpha_d}{S_a}\tilde{A}_1(s) \tag{11.452}$$

$$\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial \tilde{p}}{\partial R}\right) - \frac{s}{c}\tilde{p} = -\frac{s\alpha_e}{c}\tilde{T} - \frac{2s\alpha}{S}\tilde{A}_1(s) \tag{11.453}$$

The general solutions of the above are

$$\tilde{T} = C_1\frac{\sinh(\lambda_1\sqrt{s}R)}{R} + C_2\frac{\sinh(\lambda_2\sqrt{s}R)}{R} - \frac{2(\alpha_d S + \alpha\alpha_p S_a)}{SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1(s) \tag{11.454}$$

$$\tilde{p} = \frac{C_1(c-c_a-C_Q)}{2c\alpha_p}\frac{\sinh(\lambda_1\sqrt{s}R)}{R} + \frac{C_2(c-c_a+C_Q)}{2c\alpha_p}\frac{\sinh(\lambda_2\sqrt{s}R)}{R} - \frac{2(\alpha S_a + \alpha_d\alpha_e S)}{SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1 \tag{11.455}$$

Utilizing the temperature and pore pressure boundary conditions in (11.448), we can solve for the two constants C_1 and C_2 as

$$C_1 = \frac{R_o}{2sC_Q SS_a(1-\alpha_p\alpha_e)\sinh(\lambda_1\sqrt{s}R_o)} \times \left\{2s\left[(c-c_a+C_Q)(S\alpha_d+S_a\alpha\alpha_p)-2c\alpha_p(S_a\alpha+S\alpha_d\alpha_e)\right]\tilde{A}_1(s) + SS_aT_1(c-c_a+C_Q)(1-\alpha_p\alpha_e)\right\} \tag{11.456}$$

$$C_2 = -\frac{R_o}{2sC_Q SS_a(1-\alpha_p\alpha_e)\sinh(\lambda_2\sqrt{s}R_o)} \times \left\{2s\left[(c-c_a-C_Q)(S\alpha_d+S_a\alpha\alpha_p)-2c\alpha_p(S_a\alpha+S\alpha_d\alpha_e)\right]\tilde{A}_1(s) + SS_aT_1(c-c_a-C_Q)(1-\alpha_p\alpha_e)\right\} \tag{11.457}$$

The integrals are

$$\int R^2\tilde{T}(R,s)\,dR = C_1\left[\frac{R\cosh(\lambda_1\sqrt{s}R)}{\lambda_1\sqrt{s}}-\frac{\sinh(\lambda_1\sqrt{s}R)}{\lambda_1^2 s}\right] + C_2\left[\frac{R\cosh(\lambda_2\sqrt{s}R)}{\lambda_2\sqrt{s}}-\frac{\sinh(\lambda_2\sqrt{s}R)}{\lambda_2^2 s}\right] - \frac{2R^3(\alpha_d S+\alpha\alpha_p S_a)}{3SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1(s) \tag{11.458}$$

$$\int R^2\tilde{p}(R,s)\,dR = \frac{C_1(c-c_a-C_Q)}{2c\alpha_p}\left[\frac{R\cosh(\lambda_1\sqrt{s}R)}{\lambda_1\sqrt{s}}-\frac{\sinh(\lambda_1\sqrt{s}R)}{\lambda_1^2 s}\right] + \frac{C_2(c-c_a+C_Q)}{2c\alpha_p}\left[\frac{R\cosh(\lambda_2\sqrt{s}R)}{\lambda_2\sqrt{s}}-\frac{\sinh(\lambda_2\sqrt{s}R)}{\lambda_2^2 s}\right] - \frac{2R^3(\alpha S_a+\alpha_d\alpha_e S)}{3SS_a(1-\alpha_p\alpha_e)}\tilde{A}_1 \tag{11.459}$$

The constant $\tilde{A}_1(s)$ is obtained from the radial stress boundary conditions as

$$\tilde{A}_1(s) = \frac{3SS_aT_1(1-2\nu)(1-\alpha_p\alpha_e)}{2sD} \times \left\{\lambda_2^2(c-c_a+C_Q)\left[\eta(c-c_a-C_Q)+2c\alpha_p\eta_d\right] \times\left[1-\lambda_1\sqrt{s}R_o\coth\left(\lambda_1\sqrt{s}R_o\right)\right] -\lambda_1^2(c-c_a-C_Q)\left[\eta(c-c_a+C_Q)+2c\alpha_p\eta_d\right] \times\left[1-\lambda_2\sqrt{s}R_o\coth\left(\lambda_2\sqrt{s}R_o\right)\right]\right\} \tag{11.460}$$

where

$$
\begin{aligned}
D = 3(1-2\nu)\,\{&-\lambda_2^2[(c-c_a+C_Q)(S\alpha_d+S_a\alpha\alpha_p)-2c\alpha_p(S_a\alpha+S\alpha_d\alpha_e)]\\
&\times[\eta(c-c_a-C_Q)+2c\alpha_p\eta_d][1-R_o\sqrt{s}\lambda_1\coth(\lambda_1\sqrt{s}R_o)]\\
&+\lambda_1^2[(c-c_a-C_Q)(S\alpha_d+S_a\alpha\alpha_p)-2c\alpha_p(S_a\alpha+S\alpha_d\alpha_e)]\\
&\times[\eta(c-c_a+C_Q)+2c\alpha_p\eta_d][1-R_o\sqrt{s}\lambda_2\coth(\lambda_2\sqrt{s}R_o)]\}\\
&-cC_QR_o^2s\alpha_p\lambda_1^2\lambda_2^2\,\{3GSS_a(1+\nu)(1-\alpha_p\alpha_e)\\
&+4(1-2\nu)[S\alpha_d(\alpha_e\eta+\eta_d)+S_a\alpha(\eta+\alpha_p\eta_d)]\}
\end{aligned}
\tag{11.461}
$$

These fully define the solution in the Laplace transform domain. Numerical Laplace inversion can be used to find solution in time.

Thermally Uncoupled Solution The thermally uncoupled solution was investigated by Selvadurai and Suvorov [62]. Here, rather than reducing the solution by taking the limit of the thermally coupled solution as presented above, we shall directly obtain it using the variation in fluid content representation. For interior domain problem, (11.401)–(11.404) become

$$
u_R = \frac{\eta}{GS}\frac{1}{R^2}\int R^2\zeta(R,t)\,dR + \frac{\eta\beta_u}{GSB}\frac{1}{R^2}\int R^2T(R,t)\,dR + B_1(t)R
\tag{11.462}
$$

$$
\begin{aligned}
\sigma_{RR} = &-\frac{4\eta}{S}\frac{1}{R^3}\int R^2\zeta(R,t)\,dR - \frac{4\eta\beta_u}{SB}\frac{1}{R^3}\int R^2T(R,t)\,dR\\
&+\frac{2G(1+\nu_u)}{1-2\nu_u}B_1(t)
\end{aligned}
\tag{11.463}
$$

$$
\begin{aligned}
\sigma_{\varphi\varphi} = &\frac{2\eta}{S}\frac{1}{R^3}\int R^2\zeta(R,t)\,dR + \frac{2\eta\beta_u}{SB}\frac{1}{R^3}\int R^2T(R,t)\,dR\\
&-\frac{2\eta}{S}\zeta - \frac{2\eta\beta_u}{SB}T + \frac{2G(1+\nu_u)}{1-2\nu_u}B_1(t)
\end{aligned}
\tag{11.464}
$$

$$
p = \frac{\zeta}{S} + \alpha_eT - 2\alpha MB_1(t)
\tag{11.465}
$$

The solution of temperature is obtained from (11.399), in the Laplace transform form as

$$\frac{s\tilde{T}}{T_1} = \frac{R_o \sinh(R\sqrt{s/\kappa_T})}{R \sinh(R_o\sqrt{s/\kappa_T})} \tag{11.466}$$

The solution of ζ is found from (11.407) as

$$\tilde{\zeta} = \frac{R_q^2 T_1 \beta_c}{(1-R_q^2)s} \frac{R_o \sinh(R\sqrt{s/\kappa_T})}{R \sinh(R_o\sqrt{s/\kappa_T})} + D_1 T_1 \frac{R_o \sinh(R\sqrt{s/c})}{R \sinh(R_o\sqrt{s/c})} \tag{11.467}$$

where

$$D_1 = -\frac{1}{(1-R_q^2)Ds} \left\{ 3R_o^2 \beta_c s + 8\eta\kappa_T \left[R_q^2 B\beta_c + \beta_u (1-R_q^2) \right] \right.$$
$$\left. \times \left[1 - R_o\sqrt{s/\kappa_T} \coth(R_o\sqrt{s/\kappa_T}) \right] \right\} \tag{11.468}$$

and

$$D = 3R_o^2 s + 8B\eta c \left[1 - R_o\sqrt{s/c} \coth(R_o\sqrt{s/c}) \right] \tag{11.469}$$

We also find

$$\tilde{B}_1 = \frac{T_1(1-2\nu_u)\left[(1-R_q^2)D_1 s - \beta_c\right]}{4\eta s(1-R_q^2)(1-\nu_u)} \tag{11.470}$$

To obtain the rest of the solutions, we find these two integrals

$$\int R^2 \tilde{T}(R,s)\, dR = \frac{R_o T_1 \kappa_T}{s^2 \sinh(R_o\sqrt{s/\kappa_T})}$$
$$\times \left[R\sqrt{s/\kappa_T} \cosh(R\sqrt{s/\kappa_T}) - \sinh(R\sqrt{s/\kappa_T}) \right] \tag{11.471}$$

$$\int R^2 \tilde{\zeta}(R,s)\, dR = \frac{R_o R_q^2 T_1 \beta_c \kappa_T}{(1-R_q^2)s^2 \sinh(R_o\sqrt{s/\kappa_T})}$$
$$\times \left[R\sqrt{s/\kappa_T} \cosh(R\sqrt{s/\kappa_T}) - \sinh(R\sqrt{s/\kappa_T}) \right]$$
$$+ \frac{R_o T_1 c D_1}{s \sinh(R_o\sqrt{s/c})} \left[R\sqrt{s/c} \cosh(R\sqrt{s/c}) - \sinh(R\sqrt{s/c}) \right] \tag{11.472}$$

The above expressions then fully define the solution in the Laplace transform domain. Following the procedures outlined in Sects. 7.10 and D.2, these expressions can be analytically inverted to the time domain. For simplicity, we shall utilize the numerical Laplace inversion to evaluate the solution. In Fig. 11.12 we present the solution of a sphere with radius of 10 cm, made of rock salt, subject to an temperature increment of 100 °K. The trend of these solutions is similar to the cylinder case shown in Fig. 11.9.

11.13 Porothermoelasticity Fundamental Solution

In this section we examine the fundamental solutions of porothermoelasticity, that is, response of the medium subject to singular forcing functions, such as a concentrated force, fluid source, and heat source, etc., in an unbounded domain. Such solutions were derived by Smith and Booker [65] using the integral transform technique. Here we shall present them using a simpler technique based on the Biot decomposition, which gives much concise results.

First, we shall rewrite the governing equations by introducing the proper forcing terms. For example the equilibrium equation (11.117) is written as

$$\sigma_{ij,j} = -F_i \tag{11.473}$$

where F_i is the body force term, and (11.118) and (11.123) become

$$\frac{\partial \zeta}{\partial t} + q_{i,i} = \gamma \tag{11.474}$$

$$\mathcal{T}_o \frac{\partial s}{\partial t} + q_{i,i} = \gamma_T \tag{11.475}$$

where γ and γ_T are respectively the fluid source and the heat source. The rest of the constitutive and governing equations are the same as those in Sects. 11.1 and 11.4.

The equilibrium equation (11.473) can be combined with the constitutive equation for stress to yield a Navier equation as demonstrated in Sect. 11.4.1. Here we adopt the form of (11.171) for the reason that will become clear below:

$$G u_{i,jj} + \left(K_b + \frac{G}{3}\right) u_{j,ji} - \alpha_b M \zeta_{,i} - \frac{\alpha_u}{m_u} s_{,i} = -F_i \tag{11.476}$$

The constitutive equations and the Navier equation given above can be used to transform the mass and energy balance equations, (11.474) and (11.475), into the following diffusion equations

$$\frac{\partial \zeta}{\partial t} - c_c \nabla^2 \zeta - \frac{c_c \beta_g}{m_d} \nabla^2 s = \frac{3 M \alpha_b \kappa}{3K_b + 4G} F_{i,i} + \gamma \tag{11.477}$$

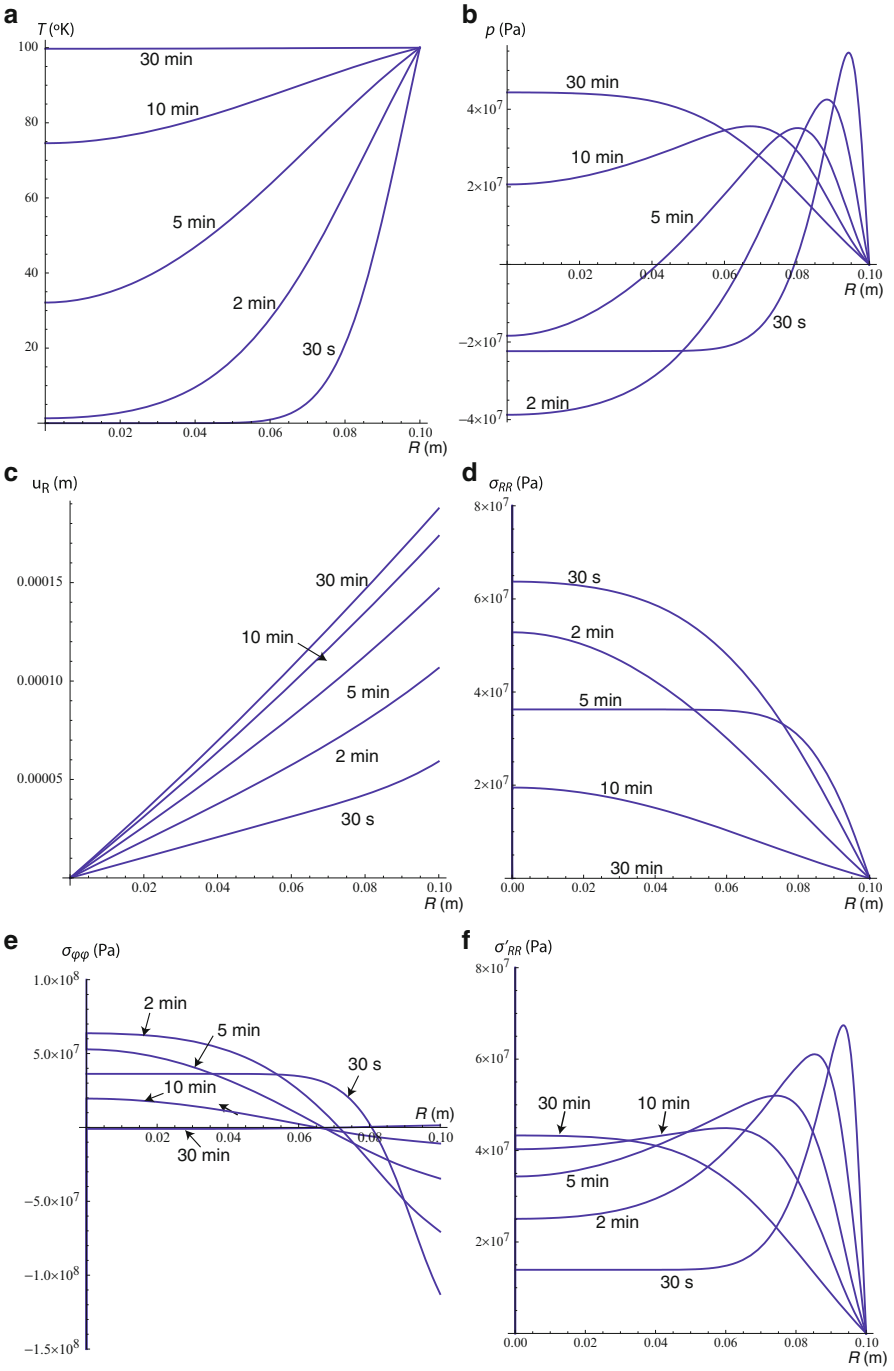


Fig. 11.12 Porothermoelastic solution of sphere problem, step rise in temperature: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial displacement u_R, (**d**) radial stress σ_{RR}, (**e**) circumferential stress $\sigma_{\varphi\varphi}$, and (**f**) Terzaghi effective radial stress σ'_{RR}

$$\frac{\partial s}{\partial t} - c_b \nabla^2 s - M c_b \beta_h \nabla^2 \zeta = \frac{3 m_d \alpha_u \kappa_T}{m_u (3K_b + 4G)} F_{i,i} + \frac{\gamma_T}{\mathcal{T}_o} \tag{11.478}$$

The above equations can be compared to (11.187) and (11.198), without the forcing function terms. We notice that using the set of variables $(u_i,\ \zeta,\ s)$, the two diffusion equations (11.477) and (11.478) are uncoupled from the Navier equation (11.476).

11.13.1 Biot Decomposition

To further decouple the set of Eqs. (11.476), (11.477), and (11.478), we introduce a variable decomposition similar to the Biot decomposition introduced in Sects. 6.6.2 and 8.9, as

$$u_i = u_i^o + \frac{\eta_b M}{G} \Phi_{,i} + \frac{\eta_u}{m_u G} \Psi_{,i} \tag{11.479}$$

in which, analogous to η and η_d, we have introduced

$$\eta_b = \frac{3\alpha_b G}{3K_b + 4G} \tag{11.480}$$

$$\eta_u = \frac{3\alpha_u G}{3K_b + 4G} \tag{11.481}$$

Substituting (11.479) into (11.476), and requiring that

$$G u^o_{i,jj} + \left(K_b + \frac{G}{3}\right) u^o_{j,ji} = -F_i \tag{11.482}$$

and

$$\zeta = \nabla^2 \Phi \tag{11.483}$$

$$s = \nabla^2 \Psi \tag{11.484}$$

we observe that (11.476) is identically satisfied. Based on these expressions, we can also write (11.477) and (11.478) into the following

$$\frac{\partial \nabla^2 \Phi}{\partial t} - c_c \nabla^4 \Phi - \frac{c_c \beta_g}{m_d} \nabla^4 \Psi = \frac{\eta_b M \kappa}{G} F_{i,i} + \gamma \tag{11.485}$$

$$\frac{\partial \nabla^2 \Psi}{\partial t} - c_b \nabla^4 \Psi - M c_b \beta_h \nabla^4 \Phi = \frac{\eta_u m_d \kappa_T}{m_u G} F_{i,i} + \frac{\gamma_T}{\mathcal{T}_o} \tag{11.486}$$

Relaxing a Laplacian operator in the above equations, we find

$$\frac{\partial \Phi}{\partial t} - c_c \nabla^2 \Phi - \frac{c_c \beta_g}{m_d} \nabla^2 \Psi = \frac{\eta_b M \kappa}{G} g_1 + g_2 \tag{11.487}$$

$$\frac{\partial \Psi}{\partial t} - c_b \nabla^2 \Psi - M c_b \beta_h \nabla^2 \Phi = \frac{\eta_u m_d \kappa_T}{m_u G} g_1 + \frac{g_3}{\mathcal{T}_o} \tag{11.488}$$

in which

$$\nabla^2 g_1 = F_{i,i} \tag{11.489}$$

$$\nabla^2 g_2 = \gamma \tag{11.490}$$

$$\nabla^2 g_3 = \gamma_T \tag{11.491}$$

With the above relaxation, the solutions are not unique, and admit an arbitrary harmonic function. This is, however, not a concern, as we are seeking a particular solution not satisfying any boundary condition.

With the proper substitutions of the Dirac delta function for these forcing functions, F_i, γ, and γ_T, we can solve the system (11.482), (11.487) and (11.488), for (u_i^o, Φ, Ψ). The displacement u_i is then obtained from (11.479). Pore pressure expression can be constructed from the constitutive equation (11.95) as

$$p = -\alpha_b M e^o + \frac{1}{S_c} \nabla^2 \Phi + \frac{\beta_g}{m_d S_c} \nabla^2 \Psi \tag{11.492}$$

where $e^o = u_{i,i}^o$. Or, e^o can be eliminated by taking divergence of (11.482) and utilizing (11.489), and we find

$$p = \frac{1}{S_c} \nabla^2 \Phi + \frac{\beta_g}{m_d S_c} \nabla^2 \Psi + \frac{\eta_b M}{G} g_1 \tag{11.493}$$

The above equation allows the derivation of pore pressure without the need of finding u_i^o. Similarly, we can derive the following equation for temperature

$$T = \frac{\beta_h M}{S_b} \nabla^2 \Phi + \frac{1}{S_b} \nabla^2 \Psi + \frac{\eta_u}{m_u G} g_1 \tag{11.494}$$

The stresses can be found from

$$\sigma_{ij} = \sigma_{ij}^o + 2\eta_b M \left(\Phi_{,ij} - \delta_{ij} \nabla^2 \Phi \right) + \frac{2\eta_u}{m_u} \left(\Psi_{,ij} - \delta_{ij} \nabla^2 \Psi \right) \tag{11.495}$$

where

$$\sigma_{ij}^o = \left(K_b - \frac{2G}{3} \right) \delta_{ij} e^o + 2G e_{ij}^o \tag{11.496}$$

11.13.2 Continuous Heat Source

We shall consider the problem of a point heat source located at $\vec{\chi}$, which is suddenly applied at time τ and then maintained at a unit magnitude throughout the time. This problem corresponds to the introduction of the following singular forcing functions:

$$\gamma_T = \delta(\vec{x} - \vec{\chi})\mathrm{H}(t - \tau); \qquad \gamma = F_i = 0 \tag{11.497}$$

The above can be substituted into the right hand sides of (11.482), (11.487) and (11.488). Due to the null right hand side of (11.482), it is obvious that

$$u_i^o = 0 \tag{11.498}$$

For the diffusion equations, we choose to solve (11.477) and (11.478) for ζ and s, and then resolve for Φ and Ψ. These two equations become

$$\frac{\partial \zeta^{hsc}}{\partial t} - c_c \nabla^2 \zeta^{hsc} - \frac{c_c \beta_g}{m_d} \nabla^2 s^{hsc} = 0 \tag{11.499}$$

$$\frac{\partial s^{hsc}}{\partial t} - c_b \nabla^2 s^{hsc} - M c_b \beta_h \nabla^2 \zeta^{hsc} = \frac{1}{\mathcal{T}_o} \delta(\vec{x} - \vec{\chi})\mathrm{H}(t - \tau) \tag{11.500}$$

where the superscript ^{hsc} marks continuous heat source solution. Apply Laplace transform to the above, we form the following set of coupled elliptic equations

$$\nabla^2 \tilde{\zeta}^{hsc} - \frac{s}{c_c} \tilde{\zeta}^{hsc} + \frac{\beta_g}{m_d} \nabla^2 \tilde{s}^{hsc} = 0 \tag{11.501}$$

$$\nabla^2 \tilde{s}^{hsc} - \frac{s}{c_b} \tilde{s}^{hsc} + M \beta_h \nabla^2 \tilde{\zeta}^{hsc} = -\frac{1}{\mathcal{T}_o c_b s} \delta(\vec{x} - \vec{\chi}) \tag{11.502}$$

The above system can be solved using the linear partial differential operator theory [35], as demonstrated in Cheng and Morohunfola [13].

We first express (11.501) and (11.502) into the matrix form

$$\begin{bmatrix} \nabla^2 - (s/c_c) & (\beta_g/m_d)\nabla^2 \\ M\beta_h \nabla^2 & \nabla^2 - (s/c_b) \end{bmatrix} \begin{Bmatrix} \tilde{\zeta}^{hsc} \\ \tilde{s}^{hsc} \end{Bmatrix} = \begin{Bmatrix} 0 \\ -\delta(\vec{x} - \vec{\chi})/\mathcal{T}_o c_b s \end{Bmatrix} \tag{11.503}$$

Applying Cramer's rule, the above can be solved as

$$\begin{vmatrix} \nabla^2 - (s/c_c) & (\beta_g/m_d)\nabla^2 \\ M\beta_h \nabla^2 & \nabla^2 - (s/c_b) \end{vmatrix} \tilde{\zeta}^{hsc} = \begin{vmatrix} 0 & (\beta_g/m_d)\nabla^2 \\ -\delta(\vec{x} - \vec{\chi})/\mathcal{T}_o c_b s & \nabla^2 - (s/c_b) \end{vmatrix} \tag{11.504}$$

$$\begin{vmatrix} \nabla^2 - (s/c_c) & (\beta_g/m_d)\nabla^2 \\ M\beta_h\nabla^2 & \nabla^2 - (s/c_b) \end{vmatrix} \tilde{s}^{hsc} = \begin{vmatrix} \nabla^2 - (s/c_c) & 0 \\ M\beta_h\nabla^2 & -\delta(\vec{x}-\vec{\chi})/\mathcal{T}_o c_b s \end{vmatrix} \tag{11.505}$$

The determinant on the left hand side can be multiplied out and factored to give the following set of decoupled partial differential equations

$$(\nabla^2 - s\mu_1^2)(\nabla^2 - s\mu_2^2)\,\tilde{\zeta}^{hsc} = \frac{\beta_g}{\mathcal{T}_o m_d c_b s}\nabla^2\delta(\vec{x}-\vec{\chi}) \tag{11.506}$$

$$(\nabla^2 - s\mu_1^2)(\nabla^2 - s\mu_2^2)\,\tilde{s}^{hsc} = -\frac{1}{\mathcal{T}_o c_b s}\left(\nabla^2 - \frac{s}{c_c}\right)\delta(\vec{x}-\vec{\chi}) \tag{11.507}$$

where $s\mu_1^2$ and $s\mu_2^2$ are eigenvalues of the system, given as

$$\mu_{1,2} = \sqrt{\frac{m_d(c_b+c_c) \pm \sqrt{m_d^2(c_b-c_c)^2 + 4m_d c_b c_c M\beta_g\beta_h}}{2c_b c_c(m_d - M\beta_g\beta_h)}} \tag{11.508}$$

We can obtain the solution of $\tilde{\Phi}$ by relaxing a Laplacian operator from (11.506), to give

$$(\nabla^2 - s\mu_1^2)(\nabla^2 - s\mu_2^2)\,\tilde{\Phi}^{hsc} = \frac{\beta_g}{\mathcal{T}_o m_d c_b s}\delta(\vec{x}-\vec{\chi}) \tag{11.509}$$

Two-Dimensional Heat Source For two-dimensional problem (a line source), solution of (11.509) can be obtained from (8.141) as

$$\tilde{\Phi}^{hsc} = -\frac{\beta_g}{2\pi\mathcal{T}_o m_d c_b s^2(\mu_1^2 - \mu_2^2)}\left[\mathrm{K}_0(\mu_1\sqrt{s}r) - \mathrm{K}_0(\mu_2\sqrt{s}r)\right] \tag{11.510}$$

Performing Laplace inversion (see Table D.1), we find

$$\begin{aligned}\Phi^{hsc} = \frac{\beta_g}{2\pi\mathcal{T}_o m_d c_b(\mu_1^2 - \mu_2^2)}\Bigg[&\frac{1}{2}te^{-\mu_1^2 r^2/4t} - \frac{1}{8}\left(\mu_1^2 r^2 + 4t\right)\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) \\ &-\frac{1}{2}te^{-\mu_2^2 r^2/4t} + \frac{1}{8}\left(\mu_2^2 r^2 + 4t\right)\mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\Bigg]\end{aligned} \tag{11.511}$$

Taking Laplacian of (11.510), we find

$$\nabla^2\tilde{\Phi}^{hsc} = -\frac{\beta_g}{2\pi\mathcal{T}_o m_d c_b s(\mu_1^2 - \mu_2^2)}\left[\mu_1^2\mathrm{K}_0(\mu_1\sqrt{s}r) - \mu_2^2\mathrm{K}_0(\mu_2\sqrt{s}r)\right] \tag{11.512}$$

Performing Laplace inversion, we obtain

$$\nabla^2\Phi^{hsc} = -\frac{\beta_g}{4\pi\mathcal{T}_o m_d c_b(\mu_1^2-\mu_2^2)}\left[\mu_1^2\,\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right)-\mu_2^2\,\mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\right] \tag{11.513}$$

We next seek the solution of (11.507). Observing the right hand side, we shall find the solution in two parts,

$$\Psi^{hsc} = \Psi_1^{hsc} + \Psi_2^{hsc} \tag{11.514}$$

where the first part is associated with the Laplacian of Dirac delta function, and the second part with the Dirac delta function. Similar to the solution of Φ^{hsc}, we obtain

$$\Psi_1^{hsc} = -\frac{m_d}{\beta_g}\,\Phi^{hsc} \tag{11.515}$$

and

$$\nabla^2\Psi_1^{hsc} = -\frac{m_d}{\beta_g}\,\nabla^2\Phi^{hsc} \tag{11.516}$$

We also find

$$\begin{aligned}\nabla^2\tilde{\Psi}_2^{hsc} &= \frac{m_d s}{\beta_g c_c}\,\Phi^{hsc}\\ &= -\frac{1}{2\pi\mathcal{T}_o c_b c_c s(\mu_1^2-\mu_2^2)}\left[\mathrm{K}_0(\mu_1\sqrt{s}r)-\mathrm{K}_0(\mu_2\sqrt{s}r)\right]\end{aligned} \tag{11.517}$$

which can be integrated twice to relax the Laplacian to give

$$\tilde{\Psi}_2^{hsc} = -\frac{1}{2\pi\mathcal{T}_o c_b c_c s^2(\mu_1^2-\mu_2^2)}\left[\frac{\mathrm{K}_0(\mu_1\sqrt{s}r)}{\mu_1^2}-\frac{\mathrm{K}_0(\mu_2\sqrt{s}r)}{\mu_2^2}\right] \tag{11.518}$$

Laplace inversion of the above gives

$$\nabla^2\Psi_2^{hsc} = -\frac{1}{4\pi\mathcal{T}_o c_b c_c(\mu_1^2-\mu_2^2)}\left[\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right)-\mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\right] \tag{11.519}$$

and

$$\begin{aligned}\Psi_2^{hsc} = \frac{1}{2\pi\mathcal{T}_o c_b c_c(\mu_1^2-\mu_2^2)}&\left[\frac{1}{2\mu_1^2}\,t\,e^{-\mu_1^2 r^2/4t}-\frac{1}{8\mu_1^2}\left(\mu_1^2 r^2+4t\right)\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right)\right.\\ &\left.-\frac{1}{2\mu_2^2}\,t\,e^{-\mu_2^2 r^2/4t}+\frac{1}{8\mu_2^2}\left(\mu_2^2 r^2+4t\right)\mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\right]\end{aligned} \tag{11.520}$$

Given (11.513), (11.516), and (11.519), we can assemble for the pore pressure solution based on (11.493)

$$p^{hsc}(r,t) = -\frac{\beta_g}{4\pi \mathcal{T}_o m_d S_c c_b c_c (\mu_1^2 - \mu_2^2)} \left[\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) - \mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right) \right] \tag{11.521}$$

in which we note that $g_1 = 0$. The temperature is found from (11.494) as

$$\begin{aligned} T^{hsc}(r,t) = & \frac{c_c \mu_1^2 (m_d - M\beta_g \beta_h) - m_d}{4\pi \mathcal{T}_o c_b c_c m_d S_b (\mu_1^2 - \mu_2^2)} \mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) \\ & - \frac{c_c \mu_2^2 (m_d - M\beta_g \beta_h) - m_d}{4\pi \mathcal{T}_o c_b c_c m_d S_b (\mu_1^2 - \mu_2^2)} \mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right) \end{aligned} \tag{11.522}$$

It is of particular interest to note that while the temperature is singular at the origin, due to the heat source, the pore pressure is not. At $r = 0$, it is in fact a constant

$$p^{hsc}(0,t) = \frac{\beta_g}{2\pi \mathcal{T}_o m_d S_c c_b c_c (\mu_1^2 - \mu_2^2)} \ln \frac{\mu_1}{\mu_2} \tag{11.523}$$

The displacement and stress can be obtained from (11.479) and (11.495), with (11.498), (11.511), (11.515), and (11.520), by differentiation.

In Fig. 11.13 we plot the fundamental solution based on the material constants of a rock salt, as given in Tables 3.1, 3.2, and 11.3. We assume a heat source of strength 628 W/m. The heat input to the medium is equivalent to that of the constant heat flux borehole problem solved in Sect. 11.8.2.

In Fig. 11.13a we plot the temperature rise from the initial temperature within 1 m distance from the heat source, at different times. We observe that the temperature is singular at the origin, as expected. The pore pressure response is plotted in Fig. 11.13b. As predicted by (11.523), the pore pressure is of finite magnitude at the origin.

Figure 11.13c shows that the fluid flux moves radially outward, but is zero at the origin. As there is no supply of fluid mass, the flow is driven by heat flux induced fluid expansion. In Fig. 11.13d the (negative) variation in fluid content is presented. We recall that a negative value of ζ means that fluid exits from the solid frame. It is observed that at smaller distance, fluid expands out of the solid frame, and it is unbounded at the origin. At large distance, fluid is absorbed into the frame.

The (negative) radial displacement is shown in Fig. 11.13e. It is of interest to note that the radial displacement moves inward, rather than outward, and it is unbounded at the origin. This is because that displacement must vanish at far field. As the solid expands, it can only move inward. The radial stress is plotted in Fig. 11.13f. It is in tension and unbounded at the origin as a consequence of the displacement field.

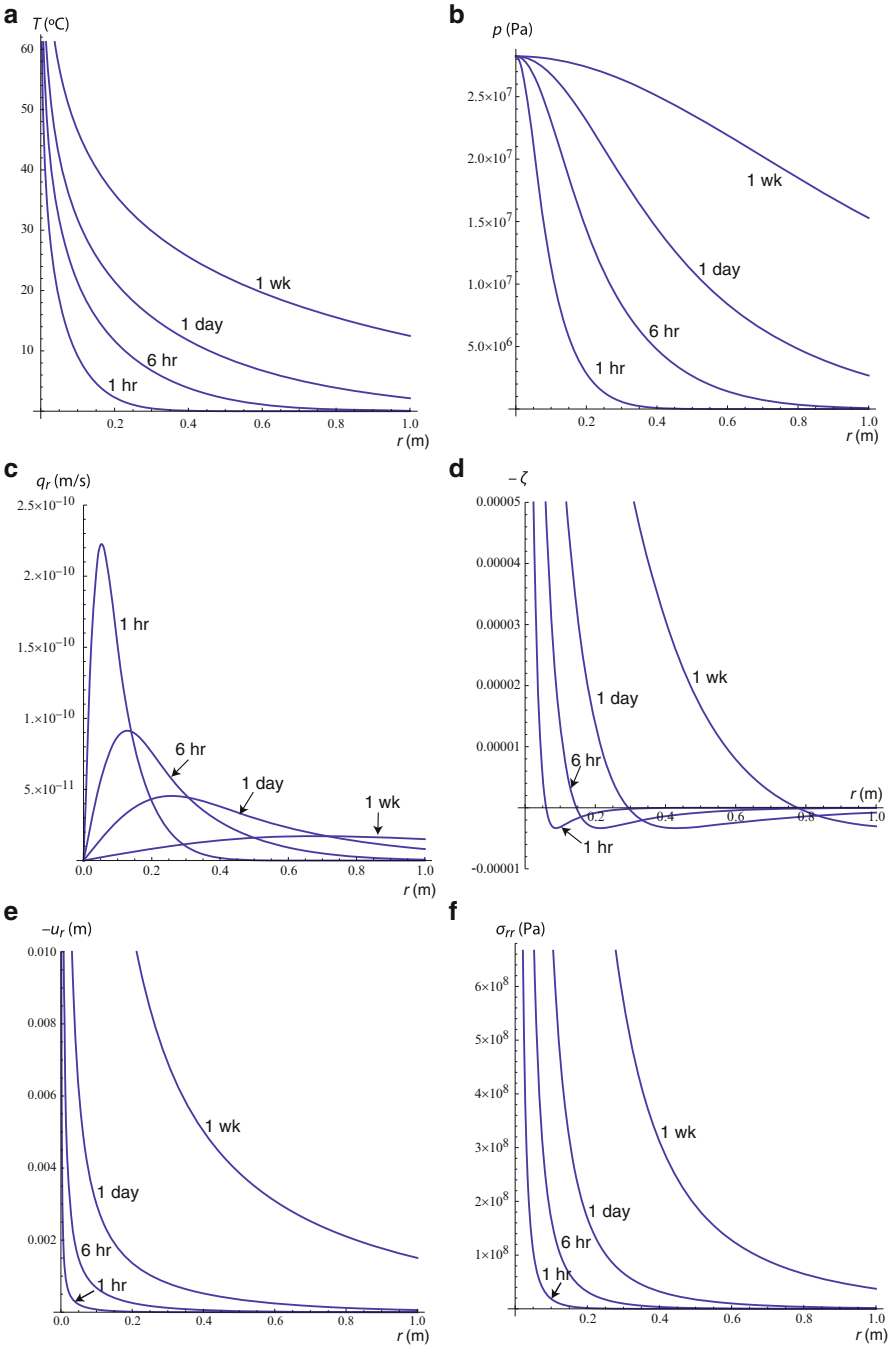


Fig. 11.13 Porothermoelastic fundamental solution of two-dimensional continuous heat source: (**a**) Temperature T, (**b**) pore pressure p, (**c**) radial fluid flux q_r, (**d**) variation in fluid content $-\zeta$, (**e**) radial displacement $-u_r$, and (**f**) radial stress σ_{rr}

Three-Dimensional Heat Source For three-dimensional heat source, solution of (11.509) can be obtained from (8.141) as

$$\tilde{\Phi}^{hsc} = -\frac{\beta_g}{4\pi \mathcal{T}_o m_d c_b s^2(\mu_1^2 - \mu_2^2) r}\left(e^{-\mu_1\sqrt{s}\,r} - e^{-\mu_2\sqrt{s}\,r}\right) \tag{11.524}$$

Performing Laplace inversion, we find

$$\begin{aligned}\Phi^{hsc} = {} & \frac{\beta_g}{4\pi \mathcal{T}_o m_d c_b(\mu_1^2 - \mu_2^2) r} \\ & \times\left[\mu_1 r\sqrt{\frac{t}{\pi}}\,e^{-\mu_1^2 r^2/4t} - \left(t + \frac{\mu_1^2 r^2}{2}\right)\operatorname{erfc}\left(\frac{\mu_1 r}{2\sqrt{t}}\right)\right. \\ & \left. -\mu_2 r\sqrt{\frac{t}{\pi}}\,e^{-\mu_2^2 r^2/4t} + \left(t + \frac{\mu_2^2 r^2}{2}\right)\operatorname{erfc}\left(\frac{\mu_2 r}{2\sqrt{t}}\right)\right]\end{aligned} \tag{11.525}$$

Taking Laplacian of (11.524), we find

$$\nabla^2\tilde{\Phi}^{hsc} = -\frac{\beta_g}{4\pi \mathcal{T}_o m_d c_b s(\mu_1^2 - \mu_2^2) r}\left(\mu_1^2 e^{-\mu_1\sqrt{s}\,r} - \mu_2^2 e^{-\mu_2\sqrt{s}\,r}\right) \tag{11.526}$$

Performing Laplace inversion, we obtain

$$\nabla^2\Phi^{hsc} = -\frac{\beta_g}{4\pi \mathcal{T}_o m_d c_b(\mu_1^2 - \mu_2^2) r}\left[\mu_1^2\operatorname{erfc}\left(\frac{\mu_1 r}{2\sqrt{t}}\right) - \mu_2^2\operatorname{erfc}\left(\frac{\mu_2 r}{2\sqrt{t}}\right)\right] \tag{11.527}$$

Similar to the two-dimensional case, we seek the solution of Ψ^{hsc} in two part. The first part Ψ_1^{hsc} is defined in the same relation as (11.515) and (11.516). The second part is associated with the Dirac delta function on the right hand side of (11.507); and we find

$$\nabla^2\tilde{\Psi}_2^{hsc} = -\frac{1}{4\pi \mathcal{T}_o c_b c_c s(\mu_1^2 - \mu_2^2) r}\left(e^{-\mu_1\sqrt{s}\,r} - e^{-\mu_2\sqrt{s}\,r}\right) \tag{11.528}$$

Integrating the above twice to relax the Laplacian, we find

$$\tilde{\Psi}_2^{hsc} = -\frac{1}{4\pi \mathcal{T}_o c_b c_c s^2(\mu_1^2 - \mu_2^2) r}\left[\frac{e^{-\mu_1\sqrt{s}\,r}}{\mu_1^2} - \frac{e^{-\mu_2\sqrt{s}\,r}}{\mu_2^2}\right] \tag{11.529}$$

Laplace inversion of the above gives

$$\nabla^2\Psi_2^{hsc} = -\frac{1}{4\pi \mathcal{T}_o c_b c_c(\mu_1^2 - \mu_2^2) r}\left[\operatorname{erfc}\left(\frac{\mu_1 r}{2\sqrt{t}}\right) - \operatorname{erfc}\left(\frac{\mu_2 r}{2\sqrt{t}}\right)\right] \tag{11.530}$$

and

$$\Psi_2^{hsc} = \frac{1}{4\pi \mathcal{T}_o c_b c_c (\mu_1^2 - \mu_2^2) r} \times \left[\frac{r}{\mu_1} \sqrt{\frac{t}{\pi}} e^{-\mu_1^2 r^2/4t} - \frac{1}{\mu_1^2} \left(t + \frac{\mu_1^2 r^2}{2} \right) \operatorname{erfc}\left(\frac{\mu_1 r}{2\sqrt{t}} \right) - \frac{r}{\mu_2} \sqrt{\frac{t}{\pi}} e^{-\mu_2^2 r^2/4t} + \frac{1}{\mu_2^2} \left(t + \frac{\mu_2^2 r^2}{2} \right) \operatorname{erfc}\left(\frac{\mu_2 r}{2\sqrt{t}} \right) \right] \tag{11.531}$$

Given (11.527), (11.516), and (11.530), we can assemble for the pore pressure solution based on (11.493)

$$p^{hsc}(r,t) = -\frac{\beta_g}{4\pi \mathcal{T}_o m_d S_c c_b c_c (\mu_1^2 - \mu_2^2) r} \left[\operatorname{erfc}\left(\frac{\mu_1 r}{2\sqrt{t}} \right) - \operatorname{erfc}\left(\frac{\mu_2 r}{2\sqrt{t}} \right) \right] \tag{11.532}$$

In this case, the pore pressure at the origin is not a constant; rather, it is inversely proportional to square root of time:

$$p^{hsc}(0,t) = \frac{\beta_g}{4\pi^{3/2} \mathcal{T}_o m_d S_c c_b c_c (\mu_1 + \mu_2)} t^{-1/2} \tag{11.533}$$

The temperature is found from (11.494) as

$$T^{hsc}(r,t) = \frac{c_c \mu_1^2 (m_d - M\beta_g \beta_h) - m_d}{4\pi \mathcal{T}_o c_b c_c m_d S_b (\mu_1^2 - \mu_2^2) r} \operatorname{erfc}\left(\frac{\mu_1 r}{2\sqrt{t}} \right) - \frac{c_c \mu_2^2 (m_d - M\beta_g \beta_h) - m_d}{4\pi \mathcal{T}_o c_b c_c m_d S_b (\mu_1^2 - \mu_2^2) r} \operatorname{erfc}\left(\frac{\mu_2 r}{2\sqrt{t}} \right) \tag{11.534}$$

The displacement and stress can be obtained from (11.479) and (11.495), with (11.498), (11.511), (11.515), and (11.520), by differentiation.

11.13.3 Continuous Fluid Source

Fundamental solution of continuous fluid source is obtained by introducing these forcing terms

$$\gamma = \delta(\vec{x} - \vec{\chi}) \mathrm{H}(t - \tau); \qquad \gamma_T = F_i = 0 \tag{11.535}$$

to the right hand sides of (11.477) and (11.478), and we find

$$\frac{\partial \zeta^{fsc}}{\partial t} - c_c \nabla^2 \zeta^{fsc} - \frac{c_c \beta_g}{m_d} \nabla^2 s^{fsc} = \delta(\vec{x} - \vec{\chi}) \mathrm{H}(t - \tau) \tag{11.536}$$

$$\frac{\partial s^{fsc}}{\partial t} - c_b \nabla^2 s^{fsc} - M c_b \beta_h \nabla^2 \zeta^{fsc} = 0 \tag{11.537}$$

where the superscript ^{fsc} marks the continuous fluid source solution.

Two-Dimensional Fluid Source By observing the symmetry between ζ and s in (11.536) and (11.537), the continuous fluid source solution can be obtained by replacing the various parameters by their counterparts; that is, $\Phi \leftrightarrow \Psi$, $c_b \leftrightarrow c_c$, $\beta_g \leftrightarrow \beta_h$, $m_d \leftrightarrow 1/M$, and $S_b \leftrightarrow S_c$. We hence obtain from (11.511) and (11.513) the following

$$\begin{aligned}\Psi^{fsc} = \frac{M\beta_h}{2\pi c_c(\mu_1^2 - \mu_2^2)} \Bigg[&\frac{1}{2} t e^{-\mu_1^2 r^2/4t} - \frac{1}{8}\left(\mu_1^2 r^2 + 4t\right) \mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) \\ &- \frac{1}{2} t e^{-\mu_2^2 r^2/4t} + \frac{1}{8}\left(\mu_2^2 r^2 + 4t\right) \mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\Bigg]\end{aligned} \tag{11.538}$$

and

$$\nabla^2 \Psi^{fsc} = -\frac{M\beta_h}{4\pi c_c(\mu_1^2 - \mu_2^2)} \left[\mu_1^2 \mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) - \mu_2^2 \mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\right] \tag{11.539}$$

The solution of Φ^{fsc} is expressed in two parts,

$$\Phi^{fsc} = \Phi_1^{fsc} + \Phi_2^{fsc} \tag{11.540}$$

where

$$\Phi_1^{fsc} = -\frac{1}{M\beta_h} \Psi^{fsc} \tag{11.541}$$

The second part is found as

$$\begin{aligned}\Phi_2^{fsc} = \frac{1}{2\pi c_b c_c(\mu_1^2 - \mu_2^2)} \Bigg[&\frac{1}{2\mu_1^2} t e^{-\mu_1^2 r^2/4t} - \frac{1}{8\mu_1^2}\left(\mu_1^2 r^2 + 4t\right) \mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) \\ &- \frac{1}{2\mu_2^2} t e^{-\mu_2^2 r^2/4t} + \frac{1}{8\mu_2^2}\left(\mu_2^2 r^2 + 4t\right) \mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\Bigg]\end{aligned} \tag{11.542}$$

and

$$\nabla^2 \Phi_2^{fsc} = -\frac{1}{4\pi c_b c_c(\mu_1^2 - \mu_2^2)} \left[\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) - \mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\right] \tag{11.543}$$

The temperature solution is then

$$T^{fsc}(r,t) = -\frac{M\beta_h}{4\pi S_b c_b c_c(\mu_1^2-\mu_2^2)}\left[\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right)-\mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right)\right] \tag{11.544}$$

and pore pressure is

$$\begin{aligned} p^{fsc}(r,t) = {} & \frac{c_b\mu_1^2(m_d - M\beta_g\beta_h)-m_d}{4\pi c_b c_c m_d S_c(\mu_1^2-\mu_2^2)}\mathrm{E}_1\left(\frac{\mu_1^2 r^2}{4t}\right) \\ & -\frac{c_b\mu_2^2(m_d - M\beta_g\beta_h)-m_d}{4\pi c_b c_c m_d S_c(\mu_1^2-\mu_2^2)}\mathrm{E}_1\left(\frac{\mu_2^2 r^2}{4t}\right) \end{aligned} \tag{11.545}$$

Three-Dimensional Fluid Source The three-dimensional continuous fluid source solutions can be obtained the same way as the two-dimensional ones, by replacing the parameters and materials constants by their corresponding part in the three-dimensional continuous heat source solutions.

11.13.4 Continuous Fluid and Heat Dipole

The dipole (a pair of source and sink pushed together) solution is obtained by taking the spatial derivative of the source solution, with a negative sign. As dipole is a vector, we can express

$$\Phi_i^{hpc} = -\Phi_{,i}^{hsc}; \qquad \Psi_i^{hpc} = -\Psi_{,i}^{hsc} \tag{11.546}$$

and

$$\Phi_i^{fpc} = -\Phi_{,i}^{fsc}; \qquad \Psi_i^{fpc} = -\Psi_{,i}^{fsc} \tag{11.547}$$

where the superscripts ^{hpc} and ^{fpc} stand for continuous heat dipole and continuous fluid dipole, respectively. We then obtain

$$p_i^{hpc} = -p_{,i}^{hsc}; \qquad T_i^{hpc} = -T_{,i}^{hsc} \tag{11.548}$$

and so forth.

11.13.5 Continuous Force

The continuous force solution corresponds to the forcing functions of

$$F_{ik} = \delta_{ik}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau); \qquad \gamma_T = 0 \tag{11.549}$$

where the subscript k marks the direction of the applied point force. According to (11.479), the solution consists of two parts, a part u_i^o satisfying the Navier equation (11.482), and a part given by the potentials Φ and Ψ. Substituting (11.549) to the right hand side of (11.482),

$$G u^o_{ik,jj} + \left(K_b + \frac{G}{3}\right) u^o_{jk,ji} = -\delta_{ik}\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau) \tag{11.550}$$

we can obtain u^o_{ik} from the well-known Kelvin solution (8.152) in the form

$$\begin{aligned} u^o_{ik} &= \frac{1}{8\pi G(1-\nu_b)}\left[\frac{x_i x_k}{r^2} - (3-4\nu_b)\delta_{ik}\ln r\right], \quad \text{(2D)} \\ &= \frac{1}{16\pi G(1-\nu_b)}\frac{1}{r}\left[\frac{x_i x_k}{r^2} + (3-4\nu_b)\delta_{ik}\right], \quad \text{(3D)} \end{aligned} \tag{11.551}$$

in which the Poisson ratio ν_b is defined based on the adiabatic undrained elastic constant

$$\nu_b = \frac{3K_b - 2G}{2(3K_b + G)} \tag{11.552}$$

The two diffusion equations (11.485) and (11.486) become

$$\frac{\partial \nabla^2 \Phi^{Fc}_k}{\partial t} - c_c \nabla^4 \Phi^{Fc}_k - \frac{c_c \beta_g}{m_d}\nabla^4 \Psi^{Fc}_k = \frac{\eta_b M \kappa}{G}\delta_{,k}(\vec{x} - \vec{\chi}) \tag{11.553}$$

and

$$\frac{\partial \nabla^2 \Psi^{Fc}_k}{\partial t} - c_b \nabla^4 \Psi^{Fc}_k - M c_b \beta_h \nabla^4 \Phi^{Fc}_k = \frac{\eta_u m_d \kappa_T}{m_u G}\delta_{,k}(\vec{x} - \vec{\chi}) \tag{11.554}$$

where the superscript ^{Fc} denotes the continuous force solution. By simple observation, we realize that Φ^{Fc}_k and Ψ^{Fc}_k can be obtained as the combination of derivatives of the fluid source and heat source solutions; in other words, as the combination of fluid dipole and heat dipole solutions. We hence find

$$\Phi^{Fc}_k = -\frac{\eta_b M \kappa}{G}\Phi^{fpc}_k - \frac{\eta_u m_d \kappa_T}{m_u G}\Phi^{hpc}_k \tag{11.555}$$

$$\Psi^{Fc}_k = -\frac{\eta_b M \kappa}{G}\Psi^{fpc}_k - \frac{\eta_u m_d \kappa_T}{m_u G}\Psi^{hpc}_k \tag{11.556}$$

From (11.489), we obtain

$$g_{1k} = \frac{1}{2\pi}\frac{x_k}{r^2}; \quad \text{(2D)}$$

$$= \frac{1}{4\pi}\frac{x_k}{r^3}; \quad \text{(3D)} \tag{11.557}$$

These expressions hence completely define the solutions u_{ik}^{Fc}, p_k^{Fc}, T_k^{Fc}, and σ_{ijk}^{Fc}, based on (11.479), (11.493), (11.494), and (11.495).

References

1. Abousleiman Y, Ekbote S (2005) Solutions for the inclined borehole in a porothermoelastic transversely isotropic medium. J Appl Mech ASME 72(1):102–114
2. Bear J, Corapcioglu MY (1981) A mathematical-model for consolidation in a thermoelastic aquifer due to hot water injection or pumping. Water Resour Res 17(3):723–736
3. Belotserkovets A, Prevost JH (2011) Thermoporoelastic response of a fluid-saturated porous sphere: an analytical solution. Int J Eng Sci 49(12):1415–1423
4. Biot MA (1956) Thermoelasticity and irreversible thermodynamics. J Appl Phys 27(3):240–253
5. Biot MA (1977) Variational Lagrangian-thermodynamics of nonisothermal finite strain mechanics of porous solids and thermomolecular diffusion. Int J Solids Struct 13(6):579–597
6. Boley BA, Weiner JH (1960) Theory of thermal stresses. Wiley, New York, 586pp
7. Booker JR, Savvidou C (1984) Consolidation around a spherical heat source. Int J Solids Struct 20(11–12):1079–1090
8. Booker JR, Savvidou C (1985) Consolidation around a point heat source. Int J Numer Anal Methods Geomech 9(2):173–184
9. Carnahan CL (1983) Thermodynamic coupling of heat and matter flows in near-field regions of nuclear waste repositories. MRS Proc 26:1023–1030
10. Carson JK, Lovatt SJ, Tanner DJ, Cleland AC (2005) Thermal conductivity bounds for isotropic, porous materials. Int J Heat Mass Transf 48(11):2150–2158
11. Chen GZ, Ewy RT (2005) Thermoporoelastic effect on wellbore stability. SPE J 10(2):121–129
12. Cheng AHD (2000) Multilayered aquifer systems—fundamentals and applications. Marcel Dekker, New York, Basel, 384pp
13. Cheng AHD, Morohunfola OK (1993) Multilayered leaky aquifer systems: I. Pumping well solution. Water Resour Res 29(8):2787–2800
14. Cheng AHD, Morohunfola OK (1993) Multilayered leaky aquifer systems: II. Boundary element solution. Water Resour Res 29(8):2801–2811
15. Coussy O (2004) Poromechanics. Wiley, Chichester, 298pp
16. Coussy O (2005) Poromechanics of freezing materials. J Mech Phys Solids 53(8):1689–1718
17. Coussy O, Monteiro PJM (2008) Poroelastic model for concrete exposed to freezing temperatures. Cem Concr Res 38(1):40–48
18. Cussler EL (2009) Diffusion: mass transfer in fluid systems, 3rd edn. Cambridge University Press, Cambridge/New York, 647pp
19. de Groot SR, Mazur P (1984) Non-equilibrium thermodynamics. Dover, New York, 526pp
20. de Marsily G (1986) Quantitative hydrogeology: groundwater hydrology for engineers. Academic, Orlando, 440pp
21. Demirel Y (2014) Nonequilibrium thermodynamics: transport and rate processes in physical, chemical and biological systems, 3rd edn. Elsevier, Amsterdam, 758pp
22. Derjaguin BV, Churaev NV, Muller VM (1987) Surface Forces. Plenum, New York, 440pp
23. Derski W, Kowalski S (1979) Equations of linear thermoconsolidation. Arch Mech 31(3):303–316
24. Dirksen C (1969) Thermo-osmosis through compacted saturated clay membranes. Soil Sci Soc Am J 33(6):821–826

25. Fourier JBJ (1822) Théorie Analytique de la Chaleur (The analytical theory of heat). Chez Firmin Didot, Père et Fils, Paris
26. Ghabezloo S, Sulem J, Saint-Marc J (2009) The effect of undrained heating on a fluid-saturated hardened cement paste. Cem Concr Res 39(1):54–64
27. Ghassemi A, Diek A (2002) Porothermoelasticity for swelling shales. J Pet Sci Eng 34 (1–4):123–135
28. Ghassemi A, Zhou X (2011) A three-dimensional thermo-poroelastic model for fracture response to injection/extraction in enhanced geothermal systems. Geothermics 40(1):39–49
29. Ghassemi A, Nygren A, Cheng AHD (2008) Effects of heat extraction on fracture aperture: a poro-thermoelastic analysis. Geothermics 37(5):525–539
30. Ghassemi A, Tao Q, Diek A (2009) Influence of coupled chemo-poro-thermoelastic processes on pore pressure and stress distributions around a wellbore in swelling shale. J Pet Sci Eng 67(1–2):57–64
31. Gonçalvès J, Trémosa J (2010) Estimating thermo-osmotic coefficients in clay-rocks: I. Theoretical insights. J Colloid Interface Sci 342(1):166–174
32. Gray DH, Mitchell JK (1967) Fundamental aspects of electro-osmosis in soils. J Soil Mech Found Div ASCE 93(SM6):209–236
33. Hashin Z, Shtrikman S (1962) A variational approach to theory of effective magnetic permeability of multiphase materials. J Appl Phys 33(10):3125–3131
34. Hollister CD, Anderson DR, Heath GR (1981) Subseabed disposal of nuclear wastes. Science 213(4514):1321–1326
35. Hörmander L (1969) Linear partial differential operators. Springer, Berlin/Heidelberg/New York, 285pp
36. Kanj M, Abousleiman Y (2005) Porothermoelastic analyses of anisotropic hollow cylinders with applications. Int J Numer Anal Methods Geomech 29(2):103–126
37. Katchalsky A, Curran PF (1967) Nonequilibrium thermodynamics in biophysics. Harvard University Press, Cambridge, 248pp
38. Kirkpatrick S (1973) Percolation and conduction. Rev Mod Phys 45(4):574–588
39. Kurashige M (1989) A thermoelastic theory of fluid-filled porous materials. Int J Solids Struct 25(9):1039–1052
40. Lachenbruch AH (1980) Frictional heating, fluid pressure, and the resistance to fault motion. J Geophys Res Solid Earth 85(B11):6097–6112
41. Landauer R (1952) The electrical resistance of binary metallic mixtures. J Appl Phys 23(7):779–784
42. Letey J, Kemper WD (1969) Movement of water and salt through a clay-water system: experimental verification of Onsager reciprocal relation. Soil Sci Soc Am J 33(1):25–29
43. Mase CW, Smith L (1985) Pore-fluid pressures and frictional heating on a fault surface. Pure Appl Geophys 122(2–4):583–607
44. Mase CW, Smith L (1987) Effects of frictional heating on the thermal, hydrologic, and mechanical response of a fault. J Geophys Res Solid Earth 92(B7):6249–6272
45. Mason EA, Wendt RP, Bresler EH (1972) Test of Onsager relation for ideal gas transport in membranes. J Chem Soc Faraday Trans II 68(11):1938–1959
46. Maxwell JC (1954) A treatise on electricity and magnetism, 3rd edn. Dover, New York, 560pp
47. McTigue DF (1986) Thermoelastic response of fluid-saturated porous rock. J Geophys Res Solid Earth 91(B9):9533–9542
48. McTigue DF (1990) Flow to a heated borehole in porous, thermoelastic rock: analysis. Water Resour Res 26(8):1763–1774
49. Miller DG (1959) Ternary isothermal diffusion and the validity of the Onsager reciprocity relations. J Phys Chem 63(4):570–578
50. Miller DG (1960) Thermodynamics of irreversible processes—the experimental verification of the Onsager reciprocal relations. Chem Rev 60(1):15–37
51. Msaad Y (2007) Comparison between hydraulic and thermal spalling in heated concrete based on numerical modeling. J Eng Mech ASCE 133(6):608–615

52. Neaupane KM, Yamabe T, Yoshinaka R (1999) Simulation of a fully coupled thermo-hydro-mechanical system in freezing and thawing rock. Int J Rock Mech Min Sci 36(5):563–580
53. Nowacki W (1986) Thermoelasticity, 2nd edn. Pergamon, Oxford/New York, 566pp
54. Onsager L (1931) Reciprocal relations in irreversible process I. Phys Rev 37(4):405–426
55. Onsager L (1931) Reciprocal relations in irreversible process II. Phys Rev 38(12):2265–2279
56. Palciauskas VV, Domenico PA (1982) Characterization of drained and undrained response of thermally loaded repository rocks. Water Resour Res 18(2):281–290
57. Powers DW, Lambert SJ, Shaffer SE, Hill LR, Weart WD (eds) (1978) Geological characterization report, Waste Isolation Pilot Plant (WIPP) site, Southeastern New Mexico, Technical report SAND78-1596, Sandia National Laboratories, Albuquerque
58. Rahman S, Grasley Z (2014) A poromechanical model of freezing concrete to elucidate damage mechanisms associated with substandard aggregates. Cem Concr Res 55:88–101
59. Rastogi RP, Singh K (1966) Cross-phenomenological coefficients. Part 5. Thermo-osmosis of liquids through cellophane membrane. Trans Faraday Soc 62:1754–1761
60. Rastogi RP, Agarwal RK, Blokhra RL (1964) Cross-phenomenological coefficients. Part 1. Studies on thermo-osmosis. Trans Faraday Soc 60:1386–1390
61. Schiffman RL (1971) A thermoelastic theory of consolidation. In: Cremers CJ, Kreith F, Clark JA (eds) Environmental and geophysical heat transfer. ASME, New York, pp 78–84
62. Selvadurai APS, Suvorov AP (2012) Boundary heating of poro-elastic and poro-elasto-plastic spheres. Proc R Soc A Math Phys Eng Sci 468(2145):2779–2806
63. Selvadurai APS, Suvorov AP (2014) Thermo-poromechanics of a fluid-filled cavity in a fluid-saturated geomaterial. Proc R Soc A Math Phys Eng Sci 470(2163):20130634
64. Skinner BJ (1966) Thermal expansion. In: Clark SP (ed) Handbook of physical constants. Memoirs (Geological Society of America), vol 97. Geological Society of America, New York, pp 75–96
65. Smith DW, Booker JR (1993) Green's functions for a fully coupled thermoporoelastic material. Int J Numer Anal Methods Geomech 17(3):139–163
66. Smith DW, Booker JR (1996) Boundary element analysis of linear thermoelastic consolidation. Int J Numer Anal Methods Geomech 20(7):457–488
67. Soler JM (2001) The effect of coupled transport phenomena in the Opalinus Clay and implications for radionuclide transport. J Contam Hydrol 53(1–2):63–84
68. Somerton WH (1992) Thermal properties and temperature-related behaviour of rock/fluid systems. Elsevier, Amsterdam, 256pp
69. Srivastava RC, Avasthi PK (1975) Nonequilibrium thermodynamics of thermo-osmosis of water through kaolinite. J Hydrol 24(1–2):111–120
70. Sugawara A, Yoshizawa Y (1961) An investigation on thermal conductivity of porous materials and its application to porous rock. Aust J Phys 14(4):468–469
71. Suzuki T, Yamashita T (2006) Nonlinear thermoporoelastic effects on dynamic earthquake ruptur. J Geophys Res Solid Earth 111(B3):B03307
72. Tao Q, Ghassemi A (2010) Poro-thermoelastic borehole stress analysis for determination of the in situ stress and rock strength. Geothermics 39(3):250–259
73. Tavman IH (1996) Effective thermal conductivity of granular porous materials. Int Commun Heat Mass Transf 23(2):169–176
74. Trémosa J, Gonçalvès J, Matray JM, Violette S (2010) Estimating thermo-osmotic coefficients in clay-rocks: II. In situ experimental approach. J Colloid Interface Sci 342(1):175–184
75. Vardoulakis I (2000) Catastrophic landslides due to frictional heating of the failure plane. Mech Cohesive-Frict Mater 5(6):443–467
76. Vardoulakis I (2002) Dynamic thermo-poro-mechanical analysis of catastrophic landslides. Géotechnique 52(3):157–171
77. Wang Y, Papamichos E (1994) Conductive heat flow and thermally induced fluid flow around a well bore in a poroelastic medium. Water Resour Res 30(12):3375–3384

78. Wiener O (1912) Die theorie des mischkörpers für das feld der stationären strömung. Erste abhandlung: Die mittelswertsätze für kraft, polarisation und energie (The theory of mixtures for the stationary flow field. First treatise: The averaging theorem for power, polarization and energy), Abhandlungen der Mathematisch-Physischen Klasse der Königlich-Sächsischen Gesellschaft der Wissenschaften 32:509–604
79. Woodside W, Messmer JH (1961) Thermal conductivity of porous media. I. Unconsolidated sands. J Appl Phys 32(9):1688–1699
80. Woodside W, Messmer JH (1961) Thermal conductivity of porous media. II. Consolidated rocks. J Appl Phys 32(9):1699–1706
81. Yeung AT, Mitchell JK (1993) Coupled fluid, electrical and chemical flows in soil. Géotechnique 43(1):121–134
82. Zimmerman RW (2000) Coupling in poroelasticity and thermoelasticity. Int J Rock Mech Min Sci 37(1–2):79–87

Chapter 12
Porochemoelasticity

If to any homogeneous mass we suppose an infinitesimal quantity of any substance to be added, the mass remaining homogeneous and its entropy and volume remaining unchanged, the increase of the energy of the mass divided by the quantity of the substance added is the potential for that substance in the mass considered.

—Josiah Willard Gibbs (1876)

The poroelasticity theory presented in Chaps. 2 through 10 models the force and energy interaction of mechanical origin between two material phases, a solid and a fluid. The thermal energy and force field are introduced in Chap. 11. In the physical world, there exist other types of energy and forces, such as those of electrical, magnetic, and chemical origins, and their coupling, such as the electromechanical (piezoelectric), electromagnetic, electrochemical, and magnetohydrodynamic forces. The simultaneous presence of these multiple physical phenomena, particularly their interactions, is known as *multiphysics*.

Depending on the engineering applications on hand and practical considerations, not all forces are present or significant, and need to be modeled. In geoscience, geophysical, and geomechanical applications, it is generally recognized that four processes, thermal (T), hydraulic (H), mechanical (M), and chemical (C), known as *THMC processes* [51–53], are more important. The poroelasticity theory considers the hydraulic and mechanical coupling (HM). The porothermoelasticity theory introduces the additional thermal coupling, and is a THM theory. In this chapter, we shall examine the THMC processes by including the chemical energy and force field and introduce the *porothermochemoelasticity* theory; though, for simplicity, we shall refer to it as the *porochemoelasticity* theory in this book.

Many geotechnical, biological, and synthetic porous media are chemically active, and exhibit swelling or shrinking behaviors when brought in contact with aqueous solutions. This phenomenon, observed in clays [5, 29, 49, 55], shales [14, 39, 45, 46], cartilage [19, 26, 62] and gels [22, 32], is caused by electric charges fixed to the solid, counteracted by corresponding charges in the fluid. These charges result in a variety of features, including swelling, chemico-osmosis, electro-osmosis, streaming potentials, streaming currents, and electrophoresis [34, 43].

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5_12

In biological and medical applications, swelling behavior is observed in cartilage, glycogalyx, cell, and skin [54]. Swelling of shales is of major problem for petroleum engineering. According to Steiger and Leung [50], shales make up more than 75 % of drilled formations and cause at least 90 % of wellbore-stability problems. If shale surrounding the wellbore swells, the wellbore diameter will be reduced, and the drill string and drill bit can become trapped, costing both time and money. Alternatively, if swollen shale disintegrates, motion of the drill string will be hindered by the soft, roughened walls of the well [35, 47]. Although the use of oil based drilling mud can reduce the chemical effect, its use is much restricted due to the environmental concern of its disposal [57]. Clay is used as liner and buffer for containment of landfill leachate and underground burial of nuclear or hazardous wastes in environmental geotechnology. The integrity of the barrier can be much affected by the swelling or shrinkage of the material.

Shales are fine-grained sedimentary rocks consisting of clay, silt, and mud. Clay minerals are basically crystalline and their properties are determined by the atomic structure of their crystals. When these minerals are exposed to a fluid with different physic-chemical properties, the microscopic changes take place, which manifest in macroscopic scale as swelling (or shrinkage) and an apparent pressure as *osmotic pressure*. Swelling mostly occurs in smectites and particularly montmorillonite, due to its expanding lattice and finely laminated structure, subject to cation exchange and water content.

In the following, we shall examine the modeling of the chemical effects through the concept of a *chemical potential*. The constitutive relations and transport laws are constructed to form governing equations that allow the mathematical solution of the various applications.

12.1 Electrochemical Effect

12.1.1 Thermodynamic Potential

In the absence of other driving forces, such as temperature, electrical potential, etc., molecules of a given species in a thermodynamic system tend to migrate from regions where they have a higher concentration to regions of lower concentration, until eventually all the differences in concentration are evened out everywhere, which we call a *chemical equilibrium*. An explanation for this *spontaneous process* can be derived from the kinetic theory and the random motion of molecules. However, it is simpler to describe the process in terms of a *force potential*, such as gravity potential, electrical potential, etc. We can consider that at a given temperature, a molecule has a higher *chemical potential* in a higher concentration area, and a lower chemical potential in a low concentration area. Just like a ball rolling downhill, moving from a higher gravitational potential to a lower one, releasing kinetic energy in the process, as molecules move, react, dissolve, melt, etc., they tend to go from a higher chemical potential region to a lower one, releasing an amount of *free energy*.

Gibbs Equation The concept of chemical potential and free energy was introduced by the great American physicist J. Willard Gibbs [16–18] (see Sect. F.11 for a biography). Gibbs free energy, originally called available energy [17], is: "*the greatest amount of mechanical work which can be obtained from a given quantity of a certain substance in a given initial state, without increasing its total volume or allowing heat to pass to or from external bodies, except such as at the close of the processes are left in their initial condition.*"

As discussed in Sect. 4.2, in the absence of kinetic, gravity, and other potential energies, the combination of the thermodynamics first law and the second law for reversible process gives us (4.49), which is repeated here for convenience

$$\delta \mathcal{U} = \delta \mathcal{W} + \mathcal{T} \delta \mathcal{S} \tag{12.1}$$

The above equation is derived for a closed system. With mass exchange associated with the chemical, electrical, and other potentials, we shall extend the analysis to open systems. This step was taken by Gibbs in his famous treatises on thermodynamics [16–18]. We shall follow his procedure by analyzing the work term, $\delta \mathcal{W}$.

Up to this point, we have defined the work as mechanical work only. To accommodate the energy transferred due to mass transfer, Gibbs considered different "forces" and work performed by these forces to the system. We may write

$$\mathcal{W} = \mathcal{W}_m + \mathcal{W}_c + \mathcal{W}_e + \dots \tag{12.2}$$

where $\mathcal{W}_m$ is the mechanical work, $\mathcal{W}_c$ is the chemical work by adding particles to the system against the chemical potential, $\mathcal{W}_e$ is the electrical work by adding electrical charges against electrical potential, etc. In this section, we shall consider $\mathcal{W}_m$ and $\mathcal{W}_c$ only, and the electrical effect will be introduced later.

Based on Gibbs' concept, we can express the variation of work as

$$\delta \mathcal{W} = \delta \mathcal{W}_m + \delta \mathcal{W}_c = -p \delta V + \sum_{k=1}^{n} \mu_k \delta N_k \tag{12.3}$$

where μ_k is the chemical potential per particle (or per mole) of the kth species (S.I. unit joule per mole), N_k is the number of particles (or moles) of the kth species, and n is the number of species that constitute the fluid. In the above, for simplicity, we have used p (pressure) times δV (change of volume) to represent mechanical work performed to a fluid, following thermodynamics literature. For elastic material, the mechanical work should be replaced by (4.51). Substituting (12.3) into (12.1), we obtain

$$\delta \mathcal{U} = -p \delta V + \mathcal{T} \delta \mathcal{S} + \sum_{k=1}^{n} \mu_k \delta N_k \tag{12.4}$$

This is *Gibbs equation*. Equation (12.4) is valid for open as well as closed and adiabatic systems.

As $\mathcal{U}$ is a state function of the state variables $(V, \mathcal{S}, N_k)$, we can express its total differential as

$$\delta\mathcal{U} = \frac{\partial\mathcal{U}}{\partial V}\delta V + \frac{\partial\mathcal{U}}{\partial\mathcal{S}}\delta\mathcal{S} + \sum_{k=1}^{n}\frac{\partial\mathcal{U}}{\partial N_k}\delta N_k \tag{12.5}$$

Based on (12.4) and (12.5), we find

$$p = -\frac{\partial\mathcal{U}}{\partial V} \tag{12.6}$$

$$\mathcal{T} = \frac{\partial\mathcal{U}}{\partial\mathcal{S}} \tag{12.7}$$

$$\mu_k = \frac{\partial\mathcal{U}}{\partial N_k} \tag{12.8}$$

In the above, we pay particular attention to (12.8), which shows that the chemical potential μ_k is a characteristic energy [3], defined as the change in internal energy due to the addition of N_k, at constant V, $\mathcal{S}$, and N_j, where $j \neq k$.

Gibbs-Duhem Equation As $\mathcal{U}$ is a function of the extensive properties $(V, \mathcal{S}, N_k)$, its value is directly proportional to the amount of mass in the system. In other words, in an equilibrated system, if we take one part of the mass containing a certain amount of internal energy, and another part with equal mass, and add them together, we will double the mass, the volume, the entropy, and the number of particles; and then the internal energy is also doubled. Mathematically, we identify such function as a *positive homogeneous function of first degree*.[1] Based on *Euler's homogeneous function theorem*,[2] we find

$$\mathcal{U} = V\frac{\partial\mathcal{U}}{\partial V} + \mathcal{S}\frac{\partial\mathcal{U}}{\partial\mathcal{S}} + \sum_{k=1}^{n}N_k\frac{\partial\mathcal{U}}{\partial N_k} \tag{12.9}$$

Substituting in (12.6), (12.7) and (12.8), we obtain the internal energy as

$$\mathcal{U} = \mathcal{T}\mathcal{S} - pV + \sum_{k=1}^{n}\mu_k N_k \tag{12.10}$$

[1] A positive homogeneous function is a real valued function with multiplicative scaling behavior. If the arguments of such function are multiplied by a positive factor, then the result is multiplied by some power of this factor. For example, given $f(x, y, z)$ as a multivariate real function, if $f(\lambda x, \lambda y, \lambda z) = \lambda^k f(x, y, z)$, where $\lambda > 0$, then f is a positive homogeneous function of kth degree. When $k = 1, f(\lambda x, \lambda y, \lambda z) = \lambda f(x, y, z), f$ is a positive homogeneous function of the first degree.

[2] Euler's homogeneous function theorem: Suppose that a real valued function $f(\vec{x})$ is continuously differentiable. Then f is a positive homogeneous function of degree k if and only if $\vec{x}\cdot\nabla f(\vec{x}) = kf(\vec{x})$. For $k = 1$, we have $f(x_1, x_2, \ldots) = x_1(\partial f/\partial x_1) + x_2(\partial f/\partial x_2) + \ldots$.

Taking variation of the above, we find

$$\delta \mathcal{U} = \mathcal{T}\delta \mathcal{S} + \mathcal{S}\delta \mathcal{T} - p\delta V - V\delta p + \sum_{k=1}^{n} \mu_k \delta N_k + \sum_{k=1}^{n} N_k \delta \mu_k \tag{12.11}$$

Comparing the above with (12.5), we find that the following condition must be satisfied

$$\mathcal{S}\delta \mathcal{T} - V\delta p + \sum_{k=1}^{n} N_k \delta \mu_k = 0 \tag{12.12}$$

This is the well-known Gibbs-Duhem equation, which describes the relationship between changes in chemical potential and the other state variables in a thermodynamic system [25, 40]. Particularly, for a system under isothermal and isobaric conditions,

$$\sum_{k=1}^{n} N_k \delta \mu_k = 0 \tag{12.13}$$

which shows that not all chemical potentials are independent.

Gibbs Free Energy Gibbs free energy $\mathcal{G}$ is defined as

$$\mathcal{G} = \mathcal{U} - \mathcal{T}\mathcal{S} + pV \tag{12.14}$$

This form of free energy is particularly useful under the conditions of constant temperature and pressure, which are conditions easiest to maintain in a chemical laboratory. Taking variation of (12.14), we obtain

$$\delta \mathcal{G} = \delta \mathcal{U} - \mathcal{T}\delta \mathcal{S} - \mathcal{S}\delta \mathcal{T} + p\delta V + V\delta p \tag{12.15}$$

Assuming the laboratory conditions of constant temperature and pressure, and using (12.1) in the above, we obtain

$$\delta \mathcal{G} = \delta \mathcal{W} + p\delta V = \delta \mathcal{W} - \delta \mathcal{W}_m \tag{12.16}$$

where the right hand side is the work done to the system (or the negative work of the system done to the environment) less that of the mechanical work. This part of work is called the *useful work* by Gibbs, which is the part of work contributed by the chemical reactions, the transport of electricity, etc. Hence Gibbs energy is related to the useful work under the above stated conditions.

Substituting (12.4) into (12.15), we find

$$\delta \mathcal{G} = -\mathcal{S}\delta \mathcal{T} + V\delta p + \sum_{k=1}^{n} \mu_k \delta N_k \tag{12.17}$$

The above expression gives another definition of the chemical potential, as compared to (12.8),

$$\mu_k = \left(\frac{\partial \mathcal{G}}{\partial N_k} \right)_{p,\mathcal{T},N_j} \tag{12.18}$$

In the above, it is reminded that the partial differentiation is taken by keeping p, $\mathcal{T}$, and N_j, where $j \neq k$, constant. This relation shows that the chemical potential μ_k is a *partial molar energy*, that is, energy per mole of the kth species, under the typical laboratory condition of constant temperature and pressure. Under these conditions, the Gibbs free energy is the sum of chemical energies of the various species

$$\delta \mathcal{G} = \mu_1 \delta N_1 + \mu_2 \delta N_2 + \cdots + \mu_n \delta N_n \tag{12.19}$$

Helmholtz Free Energy As demonstrated in Chaps. 4 and 11, for our purposes of constructing the constitutive equations, what is more useful is the Helmholtz free energy defined in (4.53). Utilizing the result derived in (4.56), we can express its variation as

$$\delta \mathcal{A} = \delta \mathcal{W} - \mathcal{S} \delta \mathcal{T} \tag{12.20}$$

Here $\mathcal{W}$ represents the total work given by (12.2), and not just mechanical work. Hence Helmholtz free energy is associates with the total work performed to the system under isothermal condition. With the existence of only mechanical and chemical work, substitution of (12.3) into the above yields

$$\delta \mathcal{A} = -p \delta V - \mathcal{S} \delta \mathcal{T} + \sum_{k=1}^{n} \mu_k \delta N_k \tag{12.21}$$

The above equation gives yet another definition of the chemical potential

$$\mu_k = \left(\frac{\partial \mathcal{A}}{\partial N_k} \right)_{V,\mathcal{T},N_j} \tag{12.22}$$

where $j \neq k$.

Intensive Properties The various energy quantities presented so far in this chapter are extensive properties. For the derivation of the continuum theory, we shall introduce the intensive properties as the extensive properties per unit volume. With these definitions, we can write (12.20) as

$$\delta \mathcal{A} = \delta \mathcal{W} - s \delta \mathcal{T} \tag{12.23}$$

where $\mathcal{A}$ and $\mathcal{W}$ are respectively the Helmholtz free energy density, and the work density, and s is the entropy density. Dividing (12.3) through by the volume V, we can express the equation for work as the following

$$\delta \mathcal{W} = -p\, e + \sum_{k=1}^{n} \mu_k \delta c_k \tag{12.24}$$

where e is the volumetric strain and

$$c_k = \frac{N_k}{V} \tag{12.25}$$

is the *molar concentration* (or *molarity*) of the kth species in the fluid, with the unit mol/m^3. The total molar concentration of the fluid is the sum of the partial concentrations of all species

$$c = \sum_{k=1}^{n} c_k \tag{12.26}$$

For a solid that can sustain shear stress and shear deformation, we may modify (12.24) to

$$\delta \mathcal{W} = \sigma_{ij} e_{ij} + \sum_{k=1}^{n} \mu_k \delta c_k \tag{12.27}$$

The Helmholtz free energy density $\mathcal{A}$ is then

$$\delta \mathcal{A} = -p\, e - s \delta \mathcal{T} + \sum_{k=1}^{n} \mu_k \delta c_k \tag{12.28}$$

This defines the chemical potential of the kth species as

$$\mu_k = \left(\frac{\partial \mathcal{A}}{\partial c_k} \right)_{e, \mathcal{T}, c_j} \tag{12.29}$$

where $j \neq k$. For a solid, we may modify (12.28) to

$$\delta \mathcal{A} = \sigma_{ij} e_{ij} - s \delta \mathcal{T} + \sum_{k=1}^{n} \mu_k \delta c_k \tag{12.30}$$

The Gibbs-Duhem equation (12.12) expressed in terms of intensive properties becomes

$$s \delta \mathcal{T} - \delta p + \sum_{k=1}^{n} c_k \delta \mu_k = 0 \tag{12.31}$$

Electrochemical Potential Up to this point the solute species are considered as electrically uncharged. This is however not the case if the solute contains electrolytes, such as salt, acid, etc., dissolved in a solvent such as water. When the solution contains charged particles, we must consider the coexisting electrical and chemical effects, namely the electrochemical effect. For this purpose, we can introduce an *electrochemical potential* for the kth ionic species as

$$\hat{\mu}_k = \mu_k + z_k \mathfrak{F} \psi \tag{12.32}$$

where μ_k is the chemical potential, z_k is the ionic valence, $\mathfrak{F}$ is Faraday constant ($= 9.65 \times 10^4$ coulomb/mole) $z_k \mathfrak{F}$ is the total charge carried by a mole of this substance, and ψ is the *electrical potential*. The ionic valence z_k may be positive, negative, or zero, depending on the charge of the species. As discussed in thermodynamics textbook [25], there is no general physical method that can isolate the chemical and electrical parts of the electrochemical potential; hence $\hat{\mu}_k$ should be regarded as the primary thermodynamic quantity for the characterization of electrochemical systems. With the above statement, all the thermodynamic relations derived in this section remain valid with μ_k replaced by $\hat{\mu}_k$. For simplicity, however, we shall continue to refer below the chemical potential, with the understanding that the results presented incorporate the electrochemical effect.

12.1.2 Clay as a Chemically Active Material

Clay minerals are found in soils and rocks. In fact, soil is a mixture of various minerals. By the composition of clay, silt, and sand particles, soil can be classified as clay, sandy clay, silty clay, loam, silt, sand, and many other soil types [7]. Shale is a fine-grained, clastic sedimentary rock that is also a mixture of clay and other minerals, especially quartz and calcite.

Clay minerals normally occur as crystals of colloidal size in which sheetlike silicate layers about 10 Å thick are stacked one above the other. Based on their chemical composition and molecular structure, clays are roughly classified into three groups: the kaolinite, montmorillonite/smectite, and illite [59]. The general structure of the kaolinite group is composed of silicate sheets (Si_2O_5) bonded to aluminum oxide/hydroxide layers called gibbsite ($Al(OH)_3$). The silicate and gibbsite (s-g) layers are tightly bonded together with only weak bonding existing between the s-g paired layers. The montmorillonite/smectite group is composed of silicate layers sandwiching a gibbsite (or brucite $Mg(OH)_2$) layer in between, in an s-g-s stacking sequence. Variable amount of water molecules would lie between the s-g-s sandwiches. The structure of the illite group is similar to the montmorillonite group with silicate layers sandwiching a gibbsite-like layer in between, in an s-g-s stacking sequence, with variable amount of water molecules lying between the s-g-s sandwiches as well as potassium ions.

The clay layers are negatively charged because of ionic substitutions at various sites within their structures and, as a result, exchangeable cations are adsorbed on their surfaces. Also, they are either fully expandable (separable by water to a distance that varies with the applied pressure), partially expandable (separable by water to a fixed distance of 5–10 Å that is essentially independent of the applied pressure), or nonexpandable, depending on the surface charge density and the size and valence of the exchangeable cations [60].

When a mass of clay crystals is placed in contact with water or an aqueous solution, water is absorbed between the layers. The interlayer in montmorillonite or smectites is expanded, and the separation between individual smectite sheets varies with the available water in the soil. Because of this, they are often referred to as "swelling clays". Soils having high concentrations of smectites can undergo as much as a 30 % volume change due to wetting and drying. Clay minerals also account for about 50–60 weight percentage of most shales, and are often responsible for many of the problems encountered when drilling through shales.

Osmosis and Osmotic Pressure *Osmosis* is the spontaneous net movement of solvent molecules through a *semipermeable membrane* into a region of higher solute concentration, in the direction that tends to equalize the solute concentrations on the two sides. Semipermeable membrane means that the membrane has the ability to block the transport of the solute of interest (such as salt ions), and to allow only the solvent molecules (such as water) to pass.

Figure 12.1 illustrates such a phenomenon by showing a container divided into two parts by a semipermeable membrane. The right side is filled with a solution containing a solute of certain concentration, and the left side is a pure solvent such as water (or a solution of lower concentration than the right). Figure 12.1a shows that initially the fluids are filled to the same level on both sides of the container, exerting equal hydraulic pressure on the membrane. The difference in the chemical potential of the two fluids creates a spontaneous thermodynamic response attempting to equalize the solute concentration on both sides. Due to the selective permeability of the membrane, the solute molecules or the ions are prohibited from migrating

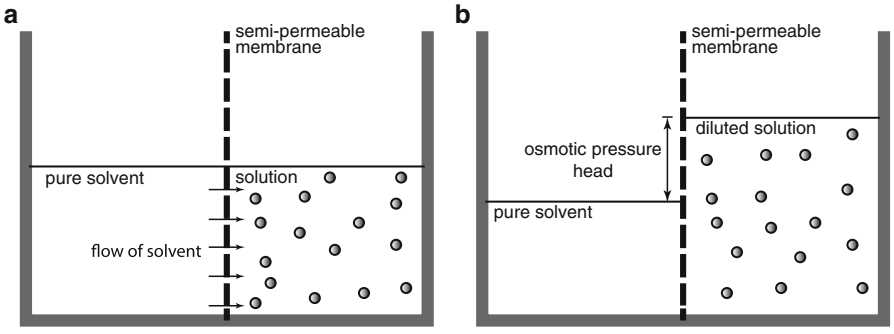


Fig. 12.1 Osmosis and osmotic pressure. (**a**) Osmosis taking place; and (**b**) system in equilibrium

from the right chamber to the left; hence the solvent molecules are propelled from the left chamber to the right. This net flux raises the fluid level on the right side and lowers that on the left. The higher fluid level creates a differential hydraulic pressure, causing a flow of solvent from right to left. Figure 12.1b shows that eventually these two flows balance each other out, and the system is in equilibrium. Under this condition, it appears that the solvent on the left side not only exerts a hydraulic pressure, but also an osmotic pressure equivalent to the extra solute column height on the right in order to maintain the force balance.

Osmosis is a vital process in biological systems, as biological membranes are semipermeable. Osmosis provides the primary means by which water is transported into and out of plant and animal cells. The osmotic force can be unexpectedly large. For example, a 1 % sugar solution can create an osmotic pressure of 2/3 atm. It is such force that allows a tree to lift its sap tens of feet above ground to its topmost twigs by the osmatic pressure at it roots [58].

Non-ideal Semipermeable Membrane Clay is made up of aggregated particles, which comprises of individual clay platelets stacked in layers. The platelets are negatively charged, thus they attract positively charged ions in between them and at the outer surfaces. The electric field created by the surface charges repels ions, such as Na^+, K^+, and Ca^{2+}, in the solution. When two such clay particles are close together in a porous medium and their electrical fields overlap, such as illustrated in Fig. 12.2a, charged ions are blocked from passage; hence the channel effectively serves as an *ideal semipermeable membrane*. However, if the two clay particles are far apart, and their electrical fields do not overlap, such as illustrated in Fig. 12.2b, then there exists a channel through which ions can pass freely. The clay particle

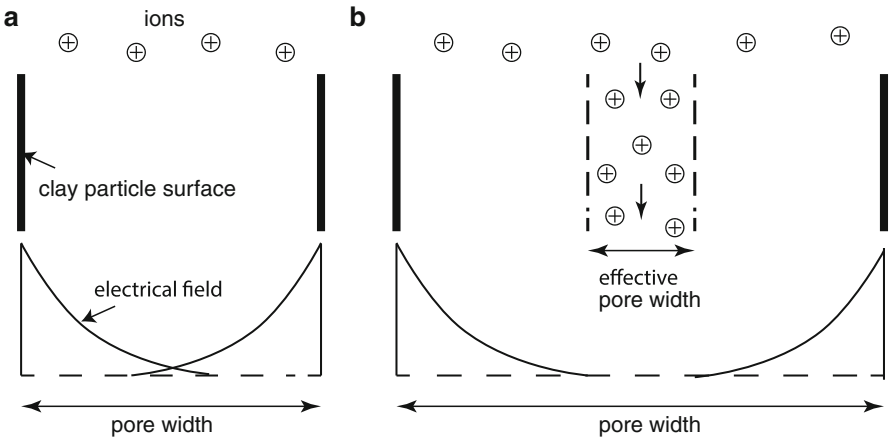


Fig. 12.2 Clay as semipermeable membrane. (**a**) Electrical field overlapping, ideal semipermeable membrane; and (**b**) electrical field non-overlapping, leaky semipermeable membrane

assembly is then a *non-ideal semipermeable membrane*, or a *leaky membrane*. We can define a membrane efficiency ω_e. The ideal membrane has the efficiency of $\omega_e = 1$, and the leaky membrane has $0 < \omega_e < 1$. When clay particles are not present, there is no membrane effect, and $\omega_e = 0$.

Clay Swelling Mechanisms There are two types of clay swelling mechanisms, the *innercrystalline swelling* and the *osmotic swelling* [31]. In the fully dry montmorillonite, the layers lie so close together that they are almost in contact. The negatively charged layers are held together by the interlayer cations and the van der Waals force at this small distance. When in contact with water, the cations hydrate and order themselves on a plane halfway between the clay layers. The act of water molecules entering the interlayer space is called *absorption*, which leads to a widening of the spacing between the layers (see Fig. 12.3). The volume of montmorillonite, for example, can double in the process of innercrystalline swelling. The hydration energy is significantly higher than the bond between the clay layers, and swelling pressures of more than 100 N/mm^2 can result.

Unlike innercrystalline swelling, which acts over small distances (up to 1 nm), osmotic swelling, which is based on the repulsion between electric double layers, can act over much larger distances. A high negative potential exists directly at the surface of the clay layer. The negatively charged clay surface must be compensated by positive ions close to the surface. The positive ion concentration decreases with increasing distance from the surface, where the concentration of negative ions increases. The cloud of ions hence forms a diffuse electric double layer. Water contained within the cloud is considered as *adsorbed*. When two such negative potential fields overlap, they repel each other, causing the swelling of clay (Fig. 12.3). This swelling stress, however, is significantly smaller than that of the innercrystalline swelling, and hardly exceeds 2 N/mm^2.

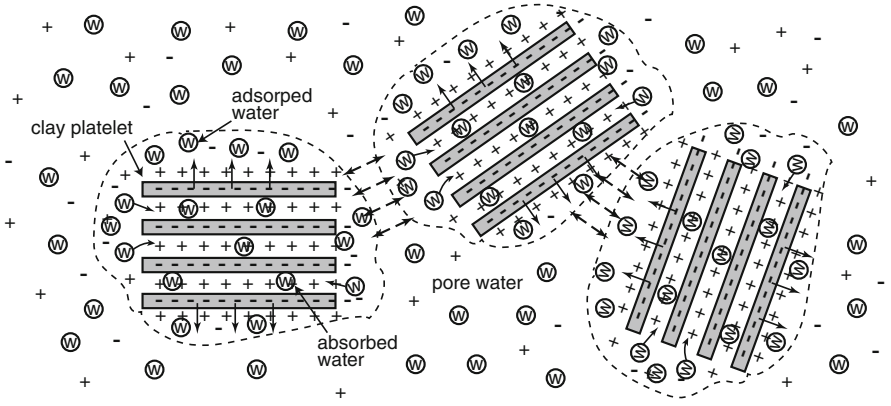


Fig. 12.3 Mechanisms for clay innercrystalline swelling and osmotic swelling

12.1.3 Chemical Potential and Osmotic Pressure for Dilute Solution

In the preceding section we have defined the chemical potential μ_k as the partial molar energy of the kth species, which can be a function of various forms of energy, such as temperature, pressure, and species concentration. In this section we shall seek an explicit expression of the chemical potential for an ideal dilute solution to these quantities, and particularly its relation to the osmotic pressure.

Chemical Potential Based on the definition of (12.18), we can differentiate the chemical potential with respect to pressure, and obtain

$$\frac{\partial \mu_k}{\partial p} = \frac{\partial}{\partial p}\left(\frac{\partial \mathcal{G}}{\partial N_k}\right) = \frac{\partial}{\partial N_k}\left(\frac{\partial \mathcal{G}}{\partial p}\right) \tag{12.33}$$

According to (12.17), we have

$$\frac{\partial \mathcal{G}}{\partial p} = V \tag{12.34}$$

where V is the volume of the mixture (solution). Hence we find

$$\frac{\partial \mu_k}{\partial p} = \frac{\partial V}{\partial N_k} = v_k \tag{12.35}$$

where we define v_k as the *partial molar volume* of species k, that is, the partial volume occupied by 1 mole of species k in the solution.

Assume that the solute volume V is an extensive property of the various particles, we can express

$$V = \sum_{k=1}^{n} N_k \frac{\partial V}{\partial N_k} = \sum_{k=1}^{n} N_k v_k \tag{12.36}$$

based on Euler's homogeneous function theorem (see Footnotes 1 and 2 in Sect. 12.1.1). Divide the above equation by the total number of moles contained in the volume

$$N = \sum_{k=1}^{n} N_k \tag{12.37}$$

we obtain the average molar volume for the solution as

$$v = \frac{V}{N} = \sum_{k=1}^{n} x_k v_k \tag{12.38}$$

where

$$x_k = \frac{N_k}{N} \tag{12.39}$$

is the *molar fraction* of species k. It is obvious that

$$\sum_{k=1}^{n} x_k = 1 \tag{12.40}$$

We can also connect the above quantities with the earlier introduced partial and total molar concentration, c_k and c, defined in (12.25) and (12.26), as

$$c_k = \frac{x_k}{v} = x_k\, c \tag{12.41}$$

$$c = \frac{1}{v} = \sum_{k=1}^{n} c_k \tag{12.42}$$

Experimental evidences indicate that in many cases the partial molar volumes v_k of the various components in a liquid mixture are approximately constant; hence we can integrate (12.35) with respect to p to obtain

$$\mu_k = p\, v_k + \mu_k^c(\mathcal{T}, N_i) \tag{12.43}$$

where μ_k^c is an integration constant that is dependent on the absolute temperature $\mathcal{T}$ and all the chemical composition N_i (that is, all the variables that were kept constant in performing the partial differentiation in (12.35)). Assume that the mixture obeys the ideal gas law, we can express

$$\mu_k^c = \mathcal{R}\mathcal{T} \ln x_k + \mu_k^o(\mathcal{T}) \tag{12.44}$$

where $\mathcal{R} = 8.3144621\,\mathrm{J/^\circ K \cdot mol}$ is the gas constant, and $\mu_k^o(\mathcal{T})$ is a constant that is dependent on the temperature only.

Consider the special case of a binary system containing a solvent and a single solute species. We can express these chemical potentials as

$$\mu_w = p\, v_w + \mathcal{R}\mathcal{T} \ln x_w + \mu_w^o(\mathcal{T}) \tag{12.45}$$

$$\mu_s = p\, v_s + \mathcal{R}\mathcal{T} \ln x_s + \mu_s^o(\mathcal{T}) \tag{12.46}$$

where the subscripts $_w$ and $_s$ respectively denote the solvent (water) and the solute. In the above form, we have isolated the chemical potential to a part that is dependent on the pressure, a part on the solute concentration, and a constant dependent on temperature. At constant temperature, we can write the variation of the above as

$$\delta\mu_w = v_w \delta p + \frac{\mathcal{RT}}{x_w}\delta x_w \tag{12.47}$$

$$\delta\mu_s = v_s \delta p + \frac{\mathcal{RT}}{x_s}\delta x_s \tag{12.48}$$

Here we recall that v_w and v_s are constant. Or, expressed into molar concentrations, and linearizing, we have

$$\delta\mu_w = v_w \delta p + \frac{\mathcal{RT}}{c_w^o}\delta c_w \tag{12.49}$$

$$\delta\mu_s = v_s \delta p + \frac{\mathcal{RT}}{c_s^o}\delta c_s \tag{12.50}$$

where c_w and c_s are the molar concentration of water and solute, respectively, and c_w^o and c_s^o are their initial values.

Chemical Activity Equations (12.45) and (12.46) are derived for an ideal solution. Real solutions approach the behavior predicted by these equations only at infinite dilution. To account for the deviation from ideality, the common practice is to introduce an *activity coefficient* γ_k to multiply x_k to give the active or effective concentration, known as the *activity* a_k, as

$$a_k = \gamma_k x_k \tag{12.51}$$

such that

$$\mu_k = p\, v_k + \mathcal{RT} \ln a_k + \mu_k^o(\mathcal{T}) \tag{12.52}$$

The activity coefficient has the property that as $x_k \to 0$, $\gamma_k \to 1$.

Osmotic Pressure Now consider the osmotic phenomenon as illustrated in Fig. 12.1, in which a solution containing a solute maintains equilibrium with an infinite reservoir of pure solvent, partitioned by an ideal semipermeable membrane. As the function of the semipermeable membrane is to block the solutes, while allowing the solvent to freely flow through, when the system reaches equilibrium, the chemical potentials of the solvent on both sides of the membrane, as well as in the membrane, are equal,

$$\delta\mu_w = 0 \tag{12.53}$$

This allows us to express

$$v_w \delta p = -\frac{\mathcal{RT}}{x_w}\delta x_w = \frac{\mathcal{RT}}{1 - x_s}\delta x_s \tag{12.54}$$

Here we recall that

$$x_w + x_s = 1 \tag{12.55}$$

We therefore see that the right hand side term of (12.54), which is dependent on the solute molar fraction x_s, plays the role of a chemical pressure that serves to balance the mechanical pressure p.

Assume that the right side chamber of Fig. 12.1b is initially without solute concentration, and corresponds to an initial pressure p_o. If the concentration is continuously increased to x_s, while the equilibrium is maintained all the time with the left side chamber of pure solvent, the pressure will increase to a value $p_o + \Pi$. We can integrate the left side and right side of (12.54) based on these limits

$$\int_p^{p_o+\Pi} v_w \, dp = \mathcal{RT} \int_0^{x_s} \frac{1}{1-x_s} \, dx_s \tag{12.56}$$

and obtain

$$\Pi = -\frac{\mathcal{RT}}{v_w} \ln(1 - x_s) \tag{12.57}$$

Here Π is known as the *osmotic pressure*. Substituting (12.57) back into (12.45), we obtain

$$\mu_w = v_w(p - \Pi) + \mu_w^o(\mathcal{T}) \tag{12.58}$$

which shows the role of Π balancing the mechanical pressure.

Dilute Solution For a dilute solution

$$x_s \ll 1; \quad x_w \approx 1; \quad \ln(1 - x_s) \approx -x_s; \quad N_w \approx N; \quad v_w \approx v \tag{12.59}$$

These allows us to approximate (12.47) as

$$\delta\mu_w \approx v_w \delta p - \mathcal{RT}_o \delta x_s \tag{12.60}$$

Or, expressed in molar concentration, (12.48) and (12.60) becomes

$$\delta\mu_w \approx v_w \delta p - v_w \mathcal{RT}_o \delta c_s \tag{12.61}$$

$$\delta\mu_s \approx v_s \delta p + \frac{\mathcal{RT}_o}{c_o} \delta c_s \tag{12.62}$$

where we have linearized (12.66) by approximating c_s by its initial value c_o. For osmotic pressure, we can express (12.57) as

$$\Pi \approx \frac{\mathcal{R}\mathcal{T}_o}{v_w} x_s \approx \mathcal{R}\mathcal{T}_o\, c_s \tag{12.63}$$

Equation (12.63) is the *van't Hoff's equation of osmotic pressure* for dilute solution. Its variation is

$$\delta\Pi \approx \frac{\mathcal{R}\mathcal{T}_o}{v_w} \delta x_s \approx \mathcal{R}\mathcal{T}_o\, \delta c_s \tag{12.64}$$

Substituting (12.64) into (12.48) and (12.60), we obtain

$$\delta\mu_w \approx v_w(\delta p - \delta\Pi) \tag{12.65}$$

$$\delta\mu_s \approx v_s \delta p + \frac{\delta\Pi}{c_o} \tag{12.66}$$

12.2 Micromechanical Analysis

The processes of chemically and electrochemically induced swelling in clay and clay-rich soils and rocks are highly complex [6, 33]. The description given in the preceding section barely touches such complexity. For application purposes, however, constitutive relations are needed for the modeling. Such relations have been attempted through a number of approaches, for example, as empirical relations [28, 30], elasto-plastic models [2, 61], mixture theory [1, 5, 23], and molecular dynamic simulations [21, 24].

For the purpose of this book, we shall examine only the "Biot-like" models, that is, models that considers the mechanical interaction between the solid and fluid phases, with the additional coupling effects of solute diffusion and the chemically induced mechanical responses. Mody and Hale [35] and Sherwood [45] appear to be the first to develop Biot-like models. These and other models [39, 56] consider the shale as a perfect semipermeable membrane, or an imperfect membrane with a given membrane efficiency, thus only the differences in chemical potential between the pore water in porous medium and the exterior water that it is in contact with plays a role in the model. However, it is known that clays and shales are not ideal membranes and do allow the transfer of ions [11, 48, 57], hence, ion transfer and associated effects should be considered.

Using arguments of irreversible thermodynamics, Heidug and Wong [20] developed a fully coupled model that coupled solute transfer in the mud/shale system with the stress and pore pressure in the formation. The model explicitly considered the "bound water" (innercrystalline water), and incorporated its chemical and

mechanical responses as a part of the solid constituent. Ghassemi and Diek [14] extended the model to admit compressible constituents. In these models, the state variable for the chemical effect is reduced and represented as the solute concentration. Sarout and Detournay [44] presented a formulation that uses the osmotic pressure as the state variable. Other models include Moyne and Murad [36] based on homogenization, Dormieux et al. [8] based on micromechanical modeling, Gajo et al. [12] and Loret et al. [27] that considered both the absorbed and adsorbed water, and Bennethum et al. [5] and Murad and Cushman [37] based on hybrid mixture theory.

In the following, we shall present a model similar to that of Heidug and Wong [20] and Ghassemi and Diek [14], but with the micromechanical analysis based on variational energy principles, as presented for the quasi-static poroelasticity theory in Chap. 4 and for porothermoelasticity in Chap. 11. Also, following Sarout and Detournay [44], osmotic pressure is introduced as a state variable.

Physicochemical Assumptions In the model to follow, we shall assume that in addition to the solid and the *free* fluid phase considered so far for poroelastic materials, there exist a fluid phase that is absorbed into the inner space of clay platelets (innercrystalline water) and a fluid phase that is adsorbed onto the surface and the surrounding of the clay aggregate (osmotic water) (see Fig. 12.3). When a dry or unsaturated clay is supplied with water, the hydration of the platelets, that is, water molecules entering the inner space, can cause a dramatic swelling of the clay. For the continuum mechanics modeling purpose, this phenomenon is in the large deformation range.

In the present study, we shall assume that the initial hydration has already taken place, and the porous material is originally saturated. Hydration can still happen when the chemical potential of the surrounding fluid changes. However, as indicated in the preceding section, the hydration pressure for the innercrystaline swelling in much greater than that of the osmotic swelling; hence it is difficult for additional water to enter the clay inner space. We anticipate the swelling is caused by the osmotic, and not by the innercrystaline hydration mechanism.

Although there does not exist a physical membrane like biological materials, the electrical field formed around a clay particle has the effect of an osmotic barrier that excludes certain ions from entering, while allows the water molecules to pass. Due to the osmotic effect, the hydraulic pressure inside the "membrane" is different from that of the free pore water outside of the influence of the electric field. The modeling of these absorbed and adsorbed water with different pressure poses challenges for a practical model. As the materials that we consider are not pure clays, and are soils and rocks with certain amount of clay content, the adsorbed water surrounding the clay particles are generally not connected to form an equilibrated fluid phase of a single pressure and chemical potential within and REV. For a meaningful continuum model, such pressure needs to be averaged over the REV just like the solid stresses, as discussed in Sect. 4.1.2. From practical considerations, it is unlikely that such averaged pressure and chemical potential can be measured. Hence we make an

important assumption based on the practicality of the model: we shall consider the absorbed and adsorbed water as a part of the solid phase in the averaging process of the micromechanics model.

The assumption that the domains of the adsorbed water do not overlap to form a connected phase also means that the porous material as a whole is not a perfect semipermeable membrane, as demonstrated in Fig. 12.2b. This suggests that the solutes can diffuse into the porous medium and the solute mass transport needs to be modeled.

For the fluid, we assume that it is a dilute solution, with a single solvent containing a number of solute species each in small concentration. There is no chemical reaction taking place among the species. For convenience, the model is sometimes simplified to contain one solute species only.

Finally, we make the important assumption in the constitutive relation modeling: we assume that chemical equilibrium within the REV is maintained at all times. This means that all the fluids in the REV, the free pore water, the absorbed and adsorbed water, as well as the solid, are in chemical equilibrium with each other. This assumption is consistent with the local pore pressure equilibrium in quasi-static poroelasticity and the local thermal equilibrium in porothermoelasticity. These assumptions do not imply the global equilibrium of these quantities. The assumption holds only if the global diffusion time is much larger than the local diffusion time needed to reach equilibrium. If such condition is not satisfied, then the local mass transfer laws among the fluids need to be established, leading to a more complex model.

Virtual Work Consider a solid phase and a fluid phase. The virtual work conducted on the solid phase can be expressed as

$$\delta \mathcal{W}_s = \overline{\sigma}^s_{ij}\, \overline{e}^s_{ij} + \sum_{k=1}^{n} \overline{\mu}^b_k\, \delta \overline{c}^b_k \tag{12.67}$$

based on (12.27). In the above, $\overline{\sigma}^s_{ij}$ and $\overline{e}^s_{ij}$ are the externally averaged solid stress and strain, incorporating the bound fluid as a part of solid, $\overline{\mu}^b_k$ is the chemical potential of the kth species of the *bound fluid*, and $\overline{c}^b_k$ is its concentration. Here we shall use the term "bound fluid" to refer to both the absorbed and adsorbed fluid. We also note that the summation $k = 1, \ldots, n$ is over all the species in the fluid. For a dilute solution, this includes the solutes as well as the solvent.

Similarly, for the virtual work done to the fluid phase, we express

$$\delta \mathcal{W}_f = -\overline{p}^f\, \overline{e}^f + \sum_{k=1}^{n} \overline{\mu}^p_k\, \delta \overline{c}^p_k \tag{12.68}$$

where the superscript ^{p} denotes the pore fluid. We note that although we assume that there exists local chemical equilibrium, we shall keep the chemical potentials of the bound fluid and pore fluid different for now, and equate them at a later stage. The total virtual work applied to the REV is then

$$\delta \mathcal{W} = (1-\phi)\,\delta \mathcal{W}_s + \phi\,\delta \mathcal{W}_f \tag{12.69}$$

where ϕ is the effective porosity, referring to the space occupied by the free pore fluid.

Helmholtz Free Energy We shall define the solid and fluid Helmholtz free energy density as functions of the following state variables.

$$\overline{\mathcal{A}}_s = \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi, \overline{\mathcal{T}}_s, \overline{c}^b_k) \tag{12.70}$$

$$\overline{\mathcal{A}}_f = \overline{\mathcal{A}}_f(\overline{\epsilon}^f, \overline{\mathcal{T}}_f, \overline{c}^p_k) \tag{12.71}$$

Comparing the above to the case of porothermoelasticity in (11.3) and (11.4), we notice the additional state variables $\overline{c}^b_k$ and $\overline{c}^p_k$, which introduce the chemical effect. We further note that the solid Helmholtz free energy becomes dependent on the chemical species by the virtue of the presence of the bound water, which is considered a part of the solid.

Following the formality of the variational energy principles, we can construct the fully coupled constitutive equations for porothermochemoelasticity. However, for simplicity, in this book we shall construct only the porochemoelasticity model by ignoring the thermal effect. Some attempts to construct the fully or partially coupled porothermochemoelastic models can be found in [15, 37, 41].

Under isothermal condition, (12.70) and (12.71) become

$$\overline{\mathcal{A}}_s = \overline{\mathcal{A}}_s(\overline{\epsilon}^s_{ij}, \phi, \overline{c}^b_k) \tag{12.72}$$

$$\overline{\mathcal{A}}_f = \overline{\mathcal{A}}_f(\overline{\epsilon}^f, \overline{c}^p_k) \tag{12.73}$$

The total Helmholtz free energy is

$$\overline{\mathcal{A}} = (1-\phi)\overline{\mathcal{A}}_s + \phi\,\overline{\mathcal{A}}_f \tag{12.74}$$

Taking the variation of the above with respect to the state variables, we obtain

$$\begin{aligned}\delta\overline{\mathcal{A}} = {} & (1-\phi)\frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s_{ij}}\delta\overline{\epsilon}^s_{ij} + (1-\phi)\frac{\partial \overline{\mathcal{A}}_s}{\partial \phi}\delta\phi + (1-\phi)\frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{c}^b_k}\delta\overline{c}^b_k \\ & + \phi\frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{\epsilon}^f}\delta\overline{\epsilon}^f + \phi\frac{\partial \overline{\mathcal{A}}_f}{\partial \overline{c}^p_k}\delta\overline{c}^p_k\end{aligned} \tag{12.75}$$

Utilizing (4.37) and (4.38), we can convert the internal strains in the above to the external strains, as

$$\delta\overline{\mathcal{A}} = (1-\phi)\frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s_{ij}}\delta\overline{e}^s_{ij} - \frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{\epsilon}^s}\delta\phi + (1-\phi)\frac{\partial \overline{\mathcal{A}}_s}{\partial \phi}\delta\phi + (1-\phi)\frac{\partial \overline{\mathcal{A}}_s}{\partial \overline{c}^b_k}\delta\overline{c}^b_k$$

$$+\phi\frac{\partial\overline{\mathcal{A}}_f}{\partial\overline{\epsilon}^f}\delta\overline{\epsilon}^f+\frac{\partial\overline{\mathcal{A}}_f}{\partial\overline{\epsilon}^f}\delta\phi+\phi\frac{\partial\overline{\mathcal{A}}_f}{\partial\overline{c}_k^p}\delta\overline{c}_k^p \tag{12.76}$$

By equating the above variation of Helmholtz energy with the virtual work, given as (12.67), (12.68) and (12.69), based on (12.20), we find the following definitions of stresses, pressure, and chemical potential

$$\overline{\sigma}_{ij}^s=\frac{\partial\overline{\mathcal{A}}_s}{\partial\overline{\epsilon}_{ij}^s} \tag{12.77}$$

$$\overline{p}^f=-\frac{\partial\overline{\mathcal{A}}_f}{\partial\overline{\epsilon}^f} \tag{12.78}$$

$$\overline{\mu}_k^b=\frac{\partial\overline{\mathcal{A}}_s}{\partial\overline{c}_k^b} \tag{12.79}$$

$$\overline{\mu}_k^p=\frac{\partial\overline{\mathcal{A}}_f}{\partial\overline{c}_k^p} \tag{12.80}$$

We also establish the following porosity equilibrium equation

$$(1-\phi)\frac{\partial\overline{\mathcal{A}}_s}{\partial\phi}-\frac{\partial\overline{\mathcal{A}}_s}{\partial\overline{\epsilon}^s}+\frac{\partial\overline{\mathcal{A}}_f}{\partial\overline{\epsilon}^f}=0 \tag{12.81}$$

or

$$(1-\phi)\frac{\partial\overline{\mathcal{A}}_s}{\partial\phi}+\overline{p}^s-\overline{p}^f=0 \tag{12.82}$$

which are the same as (4.78) and (4.79).

Quadratic Form To construct a linear theory, we expand the functional around the equilibrium point in a quadratic form

$$\begin{aligned}\overline{\mathcal{A}}_s=&\frac{1}{2}\hat{\lambda}_s\overline{\epsilon}^s\overline{\epsilon}^s+\hat{\mu}_s\overline{\epsilon}_{ij}^s\overline{\epsilon}_{ij}^s+\frac{1}{2}K_\phi\,\Delta\phi^2+K_\psi\,\overline{\epsilon}^s\,\Delta\phi+\frac{1}{2}\omega_k^b\,(\Delta\overline{c}_k^b)^2\\&+B_k^b\,\Delta\overline{c}_k^b\overline{\epsilon}^s+B_k^\psi\,\Delta\overline{c}_k^b\Delta\phi\end{aligned} \tag{12.83}$$

$$\overline{\mathcal{A}}_f=\frac{1}{2}K_f\overline{\epsilon}^f\overline{\epsilon}^f+\frac{1}{2}\omega_k^p\,(\Delta\overline{c}_k^p)^2+B_k^p\Delta\overline{c}_k^p\overline{\epsilon}^f \tag{12.84}$$

in which $\Delta\overline{c}_k^b$ and $\Delta\overline{c}_k^p$ are the change of molar concentration from an initial equilibrium state, and ω_k^b, ω_k^p, B_k^b, B_k^p, B_k^ψ are various constitutive constants. In the above, for simplicity, we have ignored the cross-species effect by dropping the $\Delta\overline{c}_i^b\Delta\overline{c}_j^b$ and $\Delta\overline{c}_i^p\Delta\overline{c}_j^p$ terms, where $i\neq j$.

Stress-Strain Relations With the Helmholtz free energy defined by the quadratic forms, we can obtain the stress and strain relations at the micromechanical level based on (12.77)–(12.80) as

$$\overline{\sigma}_{ij}^{s} = \delta_{ij}\left(K_{\alpha}\,\overline{\epsilon}^{s} + K_{\psi}\,\Delta\phi + B_{k}^{b}\,\Delta\overline{c}_{k}^{b}\right) + 2G_{s}\langle\overline{\epsilon}_{ij}^{s}\rangle \tag{12.85}$$

$$\overline{p}^{f} = -K_{f}\,\overline{\epsilon}^{f} - B_{k}^{p}\,\Delta\overline{c}_{k}^{p} \tag{12.86}$$

$$\Delta\overline{\mu}_{i}^{b} = \omega_{i}^{b}\,\Delta\overline{c}_{i}^{b} + B_{i}^{b}\,\overline{\epsilon}^{s} + B_{i}^{\psi}\,\Delta\phi \tag{12.87}$$

$$\Delta\overline{\mu}_{i}^{p} = \omega_{i}^{p}\,\Delta\overline{c}_{i}^{p} + B_{i}^{p}\,\overline{\epsilon}^{f} \tag{12.88}$$

where $\Delta\overline{\mu}_{i}^{b}$ and $\Delta\overline{\mu}_{i}^{p}$ are the incremental chemical potential from an equilibrium state. Other stress and strains quantities, such as $\overline{\sigma}_{ij}^{s}$, $\overline{p}^{f}$, $\overline{\epsilon}^{s}$, $\overline{\epsilon}^{f}$, etc., are all defined in the incremental sense, as discussed in Chap. 4. We also note in the above that summation should be performed over the repeated index k in (12.85) and (12.86) (over the chemical species), while no summation should be performed for the repeated index i in (12.87) and (12.88). Equation (12.85) can be contracted to give the solid volumetric response as

$$\overline{p}^{s} = -K_{\alpha}\,\overline{\epsilon}^{s} - K_{\psi}\,\Delta\phi - B_{k}^{b}\,\Delta\overline{c}_{k}^{b} \tag{12.89}$$

From the porosity equilibrium equation (12.82), we obtain the constitutive equation relating the solid and fluid stress differential to the porosity change as

$$\overline{p}^{s} - \overline{p}^{f} = -(1-\phi)\,K_{\phi}\,\Delta\phi - (1-\phi)\,K_{\psi}\,\overline{\epsilon}^{s} - (1-\phi)B_{k}^{\psi}\,\Delta\overline{c}_{k}^{b} \tag{12.90}$$

Strain-Stress Relations The above derived results, (12.85)–(12.90), are stress-strain relations. These relations can be solved to give the strain-stress relations as

$$\overline{\epsilon}^{s} = -\frac{\left[(1-\phi)K_{\phi} - K_{\psi}\right]\overline{p}^{s} + K_{\psi}\,\overline{p}^{f}}{(1-\phi)(K_{\alpha}K_{\phi} - K_{\psi}^{2})} - \frac{(K_{\phi}B_{k}^{b} - K_{\psi}B_{k}^{\psi})\Delta\overline{c}_{k}^{b}}{K_{\alpha}K_{\phi} - K_{\psi}^{2}} \tag{12.91}$$

$$\overline{\epsilon}^{f} = -\frac{1}{K_{f}}\left(\overline{p}^{f} + B_{k}^{p}\Delta\overline{c}_{k}^{p}\right) \tag{12.92}$$

$$\begin{aligned}\Delta\overline{\mu}_{i}^{b} = &-\frac{B_{i}^{\psi}[K_{\alpha} - (1-\phi)K_{\psi}] + B_{i}^{b}[(1-\phi)K_{\phi} - K_{\psi}]}{(1-\phi)(K_{\alpha}K_{\phi} - K_{\psi}^{2})}\overline{p}^{s} \\ &+ \frac{K_{\alpha}B_{i}^{\psi} - K_{\psi}B_{i}^{b}}{(1-\phi)(K_{\alpha}K_{\phi} - K_{\psi}^{2})}\overline{p}^{f} + \omega_{i}^{b}\Delta\overline{c}_{i}^{b} \\ &- \frac{K_{\phi}B_{i}^{b}B_{k}^{b} - K_{\psi}B_{i}^{b}B_{k}^{\psi} - K_{\psi}B_{k}^{b}B_{i}^{\psi} + K_{\alpha}B_{i}^{\psi}B_{k}^{\psi}}{K_{\alpha}K_{\phi} - K_{\psi}^{2}}\Delta\overline{c}_{k}^{b}\end{aligned} \tag{12.93}$$

$$\Delta\overline{\mu}_i^p = -\frac{B_i^p}{K_f}\overline{p}^f - \frac{B_i^p B_k^p \Delta\overline{c}_k^p}{K_f} + \omega_i^p \Delta\overline{c}_i^p \tag{12.94}$$

Again, summation is not performed over the repeated index i in (12.93) and (12.94). We also obtain

$$\Delta\phi = -\frac{\left[K_\alpha - (1-\phi)K_\psi\right]\overline{p}^s - K_\alpha\overline{p}^f}{(1-\phi)(K_\alpha K_\phi - K_\psi^2)} - \frac{(K_\alpha B_k^\psi - K_\psi B_k^b)\Delta\overline{c}_k^b}{K_\alpha K_\phi - K_\psi^2} \tag{12.95}$$

Similar to the porothermoelasticity case presented in Sect. 11.1.2, to have coefficients that have more direct physical meaning, we replace K_α by K_s, and introduce the following constitutive constants

$$\beta_k^b = -\frac{K_\phi B_k^b - K_\psi B_k^\psi}{K_\alpha K_\phi - K_\psi^2} = \frac{K_\phi B_k^b - K_\psi B_k^\psi}{K_s K_\phi} \tag{12.96}$$

$$\beta_k^p = \frac{B_k^p}{K_f} \tag{12.97}$$

$$\beta_k^\psi = -\frac{K_\alpha B_k^\psi - K_\psi B_k^b}{K_\alpha K_\phi - K_\psi^2} = \frac{(K_s K_\phi + K_\psi^2)B_k^\psi - K_\phi K_\psi B_k^b}{K_s K_\phi^2} \tag{12.98}$$

The inverse relations of the above are

$$B_k^b = \frac{(K_s K_\phi + K_\psi^2)\beta_k^b + K_\phi K_\psi \beta_k^\psi}{K_\phi} \tag{12.99}$$

$$B_k^p = K_f \beta_k^p \tag{12.100}$$

$$B_k^\psi = K_\psi \beta_k^b + K_\phi \beta_k^\psi \tag{12.101}$$

Using the above relations, we can rewrite (12.91), (12.92), and (12.95) as

$$\overline{\epsilon}^s = -\frac{\left[(1-\phi)K_\phi - K_\psi\right]\overline{p}^s + K_\psi\,\overline{p}^f}{(1-\phi)K_s K_\phi} - \beta_k^b \Delta\overline{c}_k^b \tag{12.102}$$

$$\overline{\epsilon}^f = -\frac{\overline{p}^f}{K_f} - \beta_k^p \Delta\overline{c}_k^p \tag{12.103}$$

$$\Delta\phi = -\frac{[K_s K_\phi - (1-\phi)K_\phi K_\psi + K_\psi^2]\,\overline{p}^s - (K_s K_\phi + K_\psi^2)\,\overline{p}^f}{(1-\phi)K_s K_\phi^2} - \beta_k^\psi \Delta\overline{c}_k^b \tag{12.104}$$

For the chemical potentials, (12.93) and (12.94) become

$$\Delta\overline{\mu}_i^b = -\frac{[(1-\phi)\beta_i^b + \beta_i^\psi]\overline{p}^s - \beta_i^\psi \overline{p}^f}{1-\phi} + \omega_i^b \Delta\overline{c}_i^b - \frac{K_s K_\phi \beta_i^b \beta_k^b + (K_\psi \beta_i^b + K_\phi \beta_i^\psi)(K_\psi \beta_k^b + K_\phi \beta_k^\psi)}{K_\phi} \Delta\overline{c}_k^b \tag{12.105}$$

$$\Delta\overline{\mu}_i^p = -\beta_i^p \overline{p}^f - K_f \beta_i^p \beta_k^p \Delta\overline{c}_k^p + \omega_i^p \Delta\overline{c}_i^p \tag{12.106}$$

Interpretation of Material Constants Based on (12.102), (12.103) and (12.104), we observe that if a specimen is free from stress and pore pressure, $\overline{p}^s = \overline{p}^f = 0$, and is subject to a chemical change in the pore fluid or bound fluid, these relations become

$$\overline{\epsilon}^s = -\beta_k^b \Delta\overline{c}_k^b \tag{12.107}$$

$$\overline{\epsilon}^f = -\beta_k^p \Delta\overline{c}_k^p \tag{12.108}$$

$$\Delta\phi = -\beta_k^\psi \Delta\overline{c}_k^b \tag{12.109}$$

Equation (12.107) shows that β_k^b is a chemical swelling coefficient of the solid phase corresponding to the concentration change of kth species in the bound fluid. It is analogous to a thermal expansion coefficient due to a temperature change. However, the above sign convention indicates that the solid tends to shrink due to an increase of solute concentration. Similarly, (12.108) shows that β_k^p is a swelling coefficient of the fluid phase, corresponding to the change in concentration of the kth species in pore fluid.

Equation (12.109) further demonstrates that a concentration change in bound fluid can cause porosity to change, with β_k^ψ as its swelling coefficient. This is a result of the non-uniform swelling of individual particles at the microscopic level. To simplify the model, we can invoke the ideal porous medium assumption, which states that the material is microscopically homogeneous and isotropic in response to mechanical, thermal, and chemical loads. We then anticipate that the solid swells in a self-similar geometry, and there is no porosity change. This ideal porous medium assumption necessitates the condition

$$\beta_k^\psi = 0 \tag{12.110}$$

in addition to the earlier introduced one of $K_\psi = \beta_\psi = 0$ for poroelasticity and porothermoelasticity.

For the chemical potentials in (12.105) and (12.106), the free stress and pore pressure condition leads to

$$\Delta\overline{\mu}_i^b = \omega_i^b \Delta\overline{c}_i^b - \frac{K_s K_\phi \beta_i^b \beta_k^b + (K_\psi \beta_i^b + K_\phi \beta_i^\psi)(K_\psi \beta_k^b + K_\phi \beta_k^\psi)}{K_\phi} \Delta\overline{c}_k^b \tag{12.111}$$

$$\Delta\overline{\mu}_i^p = -K_f\beta_i^p\beta_k^p\Delta\overline{c}_k^p + \omega_i^p\Delta\overline{c}_i^p \tag{12.112}$$

We notice that the chemical potential of species i is dependent on the concentration of not only species i, but also of all other species k. This is a consequence of the chemical-mechanical coupling introduced through the coefficient β_k^b, β_k^ψ, and β_k^p.

These relations can be compared to the theoretical result of an ideal solution, given as (12.47) and (12.48)

$$\delta\mu_i = v_i\delta p + \frac{\mathcal{RT}}{x_i}\delta x_i \tag{12.113}$$

Under constant pressure, we can express the above as

$$\Delta\mu_i = \frac{\mathcal{RT}}{c_i}\Delta c_i \approx \frac{\mathcal{RT}}{c_i^o}\Delta c_i \tag{12.114}$$

where c_i^o is the reference molar concentration of species i. Based on the above, and ignore the mechanical coupling terms in (12.112), we may estimate

$$\omega_i^p = \frac{\mathcal{RT}}{c_i^{po}} \tag{12.115}$$

In the above we have refrained from estimating ω_i^b using a similar formula, because $\Delta\overline{\mu}_i^b$ and $\Delta\overline{c}_i^b$ are properties of the bound fluid, considered as a part of the solid. The bound fluid is attached to the clay particles and generally is not connected with each other. The chemical potential and concentration are averaged quantities, and may not satisfy relation like (12.113); hence in general ω_i^b is a constitutive constants that should be measured within the porous medium mixture.

Ideal Porous Medium For an ideal porous medium, we observe the simplification of (12.102)–(12.106),

$$\overline{\epsilon}^s = -\frac{\overline{p}^s}{K_s} - \beta_k^b\Delta\overline{c}_k^b \tag{12.116}$$

$$\Delta\phi = -\frac{\overline{p}^s - \overline{p}^f}{(1-\phi)K_\phi} \tag{12.117}$$

$$\Delta\overline{\mu}_i^b = -\beta_i^b\overline{p}^s + \omega_i^b\Delta\overline{c}_i^b - K_s\beta_i^b\beta_k^b\Delta\overline{c}_k^b \tag{12.118}$$

$$\Delta\overline{\mu}_i^p = -\beta_i^p\overline{p}^f + \omega_i^p\Delta\overline{c}_i^p - K_f\beta_i^p\beta_k^p\Delta\overline{c}_k^p \tag{12.119}$$

while (12.103) remains the same. Of particular interest is that under the ideal porous medium assumption, (12.117) shows that the porosity variation is independent of the solute concentration, and is a linear function of the Terzaghi effective stress $P' = (1-\phi)(\overline{p}^s - \overline{p}^f)$. As porosity variation is closely tied to material failure through pore collapse, as discussed in Sect. 3.2, (12.117) suggest that solute concentration does not play a direct role in this process.

12.3 Constitutive Equation

For application purposes, the constitutive relations need to be built on the observable bulk continuum quantities, such as the total stress σ_{ij}, pore pressure p, deformation of the frame e_{ij}, and the variation in fluid content ζ. Here we shall assemble the results of micromechanical analysis, in terms of solid and fluid averaged stresses and strains, into these bulk continuum parameters.

Up to this point, we have considered two sets of chemical potentials, that of the bound fluid $\overline{\mu}_k^b$ and that of the pore fluid $\overline{\mu}_k^p$. Considering the difficulty to observe, measure, and model the chemical imbalance between the free fluid and the bound fluid, and the resulting diffusion between the two fluid phases in the microscopic scale within the REV, the current model shall adopt the *local chemical equilibrium* assumption, that is

$$\overline{\mu}_k^b = \overline{\mu}_k^p = \overline{\mu}_k \tag{12.120}$$

As discussed before, this assumption is consistent with the earlier ones involving the local pore pressure equilibrium and the local thermal equilibrium in the construction of the poroelastic and porothermoelastic constitutive equations.

Furthermore, to shorten the expressions that relate the bulk continuum parameters to the micromechanical ones, we shall assume that there exists only one solute species. In other words, we assume that the fluid is a *solution* consisting of a *solute* (assumed to be water) and a *solvent* (*diluent*). With the above assumptions we can expand (12.87) and (12.88) into four equations as follows

$$\Delta\overline{\mu}_s = \omega_s^b \,\Delta\overline{c}_s^b + B_s^b\,\overline{\epsilon}^s + B_s^\psi\,\Delta\phi \tag{12.121}$$

$$\Delta\overline{\mu}_w = \omega_w^b \,\Delta\overline{c}_w^b + B_w^b\,\overline{\epsilon}^s + B_w^\psi\,\Delta\phi \tag{12.122}$$

$$\Delta\overline{\mu}_s = \omega_s^p \,\Delta\overline{c}_s^p + B_s^p\,\overline{\epsilon}^f \tag{12.123}$$

$$\Delta\overline{\mu}_w = \omega_w^p \,\Delta\overline{c}_w^p + B_w^p\,\overline{\epsilon}^f \tag{12.124}$$

where we have used the subscript $_s$ to denote solute, and $_w$ for solvent (water). We also dropped the superscript for chemical potential as there is no longer a need to differentiate them between that of the bound fluid and of pore fluid, due to the assumption (12.120).

Finally, we recall the Gibbs-Duhem equation (12.12), which states that not all chemical potentials are independent. Under isothermal condition, we can express (12.31) for the pore fluid phase as

$$c_s^{po}\,\Delta\overline{\mu}_s + c_w^{po}\,\Delta\overline{\mu}_w - \overline{p}^f = 0 \tag{12.125}$$

where c_s^{po} and c_w^{po} are the reference mole concentration of the solute and solvent of the pore fluid.

Strain-Stress Relations Based on the relations between of the internal and external strains, (4.37) and (4.38), and the definition of the total stress (4.120), we can assemble the results in the preceding section to obtain the constitutive equations in terms of the bulk continuum variables, such as the volumetric strain of the solid frame e, the total compressive stress P, the variation in fluid content ζ, the pore pressure p, etc. To do so, we first collect the available constitutive equations as (12.86), (12.89), (12.90) and (12.121)–(12.125). We can use the relations (4.37), (4.118), and (4.120) to replace the variables $(\overline{\epsilon}^s, \overline{\epsilon}^f, \overline{p}^s)$, by $(\overline{e}^s, \zeta, P)$. We also note that by definition, $\overline{e}^s = e$ and $\overline{p}^f = p$. Hence with the eight equations shown above, we can solve for eight variables $(e, \zeta, \Delta\phi, \Delta\overline{\mu}_s, \Delta\overline{\mu}_w, \Delta\overline{c}_s^b, \Delta\overline{c}_w^b, \Delta\overline{c}_w^p)$ as functions of $(P, p, \Delta\overline{c}_s^p)$.

Of the eight constitutive equations, we select the following three for practical modeling purposes, which express the variables (e, ζ, μ_s) as functions of (P, p, c_s), as

$$e = -\frac{1}{K_c}\left(P - \alpha_c p + \alpha_\mu c_s\right) \tag{12.126}$$

$$\zeta = -\frac{\alpha_c'}{K_c}\left(P - \frac{p}{B_c}\right) - \beta_\mu c_s \tag{12.127}$$

$$\mu_s = \beta_\mu' p + \gamma_\mu c_s \tag{12.128}$$

to form a complete set of constitutive equations at the bulk continuum level. In the above, we have replaced the symbol $\Delta\overline{c}_s^p$ by c_s, and $\Delta\overline{\mu}_s$ by μ_s, for simplicity of notation. We have also introduced a set of bulk continuum constitutive constants, such as K_c, α_c, B_c, etc., which are combinations of the micromechanical constants, such as $\beta_s^b, \beta_w^b, \omega_s^b, \omega_w^b$, etc., with their relations defined by the solution of the eight equation system. We note that among the chemical potentials and concentrations, $(\Delta\overline{\mu}_s, \Delta\overline{\mu}_w, \Delta\overline{c}_s^b, \Delta\overline{c}_w^b, \Delta\overline{c}_w^p, \Delta\overline{c}_s^p)$, only c_s, the solute concentration in the pore fluid, and its chemical potential μ_s, are selected in the macroscopic model given by (12.126), (12.127) and (12.128). For practical purposes, only these quantities are readily observable and measurable.

It is of interest to observe from (12.128) that the solute chemical potential μ_s is dependent on the solute concentration c_s and the pore pressure p, but is independent of the solid stress P. With a coefficient identically zero, we have introduced eight independent constitutive constants $\{K_c, \alpha_c, \alpha_c', \alpha_\mu, B_c, \beta_\mu, \beta_\mu', \gamma_\mu\}$ in the system (12.126), (12.127) and (12.128). Unlike the constitutive equations for poroelasticity presented in Sect. 4.6.1, and those for porothermoelasticity presented in Sect. 11.1.4, these constants do not possess matrix symmetry. The reason is that the original conjugate system of variables associated with the reversible energy principle involves the pairing of generalized forces $(P, p, \Delta\overline{\mu}_s^p, \Delta\overline{\mu}_w^p, \Delta\overline{\mu}_s^b, \Delta\overline{\mu}_w^b)$ and generalized displacements $(e, \zeta, \phi\Delta\overline{c}_s^p, \phi\Delta\overline{c}_w^p, (1-\phi)\Delta\overline{c}_s^b, (1-\phi)\Delta\overline{c}_w^b)$. The constitutive constants in these six constitutive equations do observe matrix symmetry, with a total of $21 - 2 = 19$ independent constants (two are identically zero). In the current case, with the assumption of equilibrated chemical potential

between the solid and fluid phase, expressed as (12.120) for the solute and solvent, and the use of Gibbs-Duhem equation (12.125), we have eliminated three such equations. The resultant system of three equations shown as (12.126), (12.127) and (12.128) is no longer symmetrical and all eight constitutive constants are independent.

The full expressions of the bulk material constants in terms of the micromechanical constants are lengthy, and are not presented. These expressions can be simplified by using the ideal porous medium assumption based on (12.110),

$$K_\psi = \beta_s^\psi = \beta_w^\psi = 0 \tag{12.129}$$

These expressions are presented in Sect. B.8. To further simplify, we argue that for a dilute solution, the initial solute molar concentration is much smaller than that of the solvent, $c_s^{po} \ll c_w^{po}$. We shall further assume that the solute chemical potential μ_s has a weak dependence on the pore pressure; thus we can set $\beta'_\mu = 0$ in (12.128). Equation (12.128) can then be compared to the ideal dilute solution case given as (12.114), which relates chemical potential to the solute concentration only. The above assumptions are imposed by setting

$$c_s^{po} = \beta_s^p = 0 \tag{12.130}$$

in the various relations for the constitutive constants. Finally we obtain the following simplified results:

$$K_c = \frac{(1-\phi)^3 K_s K_\phi}{D_1}\left[\omega_s^b \omega_w^b - K_s(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)\right] \tag{12.131}$$

$$\alpha_c = \alpha'_c - \frac{(1-\phi)^3 K_s K_\phi \beta_w^b \omega_s^b}{c_w^{po} D_1} \tag{12.132}$$

$$\alpha'_c = 1 - \frac{(1-\phi)^3 K_\phi\, \omega_s^b \omega_w^b}{D_1} \tag{12.133}$$

$$\alpha_\mu = \frac{(1-\phi)^3 K_s K_\phi\, \beta_s^b \omega_s^p \omega_w^b}{D_1} \tag{12.134}$$

$$B_c = \frac{K_f c_w^{po}}{D_3}(\omega_w^p - K_f \beta_w^p \beta_w^p)\left\{[K_s + \phi(1-\phi)^2 K_\phi]\omega_s^b \omega_w^b - K_s^2(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)\right\} \tag{12.135}$$

$$\beta_\mu = \frac{\phi \beta_s^b \omega_w^b \omega_s^p}{K_s(\omega_s^b \omega_w^b - \beta_w^b \beta_w^b \omega_s^b - \beta_s^b \beta_s^b \omega_w^b)} \tag{12.136}$$

$$\beta'_\mu = 0 \tag{12.137}$$

$$\gamma_\mu = \omega_s^p \tag{12.138}$$

where

$$D_1 = [K_s + (1-\phi)^2 K_\phi]\omega_s^b \omega_w^b - K_s^2(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b) \tag{12.139}$$

$$\begin{aligned} D_3 = {} & K_f^2 c_w^{po} \beta_w^p \beta_w^p \times \left\{K_s^2(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b) - [K_s + \phi^2(1-\phi)^2 K_\phi]\omega_s^b \omega_w^b\right\} \\ & + \phi(1-\phi)^3 K_s K_\phi K_f \left[K_s \beta_w^p (\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b) - K_f \beta_w^b \beta_w^p \beta_w^p \omega_s^b \right. \\ & \left. - \omega_s^b (\beta_w^p \omega_w^b - \beta_w^b \omega_w^p)\right] + c_w^{po} \left\{\phi(1-\phi)^3 K_s K_\phi + K_f [K_s + \phi^2(1-\phi)^2 K_\phi]\omega_w^p\right\} \\ & \omega_s^b \omega_w^b - c_w^{po} K_s^2 [K_f + \phi(1-\phi)^3 K_\phi](\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)\omega_w^p \end{aligned} \tag{12.140}$$

By noticing from (12.137) that another constant vanishes, the system now contains seven independent constitutive constants.

Stress-Strain Relation Alternatively, we can express the constitutive equations (12.126), (12.127) and (12.128) in the stress-strain form as

$$P = -K_v e + \alpha_c M_c \zeta - K_v \beta_v c_s \tag{12.141}$$

$$p = M_c(-\alpha_c' e + \zeta + \beta_\varepsilon c_s) \tag{12.142}$$

$$\mu_s = M_c \beta_\mu'(\alpha_c' e - \zeta) + \gamma_\mu' c_s \tag{12.143}$$

The above system contains eight independent material constants $\{K_v, M_c, \alpha_c, \alpha_c', \beta_\mu', \beta_v, \beta_\varepsilon, \gamma_\mu'\}$. Comparing with (12.126), (12.127) and (12.128), we find the following relations

$$K_v = \frac{K_c}{1 - \alpha_c B_c} \tag{12.144}$$

$$M_c = \frac{K_c B_c}{\alpha_c'(1 - \alpha_c B_c)} \tag{12.145}$$

$$\beta_v = \frac{\alpha_\mu}{K_c} - \frac{B_c \alpha_c \beta_\mu}{\alpha_c'} \tag{12.146}$$

$$\beta_\varepsilon = \beta_\mu - \frac{\alpha_c' \alpha_\mu}{K_c} \tag{12.147}$$

$$\gamma_\mu' = \gamma_\mu - \frac{B_c \beta_\mu'(K_c \beta_\mu - \alpha_c' \alpha_\mu)}{\alpha_c'(1 - \alpha_c B_c)} \tag{12.148}$$

Physical Significance of Bulk Material Constants The physical significance of the above derived constitutive constants can be examined as follows. By setting pore pressure p and solute concentration c_s in (12.126) to zero, it is clear that K_c is a *drained bulk modulus under constant solute concentration*. We note that K_c is dependent only on the solid properties, which includes the bound fluid. We also observe that $K_c \neq K$, due to the chemical-mechanical coupling. It is easy to show that by setting the coupling constants in (12.131) and (B.211) to zero, $\beta_s^b = \beta_w^b = \beta_s^\psi = \beta_w^\psi = 0$, then K_c reduces to K.

We recognize that the pair of coefficients α_c and α_μ in (12.126) plays the role of effective stress coefficients. That is, a (modified) Biot effective stress can be defined as

$$P'' = P - \alpha_c p + \alpha_\mu c_s \tag{12.149}$$

such that (12.126) becomes

$$P'' = -K_c e \tag{12.150}$$

which is an effective stress law. Hence α_c is analogous to the Biot effective stress coefficient α, and α_μ may be defined as a *chemical effective stress coefficient*.

We also notice that unlike the poroelastic case, the constitutive constants in (12.126), (12.127) and (12.128) do not possess symmetry, and $\alpha_c' \neq \alpha_c$. Equations (12.132) and (12.133) show that both α_c and α_c' are solid properties, similar to the Biot effective stress coefficient α. We also observe similar bounds as $0 \leq \alpha_c \leq \alpha_c' \leq 1$, where the upper bound is reached when $K_s \to \infty$. Based on (12.142), α_c' can be determined from a drained test under constant solution concentration as

$$\alpha_c' = \frac{\zeta}{e} \tag{12.151}$$

similar to (2.60).

From (12.127) we find that under the undrained ($\zeta = 0$) and constant solute concentration ($c_s = 0$) conditions, the equation reduces to

$$p = B_c P \tag{12.152}$$

The constant B_c is then the *Skempton pore pressure coefficient under constant solute concentration*. When a specimen is free from solid stress and pore pressure, $P = p = 0$, (12.126) and (12.127) become

$$e = -\frac{\alpha_\mu}{K_c} c_s \tag{12.153}$$

$$\zeta = -\beta_\mu c_s \tag{12.154}$$

Hence α_μ / K_c and β_μ are related to the chemical swelling coefficients under free stress and pore pressure condition.

By setting $\zeta = c_s = 0$ in (12.141), it is clear that K_v is the *undrained bulk modulus under constant solute concentration*. Under the undrained and free expansion conditions, $\zeta = P = 0$, (12.141) becomes

$$e = -\beta_v c_s \tag{12.155}$$

Hence β_v is related to a chemical expansion coefficient under undrained condition.

From (12.142), with $e = c_s = 0$, we find

$$p = M_c \zeta \tag{12.156}$$

Hence M_c is similar to the Biot modulus M and is the inverse of a constant stress storage coefficient. The significance of β_ε is found by setting $e = p = 0$ in (12.142) to obtain

$$\zeta = -\beta_\varepsilon c_s \tag{12.157}$$

The coefficient β_ε is the proportionality constant for fluid leaving the porous medium frame with increasing solute concentration, under constant pore pressure and fixed porous medium volume. The pair of coefficients, γ_μ and γ'_μ in (12.128) and (12.143) are constitutive constants relating chemical potential to solute concentration, under constant stress and constant strain conditions, respectively.

Osmotic Pressure Formulation In the selection of variables for the constitutive equations, the solute concentration c_s, instead of chemical potential μ_s, is chosen as the modeling parameter because it can be measured from the extracted fluid, and maintained as a boundary condition. The main interest of the application of the porochemoelasticity theory, however, is on the mechanical responses of the porous medium, in terms of changes in stress, pore pressure, and deformation. For this purpose, it is easier to visualize the chemical effect through an analogy, by considering it as an equivalent osmotic pressure [44]; hence we shall present here such formulation.

Based on (12.64), we can rewrite (12.126), (12.127) and (12.128) into the following

$$e = -\frac{1}{K_c}\left[P - \alpha_c \left(p - \mathcal{R}_a \Pi\right)\right] \tag{12.158}$$

$$\zeta = -C_c \left(B_c P - p + \mathcal{R}_b \Pi\right) \tag{12.159}$$

$$\mu_s = -\beta'_\mu p + \gamma_\pi \Pi \tag{12.160}$$

where

$$C_c = \frac{\alpha'_c}{K_c B_c} \tag{12.161}$$

is a constant stress storage coefficient under constant solute concentration condition, similar to that defined in (2.43). We also have introduced the following parameters

$$\mathcal{R}_a = \frac{\alpha_\mu}{\alpha_c \mathcal{R} \mathcal{T}_o} \tag{12.162}$$

$$\mathcal{R}_b = \frac{\beta_\mu}{C_c \mathcal{R} \mathcal{T}_o} \tag{12.163}$$

$$\gamma_\pi = \frac{\gamma_\mu}{\mathcal{R} \mathcal{T}_o} \tag{12.164}$$

whose significance is examined below. We observe in (12.158), (12.159) and (12.160) that the conversion into osmotic pressure allows us to compare the chemical effect with the hydraulic one in the same pressure unit. Its relative strength is characterized by the two dimensionless coefficients $\mathcal{R}_a$ and $\mathcal{R}_b$. For these coefficients, we can consider the following two limits. When the porous medium is chemically inert, that is, the solute concentration in the fluid has no effect on the mechanical behavior of the solid, then $\mathcal{R}_a = \mathcal{R}_b = 0$. On the other hand, if the porous medium consists entirely of the chemically active clay particles, whose ranges of electrical influence overlap each other, such that the porous medium behaves as a perfect semipermeable membrane, we then anticipate the full osmotic effect, and $\mathcal{R}_a = \mathcal{R}_b = 1$. These limiting conditions are closely tied to the *osmotic reflection coefficient* $\mathcal{R}$ to be introduced in Sect. 12.4.2; hence we shall refer $\mathcal{R}_a$ as a *chemo-mechanical reflection coefficient*, and $\mathcal{R}_b$ a *chemo-hydraulic reflection coefficient*. In general, $\mathcal{R}_a$ and $\mathcal{R}_b$ are constants between 0 and 1, and need to be measured.

Full Constitutive Equations The constitutive equations presented above involve only the volumetric response. Assume that the chemical effect does not affect the shear property of the porous medium frame, we can incorporate the shear deformation constitutive law into (12.126) and express it as

$$\sigma_{ij} = \left(K_c - \frac{2G}{3} \right) \delta_{ij} e + 2G e_{ij} - \alpha_c \delta_{ij} p + \alpha_\mu \delta_{ij} c_s \tag{12.165}$$

which can be combined with (12.127) and (12.128) to form a complete set of constitutive equations. Alternatively, we may choose the osmotic pressure instead of solute concentration as the variable, and express (12.158) as

$$\sigma_{ij} = \left(K_c - \frac{2G}{3} \right) \delta_{ij} e + 2G e_{ij} - \alpha_c \delta_{ij} \left(p - \mathcal{R}_a \Pi \right) \tag{12.166}$$

Together with (12.159) and (12.160), these form a complete set of constitutive equations.

In yet another form, we can use the variation in fluid content instead of pore pressure on the right hand side, and present (12.141) as

$$\sigma_{ij} = \left(K_v - \frac{2G}{3} \right) \delta_{ij} e + 2G e_{ij} - \alpha_c M_c \delta_{ij} \zeta + K_v \beta_v \delta_{ij} c_s \tag{12.167}$$

Or, we can replace c_s by Π and obtain

$$\sigma_{ij} = \left(K_v - \frac{2G}{3}\right)\delta_{ij}e + 2Ge_{ij} - \alpha_c M_c \delta_{ij}\left[\zeta - C_c(\mathcal{R}_a - \mathcal{R}_b)\,\Pi\right] \tag{12.168}$$

The above stress-strain relations can also be expressed as strain-stress relations. For example, (12.126) becomes

$$e_{ij} = \left(\frac{1}{9K_c} - \frac{1}{6G}\right)\delta_{ij}\sigma_{kk} + \frac{1}{2G}\sigma_{ij} + \frac{\alpha_c}{3K_c}\delta_{ij}p - \frac{\alpha_\mu}{3K_c}\delta_{ij}c_s \tag{12.169}$$

and (12.158) takes the form

$$e_{ij} = \left(\frac{1}{9K_c} - \frac{1}{6G}\right)\delta_{ij}\sigma_{kk} + \frac{1}{2G}\sigma_{ij} + \frac{\alpha_c}{3K_c}\delta_{ij}\left(p - \mathcal{R}_a\Pi\right) \tag{12.170}$$

12.4 Balance and Transport Law

To construct the full set of governing equations, we need to introduce other physical laws, such as the balance of mass, momentum, and energy, and their transport laws. Some of these laws have been discussed in Sects. 6.1, 6.2, 11.2, and 11.3. Those that are relevant to the current isothermal, quasi-static linear porochemoelasticity theory are presented below to form a complete predictive model.

12.4.1 Balance Law

Force Equilibrium The static force equilibrium has been discussed in Sect. 6.2.2, and it is unchanged for the present model. For easy reference, the force equilibrium equation (6.61) is reintroduced below:

$$\sigma_{ij,j} = 0 \tag{12.171}$$

Fluid Mass Balance The mass balance of a single phase fluid has been discussed in Sect. 6.2.1. In the present case, the single phase fluid can contain multiple species in solution. Our interest here lies in a dilute solution, meaning that the solutes are of small concentration. Assuming that the density of the bulk fluid is not affected by the variation of the solute concentration, the same fluid mass balance (continuity) equation (6.45) applies for the bulk fluid

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \vec{q} = 0 \tag{12.172}$$

Solute Mass Balance Consider that the fluid is a solution containing multiple solute species in dilute concentration, which are chemically non-reactive. The solute mass balance can be expressed as [4]

$$\frac{\partial \phi c_k}{\partial t} = -\nabla \cdot \left(c_k \vec{q}\right) - \nabla \cdot \vec{J}_k^s - \nabla \cdot \vec{J}_k^d - f_{k\to s} + \Gamma_k \tag{12.173}$$

In the above, c_k is the mass concentration of solute species k, $\vec{J}_k^s$ is the diffusive mass flux (mass crossing per unit area of the porous medium per unit time) of species k, $\vec{J}_k^d$ is the dispersive mass flux per unit porous medium area due to the mechanical dispersion of porous medium flow [4], $f_{k\to s}$ is the rate of mass transfer of the kth species from the fluid phase to the solid phase per unit volume of porous medium, through the adsorption of solute mass to the clay or non-clay particle surfaces, or its exchange with the bound fluid of clay particles, and Γ_k is the production of mass of species k per unit volume of porous medium.

In the present model, we shall drop the mass transfer and production terms in (12.173), though these terms can easily be added back if these mechanisms exist and the laws governing their exchange are known. The first terms on the right hand side of (12.173) is the *advective transport*, and can be expressed as

$$\nabla \cdot \left(c_k \vec{q}\right) = c_k \nabla \cdot \vec{q} + \vec{q} \cdot \nabla c_k \tag{12.174}$$

The first term of the above corresponds to the gain of solute mass through the gain of fluid volume in the porous medium frame due to the compressibility of the fluid, the solid, and the frame. The second term represents the gain of solute mass through concentration gradient by replacing the lower concentration fluid by a higher concentration one. These two terms are nonlinear as they involve the product of dependent variables.

The dispersive flux term $\vec{J}_k^d$ in (12.173) can be expressed into such gradient law [4]:

$$\vec{J}_k^d = -\mathbb{D} \nabla c_k \tag{12.175}$$

where $\mathbb{D}$ is a second rank dispersion coefficient tensor with components D_{ij}. For isotropic porous medium,

$$D_{ij} = \left[a_T \delta_{ij} + (a_L - a_T) \frac{V_i V_j}{V^2} \right] V \tag{12.176}$$

in which a_L and a_T are respectively the longitudinal and transverse dispersivity, which are functions of the pore geometry, and

$$\vec{V} = \frac{\vec{q}}{\phi} \tag{12.177}$$

is the porous medium seepage velocity, with $V = |\vec{V}|$. We notice that the dispersion coefficient is proportional to the magnitude of the fluid specific flux; hence the dispersive flux term is also nonlinear.

At this point we would like to identify two types of problems: problems in which the global fluid flux is significant and the concentration transported by advection is non-negligible, and problems that are stress, deformation, and chemically driven, which induces small amount of local flow to achieve pore pressure and chemical potential equilibrium. In the latter case, we shall neglect the advection and the dispersion terms, and (12.173) becomes

$$\phi \frac{\partial c_k}{\partial t} + \nabla \cdot \vec{J}_k^s = 0 \tag{12.178}$$

In the above we assumed that porosity is not a function of time. With the mass diffusion flux $\vec{J}_k^s$ law to be introduced in the section below, the above equation is linear. Finally, considering a single solute species only, the above equation becomes

$$\phi \frac{\partial c_s}{\partial t} + \nabla \cdot \vec{J}_s = 0 \tag{12.179}$$

12.4.2 Transport Law

Consider a molecule of salt that dissociates into ν^+ cations of valence z^+, and ν^- anions of valence z^-, in a solvent, such that

$$\nu^+ z^+ + \nu^- z^- = 0 \tag{12.180}$$

For a clay that is an imperfect membrane, we can write the flux laws for the cation, anion, and solvent, driven by the gradient of fluid pressure and chemical potential. Here we denote by J_s^+ and J_s^- respectively the flux of cations and anions measured in moles, crossing a cross-sectional area of clay, per unit time. When there is no ongoing electrical current, the electron neutrality condition requires the fluxes to satisfy

$$J_s^+ z^+ + J_s^- z^- = 0 \tag{12.181}$$

Dividing (12.181) by (12.180), it is easy to show that

$$\frac{J_s^+}{\nu^+} = \frac{J_s^-}{\nu^-} = J_s \tag{12.182}$$

where J_s is the flux of salt in moles. As demonstrated by Sherwood and Craster [48], the above relations allows us to model the transport of disassociated ions as a salt, which is consistent with the single species transport model we assumed so far.

Given a binary solution that contains a solvent and a single solute, and following the presentation in Sect. 11.3.1, we can express the flux law for the binary mixture as

$$\vec{J}_w = -L_{ww}\nabla\mu_w - L_{ws}\nabla\mu_s \tag{12.183}$$

$$\vec{J}_s = -L_{sw}\nabla\mu_w - L_{ss}\nabla\mu_s \tag{12.184}$$

where $\vec{J}_w$ and $\vec{J}_s$ are respectively the flux of the solvent and the solute in mole per cross-sectional area of the porous medium per unit time, μ_w an μ_s are the chemical potential for the solvent and the solute, and L_{ww}, L_{ws}, L_{sw}, L_{ss} are the constitutive constants for the gradient law. According to Onsager's law,

$$L_{ws} = L_{sw} \tag{12.185}$$

Hence there exist three independent gradient law constants.

To accommodate the present modeling variables of pore pressure p and solute concentration c_s, we can replace the chemical potentials using (12.61) and (12.62), and obtain

$$\vec{J}_w = -L_{pp}\nabla p - L_{pc}\nabla c_s \tag{12.186}$$

$$\vec{J}_s = -L_{cp}\nabla p - L_{cc}\nabla c_s \tag{12.187}$$

where

$$L_{pp} = v_s L_{ws} + v_w L_{ww} \tag{12.188}$$

$$L_{pc} = \frac{\mathcal{RT}_o}{c_o}(L_{ws} - c_o v_w L_{ww}) \tag{12.189}$$

$$L_{cp} = v_s L_{ss} + v_w L_{ws} \tag{12.190}$$

$$L_{cc} = \frac{\mathcal{RT}_o}{c_o}(L_{ss} - c_o v_w L_{ws}) \tag{12.191}$$

where we have linearized by approximating c_s by its initial value c_o. We note that the above coefficients lack symmetry, that is, $L_{ps} \neq L_{sp}$, for the variables (J_w, J_s) and $(\nabla p, \nabla c_s)$ do not form a conjugate flux and force pair for the dissipation function.

We further note that under the condition of no pressure gradient, (12.191) reduces to a law similar to the Fick's law

$$\vec{J}_s = -L_{cc}\nabla c_s \tag{12.192}$$

where L_{cc} is the apparent mass diffusion coefficient for a chemically active porous medium at the given background solute concentration. The coefficient L_{cc} is not the *mass diffusion coefficient* of Fick's law D_m, and is smaller, for a few reasons. First, in a porous medium the flux is defined by the straight line distance between two points, while the solute ions need to travel through a tortuous path. Furthermore, $\vec{J}_s$ is not the mass flux in the fluid domain. Rather, it is the apparent mass flux across a unit area of porous medium cross section. Hence an *apparent mass diffusion coefficient for porous medium*, denoted as D_c, needs to be modulated by a *tortuosity* factor τ_t [10], as well as a porosity, and

$$D_c = \frac{\phi}{\tau_t} D_m \tag{12.193}$$

Another important effect to introduce is the semipermeable membrane nature of clay, which can partially block the diffusion of the solute ions, as illustrated in Fig. 12.2. Hence the molecular diffusion should be further modulated by a *transmission coefficient* λ_D, which is the complementary part of a *reflection coefficient* $\mathcal{R}$, both of which have the values between 0 and 1. For a transmission coefficient $\lambda_D = 1$ ($\mathcal{R} = 0$), it means that the rock is chemically inert and no selective filtration is taking place. A value of $\lambda_D = 0$ ($\mathcal{R} = 1$) means an ideal ion exclusion membrane. Hence for a chemically active porous medium, we may express its apparent mass diffusion coefficient as

$$L_{cc} = (1 - \mathcal{R}) D_c \tag{12.194}$$

To gain further insight into these gradient law coefficients, we define the following volumetric fluxes. First,

$$\vec{q} = v_w \vec{J}_w + v_s \vec{J}_s \tag{12.195}$$

is the volume of solution (solvent and solute) per cross-sectional area of porous medium per unit time. This quantity is the same as the specific discharge of porous medium flow as defined in Sect. 6.1. Second, we define a solute volume flux relative to the solvent flux, per unit area of porous medium, as

$$\vec{q}_r = \frac{v}{x_s} \vec{J}_s - \frac{v}{x_w} \vec{J}_w \approx v_w \left(\frac{1}{x_s} \vec{J}_s - \vec{J}_w \right) \tag{12.196}$$

in which we have invoked the dilute solution assumption (12.59).

It can be shown that with the approximation in (12.196), the product of the flux ($\vec{q}$, $\vec{q}_r$) and force (∇p, $\nabla \Pi$) pair forms the proper dissipation function and the following gradient laws satisfy the Onsager reciprocal relation [25, 44, 48]

$$\vec{q} = -\kappa \nabla p + \mathcal{R} \kappa \nabla \Pi \tag{12.197}$$

$$\vec{q}_r = \mathcal{R} \kappa \nabla p - \kappa_D \nabla \Pi \tag{12.198}$$

We particularly notice the symmetry of the coefficients. In the above equation, κ is the permeability coefficient (mobility) of Darcy's law, and $\mathcal{R}$ is a reflection coefficient characterizing the strength of the counter flow induced by the osmotic pressure gradient. The reflection coefficient is a dimensionless parameter, and $0 < \mathcal{R} < 1$, with the lower bound corresponding to a chemically inert rock with no selective filtration of the solute and solvent. In that case, the concentration gradient represented by the second term on the right hand side of (12.197) does not drive a solvent flow. The upper bound corresponds to the ideal ion exclusion membrane and the concentration gradient exerts a full osmotic pressure. We also notice that (12.197) and (12.198) contain three independent coefficients $\{\kappa, \mathcal{R}, \kappa_D\}$, which are consistent with (12.183) and (12.184). In fact, we find the following relations

$$\kappa = v_w^2 L_{ww} + 2v_w v_s L_{ws} + v_s^2 L_{ss} \tag{12.199}$$

$$\begin{aligned} \kappa\,\mathcal{R} &= \frac{1}{c_o}\left[c_o v_w^2 L_{ww} - (1 - c_o v_s) v_w L_{ws} - v_s L_{ss}\right] \\ &\approx \frac{1}{c_o}\left[c_o v_w^2 L_{ww} - v_w L_{ws} - v_s L_{ss}\right] \end{aligned} \tag{12.200}$$

$$\kappa_D = \frac{1}{c_o^2}\left(c_o^2 v_w^2 L_{ww} - 2c_o v_w L_{ws} + L_{ss}\right) \tag{12.201}$$

In the above we note that

$$c_o v_s \approx x_s^o \frac{v_s}{v_w} \ll 1 \tag{12.202}$$

where $x_s^o \ll 1$ is the initial molar fraction of the solute, and v_s and v_w as defined by (12.35) are of the same order of magnitude.

Through these definitions, we can solve the constitutive constants L_{ww}, L_{ws} and L_{ss} as functions of $(\kappa, \mathcal{R}, \kappa_D)$. Our final goal is to establish two useful equations, one for the specific discharge, and one for the concentration flux, expressed in terms of three selected gradient law constants. For the clear physical meaning, we select the set as the permeability, the reflection coefficient, and the apparent diffusion coefficient, as $(\kappa, \mathcal{R}, D_c)$. We hence select the pair of equations, (12.197) and (12.187), expressed in the following forms

$$\vec{q} = -\kappa \nabla p + \kappa_{pc} \nabla c_s \tag{12.203}$$

$$\vec{J}_s = \kappa_{cp} \nabla p - (1 - \mathcal{R}) D_c \nabla c_s \tag{12.204}$$

where

$$\kappa_{pc} = \mathcal{R}\kappa\mathcal{R}\mathcal{T}_o \tag{12.205}$$

$$\kappa_{cp} = \frac{(1 - \mathcal{R})\, c_o}{(1 + c_o v_s)} \kappa \;\approx\; (1 - \mathcal{R})\, c_o\, \kappa \tag{12.206}$$

12.5 Governing Equation

To form a solution system, we assemble the constitutive, balance, and transport equations, and eliminate variables among them, to arrive at partial differential equations of familiar forms.

Navier Equation Substituting the constitutive equation (12.165) into the equilibrium equation (12.171), we obtain

$$G\nabla^2\vec{u} + \left(K_c + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha_c\nabla p + \alpha_\mu\nabla c_s = 0 \tag{12.207}$$

Or, in terms of osmotic pressure, we have from (12.166)

$$G\nabla^2\vec{u} + \left(K_c + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha_c\nabla p + \mathcal{R}_a\alpha_c\nabla\Pi = 0 \tag{12.208}$$

Or, we can use ζ as variable instead of p, and obtain from (12.167)

$$G\nabla^2\vec{u} + \left(K_\nu + \frac{G}{3}\right)\nabla(\nabla\cdot\vec{u}) - \alpha_c M_c\nabla\zeta + K_\nu\beta_\nu\nabla c_s = 0 \tag{12.209}$$

Diffusion Equations Combining the continuity equation (12.172) with the transport law (12.203), we obtain

$$\frac{\partial\zeta}{\partial t} - \kappa\nabla^2 p + \mathcal{R}\kappa\mathcal{R}\mathcal{T}_o\nabla^2 c_s = 0 \tag{12.210}$$

A pore pressure diffusion equation can be obtained by substituting (12.142) into the above to eliminate ζ, and we find

$$\frac{1}{M_c}\frac{\partial p}{\partial t} - \kappa\nabla^2 p = -\alpha_c'\frac{\partial e}{\partial t} + \beta_\varepsilon\frac{\partial c_s}{\partial t} - \mathcal{R}\kappa\mathcal{R}\mathcal{T}_o\nabla^2 c_s \tag{12.211}$$

From (12.179) and (12.204), we find

$$\phi\frac{\partial c_s}{\partial t} - (1-\mathcal{R})D_c\nabla^2 c_s + (1-\mathcal{R})c_o\kappa\nabla^2 p = 0 \tag{12.212}$$

We can eliminate the $\nabla^2 p$ and $\nabla^2 c_s$ terms between (12.211) and (12.212) to obtain the following pair of standard diffusion equations, with coupling terms on the right hand side

$$\frac{\partial p}{\partial t} - M_c\kappa_p\nabla^2 p = -M_c\alpha_c'\frac{\partial e}{\partial t} - M_c\beta_\alpha\frac{\partial c_s}{\partial t} \tag{12.213}$$

$$\frac{\partial c_s}{\partial t} - D_c' \nabla^2 c_s = -\frac{\alpha_c'}{\beta_\beta}\frac{\partial e}{\partial t} - \frac{1}{M_c \beta_\beta}\frac{\partial p}{\partial t} \tag{12.214}$$

In the above we have used the following shorthand notations

$$\kappa_p = \left[1 - \frac{\mathcal{R} c_o \kappa \mathcal{R} \mathcal{T}_o}{D_c}\right]\kappa \tag{12.215}$$

$$\beta_\alpha = \frac{\mathcal{R}\phi\kappa\mathcal{R}\mathcal{T}_o}{(1-\mathcal{R})D_c} - \beta_\varepsilon \tag{12.216}$$

$$\beta_\beta = \frac{\phi}{(1-\mathcal{R})c_o} - \beta_\varepsilon \tag{12.217}$$

$$D_c' = \frac{(1-\mathcal{R})(D_c - \mathcal{R} c_o \kappa \mathcal{R}\mathcal{T}_o)}{\phi + (1-\mathcal{R})c_o\beta_\varepsilon} \tag{12.218}$$

From (12.215) and (12.218) we observe that in order not to violate the physical laws, the diffusion coefficient must be positive; hence the following physical constraint exists

$$D_c > \mathcal{R} c_o \kappa \mathcal{R}\mathcal{T}_o \tag{12.219}$$

This constraint means that in the solute concentration flux law (12.204), the hyperfiltration effect (concentration flux driven by pressure gradient, see Table 11.1) cannot be larger than the molecular diffusion effect. Equations (12.207), (12.213) and (12.214) then form a complete solution system to solve for the unknowns $(\vec{u}, p, c_s)$.

It is of interest to note that when the system reaches a steady state, then all the time derivative terms in (12.213) and (12.214) drop off, and we obtain the following conditions under equilibrium:

$$\nabla^2 p = 0 \tag{12.220}$$

$$\nabla^2 c_s = 0 \tag{12.221}$$

In the following we shall examine a few limiting cases.

Perfect Membrane Model If a porous medium behaves like a perfect semipermeable membrane that entirely blocks the solute transport, the reflection coefficient takes the value of 1. Substituting the condition into (12.217) and (12.218), we observe

$$\beta_\beta \to \infty \quad \text{and} \quad D_c' \to 0; \quad \text{as} \quad \mathcal{R} \to 1 \tag{12.222}$$

Hence (12.214) reduces to

$$\frac{\partial c_s}{\partial t} = 0 \tag{12.223}$$

suggesting that the solute is not transported, just as expected. For the pore pressure diffusion equation, we resort to the original Eq. (12.211) and obtain

$$\frac{\partial p}{\partial t} - M_c \kappa \nabla^2 p = -M_c \alpha_c' \frac{\partial e}{\partial t} - M_c \kappa \nabla^2 \Pi \tag{12.224}$$

in which we have converted the solute concentration c_s to the osmotic pressure Π. Utilizing (12.223) again, we can combine the pore pressure with osmotic pressure and express (12.224) into the following

$$\frac{\partial (p - \Pi)}{\partial t} - M_c \kappa \nabla^2 (p - \Pi) = -M_c \alpha_c' \frac{\partial e}{\partial t} \tag{12.225}$$

Particularly, under steady state, it reduces to

$$\nabla^2 (p - \Pi) = 0 \tag{12.226}$$

The above shows that the individual balance of pore pressure and solution concentration as indicated in (12.220) and (12.221) is not required, and only the combined effect needs to be balanced.

We note that the full osmotic effect assumption, $\mathcal{R} = 1$, also implies that the chemo-mechanical and chemo-hydraulic reflection coefficients introduced in (12.158) and (12.159) have the limiting value of $\mathcal{R}_a = \mathcal{R}_b = 1$. This condition allows us to express the Navier equation (12.208) into the following

$$G \nabla^2 \vec{u} + \left(K_c + \frac{G}{3} \right) \nabla (\nabla \cdot \vec{u}) - \alpha_c \nabla (p - \Pi) = 0 \tag{12.227}$$

Equations (12.225) and (12.227) form a complete solution system solving for the unknowns $(\vec{u}, p - \Pi)$. This reduced system is the same as the model introduced by Sherwood [45].

Chemically Uncoupled Model In this model we assume that the hyperfiltration effect is weak, meaning that the solute concentration flux $\vec{J}_s$ is driven only by the concentration gradient, and not by the pressure gradient. This is equivalent to setting

$$\kappa_{cp} = 0 \tag{12.228}$$

in (12.204). Or, referring to (12.206), this means that $c_o \ll 0$ such that κ_{cp} can be neglected. This assumption leads to the following reductions

$$\beta_\beta \to \infty; \quad D_c' \to \frac{1 - \mathcal{R}}{\phi}; \quad \kappa_p \to \kappa; \quad \text{as} \quad c_o \to 0 \tag{12.229}$$

The above assumption is certainly consistent with the physical constraint (12.219) introduced earlier. Equation (12.214) then reduces to

$$\frac{\partial c_s}{\partial t} - D_c' \nabla^2 c_s = 0 \tag{12.230}$$

Hence the solute transport is uncoupled from the poroelastic processes, and can be independently solved. The concentration gradient then serves as a body force term in (12.207), and its rate of change as a source term in (12.213). This model is analogous to the thermally uncoupled model for porothermoelasticity introduced in Sect. 11.4.4. This is also the same assumption used in the model of Heidug and Wong [20] and Ghassemi and Diek [13, 14].

Chemically Inert Model In this case we consider that the porous medium is chemically inert and does not block the ion transport; hence $\mathcal{R} = 0$. This assumption is also tied to the condition $\mathcal{R}_a = \mathcal{R}_b = 0$, which further leads to $\alpha_\mu = \beta_\mu = 0$, based on (12.162) and (12.163), and $\beta_\varepsilon = \beta_\alpha = 0$ and $\kappa_p = \kappa$, based on (12.147), (12.215) and (12.216). Consequently, (12.213) becomes

$$\frac{\partial p}{\partial t} - \kappa M \nabla^2 p = -\alpha M \frac{\partial e}{\partial t} \tag{12.231}$$

which is the same as that for the poroelastic case given as (6.69). The Navier equation also similarly reduces to that of the poroelasticity case.

Irrotational Displacement Field As discussed in Sect. 6.4.3, for problems under certain geometrical constraint, such as uniaxial strain, axial symmetry, or spherical symmetry, the displacement field is irrotational, and the displacement vector can be expressed as the gradient of a scalar potential. Following a similar treatment as in Sect. 6.4.3, we can integrate the Navier equation (12.207) to obtain

$$e = \frac{\eta_c}{G} p - \frac{\eta_\mu}{G} c_s + g(t) \tag{12.232}$$

where $g(t)$ is an arbitrary function of time, and we have defined

$$\eta_c = \frac{3\alpha_c G}{3K_c + 4G} \tag{12.233}$$

$$\eta_\mu = \frac{3\alpha_\mu G}{3K_c + 4G} \tag{12.234}$$

From the above, and the constitutive equation (12.165), we find that the following expression is a function of time only

$$\sigma_{kk} + 4\eta_c p - 4\eta_\mu c_s = 3K_c\, g(t) \tag{12.235}$$

And under plane strain condition, we have

$$\sigma_{kk} + 2\eta_c p - 2\eta_\mu c_s = \left(2K_c + \frac{2G}{3}\right) g(t) \tag{12.236}$$

where $\sigma_{kk} = \sigma_{xx} + \sigma_{yy}$.

Using (12.232) and (12.235) in the pore pressure diffusion equation (12.213), it can be transformed to

$$\frac{\partial p}{\partial t} - c_\alpha \nabla^2 p = -\alpha_\alpha \frac{\partial c_s}{\partial t} - \frac{\alpha_c'}{3K_c S_\alpha} \frac{d}{dt}\left(\sigma_{kk} + 4\eta_c p - 4\eta_\mu c_s\right) \tag{12.237}$$

Similarly, from the heat diffusion equation (12.214), we find

$$\frac{\partial c_s}{\partial t} - c_\beta \nabla^2 c_s = -\frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} - \frac{\alpha_c'}{3K_c S_\beta} \frac{d}{dt}\left(\sigma_{kk} + 4\eta_c p - 4\eta_\mu c_s\right) \tag{12.238}$$

In the above we have defined

$$c_\alpha = \frac{\kappa_p}{S_\alpha} \tag{12.239}$$

$$c_\beta = \frac{\beta_\beta D_c'}{S_\beta} \tag{12.240}$$

$$S_\alpha = \frac{1}{M_c} + \frac{\alpha_c' \eta_c}{G} \tag{12.241}$$

$$S_\beta = \beta_\beta - \frac{\alpha_c' \eta_\mu}{G} \tag{12.242}$$

$$\alpha_\alpha = \frac{M_c(G\beta_\alpha - \alpha_c'\eta_\mu)}{G + M_c \alpha_c' \eta_c} \tag{12.243}$$

$$\alpha_\beta = \frac{M_c(G\beta_\beta - \alpha_c'\eta_\mu)}{G + M_c \alpha_c' \eta_c} \tag{12.244}$$

For plane strain problem, the above equations can be expressed as

$$\begin{aligned}\frac{\partial p}{\partial t} - c_\alpha \nabla^2 p &= -\alpha_\alpha \frac{\partial c_s}{\partial t} \\ &\quad - \frac{3\alpha_c'}{2S_\alpha(3K_c + G)} \frac{d}{dt}\left(\sigma_{kk} + 2\eta_c p - 2\eta_\mu c_s\right)\end{aligned} \tag{12.245}$$

$$\begin{aligned}\frac{\partial c_s}{\partial t} - c_\beta \nabla^2 c_s &= -\frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} \\ &\quad - \frac{3\alpha_c'}{2S_\beta(3K_c + G)} \frac{d}{dt}\left(\sigma_{kk} + 2\eta_c p - 2\eta_\mu c_s\right)\end{aligned} \tag{12.246}$$

Or, utilizing the definitions (12.235) and (12.236), both sets of equations, (12.237) and (12.238), and (12.245) and (12.246), can be expressed as follows

$$\frac{\partial p}{\partial t} - c_\alpha \nabla^2 p = -\alpha_\alpha \frac{\partial c_s}{\partial t} - \frac{\alpha_c'}{S_\alpha}\frac{dg(t)}{dt} \tag{12.247}$$

$$\frac{\partial c_s}{\partial t} - c_\beta \nabla^2 c_s = -\frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} - \frac{\alpha_c'}{S_\beta}\frac{dg(t)}{dt} \tag{12.248}$$

12.6 Uniaxial Strain Problem

In the following sections we shall explore a few analytical solutions involving simple geometries aimed at the understanding the basic porochemoelastic phenomena. We first examine the uniaxial strain problems with the following assumptions

$$u_z = u_z(z,t); \quad p = p(z,t); \quad c_s = c_s(z,t); \quad u_x = u_y = 0 \tag{12.249}$$

which also leads to the conditions

$$\sigma_{xy} = \sigma_{yz} = \sigma_{zx} = 0 \tag{12.250}$$

12.6.1 One-Dimensional Governing Equation

With the functional dependence given in (12.249), the Navier equation (12.207) reduces to

$$\left(K_c + \frac{4G}{3}\right)\frac{\partial^2 u_z}{\partial z^2} - \alpha_c \frac{\partial p}{\partial z} + \alpha_\mu \frac{\partial c_s}{\partial z} = 0 \tag{12.251}$$

The diffusion equations are reduced from (12.213) and (12.214) to give

$$\frac{\partial p}{\partial t} - M_c \kappa_p \frac{\partial^2 p}{\partial z^2} = -M_c \alpha_c' \frac{\partial e_{zz}}{\partial t} - M_c \beta_\alpha \frac{\partial c_s}{\partial t} \tag{12.252}$$

$$\frac{\partial c_s}{\partial t} - D_c' \frac{\partial^2 c_s}{\partial z^2} = -\frac{\alpha_c'}{\beta_\beta}\frac{\partial e_{zz}}{\partial t} - \frac{1}{M_c \beta_\beta}\frac{\partial p}{\partial t} \tag{12.253}$$

The equilibrium equation (12.171) becomes

$$\frac{\partial \sigma_{zz}}{\partial z} = 0 \tag{12.254}$$

and we may conclude that

$$\sigma_{zz} = \sigma_{zz}(t) \tag{12.255}$$

Utilizing the constitutive equation (12.165) in the form

$$\sigma_{zz} = \left(K_c + \frac{4G}{3}\right) e_{zz} - \alpha_c p + \alpha_\mu c_s \tag{12.256}$$

to eliminate e_{zz} in (12.252) and (12.253), we obtain

$$\frac{\partial p}{\partial t} - c_\alpha \frac{\partial^2 p}{\partial z^2} = -\alpha_\alpha \frac{\partial c_s}{\partial t} - \frac{3M_c\alpha_c'}{4G + 3K_\nu} \frac{d\sigma_{zz}}{dt} \tag{12.257}$$

$$\frac{\partial c_s}{\partial t} - c_\beta \frac{\partial^2 c_s}{\partial z^2} = -\frac{1}{\alpha_\beta} \frac{\partial p}{\partial t} - \frac{3M_c\alpha_c'}{\alpha_\beta(4G + 3K_\nu)} \frac{d\sigma_{zz}}{dt} \tag{12.258}$$

12.6.2 Half Space Problem

Consider the problem of a half space, $z \geq 0$. For all problems discussed below, the pore pressure and solute concentration are initially in equilibrium, $p(z, 0) = c_s(z, 0) = 0$.

Chemical Loading At $t = 0^+$ the concentration at the surface $z = 0$ is instantly raised, while the stress and pore pressure remain unchanged

$$\sigma_{zz} = 0; \quad p = 0; \quad c_s = c_1 \tag{12.259}$$

Here all quantities are defined in the incremental sense. Due to the boundary condition $\sigma_{zz} = 0$, (12.257) and (12.258) simplify to the following

$$\frac{\partial p}{\partial t} - c_\alpha \frac{\partial^2 p}{\partial z^2} = -\alpha_\alpha \frac{\partial c_s}{\partial t} \tag{12.260}$$

$$\frac{\partial c_s}{\partial t} - c_\beta \frac{\partial^2 c_s}{\partial z^2} = -\frac{1}{\alpha_\beta} \frac{\partial p}{\partial t} \tag{12.261}$$

These equations are decoupled from the Navier equation, and can be solved independent of it.

We observe that the above system (12.260) and (12.261) is analogous to the one-dimensional porothermoelastic problem investigated in Sect. 11.6.1. We hence easily find the solution of the above problem as

$$\frac{c_s(z, t)}{c_1} = \frac{c_\alpha - c_\beta + C_Q}{2C_Q} \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right)$$

$$-\frac{c_\alpha - c_\beta - C_Q}{2C_Q}\,\mathrm{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right) \tag{12.262}$$

$$\frac{p(z,t)}{\alpha_\alpha c_1} = \frac{\alpha_\beta\left[(c_\alpha - c_\beta)^2 - C_Q^2\right]}{4c_\alpha \alpha_\alpha C_Q}\left[\mathrm{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right) - \mathrm{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right)\right] \tag{12.263}$$

$$\lambda_{1,2} = \sqrt{\frac{(c_\alpha + c_\beta) \pm C_Q}{2c_\alpha c_\beta}} \tag{12.264}$$

in which

$$C_Q = \sqrt{(c_\alpha - c_\beta)^2 - \frac{4c_\alpha c_\beta \alpha_\alpha}{\alpha_\beta}} \tag{12.265}$$

From (12.256) with the condition $\sigma_{zz} = 0$, we obtain

$$e_{zz} = \frac{3}{4G + 3K_c}(\alpha_c p - \alpha_\mu c_s) \tag{12.266}$$

The displacement u_z is then found by integrating the above, and

$$\begin{aligned} u_z(z,t) = {} & \frac{3c_1}{4c_\alpha C_Q(4G + 3K_c)} \\ & \times \Big\{ (c_\alpha - c_\beta + C_Q)\left[\alpha_c \alpha_\beta (c_\alpha - c_\beta - C_Q) + 2c_\alpha \alpha_\mu\right] \\ & \times \left[\frac{2}{\lambda_1}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_1^2 z^2/4t} - z\,\mathrm{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right)\right] \\ & -(c_\alpha - c_\beta - C_Q)\left[\alpha_c \alpha_\beta (c_\alpha - c_\beta + C_Q) + 2c_\alpha \alpha_\mu\right] \\ & \times \left[\frac{2}{\lambda_2}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_2^2 z^2/4t} - z\,\mathrm{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\right]\Big\} \end{aligned} \tag{12.267}$$

Another quantity of interest is the water flux induced by the chemical effect. Based on (12.203), we find

$$\begin{aligned} q_z = {} & -\frac{c_1 \kappa}{4c_\alpha C_Q \sqrt{\pi t}}\Big\{ (c_\alpha - c_\beta + C_Q)\left[\alpha_\beta (c_\alpha - c_\beta - C_Q)\right. \\ & \left. +2c_\alpha \mathcal{R}\mathcal{R}\mathcal{T}_o\right]\lambda_1 e^{-\lambda_1^2 z^2/4t} - (c_\alpha - c_\beta - C_Q)\left[\alpha_\beta (c_\alpha - c_\beta + C_Q)\right. \\ & \left. +2c_\alpha \mathcal{R}\mathcal{R}\mathcal{T}_o\right]\lambda_2 e^{-\lambda_2^2 z^2/4t}\Big\} \end{aligned} \tag{12.268}$$

The volume (per unit area) of water entering the formation is

$$\begin{aligned} V(t) &= \int_0^t q_z(0,t)\, dt \\ &= -\frac{c_1\kappa}{2c_\alpha C_Q}\sqrt{\frac{t}{\pi}}\left\{\alpha_\beta(\lambda_1-\lambda_2)[(c_\alpha-c_\beta)^2-C_Q^2]\right. \\ &\quad \left. +2c_\alpha \mathcal{R}\mathcal{R}\mathcal{T}_o[\lambda_1(c_\alpha-c_\beta+C_Q)-\lambda_2(c_\alpha-c_\beta-C_Q)]\right\} \end{aligned} \tag{12.269}$$

To gain insight into the porochemoelastic effects, it is desirable to evaluate the above solutions using a set of real data. A complete set of data prepared for this purpose, however, does not seem to exist. In the following we shall assemble such set of data by consulting various sources.

First, we shall select the porous medium as a Gulf of Mexico Shale, with its poroelastic material constants summarized in Tables 3.1 and 3.2. By examining the governing equations (12.207), (12.213) and (12.214), we find that the model requires the following material constants $\{K_c, G, \alpha_c, \alpha_c', \alpha_\mu, M_c, \kappa_p, D_c', \beta_\alpha, \beta_\beta\}$. By the assumption that the shear modulus is not affected by the chemical activities, we can select from Table 3.1 that $G = 7.6 \times 10^8$ Pa. K_c is the bulk modulus measured under the drained and constant solute concentration condition. Lacking information on its sensitivity to the concentration condition, we shall adopt $K_c \approx K = 1.1 \times 10^9$ Pa. Next we estimate the two effective stress coefficients, α_c and α_c'. Depending on the laboratory technique used to measure α reported in Table 3.2, the value could represent α_c or α_c'. Lacking additional information, we shall select $\alpha_c' = 0.968$ as reported in the table, and assume $\alpha_c = 0.95$. Other data extracted from Tables 3.1 and 3.2 include Skempton pore pressure coefficient $B_c = 0.876$, porosity $\phi = 0.3$, and intrinsic permeability $k = 1 \times 10^{-19}\,\mathrm{m}^2$. We shall assume that the shale is initially saturated with a salt solution with solute mass fraction of $x_s^o = 0.1$. We are also given the molar mass of salt as $m_s = 0.0585$ kg/mol, and water as $m_w = 0.018$ kg/mol. The density of the salt solution is $\rho_f = 1.11\times10^3$ kg/m^3, and its viscosity is $\mu_f = 3\times10^{-4}$ N·s/m^2. The gas constant is $\mathcal{R} = 8.31$ m·N/°K·mol, and the temperature is $\mathcal{T}_o = 293$ °K. Finally, without further information, we select the three reflection coefficients as $\mathcal{R} = 0.3$, and $\mathcal{R}_a = \mathcal{R}_b = 0.8$, similar to those selected in the literature [14, 38, 44].

Based on the above selected raw data, we can calculate the initial molar concentration of salt in the solution as

$$c_o = \frac{\rho_f x_s^o}{x_s^o m_s + x_w^o m_w} = 5.04 \times 10^3\,\mathrm{mol/m^3} \tag{12.270}$$

Other needed parameters are calculated from the various formulas. For example, α_μ and β_μ are related to $\mathcal{R}_a$ and $\mathcal{R}_b$, and are calculated from (12.162) and (12.163) to be $\alpha_\mu = 1.85\times10^3$ N·m/mol and $\beta_\mu = 1.96\times10^{-6}$ m^3/mol. K_ν and M_c are defined in (12.144) and (12.145), etc. The values of these given or calculated parameters are listed in Table 12.1.

Table 12.1 Input data for the half space problem based on a Gulf of Mexico shale

Given		Calculated	
G	$7.60 \times 10^8\,\mathrm{N/m^2}$	M_c	$5.93 \times 10^9\,\mathrm{N/m^2}$
K_c	$1.10 \times 10^9\,\mathrm{N/m^2}$	K_v	$6.56 \times 10^9\,\mathrm{N/m^2}$
B_c	0.876	α_μ	$1.85 \times 10^3\,\mathrm{N \cdot m/mol}$
ϕ	0.3	α_α	$2.69 \times 10^4\,\mathrm{N \cdot m/mol}$
α_c	0.95	α_β	$1.39 \times 10^5\,\mathrm{N \cdot m/mol}$
α_c'	0.968	β_μ	$1.96 \times 10^{-6}\,\mathrm{m^3/mol}$
k	$1 \times 10^{-19}\,\mathrm{m^2}$	β_ε	$3.28 \times 10^{-7}\,\mathrm{m^3/mol}$
μ_f	$3.0 \times 10^{-4}\,\mathrm{N \cdot s/m^2}$	β_α	$1.71 \times 10^{-5}\,\mathrm{m^3/mol}$
D_c	$6.0 \times 10^{-9}\,\mathrm{m^2/s}$	β_β	$8.47 \times 10^{-5}\,\mathrm{m^3/mol}$
x_s^o	0.1	c_α	$4.39 \times 10^{-7}\,\mathrm{m^2/s}$
m_s	$0.0585\,\mathrm{kg/mol}$	c_β	$1.12 \times 10^{-8}\,\mathrm{m^2/s}$
m_w	$0.018\,\mathrm{kg/mol}$	κ	$3.33 \times 10^{-16}\,\mathrm{m^4/N \cdot s}$
ρ_f	$1.11 \times 10^3\,\mathrm{kg/m^3}$	κ_p	$2.65 \times 10^{-16}\,\mathrm{m^4/N \cdot s}$
$\mathcal{R}$	$8.31\,\mathrm{N \cdot m/°K \cdot mol}$	D_c'	$1.11 \times 10^{-8}\,\mathrm{m^2/s}$
$\mathcal{T}_o$	$293\,°\mathrm{K}$	x_w^o	0.9
$\mathcal{R}$	0.3	c_o	$5.04 \times 10^3\,\mathrm{mol/m^3}$
$\mathcal{R}_a$	0.8	λ_1	$9.47 \times 10^3\,\sqrt{\mathrm{s}}$
$\mathcal{R}_b$	0.8	λ_2	$1.35 \times 10^3\,\sqrt{\mathrm{s}}$

For the current problem, we assume that the mass fraction of salt in the solution in contact with the surface of the half space is suddenly decreased from 10 % to 5 %. This means that the boundary condition for concentration in (12.259) is given by

$$c_1 = -2.52 \times 10^3\,\mathrm{mol/m^3} \tag{12.271}$$

or, equivalently, an osmotic pressure of

$$\Pi_1 = -6.13 \times 10^6\,\mathrm{N/m^2} \tag{12.272}$$

In Fig. 12.4a we plot the solute concentration c_s versus the first 10 cm of depth at various times. In order to compare this effect with the pore pressure, the solute concentration is presented in its equivalent osmotic pressure Π. We note that both the perturbed concentration and the osmotic pressure are negative. The speed of solute concentration penetration is slow due to the low molecular diffusion and permeability coefficients of the shale. Because there exists a coupling effect, this reduction in salt concentration also generates a positive pore pressure, which is plotted in Fig. 12.4b. At $t = 1\,\mathrm{h}$, the peak pressure is about 20 % of the osmotic pressure loading at the boundary.

As both the hydraulic and the osmotic pressure have an effect on the mechanical behavior of the porous medium, it is of interest to examine the effective stress, either

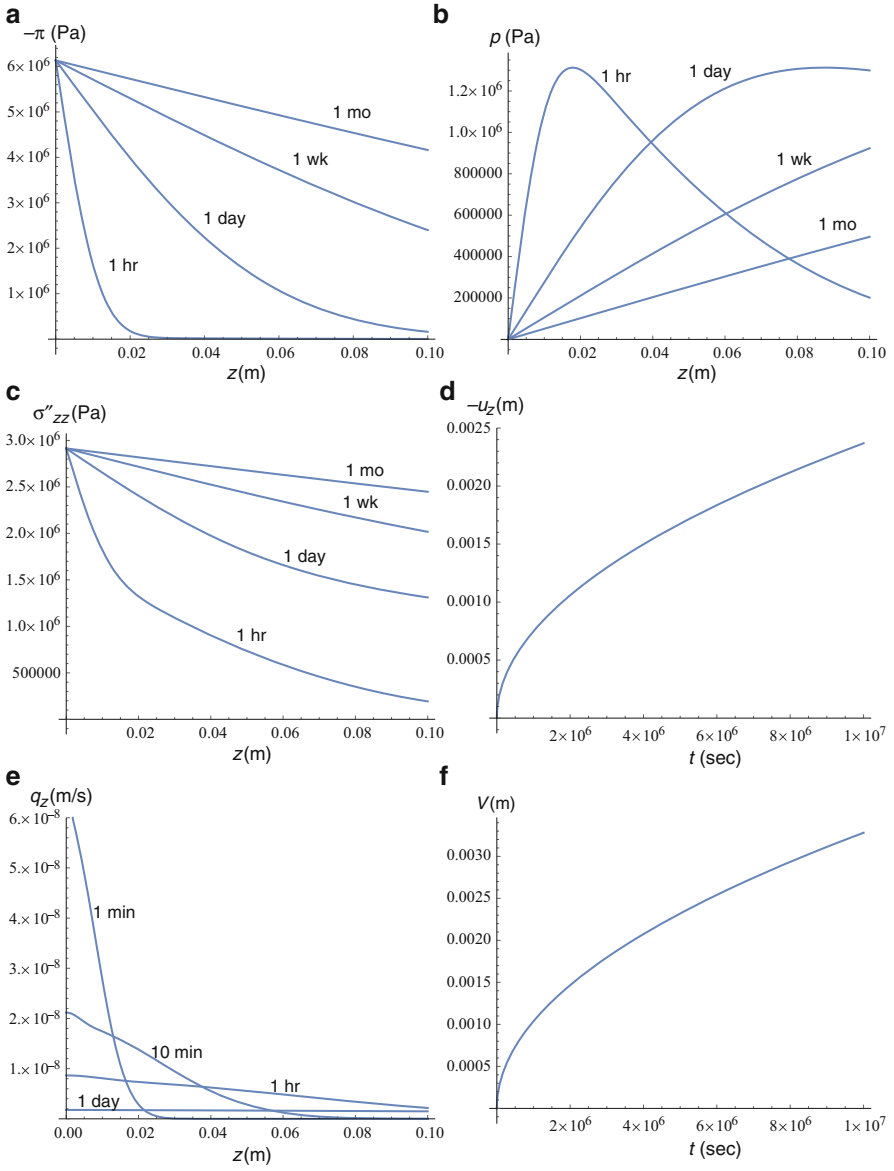


Fig. 12.4 One-dimensional half space problem with chemical loading. (**a**) Solute concentration versus depth (expressed as osmotic pressure); (**b**) pore pressure versus depth; (**c**) Biot effective stress versus depth; (**d**) vertical displacement at surface; (**e**) vertical water flux versus depth; and (**f**) volume of water entering the formation versus time

the modified Biot effective stress or the Terzaghi effective stress, depending on the phenomenon of interest. Equation (12.166) shows that the deformation of the porous medium frame is related to the modified Biot effective stress

$$\sigma''_{zz} = \sigma_{zz} + \alpha_c(p - \mathcal{R}_a \Pi) \tag{12.273}$$

while (12.117) suggests that material failure is likely to be associated with the Terzaghi effective stress

$$\sigma'_{zz} = \sigma_{zz} + p \tag{12.274}$$

In Fig. 12.4c we plot the modified Biot effective stress σ''_{zz} versus depth z at various times, with $\sigma_{zz} = 0$ as given by the boundary condition and required by equilibrium equation. We observe that the reduction in salt concentration produces an effective tensile stress that increases with time.

The vertical displacement u_z at the surface $z = 0$ is plotted in Fig. 12.4d as function of time. It shows that the porous medium swells due to the reduction in solute concentration. Figure 12.4e gives the vertical water flux q_z versus depth at various times. The reduction of solute concentration induces a flux into the formation. The volume of water entering the formation per surface area of the formation is presented as function of time in Fig. 12.4f.

Hydraulic Loading In the second problem, the loading at the surface $z = 0$ is provided by a step rise of pore pressure at $t = 0^+$, while the stress and solute concentration remain unchanged

$$\sigma_{zz} = 0; \quad p = p_1; \quad c_s = 0 \tag{12.275}$$

Solution of this problem can be obtained by observing the symmetry of the governing equations, (12.260) and (12.261), with respect to p and c_s. Hence the concentration and pressure solutions are

$$\frac{c_s(z,t)\alpha_\beta}{p_1} = \frac{\alpha_\beta\left[(c_\alpha - c_\beta)^2 - C_Q^2\right]}{4c_\beta\alpha_\alpha C_Q}\left[\operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right) - \operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right)\right] \tag{12.276}$$

$$\frac{p(z,t)}{p_1} = -\frac{c_\alpha - c_\beta - C_Q}{2C_Q}\operatorname{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right) + \frac{c_\alpha - c_\beta + C_Q}{2C_Q}\operatorname{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right) \tag{12.277}$$

And the displacement is

$$
\begin{aligned}
u_z(z,t) = & \frac{3p_1}{4c_\beta \alpha_\alpha C_Q(4G+3K_c)} \\
& \times \Big\{ -(c_\alpha - c_\beta - C_Q)\left[\alpha_\mu(c_\alpha - c_\beta + C_Q) - 2c_\beta \alpha_c \alpha_\alpha\right] \\
& \times \left[\frac{2}{\lambda_1}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_1^2 z^2/4t} - z\,\mathrm{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right)\right] \\
& + (c_\alpha - c_\beta + C_Q)\left[\alpha_\mu(c_\alpha - c_\beta - C_Q) + 2c_\beta \alpha_c \alpha_\alpha\right] \\
& \times \left[\frac{2}{\lambda_2}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_2^2 z^2/4t} - z\,\mathrm{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\right] \Big\}
\end{aligned}
\tag{12.278}
$$

Given a pressure loading of $p_1 = 2 \times 10^6$ Pa, we plot in Fig. 12.5 the solute concentration (expressed in osmotic pressure) and the pressure response as functions of depth at various times.

Mechanical Loading In this case, the boundary conditions at $z = 0$ and $t = 0^+$ are given as

$$
\sigma_{zz} = -P_1; \quad p = 0; \quad c_s = 0 \tag{12.279}
$$

The above condition is introduced into the governing equations (12.257) and (12.258) to obtain

$$
\frac{\partial p}{\partial t} - c_\alpha \frac{\partial^2 p}{\partial z^2} + \alpha_\alpha \frac{\partial c_s}{\partial t} = \frac{3P_1 M_c \alpha_c'}{4G + 3K_v}\delta(t-0) \tag{12.280}
$$

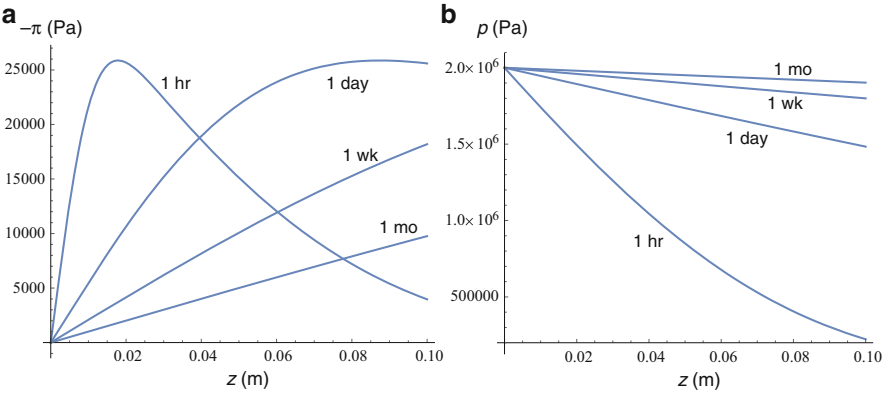


Fig. 12.5 One-dimensional half space problem with hydraulic loading. (**a**) Solute concentration versus depth (expressed as osmotic pressure); and (**b**) pore pressure versus depth

$$\frac{\partial c_s}{\partial t} - c_\beta \frac{\partial^2 c_s}{\partial z^2} + \frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} = \frac{3P_1 M_c \alpha_c'}{\alpha_\beta (4G + 3K_\nu)} \delta(t - 0) \tag{12.281}$$

To solve the above system, we can split the solution into a homogeneous part (p_h, c_h) and a particular part (p_p, c_p). First, the particular solution is easily found as

$$p_p = \frac{3P_1 M_c \alpha_c'}{4G + 3K_\nu} \mathrm{H}(t - 0) \tag{12.282}$$

$$c_p = 0 \tag{12.283}$$

We can then find the homogeneous solution, which satisfies (12.260) and (12.261) with the modified boundary conditions

$$p_h(0, t) = -\frac{3P_1 M_c \alpha_c'}{4G + 3K_\nu}; \quad c_h(0, t) = 0 \tag{12.284}$$

The solution of this problem is the same as the hydraulic loading case, (12.276) and (12.277), with different load magnitude. Combining the homogeneous and particular solutions, we obtain

$$c_s(z, t) = \frac{3P_1 M_c \alpha_c' \left[(c_\alpha - c_\beta)^2 - C_Q^2\right]}{4 c_\beta \alpha_\alpha C_Q (4G + 3K_\nu)} \left[\mathrm{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right) - \mathrm{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right)\right] \tag{12.285}$$

$$p(z, t) = \frac{3P_1 M_c \alpha_c'}{4G + 3K_\nu} \left[\frac{c_\alpha - c_\beta - C_Q}{2C_Q} \mathrm{erfc}\left(\frac{\lambda_1 z}{2\sqrt{t}}\right) - \frac{c_\alpha - c_\beta + C_Q}{2C_Q} \mathrm{erfc}\left(\frac{\lambda_2 z}{2\sqrt{t}}\right) + 1\right] \tag{12.286}$$

Given a loading magnitude of $P_1 = 2 \times 10^6$ Pa, we present in Fig. 12.6 the osmotic pressure and pore pressure versus depth at various times.

We observe from Fig. 12.6b that at the instant of the stress loading, a constant pore pressure rise of the magnitude

$$p(z, 0^+) = \frac{3P_1 M_c \alpha_c'}{4G + 3K_\nu} = \frac{3P_1 B_c K_\nu}{4G + 3K_\nu} \tag{12.287}$$

is created, due to the Skempton pore pressure effect. It then dissipates to zero, starting from the surface, due to the drained boundary condition. Figure 12.6a on the other hand shows that the solute concentration (osmotic pressure) is initially zero. A positive osmotic pressure is then generated due to the decrease in pore pressure, as the rate of pore pressure decrease serves as source term for the solute concentration diffusion equation, as shown in (12.281). The magnitude of the osmotic pressure, however, is much smaller than that of pore pressure.

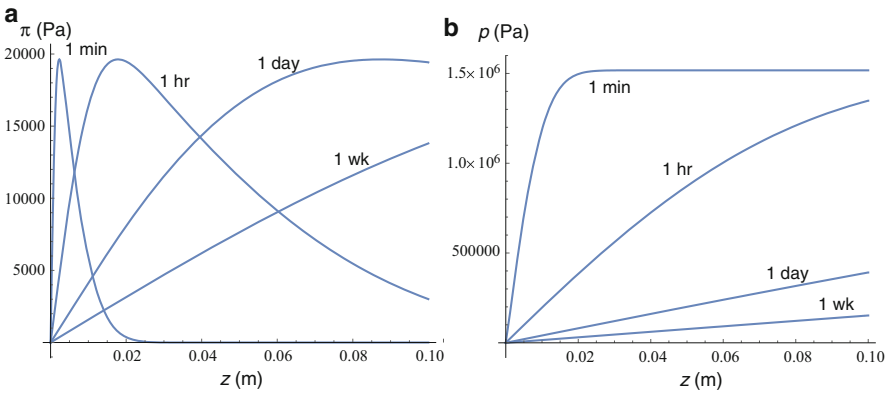


Fig. 12.6 One-dimensional half space problem with mechanical loading. (**a**) Solute concentration versus depth (expressed as osmotic pressure); and (**b**) pore pressure versus depth

12.7 Axial Symmetry Problem

In this section we examine problems possessing axial symmetry and under plane strain condition. The dependent variables have the following functional dependence on the spatial coordinates

$$u_r = u_r(r,t); \quad u_\theta = u_z = 0; \quad p = p(r,t); \quad c_s = c_s(r,t) \tag{12.288}$$

As demonstrated in Sect. 7.12, such displacement field is irrotational, which allows the simplification of solution.

12.7.1 General Solution

From (12.165) we find that the stress-strain relations are

$$\sigma_{rr} = \left(K_c - \frac{2G}{3}\right) e + 2G e_{rr} - \alpha_c p + \alpha_\mu c_s \tag{12.289}$$

$$\sigma_{\theta\theta} = \left(K_c - \frac{2G}{3}\right) e + 2G e_{\theta\theta} - \alpha_c p + \alpha_\mu c_s \tag{12.290}$$

$$\sigma_{zz} = \left(K_c - \frac{2G}{3}\right) e - \alpha_c p + \alpha_\mu c_s \tag{12.291}$$

And the Navier equation is

$$\left(K_c + \frac{4G}{3}\right)\left(\frac{\partial^2 u_r}{\partial r^2} + \frac{1}{r}\frac{\partial u_r}{\partial r} - \frac{u_r}{r^2}\right) - \alpha_c \frac{\partial p}{\partial r} + \alpha_\mu \frac{\partial c_s}{\partial r} = 0 \tag{12.292}$$

Following a similar treatment in Sect. 11.7, we find the radial displacement as

$$u_r = \frac{\eta_c}{G}\frac{1}{r}\int rp(r,t)\,dr - \frac{\eta_\mu}{G}\frac{1}{r}\int rc_s(r,t)\,dr + A_1(t)r + \frac{A_2(t)}{r} \tag{12.293}$$

Based on the constitutive equations (12.289), (12.290) and (12.291), we arrive at the following

$$\sigma_{rr} = -2\eta_c\frac{1}{r^2}\int rp(r,t)\,dr + 2\eta_\mu\frac{1}{r^2}\int rc_s(r,t)\,dr + \frac{2G}{1-2\nu_c}A_1(t) - \frac{2G}{r^2}A_2(t) \tag{12.294}$$

$$\sigma_{\theta\theta} = 2\eta_c\frac{1}{r^2}\int rp(r,t)\,dr - 2\eta_\mu\frac{1}{r^2}\int rc_s(r,t)\,dr - 2\eta_c p + 2\eta_\mu c_s + \frac{2G}{1-2\nu_c}A_1(t) + \frac{2G}{r^2}A_2(t) \tag{12.295}$$

$$\sigma_{zz} = -2\eta_c p + 2\eta_\mu c_s + \frac{4G\nu_c}{1-2\nu_c}A_1(t) \tag{12.296}$$

where

$$\nu_c = \frac{3K_c - 2G}{2(3K_c + G)} \tag{12.297}$$

is a Poisson ratio based on the constant solute concentration drained bulk modulus. The above solutions are expressed in terms of two variables p and c_s, which need to be solved from the two diffusion equations.

From (12.294) and (12.295) it is easy to show that

$$\sigma_{rr} + \sigma_{\theta\theta} + 2\eta_c p - 2\eta_\mu c_s = \left(4K_c + \frac{4G}{3}\right)A_1(t) \tag{12.298}$$

Comparing with (12.236), we can express the two diffusion equations (12.247) and (12.248) into the following

$$\frac{\partial p}{\partial t} - c_\alpha\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = -\alpha_\alpha\frac{\partial c_s}{\partial t} - \frac{2\alpha_c'}{S_\alpha}\frac{dA_1(t)}{dt} \tag{12.299}$$

$$\frac{\partial c_s}{\partial t} - c_\beta\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c_s}{\partial r}\right) = -\frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} - \frac{2\alpha_c'}{S_\beta}\frac{dA_1(t)}{dt} \tag{12.300}$$

These two equations can be used to solve for the two variables p and c_s, which in turn define the whole solution through (12.293)–(12.296).

12.7.2 Borehole Problem

Consider the plane strain borehole problems investigated in Sects. 7.14 and 11.8. The solution domain is defined within $r_o \le r < \infty$. For the solution to exist as $r \to \infty$, it is necessary that $A_1(t) = 0$ in the general solutions derived in the above section. The set of equations (12.293)–(12.296) then reduces to

$$u_r = \frac{\eta_c}{G}\frac{1}{r}\int rp(r,t)\,dr - \frac{\eta_\mu}{G}\frac{1}{r}\int rc_s(r,t)\,dr + \frac{A_2(t)}{r} \tag{12.301}$$

$$\sigma_{rr} = -2\eta_c\frac{1}{r^2}\int rp(r,t)\,dr + 2\eta_\mu\frac{1}{r^2}\int rc_s(r,t)\,dr - \frac{2G}{r^2}A_2(t) \tag{12.302}$$

$$\sigma_{\theta\theta} = 2\eta_c\frac{1}{r^2}\int rp(r,t)\,dr - 2\eta_\mu\frac{1}{r^2}\int rc_s(r,t)\,dr - 2\eta_c p + 2\eta_\mu c_s + \frac{2G}{r^2}A_2(t) \tag{12.303}$$

$$\sigma_{zz} = -2\eta_c p + 2\eta_\mu c_s \tag{12.304}$$

The diffusion equations (12.299) and (12.300) become

$$\frac{\partial p}{\partial t} - c_\alpha\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = -\alpha_\alpha\frac{\partial c_s}{\partial t} \tag{12.305}$$

$$\frac{\partial c_s}{\partial t} - c_\beta\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c_s}{\partial r}\right) = -\frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} \tag{12.306}$$

Chemical Loading First, we examine the problem that the loading is applied through a change of solute concentration in the borehole, while keeping all other conditions constants. This leads to the boundary condition that at $t = 0^+$,

$$\sigma_{rr} = 0; \quad p = 0; \quad c_s = c_1; \quad \text{at } r = r_o \tag{12.307}$$

As the solution system is the same as that for the thermoelasticity case, we can simply utilize the solution in Sect. 11.8.1 and obtain the current solution in the Laplace transform space as

$$\tilde{c}_s = \frac{c_1}{2C_Q s}\left[\frac{(c_\alpha - c_\beta + C_Q)\mathrm{K}_0(\lambda_1\sqrt{s}r)}{\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{(c_\alpha - c_\beta - C_Q)\mathrm{K}_0(\lambda_2\sqrt{s}r)}{\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{12.308}$$

$$\tilde{p} = \frac{c_1 c_\beta \alpha_\alpha}{C_Q s}\left[\frac{\mathrm{K}_0(\lambda_1\sqrt{s}r)}{\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{\mathrm{K}_0(\lambda_2\sqrt{s}r)}{\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{12.309}$$

where the parameters λ_1, λ_2 and C_Q are the same as those defined in (12.264) and (12.265). These Laplace transform solutions can be inverted to the time domain, similar to the inversion between (7.608) and (7.614). However, for the simplicity of numerical evaluation, we shall retain the Laplace transform expressions and utilize the numerical Laplace inversion for their evaluation.

To obtain the displacement and stress solutions, we carry out these two integrals

$$\int r\tilde{c}_s(r,s)\,dr = -\frac{c_1 r}{2C_Q s^{3/2}}\left[\frac{(c_\alpha - c_\beta + C_Q)\mathrm{K}_1(\lambda_1\sqrt{s}r)}{\lambda_1 \mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{(c_\alpha - c_\beta - C_Q)\mathrm{K}_1(\lambda_2\sqrt{s}r)}{\lambda_2 \mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{12.310}$$

$$\int r\tilde{p}(r,s)\,dr = -\frac{c_1 c_\beta \alpha_\alpha r}{C_Q s^{3/2}}\left[\frac{\mathrm{K}_1(\lambda_1\sqrt{s}r)}{\lambda_1 \mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{\mathrm{K}_1(\lambda_2\sqrt{s}r)}{\lambda_2 \mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{12.311}$$

The Laplace transform of $A_2(t)$ is found as

$$\tilde{A}_2 = -\frac{c_1 r_o}{2C_Q G s^{3/2}}\left\{\frac{[\eta_\mu(c_\alpha - c_\beta + C_Q) - 2\eta_c\alpha_\alpha c_\beta]\mathrm{K}_1(\lambda_1\sqrt{s}r_o)}{\lambda_1 \mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{[\eta_\mu(c_\alpha - c_\beta - C_Q) - 2\eta_c\alpha_\alpha c_\beta]\mathrm{K}_1(\lambda_2\sqrt{s}r_o)}{\lambda_2 \mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right\} \tag{12.312}$$

The above expressions, (12.310), (12.311) and (12.312), fully define the solutions (12.301)–(12.304) in the Laplace transform domain.

To examine these solutions, we assume that a borehole of radius $r_o = 10$ cm is excavated in a Gulf of Mexico shale with the material data set presented in Table 12.1. The formation is initially in equilibrium with the chemical, hydraulic, and mechanical forces. At $t = 0^+$, the chemistry of the fluid that the borehole wall is in contact with is suddenly changed, by the lowering of salt molar concentration in the amount c_1 as given in (12.271). All other conditions remain unchanged. The solution of this problem is presented in Fig. 12.7.

In Fig. 12.7a the osmotic pressure is plotted versus the radius distance for the first 20 cm of the formation, at various times. The pressure is negative due to the reduction in salt concentration. As time increases, the negative concentration diffuses further into the formation. Figure 12.7b shows that a positive pore pressure is generated, which must be reduced to zero at the borehole wall due to the boundary condition imposed. A peak pressure is observed near the wall. As time increases, it propagates into the formation with diminishing magnitude.

Figure 12.7c and d plot the radial and circumferential stresses, respectively. We observe that the radial stress vanishes at the borehole wall due to the boundary condition, but is in compression further into the formation, which increases with time. The circumferential (hoop) stress is in compression at the wall, and remains constant with time. Further into the formation, the stress is initially in tension and grows into compression as time increases.

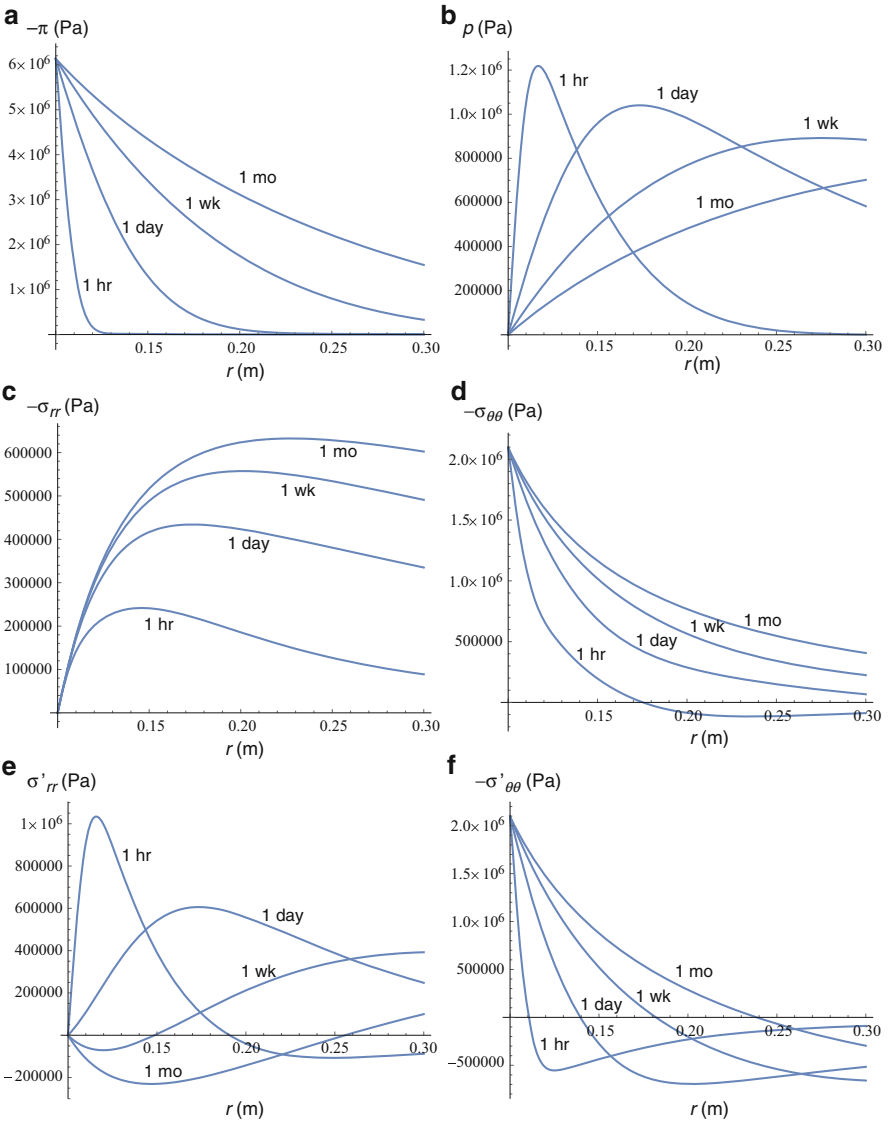


Fig. 12.7 Borehole under chemical loading. (**a**) Osmotic pressure versus radial distance at various times; (**b**) pore pressure; (**c**) radial stress; (**d**) tangential stress; (**e**) Terzaghi effective radial stress; and (**f**) Terzaghi tangential stress

For the purpose of borehole stability analysis, the Terzaghi effective stress is of interest. Figure 12.7e and f present the radial and circumferential Terzaghi effective stress, respectively. We observe that the positive pore pressure peak near the wellbore creates an effective tensile stress region. As the pore pressure peak

propagates further into the formation, the near wall region becomes compressive. For the hoop stress, there is a switch from the effective compressive stress at the wall to effective tensile stress within a short distance, which is also created by the pore pressure effect. These near wellbore changes in effective stress can cause borehole stability problems that need to be careful analyzed.

Hydraulic Loading Next we examine the problem of a step rise in pore pressure loading, with the boundary conditions

$$\sigma_{rr} = 0; \quad p = p_1; \quad c_s = 0; \quad \text{at } r = r_o \tag{12.313}$$

The solution of such problem can be obtained by observing the symmetry between the role of pore pressure and solution concentration in the pair of diffusion equations (12.305) and (12.306). We hence obtain

$$\tilde{p} = \frac{p_1}{2C_Q s}\left[\frac{(c_\alpha - c_\beta + C_Q)\mathrm{K}_0(\lambda_2\sqrt{s}r)}{\mathrm{K}_0(\lambda_2\sqrt{s}r_o)} - \frac{(c_\alpha - c_\beta - C_Q)\mathrm{K}_0(\lambda_1\sqrt{s}r)}{\mathrm{K}_0(\lambda_1\sqrt{s}r_o)}\right] \tag{12.314}$$

$$\tilde{c}_s = \frac{p_1 c_\alpha}{\alpha_\beta C_Q s}\left[\frac{\mathrm{K}_0(\lambda_1\sqrt{s}r)}{\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{\mathrm{K}_0(\lambda_2\sqrt{s}r)}{\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{12.315}$$

For the displacement and stress solutions, we find these two integrals

$$\int r\tilde{p}(r,s)\,dr = -\frac{p_1 r}{2C_Q s^{3/2}}\left[\frac{(c_\alpha - c_\beta + C_Q)\mathrm{K}_1(\lambda_2\sqrt{s}r)}{\lambda_2\mathrm{K}_0(\lambda_2\sqrt{s}r_o)} - \frac{(c_\alpha - c_\beta - C_Q)\mathrm{K}_1(\lambda_1\sqrt{s}r)}{\lambda_1\mathrm{K}_0(\lambda_1\sqrt{s}r_o)}\right] \tag{12.316}$$

$$\int r\tilde{c}_s(r,s)\,dr = -\frac{p_1 c_\alpha r}{\alpha_\beta C_Q s^{3/2}}\left[\frac{\mathrm{K}_1(\lambda_1\sqrt{s}r)}{\lambda_1\mathrm{K}_0(\lambda_1\sqrt{s}r_o)} - \frac{\mathrm{K}_1(\lambda_2\sqrt{s}r)}{\lambda_2\mathrm{K}_0(\lambda_2\sqrt{s}r_o)}\right] \tag{12.317}$$

and

$$\tilde{A}_2 = \frac{p_1 r_o}{2\alpha_\beta C_Q G s^{3/2}}\left\{\frac{[\alpha_\beta\eta_c(c_\alpha - c_\beta + C_Q) + 2\eta_\mu c_\alpha]\mathrm{K}_1(\lambda_2\sqrt{s}r_o)}{\lambda_2\mathrm{K}_0(\lambda_2\sqrt{s}r_o)} - \frac{[\alpha_\beta\eta_c(c_\alpha - c_\beta - C_Q) + 2\eta_\mu c_\alpha]\mathrm{K}_1(\lambda_1\sqrt{s}r_o)}{\lambda_1\mathrm{K}_0(\lambda_1\sqrt{s}r_o)}\right\} \tag{12.318}$$

The displacement and stresses are then defined in (12.301)–(12.304).

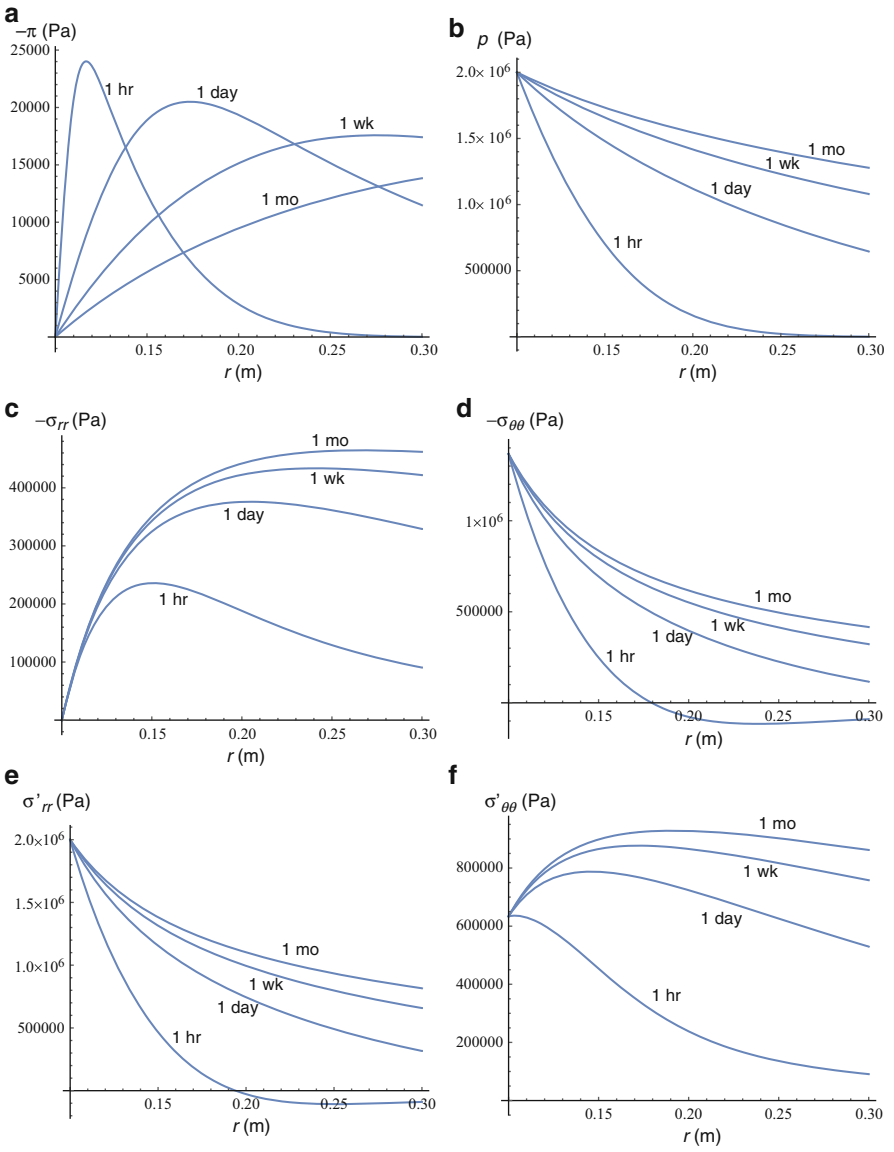


Fig. 12.8 Borehole under hydraulic loading. (**a**) Osmotic pressure versus radial distance at various times; (**b**) pore pressure; (**c**) radial stress; (**d**) tangential stress; (**e**) Terzaghi effective radial stress; and (**f**) Terzaghi tangential stress

For a pressure loading of $p_1 = 2 \times 10^6$ Pa, osmotic pressure, pore pressure, radial and circumferential stress, and the Terzaghi effective radial and circumferential stresses are plotted in Fig. 12.8.

Mechanical Loading For the problem of mechanical loading with the boundary conditions

$$\sigma_{rr} = -P_1; \quad p = 0; \quad c_s = 0; \quad \text{at } r = r_o \tag{12.319}$$

the solution is

$$u_r = \frac{P_1 r_o^2}{2Gr} \tag{12.320}$$

$$\sigma_{rr} = -\frac{P_1 r_o^2}{r^2} \tag{12.321}$$

$$\sigma_{\theta\theta} = \frac{P_1 r_o^2}{r^2} \tag{12.322}$$

Otherwise there is no pore pressure and solute concentration generated

$$p = c_s = 0 \tag{12.323}$$

as the deformation is entirely in shear strain.

For the problem of a borehole exposed to a drilling fluid that can exert radial stress, pore pressure, and solution concentration changes, the above solutions due to the chemical, hydraulic, and mechanical loading, can be superposed to find the combined effect. These solutions can be applied to borehole stability analyses, such as those found in [9, 15, 35, 39, 42, 47, 63].

12.7.3 Cylinder Problem

For the geometry of a cylinder, it is necessary to set $A_2(t) = 0$ in (12.293)–(12.296) and we find

$$u_r = \frac{\eta_c}{G}\frac{1}{r}\int rp(r,t)\,dr - \frac{\eta_\mu}{G}\frac{1}{r}\int rc_s(r,t)\,dr + A_1(t)r \tag{12.324}$$

$$\sigma_{rr} = -2\eta_c\frac{1}{r^2}\int rp(r,t)\,dr + 2\eta_\mu\frac{1}{r^2}\int rc_s(r,t)\,dr + \frac{2G}{1-2\nu_c}A_1(t) \tag{12.325}$$

$$\sigma_{\theta\theta} = 2\eta_c\frac{1}{r^2}\int rp(r,t)\,dr - 2\eta_\mu\frac{1}{r^2}\int rc_s(r,t)\,dr - 2\eta_c p + 2\eta_\mu c_s + \frac{2G}{1-2\nu_c}A_1(t) \tag{12.326}$$

$$\sigma_{zz} = -2\eta_c p + 2\eta_\mu c_s + \frac{4G\nu_c}{1-2\nu_c}A_1(t) \tag{12.327}$$

in which the pore pressure p and solute concentration c_s need to be solved from the pair of diffusion equations (12.299) and (12.300).

Chemical Loading The problem is defined by the boundary conditions given as (12.307). The diffusion equations (12.299) and (12.300) can be solved for the particular and the general solution, which are combined to give

$$\tilde{c}_s = -C_1 \mathrm{I}_0(\lambda_1\sqrt{s}r) - C_2 \mathrm{I}_0(\lambda_2\sqrt{s}r) - \frac{2\alpha_c'(\alpha_\beta S_\alpha - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.328}$$

$$\tilde{p} = \frac{C_1\alpha_\beta(c_\alpha - c_\beta - C_Q)\mathrm{I}_0(\lambda_1\sqrt{s}r)}{2c_\alpha} + \frac{C_2\alpha_\beta(c_\alpha - c_\beta + C_Q)\mathrm{I}_0(\lambda_2\sqrt{s}r)}{2c_\alpha} + \frac{2\alpha_\beta\alpha_c'(\alpha_\alpha S_\alpha - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.329}$$

where

$$C_1 = -\frac{1}{2sC_Q S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)\mathrm{I}_0(\lambda_1\sqrt{s}r_o)} \times\{2s\alpha_c'[(c_\alpha - c_\beta + C_Q)(S_\alpha\alpha_\beta - S_\beta) - 2c_\alpha(S_\alpha\alpha_\alpha - S_\beta)]\tilde{A}_1(s) + S_\alpha S_\beta c_1(c_\alpha - c_\beta + C_Q)(\alpha_\beta - \alpha_\alpha)\} \tag{12.330}$$

$$C_2 = \frac{1}{2sC_Q S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)\mathrm{I}_0(\lambda_2\sqrt{s}r_o)} \times\{2s\alpha_c'[(c_\alpha - c_\beta - C_Q)(S_\alpha\alpha_\beta - S_\beta) - 2c_\alpha(S_\alpha\alpha_\alpha - S_\beta)]\tilde{A}_1(s) + S_\alpha S_\beta c_1(c_\alpha - c_\beta - C_Q)(\alpha_\beta - \alpha_\alpha)\} \tag{12.331}$$

The integrals are

$$\int r\tilde{c}_s(r,s)\,dr = -\frac{C_1 r\mathrm{I}_1(\lambda_1\sqrt{s}r)}{\lambda_1\sqrt{s}} - \frac{C_2 r\mathrm{I}_1(\lambda_2\sqrt{s}r)}{\lambda_2\sqrt{s}} - \frac{\alpha_c' r^2(S_\alpha\alpha_\beta - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.332}$$

$$\int r\tilde{p}(r,s)\,dr = \frac{C_1\alpha_\beta r(c_\alpha - c_\beta - C_Q)\mathrm{I}_1(\lambda_1\sqrt{s}r)}{2c_\alpha\lambda_1\sqrt{s}} + \frac{C_2\alpha_\beta r(c_\alpha - c_\beta + C_Q)\mathrm{I}_1(\lambda_2\sqrt{s}r)}{2c_\alpha\lambda_2\sqrt{s}} + \frac{\alpha_\beta\alpha_c' r^2(S_\alpha\alpha_\alpha - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.333}$$

For the above solutions, we still need to resolve the constant $\tilde{A}_1(s)$. This is accomplished by utilizing the radial stress boundary condition, and we obtain

$$\begin{aligned}\tilde{A}_1(s) = & \frac{S_\alpha S_\beta c_1(1-2\nu_c)(\alpha_\beta - \alpha_\alpha)}{D} \\ & \times\{\lambda_1(c_\alpha - c_\beta - C_Q)\left[\eta_c\alpha_\beta(c_\alpha - c_\beta + C_Q) + 2c_\alpha\eta_\mu\right] \\ & \times \mathrm{I}_0(\lambda_1\sqrt{s}r_o)\mathrm{I}_1(\lambda_2\sqrt{s}r_o) \\ & -\lambda_2(c_\alpha - c_\beta + C_Q)\left[\eta_c\alpha_\beta(c_\alpha - c_\beta - C_Q) + 2c_\alpha\eta_\mu\right] \\ & \times \mathrm{I}_0(\lambda_2\sqrt{s}r_o)\mathrm{I}_1(\lambda_1\sqrt{s}r_o)\}\end{aligned} \tag{12.334}$$

where

$$\begin{aligned}D = & 2s\alpha_c'(1-2\nu_c)\left\{\lambda_2\left[(c_\alpha - c_\beta + C_Q)(S_\alpha\alpha_\beta - S_\beta) - 2c_\alpha(S_\alpha\alpha_\alpha - S_\beta)\right]\right. \\ & \times\left[\eta_c\alpha_\beta(c_\alpha - c_\beta - C_Q) + 2c_\alpha\eta_\mu\right]\mathrm{I}_0(\lambda_2\sqrt{s}r_o)\mathrm{I}_1(\lambda_1\sqrt{s}r_o) \\ & -\lambda_1\left[(c_\alpha - c_\beta - C_Q)(S_\alpha\alpha_\beta - S_\beta) - 2c_\alpha(S_\alpha\alpha_\alpha - S_\beta)\right] \\ & \left.\times\left[\eta_c\alpha_\beta(c_\alpha - c_\beta + C_Q) + 2c_\alpha\eta_\mu\right]\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\mathrm{I}_1(\lambda_2\sqrt{s}r_o)\right\} \\ & +4c_\alpha C_Q r_o\lambda_1\lambda_2 s^{3/2}\left\{GS_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)\right. \\ & \left.-\alpha_c'(1-2\nu_c)\left[S_\alpha\alpha_\beta(\eta_c\alpha_\alpha + \eta_\mu) - S_\beta(\eta_c\alpha_\beta + \eta_\mu)\right]\right\} \\ & \times\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\mathrm{I}_0(\lambda_2\sqrt{s}r_o)\end{aligned} \tag{12.335}$$

With the back substitution, the entire solutions are defined.

For a cylinder of radius 10 cm, and a chemical loading the same as the borehole problem in Sect. 12.7.2, we present in Fig. 12.9a the osmotic pressure versus the radius distance at various times. The pore pressure generated by the chemical loading is plotted in Fig. 12.9b. At small time, a peak pressure is observed near the surface, which is reduced to negative toward the center, because the center is largely in tension. As time increases, the positive pore pressure propagates inward, and then eventually all pore pressure must dissipate to zero. The radial stress is shown in Fig. 12.9c. As the outer core of the cylinder expands due to the reduction in solute concentration, a tension develops in the inner part. The magnitude of the tension first increases, and then decreases. At large time the core is uniformly expanded, and the radial stress vanishes. The circumferential stress as shown in Fig. 12.9d is more complex. It is in compression in the outer part of the cylinder and turns into tension moving inward. Again, at large time, all stresses vanish. The Terzaghi effective stresses are plotted in Fig. 12.9e and f. The effective radial stress seems to be dominated by the pore pressure behavior, while the effective circumferential stress exhibits more complex behavior.

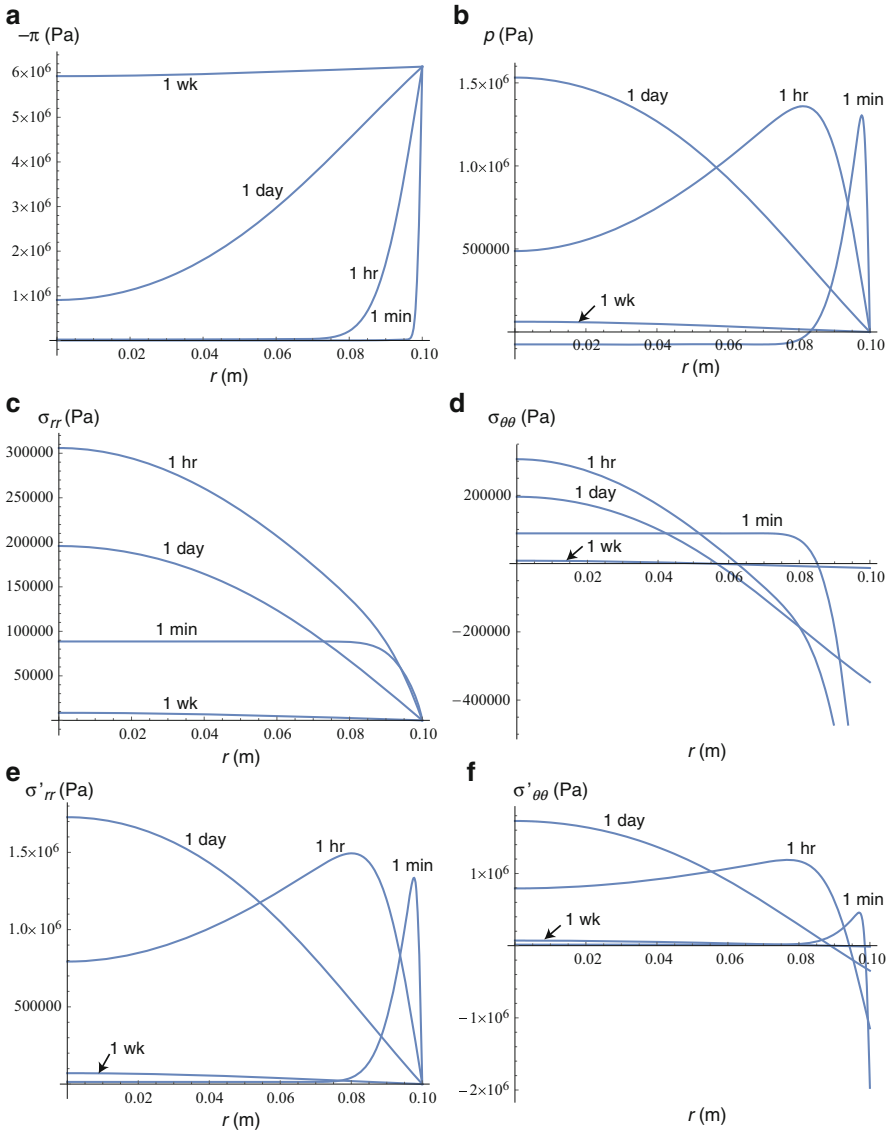


Fig. 12.9 Cylinder under chemical loading. (**a**) Osmotic pressure versus radial distance at various times; (**b**) pore pressure; (**c**) radial stress; (**d**) tangential stress; (**e**) Terzaghi effective radial stress; and (**f**) Terzaghi tangential stress

Hydraulic Loading The problem is defined by the boundary conditions (12.313). The solution can be obtained by invoking the symmetry of the diffusion equations (12.299) and (12.300), and we obtain

$$\tilde{p} = -C_1 \mathrm{I}_0(\lambda_1\sqrt{s}r) - C_2 \mathrm{I}_0(\lambda_2\sqrt{s}r) + \frac{2\alpha_\beta\alpha_c'(S_\alpha\alpha_\alpha - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.336}$$

$$\tilde{c}_s = \frac{C_1\alpha_\beta(c_\alpha - c_\beta - C_Q)\mathrm{I}_0(\lambda_1\sqrt{s}r)}{2c_\alpha} + \frac{C_2\alpha_\beta(c_\alpha - c_\beta + C_Q)\mathrm{I}_0(\lambda_2\sqrt{s}r)}{2c_\alpha} - \frac{2\alpha_c'(S_\alpha\alpha_\beta - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.337}$$

where

$$C_1 = -\frac{1}{2sC_Q S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)\mathrm{I}_0(\lambda_1\sqrt{s}r_o)} \times\{2s\alpha_c'[\alpha_\beta(c_\alpha - c_\beta - C_Q)(S_\alpha\alpha_\alpha - S_\beta) + 2c_\alpha(S_\alpha\alpha_\beta - S_\beta)]\tilde{A}_1(s) - S_\alpha S_\beta p_1(c_\alpha - c_\beta - C_Q)(\alpha_\beta - \alpha_\alpha)\} \tag{12.338}$$

$$C_2 = \frac{1}{2sC_Q S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)\mathrm{I}_0(\lambda_2\sqrt{s}r_o)} \times\{2s\alpha_c'[\alpha_\beta(c_\alpha - c_\beta + C_Q)(S_\alpha\alpha_\alpha - S_\beta) + 2c_\beta\alpha_\alpha(S_\alpha\alpha_\beta - S_\beta)]\tilde{A}_1(s) - S_\alpha S_\beta p_1(c_\alpha - c_\beta + C_Q)(\alpha_\beta - \alpha_\alpha)\} \tag{12.339}$$

The integrals are

$$\int r\tilde{p}(r,s)\,dr = -\frac{C_1 r\mathrm{I}_1(\lambda_1\sqrt{s}r)}{\lambda_1\sqrt{s}} - \frac{C_2 r\mathrm{I}_1(\lambda_2\sqrt{s}r)}{\lambda_2\sqrt{s}} + \frac{\alpha_\beta\alpha_c' r^2(S_\alpha\alpha_\alpha - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.340}$$

$$\int r\tilde{c}_s(r,s)\,dr = \frac{C_1\alpha_\beta r(c_\alpha - c_\beta - C_Q)\mathrm{I}_1(\lambda_1\sqrt{s}r)}{2c_\alpha\lambda_1\sqrt{s}} + \frac{C_2\alpha_\beta r(c_\alpha - c_\beta + C_Q)\mathrm{I}_1(\lambda_2\sqrt{s}r)}{2c_\alpha\lambda_2\sqrt{s}} - \frac{\alpha_c' r^2(S_\alpha\alpha_\beta - S_\beta)}{S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.341}$$

Finally, we find the constant $\tilde{A}_1(s)$ as

$$\tilde{A}_1(s) = \frac{S_\alpha S_\beta p_1(1 - 2\nu_c)(\alpha_\beta - \alpha_\alpha)}{D} \times\{\lambda_1(c_\alpha - c_\beta + C_Q)\left[\eta_c(c_\alpha - c_\beta - C_Q) - 2c_\beta\alpha_\alpha\eta_\mu\right]$$

$$
\begin{aligned}
&\times \mathrm{I}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{I}_1(\lambda_2\sqrt{s}r_o)\\
&-\lambda_2(c_\alpha - c_\beta - C_Q)\left[\eta_c(c_\alpha - c_\beta + C_Q) - 2c_\beta\alpha_\alpha\eta_\mu\right]\\
&\times \mathrm{I}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{I}_1(\lambda_1\sqrt{s}r_o)\}
\end{aligned}
\tag{12.342}
$$

where

$$
\begin{aligned}
D = {} & 2s\alpha_c'(1-2\nu_c)\left\{\lambda_2\left[\alpha_\beta(c_\alpha - c_\beta - C_Q)(S_\alpha\alpha_\alpha - S_\beta) + 2c_\beta\alpha_\alpha(S_\alpha\alpha_\beta - S_\beta)\right]\right.\\
&\times\left[-\eta_c(c_\alpha - c_\beta + C_Q) + 2c_\beta\alpha_\alpha\eta_\mu\right]\mathrm{I}_0(\lambda_2\sqrt{s}r_o)\,\mathrm{I}_1(\lambda_1\sqrt{s}r_o)\\
&-\lambda_1\left[\alpha_\beta(c_\alpha - c_\beta + C_Q)(S_\alpha\alpha_\alpha - S_\beta) + 2c_\beta\alpha_\alpha(S_\alpha\alpha_\beta - S_\beta)\right]\\
&\times\left.\left[-\eta_c(c_\alpha - c_\beta - C_Q) + 2c_\beta\alpha_\alpha\eta_\mu\right]\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{I}_1(\lambda_2\sqrt{s}r_o)\right\}\\
&+4c_\beta C_Q r_o\lambda_1\lambda_2 s^{3/2}\left\{GS_\alpha S_\beta\alpha_\alpha(\alpha_\beta - \alpha_\alpha)\right.\\
&\left.-\alpha_c'\alpha_\alpha(1-2\nu_c)\left[S_\beta(\eta_c + \alpha_\beta\eta_\mu) - S_\alpha\alpha_\beta(\eta_c + \alpha_\alpha\eta_\mu)\right]\right\}\\
&\times\mathrm{I}_0(\lambda_1\sqrt{s}r_o)\,\mathrm{I}_0(\lambda_2\sqrt{s}r_o)
\end{aligned}
\tag{12.343}
$$

12.8 Spherical Symmetry Problem

Problems of spherical symmetry is defined by the following

$$
u_R = u_R(R,t); \quad u_\theta = u_\varphi = 0; \quad p = p(R,t); \quad \zeta_s = \zeta_s(R,t) \tag{12.344}
$$

For stresses and strains, all normal components exist, while all shear components vanish.

12.8.1 General Solution

Following the similar procedure as in Sect. 7.9,

$$
e = \frac{\eta_c}{G}p - \frac{\eta_\mu}{G}\zeta_s + 3A_1(t) \tag{12.345}
$$

from which we can construct the following general solutions

$$
u_R = \frac{\eta_c}{G}\frac{1}{R^2}\int R^2 p(R,t)\,dR - \frac{\eta_\mu}{G}\frac{1}{R^2}\int R^2\,\zeta_s(R,t)\,dR + A_1(t)R + \frac{A_2(t)}{R^2} \tag{12.346}
$$

$$\sigma_{RR} = -4\eta_c \frac{1}{R^3}\int R^2 p(R,t)\, dR + 4\eta_\mu \frac{1}{R^3}\int R^2 c_s(R,t)\, dR + \frac{2G(1+\nu_c)}{1-2\nu_c}A_1(t) - \frac{4G}{R^3}A_2(t) \tag{12.347}$$

$$\sigma_{\varphi\varphi} = 2\eta_c \frac{1}{R^3}\int R^2 p(R,t)\, dR - 2\eta_\mu \frac{1}{R^3}\int R^2 c_s(R,t)\, dR - 2\eta_c p + 2\eta_\mu c_s + \frac{2G(1+\nu_c)}{1-2\nu_c}A_1(t) + \frac{2G}{R^3}A_2(t) \tag{12.348}$$

From the above, it is clear that

$$\sigma_{RR} + 2\sigma_{\varphi\varphi} + 4\eta_c p - 4\eta_\mu c_s = \frac{6G(1+\nu_c)}{1-2\nu_c}A_1(t) \tag{12.349}$$

The pore pressure and solute concentration are solved from these diffusion equations

$$\frac{\partial p}{\partial t} - c_\alpha \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2 \frac{\partial p}{\partial R}\right) = -\alpha_\alpha \frac{\partial c_s}{\partial t} - \frac{3\alpha_c'}{S_\alpha}\frac{dA_1(t)}{dt} \tag{12.350}$$

$$\frac{\partial c_s}{\partial t} - c_\beta \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2 \frac{\partial c_s}{\partial R}\right) = -\frac{1}{\alpha_\beta}\frac{\partial p}{\partial t} - \frac{3\alpha_c'}{S_\beta}\frac{dA_1(t)}{dt} \tag{12.351}$$

12.8.2 Cavity Problem

The cavity problem is an exterior domain problem. The boundedness condition at $R \to \infty$ requires us to set $A_1(t) = 0$ in the above equations. For the set of boundary conditions

$$\sigma_{RR} = 0; \quad p = 0; \quad c_s = c_1; \quad \text{at } R = R_o \tag{12.352}$$

the solution of (12.350) and (12.351) is found as the following

$$p(R,T) = \frac{c_1 c_\beta \alpha_\alpha R_o}{C_Q R}\left\{\operatorname{erfc}\left[\frac{\lambda_1(R-R_o)}{2\sqrt{t}}\right] - \operatorname{erfc}\left[\frac{\lambda_2(R-R_o)}{2\sqrt{t}}\right]\right\} \tag{12.353}$$

$$c_s(R,t) = \frac{c_1 R_o}{2C_Q R}\left\{(c_\alpha - c_\beta + C_Q)\operatorname{erfc}\left[\frac{\lambda_1(R-R_o)}{2\sqrt{t}}\right] - (c_\alpha - c_\beta - C_Q)\operatorname{erfc}\left[\frac{\lambda_2(R-R_o)}{2\sqrt{t}}\right]\right\} \tag{12.354}$$

The integrals are

$$\int R^2 p(R,t)\, dR = -\frac{c_1 c_\beta \alpha_\alpha R_o}{C_Q}\left\{ \frac{R+R_o}{\lambda_1}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_1 (R-R_o)^2/4t} - \left[\frac{R^2-R_o^2}{2} - \frac{t}{\lambda_1^2}\right] \operatorname{erfc}\left[\frac{\lambda_1 (R-R_o)}{2\sqrt{t}}\right] - \frac{R+R_o}{\lambda_2}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_2 (R-R_o)^2/4t} + \left[\frac{R^2-R_o^2}{2} - \frac{t}{\lambda_2^2}\right] \operatorname{erfc}\left[\frac{\lambda_2 (R-R_o)}{2\sqrt{t}}\right]\right\} \tag{12.355}$$

$$\int R^2 c_s(R,t)\, dR = \frac{c_1 R_o}{2C_Q} \times \left\{ (c_\alpha - c_\beta + C_Q)\left(-\frac{R+R_o}{\lambda_1}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_1 (R-R_o)^2/4t} + \left[\frac{R^2-R_o^2}{2} - \frac{t}{\lambda_1^2}\right] \operatorname{erfc}\left[\frac{\lambda_1 (R-R_o)}{2\sqrt{t}}\right]\right) - (c_\alpha - c_\beta - C_Q)\left(-\frac{R+R_o}{\lambda_2}\sqrt{\frac{t}{\pi}}\, e^{-\lambda_2 (R-R_o)^2/4t} + \left[\frac{R^2-R_o^2}{2} - \frac{t}{\lambda_2^2}\right] \operatorname{erfc}\left[\frac{\lambda_2 (R-R_o)}{2\sqrt{t}}\right]\right)\right\} \tag{12.356}$$

The function $A_2(t)$ is given by

$$A_2(t) = -\frac{R_o c_1}{2GC_Q \lambda_1^2 \lambda_2^2}\sqrt{\frac{t}{\pi}}\left\{ (\lambda_1 - \lambda_2)\left[\eta_\mu (c_\beta - c_\alpha) - 2c_\beta \alpha_\alpha \eta_c\right] \times \left[2R_o \lambda_1 \lambda_2 + \sqrt{\pi t}(\lambda_1 + \lambda_2)\right] + C_Q \eta_\mu \left[2R_o \lambda_1 \lambda_2 (\lambda_1 + \lambda_2) + \sqrt{\pi t}(\lambda_1^2 + \lambda_2^2)\right]\right\} \tag{12.357}$$

The displacement and stresses are given as (12.346), (12.347) and (12.348).

12.8.3 Sphere Problem

The geometry of a sphere requires that $A_2(t) = 0$ in the equations presented in Sect. 12.8.1. The particular and general solutions of (12.350) and (12.351) are obtained in the Laplace transform domain as

$$\tilde{c}_s = -C_1 \frac{\sinh(\lambda_1 \sqrt{s} R)}{R} - C_2 \frac{\sinh(\lambda_2 \sqrt{s} R)}{R} - \frac{2\alpha_c'(\alpha_\beta S_\alpha - S_\beta)}{S_\alpha S_\beta (\alpha_\beta - \alpha_\alpha)} \tilde{A}_1(s) \tag{12.358}$$

$$\begin{aligned} \tilde{p} = {} & \frac{C_1(c_\alpha - c_\beta - C_Q)}{2c\alpha_p} \frac{\sinh(\lambda_1 \sqrt{s} R)}{R} \\ & + \frac{C_2 \alpha_\beta (c_\alpha - c_\beta + C_Q)}{2c_\alpha} \frac{\sinh(\lambda_2 \sqrt{s} R)}{R} + \frac{2\alpha_\beta \alpha_c'(\alpha_\alpha S_\alpha - S_\beta)}{S_\alpha S_\beta (\alpha_\beta - \alpha_\alpha)} \tilde{A}_1(s) \end{aligned} \tag{12.359}$$

For the boundary condition of chemical loading, that is, $c = c_1$ at $R = R_o$, we find

$$\begin{aligned} C_1 = {} & -\frac{R_o}{2sC_Q S_\alpha S_\beta (\alpha_\beta - \alpha_\alpha) \sinh(\lambda_1 \sqrt{s} R_o)} \\ & \times \{2s\alpha_c' \left[(c_\alpha - c_\beta + C_Q)(S_\alpha \alpha_\beta - S_\beta) - 2c_\alpha (S_\alpha \alpha_\alpha - S_\beta) \right] \tilde{A}_1(s) \\ & + c_1 S_\alpha S_\beta (c_\alpha - c_\beta + C_Q)(\alpha_\beta - \alpha_\alpha)\} \end{aligned} \tag{12.360}$$

$$\begin{aligned} C_2 = {} & \frac{R_o}{2sC_Q S_\alpha S_\beta (\alpha_\beta - \alpha_\alpha) \sinh(\lambda_2 \sqrt{s} R_o)} \\ & \times \{2s\alpha_c' \left[(c_\alpha - c_\beta - C_Q)(S_\alpha \alpha_\beta - S_\beta) - 2c_\alpha (S_\alpha \alpha_\alpha - S_\beta) \right] \tilde{A}_1(s) \\ & - c_1 S_\alpha S_\beta (c_\alpha - c_\beta - C_Q)(\alpha_\beta - \alpha_\alpha)\} \end{aligned} \tag{12.361}$$

The integrals are

$$\begin{aligned} \int R^2 \tilde{c}_s(R,s)\, dR = {} & -C_1 \left[\frac{R \cosh(\lambda_1 \sqrt{s} R)}{\lambda_1 \sqrt{s}} - \frac{\sinh(\lambda_1 \sqrt{s} R)}{\lambda_1^2 s} \right] \\ & - C_2 \left[\frac{R \cosh(\lambda_2 \sqrt{s} R)}{\lambda_2 \sqrt{s}} - \frac{\sinh(\lambda_2 \sqrt{s} R)}{\lambda_2^2 s} \right] \\ & - \frac{2R^3 \alpha_c'(\alpha_\beta S_\alpha - S_\beta)}{3S_\alpha S_\beta (\alpha_\beta - \alpha_\alpha)} \tilde{A}_1(s) \end{aligned} \tag{12.362}$$

$$\begin{aligned} \int R^2 \tilde{p}(R,s)\, dR = {} & \frac{C_1 \alpha_\beta (c_\alpha - c_\beta - C_Q)}{2c_\alpha} \\ & \times \left[\frac{R \cosh(\lambda_1 \sqrt{s} R)}{\lambda_1 \sqrt{s}} - \frac{\sinh(\lambda_1 \sqrt{s} R)}{\lambda_1^2 s} \right] \\ & + \frac{C_2 \alpha_\beta (c_\alpha - c_\beta + C_Q)}{2c_\alpha} \left[\frac{R \cosh(\lambda_2 \sqrt{s} R)}{\lambda_2 \sqrt{s}} - \frac{\sinh(\lambda_2 \sqrt{s} R)}{\lambda_2^2 s} \right] \end{aligned}$$

$$+\frac{2R^3\alpha_\beta\alpha_c'(\alpha_\alpha S_\alpha - S_\beta)}{3S_\alpha S_\beta(\alpha_\beta - \alpha_\alpha)}\tilde{A}_1(s) \tag{12.363}$$

The constant $\tilde{A}_1(s)$ is given by

$$\begin{aligned}\tilde{A}_1(s) = &\frac{3c_1 S_\alpha S_\beta(1-2\nu_c)(\alpha_\beta - \alpha_\alpha)}{2sD}\\ &\times\left\{\lambda_2^2(c_\alpha - c_\beta + C_Q)\left[\eta_c\alpha_\beta(c_\alpha - c_\beta - C_Q) + 2c_\alpha\eta_\mu\right]\right.\\ &\times\left[1 - \lambda_1\sqrt{s}R_o\coth\left(\lambda_1\sqrt{s}R_o\right)\right]\\ &-\lambda_1^2(c_\alpha - c_\beta - C_Q)\left[\eta_c\alpha_\beta(c_\alpha - c_\beta + C_Q) + 2c_\alpha\eta_\mu\right]\\ &\left.\times\left[1 - \lambda_2\sqrt{s}R_o\coth\left(\lambda_2\sqrt{s}R_o\right)\right]\right\}\end{aligned} \tag{12.364}$$

where

$$\begin{aligned}D = &3(1-2\nu_c)\left\{-\lambda_2^2\alpha_c'[(c_\alpha - c_\beta + C_Q)(S_\alpha\alpha_\beta - S_\beta) - 2c_\alpha(S_\alpha\alpha_\alpha - S_\beta)]\right.\\ &\times[\eta_c\alpha_\beta(c_\alpha - c_\beta - C_Q) + 2c_\alpha\eta_\mu][1 - R_o\sqrt{s}\lambda_1\coth(\lambda_1\sqrt{s}R_o)]\\ &+\lambda_1^2\alpha_c'[(c_\alpha - c_\beta - C_Q)(S_\alpha\alpha_\beta - S_\beta) - 2c_\alpha(S_\alpha\alpha_\alpha - S_\beta)]\\ &\left.\times[\eta_c\alpha_\beta(c_\alpha - c_\beta + C_Q) + 2c_\alpha\eta_\mu][1 - R_o\sqrt{s}\lambda_2\coth(\lambda_2\sqrt{s}R_o)]\right\}\\ &+c_\alpha C_Q R_o^2 s\lambda_1^2\lambda_2^2\left\{3GS_\alpha S_\beta(1+\nu_c)(\alpha_\beta - \alpha_\alpha)\right.\\ &\left.-4\alpha_c'(1-2\nu_c)[S_\alpha(\alpha_\alpha\eta_c + \eta_\mu) - S_\beta(\alpha_\beta\eta_c + \eta_\mu)]\right\}\end{aligned} \tag{12.365}$$

The displacement and stresses are defined in (12.346), (12.347) and (12.348).

References

1. Achanta S, Cushman JH, Okos MR (1994) On multicomponent, multiphase thermomechanics with interfaces. Int J Eng Sci 32(11):1717–1738
2. Alonso EE, Vaunat J, Gens A (1999) Modelling the mechanical behaviour of expansive clays. Eng Geol 54(1–2):173–183
3. Baierlein R (2001) The elusive chemical potential. Am J Phys 69(4):423–434
4. Bear J, Cheng AHD (2010) Modeling groundwater flow and contaminant transport. Springer, Dordrecht/London, 834pp
5. Bennethum L, Murad M, Cushman J (2000) Macroscale thermodynamics and the chemical potential for swelling porous media. Transp Porous Media 39(2):187–225
6. Bronick CJ, Lal R (2005) Soil structure and management: a review. Geoderma 124(1–2):3–22
7. Das BM (1997) Advanced soil mechanics, 2nd edn. Taylor & Francis, London/New York, 457pp
8. Dormieux L, Lemarchand E, Coussy O (2003) Macroscopic and micromechanical approaches to the modelling of the osmotic swelling in clays. Transp Porous Media 50(1–2):75–91

9. Ekbote S, Abousleiman Y (2006) Porochemoelastic solution for an inclined borehole in a transversely isotropic formation. J Eng Mech ASCE 132(7):754–763
10. Epstein N (1989) On tortuosity and the tortuosity factor in flow and diffusion through porous media. Chem Eng Sci 44(3):777–779
11. Fritz SJ (1986) Ideality of clay membranes in osmotic processes–a review. Clays Clay Miner 34(2):214–223
12. Gajo A, Loret B, Hueckel T (2002) Electro-chemo-mechanical couplings in saturated porous media: elastic-plastic behaviour of heteroionic expansive clays. Int J Solids Struct 39(16):4327–4362
13. Ghassemi A, Diek A (2002) Porothermoelasticity for swelling shales. J Pet Sci Eng 34(1–4):123–135
14. Ghassemi A, Diek A (2003) Linear chemo-poroelasticity for swelling shales: theory and application. J Pet Sci Eng 38(3–4):199–212
15. Ghassemi A, Tao Q, Diek A (2009) Influence of coupled chemo-poro-thermoelastic processes on pore pressure and stress distributions around a wellbore in swelling shale. J Pet Sci Eng 67(1–2):57–64
16. Gibbs JW (1873) Graphical methods in the thermodynamics of fluids. Trans Conn Acad Arts Sci 2:309–342
17. Gibbs JW (1873) A method of geometrical representation of the thermodynamic properties of substances by means of surfaces. Trans Conn Acad Arts Sci 2:382–404
18. Gibbs JW (1876/1878) On the equilibrium of heterogeneous substances. Trans Conn Acad Arts Sci 3:108–248, 343–524
19. Gu WY, Lai WM, Mow VC (1998) A mixture theory for charged hydrated soft tissues containing multi-electrolytes: passive transport and swelling behaviors. J Biomech Eng ASME 120(2):169–180
20. Heidug WK, Wong SW (1996) Hydration swelling of water-absorbing rocks: a constitutive model. Int J Numer Anal Methods Geomech 20(6):403–430
21. Hensen EJM, Smit B (2002) Why clays swell. J Phys Chem B 106(49):12664–12667
22. Hong W, Liu Z, Suo Z (2009) Inhomogeneous swelling of a gel in equilibrium with a solvent and mechanical load. Int J Solids Struct 46(17):3282–3289
23. Huyghe JM, Janssen JD (1997) Quadriphasic mechanics of swelling incompressible porous media. Int J Eng Sci 35(8):793–802
24. Karaborni S, Smit B, Heidug W, Urai J, van Oort E (1996) The swelling of clays: molecular simulations of the hydration of montmorillonite. Science 271(5252):1102–1104
25. Katchalsky A, Curran PF (1967) Nonequilibrium thermodynamics in biophysics. Harvard University Press, Cambridge, 248pp
26. Lai WM, Hou JS, Mow VC (1991) A triphasic theory for the swelling and deformation behaviors of articular cartilage. J Biomech Eng ASME 113(3):245–258
27. Loret B, Hueckel T, Gajo A (2002) Chemo-mechanical coupling in saturated porous media: elastic-plastic behaviour of homoionic expansive clays. Int J Solids Struct 39(10):2773–2806
28. Low PF (1980) The swelling of clay. 2. Montmorillonites. Soil Sci Soc Am J 44(4):667–676
29. Low PF (1987) Structural component of the swelling pressure of clays. Langmuir 3(1):18–25
30. Low PF, Margheim JF (1979) Swelling of clay. 1. Basic concepts and empirical equations. Soil Sci Soc Am J 43(3):473–481
31. Madsen FT, Müller-Vonmoos M (1989) The swelling behaviour of clays. Appl Clay Sci 4(2):143–156
32. Marcombe R, Cai S, Hong W, Zhao X, Lapusta Y, Suo Z (2010) A theory of constrained swelling of a pH-sensitive hydrogel. Soft Matter 6(4):784–793
33. McBride MB (1997) A critique of diffuse double layer models applied to colloid and surface chemistry. Clays Clay Miner 45(4):598–608
34. Mitchell JK, Soga K (2005) Fundamentals of soil behavior, 3rd edn. Wiley, New York, 592pp
35. Mody FK, Hale AH (1993) Borehole-stability model to couple the mechanics and chemistry of drilling-fluid/shale interactions. J Pet Technol 45(11):1093–1101

36. Moyne C, Murad MA (2002) Electro-chemo-mechanical couplings in swelling clays derived from a micro/macro-homogenization procedure. Int J Solids Struct 39(25):6159–6190
37. Murad MA, Cushman JH (2000) Thermomechanical theories for swelling porous media with microstructure. Int J Eng Sci 38(5):517–564
38. Nguyen V, Abousleiman Y (2005) Analysis of Biot's poroelastic parameters in chemically active porous media. In: Abousleiman YN, Cheng AHD, Ulm F-J (eds) Poromechanics III—Biot Centennial (1905–2005), Proceedings of the 3rd Biot conference on poromechanics. Balkema, Leiden, 695–703
39. Nguyen VX, Abousleiman YN (2010) Incorporating electrokinetic effects in the porochemoelastic inclined wellbore formulation and solution. Anais Da Academia Brasileira De Ciencias 82(1):195–222
40. Rogers DW (1962) Deriving the Gibbs-Duhem equation. J Chem Educ 39(10):527–528
41. Roshan H, Oeser M (2012) A non-isothermal constitutive model for chemically active elastoplastic rocks. Rock Mech Rock Eng 45(3):361–374
42. Roshan H, Rahman SS (2011) A fully coupled chemo-poroelastic analysis of pore pressure and stress distribution around a wellbore in water active rocks. Rock Mech Rock Eng 44(2):199–210
43. Sachs JR, Grodzinsky AJ (1989) An electromechanically coupled poroelastic medium driven by an applied electric-current—surface detection of bulk material properties. Physicochem Hydrodyn 11(4):585–614
44. Sarout J, Detournay E (2011) Chemoporoelastic analysis and experimental validation of the pore pressure transmission test for reactive shales. Int J Rock Mech Min Sci 48:759–772
45. Sherwood JD (1993) Biot poroelasticity of a chemically active shale. Proc R Soc A–Math Phys Eng Sci 440(1909):365–377
46. Sherwood JD (1994) A model for the flow of water and ions into swelling shale. Langmuir 10(7):2480–2486
47. Sherwood JD, Bailey L (1994) Swelling of shale around a cylindrical wellbore. Proc R Soc Lond Ser A–Math Phys Eng Sci 444(1920):161–184
48. Sherwood JD, Craster B (2000) Transport of water and ions through a clay membrane. J Colloid Interface Sci 230(2):349–358
49. Sposito G, Skipper NT, Sutton R, Park SH, Soper AK, Greathouse JA (1999) Surface geochemistry of the clay minerals. Proc Natl Acad Sci 96(7):3358–3364
50. Steiger RP, Leung PK (1992) Quantitative determination of the mechanical properties of shales. SPE Drill Eng 7(3):181–185
51. Taron J, Elsworth D (2009) Thermal-hydrologic-mechanical-chemical processes in the evolution of engineered geothermal reservoirs. Int J Rock Mech Min Sci 46(5):855–864
52. Taron J, Elsworth D, Min K-B (2009) Numerical simulation of thermal-hydrologic-mechanical-chemical processes in deformable, fractured porous media. Int J Rock Mech Min Sci 46(5):842–854
53. Tsang C-F (2009) Introductory editorial to the special issue on the DECOVALEX-THMC project. Environ Geol 57(6):1217–1219
54. van Meerveld J, Molenaar MM, Huyghe JM, Baaijens FPT (2003) Analytical solution of compression, free swelling and electrical loading of saturated charged porous media. Transp Porous Media 50(1–2):111–126
55. van Olphen H (1977) An introduction to clay colloid chemistry, 2nd edn. Wiley, New York, 318pp
56. van Oort E (2003) On the physical and chemical stability of shales. J Pet Sci Eng 38(3–4):213–235
57. van Oort E, Hale AH, Mody FK, Roy S (1996) Transport in shales and the design of improved water-based shale drilling fluids. SPE Drill Complet 11(3):137–146
58. van't Hoff JH (1901) Osmotic pressure and chemical equilibrium, Nobel Prize Lecture
59. Velde B (1992) Introduction to clay minerals: chemistry, origins, uses and environmental significance. Springer, Dordrecht, 198pp

60. Viani BE, Low PF, Roth CB (1983) Direct measurement of the relation between interlayer force and interlayer distance in the swelling of montmorillonite. J Colloid Interface Sci 96(1):229–244
61. Wheeler SJ, Sivakumar V (1995) An elasto-plastic critical state framework for unsaturated soil. Géotechnique 45(1):35–53
62. Wilson W, van Donkelaar CC, Huyghe JM (2005) A comparison between mechano-electrochemical and biphasic swelling theories for soft hydrated tissues. J Biomech Eng 127(1):158–165
63. Yu M, Chenevert ME, Sharma MM (2003) Chemical-mechanical wellbore instability model for shales: accounting for solute diffusion. J Pet Sci Eng 38(3–4):131–143

Appendix A
List of Symbols

Roman Letter

A	action
$\mathcal{A}, \mathcal{A}_s, \mathcal{A}_f$	Helmholtz free energy, and for solid and fluid phase
$\mathcal{A}, \mathcal{A}_s, \mathcal{A}_f$	Helmholtz free energy density, and for solid and fluid phase
$\overline{\mathcal{A}}, \overline{\mathcal{A}}_s, \overline{\mathcal{A}}_f$	REV averaged Helmholtz free energy density, and strain energy density function, and for solid and fluid phase
a^f, a^f_{ij}	fluid apparent mass constitutive constant scalar and tensor
a_k	chemical activity of species k
a_L, a_T	longitudinal and transverse dispersivity
$B, \vec{B}, B_{ij}$	Skempton pore pressure coefficient B scalar, vector (6×1), and tensor (3×3)
B_c	Skempton pore pressure coefficient under constant solute concentration
B_s, B_f, B_ψ	constitutive constants for porothermoelastic quadratic form
B^b_k, B^p_k, B^ψ_k	constitutive constants for porochemoelastic quadratic form for species k
B^b_s, B^p_s, B^ψ_s	constitutive constants for porochemoelastic quadratic form for solute
B^b_w, B^p_w, B^ψ_w	constitutive constants for porochemoelastic quadratic form for solvent (water)
$\mathfrak{B}, \mathfrak{B}^*$	generalized boundary normal derivative operator and its adjoint operator
$b, \mathbf{b}, b_{ij}$	$(= \mathbf{k}^{-1})$ resistivity scalar and tensor
b_{ij}	$\{i, j\} = \{T, p\}$, cross resistivity coefficient
C	constant stress storage coefficient
C_c	constant stress storage coefficient under constant solute concentration
C_{ijkl}	drained compliance tensor
$\boldsymbol{C}, C_{ij}$	drained compliance tenor engineering notation (6×6)

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

C^s_{ijkl}	compliance tensor of the solid constituent
C^u_{ijkl}	undrained compliance tensor
C_{bp}, C_{pp}	compliance tensor coefficient
C_{bc}	$(= 1/K = C_K)$ drained frame compressibility
C_{pc}	$(= 1/K_p)$ pore volume compressibility
C_K	$(= 1/K)$ drained frame compressibility
C_u	$(= 1/K_u)$ undrained frame compressibility
C'_s	$(= 1/K'_s)$ unjacketed frame compressibility
C''_s	$(= 1/K''_s)$ unjacketed pore volume compressibility
C_s	$(= 1/K_s)$ solid compressibility
C_φ	$(= 1/K_\varphi)$ porosity compressibility
C_ϕ	$(= 1/K_\phi)$ porosity compressibility
C_f	$(= 1/K_f)$ fluid compressibility
C^*_K	$(= 1/K^* = C_{iijj})$ generalized drained frame compressibility
C_a, C^a_{ij}	added mass coefficient scalar and tensor
c_{ij}	$(= C^s_{ijkk})$ contracted solid compliance tensor
c^s_{ij}	micromechanical solid compliance tensor
c	consolidation coefficient $[\mathrm{L}^2/\mathrm{T}]$
c_c	adiabatic undrained consolidation coefficient
c_α	consolidation coefficient under constant concentration
c_β	concentration diffusivity coefficient under constant pressure
c_a	entropy diffusivity coefficient
c_b	entropy diffusivity coefficient
c_v	specific heat at constant strain
c_d	drained specific heat at constant strain
c_u	undrained specific heat at constant strain
c_f	specific heat of fluid phase at constant pressure
c_s	specific heat of solid phase at constant stress
c_p	specific heat at constant stress
c'	cohesion
$\overline{c}$	average crack half length
c	total molar concentration
c_k	molar concentration (molarity) of species k
c_s, c_w	molar concentration of solute and solvent (water)
c^o_s, c^o_w	initial molar concentration of solute and solvent (water)
c^o	$(= c^o_s)$ initial molar concentration of solute
$\overline{c}^p_k, \overline{c}^b_k$	volume averaged molar concentration of species k in pore fluid and bound fluid
$\overline{c}^b_s, \overline{c}^b_w$	volume averaged molar concentration of solute and solvent (water) in bound fluid
$\overline{c}^p_s, \overline{c}^p_w$	volume averaged molar concentration of solute and solvent (water) in pore fluid
c_s	$(= \Delta\overline{c}^p_s)$ molar concentration of solute in pore fluid

c_s^{po}, c_w^{po}	reference molar concentration of the solute and solvent of the pore fluid
$\mathcal{D}$	dissipation function (rate of energy dissipation)
$\mathcal{D}, \mathcal{D}_s, \mathcal{D}_f$	dissipation density function, and for solid and fluid phase
$\hat{D}$	relative fluid to solid displacement discontinuity
$\mathbb{D}, D_{ij}$	dispersion coefficient tensor
D_m	mass diffusion coefficient
D_c, D_c'	apparent mass diffusion coefficient
d	mean grain size
$\hat{d}, \hat{d}_i, \hat{d}_{ij}$	displacement discontinuity scalar, vector and tensor
$\hat{d}_n, \hat{d}_s$	normal and shear displacement discontinuity
E, E_i	Young's modulus scalar and vector (anisotropic)
E_u	undrained Young's modulus
E_d	drained dynamic modulus
E_{du}	undrained dynamic modulus
E_{ijkl}	displacement discontinuity forcing function
E_{ij}	eigenstrain tensor
$\mathbb{E}_1, \mathbb{E}_2$	evolutional function equivalent to Young's modulus in plate bending problem
$\mathcal{E}, \mathcal{E}_s, \mathcal{E}_f$	internal energy, and for solid and fluid phase
$\mathcal{E}$	McNamee-Gibson displacement function
e	$(= e_{ii})$ volumetric strain, positive for dilatation
e_{ij}, $\vec{e}$	total strain tensor (3×3) and vector (engineering notation) (6×1)
e_{ij}^p	pore strain tensor
e_{ij}^s	solid strain tensor
e^f	fluid strain
$\overline{e}_{ij}^s$	solid external strain tensor
$\overline{e}^f$	fluid external strain
$\langle e_{ij} \rangle$	deviatoric strain tensor
$\langle \overline{e}_{ij}^s \rangle$	solid external deviatoric strain tensor
e_{ij}^o	elastic strain tensor of Biot decomposition
e_{ij}^*	total strain tensor (including eigenstrain)
$\vec{F}, F_i$	total body force vector
$F(\kappa_\mu)$	viscosity correction factor
$\mathcal{F}$	dissipation energy function (dissipated energy)
$\mathcal{F}$	dissipation energy density function
$\mathfrak{F}$	Faraday constant
f	$(= \omega/2\pi)$ frequency (in Hz)
$\vec{f}, f_i$	fluid body force vector
G, G_{ij}	shear modulus scalar and tensor
G_s	shear modulus of solid constituent
$\mathcal{G}$	Gibbs free energy
g	gravitational acceleration

$\vec{H}, H_i$	entropy displacement vector
h	piezometric head [L]
h^*	Hubbert's potential
$\hbar$	heat density (per unit volume)
$\vec{J}_T$	heat flux
$\vec{J}_p$	fluid volume flux
$\vec{J}_c$	solute mass flux
$\vec{J}_e$	electrical current
$\vec{J}_s, \vec{J}_w$	flux of solvent and solute (water) per cross-sectional area of porous medium
$\vec{J}_k^s, \vec{J}_k^d$	diffusive and dispersive mass flux of species k per unit area of porous medium per unit time
$\mathbf{K}, \mathsf{K}_{ij}$	hydraulic conductivity tensor [L/T]
K	drained bulk modulus
K_u	undrained bulk modulus
K_p	pore volume bulk modulus
K_s	solid bulk modulus
K_s'	unjacketed frame bulk modulus
K_s''	unjacketed pore volume bulk modulus
K_f	fluid bulk modulus
K_α	solid bulk modulus for porosity preserving deformation
K_φ	micromechanical porosity bulk modulus
K_ϕ	intrinsic porosity bulk modulus
K_ψ	micro inhomogeneity and anisotropy bulk modulus
K^*	generalized drained bulk modulus
K_a	adiabatic drained bulk modulus
K_b	adiabatic undrained bulk modulus
K_d	adiabatic bulk modulus
K_c	constant solute concentration drained bulk modulus
K_v	constant solute concentration undrained bulk modulus
$\mathcal{K}$	kinetic energy
$\mathcal{K}, \mathcal{K}_s, \mathcal{K}_f$	kinetic energy density, and for solid and fluid phase
k, $\mathbf{k}$	intrinsic permeability scalar and tensor [L^2]
$\mathbf{k}^*$	dimensionless intrinsic permeability tensor
k_T	thermal conductivity
k_{pT}	thermo-osmosis coefficient
k_{Tp}	mechano-caloric coefficient
k_f	fluid kinetic energy per unit mass due to local velocity fluctuation
k_s	solid kinetic energy per unit mass due to local velocity fluctuation
L	wavelength
L_c	characteristic length
L_{ij}	$\{i,j\} = \{T, p, c, e\}$, Onsager law coefficient

L_{ij}	$\{i,j\} = \{s,w\}$, diffusion coefficient for flux law between solute and solvent (water)
L_{ij}	$\{i,j\} = \{p,c\}$, diffusion coefficient for flux law between pore pressure and solute concentration
$\mathcal{L}$	Lagrangian
$\mathcal{L}$	Lagrangian density
$\mathscr{L}$	Laplace transform
$\mathfrak{L}, \mathfrak{L}^*$	linear partial differential operator and its adjoint operator
$\ell, \boldsymbol{\ell}$	wave number scaler and vector, real and complex
ℓ_r, ℓ_i	real and imaginary part of complex wave number
M	Biot modulus
M_{sq}	complex Biot modulus for squirt flow
M_c	constant concentration Biot modulus
M_{ijkl}	drained elastic modulus tensor
$\boldsymbol{M}, M_{ij}$	drained elastic modulus tenor engineering notation (6×6)
M^u_{ijkl}	undrained elastic modulus tensor
$\mathcal{M}$	moment
m_α, m_β	constitutive constants for porothermoelastic quadratic form
m_d	$(= c_d/\mathcal{T}_o)$ drained thermoelastic constitutive constant
m_u	$(= c_u/\mathcal{T}_o)$ undrained thermoelastic constitutive constant
m_f	$(= c_f/\mathcal{T}_f^o)$ thermoelastic constitutive constant for fluid phase
m_s	$(= c_s/\mathcal{T}_s^o)$ thermoelastic constitutive constant for solid phase
m	$(= c_p/\mathcal{T}_o)$ thermoelastic constitutive constant at constant stress
m_f	fluid mass
N	total number of particles or moles of atoms or molecules
N_k	number of particles or moles of kth species
N_s, N_w	number of particles of solute and solvent (water)
$\vec{n}$, n_i	unit outward normal vector
P	$(= -\sigma_{ii}/3)$ mean total compressive stress
P_o	far-field mean total compressive stress, or loading magnitude
P'	$(= P - p)$ Terzaghi effective compressive stress
P''	$(= P - \alpha p)$ Biot effective compressive stress
p	pore pressure
$\overline{p}^f$	fluid external pressure
$\overline{p}^s$	$(= -\overline{\sigma}^s_{ii}/3)$ solid external pressure
p_b	borehole breakdown pressure
Q	volume of injected fluid per unit porous medium volume (dimensionless)
Q	heat energy
$\mathcal{Q}$	McNamee-Gibson displacement function
Q^{-1}	inverse of quality factor
$\vec{q}$, q_i	fluid specific discharge vector [L/T]
q	$(= q_i n_i)$ normal specific discharge

$\vec{q},\ q_i$	heat flux vector
$\vec{q}_r$	solute to solvent relative flux
R	$(= \lvert\vec{\chi} - \vec{x}\rvert)$ radial distance, 3D
$\mathcal{R}$	gas constant
$\mathcal{R}$	osmotic reflection coefficient
$\mathcal{R}_a$	chemo-mechanical reflection coefficient
$\mathcal{R}_b$	chemo-hydraulic reflection coefficient
R_q	$(= \sqrt{c/\kappa_T})$ square root of ratio of consolidation coefficient and the heat diffusivity coefficient
R_{p1}, R_{p2}, R_s	Amplitude ratio of reflected or refracted wave to incident wave
r	$(= \lvert\vec{\chi} - \vec{x}\rvert)$ radial distance, 2D or 3D
S	storage coefficient (constant stress uniaxial strain)
S_a	adiabatic drained thermal storage coefficient
S_b	adiabatic storage coefficient for entropy diffusion equation
S_c	adiabatic storage coefficient for variation in fluid content diffusion equation
S_e	$(= 1/M)$ constant strain storage coefficient
S_σ	$(= C)$ constant stress storage coefficient
S_α	constant concentration storage coefficient
S_β	constant pressure storage coefficient for solute diffusion
S_s	specific storativity [1/L]
S_o	far-field deviatoric stress
S_r	degree of saturation
$\mathcal{S}$	McNamee-Gibson displacement function
$\mathcal{S}$	entropy
$\mathcal{S}_s$	specific surface (per porous medium volume)
$\mathcal{S}_v$	$(= (1-\phi)\mathcal{S}_s)$ specific surface (per solid volume)
s	Laplace transform parameter
s^*	dimensionless Laplace transform parameter
$\hat{s}, \hat{s}_i, \hat{s}_{ij}$	stress discontinuity scalar, vector, and tensor
$\overline{s}^f$	fluid internal stress
$\overline{s}^s_{ij}$	solid internal stress tensor
$\overline{s}^s$	$(= \overline{s}^s_{ii}/3)$ solid internal average stress
s	entropy density
$\overline{s}_s, \overline{s}_f$	solid and fluid volume averaged entropy density
T	incremental temperature
$\overline{T}, \overline{T}_s, \overline{T}_f$	volume averaged incremental temperature, and for solid and fluid phase
$\mathcal{T}$	absolute temperature
$\overline{\mathcal{T}}, \overline{\mathcal{T}}_s, \overline{\mathcal{T}}_f$	volume averaged absolute temperature, and for solid and fluid phase
$\mathcal{T}_o, \mathcal{T}_s^o, \mathcal{T}_f^o$	reference absolute temperature, and for solid and fluid phase
$\mathcal{T}$	wave period
t	time

t^*	dimensionless time
t_c	characteristic time
t_i	surface traction vector
$\mathcal{U}$	internal energy
$\mathcal{U}, \mathcal{U}_s, \mathcal{U}_f$	strain energy density function, and for solid and fluid phase
$\vec{U}, U_i$	fluid displacement vector
$\vec{u}, u_i$	solid frame displacement vector
u_i^s, u_i^f	solid and fluid displacement vector (microscopic)
u_i^{fe}	fluid displacement vector extended to solid space
u_i^{se}	solid displacement vector extended to fluid space
u_i^o	elastic displacement vector of Biot decomposition
V	$(= V_s + V_p)$ total volume of solid frame, or total volume of solution
V_f	$(= V_p)$ fluid volume
V_s	solid volume
V_p	pore volume
ΔV	change of volume of solid frame
ΔV_f	change of fluid volume of the original group of fluid mass
$\Delta V_f'$	change of fluid volume with respect to porous frame
ΔV_s	change of solid volume
ΔV_p	change of pore volume
$\vec{V}$	$(= \vec{q}/\phi)$ seepage velocity
$\mathcal{V}$	potential energy
$\mathcal{V}$	potential energy density
v_i^s, v_i^f	solid and fluid velocity vector
$\overline{v_i}^-$	relative fluid to solid velocity vector
v_{pe}, v_{se}	elastic compressional and shear wave velocity
v_{p1}, v_{p2}, v_s	first and second compressional wave and shear wave velocity
v_w	wave velocity in water
v_{ph}	phase velocity
v_{gr}	group velocity
v	average molar volume
v_k	partial molar volume of species k
v_s, v_w	partial molar volume of solute and solvent (water)
$\mathcal{W}, \mathcal{W}_s, \mathcal{W}_f$	total work, and of the solid and fluid constituent
$\mathcal{W}_m, \mathcal{W}_c$	mechanical and chemical work
$\mathcal{W}_{\text{irr}}$	irreversible work
$\mathcal{W}, \mathcal{W}_s, \mathcal{W}_f$	work density (per unit volume), and of solid and fluid constituent
$\vec{w}, w_i$	$(= \phi(U_i - u_i))$ specific relative fluid to solid displacement vector
w	$(= w_i n_i)$ normal specific relative fluid to solid displacement
$\vec{X}_T$	temperature gradient driving force
$\vec{X}_p$	pore pressure gradient driving force
$\vec{X}_c$	solute mass concentration (chemical potential) gradient driving force

$\vec{X}_e$	electrical potential gradient driving force
x_k	molar fraction of species k
x_s, x_w	molar fraction of solute and solvent (water)
z^+, z^-	valence of cations and anions

Greek Letter

$\alpha, \vec{\alpha}, \alpha_{ij}$	Biot effective stress coefficient scalar, vector, and tensor
α_a	adiabatic drained Biot effective stress coefficient
α_b	adiabatic undrained Biot effective stress coefficient
α_T	$(= 3K\beta_l)$ thermoelastic effective stress coefficient
α_d	$(= K\beta_d)$ drained thermoelastic effective stress coefficient
α_u	$(= K_u\beta_u)$ undrained thermoelastic effective stress coefficient
α_e	thermoelastic effective stress coefficient in pore pressure diffusion equation
α_p, α_t	poroelastic effective stress coefficient in thermal diffusion equation
α_c, α_c'	constant solute concentration effective stress coefficient
α_μ	chemical effective stress coefficient
$\alpha_\alpha, \alpha_\beta$	chemical effective stress coefficients in pressure and solution diffusion equation
α_l	wave attenuation coefficient $[\mathrm{L}^{-1}]$
$\alpha_{p1}', \alpha_{p2}', \alpha_s'$	complex wave reflection or refraction angle for P1, P2 and S wave
β	pore volume effective stress coefficient
β_l	coefficient of linear thermal expansion
β_d	drained coefficient of volumetric thermal expansion for porous medium frame
β_u	undrained coefficient of volumetric thermal expansion for porous medium frame
β_v	coefficient of volumetric thermal expansion for variation in fluid content
β_e	coefficient of volumetric thermal expansion for variation in fluid content at constant frame volume
β_f	coefficient of volumetric thermal expansion for fluid
β_s	coefficient of volumetric thermal expansion for solid
β_ψ	coefficient of volumetric thermal expansion for porosity
β_b, β_c	thermal expansion coefficient
β_g, β_h	thermal expansion coefficient
β_k^b, β_k^ψ	solid and porosity swelling coefficient of species k in bound water
β_k^p	fluid swelling coefficient of species k in pore fluid
β_μ, β_μ'	chemical swelling coefficients under free stress and pore pressure condition

β_v	chemical swelling coefficients under free stress and undrained condition
β_ε	chemical swelling coefficients under constant frame and drained condition
$\beta_\alpha, \beta_\beta$	chemical swelling coefficients
β_{ij}	$(= \phi^2 \mu_f b_{ij})$ resistant coefficient tensor
$\Gamma, \Gamma_s, \Gamma_f$	total, solid, and fluid external surface
Γ_k	production of mass of species k per unit volume of porous medium
γ	fluid source intensity, volume of injected fluid per unit porous medium volume per unit time [1/T]
γ_T	heat source
γ_b	coefficient of fluid content
γ_s, γ_f	specific weight of solid and fluid
γ_{be}	barometric efficiency
γ_{te}	tidal efficiency
γ_{ij}	$(= 2e_{ij})$ shear strain (engineering notation)
γ_k	chemical activity coefficient of species k
γ_μ, γ'_μ	constitutive constant for chemical potential under constant stress and constant strain condition
γ_π	constitutive constant for chemical potential associated with osmotic pressure
δ_b	$(= 1/K'_s)$ unjacketed compressibility
δ_ℓ	boundary layer thickness
δ_p	pore geometry factor
δ', δ''	Coulomb shear and volumetric damping coefficient
ϵ	crack density parameter
$\overline{\epsilon}^s, \overline{\epsilon}^f$	solid fluid internal volumetric strain
$\overline{\epsilon}^s_{ij}$	solid internal strain tensor
$\langle \overline{\epsilon}^s_{ij} \rangle$	solid internal deviatoric strain tensor
η	poroelastic stress coefficient
η_d	thermoelastic stress coefficient
η_b	adiabatic undrained poroelastic stress coefficient
η_u	adiabatic undrained thermoelastic stress coefficient
$\theta_{p1}, \theta_{p2}, \theta_s$	incident wave angle for P1, P2 and S wave
$\theta'_{p1}, \theta'_{p2}, \theta'_s$	reflected or refracted wave phase angle for P1, P2 and S wave
$\overline{\theta}$	Cauchy principal value integration coefficient
ζ	variation in fluid content, positive for fluid entering the frame
$\kappa, \boldsymbol{\kappa}, \kappa_{ij}$	$(= k/\mu_f)$ permeability (mobility) coefficient scalar and tensor [L^4/FT]
κ_T	heat diffusivity coefficient
κ_{Tp}	mechano-caloric diffusivity coefficient
κ_D	permeability coefficient of osmotic pressure
κ_{pc}, κ_{cp}	osmotic-hydraulic permeability coefficient
κ_p	constant concentration permeability coefficient for pressure

κ_b	$(= 1/K)$ jacketed compressibility
λ	Lame constant
$\hat{\lambda}_s$	Lame constant of solid phase
λ_u	undrained Lame constant
λ_b	adiabatic undrained Lame constant
λ_D	osmotic transmission coefficient
λ_f	fluid volumetric viscosity
μ	$(= G)$ Lame constant
$\hat{\mu}_s$	$(= G_s)$ Lame constant of solid phase
μ_k	chemical potential of the kth species
μ_s, μ_w	chemical potential for solute and solvent (water)
μ_s	$(= \Delta\overline{\mu}_s)$ chemical potential for solute in pore fluid
$\overline{\mu}_k^p, \overline{\mu}_k^b$	volume averaged chemical potential of species k in pore water and bound water
$\overline{\mu}_k$	$(= \overline{\mu}_k^p = \overline{\mu}_k^b)$ volume averaged chemical potential of species k
$\overline{\mu}_s, \overline{\mu}_w$	volume averaged chemical potential of solute and solvent (water)
$\hat{\mu}_k$	electrochemical potential of the kth species
μ_k^c	integration constant for species k dependent on $\mathcal{T}$ and N_i, $i \neq k$
μ_k^o	integration constant for species k dependent on $\mathcal{T}$ only
μ_s^o, μ_w^o	integration constant for solute and solvent (water) dependent on $\mathcal{T}$ only
$\mu(\chi)$	double layer potential distribution density
$\mu_i(\chi)$	displacement discontinuity distribution density
μ_f	fluid dynamic viscosity
μ_a	fluid apparent viscosity
μ	viscoelastic viscosity
μ_K, μ_G	viscosity for the porous frame and shear modulus
μ_s, μ_ϕ	viscosity for solid phase and porosity
ν, ν_{ij}	drained Poisson ratio scalar and tensor
ν_u	undrained Poisson ratio
ν_b	adiabatic undrained Poisson ratio
ν_s	Poisson ratio of solid phase
ν_c	constant solute concentration drained Poisson ratio
ν^+, ν^-	number of cations and anions
Π	osmotic pressure
ρ, ρ_s, ρ_f	total, solid, and fluid density
ρ_a, ρ_{ij}^a	added mass density scalar and tensor
Σ_s, Σ_f	solid and fluid surface
Σ_i	solid/fluid interface
σ	entropy production
$\sigma_{ij}, \vec{\sigma}$	total stress tensor (3×3) and vector (engineering notation) (6×1)
σ'_{ij}	Terzaghi effective stress tensor
σ''_{ij}	Biot effective stress tensor

σ_n	normal stress
σ_{ij}^s	solid stress tensor
$\overline{\sigma}_{ij}^s$	solid external stress tensor
$\langle\sigma_{ij}\rangle$	deviatoric stress tensor
σ_{ij}^o	elastic stress tensor of Biot decomposition
$\sigma(\chi)$	single layer potential distribution density
$\sigma_i(\chi)$	stress discontinuity distribution density
τ	shear stress
τ_t	tortuosity
Φ	porous medium geometric factor
ϕ	porosity
ϕ_a	areal porosity
ϕ_v	volumetric porosity
ϕ'	friction angle
Ψ_i	Boussinesq-Papkovitch function
ψ	Boussinesq-Papkovitch function
$\Omega, \Omega_s, \Omega_f$	total, solid, and fluid volume
$\vec{\Omega}$	rotation vector of displacement
$\mathbf{\Omega}_s, \mathbf{\Omega}_f$	rotation vector of displacement of solid and fluid
$\mathbf{\Omega}_r$	relative fluid to solid displacement rotation vector
ω	angular frequency (in radian)
ω^*	dimensionless frequency
ω_c	characteristic frequency
ω_e	membrane efficiency
ω_k^b, ω_k^p	constitutive constants for porochemoelastic quadratic form

Mathematical Function

ber	real part of Kelvin function of the first kind
bei	imaginary part of Kelvin function of the first kind
ber$'$	derivative of ber
bei$'$	derivative of bei
E	complete elliptic integral of the second kind
E_1	elliptic integral
erf	error function
erfc	$(= 1 - \mathrm{erf})$ complimentary error function
$\mathrm{H}(t - t')$	Heaviside unit step function
$\mathrm{H}_0^{(1)}$	Hankel function of the first kind of order zero
I_i	modified Bessel function of the first kind of order i
J_i	Bessel function of the first kind of order i
K_i	modified Bessel function of the second kind of order i

K	complete elliptic integral of the first kind
Y_i	Bessel function of the second kind of order i
$\delta(x, x')$	Dirac delta function
δ_{ij}	Kronecker delta
ε_{ij}	Levi-Cevita permutation symbol
ϑ_{ijk}	permutation symbol

Appendix B
Poroelastic Constants

B.1 Relations Among Bulk Poroelastic Constants

For isotropic materials, there exist only four independent poroelastic constants (see Sect. 2.3). For different purposes, they can be expressed into different forms. In the following, we present the conversion of constants from one set to the other.

$$\lambda = \frac{2G\nu}{1-2\nu} = K - \frac{2G}{3} \tag{B.1}$$

$$\lambda_u = \frac{2G\nu_u}{1-2\nu_u} = K_u - \frac{2G}{3} \tag{B.2}$$

$$E = 2G(1+\nu) = \frac{9KG}{3K+G} \tag{B.3}$$

$$E_u = 2G(1+\nu_u) = \frac{9K_uG}{3K_u+G} \tag{B.4}$$

$$K = \frac{2G(1+\nu)}{3(1-2\nu)} = K_u - \alpha^2 M = K_u(1-\alpha B) = \frac{\alpha}{BC} = \frac{\alpha K_p}{\phi} \tag{B.5}$$

$$K_u = \frac{2G(1+\nu_u)}{3(1-2\nu_u)} = K + \alpha^2 M = \frac{\alpha M}{B} = \frac{K}{1-\alpha B} \tag{B.6}$$

$$K_p = \frac{\phi K}{\alpha} = \frac{\phi}{CB} \tag{B.7}$$

$$\begin{aligned}\nu &= \frac{3K-2G}{2(3K+G)} = \frac{2G\nu_u - \alpha^2 M(1-2\nu_u)}{2[G-\alpha^2 M(1-2\nu_u)]} \\ &= \frac{9\nu_u - 2GCB^2(1+\nu_u)}{9+2GCB^2(1+\nu_u)} = \frac{GS\nu_u - 2\eta^2(1-\nu_u)}{GS-2\eta^2(1-\nu_u)}\end{aligned} \tag{B.8}$$

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

$$\begin{aligned}\nu_u &= \frac{3K_u - 2G}{2(3K_u + G)} = \frac{2G\nu + \alpha^2 M(1-2\nu)}{2\left[G + \alpha^2 M(1-2\nu)\right]} \\ &= \frac{9\nu + 2GCB^2(1+\nu)}{9 - 2GCB^2(1+\nu)} = \nu + \frac{9\alpha^2 GM}{2(G+3K)\left[G + 3\left(K + \alpha^2 M\right)\right]} \end{aligned} \tag{B.9}$$

$$\begin{aligned}\alpha &= KCB = \frac{K_u CB}{1 + K_u CB^2} = \frac{K_u B}{M} = \frac{K_u - K}{K_u B} = \frac{2\eta(1-\nu)}{1-2\nu} \\ &= \frac{\eta}{3G}(3K + 4G) = \frac{3(\nu_u - \nu)}{B(1-2\nu)(1+\nu_u)} = \frac{\phi K}{K_p} \end{aligned} \tag{B.10}$$

$$\eta = \frac{3\alpha G}{3K + 4G} = \frac{\alpha(1-2\nu)}{2(1-\nu)} \tag{B.11}$$

$$\begin{aligned}B &= \frac{\alpha M}{K_u} = \frac{\alpha M}{K + \alpha^2 M} = \frac{\alpha}{CK} = \frac{K_u - K}{\alpha K_u} = \frac{3(\nu_u - \nu)}{\alpha(1-2\nu)(1+\nu_u)} \\ &= \frac{3(\nu_u - \nu)}{2\eta(1-\nu)(1+\nu_u)} = \frac{\phi}{K_p C} \end{aligned} \tag{B.12}$$

$$\begin{aligned}M &= \frac{1}{C} + K_u B^2 = \frac{1}{C(1 - KCB^2)} = \frac{K_u B}{\alpha} = \frac{K_u}{KC} = \frac{K_u - K}{\alpha^2} \\ &= \frac{K}{KC - \alpha^2} = \frac{BK}{\alpha(1-\alpha B)} = \frac{(1+\nu_u)(1-2\nu)}{C(1+\nu)(1-2\nu_u)} = \frac{2GB(1+\nu_u)}{3\alpha(1-2\nu_u)} \\ &= \frac{2G(\nu_u - \nu)}{\alpha^2(1-2\nu_u)(1-2\nu)} = \frac{G(1-2\nu)(\nu_u - \nu)}{2\eta^2(1-\nu)^2(1-2\nu_u)} = \frac{G}{GS - \alpha\eta} \\ &= \frac{(1-\nu_u)(1-2\nu)}{S(1-\nu)(1-2\nu_u)} \end{aligned} \tag{B.13}$$

$$\begin{aligned}S &= \frac{3K_u + 4G}{M(3K + 4G)} = \frac{1}{M} + \frac{3\alpha^2}{3K + 4G} = \frac{1}{M} + \frac{\alpha^2(1-2\nu)}{2G(1-\nu)} \\ &= \frac{1}{M} + \frac{\alpha\eta}{G} = \frac{\alpha^2(1-2\nu)^2(1-\nu_u)}{2G(1-\nu)(\nu_u - \nu)} = \frac{2\eta^2(1-\nu)(1-\nu_u)}{G(\nu_u - \nu)} \\ &= \frac{(1-\nu_u)(1-2\nu)}{M(1-\nu)(1-2\nu_u)} = C - \frac{4\alpha^2 G}{K(3K + 4G)} \\ &= \frac{C(1-\nu_u)(1+\nu)}{(1-\nu)(1+\nu_u)} \end{aligned} \tag{B.14}$$

$$\begin{aligned}C &= \frac{K + \alpha^2 M}{MK} = \frac{K_u}{M(K_u - \alpha^2 M)} = \frac{K_u}{MK} = \frac{\alpha}{KB} = \frac{\phi}{K_p B} \\ &= \frac{(1+\nu_u)(1-2\nu)}{M(1+\nu)(1-2\nu_u)} = \frac{\alpha^2(1-2\nu)^2(1+\nu_u)}{2G(1+\nu)(\nu_u - \nu)} \end{aligned} \tag{B.15}$$

$$c = \frac{\kappa}{S} = \frac{2\kappa G(1-\nu)(\nu_u - \nu)}{\alpha^2(1-2\nu)^2(1-\nu_u)} = \frac{\kappa G(\nu_u - \nu)}{2\eta^2(1-\nu)(1-\nu_u)}$$
$$= \frac{\kappa M(3K+4G)}{3K_u + 4G} \tag{B.16}$$

Limiting Case For the soil mechanics case, with incompressible solid and incompressible fluid, we find

$$K_u \to \infty \tag{B.17}$$
$$\nu_u = \frac{1}{2} \tag{B.18}$$
$$\alpha = 1 \tag{B.19}$$
$$B = 1 \tag{B.20}$$
$$K_p = \phi K \tag{B.21}$$
$$C = \frac{1}{K} \tag{B.22}$$
$$M \to \infty \tag{B.23}$$
$$S = \frac{3}{3K+4G} = \frac{1-2\nu}{2G(1-\nu)} = \frac{\eta}{G} \tag{B.24}$$
$$\eta = \frac{3G}{3K+4G} = \frac{1-2\nu}{2(1-\nu)} \tag{B.25}$$
$$c = \frac{2\kappa G(1-\nu)}{1-2\nu} = \kappa\left(K + \frac{4G}{3}\right) = \frac{\kappa G}{\eta} \tag{B.26}$$

B.2 Relations Among Bulk and Micromechanical Constants

Based on the analysis in Sect. 3.1, there exist four independent Rice-Cleary micromechanical constant [6], K, K_s', K_s'', and K_f, together with porosity ϕ; and related constants are K_φ and β. Their association with the bulk material constants is presented in the following.

$$\frac{1}{K} = \frac{1}{K_\varphi} + \frac{1}{(1-\phi)K_s'} = \frac{1}{K_s'} + \frac{\phi}{K_p} \tag{B.27}$$
$$\frac{1}{K_p} = \frac{1}{\phi}\left(\frac{1}{K} - \frac{1}{K_s'}\right) = \frac{1}{\phi K_\varphi} + \frac{1}{(1-\phi)K_s'} \tag{B.28}$$

$$K_u = K + \frac{K_f K_s''(K_s' - K)^2}{K_f K_s''(K_s' - K) + \phi K_s'^2(K_s'' - K_f)} \tag{B.29}$$

$$\alpha = 1 - \frac{K}{K_s'} \tag{B.30}$$

$$\beta = 1 - \frac{K_p}{K_s''} = 1 - \frac{\phi K K_s'}{K_s''(K_s' - K)} \tag{B.31}$$

$$B = 1 - \frac{\phi K K_s'(K_s' - K_f)}{K_f K_s''(K_s' - K) + \phi K K_s'(K_s' - K_f)} \tag{B.32}$$

$$C = \frac{\phi}{K_f} + \frac{1}{K} - \frac{1}{K_s'} - \frac{\phi}{K_s''} \tag{B.33}$$

$$M = \frac{K_f K_s'^2 K_s''}{K_f K_s''(K_s' - K) + \phi K_s'^2(K_s'' - K_f)} \tag{B.34}$$

$$\frac{1}{M} = \frac{\phi}{K_f} + \frac{1}{K_s'} - \frac{\phi}{K_s''} - \frac{K}{K_s'^2} \tag{B.35}$$

Ideal Porous Medium For ideal porous medium, $K_s' = K_s'' = K_s$, the above relations simplify to:

$$\frac{1}{K} = \frac{1}{K_\varphi} + \frac{1}{(1-\phi)K_s} = \frac{1}{K_s} + \frac{\phi}{K_p} \tag{B.36}$$

$$\frac{1}{K_p} = \frac{1}{\phi}\left(\frac{1}{K} - \frac{1}{K_s}\right) = \frac{1}{\phi K_\varphi} + \frac{1}{(1-\phi)K_s} \tag{B.37}$$

$$K_u = K + \frac{K_f(K_s - K)^2}{K_f(K_s - K) + \phi K_s(K_s - K_f)} \tag{B.38}$$

$$\alpha = 1 - \frac{K}{K_s} \tag{B.39}$$

$$\beta = 1 - \frac{K_p}{K_s} = 1 - \frac{\phi K}{K_s - K} \tag{B.40}$$

$$B = 1 - \frac{\phi K(K_s - K_f)}{K_f(K_s - K) + \phi K(K_s - K_f)} \tag{B.41}$$

$$C = \frac{\phi}{K_f} + \frac{1}{K} - \frac{1+\phi}{K_s} \tag{B.42}$$

$$M = \frac{K_f K_s^2}{K_f(K_s - K) + \phi K_s(K_s - K_f)} \tag{B.43}$$

$$\frac{1}{M} = \frac{\phi}{K_f} + \frac{1-\phi}{K_s} - \frac{K}{{K_s}^2} \tag{B.44}$$

Limiting Cases For the limiting case $(K_s', K_s'') \to \infty$, we obtain

$$K = \frac{K_p}{\phi} = K_\varphi \tag{B.45}$$

$$K_u = K + \frac{K_f}{\phi} \tag{B.46}$$

$$\alpha = 1 \tag{B.47}$$

$$\beta = 1 \tag{B.48}$$

$$B = 1 - \frac{\phi K}{K_f + \phi K} \tag{B.49}$$

$$C = \frac{\phi}{K_f} + \frac{1}{K} \tag{B.50}$$

$$M = \frac{K_f}{\phi} \tag{B.51}$$

Furthermore, if $(K_s', K_s'', K_f) \to \infty$, we obtain

$$K_u \to \infty \tag{B.52}$$

$$B = 1 \tag{B.53}$$

$$C = \frac{1}{K} \tag{B.54}$$

$$M \to \infty \tag{B.55}$$

For the limiting case $K_f \to \infty$, we obtain

$$K_u = K + \frac{K_s''(K_s' - K)^2}{K_s'(K_s'' - \phi K_s') - K_s'' K} \tag{B.56}$$

$$B = 1 + \frac{\phi K K_s'}{K_s'' K_s' - K_s'' K - \phi K K_s'} \tag{B.57}$$

$$C = \frac{1}{K} - \frac{1}{K_s'} - \frac{\phi}{K_s''} \tag{B.58}$$

$$M = \frac{{K_s'}^2 K_s''}{K_s'' K_s' - K_s'' K - \phi {K_s'}^2} \tag{B.59}$$

For the limiting case $K_f \to \infty$, together with ideal porous medium assumption, $K_s' = K_s'' = K_s$:

$$K_u = K + \frac{(K_s - K)^2}{(1-\phi)K_s - K} \tag{B.60}$$

$$B = 1 + \frac{\phi K}{K_s - (1+\phi)K} \tag{B.61}$$

$$C = \frac{1}{K} - \frac{1+\phi}{K_s} \tag{B.62}$$

$$M = \frac{{K_s}^2}{(1-\phi)K_s - K} \tag{B.63}$$

For the limiting case $K_f \to 0$, and ideal porous medium ($K_s' = K_s'' = K_s$),

$$K_u \approx K + \frac{(K_s - K)^2 K_f}{\phi {K_s}^2} \tag{B.64}$$

$$B \approx \frac{(K_s - K)K_f}{\phi K K_s} \tag{B.65}$$

$$C \approx \frac{\phi}{K_f} \tag{B.66}$$

$$M \approx \frac{K_f}{\phi} \tag{B.67}$$

B.3 Relations Among Bulk and Intrinsic Constants

As demonstrated in Sects. 4.4 and 4.5, for macroscopically isotropic material, there exist four intrinsic micromechanical material constants, K_s, K_ϕ, K_ψ, and K_f, associated with volumetric deformation. Alternatively, K_s can be replaced by K_α. The bulk material constants can be expressed in terms of the intrinsic constants as:

$$K_\alpha = K_s + \frac{K_\psi^2}{K_\phi} \tag{B.68}$$

$$K = \frac{(1-\phi)^3 K_s K_\phi^2}{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2} = \frac{(1-\phi)^3 (K_\alpha K_\phi - K_\psi^2)}{K_\alpha + (1-\phi)^2 K_\phi - 2(1-\phi)K_\psi} \tag{B.69}$$

$$K_p = \frac{\phi(1-\phi)^3 K_s K_\phi^2}{K_s K_\phi + [(1-\phi)K_\phi - K_\psi][\phi(1-\phi)K_\phi - K_\psi]}$$
$$= \frac{\phi(1-\phi)^3(K_\alpha K_\phi - K_\psi^2)}{K_\alpha + \phi(1-\phi)^2 K_\phi - (1-\phi^2)K_\psi} \tag{B.70}$$

$$K_u = \frac{K_f\{K_s K_\phi + [\phi(1-\phi)K_\phi - K_\psi]^2\} + \phi(1-\phi)^3 K_s K_\phi^2}{(1-\phi)K_f K_\phi + \phi\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2\}}$$
$$= \frac{K_f[K_\alpha + \phi^2(1-\phi)^2 K_\phi - 2\phi(1-\phi)K_\psi] + \phi(1-\phi)^3(K_\alpha K_\phi - K_\psi^2)}{(1-\phi)K_f + \phi[K_\alpha + (1-\phi)^2 K_\phi - 2(1-\phi)K_\psi]} \tag{B.71}$$

$$\alpha = 1 - \frac{(1-\phi)^2 K_\phi[(1-\phi)K_\phi - K_\psi]}{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2}$$
$$= 1 - \frac{(1-\phi)^2[(1-\phi)K_\phi - K_\psi]}{K_\alpha + (1-\phi)^2 K_\phi - 2(1-\phi)K_\psi} \tag{B.72}$$

$$B = \frac{K_f\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi][\phi(1-\phi)K_\phi - K_\psi]\}}{K_f\{K_s K_\phi + [\phi(1-\phi)K_\phi - K_\psi]^2\} + \phi(1-\phi)^3 K_s K_\phi^2}$$
$$= \frac{K_f[K_\alpha + \phi(1-\phi)^2 K_\phi - (1-\phi^2)K_\psi]}{K_f[K_\alpha + \phi^2(1-\phi)^2 K_\phi - 2\phi(1-\phi)K_\psi] + \phi(1-\phi)^3(K_\alpha K_\phi - K_\psi^2)} \tag{B.73}$$

$$C = \frac{\phi}{K_f} + \frac{K_s + [\phi(1-\phi)K_\phi - K_\psi]^2}{(1-\phi)^3 K_s K_\phi^2}$$
$$= \frac{\phi}{K_f} + \frac{K_\alpha + \phi^2(1-\phi)^2 K_\phi - 2\phi(1-\phi)K_\psi}{(1-\phi)^3(K_\alpha K_\phi - K_\psi^2)} \tag{B.74}$$

$$M = \frac{K_f\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2\}}{(1-\phi)K_f K_\phi + \phi\{K_s K_\phi + [(1-\phi)K_\phi - K_\psi]^2\}}$$
$$= \frac{K_f[K_\alpha + (1-\phi)^2 K_\phi - 2(1-\phi)K_\psi]}{(1-\phi)K_f + \phi[K_\alpha + (1-\phi)^2 K_\phi - 2(1-\phi)K_\psi]} \tag{B.75}$$

Ideal Porous Medium For ideal porous medium, $K_\psi = 0$, the above relations simplify to:

$$K_\alpha = K_s \tag{B.76}$$

$$K = \frac{(1-\phi)^3 K_s K_\phi}{K_s + (1-\phi)^2 K_\phi} \tag{B.77}$$

$$\frac{1}{K} = \frac{1}{(1-\phi)K_s} + \frac{1}{(1-\phi)^3 K_\phi} \tag{B.78}$$

$$K_p = \frac{\phi(1-\phi)^3 K_s K_\phi}{K_s + \phi(1-\phi)^2 K_\phi} \tag{B.79}$$

$$\frac{1}{K_p} = \frac{1}{(1-\phi)K_s} + \frac{1}{\phi(1-\phi)^3 K_\phi} \tag{B.80}$$

$$K_u = \frac{K_f[K_s + \phi^2(1-\phi)^2 K_\phi] + \phi(1-\phi)^3 K_s K_\phi}{(1-\phi)K_f + \phi[K_s + (1-\phi)^2 K_\phi]} \tag{B.81}$$

$$\alpha = 1 - \frac{(1-\phi)^3 K_\phi}{K_s + (1-\phi)^2 K_\phi} \tag{B.82}$$

$$B = 1 - \frac{\phi(1-\phi)^3 K_\phi (K_s - K_f)}{K_f[K_s + \phi^2(1-\phi)^2 K_\phi] + \phi(1-\phi)^3 K_s K_\phi} \tag{B.83}$$

$$C = \frac{\phi}{K_f} + \frac{1}{(1-\phi)^3 K_\phi} + \frac{\phi^2}{(1-\phi)K_s} \tag{B.84}$$

$$M = \frac{K_f[K_s + (1-\phi)^2 K_\phi]}{(1-\phi)K_f + \phi[K_s + (1-\phi)^2 K_\phi]} \tag{B.85}$$

Limiting Cases For the limiting case $K_s \to \infty$, we obtain

$$K = (1-\phi)^3 K_\phi \tag{B.86}$$

$$K_p = \phi(1-\phi)^3 K_\phi \tag{B.87}$$

$$K_u = \frac{K_f}{\phi} + (1-\phi)^3 K_\phi = \frac{K_f}{\phi} + K \tag{B.88}$$

$$\alpha = 1 \tag{B.89}$$

$$B = 1 - \frac{\phi(1-\phi)^3 K_\phi}{K_f + \phi(1-\phi)^3 K_\phi} = 1 - \frac{\phi K}{K_f + \phi K} \tag{B.90}$$

$$C = \frac{\phi}{K_f} + \frac{1}{(1-\phi)^3 K_\phi} = \frac{\phi}{K_f} + \frac{1}{K} \tag{B.91}$$

$$M = \frac{K_f}{\phi} \tag{B.92}$$

Furthermore, if both $(K_s,\ K_f) \to \infty$, we obtain

$$K_u \to \infty \tag{B.93}$$

$$B = 1 \tag{B.94}$$

$$C = \frac{1}{(1-\phi)^3 K_\phi} = \frac{1}{K} \tag{B.95}$$

$$M \to \infty \tag{B.96}$$

For the limiting case $K_f \to \infty$, we obtain

$$K_u = \frac{K_s}{1-\phi} + \phi^2(1-\phi)K_\phi - 2\phi K_\psi \tag{B.97}$$

$$B = 1 + \frac{(1-\phi)^2[\phi(1-\phi)K_\phi - K_\psi]}{K_s + \phi(1-\phi)[\phi(1-\phi)K_\phi - 2K_\psi]} \tag{B.98}$$

$$C = \frac{K_s + \phi^2(1-\phi)^2 K_\phi - 2\phi(1-\phi)K_\psi}{(1-\phi)^3(K_s K_\phi - K_\psi^2)} \tag{B.99}$$

$$M = \frac{K_s}{1-\phi} + (1-\phi)K_\phi - 2K_\psi \tag{B.100}$$

For the limiting case $K_f \to \infty$, and ideal porous medium, $K_\psi = 0$:

$$K_u = \frac{K_s}{1-\phi} + \phi^2(1-\phi)K_\phi \tag{B.101}$$

$$B = 1 + \frac{\phi(1-\phi)^3 K_\phi}{K_s + \phi^2(1-\phi)^2 K_\phi} \tag{B.102}$$

$$C = \frac{\phi^2(1-\phi)}{K_s} + \frac{1}{(1-\phi)K_\phi} \tag{B.103}$$

$$M = \frac{K_s}{1-\phi} + (1-\phi)K_\phi \tag{B.104}$$

For the limiting case $K_f \to 0$, and ideal porous medium ($K_\psi = 0$)

$$K_u \approx \frac{(1-\phi)^3 K_s K_\phi}{K_s + (1-\phi)^2 K_\phi} + \frac{[K_s + \phi(1-\phi)^2 K_\phi]^2}{\phi[K_s + (1-\phi)^2 K_\phi]^2} K_f \tag{B.105}$$

$$B \approx \frac{K_f[K_s + \phi(1-\phi)^2 K_\phi]}{\phi(1-\phi)^3 K_s K_\phi} \tag{B.106}$$

$$C \approx \frac{\phi}{K_f} \tag{B.107}$$

$$M \approx \frac{K_f}{\phi} \tag{B.108}$$

B.4 Relations Among Micromechanical and Intrinsic Constants

Relations among the micromechanical constants K_s', K_s'', K_φ, β, and the intrinsic constants K_s (or K_α), K_ϕ, K_ψ, are presented below. Relations for K are contained in Sect. B.3.

$$K_s' = \frac{(1-\phi)K_sK_\phi}{(1-\phi)K_\phi - K_\psi} = \frac{(1-\phi)(K_\alpha K_\phi - K_\psi^2)}{(1-\phi)K_\phi - K_\psi} \tag{B.109}$$

$$K_s'' = \frac{\phi(1-\phi)K_sK_\phi}{\phi(1-\phi)K_\phi - K_\psi} = \frac{\phi(1-\phi)(K_\alpha K_\phi - K_\psi^2)}{\phi(1-\phi)K_\phi - K_\psi} \tag{B.110}$$

$$K_\varphi = \frac{(1-\phi)^3K_sK_\phi^2}{K_sK_\phi - (1-\phi)K_\phi K_\psi + K_\psi^2} = \frac{(1-\phi)^3(K_\alpha K_\phi - K_\psi^2)}{K_\alpha - (1-\phi)K_\psi} \tag{B.111}$$

$$\begin{aligned}\beta &= 1 - \frac{(1-\phi)^2K_\phi[\phi(1-\phi)K_\phi - K_\psi]}{K_sK_\phi + [(1-\phi)K_\phi - K_\psi][\phi(1-\phi)K_\phi - K_\psi]} \\ &= 1 - \frac{(1-\phi)^2[\phi(1-\phi)K_\phi - K_\psi]}{K_\alpha + \phi(1-\phi)^2K_\phi - (1-\phi^2)K_\psi}\end{aligned} \tag{B.112}$$

$$K_s = \frac{(1-\phi)K_s'K_s''}{K_s'' - \phi K_s'} \tag{B.113}$$

$$K_\alpha = \frac{K_s'[(1-\phi)^2K_s'K_s'' - \phi KK_s' - (1-2\phi)KK_s'']}{(1-\phi)[K_s'(K_s'' - \phi K_s') - KK_s'']} \tag{B.114}$$

$$K_\phi = \frac{KK_s'(K_s'' - \phi K_s')}{(1-\phi)^3[K_s'(K_s'' - \phi K_s') - KK_s'']} \tag{B.115}$$

$$K_\psi = \frac{\phi KK_s'(K_s'' - K_s')}{(1-\phi)^2[K_s'(K_s'' - \phi K_s') - KK_s'']} \tag{B.116}$$

Ideal Porous Medium For ideal porous medium, $K_\psi = 0$ and $K_s' = K_s'' = K_s$, and

$$K_\varphi = (1-\phi)^3K_\phi \tag{B.117}$$

$$\beta = 1 - \frac{\phi(1-\phi)^3K_\phi}{K_s + \phi(1-\phi)^2K_\phi} \tag{B.118}$$

$$K_\phi = \frac{KK_s}{(1-\phi)^2[(1-\phi)K_s - K]} \tag{B.119}$$

Limiting Case For the limiting case $K_s \to \infty$, we obtain

$$\beta = 1 \tag{B.120}$$

$$K_\phi = \frac{K}{(1-\phi)^3} \tag{B.121}$$

B.5 Relations Among Biot-Willis and Micromechanical Constants

Here we give a summary of relations already established in Sect. 3.5, between the Biot-Willis laboratory constants and the Rice-Cleary micromechanical constants.

$$\kappa_b = \frac{1}{K} \tag{B.122}$$

$$\delta_b = \frac{1}{K_s'} \tag{B.123}$$

$$\gamma_b = \phi\left(\frac{1}{K_f} - \frac{1}{K_s''}\right) \tag{B.124}$$

B.6 Biot Notations

To allow easy comparison between the constants and notations used by Biot in his publications and those introduced in this book, we shall establish their relations below.

Biot 1941 The constitutive constants H, R and Q (two of them are independent) introduced by Biot in 1941 [1] are here denoted as H', R' and Q' to avoid confusion, as these same symbols were redefined by Biot in 1955 [2], as well as in subsequent papers [3–5]. The equivalence between $\{H', R', Q'\}$ and bulk material constants can be expressed as

$$Q' = M \tag{B.125}$$

$$H' = \frac{K}{\alpha} \tag{B.126}$$

$$R' = \frac{BK}{\alpha} \tag{B.127}$$

Biot 1955 The Biot [2] four independent constants A, N, Q, R, were discussed in detail by Biot and Willis [5]. They are related to the current bulk material constants as

$$A = \lambda + M(\alpha - \phi)^2 \tag{B.128}$$

$$N = G \tag{B.129}$$

$$Q = \phi M(\alpha - \phi) \tag{B.130}$$

$$R = \phi^2 M \tag{B.131}$$

Biot further defined

$$P = A + 2N = \lambda + 2G + M(\alpha - \phi)^2 \tag{B.132}$$

$$S = A - \frac{Q^2}{R} = \lambda \tag{B.133}$$

A useful formula is

$$\alpha = \frac{\phi(Q + R)}{R} \tag{B.134}$$

Biot 1956 For poroelastodynamics, Biot [4] introduced

$$H = P + R + 2Q = \lambda_u + 2G \tag{B.135}$$

B.7 Porothermoelasticity Constants

Relations Among Bulk Constants

$$\beta_b = \beta_e - \frac{\alpha_u \eta}{MGS} = \beta_e - \frac{K_u \eta}{MGS}\beta_u \tag{B.136}$$

$$\begin{aligned}\beta_c &= \beta_e - \frac{3\alpha K}{3K + 4G}\beta_d = \beta_e - \frac{K\eta}{G}\beta_d = \beta_e - \frac{\alpha_d\, \eta}{G} \\ &= \beta_e - \frac{\alpha \eta_d}{G} = \beta_v + \frac{4}{3}\eta\beta_d\end{aligned} \tag{B.137}$$

$$\beta_e = \alpha\beta_d + \beta_v \tag{B.138}$$

$$\beta_g = \beta_e - \frac{3\alpha_b(m_d\alpha_u - Mm_u\alpha_b\beta_e)}{m_d(3K_d + 4G)} \tag{B.139}$$

$$\beta_h = \beta_e + \frac{3\alpha_u(\alpha_u\beta_e - \alpha_b m_u)}{m_u(3K_u + 4G)} \tag{B.140}$$

$$\beta_u = \beta_d + B\beta_v \tag{B.141}$$

$$\alpha_a = \alpha + \frac{\alpha_d\beta_e}{m_d} \tag{B.142}$$

$$\alpha_b = \alpha + \frac{\alpha_u \beta_v}{m_u} \tag{B.143}$$

$$\alpha_d = K\beta_d = \frac{2\eta_d(1-\nu)}{1-2\nu} = \frac{\alpha\,\eta_d}{\eta} \tag{B.144}$$

$$\alpha_u = K_u\beta_u = \frac{M\beta_u}{B}\alpha = K\beta_d + \alpha M\beta_e = K_u\beta_d + \alpha M\beta_v \tag{B.145}$$

$$\alpha_e = \frac{\beta_c}{S} \tag{B.146}$$

$$\alpha_t = \frac{\beta_b}{S_b} \tag{B.147}$$

$$\alpha_p = \frac{\beta_c}{S_a} \tag{B.148}$$

$$m_u = m_d - M\beta_e^2 \tag{B.149}$$

$$K_a = K + \frac{\alpha_d^2}{m_d} \tag{B.150}$$

$$K_b = K_u + \frac{\alpha_u^2}{m_u} \tag{B.151}$$

$$K_d = K_b - \frac{Mm_u\alpha_b^2}{m_d} \tag{B.152}$$

$$\eta_d = \frac{3\alpha_d\,G}{3K+4G} = \frac{\eta\,\alpha_d}{\alpha} = \frac{\alpha_d(1-2\nu)}{2(1-\nu)} \tag{B.153}$$

$$\eta_b = \frac{3\alpha_b G}{3K_b+4G} \tag{B.154}$$

$$\eta_u = \frac{3\alpha_u G}{3K_b+4G} \tag{B.155}$$

$$S_a = \frac{m_d(3K_a+4G)}{3K+4G} = m_d + \frac{3\alpha_d^2}{3K+4G} = m_d + \frac{\alpha_d\,\eta_d}{G} \tag{B.156}$$

$$S_b = \frac{m_u(3K_b+4G)}{3K_u+4G} = m_u + \frac{3\alpha_u^2}{3K_u+4G} = m_u + \frac{\alpha_u\eta\beta_u}{GSB} \tag{B.157}$$

$$S_c = \frac{m_u(3K_b+4G)}{m_d M(3K_d+4G)} \tag{B.158}$$

$$c_a = \frac{m_d\kappa_T}{S_a} \tag{B.159}$$

$$c_b = \frac{m_d\kappa_T}{S_b} \tag{B.160}$$

$$c_c = \frac{\kappa}{S_c} \tag{B.161}$$

$$\kappa_T = \frac{k_T}{c_d} = \frac{k_T}{m_d \mathcal{T}_o} \tag{B.162}$$

Limits for Thermally Uncoupled Model

$$\left(\frac{\alpha_d}{m_d},\ \frac{\beta_e}{m_d},\ \alpha_p\right) \to 0 \tag{B.163}$$

$$(\alpha_a,\ \alpha_b) \to \alpha \tag{B.164}$$

$$(m_u,\ S_a) \to m_d \tag{B.165}$$

$$K_a \to K \tag{B.166}$$

$$K_b \to K_u \tag{B.167}$$

$$c_a \to \kappa_T \tag{B.168}$$

Relation among Bulk and Intrinsic Constants

$$\beta_d = \beta_s + \frac{\beta_\psi}{1-\phi} \tag{B.169}$$

$$\beta_v = \phi\,(\beta_f - \beta_s) - \frac{\beta_\psi}{1-\phi} \tag{B.170}$$

$$\beta_u = (1-\phi B)\beta_s + \phi B \beta_f + \frac{1-B}{1-\phi}\beta_\psi \tag{B.171}$$

$$\beta_e = (\alpha-\phi)\beta_s + \phi\beta_f - \frac{1-\alpha}{1-\phi}\beta_\psi \tag{B.172}$$

$$m_d = m - K\left(\beta_s + \frac{\beta_\psi}{1-\phi}\right)^2 \tag{B.173}$$

$$m_u = m - M\left[(\alpha-\phi)\beta_s + \phi\beta_f - \frac{1-\alpha}{1-\phi}\beta_\psi\right]^2 \tag{B.174}$$

with K, α, B, and M defined in (B.69), (B.72), (B.73), and (B.75).

For ideal porous medium

$$\beta_d = \beta_s \tag{B.175}$$

$$\beta_v = \phi\,(\beta_f - \beta_s) \tag{B.176}$$

$$\beta_u = (1-\phi B)\beta_s + \phi B \beta_f \tag{B.177}$$

$$\beta_e = (\alpha-\phi)\beta_s + \phi\beta_f \tag{B.178}$$

$$m_d = m - K\beta_s^2 \tag{B.179}$$

$$m_u = m - M\left[(\alpha - \phi)\beta_s + \phi\beta_f\right]^2 \tag{B.180}$$

with K, α, B, and M defined in (B.77), (B.82), (B.83), and (B.85).

B.8 Porochemoelasticity Constants

Relations Among Bulk Constants

$$K_v = \frac{K_c}{1 - \alpha_c B_c} = K_c + M_c \alpha_c \alpha_c' \tag{B.181}$$

$$M_c = \frac{K_c B_c}{\alpha_c'(1 - \alpha_c B_c)} \tag{B.182}$$

$$\nu_c = \frac{3K_c - 2G}{2(3K_c + G)} \tag{B.183}$$

$$C_c = \frac{\alpha_c'}{K_c B_c} \tag{B.184}$$

$$B_c = \frac{\alpha_c' M_c}{K_v} = \frac{\alpha_c' M_c}{K_c + \alpha_c \alpha_c' M_c} = \frac{\alpha_c'}{C_c K_c} = \frac{K_v - K_c}{\alpha_c K_v} \tag{B.185}$$

$$\alpha_a = \alpha + \frac{\alpha_d \beta_e}{m_d} \tag{B.186}$$

$$\alpha_b = \alpha + \frac{\alpha_u \beta_e}{m_u} \tag{B.187}$$

$$\alpha_d = K\beta_d \tag{B.188}$$

$$\alpha_e = \frac{\beta_c}{S} \tag{B.189}$$

$$\alpha_p = \frac{\beta_c}{S_a} \tag{B.190}$$

$$\alpha_u = K_u \beta_u \tag{B.191}$$

$$\alpha_\alpha = \frac{M_c[\beta_\alpha(4G + 3K_c) - 3\alpha_c'\alpha_\mu]}{4G + 3K_v} = \frac{M_c(G\beta_\alpha - \alpha_c'\eta_\mu)}{G + M_c\,\alpha_c'\eta_c} \tag{B.192}$$

$$\alpha_\beta = \frac{M_c[\beta_\beta(4G + 3K_c) - 3\alpha_c'\alpha_\mu]}{4G + 3K_v} = \frac{M_c(G\beta_\beta - \alpha_c'\eta_\mu)}{G + M_c\,\alpha_c'\eta_c} \tag{B.193}$$

$$\eta_c = \frac{3\alpha_c G}{3K_c + 4G} \tag{B.194}$$

$$\eta_\mu = \frac{3\alpha_\mu G}{3K_c + 4G} = \frac{\eta_c \alpha_\mu}{\alpha_c} \tag{B.195}$$

$$\beta_e = \alpha\beta_d + \beta_v \tag{B.196}$$

$$\beta_v = \frac{\alpha_\mu}{K_c} - \frac{B_c \alpha_c \beta_\mu}{\alpha_c'} \tag{B.197}$$

$$\beta_\varepsilon = \frac{\alpha_c' \alpha_\mu}{K_c} - \beta_\mu \tag{B.198}$$

$$\beta_\beta = \frac{\phi}{(1-\mathcal{R})c_o} - \beta_\varepsilon \tag{B.199}$$

$$\beta_\alpha = \beta_\varepsilon + \frac{\mathcal{R}\kappa\mathcal{R}\mathcal{T}_o\phi}{(1-\mathcal{R})D_c - \mathcal{R}c_o\kappa\mathcal{R}\mathcal{T}_o} \tag{B.200}$$

$$\gamma_\mu' = \gamma_\mu - \frac{B_c\beta_\mu'(K_c\beta_\mu - \alpha_c'\alpha_\mu)}{\alpha_c'(1-\alpha_c B_c)} \tag{B.201}$$

$$\gamma_\pi = \frac{\gamma_\mu}{\mathcal{R}\mathcal{T}_o} \tag{B.202}$$

$$\kappa_p = \left[1 + \frac{\mathcal{R}^2 c_o \kappa \mathcal{R}\mathcal{T}_o}{(1-\mathcal{R})D_c - \mathcal{R}c_o\kappa\mathcal{R}\mathcal{T}_o}\right]\kappa \tag{B.203}$$

$$D_c' = \frac{\phi\,(1-\mathcal{R})(D_c - \mathcal{R}c_o\kappa\mathcal{R}\mathcal{T}_o)}{\phi - \mathcal{R}c_o\beta_\varepsilon} \tag{B.204}$$

$$c_\alpha = \frac{\kappa_p}{S_\alpha} \tag{B.205}$$

$$c_\beta = \frac{\beta_\beta D_c'}{S_\beta} \tag{B.206}$$

$$S_\alpha = \frac{4G + 3K_v}{M_c(4G + 3K_c)} = \frac{1}{M_c} + \frac{\alpha_c'\eta_c}{G} \tag{B.207}$$

$$S_\beta = \beta_\beta - \frac{3\alpha_c'\alpha_\mu}{4G + 3K_c} = \beta_\beta - \frac{\alpha_c'\,\eta_\mu}{G} \tag{B.208}$$

$$\mathcal{R}_a = \frac{\alpha_\mu}{\alpha_c\mathcal{R}\mathcal{T}_o} \tag{B.209}$$

$$\mathcal{R}_b = \frac{\beta_\mu}{C_c\mathcal{R}\mathcal{T}_o} \tag{B.210}$$

Relation Among Bulk and Intrinsic Constants Full expressions too lengthy to list.

For ideal porous medium, $K_\psi = \beta_s^\phi = \beta_w^\phi = 0$,

$$K_c = \frac{(1-\phi)^3 K_s K_\phi}{D_1}\left[\omega_s^b\,\omega_w^b - K_s(\beta_w^b\beta_w^b\,\omega_s^b + \beta_s^b\beta_s^b\,\omega_w^b)\right] \tag{B.211}$$

$$\alpha_c = \alpha_c' - \frac{K_s K_\phi (1-\phi)^3}{D_1 D_2} \left\{ [(1 + c_s^{po} \beta_s^p) \beta_w^b \, \omega_s^b - c_w^{po} \beta_s^b \beta_s^p \, \omega_w^b] \, \omega_w^p \right.$$
$$\left. - K_f \beta_w^p (\beta_w^b \beta_w^p \, \omega_s^b + \beta_s^b \beta_s^p \, \omega_w^b) \right\} \tag{B.212}$$

$$\alpha_c' = 1 - \frac{(1-\phi)^3 K_\phi \, \omega_s^b \, \omega_w^b}{D_1} \tag{B.213}$$

$$\alpha_\mu = \frac{(1-\phi)^3 K_s K_\phi}{D_1 D_2} \left[\omega_s^p \, \omega_w^p - K_f (\beta_w^p \, \beta_w^p \, \omega_s^p + \beta_s^p \, \beta_s^p \, \omega_w^p) \right]$$
$$\times (c_w^{po} \beta_s^b \, \omega_w^b - c_s^{po} \beta_w^b \, \omega_s^b) \tag{B.214}$$

$$B_c = \frac{K_f}{D_3} [c_w^{po} \omega_w^p - K_f \beta_w^p (c_s^{po} \beta_s^p + c_w^{po} \beta_w^p)]$$
$$\times \left\{ [K_s + \phi (1-\phi)^2 K_\phi] \, \omega_s^b \, \omega_w^b - K_s^2 (\beta_w^b \beta_w^b \, \omega_s^b + \beta_s^b \beta_s^b \, \omega_w^b) \right\} \tag{B.215}$$

$$\beta_\mu = \frac{\phi}{D_4} \left\{ -(c_w^{po} \beta_s^p \, \omega_w^p - c_s^{po} \beta_w^p \, \omega_s^p) [\, \omega_s^b \, \omega_w^b - K_s (\beta_w^b \, \beta_w^b \, \omega_s^b + \beta_s^b \, \beta_s^b \, \omega_w^b)] \right.$$
$$\left. + (c_w^{po} \beta_s^b \, \omega_w^b - c_s^{po} \beta_w^b \, \omega_s^b) [\, \omega_s^p \, \omega_w^p - K_f (\beta_w^p \, \beta_w^p \, \omega_s^p + \beta_s^p \, \beta_s^p \, \omega_w^p)] \right\} \tag{B.216}$$

$$\beta_\mu' = \frac{\beta_s^p}{D_2} (c_w^{po} \omega_w^p + K_f \beta_w^p) \tag{B.217}$$

$$\gamma_\mu = \frac{c_w^{po}}{D_2} \left[\omega_s^p \, \omega_w^p - K_f (\beta_w^p \beta_w^p \, \omega_s^p + \beta_s^p \beta_s^p \, \omega_w^p) \right] \tag{B.218}$$

where

$$D_1 = [K_s + (1-\phi)^2 K_\phi] \, \omega_s^b \, \omega_w^b - K_s^2 (\beta_w^b \beta_w^b \, \omega_s^b + \beta_s^b \beta_s^b \, \omega_w^b) \tag{B.219}$$

$$D_2 = c_w^{po} \, \omega_w^p - K_f \beta_w^p (c_s^{po} \beta_s^p + c_w^{po} \beta_w^p) \tag{B.220}$$

$$D_3 = K_f^2 \beta_w^p (c_s^{po} \beta_s^p + c_w^{po} \beta_w^p) \left\{ K_s^2 (\beta_w^b \beta_w^b \, \omega_s^b + \beta_s^b \beta_s^b \, \omega_w^b) \right.$$
$$\left. - [K_s + \phi^2 (1-\phi)^2 K_\phi] \, \omega_s^b \, \omega_w^b \right\}$$
$$- \phi (1-\phi)^3 K_s K_\phi K_f \left[K_s \beta_w^p (\beta_w^b \beta_w^b \, \omega_s^b + \beta_s^b \beta_s^b \, \omega_w^b) \right.$$
$$- K_f \beta_w^p (\beta_w^b \beta_w^p \, \omega_s^b + \beta_s^b \beta_s^p \, \omega_w^b)$$
$$\left. + \beta_s^p (c_s^{po} \beta_w^b \, \omega_s^b - c_w^{po} \beta_s^b \, \omega_w^b) \, \omega_w^p - \omega_s^b (\beta_w^p \, \omega_w^b - \beta_w^b \, \omega_w^p) \right]$$
$$+ c_w^{po} \left\{ \phi (1-\phi)^3 K_s K_\phi + K_f [K_s + \phi^2 (1-\phi)^2 K_\phi] \omega_w^p \right\} \omega_s^b \, \omega_w^b$$
$$- c_w^{po} K_s^2 [K_f + \phi (1-\phi)^3 K_\phi] (\beta_w^b \beta_w^b \, \omega_s^b + \beta_s^b \beta_s^b \, \omega_w^b) \, \omega_w^p \tag{B.221}$$

$$D_4 = D_2 [\, \omega_s^b \, \omega_w^b - K_s (\beta_w^b \beta_w^b \, \omega_s^b + \beta_s^b \beta_s^b \, \omega_w^b)] \tag{B.222}$$

For dilute solution, $c_s^{po} = \beta_s^p = 0$,

$$K_c = \frac{(1-\phi)^3 K_s K_\phi}{D_1}\left[\omega_s^b \omega_w^b - K_s(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)\right] \tag{B.223}$$

$$\alpha_c = \alpha_c' - \frac{(1-\phi)^3 K_s K_\phi \beta_w^b \omega_s^b}{c_w^{po} D_1} \tag{B.224}$$

$$\alpha_c' = 1 - \frac{(1-\phi)^3 K_\phi \omega_s^b \omega_w^b}{D_1} \tag{B.225}$$

$$\alpha_\mu = \frac{(1-\phi)^3 K_s K_\phi \beta_s^b \omega_s^p \omega_w^b}{D_1} \tag{B.226}$$

$$B_c = \frac{K_f c_w^{po}}{D_3}(\omega_w^p - K_f \beta_w^p \beta_w^p)\left\{[K_s + \phi(1-\phi)^2 K_\phi]\omega_s^b \omega_w^b - K_s^2(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)\right\} \tag{B.227}$$

$$\beta_\mu = \frac{\phi \beta_s^b \omega_w^b \omega_s^p}{\omega_s^b \omega_w^b - K_s(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)} \tag{B.228}$$

$$\beta_\mu' = 0 \tag{B.229}$$

$$\gamma_\mu = \omega_s^p \tag{B.230}$$

where

$$D_1 = [K_s + (1-\phi)^2 K_\phi]\omega_s^b \omega_w^b - K_s^2(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b) \tag{B.231}$$

$$\begin{aligned} D_3 = {} & K_f^2 c_w^{po} \beta_w^p \beta_w^p \left\{K_s^2(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b) - [K_s + \phi^2(1-\phi)^2 K_\phi]\omega_s^b \omega_w^b\right\} \\ & - \phi(1-\phi)^3 K_s K_\phi K_f \left[K_s \beta_w^p(\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b) - K_f \beta_w^b \beta_w^p \beta_w^p \omega_s^b - \omega_s^b(\beta_w^p \omega_w^b - \beta_w^b \omega_w^p)\right] \\ & + c_w^{po}\left\{\phi(1-\phi)^3 K_s K_\phi + K_f[K_s + \phi^2(1-\phi)^2 K_\phi]\omega_w^p\right\}\omega_s^b \omega_w^b \\ & - c_w^{po} K_s^2[K_f + \phi(1-\phi)^3 K_\phi](\beta_w^b \beta_w^b \omega_s^b + \beta_s^b \beta_s^b \omega_w^b)\omega_w^p \end{aligned} \tag{B.232}$$

References

1. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2):155–164
2. Biot MA (1955) Theory of elasticity and consolidation for a porous anisotropic solid. J Appl Phys 26(2):182–185
3. Biot MA (1956) General solutions of the equations of elasticity and consolidation for a porous material. J Appl Mech ASME 78:91–96

4. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid.1. Low-frequency range. J Acoust Soc Am 28(2):168–178
5. Biot MA, Willis DG (1957) The elastic coefficients of the theory of consolidation. J Appl Mech ASME 24:594–601
6. Rice JR, Cleary MP (1976) Some basic stress diffusion solutions for fluid-saturated elastic porous media with compressible constituents. Rev Geophys 14(2):227–241

Appendix C
Cartesian and Curvilinear Coordinate System

For convenience of reference, the stress, strain, constitutive equations, and governing equations are expressed in Cartesian, spherical, and cylindrical coordinate systems in the following.

C.1 Cartesian Coordinate System

In Cartesian coordinates (x, y, z), the gradient operator is [1]

$$\nabla p = \frac{\partial p}{\partial x}\hat{i} + \frac{\partial p}{\partial y}\hat{j} + \frac{\partial p}{\partial z}\hat{k} \tag{C.1}$$

the divergence operator is

$$\nabla \cdot \vec{u} = e = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \tag{C.2}$$

the curl operator is

$$\nabla \times \vec{u} = 2\,\vec{\Omega} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ u_x & u_y & u_z \end{vmatrix} \tag{C.3}$$

and the Laplacian operator is given by

$$\nabla^2 p = \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} \tag{C.4}$$

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

In the above, the volumetric strain e is defined as

$$e = e_{xx} + e_{yy} + e_{zz} \tag{C.5}$$

and the strains are

$$e_{xx} = \frac{\partial u_x}{\partial x} \tag{C.6}$$

$$e_{yy} = \frac{\partial u_y}{\partial y} \tag{C.7}$$

$$e_{zz} = \frac{\partial u_z}{\partial z} \tag{C.8}$$

$$e_{xy} = e_{yx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) \tag{C.9}$$

$$e_{yz} = e_{zy} = \frac{1}{2}\left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y}\right) \tag{C.10}$$

$$e_{zx} = e_{xz} = \frac{1}{2}\left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z}\right) \tag{C.11}$$

The rotation vector is

$$\vec{\Omega} = \Omega_x \hat{i} + \Omega_y \hat{j} + \Omega_z \hat{k} \tag{C.12}$$

with

$$\Omega_x = \frac{1}{2}\left(\frac{\partial u_z}{\partial y} - \frac{\partial u_y}{\partial z}\right) \tag{C.13}$$

$$\Omega_y = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} - \frac{\partial u_z}{\partial x}\right) \tag{C.14}$$

$$\Omega_z = \frac{1}{2}\left(\frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y}\right) \tag{C.15}$$

Stress strain relations are

$$\sigma_{ij} = 2Ge_{ij} + \frac{2G\nu}{1-2\nu}\delta_{ij}e - \alpha\delta_{ij}p; \quad i,j = (x,y,z) \tag{C.16}$$

The equilibrium equations are

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = 0 \tag{C.17}$$

$$\frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = 0 \tag{C.18}$$

$$\frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = 0 \tag{C.19}$$

Navier equations are obtained by substituting the constitutive equations into the equilibrium equations.

C.2 Cylindrical Coordinate System

In cylindrical coordinates (r, θ, z), the gradient operator is [1]

$$\nabla p = \frac{\partial p}{\partial r}\,\hat{e}_r + \frac{1}{r}\frac{\partial p}{\partial \theta}\,\hat{e}_\theta + \frac{\partial p}{\partial z}\,\hat{e}_z \tag{C.20}$$

the divergence operator is

$$\nabla \cdot \vec{u} = e = \frac{1}{r}\frac{\partial\, r u_r}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} \tag{C.21}$$

the curl operator is

$$\nabla \times \vec{u} = 2\,\vec{\Omega} = \frac{1}{r}\begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \partial/\partial r & \partial/\partial \theta & \partial/\partial z \\ u_r & r u_\theta & u_z \end{vmatrix} \tag{C.22}$$

and the Laplacian operator is given by

$$\nabla^2 p = \frac{1}{r}\left[\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\frac{1}{r}\frac{\partial p}{\partial \theta}\right) + \frac{\partial}{\partial z}\left(r\frac{\partial p}{\partial z}\right)\right] \tag{C.23}$$

In the above, the volumetric strain e is defined as

$$e = e_{rr} + e_{\theta\theta} + e_{zz} \tag{C.24}$$

and the strains are

$$e_{rr} = \frac{\partial u_r}{\partial r} \tag{C.25}$$

$$e_{\theta\theta} = \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} \tag{C.26}$$

$$e_{zz} = \frac{\partial u_z}{\partial z} \tag{C.27}$$

$$e_{r\theta} = \frac{1}{2}\left(\frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} + \frac{1}{r}\frac{\partial u_r}{\partial \theta}\right) \tag{C.28}$$

$$e_{rz} = \frac{1}{2}\left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}\right) \tag{C.29}$$

$$e_{\theta z} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z}\right) \tag{C.30}$$

The rotation vector is

$$\vec{\Omega} = \Omega_r\,\hat{e}_r + \Omega_\theta\,\hat{e}_\theta + \Omega_z\,\hat{e}_z \tag{C.31}$$

with

$$\Omega_r = \frac{1}{2r}\left(\frac{\partial u_z}{\partial \theta} - \frac{\partial r u_\theta}{\partial z}\right) \tag{C.32}$$

$$\Omega_\theta = \frac{1}{2r}\left(\frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r}\right) \tag{C.33}$$

$$\Omega_z = \frac{1}{2r}\left(\frac{\partial\, r u_\theta}{\partial r} - \frac{\partial u_r}{\partial \theta}\right) \tag{C.34}$$

Stress strain relations are

$$\sigma_{ij} = 2Ge_{ij} + \frac{2G\nu}{1-2\nu}\delta_{ij}e - \alpha\delta_{ij}p; \quad i,j = (r,\theta,z) \tag{C.35}$$

The equilibrium equations are

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{1}{r}\frac{\partial \sigma_{r\theta}}{\partial \theta} + \frac{\partial \sigma_{rz}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0 \tag{C.36}$$

$$\frac{\partial \sigma_{r\theta}}{\partial r} + \frac{1}{r}\frac{\partial \sigma_{\theta\theta}}{\partial \theta} + \frac{\partial \sigma_{\theta z}}{\partial z} + \frac{2\sigma_{r\theta}}{r} = 0 \tag{C.37}$$

$$\frac{\partial \sigma_{rz}}{\partial r} + \frac{1}{r}\frac{\partial \sigma_{\theta z}}{\partial \theta} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\sigma_{rz}}{r} = 0 \tag{C.38}$$

Navier equations are obtained by substituting the constitutive equations into the equilibrium equations.

C.3 Spherical Coordinate System

In spherical coordinates (R, θ, φ), the gradient operator is [1]

$$\nabla p = \frac{\partial p}{\partial R}\,\hat{e}_R + \frac{1}{R}\frac{\partial p}{\partial \theta}\,\hat{e}_\theta + \frac{1}{R\sin\theta}\frac{\partial p}{\partial \varphi}\,\hat{e}_\varphi \tag{C.39}$$

the divergence operator is

$$\nabla \cdot \vec{u} = e = \frac{1}{R^2 \sin\theta}\left[\frac{\partial}{\partial R}(R^2 u_R \sin\theta) + \frac{\partial}{\partial \theta}(R\,u_\theta \sin\theta) + \frac{\partial}{\partial \varphi}(R\,u_\varphi)\right] \tag{C.40}$$

the curl operator is

$$\nabla \times \vec{u} = 2\,\vec{\Omega} = \frac{1}{R^2 \sin\theta}\begin{vmatrix} \hat{e}_R & R\,\hat{e}_\theta & R\sin\theta\,\hat{e}_\varphi \\ \partial/\partial R & \partial/\partial\theta & \partial/\partial\varphi \\ u_R & R\,u_\theta & R\sin\theta\,u_\varphi \end{vmatrix} \tag{C.41}$$

and the Laplacian operator is given by

$$\nabla^2 p = \frac{1}{R^2\sin\theta}\left[\frac{\partial}{\partial R}\left(R^2\sin\theta\,\frac{\partial p}{\partial R}\right) + \frac{\partial}{\partial\theta}\left(\sin\theta\,\frac{\partial p}{\partial\theta}\right) + \frac{\partial}{\partial\varphi}\left(\frac{1}{\sin\theta}\,\frac{\partial p}{\partial\varphi}\right)\right] \tag{C.42}$$

In the above, the volumetric strain e is defined as

$$e = e_{RR} + e_{\theta\theta} + e_{\varphi\varphi} \tag{C.43}$$

and the strains are

$$e_{RR} = \frac{\partial u_R}{\partial R} \tag{C.44}$$

$$e_{\theta\theta} = \frac{1}{R}\frac{\partial u_\theta}{\partial\theta} + \frac{u_R}{R} \tag{C.45}$$

$$e_{\varphi\varphi} = \frac{1}{R\sin\theta}\frac{\partial u_\varphi}{\partial\varphi} + \frac{u_R}{R} + \frac{u_\theta\cot\theta}{R} \tag{C.46}$$

$$e_{R\theta} = \frac{1}{2}\left(\frac{\partial u_\theta}{\partial R} - \frac{u_\theta}{R} + \frac{1}{R}\frac{\partial u_R}{\partial\theta}\right) \tag{C.47}$$

$$e_{R\varphi} = \frac{1}{2}\left(\frac{1}{R\sin\theta}\frac{\partial u_R}{\partial\varphi} + \frac{\partial u_\varphi}{\partial R} - \frac{u_\varphi}{R}\right) \tag{C.48}$$

$$e_{\theta\varphi} = \frac{1}{2}\left(\frac{1}{R\sin\theta}\frac{\partial u_\theta}{\partial\varphi} + \frac{1}{R}\frac{\partial u_\varphi}{\partial\theta} - \frac{u_\varphi\cot\theta}{R}\right) \tag{C.49}$$

The rotation vector is

$$\vec{\Omega} = \Omega_R\,\hat{e}_R + \Omega_\theta\,\hat{e}_\theta + \Omega_\varphi\,\hat{e}_\varphi \tag{C.50}$$

with

$$\Omega_R = \frac{1}{2R^2\sin\theta}\left(\frac{\partial R\sin\theta\,u_\varphi}{\partial\theta} - \frac{\partial R\,u_\theta}{\partial\varphi}\right) \tag{C.51}$$

$$\Omega_\theta = \frac{1}{2R\sin\theta}\left(\frac{\partial u_R}{\partial\varphi} - \frac{\partial R\sin\theta\,u_\varphi}{\partial R}\right) \tag{C.52}$$

$$\Omega_\varphi = \frac{1}{2R}\left(\frac{\partial\,R\,u_\theta}{\partial R} - \frac{\partial u_R}{\partial\theta}\right) \tag{C.53}$$

Stress strain relations are

$$\sigma_{ij} = 2Ge_{ij} + \frac{2G\nu}{1-2\nu}\delta_{ij}e - \alpha\delta_{ij}p; \quad i,j = (R,\theta,\varphi) \tag{C.54}$$

The equilibrium equations are

$$\frac{\partial\sigma_{RR}}{\partial R} + \frac{1}{R}\frac{\partial\sigma_{R\theta}}{\partial\theta} + \frac{1}{R\sin\theta}\frac{\partial\sigma_{R\varphi}}{\partial\varphi} + \frac{1}{R}\left(2\,\sigma_{RR} - \sigma_{\theta\theta} - \sigma_{\varphi\varphi} + \sigma_{R\theta}\cot\theta\right) = 0 \tag{C.55}$$

$$\frac{\partial\sigma_{R\theta}}{\partial R} + \frac{1}{R}\frac{\partial\sigma_{\theta\theta}}{\partial\theta} + \frac{1}{R\sin\theta}\frac{\partial\sigma_{\theta\varphi}}{\partial\varphi} + \frac{1}{R}\left[(\sigma_{\theta\theta} - \sigma_{\varphi\varphi})\cot\theta + 3\,\sigma_{R\theta}\right] = 0 \tag{C.56}$$

$$\frac{\partial\sigma_{R\varphi}}{\partial R} + \frac{1}{R}\frac{\partial\sigma_{\theta\varphi}}{\partial\theta} + \frac{1}{R\sin\theta}\frac{\partial\sigma_{\varphi\varphi}}{\partial\varphi} + \frac{1}{R}\left(2\,\sigma_{\theta\varphi}\cot\theta + 3\,\sigma_{R\varphi}\right) = 0 \tag{C.57}$$

Navier equations are obtained by substituting the constitutive equations into the equilibrium equations.

Reference

1. Aris R (1990) Vectors, tensors and the basic equations of fluid mechanics. Dover, New York, 320pp

Appendix D
Laplace Transform and Inversion

D.1 Laplace Transform

Laplace transform is a powerful tool that can reduce a differential operator into an algebraic operator, for the solution of partial or ordinary differential equations. It is normally applied to the time variable to resolve the time derivative. Given a function of time, $f(t)$, its Laplace transform $\tilde{f}(s)$ is defined as

$$\mathscr{L}\{f(t)\} = \tilde{f}(s) = \int_0^\infty f(t)\, e^{-st}\, dt \tag{D.1}$$

where s is the *Laplace transform parameter* and is a real variable. The inverse transformation is given by the *Bromwich integral* in the complex s plane as

$$\mathscr{L}^{-1}\{\tilde{f}(s)\} = f(t) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} \tilde{f}(s)\, e^{st}\, ds \tag{D.2}$$

where the integration is performed along the vertical line $\Re(s) = \gamma$ such that γ is greater than the real part of all singularities (poles) of $\tilde{f}(s)$.

As textbooks on Laplace transform [3, 6] are readily available, there is no intention to give an introduction to Laplace transform in this appendix. We shall, however, summarize a few essential formulae that are used in this book to obtain solutions of poroelastic boundary values problems.

The Laplace transform can remove a time derivative and turn it into an algebraic form as

$$\mathscr{L}\left\{\frac{\partial f(t)}{\partial t}\right\} = s\tilde{f}(s) - f(0) \tag{D.3}$$

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

where $f(0)$ is the initial value. The order of Laplace transform and differentiation is interchangeable, such that

$$\mathscr{L}\left\{\frac{\partial f(x,t)}{\partial x}\right\} = \frac{\partial\, \mathscr{L}\{f(x,t)\}}{\partial x} = \frac{\partial \tilde{f}(x,s)}{\partial x} \tag{D.4}$$

With (D.3) and (D.4), we can transform the diffusion equation (7.372) from a partial differential equation to an ordinary differential equation (7.374), as shown in Sect. 7.10.

To simplify the presentation of the solution, it is often desirable to normalize the variables and the equations to *dimensionless* forms. For example, the time variable t is typically combined with a diffusion coefficient c (dim. $[L^2/T]$) and a length scale L to form a dimensionless time

$$t^* = \frac{ct}{L^2} \tag{D.5}$$

We can either perform such nondimensionalization on the original differential equation, and then perform Laplace transform regarding t^* as t, or we can work on the Laplace transformed equation and define a dimensionless Laplace transform parameter

$$s^* = \frac{L^2 s}{c} \tag{D.6}$$

to rearrange the equation into dimensionless form. As the inverse Laplace transform is performed on the variable s, the following formula is handy to transform the equation in terms of s^*, such as (7.379), into an equation in terms of t^*, such as (7.395):

$$\mathscr{L}^{-1}\left\{\tilde{f}(as)\right\} = \frac{1}{a} f\left(\frac{t}{a}\right) \tag{D.7}$$

There is a powerful property of the Laplace transform, known as the *convolutional theorem*. The theorem takes the form

$$\mathscr{L}^{-1}\left\{\tilde{f}_1(s)\tilde{f}_2(s)\right\} = \int_0^t f_1(t-\tau) f_2(\tau)\, d\tau \tag{D.8}$$

The integral on the right hand side is a *convolutional integral*. Equation (D.8) can be used to find the Laplace inverse of the product of $\tilde{f}_1(s)$ and $\tilde{f}_2(s)$, if the individual inverse transforms, $f_1(t)$ and $f_2(t)$, are known. Or, this theorem in its inverse form can be used to find the Laplace transform of the convolutional integral, which can result from the application of *Duhamel principle of superposition*, in the following form

$$\mathscr{L}\left\{\int_0^t f_1(t-\tau) f_2(\tau)\, d\tau\right\} = \tilde{f}_1(s)\tilde{f}_2(s) \tag{D.9}$$

Finally, we present a short table of Laplace transform and inversion [8] for formulae used in this book, as Table D.1.

Table D.1 A short table of Laplace transform and inversion [3]

$f(t)$	$\tilde{f}(s)$
1	s^{-1}
$2(t/\pi)^{1/2}$	$s^{-3/2}$
t	s^{-2}
$\mathrm{H}(t-t_o)$	$s^{-1}\exp(-st_o)$
$\mathrm{H}(t-0)$	s^{-1}
$\delta(t-t_o)$	$\exp(-st_o)$
$\delta(t-0)$	1
$\exp(at)$	$1/(s-a)$
$t\exp(-t)$	$1/(s+1)^2$
$\gamma-\ln t$	$(\ln s)/s$
$\mathrm{erfc}(a/2\sqrt{t})$	$\exp(-a\sqrt{s})/s$
$2\sqrt{t/\pi}\,\exp(-a^2/4t)-a\,\mathrm{erfc}(a/2\sqrt{t})$	$\exp(-a\sqrt{s})/s^{3/2}$
$-a\sqrt{t/\pi}\,\exp(-a^2/4t)$ $+(t+a^2/2)\,\mathrm{erfc}(a/2\sqrt{t})$	$\exp(-a\sqrt{s})/s^2$
$[2\sqrt{t/\pi}\,(a^2+4t)\,\exp(-a^2/4t)$ $-(a^3+6at)\,\mathrm{erfc}(a/2\sqrt{t})]/6$	$\exp(-a\sqrt{s})/s^{5/2}$
$-b\exp(b^2t+ab)\,\mathrm{erfc}[b\sqrt{t}+(a/2\sqrt{t})]$ $+\exp(-a^2/4t)$	$\exp(-a\sqrt{s})/(b+\sqrt{s})$
$\{-\exp(b^2t+ab)\,\mathrm{erfc}[b\sqrt{t}+(a/2\sqrt{t})]$ $+\mathrm{erfc}(a/2\sqrt{t})\}/b$	$\exp(-a\sqrt{s})/[s(b+\sqrt{s})]$
$\exp(-a^2/4t)/2t$	$\mathrm{K}_0(a\sqrt{s})$
$\mathrm{E}_1(a^2/4t)/2$	$\mathrm{K}_0(a\sqrt{s})/s$
$[-4t\exp(-a^2/4t)$ $+(a^2+4t)\mathrm{E}_1(a^2/4t)]/8$	$\mathrm{K}_0(a\sqrt{s})/s^2$
$1/\sqrt{\pi t}-a\exp(a^2t)\,\mathrm{erfc}(a\sqrt{t})$	$1/(\sqrt{s}+a)$
$\exp(-b^2/4t)/\sqrt{\pi t}-a\exp(ab+a^2t)\times$ $\mathrm{erfc}[(b/2\sqrt{t})+a\sqrt{t}]$	$\exp(-b\sqrt{s})/(\sqrt{s}+a)$
$\sum_{n=1}^{m}[p(a_n)/q'(a_n)]\exp(a_nt)$	$p(s)/q(s);\ q(s)=(s-a_1)(s-a_2)$ $\cdots(s-a_m)$

D.2 Inverse Laplace Transform

To obtain the solution in the time domain, we can carry out the inverse Laplace transform by performing the line integral on the complex s plane as given by (D.2). The integration can in fact be performed on the closed semi-circular contour as shown in Fig. D.1, in which the semi-circle is to be pushed to infinity. For the problems investigated here, if the integrand decays fast enough as $s \to \infty$ for the integration on the open semi-circle to vanish, the closed contour integral is equivalent to the line integral in (D.2).

According to the *Cauchy integral formula*, if $f(z)$ is *analytic* (meaning it is continuous and without singularities) in the area enclosed by a closed contour C in the complex plane $z = x + iy$, then

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a}\, dz = f(a) \tag{D.10}$$

if the point a is contained in C.

On the other hand, if $f(z)$ is analytic in C, except at a number of singularity points a_i, $i = 1, 2, \ldots, n$, known as *poles*, around which $f(z)$ can be expanded into a Laurent series, then the *Cauchy residue theorem* states

$$\frac{1}{2\pi i} \oint_C f(z)\, dz = \sum_{i=1}^{n} b_i \tag{D.11}$$

where b_i are the coefficients associated with the $1/(z-a_i)$ term of the Laurent series around a_i.

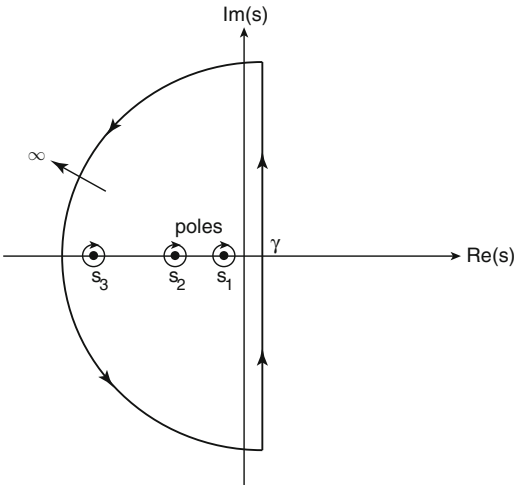


Fig. D.1 Integration contour for inverse Laplace transform

If the integrand can be expressed as the quotient of two analytic functions, $f(z)$ and $g(z)$, and $g(z)$ contains zeros at a_i, that is, the complex roots of the equation $g(z) = 0$, then the following formula exists [7]

$$\frac{1}{2\pi i}\oint_C \frac{f(z)}{g(z)}\,dz = \sum_{i=1}^{n}\frac{f(a_i)}{g'(a_i)} \tag{D.12}$$

where $g'(z)$ is the derivative of $g(z)$, and $g'(a_i)$ is the coefficient of the Taylor series expansion term $(z-a_i)$ around the point a_i. In the above, we assume that $g'(a) \neq 0$. If $g(a) = g'(a) = 0$, but $g''(a) \neq 0$, then the residual should come from the term associated with $(z-a_i)^2$, so on and so forth.

Finally, we come to the Laplace inverse transform of the function $\tilde{f}(s)$, which can be expressed as the ratio of two analytic functions $\tilde{f}(s) = h(s)/g(s)$. If $g(s)$ contains zeros at s_i, $i = 1, 2, \ldots, n$, then the Laplace inversion becomes

$$f(t) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty}\tilde{f}(s)\,e^{st}\,ds = \frac{1}{2\pi i}\oint_C \frac{h(s)}{g(s)}\,e^{st}\,ds = \sum_{i=1}^{n}\frac{h(s_i)}{g'(s_i)}\,e^{s_i t} \tag{D.13}$$

where C is the closed semi-circle contour in Fig. D.1.

D.3 Approximate Inverse Laplace Transform

As not all Laplace transform expressions can be analytically inverted, approximate inversion based on numerical algorithms is useful. Here we shall give a brief discussion and present one particular method for ease of use.

There exist a large number of approximate inverse Laplace transform algorithms [9, 10]. With issues of computational efficiency and accuracy in mind, a handful of those techniques have been tested and compared side by side [2, 4, 5]. One of the methods widely in use for engineering purposes is the *Stehfest method* [11, 12]. All approximate inverse Laplace transform results found in this book are obtained using this algorithm; hence a brief presentation is given below.

We denote $F(s)$ the Laplace transform of the function $f(t)$. The Stehfest inversion seeks to find the approximate value of $f(t)$ at a given t by sampling $F(s)$ at a set of discrete s values. The resultant series approximation is as follows:

$$f(t) \approx \frac{\ln 2}{t}\sum_{i=1}^{n} c_i\,F\left(\frac{i\ln 2}{t}\right) \tag{D.14}$$

where the coefficients c_i are given by

$$c_i = (-1)^{i+\frac{n}{2}} \sum_{k=\lfloor \frac{i+1}{2} \rfloor}^{\min(i,\frac{n}{2})} \frac{k^{\frac{n}{2}}\,(2k)!}{(\frac{n}{2}-k)!\,k!\,(k-1)!\,(i-k)!\,(2k-i)!} \tag{D.15}$$

In the above, n is the number of terms in the series, which must be even, and the brackets $\lfloor\ \rfloor$ is the *floor function* that gives the greatest integer not larger than its argument (e.g., $\lfloor \frac{3}{2} \rfloor = 1$).

Some caution needs to be exercised in using the above, and in fact in all approximate Laplace inverse algorithms. It is important to recognize that the inverse Laplace transform is inherently unstable, that is, a small perturbation of the data in the inversion space (values of $F(s)$) can lead to large error in $f(t)$ [1]. For example, one should not attempt to numerically invert a Laplace transform of a periodic function in time (such as $\sin t$), or functions that have spike-like behavior, such as a (Dirac) delta convergence sequence. However, for functions that have a steady state limit as $t \to \infty$, most approximate inversion methods should perform well.

For the Stehfest method, we should note that the series presented in (D.14) is an asymptotic series, and not a convergent series. The series is made of alternating (positive and negative) terms in pairs that get ever larger as the number of terms increases. With finite precision computation, the result becomes unstable when the number of terms becomes too large. So there exist an optimal number of terms for a stable result, which can be obtained only empirically. Generally speaking, with double precision computation, 6–10 terms in the series can give good "engineering" results, with error less than 1%. For better results, up to 20 terms may be tolerated, but one should watch for instability of the result in that range. For even higher accuracy inversion, high precision computation, such as quadruple precision, is recommended.

References

1. Bellman R, Kalaba RE, Lockett JA (1966) Numerical in version of the Laplace transform: applications to biology, economics, engineering, and physics. American Elsevier, New York, 211pp
2. Cheng AHD, Sidauruk P, Abousleiman Y (1994) Approximate inversion of the Laplace transform. Math J 4(2):76–82
3. Churchill RV (1972) Operational mathematics. McGraw-Hill, New York, 3rd edn, 481pp
4. Cost TL (1964) Approximate Laplace transform inversions in viscoelastic stress analysis. AIAA J 2(12):2157–2166
5. Davies B, Martin B (1979) Numerical inversion of the Laplace transform–survey and comparison of methods. J Comput Phys 33(1):1–32
6. LaPage WR (2010) Complex variables and the Laplace transform for engineers. Dover, New York, 512pp
7. Mei CC (1997) Mathematical analysis in engineering. Cambridge University Press, Cambridge/New York, 461pp

8. Oberhettinger F, Badii L (1973) Tables of Laplace transforms. Springer, Berlin/New York, 428pp
9. Piessens R (1975) A bibliography on numerical inversion of the Laplace transform and applications. J Comput Appl Math 1(2):115–128
10. Piessens R, Dang NDP (1976) A bibliography on numerical inversion of the Laplace transform and applications: a supplement. J Comput Appl Math 2(3):225–228
11. Stehfest H (1970) Numerical inversion of Laplace transforms. Commun ACM 13(1):47–49
12. Stehfest H (1970) Correction. Commun ACM 13(10):624

Appendix E
Poroelasticity Fundamental Solution

A comprehensive list of Green's functions for quasi-static isotropic poroelasticity is given below. The following abbreviated notations are utilized:

$$r = \left|\vec{x} - \vec{\chi}\right| \tag{E.1}$$

$$r_{,i} = \frac{\partial r}{\partial x_i} = \frac{x_i - \chi_i}{r} \tag{E.2}$$

$$\xi = \frac{r}{\sqrt{4c\,(t-\tau)}} \tag{E.3}$$

E.1 Continuous Fluid Source

$\gamma = \delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau)$
2D

$$u_i^{sc} = \frac{\eta}{8\pi G\kappa} r\, r_{,i} \left[\mathrm{E}_1(\xi^2) + \frac{1 - e^{-\xi^2}}{\xi^2}\right] \tag{E.4}$$

$$\sigma_{ij}^{sc} = \frac{\eta}{4\pi\kappa}\left[(\delta_{ij} - 2r_{,i}r_{,j})\frac{1 - e^{-\xi^2}}{\xi^2} - \delta_{ij}\mathrm{E}_1(\xi^2)\right] \tag{E.5}$$

$$p^{sc} = \frac{1}{4\pi\kappa}\mathrm{E}_1(\xi^2) = \frac{1}{S}\zeta^{sc} \tag{E.6}$$

$$w_i^{sc} = \frac{1}{8\pi c} r\, r_{,i}\left[\frac{e^{-\xi^2}}{\xi^2} - \mathrm{E}_1(\xi^2)\right] \tag{E.7}$$

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

$$q_i^{sc} = \frac{1}{2\pi}\frac{r_{,i}}{r}e^{-\xi^2} = w_i^{si} = \kappa\, p_i^{pc} = c\,\zeta_i^{pc} = w_i^{lc} = p_i^{fc}$$
$$= \frac{1}{S}\zeta_i^{fc} = -\frac{G}{\eta}\zeta_i^{Fc} \tag{E.8}$$
$$\zeta^{sc} = Sp^{sc} \tag{E.9}$$

3D

$$u_i^{sc} = \frac{\eta}{16\pi G\kappa} r_{,i}\left[2\,\mathrm{erfc}(\xi) + \frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right] \tag{E.10}$$
$$\sigma_{ij}^{sc} = \frac{\eta}{8\pi\kappa}\frac{1}{r}\left\{(\delta_{ij} - 3r_{,i}r_{,j})\left[\frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right] - 2(\delta_{ij} + r_{,i}r_{,j})\,\mathrm{erfc}(\xi)\right\} \tag{E.11}$$
$$p^{sc} = \frac{1}{4\pi\kappa}\frac{\mathrm{erfc}(\xi)}{r} = \frac{1}{S}\zeta^{sc} \tag{E.12}$$
$$w_i^{sc} = \frac{1}{16\pi c} r_{,i}\left[\frac{\mathrm{erfc}(\xi)}{\xi^2} - 2\,\mathrm{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right] \tag{E.13}$$
$$q_i^{sc} = \frac{1}{4\pi}\frac{r_{,i}}{r^2}\left[\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2} + \mathrm{erfc}(\xi)\right] = w_i^{si} = \kappa\, p_i^{pc} = c\,\zeta_i^{pc}$$
$$= w_i^{lc} = p_i^{fc} = \frac{1}{S}\zeta_i^{fc} = -\frac{G}{\eta}\zeta_i^{Fc} \tag{E.14}$$
$$\zeta^{sc} = Sp^{sc} \tag{E.15}$$

E.2 Instantaneous Fluid Source

$\gamma = \delta(\vec{x} - \vec{\chi})\,\delta(t - \tau)$
2D

$$u_i^{si} = \frac{\eta}{2\pi GS}\frac{r_{,i}}{r}(1 - e^{-\xi^2}) = p_i^{Fc} = u_i^{lc} \tag{E.16}$$
$$\sigma_{ij}^{si} = \frac{\eta}{\pi S}\frac{1}{r^2}\left[(\delta_{ij} - 2r_{,i}r_{,j})(1 - e^{-\xi^2}) - 2(\delta_{ij} - r_{,i}r_{,j})\xi^2 e^{-\xi^2}\right]$$
$$= \sigma_{ij}^{lc} = p_{ij}^{dc} \tag{E.17}$$
$$p^{si} = \frac{1}{\pi S}\frac{1}{r^2}\xi^2 e^{-\xi^2} = \frac{1}{S}\zeta^{si} = p^{lc} = \frac{1}{S}\zeta^{lc} \tag{E.18}$$

$$w_i^{si} = q_i^{sc} = \kappa p_i^{pc} = c\,\zeta_i^{pc} = w_i^{lc} = p_i^{fc} = \frac{1}{S}\zeta_i^{fc} \tag{E.19}$$

$$\begin{aligned} q_i^{si} &= \frac{2c}{\pi}\frac{r_{,i}}{r^3}\xi^4 e^{-\xi^2} = \kappa p_i^{pi} = c\,\zeta_i^{pi} = q_i^{lc} = w_i^{li} = p_i^{fi} \\ &= \frac{1}{S}\zeta_i^{fi} = -\frac{G}{\eta}\zeta_i^{Fi} \end{aligned} \tag{E.20}$$

$$\zeta^{si} = Sp^{si} = Sp^{lc} = \zeta^{lc} \tag{E.21}$$

3D

$$u_i^{si} = \frac{\eta}{4\pi GS}\frac{r_{,i}}{r^2}\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\,\xi e^{-\xi^2}\right] = p_i^{Fc} = u_i^{lc} \tag{E.22}$$

$$\begin{aligned} \sigma_{ij}^{si} = \frac{\eta}{2\pi S}\frac{1}{r^3}\Bigg\{&(\delta_{ij} - 3r_{,i}r_{,j})\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\,\xi e^{-\xi^2}\right] \\ &- \frac{4}{\sqrt{\pi}}(\delta_{ij} - r_{,i}r_{,j})\xi^3 e^{-\xi^2}\Bigg\} \\ = \sigma_{ij}^{lc} = p_{ij}^{dc} \end{aligned} \tag{E.23}$$

$$p^{si} = \frac{1}{\pi^{3/2}S}\frac{1}{r^3}\xi^3 e^{-\xi^2} = \frac{1}{S}\zeta^{si} = p^{lc} = \frac{1}{S}\zeta^{lc} \tag{E.24}$$

$$w_i^{si} = q_i^{sc} = \kappa p_i^{pc} = c\,\zeta_i^{pc} = w_i^{lc} = p_i^{fc} = \frac{1}{S}\zeta_i^{fc} \tag{E.25}$$

$$\begin{aligned} q_i^{si} &= \frac{2c}{\pi^{3/2}}\frac{r_{,i}}{r^4}\xi^5 e^{-\xi^2} = \kappa p_i^{pi} = c\,\zeta_i^{pi} = q_i^{lc} = w_i^{li} \\ &= p_i^{fi} = \frac{1}{S}\zeta_i^{fi} = -\frac{G}{\eta}\zeta_i^{Fi} \end{aligned} \tag{E.26}$$

$$\zeta^{si} = Sp^{si} = Sp^{lc} = \zeta^{lc} \tag{E.27}$$

E.3 Continuous Fluid Dipole

$\gamma_k = -\delta(\vec{x} - \vec{\chi})_{,k}\,\mathrm{H}(t - \tau)$

2D

$$\begin{aligned} u_{ik}^{pc} &= \frac{\eta}{8\pi G\kappa}\left[(2r_{,i}r_{,k} - \delta_{ik})\frac{1 - e^{-\xi^2}}{\xi^2} - \delta_{ik}\mathrm{E}_1(\xi^2)\right] \\ &= \frac{1}{\kappa}u_{ik}^{fc} = -\frac{\eta}{\kappa GS}w_{ik}^{fc} = \frac{1}{\kappa}w_{ik}^{Fc} \end{aligned} \tag{E.28}$$

$$\sigma_{ijk}^{pc} = \frac{\eta}{2\pi\kappa}\frac{1}{r}\left[(\delta_{ik}r_{,j} + \delta_{jk}r_{,i} + \delta_{ij}r_{,k} - 4r_{,i}r_{,j}r_{,k})\frac{1-e^{-\xi^2}}{\xi^2} + 2(r_{,i}r_{,j}r_{,k} - \delta_{ij}r_{,k})e^{-\xi^2}\right]$$

$$= \frac{1}{\kappa}\sigma_{ijk}^{fc} = \frac{1}{\kappa}w_{kji}^{dc} \tag{E.29}$$

$$p_k^{pc} = \frac{1}{\kappa}q_k^{sc} = \frac{1}{\kappa}w_k^{si} = \frac{c}{k}\zeta_k^{pc} = \frac{1}{\kappa}w_k^{lc} = \frac{1}{\kappa}p_k^{fc}$$

$$= \frac{1}{\kappa S}\zeta_k^{fc} = -\frac{G}{\eta\kappa}\zeta_k^{Fc} \tag{E.30}$$

$$w_{ik}^{pc} = \frac{1}{8\pi c}\left[(2r_{,i}r_{,k} - \delta_{ik})\frac{e^{-\xi^2}}{\xi^2} + \delta_{ik}\mathrm{E}_1(\xi^2)\right] \tag{E.31}$$

$$q_{ik}^{pc} = \frac{1}{2\pi}\frac{1}{r^2}\left[2r_{,i}r_{,k}(1+\xi^2) - \delta_{ik}\right]e^{-\xi^2} = w_{ik}^{pi} \tag{E.32}$$

$$\zeta_k^{pc} = \frac{1}{c}q_k^{sc} = \frac{1}{c}w_k^{si} = Sp_k^{pc} = \frac{1}{c}w_k^{lc} = \frac{1}{c}p_k^{fc}$$

$$= \frac{1}{\kappa}\zeta_k^{fc} = -\frac{G}{\eta c}\zeta_k^{Fc} \tag{E.33}$$

3D

$$u_{ik}^{pc} = \frac{\eta}{16\pi G\kappa}\frac{1}{r}\left\{(3r_{,i}r_{,k} - \delta_{ik})\left[\frac{\mathrm{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right] + 2(r_{,i}r_{,k} - \delta_{ik})\mathrm{erfc}(\xi)\right\}.$$

$$= \frac{1}{\kappa}u_{ik}^{fc} = -\frac{\eta}{\kappa GS}w_{ik}^{fc} = \frac{1}{\kappa}w_{ik}^{Fc} \tag{E.34}$$

$$\sigma_{ijk}^{pc} = \frac{\eta}{8\pi\kappa}\frac{1}{r^2}\left\{(\delta_{ij}r_{,k} + \delta_{ik}r_{,j} + \delta_{jk}r_{,i} - 5r_{,i}r_{,j}r_{,k}) \times\left[\frac{3\,\mathrm{erf}(\xi)}{\xi^2} - \frac{6}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi} + 2\,\mathrm{erfc}(\xi)\right] + 4(r_{,i}r_{,j}r_{,k} - \delta_{ij}r_{,k})\left[\mathrm{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right]\right\}$$

$$= \frac{1}{\kappa}\sigma_{ijk}^{fc} = \frac{1}{\kappa}w_{kji}^{dc} \tag{E.35}$$

$$p_k^{pc} = \frac{1}{\kappa} q_k^{sc} = \frac{1}{\kappa} w_k^{si} = \frac{1}{S} \zeta_k^{pc} = \frac{1}{\kappa} w_k^{lc} = \frac{1}{\kappa} p_k^{fc}$$
$$= \frac{1}{\kappa S} \zeta_k^{fc} = -\frac{G}{\eta \kappa} \zeta_k^{Fc} \tag{E.36}$$

$$w_{ik}^{pc} = \frac{1}{16\pi c} \frac{1}{r} \left\{ (3r_{,i}r_{,k} - \delta_{ik}) \left[\frac{\operatorname{erfc}(\xi)}{\xi^2} + \frac{2}{\sqrt{\pi}} \frac{e^{-\xi^2}}{\xi} \right] + 2(\delta_{ik} - r_{,i}r_{,k}) \operatorname{erfc}(\xi) \right\} \tag{E.37}$$

$$q_{ik}^{pc} = \frac{1}{4\pi} \frac{1}{r^3} \left\{ (3r_{,i}r_{,k} - \delta_{ik}) \left[\operatorname{erfc}(\xi) + \frac{2}{\sqrt{\pi}} \xi e^{-\xi^2} \right] + \frac{4}{\sqrt{\pi}} r_{,i}r_{,k} \xi^3 e^{-\xi^2} \right\}$$
$$= w_{ik}^{pi} \tag{E.38}$$

$$\zeta_k^{pc} = \frac{1}{c} q_k^{sc} = \frac{1}{c} w_k^{si} = S p_k^{pc} = \frac{1}{c} w_k^{lc} = \frac{1}{c} p_k^{fc} = \frac{1}{\kappa} \zeta_k^{fc} = -\frac{G}{\eta c} \zeta_k^{Fc} \tag{E.39}$$

E.4 Instantaneous Fluid Dipole

$\gamma_k = \delta(\vec{x} - \vec{\chi})_{,k}\, \delta(t - \tau)$
2D

$$u_{ik}^{pi} = \frac{\eta}{2\pi GS} \frac{1}{r^2} \left[(2r_{,i}r_{,k} - \delta_{ik})(1 - e^{-\xi^2}) - 2r_{,i}r_{,k}\xi^2 e^{-\xi^2} \right]$$
$$= -\frac{\eta}{\kappa GS} q_{ik}^{fc} = \frac{1}{\kappa} u_{ik}^{fi} = -\frac{\eta}{\kappa GS} w_{ik}^{fi} = \frac{1}{\kappa} q_{ik}^{Fc} = \frac{1}{\kappa} w_{ik}^{Fi} \tag{E.40}$$

$$\sigma_{ijk}^{pi} = \frac{2\eta}{\pi S} \frac{1}{r^3} \left[(\delta_{ik}r_{,j} + \delta_{jk}r_{,i} + \delta_{ij}r_{,k} - 4r_{,i}r_{,j}r_{,k})(1 - e^{-\xi^2} - \xi^2 e^{-\xi^2}) + 2(r_{,i}r_{,j}r_{,k} - \delta_{ij}r_{,k})\xi^4 e^{-\xi^2} \right]$$
$$= \frac{1}{\kappa} \sigma_{ijk}^{fi} = \frac{1}{\kappa} q_{kji}^{dc} = \frac{1}{\kappa} w_{kji}^{di} \tag{E.41}$$

$$p_k^{pi} = \frac{1}{\kappa} q_k^{si} = \frac{1}{S} \zeta_k^{pi} = \frac{1}{\kappa} q_k^{lc} = \frac{1}{\kappa} w_k^{li} = \frac{1}{\kappa} p_k^{fi} = \frac{1}{\kappa S} \zeta_k^{fi} = -\frac{G}{\eta\kappa} \zeta_k^{Fi} \tag{E.42}$$

$$w_{ik}^{pi} = q_{ik}^{pc} \tag{E.43}$$

$$q_{ik}^{pi} = \frac{2c}{\pi} \frac{1}{r^4} (2 r_{,i} r_{,k} \xi^2 - \delta_{ik}) \, \xi^4 e^{-\xi^2} \tag{E.44}$$

$$\zeta_k^{pi} = \frac{1}{c} q_k^{si} = S p_k^{pi} = \frac{1}{c} q_k^{lc} = \frac{1}{c} w_k^{li} = \frac{1}{c} p_k^{fi} = \frac{1}{\kappa} \zeta_k^{fi} = -\frac{G}{\eta c} \zeta_k^{Fi} \tag{E.45}$$

3D

$$\begin{aligned} u_{ik}^{pi} &= \frac{\eta}{4\pi GS} \frac{1}{r^3} \left\{ (3 r_{,i} r_{,k} - \delta_{ik}) \left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}} \xi e^{-\xi^2} \right] - \frac{4}{\sqrt{\pi}} r_{,i} r_{,k} \xi^3 e^{-\xi^2} \right\} \\ &= -\frac{\eta}{\kappa GS} q_{ik}^{fc} = \frac{1}{\kappa} u_{ik}^{fi} = -\frac{\eta}{\kappa GS} w_{ik}^{fi} = \frac{1}{\kappa} q_{ik}^{Fc} = \frac{1}{\kappa} w_{ik}^{Fi} \end{aligned} \tag{E.46}$$

$$\begin{aligned} \sigma_{ijk}^{pi} = \frac{\eta}{2\pi S} \frac{1}{r^4} \Bigg\{ & \frac{8}{\sqrt{\pi}} (r_{,i} r_{,j} r_{,k} - \delta_{ij} r_{,k}) \xi^5 e^{-\xi^2} \\ & + (\delta_{ij} r_{,k} + \delta_{ik} r_{,j} + \delta_{jk} r_{,i} - 5 r_{,i} r_{,j} r_{,k}) \\ & \times \left[3 \operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}} (3 + 2\xi^2) \xi e^{-\xi^2} \right] \Bigg\} \end{aligned}$$

$$= \frac{1}{\kappa} \sigma_{ijk}^{fi} \tag{E.47}$$

$$p_k^{pi} = \frac{1}{\kappa} q_k^{si} = \frac{1}{S} \zeta_k^{pi} = \frac{1}{\kappa} q_k^{lc} = \frac{1}{\kappa} w_k^{li} = \frac{1}{\kappa} p_k^{fi} = \frac{1}{\kappa S} \zeta_k^{fi} = -\frac{G}{\eta\kappa} \zeta_k^{Fi} \tag{E.48}$$

$$w_{ik}^{pi} = q_{ik}^{pc} \tag{E.49}$$

$$q_{ik}^{pi} = \frac{2c}{\pi^{3/2}} \frac{1}{r^5} (2 r_{,i} r_{,k} \xi^2 - \delta_{ik}) \xi^5 e^{-\xi^2} \tag{E.50}$$

$$\zeta_k^{pi} = \frac{1}{c} q_k^{si} = S p_k^{pi} = \frac{1}{c} q_k^{lc} = \frac{1}{c} w_k^{li} = \frac{1}{c} p_k^{fi} = \frac{1}{\kappa} \zeta_k^{fi} = -\frac{G}{\eta c} \zeta_k^{Fi} \tag{E.51}$$

E.5 Continuous Fluid Dilatation

$Q = \delta(\vec{x} - \vec{\chi}) \, \mathrm{H}(t - \tau)$
2D, 3D

$$\left(u_i^{lc}, \sigma_{ij}^{lc}, p^{lc}, w_i^{lc}, q_i^{lc}, \zeta^{lc} \right) = \left(u_i^{si}, \sigma_{ij}^{si}, p^{si}, w_i^{si}, q_i^{si}, \zeta^{si} \right) \tag{E.52}$$

E.6 Instantaneous Fluid Dilatation

$Q = \delta(\vec{x} - \vec{\chi})\,\delta(t - \tau)$

2D

$$\begin{aligned} u_i^{li} &= \frac{\eta}{2\pi GS}\frac{r_{,i}}{r}\left[\delta(t-\tau) - 4c\frac{1}{r^2}\xi^4 e^{-\xi^2}\right] \\ &= p_i^{Fi} \end{aligned} \tag{E.53}$$

$$\begin{aligned} \sigma_{ij}^{li} = \frac{\eta}{\pi S}\,\frac{1}{r^2}\Bigg\{ & (\delta_{ij} - 2r_{,i}r_{,j})\delta(t-\tau) \\ & +4c\frac{1}{r^2}\left[\delta_{ij}(1-2\xi^2) + 2r_{,i}r_{,j}\xi^2\right]\xi^4 e^{-\xi^2}\Bigg\} \\ = p_{ij}^{di} \end{aligned} \tag{E.54}$$

$$p^{li} = \frac{4c}{\pi S}\frac{1}{r^4}(\xi^2 - 1)\xi^4 e^{-\xi^2} \;=\; \frac{1}{S}\zeta^{li} \tag{E.55}$$

$$w_i^{li} = q_i^{si} \;=\; \kappa p_i^{pi} \;=\; c\,\zeta_i^{pi} \;=\; q_i^{lc} \;=\; p_i^{fi} \;=\; \frac{1}{S}\zeta_i^{fi} \;=\; -\frac{G}{\eta}\zeta_i^{Fi} \tag{E.56}$$

$$q_i^{li} = \frac{8c^2}{\pi}\frac{r_{,i}}{r^5}(\xi^2 - 2)\,\xi^6 e^{-\xi^2} \tag{E.57}$$

$$\zeta^{li} = Sp^{li} \tag{E.58}$$

3D

$$\begin{aligned} u_i^{li} &= \frac{\eta}{4\pi GS}\frac{r_{,i}}{r^2}\left[\delta(t-\tau) - \frac{8c}{\sqrt{\pi}}\frac{1}{r^2}\xi^5 e^{-\xi^2}\right] \\ &= p_i^{Fi} \end{aligned} \tag{E.59}$$

$$\begin{aligned} \sigma_{ij}^{li} = \frac{\eta}{2\pi S}\,\frac{1}{r^3}\Bigg\{ & (\delta_{ij} - 3r_{,i}r_{,j})\delta(t-\tau) \\ & +\frac{16c}{\sqrt{\pi}}\frac{1}{r^2}\left[\delta_{ij}(1-\xi^2) + r_{,i}r_{,j}\xi^2\right]\xi^5 e^{-\xi^2}\Bigg\} \\ = p_{ij}^{di} \end{aligned} \tag{E.60}$$

$$p^{li} = \frac{2c}{\pi^{3/2}S}\frac{1}{r^5}(2\xi^2 - 3)\xi^5 e^{-\xi^2} \;=\; \frac{1}{S}\zeta^{li} \tag{E.61}$$

$$w_i^{li} = q_i^{si} \;=\; \kappa p_i^{pi} \;=\; c\,\zeta_i^{pi} \;=\; q_i^{lc} \;=\; p_i^{fi} \;=\; \frac{1}{S}\zeta_i^{fi} \;=\; -\frac{G}{\eta}\zeta_i^{Fi} \tag{E.62}$$

$$q_i^{li} = \frac{4c^2}{\pi^{3/2}} \frac{r_{,i}}{r^6} (2\xi^2 - 5)\xi^7 e^{-\xi^2} \tag{E.63}$$

$$\zeta^{li} = Sp^{li} \tag{E.64}$$

E.7 Continuous Fluid Force

$f_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\mathrm{H}(t - \tau)$
2D, 3D

$$u_{ik}^{fc} = \kappa u_{ik}^{pc} = -\frac{\eta}{GS} w_{ik}^{fc} = w_{ik}^{Fc} \tag{E.65}$$

$$\sigma_{ijk}^{fc} = \kappa \sigma_{ijk}^{pc} = w_{kji}^{dc} \tag{E.66}$$

$$p_k^{fc} = q_k^{sc} = w_k^{si} = \kappa p_k^{pc} = c\,\zeta_k^{pc} = w_k^{lc} = \frac{1}{S}\zeta_k^{fc} = -\frac{G}{\eta}\zeta_k^{Fc} \tag{E.67}$$

$$w_{ik}^{fc} = -\frac{\kappa GS}{\eta} u_{ik}^{pc} = -\frac{GS}{\eta} u_{ik}^{fc} = -\frac{GS}{\eta} w_{ik}^{Fc} \tag{E.68}$$

$$q_{ik}^{fc} = -\frac{\kappa GS}{\eta} u_{ik}^{pi} = -\frac{GS}{\eta} u_{ik}^{fi} = w_{ik}^{fi} = -\frac{GS}{\eta} q_{ik}^{Fc} = -\frac{GS}{\eta} w_{ik}^{Fi} \tag{E.69}$$

$$\zeta_k^{fc} = Sq_k^{sc} = Sw_k^{si} = \kappa Sp_k^{pc} = \kappa \zeta_k^{pc} = Sw_k^{lc} = Sp_k^{fc} = -\frac{GS}{\eta}\zeta_k^{Fc} \tag{E.70}$$

E.8 Instantaneous Fluid Force

$f_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\delta(t - \tau)$
2D

$$u_{ik}^{fi} = \kappa u_{ik}^{pi} = -\frac{\eta}{GS} q_{ik}^{fc} = -\frac{\eta}{GS} w_{ik}^{fi} = q_{ik}^{Fc} = w_{ik}^{Fi} \tag{E.71}$$

$$\sigma_{ijk}^{fi} = \kappa \sigma_{ijk}^{pi} = q_{kji}^{dc} = w_{kji}^{di} \tag{E.72}$$

$$p_k^{fi} = q_k^{si} = \kappa p_k^{pi} = c\,\zeta_k^{pi} = q_k^{lc} = w_k^{li} = \frac{1}{S}\zeta_k^{fi} = -\frac{G}{\eta}\zeta_k^{Fi} \tag{E.73}$$

$$w_{ik}^{fi} = -\frac{\kappa GS}{\eta} u_{ik}^{pi} = q_{ik}^{fc} = -\frac{GS}{\eta} u_{ik}^{fi} = -\frac{GS}{\eta} q_{ik}^{Fc} = -\frac{GS}{\eta} w_{ik}^{Fi} \tag{E.74}$$

$$q_{ik}^{fi} = -\frac{\kappa}{2\pi}\frac{1}{r^2}\left[(2r_{,i}r_{,k} - \delta_{ik})\delta(t-\tau) + 4c\frac{1}{r^2}(\delta_{ik} - 2r_{,i}r_{,k}\xi^2)\,\xi^4 e^{-\xi^2}\right] \tag{E.75}$$

$$\zeta_k^{fi} = Sq_k^{si} = \kappa Sp_k^{pi} = \kappa\zeta_k^{pi} = Sq_k^{lc} = Sw_k^{li} = Sp_k^{fi} = -\frac{GS}{\eta}\zeta_k^{Fi} \tag{E.76}$$

3D

$$u_{ik}^{fi} = \kappa u_{ik}^{pi} = -\frac{\eta}{GS}q_{ik}^{fc} = -\frac{\eta}{GS}w_{ik}^{fi} = q_{ik}^{Fc} = w_{ik}^{Fi} \tag{E.77}$$

$$\sigma_{ijk}^{fi} = \kappa\sigma_{ijk}^{pi} = q_{kji}^{dc} = w_{kji}^{di} \tag{E.78}$$

$$p_k^{fi} = q_k^{si} = \kappa p_k^{pi} = c\zeta_k^{pi} = q_k^{lc} = w_k^{li} = \frac{1}{S}\zeta_k^{fi} = -\frac{G}{\eta}\zeta_k^{Fi} \tag{E.79}$$

$$w_{ik}^{fi} = -\frac{\kappa GS}{\eta}u_{ik}^{pi} = q_{ik}^{fc} = -\frac{GS}{\eta}u_{ik}^{fi} = -\frac{GS}{\eta}q_{ik}^{Fc} = -\frac{GS}{\eta}w_{ik}^{Fi} \tag{E.80}$$

$$q_{ik}^{fi} = -\frac{\kappa}{4\pi}\frac{1}{r^3}\Big[(3r_{,i}r_{,k} - \delta_{ik})\delta(t-\tau) + \frac{8c}{\sqrt{\pi}}\frac{1}{r^2}(\delta_{ik} - 2r_{,i}r_{,k}\xi^2)\xi^5 e^{-\xi^2}\Big] \tag{E.81}$$

$$\zeta_k^{fi} = Sq_k^{si} = \kappa Sp_k^{pi} = \kappa\zeta_k^{pi} = Sq_k^{lc} = Sw_k^{li} = Sp_k^{fi} = -\frac{GS}{\eta}\zeta_k^{Fi} \tag{E.82}$$

E.9 Continuous Fluid Dodecapole

$\gamma = \nabla^2\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau)$

2D

$$u_i^{oc} = -\frac{\eta}{2\pi G\kappa}\frac{r_{,i}}{r}e^{-\xi^2} \tag{E.83}$$

$$\sigma_{ij}^{oc} = -\frac{\eta}{\pi\kappa}\frac{1}{r^2}\left[\delta_{ij}(1+2\xi^2) - 2r_{,i}r_{,j}(1+\xi^2)\right]e^{-\xi^2} + \frac{2\eta}{\kappa}\delta_{ij}\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau) \tag{E.84}$$

$$p^{oc} = -\frac{1}{\pi\kappa}\frac{1}{r^2}\xi^2 e^{-\xi^2} - \frac{1}{\kappa}\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau) \tag{E.85}$$

$$w_i^{oc} = \frac{1}{2\pi c}\frac{r_{,i}}{r}e^{-\xi^2} + \frac{\partial\delta(\vec{x} - \vec{\chi})}{\partial x}(t-\tau)\,\mathrm{H}(t-\tau) \tag{E.86}$$

$$q_i^{oc} = \frac{2}{\pi}\frac{r_{,i}}{r^3}\xi^2 e^{-\xi^2} + \frac{\partial\delta(\vec{x} - \vec{\chi})}{\partial x}\mathrm{H}(t-\tau) \tag{E.87}$$

$$\zeta^{oc} = Sp^{oc} \tag{E.88}$$

3D

$$u_i^{oc} = -\frac{\eta}{4\pi G\kappa}\frac{r_{,i}}{r^2}\left[\operatorname{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\,\xi e^{-\xi^2}\right] \tag{E.89}$$

$$\sigma_{ij}^{oc} = -\frac{\eta}{2\pi\kappa}\frac{1}{r^3}\left\{\left[\delta_{ij}(1+2\xi^2) - r_{,i}r_{,j}(3+2\xi^2)\right]\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2} + (\delta_{ij} - 3r_{,i}r_{,j})\operatorname{erfc}(\xi)\right\} + \frac{2\eta}{\kappa}\delta_{ij}\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau) \tag{E.90}$$

$$p^{oc} = -\frac{1}{\pi^{3/2}\kappa}\frac{1}{r^3}\,\xi^3 e^{-\xi^2} - \frac{1}{\kappa}\,\delta(\vec{x}-\vec{\chi})\,\mathrm{H}(t-\tau) \tag{E.91}$$

$$w_i^{oc} = \frac{1}{4\pi c}\frac{r_{,i}}{r^2}\left[\frac{2}{\sqrt{\pi}}\,\xi\, e^{-\xi^2} + \operatorname{erfc}(\xi)\right] + \frac{\partial\delta(\vec{x}-\vec{\chi})}{\partial x}(t-\tau)\,\mathrm{H}(t-\tau) \tag{E.92}$$

$$q_i^{oc} = \frac{2}{\pi^{3/2}}\frac{r_{,i}}{r^4}\xi^5 e^{-\xi^2} + \frac{\partial\delta(\vec{x}-\vec{\chi})}{\partial x}\,\mathrm{H}(t-\tau) \tag{E.93}$$

$$\zeta^{oc} = Sp^{oc} \tag{E.94}$$

E.10 Continuous Total Force

$F_{ik} = \delta_{ik}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau)$

2D

$$u_{ik}^{Fc} = \frac{1}{8\pi G(1-\nu_u)}\left[r_{,i}r_{,k} - (3-4\nu_u)\delta_{ik}\ln r\right]\mathrm{H}(t-\tau) + \frac{\eta^2}{8\pi G^2 S}\left[(\delta_{ik} - 2r_{,i}r_{,k})\frac{1-e^{-\xi^2}}{\xi^2} + \delta_{ik}\mathrm{E}_1(\xi^2)\right] \tag{E.95}$$

$$\sigma_{ijk}^{Fc} = \frac{1}{4\pi(1-\nu_u)}\frac{1}{r}\left[(1-2\nu_u)(\delta_{ij}r_{,k} - \delta_{jk}r_{,i} - \delta_{ik}r_{,j}) - 2r_{,i}r_{,j}r_{,k}\right]\mathrm{H}(t-\tau) + \frac{\eta^2}{2\pi GS}\frac{1}{r}\left[(4r_{,i}r_{,j}r_{,k} - \delta_{ik}r_{,j} - \delta_{jk}r_{,i} - \delta_{ij}r_{,k})\frac{1-e^{-\xi^2}}{\xi^2} + 2(\delta_{ij}r_{,k} - r_{,i}r_{,j}r_{,k})e^{-\xi^2}\right] = u_{kji}^{dc} \tag{E.96}$$

$$p_k^{Fc} = u_k^{si} = u_k^{lc} \tag{E.97}$$

$$w_{ik}^{Fc} = \kappa u_{ik}^{pc} = u_{ik}^{fc} = -\frac{\eta}{GS}w_{ik}^{fc} \tag{E.98}$$

$$q_{ik}^{Fc} = \kappa u_{ik}^{pi} = -\frac{\eta}{GS} q_{ik}^{fc} = u_{ik}^{fi} = -\frac{\eta}{GS} w_{ik}^{fi} = w_{ik}^{Fi} \tag{E.99}$$

$$\begin{aligned}\zeta_k^{Fc} &= -\frac{\eta}{G} q_k^{sc} = -\frac{\eta}{G} w_k^{si} = -\frac{\eta\kappa}{G} p_k^{pc} = -\frac{\eta c}{G} \zeta_k^{pc} = -\frac{\eta}{G} w_k^{lc} \\ &= -\frac{\eta}{G} p_k^{fc} = -\frac{\eta}{GS} \zeta_k^{fc}\end{aligned} \tag{E.100}$$

3D

$$\begin{aligned}u_{ik}^{Fc} = {} & \frac{1}{16\pi G(1-\nu_u)} \frac{1}{r} \left[r_{,i} r_{,k} + (3 - 4\nu_u)\delta_{ik} \right] \mathrm{H}(t-\tau) \\ & + \frac{\eta^2}{16\pi G^2 S} \frac{1}{r} \left\{ (\delta_{ik} - 3 r_{,i} r_{,k}) \left[\frac{\operatorname{erf}(\xi)}{\xi^2} - \frac{2}{\sqrt{\pi}} \frac{e^{-\xi^2}}{\xi} \right] \right. \\ & \left. + 2(\delta_{ik} - r_{,i} r_{,k}) \operatorname{erfc}(\xi) \right\}\end{aligned} \tag{E.101}$$

$$\begin{aligned}\sigma_{ijk}^{Fc} = {} & \frac{1}{8\pi(1-\nu_u)} \frac{1}{r^2} \left[(1 - 2\nu_u)(\delta_{ij} r_{,k} - \delta_{jk} r_{,i} - \delta_{ik} r_{,j}) \right. \\ & \left. - 3 r_{,i} r_{,j} r_{,k} \right] \mathrm{H}(t-\tau) \\ & + \frac{\eta^2}{8\pi GS} \frac{1}{r^2} \left\{ 4(\delta_{ij} r_{,k} - r_{,i} r_{,j} r_{,k}) \left[\operatorname{erfc}(\xi) + \frac{2}{\sqrt{\pi}} \xi e^{-\xi^2} \right] \right. \\ & + (5 r_{,i} r_{,j} r_{,k} - \delta_{ij} r_{,k} - \delta_{ik} r_{,j} - \delta_{jk} r_{,i}) \\ & \left. \times \left[\frac{3\operatorname{erf}(\xi)}{\xi^2} - \frac{6}{\sqrt{\pi}} \frac{e^{-\xi^2}}{\xi} + 2\operatorname{erfc}(\xi) \right] \right\} = u_{kji}^{dc}\end{aligned} \tag{E.102}$$

$$p_k^{Fc} = u_k^{si} = u_k^{lc} \tag{E.103}$$

$$w_{ik}^{Fc} = \kappa u_{ik}^{pc} = u_{ik}^{fc} = -\frac{\eta}{GS} w_{ik}^{fc} \tag{E.104}$$

$$q_{ik}^{Fc} = \kappa u_{ik}^{pi} = -\frac{\eta}{GS} q_{ik}^{fc} = u_{ik}^{fi} = -\frac{\eta}{GS} w_{ik}^{fi} = w_{ik}^{Fi} \tag{E.105}$$

$$\begin{aligned}\zeta_k^{Fc} &= -\frac{\eta}{G} q_k^{sc} = -\frac{\eta}{G} w_k^{si} = -\frac{\eta\kappa}{G} p_k^{pc} = -\frac{\eta c}{G} \zeta_k^{pc} = -\frac{\eta}{G} w_k^{lc} \\ &= -\frac{\eta}{G} p_k^{fc} = -\frac{\eta}{GS} \zeta_k^{fc}\end{aligned} \tag{E.106}$$

E.11 Instantaneous Total Force

$F_{ik} = \delta_{ik}\delta(\vec{x} - \vec{\chi})\delta(t - \tau)$

2D

$$u_{ik}^{Fi} = \frac{1}{8\pi G(1-\nu_u)}\delta(t-\tau)\left[r_{,i}r_{,k} - (3-4\nu_u)\delta_{ik}\ln r\right] + \frac{\eta^2 c}{2\pi G^2 S}\frac{1}{r^2}\left[(\delta_{ik} - 2r_{,i}r_{,k})(1-e^{-\xi^2}) + 2r_{,i}r_{,k}\xi^2 e^{-\xi^2}\right] \tag{E.107}$$

$$\sigma_{ijk}^{Fi} = \frac{1}{4\pi(1-\nu_u)}\delta(t-\tau)\frac{1}{r}\left[(1-2\nu_u)(\delta_{ij}r_{,k} - \delta_{jk}r_{,i} - \delta_{ik}r_{,j}) - 2r_{,i}r_{,j}r_{,k}\right] + \frac{2\eta^2 c}{\pi GS}\frac{1}{r^3}\left[(4r_{,i}r_{,j}r_{,k} - \delta_{ik}r_{,j} - \delta_{jk}r_{,i} - \delta_{ij}r_{,k}) \times(1 - e^{-\xi^2} - \xi^2 e^{-\xi^2}) - 2(r_{,i}r_{,j}r_{,k} - \delta_{ij}r_{,k})\xi^4 e^{-\xi^2}\right] = u_{kji}^{di} \tag{E.108}$$

$$p_k^{Fi} = u_k^{li} \tag{E.109}$$

$$w_{ik}^{Fi} = \kappa u_{ik}^{pi} = -\frac{\eta}{GS}q_{ik}^{fc} = u_{ik}^{fi} = -\frac{\eta}{GS}w_{ik}^{fi} = q_{ik}^{Fc} \tag{E.110}$$

$$q_{ik}^{Fi} = \frac{\eta c}{2\pi G}\frac{1}{r^2}\left[(2r_{,i}r_{,k} - \delta_{ik})\delta(t-\tau) + 4c\frac{1}{r^2}(\delta_{ik} - 2r_{,i}r_{,k}\xi^2)\xi^4 e^{-\xi^2}\right] \tag{E.111}$$

$$\zeta_k^{Fi} = -\frac{\eta}{G}q_k^{si} = -\frac{\eta\kappa}{G}p_k^{pi} = -\frac{\eta c}{G}\zeta_k^{pi} = -\frac{\eta}{G}q_k^{lc} = -\frac{\eta}{G}w_k^{li} = -\frac{\eta}{G}p_k^{fi} = -\frac{\eta}{GS}\zeta_k^{fi} \tag{E.112}$$

3D

$$u_{ik}^{Fi} = \frac{1}{16\pi G(1-\nu_u)}\delta(t-\tau)\frac{1}{r}\left[r_{,i}r_{,k} + (3-4\nu_u)\delta_{ik}\right] + \frac{\eta^2 c}{4\pi G^2 S}\frac{1}{r^3}\left\{(\delta_{ik} - 3r_{,i}r_{,k})\left[\operatorname{erf}(\xi) - \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right] + \frac{4}{\sqrt{\pi}}r_{,i}r_{,k}\xi^3 e^{-\xi^2}\right\} \tag{E.113}$$

$$\sigma_{ijk}^{Fi} = \frac{1}{8\pi(1-\nu_u)}\delta(t-\tau)\frac{1}{r^2}\left[(1-2\nu_u)(\delta_{ij}r_{,k}-\delta_{jk}r_{,i}-\delta_{ik}r_{,j})\right.$$
$$\left.-3r_{,i}r_{,j}r_{,k}\right]+\frac{\eta^2 c}{2\pi GS}\frac{1}{r^4}\left\{\frac{8}{\sqrt{\pi}}(\delta_{ij}r_{,k}-r_{,i}r_{,j}r_{,k})\xi^5 e^{-\xi^2}\right.$$
$$-(\delta_{ij}r_{,k}+\delta_{ik}r_{,j}+\delta_{jk}r_{,i}-5r_{,i}r_{,j}r_{,k})$$
$$\left.\left[3\,\mathrm{erf}(\xi)-\frac{2}{\sqrt{\pi}}(3+2\xi^2)\xi e^{-\xi^2}\right]\right\}$$
$$= u_{kji}^{di} \tag{E.114}$$

$$p_k^{Fi} = u_k^{li} \tag{E.115}$$

$$w_{ik}^{Fi} = \kappa u_{ik}^{pi} = -\frac{\eta}{GS}q_{ik}^{fc} = u_{ik}^{fi} = -\frac{\eta}{GS}w_{ik}^{fi} = q_{ik}^{Fc} \tag{E.116}$$

$$q_{ik}^{Fi} = \frac{\eta c}{4\pi G}\frac{1}{r^3}\left[(3r_{,i}r_{,k}-\delta_{ik})\delta(t-\tau)\right.$$
$$\left.+\frac{8c}{\sqrt{\pi}}\frac{1}{r^2}(\delta_{ik}-2r_{,i}r_{,k}\xi^2)\xi^5 e^{-\xi^2}\right] \tag{E.117}$$

$$\zeta_k^{Fi} = -\frac{\eta}{G}q_k^{si} = -\frac{\eta\kappa}{G}p_k^{pi} = -\frac{\eta c}{G}\zeta_k^{pi} = -\frac{\eta}{G}q_k^{lc} = -\frac{\eta}{G}w_k^{li}$$
$$= -\frac{\eta}{G}p_k^{fi} = -\frac{\eta}{GS}\zeta_k^{fi} \tag{E.118}$$

E.12 Continuous Solid Quadrupole and Hexapole

$F_{ik,k} = \delta_{,i}\mathrm{H}(t-\tau)$
2D

$$u_i^{qc} = -\frac{1-2\nu_u}{4\pi G(1-\nu_u)}\frac{r_{,i}}{r}\mathrm{H}(t-\tau)-\frac{\eta^2}{2\pi G^2 S}\frac{r_{,i}}{r}e^{-\xi^2} \tag{E.119}$$

$$\sigma_{ij}^{qc} = -\frac{1-2\nu_u}{2\pi(1-\nu_u)}\frac{1}{r^2}(\delta_{ij}-2r_{,i}r_{,j})-\frac{\nu}{1-\nu}\delta_{ij}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau)$$
$$-\frac{\eta^2}{\pi GS}\frac{1}{r^2}\left[\delta_{ij}(1+2\xi^2)-2r_{,i}r_{,j}(1+\xi^2)\right]e^{-\xi^2} \tag{E.120}$$

$$p^{qc} = -\frac{\eta}{\pi GS}\frac{1}{r^2}\xi^2 e^{-\xi^2} \tag{E.121}$$

$$w_i^{qc} = \frac{\eta}{2\pi G}\frac{r_{,i}}{r}e^{-\xi^2} \tag{E.122}$$

$$q_i^{qc} = \frac{\eta c}{\pi G}\frac{r_{,i}}{r^2}\xi^2 e^{-\xi^2} \tag{E.123}$$

$$\zeta^{qc} = -\frac{\eta}{\pi G}\frac{1}{r^2}\xi^2 e^{-\xi^2}-\frac{\eta}{G}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) \tag{E.124}$$

3D

$$u_i^{qc} = -\frac{1-2\nu_u}{8\pi G(1-\nu_u)}\frac{r_{,i}}{r^2}\mathrm{H}(t-\tau) - \frac{\eta^2}{4\pi G^2 S}\frac{r_{,i}}{r^2}\left[\mathrm{erfc}(\xi) + \frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right] \tag{E.125}$$

$$\sigma_{ij}^{qc} = -\frac{1-2\nu_u}{4\pi(1-\nu_u)}\frac{1}{r^3}(\delta_{ij} - 3r_{,i}r_{,j}) - \frac{\nu}{1-\nu}\delta_{ij}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) - \frac{\eta^2}{2\pi GS}\frac{1}{r^3}\left\{\left[\delta_{ij}(1+2\xi^2) - r_{,i}r_{,j}(3+2\xi^2)\right]\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2} + (\delta_{ij} - 3r_{,i}r_{,j})\,\mathrm{erfc}(\xi)\right\} \tag{E.126}$$

$$p^{qc} = -\frac{\eta}{\pi^{3/2}GS}\frac{1}{r^3}\xi^3 e^{-\xi^2} \tag{E.127}$$

$$w_i^{qc} = \frac{\eta}{4\pi G}\frac{r_{,i}}{r^2}\left[\frac{2}{\sqrt{\pi}}\xi\, e^{-\xi^2} + \mathrm{erfc}(\xi)\right] \tag{E.128}$$

$$q_i^{qc} = \frac{\eta c}{\pi^{3/2}G}\frac{r_{,i}}{r^3}\xi^3 e^{-\xi^2} \tag{E.129}$$

$$\zeta^{qc} = -\frac{\eta}{\pi^{3/2}G}\frac{1}{r^3}\xi^3 e^{-\xi^2} - \frac{\eta}{G}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) \tag{E.130}$$

E.13 Continuous Solid Dilatation

$E_{ij} = (\ln r/2\pi)_{,ij}\mathrm{H}(t-\tau)$
2D

$$u_i^{cc} = \frac{1}{2\pi}\frac{r_{,i}}{r}\mathrm{H}(t-\tau) \tag{E.131}$$

$$\sigma_{ij}^{cc} = \frac{G}{\pi}\frac{1}{r^2}(\delta_{ij} - 2r_{,i}r_{,j}) + \frac{2G\nu_u}{1-2\nu_u}\delta_{ij}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) \tag{E.132}$$

$$p^{cc} = w_i^{cc} = q_i^{cc} = \zeta^{cc} = 0 \tag{E.133}$$

$E_{ij} = -(1/4\pi r)_{,ij}\mathrm{H}(t-\tau)$
3D

$$u_i^{cc} = \frac{1}{4\pi}\frac{r_{,i}}{r^2}\mathrm{H}(t-\tau) \tag{E.134}$$

$$\sigma_{ij}^{cc} = \frac{G}{2\pi}\frac{1}{r^3}(\delta_{ij} - 3r_{,i}r_{,j}) + \frac{2G\nu_u}{1-2\nu_u}\delta_{ij}\delta(\vec{x}-\vec{\chi})\mathrm{H}(t-\tau) \tag{E.135}$$

$$p^{cc} = w_i^{cc} = q_i^{cc} = \zeta^{cc} = 0 \tag{E.136}$$

E.14 Continuous Displacement Discontinuity

$E_{ijkl} = -\frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\delta(\vec{x} - \vec{\chi})\,\mathrm{H}(t-\tau)$

2D

$$u^{dc}_{ikl} = \sigma^{Fc}_{lki} \tag{E.137}$$

$$\begin{aligned}\sigma^{dc}_{ijkl} = {} & \frac{G}{2\pi\,(1-\nu_u)}\frac{1}{r^2}\mathrm{H}(t-\tau)\left[8r_{,i}r_{,j}r_{,k}r_{,l} - (1-2\nu_u)\right.\\ & \times\left(\delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl} - \delta_{ij}\delta_{kl} + 2\delta_{kl}r_{,i}r_{,j} + 2\delta_{ij}r_{,k}r_{,l}\right)\\ & \left.-2\nu_u\left(\delta_{ij}\delta_{kl} + \delta_{lj}r_{,i}r_{,k} + \delta_{il}r_{,j}r_{,k} + \delta_{jk}r_{,i}r_{,l} + \delta_{ik}r_{,j}r_{,l}\right)\right]\\ & + \frac{\eta^2}{\pi S}\frac{1}{r^2}\Bigg[\left(4\delta_{kj}r_{,i}r_{,l} + 4\delta_{ki}r_{,l}r_{,j} + 4\delta_{kl}r_{,i}r_{,j} + 4\delta_{lj}r_{,i}r_{,k} + 4\delta_{li}r_{,k}r_{,j}\right.\\ & + 4\delta_{ij}r_{,k}r_{,l} - \delta_{ij}\delta_{kl} - \delta_{kj}\delta_{il} - \delta_{ki}\delta_{jl} - 24r_{,i}r_{,j}r_{,k}r_{,l}\Big)\frac{1-e^{-\xi^2}}{\xi^2}\\ & + 2\left(2\delta_{ij}\delta_{kl} - 3\delta_{kl}r_{,i}r_{,j} - \delta_{lj}r_{,i}r_{,k} - \delta_{il}r_{,k}r_{,j}\right.\\ & \left.\left.-\delta_{kj}r_{,i}r_{,l} - \delta_{ki}r_{,l}r_{,j} - 3\delta_{ij}r_{,l}r_{,k} + 8r_{,i}r_{,j}r_{,k}r_{,l}\right)e^{-\xi^2}\right.\\ & \left.+ 4\left(\delta_{ij}\delta_{kl} - \delta_{kl}r_{,i}r_{,j} - \delta_{ij}r_{,l}r_{,k} + r_{,i}r_{,j}r_{,k}r_{,l}\right)\xi^2 e^{-\xi^2}\right]\end{aligned} \tag{E.138}$$

$$p^{dc}_{kl} = \sigma^{si}_{kl} = \sigma^{lc}_{kl} \tag{E.139}$$

$$w^{dc}_{ikl} = \kappa\sigma^{pc}_{lki} = \sigma^{fc}_{lki} \tag{E.140}$$

$$q^{dc}_{ikl} = \kappa\sigma^{pi}_{lki} = \sigma^{fi}_{lki} = w^{di}_{ikl} \tag{E.141}$$

$$\zeta^{dc}_{kl} = \frac{\eta}{\pi}\frac{1}{r^2}\left[2r_{,k}r_{,l} - \delta_{kl} - 2(\delta_{kl} - r_{,k}r_{,l})\xi^2\right]e^{-\xi^2} \tag{E.142}$$

3D

$$u^{dc}_{ikl} = \sigma^{Fc}_{lki} \tag{E.143}$$

$$\begin{aligned}\sigma^{dc}_{ijkl} = {} & \frac{G}{4\pi\,(1-\nu_u)}\frac{1}{r^3}\mathrm{H}(t-\tau)\left[15r_{,i}r_{,j}r_{,k}r_{,l} - (1-2\nu_u)\right.\\ & \times\left(\delta_{kj}\delta_{il} + \delta_{ki}\delta_{jl} - \delta_{ij}\delta_{kl} + 3\delta_{kl}r_{,i}r_{,j} + 3\delta_{ij}r_{,k}r_{,l}\right)\\ & \left.-\nu_u\left(2\delta_{ij}\delta_{kl} + 3\delta_{lj}r_{,i}r_{,k} + 3\delta_{il}r_{,k}r_{,j} + 3\delta_{kj}r_{,i}r_{,l} + 3\delta_{ki}r_{,l}r_{,j}\right)\right]\\ & + \frac{\eta^2}{4\pi S}\frac{1}{r^3}\left[\frac{8}{\sqrt{\pi}}\left(-2\delta_{ij}\delta_{kl} + 4\delta_{kl}r_{,i}r_{,j} + 4\delta_{ij}r_{,k}r_{,l} + \delta_{lj}r_{,i}r_{,k}\right.\right.\\ & \left.\left.+\delta_{il}r_{,k}r_{,j} + \delta_{kj}r_{,i}r_{,l} + \delta_{ki}r_{,l}r_{,j} - 10r_{,i}r_{,j}r_{,k}r_{,l}\right)\xi e^{-\xi^2}\right.\end{aligned}$$

$$
\begin{aligned}
&+\frac{16}{\sqrt{\pi}}\left(-\delta_{ij}\delta_{kl}+\delta_{kl}r_{,i}r_{,j}+\delta_{ij}r_{,k}r_{,l}-r_{,i}r_{,j}r_{,k}r_{,l}\right)\xi^3 e^{-\xi^2}\\
&+3\left(\delta_{lj}\delta_{ki}+\delta_{kj}\delta_{il}+\delta_{ji}\delta_{kl}-5\delta_{lj}r_{,i}r_{,k}-5\delta_{il}r_{,k}r_{,j}-5\delta_{kj}r_{,i}r_{,l}\right.\\
&\left.-5\delta_{ki}r_{,l}r_{,j}-5\delta_{ij}r_{,k}r_{,l}-5\delta_{kl}r_{,i}r_{,j}+35r_{,i}r_{,j}r_{,k}r_{,l}\right)\\
&\times\left(\frac{\operatorname{erf}(\xi)}{\xi^2}-\frac{2}{\sqrt{\pi}}\frac{e^{-\xi^2}}{\xi}\right)\\
&+2\left(\delta_{kj}\delta_{il}+\delta_{ki}\delta_{jl}-3\delta_{ij}\delta_{kl}+3\delta_{kl}r_{,i}r_{,j}+3\delta_{ij}r_{,k}r_{,l}\right.\\
&-3\delta_{il}r_{,k}r_{,j}-3\delta_{kj}r_{,i}r_{,l}-3\delta_{ki}r_{,l}r_{,j}-3\delta_{lj}r_{,i}r_{,k}\\
&\left.+15r_{,i}r_{,j}r_{,k}r_{,l}\right)\operatorname{erfc}(\xi)\Bigg]
\end{aligned}
\tag{E.144}
$$

$$
p^{dc}_{kl}=\sigma^{si}_{kl}\;=\;\sigma^{lc}_{kl} \tag{E.145}
$$

$$
w^{dc}_{ikl}=\kappa\sigma^{pc}_{lki}\;=\;\sigma^{fc}_{lki} \tag{E.146}
$$

$$
q^{dc}_{ikl}=\kappa\sigma^{pi}_{lki}\;=\;\sigma^{fi}_{lki}\;=\;w^{di}_{ikl} \tag{E.147}
$$

$$
\begin{aligned}
\zeta^{dc}_{kl}=\frac{\eta}{2\pi}\frac{1}{r^3}\Bigg\{&(3r_{,k}r_{,l}-\delta_{kl})\left[\operatorname{erfc}(\xi)+\frac{2}{\sqrt{\pi}}\xi e^{-\xi^2}\right]\\
&-\frac{4}{\sqrt{\pi}}(\delta_{kl}-r_{,k}r_{,l})\,\xi^3 e^{-\xi^2}\Bigg\}
\end{aligned}
\tag{E.148}
$$

E.15 Instantaneous Displacement Discontinuity

$E_{ijkl}=-\frac{1}{2}(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})\delta(\vec{x}-\vec{\chi})\delta(t-\tau)$
2D

$$
u^{di}_{ikl}=\sigma^{Fi}_{lki} \tag{E.149}
$$

$$
\begin{aligned}
\sigma^{di}_{ijkl}=&\frac{G}{2\pi(1-\nu_u)}\delta(t-\tau)\frac{1}{r^2}\left[8r_{,i}r_{,j}r_{,k}r_{,l}\right.\\
&-(1-2\nu_u)\left(\delta_{il}\delta_{jk}+\delta_{ik}\delta_{jl}-\delta_{ij}\delta_{kl}+2\delta_{kl}r_{,i}r_{,j}+2\delta_{ij}r_{,k}r_{,l}\right)\\
&\left.-2\nu_u\left(\delta_{ij}\delta_{kl}+\delta_{lj}r_{,i}r_{,k}+\delta_{il}r_{,j}r_{,k}+\delta_{jk}r_{,i}r_{,l}+\delta_{ik}r_{,j}r_{,l}\right)\right]\\
&+\frac{4\eta^2 c}{\pi S}\frac{1}{r^4}\Bigg\{\left(4\delta_{kj}r_{,i}r_{,l}+4\delta_{ki}r_{,l}r_{,j}+4\delta_{kl}r_{,i}r_{,j}+4\delta_{lj}r_{,i}r_{,k}\right.\\
&+4\delta_{li}r_{,k}r_{,j}+4\delta_{ij}r_{,k}r_{,l}-\delta_{ij}\delta_{kl}-\delta_{kj}\delta_{il}-\delta_{ki}\delta_{jl}\\
&\left.-24r_{,i}r_{,j}r_{,k}r_{,l}\right)\left[1-(1+\xi^2)e^{-\xi^2}\right]
\end{aligned}
$$

$$+2\left(6r_{,i}r_{,j}r_{,k}r_{,l}-\delta_{kl}r_{,i}r_{,j}-\delta_{lj}r_{,i}r_{,k}-\delta_{li}r_{,k}r_{,j}-\delta_{jk}r_{,i}r_{,l}\right.$$
$$-\delta_{ki}r_{,l}r_{,j}-\delta_{ij}r_{,k}r_{,l}\left)\xi^4e^{-\xi^2}\right.$$
$$\left.+4\left(r_{,i}r_{,j}r_{,k}r_{,l}+\delta_{ij}\delta_{kl}-\delta_{ij}r_{,k}r_{,l}-\delta_{kl}r_{,i}r_{,j}\right)\xi^6e^{-\xi^2}\right\} \tag{E.150}$$

$$p^{di}_{kl}=\sigma^{li}_{kl} \tag{E.151}$$

$$w^{di}_{ikl}=\kappa\sigma^{pi}_{lki}\;=\;\sigma^{fi}_{lki}\;=\;q^{dc}_{ikl} \tag{E.152}$$

$$q^{di}_{ikl}=\frac{2\eta c}{\pi}\delta(t-\tau)\frac{1}{r^3}(\delta_{il}r_{,k}+\delta_{kl}r_{,i}+\delta_{ik}r_{,l}-4r_{,i}r_{,k}r_{,l})$$
$$-\frac{8\eta c^2}{\pi}\frac{1}{r^5}\left[(\delta_{il}r_{,k}+\delta_{ik}r_{,l}-3\delta_{kl}r_{,i})\right.$$
$$\left.+2(\delta_{kl}r_{,i}-r_{,i}r_{,k}r_{,l})\xi^2\right]\xi^6e^{-\xi^2} \tag{E.153}$$

$$\zeta^{di}_{kl}=\frac{4\eta c}{\pi}\frac{1}{r^4}\left[\delta_{kl}(1-2\xi^2)+2r_{,k}r_{,l}\xi^2\right]\xi^4e^{-\xi^2} \tag{E.154}$$

3D

$$u^{di}_{ikl}=\sigma^{Fi}_{lki} \tag{E.155}$$

$$\sigma^{di}_{ijkl}=\frac{G}{4\pi\,(1-\nu_u)}\frac{1}{r^3}\delta(t-\tau)\left[15r_{,i}r_{,j}r_{,k}r_{,l}\right.$$
$$-(1-2\nu_u)\left(\delta_{kj}\delta_{il}+\delta_{ki}\delta_{jl}-\delta_{ij}\delta_{kl}+3\delta_{kl}r_{,i}r_{,j}+3\delta_{ij}r_{,k}r_{,l}\right)$$
$$\left.-2\nu_u\left(\delta_{ij}\delta_{kl}+\delta_{lj}r_{,i}r_{,k}+\delta_{il}r_{,j}r_{,k}+\delta_{jk}r_{,i}r_{,l}+\delta_{ik}r_{,j}r_{,l}\right)\right]$$
$$+\frac{\eta^2c}{\pi S}\frac{1}{r^5}\left\{8\left(7r_{,i}r_{,j}r_{,k}r_{,l}-\delta_{ij}\delta_{kl}-\delta_{kl}r_{,i}r_{,j}-\delta_{ij}r_{,k}r_{,l}\right.\right.$$
$$-\delta_{lj}r_{,i}r_{,k}-\delta_{il}r_{,k}r_{,j}-\delta_{kj}r_{,i}r_{,l}-\delta_{ki}r_{,l}r_{,j}\left)\xi^5e^{-\xi^2}\right.$$
$$+16\left(r_{,i}r_{,j}r_{,k}r_{,l}+\delta_{ij}\delta_{kl}-\delta_{kl}r_{,i}r_{,j}-\delta_{ij}r_{,k}r_{,l}\right)\xi^7e^{-\xi^2}$$
$$-\left(\delta_{lj}\delta_{ki}+\delta_{kj}\delta_{il}+\delta_{ji}\delta_{kl}-5\delta_{lj}r_{,i}r_{,k}-5\delta_{il}r_{,k}r_{,j}-5\delta_{kj}r_{,i}r_{,l}\right.$$
$$\left.-5\delta_{ki}r_{,l}r_{,j}-5\delta_{ij}r_{,k}r_{,l}-5\delta_{kl}r_{,i}r_{,j}+35r_{,i}r_{,j}r_{,k}r_{,l}\right)$$
$$\left.\times\left[3\,\mathrm{erf}(\xi)-\frac{2}{\sqrt{\pi}}(3+2\xi^2)\xi e^{-\xi^2}\right]\right\} \tag{E.156}$$

$$p^{di}_{kl}=\sigma^{li}_{kl} \tag{E.157}$$

$$w^{di}_{ikl}=\kappa\sigma^{pi}_{lki}\;=\;\sigma^{fi}_{lki}\;=\;q^{dc}_{ikl} \tag{E.158}$$

$$q_{ikl}^{di} = \frac{3\eta c}{2\pi}\delta(t-\tau)\frac{1}{r^4}(\delta_{il}r_{,k} + \delta_{kl}r_{,i} + \delta_{ik}r_{,l} - 5r_{,i}r_{,k}r_{,l})$$
$$-\frac{8\eta c^2}{\pi^{3/2}}\frac{1}{r^6}\left[(\delta_{il}r_{,k} + \delta_{ik}r_{,l} - 4\delta_{kl}r_{,i})\right.$$
$$\left.+2(\delta_{kl}r_{,i} - r_{,i}r_{,k}r_{,l})\xi^2\right]\xi^7 e^{-\xi^2} \tag{E.159}$$

$$\zeta_{kl}^{di} = \frac{8\eta c}{\pi^{3/2}}\frac{1}{r^5}\left[\delta_{kl}(1-\xi^2) + r_{,k}r_{,l}\xi^2\right]\xi^5 e^{-\xi^2} \tag{E.160}$$

Appendix F
Biography

F.1 Leonardo da Vinci (1452–1519)

Leonardo da Vinci, the scientist, inventor, and artist, was born an illegitimate son of a notary and a peasant girl, at a small town Vinci, in Tuscany, Italy. At age 14, he apprenticed with a famous artist in Florence to be a painter. In 1482, he entered into the service of the Duke of Milan. During the 17 years in Milan, Leonardo not only did painting and sculpting, but also designed churches, fortresses, canals, flying machines, tanks, various war devices, and submarines.

Unlike many of his contemporaries, da Vinci took the fresh approach in science by actually observing nature and asking deceptively simple scientific questions like: "How do birds fly?" "What causes cyclones?" As his curiosity took him in ever wilder directions, da Vinci always used this method of scientific inquiry: close observation, repeated testing of the observation, and precise illustration of the subject or phenomenon with brief explanatory notes. The result was volumes of remarkable notes on an amazing variety of topics, from the nature of the sun, moon and stars to the formation of fossils. During his lifetime, the range of topics of his inquiry included anatomy, zoology, botany, geology, optics, aerodynamics, and hydrodynamics, among others.

Between 1490 and 1495 da Vinci developed his habit of recording his studies in meticulously illustrated notebooks. These studies and sketches were collected into various codices and manuscripts, including the *Codex Leicester* [41], in which the text was written in its mirror image. In this codex, he presented his view that the Earth is a circulatory system like human body:

> The Earth is a living body. Its soul is its ability to grow. This soul, which also provides the Earth with its bodily warmth, is located in the inner fires of the Earth, which emerge at several places as baths, sulfur mines or volcanoes. Its flesh is the soil, its bones are the strata of rock, its cartilage is the tufa, its blood is the underground streams, the reservoir of blood around its heart is the ocean, the systole and diastole of the blood in the arteries and veins appear on the Earth as the rising and sinking of the oceans.

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

Other ideas and observations relating to porous medium include, on soil foundation settlement: [42, 43]

> As to the foundations of the component parts of temples and other public buildings, the depths of the foundations must bear the same proportions to each other as the weight of material which is to be placed upon them.
>
> Every part of the depth of Earth in a given space is composed of layers, and each layer is composed of heavier or lighter materials, the lowest being the heaviest. And this can be proved, because these layers have been formed by the sediment from water carried down to the sea, by the current of rivers which flow into it. The heaviest part of this sediment was that which was first thrown down, and so on by degrees; and this is the action of water when it becomes stagnant, having first brought down the mud whence it first flowed. And such layers of soil are seen in the banks of rivers, where their constant flow has cut through them and divided one slope from the other to a great depth; where in gravelly strata the waters have run off, the materials have, in consequence, dried and been converted into hard stone, and this happened most in what was the finest mud; whence we conclude that every portion of the surface of the Earth was once at the center of the Earth, and viceversa &c.
>
> The heaviest part of the foundations of buildings settles most, and leaves the lighter part above it separated from it.
>
> And the soil which is most pressed, if it be porous yields most.

on slope stability and landslide:

> Parallel fissures constantly occur in buildings which are erected on a hill side, when the hill is composed of stratified rocks with an oblique stratification, because water and other moisture often penetrates these oblique seams carrying in greasy and slippery soil; and as the strata are not continuous down to the bottom of the valley, the rocks slide in the direction of the slope, and the motion does not cease till they have reached the bottom of the valley, carrying with them, as though in a boat, that portion of the building which is separated by them from the rest. The remedy for this is always to build thick piers under the wall which is slipping, with arches from one to another, and with a good scarp and let the piers have a firm foundation in the strata so that they may not break away from them.
>
> In order to find the solid part of these strata, it is necessary to make a shaft at the foot of the wall of great depth through the strata; and in this shaft, on the side from which the hill slopes, smooth and flatten a space one palm wide from the top to the bottom; and after some time this smooth portion made on the side of the shaft, will show plainly which part of the hill is moving.

and on porous bone and sound transmission in inner ear:

> In the same way as to the sense of hearing, it would have sufficed if the voice had merely sounded in the porous cavity of the indurated portion of the temporal bone which lies within the ear, without making any farther transit from this bone to the common sense, where the voice confers with and discourses to the common judgment.

F.2 Robert Hooke (1635–1703)

Robert Hooke was perhaps the single greatest experimental scientist of the seventeenth century. His interests spanned from physics and astronomy, to chemistry, biology, and geology, to architecture and naval technology. He collaborated or

competed with scientists as diverse as Robert Boyle (1627–1691), Christian Huygens (1629–1695), Antonie van Leeuwenhoek (1632–1723), Christopher Wren (1632–1723), and Isaac Newton (1643–1727).

Robert Hooke was born on 18 July 1635 in the small town of Freshwater, on the Isle of Wight, England [62]. His father, John Hooke, was the curator of the church of Freshwater. Robert Hooke was born in poor health, and was expected to die; but he hung on and gradually grew stronger. His father hoped that he might train Robert for a career in the church, as he did for Robert's three brothers. However, study gave Robert headaches; hence John gave up on his son's education. Left to follow his own inclination, Robert developed an aptitude for building mechanical devices. In fact, throughout his life, Robert had developed the ability to master other people's skills without formal teaching. For example, at young age, he watched John Hoskins, a famous painter of miniatures, at work, and decided that he could do it himself. Later, his skill with drawing became an invaluable tool, when he needed to communicate his mechanical and microscopical discoveries, as well as in his career as an architect and surveyor.

After his father passed away, Robert went to London, at age of 13. He first entered the Westminster School, and from there went to Oxford, where some of the best scientists in England were working at the time. Hooke impressed them with his skills at designing experiments and building equipment, and soon became an assistant to Robert Boyle in his study the physics of gases. In 1662 Hooke was named Curator of Experiments of the newly formed Royal Society of London—meaning that he was responsible for demonstrating new experiments at the Society's weekly meetings. He later became Gresham Professor of Geometry at Gresham College, London. He died in London on March 3, 1703.

Among other accomplishments, Hooke invented the universal joint, the iris diaphragm, an early prototype of the respirator; invented or improved meteorological instruments such as the barometer, anemometer, and hygrometer; and so on. He invented the anchor escapement and the balance spring, which made accurate clocks possible, and fought with Christiaan Huygens on the priority and patent of spring driven watches. He also disputed with Isaac Newton on the priority claim of the inverse square law of gravitation force, as well as the nature of light (suggesting that it is a wave, rather than particle, as Newton proposed). He served as Chief Surveyor and worked with Christopher Wren to rebuild London after the Great Fire of 1666.

In *De Potentia Restitutivâ, or of Spring* [61], Hooke wrote:

> Take a wire string of 20, or 30, or 40 ft long, and fasten the upper part thereof to a nail, and to the other end fasten a Scale to receive the weights: Then with a pair of Compasses take the distance of the bottom of the scale from the ground or floor underneath, and set down the said distance, then put in weights into the scale and measure the several stretchings of the said string, and set them down. Then compare the several stretchings of the said string, and you will find that they will always bear the same proportions one to the other that the weights do that made them.

Hooke also described experiments with helical springs, with watch springs coiled into spirals, and also with

> a piece of dry wood that will bend and return, if one end thereof be fixed in a horizontal posture, and to the other end be hanged weights to make it bend downwards.

This suggested linear relation between the force and the deformation is the so-called *Hooke's Law* [80].

F.3 Jean Baptiste Joseph Fourier (1768–1830)

Joseph Fourier was born in Auxerre, France, the ninth of the twelve children of his father's second marriage. One of his letters showed that he really wanted to make a major impact in mathematics: "*Yesterday was my 21st birthday; at that age Newton and Pascal had already acquired many claims to immortality.*" In 1790 Fourier became a teacher at the Benedictine College, where he had studied earlier. Soon after, he was entangled in the French Revolution and joined the local revolutionary committee. He was arrested in 1794, and almost went to the guillotine. Only the political changes resulted in his being released. In 1794 Fourier was admitted to the newly established Ecole Normale in Paris, where he was taught by Joseph-Louis Lagrange (1736–1813), Pierre-Simon Laplace (1749–1827), and Gaspard Monge (1746–1818). In 1797 he succeeded Lagrange in being appointed to the Chair of Analysis and Mechanics.

In 1798, Fourier joined Napoleon's army in its invasion of Egypt as a scientific advisor. It was there that he recorded many observations that later led to his work in heat diffusion. Fourier returned to Paris in 1801. Soon Napoleon appointed him as the Prefect of Isére, headquartered at Grenoble. Among his achievements in this administrative position included the draining of swamps of Bourgoin and the construction of a new highway between Grenoble and Turin. Some of his most important scientific contributions came during this period (1802–1814). In 1807 he completed his memoir *On the Propagation of Heat in Solid Bodies* in which he not only expounded his idea about heat conduction, which was later called Fourier's Law, but also outlined his new method of mathematical analysis, which we now call Fourier analysis. This memoir however was never published, because one of its examiner, Lagrange, objected to his use of trigonometric series to express initial temperature.

Fourier was elected to the Académie des Sciences in 1817. In 1822 he published *The Analytical Theory of Heat* [47], ten years after its winning the Institut de France competition of the Grand Prize in Mathematics in 1812. The judges however criticized that he had not proven the completeness of the trigonometric (Fourier) series. The proof will come years later by *Johann Peter Gustav Lejeune Dirichlet* (1805–1859).

F.4 Thomas Young (1773–1829)

Thomas Young was born to a Quaker family, as the eldest of ten children. At the age of fourteen, Young not only had learned Greek and Latin, but also was acquainted with a dozen modern languages. From 1787–1792, he earned his living by working as tutor for a rich family. In 1792, Young began to study medicine, first in London, and then in Edinburgh. In 1796, he obtained a Doctor degree in medicine from Göttingen. He started to practice as a physician in London in 1799. Young published many of his first academic articles anonymously to protect his reputation as a physician.

In 1801 Young was appointed professor of natural philosophy at the Royal Institution. In two years he delivered 91 lectures. These lectures were published in 1807 [87], which covered a broad range of subjects and contained a number of anticipations of later theories, such as Maxwell's equation and Einstein's special relativity. In 1802, he was appointed foreign secretary of the Royal Society, of which he had been elected a fellow in 1794. He resigned his professorship in 1803, fearing that its duties would interfere with his medical practice. Thomas Young died in London on 10 May 1829 [72].

Like Leonardo da Vinci and Robert Hooke, Thomas Young was a renaissance man, whose expertise spans a significant number of different subject areas [74]. He had made notable scientific contributions to the fields of vision, light, solid mechanics, energy, physiology, language, musical harmony and Egyptology. In Young's own judgment, of his many achievements the most important was to establish the wave theory of light, contradicting the particle theory of Newton. Through theory and experiment, he had investigated the reflection, refraction, and interference of visible light, and measured its wavelengths.

Young has also been called the founder of physiological optics. In his lectures [87] he presented the hypothesis that color perception depends on the presence in the retina of three kinds of nerve fibers which respond respectively to red, green and violet light. Young developed the theory of capillary phenomena on the principle of surface tension, which was further developed by Laplace to become the Young-Laplace equation. In physiology Young made an important contribution to hemodynamics. In an *Encyclopedia Britannica* article "Languages", Young compared the grammar and vocabulary of 400 languages. In a separate work in 1813, he introduced the term Indo-European languages. Young was also one of the first who deciphered Egyptian hieroglyphs. In 1814 Young had translated the demotic text of the Rosetta Stone, an ancient Egyptian granodiorite stele inscribed with a decree issued at Memphis, Egypt, in 196 BC.

In his lectures [87], Young described the deformation of prismatic bars, and introduced the notion of modulus of elasticity (Young's modulus), as the proportionality constant between the stress and the strain. Prior to Young's contribution, engineers who applies Hooke's law in the form of $F = kx$, to identify the deformation x of a body subject to a force F, relied on a constant k, which is dependent on

both the geometry (the cross-sectional area and the length of prismatic bar) and material under consideration. Young's modulus depends only on the material. Young observed similar elastic deformation for shear force and torsion. Describing the experiments of tension and compression of bars, Young drew the attention to the fact that longitudinal deformation is always accompanied by some change in the lateral dimensions (Poisson's ratio). Introducing Hooke's law, he observed that after certain limit, the deformation became inelastic, and resulted in permanent set. He also presented results of deflection of beams, and buckling, and looked at the fracture created by impact force [80].

F.5 Claude Louis Marie Henri Navier (1785–1836)

Navier's father was a lawyer, who was a member of the National Assembly in Paris during the time of the French Revolution. He died when Navier was only eight years old. Navier was left to the care of his granduncle Emiland Gauthey, who was considered the leading civil engineer in France at that time. In 1802, Navier passed the competitive entrance exam to enter École Polytechnique, where he was taught analysis by Jean Baptiste Joseph Fourier (1768–1830). In 1804 Navier entered the École des Ponts et Chaussées (School of Bridges and Highways) and graduated as one of the top students.

It was not long after Navier's graduation that his granduncle died, and Navier was asked to take on the task of editing Gauthey's works on bridges and channels, which was published in three volumes. To bring the work up to date, Navier added editorial notes in many places, which provided the theoretical mechanics basis to the empirical work.

Navier started teaching at the École des Ponts et Chaussées in 1819 and took charge of the applied mechanics courses. In 1824, he was elected a member of Académie des Sciences, and became professor of calculus and mechanics in 1830. For his engineering contributions, He was known as the first to introduce theoretical analysis of suspension bridges. His report "*Memorandum of Suspension Bridges*" [69] was for a long time the most important book for the design of suspension bridges [73, 80].

Navier was the first to investigate the general equations of equilibrium and vibrations of elastic solids and present them in the differential equation form [68]. He set out from the hypothesis that the elastic reactions arise from variations in the intermolecular forces. He assumed that the force between two molecules, whose distance is slightly increased, is proportional to the product of the increase in the distance and some function of the initial distance. His method consists in forming an expression for the component in any direction of all the forces, which act upon a displaced molecule, and thence the equations of motion of the molecule. The equations are thus obtained in terms of the displacements of the molecule. The

solid is assumed to be isotropic, and the equation obtained contains a single elastic constant in this form [67, 80]:

$$C\left(\nabla^2\vec{u} + 2\nabla e\right) + \vec{F} = 0 \tag{F.1}$$

Navier also developed the equation of longitudinal vibration. This equation however only predicts the longitudinal wave. It takes a second elastic constant in order to support a transverse wave. The correct form of the above equation, known as the Navier-Cauchy equation, containing two elastic constants, rather than one, was later obtained by Cauchy (see Sect. F.6).[1]

Navier's lecture note at École des Ponts et Chaussées [70], first published in 1826, made a great advancement in elasticity. The state-of-the-art of mechanic of materials until that time focused on the ultimate strength (failure) of materials and structures. Navier stressed the importance of the *elastic* analysis, which can be applied to functioning structures. In that book, he established the differential equation for the deflection of beams, developed method of analyzing indeterminate structures, presented theory of thin shells, and treated retaining walls, arches, plates, and trusses. However, the role of shear stress and shear strength was not well understood at that time, not until after the death of Navier [80].

Navier also derived the Navier-Stokes equations of fluid dynamics, although he did not fully understand the physics behind it, that is, he did not recognizing the role of shear stress in a fluid. Later George Gabriel Stokes (1819–1903) looked into the internal friction in fluids in motion and deduced the same equation of motion that carries both of their names. Navier was also the first to investigate the statically indeterminate system by examining the bending of a beam resting on supports [70].

F.6 Augustin-Louis Cauchy (1789–1857)

Cauchy was born in Paris during the difficult time of French Revolution. Cauchy's father was active in the education of young Augustin-Louis. Laplace and Lagrange were frequent visitors at the Cauchy family home, and Lagrange particularly took interest in Cauchy's mathematical ability.

In 1807 Cauchy entered École des Ponts et Chaussées to study engineering. At the age of 20, he was appointed as a Junior Engineer to work on the construction of Port Napoléon in Cherbourg. In 1812, he became ill and decided to return to Paris to seek a teaching position. His initial attempts in seeking academic appointment were

[1] In modern day Molecular Dynamics simulation, it requires not only a potential that is a function of distance between two atoms, which mimics the attractive and repulsive van der Waals forces, but also a potential that is a function of the angle between two molecular bonds, in order to create the elastic effects of both a bulk modulus and a shear modulus. Navier's assumption was equivalent to the former, but not the latter.

unsuccessful, although he continued to publish important pieces of mathematical work. In 1814 he published the memoir on definite integrals that later became the basis of his theory of complex functions. In 1815 Cauchy was appointed Assistant Professor of Analysis at the École Polytechnique. In 1816 he won the Grand Prix of the Académie des Sciences for a work on waves, and was later admitted to the Académie. In 1817, he was appointed Chair of Mathematical Physics at the Collège de France.

Cauchy was staunchly Catholic and was politically a royalist. By 1830 the political events in Paris forced him to leave Paris for Switzerland. He soon lost all his positions in Paris. In 1831 Cauchy went to Turin and later accepted an offer to become a Chair of Theoretical Physics. In 1833 Cauchy went from Turin to Prague, and returned to Paris in 1838. He regained his position at the Académie but not his teaching positions because he had refused to take an oath of allegiance to the new regime. Due to his political and religious views, he continued to have difficulty in getting appointment.

Cauchy was probably next to Leonhard Euler (1707–1783) the most published author in mathematics, having produced five textbooks and over 800 articles. Cauchy and his contemporary Carl Friedrich Gauss (1777–1855) were credited for introducing rigor into modern mathematics. The formulation of elementary calculus in modern textbooks is essentially what Cauchy expounded in his three great treatises [34–36]. Cauchy was also credited for setting the mathematical foundation for complex variable and elasticity.

For elasticity, Cauchy first tried Navier's molecular hypothesis, and then abandoned it [80]. Considering elastic solid as a continuum, he had proven the theorems (1) that the stress at any point can be expressed by means of three principal stresses, on three planes at right angles to each other, and normal to the planes across which they act, and (2) that the strain at any point can be reduced to three principal extensions, of three mutually perpendicular line-elements. He made assumptions that (1) the principal stresses are linear functions of the principal extensions, and (2) in an isotropic solid, the principal planes of stress are normal to the principal axes of extension. These assumptions led to the equations of equilibrium of an isotropic solid with two constants [67], which can be expressed as

$$\mu\nabla^2\vec{u} + (\lambda + \mu)\nabla e + \vec{F} = 0 \tag{F.2}$$

known as the Navier-Cauchy equation.

F.7 George Green (1793–1841)

George Green was virtually unknown as a physicist-mathematician during his lifetime. His most influential work, known as Green's theorem, was discovered posthumously by William Thomson (Lord Kelvin).

As the son of a semi-literate, but well-to-do Nottingham baker and miller, Green was sent to a private academy at the age of eight, and left school at nine. This was the only formal education that he received until adulthood. For the next twenty years after leaving primary school, no one knew how, and from whom, Green could have acquainted himself to the advanced mathematics of his day in a backwater place like Nottingham. Even the whole country of England in those days was scientifically depressed as compared to the continental Europe. Hence it was a mystery how Green could have produced as his first publication such a masterpiece without any guidance [32, 33, 57].

The next time there existed a record about Green was in 1823. At the age of thirty, he joined the Nottingham Subscription Library as a subscriber. In the library he had some access to books and journals. The next five years was not easy for Green; he had to work full time in the mill, had two daughters born (he had seven children with Jane Smith, but never married her, due to social barrier), and his mother died in 1825. Despite these difficulties in life and his flimsy mathematical background, in 1828 he produced one of the most important mathematical works of all times—*An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism* [58]. In the introduction section, Green stated:

> The object of this Essay is to submit to Mathematical Analysis the phenomena of the equilibrium of the Electric and Magnetic Fluids, and to lay down some general principles equally applicable to perfect and imperfect conductors; but, before entering upon the calculus, it may not be amiss to give a general idea of the method that has enabled us to arrive at results, remarkable for their simplicity and generality, which it would be very difficult if not impossible to demonstrate in the ordinary way.

In the "method that has enabled us to arrive at results", Green introduced the concept of electric and magnetic potentials, whose derivatives produce the physical quantity of force, and the phrase *potential function*. He derived the three Green's identities, as shown in the modern notations below:

$$\int\int\int \left(\phi\nabla^2\psi + \nabla\phi\cdot\nabla\psi\right)\, dV = \int\int \phi\frac{\partial\psi}{\partial n}\, dS \tag{F.3}$$

$$\int\int\int \left(\phi\nabla^2\psi - \psi\nabla^2\phi\right)\, dV = \int\int \left(\phi\frac{\partial\psi}{\partial n} - \psi\frac{\partial\phi}{\partial n}\right)\, dS \tag{F.4}$$

$$\phi = \frac{1}{4\pi}\int\int \left[\frac{1}{r}\frac{\partial\phi}{\partial n} - \phi\frac{\partial\,(1/r)}{\partial n}\right]\, dS \tag{F.5}$$

where dS denotes a surface integral, and dV a volume integral. Particularly, the third identity (F.5) gives the general solution of the so-called *Dirichlet problem*, namely the solution of Laplace equation with Dirichlet-type boundary condition. The singular solution of a partial differential equation, such as the $1/r$ in (F.5), has been called *Green's function*.

As Green did not feel that he had the proper educational credential and social status to submit the Essay to the customary publication channels at that time, the journals of the *Royal Society* and the *Cambridge Philosophical Society*, he privately

published the Essay through a Nottingham publisher. The essay had 51 subscribers, who were mostly local doctors, schoolmasters, clergymen, and owners of local factories, and could hardly understand a word of it. Green was disappointed by the lack of response. One subscriber, Sir Edward Bromhead, however, was impressed by Green's prowess in mathematics. He offered to refer Green's future papers to the Societies and recommended Green to attend Cambridge University.

Several years later, Green finally enrolled at Caius College of Cambridge University at the age of forty. From 1833 to 1836, Green wrote three more papers, two on electricity published by the Cambridge Philosophical Society, and one on hydrodynamics published by the Royal Society of Edinburgh. After graduating in 1837, he stayed at Cambridge for a few years to work on his own mathematics and to wait for an appointment. From 1838 to 1839 he had two papers in hydrodynamics, two papers on reflection and refraction of light, and two papers on sound [59]. In 1839, he was elected to a Parse Fellowship at Caius College, a junior position. Due to poor health, he had to return to Nottingham in 1840. He died in 1841 at the age of forty-seven. At the time of his death, his Essay, never formally published, was virtually unknown and lost.

At the year of Green's death, William Thomson (Lord Kelvin) was admitted to Cambridge. While studying the subject of electricity as a part of preparation for his Senior Wrangler exam, he first noticed the existence of Green's paper in a footnote of a paper by Robert Murphy. He started to look for a copy, but no one knew about it. After his graduation in 1845, and before his departure to France to enrich his education, he mentioned it to his tutor William Hopkins. It so happened that Hopkins had three copies. Thomson was immediately excited about what he had read in the paper. He brought the article to Paris and showed it to Jacques Charles François Sturm and Joseph Liouville, leading mathematicians at that time. Later Thomson republished Green's essay, rescuing it from sinking into permanent obscurity.

Green's 1828 essay had profoundly influenced Thomson and James Clerk Maxwell in their study of electrodynamics and magnetism, and was considered the founder of the 'Cambridge School' of natural philosophers, of which Thomson, Stokes, Rayleigh, and Maxwell were the most illustrious members in the later half of the nineteenth century [86]. The methodology has also been applied to many other fields of physics such as acoustics, elasticity, and hydrodynamics. During the bicentennial celebration of Green's birth in 1993, physicists Julian Schwinger and Freeman Dyson delivered speeches on the role of Green's functions in the development of 20th century quantum electrodynamics.

F.8 Gabriel Lamé (1795–1870)

Lamé was born in Tours, France. He and his colleague Benoît Paul Émile Clapeyron (1799–1864) both studied at École Polytechnique, and later graduated at École des Mines. In 1820, right after graduation, Lamé and Clapeyron were sent by the French government to Saint Petersburg to teach at the Institut et Corps du

Genie des Voies de Communication (Institute of Corps of Engineers of Ways of Communication), as a part of the plan of Russian Emperor Alexander I to import Western Europe scientific knowledge and its applications to military techniques and industrial development to Russia.

During their 12 year stay in Russia, Lamé and Clapeyron not only taught, but also supervised the construction of bridges and roads. They also wrote the important memoir "*Memorandum on the inner equilibrium of homogeneous solid*" [64]. In this work they demonstrated the equivalence between Navier's molecular and Cauchy's continuum approach in deriving the equilibrium equation, presented the graphical way of representing the state of stress at a point, later known as Lamé's stress ellipsoid, derived the solutions of internally and externally pressurized hollow cylinder and sphere, and simple torsion of a shaft, and for the first time, introduced cylindrical coordinates in their investigation of an infinite cylinder. This work contained all the known theoretical solutions dealing with the deformation of isotropic elastic materials at that time. It also became the basis for the famous book by Lamé in 1852 [63], "*Lectures on the mathematical theory of elasticity of solid bodies*", which is considered the first book on elasticity [80]. In this book, Lamé recognized that Cauchy's two constants version of elasticity is the correct version, over Navier's [81, 82].

In 1832, the relation between the French and Russian governments deteriorated, and Lamé returned to Paris. in the same year Lamé was elected professor of physics at École Polytechnique. Subsequently, he held various engineering posts and professorships in Paris, including a tour as the chief engineer of mines, during which he led the construction of the first two railroads of France, from Paris to Versailles and Paris to St. Germain, and chair of mathematical physics and probability at the Sorbonne. He was elected to the Académie des Sciences in 1843.

In addition to being a prominent engineer, elasticity, Lamé was also an accomplished mathematician. He contributed in differential geometry, curvilinear coordinates, and number theory. He also tackled the Fermat's last theorem with $n = 7$. In fact, Lamé was one of the few mathematicians who were ever praised by Gauss, the greatest mathematician.

F.9 Henry-Philter Gaspard Darcy (1803–1858)

Darcy was born in Dijon, France. His father was a minor civil servant, and died when Darcy was 14. His mother rose to the task of insuring his and his brother's education. She had only a small city pension, but was able to borrow money for tutors and obtained city scholarships for her sons to attend college. In 1821, at the age 18, Darcy entered École Polytechnique, Paris. Two years later Darcy was admitted to École des Ponts et Chaussées. After graduation, he was assigned by the Imperial Corps des Ponts et Chaussées to a position in Jura, and then transferred to Dijon [27, 48].

In 1828, Darcy was assigned to a deep well drilling project that found water but could not provide an adequate supply for the town. Under his own initiative, Henry set out to provide a clean, dependable water supply to the city from more conventional surface water sources. The effort eventually produced a system that delivered 8 m^3/min of water from the Rosolic Spring through 12.7 km of covered aqueduct to an enclosed 5,700 m^3 reservoir located near the Porte Guillaume and another reservoir at Montmusard. Pressurized distribution lines totaling 28,000 m were laid in underground galleries and provided water to major buildings and 142 public street hydrants spaced 100 m apart throughout the city. The entire system was enclosed and gravity driven, thus it required no pumps or filters [29]. For this work, Darcy was awarded the Legion of Honor.

By 1848 Darcy was Chief Engineer for the Department of Côte-d'Or, but at the beginning of the Second Republic, due to political pressures, he was forced to leave Dijon. He was however soon appointed to the higher position of Chief Director for Water and Pavements, Paris. In 1855 due to failing health, he resigned from normal duties but was allowed to return to Dijon to carry out research of his own interest. In 1856 Darcy published "*Les Fontaines Publiques de la Ville de Dijon*" [39], with the famous Appendix D, in which he conducted sand column experiments for water filtering that established Darcy's law. In 1857, Darcy published "*Recherches Expérimentales Relatives au Mouvement de l'eau Dans les Tuyaux*" [40]. This presented his research on pipe flow resistance and was the first work to suggest the existence of the boundary layer in fluid flow. This publication provides the basis for his credit in the Darcy-Weisbach equation, even though Weisbach published before him in 1845. Darcy died of pneumonia in 1858 at age 54 in Paris.

F.10 Achille Ernest Oscar Joseph Delesse (1817–1881)

Delesse was a French geologist and mineralogist. Born in Metz in 1817, he was orphaned at young age and was raised by his two aunts. He was admitted to École Polytechnique at age 20, and graduated first in class two years after. As a student in mining engineering, he received the assignment of cataloging a collection of geological maps, which initiated his life interest in mineralogy. He entered École des Mines for his doctoral work, and conducted his studies by traveling to England, Germany, Poland, and Ireland. His early work on rocks earned him an appointment as Professor of Mineralogy and Geology at the University of Besançon in 1845. At the same time, he worked as a mining engineer for the city. He returned to Paris as a substitute teacher of geology at the Sorbonne and was appointed Associate Professor of Geology at the École Normale Superieure. He was also appointed Inspector of Quarries of the Seine [44].

Delesse's career had a brilliant start. However, it became stagnant during the Second Empire for political reasons. Delesse was a staunch Republican and he did not hide it. In 1864 he lost his appointment as Inspector. To compensate for the termination, he was called to Ecole des Mines de Paris where he created a course in

agriculture, drainage and irrigation. After the fall of the Empire, the Third Republic accelerated his career. Delesse finally became Chief Engineer of Mines in 1870, at age of 53, which is very late for a brilliant graduate of the Polytechnic. He became Inspector General of Mines in 1878, in charge of the southeast of France. In 1875 he was the President of the First International Congress of Geography. In 1877 he was appointed Professor of Mineralogy and Geology at the National Agronomic Institute. In 1879, his career culminated in the election to the Académie des Sciences. Unfortunately, it was not long that he could enjoy this honor. Saddened by the loss of his father, a fatigue of overtook him. His health rapidly declined, and he passed away in 1881, aged 64 [28].

Delesse was an excellent chemist. He analyzes thousands of rocks, soils, and water and cataloged them with infinite patience. He made a fundamental contribution to the understanding of metamorphism through the petrography of igneous and metamorphic rocks, especially microcrystalline. His book on the lithology of the seabed [45] is still regarded as the starting point of modern oceanography [28]. Delesse made fundamental scientific impact in many other areas. For example, his research on the nitrogen content of the rocks is considered the beginning of the organic geochemistry. Delesse was among the first to practice the method of mapping at various scales, including at national level and in three different disciplines such as geology, agronomy and hydrogeology. In addition, he had introduced the economic factor, such as the cash income of the land and contour of underground deposits.

For the purpose of this book, his argument that the volume fraction of a mineral in a rock can be determined from its surface fraction, known as Delesse's law, played an important role in the volume averaging theory of porous medium presented in Chap. 4.

F.11 Josiah Willard Gibbs (1839–1903)

Josiah Willard Gibbs was an American theoretical physicist, chemist, and mathematician. He was born in New Haven, Connecticut, in 1839, and died in the same city, in 1903 [31]. Quoting from the science writer Bryson [30], "*Gibbs is perhaps the most brilliant person that most people have never heard of. Modest to the point of near invisibility, he passed virtually the whole of his life, apart from three years spent studying in Europe, within a three-block area bounded by his house and the Yale campus in New Haven, Connecticut.*"

In 1863, Gibbs received the first Ph.D. degree in engineering granted in the U.S., for a thesis entitled "On the form of the teeth of wheels in spur gearing," in which he used geometrical techniques to investigate the optimum design for gears. This was also only the fifth Ph.D. granted in the US in any subject. In 1866, Gibbs embarked on his trip to Europe, which he spent in Paris, Berlin and Heidelberg. In the universities, he attended lectures by some of the most prominent scientists at that time, such as mathematicians Joseph Liouville, Karl Weierstrass and Leopold

Kronecker, Chemists Gustav Magnus and Robert Bunsen, and physicists Gustav Kirchhoff and Hermann von Helmholtz. As the American science community was in the backwater in those days, this only trip that Gibbs ventured outside the small town in Connecticut transformed his scope in science.

After returning from Europe, Gibbs was appointed Professor of Mathematical Physics, the first such professorship in the United States and a position he held for the rest of his life. During the first ten years, Yale did not even pay him a salary. (As the only son from his parents, he inherited a modest fortune when he was young.) While at Yale, his courses attracted an average of slightly over one student a semester. His written work was difficult to follow and employed a private form of notation that many found incomprehensible. But buried among his arcane formulations were insights of the loftiest brilliance.

Rather surprisingly his appointment to the chair at Yale came before he had published any work. Perhaps it is also surprising that Gibbs did not publish his first work until 1873 when he was 34 years old. Gibbs's important 1873 papers were "*Graphical methods in the thermodynamics of fluids*" [51] and "*A method of geometrical representation of the thermodynamic properties of substances by means of surfaces*" [52]. In 1876 Gibbs published the first part of the work, for which he is most famous, "*On the equilibrium of heterogeneous substances*," publishing the second part of this work in 1878 [53]. In this series of papers, he elucidated the thermodynamic principles of *nearly everything* [30], "on gases, mixtures, surfaces, solids, phase changes, chemical reactions, electrochemical cells, sedimentation, and osmosis" [38]. Gibbs's Equilibrium has been called "the Principia of thermodynamics," but Gibbs chose to publish these landmark observations in the *Transactions of the Connecticut Academy of Arts and Sciences*, a journal that managed to be obscure even in Connecticut, which is why both Max Planck and Einstein had repeated Gibbs's work in their early part of career without knowing its existence.

Actually Gibbs's work did not go totally unnoticed. His work had impressed James Clerk Maxwell so much that he constructed a three dimensional model of Gibbs's thermodynamic surface and sent the model to Gibbs. Unfortunately, the prospects of collaboration between Maxwell and Gibbs were cut short by Maxwell's early death in 1879, aged 48. The joke later circulated in New Haven that "only one man lived who could understand Gibbs's papers, that was Maxwell, and now he is dead" [75].

Together with Maxwell and Ludwig Boltzmann, Gibbs is considered as a founder of *statistical mechanics* [54]. He coined the term to identify the branch of theoretical physics that accounts for the observed thermodynamic properties of systems in terms of the statistics of large ensembles of particles .

Gibbs's work on vector analysis was also of major importance in mathematics. British scientists, including Maxwell, had relied on William Rowan Hamilton's quaternions in order to express the dynamics of physical quantities. He proposed defining distinct dot and cross products for pairs of vectors and introduced the now common notation for them. He was also largely responsible for the development of the vector calculus techniques still used today in electrodynamics and fluid mechanics.

F.12 Karl von Terzaghi (1883–1963)

Karl von Terzaghi, generally known as the father of soil mechanics, was born in 1883 as the first child of Army Lieutenant-Colonel Anton von Terzaghi and Amalia Eberle in Prague. Upon Anton's retirement from the army, the family moved to Graz, Austria. Anton passed away when Karl was 7 years old, and his maternal grandfather Karl Eberle became the head of the Terzaghi household. Karl Terzaghi was strongly influenced in his character development by his grandfather, who was a graduate of the Vienna Technical University, a successful mechanical engineer, and later in his life, an organizer of international banking.

From age 10 to 17, Terzaghi attended military boarding schools. Although his family anticipated that he pursue a military officer career, Terzaghi however discovered that he much preferred science to "mindless soldier games". In 1900, Terzaghi entered the Technical University in Graz to study mechanical engineering. At the university he very quickly lost interest in the lectures of mechanical engineering, which were purely professional. He rarely attended classes; rather he took the advantage of the various science and humanistic courses offered in various other faculties of the university, and pursued with broad interest in philosophy, experimental psychology, art history, plant physiology, astronomy, and especially geology [55].

At one of the university dinner gathering Terzaghi found himself sat next to Ferdinard Wittenbauer, his teacher of applied mechanics. When questioned why he did not attend classes, Terzaghi responded that he would rather read the "prescriptions" for himself. Wittenbauer challenged Terzaghi to read the original text of Lagrange's *Analytical Mechanics* in French. The high pace of developing ideas of Lagrange's mind offered Terzaghi for the first time the "unforgettable revelation" of "science in the making". Wittenbauer became Terzaghi's mentor and a "fatherly friend and confessor" [55].

Terzaghi graduated from Graz with honors in 1904. After fulfilling a compulsory one year military service, he returned to the university for one year and combined the study of geology with courses on subjects such as highway and railway engineering. His first job was as a junior design engineer for the firm Adol Baron Pittle, Vienna. The firm was becoming more involved in the relatively new field of hydroelectric power generation, and Karl became involved in the geological problems the firm faced. By 1908, he was managing a construction site, workers, and the design and construction of steel reinforced structures. He embarked on ambitious and challenging project to construct a hydroelectric dam in Croatia and St. Petersburg. During six months in Russia, he developed some novel graphical methods for the design of industrial tanks, which he submitted as a thesis for his PhD at the university. His growing list of achievements began to open more opportunity to him. He resolved to go to the United States of America, which he did in 1912. In the US, on his own, he undertook an engineering tour of major dam construction sites in the West.

When World War I broke out, Terzaghi found himself drafted into the army as an officer directing a 250 man engineering battalion, which further increased to leading 1000 men. After a short stint managing an airfield, he became a professor in the Royal Ottoman College of Engineering in Istanbul (now Istanbul Technical University). There he began a very productive period, in which he began his lifelong work of bringing true engineering understanding to soil as an engineering material whose properties could be measured in standardized ways. He set up a laboratory using only the most rudimentary of equipment, and began his revolution. His measurements and analysis of the force on retaining walls were first published in English in 1919, and was quickly recognized as an important new contribution to the scientific understanding of the fundamental behavior of soils.

At the end of the war, he was forced to resign his post at Ottoman College, but found a new post at Robert College in Istanbul. There he again constructed a laboratory out of the simplest equipment. This time he studied various experimental and quantitative aspects of the permeability of soils to water and was able to work out some theories to explain the observations. In 1924 he published much of his research in *Erdbaumechanik* [77], which revolutionized the field to great acclaim. It resulted in a job offer from the Massachusetts Institute of Technology (MIT), which he immediately accepted.

One of his first tasks in the USA was to bring his work to the attention of engineers. This he proceeded to do by writing a series of articles for the *Engineering News-Record* [78]. He once more set up a new laboratory geared to making measurements on soils with instruments of his own devising. He entered a new phase of prolific publication and a rapidly growing and lucrative involvement as an engineering consultant on many large-scale projects.

In 1928, Terzaghi returned to Europe and accepted a chair at the Vienna Technical University. His teaching workload was relatively light, so he continued his experimental investigations, and became especially interested in the problems of the settling of foundations, and of grouting. In 1936, Terzaghi's pioneer work was publicly recognized when the first International Conference on Soil Mechanics was held at Harvard University in his honor.

The political turmoil in Austria began to interfere with his work, and in 1938 Terzaghi emigrated to the United States and took up a post at Harvard University. Before the end of the war, he consulted on the Chicago Subway system, and the Newport News shipways construction, among others. He remained as a part-timer at Harvard University until his retirement in 1953 at the mandatory age of 70. He died in 1963.

Goodman [56] commented: "Among Terzaghi's publications, reports and lectures, one finds seminal contributions across a wide terrain, including: classification methods for soils and rocks; capillary phenomena in soils; the theory and documentation of consolidation and settlement; piping and its prevention; design and construction of earth, rock and concrete dams on all kinds of foundations; anchorages for suspension bridges in soils; field and laboratory measurement of pore pressures and soil properties; use of flow nets in two and three dimensions; design of drainage wells and tunnels; design to avoid scour of river and waterfront structures;

earth pressure variations on walls and bulkheads; engineering in terrain underlain by permafrost; pile foundations; soil improvement by compaction, pile-driving, grouting and incorporation of geotextiles; soil and rock tunneling; engineering geology; sinkhole formation and collapse; regional subsidence due to oil-field operations; and landslides."

F.13 Yacov Il'ich Frenkel (1894–1952)

Yakov (Jacob) Frenkel was born on February 10, 1894 in the southern Russian city Rostov-on-Don. Since his early years he showed a talent in music, fine arts, and science. Being a student in the May Gymnasium at St. Petersburg, Frenkel wrote a 300-page exposition paper on the origin of the Earth's magnetism and atmospheric electricity. The paper was shown to Abram Fedorovich Ioffe, then a professor at St. Petersburg Polytechnic Institute, and the two met for the first time. In 1913 Frenkel graduated from gymnasium with the highest honor (Gold Medal).

Worrying that he might not be admitted to the university (in Tsar's time, admission quota existed for persons of the Jewish religion), he briefly toured the US for the opportunity in North American universities. Fortunately, he was accepted by the Physics and Mechanics Department at the St. Petersburg University, where he completed the six-year course study in three years.

The physics school in St. Petersburg was associated with A. F. Ioffe, who was an excellent scientist, teacher, and scientific manager—he was "Daddy Ioffe" as the first generation of Russian physicists after the Revolution affectionately called him. In 1916, Ioffe organized the first of the seminars that later became the trademark of Russian physics. Frenkel took part in the Ioffe seminar from its very beginning together with Lev Davidovich Landau, Nikolai Nikolayevich Semenov, and Pyotr Leonidovich Kapitsa, all Nobel Prize winners later in their career.

Right after the Revolution, in 1918, Frenkel left St. Petersburg and took part in the organization of Tavrichesky University in Yalta, Crimea. Living conditions in Crimea at that time were terrible. Frenkel was also jailed for two months during that time for political reasons. In 1921 Frenkel returned to St. Petersburg and started to work in St. Petersburg Polytechnic Institute. In the four years between 1922 and 1925, he published five monographs: *The Structure of Matter* in 3 parts, *The Theory of Relativity*, *Electrical Theory of Solid Bodies*, a popular science book *Electricity and Matter*, and a textbook *Course of Vector and Tensor Analyses*. He was merely thirty years old.

In 1925, upon the recommendation of Paul Ehrenfest, Frenkel was awarded a scholarship by the Rockefeller Foundation to visit Western Europe. He spent almost a year in Germany, France, and England. He worked with Max Born in Göttingen and was acquainted with Einstein in Berlin. It was also during this time that he published the first volume of his two-volume book *Electrodynamics*. In 1927 Frenkel was one of the participants in the International Physics Congress in Como, Italy. He presented his work on the quantum theory of electroconductivity

of metals, which became an important part of the modern solid state physics. Frenkel in 1927 also calculated the theoretical stiffness of metals from its crystalline structure and showed that the stiffness was orders of magnitude greater than that of real metals. One of the major successes of Frenkel during these years was the explanation of the effect of spontaneous magnetization of ferromagnetic materials and the development of the theory of magnetic domain. In 1926 Frenkel introduced the key idea of defects of crystalline structure, known as Frenkel defects.

In 1929 Frenkel was elected a Corresponding Member of the Academy of Sciences of the USSR, among the first group of young scientists admitted to the Academy after the Revolution. In 1930, Frenkel received an invitation from the University of Minnesota to spend a year in the US as a Visiting Professor. During his visit to the US, Frenkel developed the theory of light absorption in solid bodies. In these papers he introduced the idea and the term "exciton" for the description of the waves of excitation. Frenkel in 1936 introduced the original theory of fission of atomic nuclei. In 1939, several months ahead of Niels Bohr and John Wheeler, Frenkel proposed the so called "drop model" of the atomic nucleus.

Geophysics was a field of Frenkel's early interest, and in 1944, Frenkel returned to it. He frequently visited the Institute of Theoretical Geophysics in Moscow traveling from Kazan'. In the Institute, he became interested in the work of A. G. Ivanov, who had discovered in 1939 that the propagation of seismic waves in soil was accompanied by the appearance of a potential difference between electrode probes inserted into the ground. Ivanov considered the phenomenon as being caused by the pressure difference between two points in wet soil resulting from the propagation of longitudinal waves. Frenkel took a different view. He correctly recognized the need to model the wet soil as a two-phase material. He formulated the continuum mechanical theory; and by mathematical analysis set up the characteristic equations. He then stated [49]: "*We shall not write down the expressions for its roots and shall only remark that . . . one of them corresponds to waves with a very small damping, and the other to waves with a very large damping*." Hence Frenkel discovered the existence of the second compressional wave. However, Frenkel's interest was in the seismoelectric effect; hence he dismissed the mechanical effect as unimportant by stating: "*The waves of the second kind are thus really non-existent*," because it would not propagate to a large distance. For the electrical effect, Frenkel recognized the presence of electrolytes in liquid, which created electric double layers on the interface of soil particles and the liquid. The second wave, despite its non-propagating nature, allowed the solid and fluid phases to move relative to each other. The relative movement of electrical charges created local electrical currents, whose variation in turn caused the emission of electromagnetic waves.

After the war, Frenkel returned to Leningrad and resumed his teaching and research at the Polytechnic Institute. In 1945, Frenkel was honored with the Labor Red Banner Order. Two years later, his Kinetic Theory of Liquids was awarded the First Grade State Prize. However, at that time there existed the first hint of a change in the socialism policy; and the first gust of cold wind soon reached Frenkel. The ensuing political persecution affected not only Frenkel, but also many other prominent scientists. Frenkel's work was criticized for not contributing to

the construction of the society of great socialism. His contributions in quantum mechanics and the theory of relativity were labeled as servility to Western science. Frenkel's work was greatly affected and his health deteriorated toward the end of his life. He died in 1952, not quite 58 years old.

In 1994, the Physico-Technical Institute of the now St. Petersburg Polytechnic State University, where Frenkel worked from 1921 to 1952, celebrated his centennial. Frenkel's memorial plaque now adorns the wall of the main building of the Institute [50, 66].

F.14 Maurice Anthony Biot (1905–1985)

Biot was born in Antwerp, Belgium on May 25th 1905. The war in 1914–1918 and the siege of Antwerp caused the Biot family to travel first to London, then Paris, and finally settling in Chambéry, France. These moves matured the young Biot and exposed him to several languages. Later returning to Antwerp, Biot concluded his secondary school. In 1924 he was admitted at the Université catholique de Louvain (UCL). It was at this time that Biot showed his insatiable appetite for knowledge. While pursuing his studies in Engineering, Biot was awarded a Bachelor degree in Philosophy, and also attended courses in Economics. He obtained a Mining Engineering degree in 1929, and an Electrical Engineering degree in 1930. (See Figs. F.1 and F.2 for photos of Biot at various stages of life.) Biot was awarded a Doctor of Science degree in 1931.

The sponsorship of the Belgian American Educational Foundation allowed Biot to spend the next two years in the US at the California Institute of Technology (Caltech). It was at Caltech where he first met and worked with Theodore von Kármán, who had arrived in the US in 1929. Biot acquired a Ph.D. in Aeronautical Sciences in 1932 by defending his dissertation on "Transient oscillations in elastic system" [3] in which he introduced the concept of response spectrum for earthquake analysis. The methodology brought great simplifications to the analysis of structures under transient loading and has since been used as a tool in earthquake-proof design. It was during the same period that he published his first papers on a new approach to the nonlinear theory of elasticity accounting for the effect of initial stress.

At the beginning of 1934, he obtained a fellowship from the Belgian Scientific Foundation which allowed him to pay short but interesting visits at the universities of Delft, Louvain, Cambridge, Zürich, and Göttingen. The few weeks spent at Göttingen made him aware of the remarkable advances of Germany in the field of Physics and Aeronautics. The atmosphere surrounding the new Nazi regime electrified the German students. Upon his return, Biot saw this as an ill omen but was regarded as a pessimist.

In 1934, Biot started his academic career as a teacher of applied mathematics at Harvard University. In June 1935, he returned to Pasadena as an Advanced Fellow of the Belgian American Educational Foundation. By 1936, Biot was elected to the faculty at his Alma Mater UCL, where he taught Elasticity and Analytical

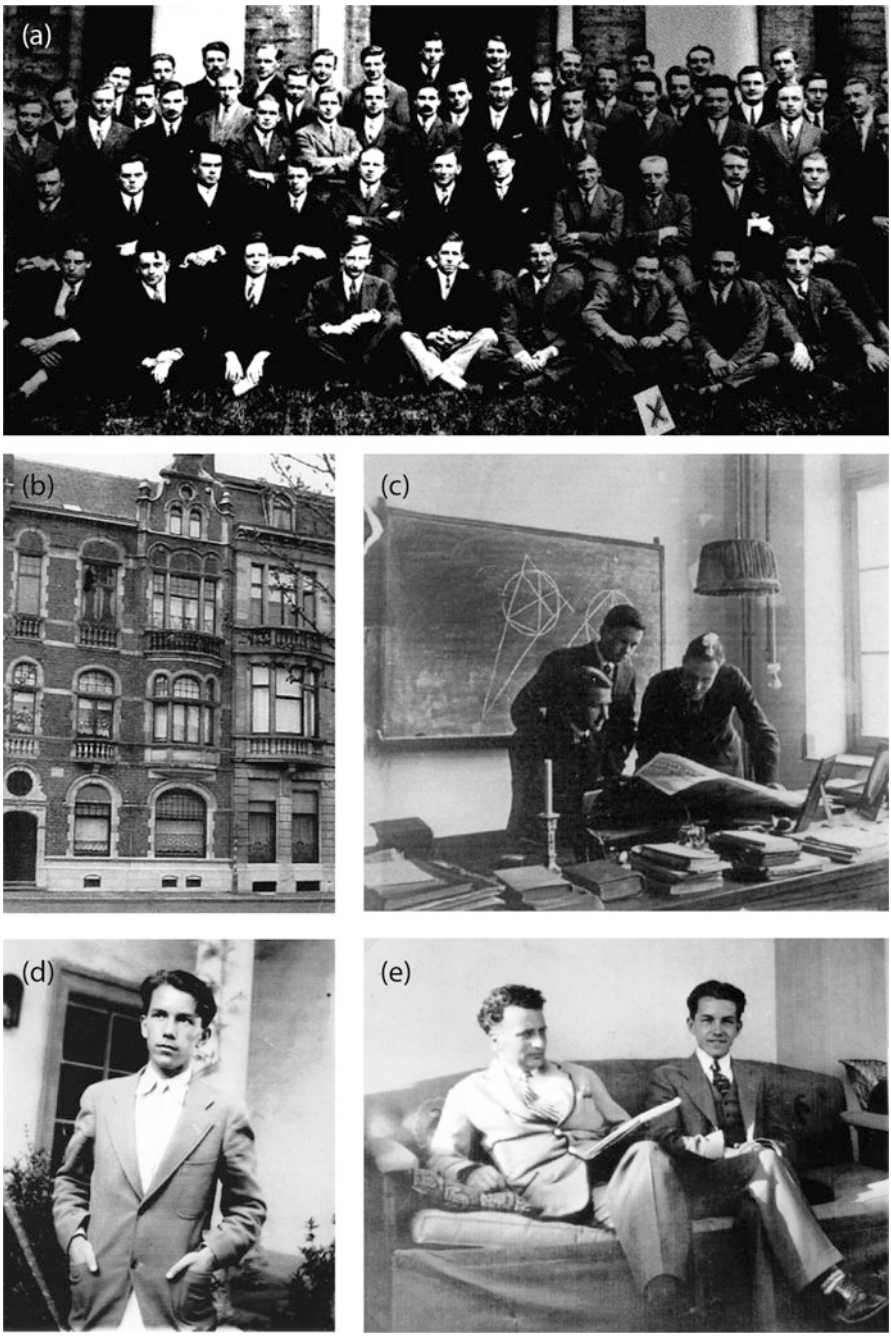


Fig. F.1 (**a**) Biot (marked with X) earning Mining Engineering degree, Louvain (1929); (**b**) Family house, Louvain (1924–1940); (**c**) Science preparatory school in Brussels (standing in the back) (1923); (**d**) Arriving at Caltech at age 26 (1931); and (**e**) At von Kármán's house (1932) (All photos courtesy of Mme Biot)



Fig. F.2 (**a**) Lieutenant Commander, heading the Structural Section of the Bureau of Aeronautics, Washington D.C. (1943); (**b**) Crossing the Rhine as a member of the Combined Allied Field Task (1944); (**c**) London Airport (1944); (**d**) At 300 Central Park West apartment, New York (1964); (**e**) Biot (right) with G.I. Taylor in Madrid IUTAM Symposium (1955); and (**f**) At 117 Avenue Paul Hymans apartment, Bruxelles (All photos courtesy of Mme Biot)

Mechanics. From 1937 to 1946, Biot was an Assistant Professor of Theoretical Mechanics and Physical Mathematics at Columbia University.

It was in 1940 that the monograph *Mathematical Methods in Engineering* [85] was written with von Kármán. Its translation into nine languages is evidence of its influence on several generations of engineers. Later in his career he wrote two more books: *Mechanics of Incremental Deformations* [19] and *Variational Principles in Heat Transfer* [21].

Unfortunately, war was brewing in Europe. Poland was overrun in 1939, and then came Holland and Belgium in 1940. France requested an armistice a few days later. Caltech at that time had undertaken a research contract with the OSRD (Office of Scientific Research and Development). Robert Millikan, President of Caltech, requested that Columbia grant a leave of absence to Biot, so that his special knowledge on vibration problems in high-speed aircraft can be utilized. It was during this time that Biot developed a three-dimensional theory of aircraft flutter.

When US entered the war, Biot decided to enlist. Assigned to the US Navy he was send to Key West where his hydro-mechanical knowledge was needed in anti-submarine warfare. In 1943, as Lieutenant Commander, he headed the Structural Dynamics Section of the Bureau of Aeronautics, US Navy, in Washington, D.C. As member of the CAFT (Combined Allied Field Task) he was in London on D-Day. He crossed the English Channel and joined the US 6th Army as the Allied pushed to the outskirts of Paris. As the US Army advanced in south Germany, Biot's mission was to gather scientific intelligence on aircraft and missile technology. He was responsible for the capture of a number of research institutes complete with files and personnel.

After the war, Biot returned to Columbia. In 1946 Brown University offered him the position of Professor in Applied Physics and Sciences, which he held until 1952. By 1951, Biot had produced a large number of scientific works for Shell Development Co., Cornell Aeronautical Laboratory, and for the US Air Force. After 1952 Biot worked largely alone as a consultant for various governmental agencies and industrial laboratories. From 1969 to 1982, Biot was a consultant for Mobil Research and Development Corporation in Dallas, in the area of Rock Mechanics.

Relocated in Brussels since 1970, Biot continued his research till his last day. It was on one of his last trips to the US that Biot felt the early signs of his illness, which would suddenly deprive him of his life on the 12th of September 1985, at the age of eighty.

Biot's interest in the mechanics of porous media dated back to 1935. Inspired by Terzaghi's 1923 work on the consolidation of clay [76], Biot published his first paper on generalized consolidation theory [6]. This publication was followed by a series of papers in the Journal of Applied Physics, starting with the classical paper entitled "*General theory of three-dimensional consolidation*" in 1941 [8], which established a rigorous, coupled theory for the mechanical response of a porous elastic material in the quasi-static range. The dynamics, or acoustics, theory was established by several papers in the Journal of the Acoustical Society of America [13, 14]. These papers constitute the foundations of what is now referred to as Biot's theory of poroelasticity.

In addition to poroelasticity, the work and original contributions that distinguished Biot's career cover an unusually broad range of science and technology, including applied mechanics, sound, heat, thermodynamics, aeronautics, geophysics, earthquake engineering [84], and electromagnetism.

As early as 1932, working under von Kármán, Biot was the first to analyze the dynamics problem of how to estimate the maximum response of oscillators to transient excitation in his dissertation [3–5]. In the subsequent 10 years [9, 10], the method developed into what is known in present day as the *response spectrum method* in earthquake engineering [83].

Biot's interest in the non-linear effects of initial stress and the inelastic behavior of solids [7, 18, 19] led to his mathematical theory of folding of stratified rocks [15, 17]. He verified his approach in the laboratory and successfully applied it to explain the dominant features of geological structures [26].

Biot also made significant contribution in the irreversible thermodynamics by using the variational approach [12, 25], and was the first to introduce the dissipation function and the minimum dissipation principle in thermodynamics to account for the dissipation phenomenon [11, 16, 21]. The variational principle allowed the new formulation of viscoelasticity [11], heat transfer [20], mass diffusion [23], chemical reaction [22], and thermorheology [24]. Biot's contribution is recognized in a quote by Nowacki in his book *Thermoelasticity* [71]: "*the real development of the theory occurred in the last twenty years. The starting point was the paper by M.A. Biot [12] in which on the basis of the thermodynamics of irreversible processes the basic relations and equations were derived and the variational theorems of thermoelasticity were formulated.*"

The honors that Biot received during his lifetime included the Timoshenko Medal of the American Society of Mechanical Engineers (ASME), the von Kármán Medal of the American Society of Civil Engineers (ASCE), and an Honorary Fellow of the Acoustical Society of America. He was also a member of the US National Academy of Engineering [37, 46].

To honor Biot's pioneering contribution, a Biot Conference on Poromechanics was convened in 1998 [79] in his home country Belgium. It subsequently held once every four years [1, 2, 60, 65] in locations in the US, France, and Austria. The ASCE established a Maurice A. Biot Medal in 2003 to annually award "*an individual who has made outstanding research contributions to the mechanics of porous materials*."

References

1. Abousleiman Y, Cheng AHD, Ulm FJ (eds) (2005) Poromechanics III–Biot centennial (1905–2005). Balkema, Leiden, 828pp
2. Auriault JL, Geindreau C, Royer P, Bloch JF, Boutin C, Lewandowska L (eds) (2002) Poromechanics II. Balkema, Lisse/Exton, 955pp
3. Biot MA (1932) Transient oscillations in elastic system. Doctoral thesis, California Institute of Technology

4. Biot MA (1933) Theory of elastic systems vibrating under transient impulse with an application to earthquake-proof buildings. Proc Natl Acad Sci 19(2):262–268
5. Biot MA (1934) Theory of vibration of buildings during earthquakes. Zeitschrift für Angewandte Mathematik und Mechanik 14(4):213–223
6. Biot MA (1935) Le problème de la consolidation des matières argileuses sous une charge (The problem of consolidation of clay material under load). Annales de la Société Scientifique de Bruxelles, Serie B 55:110–113
7. Biot MA (1939) Non-linear theory of elasticity and the linearized case for a body under initial stress. Philos Mag 27(183):468–489
8. Biot MA (1941) General theory of three-dimensional consolidation. J Appl Phys 12(2):155–164
9. Biot MA (1941) A mechanical analyzer for the prediction of earthquake stresses. Bull Seismol Soc Am 31(2):151–171
10. Biot MA (1942) Analytical and experimental methods in engineering seismology. Proc Am Soc Civil Eng 68:365–409
11. Biot MA (1955) Variational principles in irreversible thermodynamics with application to viscoelasticity. Phys Rev 97(6):1463–1469
12. Biot MA (1956) Thermoelasticity and irreversible thermodynamics. J Appl Phys 27(3):240–253
13. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid.1. Low-frequency range. J Acoust Soc Am 28(2):168–178
14. Biot MA (1956) Theory of propagation of elastic waves in a fluid-saturated porous solid. 2. Higher frequency range. J Acoust Soc Am 28(2):179–191
15. Biot MA (1957) Folding instability of a layered viscoelastic medium under compression. Proc R Soc Lond Ser A – Math Phys Sci 242(1231):444–454
16. Biot MA (1959) New thermomechanical reciprocity relations with application to thermal stress analysis. J Aerosp Sci 26(7):401–408
17. Biot MA (1961) Theory of folding of stratified viscoelastic media and its implications in tectonics and orogenesis. Geol Soc Am Bull 72(11):1595–1620
18. Biot MA (1963) Internal buckling under initial stress in finite elasticity. Proc R Soc Lond Ser A – Math Phys Sci 273(135):306–328
19. Biot MA (1965) Mechanics of incremental deformations. Wiley, New York/London, 504pp
20. Biot MA (1966) Generalized variational principles for convective heat transfer and irreversible thermodynamics. J Math Mech 15(2):177–186
21. Biot MA (1970) Variational principles in heat transfer. Oxford University Press, Oxford, 185pp
22. Biot MA (1977) New fundamental concepts and results in thermodynamics with chemical applications. Chem Phys 22(2):183–198
23. Biot MA (1977) Variational Lagrangian-thermodynamics of nonisothermal finite strain mechanics of porous solids and thermomolecular diffusion. Int J Solids Struct 13(6):579–597
24. Biot MA (1981) Generalized Lagrangian thermodynamics of thermorheology. J Therm Stress 4(3–4):293–320
25. Biot MA (1984) New variational-Lagrangian irreversible thermodynamics with application to viscous flow, reaction-diffusion, and solid mechanics. Adv Appl Mech 24:1–91
26. Biot MA, Odé H, Roever WL (1961) Experimental verification of the theory of folding of stratified viscoelastic media. Geol Soc Am Bull 72(11):1621–1631
27. Biswas AK (1970) History of hydrology. North-Holland, Amsterdam, 336pp
28. Boulain J, Trichet J (1997) Delesse Achille (1817–1881) et ses cartes thématiques (Delesse Achille (1817–1881) and thematic maps), Travaux du Comité Français d'Histoire de la Géologie, Troisième série–T.XI
29. Brown GO (2002) Henry Darcy and the making of a law. Water Resour Res 38(7):12pp
30. Bryson WM (2004) A short history of nearly everything. Broadway Books, New York, 544pp
31. Bumstead HA (1903) Josiah Willard Gibbs. Am J Sci Ser 4 16(4):187–202
32. Cannell DM (1999) George Green: an enigmatic mathematician. Am Math Mon 106(2):136–151

33. Cannell DM (2001) George Green: mathematician and Physicist, 1793–1841. The background to his life and work, 2nd edn. SIAM, Philadelphia, 316pp
34. Cauchy A-L (1821) Cours d'analyse de l'École royale polytechnique. 1. analyse algébrique (Course of analysis at Royal Polytechnic School. 1. Algebraic Analysis). Imprimerie Royale, Paris, 576pp
35. Cauchy A-L (1823) Résumé des leçons données à l'École Royale Polytechnique, sur le calcul infinitésimal (Lecture note at the Royal Polytechnic School, the calculus). Imprimerie Royale, Paris, 172pp
36. Cauchy A-L (1829) Leçon sur le calcul différentiel (Lesson on calculus). Imprimerie de Béthune, Paris, 289pp
37. Cheng AHD, Detournay E, Abousleiman Y (1998) Poroelasticity—Maurice A. Biot memorial issue: preface. Int J Solids Struct 35(34–35):4513–4516
38. Cropper WH (2001) Great physicists: the life and times of leading physicists from Galileo to Hawking. Oxford University Press, 500pp
39. Darcy H (1856) Les Fontaines Publiques de la Ville de Dijon (The public fountains of the city of Dijon). Dalmont, Paris, 647pp
40. Darcy H (1857) Recherches Expérimentales Relatives au Mouvement de l'eau Dans les Tuyaux (Experimental research relating to the movement of water in pipes). Mallett-Bachelier, Paris, 268pp
41. da Vinci L (1506–1510) Codex Leicester, (unpublished manuscript).
42. da Vinci L (1970) The notebooks of Leonardo da Vinci, vol 1 (Trans Richter JP, 1888). Dover, New York, 396pp
43. da Vinci L (1970) The notebooks of Leonardo da Vinci, vol 2 (Trans Richter JP, 1888). Dover, 514pp
44. de Lapparent A (1897) Achille Ernest Oscar Joseph Delesse (1817–1881). In Centenary book (Ecole Polytechnique), vol 1. Gauthier-Villars, Paris
45. Delesse A (1872) Carte Lithologique des Mers de France et des Mers Principales de Globe (Lithological map of the seas of France and the main seas of the globe). Lacroix, Paris, 479pp
46. Delmer A, Jaumotte A (1990) Note on Maurice Anthony Biot. Académie Royale de Belgique, Brussels
47. Fourier JBJ (1822) Théorie Analytique de la Chaleur (The analytical theory of heat), Chez Firmin Didot. Père et Fils, Paris
48. Freeze RA (1994) Henry Darcy and the fountain of Djion. Ground Water 35(1):23–30
49. Frenkel J (1944) On the theory of seismic and seismoelectric phenomena in moist soil. J Phys USSR 13(4):230–241
50. Frenkel VY (1996) Yakov Il'ich Frenkel, his work, life and letters. Birkhäuser, Basel, 323pp
51. Gibbs JW (1873) Graphical methods in the thermodynamics of fluids. Trans Conn Acad Arts Sci 2:309–342
52. Gibbs JW (1873) A method of geometrical representation of the thermodynamic properties of substances by means of surfaces. Trans Conn Acad Arts Sci 2:382–404
53. Gibbs JW (1876, 1878) On the equilibrium of heterogeneous substances. Trans Conn Acad Arts Sci 3:108–248, 343–524
54. Gibbs JW (1902) Elementary principles in statistical mechanics, developed with especial reference to the rational foundation of thermodynamics. Charles Scribner's Sons, New York/Edward Arnold, London, 207pp
55. Goodman RE (1999) Karl Terzaghi: the engineer as artist. ASCE, Reston, 340pp
56. Goodman RE (2002) Karl Terzaghi's legacy in geotechnical engineering. Geo-Strata, GeoInstitute, ASCE
57. Grattan-Guinness I (1995) Why did George Green write his essay of 1828 on electricity and magnetism? Am Math Mon 102(5):387–396
58. Green G (1828) An essay on the application of mathematical analysis to the theories of electricity and magnetism. T. Wheelhouse, Nottingham
59. Green G (1871) Mathematical papers of George Green. N.M. Ferrers (Reprinted by Chelsea, New York, 1970)

60. Hellmich C, Pichler B, Adam D (eds) (2013) Poromechanics V. ASCE, Reston, 2565pp
61. Hooke R (1678) De Potentia Restitutivâ, Or of spring, explaining the power of springy bodies. Martyn, London, 56pp
62. Inwood S (2003) The forgotten genius: the biography of Robert Hooke 1635–1703. McAdam/Cage, San Francisco, 482pp
63. Lamé G (1852) Leçons sur la théorie mathématique de l'élasticité des corps solides (Lectures on the mathematical theory of elasticity of solid bodies). Bachelier, Paris, 335pp
64. Lamé G, Clapeyron BPE (1833) Mémoire sur l'équilibre intérieur des corps solides homogènes (Memorandum on the inner equilibrium of homogeneous solid). *Mémoires présentés par divers savans* 4:465–562
65. Ling HI, Smyth A, Betti R (eds) (2009) Poromechanics IV. DEStech Publications, Lancaster, 1151pp
66. Lopatnikov SL, Cheng AHD (2005) If you ask a physicist from any country: a tribute to Yacov Il'ich Frenkel. J Eng Mech ASCE 131(9):875–878
67. Love AEH (1892) A treatise on the mathematical theory of elasticity, vol 1. Cambridge University Press, Cambridge , 354pp
68. Navier C-L (1821) Memoire sur les lois de l'équilibre et du mouvement des corps solides élastiques (Memorandum on the laws of equilibrium and motion of elastic solids). Mem Acad Sci Inst Fr 2:375–393
69. Navier C-L (1823) Rapport à Monsieur Becquey, ... Directeur Général des Ponts; ... et Mémoire sur les Ponts Suspendus (Report to Mr. Becquey ... Director General of Bridges, ... Memorandum of Suspension Bridges).
70. Navier C-L (1864) Résumé des leçons données à l'École des ponts et chaussées sur l'application de la mécanique à l'établissement des constructions et des machines (Lecture note at the school of bridges and highways on the application of mechanics to the establishment of structures and machines), 3rd edn. Dunod, Paris, 851pp
71. Nowacki W (1986) Thermoelasticity, 2nd edn. Pergamon, Oxford/New York, 566pp
72. Peacock G (1855) Life of Thomas Young, M.D., F.R.S., &c., and one of the eight foreign associates of the national institute of France. John Murray, London, 514pp
73. Picon A (1988) Navier and the introduction of suspension bridges in France. Constr Hist 4:21–34
74. Robinson A (2006) The last man who knew everything: Thomas Young, the anonymous genius who proved Newton wrong and deciphered the Rosetta Stone, among other surprising feats. Plume, New York, 304pp
75. Rukeyser M (1988) Willard Gibbs. Ox Bow Press, Woodbridge, 465pp
76. Terzaghi K (1923) Die Berechnung der Durchlässigkeitsziffer des Tones aus dem Verlauf der hydrodynamischen Spannungserscheinungen (A method of calculating the permeability of clay from the history of hydrodynamic stress variation). Sitzungsber. d. Akad. d. Wiss., Wien Math.-Naturwiss. Kl., Abt. IIa 132(3/4):125–138
77. Terzaghi K (1925) Erdbaumechanik auf Bodenphysikalischer Grundlage (Soil mechanics on soil physical basis). F. Deuticke, Leipzig u. Wien, 399pp
78. Terzaghi K (1926), Principles of soil mechanics: I-phenomena of cohesion of clay, II-compressive strength of clay, III-determination of permeability of clay, IV-settlement and consolidation of clay, V-physical differences between sand and clay, VI-elastic behavior of sand and clay, VII-friction in soil and in clay, VIII-future development and problems. Eng News-Record 95(19–23, 25–27):742–746, 796–800, 832–836, 874–878, 912–915, 987–990, 1026–1029, 1064–1068
79. Thimus JF, Abousleiman Y, Cheng AHD, Coussy O, Detournay E (eds) (1998) Poromechanics–a tribute to Maurice A. Biot. Balkema, Rotterdam/Brookfield, 648pp
80. Timoshenko SP (1983) History of strength of materials. Dover, 452pp
81. Todhunter I (1886) A history of the theory of elasticity and of the strength of materials from Galilei to the present time, vol I: Galilei to Saint-Venant (1639–1850). Cambridge University Press, Cambridge, 924pp

82. Todhunter I (1893) A history of the theory of elasticity and of the strength of materials from Galilei to the present time, vol II: Saint-Venant to Lord Kelvin. Cambridge University Press, Cambridge, 762pp
83. Trifunac MD (2006) Biot response spectrum. Soil Dyn Earthq Eng 26(6–7):491–500
84. Trifunac MD (2011) Maurice Anthony Biot: a pioneer of earthquake engineering. Ingegneria Sismica 38(1):7–20
85. von Kármán T, Biot MA (1940) Mathematical methods in engineering, an introduction to the mathematical treatment of engineering problems. McGraw-Hill, New York/London, 505pp
86. Whittaker E (1910) History of the theories of Aether and electricity. Nelson, London.
87. Young T (1807) A course of lectures on natural philosophy and the mechanical arts, 2 vols. Joseph Johnson, London, 738pp

Index

A
action, 477
activity, 716
  coefficient, 716
added mass, *see* apparent mass
adjoint operator, 398
advection, 617, 735
aftershock, 32
anelastic strain recovery, 43, 592
angle of internal friction, 30, 361
anisotropy
  general, 172, 184
  monoclinic, 172
  orthotropy, 172, 177
    plane strain, 251, 276
  transverse isotropy, 172, 179, 235, 276, 281
anti-plane problem, 257
apparent mass, 478
  Basset, 491
  constitutive constant, 479
  density, 480
attenuation coefficient, 509
  per wave cycle, 509
axial symmetry, 259, 316, 338, 385, 438, 634, 650, 743, 754

B
backstress, 36
barometric efficiency, 41
Basset force, 491
beer can experiment, 31
BEM, *see* boundary element method
Biot decomposition, 220, 432, 686
Biot function, 219, 222
Biot modulus, 73, 79, 146, 582, 595, 732
  complex, 514
Biot, Maurice Anthony, 4, 83, 857
BISQ (Biot-squirt) model, 514, 595
body force, 118, 204, 401
borehole
  breakdown pressure, 24, 365
  breakout, 25
  delayed failure, 24, 369
  ellipticity, 348
  excavated, 358, 367
  failure, 22
    compressive, 22
    shear, 25, 361, 367
    tensile, 359, 364, 370
  fluid injection and extraction, 363
  pressurized, 370
  stability analysis, 367
bound fluid, 720, 727, 730
boundary condition, 213
  Dirichlet, 214, 632
  displacement, 215
  fluid contact, 216
  flux, 215
  free surface, 216
  impermeable, 216
  interface, 218
  Neumann, 214, 632
  permeable, 216
  plate
    flexible, 216
    rigid, 216
  poroelastodynamics, 495
  porothermoelasticity, 632
  pressure, 215

© Springer International Publishing Switzerland 2016

A.H.-D. Cheng, *Poroelasticity*, Theory and Applications of Transport in Porous Media 27, DOI 10.1007/978-3-319-25202-5

boundary condition (*cont.*)
rigid surface, 216
Robin, 214
traction, 215
boundary element method, 409, 412
direct method, 412
indirect method, 412, 420
boundary integral equation, 409
boundary integral equation method, *see* boundary element method
boundary layer, 481, 486
thickness, 10, 247, 292, 487
boundary value problem, 213
well-posed, 214
Boussinesq-Papkovitch function, 218
breakdown pressure, 334
bulk continuum model, 63, 69, 88, 142, 171, 606, 611, 728
bulk modulus, 68
adiabatic, 20
drained, 6, 13, 73, 76, 85, 103, 145, 151
adiabatic, 613
complex, 516
constant solute concentration, 730
isothermal, 607
effective, 93
fluid, 8, 88, 137, 156
generalized, 184
isothermal, 20
microinhomogeneity and microanisotropy, 141, 147
pore volume, 7, 85, 104, 147, 152
porosity, 17, 87, 93, 139
solid, 7, 15, 90, 160, 184
undrained, 6, 70, 73, 76, 104, 146, 153
adiabatic, 613
constant solute concentration, 731
unjacketed frame, 86, 90, 105, 146
unjacketed pore volume, 86, 90, 105, 146, 147
viscoelastic, 583

C

capillary tube model, *see* conceptual flow model
carbon sequestration, 17, 45, 48
Cauchy integral formula, 816
Cauchy principal value, 409, 413, 418
Cauchy residue theorem, 300, 321, 816
Cauchy, Augustin-Louis, 64, 68, 845
characteristic frequency, 10, 492, 503
characteristic length, 9
squirt flow, 514
characteristic time, 10
chemical potential, 705, 714
chemical swelling coefficient, 725
Clapeyron formula, 71
closed system, 125, 616
coal cavitation, 34, 360
cohesion, 30, 361
compatibility equation
Beltrami-Michell, 207, 277, 628
St. Venant, 206
generalized plane strain, 255
plane strain, 250
complete mode conversion, 551
complete solution, 219, 427
compliance coefficient, 98, 173
drained, 175
undrained, 175
compressibility, 5, 98, 735
drained, 99, 153
fluid, 38, 67, 79, 88, 103, 163, 183
unsaturated, 39, 155, 291, 294, 512
frame, 98, 162
generalized, 184
jacketed, 103, 163
pore volume, 14, 98, 99, 153
porosity, 101, 162
solid, 12, 99, 100, 149, 162, 183
undrained, 8
unjacketed, 105, 165
conceptual flow model
capillary tube, 488
parallel plates, 488
consolidation coefficient, 9, 19, 80, 205, 281
Terzaghi, 205
transversely isotropic, 236
viscoelastic, 586
consolidation problem
Terzaghi, 9, 12, 21, 236
Terzaghi-Rendulic, 21, 26
constitutive equation, 61, 67, 203
anisotropic, 171
bulk continuum, 69
elasticity
anisotropic, 172
isotropic, 67
entropy, 612
intrinsic model, 133
isotropic, 69
micromechanics, 84, 88, 181
nonlinear, 162
porochemoelasticity, 723
porosity variation, 86
porothermoelasticity, 600
poroviscoelasticity, 576, 579

solid strain, 86
temperature, 613
transverse isotropy, 235
continuity equation, 199, 493
continuum, 64
convolutional
integral, 814
theorem, 411, 578, 814
coordinate system
Cartesian, 807
cylindrical, 316, 809
moving, 371
spherical, 295, 811
correspondence principle, 578
Coulomb friction, 515
coupling parameter
poroelastic, 635
thermoelastic, 635
crack density parameter, 93

D

da Vinci, Leonardo, 1, 42, 839
dam-reservoir system, 40
Darcy's law, 61, 190, 620, 850
Darcy, Henry, 189, 849
deformation mode, 138
degree of saturation, 155, 292, 513
Delesse's law, 116, 123, 143, 851
Delesse, Achille, 113, 116, 850
diffusion equation, 206, 276
axial strain, 261
axial symmetry, 317, 339
biharmonic, 219, 433, 697
coupled, 22, 205, 234
entropy, 19, 631, 684
frequency domain, 246
heat, 631, 636
irrotational displacement, 211
Laplace transform, 320
orthotropy, 208
plane strain, 252
porochemoelasticity, 740
axial symmetry, 755
spherical symmetry, 767
uniaxial strain, 745
porothermoelasticity, 634
axial symmetry, 651
complete model, 637
spherical symmetry, 671
thermally coupled, 630
thermally uncoupled, 636
uniaxial strain, 638
pure bending, 263, 266
spherical symmetry, 296, 306
transverse isotropy, 209, 235
uncoupled, 9, 21, 212
uniaxial strain, 237
variation in fluid content, 19, 205, 320, 636, 684
dilute solution assumption, 717, 729
dipole, *see* fundamental solution
Dirichlet problem, 215, 398, 412
dislocation, 413
edge, 467
slip, 33
dislocation method, 420
dispersion coefficient, 735
dispersivity, 735
displacement
fluid, 143
solid, 67, 143
displacement discontinuity, 33, 397
method, 413, 418
normal, 421
shear, 421
surface, 419
tensor, 406
displacement function
3D, 225
axisymmetric, 224, 385
full poroelasticity model, 222
McNamee-Gibson, 222, 232
plane strain, 222, 379
dissipation energy function, 129, 197
density, 197, 622
dissipation function, 129, 484, 738
density, 194, 241
Rayleigh, 477
double layer
method, 412, 413, 420
potential, 412
drained condition, 5, 12, 75, 87, 103, 120, 151, 163, 174, 181, 209
drained test, 103, 166
Duhamel principle of superposition, 244, 330, 579, 814
dynamic compatibility condition, 503
dynamic modulus, 501
complex, 517

E

edge dislocation method, 421
effective modulus, 92, 96, 125
Budiansky and O'Connell, 94
Mackenzie model, 92, 101
Walsh model, 93

effective stress, 12, 13, 32, 44, 85, 98, 173, 290, 361, 607
Biot, 15, 77, 644
capillary, 31
chemical, 749
Fillunger, 13
mixture theory, 15
pore collapse, 16
Terzaghi, 8, 13, 16, 17, 77, 86, 91, 92, 99, 100, 139, 342, 360, 366, 526, 615, 657, 668, 674, 678, 726, 751, 758, 763
thermal, 20, 44
effective stress coefficient, 13, 77, 86
adiabatic drained, 613
adiabatic undrained, 613
Biot, 15, 73, 77, 85, 90, 104, 145, 153
tensor, 174
chemical, 731, 748
pore volume, 85, 153
thermoelastic
drained, 607
undrained, 612
viscoelastic, 581, 584
eigenstrain, 405, 429, 465
Einstein summation, 67
elastic aquifer model, 154
elastic modulus tensor
drained, 174
undrained, 174
electrical potential, 710
electrochemical potential, 710
electroseismic effect, 46
energy diagram, 157
energy dissipation, 241, 294
enhanced geothermal systems, 37
entropy, 126
density, 19, 528, 601, 609, 709
fluid, 604
solid, 604
diffusivity coefficient, 631
displacement vector, 622
production, 619
equilibrium equation, 117, 202, 734
axial symmetry, 316
Cartesian coordinate, 808
cylindrical coordinate, 810
dynamic, 476, 493
plane strain, 251
pure bending, 265
solid and fluid phase, 120
dynamic, 485
spherical coordinate, 812
spherical symmetry, 295
uniaxial strain, 234, 639, 745
warping, 258
with body force, 401, 684
Euler's homogeneous function theorem, 706, 714
exact differential, 70, 99
existence and uniqueness, 213
elasticity, 214
integral equation, 411
poroelasticity, 214
potential problem, 214
extensive property, 126, 197, 706

F

failure criterion
Drucker-Prager, 367
Mohr-Coulomb, 25, 30, 361
fault
overthrust, 31
shear, 176, 231
slippage, 30
field equation, 189, 218
Fillunger, Paul, 13
first principles, 62
fluid content coefficient, 105
flux
diffusive, 735
dispersive, 735
fluid, 199, 618
heat, 616
relative solute to solvent, 738
solute, 618, 737
solvent, 737
Fourier transform, 380
cosine, 381
Fourier's law, 620
Fourier, Joseph, 599, 842
fracking, *see* hydraulic fracturing
frame of reference
Eulerian, 97, 117
Lagrangian, 97, 117
free energy
density, 131, 197, 611, 622, 709
Gibbs, 705, 707
Helmholtz, 127, 130, 157, 197, 483, 600, 622, 708, 721
density, 709
Frenkel, Yacov Il'ich, 475, 855
frequency domain, 246, 286, 494, 529, 545, 594
fundamental solution, 232, 397, 406
bi-Helmholtz, 426
biharmonic, 425

diffusion equation, 425
dislocation, 406
opening, 232
shear, 232
displacement discontinuity, 232, 401, 406, 413, 465, 835
double force, 406
edge dislocation, 467
elasticity
center of dilatation, 429
double force, 428
point force, 426
quadrupole and hexapole, 405
elastodynamics, 531
fluid dilatation, 232, 406, 450, 826
fluid dipole, 232, 406, 412, 445, 454, 468, 823
fluid dodecapole, 406, 454, 829
fluid force, 232, 406, 453
fluid source, 232, 397, 401, 406, 411, 436, 445, 450, 821, 828
heat source, 528, 691
Helmholtz equation, 426
Laplace equation, 424
modified Helmholtz equation, 425
point force, 232, 406, 456, 830
poroelastodynamics, 529
fluid source, 542, 544
point force, 536, 544
porothermoelasticity, 684
fluid dipole, 696
fluid source, 694
heat dipole, 696
heat source, 688
point force, 696
quadrupole and hexapole, 406, 461, 833
solid dilatation, 406, 463, 834

G

gas constant, 715
Gassmann equation, 91
Gassmann model, *see* micromechanics model
Gauss (divergence) theorem, 118
generalized plane strain, 254
axial strain, 261
bending, 262
pure shear, 257
torsion, 260
warping, 257
Gibbs equation, 706
Gibbs, Josiah Willard, 703, 851
Gibbs-Duhem equation, 706, 709, 727
governing equation, 62, 189
elastodynamics, 497
porochemoelasticity, 740
chemically uncoupled, 742
perfect membrane, 741
poroelasticity, 19, 203
axial strain, 261
axial symmetry, 316
degenerated, 209
orthotropy, 208
plane strain, 250
pure bending, 263
spherical symmetry, 295
strain and rotation, 253
transverse isotropy, 209
uniaxial, 234
warping, 258
with singular forcing term, 431
poroelastodynamics, 493
u-p formulation, 495
frequency domain, 494
time domain, 494
porothermoelasticity
complete model, 637
thermally coupled model, 627
thermally uncoupled model, 636
poroviscoelasticity, 580
thermoelasticity, 19
thermoelastodynamics, 527
grain-scale local flow, 513, 583
granular material, *see* porous material
Green's function, 232, 847
free space, 397, 406
Green's identity, 398, 404, 847
generalized, 399
Green, George, 397, 846
group velocity, 508

H

Hadamard finite part, 413, 420
Hankel transform, 385
harmonic relation, 207, 628
Hashin-Shtrikman bound, 16, 96, 625
heat density, 616
Helmholtz decomposition, 210, 498, 531
hereditary integral, 579
homogenization theory, 114, 192, 486, 719
Hooke's law, 61, 69, 842
generalized, 85, 137, 172, 181
Hooke, Robert, 61, 840
Hubbert's potential, 191
hydraulic conductivity, 63, 190
hydraulic fracturing, 24, 35, 37, 365
hyperfiltration, 619, 741

I
ideal porous medium, 13, 84, 89, 99, 140, 144, 147, 151, 183, 725
  porochemoelasticity, 726, 729
  porothermoelasticity, 605
in situ stress measurement, 25, 367
incompressible
  constituent model, 12, 15, 154, 205, 206, 210, 222, 226, 240, 290, 608, 789
  fluid, 8, 154
  solid, 12, 78, 152, 153, 160
induced seismicity, 32
influence function, 244, 330, 419
insular air saturation, 155
integral equation
  Fredholm, 411
    elasticity, 413
    first kind, 412
    poroelasticity, 413
    potential problem, 411
    second kind, 412
  Galerkin, 422
  hypersingular, 420
  null-field, 409
  Somigliana, 404, 413
    elasticity, 405
    poroelasticity, 405
    poroelastodynamics, 545
  strongly singular, 418
  Volterra
    first kind, 413
  weakly singular, 418
intensive property, 197, 708
intrinsic material constant, 137, 146, 163, 581, 593, 602
irreversible
  process, 128, 196, 484, 621
  thermodynamics, 129, 718, 861
  work, 129, 197, 622
irrotational displacement, 210, 219, 220, 235, 296, 316, 430, 435, 533, 634, 651, 743, 754
isolated system, 126, 618
isothermal condition, 20, 126, 131, 599, 707, 721, 727
isotropy, 67, 180, 233, 252
  macroscopic, 134
  microscopic, 134

J
jacketed test, 102, 164, 585

K
Kelvin solution, 405, 427, 457, 697
Kozeny-Carman equation, 101, 195

L
Lagrange's equation of motion, 477
Lagrangian, 477
Lamé constant, 69
  undrained, 73, 494
Lamé problem, 22, 339
Lamé, Gabriel, 69, 848
Laplace transform, 266, 297, 312, 320, 340, 380, 411, 578, 580, 591, 640, 756, 813
  equivalency to Fourier transform, 594
  inverse, 813
    Stehfest method, 270, 334, 817
  table of, 815
Laplace-Carson transform, 579
leakoff, 25, 36
length scale, 9, 10, 287, 291
liquefaction, 40
local chemical equilibrium, 727
local pressure equilibrium, 121
local thermal equilibrium, 606
local thermodynamic equilibrium, 127
loss modulus, 595
Love wave, 547

M
Mackenzie model, *see* effective modulus
Mandel-Cryer effect, 26, 230, 274, 297
mass conservation, 199
  solute, 735
mass diffusion coefficient, 738
  apparent, 738
material stability, 157
mechano-caloric
  coefficient, 620
  effect, 619, 621
membrane efficiency, 713
microhomogeneity, 84, 139, 147
microisotropy, 84, 139, 147
micromechanics model, 63, 84
  Biot-Willis, 84, 88
  Gassmann, 83, 89, 140
  Mackenzie, 97
  Rice-Cleary, 86, 146
mixture theory, 13, 114
  father of, 13

mobility coefficient, *see* permeability coefficient
Mohr-Coulomb, *see* failure criterion
molar
  concentration, 709
  fraction, 715
  volume, 714
molarity, *see* molar concentration
monoclinic, *see* anisotropy

N
Navier equation, 204, 220, 276, 846
  axial symmetry, 317
  elasticity, 210, 219
    with body force, 432, 697
  general solution, 219
  irrotational displacement, 211
  one-dimensional, 234
  orthotropy, 208
    plane strain, 251
  porochemoelasticity, 740
    axial symmetry, 754
    uniaxial strain, 745
  porothermoelasticity, 627, 638
    axial symmetry, 650
    irrotational, 634
    with body force, 684
  spherical symmetry, 295
  transverse isotropy, 209
    uniaxial strain, 235
  with body force, 426
Navier, Claude-Louis, 68, 573, 844
Navier-Cauchy equation, *see* Navier equation
Navier-Stokes equation, 192, 845
Neumann problem, 215
Newton's law of viscosity, 61
nonequilibrium thermodynamics, 127, 196, 618
nonlinearity
  geometric, 97, 135, 160
  material, 98, 160
  structural, 160
Noordbergum effect, 27
nucleus of strain, 401, 413, 432

O
Onsager reciprocal relation, 618, 737
orthotropy, *see* anisotropy
osmosis, 711
osmotic pressure, 47, 714, 716, 717, 732
outburst, 33

P
Papkovitch-Neuber function, 218, 427
parallel plate model, *see* conceptual flow model
permeability
  as function of porosity, 101
  as function of stress, 101
  coefficient, 19, 80, 196, 198, 486, 493
  dynamic, 486, 490
    conceptual model, 486
  effect, 22, 37, 103, 291, 524, 559, 587, 600
  intrinsic, 44, 101, 105, 193
    dimensionless, 193
  tensor, 191
phase velocity, 504
phenomenological approach, 62, 113
physical approach, 63, 191
physically-based approach, *see* physical approach
piezometric head, 81, 190
pinch-off test, 35
pipeline flotation, 39
plan stress, 269
plane strain, 11, 21, 207, 211, 230, 250, 261, 353, 547
Poisson ratio, 69
  adiabatic undrained, 697
  constant concentration drained, 755
  drained, 77
  undrained, 11, 77
pore
  dead end, 116
  interconnected, 116
  non-interconnected, 116
pore collapse, 16, 91, 726
pore pressure, 65
pore pressure spalling, 45
porochemoelastic problem
  borehole, 756
  cavity, 767
  cylinder, 761
  half space, 746
  sphere, 768
porochemoelasticity, 47, 703
poroelastic constant
  table of, 105
poroelastic problem
  anti-plane shear, 356
  axial strain, 261, 328
  axisymmetric, 230
  beam, 231
  borehole, 22, 231, 336, 356
    inclined, 350
  consolidation, 229

poroelastic problem (*cont.*)
contact, 231
Cryer problem, 297
cylinder, 43, 230, 319, 356
cylinder retrieval, 356
finite thickness layer, 230
fracture, 35, 231
generalized plane strain, 254
half space, 229, 230
axial symmetry, 385
plane strain, 381
hollow cylinder, 230, 331
hollow sphere, 230
Mandel problem, 274
moving load, 371
one-dimensional, 229, 233, 236
pore pressure meter, 310
propagating pressure wave, 284
pure bending, 262
plate, 264
pure shear, 257
seabed, 38, 231
sphere, 230
spherical cavity, 231, 306
subsidence, 232
torsion, 260
uniaxial strain, 18, 233, 236
uniaxial stress, 355
warping, 257
poroelastic stress, 25
coefficient, 21, 207, 628
poroelastodynamic problem
finite thickness layer, 521
half space, 520
one-dimensional, 517
poroelastodynamics, 45, 475
poromechanics, 1
porosity, 84, 115, 195
areal, 66, 116
variation, 86, 123
volumetric, 116
porosity equilibrium equation, 132, 135, 601, 722
linearized, 135, 602, 723
nonlinear, 161
porothermochemoelasticity, 703
porothermoelastic constant
table of, 633
porothermoelastic problem
borehole, 653
cavity, 670
cylinder, 661
heating of half space, 639
sphere, 680
porothermoelasticity, 44, 599
porous material, 1
artery, 42
articular cartilage, 3, 42
bone, 3, 42
clay, 47
example of, 2
fractured rock, 3
granular material, 2, 5, 14, 91, 138, 152, 515
metal foam, 3
nanoporous, 4, 47
nanoporous alumina, 3
pervious concrete, 3
polyurethane foam, 3
porous solid, 2, 5, 14
sand, 3
sandstone, 3
shale, 47
volcanic rock, 3
porous solid, *see* porous material
porous solid frame, 6, 66, 199
poroviscoelastic problem
borehole, 585
cylinder, 588
poroviscoelasticity, 43, 329, 573
poroviscoelastodynamics, 594
positive homogeneous function, 706
potential problem, 214
vector, 214
principle of least action, 477

Q

quadratic form, 71, 134, 158, 198, 484, 602, 611, 623, 722
quality factor
inverse of, 509

R

Rayleigh wave, 547
reciprocal theorem
Betti-Maxwell, 71, 85, 172, 399
elasticity, 399
elastodynamics, 399
Green's, 398
poroelasticity, 72, 173, 400, 404, 469
poroelastodynamics, 545
thermoelasticity, 543
reciprocal work theorem, 70, 610
reflection, 547
angle, 550
critical incident angle, 553

dissipative medium, 556
free surface, 549
impermeable surface, 560
interface, 562
non-dissipative medium, 549
reflection coefficient, 551, 566
reflection coefficient (osmotic), 738
chemo-hydraulic, 733
chemo-mechanical, 733
refraction, 547
angle, 565
critical incident angle, 547
refraction coefficient, 566
representative elementary volume, 64, 97, 115, 127, 192, 196, 478, 504, 616, 719
reservoir impoundment, 32
resistivity coefficient, 194, 403, 486
resonance, 523, 595
frequency, 523
Reuss bound, 95
REV, *see* representative elementary volume
reverse consolidation, 348
reverse groundwater level fluctuation, 28
reversible process, 71, 126
Rhade effect, 28
Riemann-Stieltjes integral, 579
rotation vector, 253, 808

S

secondary consolidation, 583
seepage velocity, 736
seismoelectric effect, 856
first kind (J-effect), 45
second kind (E-effect), 45, 476
seismoelectromagnetic effect, 46
semilinear model, 135, 160
semipermeable membrane, 711
non-ideal, 712
shear band, 33
shear modulus, 68, 73
bound of, 96
complex, 516
effective, 93, 95
frame, 76
orthotropy, 178
solid, 92
transverse isotropy, 179
viscoelastic, 582
single layer
method, 412
potential, 412
singularity
hypersingular, 413
strong (non-integrable), 413
weak (integrable), 412
Skempton pore pressure coefficient
A, 78, 176
B, 11, 74, 78, 91, 104, 145, 156, 185, 594
constant concentration, 731
tensor, 175
Skempton pore pressure effect, 10, 26, 78, 235, 237, 279, 281, 302, 311, 334, 459, 526, 592, 753
plane strain, 11
uniaxial strain, 11, 41, 234
slope stability, 30
submarine, 40
Snell's law, 551
soil mechanics
father of, 13
soil mechanics model, *see* incompressible constituent model
solid frame, *see* porous solid frame
specific discharge, 190, 197, 497, 738, 739
specific heat
constant strain, 19, 528
drained, 610
undrained, 610
constant stress, 609
fluid, 605
solid, 605
specific storativity, 18, 41, 80
specific surface, 195
spherical symmetry, 295, 438, 634, 668, 743, 766
spring analogy
parallel, 8, 11, 153
series, 7, 151
squirt flow, 512, 583
state function, 127, 130, 706
Stoneley wave, 547
storage coefficient, 18, 41, 80
adiabatic drained thermal, 629
constant strain, 18, 79, 174
constant stress, 18, 74, 80, 145, 175
uniaxial, 80
storage modulus, 595
strain, 64, 66
deviatoric, 74
fluid
external, 122
internal, 121
solid
external, 122
internal, 121
total, 67, 142, 181

strain energy, 70
density function, 131
elastic, 70
poroelastic, 72
strain-stress relation
anisotropic, 175
elasticity, 173
internal strain, 136
orthotropic, 251
porochemoelasticity, 723
poroelasticity, 73
porothermoelasticity, 603, 613
stress, 64
deviatoric, 74
fluid, 65
internal, 121
sign convention, 65
solid, 64
external, 122
internal, 121
solid partial, 65
total, 63, 65, 69, 71, 76, 83, 142, 181, 201, 485, 727
stress cloud, 368
stress discontinuity method, 415
stress function, 231
stress-strain relation
anisotropic, 173
axial symmetry, 316
elasticity, 69
internal strain, 134
isotropy, 252
orthotropy, 251
porochemoelasticity, 723
poroelasticity, 73
porothermoelasticity, 608
spherical symmetry, 295
subsidence, 29
swelling, 47
innercrystalline, 47, 713
osmotic, 47, 713

T
Terzaghi, Karl von, 9, 13, 229, 853
theoretical approach, 62
thermal conductivity, 19, 528
bound
effective porous medium, 626
Hashin and Shtrikman, 625
Maxwell-Eucken, 625
Wiener, 625
effective, 624
thermal expansion coefficient, 44
linear, 19, 528, 604
porosity, 604
variation in fluid content, 608
constant frame volume, 609
volumetric
drained, 607
fluid, 604
solid, 604
undrained, 608
thermal fracturing, 45
thermal stress, 20
theory of, 20, 599, 635
thermo-osmosis
coefficient, 620
effect, 619
thermodynamics
first law, 125
second law
irreversible, 128
reversible, 126
thermoelastic stress coefficient, 628
thermoelasticity analogy, 18, 24, 36, 44, 232, 527, 530
THMC process, 703
tidal efficiency, 11
tide
atmospheric, 41
earth, 40
ocean, 40
time scale, 9, 44, 121, 577, 583, 585, 606
tortuosity, 195, 481, 490, 738
total differential, 127, 131, 706
transmission coefficient (osmotic), 738
transverse isotropy, *see* anisotropy

U
underground coal gasification, 33
undrained condition, 5, 9, 75, 79, 144, 153, 174, 209, 234, 614
undrained test, 5, 104
unjacketed test, 86, 89, 99, 102, 104, 137, 140, 148, 160, 165, 183, 583
unsaturated, 155, 291, 294

V
van't Hoff's equation, 718
variation in fluid content, 19, 67, 88, 143, 182, 199, 201, 607, 613
variational formulation, 129, 198, 477
virtual work, 130, 720
viscoelastic model
dashpot, 575

general, 577
Kelvin-Voigt, 576
Maxwell, 576
spring, 575
standard linear solid, 576
Zener, 576
viscoelasticity, 43, 575
viscosity
apparent, 488
correction factor, 487
dynamic, 105, 191
Voigt bound, 95
volume averaging, 191

W
wave
compressional
first kind, *see* wave, fast
second kind, *see* wave, slow
dispersive, 46
fast, 46, 501
homogeneous, 557
inhomogeneous, 557
normal incidence, 551
shear, 46
slow, 46, 501
wave amplitude ratio, 551
wave frequency, 38, 284
wave number, 38, 284
complex, 504, 518, 548
vector, 548
wave propagation angle
equal amplitude, 556
equal phase, 556
wave velocity
complex, 518
elastic, 498
fast, 502
shear, 500
slow, 502
Wood's equation, 155, 512

Y
Young's modulus, 61, 68, 844
Young, Thomas, 61, 171, 843